MANAGEMENT INFORMATION SYSTEMS

MANAGING THE DIGITAL FIRM

18E

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Managing the Digital Firm

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Preface

Management Information Systems: Managing the Digital Firm is a leading textbook and courseware package with the most up-to-date and comprehensive overview of information systems used by businesses today. After using this learning package, students will be able to participate in, and even lead, management discussions of information systems, and understand how to use information technology in their future jobs to achieve bottom-line business results. Regardless of whether students are accounting, finance, management, operations management, marketing, or information systems majors, the knowledge and information in this book will be valuable throughout their business careers.

New to This Edition

Please note that all eText assessments, videos, and other interactive media items are only available in the Pearson+ eText, which is available both for standalone purchase as well as part of a MyLab subscription.

Management Information Systems: Managing the Digital Firm, 18th edition, has been thoroughly updated to cover the latest industry and technology changes that impact the course and to provide new interactive learning opportunities. In addition, chapter material is now organized around new, more sharply focused learning objectives. The major sections of each chapter are divided into shorter, more easily "digestible" chunks, each followed by a *Check Your Understanding* question (in the eText only), aiding student comprehension.

The Laudon texts are known for their outstanding real-world case studies, which describe how well-known businesses are using IT to solve problems and achieve objectives. Students are often asked to analyze a business problem and propose alternative solutions. The Laudon texts also provide hands-on MIS software and management decision-making problems that are based on real-world companies and business scenarios.

The text and learning package has a strong career focus, which incentivizes students to learn by showing exactly how each chapter will help them prepare for future jobs. In addition to the Careers section in each chapter, MyLab MIS features Career Resources, including how to incorporate MIS knowledge into resumes, cover letters, and job interviews.

New and Updated Topics

The 18th edition features three engaging Case Studies per chapter focusing on well-known companies and MyLab MIS includes an additional Video Case Study. There are also numerous new in-text examples of companies using IT applications. The text, figures, tables, and Case Studies are based on the latest-available sources from industry and MIS research.

There is new and/or updated content throughout the text on the following topics:

- **Artificial intelligence:** AI is the latest information technology that is in the process of transforming business. Public attention is riveted on AI because of its enormous potential for improving how companies conduct business, as well as many aspects of everyday life. The text has been extensively updated to cover developments in AI as follows.
  - AI is introduced in Chapter 1, as one of the major technology developments impacting MIS.
  - AI is covered throughout the text as it relates to topics in each chapter, including numerous case studies that feature how companies are using AI.
  - Chapter 11 has been rewritten to focus solely on AI, including new and expanded coverage of machine learning, neural networks and deep learning, generative AI, intelligent agents, natural language processing, computer vision systems, robotics, genetic algorithms, and fuzzy logic systems.

- **Cloud computing:** More coverage of cloud computing has been added throughout because of its importance in the MIS world.
  - Chapter 1 introduces cloud computing as one of the major technology developments students should be familiar with.
  - Chapter 5 has updated and expanded coverage with more detail on types of cloud services, private and public clouds, hybrid clouds, and managing cloud services.
  - Chapter 6 covers cloud databases.
  - Chapter 8 reviews issues related to cloud security.
  - Chapter 9 covers cloud-based CRM and ERP systems.
  - Chapter 10 discusses the use of the cloud in e-commerce.
  - Chapter 11 includes a discussion of cloud-based AI.
  - Chapter 13 discusses the use of cloud software services in creating information systems.
  - Case studies on cloud computing are included in Chapters 5, 6, 7, 9, 14, and 15.
- **The Internet of Things (IoT), big data, data analytics, and business intelligence:**
  - Chapter 1 introduces the importance of IoT, big data, and data analytics, and their relationship to cloud computing and AI.
  - Emphasis on data analytics has been expanded with the addition of Tableau projects in Chapters 2, 3, 6, and 12.
  - Case studies on data analytics and big data can be found in Chapters 1, 6, 7, 9, 10, and 12.
  - Case Studies on IoT are included in Chapters 1, 7, and 12.
- **Sustainability:** We have added new coverage of how information systems promote sustainability goals. Case studies on sustainability can be found in Chapters 2, 3, 5, 6, and 9.

The following are additional highlights of further new and/or updated coverage within the chapters:

Chapter 1
- How information systems are transforming business and careers
- Key challenges in MIS (new table)
- Environmental impact, sustainability, governance and Responsible AI as strategic business objectives

Chapter 2
- AI and collaboration; human-AI collaboration
- Knowledge management systems (updated/moved from Chapter 11)
- Knowledge work and knowledge worker (updated/moved from Chapter 11)

Chapter 3
- AI and competitive strategy
- AI and the value chain
- AI and Porter's Competitive Forces Model

Chapter 4
- Ethical risks of relying on AI
- Governance codes; Responsible AI
- Technological protections for privacy (new table)
- GDPR and AI
- AI challenges to privacy and intellectual property rights
- Liability issues: Software; AI; EU AI Act Big Tech and Digital Markets Act (EU)
- Quality of life: Bias in AI; impact of AI on jobs; social media mental health risks; metaverse mental health risks

Chapter 5
- Quantum computing
- Green computing
- Virtualization and cloud computing
- Cloud AI services
- IT infrastructure for AI
- Edge AI

Chapter 6
- Blockchain
- Big data and AI
- Data lakes
- Cloud-based data analytics platforms: Microsoft Power BI, Google Looker
- Sentiment analysis using AI
- AI and data management

Chapter 7
- AI functionality for UC systems
- Impact of AI on search
- Impact of generative AI on future of web
- Web3
- Metaverse
- 5G and private 5G networks (case study)
- 6G

Chapter 8

- Threat actors
- 5G security issues
- Metaverse security issues
- New types of attacks: LOTL and APT; credential stuffing; software supply chain
- AI security issues
- Security tools; IAM software; zero trust framework; multifactor and two-factor authentication; web application and next generation firewalls; intrusion prevention systems; SOCs
- Security protocols: WPA3; TLS; HTTPS; HSTS
- Impact of quantum computing on security
- IoT, cloud computing, and mobile platform security challenges

Chapter 9

- AI and ERP systems
- AI and supply chain management
- AI and CRM systems
- Composable ERP
- AI and enterprise applications

Chapter 10

- Impact of AI on e-commerce
- New e-commerce revenue models
- Rise of social e-commerce

Chapter 11

- AI and human intelligence
- Evolution of AI and major branches of AI
- Machine learning
- Supervised and unsupervised learning
- Neural networks and deep learning
- Generative AI
- Intelligent agents, chatbots, virtual assistants, and bots
- Natural language processing
- Computer vision systems
- Robotics: robotic applications, industrial robots, collaborative robots, service robots, humanoid robots, medical robots, drones and autonomous cars
- Genetic algorithms
- Fuzzy logic systems
- AI challenges: hallucinations, lack of transparency, bias, security, job loss, environmental impact, misinformation and disinformation, privacy, intellectual property
- Responsible AI

Chapter 12

- Data analytics, business intelligence, and AI
- Risks introduced by use of AI in decision making
- Tableau
- Predictive and prescriptive analytics AR, VR, and the metaverse as data visualization tools

Chapter 13

- Open-source development: examples, advantages, disadvantages, and security issues
- Democratization of software development: low code, no code, and citizen development
- Impact of AI on software development
- Hybrid mobile apps; adaptive web design

Chapter 14

- Software to help make technology decisions: Technology business management (TBM) framework and IT financial (ITFM) tools
- Project management methodologies: Waterfall method, agile project management; Kanban boards
- Technical debt
- AI and project management software tools

Chapter 15

- Use of AI to deal with global supply chain issues
- Use of AI for software localization

Content Features

The Core text provides an overview of fundamental MIS concepts using an integrated framework for describing and analyzing information systems. This framework shows information systems composed of management, organizations, and technology elements and is reinforced in student projects and case studies. The Core text consists of 15 chapters with hands-on projects covering the most essential topics in MIS.

Chapter Organization

Each chapter contains the following elements:

- List of Student Learning Objectives
- List of the Case Studies for each chapter
- Opening Case Study that describes a real-world organization to establish the theme and importance of the chapter
- Diagram analyzing the Opening Case Study in terms of the management, organization, and technology model used throughout the text

- "Spotlight on" Case Study focusing on a management, organization, or technology topic, with Case Study Questions
- Careers section showing students how to the chapter will help them in career preparation
- Chapter Case Study at the end of each chapter that enables students to apply chapter concepts
- Review Summary keyed to the Student Learning Objectives
- List of Key Terms that students can use to review concepts
- Review questions for students to test their comprehension of chapter material
- Discussion questions raised by the broader themes of the chapter
- Using AI Project that asks students to use a free generative AI tool to solve common business problems
- Series of Hands-On MIS Projects consisting of a Management Decision Problem, a hands-on application software project, and a project to develop Internet skills
- Collaboration and Teamwork Project to develop teamwork and presentation skills with options for using open-source collaboration tools

Student Learning-Focused

Student Learning Objectives are organized to focus student attention. Each major chapter section is based on a Learning Objective and each chapter concludes with a Review Summary and Review Questions organized around these Learning Objectives.

Key Features

We have enhanced the text to make it more interactive, leading edge, and appealing to both students and instructors.

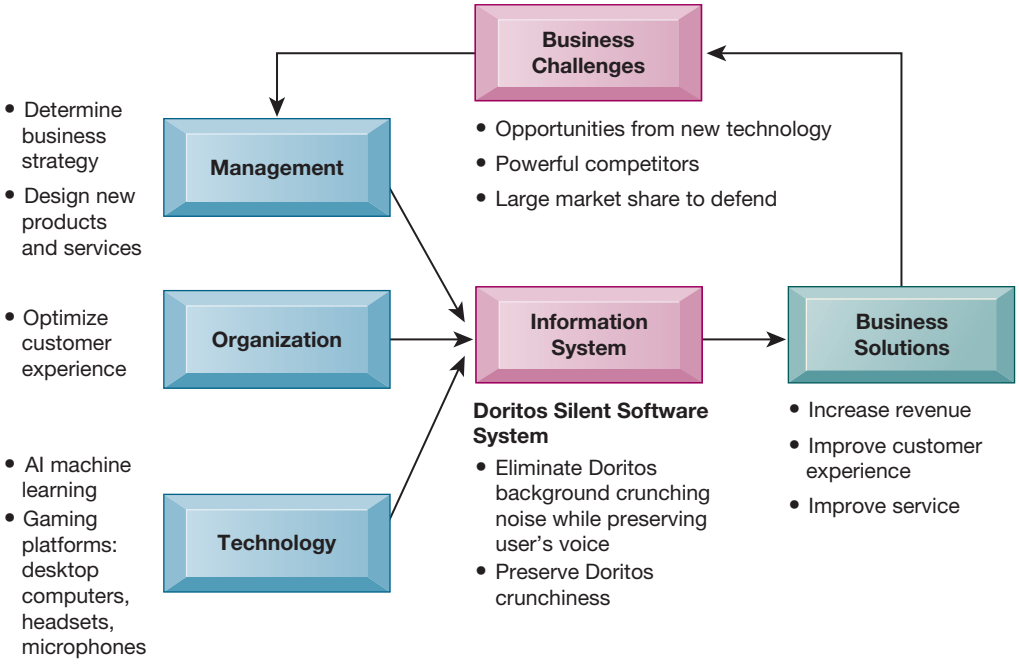
Business-Driven with Real-World Business Cases and Examples

Using real-world companies throughout helps students see the direct connection between information systems and business performance. The in-text examples and case studies describe the main business objectives driving the use of information systems and technologies in corporations all over the world: operational excellence, new products and services, customer and supplier intimacy, improved decision making, competitive advantage, survival, and promoting ESG leadership. In-text examples and case studies show students how specific companies use information systems to achieve these objectives.

Each chapter contains three major case studies focusing on real-world organizations that will be familiar to students, such as UPS, Capital One, PepsiCo, Spotify, Meta, the National Football League, Chewy, Lemonade, Reddit, Apple, Land O'Lakes, Signet Jewelers, Southwest Airlines, Hormel Foods, and Wyndham, among others.

The *Opening Case Study* sets the stage for the chapter. It describes a real-world organization facing a business challenge related to the themes and topics of the chapter, followed by a diagram that analyzes the case in terms of the management, organization, and technology model used throughout the text. It also includes a Career Benefit box which highlights how employees of the relevant company might benefit as a result.

The *Spotlight on Case Study* within each chapter highlights a management, organization, or technology topic related to the content of the chapter. The *Chapter Case Study* at the end of chapter is a more extensive case that pulls together chapter concepts and issues. Each case concludes with case study questions that encourage students to apply chapter concepts to real-world companies in class discussions, student presentations, or writing assignments.



A diagram accompanying each chapter-opening case graphically illustrates how management, organization, and technology elements work together to create an information system solution to the business challenges discussed in the case.

Spotlight on Technology Case Study

UPS Competes Globally with Information Technology

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End of Chapter Hands-On Text Activities

Real-world business scenarios and data help students to learn firsthand what MIS is all about. These projects heighten student involvement in this exciting subject.

- **NEW Using AI Projects:** An AI project in every chapter gives students experience using free generative AI tools such as ChatGPT, Google Gemini, and Microsoft Copilot to solve common business problems. These are also available as assignments in MyLab MIS.
- **Hands-On MIS Projects:** Every chapter includes a Hands-On MIS Projects section containing three types of projects: a Management Decision Problem; a hands-on application software exercise using Microsoft Excel, Access, or web page and blog creation tools; and a project that develops Internet business skills for accessing information, conducting research, and performing online calculations and analysis. Students practice solving real-world problems to learn how to achieve operational excellence and enhance decision making. The Dirt Bikes USA running case provides additional hands-on projects for each chapter.

- **NEW Tableau Projects:** Several chapters also include a Tableau project.
- **Collaboration and Teamwork Projects:** Each chapter features a collaborative project that encourages students working in teams to use Google Drive, Google Sites, Google Docs, or other open-source collaboration tools.

MyLab MIS

To improve student results, we recommend pairing this textbook with **MyLab MIS**, the teaching and learning platform that combines respected content with personalized engagement to help student see real results, while developing skills that future employers are seeking in their candidates. From **Video Case Studies** and **real-world decision simulations** to **Using AI Projects** and **Dynamic Study Modules**, MyLab MIS helps you channel your teaching style and develop confident learners. It's the power of MyLab.

What's Included

Pearson+ eText:

The course eText is accessed directly in your MyLab course. You and your students can use the eText to take notes, make highlights, and review flashcards—and even listen to your book at home or on the go with the Pearson+ mobile app.

- **NEW eText AI Study Tools** give students unlimited opportunity to achieve proficiency and master key course concepts—anytime, anywhere. Students engage with the study tools for **personalized support**, **simplified explanations,** and **productive guided practice** with an expert chatbot.
- **MIS Headlines** start each chapter with a monthly news article to keep you and your students up to date with the technology industry. Each article comes with two discussion questions you can use in class or assign through your LMS.
- **Check Your Understanding Questions** at the end of each learning objective enable students to check their understanding of the material in the section and provides feedback on answer choices.
- Figure animations walk students through important concepts in each chapter (40 in total) using a contemporary animation platform.

NEW Using AI Projects: Provide ready-to-assign activities to teach students how to use generative AI effectively. The projects demonstrate the power of free AI tools and teach the essential skills needed to use AI in school and work. The projects come with grading guidance and rubrics.

Pre-built Courses and Assignments: Ease your workload, save time, and increase efficiency with MyLab's pre-built courses and assignments. Each course has a foundation of interactive course-specific content—by authors who are experts in their field—to tailor and adapt as you see fit.

NEW Interactive Reading Assignments: Allow you to assign eText readings and gradable questions in a seamless experience with other video and low stakes-assessments. Increase student engagement with the eText and fundamental concepts of MIS.

NEW MyLab AI-powered Study Tool: Our AI-guided practice draws on Pearson content to turn questions during reading and homework mistakes into teachable moments. Improves independent learning.

Updated Scenario-Based Decision Simulations: Foster critical decision-making skills and engage students with the content. Students play various roles and make business decisions. The simulations adapt to student's choices and take them on a path based on their decisions with feedback along the way and grading details at the end.

NEW Forage Job Simulations: These immersive simulations help students gain valuable job experience with some of the world's top employers. Each simulation showcases a day in the life of a particular role. Students complete a series of task-based scenarios designed to simulate the actual work the company does. Instructors choose which job simulations to assign from the library of offerings.

NEW Video Case Study Assignments: A brand new collection of Video Case Studies (one per chapter) available in MyLab MIS focus on popular companies to cover key concepts and experiences in the MIS world, illustrating how real-world businesses and managers are using information technology and systems. Each Video Case Study is accompanied by three auto-graded true/false or multiple-choice questions and two or three manually graded essay questions.

NEW Prompt Writing Module to teach students the essential skills they need to use AI efficiently.

Chapter Warm Ups, Chapter Quizzes: These objective-based quizzes evaluate comprehension and provide a variety of assignments to prepare and assess students.

Dynamic Study Modules: Help your students stay on track and achieve a higher level of subject-matter mastery. Each module poses a series of questions about a course topic that adapts to each student's performance and offers personalized, targeted feedback to help reinforce their mastery.

Classroom Response Tool (Learning Catalytics): Help to increase student engagement and enhance their learning in the classroom. This interactive student response tool allows you to deploy questions and surveys, and assess student comprehension. It also equips you with real-time data to help adjust your instructional strategy on the fly.

MyLab MIS + Exploring 365

If you also teach Excel, Access, Word, or PPT in your MIS course—we offer a full solution in one combined MyLab MIS + Exploring 365. With this combined course, you and your students have full access to the eTexts, videos, auto-graded application projects, and training simulations from the Exploring 365 series. You select what you want to assign based on your course requirements.

Developing Career Skills

For students to succeed in a rapidly changing job market, they should be aware of their career options and how to go about developing a variety of skills. With MyLab MIS and *Management Information Systems 18e*, we focus on these skills in the following ways.

Career Resources

Every student who reads this text wants to know: How will this book help my career? The Careers section at the end of each chapter shows students how they can use the information in the chapter and in MyLab MIS as tools for career-building. There students will find a real-world scenario featuring a job related to the topics covered in that chapter. A table further describes the roles and responsibilities of the job, the IT knowledge and skills required, as well as a potential career path, including salary information. We also include a Career Benefit box after each Opening Case Study to show the career benefit obtained by the people who worked on the solution to the business problem described in the case.

The following are the careers featured in the Careers section of each chapter.

Additional Career Resources can be found in MyLab MIS including job-hunting guides and instructions on how to build a Digital Portfolio. The portfolio can be included in a resume or job application or used as a learning assessment tool for instructors.

Chapter	Job
1. Information Systems in Global Business Today	Sales Representative
2. Global E-business and Collaboration	Customer Success Analyst
3. Information Systems, Organizations, and Strategy	Business Development Representative
4. Ethical and Social Issues in Information Systems	Privacy Analyst
5. IT Infrastructure and Emerging Technologies	AWS Trainer
6. Foundations of Business Intelligence: Databases and Information Management	Data Quality Analyst
7. Telecommunications, the Internet, and Wireless Technology	Network Specialist
8. Securing Information Systems	Cybersecurity/IT Security Specialist
9. Achieving Operational Excellence and Customer Intimacy: Enterprise Applications	Customer Relationship Manager
10. E-commerce: Digital Markets, Digital Goods	E-Commerce Analyst
11. Managing Knowledge and Artificial Intelligence	AI Prompt Engineer
12. Enhancing Decision Making with Data Analytics and Business Intelligence	Data Analyst
13. Building Information Systems	Process Analyst
14. Making the Business Case for Information Systems and Managing Projects	IT Project Manager
15. Managing Global Systems	Global Supply Chain Analyst

Case Studies

The following companies and topics are covered in the book's Case Studies:

Case Study	Companies	Topics
1. Information Systems in Global Business Today		
Opening Case Study: ConocoPhillips Drills Down with Digital Technology	ConocoPhillips	Digital technologies; 3D printing; IoT; big data; data analytics; data visualization; automation; AI; machine learning
Spotlight on Technology Case Study: UPS Competes Globally with Information Technology	UPS; FedEx	Information technologies; information systems; AI
Chapter Case Study: Capital One: A Digital Transformation in the Making	Capital One; Amazon Web Services	Digital transformation; big data; data analytics; cloud computing; agile software development; sustainability; mobile apps; AI; machine learning; chatbots
2. Global E-business and Collaboration		
Opening Case Study: Microsoft Teams and AI Helps Toyota Motor North America Do Even Better	Toyota Motor North America; Microsoft	Teamwork and collaboration tools; Microsoft Teams; generative AI; chatbots
Spotlight on Technology Case Study: AI Accelerates Drug Discovery and Development	Johnson & Johnson; Merck	Human-AI collaboration
Chapter Case Study: How Much Do Virtual Meetings Help Collaboration	Zoom; Microsoft	Impact of virtual collaboration technologies; Microsoft Teams; Zoom
3. Information Systems, Organizations, and Strategy		
Opening Case Study: PepsiCo Tames the Doritos Crunch	PepsiCo	AI; machine learning; business strategy
Spotlight on Management Case Study: Spotify's Podcasting Strategy Stumbles	Spotify	Streaming music services; podcasting; business strategy
Chapter Case Study: AI Redesigns the Fashion Industry	H&M; Zara: Lenzing Group; Thread.com; StitchFix; Shopify; Nike	AI; machine learning; sustainability; personalization; generative AI

Case Study	Companies	Topics
4. Ethical and Social Issues in Information Systems		
Opening Case Study: Apps That Track: A Double-Edged Sword	Apple; Google; AccuWeather; Kochava; Outlogic; Verizon; T-Mobile	Mobile devices; privacy; ethics
Spotlight on Organizations Case Study: Section 230 Under Fire	Big Tech companies	Social media; liability; ethics
Chapter Case Study: Meta's Many Ethical Challenges	Meta	Ethics; privacy; social media; AI
5. IT Infrastructure and Emerging Technologies		
Opening Case Study: QRyde Rides High with the Cloud and AI	QRyde; Oracle	IT infrastructure; legacy systems; cloud computing; AI; mobile apps
Spotlight on Technology Case Study: AI Chips	Nvidia; OpenAI; Google; Microsoft; Amazon	IT infrastructure; AI
Chapter Case Study: Are AI and Cloud Computing Sustainable?	Microsoft; Amazon Web Services; Google; Meta	IT infrastructure; AI; cloud computing; data centers; sustainability
6. Foundations of Business Intelligence: Databases and Information Management		
Opening Case Study: Australia's Royal Flying Doctor Service Files with Oracle Autonomous Database	Royal Flying Doctor Service; Oracle	Cloud computing; databases; data management; mobile devices
Spotlight on Technology Case Study: Facial Databases: Do You Know Who's Using Your Face?	Clearview AI	AI; databases; privacy; ethics
Chapter Case Study: Pursuing Sustainability with Blockchain	Unilever; Green Token; Newlight Technologies; IBM	Blockchain; sustainability; software as a service; ethics
7. Telecommunications, the Internet, and Wireless Technology		
Opening Case Study: From the Football Field to the Cloud and Back Again: The NFL Partners with AWS	NFL; Amazon Web Services	IoT; cloud computing; data analytics; AI; machine learning; 3D visualization
Spotlight on Technology Case Study: Cleveland Clinic Turns to Private 5G Networks for Its Newest Hospital	Cleveland Clinic; Verizon; Ericsson	Wireless networks; 5G
Chapter Case Study: AI and the Future of Search	Google; OpenAI; Microsoft; Perplexity AI	Search technology; AI; GenAI; chatbots; ethics
8. Securing Information Systems		
Opening Case Study: Ransomware Attack Paralyzes U.S. Healthcare Payment Systems	Change Healthcare; UnitedHealth Group	Ransomware; threat actors; information systems security infrastructure; privacy; ethics
Spotlight on Organizations Case Study: Uptime Institute: Understanding and Preventing System Outages	Uptime Institute	Digital resiliency; data centers
Chapter Case Study: Social Engineering Creates Chaos for MGM Resorts	MGM Resorts	Social engineering; threat actors; information systems security
9. Achieving Operational Excellence and Customer Intimacy: Enterprise Applications		
Opening Case Study: TrueBlue Speeds Ahead with Enterprise Applications	TrueBlue; Oracle	Enterprise applications; cloud computing; legacy systems
Spotlight on Organizations Case Study: BYK Tames Its Supply Chains	BYK; SAP	Enterprise applications; supply chain management systems; cloud computing
Chapter Case Study: MillerKnoll Uses Salesforce and AI to Transform Its Business Strategy	MillerKnoll; Salesforce	Enterprise applications; CRM systems: cloud computing: AI; data analytics; sustainability

Case Study	Companies	Topics
10. E-commerce: Digital Markets, Digital Goods		
Opening Case Study: Chewy Blends B2C and B2B E-commerce	Chewy	E-commerce business models; business strategy
Spotlight on Organizations Case Study: Social Listening with Sprout Social	Sprout Social; Goally; TikTok	Social networks; social listening
Chapter Case Study: Lemonade Seeks to Disrupt the Insurance Industry with AI	Lemonade	Insurtech; AI; chatbots; big data; data analytics; mobile apps; digital infrastructure
11. Artificial Intelligence		
Opening Case Study: Computer Vision AI Helps Reddit Become More Accessible	Reddit; Microsoft	AI: computer vision systems;
Spotlight on Management Case Study: Apple's AI Strategy: AI in Every Pocket	Apple	AI; mobile devices; business strategy
Chapter Case Study: Is AI Coming for Your Job?		AI; ethics
12. Enhancing Decision Making with Data Analytics and Business Intelligence		
Opening Case Study: Fulton County Schools Power Up Data-Driven Decision Making with Power BI	Fulton County Schools; Microsoft	Data analytics; data visualization tools; decision making
Spotlight on Technology Case Study: Land O'Lakes Seeks to Reshape the Future of Farming with Predictive Analytics and AI	Land O'Lakes; Microsoft	Data analytics; IoT; big data; predictive analytics; AI
Chapter Case Study: Big Data Analytics and AI: A New Way of Combating Natural Disasters	NASA; California Wildlife Threat and Intelligence Integration Center; NOAA; USGS	Data analytics; IoT; big data; AI
13. Building Information Systems		
Opening Case Study: Signet Jewelers Sparkles with New Virtual Sales Systems	Signet Jewelers	Systems development; business strategy; AI; virtual collaboration tools; e-commerce
Spotlight on Management Case Study: Meet the Citizen Developers	Chevron; ING bank; AT&T; Shell	Citizen software development; AI; machine learning
Chapter Case Study: Process Mining with Celonis	Celonis; Deutsche Telekom Services Europe; Avnet; Kraft Heinz	Business process redesign; business process mining; execution management systems; AI; machine learning; digital twins
14. Making the Business Case for Information Systems and Managing Projects		
Opening Case Study: BigCommerce Turns to Wrike for Better Project Management Tools	BigCommerce; Wrike	Project management tools; cloud computing; software as a service
Spotlight on Management Case Study: Happy Feet International Overcomes Change Management and Implementation Issues	Happy Feet International; Kuebix; Oracle NetSuite	Legacy systems; implementation; change management; enterprise applications
Chapter Case Study: Southwest Airlines Invests in Digital Transformation to Navigate Future Turbulence	Southwest Airlines	Digital transformation; legacy systems; technical debt; project management; cloud computing
15. Managing Global Systems		
Opening Case Study: Hormel Foods Implements Oracle Cloud Applications to Unify Its Systems Worldwide	Hormel Foods; Oracle	Global information systems; enterprise applications; legacy systems
Spotlight on Organizations Case Study: The Rise of Digital Nationalism		Internet; data localization policies; social and political issues
Chapter Case Study: Wyndham Centralizes Its Information Systems to Better Serve Its Worldwide Hotels	Wyndham Hotel & Resorts; Salesforce; Oracle; Okta	Global information systems; enterprise applications; legacy systems; cloud computing; AI

New Video Case Studies

In addition to the case studies that appear within the text, there is one Video Case Study Assignment per chapter in MyLab MIS. The Video Case Study Assignments are all new to this edition and based on Pearson's collection of business and technology video clips, as well as videos available on official company YouTube channels. These Video Case Study Assignments require students to view the video, read the case study, and answer multiple choice and essay questions to confirm their comprehension.

Chapter	Video Case Study
1. Information Systems in Global Business Today	Twilio Drives Customer Engagement with AI
2. Global E-business and Collaboration	What's Up with WhatsApp
3. Information Systems, Organizations, and Strategy	Delivery Drones: Walmart's New Competitive Weapon
4. Ethical and Social Issues in Information Systems	Trustworthy AI
5. IT Infrastructure and Emerging Technologies	D-Wave Makes Waves with Quantum Computing
6. Foundations of Business Intelligence: Databases and Information Management	Alation Helps Companies Develop AI That Can Be Trusted
7. Telecommunications, the Internet, and Wireless Technology	Low Earth Orbit (LEO) Satellite Internet Systems: The Competition Heats Up
8. Securing Information Systems	Akamai Helps Companies Stay Secure
9. Achieving Operational Excellence and Customer Intimacy: Enterprise Applications	Forethought: AI-Driven Customer Support
10. E-commerce: Digital Markets, Digital Goods	Shopify AI
11. Artificial Intelligence	Boston Dynamics' Spot Robot: "Athletic Intelligence"
12. Enhancing Decision Making with Data Analytics and Business Intelligence	Using Snowflake for Data Analysis
13. Building Information Systems	Will GitHub Copilot Make Every Company a Software Company?
14. Making the Business Case for Information Systems and Managing Projects	Smartsheet Helps ADP to Manage Projects
15. Managing Global Systems	Reslinc Helps Companies Develop Global Supply Chain Resilience

Integrating Business with Technology

By completing the projects in this text, students will be able to demonstrate generative AI skills, business knowledge, application software proficiency, and Internet skills, as well as analytical, writing, presentation, and teamwork and collaboration skills. These projects can be used by instructors as learning assessment tools and by students as demonstrations of their business, software, and problem-solving skills to future employers. Here are some of the skills and competencies students using this text will be able to demonstrate:

- **Generative AI skills:** The ability to use generative AI tools to answer common business questions and solve common business problems.
- **Business Application skills:** The ability to use business and software skills in real-world business applications and demonstrate business knowledge and proficiency in spreadsheet, database, Tableau, and web page/blog creation tools.

- **Internet skills:** The ability to use online tools to access information, conduct research, and perform online calculations and analysis.
- **Analytical, writing, presentation, and teamwork and collaboration skills:** The ability to research a specific topic, analyze a problem, think creatively, suggest a solution, and prepare a clear written or oral presentation of the solution, working either individually or with others in a group.

Generative AI Skills

Select a business name	Chapter 1
Create brand images	Chapter 2
Analzye geographic and demographic factors	Chapter 3
Consider social, ethical, and legal issues in establishing an online presence	Chapter 4
Select an information system	Chapter 5
Perform sentiment analysis	Chapter 6
Identify and compare Internet Service Providers	Chapter 7
Devise an information systems security plan	Chapter 8
Identify customer touchpoints and improve customer experience	Chapter 9
Create an e-commerce presence	Chapter 10
Write a press release	Chapter 11
Identify business decisions	Chapter 12
Identify business processes and process efficiencies	Chapter 13
Identify change management problems	Chapter 14
Understand global supply chain issues	Chapter 15

Business Application Skills

Business Skills	Software Skills	Chapter
Finance and Accounting		
Payroll accounting	Worksheet formatting Spreadsheet formulas SUM, AVERAGE functions Absolute and relative addressing	Chapter 1
Financial statement analysis	Spreadsheet formulas Spreadsheet charts	Chapter 2*
Budgeting	Spreadsheet formulas Spreadsheet charts Spreadsheet assumptions	Chapter 2
Total Cost of Ownership (TCO) analysis	Spreadsheet formulas SUM	Chapter 5
Technology Rent vs. Buy Decision	Spreadsheet formulas	Chapter 5*
Analyze wireless services and costs	Spreadsheet formulas	Chapter 7
Financial statement analysis	Spreadsheet formulas Spreadsheet downloading and formatting	Chapter 10
Capital budgeting	Spreadsheet formulas and functions	Chapter 14*
Data analytics and visualization	Tableau	Chapter 2 Chapter 3 Chapter 6 Chapter 12

Human Resources		
Analyzing security events	Spreadsheet sorting, data filtering	Chapter 8
Employee training and skills tracking	Database design Database querying and reporting	Chapter 13*
Internal job postings	Database and web page design	Chapter 15
Manufacturing and Production		
Inventory management	Database design, querying, and reporting	Chapter 6
Analyzing supplier performance and pricing	Spreadsheet date functions Data filtering, AVERAGE function	Chapter 9
Bill of materials cost sensitivity analysis	Spreadsheet data tables Spreadsheet formulas	Chapter 12*
Sales and Marketing		
Sales trend analysis	Data analytics Database querying and reporting Database design	Chapter 3
Blog creation and design	Blog creation tools	Chapter 4
Customer profiling	Database design Database querying and reporting	Chapter 6*
Customer reservation system	Database querying and reporting	Chapter 13

* Indicates skill in Dirt Bikes Running Case in MyLab MIS

Internet Skills

Internet Skills	
Research jobs and career development	Chapter 1
Plan efficient transportation routes	Chapter 2
Research and evaluate websites	Chapter 3
Analyze web browser privacy protection	Chapter 4
Research travel costs	Chapter 5
Search online databases for products and services	Chapter 6
Conduct business research using online search tools and generative AI	Chapter 7*
Research and evaluate security outsourcing services	Chapter 8
Research and evaluate supply chain management services	Chapter 9*
Research and evaluate e-commerce hosting services	Chapter 10
Improve web searches with generative AI	Chapter 11
Compare product price, features, and availability using shopping bots	Chapter 11
Using an online DSS for financial planning	Chapter 12
Analyze website design	Chapter 13
Research buying and financing a home	Chapter 14
Research international marketing and product pricing	Chapter 15*

* Indicates skill in Dirt Bikes Running Case in MyLab MIS

Analytical, Writing, Presentation, and Teamwork and Collaboration Skills*

Business Problem	Chapter
Select team collaboration tools	Chapter 1
Management analysis and information systems recommendations	Chapter 1*
Identify management decisions and systems	Chapter 2
Conduct a strategic business analysis	Chapter 3*
Formulate a corporate code of ethics	Chapter 4
Develop a privacy policy	Chapter 4*
Research and evaluate server and mobile operating systems	Chapter 5
Identify entities and attributes in an online database	Chapter 6
Research and evaluate smartphones	Chapter 7
Research and evaluate security software tools	Chapter 8
Disaster recovery planning	Chapter 8*
Research and evaluate enterprise application vendors	Chapter 9
Perform a competitive analysis	Chapter 10
Develop an e-commerce strategy	Chapter 10*
Investigate the use of AI	Chapter 11
Investigate the use of data-driven analytics	Chapter 12
Prepare website design specifications	Chapter 13
Identify implementation problems	Chapter 14
Identify technologies for global business strategies	Chapter 15

* Indicates skill in Dirt Bikes Running Case in MyLab MIS

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Instructor's Manual	Includes: • Teaching suggestions • Sample answers to Case Study, Review, Discussion, and Video Case Questions • Solutions/solution descriptions for projects
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Kenneth C. Laudon (B.A., Stanford University; Ph.D., Columbia University) was a Professor of Information Systems at New York University's Stern School of Business, where he taught courses on Managing the Digital Firm, Information Technology and Corporate Strategy, Professional Responsibility (Ethics), and Electronic Commerce and Digital Markets for more than 30 years. He also authored numerous books dealing with e-commerce, information technology, information systems, organizations, and society, as well as more than forty articles concerned with the social, organizational, and management impacts of information systems, privacy, ethics, and digital technology. Ken's work has had a lasting influence on the MIS field and although he has passed away, his ideas continue to infuse the content of this text.

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Part One

Organizations, Management, and the Networked Enterprise

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Chapter 1: Information Systems in Global Business Today

Chapter 2: Global E-business and Collaboration

Chapter 3: Information Systems, Organizations, and Strategy

Chapter 4: Ethical and Social Issues in Information Systems

PART ONE introduces the major themes of this book, raising a series of important questions: What is an information system, and what are its management, organization, and technology dimensions? Why are information systems so essential in businesses today? Why are systems for collaboration and social business so important? How can information systems help businesses become more competitive? What broader ethical and social issues do widespread use of information systems raise?

1

Information Systems in Global Business Today

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Learning Objectives

After completing this chapter, you will be able to:

1.1 Understand how information systems transform business and careers.

1.2 Describe key challenges and opportunities created by new MIS technologies.

1.3 Describe key features of a digital firm.

1.4 Describe the business objectives of information systems.

1.5 Explain what an information system is and how it works.

1.6 Describe organizational, management, and technology dimensions of information systems.

1.7 Understand the importance of IoT, big data, cloud computing, and AI.

1.8 Understand why complementary assets are essential.

1.9 Describe different approaches used to study information systems.

1.10 Understand how this book prepares you for the future.

1.11 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** ConocoPhillips Drills Down with Digital Technology
- **Spotlight on Technology Case Study:** UPS Competes Globally with Information Technology
- **Chapter Case Study:** Capital One: A Digital Transformation in the Making

Opening Case Study

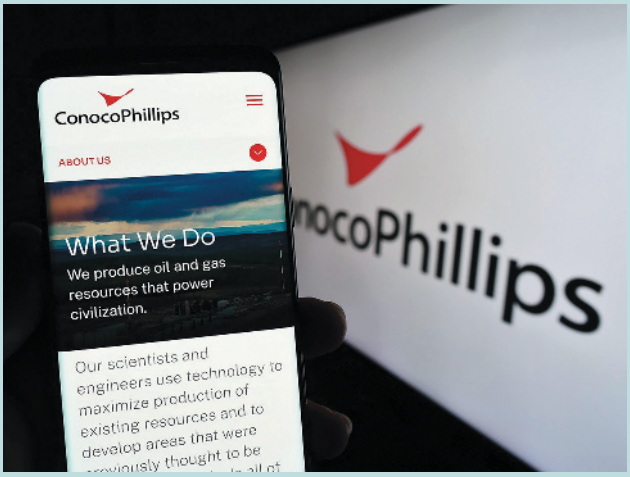
ConocoPhillips Drills Down with Digital Technology

ConocoPhillips is one of the world's largest independent oil and gas exploration and production companies. Its Kuparuk River oil field, located on Alaska's remote North Slope, has been in operation since 1981 and has produced more than 2.5 billion barrels of oil thus far. The ability to continue to recover oil from existing oil fields such as Kuparuk using existing infrastructure in a cost-effective manner is an important factor in minimizing the oil industry's environmental footprint, while continuing to support energy needs as alternative energy sources are being developed. ConocoPhillips is pursuing a variety of innovations in digital technology to do so.

For instance, in addition to issues arising from extremely cold weather, working on the Kuparuk oil field creates high costs and major supply chain challenges because it is so far from most sources of essential parts, such as for gas turbines. Gas turbines compress the natural gas that comes out of oil wells. A key component is a burner plug that enables fuel to be mixed with compressed air. The burner plugs in the gas turbines used in Kuparuk aren't commercially available anymore. When the plugs wear out, they must be made by local machine shops employing traditional manufacturing processes, which can take at least 30 weeks (and even longer for other critical parts, such as choke valves). Keeping extra parts in inventory "just in case" drives up warehousing costs.

ConocoPhillips was able to solve this problem using 3D printing, sometimes also referred to as additive manufacturing. 3D printing creates a physical object from a digital design, laying down thin layers of liquid or powdered plastic, metal, or cement, and then fusing the layers together. It enables companies to simulate many designs for various parts and identify the best configuration before manufacturing a part. These configurations are converted into a print file, which can be shared with additive manufacturing partners and then "printed" into three-dimensional objects.

With 3D printing, ConocoPhillips can replace burner plugs in two to three weeks instead of 30, and reduce the time to produce choke valves from 45 weeks to 5 or less. The 3D parts perform just as well, if not better, than those manufactured by traditional means. For instance, 3D printing technology makes it possible to print choke valves with complex geometries that reduce the amount of fuel consumed and accompanying carbon emissions. And because 3D printing enables ConocoPhillips to create parts on demand and takes place closer to the point of use, it improves inventory management and reduces transportation costs associated with those parts as well as warehousing costs. As 3D printing continues to advance, ConocoPhillips expects to use it in other areas around the world, enabling it to replace physical inventory with digital inventory, advance environmental and sustainability metrics, and improve efficiency.



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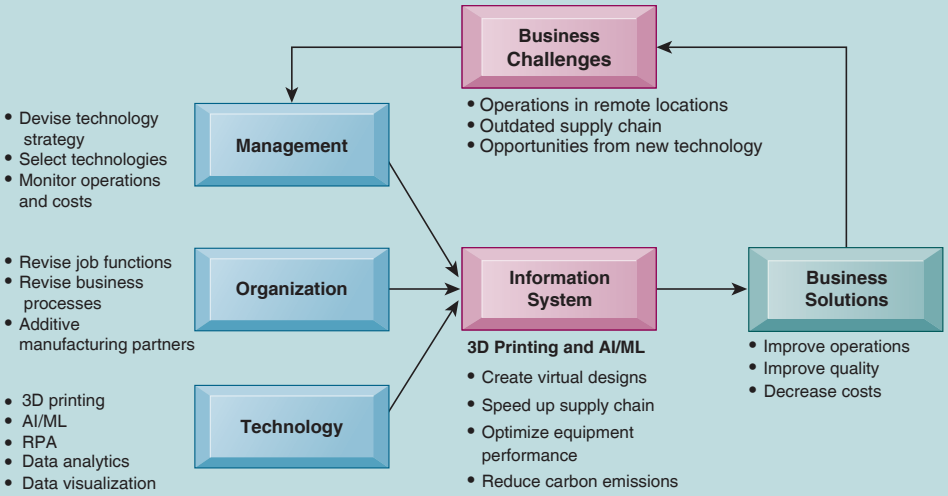
ConocoPhillips is also using a host of other innovative digital technologies, such as data analytics, data visualization, automation, and artificial intelligence/machine learning (AI/ML) to improve operational performance. It uses data analytics to harness the vast amounts of data (known as big data) generated from its business, including from Internet of Things (IoT) devices such as sensors. Data visualization tools, such as digital twin technology, enable ConocoPhillips to create a virtual representation of physical objects, allowing it to test various scenarios virtually. It employs software bot-driven robotic process automation (RPA) to automate specific repetitive tasks to free humans for more complex work. And increasingly, it is using AI/ML algorithms to optimize equipment performance and extend machine life. Other ConocoPhillips ML applications include optimizing seismic imaging, reservoir simulations, and well operations.

Going forward, ConocoPhillips's digital mission is to improve the efficiency of its operations, lower its



supply costs, reduce carbon emissions, and improve its safety performance. Innovation is a core value. The company believes that when combined with the skills of its workforce, digital technologies are critical to making its assets and processes more intelligent and automated.

SOURCES: "Digitizing Our Business," Conocophillips.com, accessed November 18, 2024. Michael Bertha, "ConocoPhillips Enlists 3D Printing for Supply Efficiencies on Alaska's North Slope," Cio.com, December 14, 2023; "Kuparak 1981—2021," ConocoPhillips Alaska and Petroleum News, March 2022.

ConocoPhillips' experience dealing with parts shortages, antiquated business processes, and supply chain issues shows how essential information systems are today. Forward-looking multinational firms such as ConocoPhillips are struggling to keep costs down, improve decision making, protect the environment, and find better ways of doing business, even in remote locations with harsh climates. ConocoPhillips addressed this problem by deploying leading-edge information technology, including 3D printing and AI/ML.

The Opening Case Study diagram above calls attention to important points raised by this case and this chapter. ConocoPhillips needs to extract oil and natural gas from numerous locations around the world, including remote locations with harsh climates such as Alaska's North Slope. The company's supply chain was not very resilient and hampered by the inability to obtain critical parts on short notice. 3D printing enabled ConocoPhillips to obtain parts more quickly, reducing transportation and warehouse costs, and to also improve their design. AI/ML and other innovative digital technologies have helped the company improve operational efficiency, optimize machine performance, extend equipment life, as well as reduce carbon emissions.

It is important to note that deploying new information technology has changed the way ConocoPhillips runs its business. To effectively use new digital tools, the company had to redesign jobs and business processes for manufacturing and obtaining parts. These changes had to be carefully planned to make sure they enhanced efficiency, service, and profitability.

Case Study Questions

1. Identify the problem described in this case. What areas of the business did it affect?
2. How did ConocoPhillips benefit from using 3D printing and AI/ML?
3. How did these new technologies change the way ConocoPhillips ran its business?

CAREER BENEFIT

Employees, managers, and IT specialists who worked on this problem gained recognition for selecting, testing, and implementing 3D printing and AI/ML systems, and for finding innovative uses for these technologies that can further improve business performance. ConocoPhillips employees and managers obtained career benefits by assisting with the adoption of new technologies and using them in their jobs to improve productivity and quality.

1.1 Understand How Information Systems Transform Business and Careers

The way businesses operate is changing at an increasingly rapid pace due to new technologies, particularly AI. According to research firm Gartner, firms spent more than $20 trillion on information technology (IT) and IT services worldwide between 2019 and 2023, and are expected to spend an additional $5 trillion in 2024 (Gartner, 2024). You can see the results of this spending around you every day.

How Information Systems Are Transforming Business

Today, information systems that incorporate the Internet and mobile devices have become essential tools of business. In 2024, around 2.9 billion people worldwide are expected to purchase something online, and more than two-thirds of all retail e-commerce revenues will be generated by people using a mobile device. What this means is that if a business isn't using information systems connected to the Internet and mobile apps, chances are it isn't being as effective as it could be.

Information systems that use sophisticated data analytics to analyze massive amounts of data are enabling businesses to track and respond to rapidly changing customer demand, reduce inventories to the lowest possible levels, and achieve higher levels of operational efficiency. As a result, supply chains have become faster-paced, with companies of all sizes depending on the delivery of just-in-time inventory to stay competitive. If a company is not employing these kinds of supply chain information systems, chances are it is not as efficient as it could be. The importance of supply chains and supply chain management has been highlighted even further by supply chain disruptions introduced by the Covid-19 pandemic.

Information systems have driven the development of both social networks and social media. In 2024, there are expected to be almost 4 billion people around the world who use social networks, representing about 50 percent of the total global population. Customers are using Facebook, Instagram, TikTok, and other forms of social media to talk to each other about a company's products and services. This means that companies need a solid online customer relationship program in place, and they need to keep track of what customers are saying about them.

Advertising is another business that has been transformed by the development of information systems. Digital advertising spending worldwide is projected to reach more than $900 billion by 2027, representing more than 75 percent of total media ad spending. More than three-quarters of this spending will be for mobile advertising. Companies need to make sure that their advertising is reaching web and mobile customers.

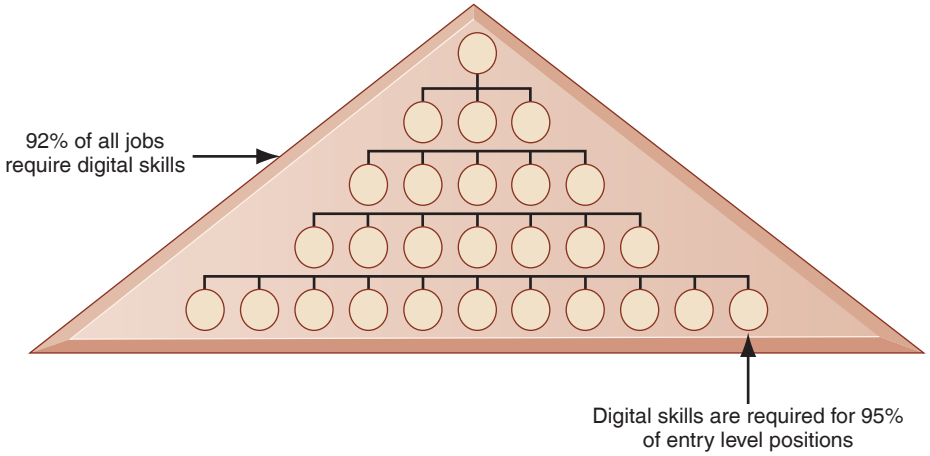
The data generated by billions of Internet-linked devices, e-commerce, and social media is driving the explosive growth of digital information. According to some estimates, almost 330 million terabytes of data are created every day. Retention of all the data created by businesses as well as consumers on social media and the Internet has raised privacy concerns and has led to the passage of laws about how such data must be safeguarded. Various U.S. laws require companies to meet certain requirements for storing financial, health, and occupational information. If they don't, the entire business may be at risk.

Artificial intelligence (AI) is the latest information technology that is in the process of transforming business. Public attention is riveted on AI because of its enormous potential for improving how companies conduct business, as well as many aspects of everyday life. If used properly, AI-driven information systems can lower costs, reduce risks, speed time to market, and create superior products and services, providing a competitive advantage (Emarketer, 2024, 2023; Duarte, 2023).

The Transformation of Jobs: What All This Means for You

These technology-driven changes are transforming jobs and careers—how you work, where you work, and how well you will be compensated. Many millions of new jobs have been created, not only for IT specialists but also for business professionals who now use IT tools to

Figure 1.1 High Demand for Digital Skills



accomplish their work. Employees across all industries are seeing more and more technology tools incorporated into their workflows.

A 2023 study by the National Skills Coalition investigated the demand for digital skills in the U.S. labor market. The report analyzed 43 million jobs from 2021 job postings and found strong demand for digital skills across every industry and in almost every occupation, even those that only required limited education or for entry-level positions. The report further found that workers who qualify for jobs that require even *one* digital skill can earn an average of 23 percent more than in a job that does not require any digital skills. Moving from a job that does not require any digital skills to one requiring at least three can increase pay by an average of 45 percent. Figure 1.1 illustrates these findings (National Skills Coalition, 2023). The conclusion is obvious: People who can understand and work with technology are at a great advantage. Do you have the career skills to thrive and succeed in this fast-paced, digital-first world of business?

Knowing how to use technology tools to perform your current job may be sufficient right now for today's workplace, but gaining a deeper understanding of how technology works and is evolving is also essential for getting ahead. For instance, the increasing use of AI tools will be transforming many jobs over the next five to ten years. The last section of each chapter in this book will help you understand how what you have learned in the chapter can translate into a higher paying job and career advancement. These recommendations are not limited to IT careers—they also apply to most career paths in this tech-driven world. Additional career resources are available in MyLab MIS.

1.2 Describe Key Challenges and Opportunities Created by New MIS Technologies

In this book, we focus primarily on management information systems (MIS), which can be broadly defined as information systems used to help optimize firm performance. The technologies behind management information systems are changing and evolving every day. This creates both challenges and opportunities, which are what make management information systems one of the most exciting areas in business today. Globalization, which in large part has been driven by changes in technology, brings with it its own set of challenges and opportunities.

Key Challenges in Management Information Systems

Table 1.1 summarizes the major challenges that technology changes have created in using management information systems. No matter what your role is in a business, you are likely to encounter some, if not all, of these challenges at some point in your career.

Table 1.1 Key Challenges in Management Information Systems

Change	Business Challenge
Technology	
Cloud computing: Companies are increasingly turning to cloud computing, a system for storing, managing, and processing data using an Internet-connected network of remote computing centers, to perform tasks traditionally performed at corporate data centers. Major business applications can be delivered online as an Internet service (software as a service [SaaS]).	What are the costs and benefits of cloud computing and how much of the firm's IT infrastructure should be moved to cloud providers?
Big data and the Internet of Things (IoT): Businesses are looking for insights in huge volumes of data from web traffic, email messages, social media content, and Internet-connected devices (sensors). More powerful data analytics and interactive dashboards can provide real-time performance information to managers to enhance decision making.	Does your firm have the ability to analyze and use big data?
Artificial intelligence (AI): Computer systems can perform tasks normally requiring human intelligence, such as speech recognition, visual perception, recognizing patterns, making predictions, and learning from experience, with the potential for dramatically improving operations and decision making.	Where can your company use AI, and where can you find the expertise to implement it? What benefits can you expect? How much will it cost?
The mobile platform: Business and personal computing is increasingly moving to smartphones, tablet computers, car infotainment systems, and wearable devices. Thousands of applications are now available to support collaboration, coordination of work, communication with colleagues and customers, and online purchases on mobile devices. Almost 90 percent of Internet users use a mobile device to access the Internet at least some of the time.	Is your firm making the best use of mobile capabilities for its employees and customers? How could your firm improve? What are the costs and benefits?
Management	
Return on investment (ROI): Although firms spend millions on information systems and services, they typically have little understanding of how much benefit they receive.	How can your firm measure and understand the benefit it is receiving from IT expenditures? Are there alternative sources of these services that would cost less?
Online collaboration and social networks: Millions of business professionals use Google Apps, Microsoft 365, Slack, and Zoom to support online meetings, project management, blogs, personal profiles, and online communities.	Is your firm making a coordinated effort to use new technologies to improve coordination, collaboration, and knowledge sharing? Which of the many alternatives should it be using?
Organizations	
Security and privacy: Security lapses and protecting customer privacy are major public issues that affect all businesses.	How does your company know its data are secure? How much does it spend on security now? What privacy policies does your firm have in place? How should the firm expand its privacy protections as new laws emerge?

Change	Business Challenge
Social business: Businesses use social network platforms, including Facebook, X (formerly Twitter), TikTok, Instagram, and internal corporate social tools, to deepen interactions with employees, customers, and suppliers.	How is your company using social business tools? Is your company getting real value from these platforms?
Remote and hybrid work: The Internet, cloud computing, smartphones, and tablet computers make it possible for growing numbers of people to work away from the traditional office. This capability proved critical during the height of the Covid-19 pandemic, when more than 70 percent of workers worked from home.	Going forward, what will your firm's policies and technologies be for remote work, and what are the challenges and opportunities?

Globalization Challenges and Opportunities

Globalization
The flow of goods, services, capital, people, and ideas across international boundaries.

Globalization is defined as the flow of goods, services, capital, people, and ideas across international boundaries. A growing percentage of the economies of the United States and other countries depend on imports and exports. According to the World Trade Organization, world trade of goods and services constitutes almost 60 percent of overall global economic activity (World Trade Organization, 2023). More than 35 percent of the revenue of U.S. Fortune 500 companies is generated outside of the United States.

The level of globalization in the world today has been made possible by advances in information technology. The emergence of the Internet into a full-blown international communications system has drastically reduced the costs of operating and transacting on a global scale. Communication between a factory floor in Mumbai, India and a distribution center in Rapid City, South Dakota, is now instant and virtually free. Customers can now shop in a worldwide marketplace, obtaining price and quality information reliably 24 hours a day. Firms producing goods and services on a global scale can achieve extraordinary cost reductions by finding low-cost suppliers and managing production facilities in other countries.

Globalization gives businesses the opportunity to expand into new markets, reach international buyers, and increase revenue. However, it also presents challenges for businesses. The challenge for businesses is to understand the potential impact that globalization can have on markets for goods and services. For instance, businesses can learn how to profit from the lower costs available in world markets and the chance to serve a marketplace with billions of customers. Globalization can provide opportunities to develop more profitable products and services. On the other hand, globalization can create increased competition from other companies who may be more successful in leveraging the benefits of globalization.

What does globalization mean for you and your career? It's not just goods that move across borders, but jobs as well. Some of these jobs are high-level positions that pay well and require a college degree. Between 1998 and 2021, the United States lost 5 million manufacturing jobs to offshore producers. U.S. multinational companies employ more than 14 million people outside the United States, many of them in service jobs in information technology, customer call centers, human resources, financial services, consulting, engineering, architecture, and even medical services, such as radiology. On the plus side, the U.S. economy continues to create millions of new jobs, and some of those can be attributed to globalization. The challenge for you as a student and in your future career is to develop high-level skills through education and on-the-job experience that cannot be replicated at a lower cost abroad (Scot et al., 2022).

Digital firm
Organization where nearly all significant business processes and relationships with customers, suppliers, and employees are digitally enabled, and key corporate assets are managed through digital means.

1.3 Describe Key Features of a Digital Firm

The changes we have just described have created the conditions for the emergence of "digital firms." A **digital firm** is one in which the organization's significant business relationships with customers, suppliers, and employees are digitally enabled and corporate assets

are managed through digital means. *Core business processes* are accomplished through digital networks spanning the entire organization or linking multiple organizations.

Business processes refer to the set of logically related tasks and behaviors that organizations develop over time to produce specific business results and the unique manner in which these activities are organized and coordinated. Developing a new product, generating and fulfilling an order, creating a marketing plan, and hiring an employee are examples of business processes, and the ways organizations accomplish their business processes can be a source of competitive strength. More detail about business processes can be found in Chapters 2 and 3.

Business processes
The unique ways in which organizations coordinate and organize work activities, information, and knowledge to produce a product or service.

Key corporate assets—intellectual property, core competencies, and financial and human assets—are managed through digital means. In a digital firm, the information required to support key business decisions is available at all times and anywhere in the firm.

A digital firm can sense and respond to its environment far more rapidly than a traditional firm, giving it more flexibility to survive in turbulent times. Digital firms offer extraordinary opportunities for more flexible global organization and management. In digital firms, both time shifting and space shifting are the norm. *Time shifting* refers to business being conducted continuously, 24/7, rather than in narrow "workday" time bands of 9 a.m. to 5 p.m. *Space shifting* means that work takes place in a global workshop as well as within national boundaries. Work is accomplished physically wherever in the world it is best accomplished using digital tools and networks to link information and people.

Firms often undertake digital transformation initiatives in an effort to become a digital firm. **Digital transformation** refers to large-scale change efforts to capture the benefits of digital technology, driving fundamental changes in how businesses operate and how they deliver value to customers. Digital transformation is an ongoing process. Firms engaging in digital transformation move progressively closer to becoming digital firms. Cisco Systems, Caterpillar, and 3M are examples of businesses that are taking an aggressive "digital first" approach, using the Internet and information technology to drive nearly every aspect of their business. Most companies are not fully digital, but they are moving toward closer digital ties to suppliers, customers, and employees.

Digital transformation
Large-scale change efforts to capture the benefits of digital technology, driving fundamental changes in how businesses operate and how they deliver value to customers.

Digital transformation is a complex endeavor requiring enterprises to rethink their business models, change their organizational structures and practices, become more data-driven, and embrace a culture of continuous change.

1.4 Describe the Business Objectives of Information Systems

There is a growing interdependence between a firm's ability to use information technology and its ability to implement corporate strategies and achieve corporate goals. What a business would like to do in five years often depends on what its systems will be able to do. Examine Figure 1.2, which illustrates this relationship. Changes in strategy, rules, and business processes increasingly require changes in hardware, software, data management, and networking and telecommunications technologies. Increasing market share, becoming the high-quality or low-cost producer, developing new products, and increasing employee productivity depend more and more on the kinds and quality of information systems in the organization. The more you understand about this relationship, the more valuable you will be in the business world.

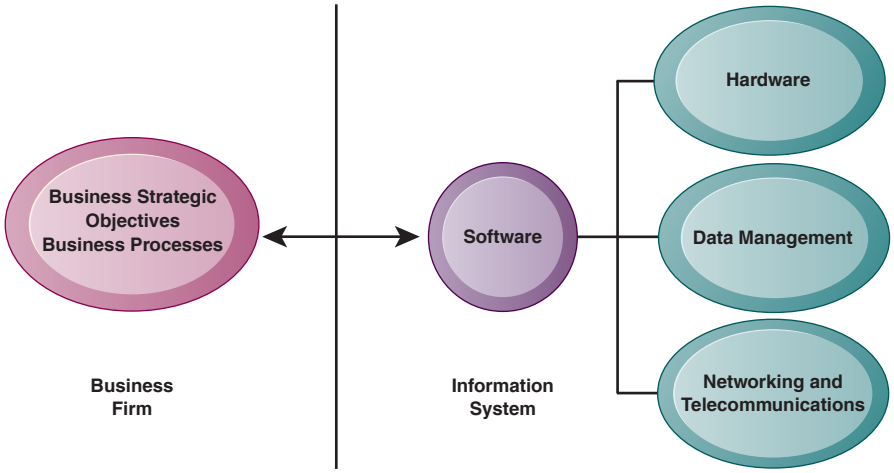
Business firms invest heavily in information systems to achieve six strategic business objectives: operational excellence, new products, services, and business models, customer and supplier intimacy, improved decision making, competitive advantage, and survival. In addition, companies today are increasingly concerned about the environmental impact of their operations and sustainability, as well as proper governance of the organization and the responsible use of AI. Such concerns are becoming a strategic business imperative, and information systems are central in enabling companies to meet their goals.

Operational Excellence

Businesses continuously seek to improve the efficiency of their operations to achieve higher profitability. Information systems and technologies are some of the most important tools available to managers for achieving higher levels of efficiency and productivity in business operations, especially when coupled with changes in business practices and management behavior.

Figure 1.2 The Interdependence of Organizations and Information Systems

Today, a firm's information systems and its business capabilities have become interdependent. Often, what the organization would like to do depends on what its systems will permit it to do.



Walmart, the largest retailer on Earth, exemplifies the power of information systems coupled with sophisticated business practices and supportive management to achieve world-class operational efficiency. In fiscal 2024, Walmart generated $648 billion in sales—largely because of its Retail Link and Global Replenishment System, which digitally links its suppliers to every one of Walmart's stores worldwide. As soon as a customer purchases an item, the supplier monitoring the item knows to ship a replacement to the shelf. Walmart is the most efficient retail store in its industry.

New Business Models, Products, and Services

Business model

Describes how a company produces, delivers, and sells a product or service in order to create value.

Information systems and technologies are a major enabling tool for firms to create new business models, as well as new products and services. A **business model** describes how a company produces, delivers, and sells a product or service to generate revenue. For instance, Apple transformed a music distribution business model based at the time on the sale of physical products such as vinyl records, tapes, and CDs into a legal Internet-based digital download distribution model based on its own iTunes store.

Customer and Supplier Intimacy

When a business really knows its customers and serves them well, the way they want to be served, its customers generally respond by returning and purchasing more. This raises revenues and profits. Likewise, with suppliers, the more a business engages its suppliers, the better the suppliers can provide vital inputs. This lowers the business's costs. How to really know your customers or suppliers is a central problem for businesses.

High-end hotel chains such as Ritz-Carlton exemplify the use of information systems and technologies to achieve customer intimacy. Every time new prospects share their information or a guest checks in to a Ritz hotel, their information is stored in a cloud-based relationship management system. When that guest returns to the hotel or logs into their online account, the system cross-references their data profile and tailors that customer's guest experience to their exact preferences (such as room temperature, check-in and checkout times, and room location) and history. This highly personalized experience makes the customer feel valued, included, and part of the Ritz brand. The hotels also analyze their customer data to identify their best customers and develop individualized marketing campaigns based on customers' preferences.

Hong Kong-headquartered TAL Apparel, one of the world's largest contract apparel manufacturers, exemplifies the use of information systems to enable supplier intimacy. TAL Apparel produces shirts, blouses, knits, pants, outerwear, and suits for many leading garment brands, including Bonobos and L.L.Bean. TAL manufactures one out of every six dress shirts sold in the United States. The company also helps its clients manage their supply chains. Every time its clients sell a dress shirt, for example, the record of the sale appears immediately on TAL's computers in Hong Kong. TAL runs

the numbers through a computer model it developed and decides how many replacement shirts to make and in what styles, colors, and sizes. TAL then sends the shirts to the retailer. TAL's systems reduce inventory costs and ensure that what customers want is actually on retailers' shelves.

Improved Decision Making

Many business managers operate in an information fog, never really having the right information at the right time to make an informed decision. Instead, managers rely on forecasts, best guesses, and luck. The result is over or underproduction of goods and services, misallocation of resources, and poor response times. These poor outcomes raise costs and lose customers. Information systems and technologies have now made it possible for managers to use real-time data from the marketplace when making decisions.

For instance, Coca-Cola Bottling Company Consolidated uses a series of business intelligence dashboards that display metrics like delivery operations, budget, and profitability, consolidating data from hundreds of disparate sources. Leadership dashboards focus on strategy, business growth, and profitability. Sales teams can now access mobile dashboards on iPads in the field, increasing timeliness to sell more products. Field sales dashboards allow sales teams to see a full overview of customer portfolios and track sales quotas.

Competitive Advantage

When firms achieve one or more of these business objectives—operational excellence, new products, services, and business models, customer/supplier intimacy, and improved decision making—chances are they have already achieved a competitive advantage. Doing things better than competitors, charging less for superior products, and responding to customers and suppliers in real time all add up to higher sales and profits that competitors cannot match. Apple, Walmart, and UPS are industry leaders because they know how to use information systems for this purpose.

Survival

Business firms also invest in information systems and technologies because they are necessities of doing business. Sometimes these necessities are driven by industry-level changes. For instance, after Citibank introduced the first automated teller machines (ATMs) in the New York region to attract customers through higher service levels, its competitors rushed to provide ATMs to their customers to keep up with Citibank. Today, virtually all banks in the United States have regional ATMs and link to national and international ATM networks, such as Cirrus. Providing ATM services to retail banking customers is simply a requirement of being in and surviving in the retail banking business.

Many federal and state statutes and regulations create a legal duty for companies and their employees to retain records, including digital records. For instance, the Toxic Substances Control Act, which regulates the exposure of U.S. workers to more than 75,000 toxic chemicals, requires firms to retain records on employee exposure for 30 years. The Sarbanes–Oxley Act, which was intended to improve the accountability of public firms and their auditors, requires public companies to retain audit working papers and records, including all email messages, for five years. The Dodd–Frank Act requires financial service firms to expand their public reporting greatly on derivatives and other financial instruments. Firms turn to information systems and technologies to provide the capability to respond to information retention and reporting requirements.

Environmental Impact, Sustainability, Governance, and Responsible AI

Over the past decade, there has been a growing awareness of the important role that businesses play and the impact that they have on society. With that growing awareness, there has come pressure from employees, customers, investors, and governmental authorities with respect to business conduct in three primary areas: sustainability and environmental impact, corporate governance, and increasingly, the responsible use of AI.

Sustainability and environmental concerns encompass aspects such as energy use, carbon footprint, recycling practices, pollution, and natural resource conservation. Governance criteria primarily relate to the actions of the company's leadership, executive pay, internal controls, and shareholder rights. Responsible AI initiatives aim to provide guidelines to ensure companies use AI technologies in an ethical manner. Many aspects of these concerns are intertwined with one another.

Showing leadership in these areas can add business value by facilitating growth, reducing costs, minimizing regulatory and legal issues, increasing employee productivity, and optimizing assets and investments. For instance, many consumers are no longer willing to support companies that don't prioritize sustainability.

Information systems are essential for enabling companies to achieve these sorts of goals. For instance, Crowley, a global logistics, marine, and energy solutions company, worked with consulting firm PwC and Salesforce to implement Salesforce Net Zero Cloud. Salesforce Net Zero Cloud quickly analyzes and reports reliable environmental data from a variety of disparate sources. Salesforce Net Zero Cloud helps Crowley measure and analyze its carbon emissions and those of its suppliers. Crowley expects to reduce overall emissions by 4.2 million metric tons of greenhouse gases per year, the equivalent of removing more than 900,000 cars from the road annually (PWC, 2022). We discuss other examples of information systems being used for these purposes throughout the text.

1.5 Explain What an Information System Is and How It Works

So far, we've used the terms *information systems* and *information technologies* informally without defining them. **Information technology (IT)** consists of all the hardware and software that a firm needs to use to achieve its business objectives. This includes not only computers, storage technology, and mobile devices, but also software such as the Windows and iOS operating systems, the Microsoft 365 productivity suite, and the many thousands of computer programs that can be found in a typical large firm. Information systems are more complex and can be understood best by looking at them from both a technology perspective and a business perspective.

Information technology (IT)
All the hardware and software technologies a firm needs to achieve its business objectives.

Information system (IS)
Interrelated components working together to collect, process, store, and distribute information to support decision making, coordination, control, analysis, and visualization in an organization.

Information
Data that have been shaped into a form that is meaningful and useful to humans.

Data
Streams of raw facts representing events occurring in organizations or the physical environment before they have been organized and arranged into a form that people can understand and use.

Input
The capture or collection of raw data from within the organization or from its external environment for processing in an information system.

Processing
The conversion, manipulation, and analysis of raw input into a form that is more meaningful to humans.

What Is an Information System

An **information system (IS)** can be defined technically as a set of interrelated components that collect (or retrieve), process, store, and distribute information to support decision-making, coordinating, and control in an organization. In addition, information systems may also help managers and workers analyze problems, visualize complex subjects, and create new products.

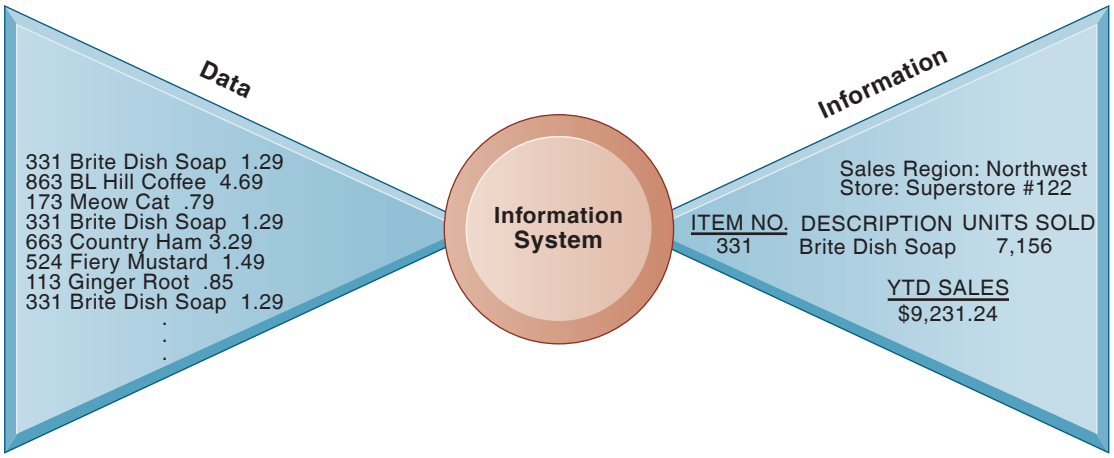
Information systems contain information about significant people, places, and things within the organization or in the environment surrounding it. By **information**, we mean data that have been shaped into a form that is meaningful and useful to humans. **Data**, in contrast, are streams of raw facts representing events occurring in organizations or the physical environment before they have been organized and arranged into a form that people can understand and use.

A brief example contrasting information and data may prove useful. Examine Figure 1.3, which provides an example of an information system used by a supermarket. Supermarket checkout counters scan barcodes on products. The barcodes contain data about the product, such as a numerical identifier, name, and price ("331 Brite Dish Soap, 1.29"), shown in the left portion of the diagram. An information system enables these pieces of data to be totaled and analyzed to provide meaningful information, displayed in the right portion of the diagram, such as the total number of bottles of dish detergent sold at a particular store, which brands of dish detergent were selling the most rapidly at that store or sales territory, or the total amount spent on that brand of dish detergent at that store or sales region.

Three activities in an information system produce the information that organizations need to make decisions, control operations, analyze problems, and create new products or services. These activities are input, processing, and output. Examine Figure 1.4, which illustrates how these activities work together within an information system. **Input** captures or collects raw data from within the organization or from its external environment. **Processing** converts this raw

Figure 1.3 Data and Information

Raw data from a supermarket checkout counter can be processed and organized to produce meaningful information, such as the total unit sales of dish detergent or the total sales revenue from dish detergent for a specific store or sales territory.



input into a meaningful form by classifying, arranging, and performing calculations on it. **Output** transfers the processed information to the people who will use it or to the activities for which it will be used. Information systems also provide **feedback**, which is output that is returned to appropriate members of the organization to help them evaluate or correct the input stage. Figure 1.4 also illustrates that an organization's information systems do not operate within a vacuum: they are impacted by various elements in the environment that surround them, such as the organization's suppliers, customers, competitors, and stockholders as well as governmental agencies that regulate the organization.

Output

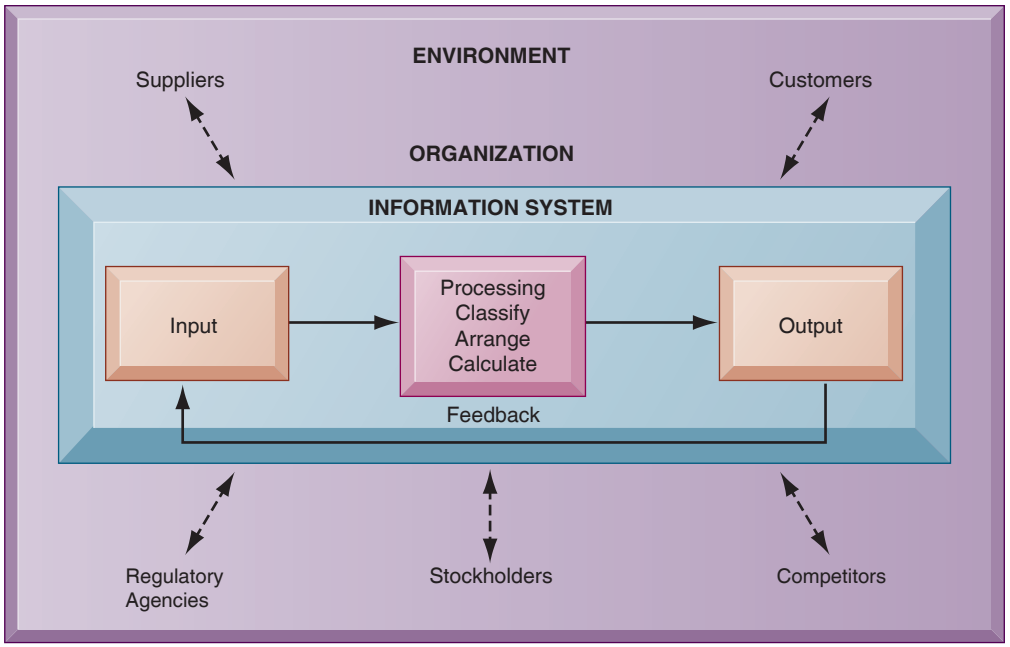
The distribution of processed information to the people who will use it or to the activities for which it will be used.

Feedback

Output that is returned to the appropriate members of the organization to help them evaluate or correct input.

Figure 1.4 Functions of an Information System

An information system contains information about an organization and its surrounding environment. Three basic activities—input, processing, and output—produce the information organizations need. Feedback is output returned to appropriate people or activities in the organization to evaluate and refine the input. Actors in the external environment, such as customers, suppliers, competitors, stockholders, and regulatory agencies, interact with the organization and its information systems.



Although computer-based information systems use computer technology to process raw data into meaningful information, there is a sharp distinction between a computer and a computer program, and an information system. Computers and related software programs are the technical foundation, the tools and materials of modern information systems. Computers provide the equipment for storing and processing information. Computer programs, or software, are sets of operating instructions that direct and control computer processing. Knowing how computers and computer programs work is important in designing solutions to organizational problems, but computers are only part of an information system.

A house is an appropriate analogy. Houses are built with hammers, nails, and wood, but these alone do not make a house. The architecture, design, setting, landscaping, and all of the decisions that lead to the creation of these features are part of the house and are crucial for solving the problem of putting a roof over one's head. Computers and programs are the hammer, nails, and lumber of computer-based information systems, but alone they cannot produce the information a particular organization needs. To understand information systems, you must understand the problems they are designed to solve, their architectural and design elements, and the organizational processes that lead to these solutions.

Information systems literacy
Broad-based understanding of information systems that includes behavioral knowledge about organizations and individuals using information systems as well as technical knowledge about computers.

Computer literacy
Knowledge about information technology and understanding how computer-based technologies work.

Management information systems (MIS)
The study of information systems, focusing on their use in business and management. Also refers to specific category of information system serving middle management.

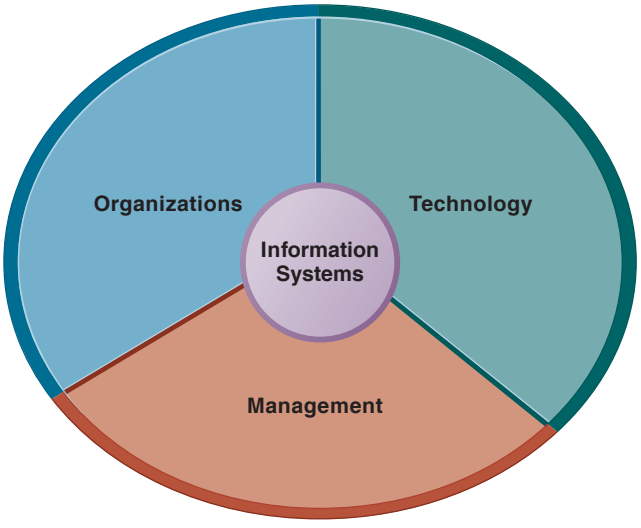
It Isn't Simply Technology: The Role of Management and Organizations

To understand information systems fully, you need to be aware of the broader organizational, managerial, and information technology dimensions of systems and their power to provide solutions to challenges and problems in the business environment. Examine Figure 1.5, which highlights the interconnected nature of these dimensions, showing that organizations, technology, and management are all interrelated components underlying the effective use of information systems. We refer to this broader understanding of information systems, which encompasses an understanding of the managerial and organizational dimensions of systems as well as the technical dimensions of systems, as **information systems literacy**. Information systems literacy includes a behavioral as well as a technical approach to studying information systems. **Computer literacy**, in contrast, focuses primarily on knowledge about information technology and understanding how computer-based technologies work.

The field of **management information systems (MIS)** tries to achieve this broader information systems literacy. MIS deals with behavioral issues as well as technical issues surrounding the development, use, and impact of information systems that managers and employees in the firm use. (Note that the term *management information systems* also refers to a specific category of information systems, as previously noted in Section 1.2 and further described in later chapters.)

Figure 1.5 Information Systems Are More Than Computers

Using information systems effectively requires an understanding of the organization, management and information technology shaping the systems. An information system provides a solution to important business problems or challenges facing the firm.



1.6 Describe Organizational, Management, and Technology Dimensions of Information Systems

Let's now further examine each of the organizational, management, and technology dimensions of information systems.

Organizations

Information systems are an integral part of organizations. Although we tend to think about information technology changing organizations and business firms, it is, in fact, a two-way street. The history and culture of business firms also affect how the technology is used and how it should be used. To understand how a specific business firm uses information systems, you need to know something about the structure, history, and culture of the company.

Organizations have a structure that is composed of different levels and specialties. Their structures reveal a clear-cut division of labor. A business firm is organized as a hierarchy, or a pyramid structure, of rising authority and responsibility. The upper levels of the hierarchy consist of managerial, professional, and technical employees, whereas the lower levels consist of operational personnel.

Examine Figure 1.6, which illustrates this hierarchy in the form of a pyramid. **Senior management**, which makes long-range strategic decisions about products and services as well as ensures financial performance of the firm, is at the top of the pyramid. **Middle management,** which carries out the programs and plans of senior management, forms the middle layer, and **operational management** at the base of the pyramid is responsible for monitoring the daily activities of the business. **Knowledge workers**, such as engineers, scientists, or architects, design products or services and create new knowledge for the firm,

Senior management
People occupying the topmost position in an organization's hierarchy and who are responsible for making long-range decisions.

Middle management
People in the middle of an organization's hierarchy and who are responsible for carrying out the plans and goals of senior management.

Operational management
People who are responsible for monitoring the day-to-day activities of an organization.

Knowledge workers
People such as engineers, scientists, or architects who design products or services and create knowledge for the organization.

Figure 1.6 Levels in a Firm

Business organizations are hierarchies consisting of three principal levels: senior management, middle management, and operational management. Information systems serve each of these levels. Scientists and knowledge workers often work with middle management.

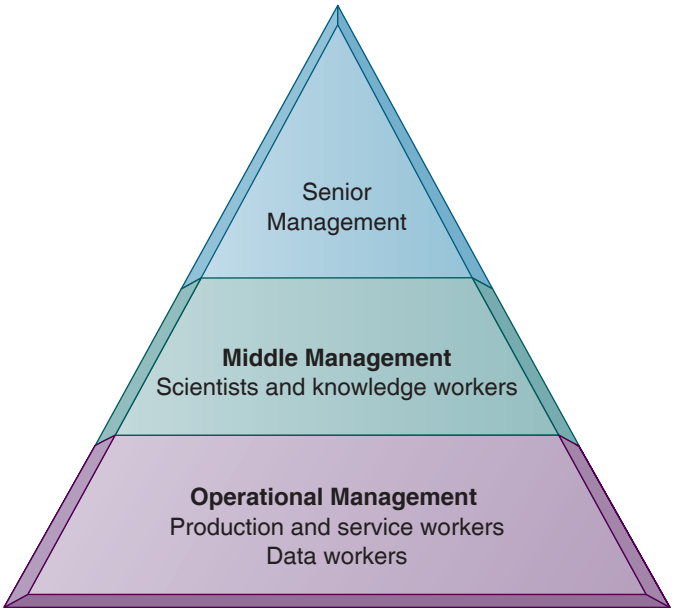


Table 1.2 Major Business Functions

Function	Purpose
Sales and marketing	Selling the organization's products and services
Manufacturing and production	Producing and delivering products and services
Finance and accounting	Managing the organization's financial assets and maintaining the organization's financial records
Human resources	Attracting, developing, and maintaining the organization's labor force; maintaining employee records

Data workers
People such as clerks or bookkeepers who assist with administrative work at all levels of a firm.

Production and service workers
People who actually produce the products or services of the organization.

Business functions
Specialized tasks performed in a business organization, such as manufacturing and production, sales and marketing, finance and accounting, and human resources.

Culture
Fundamental set of assumptions, values, and ways of doing things that has been accepted by most members of an organization.

frequently working with middle management. Operational management typically oversees **data workers**, such as bookkeepers or clerks, who assist with administrative work at all levels of the firm, along with **production and service workers** who actually produce the product and deliver the service.

Experts are employed and trained for different **business functions**, specialized tasks performed by business organizations, such as sales and marketing, manufacturing and production, finance and accounting, and human resources (see Table 1.2). The firm builds information systems to serve these different specialties and levels of the firm. Chapters 2 and 3 provide more detail on these business functions and organizational levels and the ways in which information systems support them.

An organization accomplishes and coordinates work through this structured hierarchy and through its business processes, which are logically related tasks and behaviors for accomplishing work. Developing a new product, fulfilling an order, and hiring a new employee are examples of business processes.

Most organizations' business processes include formal rules that have been developed over a long time for accomplishing tasks. These rules guide employees in a variety of procedures, from writing an invoice to responding to customer complaints. Some of these business processes have been written down, but others are informal work practices, such as a requirement to return telephone calls from co-workers or customers that are not formally documented. Information systems automate many business processes. For instance, how a customer receives credit or how a customer is billed is often determined by an information system that incorporates a set of formal business processes.

Each organization has a unique **culture**, or fundamental set of assumptions, values, and ways of doing things, that has been accepted by most of its members. Parts of an organization's culture can usually be found embedded in its information systems. For instance, customer service is a top priority for UPS (United Parcel Service), and this aspect of its organizational culture is reflected in the company's package tracking systems. We discuss UPS's information systems further in the Spotlight on Technology Case Study, *UPS Completes Globally with Information Technology*, later in the chapter.

Different levels and specialties in an organization create different interests and points of view. These views often conflict, which can lead to organizational politics. Information systems come out of this cauldron of differing perspectives, conflicts, compromises, and agreements that are a natural part of all organizations.

Management

Business firms require many kinds of skills and people, including managers as well as rank-and-file employees. The job of managers is to make sense of the many situations organizations face, make decisions, and formulate action plans to solve organizational problems. Managers perceive business challenges in the environment. They set the organizational strategy for responding to those challenges, and they allocate the human and financial resources to coordinate the work and achieve success. Throughout, they must exercise responsible leadership.

However, managers must do more than manage what already exists. They must also create new products and services, and even recreate the organization from time to time. A substantial part of management responsibility is creative work driven by new knowledge and information. Information systems can play a powerful role in helping managers develop novel solutions to a broad range of problems. Chapter 12 covers management decision-making in detail.

Technology

Information technology is one of many tools managers use to cope with change and complexity. Information technology includes many different components. **Computer hardware** consists of the physical equipment used for input, processing, and output activities in an information system. It includes the following: computers of various sizes and shapes (including mobile devices); various input, output, and storage devices; and networking devices that link computers.

Computer software consists of the detailed preprogrammed instructions that control and coordinate the computer hardware components in an information system. Chapter 5 describes the software and hardware platforms firms use today in greater detail.

Data management technology consists of the software governing the organization of data on physical storage media. More detail on data organization and access methods can be found in Chapter 6.

Networking and telecommunications technology, consisting of both physical devices and software, links the various pieces of hardware and transfers data from one physical location to another. Computers and communications equipment can be connected in networks for sharing voice, data, images, sound, and video. A **network** links two or more computers to share data or resources, such as a printer.

The world's largest and most widely used network is the **Internet**, a global network of networks that uses universal standards (described further in Chapter 7) to connect millions of networks in more than 230 countries and territories around the world. As of 2024, there were 4.8 billion Internet users worldwide, almost 60 percent of the global population (Emarketer, 2023c).

The Internet provides a universal technology platform on which to build new products, services, strategies, and business models. This same technology platform has internal uses, providing connectivity to link different systems and networks within the firm. An internal corporate network based on Internet technology and standards is called an **intranet**. A private intranet extended to authorized users outside the organization is called an **extranet**. Firms use extranets to coordinate their activities with other firms for making purchases, collaborating on design, and performing other interorganizational work. For most firms today, using Internet technology is a business necessity and a competitive advantage.

The **World Wide Web** (today, normally referred to as just the "web") is an Internet service that uses universally accepted standards for storing, retrieving, formatting, and displaying information in a page format on the Internet. Web pages contain text, graphics, animations, sound, and video, and are linked to other web pages. By clicking highlighted words or buttons on a web page, you can link to related pages to find additional information and links to other locations on the web. The web now serves as the foundation for many information systems.

All of these technologies, along with the people required to run and manage them, represent resources that can be shared throughout the organization and constitute the firm's **information technology infrastructure (IT infrastructure)**. The IT infrastructure provides the foundation, or *platform*, on which the firm can build its specific information systems. Each organization must carefully design and manage its IT infrastructure so that it has the set of technology services it needs for the work it wants to accomplish with information systems. Part II of this text examines each major technology component of information technology infrastructure and shows how they all work together to create the technology platform for the organization.

The Spotlight on Technology Case Study, *UPS Competes Globally with Information Technology*, describes some of the typical technologies used in information systems today. UPS invests heavily in information systems technology to make its business more efficient and customer-oriented. It uses an array of information technologies, including barcode scanning systems, wireless networks, large mainframe computers, handheld computers, the Internet, and many pieces of software for tracking packages, calculating fees, maintaining customer accounts, and managing logistics. As you read this case, try to identify the problem this company was facing, what alternative solutions were available to management, and how well the chosen solution worked.

Computer hardware

Physical equipment used for input, processing, and output activities in an information system.

Computer software

Detailed, preprogrammed instructions that control and coordinate the work of computer hardware components in an information system.

Data management technology

Software governing the organization of data on physical storage media.

Networking and telecommunications technology

Physical devices and software that link various computer hardware components and transfer data from one physical location to another.

Network

The linking of two or more computers to share data or resources, such as a printer.

Internet

Global network of networks using universal standards to connect millions of different networks.

Intranet

An internal network based on Internet technology and standards.

Extranet

Private intranet that is accessible to authorized outsiders.

World Wide Web

A Internet service that uses universally accepted standards for storing, retrieving, formatting, and displaying information in a page format on the Internet; usually referred to as just the "web".

Information technology infrastructure (IT infrastructure)

Computer hardware, software, data, storage technology, and networks providing a portfolio of shared IT resources for the organization.

Spotlight on Technology Case Study

UPS Competes Globally with Information Technology

United Parcel Service (UPS) started out in 1907 in a closet-sized basement office. Jim Casey and Claude Ryan—two teenagers from Seattle with two bicycles and one phone—promised the "best service and lowest rates." UPS has used this formula successfully for more than a century to become the world's largest ground and air package-delivery company. It's a global enterprise with more than 500,000 employees, more than 125,000 delivery vehicles, and 557 aircraft.

Today, UPS delivers close to 25 million packages each day in more than 220 countries and territories. The firm has been able to maintain leadership in small-package delivery services despite stiff competition from FedEx, the US Postal Service, and Amazon by investing heavily in advanced information technology. UPS spends $1 billion annually on IT to maintain a high level of customer service while keeping costs low and streamlining its overall operations.

It all starts with the scannable bar-coded label attached to a package, which contains detailed information about the sender, the destination, and when the package should arrive. Customers can download and print their own labels using special software provided by UPS or by accessing the UPS website or mobile app. Before the package is even picked up, information from the "smart" label is transmitted to one of UPS's computer centers in Mahwah, New Jersey, or Alpharetta, Georgia, and sent to the distribution center nearest its final destination.

The first thing a UPS driver picks up each day is a handheld computer called a Delivery Information Acquisition Device (DIAD), which can access a cell phone network. As soon as drivers log on, their route is downloaded onto the DIAD. The DIAD also automatically captures customers' signatures along with pickup and delivery information. Package tracking information is then transmitted to UPS's computer network for storage and processing. From there, the information can be accessed worldwide to provide proof of delivery to customers or to respond to customer queries. UPS processes an average of 743.5 million tracking requests each day.

Through its automated package tracking system, UPS can monitor and even reroute packages throughout the delivery process. At various points along the route from sender to receiver, barcode devices scan shipping information on the package label and feed data about the progress of the package into the central computer. Customer service representatives are able to check the status of any package from desktop computers. UPS customers can also access this information from the company's website using their own computers or mobile devices.

© Ground Picture/Shutterstock

UPS created an AI application called DeliveryDefense designed to tackle the growing problem of shipment loss. The application examines characteristics such as location, loss frequency, returns volume, and delivery attempts, grouping them based on similarities. The application predicts the likelihood of a successful delivery, enabling shippers and their customers to reroute at-risk shipments to The UPS Store or a UPS Access Point. The solution can also be built into a customer's ordering platform to improve their delivery outcomes and also helps UPS customers deter package theft. For example, DeliveryDefense helps gold and silver dealer Texas Precious Metals identify addresses that are likely targets for "porch pirates."

An AI-powered app called Network Planning Tools (NPT) helps UPS reroute packages away from bad storms and other trouble spots in its global network. NPT also helps the company to move unusually large shipments efficiently. AI machine learning algorithms handle the massive amounts of data required to move packages around the world and on time, regardless of the weather.

UPS's On-Road Integrated Optimization and Navigation (ORION) is an AI platform for planning and optimizing the routes taken by UPS drivers, based on advanced algorithms, AI machine learning, and deep learning (which is largely self-taught and can recognize patterns and make predictions from very large datasets). UPS vehicles are equipped with systems that capture logistical data. Dispatchers at UPS distribution centers use ORION routing software to create the most efficient delivery route for each driver, taking into account traffic, weather conditions, and the location

of each stop. In a network with 66,000 routes in the United States alone, shaving even one mile off each driver's daily route translates into big savings in time, fuel consumption, miles driven, and carbon emissions. Since ORION's initial deployment, it has saved UPS about 100 million miles and 10 million gallons of fuel per year.

SOURCES: UPS, "Company Facts," About.ups.com, accessed November 18, 2024; Anastasia Morozova, "Chatbots in Logistics and Supply Chain: How Do Companies Like UPS Benefit from Chatbots," Jasoren.com, accessed November 18, 2024; Zach Williamson, "UPS Saving Millions at the Pump, Emphasizes Importance of Planning Ahead," 3newsnow.com, July 18, 2022.

Case Study Questions

1. What are the inputs, processing, and outputs of UPS's package tracking system?
2. What technologies are used by UPS? How are these technologies related to UPS's business strategy? What role is played by AI?
3. What strategic business objectives do UPS's information systems address?
4. What would happen if UPS's information systems were not available?

Let's identify the organization, management, and technology elements in the UPS package tracking system we have just described. The organization element anchors the package tracking system in UPS's sales and production functions (the main product of UPS is a service—package delivery). It specifies the required procedures for identifying packages with both sender and recipient information, taking inventory, tracking the packages en route, and providing package status reports for UPS customers and customer service representatives.

The system must also provide information to satisfy the needs of managers and workers. UPS drivers need to be trained in both package pickup and delivery procedures, as well as how to use the package tracking system so that they can work efficiently and effectively. UPS customers may need some training to use UPS in-house package tracking software, the UPS website, or mobile apps.

UPS's management is responsible for monitoring service levels and costs, and for promoting the company's strategy of combining low cost and superior service. Management decided to use automation to increase the ease of sending a package via UPS and checking its delivery status, thereby reducing delivery costs and increasing sales revenues.

The technology supporting this system consists of handheld computers, bar code scanners, wired and wireless communications networks, desktop computers, UPS's central computer, storage technology for the package delivery data, UPS in-house package tracking software (and other software driving operations such as ORION), and software to access the web. The result is an information system solution to the business challenge of providing a high level of service with low prices in the face of mounting competition.

1.7 Understand the Importance of IoT, Big Data, Cloud Computing, and AI

Among all of these technologies, four developments are especially important today. They are the Internet of Things (IoT), big data, cloud computing, and artificial intelligence (AI). We will take a more detailed look at these in the following section.

The **Internet of Things (IoT)** describes a network of physical objects—"things"—that are embedded with sensors, software, and other technologies enabling them to connect and exchange data with other devices and systems via the Internet. These devices range from ordinary household objects to sophisticated industrial tools. The number of connected IoT devices is projected to reach nearly 22 billion by 2025 (Vailshery, 2023). Digitally linking these devices gives them enough intelligence to make simple decisions and remember particular patterns and routines to be carried out without any human involvement. IoT data is also a major source of data for systems with more sophisticated functionality such as forecasts, predictions, and optimizations.

These connected IoT devices, as well as data from websites and other sources, generate **big data**: large quantities of data (typically characterized by their volume, variety, and the velocity at which they are collected) that can provide useful information but that are often too vast and complex to capture, store, and process using traditional methods.

Internet of Things (IoT)

Network of physical objects (things) embedded with sensors, software, and other technologies enabling them to connect and exchange data with other devices and systems via the Internet.

Big data

Data sets typically characterized by the 3Vs (the extreme volume of data, the wide variety of data, and the velocity at which data must be processed) and which are beyond the ability of a typical DBMS to capture, store, and analyze.

Cloud computing
Model of computing in which processing, storage, software, and other services are provided as a shared pool of resources that can be accessed via the Internet.

Artificial intelligence (AI)
Effort to develop computer systems that are able to perform tasks normally requiring human intelligence.

Machine learning (ML)
Type of AI focused on building computer systems that can learn and improve on their own by using mathematical models (algorithms) to identify patterns and relationships in very large amounts of data.

Generative AI (GenAI)
Tools such as ChatGPT, Google's Gemini, and Microsoft Copilot that are able to identify patterns and structures within existing data to generate new and original content.

Cloud computing is a model of computing in which processing, storage, software, and other services are provided as a shared pool of resources that can be accessed via the Internet. Organizations and individuals can access these resources any time and at any place using an Internet-connected device such as a desktop, laptop, or mobile device. Cloud computing makes it possible for organizations to acquire much more analytical and processing power to expand the reach of their systems. It provides a more robust IT infrastructure than what is typically available on-site for artificial intelligence applications and the management and analysis of big data.

Artificial intelligence (AI) refers to computer systems that are able to perform tasks normally requiring human intelligence, such as speech recognition, visual perception, recognizing patterns, making predictions, writing essays, and learning from past experience.

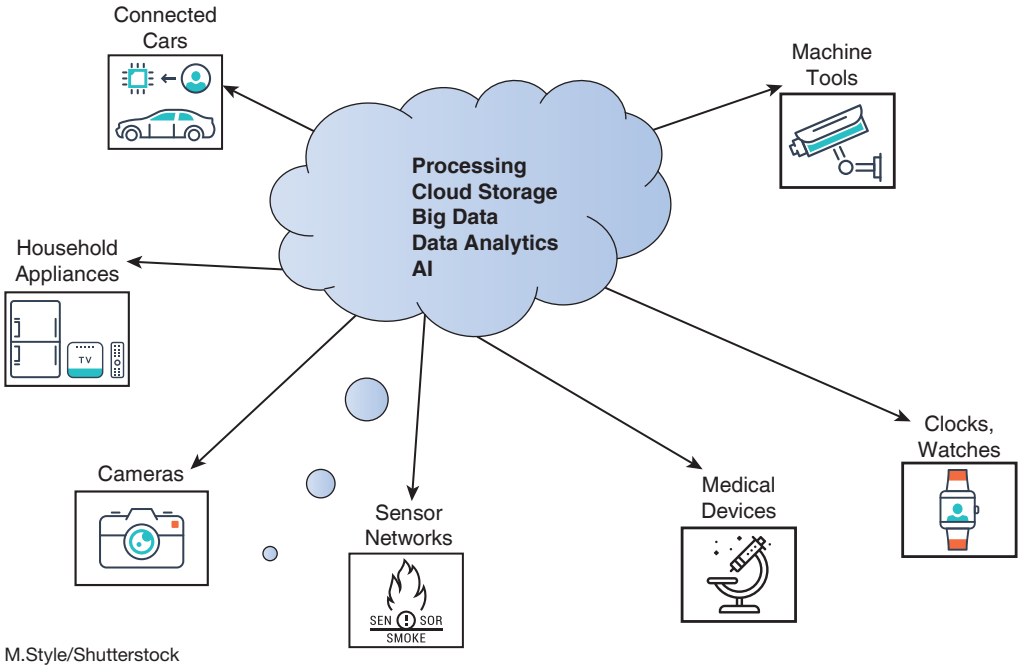
Machine learning (ML) is a type of AI that is focused on building computer systems that can learn and improve on their own. Machine learning uses mathematical models (algorithms) to identify patterns and relationships in very large amounts of data.

Although AI today cannot replicate everything humans do, it is still having a transformative impact. AI is revolutionizing everyday operations and decision-making in industries such as manufacturing, transportation, healthcare, retail, financial services, and education. Search engines, self-driving cars, industrial robots, personalized recommendation systems for products and services, fraud detection systems, and smart homes and cities are just some of the applications where AI can substitute for and even extend the power of humans. Decisions which used to take months because of manual processing and data inputs are now carried out in a few seconds with machines that measure, identify patterns, make predictions, understand and communicate in natural language, categorize requirements, and indicate risks with phenomenal speed, accuracy, and precision.

Generative AI (GenAI) tools such as ChatGPT, Google's Gemini (formerly called Bard), and Microsoft's Copilot are able to identify patterns and structures within existing data to generate new and original content, including textual responses to questions, articles, essays, fiction, and poetry, new software programs, musical compositions, and successful answers to test questions. AI has become such an important topic that we have an entire chapter devoted to AI and AI applications. We will also provide many examples of how organizations are using all of these technologies throughout the text.

Examine Figure 1.7, which illustrates the relationship between cloud computing, big data, AI, and IoT. It shows different kinds of IoT devices, including consumer appliances, sensors, medical devices, and connected cars, which generate millions and trillions of data points: big data.

Figure 1.7 Cloud Computing, Big Data, Artificial Intelligence, and the Internet of Things (IoT)
Using remote cloud computing centers has made it possible to link large numbers of many different kinds of devices via the Internet so that they can exchange data. Cloud computing has also enabled organizations to create more robust IT infrastructures for managing and analyzing big data and AI applications.



M.Style/Shutterstock

The devices transmit data via the Internet to a remote cloud computing center for storage, processing, and analysis. The results might include analysis of X-ray images for signs of cancer, directions for a connected car to turn right, or predictions of when a part in a computer-controlled machine tool might fail before the event occurs. Cloud computing centers have the affordable high-performance computing power to process big data and AI applications efficiently and make big data available to AI systems.

1.8 Understand Why Complementary Assets Are Essential

Managers and business firms invest in information technology and systems because they provide real economic value to the business. The decision to build or maintain an information system assumes that the returns on this investment will be superior to other investments in buildings, machines, or other assets. These superior returns will be expressed as increases in productivity, increases in revenues (which will increase the firm's stock market value), or perhaps as superior long-term strategic positioning of the firm in certain markets (which will produce superior revenues in the future).

We can see that from a business perspective, an information system is an important instrument for creating value for the firm. Information systems enable the firm to increase its revenue or decrease its costs by providing information that helps managers make better decisions or that improves the execution of business processes. For example, the information system for analyzing supermarket checkout data illustrated in Figure 1.3 can increase firm profitability by helping managers make better decisions as to which products to stock and promote in retail supermarkets.

The business perspective calls attention to the organizational and managerial nature of information systems. An information system represents an organizational and management solution, based on information technology, to a challenge or problem posed by the environment. Every chapter in this book begins with a short case study that illustrates this concept. The Opening Case Study diagram illustrates the relationship between a business challenge and resulting management and organizational decisions to use IT as a solution to challenges generated by the business environment. You can use this diagram as a starting point for analyzing any information system or information system problem you encounter.

Review the Opening Case Study diagram. The diagram shows how ConocoPhillips tackled business challenges created by operations in remote locations and outdated equipment. The diagram also illustrates the management, organization, and technology elements that together helped ConocoPhillips create an information system solution using 3D printing and AI/ML for those challenges.

Complementary Assets: Organizational Capital and the Right Business Model

Awareness of the organizational and managerial dimensions of information systems can help us understand why some firms achieve better results from their information systems than others. Studies of returns from information technology investments show that there is considerable variation in the returns firms receive. Some firms invest a great deal and receive a great deal; others invest an equal amount and receive few returns. Still, other firms invest little and receive much, whereas others invest little and receive little. This suggests that investing in information technology does not by itself guarantee good returns. What accounts for this variation among firms?

The answer lies in the concept of complementary assets. Information technology investments alone cannot make organizations and managers more effective unless they are accompanied by supportive values, structures, and behavior patterns in the organization and other complementary assets. Business firms need to change how they do business before they can really reap the advantages of new information technologies.

Table 1.3 Complementary Organizational, Managerial, and Social Assets Required to Optimize Returns From Information Technology Investments

Organizational assets	Supportive organizational culture that values efficiency and effectiveness
	Appropriate business model
	Efficient business processes
	Decentralized authority
	Distributed decision-making rights
	Strong IS development team
Managerial assets	Strong senior management support for technology investment and change
	Incentives for management innovation
	Teamwork and collaborative work environments
	Training programs to enhance management decision skills
	Management culture that values flexibility and knowledge-based decision making
Social assets	The Internet and telecommunications infrastructure
	IT-enriched educational programs raising labor force computer literacy
	Standards (both government and private sector)
	Laws and regulations creating fair, stable market environments
	Technology and service firms in adjacent markets to assist implementation

Complementary assets
Additional assets required to derive value from a primary investment.

Complementary assets are those assets required to derive value from a primary investment. For instance, to realize value from automobiles, substantial complementary investments in highways, roads, gasoline stations, repair facilities, and a legal regulatory structure are required to set standards and control drivers.

Research indicates that firms that support their technology investments with investments in complementary assets, such as new business models, new business processes, management behavior, organizational culture, or training, receive superior returns, whereas firms failing to make these complementary investments receive less or no returns on their information technology investments. The desire of firms to reap benefits from new AI technologies has raised renewed interest in this topic (Berg et al., 2023; Brynjolfsson et al., 2020; Laudon, 1974).

Table 1.3 lists the major complementary investments that firms need to make to realize value from their information technology investments. Some of this investment involves tangible assets, such as buildings, machinery, and tools. However, the value of investments in information technology depends to a large extent on complementary investments in management and organization.

Key organizational complementary investments are a supportive business culture that values efficiency and effectiveness, an appropriate business model, efficient business processes, decentralization of authority, highly distributed decision rights, and a strong information system (IS) development team.

Important managerial complementary assets include strong senior management support for change, incentive systems that monitor and reward individual innovation, an emphasis on teamwork and collaboration, training programs, and a management culture that values flexibility and knowledge.

Important social investments (not made by the firm but by the society at large, other firms, governments, and other key market actors) are the Internet and the supporting Internet culture, educational systems, network and computing standards, regulations and laws, and the presence of technology and service firms.

Throughout the book, we emphasize a framework of analysis that considers technology, management, and organizational assets and their interactions. Perhaps the single most important theme in the book, reflected in case studies and exercises, is that managers need to consider the broader organization and management dimensions of information systems to understand current problems as well as to derive substantial above-average

returns from their information technology investments. As you will see throughout the text, firms that can address these related dimensions of the IT investment are, on average, richly rewarded.

1.9 Describe Different Approaches Used to Study Information Systems

The study of information systems is a multidisciplinary field. No single theory or perspective dominates. Examine Figure 1.8, which illustrates the major disciplines that contribute problems, issues, and solutions in the study of information systems. In general, the field can be divided into technical and behavioral approaches. Information systems are sociotechnical systems. Though they are composed of machines, devices, and "hard" physical technology, they require substantial social, organizational, and intellectual investments to make them work properly.

Technical Approach

The technical approach to information systems emphasizes mathematically-based models to study information systems, as well as the physical technology and formal capabilities of these systems. The disciplines that contribute to the technical approach are computer science, management science, and operations research.

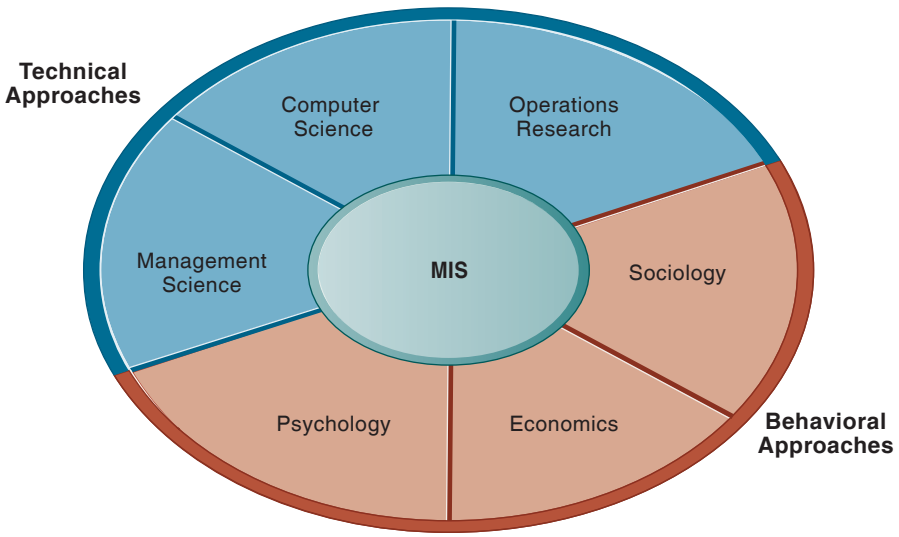
Computer science is concerned with establishing theories of computability, methods of computation, and methods of efficient data storage and access. Management science emphasizes the development of models for decision-making and management practices. Operations research focuses on mathematical techniques for optimizing selected parameters of organizations, such as transportation, inventory control, and transaction costs.

Behavioral Approach

An important part of the information systems field is concerned with behavioral issues that arise in the development and long-term maintenance of information systems. Issues such as strategic business integration, design, implementation, utilization, and management cannot be

Figure 1.8 Contemporary Approaches to Information Systems

The study of information systems deals with issues and insights contributed from technical and behavioral disciplines.



explored usefully with the models used in the technical approach. Other behavioral disciplines contribute important concepts and methods.

For instance, sociologists study information systems with an eye toward how groups and organizations shape the development of systems and also how systems affect individuals, groups, and organizations. Psychologists study information systems with an interest in how human decision makers perceive and use formal information. Economists study information systems with an interest in understanding the production of digital goods, the dynamics of digital markets, and how new information systems change the control and cost structures within the firm.

The behavioral approach does not ignore technology. Indeed, information systems technology is often the stimulus for a behavioral problem or issue. But the focus of this approach is generally not on technical solutions. Instead, it concentrates on changes in attitudes, management and organizational policy, and behavior.

Approach of This Text: Sociotechnical Systems

Throughout this book, you will find a rich story with four main actors: suppliers of hardware and software (the technologists); business firms making investments and seeking to obtain value from the technology; managers and employees seeking to achieve business value (and other goals); and the contemporary legal, social, and cultural context (the firm's environment). Each of these actors plays an important role in management information systems.

The study of management information systems (MIS) arose to focus on the use of computer-based information systems in business firms and government agencies. MIS combines the work of computer science, management science, and operations research with a practical orientation toward developing system solutions to real-world problems and managing information technology resources. It is also concerned with behavioral issues surrounding the development, use, and impact of information systems, which are typically discussed in the fields of sociology, economics, and psychology.

Our experience as academics and practitioners leads us to believe that no single approach effectively captures the reality of information systems. The successes and failures of information systems are rarely all technical or all behavioral. Our best advice to students is to understand the perspectives of many disciplines. Indeed, the challenge and excitement of the information systems field is that it requires an appreciation and tolerance of many different approaches.

Sociotechnical view

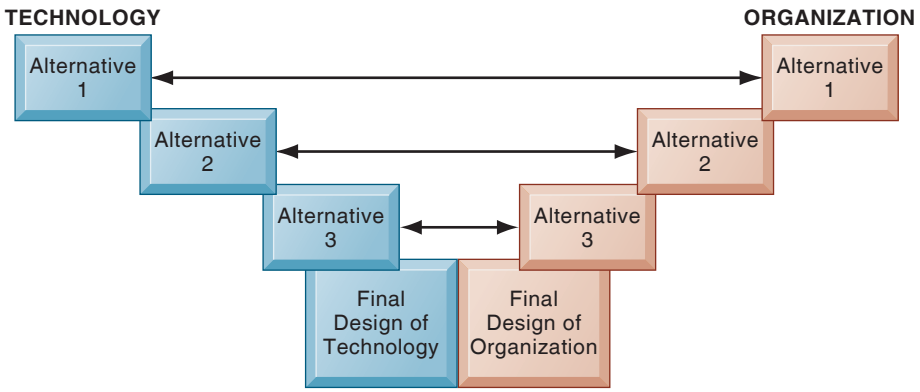
Viewing systems as being composed of both technical and social elements.

The view we adopt in this book is best characterized as the **sociotechnical view** of systems. In this view, optimal organizational performance is achieved by jointly optimizing both the social and technical systems used in production. Adopting a sociotechnical systems perspective helps to avoid a purely technological approach to information systems. For instance, the fact that information technology is rapidly declining in cost and growing in power does not necessarily or easily translate into productivity enhancement or bottom-line profits. The fact that a firm has recently installed an enterprise-wide financial reporting system does not necessarily mean that it will be used or used effectively. Likewise, the fact that a firm has recently introduced new business procedures and processes does not necessarily mean employees will be more productive in the absence of investments in new information systems to enable those processes.

In this book, we stress the need to optimize the firm's performance as a whole. Both the technical and behavioral components need attention. This means that technology must be changed and designed in such a way as to fit organizational and individual needs. Sometimes, the technology may have to be "de-optimized" to accomplish this fit. For instance, mobile device users adapt this technology to their personal needs, and as a result, manufacturers quickly seek to adjust the technology to conform to user expectations. Organizations and individuals must also be changed through training, learning, and planned organizational change to allow the technology to operate and prosper. Examine Figure 1.9, which illustrates this process of mutual adjustment in a sociotechnical system.

Figure 1.9 A Sociotechnical Perspective on Information Systems

In a sociotechnical perspective, the performance of a system is optimized when both the technology and the organization mutually adjust to one another until a satisfactory fit is obtained.



1.10 Understand How This Book Prepares You for the Future

Business firms are looking for candidates who have a broad range of problem-solving skills: the ability to read, write, and present ideas, as well as the technical and business expertise required for specific tasks. Regardless of your major or your future occupation, information systems and technologies will play an important and expanding role in your day-to-day work and your career. Your career opportunities and your compensation will, in part, depend on your ability to help business firms use information technology to achieve their objectives.

This book is explicitly designed to prepare you for your future career. It provides the necessary knowledge and foundational concepts for understanding the role of information systems in organizations. You will be able to use this knowledge to identify opportunities for increasing the effectiveness and profitability of a business. You will learn how to use information systems to improve operations, create new products and services, improve decision making, increase customer intimacy, and promote competitive advantage.

You'll also learn how to use technology to produce business value. This book will help you understand and demonstrate the value of an IT solution for the business rather than just its technical details. The ability to question and analyze the value a solution provides to the business is very important. You should be able to help your organization answer the question, "Why are we doing this, and what's the value to the business?"

Understanding Business Applications of Technology

This book enhances your ability to use information systems to solve problems that you will encounter on the job. You will learn how to analyze and define a business problem, and how to design an appropriate information system solution. Using technology successfully in your job will make you more productive in general. The information in this book is valuable for both future IT and non-IT specialists.

It has been said that in the business world, problems are not just problems; they are also opportunities—opportunities for the organization to refocus and improve, and also opportunities for you to demonstrate your IT and business expertise and problem-solving skills. Assume more responsibility at work, be willing to assist and educate your colleagues, show initiative, offer to help with presentations, and take on assignments where you will learn something new. Gaining experience from dealing with these challenges can help grow your professional knowledge and expertise, which will lead to career advancement.

The following features of the text and the accompanying learning package reinforce problem-solving and career orientation.

A Framework for Describing and Analyzing Information Systems

The text provides you with a framework for describing and analyzing IT-related problems you'll encounter in business. You'll learn how to examine the management, organizational, and technology dimensions of information systems and develop solutions. The framework is used repeatedly throughout the text to help you understand the role of information systems in business and analyze information systems problems.

A Model for Problem Solving

You will learn how to identify a business problem, design alternative solutions, choose the correct solution, and implement the solution. You will be asked to use this problem-solving method to answer the questions posed by the case studies in each chapter. Chapters 13 and 14 will show you how to use this approach to design and build new information systems and determine their business value.

AI and Hands-On MIS Projects for Stimulating Critical Thinking and Problem Solving

Each chapter concludes with a new AI project as well as a series of hands-on MIS projects to sharpen your critical-thinking and problem-solving skills. The hands-on MIS projects include a Management Decision Problem, an application software problem, a project to help build online skills, and a teamwork and collaboration project.

Career Resources

To make sure you know how the text is useful for your future business career, we've added a full set of career resources to help you with career development and job hunting.

- The last major section of each chapter, titled "Understand How the Information in This Chapter Can Help Your Career" includes a real-world scenario showing how a company solved an IT-related business problem related to the concepts in the chapter, as well as the job roles involved in doing so.
- We also provide descriptions of jobs requiring the use of IT, along with typical career paths and salary ranges for each position. We include non-IT as well as IT jobs.

Additional career resources can be found in MyLab MIS. These include:

- **Digital Portfolio:** A template for preparing a digital portfolio to demonstrate the business knowledge, AI skills, application software skills, Internet skills, and analytical skills you have acquired in this course. You can include this portfolio in your résumé or job applications. Your professors can also use the portfolio to assess the skills you have learned.
- **Job-Hunting Tips:** Here are some tips to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, and prepare for job interviews.

1.11 Understand How the Information in This Chapter Can Help Your Career

The information you have learned about in Chapter 1 can help you advance your career by helping you to understand that an information system has managerial and organizational components, as well as technical components, and that all of these components need to work together to provide value for the firm. It also demonstrates why information systems are so essential for running a business today. This chapter is a starting point. You can find additional information on these topics in the rest of the book, as well as in MyLab MIS.

Real-World Scenario

Faciliteq Rebuilds Its Sales Team

Faciliteq is a specialty contractor and distributor that provides commercial office interiors such as modular walls, raised access flooring, and office furniture. The company has offices in Las Vegas, Denver and Phoenix. Until recently, the company was unable to build a sales team across several market areas with diverse targets. The massive amounts of data it was collecting were spread across various locations and different systems, including spreadsheets and separate databases housing inconsistent and conflicting data.

Faciliteq's sales teams had to work with redundant contacts, overlapping decision making and lack of proper communication while following up with the prospects. New salespeople were confused about their roles and responsibilities.

To solve this problem, Faciliteq implemented Salesmate CRM. This solution eliminated overlapping communication among sales reps, who are now able to track all the activities across various stages of their sales pipeline along with the dollar volume at each state, and make more accurate deal forecasts. The system now automatically updates records; assigns and rotates deals; scores leads; and sends out emails to fellow workers and clients.

Job Description and Career Path: Sales Representative

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Sales representative	A sales representative is the principal point of contact between a business and its customers. Sales representatives need to have a strong understanding of the business's products and services. Job responsibilities include: • Servicing customer accounts, ensuring current customers have the right products and services • Identifying prospects and customer leads, makes sales pitches to prospects and customers • Making presentations and demos. • Monitoring orders. • Working in a team.	• CRM systems for identifying contacts, leads, prospects, and profitable customers, team coordination, generating sales quotes, tracking orders, sales analytics, customer account management. • Microsoft Office (spreadsheets for budget and commissions calculations, PowerPoint product presentations, Teams (messaging, online meetings) • Facility with social media	• Sales representative: Base salary, $40,000–$45,000; commission, $2,000–$27,000 • Account manager: Base salary, $55,000–$58,000; commission, $2,000–$37,000 • Sales manager: Base salary, $70,000–$75,000; commission, $5,000–$50,000 • Director of Sales: Base salary, $120,000–$125,000; commission, $8,000–$75,000 Salaries can vary based on factors like location, industry, and company size.

Chapter Case Study

Capital One: A Digital Transformation in the Making

Founded in 1994, Virginia-based Capital One has become a diversified financial holding company offering an array of consumer, retail banking, and credit card services. The company has brought innovation to banking, creating new digital products and services that make the banking experience more personalized and pleasant. The company's intensive use of IT and data-driven insights have played a critical role in fueling its growth, and it continues to innovate solutions and experiences across physical and digital platforms.

Capital One was an early mover in using big data and analytics to understand consumer spending patterns. The company's ability to relentlessly analyze customer preferences and behavior to create products and services personalized for each customer continues to distinguish it from competitors. Capital One credit cards are not just credit cards. Each credit card is a finely tuned instrument for eliciting as much revenue as possible from every individual.

For example, the company may use internal and external information to offer a loan in real-time to a customer in addition to supplying that person's FICO (credit) score, which gives it an edge over its competition. Or, upon review of a customer's spending habits and financial goals, Capital One might recommend a change such as switching to a card that earns fewer rewards but has no annual fee. There are cards that emphasize cash back, charge low fees, build credit scores, and earn luxury travel perks along with options tailored to students, small business owners, or people who spend heavily on groceries and restaurants. So far, the strategy has worked: Capital One is snaring many new young subscribers to banking.

The company was ripe for digital transformation. Digital technology (both online and mobile) is so central to Capital One strategy that the company had to change the way it operated into an organization that delivers like a technology company rather than a traditional bank. This "digital-first" strategy is based on intensive use of analytics, investment in digital talent, structuring the company's IT workforce for rapid software development, and deployment of new innovative personalized banking. Capital One's digital transformation required recruiting new employees with the talent and skills required to build personalized apps, adopting cloud-based IT infrastructure, and implementing agile software development. The company redesigned business processes to make them more digital and customer-oriented and hired new executives to spearhead the changes.

To outpace competition from large powerful banks such as Citibank, Wells Fargo, and Bank of America, Capital One needed to deliver new technology products faster and better. It needed more employees with mobile application development and software engineering skills to write the bank's own unique software, instead of relying on generic third-party apps available to any company. It needed IT teams to work closely with its own business units to fine-tune its information systems and products. And it wanted to run the company on its own flexible IT infrastructure to enable it to quickly develop new applications. Capital One has flipped from outsourcing 70 percent to 75 percent of its technology platform to now operating 70 percent to 75 percent in-house, with a total IT workforce of over 12,000 employees and contractors. The company has been able to build 80 percent of the nearly 2,000 applications it now runs in the cloud from the ground up.

There were also environmental benefits: By moving to cloud computing, the company recycled 41 tons of copper and 62 tons of steel and removed a total of 13.5 million feet of cable from its data centers. It also anticipates cloud migration to save 10 megawatts of power, equivalent to 650,000 LED light bulbs, per year,

Capital One creates its own customer-facing software, such as its online website Capitalone.com, where customers log in to manage their bank and credit card accounts. Capitalone.com is based on a heavily customized, off-the-shelf software product, but its IT department is building an in-house version. The company has similar plans for its smartphone and tablet apps and for an online portal for commercial loan customers.

In November 2020, Capital One announced the closure of all of its on-premises data centers, and was the first U.S. bank to do so. It moved applications and systems to Amazon Web Services (AWS). This cloud-first policy helped Capital One move away from routine technology maintenance to focus on accelerating customer-centric innovation, using AI and machine learning (ML) to turn very large quantities of data into insights.

According to Chris Nims, Capital One senior vice president of cloud and productivity engineering, switching to a cloud IT infrastructure has enabled both instant provisioning of technology resources and more rapid innovation. The company is able to manage big data at a much larger scale and take advantage of AI and ML to deliver enhanced customer experiences.

Migrating to the cloud as quickly and aggressively as Capital One did is unusual among very large Fortune 100 companies, especially those in the heavily regulated financial services industry. However, since its founding, Capital One has defined itself as a disrupter with a mission of changing its industry. Capital One maintains a national banking presence in the United States with close to 56,000 U.S. employees and more than 100 million customers. It operates primarily as a "digital bank," maintaining about 300 branches in only a few states. The company provides over 70,000 fee-free ATMs across the United States for its customers, as well as 50 Capital One Cafés, coffee shops in U.S. cities that offer access to Capital One banking products, customer support, and financial tools and workshops. The cafés serve as community hubs where visitors can connect to free wireless Internet, use the space to work or study, or just grab a cup of coffee.

With banking customers increasingly expecting customized experiences delivered instantly in real time, the Capital One management team realized that the company would need to use the power of big data and ML to enhance the banking experience. Like other banks, Capital One uses ML for classic applications such as credit decisions, and fraud and money laundering detection. But it has gone much further in finding innovative uses for AI tools for every facet of its business, including customer-facing personal assistants, call center operations, and creating more resilient, reliable, and customer-friendly mobile apps.

For example, Capital One launched a natural language chatbot called Eno to answer questions within the Capital One mobile app. Since then, Eno has acquired additional capabilities. It helps customers check their bank balances, track purchases, pay bills online, understand their spending habits, and proactively monitor their accounts to keep them secure. Eno now has tools for fraud detection, automatically alerting customers about potential frauds incidents in real-time as they occur, and if required, locking their card. Eno communicates these decisions proactively and allows customers to respond in their own terms and words. As more and more customers use this tool, they are providing Eno with additional data for training, which in turn improves Eno's algorithms, reducing fraud even further.

Since Capital One's mobile app is an essential part of the firm's strategy for delivering seamless digital services to its customers, the firm used ML to shorten the time to solve mobile app performance problems. It created an ML-based intelligence layer atop its various tools monitoring mobile app failures. The ML layer integrated the data produced by individual monitoring tools and linked the data with appropriate descriptions to help engineers identify a failed component and the reason for its failure. This enabled Capital One to cut the turnaround time for dealing with mobile app failure incidents by up to 50 percent. For Capital One, this was a major competitive advantage.

Capital One's foray into AI required some organizational changes. The company created two larger ML teams for its credit cards line of business. Company executives say that its focus on ML is not just on fraud detection or credit decisions but improving all aspects of customer interaction and operations. As CEO Rob Alexander noted, every time a decision is made, it creates an opportunity to use information technology, including ML, to learn more about customers—which customers to cultivate, what products and rewards to offer them, what spending limits to put in place, and how to identify fraud. It is an ongoing process of continuous improvement.

Digital transformation isn't just about changing technology. To become a technology-driven company like Capital One also requires major changes in organizational processes, culture, leadership style, and mindset. Everyone in the organization needs to cultivate behaviors and attitudes that support the company's business objectives, new ways of working, and culture of continuous improvement. Capital One has redesigned its business for the digital world.

SOURCES: "Technology at Capital One," Capitalone.com, accessed August 9, 2024; Emily Sherman, "Best Capital One Credit Cards," Wall Street Journal, May 28, 2024; John Moore, "Capital One Machine Learning Taps MLOps," Techtarget.com, February 1, 2023; Salim Syed, "How To Succeed in a Data-Driven Digital Transformation," Capitalone.com, June 21, 2022; Deloitte, "At Capital One, a New Era in Customer Experience," CMO Today, Deloitte.wsj.com, January 24, 2022; "Capital One Completes Migration from Data Centers to AWS, Becomes First US Bank to Announce Going All In on the Cloud," Aws.amazon.com, 2020; Tom Davenport and Randy Bean, "From Analytics First to AI First at Capital One," Forbes.com, July 10, 2019.

Case Study Questions

1. Why was Capital One a good candidate for digital transformation?
2. How did Capital One change its business model to make it technology-driven? What management, organizational and technology issues did it address?
3. What role does AI play at Capital One?
4. How does Capital One differ from a traditional bank?
5. Visit the Capital One website and identify all the personalization options that are available for Capital One credit cards.

Review Summary

1.1 Understand how information systems transform business and careers.

Information systems are a foundation for conducting business today. Information systems that incorporate the Internet and mobile devices have become essential tools of business. Information systems that use sophisticated data analytics to analyze massive amounts of data are enabling businesses to track and respond to rapidly changing customer demand, reduce inventories to the lowest possible levels, and achieve higher levels of operational efficiency. Information systems have driven the development of both social networks and social media. Advertising is another business that has been transformed by the development of information systems. AI is the latest information technology that is in the process of transforming business. These technology-driven changes are also transforming jobs and careers. Employees across all industries are seeing more and more technology tools incorporated into their workflows.

1.2 Describe key challenges and opportunities created by new MIS technologies.

What makes management information systems one of the most exciting topics in business today are the challenges and opportunities created by the continual change in technology, management's use of the technology, and the impact this has on business success. Key technology challenges include cloud computing, big data, IoT, AI, and the mobile platform. Key management challenges include return on investment, online collaboration, and social networks. Key organizational challenges include security and privacy, social business, and remote and hybrid work. The level of globalization today, which has been made possible by advances in information technology, introduces its own set of challenges, such as the potential loss of jobs to offshore companies and the need for businesses to understand the potential impact of globalization on the market for its goods and services, as well as opportunities to reduce costs and increase revenues.

1.3 Describe key features of a digital firm.

A digital firm is one in which the organization's significant business relationships with customers, suppliers, and employees are digitally enabled and mediated. Core business processes are accomplished through digital networks spanning the entire organization or linking multiple organizations. Key corporate assets are managed through digital means. A digital firm can sense and respond to its environment far more rapidly than a traditional firm, giving it more flexibility to survive in turbulent times. Firms often undertake digital transformation initiatives in the effort to become a digital firm.

1.4 Describe the business objectives of information systems.

Businesses use information systems to achieve six major objectives: operational excellence, new products, services, and business models, customer/supplier intimacy, improved decision making, competitive advantage, and day-to-day survival. In addition, companies today are increasingly concerned about the environmental impact of their operations and sustainability, as well as proper governance of the organization and the responsible use of AI. Such concerns are becoming a strategic business imperative, and information systems are central in enabling companies to meet their goals.

1.5 Explain what an information system is and how it works.

From a technical perspective, an information system collects, stores, and disseminates information from an organization's environment and internal operations to support organizational functions and decision-making, communication, coordination, control, analysis, and visualization. Information systems transform raw data into useful information through three basic activities: input, processing, and output. From a business perspective, an information system provides a solution to a problem or challenge facing a firm.

1.6 Describe the organizational, management, and technology dimensions of information systems.

An information system represents a combination of organization, management, and technology elements. The organizational dimension of information systems involves issues such as the organization's hierarchy, functional specialties, business processes, culture, and political interest groups. The management dimension of information systems involves issues such as training, job attitudes, and management behavior. The technology dimension consists of computer hardware, software, data management technology, and networking/telecommunications technology, including the Internet.

1.7 Understand the importance of IoT, big data, cloud computing, and AI.

The Internet of Things (IoT) is a network of Internet-connected devices (sensors) which, along with data from web traffic, email messages, and social media content, generates very large quantities of data (big data) that can be analyzed to improve operations and decision making. Cloud computing, using remote computing centers linked via the Internet, provides a source of computing power large enough to handle big data and AI applications. AI systems can perform tasks normally requiring human intelligence, such as speech recognition, visual perception, recognizing patterns, making predictions, generating content, and learning from experience.

1.8 Understand why complementary assets are essential.

In order to obtain meaningful value from information systems, organizations must support their technology investments with appropriate complementary investments in organizations and management. These complementary assets include new business models and business processes, supportive organizational culture and management behavior, and appropriate technology standards, regulations, and laws. New information technology investments are unlikely to produce high returns unless businesses make the appropriate managerial and organizational changes to support the technology.

1.9 Describe different approaches used to study information systems.

The study of information systems deals with issues and insights contributed from technical and behavioral disciplines. The disciplines that contribute to the technical approach, focusing on formal models and capabilities of systems, are computer science, management science, and operations research. The disciplines contributing to the behavioral approach, focusing on the design, implementation, management, and business impact of systems, are psychology, sociology, and economics. A sociotechnical view of systems considers both technical and social features of systems and solutions that represent the best fit between them.

Key Terms

Artificial intelligence (AI), 20
Big data, 19
Business functions, 16
Business model, 10
Business processes, 9
Cloud computing, 20
Complementary assets, 22
Computer hardware, 17
Computer literacy, 14
Computer software, 17
Culture, 16
Data, 12
Data management technology, 17
Data workers, 16
Digital firm, 8
Digital transformation, 9
Extranet, 17
Feedback, 13
Generative AI (GenAI), 20
Globalization, 8
Information, 12
Information system (IS), 12
Information systems literacy, 14
Information technology (IT), 12
information technology infrastructure (IT infrastructure), 17
Input, 12
Internet, 17
Internet of Things (IoT), 19
Intranet, 17
Knowledge workers, 15
Machine learning (ML), 20
Management information systems (MIS), 14
Middle management, 15
Network, 17
Networking and telecommunications technology, 17
Operational management, 15
Output, 13
Processing, 12
Production and service workers, 16
Senior management, 15
Sociotechnical view, 24
World Wide Web, 17

Review Questions

1.1 Understand how information systems transform business and careers.

- Identify three ways information systems are transforming business.
- Describe the demand for digital skills in the U.S. labor market.

1.2 Describe key challenges and opportunities created by new MIS technologies.

- Describe the key challenges created by new technologies used in management information systems.
- Define globalization, and describe challenges and opportunities it creates.

1.3 Describe key features of a digital firm.

- Describe the characteristics of a digital firm.
- Describe digital transformation.

1.4 Describe the business objectives of information systems.

- List and describe six reasons why information systems are so important for businesses today.
- Explain what governance relates to, and why it has become important for a company to show leadership with respect to governance issues, as well as sustainability and environmental impact.

1.5 Explain what information system is and how it works.

- Define an information system and describe the activities it performs.
- Explain how data differs from information.
- Define information system literacy and explain how it differs from computer literacy.

1.6 Describe the organizational, management, and technology dimensions of information systems.

- List and describe the organizational, management, and technology dimensions of information systems.
- Explain how the Internet and the World Wide Web are related to the other technology components of information systems.
- Describe the role IT infrastructure plays in a firm.

1.7 Understand the importance of IoT, big data, cloud computing, and AI.

- Define the Internet of Things, and explain why it is important.
- Define big data and explain why it is important.
- Define cloud computing and explain why it is important.
- Define artificial intelligence and explain why it is important.

1.8 Understand why complementary assets are essential.

- Define complementary assets and describe their relationship to information technology.
- Describe the complementary social, managerial, and organizational assets required to optimize returns from information technology investments.

1.9 Describe different approaches used to study information systems.

- List and describe each discipline that contributes to a technical approach to information systems.
- List and describe each discipline that contributes to a behavioral approach to information systems.
- Describe the socio-technical perspective on information systems.

Discussion Questions

1.10 Information systems are too important to be left to computer specialists. Do you agree? Why or why not?

1.11 If you were setting up the website for a new minor league baseball team, what management, organizational, and technology issues might you encounter?

1.12 What are some of the organizational, managerial, and social complementary assets that help make UPS's information systems so successful?

Using AI Project

1.13 AI Project: Selecting a Business Name

You have been hired as a consultant by two friends who want to open a retail store selling organic pet food and treats, as well as pet toys and clothing made from organic materials. They have asked you to advise them on a good name for their business. Using Microsoft's Copilot, Google's Gemini, or ChatGPT, develop a prompt that suggests some appropriate names for the business. Select the name that you believe to be the best and explain why this is a good choice. Save a copy of your prompt, the output of the chat session, and your decision and rationale with respect to your choice of name. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience in analyzing a visitor management problem at a busy commercial and industrial complex, using spreadsheet software to improve payroll reporting operations, using the Internet to research job requirements, and choosing team collaboration tools.

For student files please visit www.pearsonhighered.com/laudon or MyLab MIS.

1.14 Management Decision Problem

The Brooklyn Navy Yard (BNY) is a shipyard and industrial complex located in Brooklyn, New York. It encompasses a 300-acre waterfront site for more than 450 businesses employing more than 11,000 people and generating more than $2.5 billion per year in economic impact for New York City. The BNY needs to keep track of its thousands of visitors and nearly 10,000 employees per day, but it had been unable to do so efficiently and safely. Its internally hosted web portal for managing visitors was slow and not very user-friendly and was unable to scale to handle the many thousands more visitors projected for the future. Most BNY tenants were not pre-registering their visitors or delivery drivers, causing long lines and traffic at the security gates. Security checks were performed manually by checking IDs

against a printed watch list. BNY was unable to create a welcoming experience for visitors and employees. Identify BNY's problem and analyze its business impact. What business functions were affected? Is it a management problem, an organizational problem, or a technology problem? Suggest an information system solution and identify some implementation challenges that the solution would have to address.

1.15 Using Spreadsheet Software to Create a Payroll Register

Spreadsheet software skills: Formulas, absolute and relative addressing, worksheet formatting, SUM, AVERAGE.

Business skills: Payroll accounting

In this project, you will create a payroll register for the eight senior executives of a machine tool company with 8,500 employees. All employees are paid through the firm's automated payroll system except for the eight members of the company's executive steering committee, including the CEO. The payroll system for the steering committee is produced separately on a spreadsheet because it contains highly confidential information about senior executives' salaries. This group of executives is paid monthly.

The MIS18_Project 1.15 Student File displays the names of the executive steering committee members, their Social Security numbers, and annual salaries. Develop a worksheet that creates a payroll register report for these employees for the first month of the year (January 2025). The worksheet should automatically calculate monthly gross pay, year-to-date gross pay, net pay, and all deductions. These deductions include federal income tax withholding (25 percent of gross pay), state income tax withholding (5 percent of gross pay), FICA (Social Security) (6.2 percent of gross pay), Medicare (1.45 percent of gross pay), group health insurance ($600 per month), and monthly contributions to a profit-sharing plan (5 percent of gross pay). It should also provide totals for each of these categories for the current pay period. Create an "Assumptions" section for all deductions and other variables in the upper left of the worksheet so you can easily make changes in deductions and formulas using the addressing function of the spreadsheet software. Listing variables also allows all assumptions to be clearly visible and reported. Use formulas to calculate monthly gross pay, year-to-date gross pay, all deductions, and net pay. Be sure these formulas reference the appropriate cells in the "Assumptions" section of your worksheet (e.g., A56) rather than actual values (e.g., 20 percent). Use the SUM function to provide totals for gross pay, net pay, and each deduction category. Use the AVERAGE function to calculate the average annual salary paid to these top eight executives.

1.16 Using The Internet to Research Jobs

Software skills: Online research

Business skills: Job searching

Visit a job search website/app such as Monster.com or Indeed.com. Spend some time examining jobs for accounting, finance, sales, marketing, and human resources. Find two or three descriptions of jobs that require some information systems knowledge. What information systems knowledge do these jobs require? What do you need to do to prepare for these jobs? Write a one- to two-page report summarizing your findings.

1.17 Collaboration and Teamwork Project: Selecting Team Collaboration Tools

Form a team with two or more of your classmates and review the capabilities of Google Drive and Google Sites for your team collaboration work. Compare the capabilities of these two tools for storing team documents, project announcements, source materials, work assignments, illustrations, presentations, and web pages of interest. Learn how each works with Google Docs. Explain why Google Drive or Google Sites is more appropriate for your team. If possible, use Google Docs to brainstorm and develop a presentation of your findings for the class. Organize and store your presentation using the Google tool you have selected.

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2

Global E-business and Collaboration

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Learning Objectives

After completing this chapter, you will be able to:

2.1 Understand how business processes relate to information systems.

2.2 Explain how information systems serve different management groups.

2.3 Explain how enterprise applications improve firm performance.

2.4 Understand the importance of collaboration and social business systems.

2.5 Describe collaboration tools.

2.6 Describe social business tools.

2.7 Understand how to evaluate collaboration and social business tools.

2.8 Describe knowledge management systems.

2.9 Describe the role of the information systems function.

2.10 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** Microsoft Teams and AI Help Toyota Motor North America Do Even Better
- **Spotlight on Technology Case Study:** AI Accelerates Drug Discovery and Development
- **Chapter Case Study:** How Much Do Virtual Meetings Help Collaboration?

Opening Case Study

Microsoft Teams and AI Help Toyota Motor North America Do Even Better

Toyota Motor North America (TMNA) oversees all operations of the Toyota Motor Corporation in the United States, Canada, and Mexico. Its operations include research and development, manufacturing, sales, marketing, and after-sales and corporate functions. Originally, TMNA operated separate headquarters for sales and manufacturing, but in 2015, brought them together into one headquarters in Plano, Texas, which now oversees 14 sales regions, 14 manufacturing plants, 1,500 dealerships, and more than 49,000 employees.

Toyota is known for its reliable vehicles and its manufacturing principles and management philosophy of continuous improvement and respect for people. The Toyota Way always looks for ways to improve operations, innovate, and collaborate. New digital tools are empowering employees to work more efficiently by promoting teamwork, gradual improvement of everyday business processes, and increased engagement.

As Jason Ballard, TMNA's general manager for infrastructure and operations services notes, Toyota has been very focused on creating a common technology stack so that its team members can be more efficient. (A technology stack consists of all the technology services used to build and run an application or website.) For instance, prior to the Covid-19 pandemic, it had begun the process of slowly implementing Microsoft Teams via a pilot project. Microsoft Teams is a business communication and collaboration platform developed by Microsoft as part of the Microsoft 365 family of products. Teams has capabilities for workspace chat, audio and videoconferencing, screen sharing, online meetings, file storage, and communication through the public switched telephone network. Teams has channels, which are conversation boards for teammates. All team members can view and add to different conversations for which they are authorized. Chat can take place between teams, groups, and individuals.

When the Covid-19 pandemic shut TMNA's factories and forced office workers to work from home, TMNA quickly accelerated the deployment of Microsoft Teams to its entire workforce to keep the company running in a safe work environment. Adopting Teams helped TMNA create a more uniform and secure work environment, replacing ad hoc videoconferencing tools such as Zoom that had been adopted unofficially by different groups in the organization with a single software platform. TMNA's implementation of Teams included Teams Phone, which enables employees to make, receive, and transfer phone calls from landlines and mobile devices over the public switched telephone network, all within Teams. IT support employees can now contact the right people to respond to problems more quickly. Previously, they struggled to reach colleagues using a global address book.

After a large number of office-based employees started working remotely, TMNA used Teams for all of its meetings and collaboration activities. For example, a global meeting on enterprise resource planning used Teams live events to bring together Toyota representatives from six continents to share presentations and collaborate on best practices. Before adopting Teams, it would have taken TMNA months to schedule a meeting with representatives from every continent, and the cost would have been very high.

Employees throughout the company used Teams to work together virtually and drive productivity. Before the Covid-19 pandemic, high-level managers inspected manufacturing plants in person. After the onset of the pandemic, inspections went virtual. Senior VPs and executives had daily Teams meetings with plant floor members where they could actually see what was taking place on the factory floor. Each plant had a cart with an iPad

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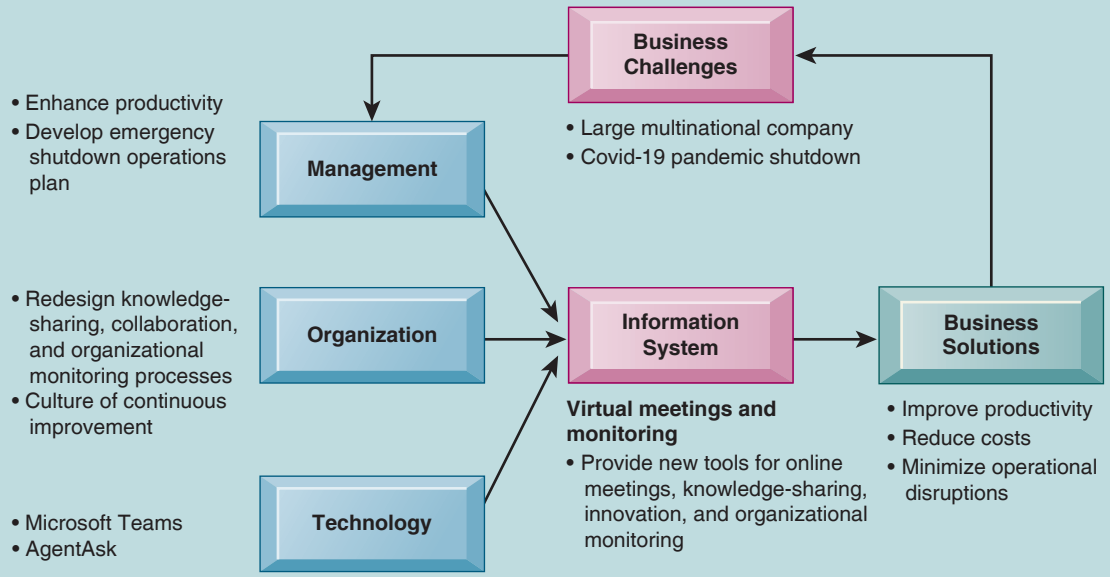


and external headset that was moved around. For instance, TMNA used Teams meetings to show a new process that had been implemented in one of its plants in action for the entire executive team, which felt the Teams presentation was more efficient and cost-effective than an in-person meeting.

TMNA has also recently integrated a generative AI tool, named AgentAsk, into Teams. AgentAsk is a ChatGPT-like tool that can help employees find answers to common questions and issues. Created in partnership with AI platform specialist Moveworks, AgentAsk was able to be integrated into Teams within a matter of days. Use of AgentAsk initially focused on IT service desk requests, where its "mean time to repair" (i.e., solve an issue) is 11.4 minutes, compared to an industry average of three days. TMNA is now encouraging employees to use AgentAsk for other purposes, and they are increasingly turning to it for questions that involve finance, legal, travel and expenses, and more. For example, TMNA general manager Ballard reports that in the past, he'd had to spend a lot of time accessing various enterprise systems to identify the number of outstanding approvals he had to handle. Now, in contrast, he can just access AgentAsk through Microsoft Teams and get the answer in a matter of seconds. TMNA has found that the combination of AgentAsk with Teams has given employees and service desk agents back more than 70,000 hours of productivity in the past year.

SOURCES: Toyota Motor North America, "Toyota Motor North America Reports 2023 U.S. Sales Results," Prnewswire.com, January 3, 2024; Thor Olavsrud, "Toyota Transforms IT Service Desk with Gen AI," Cio.com, December 8, 2023; Grant McWhirter, "Toyota Motor North America Migrates 40,000 to Microsoft Teams in Three Weeks," Digital-first.com.au, September 15, 2022; "Driving Employee Engagement with Microsoft Teams Accelerates Business at Toyota Motor North America," Customers.microsoft.com, accessed July 7, 2022.

TMNA's experience illustrates how much organizations today rely on information systems to improve their performance and remain competitive. It also shows how much systems supporting collaboration and teamwork make a difference in an organization's ability to innovate, execute, grow, profit, and survive.

The Opening Case Study diagram above calls attention to important points raised by this case and this chapter. TMNA is known for its focus on quality, efficiency, continuous improvement, and employee well-being. Corporate culture encourages everyone in the company to do their best and to do better every day. The challenge was to continue to operate a very large multinational company following these principles when the Covid-19 pandemic forced factories and offices to shut down worldwide. This prevented employees and managers in this far-flung company from freely sharing information and performing job tasks, impacting production, sales, and earnings.

TMNA management found that the best solution was to deploy new technology to help the entire workforce to operate remotely. The company adopted Microsoft Teams communication and collaboration tools to increase employee collaboration and engagement, and to monitor operations from afar. Virtual meeting and messaging tools actively engaged employees and enabled them to obtain more knowledge from colleagues and managers. TMNA also integrated a new generative AI tool, AgentAsk, into its Microsoft Teams platform. There is now more effective sharing of employee knowledge and enhanced productivity.

New technology alone would not have solved TMNA's problem. To make the solution effective, TMNA had to change its business processes for knowledge dissemination and collaborative work, and the new technology made these changes possible.

Case Study Questions

1. How did Microsoft Teams make it possible for TMNA workers to work remotely?
2. How effective was Teams in promoting collaboration and employee engagement?
3. How has AgentAsk improved productivity at TMNA?

CAREER BENEFIT

Working on this project provided TMNA's IT staff with opportunities to develop expertise in team communication, collaboration, and AI tools, making them more qualified for higher paying jobs at TMNA or other companies. Non-IT staff also benefited from contributing to the development of the solution as well as developing expertise in using communication, collaboration, and generative AI tools.

2.1 Understand How Business Processes Relate to Information Systems

In order to operate, businesses must deal with many different pieces of information about suppliers, customers, employees, invoices, payments, and of course their products and services. They must organize work activities that use this information to operate efficiently and enhance the overall performance of the firm. Information systems make it possible for firms to manage all their information, make better decisions, and improve the execution of their business processes.

Business Processes

Business processes, which were introduced in Chapter 1, refer to the way work is organized, coordinated, and focused to produce a valuable product or service. Business processes are the collection of activities required to produce a product or service. These activities are supported by flows of material, information, and knowledge among the participants in business processes. Business processes also refer to the unique ways in which organizations coordinate work, information, and knowledge, and the ways in which management chooses to coordinate work.

The performance of a business firm largely depends on how well its business processes are designed and coordinated. A company's business processes can be a source of competitive strength if they enable the company to innovate or to execute better than its rivals. Business processes can also be liabilities if they are based on inefficient ways of working that impede organizational responsiveness and efficiency. The Opening Case Study describing TMNA's improvements in knowledge-sharing processes clearly illustrates these points, as do many of the other cases in this text.

Every business can be seen as a collection of business processes. Some of these processes are part of larger, encompassing processes. Many business processes are tied to a specific functional area. For example, the sales and marketing function is responsible for identifying customers, and the human resources function is responsible for hiring employees. Table 2.1 describes some typical business processes for each of the functional areas of business.

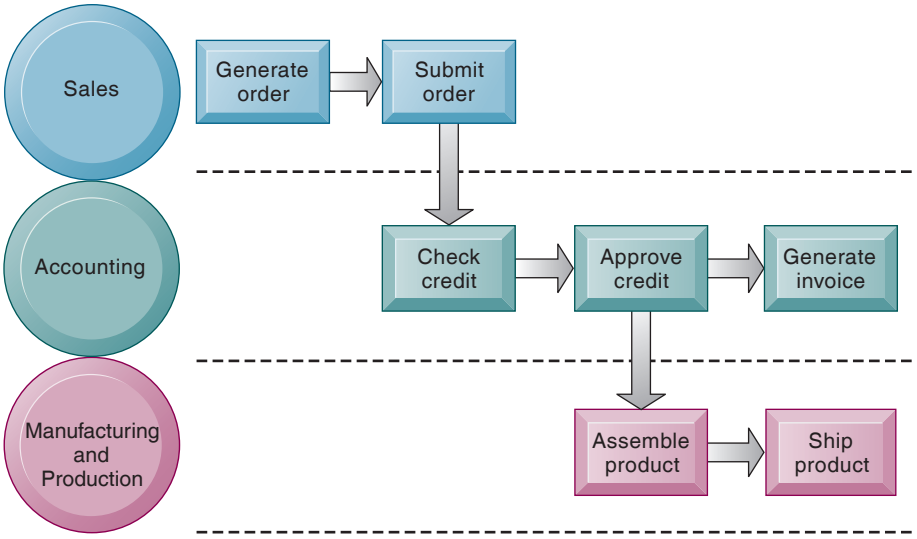
Other business processes cross many functional areas and require coordination across departments. Examine Figure 2.1, which illustrates the seemingly simple business process of fulfilling a customer order for a physical product. When the sales department receives a customer order, it generates and then submits a sales order to accounting to ensure that the customer can pay for the order by either a credit verification or a request for immediate payment prior to shipping. Once the customer credit is established, the production department has to pull the product from inventory or produce the product. Next, the product needs to be shipped (which may require working with a logistics firm such as UPS or FedEx). The accounting department then generates a bill or invoice and sends a notice to the customer, indicating that the product has shipped. The sales department has to be notified of the shipment and prepare to support the customer by answering calls or fulfilling warranty claims.

Table 2.1 Examples of Functional Business Processes

Functional Area	Business Process
Manufacturing and production	Assembling the product
	Checking for quality of the product
	Producing bills of materials
Sales and marketing	Identifying customers
	Making customers aware of the product
	Selling the product
Finance and accounting	Paying creditors
	Creating financial statements
	Managing cash accounts
Human resources	Hiring employees
	Evaluating employees' job performance
	Enrolling employees in benefits plans

Figure 2.1 The Order Fulfillment Process

Fulfilling a customer order involves a complex set of steps that requires the close coordination of the sales, accounting, and manufacturing functions.



What at first appears to be a simple process—fulfilling an order—turns out to be a very complicated series of business processes that require the close coordination of major functional groups in a firm. Moreover, to perform all these steps efficiently in the order fulfillment process requires the rapid flow of a great deal of information within the firm, with business partners such as delivery firms, and with the customer. The particular order fulfillment process we have just described is not only *cross-functional*, it is also *interorganizational* because it includes interactions with delivery firms and customers who are outside the boundaries of the organization. Ordering raw materials or components from suppliers would be another interorganizational business process.

How Information Technology Enhances Business Processes

Exactly how do information systems enhance business processes? Information systems automate many steps in business processes that were formerly performed manually, such as checking a client's credit or generating an invoice and shipping order. Today, however, information technology can do much more. New technology can actually change the flow of information,

making it possible for many more people to access and share information, replacing sequential steps with tasks that can be performed simultaneously and eliminating delays in decision making. It can even transform the way the business works and drive new business models. Ordering a book online from Amazon.com and streaming a song from Apple Music are new business processes based on new business models that are inconceivable without information technology.

That's why it's so important to pay close attention to business processes, both in your information systems course and in your future career. By analyzing business processes, you can achieve a clear understanding of how a business actually works. Moreover, by conducting a business process analysis, you will also begin to understand how to change the business to make it more efficient or effective. Throughout this book, we examine business processes with a view to understanding how they might be changed, or replaced, by using information technology to achieve greater efficiency, innovation, and customer service.

2.2 Explain How Information Systems Serve Different Management Groups

Now that you understand business processes, it is time to look more closely at how information systems support the business processes of a firm. Because there are different interests, specialties, and levels in an organization, there are different kinds of systems. No single system can provide all the information an organization needs.

A typical business organization will have systems supporting processes for each of the major business functions: sales and marketing, manufacturing and production, finance and accounting, and human resources. Functional systems that operate independently of each other are becoming outdated because they cannot easily share information to support cross-functional business processes. They are being replaced with large-scale cross-functional systems that integrate the activities of related business processes and organizational units. We describe these integrated cross-functional applications later in this section.

A typical firm will also have different systems supporting the decision-making needs of each of the main management groups described in Chapter 1. Operational management, middle management, and senior management each use specific types of systems to support the decisions they must make to run the company. They include transaction processing systems and systems for business intelligence. Let's look at these systems and the types of decisions they support.

Transaction Processing Systems

Transaction processing system (TPS)

Information system that performs and records the daily routine transactions necessary to conduct business.

Operational managers need systems that keep track of the elementary activities and transactions of the organization, such as sales, receipts, cash deposits, payroll, credit decisions, and the flow of materials in a factory. A **transaction processing system (TPS)** provides this kind of information. A TPS is an information system that performs and records the daily routine transactions necessary to conduct business, such as sales order entry, payroll, employee record keeping, and shipping.

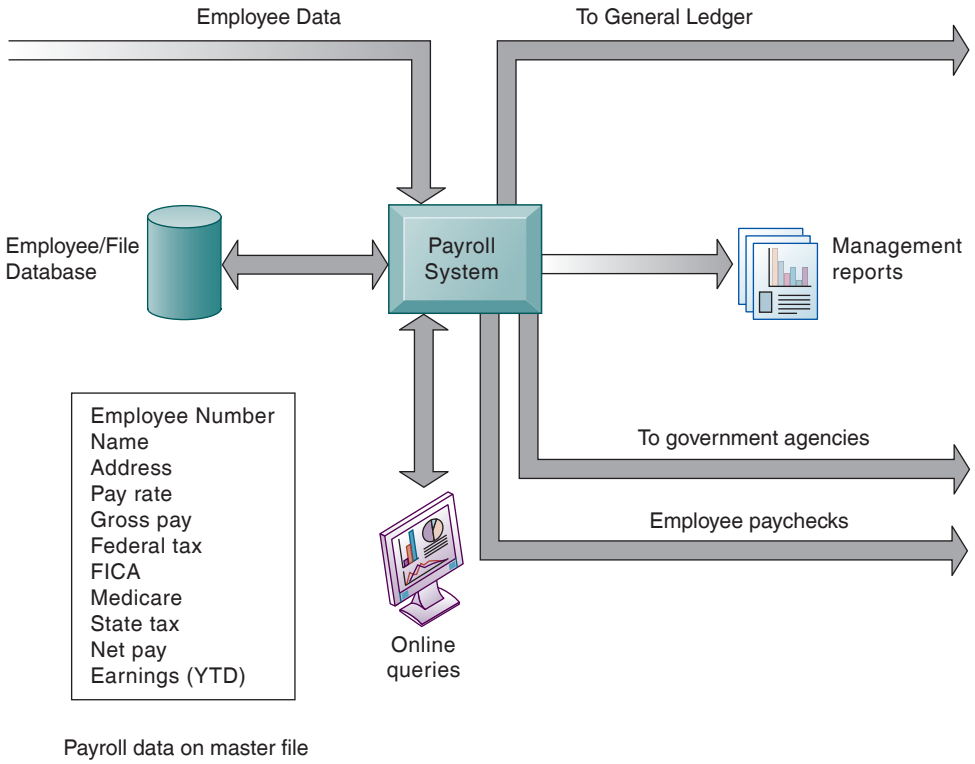
The principal purpose of systems at this level is to answer routine questions and to track the flow of transactions through the organization. How many parts are in inventory? What happened to Mr. Garcia's payment? To answer these kinds of questions, information generally must be easily available, current, and accurate.

At the operational level, tasks, resources, and goals are predefined and highly structured. The decision to grant credit to a customer, for instance, is made by a lower-level supervisor according to predefined criteria. All that must be determined is whether the customer meets the criteria.

Managers need TPS to monitor the status of internal operations and the firm's relations with the external environment. TPS are also major producers of information for the other systems and business functions. Examine Figure 2.2, which illustrates a TPS for payroll processing. A payroll system keeps track of money paid to employees. An employee time sheet with employee data such as the employee's name, identification number, and number of hours worked per week represents a single transaction for this system. Once this transaction is input

Figure 2.2 A Payroll TPS

A TPS for payroll processing captures employee payment transaction data (such as a timecard). System outputs include online and hard copy reports for management and employee paychecks.



in the payroll system, it updates the system's master employee file (or database—see Chapter 6) that permanently maintains employee payroll data for the organization. The data in the system can typically be accessed and queried online and, in addition to being used to generate employee paychecks, can be combined in different ways to create reports of interest to management and government agencies. For instance, payroll data can be used to supply employee payment history data for insurance, pension, and other benefits calculations to the firm's human resources function and employee payment data to government agencies such as the U.S. Internal Revenue Service and Social Security Administration. The payroll system, along with other accounting TPS, also supplies data to the company's general ledger system, which is responsible for maintaining records of the firm's income and expenses and for producing reports such as income statements and balance sheets.

TPS failure for even a short period of time can have a significant negative impact on a firm as well as on other firms linked to it. For example, Southwest Airlines travelers faced long lines and delays after a computer system failure grounded all Southwest flights on the morning of April 18, 2023. Approximately 1,700 flights nationwide were affected.

Systems for Business Intelligence

Firms also have business intelligence systems that focus on delivering information to support management decision-making. **Business intelligence (BI)** is a contemporary term for data analytics and other software tools for organizing, analyzing, and providing access to data to help managers and other enterprise users make more informed and thus better business decisions. BI systems address the decision-making needs of all levels of management. This section provides a brief introduction to business intelligence. You'll learn more about this topic in Chapters 6 and 12.

Business intelligence (BI)
Applications and technologies to help users make more informed and thus better business decisions.

Management Information Systems

BI systems for middle management help with monitoring, controlling, decision-making, and administrative activities. In Chapter 1, we defined management information systems as the study of information systems in business and management. As noted in Chapter 1, the term

Management information systems (MIS)

The study of information systems, focusing on their use in business and management. Also refers to specific category of information system serving middle management.

management information systems (MIS) also designates a specific category of information systems serving middle management. MIS provide middle managers with reports about the organization's current performance. Managers use this information to monitor and control the business and predict future performance.

MIS summarize and report on the company's basic operations using data supplied by TPS. The basic transaction data from TPS are compressed and usually presented in reports that are produced on a regular schedule. Today, many of these reports are delivered online. Examine Figure 2.3, which shows how a typical MIS obtains transaction-level order, production, and accounting data from order processing, manufacturing resource planning, and general ledger TPS, respectively. The MIS transforms this data into MIS files such as sales data, unit production cost data, product change data, and expense data that managers can access via online displays and dashboards and use to generate various reports.

Next, examine Figure 2.4, which shows a sample report from the system illustrated in Figure 2.3. MIS typically provide answers to routine questions that have been specified in advance and have a predefined procedure for answering them. In this instance, the MIS report

Figure 2.3 How Management Information Systems Obtain Their Data from the Organization's TPS

In the system illustrated by this diagram, three TPS supply summarized transaction data to the MIS reporting system at the end of a specified time period. Managers gain access to the organizational data through the MIS, which provides them with the appropriate reports.

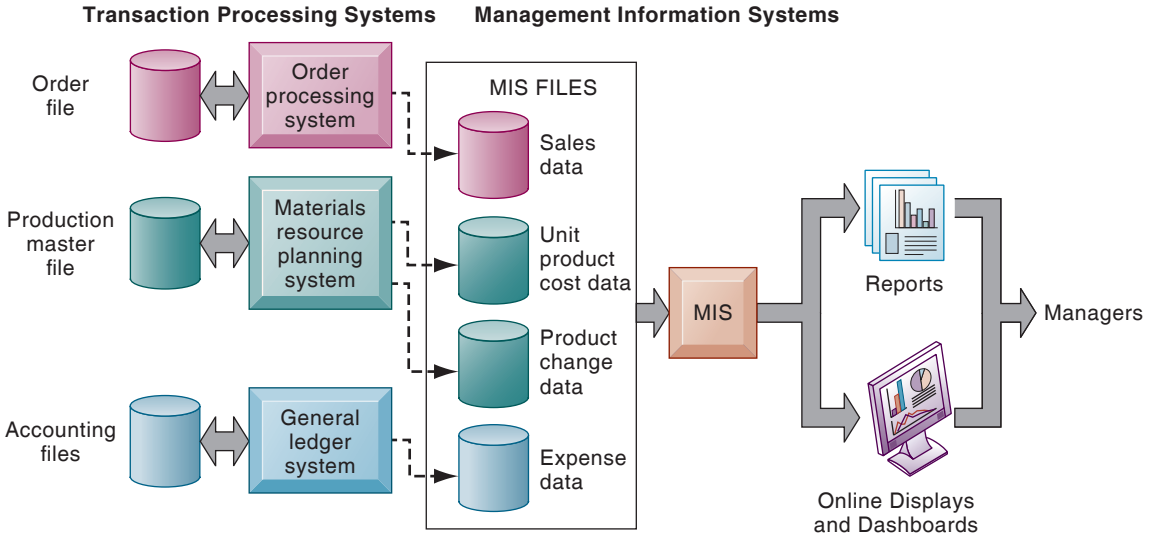


Figure 2.4 SAMPLE MIS Report

This report, showing summarized annual sales data, was produced by the MIS in Figure 2.3.

Consolidated Consumer Products Corporation Sales by Product and Sales Region: 2024

PRODUCT CODE	PRODUCT DESCRIPTION	SALES REGION	ACTUAL SALES	PLANNED	ACTUAL versus PLANNED
4469	Carpet Cleaner	Northeast	4,066,700	4,800,000	0.85
		South	3,778,112	3,750,000	1.01
		Midwest	4,867,001	4,600,000	1.06
		West	4,003,440	4,400,000	0.91
	TOTAL		16,715,253	17,550,000	0.95
5674	Room Freshener	Northeast	3,676,700	3,900,000	0.94
		South	5,608,112	4,700,000	1.19
		Midwest	4,711,001	4,200,000	1.12
		West	4,563,440	4,900,000	0.93
	TOTAL		18,559,253	17,700,000	1.05

compares total annual sales figures for specific products (carpet cleaner and room freshener) to planned targets. These systems generally are not flexible and have little analytical capability. Most MIS use simple routines, such as summaries and comparisons, as opposed to sophisticated mathematical models or statistical techniques.

Decision-Support Systems

Decision-support system (DSS)
Information system that combines data and sophisticated analytical models or data analysis tools to support decision making.

Other types of BI systems support more nonroutine decision-making. A **decision-support system (DSS)** is an information system that combines data and sophisticated analytical models or data analysis tools to support decision making. DSS focus on problems that are unique and rapidly changing, for which the procedure for arriving at a solution may not be fully predefined in advance. They try to answer questions such as these: What would be the impact on production schedules if we were to double sales in the month of December? What would happen to our return on investment if a factory schedule were delayed for six months?

Although DSS use internal information from TPS and MIS, they often bring in information from external sources, such as current stock prices or product prices of competitors. Super-user managers and business analysts who want to use sophisticated analytics and models to analyze data employ these systems.

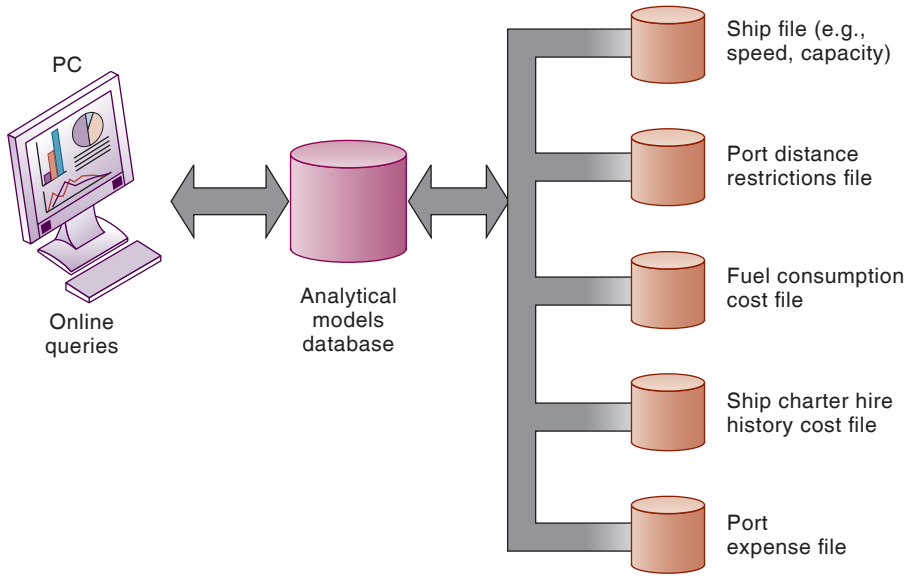
Examine Figure 2.5, which illustrates an interesting, highly specific, and powerful DSS used by a large global shipping company that primarily carries bulk cargoes of coal, oil, ores, and finished products for its parent company. The firm owns some vessels, charters others, and bids for shipping contracts in the open market to carry general cargo. The DSS calculates financial and technical voyage details. Financial calculations include ship/time costs (fuel, labor, capital), freight rates for various types of cargo, and port expenses. Technical details include myriad factors, such as ship cargo capacity, speed, port distances, fuel and water consumption, and loading patterns (location of cargo for different ports).

The system can answer questions such as the following: Given a customer delivery schedule and an offered freight rate, which vessel should be assigned at what rate to maximize profits? What is the optimal speed at which a particular vessel can optimize its profit and still meet its delivery schedule? What is the optimal loading pattern for a ship bound for San Diego, California from Malaysia? The system operates on a powerful desktop computer, providing a system of menus that makes it easy for users to enter data or obtain information.

The DSS we have just described draws heavily on models. Other BI systems are more data-driven, focusing instead on extracting useful information from big data, including the Internet of Things (IoT). For example, Starbucks collects individual purchase data on millions of customers

Figure 2.5 Voyage-Estimating Decision Support System

This DSS operates on a powerful desktop computer. Managers who must develop bids on shipping contracts use it daily.



captured by its mobile app and loyalty card program. The company analyzes the data to learn more about customers' beverage preferences, product preferences, and usage patterns. This information helps Starbucks managers make better decisions about how to target marketing campaigns and individual app and email offers for what customers will likely prefer.

Executive Support Systems

BI systems also address the decision-making needs of senior management. Senior managers need systems that focus on strategic issues and long-term trends, both in the firm and in the external environment. They are concerned with questions such as: What will employment levels be in five years? What are the long-term industry cost trends? What products should we be making in five years?

Executive support system (ESS)

Information system designed to help senior managers make nonroutine decisions that require judgment, evaluation and insight.

Portal

Web interface for presenting integrated personalized business content from a variety of sources. Also refers to a website that acts as initial point of entry to the web with an integrated package of content and services.

An **executive support system (ESS)** helps senior management make these decisions. An ESS can be used to address nonroutine decisions requiring judgment, evaluation, and insight because there is no agreed-on procedure for arriving at a solution. An ESS presents graphs and data from many sources through an interface that is easy for senior managers to use. Often, the information is delivered to senior executives through a **portal**, which uses a web interface to present integrated personalized business content from a variety of sources.

ESS are designed to incorporate data about external events such as new tax laws or competitors, but they also draw summarized information from internal MIS and DSS. They filter, compress, and track critical data, displaying the data of greatest importance to senior managers. Increasingly, such systems include BI analytics for analyzing trends, forecasting, and drilling down to data at greater levels of detail.

For example, Sabre Airline Solutions, which provides booking tools, revenue management, and other technology for airlines, hotels and travel companies, turned to BI technology for real-time data on customer behavior and better support for management decision-making. Sabre executive dashboards provide a single point of access for critical information relevant to Sabre's operations, with a 360-degree overview of business health, reservations, operational performance, and ticketing, presented in user-friendly displays. The dashboards display key performance metrics such as financial performance, customer satisfaction, operational efficiency, and market share. The dashboards enable executives to track progress toward business goals. Executives can customize their view of the data based on their specific information requirements, and the data can be sliced and diced based on these views. For example, sales

A BI dashboard delivers comprehensive and accurate information for decision-making, often using a single screen. The graphical overview of key performance indicators helps managers quickly spot areas that need attention.

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dashboards for salespeople, sales managers, and C-level executives can provide views by state or country on a bubble map, sales by date, sales by customer, sales by salesperson, sales by item, and sales by customer (Sabre, 2024).

Sabre's executive dashboard is an example of a **BI dashboard**, which displays on a single screen graphs and charts of key performance indicators for managing a company. BI dashboards have become popular tools for management decision-makers. Tableau is a popular business intelligence and analytics software tool available from Salesforce that enables users to create BI dashboards and other data visualizations. We discuss Tableau further in Chapter 12, and also provide projects that will help you learn more about Tableau at the end of this chapter and in Chapters 3, 6, and 12.

BI dashboard

Displays on a single screen graphs and charts of key performance indicators for managing a company.

2.3 Explain How Enterprise Applications Improve Firm Performance

Reviewing all the types of systems just described, you might wonder how a business can manage all the information in these differing systems. You might also wonder how costly it is to maintain so many systems. You might also wonder how all these systems can share information and how managers and employees can coordinate their work. In fact, these are all important questions for businesses today.

Enterprise Applications

Getting the different kinds of systems in a company to work together has proven to be a major challenge. Typically, companies are built through both normal organic growth and through the acquisition of smaller firms. Over time, companies end up with a collection of systems, most of them older, and face the challenge of getting them all to communicate with one another and work together as one corporate system. There are several solutions to this problem.

One solution is to implement **enterprise applications**, which are systems that can coordinate activities, decisions, and knowledge across many different functions, levels, and business units in a firm. Enterprise applications help businesses become more flexible and productive by coordinating their business processes more closely and integrating groups of processes so they focus on efficient management of resources and customer service.

Enterprise applications

Systems that can coordinate activities, decisions, and knowledge across many different functions, levels, and business units in a firm. Include enterprise systems, supply chain management systems, customer relationship management systems, and knowledge management systems.

Examine Figure 2.6, which illustrates the interrelationship that the four major types of enterprise applications—enterprise systems, supply chain management systems, customer relationship management systems, and knowledge management systems—have with the business functions and processes of a firm. Each of these enterprise applications integrates a related set of functions and business processes to enhance the performance of the organization as a whole. Figure 2.6 shows that the architecture for these enterprise applications encompasses processes spanning the entire organization and, in some cases, extending beyond the organization to customers, suppliers, and other key business partners.

Enterprise Systems

Firms use an **enterprise system**, also known as an *enterprise resource planning* (*ERP*) *system*, to integrate business processes in manufacturing and production, finance and accounting, sales and marketing, and human resources into a single software system. Information that was previously fragmented in many systems is stored in a single comprehensive data repository where it can be used by many parts of the business.

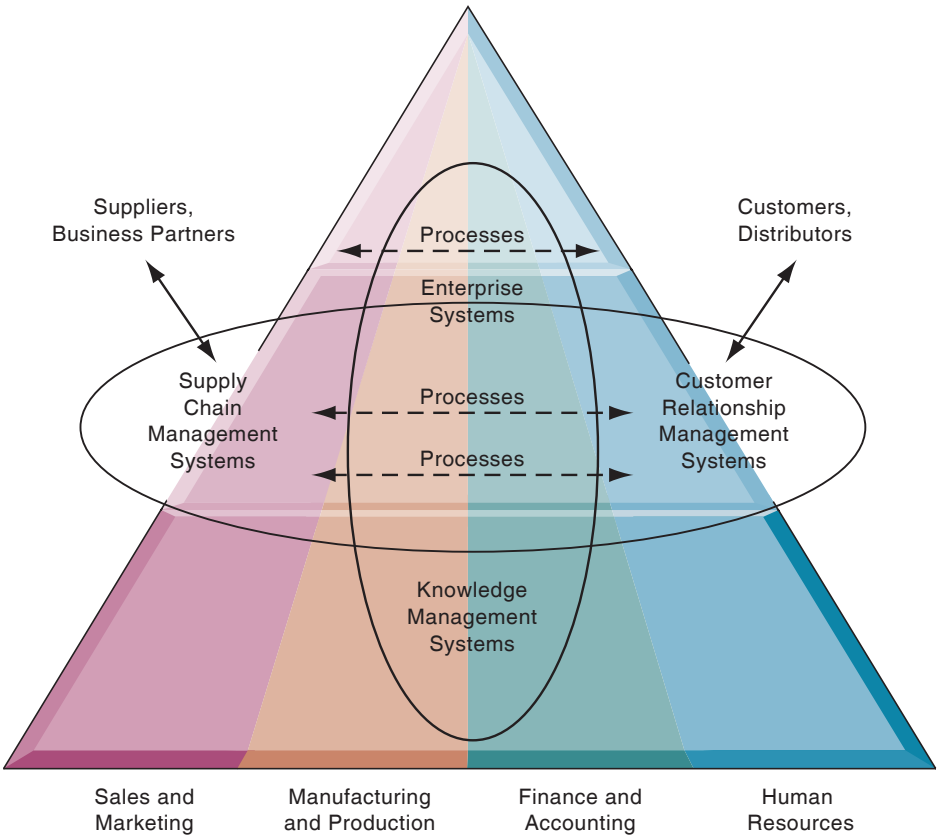
Enterprise system

Integrated enterprise-wide information systems that enable firms to coordinate key business processes in manufacturing and production, finance and accounting, sales and marketing and human resources. Also known as an enterprise resource planning (ERP) system.

For example, when a customer places an order, the order data flow automatically to other parts of the company that they affect. The order transaction triggers the warehouse to pick the ordered products and schedule shipment. The warehouse informs the factory to replenish whatever has been depleted. The accounting department is notified to send the customer an invoice. Customer service representatives track the progress of the order through every step to inform customers about the status of their orders. Managers can use firmwide information to make more precise and timely decisions about daily operations and longer-term planning.

Figure 2.6 Enterprise Application Architecture

Enterprise applications automate processes that span multiple business functions and organizational levels and may extend outside the organization.



Supply chain management system (SCM system)

Information system that automates the flow of information between a firm and its suppliers in order to optimize the planning, sourcing, manufacturing, and delivery of products and services.

Interorganizational system

Information system that automates the flow of information across organizational boundaries and links a company to its customers, distributors, or suppliers.

Customer relationship management system (CRM system)

Information system that enables a firm to coordinate all the business processes that deal with customers in sales, marketing, and service to optimize revenue, customer satisfaction, and customer retention.

Knowledge management system (KMS)

Information system that supports the creation, capture, storage, and dissemination of firm expertise and knowledge.

Supply Chain Management Systems

Firms use a **supply chain management system (SCM system)** to help manage relationships with their suppliers. These systems help suppliers, purchasing firms, distributors, and logistics companies share information about orders, production, inventory levels, and delivery of products and services so that they can source, produce, and deliver goods and services efficiently. The ultimate objective is to get the right number of their products from their source to their point of consumption in the shortest time and at the lowest cost. These systems increase firm profitability by lowering the costs of moving and making products and by enabling managers to make better decisions about how to organize and schedule sourcing, production, and distribution.

SCM systems are one type of **interorganizational system** because they automate the flow of information across organizational boundaries. You will find examples of other types of interorganizational information systems throughout this text because such systems make it possible for firms to link to their customers, distributors, and suppliers.

Customer Relationship Management Systems

Firms use a **customer relationship management system (CRM system)** to help manage their relationships with their customers. A CRM system enables a firm to coordinate all the business processes that deal with customers in sales, marketing, and service to optimize revenue, customer satisfaction, and customer retention. This information helps firms identify, attract, and retain the most profitable customers; provide better service to existing customers; and increase sales.

Knowledge Management Systems

Some firms perform better than others because they have better knowledge about how to create, produce, and deliver products and services. A **knowledge management system (KMS)** supports the creation, capture, storage, and dissemination of firm expertise and knowledge. It collects relevant

knowledge and experience in the firm, as well as links to external sources of knowledge and makes that knowledge available wherever and whenever it is needed to improve business processes and management decisions. We examine enterprise systems, SCM systems, and CRM systems in greater detail in Chapter 9. We discuss knowledge management in more depth in Section 2.8.

Intranets and Extranets

Enterprise applications create deep-seated changes in the way the firm conducts its business, offering many opportunities to integrate important business data into a single system. They are often costly and difficult to implement. Intranets and extranets deserve to be mentioned here as alternative tools for increasing integration and expediting the flow of information within the firm and with customers and suppliers.

As you learned in Chapter 1, intranets are simply internal company websites that are accessible only by employees. The term *intranet* refers to an internal network in contrast to the Internet, which is a public network linking organizations and other external networks. Intranets use the same technologies and techniques as the Internet, and they are often simply a private access area of a company website. Extranets are company websites that are accessible to authorized vendors and suppliers and are often used to coordinate the movement of supplies to the firm's production apparatus.

For example, Marigold (formerly CM Group), which provides marketing technology platforms, uses a company-wide intranet to connect its 900 employees across four continents. The intranet replaces multiple communication systems and provides a single system for seamless information sharing and real-time collaboration. Employees now have easier access to the information they need to do their jobs (Marigold, 2024).

2.4 Understand the Importance of Collaboration and Social Business Systems

With all of these systems and information, you might wonder how it is possible to make sense of them. How do people working in firms pull it all together, work toward common goals, and coordinate plans and actions? In addition to the types of systems we have just described, businesses need special systems to support collaboration and teamwork.

What Is Collaboration?

Collaboration is working with others to achieve shared and explicit goals. Collaboration focuses on task or mission accomplishment and usually takes place in a business or other organization, and between businesses. You collaborate with a colleague in Tokyo who has expertise on a topic about which you know nothing. You collaborate with many colleagues in publishing a company blog. If you're in a law firm, you collaborate with members of an accounting firm in servicing the needs of a client with tax problems.

Collaboration
Working with others to achieve shared and explicit goals.

Collaboration can be short-lived, lasting a few minutes, or longer-term, depending on the nature of the task and the relationship among participants. Collaboration can be one-to-one or many-to-many.

Employees may collaborate in informal groups that are not a formal part of the business firm's organizational structure, or they may be organized into formal teams. A **team** is a group that has been assigned a specific mission by someone in the business. Team members need to collaborate on the accomplishment of specific tasks and collectively achieve the team mission. The team mission might be to "win a specific deal" or "increase online sales by 10 percent." Teams are often short-lived, depending on the problems they tackle, and the length of time needed to find a solution and accomplish the mission.

Team
Formal group whose members collaborate to achieve specific goals.

Collaboration and teamwork are more important today than ever, for the following reasons.

- *Changing nature of work.* The nature of work has changed from the days of traditional factory manufacturing and office work, where each stage in the production process occurred

independently of one another and was coordinated by supervisors. Work was organized into silos. Within a silo, work passed from one machine tool station to another, from one desktop to another, until the finished product was completed. Today, jobs of all kinds require much closer coordination and interaction among the parties involved in producing the service or product. Even in factories, workers today often work in production groups, or pods.

- *Growth of professional work.* "Interaction" jobs tend to be professional jobs in the service sector that require close coordination and collaboration. Professional jobs require substantial education and the sharing of information and opinions to get work done. Each actor on the job brings specialized expertise to the problem, and all the actors need to take one another into account in order to accomplish the job.
- *Changing organization of the firm.* For most of the industrial age, managers organized work in a hierarchical fashion. Orders came down the hierarchy, and responses moved back up the hierarchy. Today, work is organized into groups and teams, and the members are expected to develop their own methods for accomplishing the task. Senior managers observe and measure results but are much less likely to issue detailed orders or operating procedures. In part, this is because expertise and decision-making power have been pushed down in organizations.
- *Changing scope of the firm.* The work of the firm has changed from a single location to multiple locations—offices or factories throughout a region, a nation, or even around the globe. For instance, Henry Ford developed the first mass-production automobile plant at a single Dearborn, Michigan, factory. In 2024, Ford had 177,000 employees at more than 65 manufacturing plants worldwide. With this kind of global presence, the need for close coordination of design, production, marketing, distribution, and service obviously takes on new importance and scale. Large global companies need to have teams working on a global basis.
- *Emphasis on innovation.* Although we tend to attribute innovations in business and science to great individuals, these great individuals are most likely working with a team of brilliant colleagues. Think of Bill Gates and Steve Jobs (founders of Microsoft and Apple), both of whom are highly regarded innovators, and both of whom built strong collaborative teams to nurture and support innovation in their firms. Their initial innovations derived from close collaboration with colleagues and partners. Innovation, in other words, is a group and social process, and most innovations derive from collaboration among individuals in a lab, a business, or a government agency. Strong collaborative practices and technologies are believed to increase the rate and quality of innovation.
- *Changing culture of work and business.* It is widely believed that diverse teams produce better outputs faster than individuals working on their own. Popular notions of the crowd ("crowdsourcing" and the "wisdom of crowds") also provide cultural support for collaboration and teamwork.

What Is Social Business?

Social business

Use of social network platforms, such as Facebook, X (formerly Twitter), as well as internal corporate social tools, to engage employees, customers, and suppliers.

Many firms today enhance collaboration by embracing **social business**—the use of social network platforms, such as Facebook and X (formerly Twitter), as well as internal corporate social business tools—to engage their employees, customers, as well as suppliers. These tools enable workers to set up profiles, form groups, and "follow" each other's status updates. The goal of social business is to deepen interactions with groups inside and outside the firm to expedite and enhance information sharing, innovation, and decision making.

A key word in social business is *conversations*. Customers, suppliers, employees, managers, and even oversight agencies continually have conversations about firms, often without the knowledge of the firm or its key actors (employees and managers).

Supporters of social business argue that if firms could tune in to these conversations, they would strengthen their bonds with consumers, suppliers, and employees, increasing their emotional involvement in the firm.

All of this requires a great deal of information transparency. People need to share opinions and facts with others quite directly, without intervention from executives or others. Employees get to know directly what customers and other employees think. Suppliers will

Table 2.2 Applications of Social Business

Application	Description
Social networks	Connect through personal and business profiles
Crowdsourcing	Harness collective knowledge to generate new ideas and solutions
Shared workspaces	Coordinate projects and tasks; co-create content
Blogs and wikis	Publish and rapidly access knowledge; discuss opinions and experiences
Social commerce	Share opinions about purchasing on social platforms
File sharing	Upload, share, and comment on photos, videos, audio, text documents
Social marketing	Use social media to interact with customers; derive customer insights
Communities	Discuss topics in open forums; share expertise

learn very directly the opinions of supply chain partners, and even managers presumably will learn more directly from their employees how well they are doing. Nearly everyone involved in the creation of value will know much more about everyone else.

If such an environment could be created, it is likely to drive operational efficiencies, spur innovation, and accelerate decision making. If product designers can learn directly about how their products are doing in the market in real time, based on consumer feedback, they can speed up the redesign process. If employees can use social connections inside and outside the company to capture new knowledge and insights, they will be able to work more efficiently and solve more business problems.

Table 2.2 describes important applications of social business inside and outside the firm. This chapter focuses on enterprise social business—its internal corporate uses. Chapters 7 and 10 describe social business applications relating to customers and suppliers outside the company.

Business Benefits of Collaboration and Social Business

There is a general belief among both business and academic communities that the more a business firm is "collaborative," the more successful it will be, and that collaboration within and among firms is more essential than in the past. *MIT Sloan Management Review*'s research found that a focus on collaboration is central to how digitally advanced companies create business value and establish competitive advantage (Kiron, 2017).

Table 2.3 summarizes some of the benefits of collaboration and social business that have been identified.

Table 2.3 Business Benefits of Collaboration and Social Business

Benefit	Rationale
Productivity	People interacting and working together can capture expert knowledge and solve problems more rapidly than the same number of people working in isolation from one another. There will be fewer errors.
Quality	People working collaboratively can communicate errors and corrective actions faster than if they work in isolation. Collaborative and social technologies help reduce time delays in design and production.
Innovation	People working collaboratively can come up with more innovative ideas for products, services, and administration than the same number working in isolation from one another. There are advantages to diversity and the "wisdom of crowds."
Customer service	People working together using collaboration and social business tools can solve customer complaints and issues faster and more effectively than if they were working in isolation from one another.
Financial performance (profitability, sales, and sales growth)	As a result of all of the above, collaborative firms have the potential for superior sales, sales growth, and financial performance.

Building a Collaborative Culture and Business Processes

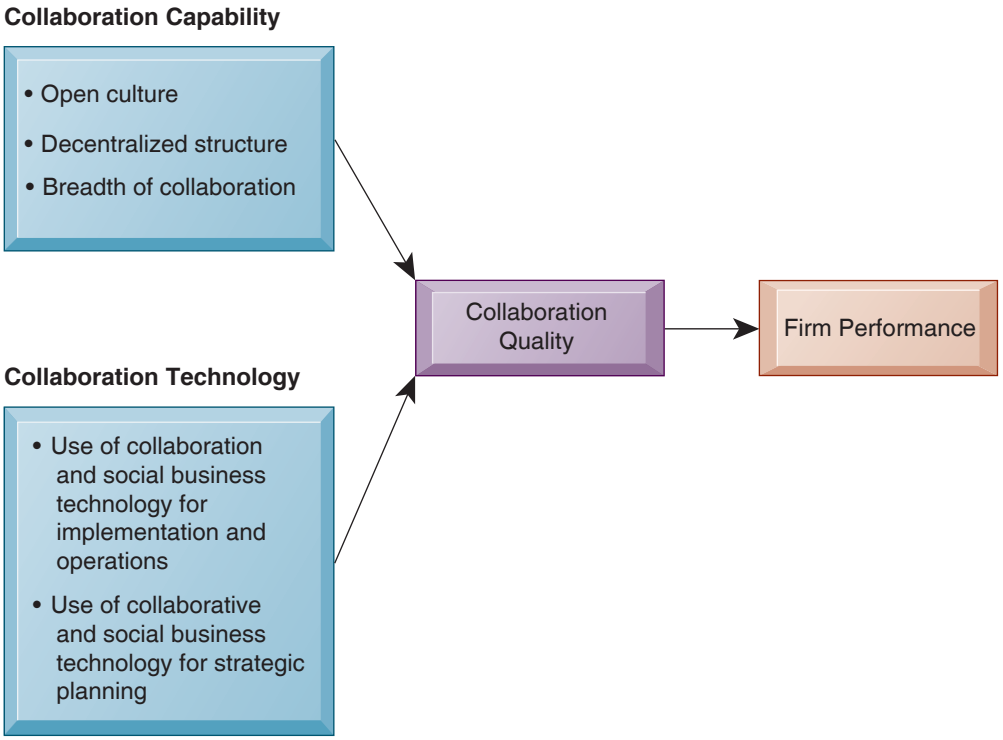
However, not all collaboration efforts succeed. A joint study of 1,100 companies by the Institute for Corporate Productivity and Rob Cross, professor of Global Business at Babson College, found that companies that promoted collaborative work were more than five times more likely to be high-performing. But it also found that among the companies studied, two-thirds of which claimed collaboration to be an explicit organizational value, only a minority were able to achieve good results. Collaboration won't take place spontaneously in a business firm, especially in the absence of supportive culture or business processes, such as how information flows throughout the organization, how decisions are made, the physical design of the workplace, and how employees are rewarded (Samdahl, 2017). Collaboration tools and technologies must be appropriate for the tasks at hand.

Review Table 2.3. Then examine Figure 2.7, which introduces the elements that impact the quality of collaboration within an organization and the organization's ability to use collaboration to enhance firm performance. These elements include the culture and business processes of the firm and the breadth with which it employs collaboration as well as the collaboration and social business technologies used by the firm. We explore each of these elements in further detail in the following sections.

With a collaborative organizational culture and business processes, senior managers are responsible for achieving results but rely on teams of employees to achieve and implement the results. Policies, products, designs, processes, and systems are much more dependent on teams at all levels of the organization to devise, to create, and to build. Teams are rewarded for their performance, and individuals are rewarded for their performance in a team. The function of middle managers is to build the teams, coordinate their work, and monitor their performance. The business culture and business processes are more "social." In a collaborative culture, senior management establishes collaboration and teamwork as vital to the organization.

Figure 2.7 Requirements for Collaboration

Successful collaboration requires appropriate business processes and organizational culture, along with appropriate collaboration technology.



2.5 Describe Collaboration Tools

A collaborative, team-oriented culture won't produce benefits without information systems in place to enable collaboration, group decision making, and social business. Some of these tools are expensive, but others are available online for free (or with premium versions for a modest fee). Let's look more closely at some of these tools.

Email and Instant Messaging Platforms

Email, text messaging, and instant messaging (IM) platforms and apps are major communication and collaboration tools for interactive jobs. Their software operates on computers and mobile devices and includes features for sharing files, as well as transmitting messages. Many IM systems allow users to engage in real-time conversations with multiple participants simultaneously. In recent years, email use has declined, with messaging and social media becoming preferred channels of communication.

Slack is a messaging platform for businesses that provides a single place for messaging, tools, and files. It has capabilities for integrated file sharing, video and phone calls, screen sharing directly through its platform, and the ability to invite others from outside the organization to collaborate. Teams can work in dedicated spaces called channels that bring together the right people and information.

Collaboration Platforms

There are now suites of software products providing multifunction platforms for collaboration among teams of employees who work together from many different locations. The most widely used are Internet-based audioconferencing and videoconferencing systems; cloud collaboration services such as Google's online services and tools; and corporate collaboration systems such as Microsoft Teams and Microsoft SharePoint.

Virtual Meeting Systems

To reduce travel expenses and help people in different locations to meet and collaborate, many companies, both large and small, have adopted videoconferencing and web conferencing technologies. More companies are using these systems to enable their employees to work remotely as well as in the office.

A videoconference allows individuals at two or more locations to communicate simultaneously through two-way video and audio transmissions. High-end videoconferencing systems feature **telepresence** technology, an integrated audio and visual environment that allows people to give the appearance of being present at a location other than their actual physical location. Free or low-cost Internet-based systems such as Microsoft Teams and Amazon Chime are lower quality but improving, and they are very useful for smaller companies. Apple's FaceTime is appropriate for one-to-one or small group videoconferencing. Zoom has emerged as a powerful videoconferencing tool with many features of high-end systems but is much less expensive and easier to use. Some of these tools are available on mobile devices.

Telepresence

Technology that allows people to give the appearance of being present at a location other than their actual physical location.

Cloud Collaboration Services

Cloud collaboration allows employees to work together and collaborate on documents and other file types which might be stored outside the company. Cloud collaboration happens when a user creates/uploads a file online and then shares access to other individuals. All changes made are saved and synced to the cloud, so all users always have access to the same, updated version of the document.

Google and Microsoft Cloud Services

Google offers many online tools and services, and some are suitable for collaboration. They include Google Drive, Google Docs, Google Workspace, and Google Sites. Most are free of charge.

Google Drive is a file storage and synchronization service for cloud storage, file sharing, and collaborative editing. These web-based online file-sharing services allow users to upload files to secure online storage sites from which the files can be shared with others. Microsoft OneDrive and Dropbox are other leading cloud storage services. They feature both free and paid services, depending on the amount of storage space and administration required. Users can synchronize their files stored online with their local PCs and other devices, with options for making the files private or public and for sharing them with designated contacts.

Google Drive and Microsoft OneDrive are integrated with tools for document creation and sharing. OneDrive provides online storage for Microsoft 365 documents and other files and works with Microsoft 365 apps, both installed and on the web. Google Drive is integrated with Google Docs, Sheets, and Slides (often called Google Docs), a suite of productivity applications that offer collaborative editing on documents, spreadsheets, and presentations. Google's cloud-based productivity suite for businesses, called Google Workspace, also works with Google Drive. Google Sites allows users to quickly create online team-oriented websites where multiple people can collaborate and share files.

Microsoft Teams and Microsoft SharePoint

Microsoft Teams, profiled in the Opening Case Study, is a communication and collaboration platform that allows people to have conversations, share files, and collaborate with others on their team in real time. Microsoft Teams is often integrated with Microsoft SharePoint. SharePoint is a collaboration and document management service that helps organizations share and manage content, knowledge, and applications. On-premises and cloud versions are available. SharePoint has a web-based interface and close integration with productivity tools such as Microsoft 365. SharePoint software makes it possible for employees to share their documents and collaborate on projects using Microsoft documents as the foundation.

SharePoint can be used to host internal websites that organize and store information in one central workspace to enable teams to coordinate work activities, collaborate on and publish documents, maintain task lists, implement workflows, and share information via wikis and blogs. Users can control versions of documents and document security. Because SharePoint stores and organizes information in one place, users can find relevant information quickly and efficiently while working closely together on tasks, projects, and documents. Enterprise search tools help locate people, expertise, and content. SharePoint also includes social tools.

Artificial Intelligence (AI) and Collaboration

Human-AI collaboration

Refers to interactions between humans and artificial intelligence systems that combine their respective complementary strengths to achieve shared goals.

Human-AI collaboration refers to interactions between humans and artificial intelligence systems that combine their respective complementary strengths to achieve shared goals. Rather than working in isolation, humans and AI collaborate synergistically, leveraging each other's capabilities to produce outcomes that are better than those from humans or AI individually working on their own.

AI technology can assist humans in a wide range of tasks that facilitate team collaboration, such as scheduling and managing meetings, transcribing meeting notes, and dictating emails. AI can also assist team collaborators with information gathering and data crunching, streamlining workflows, and freeing humans for higher-level tasks that require creative thinking and judgment. As AI technology has become more sophisticated and powerful, it can also pair with humans more closely to generate new ideas or solutions together. Table 2.4 describes some of these applications for human-AI collaboration.

The Spotlight on Technology Case Study, *AI Accelerates Drug Discovery and Development*, describes recent efforts at human-AI collaboration for drug discovery. The pharmaceutical industry faces mounting pressure to discover and develop new drugs faster and at a lower cost. AI has become a promising tool for overcoming long-standing challenges such as high failure rates and lengthy development periods.

Table 2.4 Examples of Human-AI Collaboration Applications

Task	Example
Medical diagnoses	AI algorithms can analyze medical images, including X-rays, MRIs, ultrasounds, CT scans, electrocardiograms, and echocardiograms to detect potential abnormalities and assist healthcare providers in identifying and diagnosing diseases more accurately and quickly. Radiologists collaborate with AI algorithms to validate the findings, interpret complex cases, and make final diagnoses. Johnson & Johnson is working with Anumana and Mayo Clinic to develop AI algorithms that can detect pulmonary hypertension early on and has a similar initiative with Ultromics Ltd. and Atman Health focused on cardiac amyloidosis.
Aggregating and synthesizing reviews to improve product appeal	CarMax's search optimization team employed the AI language model GPT-3.5 to aggregate and synthesize 5,000 online customer reviews for requested makes and models. GPT-3.5 synthesized the survey information and produced descriptions that, after human review, were placed on the company's website.
Selecting financial instruments to buy and sell	Traders at JP Morgan Chase work with a ChatGPT-like model that analyzed speeches from the Federal Reserve over the last 25 years to try to decipher trading signals and gain an advantage.
Human-AI collaboration for customized manufacturing	BionicWorkplace's intelligent collaborative workstation works with humans in customized, short manufacturing runs. The BionicWorkplace helps workers perform manufacturing tasks more efficiently and safely. The workstation is equipped with a bionic arm, various assistance systems, and tools. It employs sensors and camera systems to register the positions of the human operator, components, and tools, and uses this information to derive the optimal sequence of operations to be carried out between the human and the robot to manufacture the target item. The system continuously learns from each action. Humans can intuitively control the BionicWorkplace with gestures, touch, or speech.

Spotlight on Technology Case Study

AI Accelerates Drug Discovery and Development

One of most promising applications for collaborative AI is drug discovery and development. Drug discovery is the process by which new medications are discovered. Drug development refers mainly to the complete process of bringing newly discovered drugs to the market. In the past, drugs were discovered by identifying the active ingredient from traditional remedies or by happenstance. More recently, drug discovery focuses on chemical libraries of synthetic small molecules, natural products, or extracts that have been screened in intact cells or whole organisms to identify substances that have a desirable therapeutic effect.

Researchers go through numerous rounds of tests to identify those molecules with the potential to become an effective compound. The researchers often do not know which chemical structures will have the desired biological impact, making the process of refining a promising compound into a viable drug candidate extremely expensive and time-consuming. It night take 10–15 years or more to develop a new drug that can be licensed for use.

It is very rare to discover a molecule, either natural or artificial, that is effective against a protein target implicated in a disease. Only a small percentage of promising compounds move into clinical trials and an even smaller percentage become approved medications. Thousands of compounds are tested in the laboratory to find one that may act on specific genes or proteins involved in a disease. Once a promising compound has been found, the chemical structure is used as a starting point to make a drug that has the most benefit and the least harm.

AI technology has proved useful for drug discovery in a number of ways. It can help researchers predict the properties of a potential compound, preventing work on compounds that are unlikely to be effective. It can cut down on repetitive tasks, such as manual analysis of thousands of microscopic cell images. AI has also become a powerful collaborative tool for generating ideas for entirely new compounds, accelerating the discovery of effective new drugs.

AI is enabling researchers to develop more targeted treatments, driving progress toward precision medicine. For example, in oncology, an AI algorithm can be applied to digitized images of biopsies to help identify subtle differences between tumors, pointing to the presence of genetic mutations in a subset of patients. Researchers can use these findings to develop medicines specifically designed for that subset of patients. AI algorithms can predict the toxicity of promising compounds based on their chemical structure, reducing the number of compounds that need to be tested experimentally. AI can also be used to design more selective drugs with fewer side effects.

One of the biggest challenges of clinical trials is efficiently recruiting and enrolling patients that meet the selection criteria. Advanced analytics help researchers find locations and healthcare organizations where diverse patients who are representative of the entire U.S. population are more likely to be treated. Researchers can then recruit eligible patients from those study sites into clinical trials. This is critical, given the importance of ensuring medicines are studied in diverse patient populations that are representative of those impacted by diseases.

For instance, at Johnson & Johnson, researchers are applying AI and machine learning (ML) algorithms to large, anonymized datasets to locate clinical research sites serving patients with more diverse healthcare and socioeconomic backgrounds. These patients could potentially benefit from medicines being studied and would make clinical studies more representative and balanced. The clinical trial operations team can then work to determine the likelihood of enrolling the newly identified sites into their clinical trials.

Merck & Co., another major pharmaceutical manufacturer, is exploring ways teams across its organization can use AI. Merck is working with Variational AI's Enki Platform to generate novel small molecules that match its target product profiles (TPPs). A target product profile defines the desired characteristics of a product. Enki can generate small molecules in response to TPPs prompts (similar to the way DALL-E and Midjourney create images in response to text prompts). The user picks targets they want to hit, as well as those they want to miss, and chooses other attributes. Enki then generates molecules that match the TPPs. Merck has also partnered with BenchSci, Atomwise, C4 Therapeutics, and ACMED on other AI drug discovery and development initiatives.

Many AI applications for drug discovery are based on companies' proprietary AI tools for discovering and developing their own assets. Using the Enki Platform, chemists do not need to develop their own generative AI models, but can now submit their TPPs and get novel, diverse, selective, and synthesizable lead-like structures in days to move quickly into lead optimization.

SOURCES: AI in Drug Discovery," Emdgroup.com, accessed August 26, 2024; Nick Paul Taylor, "Merck Finds Drug Discovery DALL-E, Becoming Early User of Generative AI Tool," Fiercebiotech.com, January 25, 2024; Variational AI, "Variational AI Announces Generative AI Project with Merck," Variational.ai, January 25, 2024; Ashley Welch, "Artificial Intelligence is Helping Revolutionize Healthcare as We Know It," Jnj.com, September 13, 2023.

Case Study Questions

1. Why is drug discovery appropriate for AI applications?
2. Describe how AI applications have improved drug discovery.

2.6 Describe Social Business Tools

Many of the collaboration tools and technologies we have just described include some capabilities for supporting social business. However, there are also more specialized tools for this purpose, including enterprise social networking tools and virtual world and metaverse platforms.

Enterprise Social Networking Tools

Enterprise social networking tools create business value by connecting the members of an organization through profiles, updates, and notifications, similar to Facebook features but tailored to internal corporate uses. Examples include Salesforce Chatter, and Microsoft Viva Engage. Table 2.5 provides more detail about these internal social capabilities.

Table 2.5 Enterprise Social Networking Software Capabilities

Capability	Description
Profiles	Ability to set up member profiles describing who individuals are, educational background, and interests. Includes work-related associations and expertise (skills, projects, teams).
Content sharing	Share, store, and manage content including documents, presentations, images, and videos.
Feeds and notifications	Real-time information streams, status updates, and announcements from designated individuals and groups.
Groups and team workspaces	Establish groups to share information, collaborate on documents, and work on projects with the ability to set up private and public groups and to archive conversations to preserve team knowledge.
Tagging and social bookmarking	Indicate preferences for specific pieces of content, similar to the Facebook Like button. Tagging lets people add keywords to identify content they like.
Permissions and privacy	Ability to make sure private information stays within the right circles, as determined by the nature of relationships. In enterprise social networks, there is a need to establish who in the company has permission to see what information.

Virtual Worlds and the Metaverse

Virtual worlds are online, 3D environments populated by residents who have built graphical representations of themselves known as avatars. Real-world people represented by avatars meet, interact, and exchange ideas at these virtual locations using gestures, chatbot conversations, and voice communication, making virtual worlds potential online communication and collaboration tools. An early example of this is Second Life. Today, interest in virtual worlds is focused on the metaverse, which we describe more fully in Chapters 4 and 7. The metaverse is a nascent network of 3D virtual worlds focused on social connection where people can interact, do business, and forge social connections through their virtual avatars.

Virtual worlds
Online 3D environments populated by residents who have built graphical representations of themselves known as avatars.

The U.S. and Canadian businesses of the multinational professional services network KPMG have launched a "collaboration hub" in the metaverse. The hub is a private place where employees, partners, and clients can conduct virtual team meetings and share ideas using virtual whiteboards and other tools. In the metaverse, they are also able to conduct training sessions, job interviews, and employee evaluations, along with meetings to demonstrate KPMG's services and capabilities (Strack, 2022).

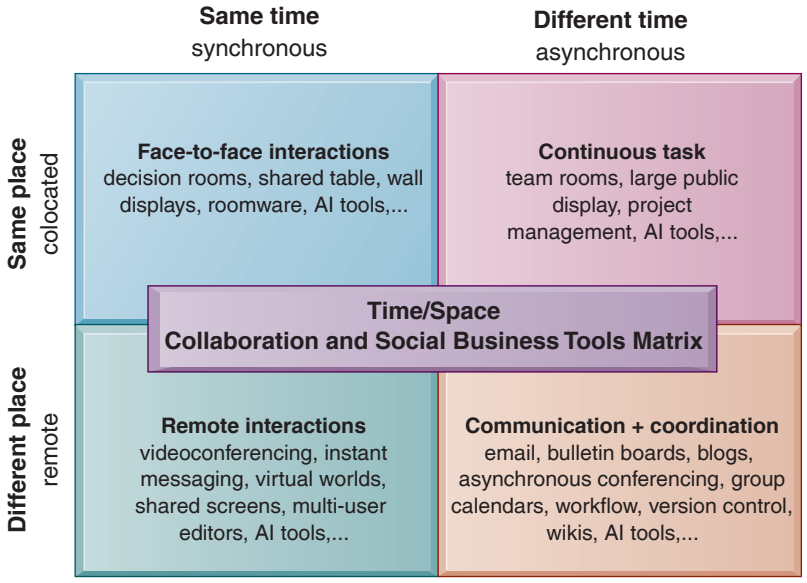
2.7 Understand How to Evaluate Collaboration and Social Business Tools

With so many collaboration and social business tools and services available, how do you choose the right technology for your firm? You need a framework for understanding just what problems these tools are designed to solve. Examine Figure 2.8, which illustrates a time/space collaboration and social business tool matrix developed by a number of collaborative-work scholars that can be helpful in addressing this issue.

The time/space matrix focuses on two dimensions of the collaboration problem: time and space (place or location). For instance, you need to collaborate with people in different time zones, and you cannot all meet at the same time. Midnight in New York is 10:30 a.m. in Mumbai, so this makes it difficult to have a videoconference (the people in New York are too tired). Time is clearly an obstacle to collaboration on a global scale.

Figure 2.8 The Time/Space Collaboration and Social Business Tool Matrix

Collaboration and social business technologies can be classified in terms of whether they support interactions at the same or different times or places, and whether these interactions are remote or co-located.



Place (location) also inhibits collaboration in large global or even national and regional firms. Assembling people for a physical meeting is made difficult by the physical dispersion of distributed firms (firms with more than one location), the cost of travel, and the time limitations of managers.

The collaboration and social business technologies we have just described are ways of overcoming the limitations of time and space. Using this time/space framework will help you choose the most appropriate collaboration and teamwork tools for your firm. Note that some tools are applicable in more than one time/place scenario. For example, collaboration suites such as Microsoft Teams have capabilities for both synchronous (IM, meeting tools) and asynchronous (email, wikis, document editing) interactions.

Here's a "to-do" list to get started. If you follow these six steps, you should be led to investing in the correct collaboration and social business tools for your firm at a price the firm can afford and within its risk tolerance.

1. What are the collaboration challenges facing the firm in terms of time and space? Locate your firm in the time/space matrix. Your firm can occupy more than one cell in the matrix. Different collaboration tools will be needed for each situation.
2. Within each cell of the matrix where your firm faces challenges, exactly what kinds of solutions are available? Make a list of vendor products.
3. Analyze each of the products in terms of the product's cost and benefits to your firm. Be sure to include the costs of training in your cost estimates and the costs of involving the information systems department, if needed.
4. Identify the risks to security and vulnerability involved with each of the products. Is your firm willing to put proprietary information into the hands of external service providers over the Internet? Is your firm willing to expose its important operations to systems controlled by other firms? What are the financial risks facing your vendors? Will they be here in three to five years? What would be the cost of making a switch to another vendor in the event the vendor firm fails?
5. Seek the help of potential users to identify implementation and training issues. Some of these tools are easier to use than others.
6. Make your selection of candidate tools and invite the vendors to make presentations.

In addition to selecting the right digital tools for communication and collaboration, organizations need to make sure the tools actually support productive ways of working. Access to technology alone often can't fix poorly designed business processes and entrenched organizational behaviors.

2.8 Describe Knowledge Management Systems

Collaboration and knowledge-sharing will be less effective and efficient if the requisite knowledge for collaboration, decision-making, and ongoing operations is not available. Knowledge management systems are a specific capability for addressing this need. **Knowledge management** refers to the set of business processes developed in an organization to create, store, transfer, and apply knowledge.

Knowledge management
The set of processes developed in an organization to create, gather, store, maintain, and disseminate the firm's knowledge.

Knowledge-based core competencies of firms—the two or three things that an organization does best—are key organizational assets. Knowing how to do things effectively and efficiently in ways that other organizations cannot duplicate is a primary source of profit and competitive advantage that cannot be easily purchased by competitors in the marketplace.

For instance, having a unique build-to-order production system constitutes a form of knowledge and perhaps a unique asset that other firms cannot easily copy. With knowledge, firms become more efficient and effective in their use of scarce resources. Without knowledge, firms become less efficient and less effective in their use of resources, and ultimately fail.

Types of Knowledge

Businesses today need to organize and manage both structured and semi-structured knowledge assets. **Structured knowledge** is explicit knowledge that exists in formal documents such as reports as well as in formal rules that organizations derive by observing experts and their decision-making behaviors. According to experts, however, at least 80 percent of an organization's business content is unstructured or semi-structured—information in folders, messages, memos, proposals, emails, graphics, chat room exchanges, slide presentations, and even videos created in different formats and stored in many locations. In still other cases, there is no formal or digital information of any kind, and the knowledge resides in the heads of employees. This is called **tacit knowledge,** whereas knowledge that has been documented is called **explicit knowledge.** Knowledge management systems feature capabilities for storing and searching for all of these different types of knowledge and for locating employee expertise within the firm.

Structured knowledge
Knowledge in the form of structured documents and reports.

Tacit knowledge
Expertise and experience of organizational members that has not been formally documented.

Explicit knowledge
Knowledge that has been documented.

Types of Knowledge Management Systems

There are three major types of knowledge management systems: enterprise-wide knowledge management systems, knowledge work systems, and "intelligent" techniques. Examine Figure 2.9, which illustrates these three major types.

Enterprise-wide knowledge management systems include general-purpose firm-wide systems that collect, store, distribute, and apply digital content and knowledge. A **knowledge work system (KWS)** is a specialized information system that aids engineers, scientists, and other knowledge workers in the discovery, creation, and integration of new knowledge into an organization. Knowledge management also includes a diverse group of **intelligent techniques**, such as data mining and AI tools such as machine learning, deep learning neural networks, intelligent natural language processing, computer vision systems, robotics, genetic algorithms, fuzzy logic, and expert systems. These techniques have different objectives, from a focus on discovering knowledge (data mining, machine learning, deep learning neural networks, computer vision systems) to discovering optimal solutions for problems (genetic algorithms), to software "agents" and robots that can use learned or built-in knowledge to perform specific tasks or actions, to distilling knowledge in the form of rules for a computer program (expert systems). We explore enterprise-wide knowledge management systems and KWS in more depth in the following sections. Chapters 11 and 12 provide more detail about intelligent techniques.

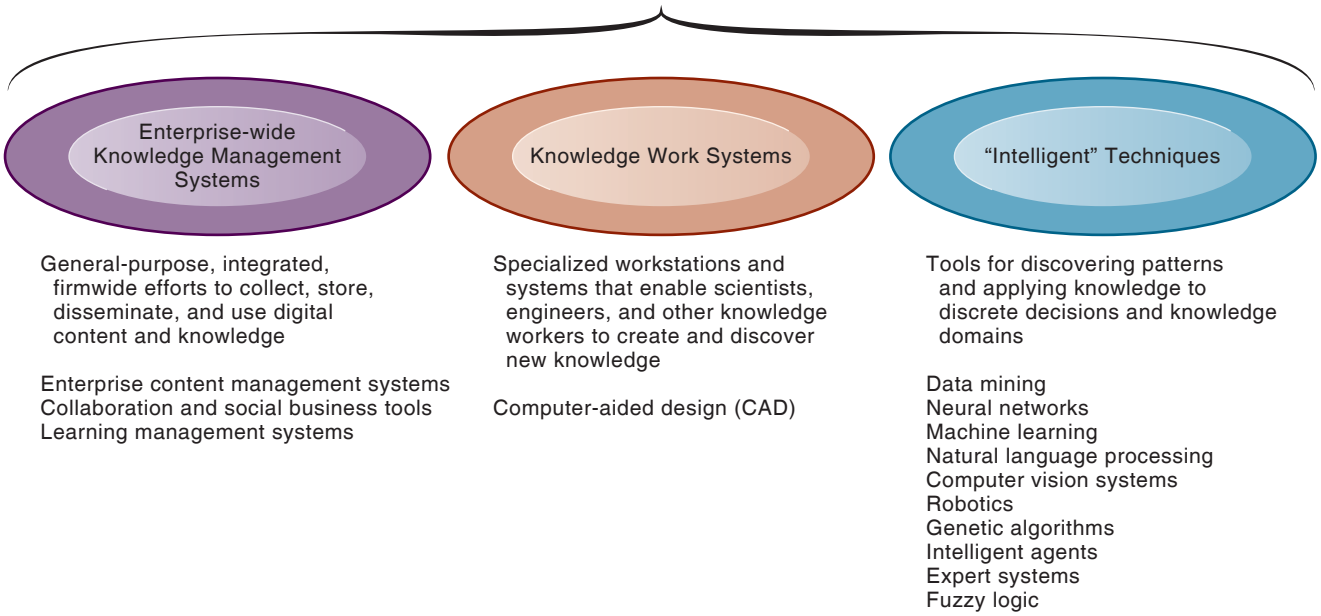
Enterprise-wide knowledge management systems
General-purpose, firmwide systems that collect, store, distribute, and apply digital content and knowledge.

Knowledge work system (KWS)
Information system that aids knowledge workers in the discovery, creation and integration of new knowledge into the organization.

Intelligent techniques
Technologies such as data mining and AI tools that aid human decision makers by capturing individual and collective knowledge, discovering patterns and behaviors in large quantities of data, and generating solutions to problems that are too large and complex for humans to solve on their own.

Figure 2.9 Major Types of Knowledge Management Systems

There are three major types of knowledge management systems, and each can be broken down further into more specialized types of knowledge management systems.



Enterprise-wide Knowledge Management Systems

Systems for enterprise-wide knowledge management include enterprise content management systems, as well as supporting technologies such as collaboration and social business tools, and systems for managing various types of employee learning and training.

Enterprise Content Management Systems

Enterprise content management system (ECM system)

Information system that helps organizations collect and manage structured, semi-structured and unstructured types of information, providing a unified repository of documents, reports, presentations, and best practices.

Taxonomy

Method of classifying things according to a predetermined system.

Wiki

Collaborative website that makes it easy for users to add, delete, or modify content, without any knowledge of web page development or programming techniques.

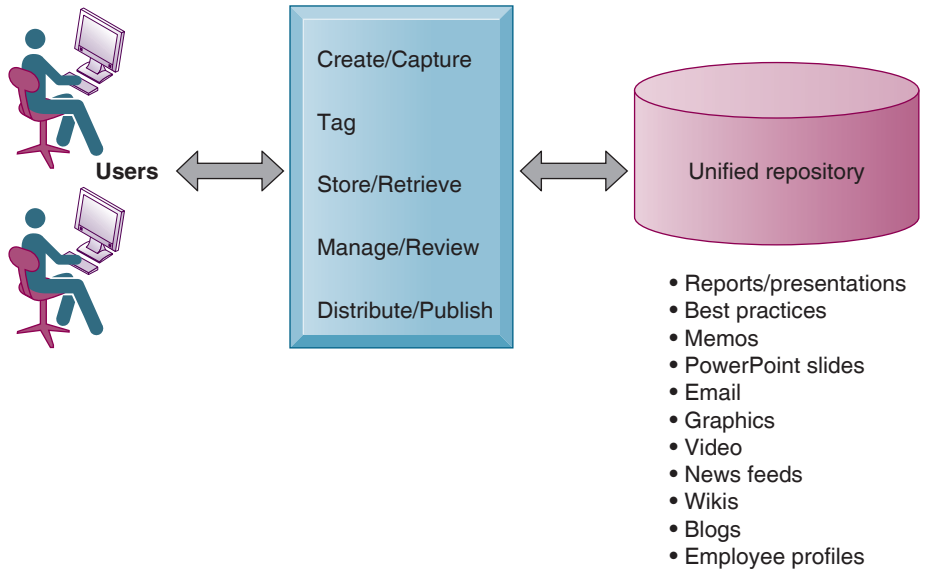
Organizations can use an **enterprise content management system (ECM system)** to help collect and manage structured, semi-structured, and unstructured types of information. A key problem in managing knowledge is the creation of an appropriate classification scheme, or **taxonomy**, to organize information into meaningful categories so that it can be easily accessed. Once the categories for classifying knowledge have been created, each knowledge object needs to be "tagged," or classified, so that it can be easily retrieved. ECM systems have capabilities for knowledge capture, tagging, storage, retrieval, distribution, and preservation. ECM systems interface with corporate databases and content repositories and can be used to create an enterprise knowledge portal that provides a single point of access to information resources. OpenText, IBM, and Oracle are leading vendors of ECM system software.

Examine Figure 2.10, which illustrates how an ECM system functions. Such systems include a unified repository of documents, reports, presentations, and best practices, as well as capabilities for collecting and organizing semi-structured knowledge such as email and unstructured knowledge such as images, audio files, and videos.

Major ECM systems also enable users to access external sources of information, such as news feeds and research, and to communicate by email, chat/IM, discussion groups, and videoconferencing. They have started to incorporate blogs, wikis, and other enterprise social networking tools. A **wiki** is a type of collaborative website that makes it easy for users to contribute and edit text content and graphics without any knowledge of web page development or programming techniques. (The most well-known public wiki is Wikipedia, the largest collaboratively edited reference project in the world.) Wikis are very useful tools for storing and sharing corporate knowledge and insights.

Figure 2.10 An Enterprise Content Management System

An enterprise content management system has capabilities for classifying, organizing, and managing structured, semi-structured, and unstructured knowledge and making it available throughout the enterprise.



Systems for Locating and Sharing Expertise

Some of the knowledge that businesses need is not in the form of a digital document but instead is tacit knowledge that resides in the memory of individual experts in the firm. Contemporary ECM systems, along with the systems for collaboration and social business described in Sections 2.5 and 2.6, have capabilities for locating experts and tapping their knowledge. These include online directories of corporate experts and their profiles, with details about their job experience, projects, publications, and educational degrees, as well as repositories of expert-generated content. Specialized search and collaboration tools make it easier for employees to find the appropriate expert in an organization. For knowledge resources outside the firm, social networking and social business tools enable users to bookmark web pages of interest, tag these bookmarks with keywords, and share the tags and web page links with other people.

Learning Management Systems

Companies need ways to keep track of and manage employee learning and to integrate it more fully into their knowledge management and other corporate systems. A **learning management system (LMS)** provides tools for the management, delivery, tracking, and assessment of various types of employee learning and training.

Learning management system (LMS)

Tools for the management, delivery, tracking, and assessment of various types of employee learning.

Contemporary LMS support multiple modes of learning, including downloadable videos, web-based classes, live instruction in classes or online, and group learning in online forums and chat sessions. The LMS consolidates mixed-media training, automates the selection and administration of courses, assembles and delivers learning content, and measures learning effectiveness.

Major Types of Knowledge Work Systems

The enterprise-wide knowledge management systems just described provide a wide range of capabilities that can be used by many if not all the workers and groups in an organization. Firms also have specialized systems for knowledge workers to help them create new knowledge and to ensure that this knowledge is properly integrated into the business.

Knowledge Workers and Knowledge Work

Knowledge workers, introduced in Chapter 1, include researchers, designers, architects, scientists, and engineers who primarily create knowledge and information for the organization. Knowledge workers usually have high levels of education and memberships in professional organizations and are often asked to exercise independent judgment as a routine aspect of their work. For example, knowledge workers create new products or find ways of improving existing ones. Knowledge workers perform key roles that are critical to the organization and to the managers who work within the organization.

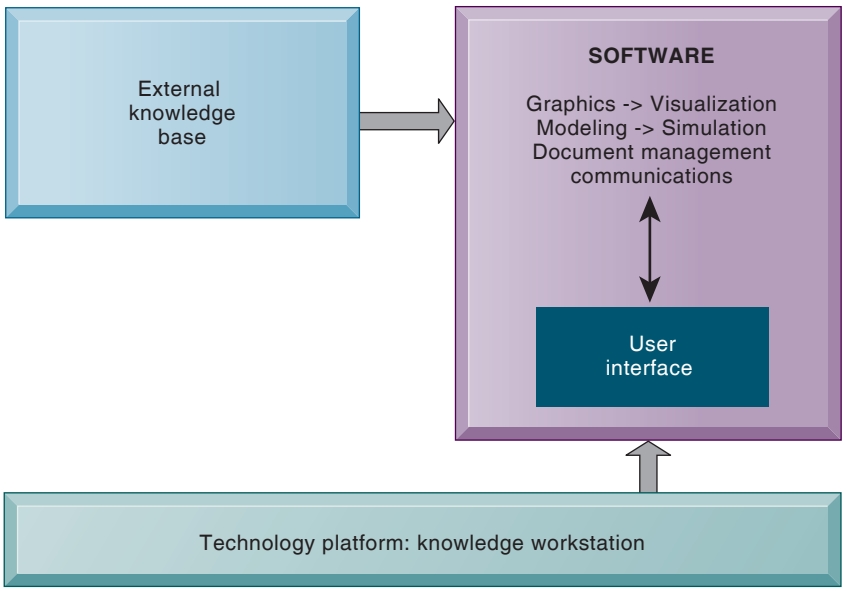
Requirements of Knowledge Work Systems

Examine Figure 2.11, which summarizes the requirements of KWS. Most knowledge workers rely on tools such as word processing, messaging, email, videoconferencing, collaboration, and scheduling systems, which are designed to increase worker productivity in the office. However, knowledge workers also require highly specialized KWS with powerful graphics, analytical tools, and communications and document management capabilities. For example, the knowledge workstation for a graphics designer typically includes a more powerful central processing unit, and more RAM (random access memory) than a typical desktop computer, with additional secondary storage for large files and a dedicated graphics card or graphics processing unit (GPU), which can significantly improve performance when working with large files and complex designs.

These systems require sufficient computing power to handle the sophisticated graphics or complex calculations necessary for such knowledge workers as scientific researchers, engineers, and product designers. Because knowledge workers need knowledge from the external world, these systems also must give the worker quick and easy access to external databases. They typically feature user-friendly interfaces that enable users to perform needed tasks without having to spend a great deal of time learning how to use the system. Knowledge workers are highly paid—wasting a knowledge worker's time is simply too expensive.

Figure 2.11 Requirements of Knowledge Work Systems

Knowledge work systems require strong links to external knowledge bases in addition to specialized hardware and software.



2.9 Describe the Role of the Information Systems Function

We've seen that businesses need information systems to operate today and that they use many kinds of systems, but who is responsible for running these systems? End users manage their systems from a business standpoint, but managing the technology requires a special information systems function. In all but the smallest of firms, the **information systems department** is the formal organizational unit responsible for information technology services within the organization. The information systems department is responsible for maintaining the hardware, software, data storage, and networks that comprise the firm's IT infrastructure. We describe IT infrastructure in detail in Chapter 5.

Information systems department

The formal organizational unit that is responsible for information technology services within the organization.

The Information Systems Department

The information systems department consists of specialists such as programmers, systems analysts, and information systems managers. **Programmers** are highly trained technical specialists who write the software instructions for computers. **Systems analysts** constitute the principal liaisons between the information systems groups and the rest of the organization. It is the systems analyst's job to translate business problems and requirements into information requirements and systems. **Information systems managers** are leaders of teams of programmers and analysts, project managers, physical facility managers, telecommunications managers, or database specialists. They are also managers of computer operations and data entry staff. External specialists, such as hardware vendors and manufacturers, software firms, and consultants, also frequently participate in the day-to-day operations and long-term planning of information systems.

Programmers

Highly trained technical specialists who write computer software instructions.

Systems analysts

Specialists who translate business problems and requirements into information requirements and systems, acting as liaison between the information systems department and the rest of the organization.

Information systems managers

Leaders of the teams of various specialists in the information systems department.

In many companies, the information systems department is headed by a **chief information officer (CIO)**. The CIO is a senior manager who oversees the use of information technology and information systems in the firm. Today's CIOs are expected to have a strong business background, as well as information systems expertise, and to play a leadership role in integrating technology with the firm's business strategy. Large firms today also have positions for a chief security officer, chief knowledge officer, chief data officer, chief AI officer, and chief privacy officer, all of whom work closely with the CIO.

Chief information officer (CIO)

Senior manager in charge of the information technology and information systems in the firm.

The **chief security officer (CSO)** is in charge of information systems security for the firm and is responsible for enforcing the firm's information security policy (see Chapter 8). (Where information systems security is separated from physical security, this position is sometimes called the chief information security officer [CISO]). The CSO is responsible for educating and training users and information systems specialists about security, keeping management aware of security threats and breakdowns, and maintaining the tools and policies chosen to implement security.

Chief security officer (CSO)

In charge of information systems security and responsible for enforcing the firm's information security policy.

Information systems security and the need to safeguard personal data have become so important that companies collecting vast quantities of personal data have established positions for a **chief privacy officer (CPO)**. The CPO is responsible for ensuring that the company complies with data privacy laws and regulations.

Chief privacy officer (CPO)

Responsible for ensuring the company complies with existing data privacy laws and regulations.

The **chief knowledge officer (CKO)** is responsible for the firm's knowledge management program. The CKO helps design programs and systems to find new sources of knowledge or to make better use of existing knowledge in organizational and management processes.

Chief knowledge officer (CKO)

Senior executive in charge of the organization's knowledge management program.

Chief data officer (CDO)
Responsible for enterprise-wide governance and utilization of information to maximize the value the organization can realize from its data.

Chief AI officer (CAIO)
Responsible for overseeing a company's overall strategy, acquisition, implementation, and monitoring of AI technology.

End users
Representatives of departments outside the information systems group for whom applications are developed.

The **chief data officer (CDO)** is responsible for enterprise-wide governance and usage of information to maximize the value an organization can realize from its data. The CDO ensures that the firm is collecting appropriate data to serve its needs, deploying appropriate technologies for analyzing the data, and using the results to support business decisions. This position arose to deal with the massive amounts of data that organizations are now generating and collecting (see Chapter 6).

An emerging addition to the C-suite is the **chief AI officer (CAIO**). The CAIO is a senior executive responsible for overseeing a company's overall strategy, acquisition, implementation, and monitoring of AI technology. This position requires a deep understanding of the business, expertise in AI and related technologies, and regulatory awareness.

End users are representatives of departments outside the information systems group for whom applications are developed. These users are playing an increasingly large role in the design and development of information systems.

In the early years of computing, the information systems group was composed mostly of programmers who performed highly specialized but limited technical functions. Today, a growing proportion of staff members are systems analysts and network specialists, with the information systems department acting as a powerful change agent in the organization. The information systems department suggests new business strategies and new information-based products and services and coordinates both the development of the technology and the planned changes in the organization.

Organizing the Information Systems Function

IT governance
Strategy and policies for using information technology within an organization, specifying the decision rights and framework for accountability to ensure that the use of information technology supports the organization's strategies and objectives.

The question of how the information systems department should be organized is part of the larger issue of IT governance. **IT governance** includes the strategy and policies for using information technology within an organization. It specifies the decision rights and framework for accountability to ensure that the use of information technology supports the organization's strategies and objectives. How much should the information systems function be centralized? What decisions must be made to ensure effective management and use of information technology, including the return on IT investments? Who should make these decisions? How will these decisions be made and monitored? Firms with superior IT governance will have clearly thought out the answers.

2.10 Understand How the Information in This Chapter Can Help Your Career

The information in Chapter 2 can help you advance your career by helping you understand the different types of systems you are likely to encounter and their role in the business. Chapter 2 also describes the increasingly important business role of collaboration, teamwork, and knowledge work, and the tools that can make this work more effective. You'll also be able to identify the typical issues and problems you are likely to encounter when engaging in collaborative work and ways of dealing with them.

Real-World Scenario

Customer Success Analysts Help Princess Cruises Maximize Revenue

Princess Cruises, considered one of the most recognized cruise lines in the world, uses a variety of retail promotions to attract customers and remain competitive in the market. However, Princess needed a measurable way to verify which promotion offers had the most appeal to new and existing customers. Princess enlisted the services of Hanover Research, a marketing and data analytics firm with expertise in customer experience analysis. Hanover's team of data analysts worked with customer success analysts and marketing staff from Princess to identify which special offers, such as on-board credit or free gratuities, and which promotional campaigns were most appealing to customers. Their findings showed Princess the optimal combination of offers to maximize revenue.

Job Description and Career Path: Customer Success Analyst

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Customer Success Analyst	Responsible for ensuring that customers have a positive experience with a company's products or services. Includes analyzing customer data and feedback to identify improvements that will produce a better customer experience. Job responsibilities include: • Analyzing customer data on customer satisfaction levels, churn, retention rates. • Conducting surveys and customer interviews to identify areas for improving customer experience and engagement. • Collaborating with various teams to address customer needs and resolve issues. • Communicating frequently with customers to keep them informed of product updates and features.	• Knowledge of Microsoft 365 tools: Excel, Word, PowerPoint, Teams • Experience with Salesforce CRM software • Experience with data analytics and business intelligence platforms such as Tableau	• Customer success analyst (0–2 years' experience): $45,000–$60,000 • Customer success analyst II (2–4 years' experience): $60,000–$75,000 • Customer success manager $80,000–$100,000 • Senior customer success manager: $125,000–$140,000 Salaries can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

How Much Do Virtual Meetings Help Collaboration?

During the Covid-19 pandemic, many organizations turned to virtual meeting systems such as Zoom and Microsoft Teams so that employees could perform their jobs at home. Virtual meeting software has helped employees to work and collaborate remotely and gives managers a tool for managing employees from afar.

Although many employees have returned to the office, almost 13 percent of full-time U.S. employees still work from home, and around 28 percent of employees have adapted to a hybrid work model in which they spend some days in the office and work some days remotely. However, as working virtually takes root in the workplace, organizations need to pay closer attention to the risks and rewards of collaborative technologies. Is virtual meeting technology good or bad for employees and managers, and how much does the technology help or hinder the collaboration process?

Since many companies went completely remote in 2020, the number of meetings per day has increased. It looks like virtual meetings—and lots of them—are here to stay, and not everyone is happy. Even before the Covid-19 pandemic, more than 70 percent of managers thought meetings, whether virtual or face-to-face, were costly and unproductive. Studies show that spending too much time in meetings negatively affects people's psychological, physical, and mental well-being. Having too many meetings diverts workers during their most productive hours, interrupts people's train of thought, and detracts from effective collaboration. Regardless of company size, both the number of collaborative demands and the diversity of them are overwhelming people. This is a serious problem.

In addition to the problems caused by the frequency and dynamics of virtual meetings, there are problems caused by the nature of the technology itself. Recent research has found that videoconferencing technology can actually make it harder for people to collaborate. Dr. Anita Williams Wooley, associate professor of organizational behavior and theory at Carnegie Mellon's Tepper School of Business, compared how well two-person teams of unacquainted workers in different locations using videoconferencing and similar teams communicating only with audio tools were able to perform a collaborative task. Wooley and her fellow researchers found that the teams communicating via only audio means scored higher than those using videoconferencing. These teams were more successful in synchronizing their vocal cues and speaking turns. Wooley believes communicating via video can draw attention away from audio cues in many cases, reducing capacity to carry out the work itself.

Meeting virtually creates more complexity for both meeting leaders and attendees. People aren't used to the unnatural lack of nonverbal cues, prolonged eye contact, or overload of faces to process on the screen. Seeing our own faces as we talk or listen, and awareness of how we appear online is stressful. The amount of effort required to process all of these stimuli while simultaneously thinking and communicating is fatiguing. The term "Zoom fatigue" is now part of our vocabulary. Constant connectedness and overuse of virtual communication platforms, particularly videoconferencing, cause many people to experience feelings of tiredness, anxiety, or burnout.

During 2020, doctoral students Katie Kavanagh, Nicole Voss, and Liana Kreamer, along with Professor Steven G. Rogelberg, all from the University of North Carolina at Charlotte, collected data from 150 employees from a range of industries in the United States and Europe to learn about their experiences in virtual meetings. The vast majority of those surveyed reported feeling fatigued and drained during and after their virtual meetings—more so than with in-person meetings. In addition to feeling exhausted from the number and scheduling of meetings, survey respondents varied widely in their explanations for why this was so. Different people had different virtual meeting preferences. Some employees reported, for example, that they found virtual meetings more fatiguing than in-person meetings because there are fewer social cues, while others appreciated the detached virtual meeting style. Managers need to take all of these factors into account when using virtual meeting technology.

Virtual collaboration also introduces risks of isolation, exclusion, surveillance, and self-censorship. Heavy reliance on virtual meeting technology in the manager-employee relationship can increase isolation between managers and their employees. This isolation becomes amplified when there is an upsurge in videoconferencing fatigue, or when technology cannot adequately replace less formal social interactions in remote work environments. Diminishing opportunities for spontaneous face-to-face social interaction widen the manager-employee gap.

Virtual meeting technology may also hinder some employees' ability to fully participate in organizational life. There may be generational and personality differences in facility with technology and work behaviors that affect attitudes toward virtual meetings and levels of participation. The virtual meeting platform will showcase people who actively participate in virtual team activities, while unintentionally excluding people perceived as not participating or those who lack the technology skills to do so. Technology has the power to reshape "in" and "out" groups in the work environment.

Collaboration technologies can be used by companies and managers to monitor employees. Many collaboration platforms track user activity and the duration of time spent on them. The data can tell managers about individual user engagement and can potentially be used to make judgments about employee productivity, even though this is not the technology's main purpose. For instance, Microsoft Teams'

capability to provide detailed automated user analytics can be used to show how people are working together. It is important to inform employees exactly what data are collected by the technologies they use, how they are collected, and for what purpose. Research shows that negative perceptions of managerial surveillance lower trust in the manager–employee relationship, adversely affecting employee performance and job satisfaction.

Collaboration technologies create a sense of openness where managers and employees can easily text, call, see each other, and freely share information. However, this openness may also make some people feel they will be judged if they say the wrong thing. An employee may hold back on sharing their thoughts in a recorded online meeting if they're not confident about sounding intelligent. Collaboration technologies can create conditions for self-censorship, even if the technology is perceived to be user-friendly. This may limit opportunities to build closer connections with managers and other employees.

Although each organization has its own specific needs, recommendations for making virtual meetings more effective and less tiring include the following:

- Cancel unnecessary meetings and make necessary meetings shorter.
- Use breakout rooms for problem-solving, discussions, and social interactions.
- Use asynchronous meetings, such as by creating a shared Google Doc for employees to contribute to throughout the day.
- Build in breaks during long meetings and in between back-to-back meetings.
- Moderate and facilitate virtual meetings more actively, moving topics along when needed, and ensuring that everyone has an opportunity to contribute.
- Turn off "self-view" if possible on your meeting platform and make camera use optional for some meetings.
- Implement meeting-free blocks of time or days.

This last recommendation is especially important. Although building trust and achieving team cohesion rely on frequent, quality interactions, increasing virtual meetings is no longer the best way to accomplish this. As a result, many organizations, including Meta, Clorox, Slack, Asana, and Twilio, have adopted no-meeting days, during which people work according to their own rhythms and collaborate with others at a pace and on a schedule that is not forced.

Professor Ben Laker, University of Reading Henley Business School, along with human resource management professors Vijay Pereira, Pawan Budhwar, and Ashish Malik, surveyed 76 companies, with more than 1,000 employees each and operations in more than 50 countries. These companies had introduced from one to five no-meeting days per week (including one-on-one meetings) during 2021. Laker and his colleagues examined data comparing employee stress levels before and after a reduction in meetings and assessed the impact on productivity, collaboration, and engagement. Nearly half (47 percent) of the companies Laker and his team studied reduced meetings by 40 percent by introducing two no-meeting days per week. Thirty-five percent instituted three no-meeting days, and 11 percent implemented four. The remaining 7 percent eradicated meetings entirely.

The subsequent impact of introducing meeting-free days was profound. Regardless of the number of meeting-free days instituted, autonomy, communication, engagement, and satisfaction all improved, resulting in decreased stress and a rise in productivity. The best results were achieved at companies that had three meeting-free days per week. Eliminating 60 percent of meetings, equivalent to three days per week, increased cooperation by 55 percent and lowered the risk of stress by 57 percent. Workers replaced meetings with better ways of connecting one-on-one, at a pace suitable for them.

SOURCES: Cloey Callahan, "How Asana and Slack's Meeting Purges Have Paid Off," Digiday.com, February 14, 2024; Burcu Bulgurcu et al., "Are Enterprise Social Platforms All Talk?" *MIT Sloan Management Review*, February 14, 2024; Benjamin Laker, "New Year, New Approach: Resolving the Meeting Load Paradox in 2024," Forbes.com, January 2, 2024; Katherine Haan, "Remote Work Statistics and Trends," Forbes.com, June 12, 2023; Tracy Patch, "8 Common Virtual Meeting Challenges and How to Overcome Them," Switchboard.app, June 9, 2023; Ben Laker et al., "The Surprising Impact of Meeting-Free Days," *MIT Sloan Management Review*, January 18, 2022; Lebene Soga et al., "Design Your Work Environment to Manage Unintended Tech Consequences," *MIT Sloan Management Review*, September 20, 2021; Rob Cross, "Easing the Invisible Burdens of Collaboration," *MIT Sloan Management Review*, September 14, 2021; Heidi Mitchell, "How Videoconferences Make It Harder for Employees to Collaborate," *Wall Street Journal*, June 10, 2021; Katie Kavanagh et al., "How to Combat Virtual Meeting Fatigue," *MIT Sloan Management Review*, March 30, 2021.

Case Study Questions

1. How do virtual meeting technologies enhance collaboration?
2. How do virtual meeting technologies hinder collaboration? Identify the management, organization, and technology factors that contribute to this problem.
3. What can organizations do to improve the effectiveness of collaboration technology?
4. If a company with 500 employees wants to implement virtual meeting technology to support remote work and collaboration, what issues should be considered?

Review Summary

2.1 Understand how business processes relate to information systems.

A business process is a logically related set of activities that defines how specific business tasks are performed, and it represents a unique way in which an organization coordinates work, information, and knowledge. Managers need to pay attention to business processes because they determine how well the organization can execute its business. Business processes may also be a source of strategic advantage. There are business processes specific to each of the major business functions, but many business processes are cross-functional. Information systems automate parts of business processes, and they can help organizations redesign and streamline these processes.

2.2 Explain how information systems serve different management groups.

Operational management is served by transaction processing systems (TPS), such as payroll or order processing, that track the flow of the daily routine transactions necessary to conduct business. Management information systems (MIS) produce reports serving middle management by condensing information from TPS. Decision support systems (DSS) support management decisions that are unique and rapidly changing using advanced analytical models. MIS and DSS provide business intelligence that helps managers and enterprise employees make more informed decisions. BI systems serve multiple levels of management and also include executive support systems (ESS) for senior management that provide data in the form of graphs, charts, and dashboards delivered via portals using many sources of internal and external information.

2.3 Explain how enterprise applications improve firm performance.

Enterprise applications are systems that can coordinate activities, decisions, and knowledge across many different functions, levels, and business units in a firm. Enterprise systems, also known as enterprise resource planning (ERP) systems, integrate business processes in manufacturing and production, finance and accounting, sales and marketing, and human resources into a single software system. Supply chain management (SCM) systems help the firm manage its relationship with suppliers to optimize the planning, sourcing, virtual manufacturing, and delivery of products and services. Customer relationship management (CRM) systems coordinate the business processes surrounding the firm's customers. Knowledge management systems (KMS) enable firms to optimize the creation, sharing, and distribution of knowledge. Intranets and extranets are private corporate networks based on Internet technology. Intranets are internal corporate networks. Extranets make portions of private corporate intranets available to outsiders.

2.4 Understand the importance of collaboration and social business systems.

Collaboration means working with others to achieve shared and explicit goals. Social business is the use of internal and external social network platforms to engage employees, customers, and suppliers, and it can enhance collaborative work. Collaboration and social business have become increasingly important in business because of globalization, the decentralization of decision-making, and growth in jobs where interaction is the primary value-adding activity. Collaboration and social business enhance innovation, productivity, quality, and customer service.

2.5 Describe collaboration tools.

Tools and technologies for collaboration include email and instant messaging, virtual meeting systems (such as Zoom and Microsoft Teams), cloud-based file-sharing services, corporate collaboration platforms such as Microsoft SharePoint, and AI technology.

2.6 Describe social business tools.

Tools and technologies for social business include enterprise social networking tools such as Salesforce Chatter and Microsoft's Viva Engage, as well as virtual world and metaverse platforms.

2.7 Understand how to evaluate collaboration and social business tools.

A six-step "to-do" list that uses a time/space collaboration and social business tool matrix can be helpful in evaluating collaboration and social business tools.

2.8 Describe knowledge management systems.

Knowledge management refers to the set of business processes developed in an organization to create, store, transfer, and apply knowledge. Businesses today need to organize and manage both structured, semi-structured, and unstructured knowledge. There are three major types of knowledge management systems: enterprise-wide knowledge management systems, knowledge work systems, and "intelligent" techniques. Enterprise-wide knowledge management systems include enterprise content management systems, collaboration and social business tools, and learning management systems. Knowledge work systems (KWS) are specialized systems built for engineers, scientists, and other knowledge workers charged with discovering and creating new knowledge for an organization. Intelligent techniques include a diverse group of tools, such as data mining and various AI technologies.

2.9 Describe the role of the information systems function.

The information systems department is the formal organizational unit responsible for information technology services. It is responsible for maintaining the hardware, software, data storage, and networks that comprise the firm's IT infrastructure. The department consists of specialists, such as programmers, systems analysts, and information systems managers, and is often headed by a chief information officer (CIO). Other important IT management positions include the chief security officer (CSO), chief privacy officer (CPO), chief knowledge officer (CKO), chief data officer (CDO), and chief AI officer (CAIO). IT governance includes the strategy and policies for using information technology within an organization.

Key Terms

BI dashboard, 45
Business intelligence (BI), 41
Chief AI officer (CAIO), 62
Chief data officer (CDO), 62
Chief information officer (CIO), 61
Chief knowledge officer (CKO), 61
Chief privacy officer (CPO), 61
Chief security officer (CSO), 61
Collaboration, 47
Customer relationship management system (CRM system), 46
Decision-support system (DSS), 43
End users, 62
Enterprise applications, 45
Enterprise content management system (ECM system), 58
Enterprise system, 45
Enterprise-wide knowledge management systems, 57
Executive support system (ESS), 44
Explicit knowledge, 57
Human-AI collaboration, 52
Information systems department, 61
Information systems managers, 61
Intelligent techniques, 57
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IT governance, 62
Knowledge management, 57
Knowledge management system (KMS), 46
Learning management system (LMS), 59
Knowledge work system (KWS), 57
Management information system (MIS), 42
Portal, 44
Programmers, 61
Social business, 48
Structured knowledge, 57
Supply chain management system (SCM system), 46
Systems analysts, 61
Tacit knowledge, 57
Taxonomy, 58
Team, 47
Telepresence, 51
Transaction processing system (TPS), 40
Virtual worlds, 55
Wiki, 58

Review Questions

2.1 Understand how business processes relate to information systems.

- Define business processes and describe the role they play in organizations.
- Describe the relationship between information systems and business processes.

2.2 Explain how information systems serve different management groups.

- Define business intelligence systems.
- Describe the characteristics of transaction processing systems (TPS) and the role they play in a business.
- Describe the characteristics of management information systems (MIS), decision-support systems (DSS), and executive support systems (ESS) and explain how each type of system helps managers make decisions.

2.3 Explain how enterprise applications improve firm performance.

- Explain how enterprise applications improve firm performance.
- Define enterprise systems, supply chain management (SCM) systems, customer relationship management (CRM) systems, and knowledge management systems (KMS), and describe their business benefits.

2.4 Understand the importance of collaboration and social business systems.

- Define collaboration and social business, and explain why they have become so important in business today.
- List and describe the business benefits of collaboration and social business.
- Describe a supportive organizational culture for collaboration.

2.5 Describe collaboration tools.

- List and describe the various types of collaboration tools and technologies.
- Explain how AI technology is being used to aid collaboration.

2.6 Describe social business tools.

- List and describe the various types of social business tools and technologies.

2.7 Understand how to evaluate collaboration and social business tools.

- Describe the time/space collaboration and social business tool matrix.
- Identify the process that should be followed to help decide which collaboration or social business tool should be evaluated.

2.8 Describe knowledge management systems.

- Describe how knowledge management systems help organizations make better use of their structured, semi-structured, and unstructured knowledge, as well as their tacit knowledge assets.
- List and describe the three types of systems for knowledge management.

2.9 Describe the role of the information systems function.

- Describe how the information systems function supports a business.
- Describe the roles that programmers, systems analysts, information systems managers, chief information officer (CIO), chief security officer (CSO), chief data officer (CDO), chief privacy officer (CPO), chief AI officer (CAIO), and chief knowledge officer (CKO) play.

Discussion Questions

2.10 How could information systems be used to support the order fulfillment process illustrated in Figure 2.1? What are the most important pieces of information these systems should capture? Explain your answer.

2.11 Identify the steps that are performed in the process of selecting and checking out a book from your college library and the information that flows among these activities. Diagram the process. Are there any ways this process could be adjusted to improve the performance of your library? Diagram the improved process.

2.12 Use the Time/Space Collaboration and Social Business Tool Matrix to classify the collaboration and social technologies Toyota Motor North America uses.

Using AI Project

2.13 AI Project: Selecting a Brand Image

You believe the retail pet store described in the AI Project in Chapter 1 will be even more appealing and memorable if it has a brand image. Using Microsoft Copilot, Google Gemini, or ChatGPT, create a prompt or series of prompts to generate a brand image for the store using the name you selected in Chapter 1. Save a copy of your prompts and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience in analyzing opportunities to improve business processes with new information system applications; using spreadsheet software to develop a town budget; using Tableau business intelligence and analytics software, and using Google Maps to plan efficient transportation routes.

For student files please visit www.pearsonhighered.com/laudon or MyLab MIS.

2.14 Management Decision Problem

The U.S. Census, conducted every 10 years, counts every resident in the United States. The results of the census are used to determine the number of seats each state has in the U.S. House of Representatives, Electoral College votes, and an estimated $1.5 trillion each year in federal funding for the states for healthcare, education, infrastructure improvements, and more.

In the past, all census data were collected by paper surveys, transported to the U.S. Census Bureau, and input manually into the Bureau's information systems. The 2020 census was primarily conducted online, with about 60 percent of the U.S. population of 330 million filling out a form on their Internet-linked smartphones, tablets, or personal computers. The Bureau also collected census responses over the phone and on paper forms. An unprecedented amount of data had to be collected, stored, secured, and interpreted. What are some of the management, organizational, and technology issues that the Census Bureau had to address in modernizing a massive, important paper-based system?

2.15 Using a Spreadsheet to Create a Town Budget

Software skills: Spreadsheet charts, formulas, and assumptions

Business skills: Preparing a town budget

In this exercise, you will learn how to use spreadsheet software to improve management decisions about a budget and budget projections, in this case for a town. The town of ClearSkies, with a population of 9,042, is located in Oklahoma. Each November, the town manager works with the mayor and the board of trustees to develop the town's budget for the forthcoming year. The town's main source of revenue is the local property tax, but it also receives some aid from the state government and some revenue from miscellaneous licenses and fees.

The town leaders limit tax increases to 3 percent or less. They fear revenues may be declining because the state wants to cut the state aid it provides to local governments. Anticipating continued cutbacks in state aid, the town manager would like to develop preliminary budgets for the next two years to see if they can be supported by anticipated revenues. If planned expenditures exceed revenues, the town manager, the mayor, and the trustees must develop an alternative budget that is balanced.

The MIS18_Project 2.15_Student File shows the receipts and disbursements for the town in 2025. The 2025 budget can be used as the basis for projecting the town budget for the next two years. In projecting the next two years' budgets, the town manager wants to use the following assumptions: She expects the state will reduce its aid to the town by 6 percent each year. Historically, expenditures for employee salaries and benefits have been rising 5 percent annually. She expects all other expenditures to rise at a rate of 6 percent annually and miscellaneous receipts to rise 4 percent annually. Tax increases should be 3 percent per year. The town's expenditure for debt service to pay off previous loans will

remain constant. Can the town balance its budget if it keeps its promise not to raise taxes beyond 3 percent or go further into debt?

In developing the town's budget projections, create an Assumptions section of the worksheet to identify factors in your calculations of receipts and disbursements. Make sure formulas reference cells in the Assumptions section wherever possible. Calculate the receipts and disbursements for each of the categories on the worksheet for 2025 through 2027.

Calculate total receipts and total disbursements. The problem has been simplified so that all revenue is collected in the year it is due. Complete the worksheet by subtracting total receipts from total disbursements in 2025 through 2027.

Then, write a brief analysis of the projected budgets. If these budgets don't balance, what steps would you recommend be taken by the town? Revise your worksheet to incorporate your recommendations and save it under another name. Keep revising the worksheet until you have developed a balanced budget. Is there some way for the town to come up with a balanced budget without raising taxes?

The town manager wants to highlight the portion of salary expenditures represented by the town court, and the best way to do that is visually in a pie chart. Create a pie chart showing the various categories of salary disbursements for 2025.

2.16 Getting to Know Tableau

This chapter introduced you to various systems for business intelligence, including Tableau, a popular business intelligence and analytics software tool available from Salesforce. In this project, you will start to become familiar with Tableau. You will begin by researching the following versions of Tableau: Tableau Public, Tableau Desktop, and Tableau Cloud. Gather information about the key features, unique functionalities, similarities and differences, target audiences, pricing models, and advantages and disadvantages of each of these versions, and when it might be better to use one version as compared to another. Then, write a report or create a presentation that summarizes your research. Use Tableau to create at least three data visualizations to illustrate your findings and integrate them into your report or presentation. If you don't have access to Tableau Desktop, ask your instructor, or alternatively, you can create a free Tableau Public account.

2.17 Using Google Maps to Plan Efficient Transportation Routes

In this exercise, you will use Google Maps to map out transportation routes for a business and select the most efficient route. You have just started working as a dispatcher for Cross-Country Transport, a new trucking and delivery service based in Cleveland, Ohio. Your first assignment is to plan a delivery of office equipment and furniture from Elkhart, Indiana (at the corner of E. Indiana Ave. and Prairie Street), to Hagerstown, Maryland (corner of Eastern Blvd. N. and Potomac Ave.). To guide your trucker, you need to know the most efficient route between the two cities. Use Google Maps to find the route that is the shortest distance between the two cities and also to find the route that takes the least time. Compare the results. Which route should Cross-Country use?

2.18 Collaboration and Teamwork Project: Identifying Management Decisions and Systems

With a team of two or more other students, search business publications on the web to find a description of a manager of a corporation. Gather information about what the manager does and the role the manager plays in the company. Identify the organizational level and business function where this manager works. Make a list of the kinds of decisions this manager has to make and the kind of information that the manager would need for those decisions. Suggest how information systems could supply this information. If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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3

Information Systems, Organizations, and Strategy

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Learning Objectives

After completing this chapter, you will be able to:

3.1 Discuss technical and behavioral views of organizations.

3.2 Discuss features of organizations that affect information systems.

3.3 Describe the impact of information systems on organizations.

3.4 Discuss Porter's competitive forces model.

3.5 Discuss the value chain model.

3.6 Discuss synergies, core competencies, and network economics.

3.7 Understand how AI can be used in competitive strategy.

3.8 Describe the challenges posed by strategic information systems.

3.9 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** PepsiCo Tames the Doritos Crunch
- **Spotlight on Management Case Study:** Spotify's Podcasting Strategy Stumbles
- **Chapter Case Study:** AI Redesigns the Fashion Industry

Opening Case Study

PepsiCo Tames the Doritos Crunch

PepsiCo is a global leader in the food and beverage industry, noted for consumer favorites such as Pepsi Cola, Gatorade, Fritos, Ruffles potato chips, and Doritos. Although PepsiCo's business strategy emphasizes creating snacks or beverages attuned to changing consumer tastes and lifestyles, it also focuses on ways of making the entire experience of consuming its food products more enjoyable. How PepsiCo improved the customer experience for its highly popular Doritos-flavored tortilla chips is a case in point.

Doritos is one of PepsiCo's top-selling brands, and management believes it has huge upside growth potential worldwide. In 2023, Doritos was the leading tortilla chips brand in the United States, with more than 1.1 billion units sold. Doritos is also a top snacking choice for PC gamers. PepsiCo market researchers estimated that 85 percent of U.S. gamers consumed Doritos during the three-month period they studied, and it was the gamers' snack of choice. However, nearly a third of gamers reported that other people's crunching distracted them and impacted their performance. Gaming is a phenomenally popular pastime, with 3.25 million players worldwide, including roughly 65 percent of the U.S. population. This is a market that must be taken very seriously.

There was fierce consumer backlash when PepsiCo tried to modify the Doritos recipe to eliminate Doritos' characteristic crunch sound. Doritos fans loved the crunch sound as much as Doritos' taste and did not want less crunchy chips. Could they have their crunch and eat it too without alienating those around them? Was this possible? Yes, thanks to artificial intelligence (AI) technology.

In November 2023, PepsiCo introduced Doritos Silent, a groundbreaking AI crunch-cancellation software tool that lets PC gamers enjoy their favorite chips without distracting other players. The crunch-cancellation software is able to retain the speaking voice of the person eating Doritos, while silencing crunching sounds so that they can't be heard by fellow game participants.

To create Doritos Silent, developers used AI machine learning (AI/ML) to analyze more than 5,000 different crunch sounds. Approximately 500 people were enlisted to crunch Doritos into a microphone to teach an AI model what Doritos crunching sounds like, so it could isolate it. Sounds of people talking were mixed with the crunch sounds so that the AI model could "learn" how to distinguish the sounds of crunching and voicing. Doritos Silent was trained exclusively on Doritos, but it works with other crunch sounds such as potato chips, crackers, raw vegetables, and other snacking foods, and is compatible with all types of desktop computers. PepsiCo has assured customers that no crunch will go undetected.

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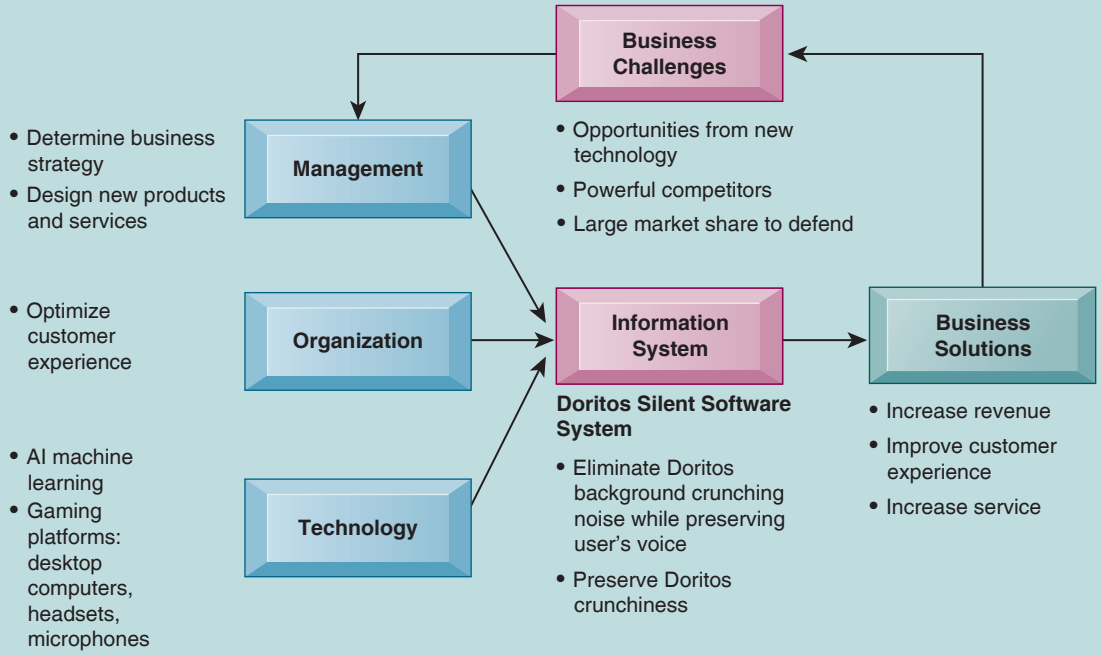


SOURCES: Pepsico.com, accessed August 17, 2024; Doritos.co.uk, accessed August 17, 2024; Sydney Page, "The Sound of Crunching Chips Is Annoying: Doritos Has Made a Silencer," *Washington Post*, November 23, 2023; Christopher Doering, "PepsiCo's Doritos Brand Creates Technology to 'Silence' Its Crunch," Fooddive.com, November 2, 2023; PepsiCo, "Doritos Silent Brings Crunch-canceling Tech to PC Gamers Around the World," Pepisco.com, November 1, 2023.

PepsiCo's experience with improving its strategy with its Doritos brand illustrates some of the ways that information systems help businesses compete, as well as the challenges of finding the right business strategy and how to use technology in that strategy. The food and beverage industry is an extremely crowded and competitive playing field. Even though Doritos is the top tortilla chip snack in the United States, PepsiCo must continually be on the lookout for ways to improve Doritos and its customer experience to stay ahead of the pack. That includes meeting the needs of large market segments such as gamers, who love Doritos chips' flavor, but abhor the crunching sound that fellow gamers make when eating them.

The Opening Case Study diagram above calls attention to important points raised by this case and this chapter. PepsiCo's business strategy focuses on creating snacks or beverages attuned to changing consumer tastes and lifestyles. It must not only focus on how its products taste, but also on the entire customer experience of using the products. PepsiCo used AI/ML to create Doritos Silent software that eliminates background crunching noises that gamers found annoying, while at the same time enabling PepsiCo to preserve Doritos' taste and texture. This helped PepsiCo maintain the popularity of Doritos among a very important market segment (gamers) and grow sales.

Case Study Questions

1. What competitive strategies is PepsiCo using for its Doritos brand?
2. Why is managing the customer experience important for those strategies?
3. What role did AI play in making Doritos more competitive?

CAREER BENEFIT

Developing Doritos Silent was a high-visibility project because there were such high expectations from and so much interest in AI. Developing a successful AI solution for a tricky and important problem would undoubtedly boost the careers of both IT and non-IT staff and give them a great deal of visibility throughout the organization. All of the people working on Doritos Silent would be more in demand internally and also at other organizations because of the AI expertise and experience they acquired.

3.1 Discuss Technical and Behavioral Views of Organizations

Information systems and organizations influence one another. Information systems are built by managers to serve the interests of the business firm. At the same time, the organization must be aware of and open to the influences of information systems to benefit from new technologies.

Examine Figure 3.1, which illustrates the complexity of the relationship between information technology (IT) and organizations. The relationship is influenced by many mediating factors, including the organization's structure, business processes, politics, culture, surrounding environment, and management decisions. You will need to understand how information systems can change social and work life in your firm. You will not be able to design new systems successfully or understand existing systems without understanding your own business organization.

As a manager, you will be the one to decide which systems will be built, what they will do, and how they will be implemented. You may not be able to anticipate all of the consequences of these decisions. Some of the changes that occur in business firms because of new IT investments cannot be foreseen and have results that may or may not meet your expectations. Who would have imagined 25 years ago, for instance, that email and instant messaging would become a dominant form of business communication and that some managers would be inundated with more than 200 email messages each day?

What Is an Organization?

Organization

A stable, formal social structure that takes resources from the environment and processes them to produce outputs.

An **organization** is a stable, formal social structure that takes resources from the environment and processes them to produce outputs. Examine Figure 3.2, which illustrates how an organization is defined from a technical perspective. This technical definition focuses on three elements of an organization. Capital and labor are primary production factors provided by the environment. The organization (the firm) transforms these inputs into products and services in a production function. The products and services are consumed by the environment in return for supplying inputs.

Figure 3.1 The Two-Way Relationship Between Organizations and Information Technology

This complex two-way relationship is mediated by many factors, not the least of which are the decisions made—or not made—by managers. Other factors mediating the relationship include the organizational culture, structure, politics, business processes, and environment.

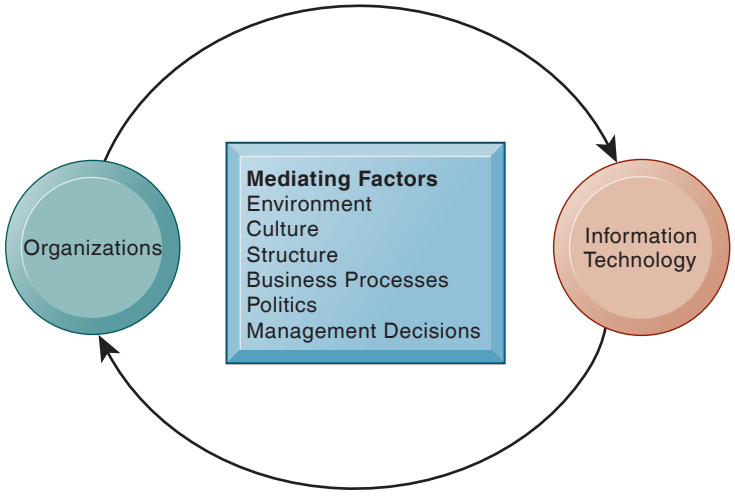
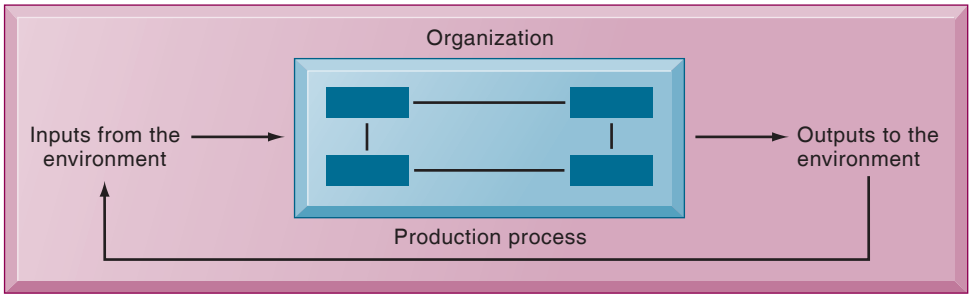


Figure 3.2 The Technical Microeconomic Definition of the Organization

In the microeconomic definition of organizations, capital and labor (the primary production factors provided by the environment) are transformed by the firm through the production process into products and services (outputs to the environment). The products and services are consumed by the environment, which supplies additional capital and labor as inputs in the feedback loop.



An organization is more stable than an informal group (such as a group of friends that meets every Friday for lunch) in terms of longevity and routineness. Organizations are formal legal entities with internal rules and procedures that must abide by laws. Organizations are also social structures because they are collections of social elements, much as a machine has a structure—a particular arrangement of valves, cams, shafts, and other parts.

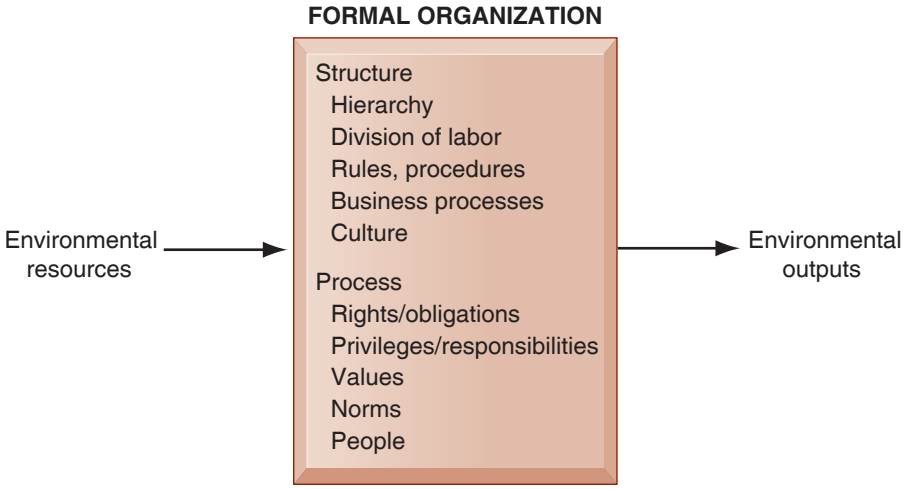
This definition of organizations is powerful and simple, but it is not very descriptive or even predictive of real-world organizations. A more realistic behavioral definition of an organization is a collection of rights, privileges, obligations, and responsibilities delicately balanced over a period of time through conflict and conflict resolution. Examine Figure 3.3, which illustrates the definition of an organization from a behavioral perspective.

In this behavioral view of the firm, people who work in organizations develop customary ways of working; they gain attachments to existing relationships; and they make arrangements with subordinates and superiors about how work will be done, the amount of work that will be done, and under what conditions work will be done. Most of these arrangements and feelings are not discussed in any formal rulebook.

How do these definitions of organizations relate to information systems technology? A technical view of organizations encourages us to focus on how inputs are combined to create outputs when technology changes are introduced into the company. The firm is seen as infi-

Figure 3.3 The Behavioral View of Organizations

The behavioral view of organizations emphasizes group relationships, values, and structures.



nitely malleable, with capital and labor substituting for each other quite easily. But the more realistic behavioral definition of an organization suggests that building new information systems, or rebuilding old ones, involves much more than a technical rearrangement of machines or workers—that some information systems change the organizational balance of rights, privileges, obligations, responsibilities, and feelings that have been established over a long period of time.

Changing these elements can take a long time, be very disruptive, and require more resources to support training and learning. For instance, the length of time required to implement a new information system effectively is much longer than usually anticipated simply because there is a lag between implementing a technical system and teaching employees and managers how to use the system.

Technological change requires changes in who owns and controls information, who has the right to access and update that information, and who makes decisions about whom, when, and how. This more complex view forces us to look at the way work is designed and the procedures used to achieve outputs.

The technical and behavioral definitions of organizations are not contradictory. Indeed, they complement each other: The technical definition tells us how firms in competitive markets combine capital, labor, and information technology, whereas the behavioral model takes us inside the individual firm to see how that technology affects the organization's inner workings. Section 3.2 describes how each of these definitions of organizations can help explain the relationships between information systems and organizations.

3.2 Discuss Features of Organizations that Affect Information Systems

All modern organizations share certain characteristics. They are bureaucracies with clear-cut divisions of labor and specialization. Organizations arrange specialists in a hierarchy of authority in which everyone is accountable to someone, and authority is limited to specific actions governed by abstract rules or procedures. These rules create a system of impartial and universal decision making. Organizations try to hire and promote employees based on technical qualifications and professionalism (not personal connections). The organization is devoted to the principle of efficiency: maximizing output using limited inputs. Other features of organizations include their business processes, organizational culture, organizational politics, surrounding environments, structure, goals, constituencies, and leadership styles. All of these features affect the kinds of information systems used by organizations.

Routines and Business Processes

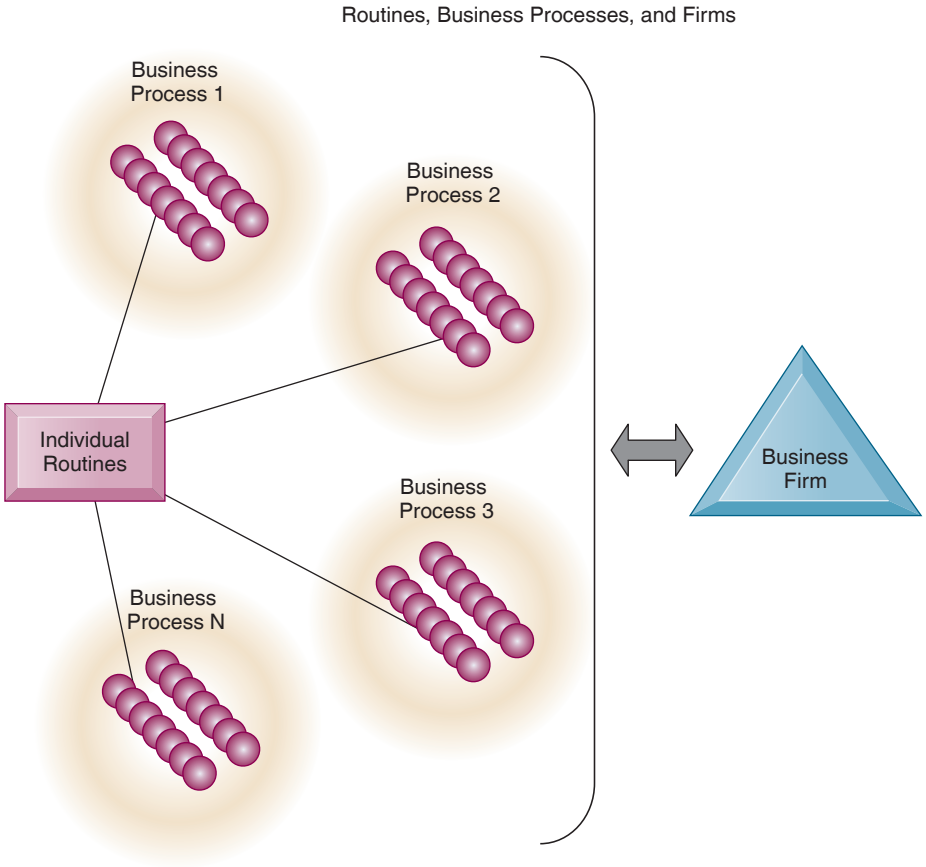
Routines

Precise rules, procedures, and practices that have been developed to cope with expected situations. Sometimes called standard operating procedures.

All organizations, including business firms, become very efficient over time because individuals in the firm develop routines for producing goods and services. **Routines**—sometimes called *standard operating procedures*—are precise rules, procedures, and practices that have been developed to cope with virtually all expected situations. As employees learn these routines, they become highly productive and efficient, and the firm is able to reduce its costs over time as efficiency increases. For instance, when you visit a doctor's office, receptionists have a well-developed set of routines for gathering basic information from you, nurses have a different set of routines for preparing you for an interview with a doctor, and the doctor has a well-developed set of routines for diagnosing you. *Business processes*, which we introduced in Chapters 1 and 2, are collections of such routines. Examine Figure 3.4, which shows that a firm is a collection of these business processes.

Figure 3.4 Routines, Business Processes, and Firms

All organizations are composed of individual routines and behaviors, a collection of which make up a business process. A collection of business processes makes up the business firm. New information system applications require that individual routines and business processes change to achieve high levels of organizational performance.



Organizational Politics

People in organizations occupy different positions with different specialties, concerns, and perspectives. As a result, they naturally have divergent viewpoints about how resources, rewards, and punishments should be distributed. These differences matter to both managers and employees, and they result in political struggles for resources, competition, and conflict within every organization. Political resistance is one of the great difficulties of bringing about organizational change—especially the development of new information systems. Virtually all large information systems investments by a firm that bring about significant changes in strategy, business objectives, business processes, and procedures become politically charged events. Managers who know how to work with the politics of an organization will be more successful than less-skilled managers in implementing new information systems.

Organizational Culture

All organizations have bedrock, unassailable, unquestioned (by the members) assumptions that define their goals and products. Organizational culture encompasses this set of assumptions about what products the organization should produce and how, where, and for whom it should produce them. Generally, these cultural assumptions are taken totally for granted and are rarely publicly announced or discussed. Business processes—the actual way business firms produce value—are usually ensconced in the organization's culture.

You can see organizational culture at work by looking around your university or college. Some bedrock assumptions of university life are that professors know more than students, the reason students attend college is to learn, and classes follow a regular schedule. Organizational culture is a powerful unifying force that restrains political conflict and promotes common understanding, agreement on procedures, and common practices. If we all share the same basic cultural assumptions, agreement on other matters is more likely.

At the same time, organizational culture is a powerful restraint on change, especially technological change. Most organizations will do almost anything to avoid making changes in basic assumptions. Any technological change that threatens commonly held cultural assumptions usually meets a great deal of resistance. However, there are times when the only sensible way for a firm to move forward is to employ a new technology that directly opposes an existing organizational culture. When this occurs, the technology is often stalled while the culture slowly adjusts.

Organizational Environments

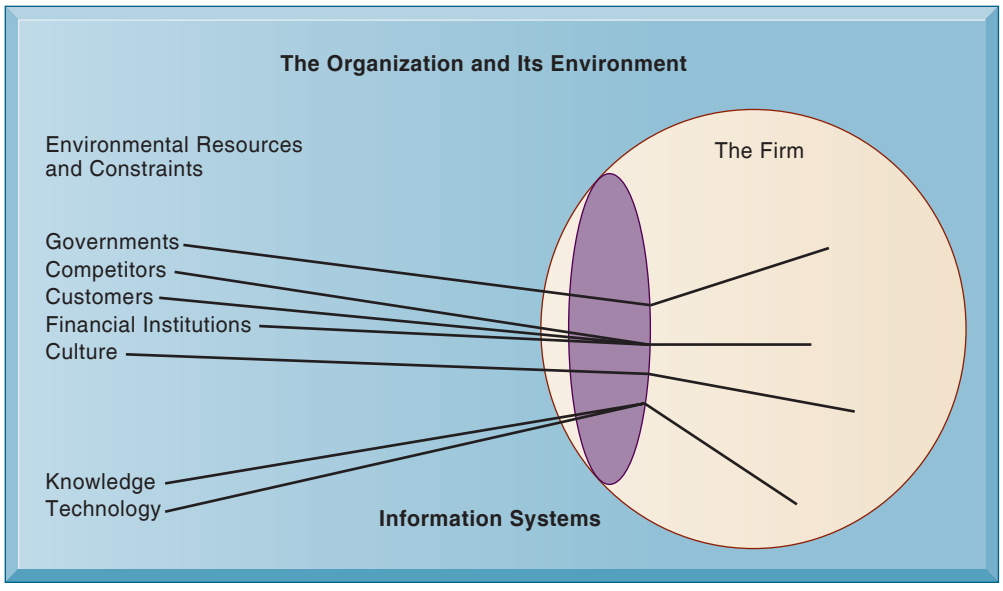
Organizations reside in environments from which they draw resources and to which they supply goods and services. Organizations and environments have a reciprocal relationship. On the one hand, organizations are open to and dependent on the social and physical environment that surrounds them. Without financial and human resources—people willing to work reliably and consistently for a set wage or revenue from customers—organizations could not exist. Organizations must respond to legislative and other requirements imposed by government as well as the actions of customers and competitors. On the other hand, organizations can influence their environments. For example, business firms sometimes form alliances with other businesses to influence the political process.

Examine Figure 3.5, which illustrates the role of information systems in helping organizations perceive changes in their environments and also in helping organizations act on their environments. Information systems are key instruments for *environmental scanning*, helping managers identify external changes that might require an organizational response.

Environments generally change much faster than organizations. New technologies, new products, and changing public tastes and values (many of which result in new government regulations) put strains on any organization's culture, politics, and people. Most organizations are unable to quickly adapt to a rapidly changing environment. Inertia built into an organization's

Figure 3.5 Environments and Organizations Have a Reciprocal Relationship

Environments shape what organizations can do, but organizations can influence their environments and decide to change environments altogether. Information technology plays a critical role in helping organizations perceive environmental change and in helping organizations act on their environment.



standard operating procedures, the political conflict raised by changes to the existing order, and the threat to closely held cultural values inhibit organizations from making significant changes. Young firms typically lack resources to sustain even short periods of troubled times. It is not surprising that less than 50 percent of the *Fortune* 500 companies in 2000 still exist today.

Disruptive Technologies: Riding the Wave

Sometimes, a technology and resulting business innovation come along to radically change the business landscape and environment. These innovations are loosely called "disruptive" (Christensen, 2003; Christensen, Raynor, and McDonald, 2015). What makes a technology disruptive? **Disruptive technologies** are those that substitute for and significantly reduce the market share of existing products, services, and business models. The car substituted for the horse-drawn carriage, the word processor for typewriters, streaming music services for portable CD players, and digital photography for process film photography. Table 3.1 describes just a few disruptive technologies from the past.

Disruptive technologies

Technologies that substitute for and significantly reduce the market share of existing products, services, and business models.

In some cases, entire industries are put out of business. In other cases, disruptive technologies expand the market, usually with less functionality and much less cost than existing products. Eventually, they turn into low-cost competitors for whatever was sold before. Disk drives are an example: Small hard disk drives used in PCs extended the market for disk drives by offering cheap digital storage for small files.

Some firms are able to create these technologies and ride the wave to profits; others learn quickly and adapt their business; still others are obliterated because their products, services, and business models become obsolete. They may be very efficient at doing what no longer needs to be done! There are also cases where no firms benefit, and all the gains go to consumers (firms fail to capture any profits). Moreover, not all change or technology is disruptive. Managers of existing businesses often do make the right decisions and find ways to continue competing. Disruptive technologies are tricky. Firms that invent disruptive technologies as "first movers" do not always benefit if they lack the resources to exploit the technology or fail to see the opportunity. The MITS Altair 8800 is widely regarded as the first PC, but its inventors did not take advantage of their first mover status. Second movers, so-called fast followers, such as IBM and Microsoft, reaped the rewards. Citibank's ATMs revolutionized retail banking, but they were copied by other banks. Now all banks use ATMs, with the benefits going mostly to the consumers.

Table 3.1 Disruptive Technologies: Winners and Losers

Technology	Description	Winners and Losers
Microprocessor chips (1971)	Thousands and eventually millions of transistors on a silicon chip	Microprocessor firms win (Intel, Texas Instruments), while transistor firms (GE) decline.
Personal computers (1975)	Small, inexpensive, but fully functional desktop computers	PC manufacturers (HP, Apple, IBM) and chip manufacturers prosper (Intel), while mainframe (IBM) and minicomputer (DEC) firms lose.
World Wide Web (1989)	A global database of digital files and "pages" instantly available	Owners of online content and news benefit, while traditional publishers (newspapers, magazines, and broadcast television) lose.
Internet music, video, TV services (1998)	Repositories of downloadable music, video, and TV broadcasts on the web	Owners of Internet platforms, telecommunications providers owning Internet backbone (ATT, Verizon), and local Internet service providers win, while content owners and physical retailers (Tower Records, Blockbuster) lose.
Software as web service	Using the Internet to provide remote access to online software	Online software services companies (Salesforce.com) win, while traditional "boxed" software companies (Microsoft, SAP, Oracle) lose or adapt.
Artificial intelligence	Using machine learning and generative AI to make decision making more rapid and accurate and to create new knowledge and content	Organizations providing AI technology and services such as NVIDIA and organizations with AI expertise win, while providers and users of less advanced technologies, along with book, film, music, and other content creators and providers, lose.

Table 3.2 Organizational Structures

Organizational Type	Description	Examples
Entrepreneurial structure	Young, small firm in a fast-changing environment. It has a simple structure and is managed by an entrepreneur serving as its single chief executive officer.	Small start-up business
Machine bureaucracy	Large bureaucracy existing in a slowly changing environment, producing standard products. It is dominated by a centralized management team and centralized decision making.	Midsize manufacturing firm
Divisionalized bureaucracy	Combination of multiple machine bureaucracies, each producing a different product or service, all topped by one central headquarters.	*Fortune* 500 firms, such as General Motors
Professional bureaucracy	Knowledge-based organization where goods and services depend on the expertise and knowledge of professionals. Dominated by department heads with weak centralized authority.	Law firms, school systems, hospitals
Adhocracy	Task force organization that must respond to rapidly changing environments. Consists of large groups of specialists organized into short-lived multidisciplinary teams and has weak central management.	Consulting firms, such as the RAND Corporation

Organizational Structure

All organizations have a structure or shape. Mintzberg's classification, described in Table 3.2, identifies five basic kinds of organizational structure (Mintzberg, 1971).

The kind of information systems you find in a business firm—and the nature of problems with these systems—often reflects the type of organizational structure. For instance, in a professional bureaucracy such as a hospital, it is not unusual to find parallel patient record systems operated by the administration, another by doctors, and another by other professional staff such as nurses and social workers. In small entrepreneurial firms, you will often find poorly designed systems developed in a rush that quickly outgrow their usefulness. In huge multidivisional firms operating in hundreds of locations, you will frequently find there is not a single integrating information system, but instead each locale or each division has its own set of information systems.

Other Organizational Features

Organizations have goals and use different means to achieve them. Some organizations have coercive goals (e.g., prisons); others have utilitarian goals (e.g., businesses). Still others have normative goals (universities, religious groups). Organizations also serve different groups or have different constituencies, some primarily benefiting their members, others benefiting clients, stockholders, or the public. The nature of leadership differs greatly from one organization to another—some organizations may be more democratic or authoritarian than others. Another way organizations differ is by the tasks they perform and the technology they use. Some organizations perform primarily routine tasks that can be reduced to formal rules that require little judgment (such as manufacturing auto parts), whereas others (such as consulting firms) work primarily with nonroutine tasks.

3.3 Describe the Impact of Information Systems on Organizations

Information systems have become integral, online, interactive tools deeply involved in the minute-to-minute operations and decision making of large organizations. Over the past decade, information systems have fundamentally altered the economics of organizations and greatly increased the possibilities for organizing work. Theories and concepts from economics and sociology help us understand the changes brought about by IT.

Economic Impacts

From the point of view of economics, IT changes both the relative costs of capital and the costs of information. Information systems technology can be viewed as a factor of production that can be substituted for traditional capital and labor. As the cost of information technology decreases, it is substituted for labor, which historically has been a rising cost. Hence, information technology should result in a decline in the number of middle managers and clerical workers as information technology substitutes for their labor.

As the cost of information technology decreases, it also substitutes for other forms of capital, such as buildings and machinery, which remain relatively expensive. Hence, over time, we should expect managers to increase their investments in IT because of its declining cost relative to other capital investments.

IT also affects the cost and quality of information and changes the economics of information. Information technology helps firms contract in size because it can reduce **transaction costs**—the costs incurred when a firm buys on the marketplace what it cannot make itself. According to **transaction cost theory**, firms and individuals seek to economize on transaction costs, much as they do on production costs. Using markets is expensive because of costs such as locating and communicating with distant suppliers, monitoring contract compliance, buying insurance, obtaining information on products, and so forth (Coase, 1937; Williamson, 1985). Traditionally, firms have tried to reduce transaction costs through vertical integration, by getting bigger, hiring more employees, and buying their own suppliers and distributors, as both General Motors and Ford used to do.

Transaction costs

Costs of participating in a market; for example when a firm buys in the marketplace what it cannot make itself, or a consumer expends time and resources.

Transaction cost theory

Economic theory that firms and individuals seek to economize on transaction costs. Traditionally this was done vertical integration, with firms becoming larger, but today information technology enables firms to do so while decreasing the number of employees.

Information technology, especially the use of networks, can help firms lower the cost of market participation (transaction costs), making it worthwhile for firms to contract with external suppliers instead of using internal sources. As a result, firms can shrink in size (numbers of employees) because it is far less expensive to outsource work to a competitive marketplace rather than hire employees.

For instance, by using computer links to external suppliers, automakers such as Ford, Toyota, and Honda can achieve economies by obtaining more than 70 percent of their parts from those external suppliers. Information systems make it possible for companies such as Apple to outsource assembly of iPhones to contract manufacturers such as Foxconn instead of making their products themselves.

As transaction costs decrease, firm size (the number of employees) should shrink because it becomes easier and cheaper for the firm to contract for the purchase of goods and services in the marketplace rather than to make the product or offer the service itself. Firm size can stay constant or contract even as the company increases its revenues. For example, when Eastman Chemical Company split off from Kodak in 1994, it had $3.3 billion in revenue and 24,000 full-time employees. In 2023, it generated $9.2 billion in revenue with only 14,000 employees.

Information technology can also reduce internal management costs. According to **agency theory**, the firm is viewed as a "nexus of contracts" among self-interested individuals rather than as a unified, profit-maximizing entity (Jensen and Meckling, 1992). A principal (owner) employs "agents" (employees) to perform work on the principal's behalf. However, agents need constant supervision and management; otherwise, they will tend to pursue their own interests rather than those of the owners. As firms grow in size and scope, agency costs or coordination costs rise because owners must expend more and more effort supervising and managing employees.

Agency theory

Economic theory that views the firm as a nexus of contracts among self-interested individuals who must be supervised and managed.

Information technology, by reducing the costs of acquiring and analyzing information, permits organizations to reduce agency costs because it becomes easier for managers to oversee a greater number of employees. By reducing overall management costs, information technology enables firms to increase revenues while shrinking the number of middle managers and clerical workers. We have seen examples in earlier chapters where information technology expanded the power and scope of small organizations by enabling them to perform coordinated activities such as processing orders or keeping track of inventory with very few clerks and managers.

Because IT reduces both agency and transaction costs for firms, we should expect firm size to shrink over time as more capital is invested in IT. Firms should have fewer managers, and we expect to see revenue per employee increase over time.

Organizational and Behavioral Impacts

Theories based in the sociology of complex organizations also provide some understanding of how and why firms change with the implementation of new IT applications.

IT Flattens Organizations

Large bureaucratic organizations, which primarily developed before the computer age, are often inefficient, slow to change, and less competitive than newly created organizations. Some of these large organizations have downsized, reducing the number of employees and the number of levels in their organizational hierarchies.

Behavioral researchers have theorized that information technology facilitates flattening of hierarchies by broadening the distribution of information to empower lower-level employees and increase management efficiency. Examine Figure 3.6, which illustrates this organizational flattening. IT pushes decision-making rights lower in the organization because lower-level employees receive the information they need to make decisions without supervision. (This empowerment is also possible because of higher educational levels among the workforce, which give employees more capabilities to make intelligent decisions.) Because managers now receive so much more accurate information on time, they become much faster at making decisions, so fewer managers are required. Management costs decline as a percentage of revenues, and the hierarchy becomes much more efficient.

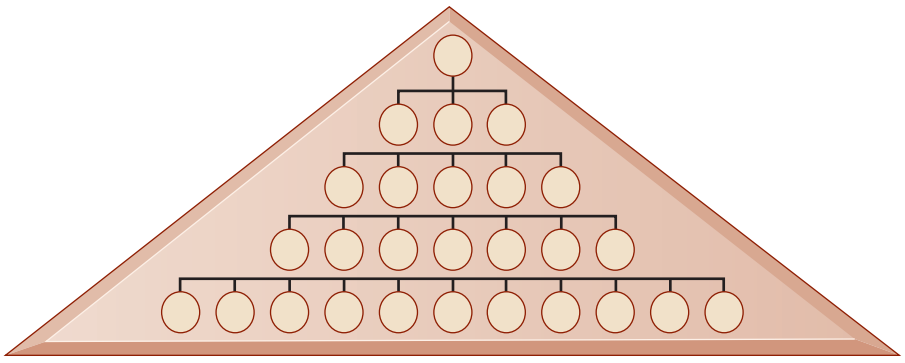
These changes mean that the management span of control has also been broadened, enabling high-level managers to manage and control more workers spread over greater distances. Many companies have eliminated thousands of middle managers as a result of these changes.

Understanding Organizational Resistance to Change

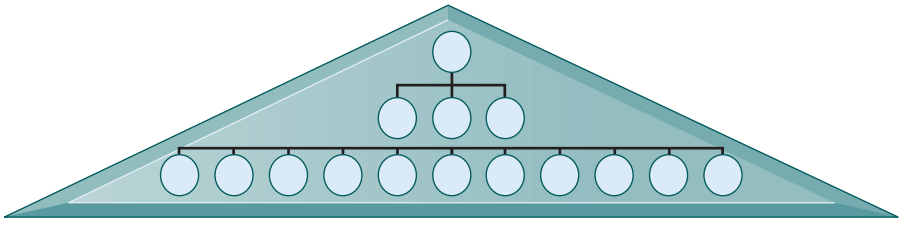
Information systems inevitably become bound up in organizational politics because they influence access to a key resource—namely, information. Information systems can affect who does what to whom, when, where, and how in an organization. Many new information systems

Figure 3.6 Flattening Organizations

Information systems can reduce the number of levels in an organization by providing managers with information to supervise larger numbers of workers and by giving lower-level employees more decision-making authority.



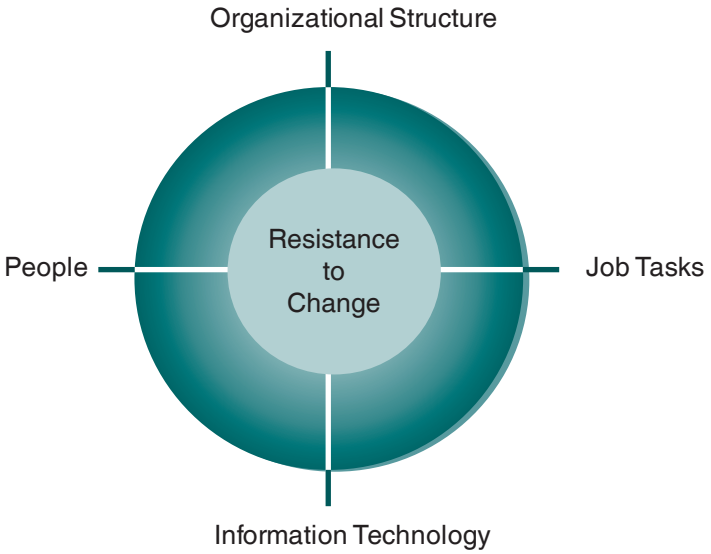
A traditional hierarchical organization with many levels of management



An organization that has been "flattened" by removing layers of management

Figure 3.7 Organizational Resistance to Information System Innovations

Implementing information systems has consequences for task arrangements, structures, and people. According to this model, to implement change, all four components must be changed simultaneously.



require changes in personal routines that can be painful for those involved and require retraining and additional effort that may or may not be compensated. Because information systems potentially change an organization's structure, culture, business processes, and strategy, there is often considerable resistance to them when they are introduced.

There are several ways to visualize organizational resistance. Research on organizational resistance to innovation suggests that four factors are paramount: the nature of the IT innovation, the organization's structure, the culture of people in the organization, and the tasks affected by the innovation. Examine Figure 3.7, which illustrates the interaction of these four factors. Here, changes in technology are absorbed, interpreted, deflected, and defeated by organizational task arrangements, structures, and people. In this model, the only way to bring about change is to change the technology, tasks, structure, and people simultaneously. Other authors have spoken about the need to "unfreeze" organizations before introducing an innovation, quickly implementing it, and "refreezing" or institutionalizing the change (Kolb and Frohman, 1970).

Because organizational resistance to change is so powerful, many information technology investments flounder and do not increase productivity. Indeed, research on project implementation failures demonstrates that the most common reason for the failure of large projects to reach their objectives is not the failure of the technology, but organizational and political resistance to change. Chapter 14 treats this issue in detail. Therefore, as a manager involved in future IT investments, your ability to work with people and organizations is just as important as your technical awareness and knowledge.

Implications for the Design and Understanding of Information Systems

To deliver genuine benefits, information systems must be built with a clear understanding of the organization in which they will be used. In our experience, the central organizational factors to consider when planning a new system are the following:

- The environment in which the organization must function
- The structure of the organization: hierarchy, specialization, routines, and business processes
- The organization's culture and politics
- The type of organization and its style of leadership

- The principal interest groups affected by the system and the attitudes of workers who will be using the system
- The kinds of tasks, decisions, and business processes that the information system is designed to assist

3.4 Discuss Porter's Competitive Forces Model

In almost every industry you examine, you will find that some firms do better than most others. There's almost always a standout firm. In the automotive industry, Toyota is considered a superior performer. In pure online retail, Amazon is the leader; in offline retail, Walmart, the largest retailer on earth, is the leader. In web search, Google is considered the leader.

Firms that "do better" than others are said to have a competitive advantage over others: They either have access to special resources that others do not, or they are able to use commonly available resources more efficiently—usually because of superior knowledge and information assets. In any event, they do better in terms of revenue growth, profitability, or productivity growth (efficiency), all of which ultimately in the long run translate into higher stock market valuations than their competitors.

But why do some firms do better than others, and how do they achieve competitive advantage? How can you analyze a business and identify its strategic advantages? How can you develop a strategic advantage for your own business? And how do information systems contribute to strategic advantages? One answer to that question is Michael Porter's competitive forces model.

Porter's Competitive Forces Model

Competitive forces model

Most widely used model for understanding competitive advantage. Views the strategic position of the firm as being determined not only by competition with its traditional direct competitors, but also four other forces in the firm's environment: new market entrants, substitute products and services, customers, and suppliers.

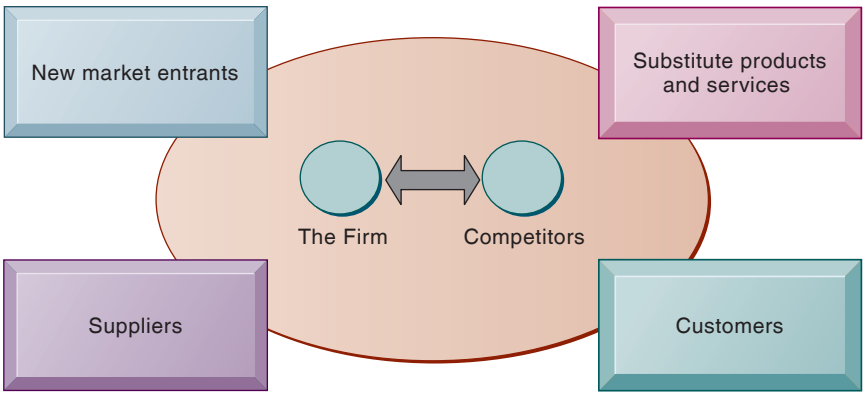
Arguably, the most widely used model for understanding competitive advantage is Michael Porter's **competitive forces model**. Examine Figure 3.8, which illustrates how this model provides a general view of the firm, its competitors, and the firm's environment. Earlier in this chapter, we described the importance of a firm's environment and the dependence of firms on environments. Porter's model is all about the firm's general business environment. In this model, five competitive forces shape the fate of the firm.

Traditional Competitors

All firms share market space with other competitors who are continuously devising new, more-efficient ways to produce by introducing new products and services, and attempting to attract customers by developing their brands and imposing switching costs on their customers.

Figure 3.8 Porter's Competitive Forces Model

In Porter's competitive forces model, firm strategy and the strategic position of the firm are determined not only by competition with its traditional direct competitors, but also by four other forces in the firm's environment: new market entrants, substitute products and services, customers, and suppliers.



New Market Entrants

In a free economy with mobile labor and financial resources, new companies are always entering the marketplace. In some industries, there are very low barriers to entry, whereas in other industries, entry is very difficult. For instance, it is fairly easy to start a pizza business or just about any small retail business, but it is much more expensive and difficult to enter the computer chip business, which has very high capital costs and requires significant expertise and knowledge that are hard to obtain. New companies have several possible advantages: They are not locked into old plants and equipment, they often hire younger workers who are less expensive and perhaps more innovative, they are not encumbered by old worn-out brand names, and they are "more hungry" (more highly motivated) than traditional occupants of an industry. These advantages are also their weaknesses: They depend on outside financing for new plants and equipment, which can be expensive; they have a less-experienced workforce; and they have little brand recognition.

Substitute Products and Services

In just about every industry, there are substitutes that your customers might use if your prices become too high. New technologies create new substitutes all the time. Electric batteries can substitute for gasoline or diesel in cars and trucks, and wind, solar, and hydro power for industrial electricity generation. Likewise, Internet and wireless telephone service can substitute for traditional telephone service. And, of course, streaming television services that allow you to stream television and movies have become a substitute for traditional broadcast and pay cable TV services. The more substitute products and services in your industry, the less you can control pricing and the lower your profit margins.

Customers

A profitable company depends, in large measure, on its ability to attract and retain customers (while denying them to competitors) and charge high prices. The power of customers grows if they can easily switch to a competitor's products and services, or if they can force a business and its competitors to compete on price alone in a transparent marketplace where there is little **product differentiation** (a competitive strategy for creating brand loyalty by developing new and unique products and services that are not easily duplicated by competitors), and all prices are known instantly (such as on the Internet). For instance, in the used college textbook market on the Internet, students (customers) can find multiple suppliers of just about any current college textbook. In this case, online customers have extraordinary power over used-book firms.

Product differentiation

Competitive strategy for creating brand loyalty by developing new and unique products and services that are not easily duplicated by competitors.

Suppliers

The market power of suppliers can have a significant impact on firm profits, especially when the firm cannot raise prices as fast as suppliers can. The more alternative suppliers a firm has, the greater control it can exercise over suppliers in terms of price, quality, and delivery schedules. For instance, manufacturers of laptop computers almost always have multiple competing suppliers of key components, such as keyboards, solid-state, and display screens.

Information Systems Strategies for Dealing with Competitive Forces

What is a firm to do when it is faced with all these competitive forces? And how can the firm use information systems to counteract some of these forces? How do you prevent substitutes and inhibit new market entrants? There are four generic strategies, each of which is often enabled by using information technology and systems: low-cost leadership, product differentiation, focus on market niche, and strengthening customer and supplier intimacy.

Supermarkets and large retail stores such as Walmart use sales data captured at the checkout counter to determine which items have sold and need to be reordered. Walmart's continuous replenishment system transmits orders to restock directly to its suppliers. The system enables Walmart to keep costs low while fine-tuning its merchandise to meet customer demands.

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Low-Cost Leadership

Use information systems to achieve the lowest operational costs and the lowest prices. The classic example is Walmart. By keeping prices low and shelves well stocked using a legendary inventory replenishment system, Walmart became the leading retail business in the United States. Walmart's continuous replenishment system sends orders for new merchandise directly to suppliers as soon as consumers pay for their purchases at the cash register. Point-of-sale terminals record the bar code of each item passing the checkout counter and send a purchase transaction directly to a central information system at Walmart headquarters. The information system collects the orders from all Walmart stores and transmits them to suppliers. Suppliers can also access Walmart's sales and inventory data using web technology.

Because the system replenishes inventory with lightning speed, Walmart does not need to spend much money on maintaining large inventories of goods in its own warehouses. The system also enables Walmart to adjust purchases of store items to meet customer demands. By using systems to keep operating costs low, Walmart pays a lower percentage of sales revenue for overhead than most firms in the retail industry.

Efficient customer response system

Information system that directly links consumer behavior back to distribution and production, and supply chains.

Walmart's continuous replenishment system is also an example of an efficient customer response system. An **efficient customer response system** directly links consumer behavior to distribution and production and supply chains. Walmart's continuous replenishment system provides such an efficient customer response.

Product Differentiation

Use information systems to enable new products and services or improve customer convenience in using your existing products and services. Big Tech firms like Google, Meta (Facebook), Amazon, Apple, and others are pouring billions of dollars into research and deployment of new services, and enhancements to their most valuable services and products in order to differentiate them from potential competitors. For instance, Apple has continued to differentiate its mobile devices with nearly annual introductions of new iPhones and iPad models. Crayola is creating new digital products and services (described in Section 3.9) that enable children to use digital tools for drawing, designing, and other creative tasks that used to require hand-held crayons. The continual stream of innovations flowing from these and other technology-driven companies ensures their products are unique and difficult to copy.

Manufacturers and retailers are using information systems to create products and services that are customized and personalized to fit the precise specifications of individual customers. For example, Baume, an offshoot of the luxury brand Baume et Mercier, uses recycled, upcycled, and sustainable materials to craft distinctive watches. With Baume's mobile-optimized website, which is built on Salesforce Commerce Cloud, customers can pick and mix different watch elements, including dial style, casing finish, strap material, and engraved messages. Customers can choose from 2,160 variations to create their own unique product. This ability to offer individually tailored products or services is called **mass customization**.

Mass customization
The capacity to offer individually tailored products or services on a large scale.

More and more companies are differentiating their products not just by the features of the products themselves, but also by the entire experience of buying and using the product. This is called the "customer experience," and **customer experience management**, which involves management of all the interactions between a customer and a company throughout their entire business relationship, has become incorporated into competitive strategy. The chapter-opening case on how PepsiCo used AI to improve the experience of eating Doritos is an example.

Customer experience management
Management of all the interactions between a customer and a company throughout their entire business relationship.

Table 3.3 lists a number of companies that developed IT-based products and services that other firms found difficult to copy—or at least took a long time to copy.

Focus on Market Niche

Use information systems to enable a specific market focus and serve this narrow target market better than competitors. Information systems support this strategy by producing and analyzing data for finely tuned sales and marketing techniques. Information systems enable companies to analyze customer buying patterns, tastes, and preferences closely so that they efficiently pitch advertising and marketing campaigns to smaller and smaller target markets.

The data come from a range of sources—credit card transactions, demographic data, purchase data from checkout counter scanners at supermarkets and retail stores, and data collected when people access and interact with websites. Sophisticated software tools find patterns in these large pools of data and infer rules from them to guide decision making. This has become one of the most popular applications today for AI/ML. Analysis of such data drives one-to-one marketing that creates personal messages based on individualized preferences. For example, the Charlotte Hornets of the NBA maintain detailed records on their millions of fans that include real-time data on buying tickets online and at games, purchasing team gear in-store and online, and eating and drinking at games. The team combines these fan records to create a single profile for each fan. The Hornets use these profiles to build a more personal relationship with fans, which can lead to a better fan experience, more targeted marketing, and more sales. Contemporary customer relationship management (CRM) systems augmented by AI tools feature analytical capabilities for this type of intensive data analysis.

Credit card companies are able to use this strategy to predict their most profitable cardholders. The companies gather vast quantities of data about consumer purchases and other behaviors and mine these data—often using AI/ML—to construct detailed profiles that identify cardholders who might be good or bad credit risks. We discuss the tools and technologies for data analysis in detail in Chapters 6, 11, and 12, and there are other real-world examples throughout the text.

Table 3.3 IT-Enabled New Products and Services Providing Competitive Advantage

Product or Service	Description
One-click shopping: Amazon	Amazon's patent on one-click shopping gave it a significant competitive advantage over other online retailers in the early years of e-commerce.
Online music: Apple iTunes Store	Apple was the first firm to develop a legal online marketplace for digital music from all five major record labels, and quickly became the largest music vendor in the United States.
Golf club customization: Ping	Customers can have golf clubs custom-fit and built to their individual specifications.
Online person-to-person payment: PayPal	PayPal enables the transfer of money between individual bank accounts and between bank accounts and credit card accounts.

Table 3.4 Four Basic Competitive Strategies

Strategy	Description	Example
Low-cost leadership	Use information systems to produce products and services at a lower price than competitors while enhancing quality and level of service	Walmart
Product differentiation	Use information systems to differentiate products, and enable new services and products	Uber, Nike, Apple
Focus on market niche	Use information systems to enable a focused strategy on a single market niche; specialize	Hilton Hotels, Harrah's
Customer and supplier intimacy	Use information systems to develop strong ties and loyalty with customers and suppliers	Toyota Corporation, Amazon

Strengthen Customer and Supplier Intimacy

Use information systems to tighten linkages with suppliers and develop intimacy with customers. Toyota, Ford, and other automobile manufacturers use information systems to facilitate direct access by suppliers to production schedules and even permit suppliers to decide how and when to ship supplies to their factories. This allows suppliers more lead time in producing goods. On the customer side, Amazon keeps track of user preferences and can recommend products purchased by others to its customers. Strong linkages to customers and suppliers increase **switching costs** (the expense incurred in lost time and expenditure of resources when changing from one supplier or system to a competing supplier or system) and loyalty to your firm.

Switching costs

The expense incurred in lost time and expenditure of resources when changing from one supplier or system to a competing supplier or system.

Table 3.4 summarizes the competitive strategies we have just described. Some companies focus on one of these strategies, but you will often see companies pursuing several of them simultaneously. For example, Walmart has traditionally pursued a low-cost leadership strategy, but it also offers unique new services such as drone package delivery and more targeted marketing.

The Internet's Impact on Competitive Advantage

Because of the Internet, the traditional competitive forces are still at work, but competitive rivalry has become much more intense (Porter, 2001). Internet technology is based on universal standards that any company can use, making it easy for rivals to compete on price alone and for new competitors to enter the market. Because information is available to everyone, the Internet raises the bargaining power of customers, who can quickly find the lowest-cost provider on the web. Profits have been dampened. Table 3.5 summarizes some of the potentially negative impacts identified by Porter of the Internet on business firms.

The Internet has nearly destroyed some industries and has severely threatened more. For instance, the printed encyclopedia industry and the travel agency industry have been nearly decimated by the availability of substitutes over the Internet. Likewise, the Internet has had a significant impact on the retail, music, book, retail brokerage, software, telecommunications, and newspaper industries.

Table 3.5 Impact of the Internet on Competitive Forces and Industry Structure

Competitive Force	Impact of the Internet
Substitute products or services	Enables substitutes to emerge with new approaches to meeting needs and performing functions
Customers' bargaining power	Availability of global price and product information shifts bargaining power to customers
Suppliers' bargaining power	Procurement over the Internet tends to raise bargaining power over suppliers; suppliers can also benefit from reduced barriers to entry and from the elimination of distributors and other intermediaries standing between them and their users
Threat of new entrants	Reduces barriers to entry, such as the need for a sales force, access to channels, and physical assets; provides a technology for driving business processes that makes other things easier to do
Positioning and rivalry among existing competitors	Widens the geographic market, increasing the number of competitors and reducing differences among competitors; makes it more difficult to sustain operational advantages; puts pressure to compete on price

However, the Internet has also created entirely new markets; formed the basis for thousands of new products, services, and business models; and provided new opportunities for building brands with very large and loyal customer bases. Amazon, eBay, YouTube, Facebook, Expedia, and Google are examples. In this sense, the Internet has "transformed" entire industries, forcing firms to change how they do business.

Smart Products and the Internet of Things

The growing use of sensors in industrial and consumer products, often called the Internet of Things (IoT), is an excellent example of how the Internet is changing competition within industries and creating new products and services. Under Armour and other sports and fitness companies are pouring money into wearable health trackers and fitness equipment that use sensors to report users' activities to remote corporate computing centers where the data can be analyzed. John Deere tractors are loaded with field radar, GPS transceivers, and hundreds of sensors that keep track of the equipment. The result is what are referred to as "smart products"—products that are a part of a larger set of information-intensive services sold by firms.

Smart, Internet-connected products offer new functionality, greater reliability, and more intense use of products while providing detailed information that can be used to improve both the products and the customer experience. They expand opportunities for product and service differentiation. When you buy a wearable digital health product, you not only get the product itself, you also get a host of services available from the manufacturer's cloud servers. Smart products increase rivalry among firms that will either innovate or lose customers to competitors. Smart products generally raise switching costs and inhibit new entrants to a market because existing customers are trapped in the dominant firm's software environment. Finally, smart products may decrease the power of suppliers of industrial components if, as many believe, the physical product becomes less important than the software and hardware that make it run.

3.5 Discuss the Value Chain Model

Although the Porter model is very helpful for identifying competitive forces and suggesting generic strategies, it is not very specific about what exactly to do, and it does not provide a methodology to follow for achieving competitive advantages. If your goal is to achieve operational excellence, where do you start? Here's where the business value chain model is helpful.

The **value chain model** highlights specific activities in the business where competitiv e strategies can best be applied, and where information systems are most likely to have a strategic impact (Porter, 1985). This model identifies specific, critical leverage points where a firm can use information technology most effectively to enhance its competitive position. The value chain model views the firm as a series or chain of basic activities that add a margin of value to a firm's products and services. Examine Figure 3.9, which illustrates how these activities can be categorized as either primary activities or support activities.

Primary activities are most directly related to the production and distribution of the firm's products and services, which create value for the customer. Primary activities include inbound logistics, operations, outbound logistics, sales and marketing, and service. Inbound logistics includes receiving and storing materials for distribution to production. Operations transforms inputs into finished products. Outbound logistics entails storing and distributing finished products. Sales and marketing includes promoting and selling the firm's products. The service activity includes maintenance and repair of the firm's goods and services.

Support activities make the delivery of the primary activities possible and consist of activities related to the firm's organizational infrastructure (administration and management), human resources (employee recruiting, hiring, and training), technology (improving products and the production process), and procurement (purchasing input).

Value chain model

Model that views the firm as a series or chain of basic activities that add a margin of value to a firm's products or services and identifies where information systems can best be applied to achieve a competitive advantage.

Primary activities

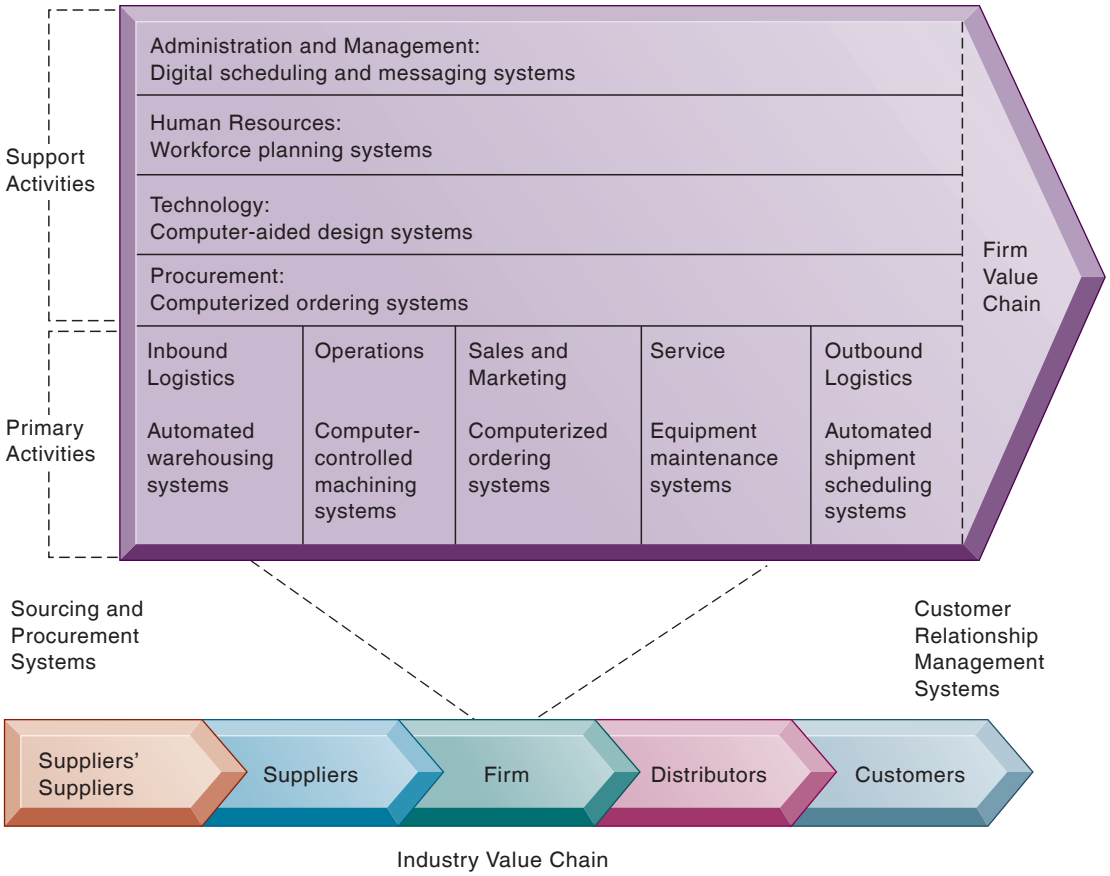
Activities most directly related to the production and distribution of a firm's products or services.

Support activities

Activities that make the delivery of a firm's primary activities possible. Consist of actitivies related to the firm's organizational infrastructure, human resources, technology, and procurement.

Figure 3.9 The Value Chain Model

This figure provides examples of systems for both primary and support activities of a firm and of its value partners that can add a margin of value to a firm's products or services.



Now you can ask at each stage of the value chain, "How can we use information systems to improve operational efficiency and improve customer and supplier intimacy?" This will force you to critically examine how you perform value-adding activities at each stage and how the business processes might be improved. For example, value chain analysis indicated that PepsiCo, described in the Opening Case Study, could use technology to improve its processes for sales, marketing, customer service, and differentiating its products.

You can also begin to ask how information systems can be used to improve the relationship with customers and with suppliers who lie outside the firm's value chain but belong to the firm's extended value chain, where they are absolutely critical to your success. Here, supply chain management systems that coordinate the flow of resources into your firm and customer relationship management systems that coordinate your sales and support employees with customers are two of the most common system applications that result from a business value chain analysis. We discuss these enterprise applications in detail later in Chapter 9. We also discuss how AI technology can be used throughout the value chain in Section 3.7 of this chapter.

Benchmarking

Involves comparing efficiency and effectiveness of business processes to strict standards and then measuring organizational performance against those standards.

Using the business value chain model will also cause you to consider benchmarking your business processes against your competitors or others in related industries and identifying industry best practices. **Benchmarking** involves comparing the efficiency and effectiveness of your business processes to strict standards and then measuring performance against those standards. Industry **best practices** are usually identified by consulting companies, research organizations, government agencies, and industry associations as the most successful solutions or problem-solving methods for consistently and effectively achieving a business objective.

Best practices

The most successful solutions or problem-solving methods for consistently and effectively achieving a business objective that have been developed by a specific organization or industry.

Once you have analyzed the various stages in the value chain at your business, you can come up with candidate applications of information systems. Then, once you have a list of candidate applications, you can decide which to develop first. By making improvements in

your own business value chain that your competitors might miss, you can achieve competitive advantage by attaining operational excellence, lowering costs, improving profit margins, and forging a closer relationship with customers and suppliers. If your competitors are making similar improvements, then at least you will not be at a competitive disadvantage—the worst of all cases!

Extending the Value Chain: The Value Web

Figure 3.9 also shows that a firm's value chain is linked to the value chains of its suppliers, distributors, and customers. After all, the performance of most firms depends not only on what goes on inside a firm but also on how well the firm coordinates with direct and indirect suppliers, delivery firms (logistics partners, such as FedEx or UPS), and, of course, customers.

How can information systems be used to achieve strategic advantage at the industry level? By working with other firms, industry participants can use information technology to develop industrywide standards for exchanging information or business transactions electronically, which force all market participants to subscribe to similar standards. Such efforts increase efficiency, making product substitution less likely and perhaps raising entry costs—thus discouraging new entrants. Also, industry members can build industrywide, IT-supported consortia, symposia, and communications networks to coordinate activities concerning government agencies, foreign competition, and competing industries.

Looking at the industry value chain encourages you to think about how to use information systems to link up more efficiently with your suppliers, strategic partners, and customers. Strategic advantage derives from your ability to relate your value chain to the value chains of other partners in the process. For instance, if you are Amazon.com, you want to build systems that:

- Make it easy for suppliers to display goods and open stores on the Amazon site.
- Make it easy for customers to pay for goods.
- Develop systems that coordinate the shipment of goods to customers.
- Develop shipment tracking systems for customers.

Internet technology has made it possible to create highly synchronized industry value chains called value webs. A **value web** is a collection of independent firms that use information technology to coordinate their value chains to produce a product or service for a market collectively. It is more customer-driven and operates in a less linear fashion than the traditional value chain.

Value web

Collection of independent firms that use information technology to coordinate their value chains to collectively produce a product or service for a market.

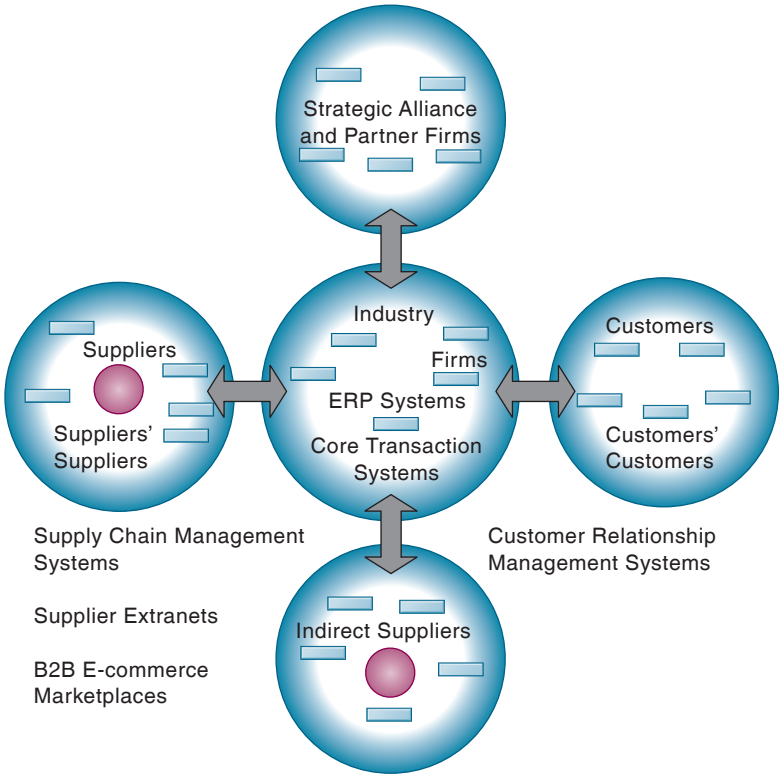
Examine Figure 3.10, which illustrates the value web as a networked system that operates in a less linear fashion than the traditional value chain displayed in Figure 3.9. It also shows that this value web synchronizes the business processes of customers, suppliers, and trading partners among different companies in an industry or in related industries. These value webs are more customer-driven, flexible, and adaptive to changes in supply and demand. Relationships can be bundled or unbundled in response to changing market conditions. Firms will accelerate time to market and to customers by optimizing their value web relationships to make quick decisions on who can deliver the required products or services at the right price and location.

3.6 Discuss Synergies, Core Competencies, and Network Economics

A large corporation is typically a collection of businesses. Often, the firm is organized financially as a collection of strategic business units, and the returns to the firm are directly tied to the performance of all the strategic business units. Information systems can improve the overall performance of these business units by promoting synergies and core competencies.

Figure 3.10 The Value Web

The value web is a networked system that can synchronize the value chains of business partners within an industry to respond rapidly to changes in supply and demand.



Synergies

Synergies develop when the output of some units can be used as inputs to other units, or when two organizations pool markets and expertise. These relationships lower costs and generate profits. Bank and financial firm mergers, such as the merger of JPMorgan Chase and Bank of New York, as well as Bank of America and Countrywide Financial Corporation, occurred precisely for this purpose.

One use of information technology in these synergy situations is to tie together the operations of disparate business units so that they can act as a whole. For example, acquiring MGM in 2022 enabled Amazon to bolster its position in the streaming entertainment industry. Information systems helped Amazon and MGM leverage their content, consolidate operations, lower costs, and increase cross-marketing of entertainment products.

Enhancing Core Competencies

Core competency

Activity at which a firm excels as an industry leader.

Yet another way to use information systems for competitive advantage is to think about ways that systems can enhance core competencies. The argument is that the performance of all business units will increase insofar as these business units develop, or create, a central core of competencies. A **core competency** is an activity for which a firm is an industry leader. Core competencies may involve being the world's best miniature parts designer, the best package delivery service, or the best thin-film manufacturer. In general, a core competency relies on knowledge that is gained over many years of practical field experience with a technology. This practical knowledge is typically supplemented with a long-term research effort and committed employees.

Any information system that encourages the sharing of knowledge across business units enhances competency. Such systems might encourage or enhance existing competencies and help employees become aware of new external knowledge; such systems might also help a business leverage existing competencies to related markets. For example, Procter & Gamble (P&G), a world leader in brand management and consumer product innovation, uses a series of

systems to enhance its core competencies by helping people working on similar problems share ideas and expertise. Employees working in research and development (R&D), engineering, purchasing, marketing, legal affairs, and business information systems around the world can share documents, reports, charts, videos, and other data from various sources online and locate employees with special expertise. P&G systems also can link to research scientists and entrepreneurs outside the company who are searching for new, innovative products worldwide.

Network-Based Strategies

Internet and networking technology have inspired strategies that take advantage of firms' abilities to create networks or network with each other. Network-based strategies include the use of network economics, a virtual company model, and business ecosystems.

Network Economics

Network economics refers to market situations where the economic value being produced depends, in large part, on the number of people using a product. For certain products and markets, the real economic value comes from the fact that other people use the product. In these situations, "network effects" are at work. For instance, what's the value of a telephone if it is not connected to millions of others? Email has value because it allows us to communicate with millions of others. Business models that are based on network effects have been highly successful on the Internet, including social networks, software, messaging apps, and on-demand companies like Uber and Airbnb.

Network economics

Refers to market situations where the economic value being produced depends in large part on the number of people using a product, and where the marginal costs of adding another participant are about zero but can create much larger marginal gains.

In traditional economics—the economics of factories and agriculture—production experiences diminishing returns. The more any given resource is applied to production, the lower the marginal gain in output, until a point is reached where the additional inputs produce no additional outputs. This is the law of diminishing returns, and it is the foundation for most of modern economics.

In some situations, the law of diminishing returns does not work. For instance, in a network, the marginal costs of adding another participant are about zero, whereas the marginal gain is much larger. The larger the number of subscribers in a telephone system or the Internet, the greater the value to all participants because each user can interact with more people. It is not much more expensive to operate a television station with 1,000 subscribers than with 10 million subscribers. The value of a community of people grows with size, whereas the cost of adding new members is inconsequential. This is referred to as a "network effect." The value of Facebook to users increases greatly as more people use the social network.

From this network economics perspective, information technology can be strategically useful. Websites can be used by firms to build communities of users—like-minded customers who want to share their experiences. This builds customer loyalty and enjoyment and builds unique ties to customers. eBay, the giant online auction site, is an example. This business is based on a network of millions of users and has built an online community by using the Internet. The more people offering products on eBay, the more valuable eBay is to everyone because more products are listed, and more competition among suppliers lowers prices. Network economics also provides strategic benefits to commercial software vendors. The value of their software and complementary software products increases as more people use them, and there is a larger installed base to justify continued use of the product and vendor support.

Virtual Company Model

Another network-based strategy uses the model of a virtual company to create a competitive business. A **virtual company** uses networks to link people, assets, and ideas, enabling it to ally with other companies to create and distribute products and services without being limited by traditional organizational boundaries or physical locations. One company can use the capabilities of another company without being organizationally tied to that company. The virtual company model is useful when a company finds it cheaper to acquire products, services, or capabilities from an external vendor or when it needs to move quickly to exploit new market opportunities and lacks the time and resources to respond on its own.

Virtual company

Organization using networks to link people, assets, and ideas, enabling it to ally with other companies to create and distribute products and services without being limited to traditional organizational boundaries or physical location.

Fashion companies such as GUESS, Ann Taylor, Levi Strauss, and Reebok enlist Hong Kong–based Li & Fung to manage production and shipment of their garments. Li & Fung handles product development, raw material sourcing, production planning, quality assurance,

and shipping. Li & Fung does not own any fabric, factories, or machines, outsourcing all of its work to a network of more than 15,000 suppliers in 40 countries all over the world. Customers place orders with Li & Fung over its private extranet. Li & Fung then sends instructions to appropriate raw material suppliers and factories where the clothing is produced. The Li & Fung extranet tracks the entire production process for each order. Working as a virtual company keeps Li & Fung flexible and adaptable so that it can design and produce the products ordered by its clients in short order to keep pace with rapidly changing fashion trends.

Business Ecosystems and Platforms

The Internet and the emergence of digital firms call for some modification of the industry competitive forces model. The traditional Porter model assumes a relatively static industry environment; relatively clear-cut industry boundaries; and a relatively stable set of suppliers, substitutes, and customers. Instead of participating in a single industry, some of today's firms participate in industry sets—collections of industries that provide related services and products that deliver value to the customer. Examine Figure 3.11, which illustrates these loosely coupled but interdependent networks of suppliers, distributors, outsourcing firms, transportation service firms, and technology manufacturers. This is called a **business ecosystem**.

Business ecosystem

Loosely coupled but interdependent networks of suppliers, distributors, outsourcing firms, transportation service firms, and technology manufacturers.

An example of a business ecosystem is the mobile Internet platform. In this ecosystem, there are four industries: device makers (Apple iPhone, Samsung, LG, and others), wireless telecommunication firms (AT&T, Verizon, T-Mobile, and others), independent software applications providers (generally small firms selling games, applications, and ring tones), and Internet service providers (who participate as providers of Internet service to the mobile platform). Each of these industries has its own history, interests, and driving forces. But these elements come together in a sometimes cooperative and sometimes competitive new mobile digital platform ecosystem, creating value for consumers that none of them could achieve acting alone.

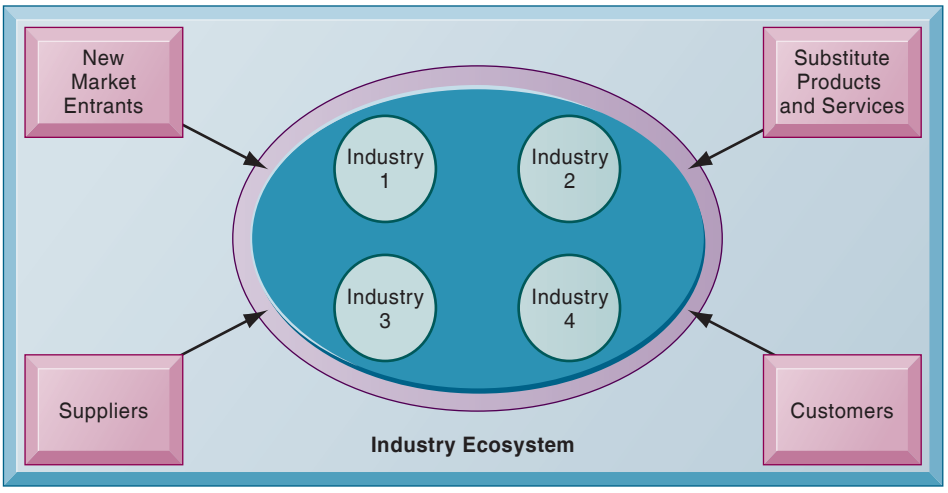
Business ecosystems typically have one or only a few keystone firms that dominate the ecosystem and create a **platform** composed of information systems, technologies, and services that thousands of other firms in different industries use to enhance their own capabilities. For instance, Facebook is a platform used by billions of people and millions of businesses to interact and share information, as well as to buy, market, and sell numerous products and services. Microsoft is another example. More firms are trying to use information systems to develop into keystone firms by building IT-based platforms that other firms can use. Alternatively, firms should consider how their information systems will enable them to become profitable niche players in the larger ecosystems created by keystone firms (Zhou, 2024; Cutolo, 2021).

Platform

Information systems, technologies, and services provided by a keystone firm in a business ecosystem that thousands of other firms in different industries use to enhance their own capabilities.

Figure 3.11 An Ecosystem Strategic Model

The digital firm era requires a more dynamic view of the boundaries among industries, firms, customers, and suppliers, with competition occurring among industry sets in a business ecosystem. In the ecosystem model, multiple industries work together to deliver value to the customer. IT plays an important role in enabling a dense network of interactions among the participating firms.



3.7 Understand How AI Can Be Used in Competitive Strategy

AI is transforming the way information systems operate and deliver value to organizations. Systems using AI technology can automate and enhance decision-making by analyzing patterns and trends in vast amounts of information, improve user experiences, open new revenue streams, optimize resource allocation and business processes, and create new products, services, and business models. You will find numerous examples of information systems that incorporate AI in every chapter of this text.

AI and the Value Chain

AI is making it easier for companies to improve performance based on their existing business models, but it is also spawning new business models and reshaping the competitive landscape. It has given birth to new products, services, and sources of value. Decision making has become much more data-driven, accurate, and precise for both operational and higher-level decisions. AI cannot make strategic decisions on its own, but it can be used to advise senior-level executives on how to pilot the firm more strategically. AI cannot fully substitute for human creativity and wisdom, but it can be a useful tool for ideation and problem-solving. Table 3.6 provides examples of how AI can add value at each step in the value chain, creating opportunities to boost competitive advantage.

AI and Porter's Competitive Forces Model

AI technology has impacted the various competitive forces identified in Porter's model, but not entirely in the same way as the Internet. There is intense pressure on firms to build products that will supply an unassailable competitive advantage. Some firms will succeed,

Table 3.6 Examples of AI Applications in the Value Chain

Value Chain Activity	AI Application
Inbound Logistics	AI in automated warehousing can maintain real-time, error-free inventory records, identify defects, and assign incoming shipments from suppliers to optimal warehouse locations.
Operations	AI identifies defects in large manufacturing runs, increasing quality while lowering costs.
Sales and Marketing	AI analyzes customer behavior to personalize marketing campaigns.
Service	AI chatbots provide quick answers to common queries from customers, sales reps, and service technicians.
Outbound Logistics	AI optimizes outbound shipment schedules, delivery routes, and route planning, reducing transportation costs.
Administration and Management	Predictive analytics can help managers anticipate demand fluctuations, optimize inventory levels, and allocate resources effectively.
Human Resources	AI algorithms predict which sourcing channels would most likely attract the right candidates for a specific job, reducing firm spending on identifying and recruiting new employees.
Technology	Firms can use AI technology to develop new products, such as noise cancelling technology for consumer products, hearing aids, earphones.
Procurement	AI can improve supplier selection by analyzing historical data, supplier performance, and market trends.

Table 3.7 AI and Porter's Competitive Forces Model

Competitive Force	Impact of AI
Substitute products or services	AI can help companies develop unique products or services that are difficult to substitute. For example, a company that uses AI to offer heavily personalized services such as fashion recommendations for individual customers may face less threat from substitutes than a company offering generic services.
Customers' bargaining power	AI can enhance the buying experience, building customer loyalty and reducing the bargaining power of buyers. For example, AI-powered shipping and delivery services or customer service systems make the experiences of purchasing and using a product more efficient and seamless, making customers less likely to switch to competitors.
Suppliers' bargaining power	AI can reduce the bargaining power of suppliers. For instance, AI-driven supply chain management systems can help companies find alternative suppliers, reducing the bargaining power of existing suppliers. However, AI can also increase supplier bargaining power by enabling suppliers to provide customers with more customized and affordable goods and services to meet their specific requirements.
Threat of new entrants	AI applications that are expensive to build and run or that require highly specialized knowledge can deter new market entrants, as can AI applications that create a loyal customer base or a unique user experience. However, the proliferation of commercially available AI platforms and open-source software has also made it easier to implement some form of AI. Some new market entrants will be able to compete with existing firms in their market space. Established players will need to devote more of their resources to developing their own AI technology and datasets.
Positioning and rivalry among existing competitors	AI can intensify competition among existing firms. Companies that successfully use AI may gain a competitive edge over other firms in their industry, forcing these firms to adopt AI or risk becoming uncompetitive.

but many won't because they lack the financial resources and know-how to work with AI tools. Equally essential are the enormous datasets required to train and build meaningful AI applications. The right data must be available, accurate, complete, error-free, and well-organized, and it must be appropriate for the problem to be solved by AI. Building successful AI applications also requires expertise with technology implementations and skillful change management (see Chapter 14). Companies that have AI expertise and the resources to pay for the extensive computing power and energy consumption of AI applications will be at an advantage. However, this will not entirely deter new market entrants who are able to take advantage of open-source software and AI platforms. AI can either increase or decrease the bargaining power of customers and suppliers, and it can be the foundation for new products and services. Table 3.7 shows how AI is impacting all aspects of Porter's competitive forces model, recasting industry structures and the rules of competition.

3.8 Describe the Challenges Posed by Strategic Information Systems

Strategic information systems often change the organization as well as its products, services, and operating procedures, driving the organization into new behavioral patterns. Successfully using information systems to achieve a competitive advantage is challenging

and requires precise coordination of technology, organizations, and management. However, having a good strategy doesn't mean it's always the right strategy or that it is being properly executed, as the Spotlight on Management Case Study, *Spotify's Podcasting Strategy Stumbles*, illustrates.

Sustaining Competitive Advantage

The competitive advantages that strategic systems confer do not necessarily last long enough to ensure long-term profitability. Because competitors can retaliate and copy strategic systems, competitive advantage is not always sustainable. Markets, customer expectations, and technology change; globalization has made these changes even more rapid and unpredictable. The Internet can make competitive advantages disappear very quickly because virtually

Spotlight on Management Case Study

Spotify's Podcasting Strategy Stumbles

Spotify is the world's largest on-demand music service, with more than 574 million active users in 180 countries. The company is known for its skillful use of technology to create ultra-personalized recommendations for songs and other audio content for its users.

Spotify doesn't have a monopoly on music streaming. Users can also find their favorite music, audiobooks, and podcasts on many other platforms. What makes Spotify stand out is its superior way of precisely serving the right audio (or other content) at the right time to each individual using its service. This ability to hyper-personalize recommendations for each Spotify user would not be possible without AI and is a major source of its competitive advantage.

Spotify uses AI in myriad ways to drive this personalization strategy. For example, Spotify's most popular feature is AI DJ, an AI-powered disc jockey that chooses what to play for users based on their specific music tastes and listening behavior. The AI DJ curates tracks for the individual user based on the data it has acquired about them, using generative AI to narrate its selections in a hyper-realistic voice. AI DJ will switch songs based on each user's real-time feedback. It can discover music you haven't listened to recently or recommend new selections based on what it thinks you'd like.

Spotify's Discover Weekly uses Spotify AI algorithms to craft a series of custom-made playlists of 30 songs each Monday for every user. Playlist selection criteria include each user's streaming history, likes and dislikes, and songs they added to the playlist. The playlist includes tracks that the user might not have heard before, but these recommendations are based on each user's search history and preferences. This capability helps introduce listeners to new tunes.

Spotify uses AI audio models to analyze data from raw audio tracks and categorize songs. This helps Spotify evaluate all songs to create recommendations, whether or not they are covered online. For example, if a new song is released by a new artist on the Spotify platform, AI natural language processing (NLP) models might not discover it if coverage online and in social media is low. However, by using song data from audio models, the platform will still be able to analyze the track and recommend it to similar users along with other more popular songs.

Spotify's business strategy is multi-faceted and includes optimizing revenue from its free/freemium business model, forming partnerships with artists, music labels, and content providers, and expanding its platform to include podcasts, audiobooks, and other audio content, in addition to personalization. This approach has diversified Spotify revenue streams, amplified user engagement, and solidified its market presence. Spotify's profitability has fluctuated, but experts believe the company is positioned to grow its earnings going forward.

This multipronged business strategy had served Spotify well, but it didn't prevent the company from stumbling when it decided to also become a content creator, with podcasts delivering that content. Spotify's podcast strategy focused on original content from its in-house studios Parcast and Gimlet. These studios had excellent staff and built-in audiences. However, as part of Spotify, the studios failed to attract large audiences or produce new hit shows. Spotify did not give their shows the right marketing or support. Keeping certain podcasts exclusive to Spotify caused their audience to drop.

Spotify's preoccupation with original content also included making expensive deals with celebrities and celebrity studios as a means of attracting new listeners. Some deals worked out, including the Batman Unburied podcast by David S. Goyer and the Case 63 scripted drama with Julianne Moore and Oscar Isaac. But many have not. Spotify's troubled celebrity deal with Prince Harry and Meghan Markle's Archewell Audio production company fell apart when

Archewell failed to deliver all of the podcasts specified in its $20 million Spotify contract. Similar deals with high-profile studios such as Barack and Michele Obama's Higher Ground also fizzled out.

All told, Spotify spent more than $1 billion to build its podcasting empire, far more than any of its rivals. It paid $286 million for the podcast studios and spent $250,000 or more per episode on exclusive shows to lure new listeners. These investments have not paid off. No one in the podcasting business is making much money. The amount of Spotify's investment relative to the size of the podcasting market was way out of line.

Spotify's management did not understand that podcasts have a fundamentally different formula for success from film, video games, books, and music. Franchises, intellectual property (IP), and name recognition can be enough to deliver a hit across many different mediums. For podcasts, however, a series from an acclaimed filmmaker, best-selling author, celebrity, or former president may barely register on podcast channels.

Spotify has now modified its podcast strategy to capitalize on its ability to sell ads to deliver more value for content creators and users. It cut podcasting staff and programming and it is upgrading its advertising tools for podcasters. The company is using listener preferences, based on music choices and listening habits, to target listeners with ads. It introduced a marketplace called Spotify Audience Network that enables advertisers to reach particular types of listeners, rather than buying spots on specific shows. To date, Spotify hasn't come up with anything unique in podcasting to make its platform stand out from competitors such as Apple, Stitcher, SiriusXM, and others. The most popular platform for podcasts is YouTube.

SOURCES: Ipshita Sen, "How AI Helps Spotify Win in the Music Streaming World," Outsideinsight.com, accessed August 17, 2024; Mike Kaput, "How Spotify Uses AI (And What You Can Learn from It)," Marketingaiinstitute.com, January 26, 2024; Anne Steele and Sarah Krouse, "Spotify's $1 Billion Podcast Bet Turns into a Serial Drama," *Wall Street Journal*, September 5, 2023; Amrita Khalid, "How Spotify's Podcast Plan Is Going Off the Rails," Theverge.com, June 21, 2023.

Case Study Questions

1. Analyze Spotify's business strategy using the value chain and competitive forces models.
2. What is the role of information technology and AI in Spotify's business strategy?
3. Why did Spotify's focus on podcasting initially stumble? What management, organization, and technology factors were involved?
4. Spotify Chief Executive Daniel Ek has said he wants Spotify to evolve from a music streaming company to being the world's largest audio company, with content including audiobooks, education, sports and news. According to Ek, podcasting has helped make the company a major podcasting platform despite its rocky start. Do you think Spotify will be able to achieve this goal? Why or why not?

all companies can use this technology. Classic strategic systems, such as American Airlines' SABRE computerized reservation system, Citibank's ATM system, and FedEx's package tracking system, benefited by being the first in their industries. Then rival systems emerged. Amazon was an e-commerce leader but now faces serious competition from Walmart, eBay, and Google. Information systems alone cannot provide an enduring business advantage. Systems originally intended to be strategic frequently become tools for survival, required by every firm to stay in business, or they may inhibit organizations from making the strategic changes essential for future success.

Aligning IT with Business Objectives

Research on IT and business performance has found that firms that can successfully align information technology with their business objectives are more likely to be profitable. When alignment is missing, the technology might work but it might not deliver an outcome the business needs (Hind et. al., 2023; Luftman et. al., 2017).

Most businesses get it wrong: Information technology takes on a life of its own and does not serve management and shareholder interests very well. Instead of businesspeople taking an active role in shaping IT to the enterprise, they ignore it or come up with unrealistic expectations of what IT can and cannot do. This is especially true for firms pursuing benefits from AI. Such firms pay a hefty price in poor performance and business results. Successful firms and managers understand what IT can do and how it works, take an active role in shaping its use, and measure its impact on revenues and profits.

Management Checklist: Performing a Strategic Systems Analysis

To align IT with the business and use information systems effectively for competitive advantage, managers need to perform a strategic systems analysis. To identify the types of systems that provide a strategic advantage to their firms, managers should ask the following questions:

1. What is the structure of the industry in which the firm is located?
   - What are some of the competitive forces at work in the industry? Are there new entrants to the industry? What is the relative power of suppliers, customers, and substitute products and services over prices?
   - Is the basis of competition quality, price, or brand?
   - What is the direction and nature of change within the industry? From where is the momentum and change coming? How much of this change is being driven by AI technology?
   - How is the industry currently using information technology? Is the organization behind or ahead of the industry in its application of information systems?
2. What are the business and industry value chains for this particular firm?
   - How is the company creating value for the customer—through lower prices and transaction costs or higher quality? Are there any places in the value chain where the business could create more value for the customer and additional profit for the company?
   - Does the firm understand and manage its business processes using the best practices available? Is it taking maximum advantage of supply chain management, customer relationship management, and enterprise systems?
   - Does the firm leverage its core competencies?
   - Is the industry supply chain and customer base changing in ways that benefit or harm the firm?
   - Can the firm benefit from strategic partnerships, value webs, ecosystems, or platforms?
   - Where in the value chain will information systems provide the greatest value to the firm?
3. Have we aligned IT with our business strategy and goals? Have we correctly articulated our business strategy and goals?
   - Is IT improving the right business processes and activities to promote this strategy?
   - Are we using the right metrics to measure progress toward those goals?
   - Are we using AI in ways that genuinely make our firm more competitive?

3.9 Understand How the Information in This Chapter Can Help Your Career

The information you have learned in Chapter 3 can help you advance your career by helping you understand how information technology impacts organizations and how you and your company should take this into account. The chapter also provides many examples of how information technology can support the various competitive strategies used in your job. Finding ways of using IT to improve your company's competitive position will benefit you throughout your career.

Real-World Scenario

Crayola Develops Its Business with Digital Technology

Crayola, based in Easton, Pennsylvania, has been a cherished brand for generations, producing high-quality, non-toxic crayons, markers, pencils, modeling clay, creative toys, and innovative art tools. Crayola's popularity dipped with the advent of computers and online learning, as digital tools started to supplant hand-held art supplies. Crayola was able to meet this challenge by transforming its business model from a manufacturer of crayons and art tools into a trusted source of tools and experiences for creative play. The company still produces crayons and traditional art tools, but it also has created a digital product line, which includes tools such as iMarker, an all-in-one digital pen, crayon, and pencil, designed for use with the Color Studio HD iPad app. Crayola also invested more in digital marketing, including online advertising, promotions, social media, and the use of business development representatives, to connect with parents and educators. Crayola closely tracks activity on its online channels through Google Analytics to make sure it is getting the most out of its marketing and ad campaigns.

Job Description and Career Path: Business Development Representative

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Business Development Representative	Sales professional who specializes in seeking new business opportunities by contacting and developing relationships with potential customers Job responsibilities include: • Lead Generation: Creating quality sales leads by employing methods such as cold email, cold calling, social selling, and networking • Qualifying leads and engaging with prospective buyers, including working with existing business accounts to identify potential opportunities • Booking appointments with prospective customers that can turn into legitimate sales opportunities • Developing sales and marketing strategies to tap into untapped markets	• Knowledge of Microsoft 365 tools: Excel, Word, PowerPoint, Teams • Experience with Salesforce CRM software • Experience with data analytics and business intelligence platforms such as Tableau	• Business Development Representative: $50,000–$60,000 • Business Development Manager: $75,000–$100,000 • Director of Business Development $125,000–$160,000 • VP Business Development: $175,000–$250,000 Salaries can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

AI Redesigns the Fashion Industry

Artificial intelligence is finding many applications across industries. In the fashion industry, it is becoming a game-changer. The ways clothing and fashion accessories are being designed, produced, marketed, and sold are undergoing profound transformation. AI is also reshaping the customer experience.

For starters, AI is revolutionizing how brands anticipate and capitalize on fashion trends. Fashion is all about keeping up with trends—what's hot, whether clothing designs capture the trends of the moment, and where the hottest fashion trends can be found. Fashion forecasters traditionally made trend predictions by observing street fashion, designer runway shows, and pop culture. These predictions were not always accurate and were highly subjective.

Today, buying and merchandising can use AI predictive analytics and trend forecasting to make smarter buying decisions that will help fashion retailers increase sales and reduce overstock. AI tools can sift through enormous amounts of data, gleaning insights from various sources such as social media posts, fashion blogs, online retail sites, retail sale transactions, and global fashion events. This data includes styles, color schemes, fabric choices, and regional fashion preferences. AI can spot patterns and predict trends with impressive accuracy.

For example, Heuritech, a Paris-based AI platform, uses machine learning to analyze more than 3 million images from social media and the web each day for unbiased trend analysis. Heuritech AI can detect 2,000 different fashion attributes, such as macro prints, colors and fabrics, and granular shapes to forecast market shifts. The system can forecast and compare the current visibility of a color in China versus what it is likely to be in the United States in ten months, identify the consumer segments that will adopt it, and compare the ideal month to release items with that color. The system allows fashion brands such as Louis Vuitton and Dior to align their collections based on actual customer preferences and needs.

AI is also being used to optimize fashion supply chains. The supply chain for the fashion industry is one of the most complex and challenging to manage. Each step, from sourcing raw materials, to managing inventory, to ensuring timely delivery, is riddled with inefficiency and waste. AI can help streamline and optimize these processes. AI algorithms can analyze historical data to predict demand more accurately, allowing better inventory management. For instance, fast fashion retail chain H&M utilizes AI to analyze store receipts and returns, helping H&M adjust the quantities of each garment for each store. This kind of data analysis reduces the chances of having too much unsold stock, which enables companies to cut their costs, while promoting sustainability. Additionally, AI can also help in managing the logistics of the supply chain. For instance, AI systems can track and analyze shipping data, ensuring that products are delivered in the most efficient manner. This also saves time and reduces carbon emissions.

Zara, another fast fashion leader noted for skillful supply chain management, has been using AI to make its fast fashion business processes even faster and more "intelligent." Zara has a vertical supply chain where it controls everything from design to display to shipping, allowing it to gather immense quantities of data at every step. The data are analyzed to identify inefficiencies, pinpoint areas of success, and create more accurate demand forecasts and predictions. Armed with all of this information, Zara can develop and deliver new items to stores in as little as one week, a significantly faster turnaround time compared to the industry average of three to six months.

Sustainability is another area in which AI is having an impact on the fashion industry. The fashion industry is believed to be the second largest polluter worldwide, generating 186 billion pounds of textile waste annually. Companies create far more products than they can sell, and the fashion industry emits about the same quantity of greenhouse gases per year as the economies of France, Germany, and the United Kingdom combined. Most clothing and other fashion apparel are generally non-biodegradable and rarely recycled, so 87 percent of all material used ends up in landfills or incinerators. Fashion companies are starting to respond to consumer demands for more sustainable fabrics, but disposing of unsold clothing remains a huge problem.

Information technology, including AI, has proved very effective for curbing waste. Companies such as H&M and Zara are integrating AI into RFID technology and clothing tags with microchips, providing full visibility of their inventory, facilitating more informed forecasting decisions, and minimizing waste from unsold clothing. Zara also has an initiative with Jetlore, a consumer behavior prediction platform that uses AI to map consumer behavior into structured predictive attributes, such as size, color, fit, or style preferences, which help the company produce and sell only goods that customers actually want.

Lenzing Group, a world-leading provider of specialty fibers for the textile and nonwoven industries, is using AI to help it work with its customers and business partners to reduce carbon emissions. It uses machine learning and predictive analytics, along with a real-time ocean shipment tracker, to provide customers with access to carbon emission data at the shipment and container level. Lenzing and its customers use this real-time data to collaboratively establish goals and adjust shipment plans to reduce carbon emissions.

Refiberd has developed AI-based sorting technology for textile-to-textile re-sorting. The system redirects up to 70 percent of textile waste toward high-value recyclers. AI-powered hyperspectral imaging identifies specific fiber composition and contaminants present in textile waste. (Hyperspectral imaging collects and processes information across the electromagnetic spectrum to obtain the spectrum for each pixel in an image. This allows for the identification of objects and materials by analyzing their unique spectral signatures.)

One of the most significant applications of AI in fashion is its ability to personalize the fashion customer experience. For example, Thread.com, a UK-based startup, uses AI in combination with human stylists to provide personalized style advice and outfit recommendations for its customers. After customers supply their sizes, style preferences, and price ranges, AI algorithms, in collaboration with human stylists, curate a personalized shopping experience for each individual customer. StitchFix provides a similar service in the United States.

Small and medium-sized fashion retailers are also using AI to personalize their offerings. Apps like EasySize help online fashion shops offer personalized size recommendations to their customers, reducing returns and increasing customer satisfaction. A machine learning algorithm analyzes shopper feedback and calibrates size and fit recommendations based on the expectations and experiences of shoppers all over the world rather than body measurements. The EasySize app is available for e-commerce stores of all sizes through integration with the Shopify e-commerce platform. EasySize often works with small and fast-growing, direct-to-consumer fashion brands who have small teams and need things to be simple, reliable, and to work as fast as possible.

It gets even better: 3DLOOK is revolutionizing the way brands understand and cater to their customers' unique body shapes and sizes. Using mobile body scanning and AI, the technology provides highly accurate body measurements. 3DLOOK collects brand-specific data on core body measurements, fitting forms, avatars, and live models, along with specific grading rules. This provides a comprehensive understanding of the brand's unique sizing and fit standards. Utilizing this data, size recommendation algorithms map the brand-specific garment data to the customer's unique body data and fit preferences. The results are a highly personalized size suggestion for a perfect fit from each garment. By using 3DLOOK's YourFit solution, brands can potentially achieve a 13 to 16 percent increase in conversion rates and a 20 to 25 percent increase in their average order value.

AI is also transforming the creative process in the fashion industry, providing new design ideas and also helping fashion designers become more creative themselves. Designers can use generative AI tools such as DALL-E for clothing and jewelry design ideas in response to text prompts, or more powerful specialized design tools to create designs based on a designer's specific style.

Hong Kong fashion technology innovator Calvin Wong introduced a designer-led AI system called AiDA that generates blueprints from initial sketches and color ideas, preserving the designer's creativity while offering suggestions for improvement. Using AI image recognition, detection, and generation, the system provides a digital mood board for storing the sketches and color ideas. From this, AiDA's algorithms generate blueprints, offering designers a canvas for adjustments and personal touches. AiDA is a supportive tool and not a replacement for designer creativity or roles. Nike is another firm using AI to design products. Nike used AI to design its ISPA Universal shoe by feeding data on three previous shoe models to an algorithm, producing an eco-friendly shoe made from materials like Bio-EVA foam derived from the renewable raw material sugarcane.

Although AI has the potential to revolutionize the fashion industry, it also poses new challenges that businesses will need to address. With AI algorithms capable of doing the jobs of designers, marketers, and other fashion professionals, some jobs will be lost. This could have a significant impact on the entire fashion industry. There is also a risk that fashion will become less individual and less creative if AI is allowed to drive design and marketing decisions. This could lead to a decline in the quality and popularity of fashion products.

SOURCES: 3dlook.com, accessed August 26, 2024; Refiberd.com, accessed August 26, 2024; Easysize.com, accessed August 26, 2024; Euritech.com, accessed August 26, 2024; Stephanie Nikolopoulos and Melissa Epifano "H&M, Zara, Fast Fashion Turn to Artificial Intelligence to Transform the Supply Chain," Thomasnet.com, October 25, 2023; "Calvin Wong Launches AI Guiding Designers from Sketch to Runway," Apparelresources.com, September 18, 2023; Lenzing AG, "Lenzing Takes Supply Chain Transparency to Next Level by Combining Real-Time Shipment Tracking and Carbon Visibility," Lenzing.com, September 18, 2023.

Case Study Questions

1. Identify the points in the fashion industry value chain that could benefit from AI.
2. How has AI changed operations and decision making at the organizations discussed in this case study?
3. What strategic business objectives is AI helping the companies in this case study pursue? Explain your answer.

Review Summary

3.1 Discuss technical and behavioral views of organizations.

Information systems and organizations influence one another. The technical view of organizations sees the organization as a stable formal social structure that takes resources from the environment and processes them to produce outputs (products and services). The behavioral view of organizations emphasizes group relationships, values, and structures. Both views are required in order to fully understand the organizational impact of information technology.

3.2 Discuss features of organizations that affect information systems.

Modern organizations are hierarchical, specialized, and impartial, using explicit routines to maximize efficiency. Organizations have their own cultures and politics arising from differences in interest groups, and they are affected by their surrounding environment. Organizations differ in goals, groups served, social roles, leadership styles, incentives, types of tasks performed, and type of structure. These features help explain differences in organizations' use of information systems. Information systems and the organizations in which they are used interact with and influence each other.

3.3 Describe the impact of information systems on organizations.

The introduction of a new information system will affect organizational structure, goals, work design, values, competition between interest groups, decision making, and day-to-day behavior. At the same time, information systems must be designed to serve the needs of important organizational groups and will be shaped by the organization's structure, business processes, goals, culture, politics, and management. Information technology can reduce transaction and agency costs, and such changes have been accentuated in organizations using the Internet. New systems disrupt established patterns of work and power relationships, so there is often considerable resistance to them when they are introduced.

3.4 Discuss Porter's competitive forces model.

In Porter's competitive forces model, the strategic position of the firm and its strategies are determined by competition with its traditional direct competitors, but they are also greatly affected by new market entrants, substitute products and services, suppliers, and customers. Information systems help companies compete by maintaining low costs, differentiating products or services, focusing on market niche, strengthening ties with customers and suppliers, and increasing barriers to market entry with high levels of operational excellence.

3.5 Discuss the value chain model.

The value chain model highlights specific activities in the business where competitive strategies and information systems will have the greatest impact. The model views the firm as a series of primary and support activities that add value to a firm's products or services. Primary activities are directly related to production and distribution, whereas support activities make the delivery of primary activities possible. A firm's value chain can be linked to the value chains of its suppliers, distributors, and customers. A value web consists of information systems that enhance competitiveness at the industry level by promoting the use of standards and industrywide consortia and by enabling businesses to work more efficiently with their value partners.

3.6 Discuss synergies, core competencies, and network economics.

Information systems achieve additional efficiencies or enhance services by tying together the operations of disparate business units. Information systems help businesses leverage their core competencies by promoting the sharing of knowledge across business units. Information systems facilitate business models based on large networks of users or subscribers that take advantage of network economics. A virtual company strategy uses networks to link to other firms so that a company can use the capabilities of other companies to build, market, and distribute products and services. In business ecosystems, multiple industries work together to deliver value to the customer. Information systems support a dense network of interactions among the participating firms. Business ecosystems typically have one or more keystone firms that dominate the ecosystem and create a platform that can be used by other firms.

3.7 Understand how AI can be used in competitive strategy.

AI is making it easier for companies to improve performance based on their existing business models, but it is also spawning new business models and reshaping the competitive landscape. It has given birth to new products, services, and sources of value. Decision making has become much more data-driven, accurate, and precise for both operational and higher-level decisions. AI cannot make strategic decisions on its own, but it can be used to advise senior-level executives on how to pilot the firm more strategically. AI cannot fully substitute for human creativity and wisdom, but it can be a useful tool for ideation and problem-solving.

3.8 Describe the challenges posed by strategic information systems.

Implementing strategic systems often requires extensive organizational change and a transition from one sociotechnical level to another. Such changes are often difficult and painful to achieve. Moreover, not all strategic systems are profitable, and they can be expensive to build. Many strategic information systems are easily copied by other firms, so that strategic advantage is not always sustainable.

Key Terms

Agency theory, 81
Benchmarking, 90
Best practices, 90
Business ecosystem, 94
Competitive forces model, 84
Core competency, 92
Customer experience management, 87
Disruptive technologies, 79
Efficient customer response system, 86
Mass customization, 87
Network economics, 93
Organization, 74
Platform, 94
Primary activities, 89
Product differentiation, 85
Routines, 76
Support activities, 89
Switching costs, 88
Transaction costs, 81
Transaction cost theory, 81
Value chain model, 89
Value web, 91
Virtual company, 93

Review Questions

3.1 Discuss technical and behavioral views of organizations.

- Define organization and compare the technical and behavioral views of organizations.

3.2 Discuss the features of organizations that affect information systems.

- Identify and describe the features of organizations that help explain differences in organizations' use of information systems.
- Define and describe disruptive technologies and how they affect organizations.

3.3 Describe the impact of information systems on organizations.

- Describe the major economic theories that help explain how information systems affect organizations.
- Describe the major behavioral theories that help explain how information systems affect organizations.
- Explain why there is considerable organizational resistance to the introduction of information systems.

3.4 Discuss Porter's competitive forces model.

- Define Porter's competitive forces model and explain how it works.
- Describe what the competitive forces model explains about competitive advantage.
- Describe four competitive strategies enabled by information systems that firms can pursue.
- Describe how information systems can support each of these competitive strategies and give examples.
- Describe how the Internet has changed competitive forces and competitive advantage.

3.5 Discuss the value chain model.

- Define and describe the value chain model.
- Explain how the value chain model can be used to identify opportunities for information systems.
- Define the value web and show how it is related to the value chain.
- Explain how the value web helps businesses identify opportunities for strategic information systems.

3.6 Discuss synergies, core competencies, and network economics.

- Describe how promoting synergies and core competencies enhance competitive advantage.
- Explain how businesses benefit from using network economics and business ecosystems.
- Define and describe a virtual company and the benefits of pursuing a virtual company strategy.

3.7 Understand how AI can be used in competitive strategy.

- Describe how AI has impacted organizational performance and decision making.
- Describe how AI can add value at various steps in the value chain.
- Describe how AI has impacted the competitive forces identified in Porter's model.

3.8 Describe the challenges posed by strategic information systems.

- List and describe the management challenges posed by strategic information systems.
- Explain how to perform a strategic systems analysis.

Discussion Questions

3.9 It has been said that there is no such thing as a sustainable strategic advantage. Do you agree? Why or why not?

3.10 It has been said that the advantage that leading-edge retailers such as Amazon and Walmart have over their competition isn't technology; it's their management. Do you agree? Why or why not?

3.11 What are some of the issues to consider in determining whether the Internet would provide your business with a competitive advantage?

Using AI Project

3.12 AI Project: Assessing Impact of Geographic and Demographic Factors

You have been hired as a consultant by two friends who want to open a retail store selling organic pet food and treats, as well as pet toys and clothing made from organic materials. They have asked you to help them understand the most important geographic and demographic factors that might affect the success of their prospective business, such as proximity to good roads and parking, or the average age and income of the population near the store location. Using Microsoft's Copilot, Google's Gemini, or ChatGPT, develop a prompt that you can use to help advise them as to the most important geographic and demographic factors for a small retail pet store business featuring organic products. Save a copy of your prompt and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience in analyzing the impact of business processes on business performance; using a database to improve sales performance; using web tools to configure and price a product; and collaborate in a team to identify strategic information systems.

For student files please visit www.pearsonhighered.com/laudon or MyLab MIS.

3.13 Management Decision Problem

Entrust Corporation, formerly Datacard Group, produces hardware and software for establishing trusted identities and the issuance of personalized financial cards, passports, driver's licenses, identification cards, and mobile devices. Entrust is one of the world's leading suppliers of payment cards. As a part of doing business with Entrust, customers—which include financial institutions and credit card bureaus, government agencies, and major corporations—supply the Shakopee, Minnesota-based company with large volumes of business data. Entrust had a predominantly manual process for distribution pricing for its secure identification and card issuance products to about 250 partners quarterly by emailing a price book of multiple spreadsheets. Creating the book involved hours of manual input by people in marketing and sales. The process of changing a price was complex, time-consuming, and people-dependent. People who received the price books had to manually type them into their own systems and check the data—a process that could take up to two days. Assess the impact of this process on Entrust's business performance, and suggest some management, organization, and technology issues that would need to be addressed by a solution.

3.14 Using Data Analytics to Analyze Sales Performance

Software skills: Database querying and reporting; database design

Business skills: Sales trend analysis

In this exercise, you will start out with raw transactional sales data and then use Microsoft Access database software to develop queries that help managers make better decisions about product pricing, sales promotions, and inventory replenishment. The MIS18_Project3.14_Student File contains a Store and Regional Sales Database developed in Microsoft Access. The database provides raw data on weekly store sales of computer equipment in various sales regions. The database includes fields for store identification number, sales region, item number, item description, unit price, units sold, and the weekly sales period when the sales were made. Use Access to develop some queries to make this information more useful for running the business. This information includes the following:

- All 24-inch monitor sales by Sales Region for Week Ending December 29, 2024, sorted within region from highest to lowest number of units sold.
- All 24-inch monitor sales by store for the East Sales Region, sorted by Store No. and Week Ending in ascending order.
- All sales of all products for the East Sales Region, sorted from highest to lowest total number of units sold by item number.

Then describe how the information in these queries could be used for management decision making.

3.15 Getting to Know Tableau

In this project, you will learn how you can use Tableau to develop a competitive strategy based on identifying market niches. Access the Superstore data set, which is a sample data set available on the Tableau Public website and also within Tableau Desktop. The data set includes data about sales, profits, and other metrics from an online retailer. Use Tableau to create a dashboard with at least four data visualizations, such as bar charts, line charts, scatter plots, and/or maps, to analyze sales and profit data across the following dimensions: Segment; Product category and subcategory; Region, State, and Zip code. Use the dashboard to help you identify one or more specific market niches that the retailer should focus on. Write a report or create a presentation summarizing your findings and explaining how the visualizations helped you to reach your conclusions. If you don't have access to Tableau Desktop, ask your instructor, or alternatively, you can create a free Tableau Public account.

3.16 Using Online Tools to Configure and Price a Product

Software skills: Online research tools

Business skills: Researching product information and pricing

In this exercise, you will use online research tools to find product information about a car of your choice and use that information to make a purchase decision. You will also evaluate two of these tools.

You are interested in purchasing a lightly used low mileage Honda CR-V (or some other car of your choice). Go to the website of CarsDirect.com and begin your investigation. Research the various used CR-V models, and choose one you prefer in terms of price, features, and safety ratings. Locate and read at least two reviews. Then visit the Carvana.com website or app. Compare the information on pricing and available inventory for the car at the Carvana website or app with that at CarsDirect. Try to locate the lowest price for the car you want in inventory along with financing and vehicle pickup and delivery. Compare the Carsdirect.com customer experience to that provided by Carvana. Which do you prefer for this type of purchase? Explain your answer.

3.17 Collaboration and Teamwork Project: Identifying Opportunities for Strategic Information Systems

With a team of two or more other students, select a company described in *The Wall Street Journal, Fortune, Forbes*, or another business publication available on the web. Visit the company's website to find additional information about that company and to see how the firm is using the web. On the basis of this information, analyze the business. Include a description of the organization's features, such as important business processes, culture, structure, and environment as well as its business strategy. Suggest strategic information systems appropriate for that particular business, including those based on Internet technology, if appropriate. If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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4

Ethical and Social Issues in Information Systems

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Learning Objectives

After completing this chapter, you will be able to:

4.1 Describe IT/IS-related ethical, social, and political issues.

4.2 Describe principles for ethical conduct.

4.3 Discuss information rights and privacy issues.

4.4 Discuss intellectual property issues.

4.5 Discuss system quality issues.

4.6 Discuss accountability and control issues.

4.7 Discuss quality of life issues.

4.8 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** Apps That Track: A Double-Edged Sword
- **Spotlight on Organizations Case Study:** Section 230 Under Fire
- **Chapter Case Study:** Meta's Many Ethical Challenges

Opening Case Study

Apps That Track: A Double-Edged Sword

How often do you check your local weather forecast? If you are like many people, you might do so multiple times a day, probably by opening a weather app on a mobile device. Few realize that this simple act can open the door to the widespread sale of data about your exact physical location to third parties, who can then use it for purposes far removed from the function of the app that originally gathered it. And it's not just weather apps that collect and sell this kind of data. Location data forms the basis for a multibillion-dollar industry, populated by largely unregulated companies that operate behind the scenes.

To figure out where you are, mobile devices use data from a variety of technologies embedded within the device, such as GPS sensors, Bluetooth, and connections to local Wi-Fi and cellular networks. Mobile apps often make use of a third-party SDK (software development kit) such as Foursquare's Pilgrim SDK, a free software tool that enables an app to collect data, or may be paid by an SDK developer to include its code within an app.

Apple has taken some steps to make privacy issues related to apps more visible. If an app uses an SDK, the developer must describe what data the SDK collects and how the data may be used. Apps on the iPhone platform must ask for your permission before tracking your activity (including your physical location). Google instituted similar policies in 2022. However, many users are so conditioned to clicking "Yes" that they don't stop to consider the ramifications. In addition, some app developers employ "dark patterns," a design tactic that encourages you to make a choice detrimental to your own interests, for example, by prompting you to enable location tracking while simultaneously suggesting that the app might not work as intended if location tracking is not enabled. As a case in point, contrary to what many people assume, weather apps do not actually "need" to take constant readings of location data to provide a local weather forecast, but these apps do not necessarily make that fact apparent.

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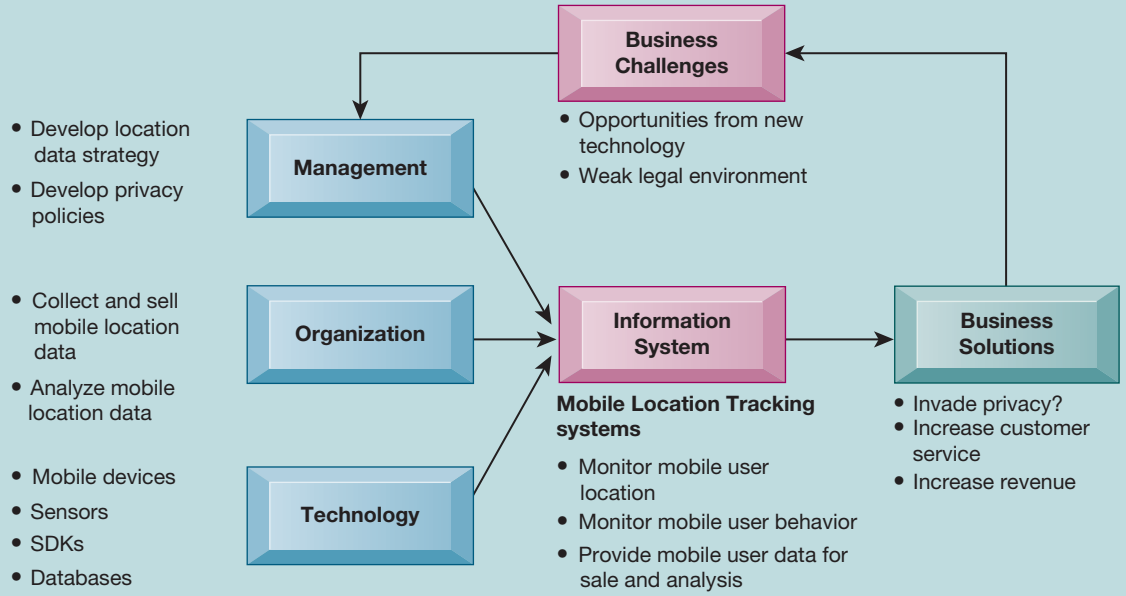
Once you agree to tracking, the app will typically refer you to a privacy policy that details how that data can be used. Research has shown that few users ever actually read such policies or understand their ramifications. Companies that sell user data often justify that doing so provides them with a necessary revenue stream. For example, AccuWeather's privacy policy notes that its ability to provide its app for free is supported by its sale of user information to third parties. Companies that sell user data also note that the information collected is usually anonymized (although studies have shown that it can be relatively easy to tie such data to a particular person). Companies that buy data may also be doing so for a variety of legitimate reasons such as for analytics, fraud detection, and advertising and marketing purposes.

You might not object to your data being used to provide you with relevant advertising. However, it's what happens to that data once it is in the hands of a data broker that is of concern. Currently, there are few laws or regulations in the United States that restrict what a data broker can do with that data. Once it has been collected, it can be sold over and over again. For instance, although the U.S. Supreme Court has held that the government needs a warrant to compel companies such as cell phone carriers to produce location data, no such restriction applies to the government's purchase of such data from data brokers.

That might be starting to change, however. Federal government agencies have begun to take action against location data brokers and telecommunications providers. For instance, in November 2023, the Federal Trade Commission (FTC) filed an amended complaint in U.S. District Court against data broker Kochava, alleging that Kochava was collecting, using, and disclosing massive amounts of precise geolocation data without user notification or consent. In February 2024, the Court ruled that the suit could proceed, endorsing the FTC's position that the unfettered data collection of sensitive geolocation data could be considered a violation of Section 5 of the FTC Act, which prohibits unfair or deceptive acts or practices in or affecting commerce. In April 2024, the FTC finalized an order prohibiting data broker Outlogic (formerly known as X-Mode) from sharing or selling to third parties, including U.S. government agencies, raw location data that Outlogic gathered from hundreds of different mobile apps and which could be used to track a person's visits to sensitive locations such as medical clinics and places of worship. In May 2024, the FTC also finalized a similar settlement with digital marketing and data broker InMarket Media.

The Federal Communications Commission (FCC) is also taking action. In April 2024, it fined AT&T, Verizon, T-Mobile, and Sprint (now owned by T-Mobile) nearly $200 million for selling real-time location information to data aggregators without their customers'



consent, albeit more than five years after those practices first came to light.

However, critics charge that these actions do not go nearly far enough and that relying on concepts like notification and user consent are a totally inadequate method of protecting privacy. There has been a new push for the enactment of a comprehensive federal data privacy law with the introduction of the proposed American Privacy Rights Act, but the prospects for its passage, although perhaps better than in the past, still remain uncertain.

SOURCES: "Collecting, Using, and Sharing Location Data," Reuters.com, June 2024; Forrester, "Proposed Federal Privacy Law Gains Momentum," Forbes.com, May 15, 2024; Joseph Jerome, "The FCC's Big Fines Over Location Data Aren't a Privacy Success Story," Techpolicy.com, May 3, 2024; Wendy Davis, "Kochava Must Face FTC Location Privacy Charges," Mediapost.com, February 5, 2024; Dell Cameron, "The Sad Truth of the FTC's 'Historic' Privacy Win," Wired.com, January 16, 2024; Alfred Ng and Jon Keegan, "Who Is Policing the Location Data Industry," Themarkup.org, February 24, 2022; Jon Keegan and Alfred Ng, "There's a Multibillion-Dollar Market for Your Phone's Location Data," Themarkup.com, September 30, 2021.

The challenges that location tracking poses to privacy, described in the Opening Case Study, show that technology can be a double-edged sword. It can be the source of benefits, including the ability to use mobile devices to access maps and driving directions or local weather and business news. At the same time, digital technology creates new opportunities for invading privacy and using information that could cause harm.

The Opening Case Study diagram above calls attention to important points that this case and chapter raise. Advances in mobile device technologies, mobile communications, and data management and analytics have created opportunities for organizations to track the locations of individuals who are using mobile devices at any given time. The organizations described in the opening case are benefiting by collecting and using location-based data to monitor user behavior so that this information can be sold for marketing and other purposes. The use of location data is also taking benefits away from individuals because such use can invade their privacy. Individuals might be subject to job discrimination or special scrutiny based on behavior patterns revealed by location tracking. There are very few privacy protections that apply to location tracking.

This case also illustrates an ethical dilemma because it shows two sets of interests at work: the interests of organizations that want to use location data to enhance profit and the interests of those who fervently believe that businesses and public organizations should not use personal data because doing so invades privacy or may harm individuals. The case further illustrates the importance of being sensitive to both the positive and negative impacts of information systems and the importance of learning how to resolve ethical dilemmas involving information systems.

Case Study Questions

1. Does analyzing location data create an ethical dilemma? Why or why not?
2. Should there be new privacy laws to protect personal data collected from mobile device users? Why or why not?

CAREER BENEFIT

Awareness of the ethical, social, and legal issues surrounding the use of information technology and information systems is a critical skill for everyone who uses them in an organization. Recognizing ethical dilemmas and understanding how to best resolve them is also a critical skill.

4.1 Describe IT/IS-related Ethical, Social, and Political Issues

In this chapter, we examine the ethical, social, and political issues raised by the use of information technology and information systems. **Ethics** refers to the principles of right and wrong that individuals, acting as free moral agents, use to make choices to guide their behavior. Information systems raise ethical questions for both individuals and societies because these systems create opportunities for intense social change and, thus, threaten existing distributions of power, money, rights, and obligations. Like other technologies, such as steam engines, electricity, and the telephone, information technology can be used to achieve social progress, but it can also be used to commit crimes and threaten cherished social values. The development of information technology will produce benefits for many and costs for others.

Ethics

Principles of right and wrong that can be used by individuals acting as free moral agents to make choices to guide their behavior.

Ethical, social, and political issues in information systems have been given additional urgency by the rise of information technologies, the Internet, and most recently, the increased use of artificial intelligence, unleashing concerns about the appropriate use of customer information, the protection of personal privacy, and the protection of intellectual property.

Other pressing ethical issues that information systems raise include setting standards to safeguard system quality that protect the safety of the individual and society, establishing accountability for the consequences of information systems, and preserving values and institutions considered essential to the quality of life in today's information society. When using information systems, it is essential to ask, "What is the ethical and socially responsible course of action?".

Ethical Challenges in Conducting Business

Before delving into the ethical, social, and political issues surrounding the use of information systems, it is important to consider the general topic of ethics in business. Organizations and their employees face numerous ethical challenges in conducting business. Table 4.1 provides a small sample of recent examples involving failures in ethical business judgment. These lapses in judgment occurred across a broad spectrum of industries.

In today's environment, employees who are convicted of violating the law face the risk of time in prison. The U.S. Department of Justice has indicated that prosecuting corporate executives for "white-collar crime" is one of the Department's highest priorities. U.S. federal sentencing guidelines mandate that federal judges impose stiff sentences based on the monetary value of the crime, the presence of a conspiracy to prevent discovery of the crime, the use of structured financial transactions to hide the crime, and failure to cooperate with prosecutors.

Although business firms in the past often paid for the legal defense of employees enmeshed in civil charges and criminal investigations, firms are now encouraged to cooperate with prosecutors to reduce charges against the firm itself. More than ever, as a manager or employee, you will have to decide for yourself what constitutes proper legal and ethical conduct.

Failures in ethical business judgment are not usually masterminded by employees of information systems departments, but information technology and information systems are often instrumental in many of these crimes. In many cases, the perpetrators of these crimes artfully used information systems to bury their actions from public scrutiny. We deal with the issue of information systems controls in Chapter 8.

Table 4.1 Recent Examples of Failures in Ethical Business Judgment

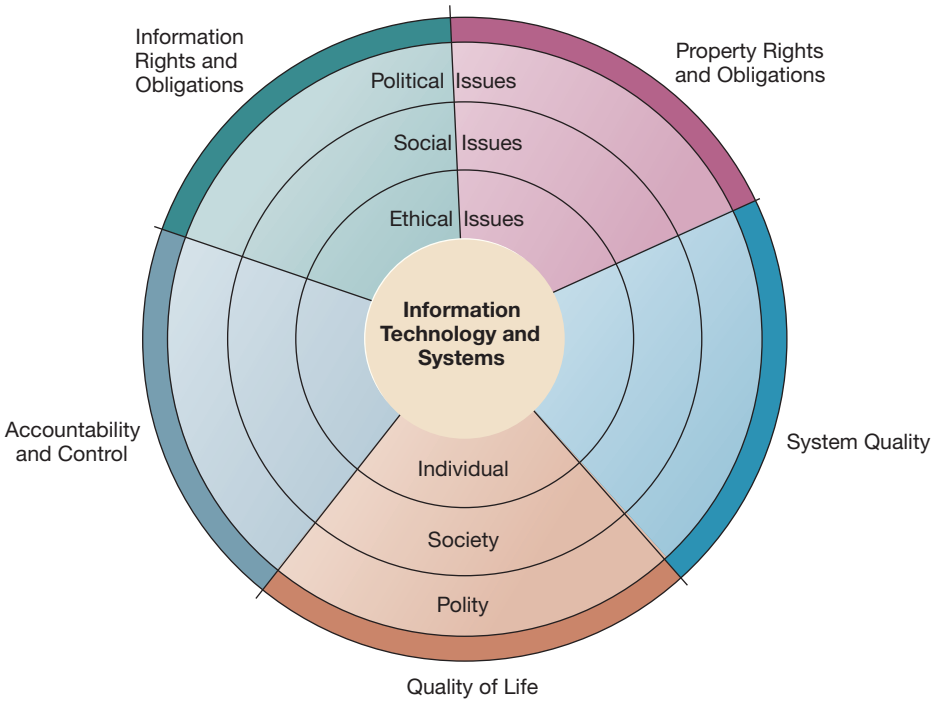
Company	Failure in Ethical Business Judgment
FX	Sam Bankman-Friedman, head of the cryptocurrency trading firm FX, was sentenced to 25 years in prison for his role in what has been described as one of the biggest financial frauds in U.S. history, leading to the collapse of FX. Bankman-Friedman's crimes were enabled by defects in internal controls, regulatory compliance, human resources, and system integrity.
Robinhood	Fintech brokerage firm Robinhood was fined a record $70 million by the Financial Industry Regulatory Authority (FINRA) for systemic supervisory failures in several critical parts of its business, including distributing false and misleading information to customers of the firm, as well as systems outages that prevented customers from placing timely trades.
Volkswagen AG	Installed "defeat-device" emissions software on more than 500,000 diesel cars in the United States and on roughly 10.5 million more worldwide that allowed the cars to meet U.S. emissions standards during regulatory testing while actually spewing unlawful levels of pollutants into the air in real-world driving. Penalties paid internationally have totalled $35 billion, with $9.5 billion going to U.S. car owners. Volkswagen sued top company executives for being complicit in the fraud and obtained settlements totalling $365 million. Criminal charges were also brought against a number of VW executives, including U.S.-based Oliver Schmidt, who was sentenced to seven years in prison and fined $400,000.
Wells Fargo	Wells Fargo employees admitted to opening millions of false accounts, manipulating terms of mortgages, and forcing auto loan customers to purchase unneeded insurance. Wells Fargo was fined $2.5 billion by the US government.

A Model for Thinking about Ethical, Social, and Political Issues

Ethical, social, and political issues are closely linked. The ethical dilemma you may face as someone who is involved in the use of information systems by your organization typically is reflected in social and political debate. Examine Figure 4.1, which illustrates one way to think about these relationships. Imagine society as a more or less calm pond on a summer day, a delicate ecosystem in partial equilibrium with individuals and with social and

Figure 4.1 The Relationship Among Ethical, Social, and Political Issues in an Information Society

The introduction of new information technology has a ripple effect, raising new ethical, social, and political issues that must be dealt with on the individual, social, and political levels. These issues have five moral dimensions: information rights and obligations, property rights and obligations, system quality, quality of life, and accountability and control.



political institutions. Individuals know how to act in this pond because social institutions (family, education, organizations) have developed well-honed rules of behavior, and these rules are supported by laws developed in the political sector that prescribe behavior and promise sanctions for violations. Now toss a rock into the center of the pond. What happens? Ripples, of course.

Imagine instead that the disturbing force is a powerful shock of new information technology and systems hitting a society more or less at rest. Suddenly, individual actors are confronted with new situations often not covered by the old rules. Social institutions cannot respond overnight to these ripples—it may take years to develop etiquette, expectations, social responsibility, or approved rules. Political institutions also require time to develop new laws and often require the demonstration of real harm before they act. In the meantime, you may have to act. You may be forced to act in a legal gray area.

We can use this model to illustrate the dynamics that connect ethical, social, and political issues. This model is also useful for identifying the main moral dimensions of the information society, which cut across various levels of action—individual, social, and political.

Five Moral Dimensions of the Information Age

The major ethical, social, and political issues that information systems raise include the following moral dimensions.

- *Information rights and obligations* What information rights do individuals and organizations possess with respect to themselves? What can these rights protect?
- *Property rights and obligations* How will traditional intellectual property rights be protected in a digital society in which tracking and accounting for ownership are difficult while ignoring such property rights is so easy?
- *System quality* What standards of data quality and system quality should we demand to protect individual rights and the safety of society?
- *Accountability and control* Who can and will be held accountable and liable for the harm done to individual and collective rights?
- *Quality of life* What values should be preserved in an information- and knowledge-based society? Which institutions should we protect from violation? Which cultural values and practices does information technology support?

We explore these moral dimensions in detail in Sections 4.3 through 4.7.

Key IT Trends That Raise Ethical Issues

Ethical issues long preceded information technology. Nevertheless, information technology has heightened ethical concerns, taxed existing social arrangements, and made some laws obsolete or insufficient. Technological trends that are responsible for many of these ethical stresses are summarized in Table 4.2.

The ever-increasing rise in computing power over the past several decades has made it possible for most organizations to use information systems for their core production processes. As a result, our dependence on systems and our vulnerability to system errors and poor data quality have increased. Social rules and laws have not yet fully adjusted to this dependence. In addition, standards for ensuring the accuracy and reliability of information systems (see Chapter 8) are not universally accepted or enforced.

Advances in data storage techniques and rapidly declining storage costs have enabled the collection of big data and have been responsible for the proliferation of databases on individuals—employees, customers, and potential customers—maintained by private and public organizations. These advances in data storage have made the routine violation of individual privacy both inexpensive and effective. Data storage systems for terabytes and petabytes of data are now available onsite or as online services for firms of all sizes to use in identifying customers.

Advances in data analysis techniques for large pools of data are another technological trend that heightens ethical concerns because organizations can find out highly detailed personal information about individuals. With contemporary data management tools (see Chapter 6), companies can assemble and combine myriad pieces of data about you stored in information systems much more easily than they could in the past.

Table 4.2 IT Trends that Raise Ethical Issues

Trend	Impact
Ever-increasing rise in computing power	More organizations depend on computer systems for critical operations and become more vulnerable to system failures.
Rapid decline of data storage costs	The steep reduction in cost enables the collection and storage of big data. Organizations can easily maintain detailed databases on individuals. There are no limits on the data collected about you.
Advances in data analysis	Companies can analyze vast quantities of data gathered on individuals to develop detailed profiles of individual behavior. Large-scale population surveillance is enabled.
Advances in networking	The cost of making data accessible from anywhere falls exponentially. Access to data becomes more difficult to control.
Proliferation of mobile devices	Mobile devices may be tracked without user consent or knowledge. The always-on device becomes a tether.
Artificial Intelligence (AI)	Increased reliance on various forms of AI in decision making substitutes data-driven calculations for human judgment. AI also poses challenges to privacy, further facilitating large-scale population surveillance, and to intellectual property rights.

Think of all the ways you generate digital information about yourself—browsing online; using mobile apps; making online and in-person retail and service purchases; interacting with the financial services industry; participating in social networks; making mobile phone calls; using a smart TV, smart speaker, and other smart home devices; driving an Internet-enabled car; as well as interacting with local, state, and federal government institutions (including courts and the police). Each one of these actions generates digital data. Put together and mined properly, this information can reveal not only your credit information but also your tastes, your associations, what you read and watch, your political interests, your driving habits, and more.

Companies purchase personal information from a variety of sources to help them more finely target their marketing campaigns. Chapters 6 and 12 describe how companies can analyze large pools of data from multiple sources to rapidly identify buying patterns of customers and make individualized recommendations. The use of computers to combine data from multiple sources and create digital dossiers of detailed information on individuals is called **profiling**. For example, Google Marketing Platform tracks the activities of visitors of participating websites and uses this information to create a profile of each online visitor, adding more detail to the profile as the visitor accesses an associated Google Marketing Platform site.

Profiling

The use of computers to combine data from multiple sources and create electronic dossiers of detailed information on individuals.

Credit card purchases can make personal information available to a wide variety of organizations. Advances in information technology facilitate the invasion of privacy.

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Over time, Google Marketing Platform can create a detailed dossier of a person's spending and online behavioral habits that is sold to companies to help them target their online ads more precisely. Advertisers can combine online consumer information with offline consumer information such as credit card purchases at stores.

LexisNexis Risk Solutions gathers data from police, criminal, and motor vehicle records; credit and employment histories; current and previous addresses; professional licenses; and insurance claims to assemble and maintain dossiers on almost every adult in the United States. The company sells this personal information to businesses and government agencies. Demand for personal data is so enormous that data broker businesses, such as Acxiom, Epsilon Data Management, CoreLogic, and LexisNexis Risk Solutions, are flourishing. The three major credit monitoring services, TransUnion, Equifax, and Experian, also maintain huge databases of personal data.

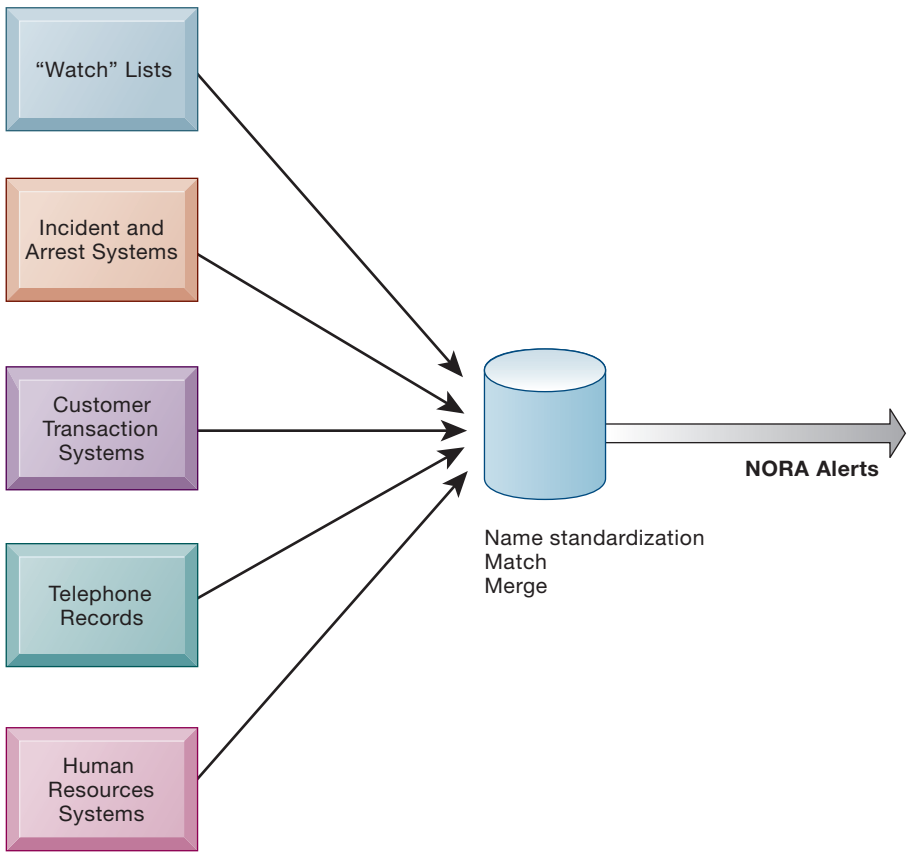
A data analysis technology called **nonobvious relationship awareness (NORA)** has given both the government and the private sector even more powerful profiling capabilities. Examine Figure 4.2, which illustrates how NORA works. NORA can take information about people from many disparate sources, such as "watch" lists, incident and arrest systems, consumer transaction systems, telephone records, and human resources systems, and correlate relationships to find obscure connections that might help identify criminals or terrorists. NORA technology processes the data and extracts information as the data are being generated so that, for example, it can—before a person gets on an airplane—instantly discover that the person at the airline ticket counter has the same phone number as a known terrorist. NORA is also often used by the casino industry as an anti-fraud technology. Although NORA technology is considered a valuable security tool, it does have privacy implications because it can provide such a detailed picture of the activities and associations of a particular individual. In 2005, IBM purchased the company that originally developed NORA technology, and now offers it under the name IBM Infosphere Identity Insight.

Nonobvious relationship awareness (NORA)

Data analysis technology that can find obscure hidden connections among people and entities by analyzing information from many different sources to correlate relationships.

Figure 4.2 Nonobvious Relationship Awareness (NORA)

NORA technology can take information about people from disparate sources and find obscure, nonobvious relationships. It might discover, for example, that an applicant for a job at a casino shares a telephone number with a known criminal and then issue an alert to the hiring manager.



Advances in networking, including the Internet, have greatly reduced the costs of accessing large quantities of data and have enabled the mining of large pools of data using desktop/laptop computers, mobile devices, and cloud servers, permitting an invasion of privacy on a scale and with a precision heretofore unimaginable. The proliferation of mobile devices, such as smartphones, that users carry everywhere means that users can potentially be tracked without their knowledge or consent.

Finally, companies and individuals are increasingly relying on AI technologies as a substitute for human judgment in decision making. However, doing so raises serious ethical issues. Human judgment relies not only on reasoning but also on other human capabilities, such as empathy, and also typically takes into account factors that inform moral and ethical behavior. AI, on the other hand, may make decisions without regard to consequences that might violate ethical and moral standards. Another issue with AI is that it reduces multidimensional reality into a more simplistic representation of the world, one that appears on the surface to be precise and undeniably correct, thus encouraging acceptance of its decisions without questioning them. AI also poses challenges to privacy and intellectual property rights. We discuss these issues further later in this chapter and in Chapter 11 (Moser, den Hond, and Lindebaum, 2022).

4.2 Describe Principles for Ethical Conduct

Ethics is a concern of humans who have freedom of choice. Ethics is about individual choice: When faced with alternative courses of action, what is the correct moral choice? What are the main features of ethical choice?

Basic Concepts: Responsibility, Accountability, Liability, and Due Process

Responsibility

Accepting the potential costs, duties, and obligations for the decisions one makes.

Accountability

Feature of systems and social institutions; mechanisms for assessing responsibility for decisions made and actions taken.

Liability

Body of laws that permit individuals to recover the damages done to them by other actors, systems, or organizations.

Due process

A process in which laws are known and understood and there is an ability to appeal to higher authorities to ensure that laws are applied correctly.

Ethical choices are decisions made by individuals who are responsible for the consequences of their actions. **Responsibility** is a key element of ethical action. Responsibility means that you accept the potential costs, duties, and obligations for the decisions you make. **Accountability** is a feature of systems and social institutions; it means that mechanisms are in place to determine who took action and who is responsible. Systems and institutions in which it is impossible to find out who took what action are inherently incapable of ethical analysis or ethical action. **Liability** extends the concept of responsibility further to the area of laws. Liability is a feature of political systems in which a body of laws is in place that permits individuals to recover the damages done to them by other actors, systems, or organizations. **Due process** is a related feature of law-governed societies and is a process in which laws are known and understood, and the ability exists to appeal to higher authorities to ensure that laws are applied correctly.

These basic concepts form the underpinning of an ethical analysis of information systems and those who manage them. First, information technologies are filtered through social institutions, organizations, and individuals. Systems do not have impacts by themselves. Whatever information system effects exist are products of institutional, organizational, and individual actions and behaviors. Second, responsibility for the consequences of technology falls clearly on the institutions, organizations, and individuals who choose to use the technology. Using information technology in a socially responsible manner means that you can and will be held accountable for the consequences of your actions. Third, in an ethical political society, individuals and others can recover damages done to them through a set of laws characterized by due process.

Ethical Analysis

When confronted with a situation that seems to present ethical issues, how should you analyze it? The following five-step process should help:

1. *Identify and describe the facts clearly.* Find out who did what to whom and where, when, and how. In many instances, you will be surprised at the errors in the initially reported

facts, and often you will find that simply getting the facts straight helps define the solution. It also helps to get the opposing parties involved in an ethical dilemma to agree on the facts.

2. *Define the conflict or dilemma and identify the higher-order values involved.* Ethical, social, and political issues always reference higher values. The parties to a dispute all claim to be pursuing higher values (e.g., freedom, privacy, protection of property, or the free enterprise system). Typically, an ethical issue involves a dilemma: two diametrically opposed courses of action that support worthwhile values. For example, the Opening Case Study, *Apps That Track: A Double-Edged Sword*, illustrates two competing values: organizational efficiency and profit versus respect for individual privacy.
3. *Identify the stakeholders.* Every ethical, social, and political issue has stakeholders: players in the game who have an interest in the outcome, who have invested in the situation, and who usually have vocal opinions. Find out the identity of these groups and what they want. This will be useful later when designing a solution.
4. *Identify the options that you can reasonably take.* You may find that none of the options satisfy all the interests involved, but that some options do a better job than others. Sometimes arriving at a good or ethical solution may not always be a balancing of consequences to stakeholders.
5. *Identify the potential consequences of your options.* Some options may be ethically correct but disastrous from other points of view. Other options may work in one instance but not in similar instances. Always ask yourself, "What if I choose this option consistently over time?"

Ethical Principles

Once your analysis is complete, what ethical principles or rules should you use to make a decision? What higher-order values should inform your judgment? Although you are the only one who can decide which among many ethical principles you will follow, and how you will prioritize them, it is helpful to consider some ethical principles with deep roots in many cultures that have survived throughout recorded history:

1. The **Golden Rule**: do unto others as you would have them do unto you. Putting yourself in the place of others and thinking of yourself as the object of the decision can help you think about fairness in decision making.
2. **Immanuel Kant's categorical imperative:** if an action is not right for everyone to take, it is not right for anyone. Ask yourself, "If everyone did this, could the organization, or society, survive?"
3. The **slippery slope rule:** if an action cannot be taken repeatedly, it is not right to take at all. An action may bring about a small change now that is acceptable, but if it is repeated, it would bring unacceptable changes in the long run. In the vernacular, it might be stated as "once started down a slippery path, you may not be able to stop."
4. The **utilitarian principle:** take the action that achieves the higher or greater value. This rule assumes you can prioritize values in a rank order and understand the consequences of various courses of action.
5. The **risk aversion principle:** take the action that produces the least harm or the least potential cost. Some actions have extremely high failure costs of very low probability (e.g., building a nuclear generating facility in an urban area) or extremely high failure costs of moderate probability (speeding and automobile accidents). Avoid actions that have extremely high failure costs; focus on reducing the probability of accidents occurring.
6. The **ethical no-free-lunch rule:** assume that virtually all tangible and intangible objects are owned by someone else unless there is a specific declaration otherwise. If something someone else has created is useful to you, it has value, and you should assume the creator wants compensation for this work.

Golden Rule

Ethical principle that involves putting oneself in the place of others and thinking of yourself as the object of a decision.

Immanuel Kant's categorical imperative

Ethical principle that states that if an action is not right for everyone to take it is not right for anyone.

Slippery slope rule

Ethical principle that if an action cannot be taken repeatedly, it is not right to take at all.

Utilitarian principle

Ethical principle that assumes one can put values in rank order of utility and understand the consequences of various courses of action.

Risk aversion principle

Ethical principle that one should take the action that produces the least harm or the least potential cost.

Ethical no-free-lunch rule

Assumption that virtually all tangible and intangible objects are owned by someone else, unless there is a specific declaration otherwise, and that the creator wants compensation for this work.

Actions that do not easily pass these rules deserve close attention and a great deal of caution. The appearance of unethical behavior may do as much harm to you and your company as actual unethical behavior.

Professional Codes of Conduct

When groups of people claim to be professionals, they take on special rights and obligations because of their special claims to knowledge, wisdom, and respect. Professional codes of conduct are promulgated by associations of professionals such as the American Medical Association (AMA), the American Bar Association (ABA), the Association of Information Technology Professionals (AITP), and the Association for Computing Machinery (ACM). These professional groups take responsibility for the partial regulation of their professions by determining entrance qualifications and competence. Codes of ethics are promises by professions to regulate themselves in the general interest of society. For example, avoiding harm to others, honoring property rights (including intellectual property), and respecting privacy are among the General Moral Imperatives of the ACM's Code of Ethics and Professional Conduct.

Governance Codes

One tool that organizations are using to help ensure ethical decision making is to create and implement governance codes, which establish guidelines and policies, as well as implementing technical guardrails with respect to conduct. For example, global consulting firm Accenture has adopted a Responsible AI Compliance program that rests on a set of core principles, such as "human by design" (focus on AI's impact on humans), fairness, transparency, safety, accountability, compliance, and sustainability. To implement these principles, they established an AI governance structure, conducted an AI risk assessment, created an AI compliance program, and established ongoing monitoring of AI (Accenture, 2024).

To help enterprises develop their own policies, the Responsible AI Institute, whose members include Amazon Web Services, Boston Consulting Group, ATB Financial, and Shell has recently introduced an AI policy template. The template is described as an "industry-agnostic, plug-and-play policy document" that allows organizations to develop policies that are aligned with both business needs and risks (Barker, 2024). We discuss Responsible AI further in Chapter 11.

Some Real-World Ethical Dilemmas

Information systems have created new ethical dilemmas in which one set of interests is pitted against another. For example, many companies use voice recognition software to reduce the size of their customer support staff by enabling computers to recognize a customer's responses to a series of automated questions. Many companies monitor what their employees are doing on the Internet to prevent them from wasting company resources on nonbusiness activities.

In each instance, you can find competing values at work, with groups lined up on either side of the debate. A company may argue, for example, that it has the right to use information systems to increase productivity and reduce the size of its workforce to lower costs and stay in business. Employees displaced by information systems may argue that employers have some responsibility for their welfare. Business owners might feel obligated to monitor employee email and Internet use to minimize drains on productivity. Employees might believe they should be able to use the Internet for short personal tasks. A close analysis of the facts can sometimes produce solutions that give each side "half a loaf." Try to apply some of the principles of ethical analysis described to each of these cases. What is the right thing to do?

4.3 Discuss Information Rights and Privacy Issues

In this section, we take a closer look at information rights, one of the moral dimensions of information systems first described in Figure 4.1. We identify ethical, social, and political issues and use real-world examples to illustrate the values involved, the stakeholders, and the options chosen.

Information Rights: Privacy and Freedom in the Internet Age

Information rights are the rights that individuals and organizations have with respect to the information that pertains to them. **Privacy** is a component of information rights and is the claim of individuals to be left alone, free from surveillance or interference from other individuals or organizations, including the state. Claims to privacy are also involved at the workplace. Millions of employees are subject to digital and other forms of high-tech surveillance. Information technology and systems threaten individual claims to privacy by making the invasion of privacy cheap, profitable, and effective.

Information rights
The rights that individuals and organizations have with respect to information that pertains to them.

Privacy
Claim of individuals to be left alone, free from surveillance or interference from other individuals or organizations, including the state; component of information rights.

Legal Protections for Privacy in the United States

The claim to privacy is protected in the United States Constitution in a variety of ways and also through various statutes. The same holds true for many other countries. In the United States, the claim to privacy is protected primarily by the First Amendment's guarantees of freedom of speech and association, the Fourth Amendment's protections against unreasonable search and seizure of one's personal documents or home, and the Fifth and Fourteenth Amendments' guarantee of due process.

Table 4.3 describes the major U.S. federal statutes that set forth the conditions for handling information about individuals in such areas as credit reporting, education, financial records, newspaper records, and electronic and digital communications. The Privacy Act of 1974 was one of the first, and arguably, the most important of these laws, regulating the federal government's collection, use, and disclosure of information.

Most U.S. and European privacy law is based on a set of principles called **Fair Information Practices (FIP)**, first set forth in a 1973 report by a federal government advisory committee (U.S. Department of Health, Education, and Welfare, 1973). FIP is a set of principles governing the collection and use of information about individuals. FIP principles are based on the notion of a mutuality of interest between the record holder and the individual. The individual has an interest in engaging in a transaction, and the record keeper—usually a business or government agency—requires information about the individual to support the transaction. After information

Fair Information Practices (FIP)
A set of principles originally set forth in 1973 that governs the collection and use of information about individuals and forms the basis of most U.S. and European privacy laws.

Table 4.3 Federal Privacy Laws in the United States

Privacy Laws That Apply to the U.S. Federal Government	Privacy Laws That Apply to Private Institutions
Privacy Act	Fair Credit Reporting Act
Privacy Protection Act	Family Educational Rights and Privacy Act
Electronic Communications Privacy Act	Right to Financial Privacy Act
Computer Matching and Privacy Protection Act	Cable Communications Policy Act
Computer Security Act	Video Privacy Protection Act
Federal Managers Financial Integrity Act	Telephone Consumer Protection Act (TCPA)
Driver's Privacy Protection Act	Health Insurance Portability and Accountability Act (HIPAA)
E-Government Act	Children's Online Privacy Protection Act (COPPA)
USA Freedom Act	Financial Services Modernization Act (Gramm-Leach-Bliley Act)

Table 4.4 Federal Trade Commission's Fair Information Practice Principles

Principle	Description
Notice/awareness (core principle)	Websites must disclose their information practices before collecting data. Includes identification of collector; uses of data; other recipients of data; nature of collection (active/inactive); voluntary or required status; consequences of refusal; and steps taken to protect confidentiality, integrity, and quality of the data.
Choice/consent (core principle)	A process enabling choice must be in place, allowing consumers to choose how their information will be used for purposes other than supporting the transaction, including internal use and transfer to third parties.
Access/participation	Consumers should be able to review and contest the accuracy and completeness of data collected about them in a timely, inexpensive process.
Security	Data collectors must take responsible steps to ensure that consumer information is accurate and secure from unauthorized use.
Enforcement	A mechanism must be in place to enforce FIP principles. This can involve self-regulation, legislation giving consumers legal remedies for violations, or federal statutes and regulations.

is gathered, the individual maintains an interest in the record, and the record may not be used to support other activities without the individual's consent. In 1998, the Federal Trade Commission (FTC) restated and extended the original FIP principles to provide guidelines for protecting online privacy and updated them further in 2010 to take into account new privacy-invading technology. Table 4.4 describes the FTC's FIP principles.

The FTC's FIP principles have been used throughout the years as guidelines to drive changes in privacy legislation. For instance, the Children's Online Privacy Protection Act (COPPA) requires websites and mobile apps to obtain parental permission before collecting information on children under the age of 13. The FTC has also added three principles to its framework for privacy. Firms should adopt privacy by design, building products and services that protect privacy; firms should increase the transparency of their data practices; and firms should require consumer consent and provide clear options to opt out of data-collection schemes. The FTC has extended its privacy policies and enforcement actions to a wide range of privacy issues across multiple industries, including behavioral targeting, geolocation tracking, social media, the Internet of Things (IoT), the mobile app ecosystem, and AI, among others (Federal Trade Commission, 2024).

Public opinion polls show an ongoing distrust of online marketers. Although there are many studies of privacy issues, there has been no significant federal legislation in recent years. A recent survey by the Pew Research Center found that more than 80 percent of those surveyed were concerned about their online privacy; most believe they have little to no control over what companies do with their data; and most also say they have little to no understanding about what companies do with their data (Pew Research, 2023).

Privacy protections have also been added to laws deregulating financial services and safeguarding the maintenance and transmission of health information about individuals. The Gramm-Leach-Bliley Act, which repeals earlier restrictions on affiliations among banks, securities firms, and insurance companies, includes some privacy protection for consumers of financial services. All financial institutions are required to disclose their policies and practices for protecting the privacy of nonpublic personal information and to allow customers to opt out of information-sharing arrangements with nonaffiliated third parties.

The Health Insurance Portability and Accountability Act (HIPAA) includes privacy protection for medical records. The law gives patients access to their personal medical records that healthcare providers, hospitals, and health insurers maintain and the right to authorize how protected information about themselves can be used or disclosed. Doctors, hospitals, and other healthcare providers must limit the disclosure of personal information about patients to the minimum amount necessary to achieve a given purpose.

General Data Protection Regulation (GDPR)

Updated framework for protecting personally identifiable information; unifies data privacy laws across the European Union, focusing on making businesses more transparent and expanding the privacy rights of data subjects.

The European General Data Protection Regulation

In 2018, the European Commission implemented the European Union's **General Data Protection Regulation (GDPR)**, which is arguably the most important privacy legislation since the FTC's FIP principles. It applies to all firms and organizations that collect, store, or

process personal information of EU citizens, and these protections apply worldwide regardless of where the processing takes place (European Commission, 2018).

The GDPR is an updated framework for protecting personally identifiable information (PII) and replaces an earlier Data Protection Directive. In Europe, privacy protection is historically much stronger than it is in the United States. In the United States, there is no federal agency charged with enforcing privacy laws, and there is no single privacy statute governing the use of PII by private organizations. Instead, privacy laws are piecemeal, sector by sector (e.g., medical privacy, educational privacy, and financial privacy laws). These are enforced by the FTC, through self-regulation by businesses, and by individuals, who must sue companies in court to recover damages.

In the EU, data protection laws are comprehensive, apply to all organizations, and are enforced by data protection agencies in each country to pursue complaints brought by citizens and to actively enforce privacy laws. The GDPR protects a wide variety of PII: basic identity information such as name, address, and ID numbers; web data such as location, IP address, cookie data, and RFID tags; health and genetic data; mobile phone number; driver's license and passport number; biometric and facial data; racial and ethnic data; political opinions; and sexual orientation.

The main objective of the GDPR is to strengthen the rights of citizens to their own personal information and to strengthen oversight of firms to ensure they implement these individual rights. A second thrust was to harmonize conflicting data protection standards among the EU member nations and create a single EU agency to implement and enforce the regulation. And third, to enforce these conditions worldwide for all organizations that operate in the EU, or process data pertaining to EU citizens, regardless of where the organization is located.

For individuals, the GDPR requires organizations to allow consumers to access all their personal information without charge within one month; delete personal data (the right to be forgotten); ensure data portability so consumers are not locked into a particular service; and guarantee the right to sue providers for damages or abuse of PII, including class action lawsuits.

Organizational requirements have been strengthened. Organizations must have a data protection officer that reports to senior management. Explicit consent must be obtained from users before the organization can collect data (positive opt-in), eliminating default opt-in processes. Organizations must publish the rationale for data collection and the length of time data are retained, and report any breaches or hacks within 72 hours. Organizations must also build privacy protections into all new systems (privacy by design); limit targeting and retargeting of individuals to audience-level anonymized data (rather than targeting based on intimate, personal profiles); and limit the collection of personal data to only that which is needed to support a task or a transaction, and then delete it shortly thereafter. Abuse of PII can be fined up to $20 million or 4 percent of the organization's global revenue, whichever is greater. Finally, the EU will enforce the GDPR requirements with non-EU countries like the United States using intergovernmental privacy shield agreements that ensure that EU data processed in non-EU nations meets GDPR standards. Privacy shield agreements are a more enforceable version of earlier safe harbor agreements. A **safe harbor** is a private self-regulating policy and enforcement mechanism that meets the objectives of government regulators and legislation but does not involve government regulation or enforcement.

Safe harbor

Private self-regulating policy and enforcement mechanism that meets the objectives of government regulations but does not involve government regulation or enforcement.

The GDPR is primarily aimed at Meta (owner of Facebook, Instagram, and WhatsApp), Google, and other online ad-based businesses that track and collect personal data from users in order to create comprehensive digital profiles and target the users with ads. Google and Meta dominate their markets in Europe, but at the same time, they are widely criticized for invading privacy and not protecting PII (see the Chapter Case Study, *Meta's Many Ethical Challenges*).

The GDPR and AI

While AI is not explicitly mentioned in the GDPR, the broad definition of "processing" under the GDPR makes it clear that the GDPR does apply to AI systems to the extent such systems include personal data at some point in their lifecycle of either development or operation. The GDPR also includes provisions with respect to automated decision-making, which are also likely to apply, given that AI systems are frequently used to make automated decisions that impact individuals. Finally, as you will learn later in the chapter, the European Parliament has also recently adopted a comprehensive act (the EU AI Act) regulating artificial intelligence.

The relationship between AI and data protection is expressly recognized in the text of the EU AI Act, which states that it is without prejudice to the GDPR (meaning that it does not override any protections the GDPR may also offer).

Challenges Posed by the Internet

The Internet poses numerous challenges for the protection of individual privacy. Companies can track a user's online search history, the websites and web pages visited, the online content accessed, and the items inspected or purchased online. Mobile apps can also track users (see the Opening Case Study, *Apps that Track: A Double-Edged Sword*). This monitoring and tracking occurs in the background without the user's knowledge. It is conducted not just by individual websites (known as first-party tracking) but by advertising networks such as Microsoft Advertising, Google's Marketing Platform, and Meta's Audience Network that are capable of tracking personal browsing behavior across thousands of websites (known as third-party tracking). In the past, website publishers and the online advertising industry have defended tracking of individuals across the web because doing so allows more relevant ads to be targeted to users, and this helps pay for the cost of providing online content. The commercial demand for this personal information is virtually insatiable. However, these practices also impinge on individual privacy.

Cookies

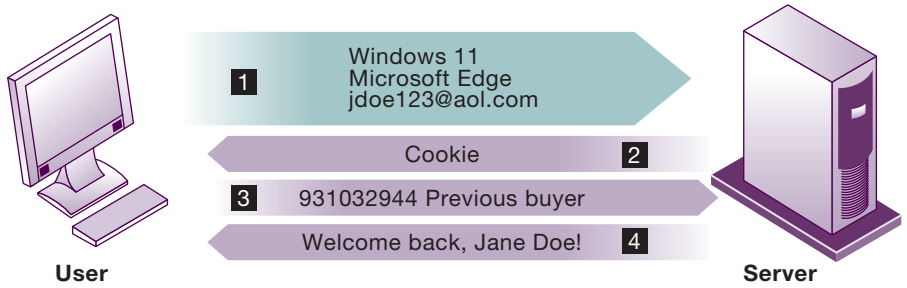
Small text file placed on a user's computer or mobile device when the user visits a website; used to identify the user and track visits to the website.

Cookies are one method used to monitor and track online users. Examine Figure 4.3, which illustrates how cookies work. When a user visits a website, the website's web server places a small text file (a "cookie") on the user's computer or mobile device. Cookies identify the visitor's web browser software, as well as other information, and track visits to the website. When the visitor returns to a site that has stored a cookie, the website software searches the visitor's computer or mobile device, finds the cookie, and knows what that person has done in the past. It may also update the cookie, depending on the activity during the visit. In this way, the site can customize its content for each visitor's interests. For example, if you purchase a book on Amazon.com and return later from the same browser, the site will welcome you by name and recommend other books of interest based on your past purchases. Cookies also make "quick checkout" options possible by allowing a site to keep track of users as they add items to a shopping cart. This type of cookie is known as a first-party cookie. First-party cookies are typically considered to be "good" cookies because they help enhance the user experience.

Websites using cookie technology cannot directly obtain visitors' names and addresses. However, if a person has registered at a site, that information can be combined with cookie data to identify the visitor. Website owners can also combine the data they have gathered from cookies and other website monitoring tools with personal data from other sources, such as

Figure 4.3 How First-Party Cookies Identify Web Visitors

Cookies are placed by a website on a visitor's computer. When the visitor returns to that website, the web server requests the ID number from the cookie and uses it to access the data stored by that server on that visitor. The website can then use these data to display personalized information.



1. The web server reads the user's web browser and determines the operating system, browser name, version number, Internet address, and other information.
2. The server transmits a tiny text file with user identification information called a cookie, which the user's browser receives and stores on the user's computer.
3. When the user returns to the website, the server requests the contents of any cookie it deposited previously in the user's computer.
4. The web server reads the cookie, identifies the visitor, and calls up data on the user.

offline data collected from surveys or paper catalog purchases, to develop very detailed profiles of their visitors. First-party cookies can be blocked and/or deleted by users, although the majority of users do not do so.

Third-party cookies operate similarly to first-party cookies but enable advertising platforms to track user behavior across websites and devices. In the past, third-party cookies were supported by all the major web browsers, but that has begun to change. Apple's Safari and Mozilla's Firefox browsers now block third-party cookies by default. Google, which had previously stated that it would also follow suit, delayed implementation several times and recently announced that it had scrapped that plan, and would retain third-party cookies after all. Instead, it said it would introduce new methods that would enable users to choose how they wanted to be tracked while using Google search products. We discuss other technology-based efforts to enhance privacy in the following section on technological solutions.

Web beacon

Tiny image invisibly embedded in email messages and web pages that is designed to monitor the behavior of the user visiting a website or sending email; also called a web bug or tracking file.

In addition to cookies, there are other tools for the surveillance of Internet users. A **web beacon**, also called a *web bug* (or simply tracking file), is a tiny image that keeps a record of users' online clickstreams. It reports data back to whoever owns the tracking file, which can be invisibly embedded in an email message or web page to monitor the behavior of the user visiting the website or receiving the email. Web beacons are placed on popular websites by third-party firms who pay the websites a fee for access to their audience. So how common is web tracking? A recent analysis by *Wired* of the top 10,000 most popular websites found that thousands of websites are sharing data with hundreds of different advertising firms, with dozens data sharing with more than 1,000 companies. For instance, publisher Dotdash Meredith, which operates popular websites such as Investopedia.com, Allrecipes.com, and People.com, share data with more than 1,600 firms. The biggest trackers were Google, Microsoft, and Meta, all of whom are in the business of selling ads to advertising firms and marketers (Burgess, 2024).

Adware can secretly install itself on an Internet user's computer by piggybacking on larger applications. Once installed, adware calls out to websites to send ads and other unsolicited material to the user. A more malicious version of adware, known as spyware, can also track the user's browsing habits. More information is available about intrusive software in Chapter 8.

More than 90 percent of global Internet users use Google Search and other Google services, making Google the world's largest collector of online user data. Whatever Google does with its data has an enormous impact on online privacy. Most experts believe that Google possesses the largest collection of personal information in the world—more data on more people than any government agency. Google uses behavioral targeting (also known as interest-based advertising) to target individuals as they move from one site to another to show them relevant ads based on their search activities and other data that Google has collected about them. For instance, one of its programs enables advertisers to target ads based on the search histories of Google users, along with any other information the user submits to Google such as age, demographics, region, and web activities (such as blogging). Google's AdSense program enables Google to help advertisers select keywords and design ads for various market segments based on search histories such as helping a clothing website create and test ads targeted at teenagers. Google displays targeted ads on YouTube and Google mobile applications, and its Google Marketing Platform ad network serves targeted ads. We further discuss behavioral targeting in Chapter 10.

Informed consent

Consent given with knowledge of all the facts needed to make a rational decision.

Opt-out

Model of informed consent permitting the collection of personal information until the consumer specifically requests that the data not be collected.

Opt-in

Model of informed consent permitting prohibiting an organization from collecting any personal information unless the individual specifically takes action to approve information collection and use.

In the United States, businesses are allowed to gather transaction information generated in the marketplace and then use that information for other marketing purposes without obtaining the **informed consent** (consent given with knowledge of all the facts needed to make a rational decision) of the individual whose information is being used. Firms that do so argue that when users agree to the sites' terms of service, they are also agreeing to allow the site to collect information about their online activities. An **opt-out** model of informed consent permits the collection of personal information until the consumer specifically requests the data not to be collected. Privacy advocates would like to see wider use of an **opt-in** model of informed consent in which a business is prohibited from collecting any personal information unless the consumer specifically takes action to approve information collection and use. Here, the default option is no collection of user information. Many websites now employ "cookie banners"—a pop-up window that allows users to accept or reject cookies from the site. Although intended as a form of opt-in informed consent, most privacy advocates feel that cookie banners are ineffective, as few people actually read the disclosures about what data are being collected before clicking the Accept button.

The online industry has preferred self-regulation to privacy legislation for protecting consumers. Members of the online advertising industry have created an industry association called the Network Advertising Initiative (NAI) to develop privacy policies to help consumers opt out of advertising network programs and to provide consumers redress from abuses. Individual firms such as Apple, Google, and Microsoft have also launched privacy initiatives in an effort to address public concern about tracking people online.

In general, most businesses do little to protect the privacy of their customers, and consumers do not do as much as they should to protect themselves. For websites and apps that depend on advertising for support, most revenue derives from selling access to customer information. The vast majority of people claim to be concerned about online privacy, but most do not read the privacy statements on websites. In addition, website privacy policies are often ambiguous about key terms and too complicated for the average consumer to understand (Laudon and Traver, 2024). What firms call a privacy policy is in fact a data use policy. The concept of privacy is associated with consumer rights, which many firms do not wish to recognize. A data use policy simply tells customers how the information will be used and does not mention rights.

Challenges Posed by AI

AI is the latest information technology that poses a threat to privacy. Many of the issues already noted above are amplified by the use of AI. Google, OpenAI, and Microsoft have also been accused of training their generative AI tools on the personal data of millions of people that was scraped from the Internet without the users' knowledge or consent.

The development of more specifically focused AI tools on a firm-wide level also raises concerns. For instance, AI applications developed by firms in the health and financial services industries may require sensitive data for training purposes. AI applications developed for use in online advertising can create very detailed profiles of individuals based on their online behavior, which also raises privacy concerns.

The collection of data for use by facial recognition systems (see Chapter 11) has also been the subject of much concern. Massive databases of people's faces collected from social networks, photo websites, dating services, and cameras placed in public spaces have been compiled by a number of companies as well as by academic researchers. Georgetown University has estimated that photos of nearly half of all U.S. adults have been entered into at least one facial recognition database. Both Meta and Google have faced lawsuits charging that their actions in amassing facial databases violate the Illinois Biometric Information Privacy Act, one of a growing number of state laws that address this issue, and which require firms to obtain user consent before collecting biometric and facial information. In another recent case, Prisma Labs, the developer of the image-generating app Lensa, has also been accused of collecting and storing the facial information of users without their consent in violation of Illinois' law.

Technological Solutions

In addition to legislation, a number of technological solutions have been developed to help protect user privacy (sometimes referred to as privacy-enhancing techniques (PETs)). Table 4.5 lists some of the most common tools that are available. For the most part, technical solutions thus far have failed to provide effective protection for online privacy, in part because most of them require users to be proactive in implementing them.

4.4 Discuss Intellectual Property Issues

Intellectual property

Products of the mind created by individuals or corporations and which may be protected by copyright, patent, trademark and trade secret law.

Today's information technologies and systems have severely challenged existing laws and social practices that protect intellectual property. **Intellectual property** refers to products of the mind created by individuals or corporations. Information technology has made it difficult to protect intellectual property because digital information can be so easily copied or distributed. Intellectual property is subject to a variety of protections under four legal traditions: copyright, patents, trademarks, and trade secrets.

Table 4.5 Technological Protections for Online Privacy

Technology	Protection
Apple App Tracking Transparency (ATT)	Requires any app that wants to track user activity and share it with other apps or websites to ask user for permission
Apple Intelligent Tracking Prevention (ITP) for Safari web browser	Monitors and disables cross-site tracking cookies and blocks trackers' ability to identify users by IP address
Google Privacy Sandbox	Originally developed by Google to replace third-party cookies; now being focused on technologies that will allow users to choose how they want to be tracked while using Google search products
Differential privacy software	Reduces the ability to merge different files and de-anonymize consumer data. In AI, this means adding "statistical noise" to the raw data to protect individual privacy.
Privacy default browsers	Eliminates tracking cookies and prevents IP tracking
Message encryption apps	Encrypts text and other data transferred using smartphones
Spyware blockers	Detects and removes spyware, adware, keyloggers, and other malware
Pop-up and ad blockers	Prevents calls to ad servers; restricts downloading of images at user request
Secure email	Email and document encryption
Anonymous remailers	Enhanced privacy protection for email
Anonymous surfing	Enhanced privacy protection for web browsing
Cookie managers	Blocks third-party cookies
Public key encryption	Enables encryption of email and documents

Copyright

Copyright
Protects creators of certain types of intellectual property against copying by others for a certain period of time, depending on several factors.

Copyright protects creators of intellectual property from having their work copied by others for a certain period of time, depending on several factors. As a general rule, copyright protection lasts for the life of the author, plus an additional 70 years after the author's death. For corporate-owned works, copyright protection generally lasts for 95 years after initial creation. Copyright protects literary, dramatic, musical, and artistic works, such as novels, poetry, songs, plays, maps, drawings, and artwork. The intent behind copyright laws has been to encourage creativity and authorship by ensuring that creative people receive the financial and other benefits of their work. Most industrial nations have their own copyright laws, and there are several international conventions and bilateral agreements through which nations coordinate and enforce their copyright laws.

In the mid-1960s, the Copyright Office began registering software programs, and in 1980, Congress passed the Computer Software Copyright Act, which provides protection for software program code and copies of the original code sold in commerce; it sets forth the rights of the purchaser to use the software, while the creator retains legal title.

Copyright protects against copying entire programs or their parts. Damages and relief are readily obtained for infringement. The drawback to copyright protection is that the underlying ideas behind a work are not protected; only their manifestation in a work is protected. A competitor can use your software, understand how it works, and build new software that follows the same concepts without infringing on a copyright.

Look-and-feel copyright infringement lawsuits are precisely about the distinction between an idea and its expression. For instance, in the early 1990s, Apple sued Microsoft and Hewlett-Packard, claiming that the defendants copied the expression of overlapping windows in Apple's Macintosh interface. The defendants countered that the idea of overlapping windows can be expressed only in a single way and, therefore, was not protectable. When ideas and their expression merge, the expression cannot be copyrighted. Ultimately, the courts agreed with the defendants.

In general, courts follow the reasoning of a 1989 case—*Brown Bag Software v. Symantec Corp*—in which the court dissected the elements of software alleged to be infringing. The court found that similar concepts, functions, general functional features (e.g., drop-down menus), and colors are not protectable by copyright law (*Brown Bag Software v. Symantec Corp.*, 1992).

Patents

Patent
Grants the owner an exclusive monopoly on the ideas behind an invention for a certain period of time, typically 20 years; designed to ensure that inventors of new machines or methods are rewarded for their labor while enabling the widespread use of their inventions.

A **patent** grants the owner an exclusive monopoly on the ideas behind an invention for a certain period of time, typically 20 years. The congressional intent behind patent law was to ensure that inventors of new machines, devices, or methods receive the full financial and other rewards of their labor, and yet to make widespread use of the invention possible by providing detailed diagrams for those wishing to use the idea under license from the patent's owner. The granting of a patent is determined by the United States Patent and Trademark Office and relies on court rulings.

The key concepts in patent law are originality, novelty, and invention. The Patent Office did not accept applications for software patents routinely until a 1981 Supreme Court decision held that computer programs could be part of a patentable process.

The strength of patent protection is that it grants a monopoly on the underlying concepts and ideas of software. The difficulty is passing stringent criteria of nonobviousness (e.g., the work must reflect some special understanding and contribution), originality, and novelty as well as years of waiting to receive protection.

Trademarks

Trademarks
Marks, symbols, or images used to distinguish products in the marketplace.

Trademarks are the marks, symbols, and images used to distinguish products in the marketplace. Trademark laws protect consumers by ensuring that they receive what they paid for. These laws also protect the investments that firms have made to bring products to market. Typical trademark infringement violations occur when one firm appropriates or pirates the marks of a competing firm. Infringement also occurs when firms dilute the value of another firm's marks by weakening the connection between a mark and the product. For instance, if a search engine firm copies the trademarked Google icon, colors, and images, it would be infringing on Google's trademarks. It would also be diluting the connection between the Google search service and its trademarks, potentially creating confusion in the marketplace.

Trade Secrets

Trade secret
Any intellectual work product used for a business purpose, provided it is not based on information in the public domain.

Any intellectual work product—a formula, device, pattern, or compilation of data—used for a business purpose can be classified as a **trade secret**, provided that it is not based on information in the public domain. Protections for trade secrets vary from state to state. In general, trade secret laws grant a monopoly on the ideas behind a work product, but the monopoly can be very tenuous.

Software that contains novel or unique elements, procedures, or compilations can be considered a trade secret. Trade secret law protects the actual ideas in a work product, not only their manifestation. To make a claim of trade secret infringement, the creator or owner must take care to bind employees and customers with nondisclosure agreements and prevent the secret from falling into the public domain.

The limitation of trade secret protection is that although virtually all software programs of any complexity contain unique elements of some sort, it is difficult to prevent the ideas in the work from falling into the public domain when the software is widely distributed.

Challenges to Intellectual Property Rights Posed by Today's Information Technologies and Systems

Today's information technologies and systems pose severe challenges to existing methods of protecting intellectual property and, therefore, create significant ethical, social, and political issues. Digital media differ from books, periodicals, and other media in terms of the former's ease of replication; ease of transmission; ease of alteration; ease of theft because of compactness; and difficulties in establishing uniqueness.

Challenges Posed by the Internet

The proliferation of digital networks, including the Internet, has made it even more difficult to protect intellectual property. Before the widespread use of digital networks, copies of software, books, magazine articles, or films had to be stored on physical media such as paper, computer

disks, or videotape, creating some hurdles to distribution. Using digital networks, information can be more widely reproduced and distributed. Software and media piracy remains an ongoing issue, with the number of visits to piracy websites across all industries, from movies, television, and music, to publishing and software, increasing to 230 billion worldwide in 2023, an increase of 6.7 percent from 2022 (MUSO, 2024).

The Internet was designed to transmit information, including copyrighted information, freely around the world. You can easily copy and distribute virtually anything to millions of people worldwide, even if they are using different types of computer systems. Information can also be illicitly copied from one place and distributed throughout other systems and networks, even though these parties do not willingly participate in the infringement.

Individuals have been illegally copying and distributing digitized music files on the Internet for several decades. File-sharing services such as Napster and, later, Grokster, Kazaa, Morpheus, Megaupload, and The Pirate Bay sprang up in the early years of the web to enable users to locate and swap digital music and video files, including those protected by copyright. Illegal file sharing became so widespread that it threatened the viability of the music recording industry and, at one point, consumed 20 percent of Internet bandwidth. The recording industry won several legal battles for shutting these services down, but it has not been able to halt illegal file sharing entirely. The motion picture and cable television industries continue to wage similar battles. Several European nations have worked with U.S. authorities to shut down illegal sharing sites, with mixed results.

However, as legitimate online music stores such as iTunes and streaming services such as Spotify, Apple Music, Amazon Prime Music, and Pandora have expanded, illegal file sharing has significantly declined. The Apple iTunes Store legitimized paying for music and entertainment and created a closed environment from which music and videos could not be easily copied and widely distributed. Amazon's Kindle also protects the rights of publishers and writers because its books cannot be copied to the Internet and distributed. Music and video streaming services also inhibit piracy because the streams cannot be easily recorded on separate devices. Since 2015, music industry revenues have more than doubled after a precipitous drop of more than 50 percent between 1999 and 2014.

The **Digital Millennium Copyright Act (DMCA)** also provides some copyright protection. The DMCA implements a World Intellectual Property Organization Treaty that makes it illegal to circumvent technology-based protections of copyrighted materials. Internet service providers (ISPs) are required to take down sites of copyright infringers they are hosting when the ISPs are notified of the infringement. Microsoft and other major software and information content firms are represented by the Software and Information Industry Association (SIIA), which lobbies for new laws and enforcement of existing laws to protect intellectual property around the world. The SIIA runs an antipiracy hotline for individuals to report piracy activities, offers educational programs to help organizations combat software piracy, and has published guidelines for employee use of software.

Digital Millennium Copyright Act (DMCA)

Makes it illegal to make, distribute, or use devices that circumvent technology-based protections of copyrighted materials.

Challenges Posed by AI

AI raises numerous issues with respect to the protection of various forms of intellectual property. We provide an overview here and also discuss these issues further in Chapter 11.

Copyright

One of the primary issues that has arisen thus far has to do with the use, without permission, by companies such as OpenAI, Google, Microsoft, and Meta of copyrighted works to train their AI models. For instance, it has been alleged that upwards of 170,000 copyrighted works were used to train the Meta LLAMA large language model. In the first major test with respect to this issue, the New York Times has filed a high-profile lawsuit against OpenAI, the developer of ChatGPT, and its partner Microsoft, charging that OpenAI scraped millions of the New York Times articles, along with those of other copyright owners, from the web without permission. OpenAI and Microsoft argue that its action constituted "fair use," a doctrine that allows the use of copyrighted material without first obtaining permission under specified circumstances. Other similar lawsuits have also been filed. For example, Getty Images, a stock photo company, has sued Stability AI, alleging that Stability AI infringed on its copyrights by using Getty's images to train its AI models.

Copyright issues also arise with respect to output generated by AI. If AI produces content that "looks like" or is in the style of a creator, such as a writer, artist, or singer, does that infringement that party's copyright? There are also instances when AI generates content that is a verbatim copy. For example, in its lawsuit against OpenAI and Microsoft, the New York Times claims that ChatGPT sometimes generates verbatim portions of New York Times articles in response to user queries. OpenAI, Microsoft, and its subsidiary GitHub are currently being sued on the grounds that GitHub Copilot, a code-generating AI application, was replicating long sections of licensed open-source code without providing credit, as required by the licenses under which they were originally published.

There is also an issue as to whether works generated by AI are eligible for copyright protection under U.S. law. In March 2023, the U.S. Copyright Office issued a policy statement declaring that works created solely by AI engines were not eligible for copyright protection. Only works that include human-authored elements are considered copyrightable. In August 2023, in the first court ruling on this issue, the court affirmed this policy, stating that human authorship is a bedrock requirement of copyright.

Patents

Questions have also arisen with respect to the use of AI in the creation of inventions. In February 2024, the USPTO issued guidance that makes clear that while an AI system cannot be listed as an inventor on a patent application, that it would consider issuing patents for inventions that were created with the assistance of an AI system, so long as the human inventor significantly contributed to the invention. The guidance includes instructions on how to determine whether the human contribution is sufficient to qualify (USPTO, 2024).

Trademarks

As with copyright, the use of generative AI raises trademark concerns, particularly in terms of the output generated. Given that AI algorithms are trained using preexisting materials, there is an increased likelihood that an AI algorithm may generate logos or designs that closely resemble existing trademarks, which if actually used in commerce, could potentially lead to consumer confusion or dilution of established brands. AI chatbots or virtual assistants might also unknowingly infringe on trademarks when responding to customer inquiries or providing product recommendations. Whether AI-generated images can be trademarked remains an open question. Thus far, the USPTO has not issued any guidance indicating that AI-generated trademarks are subject to different rules than traditionally developed trademarks.

Trade Secrets

AI poses a potential threat to the protection of trade secrets. According to research firm Cyberhaven Labs, although many companies have prohibited the use of unlicensed, non-corporate AI chatbots such as ChatGPT, Google Gemini, and Microsoft Copilot for work-related tasks, many employees are using them and sharing company legal documents, source code, and employee information in the process, potentially leading to invalidation of trade secret protection. Most users do not understand what happens to their companies' data once they share it with an unlicensed AI. So far, there have not been any high-profile reports about major company trade secrets revealed by large public AIs, but it is quite possible that this may happen in the future.

Trade secret laws are also being used by companies and public authorities to avoid having to disclose how their AI engines work, leading to what is referred to as a "black box" issue. In a world where AI increasingly is playing a pivotal role, how AI actually makes those decisions is not apparent, in part because those details constitute proprietary trade secrets. We discuss this issue further in Section 4.6.

4.5 Discuss System Quality Issues

In this and the following two sections, we take a closer look at the issues that information technologies and systems raise with respect to system quality, accountability and control, and the quality of everyday life, which, along with information rights and obligations and property rights and obligations, are part of the moral dimensions of information systems first described in Figure 4.1.

Given the centrality of information systems to the functioning of businesses and society, what is an acceptable level of system quality? If a person is injured by a machine controlled, in part, by software, who should be held accountable and, therefore, held liable? Should a social network be held accountable and liable for the posting of offensive material? What kinds of acts should be considered computer crimes? How should individuals, society, and governments deal with the potentially negative impacts of new information technologies on the quality of everyday life?

System Quality: Data Quality and System Errors

In 2023, many of the largest cloud providers experienced significant outages, which took out sites and services throughout the United States and Europe. For example, in June 2023, an Amazon Web Services (AWS) outage impacted the websites of the New York Metropolitan Transportation Authority, the Boston Globe, Associated Press, the Securities and Exchange Commission's EDGAR system, and Southwest Airlines, among others. The root cause of the issue appeared to be with a subsystem responsible for capacity management for AWS's Lambda service. In April 2023, Microsoft users faced a nearly six-hour-long issue with Microsoft 365 online applications, including Teams, SharePoint Online, and Outlook. And in February 2023, Oracle's global cloud infrastructure (OCI) suffered a multi-day outage affecting customers in numerous regions around the world arising from a performance issue in the back-end infrastructure that supports the OCI public DNS API.

Despite these high-profile incidents, outages at cloud computing services are infrequent. But as more and more firms rely on cloud providers and centralize their data and operations with a small group of cloud providers, these outages have called into question the reliability and quality of cloud services. Are these outages acceptable?

The debate raises a related but independent moral dimension: What is an acceptable, technologically feasible level of system quality? At what point should system managers say, "Stop testing, we've done all we can to perfect this software. Ship it!" Individuals and organizations may be held responsible for avoidable and foreseeable consequences, which they have a duty to recognize and correct. The gray area is that some system errors are foreseeable and correctable only at very great expense, an expense so great that pursuing this level of perfection is not feasible economically—no one could afford the product!

For example, although software companies try to debug their products before releasing them to the marketplace, they knowingly ship buggy products because the time and cost of fixing all the minor errors would prevent these products from ever being released. What if the product was not offered in the marketplace? Would social welfare as a whole falter and perhaps even decline? Carrying this further, just what is the responsibility of a producer of computer services—should the producer withdraw the product that can never be perfect, warn the user, or forget about the risk (let the buyer beware)?

Three principal sources of poor system performance are (1) software bugs and errors, (2) hardware or facility failures caused by natural or other causes, and (3) poor input data quality. Zero defects in software code of any complexity cannot be achieved, and the seriousness of the remaining bugs cannot be estimated (Choi, 2020). Hence, there is a technological barrier to perfect software, and users must be aware of the potential for catastrophic failure. The software industry has not yet arrived at testing standards that would result in software of acceptable but imperfect performance.

Although software bugs and facility catastrophes are likely to be widely reported in the press, by far the most common source of business system issues is poor data quality (see Chapter 6). Few companies routinely measure the quality of their data, but individual organizations report data error rates ranging from 0.5 to 30 percent.

4.6 Discuss Accountability and Control Issues

The use of information technology and systems may sometimes result in harm, creating questions as to who, if anyone, should be held accountable for that harm. What laws exist to control the misuse of information technology and systems?

Software and Internet Liability Issues

Poorly designed software can sometimes result in injury or economic loss. For instance, it has been reported that up to 60 percent of data breaches are caused by unpatched software vulnerabilities. Who, if anyone, is liable if a business or person is harmed by software that is defective? Should the company that developed the software be held legally liable for any harm that results from using the software? What about the individual software developers? The main concern is: What level of defects, if any, should be considered ordinary and customary and, therefore, acceptable?

Liability for software-related problems is a complicated issue from a legal standpoint. If someone suffers a personal injury or property damage because of a defective physical product, the seller, distributor, and/or manufacturer of that product can normally be held liable under the concept of strict product liability, whether any of those parties were negligent or not. Whether software is a "product" that is subject to strict product liability remains unsettled. Courts that have grappled with the question have generally found that software is not a product. This finding is important because it is typically very difficult to prove negligence, which is an alternative legal standard that might provide a basis for a lawsuit. However, when software is an essential part of a machine, such as a car, plane, or robot, it is more likely to be considered a product and potentially be subject to strict product liability laws or other types of liability. For example, Boeing has admitted to full responsibility for the 2019 crash of its 737 Max in Ethiopia because of a faulty software system.

Software vendors and software service providers typically use contracts and end-user license agreements to limit any potential liability they may have for defective software. Unfortunately, very few users actually read such agreements before using the software and, even when they do read them, most users do not understand that the agreements effectively exempt software vendors from almost all liability. Courts have typically upheld such agreements, severely impacting the ability of software users to recover for any damages they may have suffered.

The Internet has introduced additional questions surrounding the issue of liability. Should a website or social network such as Facebook or X (formerly Twitter) be held liable and accountable for the content a user posts when that content contains false, offensive, or potentially harmful material, or should it be held harmless from any liability with respect to that content? The Spotlight on Organizations Case Study, *Section 230 Under Fire* further examines this issue.

Spotlight on Organizations Case Study

Section 230 Under Fire

In 1996, the Internet and web as we know it today did not yet really exist. There was no Facebook, Twitter, Google, YouTube, or Wikipedia. Only about 20 million U.S. adults had access to the Internet, and they spent fewer than 30 minutes a month online. AOL was the world's largest Internet service provider, along with Prodigy and CompuServe. But there were a few features that would be recognizable today, including bulletin boards (forums) and blogs (although they were not yet called that). The U.S. Congress at that time was very concerned about online pornography and in that year, passed the Communications Decency Act, a law designed primarily to regulate obscenity and indecency on the Internet. Most of the law was quickly found to be unconstitutional by the Supreme Court, but one provision survived: Section 230. Congress had no inkling at the time that Section 230 would later come to be viewed as making the Internet of today possible.

Section 230 has two major provisions: Section 230(c)(1) and Section 230(c)(2). Section 230(c)(1) provides that online companies that enable the posting of user content should not be treated as a publisher or speaker of that content and, therefore, should not be held legally responsible (liable) for that content. This section of the law likens such services to the telephone system or bookstores, which are not responsible for unlawful content that they may play a part in distributing. Congress believed that such a provision would give needed support for what was, at the time, a nascent industry. Without this provision, early Internet companies might have had to spend too much time and resources moderating content to avoid being subjected to potential liability.

Section 230(c)(2) provides that online companies shall not be held liable if they choose, in good faith, to remove user content they deem offensive. This provision resulted from a lawsuit involving user content on a Prodigy bulletin board that a company alleged was defamatory. A New York state court found Prodigy liable because Prodigy exercised editorial control over bulletin board content, at times removing content it believed objectionable. The court case presented Internet firms with a difficult choice—either moderate content (but be subject to the risk of potential legal liability) or don't moderate any content (but run the risk of objectionable content that might harm their business interests). Section 230(c)(2) aimed to remove that concern.

In the years that followed the passage of Section 230, courts interpreted it as creating broad immunity with respect to any claims based on user content, even if the platform knew the content was unlawful. Doing so enabled business models based on user content, such as social media, to flourish. The law has been widely characterized as laying the foundation for the Internet we know today.

However, Section 230 has become a lightning rod in recent years, particularly in relation to social media platforms such as Facebook and X (formerly Twitter) that host a flood of divisive content, including what many believe to be harmful disinformation. In the current heightened political climate, Section 230 has been attacked from both sides. Many Republicans have called for its repeal or reform because they believe that Big Tech companies are biased against conservative viewpoints and that, enabled by Section 230(c)(2), they act to censor certain voices. These critics want to prevent Big Tech platforms from being able to unilaterally decide what content they will remove, label, or restrict. Many Democrats also want reform, but for a different reason. They contend that Section 230(c)(1), by granting Internet platforms legal immunity with respect to user content, reduces their incentive to proactively remove content that causes social harm, especially when that content can be economically valuable to them. Seemingly no one is happy with Section 230 anymore, except the technology companies it was originally designed to protect. Most of these companies want Section 230 to remain as it has always been, with some companies viewing any kind of reform effort as an existential threat to the Internet.

Some Section 230 critics focus on the amplification caused by algorithms and suggest removing liability protection when social media algorithms promote harmful content. However, it is unclear whether it would be possible to design an algorithm that amplifies only "good" content, nor is it clear as to how "good" versus "bad" content should be determined. Others have suggested that a platform's safe harbor protections be tied to the use of reasonable content moderation policies, although this raises First Amendment issues. Some lawmakers have proposed eliminating Section 230 protections for "market-dominant" Internet platforms or have said that eligibility for Section 230 safe harbors should be based on political neutrality. Still others warn that any Section 230 reform itself may have unintended and harmful side effects, pointing to a previous amendment aimed at fighting online sex trafficking that reportedly actually made prosecuting such crimes more difficult. The Internet Society warns that Section 230 impacts not just platforms such as Facebook and Google but also underlying infrastructure providers, such as cloud providers, that move content and data from one place to another.

To date, the U.S. Supreme Court has sidestepped making any rulings on the scope of Section 230, leaving the issue thus far to legislators to address. In May 2024, a new bipartisan effort was launched by lawmakers in the U.S. House of Representatives to "sunset" Section 230. Energy and Commerce Committee Chair Cathy Rodgers, a Republican, and Ranking Member Frank Pallone, a Democrat, have proposed giving Congress 18 months to create a new regulatory framework to replace Section 230. In addition to all of the existing issues with Section 230, lawmakers are particularly interested in how Section 230 might apply to content created by generative AI. However, the effort remains very contentious, and given the polarized views about Section 230, it's unclear if enough overall support for the proposal can be developed anytime soon. If and when that time comes, just as the initial passage of Section 230 helped create the Internet of today, its repeal or amendment will shape the Internet of the future.

SOURCES: Melissa Quinn, "Supreme Court Declines to Review Scope of Section 230 Liability Shield for Internet Companies," Cbsnews.com, July 2, 2024; Lauren Feiner, "Lawmakers Debate Ending Section 230 in Order to Save It," Theverge.com, May 22, 2024; Marty Swant, "WTF Is Section 230," Digiday.com, March 14, 2024; Congressional Research Service, "Section 230: A Brief Overview," Crsreports.congress.gov, February 2, 2024; Julie Pattison-Gordon, "Can Section 230 Reform Advocates Learn from Past Mistakes," Govtech.com, March 15, 2022; Marguerite Reardon, "Section 230: How It Shields Facebook and Why Congress Wants Changes," Cnet.com, October 6, 2021; Michael D. Smith and Marshall Van Alstyne, "It's Time to Update Section 230," *Harvard Business Review*, August 12, 2021; Adi Robertson, "What Will Changing Section 230 Mean for the Internet," Theverge.com, February 26, 2021; Stephen Engleberg, "Twenty-Six Words Created the Internet. What Will It Take to Save It," Propublica.com, February 9, 2021; Farhad Manjoo, "Jurassic Web," Slate.com, February 24, 2009.

Case Study Questions

1. What are the ethical, social, and political issues that Internet platforms face with respect to user content?
2. Do you think Section 230 should be repealed or amended? Why or why not?
3. In what ways does the controversy over Section 230 illustrate the difficulties of creating laws governing the use of new digital technologies?

AI Liability Issues

AI raises a number of issues with respect to the question of liability for harm. Most AI algorithms operate as a "black box," with a lack of transparency into how they actually work and arrive at decisions. It makes it difficult to hold anyone accountable for incorrect output or decisions. This is a particular concern in areas such as healthcare, where AI tools are increasingly performing a wide range of health-related tasks (Duffourc and Gerke, 2024).

EU AI Act

World's first broad-based law governing the use of AI.

In Europe, the European Parliament formally adopted the EU Artificial Intelligence Act (the EU AI Act) in March 2024. The **EU AI Act** is the world's first broad-based law governing the use of AI. At the same time, the European Parliament also formally adopted a revamped Product Liability Directive. The revisions to the EU Product Liability Directive make clear that the provider of an AI system is considered a manufacturer under the Directive, and thus can be held primarily liable for harm caused by AI systems. The EU Directive also simplifies the burden of proof on consumers in relation to complex or "black box" AI, i.e., where the consumer might have difficulties proving the product is defective because of the "complexity" of the product.

Computer Crime and Abuse

New technologies, including computers, create new opportunities for committing crime by creating new, valuable items to steal, new ways to steal them, and new ways to harm others. **Computer crime** can be generally defined as the commission of illegal acts through the use of a computer or against a computer system. Simply accessing a computer system without authorization or with intent to do harm, even accidentally, is a federal crime. The most frequent types of incidents comprise a greatest hits list of cybercrime: malware, phishing, network interruption, spyware, and denial-of-service attacks. The true cost of all computer crime is unknown, but it is estimated to be in the billions of dollars. You can find a more detailed discussion of computer crime in Chapter 8.

Computer crime

The commission of illegal acts through the use of a computer or against a computer system.

Computer abuse

The commission of acts involving a computer that may not be illegal but are considered unethical.

Spam

Unsolicited commercial email.

Computer abuse is the commission of acts involving a computer that may not be illegal but are considered unethical. The popularity of the Internet, email, and mobile devices has turned one form of computer abuse—spamming—into a serious problem for both individuals and businesses. Originally, **spam** was junk email that an organization or individual sent to a mass audience who had not expressed any interest in the product or service being marketed. Today, spam typically tries to entice users to select malicious links or may market fraudulent deals and services, outright scams, pornography, illegal or counterfeit drugs, or other objectionable products. Most spam originates from bot networks, which consist of thousands of captured devices that can initiate and relay spam messages.

Spam costs for businesses are high because of the computing and network resources and the time required to deal with a large volume of unwanted email messages, as well as the potential that they may contain malicious links. In 2023, the percentage of all email that is spam worldwide was estimated to be around 45 percent (Petrosyan, 2024).

Identity and financial-theft cybercriminals are also targeting mobile devices with spam. Mobile phone spam usually comes in the form of SMS text messages, but increasingly, users are receiving spam in their Facebook, Instagram, Messenger, and WhatsApp apps as well.

ISPs and individuals can combat spam by using spam filtering software to block suspicious email before they enter a recipient's email inbox. However, spam filters may block legitimate messages. Spammers know how to skirt filters by continually changing their email accounts, by incorporating spam messages in images, by embedding spam in email attachments and digital greeting cards, and by using other people's computers that have been hijacked by botnets (see Chapter 8). Many spam messages are sent from one country but hosted by another country.

Some countries have passed laws to outlaw spamming or restrict its use. For instance, spamming is more tightly regulated in Europe than in the United States. Unsolicited commercial messaging is banned in the European Union. Digital marketing can be targeted only to people who have given prior consent.

In the United States, the CAN-SPAM Act does not outlaw spamming but does ban deceptive email practices by requiring commercial email messages to display accurate subject lines, identify the true senders, and offer recipients an easy way to remove their names

from email lists. It also prohibits the use of fake return addresses. Although a few people have been prosecuted under this law, it has had a negligible impact on spamming, in large part because of the Internet's exceptionally poor security and the use of offshore servers and botnets.

4.7 Discuss Quality of Life Issues

As the power of information technologies has increased, the potential negative social and personal costs of information technologies and systems have also begun to mount. Many of these negative social consequences are not violations of individual rights or property rights. Nevertheless, they can be extremely harmful to individuals, societies, and political institutions. Information technologies potentially can destroy valuable elements of culture and society even as they bring benefits. They can also create serious personal health risks. If there is a balance of good and bad consequences of using information systems, whom do we hold responsible for the bad consequences? In this section, we briefly examine some of the negative social and personal consequences of systems, considering individual, social, and political responses.

Big Tech: Concentrating Economic and Political Power

In the United States, Amazon accounts for more than 40 percent of all e-commerce retail sales, with almost 95 percent of all book, music, and video sales occurring on its platform. Google accounts for around 90 percent of online searches. Almost 90 percent of all social network users use Facebook. More than 60 percent of all digital ad revenue goes to Google, Meta, and Amazon, collectively. More than 90 percent of online video viewers use YouTube (owned by Google), and more than 60 percent use Amazon Prime Video. In the office, Microsoft dominates, with around 75 percent of the world's desktop computers using Windows operating system software and Microsoft software products. Apple accounts for about 60 percent of the U.S. market in smartphones, while phones with an Android operating system (developed by Google) account for the remaining 40 percent. Regulators' attention in the United States has also turned to the artificial intelligence industry, where OpenAI, Microsoft, and Nvidia increasingly dominate with their software and AI chips.

Reviewing the above, it becomes clear that in today's world of "Big Tech" firms, oligopolies and monopolies dominate. The wealth created by these firms inevitably translates into political influence: These same firms have amassed an army of lobbyists in Washington, DC and state capitals to try to ensure that legislation, or legislative inquiries that might affect their market and tax concerns reflect their interests.

Concentrations of market power are not new in the United States or Europe. Beginning in 1890 with the Sherman Antitrust Act in the United States, and continuing through the 1960s, monopolies have been considered threats to competition and to smaller start-up businesses, generally restraining free trade. Monopolies typically achieve their size by purchasing smaller competitors, crushing competitors by developing similar products, or engaging in predatory pricing by dropping prices drastically for short periods of time to force smaller firms out of business. Big Tech firms have a well-documented history of these behaviors. But antitrust thinking changed in the 1970s to a different standard of harm: consumer welfare. In this view, bigness per se was not a danger, nor was anticompetitive behavior. Instead, price and consumer welfare became paramount. As long as consumers were not forced to pay higher prices, market power was not important and was not considered a social or economic harm. In this view, because the offerings of Meta, Google, and Amazon are either free or very low cost, there can be no harm.

Critics point out that consumer welfare is harmed in ways other than price, namely, by preventing new, innovative companies from market access or from surviving long enough to prosper as independent firms. Complaints and lawsuits originated by small start-up firms alleging anticompetitive and unfair practices, and concerns about the abuse of personal privacy by Big Tech firms, have led to a torrent of public criticism, governmental investigations and regulatory actions, and lawsuits. In August 2024, in a landmark decision, a federal district

court found Google guilty of violating antitrust law by illegally maintaining a monopoly in Internet search. Google is also facing another antitrust case with respect to its ad technology. Antitrust cases are also pending against Apple, Amazon, and Meta.

In Europe, the European Union's Digital Markets Act (DMA), which aims to prevent "gatekeeper" platforms (defined on the basis of market value and which include Alphabet [owner of Google and YouTube], Meta, Amazon, Apple, Microsoft, and ByteDance [owner of TikTok]), from exploiting their interlocking products and services to lock in their users and squash potential rivals, went into effect in March 2024. The law is expected to have a far-reaching impact on the operation of app stores, online advertising, e-commerce, and messaging services. U.S. critics of Big Tech hope the passage of the DMA will provide a roadmap for the passage of similar legislation in the United States (Peters, 2024; Satariano, 2022).

Digital Divide

Does everyone have an equal opportunity to participate in the digital age? Will the social, economic, and cultural gaps that exist in the United States and other societies be reduced by information systems technology? Or will the cleavages be increased?

These questions have not yet been fully answered because the impact of digital technologies on various groups in society has not been thoroughly studied. What is known is that there is a **"digital divide"**: a significant disparity in access to computers and the Internet, as well as many other information resources, among different racial, ethnic, and income-level groups and in rural locations. For instance, a number of studies have found that low-income groups in the United States are less likely to have computers or online Internet access even though computer ownership and Internet access have soared over the past decade. Although the gap in computer access is narrowing, higher-income families in each racial or ethnic group are still more likely to have home computers and broadband Internet access than are lower-income families in the same group. Moreover, the children of higher-income families are far more likely to use their Internet access to pursue educational goals, whereas lower-income children are much more likely to spend time on entertainment and games. This is called the "time-wasting" gap.

Digital divide

Significant disparity in access to computers and the Internet, as well as many other information resources, among different racial, ethnic, and income-level groups and in rural locations.

Left uncorrected, this digital divide could lead to a society of information haves, who are computer literate and skilled, versus a group of information have-nots, who are computer illiterate and unskilled. Narrowing this digital divide by making digital information services—including broadband Internet service—available to as many people as possible, just as basic telephone service is now, has become an important priority for federal, state, and local governments in the United States.

The use of AI can also lead to increased levels of bias. For example, if an AI system is trained on data that reflects existing societal biases, it might perpetuate or amplify these biases, leading to unfair treatment or discrimination. AI's capacity to analyze and draw patterns from large datasets can lead to the stereotyping of certain groups, leading to potential algorithmic discrimination and bias. For example, predictive policing algorithms employed by U.S. law enforcement have been shown to reinforce prevailing biases. Biased algorithms can also lead to discriminatory outcomes in loan applications and job applications. We discuss this issue further in Chapter 11.

Employment: Job Loss Due to Automation, Business Process Reengineering, and AI

The reengineering of work is typically hailed in the information systems community as a major benefit of new information technology. It is much less frequently noted that the redesign of business processes and automation have also caused millions of people to lose their jobs. On the other hand, some economists believe that new technologies create as many or more jobs than they destroy. For instance, the growth of e-commerce has led to a decline in retail sales jobs but an increase in jobs for warehouse workers, supervisors, and delivery work. These economists also believe that bright, educated workers who are displaced by technology will move to better jobs in fast-growth industries.

The increasing use of AI has raised tremendous concerns about the prospect of wide-scale loss of jobs. Experts are divided as to the ultimate impact, but it is clear that AI will have a significant impact on how business is conducted, how work is performed, and the types of jobs that will be available in the future. In the short term, companies appear to be reducing the number of people they are hiring for job roles that might ultimately be partially or totally replaced by AI, such as customer service roles. We examine issues related to AI and jobs further in Chapter 11.

Rapidity of Change: Reduced Response Time to Competition

Information systems have helped to create much more efficient national and international markets. Today's rapidly moving global marketplace has reduced the normal social buffers that permitted businesses many years ago to adjust to competition. Time-based competition has an ugly side: The business you work for may not have enough time to respond to global competitors and may be wiped out, along with your job. We run the risk of developing a just-in-time society with just-in-time jobs and just-in-time workplaces, families, and vacations. One impact of on-demand services firms such as Uber is to create just-in-time jobs with no benefits or insurance for employees.

Maintaining Boundaries: Family, Work, and Leisure

Although the idea of ubiquitous computing, remote work and telecommuting, nomad computing, mobile computing, and the do-anything-anywhere computing environment is alluring to many, it also raises a number of issues. One impact of such an environment is that it weakens the traditional boundaries that separate work from family and leisure.

The advent of information systems, coupled with the growth of knowledge-work occupations, means that more and more people are working when traditionally they would have been interacting with family and friends. The work umbrella now extends far beyond the eight-hour day into commuting time, vacation time, and leisure time. The explosive growth and use of mobile devices have only heightened the sense for many employees that they are never away from work.

Even leisure time spent on the computer threatens these close social relationships. Extensive Internet and mobile device use, even for entertainment or recreational purposes, takes people away from their family and friends. Among middle school and teenage children, it can lead to harmful antisocial behavior, such as cyberbullying.

Although some people enjoy the convenience of working at home, the do-anything-anywhere computing environment can blur the traditional boundaries between work and family time.

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Weakening these institutions poses clear-cut risks. Family and friends historically have provided powerful support mechanisms for individuals, and they act as balance points in a society by preserving private life and providing a place for people to collect their thoughts, think in ways contrary to their employer, and dream.

Physical, Mental, and Cognitive Health Risks

Repetitive stress injury (RSI)
Injury that occurs when muscle groups are forced through repetitive actions with high-impact loads or thousands of repetitions with low-impact loads.

Information technology can pose a number of physical, mental, and cognitive health risks. For instance, a common occupational injury today is **repetitive stress injury (RSI)**. RSI occurs when muscle groups are forced through repetitive actions that are often made with high-impact loads (such as playing tennis) or tens of thousands of repetitions under low-impact loads (such as working at a computer keyboard). The incidence of RSI is estimated to affect as much as one-third of the labor force and accounts for one-third of all disability cases.

Carpal tunnel syndrome (CTS)
Type of RSI in which pressure on the median nerve through the wrist's bony carpal tunnel structure produces pain.

The single-largest source of RSI is computer keyboards. The most common kind of computer-related RSI is **carpal tunnel syndrome (CTS)**, in which pressure on the median nerve through the wrist's bony structure, called a carpal tunnel, produces pain. The pressure is caused by the constant repetition of keystrokes. Symptoms of CTS include numbness, shooting pain, inability to grasp objects, and tingling. Millions of workers have been diagnosed with CTS. In the United States and most other developed countries, CTS affects an estimated 1 to 3 percent of the general population in any given year (Sevy and Varacallo, 2022).

However, RSI is avoidable. Designing workstations for a neutral wrist position (for example, using a wrist rest to support the wrist), proper monitor stands, and footrests all contribute to proper posture and reduced RSI. Ergonomically correct keyboards are also an option. These measures should be supported by frequent rest breaks and rotation of employees to different jobs.

Computer vision syndrome (CVS)
Eyestrain condition related to display screen use; symptoms include headaches, blurred vision, and dry and irritated eyes.

RSI is not the only occupational illness computers cause. Back and neck pain, leg stress, and foot pain also result from poor ergonomic designs of workstations. **Computer vision syndrome (CVS)** refers to any eyestrain condition related to display screen use with desktop computers, laptops, e-readers, smartphones, and handheld video game players. CVS affects about 90 percent of people who spend three hours or more per day using a display screen. Its symptoms, which are usually temporary, include headaches, blurred vision, and dry, irritated eyes.

Much has also been written about the challenges that the Internet, mobile apps, and social media pose for mental health, particularly that of children and young adults. The impact of the increase in screen time and time spent with digital devices during the Covid-19 pandemic has further highlighted this issue. Potential negative impacts include

Repetitive stress injury (RSI) is a leading occupational disease today. The single-largest cause of RSI is computer keyboard work.

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increased exposure to negative online social interactions (cyberbullying, feelings of missing out, and social comparisons); reduced physical activity; sleep disruption; isolation from in-person contact; and the possibility of Internet/social media addiction. The topic remains controversial, however. One recent study of 430,000 U.S. and UK adolescents found little evidence for an increased association between adolescents' technology engagement and mental health problems over the past 30 years. However, the U.S. Surgeon General has released an advisory stating that there are ample indicators that social media can pose a risk of harm to the mental health and well-being of children and adolescents, and in June 2024 announced that he will push for a warning label on social media platforms similar to those that appear on tobacco and alcohol products (Barry and Kang, 2024; APS, 2021).

Information technology also may be harming our cognitive functions or at least changing how we think and solve problems. Although the Internet has made it much easier for people to access, create, and use information, some experts believe that it is also preventing people from focusing and thinking clearly on their own. They argue that excessive use of computers and smartphones reduces intelligence. Some scholars believe that exposure to computers encourages looking up answers rather than engaging in real problem-solving. People, in this view, don't learn much while surfing the web or answering email when compared to listening, drawing, arguing, looking, and exploring. In addition, increased reliance on various forms of AI (see Chapter 11) may be impacting our power to make decisions based on our own judgments (Moser, den Hond, and Lindebaum, 2022; Carr, 2015).

As it further develops, the metaverse may also present new physical and mental health challenges. The metaverse is a visual, three-dimensional (3D) virtual reality that some believe will be at the center of the future web. It aims to move the Internet beyond 2D screens and toward an immersive 3D experience.

Information technology has become part of our lives—personally as well as socially, culturally, and politically. It is unlikely that the issues and our choices will become easier as information technology continues to transform our world. The growth of the Internet, the information economy, and AI suggests that all the ethical, social, and political issues we have described will be heightened as we move further into the first fully digital century.

4.8 Understand How the Information in This Chapter Can Help Your Career

The information in this chapter can help your career by giving you an overview of the ethical, social, and political issues that you should be aware of as you interact with information technologies and information systems. Companies are becoming increasingly aware that such issues can have a significant impact on their bottom line, and awareness of these issues is necessary for everyone in a company.

Real-World Scenario

Cementos Argos Uses Privacy Analysts to Cement Competitive Advantage

Cementos Argos is a leading multinational producer and distributor of cement, concrete, and aggregates. Cementos Argos operates in 16 countries and territories, including the United States, and Central and South America. Cementos Argos decided that it needed to improve its decision making in order to gain a competitive advantage. The company established a dedicated business analytics center, investing in skilled business analysts, data scientists, as well as privacy analysts. Privacy analysts played a crucial role in ensuring compliance with privacy laws and ethical data practices.

Job Description and Career Path: Privacy Analyst

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Privacy analyst	A privacy analyst specializes in an organization's data privacy practices. A privacy analyst is pivotal to maintaining consumer trust and upholding an organizations' reputation. Job responsibilities include: • Acting as the first point of contact for internal and external privacy-related questions • Proactively informing and involving internal stakeholders (such as Customer Service, Legal, Technology, Finance) • Supporting various departments to make sure processes and controls are aligned with privacy business strategies as well as supporting requests from data subjects and authorities • Working as part of the team in creating documentation and encouraging and facilitating privacy best practices • Providing support on privacy-related projects, improvements, and innovations, including assisting in the development and implementation of privacy processes	• Strong grasp of IT and security systems, including familiarity with data encryption, data management, access controls, cybersecurity threats, and incident response strategies • Experience with data mapping and inventory tools such as OneTrust and Trust Arc • Experience with privacy impact assessment tools, such as Privacy Designer • Experience with consent management tools, such as OneTrust Preference Choice; Cookiebot	• Junior/Entry-level: $60,000–$80,000 • Mid-level: $80,00–$100,000 • Senior: $110,000–$125,000 • Executive/Chief Privacy Officer (CPO): $250,000–$350,000 Salary can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

Meta's Many Ethical Challenges

Meta is the parent company of some of the most "popular" social networks in the world, including Facebook and Instagram. At the same time, Meta (which changed its name from Facebook in 2021) is also one of the most reviled companies in the world. This seemingly contradictory situation stems in large part from Meta's business practices and how it has responded to a variety of ethical challenges throughout the years.

Meta's stated mission is to give people the power to build community and bring the world closer together. It provides various platforms that enable people to connect and share their opinions, photos, and videos with audiences ranging from just friends and family to the public at large, to find communities, and to grow businesses. As it has grown, its platforms have also become a source of information for its users about what is happening in the world.

From just about its earliest days, Meta has been hounded by controversy over its business practices. When it originally launched Facebook in 2004, it promised its users nearly complete control over who could see their personal profiles and information. But that did not last long. Shortly after it launched, it took its first steps toward monetizing the platform to help cover its costs. It quickly turned to online advertising to generate revenue. What the online advertising industry prizes is data—specifically, personal data about people as well as data that enables companies to predict how people will think and act. It soon became clear that Facebook was generating a trove of such data, setting the stage for one of Meta's most enduring ethical challenges: the privacy of its users versus Meta's desire to generate revenue based on that data. In that battle, the desire to maximize its revenue has usually won.

The Beacon program was an early indication of this challenge. Introduced in 2007, the Beacon program broadcast Facebook users' activities on participating websites, such as their purchases, to their friends' News Feeds, without user permission. Advertisers could then run ads next to those posts. Privacy advocates blistered with criticism, and more than 50,000 users signed a petition objecting to the program. Meta believed that users would accept Beacon once they had time to get used to it. But it was wrong, and it ultimately was forced to withdraw the Beacon program.

A look back over the past 15 years shows that Meta has been repeatedly embroiled in conflicts over user privacy. It has been slapped by the Federal Trade Commission more than once: first in 2012, when it agreed to stop deceiving users about their ability to control their personal data and to stop sharing data with third parties without informing users, and then again in 2019, when it was fined a record-breaking $5 billion for clearly and intentionally violating the 2012 settlement. The 2019 settlement followed closely on the heels of what has become known as the Cambridge Analytica scandal. A UK researcher had obtained permission to use Facebook for a personality quiz app. Unbeknownst to users, the app collected data not only from those who downloaded it but also from their friends and their friends' friends. Ultimately, it created a database on more than 87 million Facebook users, which was then sold to political consulting and data analytics firm Cambridge Analytica, which used the data in an effort to influence the 2016 US presidential election. Not long thereafter, a *New York Times* exposé revealed that Meta had data-sharing partnerships with at least 60 device makers as well as selected app developers, which enabled those third parties to gain access not only to data about Facebook users but also to personal data about their Facebook friends, without their consent.

Meta has also been repeatedly sued for privacy-related violations and has paid out millions to settle those lawsuits, including $650 million in 2021 to settle claims it violated an Illinois privacy law by storing facial recognition data without first obtaining users' consent, and $725 million in 2023 to settle a federal class action lawsuit that originated from the Cambridge Analytica scandal and then expanded to target Meta's broader data-sharing practices, including granting numerous third parties access to users' Facebook content and information without consent.

Meta has also continually run afoul of the European Union's General Data Protection Regulation, which mandates more stringent privacy protections than the United States does, and in 2023 was fined €1.2 billion, a new record for financial penalties imposed under the GDPR, related to Meta's transfer of EU users' data to servers located in the United States. Meta had been previously fined €405 million for violating the GDPR by failing to protect children's data on Instagram, €390 million for compelling users to consent to personalized advertisements, and €265 million for failing to prevent hackers from stealing the personal information of more than 530 million Facebook users.

Given Meta's checkered history of privacy violations, why do so many people continue to use and share sensitive details of their lives on its social network platforms? Often, it's because users do not realize all the data that are being collected about them and the purposes for which that data may be used. Many of Facebook's features and services are enabled by default, and a study by Siegel+Gale found that Facebook's privacy policy is more difficult to comprehend than government notices or typical bank credit card agreements, which are notoriously dense.

But issues surrounding user privacy are not Meta's only ethical challenge: Within the last several years, there has been an increasing focus on Meta's role in providing a platform for misinformation and other harmful content. For instance, in 2021, the *Wall Street Journal* published an exposé based on a review of Meta documents including research reports,

online employee discussions, and internal presentations. According to the *Wall Street Journal*, Meta knew that its platforms caused harm but did little, if anything, about it. For example, Meta researchers knew that Instagram could be harmful to mental health, particularly for teenage girls, but in public statements, Meta consistently downplayed its negative effect on teens. Meta employees warned that its platforms were being used to incite ethnic violence and to facilitate criminal activity such as human trafficking in certain countries, but Meta did little to respond to those concerns on a systemic basis. Meta also knew that a change to the Facebook News Feed algorithm, which it hoped would increase user engagement, rewarded posts oriented toward outrage and sensationalism, making the Facebook environment more and more toxic. A former employee known as the Facebook whistleblower painted a picture of a company unable to control the "monster" it had created. For instance, Meta's internal documents revealed its efforts to use artificial intelligence to keep its platforms free from offensive or dangerous content have fallen far short of its expectations.

Meta's business practices with respect to its competitors have also come under fire. The FTC has sued Meta for attempting to maintain its dominant position by engaging in a series of anti-competitive actions that violated U.S. antitrust law. For example, the FTC pointed to acquisitions of rivals Instagram and WhatsApp in 2014 as predatory moves meant to crush the competition. The FTC also argued that Meta harmed rival app developers by blocking them from accessing data and tools on its platform. As of September 2024, the lawsuit remains ongoing. Meta is facing similar charges in the European Union, as well as a new investigation seeking to determine whether Meta has violated the EU's new Digital Markets Act.

Meta's latest ethical challenge involves AI. Meta has been working on its own version of a generative AI large language model, which it calls LLaMA (Large Language Model Meta AI) and in April 2024, began rolling out AI assistant tools across all of its platforms, including Facebook, Instagram, and WhatsApp. The tools appear in feeds, chats, and search, without any option to disable its use. Meta's AI-everywhere push quickly raised user hackles. For instance, as with other AI chatbots, Meta's AI can make things up or spread misinformation. For instance, in one private Facebook group focused on parenting, the chatbot posted a comment in which it claimed to have a child enrolled in a prestigious gifted and talented program. Another issue involves the use of users' public information to train its AI services (Meta claims that it does not use private posts and messages). It is not clear when Meta began scraping data from users based in the United States, nor is there an easy method of opting out. Whether Meta has also violated copyright laws in training its AI system is another issue. It reportedly considered buying the publishing house Simon & Schuster in order to gain access to longer works and trying to negotiate licenses with publishers, artists, musicians, and the news industry, but rejected both those options as taking too long. Instead, according to published reports, it relied on summarized books, essays, and other works found on the Internet, without obtaining permission, despite warnings from a lawyer about ethical concerns.

Meta, not surprisingly, generally denies most of the claims that its detractors allege, although at times it has apologized for some of its missteps. But Meta's future may rely on how it responds to its ethical challenges going forward. Over the last several years, there has been an increasing focus on ethical business practices and a growing awareness that good ethics is good business. The ESG (environmental, social, and governance) movement, which focuses on a company's performance beyond just its profit and loss, is gaining increased traction and becoming a strategic business imperative. Meta says it is taking a number of steps to respond to these challenges and has published a long list of environmental, social, and governance resources explaining how it is doing so. However, given that Meta's business model relies almost entirely on the largely unfettered use of users' personal information, and given its past history and business practices, it is unclear whether Meta can be trusted.

SOURCES: Jesus Jimenez, "Can I Opt Out of Meta's A.I. Scraping on Instagram and Facebook? Sort Of," *New York Times*, June 7, 2024; "Meta Investor Relations: Environmental Social Governance Resources," Investor.fb.com, accessed June 7, 2024; Brian X. Chen, "Meta's A.I. Assistant Is Fun to Use, but It Can't Be Trusted," *New York Times*, April 24, 2024; "Meta's New AI Assistant Wanted to Answer Users' Questions and Now It's Getting on Their Nerves," Qz.com, April 23, 2024; Michael Cader, "Meta Stole Everyone's Books Because It Was Cheap and Convenient; Considered Buying S&S, But That Was Unnecessary," Lunch.publishersmarketplace.com, April 8, 2024; Foo Yun Chee and Bart H. Meijer, "Apple, Google, Meta Targeted in EU's First Digital Markets Act Probes," Reuters.com, March 25, 2024; "Meta Platforms Inc. Form 10-K for the fiscal year ended December 31, 2023," Sec.gov, February 2, 2024; Dev Kundaliya, "Meta Fined More than €1 billion for GDPR Breach," Computing.co.uk, May 22, 2023; "The Facebook Files," *Wall Street Journal*, January 12, 2022; "FTC Alleges Facebook Resorted to Illegal Buy-or-Bury Scheme to Crush Competition after String of Failed Attempts to Innovate," Ftc.gov, August 19, 2021; "A Timeline of Trouble: Facebook's Privacy Record and Regulatory Fines," Guild.com, August 4, 2021.

Case Study Questions

1. Choose an ethical dilemma discussed in this case and perform an ethical analysis of it.
2. Will Meta be able to have a successful business model without invading users' privacy? Explain your answer.
3. What ethical issues are involved in Meta's development and use of Meta AI?
4. Investigate the ESG movement. How might it impact Meta's future?

Review Summary

4.1 Describe IT/IS-related ethical, social, and political issues.

Information technology and systems are introducing changes for which laws and rules of acceptable conduct have not yet been developed. Increasing computing power, storage, data analysis, and networking capabilities—including the Internet—expand the reach of individual and organizational actions and magnify their impacts. The increasing use of mobile devices and artificial intelligence also raises ethical issues. The ease and anonymity with which information is now communicated, copied, and manipulated in online environments pose new challenges to the protection of privacy and intellectual property. The main ethical, social, and political issues that information systems raise center on information rights and obligations, property rights and obligations, system quality, accountability and control, and quality of life.

4.2 Describe principles for ethical conduct.

Basic concepts underlying ethical decisions include responsibility, accountability, liability, and due process. When confronted with an ethical issue, a five-step ethical analysis process that involves identifying and describing the facts, defining the conflict or dilemma and identifying the higher-order values involved, identifying the stakeholders, identifying the options that can be reasonably taken, and identifying the potential consequences of the options should be used. Ethical principles that should be followed in conjunction with an ethical analysis include the Golden Rule, Immanuel Kant's categorical imperative, the slippery slope rule, the utilitarian principle, the risk aversion principle, and the ethical no-free-lunch rule. These principles should be used in conjunction with an ethical analysis.

4.3 Discuss information rights and privacy issues.

Today's information technologies and systems enable companies to gather personal data about individuals from many sources easily and analyze these data to create detailed digital profiles about individuals and their behaviors. Data flowing over the Internet can be monitored at many points. Cookies and other monitoring tools closely track the activities of website visitors and mobile app users. Not all websites and mobile apps have strong privacy protection policies, and they do not always allow for informed consent regarding the use of personal information. AI is the latest information technology to raise privacy concerns.

4.4 Discuss intellectual property issues.

Intellectual property is subject to a variety of protections under four legal traditions: copyright, patents, trademarks, and trade secrets. However, today's information technologies and systems have severely challenged existing laws and social practices that protect intellectual property.

Traditional copyright laws are insufficient to protect against piracy because digital material can be copied so easily and transmitted to many locations simultaneously over the Internet. AI is the latest information technology to create concerns about the protection of intellectual property.

4.5 Discuss system quality issues.

New information technologies are raising questions about what constitutes acceptable levels of system quality, given the centrality of information systems to the functioning of businesses and society.

4.6 Discuss accountability and control issues.

The use of information technology may sometimes result in harm, creating questions as to who, if anyone, should be held accountable for that harm. What laws exist to control the misuse of information technology? Who, if anyone, is liable if a business or person is harmed by software that is defective? Should a website or social network be held accountable for the content that a user posts? Widespread use of computers increases opportunities for computer crime and computer abuse: How should such occurrences be handled by the legal system?

4.7 Discuss quality of life issues.

Although today's information technologies and systems have been sources of efficiency and wealth, they also have some negative social consequences that impact the quality of life by, for example, enabling the concentration of economic and political power. Not everyone has an equal opportunity to participate in the digital economy, and this digital divide may be exacerbating socioeconomic disparities among different ethnic groups and social classes. The use of AI may also result in the development of bias. Jobs can be lost when computers and AI replace workers or when tasks become unnecessary in reengineered business processes. Information systems and technology can also create issues with maintaining appropriate boundaries separating family, leisure and work, as well as create physical, mental, and cognitive health risks.

Key Terms

Accountability, 116
Carpal tunnel syndrome (CTS), 136
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Computer crime, 132
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Safe harbor, 121
Slippery slope rule, 117
Spam, 132
Trade secret, 126
Trademarks, 126
Utilitarian principle, 117
Web beacon, 123

Review Questions

4.1 Describe IT/IS-related ethical, social, and political issues

- Define ethics.
- Explain how ethical, social, and political issues are connected, and give some examples.
- Describe the key technological trends that heighten ethical concerns.

4.2 Describe principles for ethical conduct.

- Define responsibility, accountability, liability, and due process.
- Describe the five steps in an ethical analysis.
- Describe six ethical principles.

4.3 Discuss information rights and privacy issues.

- Define information rights, privacy, and Fair Information Practices.
- Discuss how the regulation of privacy in Europe differs from that in the United States.
- Discuss the challenges to the protection of individual privacy posed by the Internet.
- Discuss the challenges to the protection of individual privacy posed by AI.
- Explain how informed consent, legislation, industry self-regulation, and technology tools help protect the individual privacy of Internet users.

4.4 Discuss intellectual property issues.

- Define intellectual property.
- Describe the four types of laws that protect intellectual property rights.
- Discuss the challenges to intellectual property rights posed by the Internet.
- Discuss the challenges to intellectual property rights posed by AI.

4.5 Discuss system quality issues.

- Discuss the issues involved in determining an acceptable level of system quality.
- List and describe the principal causes of system quality problems.

4.6 Discuss accountability and control issues.

- Explain why it is so difficult to hold software vendors and service providers liable for failure or injury.
- Discuss the issues that AI raises with respect to the question of liability for harm.
- Describe the difference between computer crime and computer abuse.

4.7 Discuss quality of life issues.

- Name and describe four quality-of-life impacts of information technologies and systems.
- Explain how AI may potentially contribute to the development of bias.
- Describe RSI, CTS, and CVS, and explain their relationship to information technology.

Discussion Questions

4.8 Should cloud service providers be held liable for economic injuries suffered when their systems fail?

4.9 Should companies be responsible for unemployment that their information systems cause? Why or why not?

4.10 Discuss the pros and cons of allowing companies to amass personal data for behavioral targeting.

Using AI Project

4.11 AI Project: Business Use of Social Media

You have been hired as a consultant by two friends who want to open a retail pet shop selling organic pet food and treats, as well as pet toys and clothing made from organic materials. They have asked you to advise them about the ethical, social, and legal issues they need to consider if they establish an online presence on Facebook and Instagram.

Using Microsoft's Copilot, Google's Gemini, or ChatGPT, develop a prompt to help you advise them about these issues. Save a copy of your prompt and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience in analyzing the privacy implications of using online data brokers; using blog creation tools to create a simple blog; developing a corporate policy for employee web usage; and analyzing web browser privacy.

4.12 Management Decision Problem

InfoFree's website is linked to massive databases that consolidate personal data on millions of people. Users can purchase marketing lists of consumers broken down by location, age, gender, income level, home value, home ownership, and length of residence. Do data brokers such as InfoFree raise privacy issues? Why or why not? If your name and other personal information were in this database, what limitations on access would you want in order to preserve your privacy? Consider the following data users: government agencies, your employer, private business firms, and other individuals.

4.13 Creating a Simple Blog

Software skills: Blog creation

Business skills: Blog and web page design

In this project, you'll learn how to build a simple blog of your own design using the online blog creation software available at Blogger.com or Squarespace. Pick a sport, hobby, or topic of interest as the theme of your blog. Name the blog, give it a title, and choose a template for the blog. Post at least four entries to the blog, adding a label for each posting. Edit your posts if necessary. Upload an image, such as a photo from your computer or the web, to your blog. Add capabilities for other registered users, such as team members, to comment on your blog. Briefly describe how your blog could be useful to a company selling products or services related to the theme of your blog. List the tools available in Blogger that would make your blog more useful for business, and describe the business uses of each. Save your blog and show it to your instructor.

4.14 Analyzing Web Browser Privacy

Software Skills: Web browser software

Business Skills: Analyzing web browser privacy protection features

This project will help develop your skills in using the privacy protection features of leading web browser software. Examine the privacy protection features and settings for two leading web browsers, such as Microsoft Edge, Mozilla Firefox, or Google Chrome. Make a table comparing the features of two of these browsers in terms of functions provided and ease of use.

- How do these privacy protection features protect individuals?
- How do these privacy protection features affect what businesses can do on the Internet?
- Which browser does a better job of protecting privacy? Why?

4.15 Collaboration and Teamwork Project: Developing a Corporate Code of Ethics

With two or more of your classmates, develop a corporate ethics code on privacy that addresses both employee privacy and the privacy of customers and users of the corporate website. Be sure to consider email privacy and employer monitoring of worksites as well as corporate use of information about employees concerning their off-the-job behavior (e.g., lifestyle, marital arrangements, and so forth). If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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Part Two
Information Technology Infrastructure

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Chapter 5: IT Infrastructure and Emerging Technologies

Chapter 6: Foundations of Business Intelligence: Databases and Information Management

Chapter 7: Telecommunications, the Internet, and Wireless Technology

Chapter 8: Securing Information Systems

PART TWO provides the technical foundation for understanding information systems by examining hardware, software, database, and networking technologies along with tools and techniques for security and control. This part answers questions such as: What technologies do businesses today need to accomplish their work? What do I need to know about these technologies to make sure they enhance the performance of the firm? How are these technologies incorporating artificial intelligence? What technologies and procedures are required to ensure that systems are reliable and secure?

5

IT Infrastructure and Emerging Technologies

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Learning Objectives

After completing this chapter, you will be able to:

5.1 Discuss IT infrastructure and its evolution.

5.2 Describe the components of IT infrastructure.

5.3 Discuss the mobile platform, apps, and BYOD.

5.4 Discuss quantum computing and green computing.

5.5 Discuss virtualization and cloud computing.

5.6 Describe contemporary software platforms.

5.7 Discuss alternative ways of obtaining software.

5.8 Describe IT infrastructure for AI.

5.9 Discuss IT infrastructure management challenges.

5.10 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** QRyde Rides High with the Cloud and AI
- **Spotlight on Technology Case Study:** AI Chips
- **Chapter Case Study:** Are AI and Cloud Computing Sustainable?

Opening Case Study

QRyde Rides High with the Cloud and AI

Himanshu Bhatnagar, QRyde's founder and CEO, has devoted his career to helping people with physical or mental challenges obtain access to safe, affordable, and reliable transportation. He wants people in these traditionally underserved communities to have multiple transportation options so they can obtain the medical care they need, attend school, or travel to critical appointments. QRyde has helped make his dream come true.

QRyde is a first-in-class and best-of-breed cloud-based transportation management system. It enables independent transportation companies and government transit systems to optimize current operations using technology such as mobile scheduling, automated dispatching, and artificial intelligence (AI)-based route optimization to help those organizations provide services more efficiently and economically. Educational institutions, healthcare companies, and public transit agencies use QRyde's shared ride scheduling platform for ride booking, ride cost-sharing, and bidding management services to provide mobility options to people in need.

For example, Penn Highlands Healthcare, which serves residents of Northwestern/Central and Southwestern Pennsylvania, uses QRyde to coordinate rides with regional transportation providers in a service area covering 39 counties. The QRyde technology enables residents, facilities, and caregivers to identify available seats with each transportation provider, coordinate schedules and costs, and determine the transportation service that best fits their needs. The QRyde mobile app and web portal provide even more ways to request a ride for all residents through local transportation providers and transportation network companies.

QRyde is very data and computing-intensive, with substantial logistics and data integration requirements. The system originally ran in on-premise data centers and colocation facilities. (Colocation is the practice of housing privately-owned servers and networking equipment in a third-party data center. The organization that owns the equipment is responsible for its management and upkeep.) However, the business continued to grow to the point where it was coordinating 100,000 trips per day for more than 700 organizations. It needed a more powerful, flexible IT infrastructure to do this work.

Oracle Cloud Infrastructure (OCI) provided the best solution. OCI is a platform of cloud services that enables organizations to build and run a wide range of applications in a highly available, remote high-performance environment. These services include computing services, data storage and management services, networking services, developer services, application services, and security and governance services. Organizations adopting OCI can increase their use of these services as they grow and pay for only the level of service that they need. Bhatnagar selected Oracle because of its expertise in data management services and its understanding of how to optimize the data. QRyde stores and secures data both while active and at rest.

With QRyde Cloud all stakeholders of transportation systems can perform their entire business functions on one portal. These functions include client profile management, ride booking, scheduling, dispatching, ridesharing, billing, electronic fare cards, and contract

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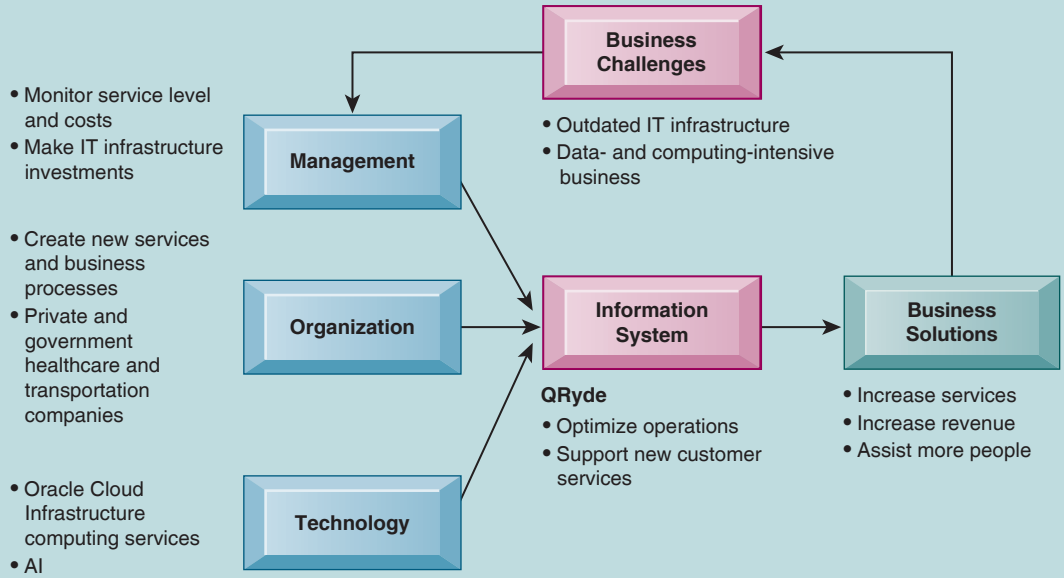


management on the same platform using text, voice, and browser-based user interfaces and experiences. QRyde Cloud removes the high-cost-of-technology barriers for communities and providers, and its cost-per-ride pricing model has helped organizations save hundreds of thousands of dollars for the optimization of transportation systems.

SOURCES: QRyde.com, accessed August 20, 2024; Phhealthcare.org, accessed August 20, 2024; "Oracle Cloud Infrastructure Platform Overview," Go.oracle.com, accessed August 20, 2024; Samiron Goswami, "QRyde's AI-Powered Approach to Public Transportation Planning," Medium.com, October 12, 2023; Justine Kavanaugh-Brown, "To Get Rides for People in Need, QRyde Taps OCI for Better Disaster Recovery, Security, and Scale," Blogs.oracle.com, November 5, 2021.

The experience of QRyde illustrates the importance of information technology infrastructure in running a business today. The right technology at the right price will improve organizational performance. QRyde was saddled with outdated hardware and software platforms that could no longer meet its computing requirements as the business grew. This prevented the company from operating efficiently and effectively and from providing its customers with the services they needed.

The Opening Case Study diagram above calls attention to important points raised by this case and this chapter. Using cloud computing for its IT infrastructure enables QRyde to increase its workload and add new capabilities such as AI for route optimization as it expands and to provide new services to the people and organizations that need them. QRyde is also able to focus more on growing the business and serving customers because it delegated the operation and management of its IT infrastructure to a cloud service provider. The company works with state-of-the-art information technology tools, pays for only the computing capacity it uses on an as-needed basis, and does not have to make extensive and costly up-front IT investments.

Case Study Questions

1. How did using cloud computing help QRyde serve its customers better?
2. What were the business benefits for QRyde of using a cloud computing infrastructure?

CAREER BENEFIT

QRyde's IT staff had the opportunity to develop expertise in important cloud computing tools and also in AI route optimization solutions. This can help them advance their careers at QRyde but also make them more qualified for higher-paying jobs at other organizations. Non-IT staff benefit from contributing to the development of such an important solution for the organization. Working with the new system helps staff to dramatically improve the efficiency and accuracy of transportation scheduling and medical services so that they can help more people in need. QRyde's solution has a wider social benefit by providing better and more accessible healthcare for underserved populations.

5.1 Discuss IT Infrastructure and Its Evolution

In Chapter 1, we defined information technology (IT) infrastructure as the shared technology resources that provide the platform for the firm's specific information system applications. A firm's IT infrastructure includes investment in hardware, software, and services—such as consulting, education, and training—that are shared across the entire firm or across entire business units in the firm. Examine Figure 5.1, which illustrates how a firm's IT infrastructure provides the foundation for serving customers, working with suppliers, managing internal firm business processes, and executing its business strategy.

Supplying firms worldwide with IT infrastructure (hardware, software, networking, and IT services) in 2024 was estimated to be a $5.06 trillion industry (Gartner, 2024). Investments in infrastructure account for between 25 and 50 percent of information technology expenditures in large firms, led by financial services firms where IT investment is well over half of all capital investment.

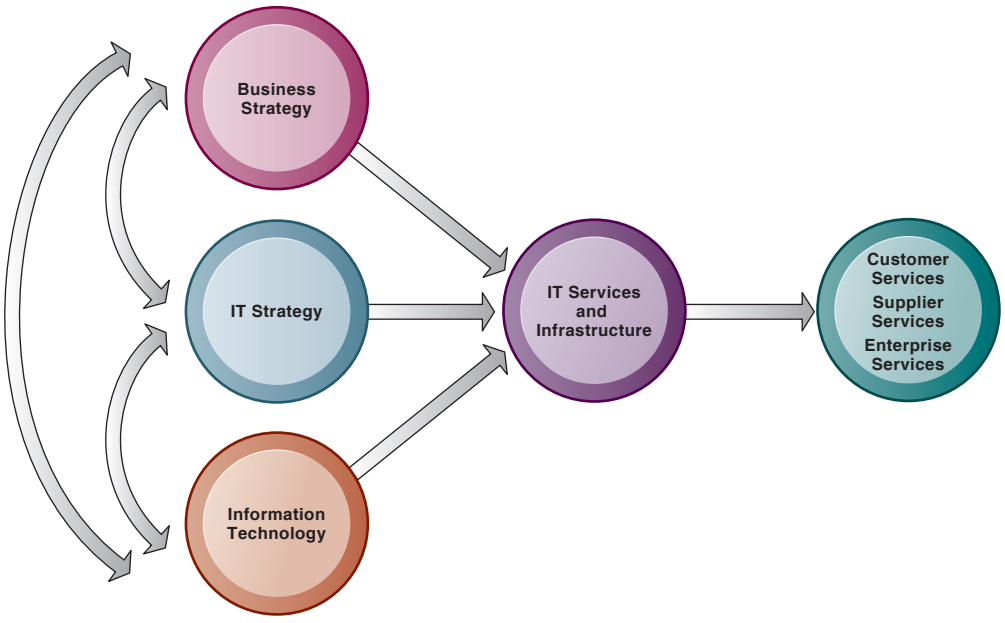
Defining IT Infrastructure

A firm's IT infrastructure consists of a set of physical devices and software applications that are required to operate the entire enterprise. But IT infrastructure also includes a set of firmwide services composed of both human and technical capabilities. These services include the following:

- Computing platforms used to provide computing services that connect employees, customers, and suppliers into a coherent digital environment, including large mainframes, midrange computers, server computers, desktop and laptop computers, mobile devices, and remote cloud computing services
- Telecommunications services that provide data, voice, and video connectivity to employees, customers, and suppliers

Figure 5.1 Connection Between the Firm, IT Infrastructure, and Business Capabilities

The services a firm can provide to its customers, suppliers, and employees are a direct function of its IT infrastructure. Ideally, this infrastructure should support the firm's business and information systems strategy. New information technologies have a powerful impact on business and IT strategies as well as the services that can be provided to customers.



- Data management services that store and manage corporate data and provide capabilities for analyzing the data
- Application software services, including online software services, that provide enterprise-wide capabilities such as enterprise resource planning, customer relationship management, supply chain management, and knowledge management systems that are shared by all business units
- Physical facilities management services that develop and manage the physical installations required for computing, telecommunications, and data management services
- IT management services that plan and develop the infrastructure, coordinate with the business units for IT services, manage accounting for the IT expenditure, and provide project management services
- IT standards services provide the firm and its business units with policies that determine which information technology will be used, when, and how
- IT education services that provide training in system use to employees and offer managers training in how to plan for and manage IT investments
- IT research and development services provide the firm with research on potential future IT projects and investments that could help the firm differentiate itself in the marketplace

Mainframe

High-performance, large-capacity computer that can process high volumes of data very quickly.

Client/server computing

A model of computing in which a client is connected in a network with one or more servers that perform common functions that the clients on the network need.

Client

Various types of devices that can request access to services or resources in a client/server network.

Servers

A computer that is dedicated to performing common functions that clients in a client/server network need, such as file storage, software applications, printing and Internet access.

Multitiered client/server architecture

Client/server network in which the work of the entire network is balanced over multiple levels of servers, depending on the kind of service being requested. Often referred to as N-tier client/server architecture.

Web server

Software that is responsible for locating and managing stored web pages and delivering them to a client in response to a request for service. Note that the term web server is also sometimes used to refer to the computer on which the web server software resides.

Evolution of IT Infrastructure

The IT infrastructure in organizations today is an outgrowth of 70 years of evolution in computing platforms. Developments in computer processing, memory chips, storage devices, networking hardware and software, and software design have exponentially increased computing power while exponentially reducing costs.

Examine Figure 5.2 which illustrates the stages in this evolution, each representing a different configuration of computing power and infrastructure elements. The five eras are general-purpose mainframe and minicomputer computing, personal computers, client/server networks, enterprise computing, and cloud and mobile computing.

Table 5.1 describes the technologies that characterize each of these eras. Technologies that characterize one era may also be used in another time period for other purposes. For example, some companies still run traditional mainframe computer-based information systems or use mainframe computers as servers supporting large websites and corporate enterprise applications (A **mainframe** is a high-performance, large-capacity computer that can process high volumes of data very rapidly.)

Client/Server Computing

The client/server computing model deserves special attention because it is the computing architecture most widely used today. **Client/server computing** is a model of computing in which a **client** (various types of devices that can request access to services or resources) is connected in a network with one or more **servers**, which are computers that are dedicated to performing common functions that the clients on the network need, such as file storage, software applications, printing, and Internet access. The Internet and cloud computing use the client/server model, as do many smaller corporate networks. Client/server computing enables businesses to distribute computing work across a series of smaller, inexpensive computers that cost much less than centralized mainframe systems. The result has been an explosion in computing power and applications throughout the firm.

The simplest client/server network consists of a client computer networked to a single server computer, with processing split between the two types of computers. This is called a *two-tiered client/server architecture*. Whereas simple client/server networks are typically found in small businesses, most larger businesses employ a more complex, **multitiered client/server architecture** (often called an *N-tier client/server architecture*) that balances the work of the network over multiple levels of servers, depending on the kind of service being requested.

Examine Figure 5.3, which shows an example of an n-tier network with two types of servers: a web server and an application server (note that the term "server" can refer to both the computer itself as well as the software that resides on the computer). The **web server** is

Figure 5.2 Eras in IT Infrastructure Evolution

Illustrated here are the typical computing configurations characterizing each of the five eras of IT infrastructure evolution.

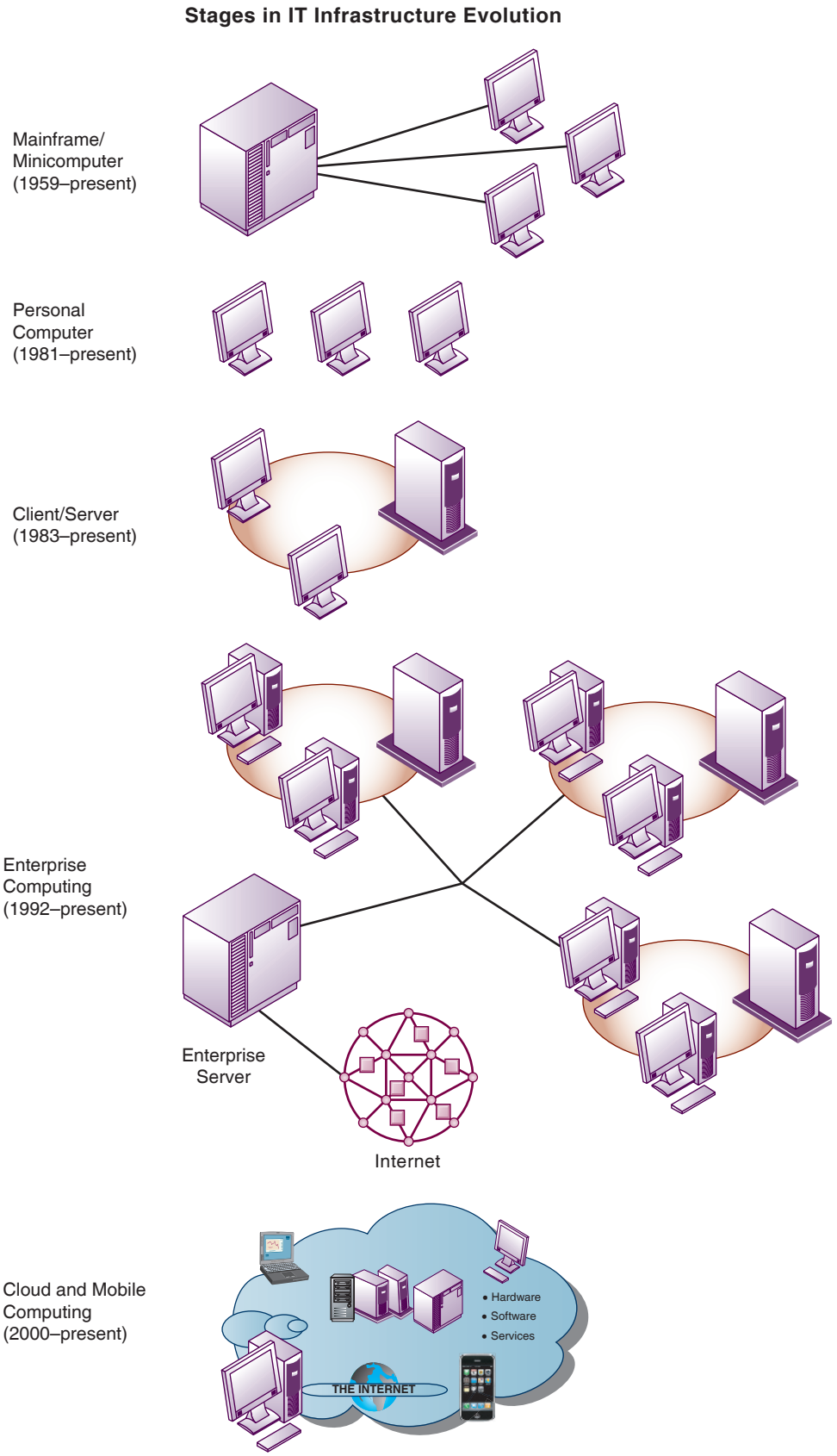
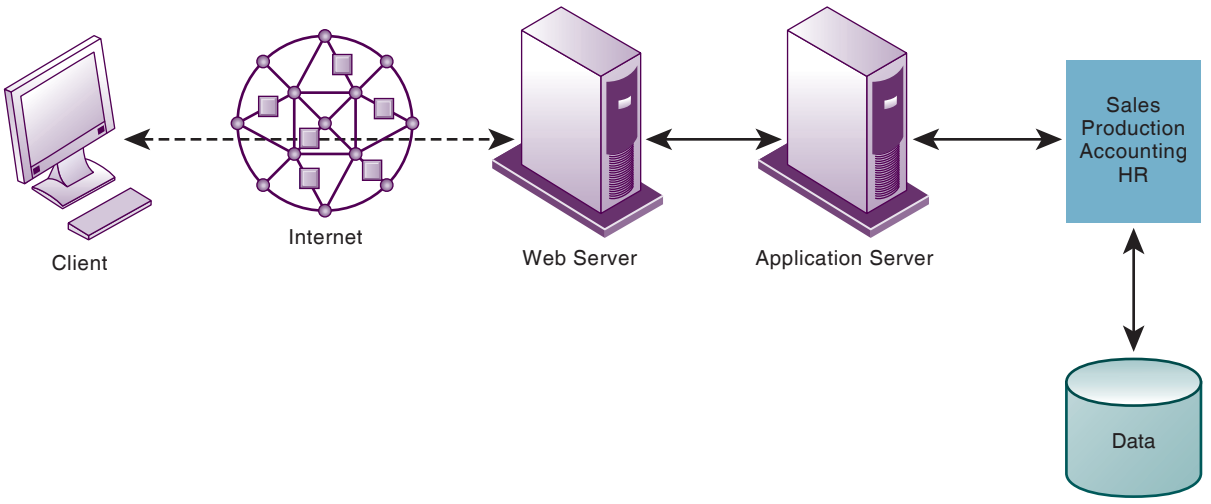


Table 5.1 Evolution of IT Infrastructure

Computing Era	Time Period	Infrastructure Description
Mainframe and Minicomputer Era	1959–present	Highly centralized computing under the control of professional programmers and systems operators, with most elements of infrastructure provided by a single vendor, the manufacturer of the hardware and the software, dominated by IBM. Smaller but powerful minicomputers support decentralized computing, customized to the specific needs of individual departments or business units. In recent years, the minicomputer has evolved into a midrange computer or midrange server and is part of a network.
Personal Computer Era	1981–present	Initially, small standalone desktop computers featuring desktop productivity software tools that were the first widely adopted by businesses and consumers. Primarily used Intel processors coupled with the Windows operating system.
Client/Server Era	1983–present	PCs called clients networked to powerful server computers with computer processing divided between both types of computers. Enables organizations to distribute computing work across the organization using a series of smaller inexpensive computers.
Enterprise Computing Era	1992–present	Network standards and protocols, including TCP/IP for the Internet, enable different pieces of computer hardware and smaller networks to link into an enterprise-wide network where information can flow freely across the organization and between the firm and other organizations.
Cloud and Mobile Computing Era	2000–present	Servers store data and software in remote cloud computing centers where they can be accessed by computers, smartphones, and other client devices linked to the Internet. Computer processing, data storage, and application development take place in cloud computing centers as services, so that subscribing companies can maintain their IT infrastructures remotely.

Figure 5.3 A Multitiered (N-Tier) Client/Server Network

In a multitiered client/server network, client requests for service are handled by different levels of servers.



Application server

Software that handles all application operations between a browser-based client and a company's back-end business applications or databases. Note that the term application server is also sometimes used to refer to the computer on which the application server software resides.

responsible for locating and managing stored web pages and delivering them to a client in response to a request for service. The **application server** handles all application operations between a browser-based client and an organization's back-end business applications or databases. So for instance, if the client requests access to data housed in a backend system (such as a product list or price information, for instance), the request is passed along to an application server. (Note that while the diagram shows the application server and the web server as two separate computers, the application server software may reside on the same computer as the web server.) Chapters 6 and 7 provide more detail on other types of software that are used in multitiered client/server architectures for e-commerce and e-business.

5.2 Describe the Components of IT Infrastructure

IT infrastructure today is composed of seven major components. Examine Figure 5.4, which illustrates these infrastructure components and the major vendors within each component category. These components constitute investments that must be coordinated with one another to provide the firm with a coherent infrastructure.

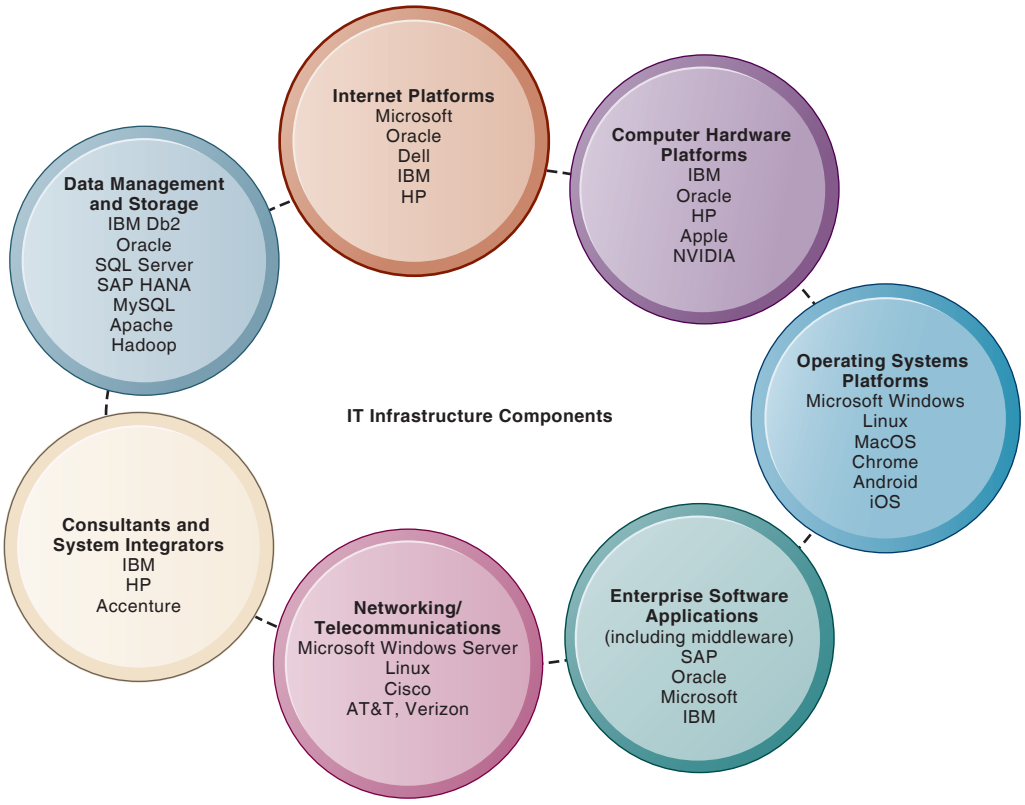
Computer Hardware Platforms

Computer hardware provides technology for computer processing, and includes mainframes, servers, desktop computers, tablets, smartphones, as well as sensors and wearable devices such as smartwatches. In the past, most computers used microprocessor chips manufactured or designed by Intel Corporation and, to a lesser extent, AMD Corporation. That is now changing, due to the surging demand for chips and other hardware components that can handle artificial intelligence (see Section 5.8). NVIDIA has emerged as a major computer hardware supplier because of its near-monopoly on Graphic Processing Units and other chips capable of handling the very large datasets required to train AI models and run AI applications.

The computer platform also changed dramatically with the introduction of mobile computing devices, from the iPod in 2001 to the iPhone in 2007 and the iPad in 2010. Worldwide, over 7 billion people use smartphones. You can think of these devices as a second computer hardware platform, one that is primarily consumer device–driven. Mobile devices are not required to perform as many tasks as traditional computers, so they consume less power and generate less heat. Processors for mobile devices are manufactured by a wide range of firms, including Apple, Samsung, and Qualcomm, using an architecture designed by ARM Holdings.

Figure 5.4 IT Infrastructure Components

There are seven major components that must be coordinated to provide the firm with a coherent IT infrastructure. Listed here are major technologies and suppliers for each component.



Mainframes have not disappeared. They continue to be used to reliably and securely handle huge volumes of transactions, for analyzing very large quantities of data, and for handling large workloads in cloud computing centers. The mainframe is still the digital workhorse for financial services, healthcare, manufacturing, and telecommunications networks that are often running software programs that are older and require a specific hardware platform. However, the number of providers has dwindled to one: IBM. IBM has also repurposed its mainframe systems so they can be used as giant servers for enterprise networks and corporate websites. A single IBM mainframe can run thousands of instances of Linux or Windows Server software and can replace thousands of smaller servers (see the discussion of virtualization in Section 5.6).

Operating system

System software that manages and controls the activities of the computer.

Microsoft Windows operating system

Microsoft's operating systems for Windows desktop computers and tablets.

Chrome OS

Google's lightweight operating system for cloud computing using a connected computer such as Google's Chromebook laptop and tablet computers.

Android

Open-source operating system for mobile devices such as smartphones and tablet computers, developed by the Open Handset Alliance led by Google; currently the most popular smartphone operating system worldwide.

iOS

Apple's operating system for the iPhone, iPad, and Apple Watch.

Multitouch

Interface that enables the use of one or more finger gestures to manipulate lists or objects on a screen without using a mouse or keyboard.

Microsoft Windows Server

Leading operating system for corporate servers in a Windows-based IT infrastructure.

Linux

Open-source operating system that is an offshoot of UNIX; is scalable, reliable, can run on many different hardware platforms and is available free or at very low cost.

UNIX

Older operating system for servers, mobile devices, and embedded systems that has been largely supplanted by Linux.

Operating System Platforms

An **operating system** is the system software that manages and controls the activities of a computer. Over 68 percent of desktop computers and tablets use some form of the **Microsoft Windows operating system**. However, there is now a much greater variety of client operating systems than in the past, with new operating systems for computing on mobile devices or cloud-connected computers.

Google's **Chrome OS** provides a lightweight operating system for cloud computing using a web-connected computer such as Google's Chromebook laptop and tablet computers. Programs and user data are stored online rather than on the user's computer and are accessed via the Chrome web browser. **Android** is an open-source operating system for mobile devices such as smartphones and tablet computers, developed by the Open Handset Alliance led by Google. It has become the most popular smartphone platform worldwide, competing with **iOS**, Apple's operating system for the iPhone, iPad, and Apple Watch. Conventional client operating system software has become more natural and intuitive by using touch technology. For instance, iOS features a **multitouch** interface, which enables the use of one or more finger gestures to manipulate lists or objects on a screen without using a mouse or keyboard.

The leading operating systems for corporate servers are Microsoft Windows Server and Linux. **Microsoft Windows Server** can provide enterprise-wide operating system and network services and appeals to organizations seeking Windows-based IT infrastructures. **Linux** is an open-source operating system that is an offshoot of **UNIX**, an older operating system for servers, mobile devices, and embedded systems. Linux is scalable, reliable, and can run on many different hardware platforms, and is available for free or at a very low cost. Linux is widely used for networks, servers, and enterprise-level tasks, and has largely supplanted UNIX. Today, 90 percent of the world's cloud computing infrastructure runs on Linux. However, UNIX elements have been incorporated into other modern operating systems, including Android and Mac OS X.

Enterprise Software Applications

Software for enterprise applications is an important component of IT infrastructure. We introduced the various types of enterprise applications in Chapter 2, and Chapter 9 provides a more detailed discussion of each. The largest providers of enterprise application software are SAP and Oracle. Also included in this category is middleware software supplied by vendors such as IBM and Oracle for achieving firmwide integration by linking the firm's existing application systems. Microsoft is attempting to move into the lower ends of this market by focusing on small and medium-sized businesses.

Data Management and Storage

Enterprise database management software is responsible for organizing and managing the firm's data so that it can be efficiently accessed and used for operations and decision making. Chapter 6 describes this software in detail. The leading database software providers are IBM (Db2), Oracle, Microsoft (SQL Server), and SAP (HANA). There are also cloud versions of these tools. MySQL is a Linux open-source relational database product now owned by Oracle Corporation, and Apache Hadoop is an open-source software framework for managing very large data sets (see Chapter 6).

Networking/Telecommunications Platforms

Network operating system software (NOS software) is software that connects multiple devices and computers on a network and allows them to share resources on the network. Leading operating systems for networks include Windows Server and Linux. Leading telecommunications equipment providers include Huawei, Nokia, Ericsson, and Cisco Systems. Telecommunications platforms are typically provided by telecommunications/telephone services companies that offer voice and data connectivity, wide area networking, wireless services, and Internet access. Leading telecommunications service providers include AT&T and Verizon. This market is exploding with new providers of cellular wireless, high-speed Internet, and Internet telephone services.

Network operating system software (NOS software)
Software that connects multiple devices and computers on a network and allows them to share resources on the network.

Internet Platforms

Internet platforms include hardware, software, and services to support a firm's website. Firms may create their own Internet platform in-house or outsource some or all of its functions to a service provider, such as a **web hosting service**, which maintains websites for fee-paying clients.

Web hosting service
Service provider that maintains websites for fee-paying clients.

The Internet created a veritable explosion in server computers, with many firms collecting thousands of small servers to run their Internet operations. There has been a steady push to reduce the number of server computers by increasing the size and power of each and by using software tools that make it possible to run more applications on a single server. Use of standalone server computers is decreasing as organizations transition to cloud computing services. The hardware server market has become increasingly concentrated in the hands of Dell, Cisco, HPE, IBM, and Oracle.

The major website and app software application development tools and suites are supplied by Microsoft (such as Microsoft Visual Studio and the Microsoft .NET development platform), Oracle, and a host of independent software developers. Chapter 7 describes the components of a firm's Internet platform in greater detail.

Consulting and System Integration Services

Today, even a large firm does not have the staff, the skills, the budget, or the necessary experience to deploy and maintain its entire IT infrastructure. Implementing a new infrastructure requires (as noted in Chapters 13 and 14) significant changes in business processes and procedures, training and education, and software integration. Leading consulting firms providing this expertise include Accenture, IBM Services, HPE, Infosys, and Wipro.

Software integration means ensuring the new infrastructure works with the firm's older, so-called legacy systems and ensuring the new elements of the infrastructure work with one another. A **legacy system** is generally an older system that continues to be used to avoid the high cost of replacing or redesigning it. Replacing legacy systems is cost-prohibitive and generally not necessary if they can be integrated into a contemporary infrastructure.

Legacy system
Generally, an older system that continues to be used to avoid the high cost of replacing or redesigning it.

In the past, technology vendors supplying infrastructure components offered purchasing firms a mixture of incompatible, proprietary, partial solutions that could not work with other vendors' products. Increasingly, vendor firms have been forced to cooperate in strategic partnerships with one another in order to keep their customers. For instance, a hardware and services provider such as IBM cooperates with all the major enterprise software providers, has strategic relationships with system integrators, and promises to work with whichever data management products its client firms wish to use (even though it sells its own database management software called Db2).

Another big change is that companies are moving more of their IT infrastructure to the cloud or to outside services, owning and managing much less on their premises. Firms' IT infrastructure will increasingly be an amalgam of components and services that are partially owned, partially rented or licensed, partially located on-site, and partially supplied by external vendors or cloud services.

5.3 Discuss the Mobile Platform, Apps, and BYOD

The exploding power of computer hardware and networking technology has dramatically changed how businesses organize their computing power, putting more of this power on networks and mobile devices and obtaining more of their computing capabilities in the form of services.

The Mobile Platform

A mobile computing platform has emerged as an alternative to desktop computers and larger computers. The iPhone and Android smartphones have taken on many functions of desktop computers, including transmitting data, surfing the web, transmitting email and instant messages, displaying digital content, and exchanging data with internal corporate systems. The mobile platform also includes small, lightweight subnotebooks such as the Google Chromebook, tablet computers such as the iPad, car infotainment systems, wearable devices, and digital e-book readers such as Amazon's Kindle with some web access capabilities.

Smartphones and tablets are becoming the primary means of accessing the Internet and are increasingly used for business computing as well as for consumer applications. For example, PepsiCo managers use iPads customized to monitor their teams' performance; pull up pricing, planograms, and contracts; and coordinate deliveries with merchandising.

Wearable computing devices are a recent addition to the mobile platform. These include smartwatches, smart glasses, smart badges, and activity trackers. Wearable computing devices have both consumer and business uses. They improve productivity by delivering information to workers in warehouses or in the field without interrupting their workflow.

Apps

App
Software program that can be accessed via the Internet.

An **app** is a software program that can be accessed via the Internet. Google refers to its online services as apps. But when we talk about apps today, most of the attention goes to the apps that have been developed for the mobile platform. It is these apps that turn smartphones and other mobile devices into general-purpose computing tools. There are now millions of apps for the iOS and Android operating systems.

Some apps, once downloaded, do not access the web, but many do, providing a streamlined non-browser pathway that enables faster access to online content and services than traditional web browsers. Today, adults in the United States spend an average of 4.5 hours a day looking at their mobile phones (Volz, 2024). As a result, many companies have redesigned their websites to mimic the look, feel, and functionality of apps.

Many apps are free or can be purchased for a small fee, much less than conventional software, which further adds to their appeal. The success of mobile platforms depends in large part on the quantity and quality of the apps they provide. Apps tie the customer to a specific platform. For example, apps that use the iOS operating system for Apple devices are not compatible with Android devices, and apps that use the Android system cannot run on Apple devices. As users add more and more apps to their mobile devices, the cost of switching to a competing mobile platform rises.

The most frequently downloaded categories of consumer apps include games, news and weather, maps/navigation, social media, music, and video/movies, as well as apps for researching and buying goods and services online. There are also apps for business users that make it possible to create and edit documents, connect to corporate systems, schedule and participate in meetings, track shipments, and dictate voice messages.

Consumerization of IT and BYOD

The popularity, ease of use, and rich array of useful applications for smartphones and tablet computers have created a groundswell of interest in allowing employees to use

their own personal computing devices in the workplace, a phenomenon popularly called **BYOD ("Bring Your Own Device")**. BYOD is one aspect of the **consumerization of IT**, in which new information technology that first emerges in the consumer market spreads into business organizations. Consumerization of IT includes not only mobile personal devices but also business uses of software services that originated in the consumer marketplace as well, such as Google Search, Gmail, Google Maps, Dropbox, and even Facebook and TikTok.

BYOD

Stands for "bring your own device" and refers to employees using their own personal computing devices in the workplace.

Consumerization of IT

New information technology originating in the consumer market that spreads to business organizations.

Consumerization of IT is forcing businesses to rethink the way they obtain and manage information technology equipment and services. Historically, at least in large firms, the IT department controlled the selection and management of the firm's hardware and software. This ensured that information systems were protected and served the purposes of the firm. Today, employees and business departments are playing a much larger role in technology selection, in many cases demanding that employees be able to use their own personal mobile devices to access the corporate network. Although consumer technologies provide new tools to foster creativity, collaboration, and productivity, they are more difficult for firms to manage and control.

When employees bring their own devices to use at work or use them to work remotely, IT departments risk losing control over the hardware. Without policies or tools for managing mobile devices, the organization can't control what apps or programs are installed, how the devices are secured, or what files are downloaded. Controlling who has access to what technology is an important method of preventing privacy breaches, hacking, and unauthorized access to corporate information. Inability to control the hardware means more vulnerabilities. That is the big trade-off with BYOD: offering employees greater flexibility while potentially exposing the company to more risk.

When employees bring their own devices to work, they may be tempted to use them on the job for entertainment or catching up with friends. It's incredibly easy for employees to get sucked into an endless black hole of text messaging, YouTube videos, and checking Facebook updates. Productivity may suffer.

BYOD requires a significant portion of corporate IT resources dedicated to managing and maintaining a large number of devices within the organization. The mobile landscape is complicated, with a variety of devices and operating systems on the market. Apple's iOS is considered a closed system and runs only on a limited number of different Apple devices. Android dominates the world smartphone market, but there are many thousands of different models of Android-based devices available around the world.

If employees are allowed to work with more than one type of mobile device and operating system, companies need an effective way to keep track of all the devices employees are using. To access company information, the company's networks must be configured to receive connections from that device. When employees make changes to their personal phone, such as switching cellular carriers, changing their phone number, or buying a new mobile device altogether, companies will need to quickly and flexibly ensure that their employees are still able to remain productive. Firms need a system that keeps track of which devices employees are using, where the device is located, whether it is being used, and what software it is equipped with. For unprepared companies, keeping track of who gets access to what data could be a nightmare.

Securing company information accessed with mobile devices becomes more challenging. If a device is stolen or compromised, companies need ways to ensure that sensitive or confidential information isn't freely available to anyone. Mobility puts assets and data at greater risk than if they were only located within company walls and on company machines. Companies often use technologies that allow them to wipe data from devices remotely or encrypt data so that if the device is stolen, it cannot be used.

5.4 Discuss Quantum Computing and Green Computing

Quantum computing and green computing are two other contemporary infrastructure trends. Quantum computing can achieve exponential increases in computing power by working with

data at the subatomic level. Green computing consists of practices and technologies for maximizing energy efficiency and minimizing environmental impact in the ways computer chips, systems, and software are designed and used.

What Is Quantum Computing?

Quantum computing
Use of principles of quantum mechanics to represent data and perform operations on the data.

Quantum computing uses the principles of quantum mechanics—the science of how matter and light behave at atomic and subatomic levels—to represent data and perform operations on that data. While conventional computers handle bits of data either as 0 or 1 but not both, quantum computing uses the subatomic qubit as its basic unit of information. A qubit can carry information in a quantum state as 0, 1, or both simultaneously. A single qubit can perform two different calculations simultaneously, two qubits can perform four, and three qubits eight, and so on, creating an exponential increase in processing power with each added qubit. Even with a relatively small number of qubits, quantum computers could outperform the most powerful supercomputers today. A quantum computer gains enormous processing power through this ability to be in multiple states at once, allowing it to solve some scientific and business problems many millions of times more rapidly than can be done with today's high-performance computers.

The current leader in quantum computing is IBM. IBM recently launched its Quantum System Two, a modular quantum computer that improves error correction by reducing decoherence, a phenomenon that leads to the loss of quantum states in a quantum system. IBM hopes to produce a 100,000-qubit quantum system by 2033. Microsoft Azure provides a comprehensive set of tools and resources for quantum computing, with access to quantum hardware, simulators, and development tools, allowing users to experiment with quantum computing tools and algorithms. Google aims to build a quantum system with 1 million qubits within a decade. Google Quantum AI's open-source library Cirq is designed for developing quantum algorithms for near-term quantum computers. Other leading companies working on quantum computing projects include Rolls-Royce, Deloitte, Roche, Fujitsu, JPMorgan Chase, and BMW.

Quantum computers require a significant amount of energy and cooling to run properly. Quantum computing hardware is mostly composed of cooling systems that keep a superconducting processor at a specific super-cooled temperature. For example, IBM has used coolant fluid to keep its quantum-ready system at -459 degrees Fahrenheit. At this super-low temperature, electrons can flow through superconductors.

Benefits and Limitations of Quantum Computing

Table 5.2 describes the benefits and also the limitations of quantum computing and helps explain why quantum computers are so challenging to work with. Most quantum computing projects today are still experimental. One exception is D-Wave, which launched what it described as the world's first commercially available quantum computer in 2011. D-Wave has pioneered in developing quantum applications that can be used in real-world work settings, as well as tools and services for building these quantum applications. D-Wave customers are developing quantum computing solutions for logistics, portfolio optimization, drug discovery, scheduling, fault detection, traffic management, and supply chain management.

Green computing
Refers to practices and technologies for designing, manufacturing, using, and disposing of computers and other IT equipment in order to minimize impact on the environment. Also referred to as green IT.

Green Computing

Green computing (also referred to as *green IT*) refers to practices and technologies for designing, manufacturing, using, and disposing of computers and other IT equipment in order to minimize their impact on the environment. Reducing computer power consumption has been a very high green priority. A **data center**, which is a facility that houses computer systems and associated components such as telecommunications, storage, security systems, and backup power supplies, can easily consume more than 100 times more power than a standard office building consumes. All this additional power consumption adds to the amount of greenhouse gases emitted and also increases corporate operating costs.

Data center
Facility that houses computer systems and associated components such as telecommunications, storage, security systems, and backup power supplies.

Table 5.2 Quantum Computing Benefits and Limitations

Quantum Computing Benefits	Description
Speed	Quantum computers are up to 158 times faster than conventional computers.
Ability to solve more complex problems	Quantum computers are designed to perform many complex calculations simultaneously. Useful for developing decryption technologies.
Simulations	Quantum computers can run more complex simulations and can simulate more intricate systems than classical computers.
Optimizations	Quantum computing's ability to process huge amounts of complex data can make artificial intelligence and machine learning much more powerful.
Quantum Computing Limitations	**Description**
Very heavy energy and resource requirements	Quantum computers require a significant amount of energy and cooling to run properly. Quantum computing cooling systems must keep a superconducting processor at a super-cooled temperature to allow electrons to flow freely. (IBM cools its quantum system to –459 degrees Fahrenheit.)
Error correction	Qubits can't use conventional error correction, which is critical in quantum computing. A single error in a calculation can affect the validity of an entire computation. Researchers are working on quantum error-correction algorithms.
Interference	The slightest disturbance in a quantum system can cause a quantum computation to collapse. A quantum computer must be totally isolated from all external interference during the computation phase.
Output observance	Retrieving output data after a quantum calculation is complete risks corrupting the data. Database search algorithms are being developed to address this problem.
Security	Quantum computing poses a profound threat to current classical encryption protocols that enable virtually all digital transactions. Over the next several years, widely used data encryption mechanisms, such as public key cryptography, could become vulnerable.

Data centers are now being designed with energy efficiency in mind, using state-of-the-art air-cooling techniques, energy-efficient equipment, and other energy-saving practices. Organizations are achieving additional efficiencies by migrating from on-premise data centers to cloud computing. Large companies such as Microsoft, Google, Meta, and Apple are starting to reduce their carbon footprints with clean energy–powered data centers, including extensive use of wind and hydropower. The chapter-ending case study provides more details on this topic, as well as the sustainability challenges posed by resource-hungry artificial intelligence (AI) applications.

Table 5.3 provides more detail about green computing practices for reducing energy usage, carbon production, and e-waste. **E-waste (electronic waste)** consists of all types of end-discarded electronic equipment, such as computers and communications equipment, household appliances, entertainment and consumer electronic equipment, and other devices powered by electricity.

E-waste (electronic waste)
All types of end-discarded electronic equipment.

Table 5.3 Examples of Green Computing Practices

Green practice	Description
Energy-efficient hardware	Energy-efficient computers and data centers that reduce power consumption and carbon emissions. This includes energy-efficient computing devices of all sizes, including smartphones, as well as energy-efficient microprocessors such as AMD Ryzen processors that minimize power consumption
Eliminating e-waste	Recycling and disposal of electronic waste according to federal, state, and local regulations
Optimizing heating, cooling, and electrical systems	Using IoT sensors and AI to collect and analyze information about power usage in a data center or a business and autonomously manage and optimize heating, cooling, and electrical systems
Using alternative energy sources and cooling methods	Using alternative energy sources, such as wind and hydroelectric power, as well as geothermal cooling and other new methods of cooling data centers.
Using and developing sustainable software	Includes sustainability and ESG reporting software, such as carbon accounting and carbon management software, and environmental management software. Also includes sustainable software engineering practices for defining, building, and running sustainable software applications that require less energy to run, using green coding practices such as optimizing algorithms, reducing code complexity, and minimizing the use of computing resources
Supporting remote work	Remote work decreases the number of people commuting to and from work, and the carbon emissions from their travel. It also reduces emissions from maintaining buildings for on-site work.
Server virtualization	Enabling multiple virtual machines to operate on a single physical server, reducing the need for multiple servers. More efficient use of server capacity decreases energy usage and CO_2 emissions.

5.5 Discuss Virtualization and Cloud Computing

Virtualization uses special software to create virtual representations of servers, storage, networks, and other physical devices on a single physical computer, providing the foundation for cloud computing.

Virtualization

Virtualization
A method of presenting a set of computing resources so that they can be accessed in ways that are not restricted by physical configuration or geographic location.

Virtualization is a method of presenting a set of computing resources (such as computing power or data storage) so that they can all be accessed in ways that are not restricted by physical configuration or geographic location. Virtualization enables a single physical resource (such as a server or a storage device) to appear to the user as multiple logical resources. For example, a server or mainframe can be configured to run many instances of an operating system (or different operating systems) so that it acts like many different machines. Each virtual server "looks" like a real physical server to software programs, and multiple virtual servers can run in parallel on a single machine. VMware is the leading virtualization software vendor for Windows and Linux servers.

Server virtualization is a common method of reducing technology costs by providing the ability to host multiple systems on a single physical machine. Most servers run at just 15 to 20 percent of capacity, and virtualization can boost server utilization rates to 70 percent or higher. Higher utilization rates translate into fewer computers required to process the same amount of work, reduced data center space to house machines, and lower energy usage. Virtualization also facilitates centralization and consolidation of hardware administration.

Virtualization also enables multiple physical resources (such as storage devices or servers) to appear as a single logical resource, as in **software-defined storage (SDS)**, which is a storage architecture that separates the software for managing data storage from the physical storage hardware. Using software, firms can pool and arrange multiple storage infrastructure resources and efficiently allocate them to meet specific application needs. SDS enables firms to replace expensive storage hardware with lower-cost commodity hardware and cloud storage hardware. There is less under- or over-utilization of storage resources.

Software-defined storage (SDS)

Storage architecture that separates the software for managing data storage from the physical storage hardware.

Cloud Computing and Virtualization

Virtualization plays a critical role in cloud computing by allowing users to create separate computing environments within the same physical instance of a resource, as described earlier in our discussion of server virtualization. The ability to abstract physical resources into logical ones improves resource allocation, deployment, and flexibility across storage, compute, and network layers of IT infrastructure. Virtualization also enables the sharing of hardware and software resources among virtual machines, which can lead to increased efficiency, cost reduction, and scalability.

It is now possible for companies and individuals to perform all of their computing work using a virtualized IT infrastructure in a remote location, as is the case with cloud computing. **Cloud computing** is a model of computing in which computer processing, storage, software, and other services are provided as a shared pool of virtualized resources over a network, primarily the Internet. These "clouds" of computing resources can be accessed on an as-needed basis from any connected device and location. Examine Figure 5.5, which illustrates the cloud computing concept and identifies the types of services it can provide.

Cloud computing

Model of computing in which computer processing, storage, software, and other services are provided as a shared pool of virtualized resources over a network, primarily the Internet.

The U.S. National Institute of Standards and Technology (NIST) defines cloud computing as having the following essential characteristics (Mell and Grance, 2009):

- **On-demand self-service:** Consumers can obtain computing capabilities such as server time or network storage as needed automatically on their own.
- **Ubiquitous network access:** Cloud resources can be accessed using standard network and Internet devices, including mobile platforms.
- **Location-independent resource pooling:** Computing resources are pooled to serve multiple users, with different virtual resources dynamically assigned according to user demand. The user generally does not know where the computing resources are located.
- **Rapid elasticity:** Computing resources can be rapidly provisioned, increased, or decreased to meet changing user demand.
- **Measured service:** Charges for cloud resources are based on the amount of resources actually used.

Cloud computing can be used to provide three different types of services:

- **Infrastructure as a service (IaaS)**: Customers use processing, storage, networking, and other computing resources from cloud service providers to run their information systems. For example, Amazon uses the spare capacity of its IT infrastructure to provide a broadly based cloud environment selling IT infrastructure services. These include its Simple Storage Service (S3) for storing customers' data and its Elastic Compute Cloud (EC2) service for running their applications. Users pay only for the amount of computing and storage resources they actually use. Examine Figure 5.6, which illustrates the range of services Amazon Web Services offers.
- **Software as a service (SaaS)**: Customers use software hosted by the vendor on the vendor's cloud infrastructure and delivered as a service over a network. Vendors typically

Infrastructure as a service (IaaS)

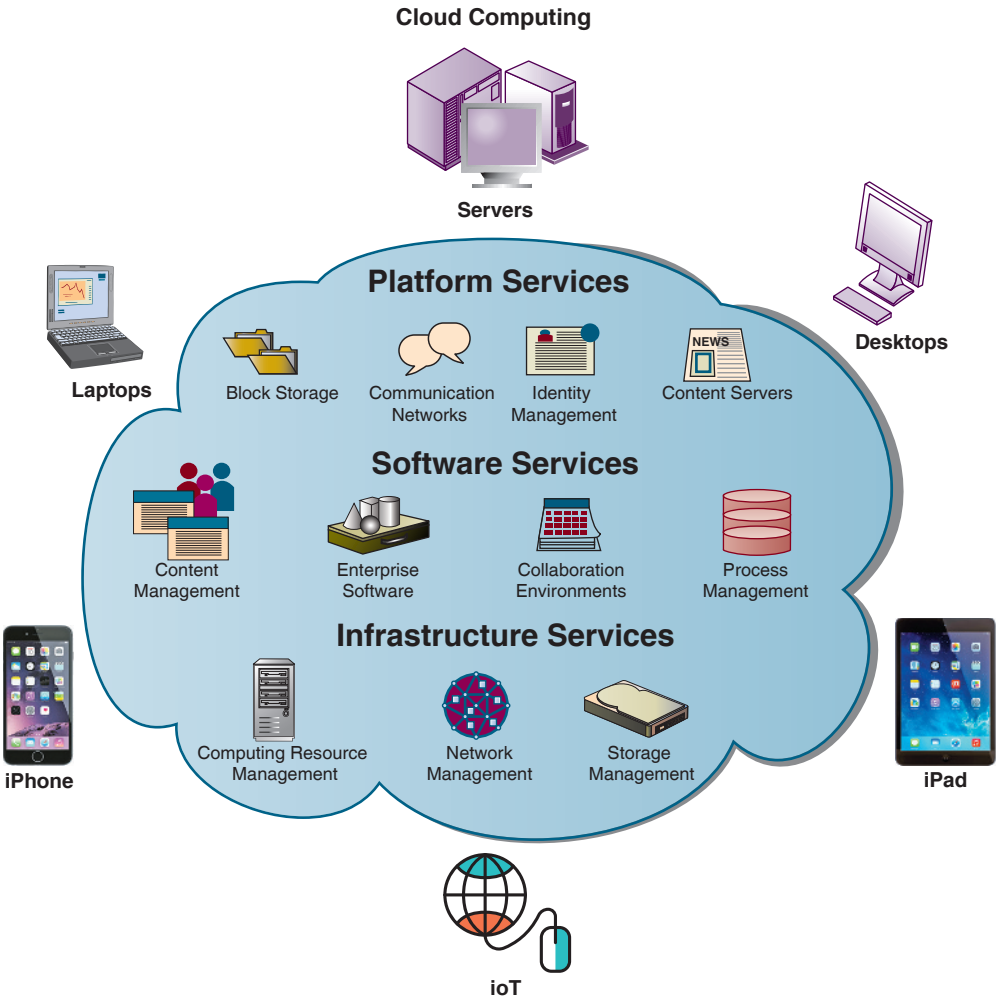
Computing infrastructure provisioned from a cloud service provider, which maintains and operates it; users pay only for the amount of computing and storage resources they actually use.

Software as a service (SaaS)

Software hosted by a vendor on the vendor's cloud infrastructure and delivered as a service on a subscription fee basis to users who access the software via a web browser; data and software are maintained on the vendors' remote servers.

Figure 5.5 Cloud Computing Platform

In cloud computing, hardware and software capabilities are a pool of virtualized resources provided over a network, often the Internet. Businesses and employees have access to applications and IT infrastructure anywhere and at any time.



Manaemedia/123RF

charge users a subscription fee, although they may also provide a pared-down free version. Users access these applications from a web browser, and the data and software are maintained on the vendors' remote servers. Leading SaaS examples include Google Workspace, Microsoft 365, and related software services for desktop productivity and collaboration. These are among the most popular software services for consumers, although they are increasingly used in business. Salesforce is a leading provider of SaaS for business. Salesforce provides customer relationship management (CRM) and other application software solutions as software services that are licensed to the user and provided over the Internet. Its sales and service clouds offer applications for improving sales and customer service.

Platform as a service (PaaS)

Platform that provides infrastructure and programming tools that enable customers to develop, run, and manage applications without having to build and maintain the infrastructure typically associated with developing an application.

- **Platform as a service (PaaS)**: Customers use infrastructure and programming tools supported by the cloud service provider to develop their own applications. For example, Microsoft offers PaaS tools and services for software development and testing as part of its Azure cloud services. Another example is Salesforce's Salesforce Platform. The Salesforce Platform gives users the ability to develop, launch, and manage applications without having to deal with the infrastructure required for creating new software. The Salesforce Platform provides a set of development tools and IT services that enable users to build new applications and run them in the cloud on Salesforce's data center infrastructure. Salesforce also lists software from other independent developers on its AppExchange, an online marketplace for third-party applications that run on the Salesforce Platform.

Figure 5.6 Amazon Web Services

Amazon Web Services (AWS) offers a collection of web services to users of its cloud platform. AWS is the world's largest provider of cloud computing services.

Cloud computing has become an affordable and sensible option for companies of all sizes, ranging from tiny Internet startups to established companies like Walmart and FedEx. According to an Edge Delta report, over 90 percent of companies worldwide use some form of cloud services, including cloud software services such as Microsoft 365 or Salesforce, and complete cloud computing infrastructures that replace on-premise data centers (Edge Delta, 2024).

Cloud computing has a number of advantages. For example, AWS provides companies with flexible computing power and data storage as well as data management, messaging, payment, and other services that can be used together or individually, as the business requires. Anyone with an Internet connection and a little bit of money can harness the same computing systems that Amazon itself uses to run its retail business. If customers provide specifications on the amount of server space, bandwidth, storage, and any other services they require, AWS can automatically allocate those resources. You don't pay a monthly or yearly fee to use Amazon's computing resources. Instead, you pay for exactly what you use. Economies of scale keep costs low.

Cloud computing also appeals to many businesses because the cloud services provider will handle all of the maintenance and upkeep of their IT infrastructures, allowing these businesses to spend more time on higher-value work. Start-up companies and smaller companies are finding that they no longer need to build their own data centers. With cloud infrastructures like Amazon's readily available, they have access to technical capability that was formerly available only to much larger businesses.

Cloud computing has some drawbacks. Unless users make provisions for storing their data locally, the responsibility for data storage and control is in the hands of the provider. Some companies worry about the security risks related to entrusting their critical data and systems to an outside vendor that also works with other companies. Companies expect their systems to be available 24/7 and do not want to suffer any loss of business capability if cloud infrastructures malfunction. Nevertheless, the trend is for companies to shift more of their computer processing and storage to some form of cloud infrastructure. Startups and small companies with limited IT resources and budgets will find public cloud services especially helpful.

Public cloud

Cloud computing model in which the cloud infrastructure is owned and maintained by a cloud service provider and made available to organizations and the general public.

There are several different cloud computing models. A **public cloud** is owned and maintained by a cloud service provider, such as Amazon Web Services, and made available to organizations and the general public. Public cloud services are often used for websites with public information and product descriptions, one-time large computing projects, developing

and testing new applications, and consumer services such as online storage of data, music, and photos. Google Drive, Dropbox, and Apple iCloud are leading examples of these consumer public cloud services.

Private cloud

Cloud computing model in which cloud infrastructure is operated solely for the benefit of a single organization.

A **private cloud** is operated solely for the benefit of a single organization. It may be managed by the organization or a third party and may be hosted either internally or externally. Like public clouds, private clouds can allocate storage, computing power, or other resources seamlessly to provide computing resources on an as-needed basis. Companies that want flexible IT resources and a cloud service model while retaining control over their own IT infrastructure are gravitating toward these private clouds.

Industry cloud platform

Cloud computing model in which specialized cloud services are provided to meet the specific requirements of a particular industry such as healthcare, finance, or manufacturing. Also known as a vertical cloud platform.

An **industry cloud platform**, also known as vertical cloud platform, provides specialized cloud services to meet the specific requirements of a particular industry, such as healthcare, finance, or manufacturing. Unlike general public clouds, which provide a broad set of tools and services for various applications that can be used by many different kinds of businesses, industry clouds are tailored to a particular sector. For example, Salesforce provides an array of industry-specific clouds, including industry clouds for government, education, health, energy and utilities, and financial services.

On-demand computing

The purchase of computing services from remote providers and paying only for computing resources actually used.

Because organizations using public clouds do not own the infrastructure, they do not have to make large investments in their own hardware and software. Instead, they purchase their computing services from remote providers and pay only for the computing resources they actually use. The term **on-demand computing** has been used to describe such services.

Although cloud computing has been touted as a less expensive yet more flexible alternative to buying and owning information technology, this isn't always the case. For large companies, paying a public cloud provider a monthly service fee for 10,000 or more employees may actually be more expensive than having the company maintain its own IT infrastructure and staff. Companies also worry about unexpected "runaway costs" from using a pay-per-use model. Integrating cloud services with existing IT infrastructures, errors, mismanagement, or unusually high volumes of web traffic will run up the bill for cloud service users. When an organization decides to transform its IT infrastructure from in-house to a cloud-based setup, applications may require a major upgrade or need to be retired and replaced.

Many large companies moving more of their computing to the cloud are unable to migrate completely. Legacy systems are the most difficult to switch over. Most midsized and large companies will gravitate toward a hybrid approach. The top cloud providers themselves—Amazon, Google, Microsoft, and IBM—use their own public cloud services for some purposes, but they continue to keep certain functions on private servers. Worries about reliability, security, and risks of change have made it difficult for them to move critical computing tasks to the public cloud.

Hybrid cloud

Computing model where firms use both their own IT infrastructure for their most essential core activities and public cloud computing for less critical systems or for additional processing capacity.

Large firms are most likely to adopt a **hybrid cloud** computing model where they use their own infrastructure for their most essential core activities and adopt public cloud computing for less critical systems or for additional processing capacity during peak business periods. Table 5.4 compares the three major types of cloud computing models. Cloud computing will gradually shift firms from having a fixed infrastructure capacity toward a more flexible infrastructure, some of it owned by the firm and some of it provided by cloud service providers.

Cloud AI Services

All of the major cloud computing vendors, such as Amazon (AWS), Microsoft Azure, Google Cloud Platform (GCP), Microsoft Azure, and Salesforce, offer a wide range of AI services. Table 5.5 compares their strengths and capabilities.

Edge Computing

Edge computing

Method of optimizing cloud computing systems by performing some data processing on a set of linked servers at the edge of the network, near the source of the data.

Having all the laptops, smartphones, tablets, wireless sensor networks, and local on-premise servers used in cloud computing systems interacting with a single central public cloud data center to process all their data can be inefficient and costly. **Edge computing** is a method of optimizing cloud computing systems by performing some data processing on a set of linked servers at the

Table 5.4 Cloud Computing Models Compared

Type of Cloud	Description	Managed by	Uses
Public cloud	Third-party service offering computing, storage, and software services to multiple customers and that is available to the public	Third-party service providers	Companies without major privacy concerns Companies seeking pay-as-you-go IT services Companies lacking IT resources and expertise
Private cloud	Cloud infrastructure operated solely for a single organization and hosted either internally or externally	In-house IT or private third-party host	Companies with stringent privacy and security requirements Companies that must have control over data sovereignty
Hybrid cloud	Combination of private and public cloud services that remain separate entities	In-house IT, private host, third-party providers	Companies requiring some in-house control of IT that are also willing to assign part of their IT infrastructures to a public cloud

Table 5.5 Major Cloud AI Services

Cloud Computing Vendor	Capabilities
Amazon Web Services (AWS)	Amazon SageMaker offers fully managed machine learning services. Includes tools for building, training, and deploying machine learning models, as well as pre-built functionalities for computer vision, natural language processing, and more.
Microsoft Azure	Microsoft Azure's AI platform focuses on enterprise solutions. Offerings include pre-trained models, cognitive services, and developer tools. Excels in integration with existing Microsoft products.
Google Cloud Platform (GCP)	GCP leverages Google's expertise in machine learning research. It is known for strong AI tools like AutoML, which simplifies model building, and TensorFlow, a popular open-source machine learning library.
Salesforce	Salesforce specializes in AI services specifically designed for the Salesforce CRM platform. These services focus on sales and customer service automation, such as lead scoring and sentiment analysis

edge of the network, near the source of the data. This reduces the amount of data flowing back and forth between local computers and other devices and the central cloud data center.

Edge computing deployments are useful when sensors or other IoT devices do not need to be constantly connected to a central cloud. For example, an oil rig in the ocean might have thousands of sensors producing large amounts of data, perhaps to confirm that systems are working properly. The data do not necessarily need to be sent over a network as soon as they are produced, so the local edge computing system could compile the data and send daily reports to a central data center or cloud for long-term storage. By only sending important data over the network, the edge computing system reduces the amount of data traversing the network.

Edge computing also reduces delays in transmitting and processing large quantities of data needed for AI applications because the data does not have to travel over a network to a remote data center or cloud for processing. This is ideal for situations where delays of milliseconds can be untenable, such as in financial services, manufacturing, or autonomous vehicles.

5.6 Describe Contemporary Software Platforms

There are three major themes in contemporary software platform evolution:

- Open-source software
- Java, HTML, and HTML5
- Web services

Open-Source Software

Open-source software
Software that provides free access to its program code, allowing users to modify the program code to make improvements or fix errors.

Open-source software is software produced by a community of thousands of programmers around the world. Open-source software is free and can be modified by users to improve the software and to come up with their own innovations. Works derived from the original code must also be free. Open-source software is, by definition, not restricted to any specific operating system or hardware technology.

Popular open-source software tools include the Linux operating system, the Apache HTTP web server, the Mozilla Firefox web browser, and the Apache OpenOffice desktop productivity suite. Google's Android mobile operating system and Chrome web browser are based on open-source tools. Because of its flexibility, stability, low cost, and capacity for continuous improvement, open-source software has become a mainstream software development tool. You can find out more about the benefits of open-source software development in the discussion of software development tools in Chapter 13.

Linux

Perhaps the most well-known open-source software is Linux, an operating system related to UNIX that we have previously discussed in Section 5.2. Linux was created by Finnish programmer Linus Torvalds and first posted on the Internet in 1991. Linux applications are now embedded in smartphones, tablet computers, and consumer electronics, and Linux has also become an important operating system for servers, mainframe computers, and supercomputers, including infrastructures for developing and running AI applications. Linux is available in free versions downloadable from the Internet or in low-cost commercial versions that include tools and support from vendors such as Red Hat. It is considered to be very reliable and resilient.

Linux can work on all the major hardware platforms and is very customizable. It can be tailored to a wide range of needs. Users can choose from a large number of customized versions of Linux to fit their specific purposes, and they can further customize the software to their liking.

Software for the Web: Java, HTML, and HTML5

Java
Operating system-independent, processor-independent, object-oriented programming language that has become a leading interactive programming environment for the web.

Java is an operating system-independent, processor-independent, object-oriented programming language created by Sun Microsystems that has become a leading interactive programming environment for the web. Java is now owned by Oracle. The Java platform has migrated into mobile phones, tablets, automobiles, music players, game machines, and set-top cable television systems serving interactive content and pay-per-view services. Java software is designed to run on any computer or computing device, regardless of the specific processor or operating system the device uses. For each of the computing environments in which Java is used, a Java Virtual Machine interprets Java programming code for that machine. In this manner, the code is written once and can be used on any machine for which there exists a Java Virtual Machine. At the enterprise level, Java is being used for more complex e-commerce and e-business applications that require communication with an organization's back-end transaction processing systems.

Web Browsers, HTML, and HTML5

A **web browser** is an easy-to-use software tool with a graphical user interface for displaying web pages and for accessing the web and other Internet resources. Microsoft Edge, Mozilla Firefox, Google Chrome, and Apple Safari browsers are examples. **Hypertext Markup Language (HTML)** is a page description language for specifying how text, graphics, video, and audio are placed on a web page and for creating dynamic links to other web pages and objects. Using these links, a user need only point at a highlighted keyword or graphic, click on it, and immediately be transported to another document.

HTML was originally designed to create and link static documents composed largely of text. Today, however, the web is much more social and interactive, and many web pages have multimedia elements—images, audio, and video. Third-party plug-in applications were required to integrate these rich media with web pages. However, these add-ons required additional programming and put strains on computer processing. The newest evolution of HTML, called **HTML5**, solves this problem by making it possible to embed images, audio, video, and other elements directly into a document without processor-intensive add-ons. HTML5 makes it easier for web pages to function across different display devices, including mobile devices as well as desktops, and it supports the storage of data offline for apps that run over the web.

JavaScript is another core technology for making web pages more dynamic and interactive. JavaScript can be used to instruct the computer on how to interact with the user upon receiving user input. The vast majority of websites use JavaScript for client-side page behavior, and all major web browsers have a built-in JavaScript engine to execute it very quickly.

Other popular programming tools for web applications include Ruby and Python. Ruby is an object-oriented programming language known for speed and ease of use in building web applications, and Python (praised for its clarity) is being used for building cloud computing applications.

Web browser

An easy-to-use software tool with a graphical user interface for displaying web pages and accessing the web and other Internet resources.

Hypertext Markup Language (HTML)

Page description language for specifying how text, graphics, video and audio are placed on a web page and for creating dynamic links to other web pages and objects.

HTML5

Newest evolution of HTML, which makes it possible to embed images, audio, video, and other elements directly into a document without add-on software.

JavaScript

Core technology for making web pages more dynamic and interactive.

Web Services

Web services refer to a set of loosely coupled software components that exchange information with each other using universal web communication standards and languages. They can exchange information between two different systems regardless of the operating systems or programming languages on which the systems are based. Different applications can use web services to communicate with each other in a standard way without time-consuming custom coding.

The foundation technology for web services is **Extensible Markup Language (XML)**. XML was developed in 1996 by the World Wide Web Consortium (W3C, the international body that oversees the development of the web) as a more powerful and flexible markup language than HTML for web pages. Whereas HTML is limited to describing how data should be presented in the form of web pages, XML can perform presentation, communication, and storage of data. In XML, a number is not simply a number; the XML tag specifies whether the number represents a price, a date, or a ZIP code. Table 5.6 illustrates some sample XML statements.

By tagging selected elements of the content of documents for their meanings, XML makes it possible for computers to manipulate and interpret their data automatically and perform

Web services

Set of loosely coupled software components that exchange information with each other using universal web communication standards and languages. Used for linking systems of different organizations or for linking disparate systems within the same organization.

Extensible Markup Language (XML)

A more powerful and flexible markup language than HTML; provides the foundation technology for web services.

Table 5.6 Examples of XML

Plain English	XML
Subcompact	<AUTOMOBILETYPE= "Subcompact">
4 passenger	<PASSENGERUNIT= "PASS">4</PASSENGER>
$16,800	<PRICE CURRENCY= "USD">$16,800</PRICE>

operations on the data without human intervention. Web browsers and computer programs, such as order processing or ERP software, can follow programmed rules for applying and displaying the data. XML provides a standard format for data exchange, enabling web services to pass data from one process to another.

Web services communicate through XML messages over standard web protocols. Companies discover and locate web services through a directory. Using web protocols, a software application can connect freely to other applications without custom programming for each different application with which it wants to communicate. Everyone shares the same standards.

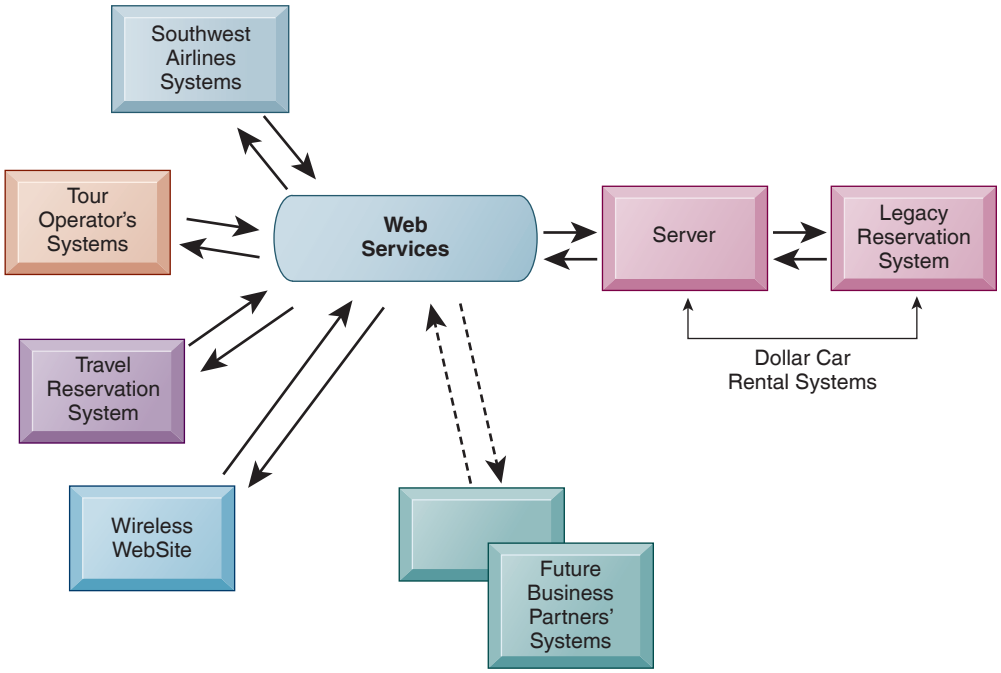
Virtually all major software vendors provide tools and entire platforms for building and integrating software applications using web services. Microsoft has incorporated web services tools into its Microsoft .NET platform.

Dollar Car Rental's systems use web services for its online booking system with Southwest Airlines' website. Although both companies' systems are based on different technology platforms, a person booking a flight on Southwest.com can reserve a car from Dollar without leaving the airline's website. Instead of struggling to get Dollar's reservation system to share data with Southwest's information systems, Dollar used Microsoft .NET web services technology as an intermediary. Reservations from Southwest are translated into web services protocols, which are then translated into formats that can be understood by Dollar's computers.

Other car rental companies have linked their information systems to airline companies' websites before. But without web services, these connections had to be built one at a time. Examine Figure 5.7 which illustrates how web services provide a standard way for Dollar's computers to "talk" to other companies' information systems without having to build special links to each one. Dollar is now expanding its use of web services to link directly to the systems of a small tour operator and a large travel reservation system as well as a mobile website. It does not have to write new software code for each new partner's information systems or each mobile device.

Figure 5.7 How Dollar Car Rental Uses Web Services

Dollar Car Rental uses web services to provide a standard intermediate layer of software to "talk" to other companies' information systems. Dollar can use this set of web services to link to other companies' information systems without having to build a separate link to each firm's systems.



5.7 Discuss Alternative Ways of Obtaining Software

Today, many business firms operate legacy systems that continue to meet a business need and that would be extremely costly to replace. But they will purchase or rent most of their new software applications from external sources.

There are three external sources for software: software packages from a commercial software vendor; cloud-based software services and tools (SaaS/PaaS); and outsourcing custom application development to an external vendor (which may or may not be offshore).

Software Packages and Software Services

A **software package** is a prewritten commercially available set of software programs that eliminates the need for a firm to write its own software programs for certain functions, such as payroll processing or order handling. In the past, software packages typically came in a box and the software had to be installed directly on a user's computer or on-premise network. Today, you're just as likely, if not more, to download the software from the vendor's website or to use the software as a cloud service delivered over the Internet and pay a subscription fee.

Software package

A prewritten, precoded, commercially available set of programs that eliminates the need to write software programs for certain functions.

Cloud-based Software Services

Cloud-based software and the data it uses are hosted on servers in data centers and can be accessed with an Internet connection and standard web browser. In addition to free or low-cost tools for individuals and small businesses provided by Google or Microsoft, enterprise software and other complex business functions are available as services from the major commercial software vendors. Instead of buying and installing software programs, subscribing companies rent the same functions from these services, with users paying either on a subscription or per-transaction basis. On-demand customer relationship management software provided by Salesforce, described earlier in this chapter, is a leading example of software as a service (SaaS).

In order to manage their relationship with an outsourcer or technology service provider, firms need a contract that includes a **service level agreement (SLA)**: a formal contract between customers and their service providers that defines the specific responsibilities of the service provider and the level of service expected by the customer. SLAs typically specify the nature and level of services provided, criteria for performance measurement, support options, provisions for security and disaster recovery, hardware and software ownership and upgrades, customer support, billing, and conditions for terminating the agreement.

Service level agreement (SLA)

Formal contract between customers and their service providers that defines the specific responsibilities of the service provider and the level of service expected by the customer.

Software Outsourcing

Software outsourcing enables a firm to contract custom software development or maintenance of existing legacy programs to outside firms, which often operate offshore in low-wage areas of the world where there are many skilled IT specialists. People.ai, a software product company specializing in B2B sales acceleration through machine learning technologies, is an example. The company needed rare engineering talent for AI algorithm development and set up a development team in Eastern Europe. People.ai delegated all back-office operations to Alcor-BPO, which sets up software R&D offices for U.S. product tech companies in Eastern Europe and Latin America. Alcor also dispatched IT recruiters to search for top IT specialists. As a result, People.ai was able to swiftly hire over 25 IT professionals and launch a fully equipped development team in just a month. The outcomes of outsourcing surpassed all expectations (Alcor, 2024).

Software outsourcing

Contracting custom software development or maintenance to an outside firm.

Offshore software outsourcing firms have primarily provided lower-level maintenance, data entry, and call center operations, although more sophisticated and experienced offshore firms, particularly in Eastern Europe and India, have been hired for new-program development, as was the case with People.ai. However, as wages offshore rise and the costs of managing offshore projects are factored in (see Chapter 13), some work that might have been sent offshore is returning to domestic companies.

5.8 Describe IT Infrastructure for AI

Developing and running AI applications require specialized hardware and software, based on the size of the dataset and the intended use of the application. Customizing hardware and software to match an organization's needs is essential for optimal performance and efficiency.

Table 5.7 describes the key hardware components for AI.

Table 5.8 describes the key software components for AI systems.

The Spotlight on Technology Case Study *AI Chips,* describes AI chips in more detail.

Table 5.7 AI Hardware Components

Component	Description
Central Processing Unit (CPU)	AI workloads require a powerful CPU to handle general computing tasks.
Graphics Processing Unit (GPU)	GPUs excel at parallel processing, making them ideal for training large AI models.
Tensor Processing Unit (TPU)	TPUs, developed by Google, are specialized chips optimized for machine learning tasks. They offer excellent performance for specific workloads.
System Memory (RAM)	Large datasets such as those in AI require very large amounts of system memory.
Data Storage and Management	Efficient data storage (databases, data lakes) and processing are essential for managing the large datasets used in AI training. Cloud storage solutions such as Amazon S3, Google Cloud Storage, and Azure Blob Storage are commonly used.

Table 5.8 AI Software Components

Component	Description
Operating system	Linux operating systems are often used because of their flexibility and compatibility with AI libraries. The open-source nature of Linux environments works well in complex installation and configuration processes required by many machine learning applications.
AI Frameworks and Libraries	Frameworks like TensorFlow, PyTorch, or Keras (which are all open source) provide tools for building and training AI models. Many developers are choosing open-source AI frameworks over proprietary software.
Version control	The GitHub platform allows software developers to create, store, manage and share their source code, with capabilities for version control.
Cloud services	Cloud platforms such as AWS, Azure, or Google Cloud are very scalable and often easier to deploy for AI than custom in-house systems.

Spotlight on Technology Case Study

AI Chips

In 1999 semiconductor maker Nvidia made history when it introduced the graphics processing unit (GPU), which is used for handling complex images and large quantities of data for graphics-related tasks in video games and other software. It turns out that Nvidia's GPUs also work really well for machine learning and generative AI applications, which must be trained using massive quantities of data in order to recognize intricate patterns, sounds, pictures, and text and to create chatbots and generate images. As a result, Nvidia is making history again as the world's leading producer of the chips powering AI applications. Nvidia-specialized AI chips underpin most of today's AI applications. It has been said that Nvidia controls as much as 80 percent of the global market for specialized AI chips. Recent generative AI applications from Microsoft, OpenAI, Amazon, and Meta could not be built without Nvidia's hardware. For instance, ChatGPT was trained using 10,000 of Nvidia's GPUs clustered together in a supercomputer belonging to Microsoft.

A GPU differs from the computer's central processing unit (CPU) in how it processes data and the number of physical processing units (cores) it uses. A CPU only has a few cores and performs serial computing – processing data one task after another. It handles all the main functions of a computer and is designed to handle a wide range of tasks quickly. A GPU has hundreds of cores and performs parallel computing, processing many calculations at once. It is more specialized for handling high-resolution images and videos concurrently. Its high throughput and ability to perform many calculations with large quantities of data simultaneously make it ideal for training large AI models.

None of the Big Tech companies like depending so heavily on Nvidia, nor does the U.S. government. The United States currently produces less than 10 percent of the world's chips and none of the most advanced logic chips, some of which power AI. The United States designs advanced logic chips, but its inability to manufacture them domestically has emerged as a national security and economic risk. The U.S. CHIPS and Science Act, or CHIPS Act, enacted in 2022, dedicates $39 billion in subsidies for creating or expanding U.S.-based semiconductor fabrication and an additional $11 billion in research and development. Leading recipients of CHIPS Act funds include Intel, Samsung, Micron Technology, and GlobalFoundries. The Biden administration anticipates that CHIPS investments for Intel and other semiconductor manufacturers will enable the United States to produce roughly 20 percent of the world's most advanced logic chips by 2030, up from the current level of zero. Producing more semiconductors domestically will help insulate U.S. chip supplies from major supply chain disruptions and dependence on Taiwan and other Asian countries that currently dominate global chip production.

The Big Tech firms are all starting to invest heavily in designing their own custom chips, which would support larger AI models and give them more control over their AI businesses. Having better computer chips could soon become a bigger competitive advantage than having better computer code. AI chips made in-house could be more easily tailored to a company's particular AI models, making its products more efficient and energy-conserving. If an iPhone, Google Pixel, or Microsoft Surface tablet can run a more powerful generative AI model and load its results a bit faster because of a custom microchip, then more customers might buy that device. That could be a game-changer.

Google has been able to train and run its flagship Gemini models with less energy and at lower cost by using custom computer processors rather than relying on Nvidia. However, many Google Cloud servers also run on Nvidia chips, and Google optimized its Gemma language model to run on Nvidia GPUs. Amazon touts its custom AI chips for their superior machine learning training performance at significantly lower costs. But the firm is also growing its partnership with Nvidia. Microsoft believes that its custom chips enhance its systems rather than replacing existing hardware powered by Nvidia.

Every company wants its own self-contained IT platform, no longer beholden to competitors' prices or external supply chain disruptions. But whether any of these cloud computing tech companies can rival or replace Nvidia remains an open issue right now, and it is highly unlikely that any of them will sever ties with Nvidia. It may well be that the dominant pattern for the future is a hybrid combination of custom computer chips and Nvidia designs.

Nvidia is currently working on even more powerful chips capable of training next-generation AI models for generative AI applications such as ChatGPT, recommender systems, and data analytics. In March 2024, CEO Jensen Huang announced a new generation of high-performance AI chips called Blackwell, which can reduce energy consumption for training large AI models by a factor of 3 to 4 compared to the previous generation of GPUs. Blackwell has 208 billion transistors and is an upgrade of the company's H100 chip, which is currently the most advanced GPU in production. Blackwell is 30 times quicker at some tasks than its predecessor. Blackwell enables connecting many GPUs so that large AI models can be trained with a

smaller carbon footprint. Among the many organizations expected to adopt Blackwell are Amazon Web Services, Dell Technologies, Google, Meta, Microsoft, OpenAI, Oracle, Tesla, and xAI.

SOURCES: Timothy Rooks, "How Nvidia's Blackwell Superchip Could Fuel an AI Revolution," Dw.com, accessed August 20, 2024; "Nvidia Blackwell Platform Arrives to Power New Era of Computing," Nvidianews.nvidia.com, March 18, 2024; Lucas Mearian, "AI Chip Shortages Continue, But There May Be an End in Sight," Computerworld.com, May 7, 2024.

Case Study Questions

1. Why are Nvidia chips so useful for AI applications?
2. Nvidia has a near-monopoly on specialized AI chips. Discuss the impact of this situation.
3. Should companies develop their own custom AI chips or use those produced by Nvidia? Explain your answer.

AI and Edge Computing

Edge artificial intelligence (Edge AI)
Refers to the deployment of AI algorithms and AI models directly on local edge devices such as smartphones, sensors or IoT devices, allowing computations to be performed close to where data is actually collected without constant reliance on central cloud infrastructure.

Edge artificial intelligence (Edge AI) refers to the deployment of AI algorithms and AI models directly on local edge devices such as smartphones, sensors, or Internet of Things (IoT) devices. This allows computations to be performed close to where data are actually collected without constant reliance on central cloud infrastructure. This reduces the workloads of cloud computers and networks. The utilization of AI CPU, GPU, and memory is greatly reduced as their workloads are distributed among edge devices.

Current applications of Edge AI include smartphones, security cameras, wearable health-monitoring accessories (e.g., smartwatches), real-time traffic updates on autonomous vehicles, connected devices, and smart home appliances. Various industries are also increasingly implementing edge AI applications to cut down costs, automate processes, improve decision-making, and optimize operations.

Generative AI technologies like the large language models that power today's conversational AI assistants and image generators require massive computational resources. However, advances in mobile hardware and technologies are reducing the need for centralized cloud processing to handle AI-powered tasks, making it possible for a growing list of smartphones to handle these complex AI tasks on their own.

5.9 Discuss IT Infrastructure Management Challenges

Creating and managing a coherent IT infrastructure raises multiple challenges: dealing with platform and technology changes (including cloud and mobile computing), management and governance, and making wise infrastructure investments.

Dealing with Platform and Infrastructure Change

Scalability
The ability of a computer, product, or system to expand to serve a larger number of users without breaking down.

Mobile device management (MDM)
Software that monitors, manages, and secures mobile devices that are deployed across multiple mobile service providers and multiple mobile operating systems used in the organization.

As firms grow, they often quickly outgrow their infrastructure. As firms shrink, they can get stuck with excessive infrastructure purchased in better times. How can a firm remain flexible if investments in IT infrastructure are fixed-cost purchases and licenses? How well does the infrastructure scale? **Scalability** refers to the ability of a computer, product, or system to expand to serve a large number of users without breaking down. New applications, mergers and acquisitions, and changes in business volume all affect computer workload and must be considered when planning hardware capacity.

Firms using mobile computing and cloud computing platforms will require new policies, procedures, and tools for managing these platforms. They will need to inventory all of their mobile devices in business use and develop policies and tools for tracking, updating, and securing them, and for controlling the data and applications that run on them. Firms often turn to **mobile device management (MDM)** software, which monitors, manages, and secures mobile devices that are deployed across multiple mobile service providers and across multiple mobile operating systems being used in the organization. MDM tools enable the IT department

to monitor mobile usage, install or update mobile software, back up and restore mobile devices, and remove software and data from devices that are stolen or lost.

Firms using cloud computing and SaaS will need to fashion new contractual arrangements with remote vendors to make sure that the hardware and software for critical applications are always available when needed and that they meet corporate standards for information security. It is up to business management to determine acceptable levels of computer response time and availability for the firm's mission-critical systems to maintain the level of business performance that is expected.

Management and Governance

A long-standing issue among information system managers and CEOs has been the question of who will control and manage the firm's IT infrastructure. Chapter 2 introduced the concept of IT governance and described some issues it addresses. Other important questions about IT governance are: Should departments and divisions have the responsibility of making their own information technology decisions, or should IT infrastructure be centrally controlled and managed? What is the relationship between central information systems management and business unit information systems management? How will infrastructure costs be allocated among business units? Each organization will need to arrive at answers based on its own needs.

Making Wise Infrastructure Investments

IT infrastructure is a major investment for the firm. If too much is spent on infrastructure, it lies idle and constitutes a drag on the firm's financial performance. If too little is spent, important business services cannot be delivered and the firm's competitors (who spent the right amount) will outperform the underinvesting firm. How much should the firm spend on infrastructure? This question is not easy to answer.

A related question is whether a firm should purchase and maintain its own IT infrastructure components or rent them from external suppliers, including those offering cloud services. The decision either to purchase your own IT assets or to rent them from external providers is called the *rent-versus-buy* decision.

Cloud computing is often a low-cost way to increase scalability and flexibility, but firms should evaluate this option carefully in light of security requirements and impact on business processes and workflows. In some instances, the cost of renting software adds up to more than purchasing and maintaining an application in-house, or firms can overspend on cloud services. Yet there are many benefits to using cloud services including significant reductions in hardware, software, human resources, and maintenance costs and access to more leading-edge technology. Moving to cloud computing allows firms to focus on their core businesses rather than technology issues.

Total Cost of Ownership of Technology Assets

The actual cost of owning technology resources (the **total cost of ownership (TCO)**) includes the original cost of acquiring and installing hardware and software as well as ongoing administration costs for hardware and software upgrades, maintenance, technical support, training, and even utility and real estate costs for running and housing the technology. The total cost of ownership (TCO) model can be used to analyze these direct and indirect costs to help firms determine the actual cost of specific technology implementations. Table 5.9 describes the most important components to consider in a TCO analysis.

Total cost of ownership (TCO)

Designates the actual total cost of owning technology resources, including the initial cost of acquiring and installing hardware and software, as well as ongoing adminisration costs for hardware and software upgrades, maintenance, technical support, training and even utility and real estate costs for running and housing the technology.

When all these cost components are considered, the TCO for a desktop computer might run up to three times the original purchase price of the equipment. Gains in productivity and efficiency from equipping employees with mobile computing devices must be balanced against increased costs from integrating these devices into the firm's IT infrastructure and from providing technical support. Other cost components include fees for wireless airtime, end-user training, help desk support, and software for special applications. Costs are higher if the mobile devices run many different applications or need to be integrated into back-end systems such as enterprise applications.

Table 5.9 Total Cost of Ownership (TCO) Cost Components

Infrastructure Component	Cost Components
Hardware acquisition	Purchase price of computer hardware equipment, including computers, storage, printers, and mobile devices
Software acquisition	Purchase or license of software for each user or for the organization as a whole
Installation	Cost to install computers and software
Training	Cost to provide training for information systems specialists and end users
Support	Cost to provide ongoing technical support, help desks, and so forth
Maintenance	Cost to upgrade the hardware and software
Infrastructure	Cost to acquire, maintain, and support related infrastructure, such as networks and specialized equipment (including storage backup units)
Downtime	Cost of lost productivity if hardware or software failures cause the system to be unavailable for processing and user tasks
Space and energy	Real estate and utility costs for housing and providing power for the technology

Hardware and software acquisition costs account for only about 20 percent of TCO, so managers must pay close attention to administration costs to understand the full cost of the firm's hardware and software. It is possible to reduce some of these administration costs through better management. Many large firms are saddled with redundant, incompatible hardware and software because their departments and divisions have been allowed to make their own technology purchases.

In addition to switching to cloud services, these firms could reduce their TCO through greater centralization and standardization of their hardware and software resources. Companies could reduce the size of the information systems staff required to support their infrastructure if the firm minimizes the number of different computer models and pieces of software that employees are allowed to use. In a centralized infrastructure, systems can be administered from a central location and troubleshooting can be performed from that location.

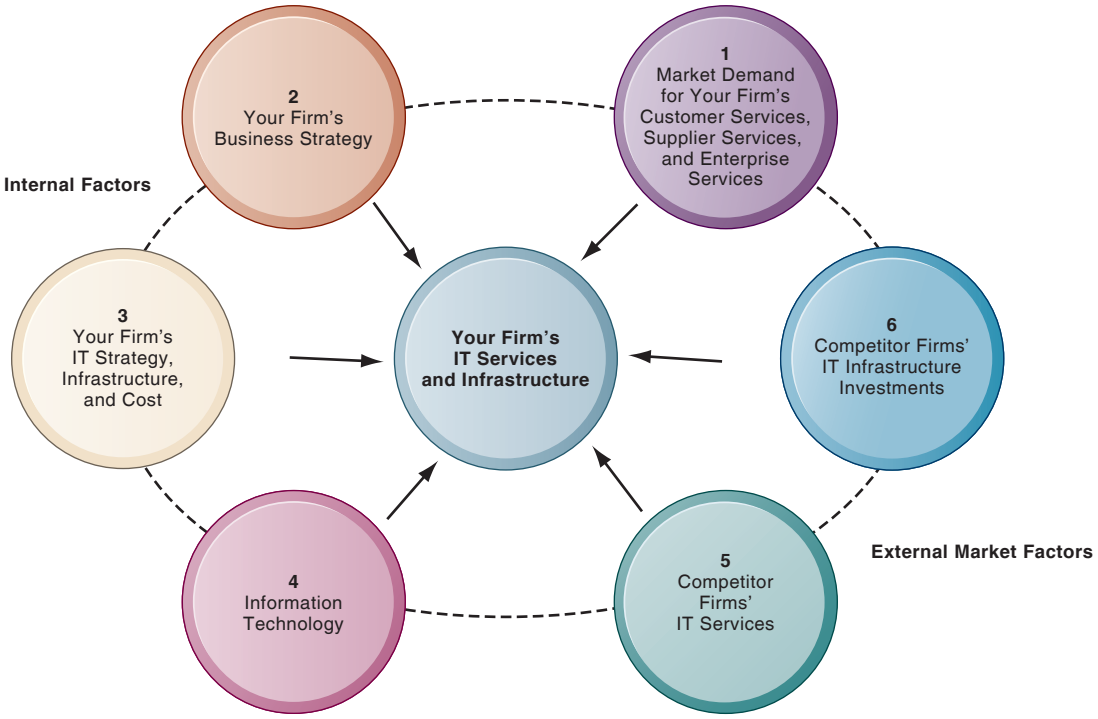
Competitive Forces Model for IT Infrastructure Investment

You can also use a competitive forces model to address the question of how much your firm should spend on IT infrastructure and how the investment should be spent. There are six factors to consider:

- **Market demand for your firm's services.** Make an inventory of the services you currently provide to customers, suppliers, and employees. Survey each group or hold focus groups to find out if the services you currently offer meet the needs of each group. For example, are customers complaining of slow responses to their queries about price and availability? Are employees complaining about the difficulty of finding the right information for their jobs? Are suppliers complaining about the difficulties of discovering your production requirements?
- **Your firm's business strategy.** Analyze your firm's five-year business strategy and try to assess what new services and capabilities will be required to achieve strategic goals.
- **Your firm's IT strategy, infrastructure, and cost.** Examine your firm's information technology plans for the next five years and assess their alignment with the firm's business plans. Determine the total IT infrastructure costs. You will want to perform a TCO analysis. If your firm has no IT strategy, you will need to devise one that takes into account the firm's five-year strategic plan.
- **Information technology assessment.** Is your firm behind the technology curve or at the bleeding edge of information technology? Both situations are to be avoided. It is usually not

Figure 5.8 Competitive Forces Model for IT Infrastructure

There are six factors you can use to answer the question "How much should our firm spend on IT infrastructure and how should that investment be spent?"



desirable to spend resources on advanced technologies that are still experimental, often expensive, and sometimes unreliable. You want to spend on technologies for which standards have been established; IT vendors are competing on cost, not design; and where there are multiple suppliers. However, you do not want to put off investment in new technologies or allow competitors to develop new business models and capabilities based on the new technologies.

- **Competitor firm services.** Try to assess what technology services competitors offer to customers, suppliers, and employees. Establish quantitative and qualitative measures to compare them to those of your firm. If your firm's service levels fall short, your company is at a competitive disadvantage. Look for ways your firm can excel at service levels.
- **Competitor firm IT infrastructure investments.** Benchmark your expenditures for IT infrastructure against your competitors. Many companies are quite public about their innovative expenditures on IT. If competing firms try to keep IT expenditures secret, you may be able to find IT investment information in public companies' SEC Form 10-K annual reports to the federal government when those expenditures affect a firm's financial results. Your firm does not necessarily need to spend as much as or more than your competitors. Perhaps it has discovered much less expensive ways of providing services, and this can lead to a cost advantage. Alternatively, your firm may be spending far less than competitors, and experiencing commensurate poor performance and losing market share.

Examine Figure 5.8, which illustrates this model.

5.10 Understand How the Information in This Chapter Can Help Your Career

The information you have learned in Chapter 5 can help you advance your career by familiarizing you with the essential concepts and vocabulary for the hardware and software technologies used by businesses today, as well as emerging IT trends. This knowledge can help you as you interview, and also on the job as you try to improve the performance of your firm.

Real-World Scenario

CloudCall Calls on AWS Trainers

CloudCall provides cloud-based telecommunications software to help companies improve the functionality and efficiency of their customer relationship management (CRM) systems. It has offices in the United States and the UK. To transition from a traditional telecommunications company to a software-as-a-service (SaaS) model powered by Amazon Web Services (AWS), the company's product and engineering group needed more knowledge about AWS services. To build employee skills and expertise, CloudCall enlisted AWS Training and Certification, which offers online programs and instructor-led training to help people build their Amazon cloud skills. Live classes with instructors teach in-demand cloud skills and best practices using a mix of presentations, discussions, and hands-on labs. Now CloudCall can provide better technical support and advanced solutions to help its customers get the most from their CRM data.

Job Description and Career Path: AWS Trainer

Job	Job Description	IT Knowledge	Career Path and Average Salaries
AWS Trainer	IT professional with a strong understanding of Amazon Web Services (AWS) who designs and delivers training programs to individuals or businesses. They help students use AWS cloud computing services effectively. Job responsibilities include: • Develop and deliver AWS training programs tailored to various audiences and experience levels. • Create course materials, including presentations, labs, and assignments. • Instruct students on AWS concepts, services, and best practices. • Conduct hands-on labs to reinforce learning and practical application. • Evaluate student progress through assessments and exams. • Stay updated on the latest AWS developments and incorporate them into training programs.	• AWS certifications relevant to the training area (such as AWS Certified Solutions Architect or AWS Certified Developer). • Experience in the IT industry, particularly cloud technologies • In-depth knowledge of AWS cloud computing services and architecture. • Experience designing and delivering technical training programs. • Excellent communication, presentation, and facilitation skills. • Ability to assess learning needs and tailor training accordingly.	• AWS Trainer • AWS Solutions Architect • Technical Trainer Lead • Cloud Consultant • Independent AWS Trainer Salaries for AWS trainers vary depending on experience, location, certifications, and the type of training provided (in-house vs. public courses). Here's a ballpark range: • Entry-level: $60,000 – $80,000 • Mid-level: $80,000 – $100,000+ • Senior-level/Lead Trainer: $100,000+ Salaries can vary based on location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

Are AI and Cloud Computing Sustainable?

Cloud computing and AI have been the source of many IT success stories. Cloud computing has been touted as a solution for cutting IT costs by eliminating the need to maintain on-premise infrastructure, making more efficient use of computing resources, and delivering flexible computing technology available any time and any place on devices of all sizes. Cloud platforms also provide the requisite computing power and resources for experimenting with new technologies and driving innovation,, including artificial intelligence (AI).

Examples of cloud computing success stories abound. For example, migrating to Microsoft Azure Cloud produced a 45 percent reduction in costs for Protocall Services, a national provider of telephonic crisis services and digital tools for behavioral health. The cloud infrastructure helped Protocall become more agile and secure, with virtually 100 percent application infrastructure uptime. According to Amazon Web Services (AWS), businesses that have migrated to the cloud with AWS reported 51 percent lower cost of operations, 62 percent more efficient IT infrastructure staff, 94 percent less unplanned downtime, and 25 percent higher developer productivity.

The move to cloud computing can reduce infrastructure, physical space, and energy usage. Computers generate a great deal of heat, so data centers must be kept cool at all times. Cooling typically accounts for 40 percent of total data center energy consumption, and up to 80 percent if the natural climate where the data center is located is warmer. A data center requires a 24/7 power supply to run servers and a cooling system to prevent overheating. Data center operators also must dispose of outdated or malfunctioning equipment, adding to e-waste. A public cloud uses resources more efficiently than an on-premise data center, where organizations may only utilize 10 percent of server computing capacity while keeping the equipment running all the time. Businesses sharing the resources of a cloud data center consume 77 percent fewer servers, 84 percent less power, and reduce carbon emissions by 88 percent compared to on-premise data centers.

Research by Capgemini, an IT consulting firm, suggests cloud computing could potentially remove 1 billion metric tons of CO^2 between 2021 and 2024. Cloud computing also contributes to sustainability through dematerialization, the replacement of high-carbon physical products with virtual equivalents. Cloud services encourage people to use virtual services like video streaming or server virtualization as opposed to resource-heavy physical products. By reducing physical products—hardware and other computing equipment—cloud computing reduces the amount of waste emanating from the disposal of these products. Approximately 50 million tons of e-waste are generated globally each year.

However, cloud computing is not completely green. Increasing use of data centers has led to even more data centers, including more in the cloud. Overall, the information technology sector is responsible for roughly 2 percent to 4 percent of global greenhouse gas emissions—about the same as the aviation industry. Data centers' global electricity use is expected to increase to between 3 percent to 13 percent of global electricity consumption by 2030. Without serious efforts to shift to clean energy sources, greenhouse gas emissions from data centers will increase at the same rate.

Data centers can't operate without air conditioning (or another type of cooling system). Power-hungry computer room air conditioners or computer room air handlers are necessities in even the most advanced data centers. In North America, most data centers draw power from "dirty" electricity grids, especially in Virginia's "data center alley," the site through which passes 70 percent of the world's Internet traffic. To cool, the cloud burns carbon. In most data centers today, cooling accounts for more than 40 percent of electricity usage.

To ensure customers' data and cloud services will be available anytime, anywhere, data centers are designed to be hyper-redundant: if one system fails, another must be on hand to take its place at a moment's notice. There are redundant power systems like diesel generators and redundant servers ready to take over computational processes should others become unavailable. In some cases, only 6 percent to 12 percent of energy consumed by data centers is devoted to active computational processes. The remainder is allocated to cooling and maintaining redundant systems and tools to prevent costly downtime.

In many data centers today, chilled water is piped through the latticework of server racks for more efficient cooling. This shift from cooling air to cooling water is an attempt to reduce carbon footprint, but it is not without cost. Weathering historic droughts and heat waves, water resources of communities in the western United States are in increasingly short supply and nearby water-cooled data centers add to the strain. Bluffdale, Utah, residents are struggling with water shortages and power outages as a result of the nearby Utah Data Center, a facility of the U.S. National Security Agency (NSA) that guzzles seven million gallons of water daily to operate. Data centers consume millions of gallons of precious Arizona water every day.

Some of the most advanced, giant "hyperscale" data centers, like those maintained by Google, Facebook (Meta), and Amazon, have pledged to transition their sites to carbon-neutral via carbon offsetting and investment in renewable energy sources like wind and solar. Amazon claims to be the world's largest renewable energy purchaser, with the goal of powering its company with 100 percent renewables

by 2025 and to become carbon net-zero by 2040. AWS has been setting up wind farms around the United States to serve as power sources for current and future data centers and established its own Amazon Wind Farm business for this purpose. Microsoft has pledged to be carbon negative by 2030 and to remove from the atmosphere all the carbon the company has ever emitted since it was founded in 1975. The company plans to have all of its data centers running on 100 percent renewable energy by 2025, with all of the energy it uses coming from carbon-free sources by 2030.

AI makes it much more difficult to achieve these objectives. Most of today's AI applications use a cloud infrastructure to provide the enormous processing power and storage capacity needed for developing, training, and using AI applications. A Deloitte study found that 70 percent of companies acquire their AI capabilities through cloud-based software, while 65 percent create AI applications using cloud services. AI's carbon footprint from its voracious energy requirements and from manufacturing the hardware it uses is huge and fast-growing. Since 2012, the most extensive AI training runs have been using exponentially more computing power, with models doubling in size as often as every three months. A single AI-powered search might require five to ten times as much computing power as a traditional one. The intense computing and data requirements of AI models will also increase data center water consumption.

As datasets and models become larger and more complex, the upsurge in energy and water use could potentially negate all the benefits from existing data center carbon reduction efforts. A study by University of Massachusetts researchers to determine how much energy is used to train popular large AI models found that training a single large language model can produce about 626,000 pounds of carbon dioxide, or the equivalent of around 300 round-trip flights between New York and San Francisco. Researchers at University of California at Riverside found that global AI demand could cause data centers to consume up to 1.1 trillion to 1.7 trillion gallons of fresh water by 2027. McKinsey predicts that data centers' electricity use will more than double by 2030.

Yet AI can be a powerful tool for promoting sustainability. AI is being used to maximize utilization of renewable energy sources like wind and solar, resulting in more environmentally friendly farming methods, as well as carbon reductions for daily living. AI is also being used to optimize logistics and reduce waste in supply chains, to monitor and enforce environmental regulations, and to optimize data center operations through machine learning algorithms that dynamically adjust temperature settings, workload distribution, and server utilization.

Tools such as the Machine Learning Emissions Calculator can help AI engineers simulate carbon emissions on AI models based on variables such as hardware, number of hours, provider, and location. Researchers at Google recommend four areas of technical best practices that reduce energy consumption and carbon emissions: model, machine, mechanization, and map optimization. They believe that following these best practices can reduce machine learning training energy by up to 100 percent and CO^2 emissions by up to 1,000 times.

SOURCES: Aws.amazon.com, accessed August 26, 2024; Niklas Sundberg, "Tackling AI's Climate Change Problem," *MIT Sloan Management Review*, Winter 2024; Martin Wong, "The Internet's Next Great Power Suck," *The Atlantic*, August 23, 2023; Alokya Kanungo, "The Green Dilemma: Can AI Fulfill Its Potential Without Harming the Environment?," Earth.org, July 18, 2023; Michael Brenner, "The Role of AI in Cloud Computing," Nutanix.com, July 13, 2023; Steven Gonzalez Montserrate, "The Staggering Ecological Impacts of Computation and the Cloud," Thereader.mitpress.mit.edu, February 14, 2022; David Kuchta, "Does Cloud Computing Help or Harm the Environment?" Treehugger.com, January 13, 2022; Kalyan Kumar, "Toward a Sustainable Transition to the Green Cloud," Cmswire.com, October 19, 2021.

Case Study Questions

1. How green is cloud computing? How much does it help or hurt the environment? Explain your answer.
2. What solutions have been proposed to make data centers greener? How effective are they?
3. Is AI harmful for the environment? Explain your answer.

Review Summary

5.1 Discuss IT infrastructure and its evolution.

IT infrastructure is the shared technology resources that provide the platform for the firm's specific information system applications. IT infrastructure includes hardware, software, and services that are shared across the entire firm. The five stages of IT infrastructure evolution are the mainframe era, the personal computer era, the client/server era, the enterprise computing era, and the cloud and mobile computing era.

5.2 Describe the components of IT infrastructure.

Major IT infrastructure components include computer hardware platforms, operating system platforms, enterprise software platforms, networking and telecommunications platforms, data management software, Internet platforms, and consulting services and systems integrators.

5.3 Discuss the mobile platform, apps, and BYOD.

Increasingly, computing is taking place on a mobile platform. Apps are software applications that run on mobile devices and are delivered over the Internet. Consumerization of IT is the business use of information technology that originated in the consumer market.

5.4 Discuss quantum computing and green computing.

Quantum computing is an emerging technology that could dramatically boost processing power through the ability to represent data in more than one state at the same time. Green computing includes practices and technologies for producing, using, and disposing of information technology hardware to minimize negative impact on the environment.

5.5 Discuss virtualization and cloud computing

Virtualization organizes computing resources so that their use is not restricted by physical configuration or geographic location. In cloud computing, firms and individuals obtain computing power and software remotely as services over a network, including the Internet, rather than purchasing and installing the hardware and software on their own computers. There are three types of cloud services: infrastructure as a service (IaaS), software as a service (SaaS), and platform as a service (PaaS), as well as three cloud computing models (public cloud, private cloud, and hybrid cloud). Edge computing helps optimize cloud computing by performing some processing on a set of linked servers at the edge of the network, near the source of the data.

5.6 Describe contemporary software platforms.

Open-source software is produced and maintained by a global community of programmers and is often downloadable for free. Linux is a powerful, resilient open-source operating system that can run on multiple hardware platforms and is used widely to run web servers. Java is an operating system– and hardware-independent programming language that is a leading interactive programming environment for the web. HTML5 makes it possible to embed images, audio, and video directly into a web document without add-on programs. Web services are loosely coupled software components based on open web standards that work with any application software and operating system. They can be used as components of web-based applications linking the systems of two different organizations or to link disparate systems of a single company.

5.7 Discuss alternative ways of obtaining software.

Much of the new software used today is not developed internally but consists of components, frameworks, and services obtained from external sources, including software packages, outsourcing custom application development, or renting online software services (SaaS).

5.8 Describe IT infrastructure for AI.

Developing and running AI applications require specialized hardware and software, based on the size of the dataset and intended use, as well as large amounts of computer memory and data storage. Important specialized AI hardware components include graphics processing units (GPUs) and tensor processing units (TPUs). Important software applications for AI include a robust operating system such as Linux and AI frameworks and libraries. Various industries are also increasingly implementing edge AI applications to reduce costs, automate processes, improve decision-making, and optimize operations.

5.9 Discuss IT infrastructure management challenges.

Major challenges include dealing with platform and infrastructure change, infrastructure management and governance, and making wise infrastructure investments. Solution guidelines include using a competitive forces model to determine how much to spend on IT infrastructure and where to make strategic infrastructure investments, and establishing the total cost of ownership (TCO) of information technology assets. The total cost of owning technology resources includes not only the original cost of computer hardware and software but also costs for hardware and software upgrades, maintenance, technical support, and training. Many firms are turning to cloud computing to reduce their IT platform costs. Firms use tools for mobile device management (MDM) to monitor, manage, and secure mobile devices that are deployed across the enterprise.

Key Terms

Android, 154
Application server, 152
App, 156
BYOD (Bring Your Own Device), 157
Chrome OS, 154
Client, 150
Client/server computing, 150
Cloud computing, 161
Consumerization of IT, 157
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Review Questions

5.1 Discuss IT infrastructure and its evolution.

- Define IT infrastructure from both a technology and a services perspective.
- List each of the eras in IT infrastructure evolution and describe its distinguishing characteristics.
- Define and describe the following: web server, multi-tiered client/server architecture.

5.2 Describe the components of IT infrastructure.

- List and describe the components of IT infrastructure that firms need to manage

5.3 Discuss the mobile platform, apps, and BYOD.

- Describe the mobile platform
- Define and give examples of apps.
- Define BYOD and explain why it is a management challenge.

5.4 Discuss quantum computing and green computing.

- Define quantum computing and explain why it is so powerful.
- Define and provide examples of green computing.

5.5 Discuss virtualization and cloud computing.

- Define virtualization and explain its relationship to cloud computing.
- Define cloud computing and describe the features of cloud computing.
- Describe the three major types of cloud services.
- Distinguish between a public cloud, private cloud, and hybrid cloud.
- Explain how edge computing can enhance cloud computing.

5.6 Describe contemporary software platforms.

- Define and describe open-source software and Linux, and explain their business benefits.
- Define Java and HTML5 and explain why they are important.

5.7 Discuss alternative ways of obtaining software.

- Name and describe the three external sources for software.

5.8 Describe IT infrastructure for AI.

- Describe important hardware technologies used in AI applications.
- Describe important software technologies used in AI applications.

5.9 Discuss IT infrastructure management challenges.

- Name and describe the management challenges posed by IT infrastructure.
- Explain how using a competitive forces model and calculating the TCO of technology assets helps firms make good infrastructure investments.

Discussion Questions

5.10 Why is selecting computer hardware and software for the organization an important management decision? What management, organization, and technology issues should be considered when selecting computer hardware and software?

5.11 Should organizations use software service providers for all their software needs? Why or why not?

5.12 What are the advantages and disadvantages of cloud computing?

Using AI Project

5.13 AI Project: Selecting Information Systems

You have been hired as a consultant by two friends who want to open a small retail store selling organic pet food and treats, as well as pet toys and clothing made from organic materials. They would like to know what kinds of information systems would be most important for a small retail store with 1 full-time employee and 3 part-time employees. They have a budget of $7000 to spend on information systems hardware, software, and implementation. They would also like to know which information system should be installed first.

Using Microsoft Copilot, Google's Gemini, or ChatGPT, develop a series of prompts to assist you in providing them with this information. Save a copy of your prompts and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience developing solutions for IT infrastructures; using spreadsheet software to calculate the total cost of ownership (TCO) of PC desktop systems; using web research to budget for a sales conference; and evaluating server and mobile operating systems.

For student files please visit www.pearsonhighered.com/laudon or MyLab MIS.

5.14 Management Decision Problem

EasyJet is the second-largest airline in the United Kingdom, operating more than 1,100 routes across more than 35 countries. A reliable and robust system for booking and managing reservations while keeping costs low is a key business requirement. EasyJet's legacy reservation system lacked the capability for customers to select their seats on a given flight when they made their reservations online. Adding this new service would require investing in an additional data center and software and modifying EasyJet's IT infrastructure. How could cloud computing services provide a solution? What management, organization, and technology factors and management decisions need to be considered when designing the solution?

5.15 Using a Spreadsheet to Evaluate Total Cost of Ownership (TCO)

Software skills: Spreadsheet formulas, SUM

Business skills: Technology pricing

This project provides an opportunity to determine the total cost of ownership (TCO) of investing in new PC desktop systems over a three-year period. You'll use spreadsheet software to compare the total three-year cost of purchasing and maintaining new PC desktop systems with Windows 11 Pro and Microsoft 365 Business for 40 employees. You can find information about specifications in the MIS18_Project 5.15 Student File.

Use your spreadsheet software to calculate TCO for investing in 40 new desktop productivity systems over a three-year period. The cost of Windows Pro is included in the desktop price. Management wants you to compare the TCO of HP and Dell desktop systems. Complete the worksheet by entering and calculating the values for one-time and recurring costs. Calculate the TCO over a three-year period and the TCO per user. What other factors should the company consider besides cost in determining the PC system option to select? Explain your answer.

5.16 Using Online Research Tools to Create a Budget

Software skills: Online research tools

Business skills: Researching transportation and lodging costs

The Foremost Composite Materials Company is planning a two-day sales conference for October 19–20, starting with a reception on the evening of October 18. The conference consists of all-day meetings that the entire sales force, numbering 120 sales representatives and their 16 managers, must attend. The company needs a hotel room for each sales representative and two common meeting rooms, one large enough to hold the entire sales force plus a few visitors (200 total) and the other able to hold half the sales force. Management has set a budget of $330,000 for the representatives' room rentals. The company would like to hold the conference in either Miami or Marco Island, Florida, at a Hilton- or Marriott-owned hotel.

Use the Hilton and Marriott websites to select a hotel in whichever of these cities would enable the company to hold its sales conference within its budget and meet its sales conference requirements. Then locate flights arriving the afternoon prior to the conference. Your attendees will be coming from Los Angeles (51), San Francisco (30), Seattle (22), Chicago (19), and Pittsburgh (14). Determine the costs of each round-trip airline ticket from these cities. When you are finished, create a budget for the conference. The budget will include the cost of each airline ticket and the room cost.

5.17 Collaboration and Teamwork Project: Evaluating Server and Mobile Operating Systems

Form a group with two or more of your classmates. Choose server or mobile operating systems to evaluate. You might research and compare the capabilities and costs of Linux versus the most recent version of the Windows operating system for servers. Alternatively, you could compare the capabilities of the Android mobile operating system with iOS for the iPhone. If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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6

Foundations of Business Intelligence: Databases and Information Management

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Learning Objectives

After completing this chapter, you will be able to:

6.1 Describe traditional file environment data management problems.

6.2 Describe DBMS and relational DBMS capabilities.

6.3 Discuss relational database design principles.

6.4 Describe non-relational, cloud, and blockchain database capabilities.

6.5 Describe tools used to analyze data in databases.

6.6 Discuss data governance, data quality assurance, and data management.

6.7 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** Australia's Royal Flying Doctor Service Flies with Oracle Autonomous Database
- **Spotlight on Technology Case Study:** Facial Databases: Do You Know Who's Using Your Face?
- **Chapter Case Study:** Pursuing Sustainability with Blockchain

Opening Case Study

Australia's Royal Flying Doctor Service Flies with Oracle Autonomous Database

Australia's Royal Flying Doctor Service (RFDS) is a national charitable health organization providing primary healthcare and 24-hour emergency services for people living in Australia's remote outback, where reliable Internet service and on-site medical care are often unavailable. The organization has more than 1,000 patient interactions a day, operates 80 aircraft, and maintains more than 300 road-based ambulances.

In the past, RFDS's staff recorded patient interactions on paper forms with up to 600 fields per interaction. Then, the information on the forms was manually entered into disparate standalone applications or maintained on paper, making data extremely difficult to organize and extract. As a result, it took significant time and effort to keep patient records updated and accurate. In remote areas with little or no Internet connectivity, health care suffered because it was not always possible to have the most recent patient data available at the right time. Can you imagine being a healthcare worker and not having access to information that could help save the life of your patient? Most healthcare workers take real-time access to medical data for granted. In addition, the lack of accessible up-to-date information resulted in unacceptably high support costs.

RFDS sought a cloud-based solution capable of supporting a countrywide patient record system that could facilitate consolidation and sharing of consistent, reliable, and accurate information in real-time across operations. It also wanted the solution to help employees analyze health and services patterns to improve patient outcomes.

RFDS selected Oracle Cloud Infrastructure (OCI) for its enterprise-wide healthcare technology platform, with Oracle Autonomous Database for transaction processing and analytics and OCI GoldenGate for replication, filtering, and transformation of data from one database to another. Oracle Autonomous Database performs routine database maintenance tasks automatically without human intervention while the system is running. It uses AI machine learning to automate database tuning, security, backups, updates, and other routine management tasks traditionally performed by database administrators and is self-repairing.

RFDS uses Oracle GoldenGate to create a digital data twin for offline mode that allows live replication of the database. GoldenGate enables RFDS doctors and healthcare workers in the outback to use a tablet to enter and retrieve the latest patient information when an Internet connection is not available. GoldenGate ensures that data captured offline are reliably synchronized with other healthcare systems when connectivity is restored. RFDS now completes its daily peak computing workloads three times faster on OCI, leading to faster and better decision-making by medical staff. By moving to the cloud, RFDS has reduced administrative workloads and lowered its data management costs by approximately 20 percent. With this architecture in place, IT staff can transition from maintenance activities to higher-value roles as technology innovators that can help the organization become a digital leader in healthcare.

SOURCES: Flyingdoctor.org.au, accessed August 26, 2024; "Royal Flying Doctor Service Launches New Digital Health Record Powered by Oracle Autonomous Database," Oracle.com, August 28, 2023; "RFDS Builds a Life-saving IT Architecture with Oracle Autonomous Database," Oracle.com, August 20, 2023.

The experience of Australia's RFDS illustrates the importance of data management. Business performance, including the ability to deliver good healthcare, depends

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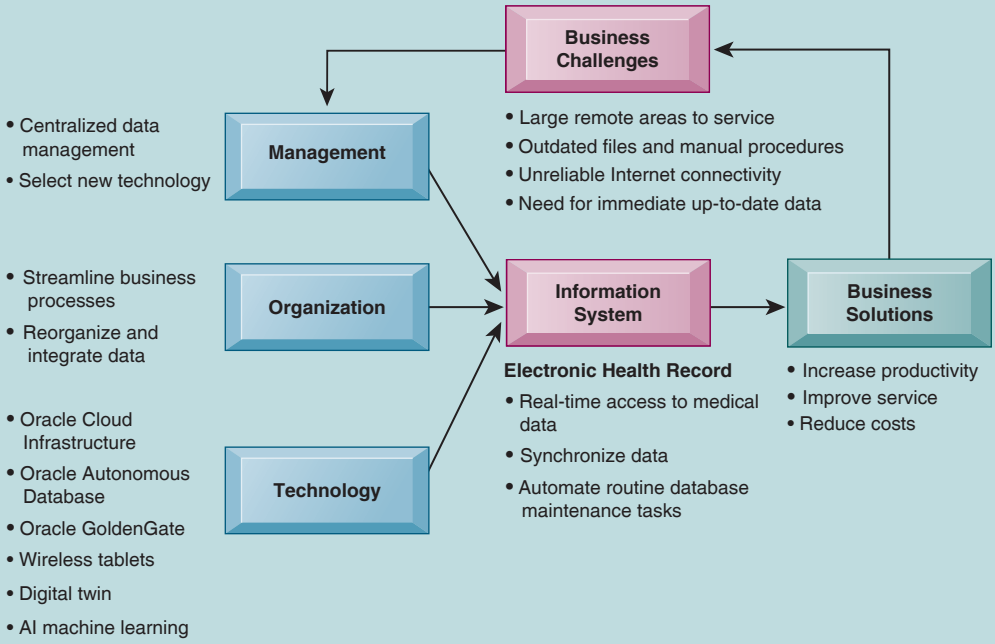


on what an organization can or cannot do with its data. Operational efficiency and decision making at RFDS were severely impaired by outdated manual processes, fragmented legacy systems, and unreliable and uneven wireless Internet connectivity. These conditions prevented RFDS doctors and staff from accessing accurate up-to-date information for patient treatment and from serving patients efficiently. How organizations store, organize, and manage their data has an enormous impact on organizational effectiveness.

The Opening Case Study diagram above calls attention to important points raised by this case. RFDS is a charitable organization providing medical care to underserved populations in Australia's remote outback. It was not always possible to use up-to-date information to treat patients in the field or access patient data online, lowering the quality of care and endangering some of the patient population.

The solution was to replace RFDS's patchwork of manual and legacy systems with an enterprise-wide cloud infrastructure (OCI) and use Oracle Autonomous Database to integrate RFDS medical data from all sources. RFDS also used Oracle GoldenGate to synchronize data created offline during treatment with the new database. RFDS staff can now use the most up-to-date and accurate information to treat patients, make decisions more rapidly, and work more efficiently.

To make this solution work, RFDS had to reorganize their data into a standard companywide format; establish rules, responsibilities, and procedures for accessing and using the data; and provide tools for making the data accessible to users online and offline.

Case Study Questions

1. What problems did using a manual patient record system create for RFDS?
2. How did switching to Oracle Autonomous Database improve how RFDS delivered healthcare?

CAREER BENEFIT

RFDS's IT staff had the opportunity to develop expertise in important cloud computing tools. This helps them advance their careers at RFDS but also makes them more qualified for higher-paying jobs at other organizations. Moving to the cloud also reduced administrative workloads so IT could focus on transitioning from routine maintenance to higher-value activities. Non-IT staff benefited from contributing to the development of such an important solution for the organization. Working with the new system helped staff to dramatically improve efficiency and accuracy in their work. RFDS's solution also had a wider social benefit in that it helped Australia provide better healthcare for underserved residents of remote areas.

6.1 Describe Traditional File Environment Data Management Problems

An effective information system provides users with accurate, timely, and relevant information. Accurate information is free of errors. Information is timely when it is available to decision makers when it is needed. Information is relevant when it is useful and appropriate for the types of work and decisions that require it. Many businesses don't have timely, accurate, or relevant information because the data in their information systems have been poorly organized and maintained. That's why data management is so essential. To understand the problem, let's first look at how information systems arrange data in computer files and then traditional file management.

File Organization Terms and Concepts

A computer system organizes data in a hierarchy that starts with bits and bytes and progresses to fields, records, files, and databases. A **bit** (or binary digit) represents the smallest unit of data a computer can handle: a value of 0 or 1. A group of bits, called a **byte**, represents a single character, which can be a letter, a number, or another symbol. A grouping of characters into a word, a group of words (such as a person's name), or a complete number (such as a person's age) is called a **field**. A group of related fields comprises a **record**; a group of records of the same type is called a **file**. A group of related files makes up a **database**.

Bit
A binary digit representing the smallest unit of data a computer can handle. It can only have one of two states, representing a value of 0 or 1.

Byte
A group of bits used to represent a single character, which can be a letter, number, or other symbol.

Field
A grouping of characters into a word, a group of words, or a complete number, such as a person's name or age.

Record
A group of related fields.

File
A group of records of the same type.

Database
A group of related files.

Entity
A person, place, thing, or event about which information is kept.

Attribute
A piece of information describing a particular entity.

Examine Figure 6.1, which illustrates the data hierarchy used to create a student database. Individual bits form the base of the hierarchy and are grouped together to form a byte. A grouping of bytes forms a field, in this case a field that identifies a specific course. A group of related fields, here a specific student's ID, course, date, and grade fields, creates a record. A grouping of similar student records creates a student course file. The student course file is then grouped with files on students' personal histories and financial backgrounds to create a student database. Databases are at the heart of all information systems because they keep track of the people, places, and things that a business must deal with on a continuing, often instant basis.

A record describes an entity. An **entity** is a person, place, thing, or event on which information is stored and maintained. Each characteristic or quality describing a particular entity is called an **attribute**. Look again at the File section of Figure 6.1. Student_ID, Course, Date, and Grade are attributes of the entity COURSE. The specific values that these attributes can have are found in the fields of the record describing the entity COURSE.

Problems with the Traditional File Environment

In most organizations, systems tended to grow independently without a companywide plan. Accounting, finance, manufacturing, human resources, and sales and marketing all developed their own systems and data files. Examine Figure 6.2, which illustrates the traditional approach to information processing. Each functional area has its own set of software applications and data files, which lead to proliferation of redundant, inconsistent data. Each application requires its own set of data files and computer program to operate. For example, the human resources functional area might have a personnel master file, a payroll file, a medical insurance file, a pension file, a mailing list file, and so forth, until tens, perhaps hundreds, of files and programs existed in the company as a whole.

As this process goes on for 5 or 10 years, the organization becomes saddled with hundreds of programs and applications that are very difficult to maintain and manage. The resulting problems include data redundancy and inconsistency, program-data dependence, inflexibility, poor data security, and an inability to share data among applications.

Figure 6.1 The Data Hierarchy

A computer system organizes data in a hierarchy that starts with the bit, which represents either a 0 or a 1. Bits can be grouped to form a byte to represent one character, number, or symbol. Bytes can be grouped to form a field, and related fields can be grouped to form a record. Related records can be collected to form a file, and related files can be organized into a database.

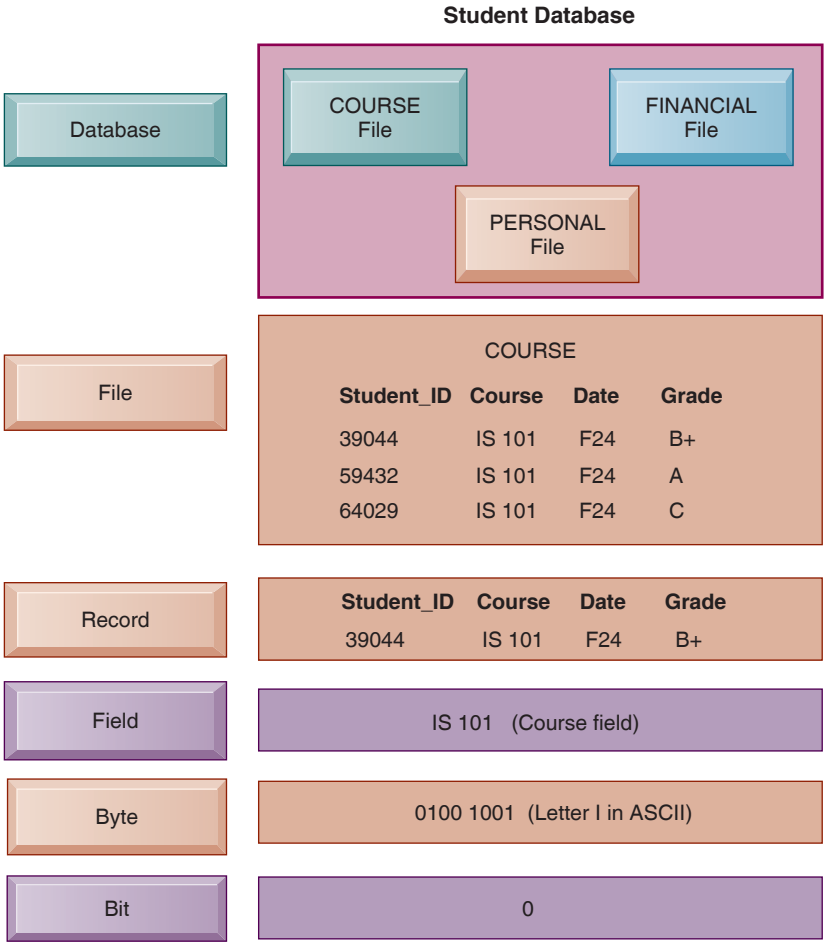
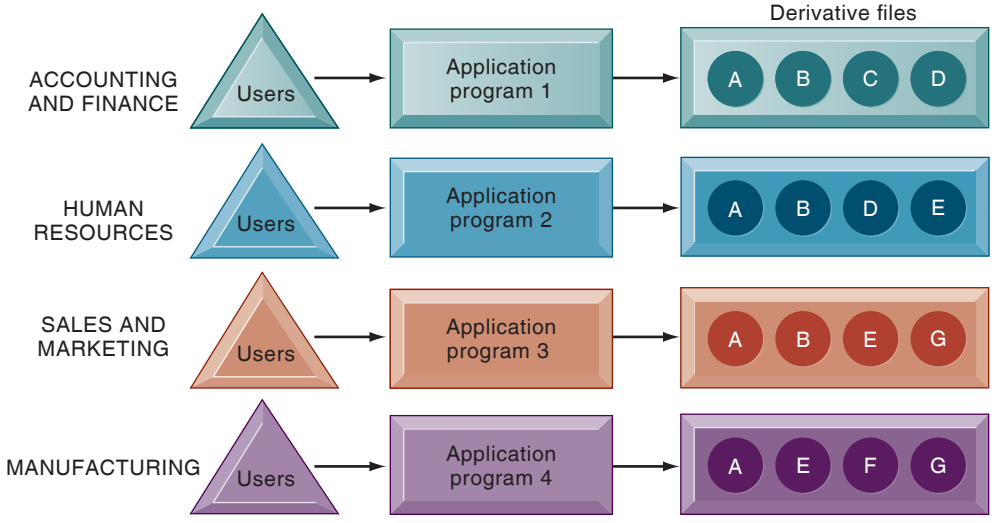


Figure 6.2 Traditional File Processing

The use of a traditional approach to file processing encourages each functional area in a corporation to develop specialized applications. Each application requires a unique data file that is likely to be a subset of a master file. These subsets of the master file lead to data redundancy and inconsistency, processing inflexibility, and wasted storage resources.



Data Redundancy and Inconsistency

Data redundancy
The presence of duplicate data in multiple data files.

Data inconsistency
The presence of different values for the same attribute, when the attribute is stored in multiple systems.

Data redundancy is the presence of duplicate data in multiple data files so that the same data are stored in more than one place or location. Data redundancy occurs when different groups in an organization independently collect the same piece of data and store it independently of each other. Data redundancy wastes storage resources and also leads to **data inconsistency**, where the same attribute may have different values. For example, in instances of the entity COURSE illustrated in Figure 6.1, the Date may be updated in some systems but not in others. The same attribute, Student_ID, might also have different names in different systems throughout the organization. Some systems might use Student_ID and others might use ID, for example.

Additional confusion can result from using different coding systems to represent values for an attribute. For instance, the sales, inventory, and manufacturing systems of a clothing retailer might use different codes to represent clothing size. One system might represent clothing size as "extra-large," whereas another might use the code "XL" for the same purpose. The resulting confusion would make it difficult for companies to create customer relationship management, supply chain management, or enterprise systems that integrate data from different sources.

Program-Data Dependence

Program-data dependence
Refers to the coupling of data stored in files and the specific programs required to update and maintain those files such that changes in the program require changes to the data.

Program-data dependence refers to the coupling of data stored in files and the specific programs required to update and maintain those files such that changes in the programs require changes to the data. Every traditional computer program must describe the location and nature of the data with which it works. In a traditional file environment, any change in a software program could require a change in the data accessed by that program. One program might be modified from a five-digit to a nine-digit ZIP code. If the original data file were changed from five-digit to nine-digit ZIP codes, then other programs that required the five-digit ZIP code would no longer work properly. Such changes could cost millions of dollars to implement properly.

Lack of Flexibility

A traditional file system can deliver routine scheduled reports after extensive programming efforts, but it cannot deliver ad hoc reports or respond to unanticipated information requirements in a timely fashion. The information required by ad hoc requests is somewhere in the system but may be too expensive to retrieve. Several programmers might have to work for weeks to put together the required data items in a new file.

Poor Security

Because there is little control or management of data, access to and dissemination of information may be out of control. Management might have no way of knowing who is accessing or even making changes to the organization's data.

Lack of Data Sharing and Availability

Because pieces of information in different files and different parts of the organization cannot be related to one another, it is virtually impossible for information to be shared or accessed in a timely manner. Information cannot flow freely across different functional areas or different parts of the organization. If users find different values for the same piece of information in two different systems, they may not want to use these systems because they cannot trust the accuracy of their data.

6.2 Describe DBMS and Relational DBMS Capabilities

Database technology cuts through many of the problems of traditional file organization. A more rigorous definition of a database is a collection of data organized to serve many applications efficiently by centralizing the data and controlling redundant data. Rather than storing data in

separate files for each application, data appear to users as being stored in only one location. A single database services multiple applications. Instead of a corporation storing employee data in separate information systems and separate files for personnel, payroll, and benefits, the corporation could create a single common human resources database.

Database Management Systems

A **database management system (DBMS)** is software that enables an organization to centralize data, manage that data efficiently, and provide access to the stored data by application programs. The DBMS acts as an interface between application programs and the physical data files. When the application program calls for a data item, such as gross pay, the DBMS finds this item in the database and presents it to the application program. Using traditional data files, the programmer would have to specify the size and format of each data element used in the program and then tell the computer where they were located.

Database management system (DBMS)
Software used to create and maintain a database and enable individual business applications to extract the data they need without having to create separate files or data definitions in the applications.

The DBMS relieves the programmer or end user from the task of understanding where and how the data are actually stored by separating the logical and physical views of the data. The *logical view* presents data, as they would be perceived by end users or business specialists, whereas the *physical view* shows how data are actually organized and structured on physical storage media.

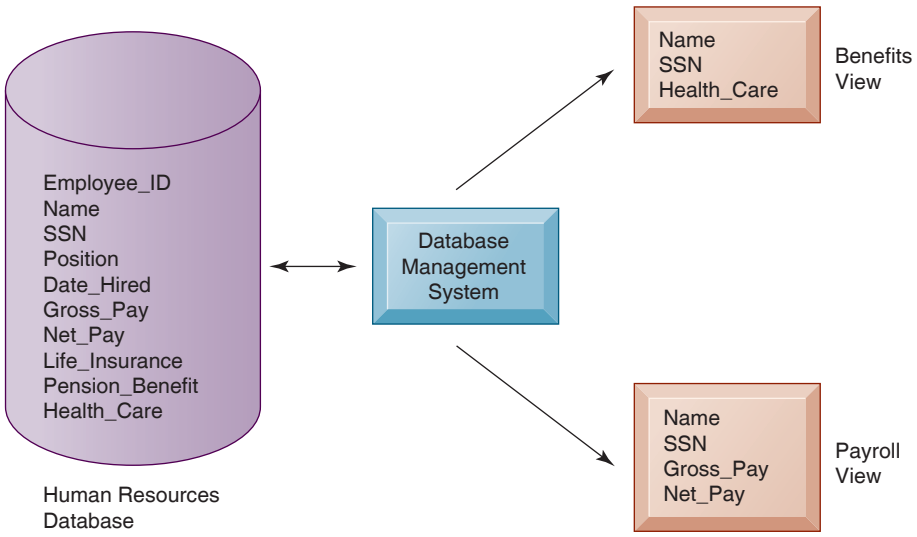
The database management software makes the physical database available for different logical views required by users. Examine Figure 6.3, which illustrates a DBMS that enables different views of the data contained within a human resources database. Different members of the human resources team need access to different types of data. For instance, a benefits specialist typically will require a view consisting of the employee's name, social security number, and health insurance coverage. A payroll department member will need data such as the employee's name, social security number, gross pay, and net pay. The data for all of these views are stored in a single database, where the organization can manage it more easily.

How a DBMS Solves the Problems of the Traditional File Environment

A DBMS reduces data redundancy and inconsistency by minimizing isolated files in which the same data are repeated. The DBMS may not enable the organization to eliminate data redundancy entirely, but it can help control redundancy. Even if the organization maintains some

Figure 6.3 Human Resources Database with Multiple Views

A single human resources database provides many different views of data, depending on the information requirements of the users. Illustrated here are two possible views, one of interest to a benefits specialist and one of interest to a member of the company's payroll department.



redundant data, using a DBMS eliminates data inconsistency because the DBMS can help the organization ensure that every occurrence of redundant data has the same values. The DBMS uncouples programs and data, enabling data to stand on their own. The description of the data used by the program does not have to be specified in detail each time a different program is written. Access and availability of information will be increased, and program development and maintenance costs reduced because users and programmers can perform ad hoc queries of the database for many simple applications without having to write complicated programs. The DBMS enables the organization to centrally manage data, their use, and security. Data sharing throughout the organization is easier because the data are presented to users as being in a single location rather than fragmented in many different systems and files.

Relational Databases and Relational DBMS

Relational database

A type of logical database model that that organizes data into two-dimensional tables with columns and rows. It can relate data stored in one table to data in another as long as the two tables share a common data element.

Relational DBMS (RDBMS)

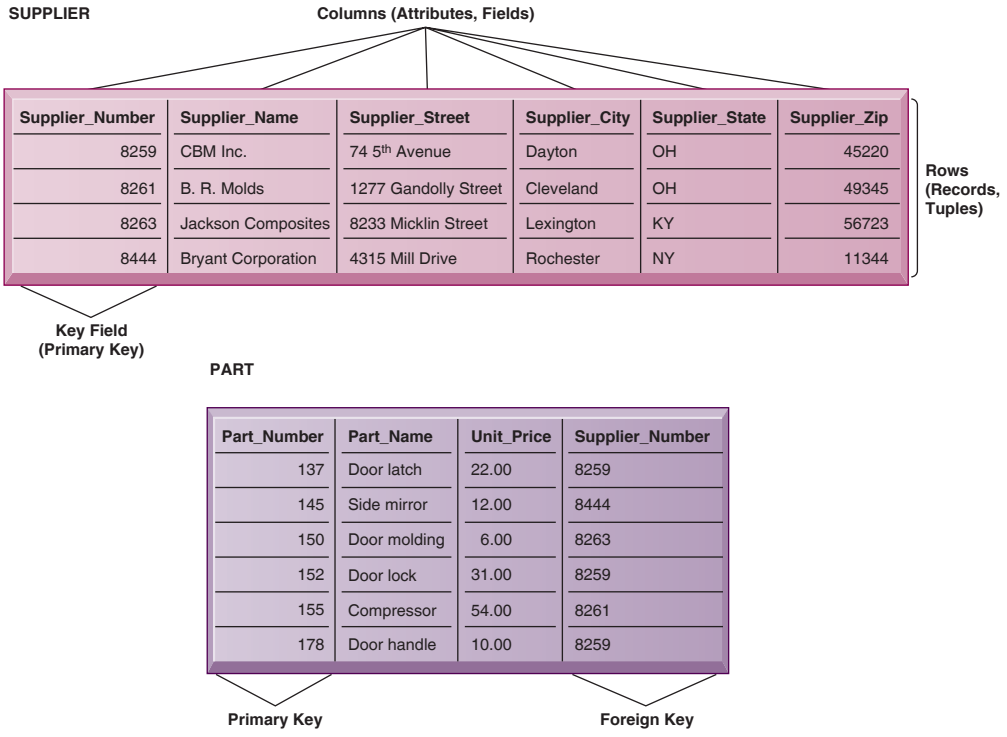
DBMS system designed to work with relational databases.

Contemporary DBMS use different database models to keep track of entities, attributes, and relationships. The most common type of database today is the **relational database**. Relational databases represent data as two-dimensional tables (called relations). Tables may be referred to as files. Each table contains data on an entity and its attributes. DBMS systems that work with relational databases are called **relational DBMS (RDBMS)** and are the most popular type of DBMS today for desktop computers as well as for larger computers and mainframes. Microsoft Access is a relational DBMS for desktop systems, whereas IBM Db2, Oracle Database, and Microsoft SQL Server are relational DBMS for large mainframes and midrange computers. MySQL is a popular open-source relational DBMS.

Examine Figure 6.4, which shows how a relational database organizes data about suppliers and parts. The database has a separate table for the entity SUPPLIER and a table for the entity PART. Each table consists of a grid of columns and rows of data. Each individual element of data for each entity is stored as a separate field, and each field represents an attribute for that entity. Fields in a relational database are also called columns. For the entity SUPPLIER, the supplier identification number, name, street, city, state, and ZIP code are stored as separate fields within the SUPPLIER table and each field represents an attribute for the entity SUPPLIER.

Figure 6.4 Relational Database Tables

A relational database organizes data in the form of two-dimensional tables. Illustrated here are tables for the entities SUPPLIER and PART showing how they represent each entity and its attributes. Supplier_Number is a primary key for the SUPPLIER table and a foreign key for the PART table.



The actual information about a single supplier that resides in a table is called a row. Rows are commonly referred to as records. Data for the entity PART have their own separate table.

The field for Supplier_Number in the SUPPLIER table uniquely identifies each record so that the record can be retrieved, updated, or sorted. It is called a **key field**. Each table in a relational database has one field that is designated as its **primary key**. The primary key is the unique identifier for all the information in any row of the table, and cannot be duplicated. Supplier_Number is the primary key for the SUPPLIER table, and Part_Number is the primary key for the PART table. Note that Supplier_Number appears in both the SUPPLIER and PART tables. In the SUPPLIER table, Supplier_Number is the primary key. When the field Supplier_Number appears in the PART table, it is called a **foreign key** and is essentially a lookup field to look up data about the supplier of a specific part.

Key field

A field in a record that uniquely identifies instances of that record so that it can be retrieved, updated, or sorted.

Primary key

Unique identifier for all the information in any row of a database table.

Foreign key

Field in a database table that enables users to find related information in another database table.

Operations of a Relational DBMS

Relational database tables can be combined easily to deliver data required by users, provided that any two tables share a common data element. Suppose you wanted to find in this database the names of suppliers who could provide part number 137 or part number 150. You would need information from two tables: the SUPPLIER table and the PART table. Note that these two tables have a shared data element: Supplier_Number.

In a relational DBMS, three basic operations are used to develop useful sets of data: select, join, and project. The *select* operation creates a subset consisting of all records in the file that meet stated criteria. In other words, *select* creates a subset of rows that meet certain criteria. The *join* operation combines relational tables to provide the user with more information than is available in individual tables. The *project* operation creates a subset consisting of columns in a table, permitting the user to create new tables that contain only the information required. Examine Figure 6.5, which illustrates how these three basic operations can be used to find the names of suppliers who provide part number 137 or part number 150. To do so, you need to select records (rows) from the PART table where the Part_Number equals 137 or 150. Next, you need to join the now-shortened PART table (only parts 137 or 150 will be presented) and the SUPPLIER table into a single new table. Finally, you extract from the new table only the following columns: Part_Number, Part_Name, Supplier_Number, and Supplier_Name.

Capabilities of Database Management Systems

A DBMS includes capabilities and tools for organizing, managing, and accessing the data in the database. The most important are its data definition language, data dictionary, and data manipulation language.

DBMS have a **data definition** capability to specify the structure of the content of the database to be specified. This capability is used to create database tables and to define the characteristics of the fields in each table. This information about the database can be documented in a data dictionary. A **data dictionary** is a tool that stores definitions of data elements and their characteristics.

Examine Figure 6.6, which illustrates the rudimentary data dictionary capabilities of Microsoft Access. The dictionary displays information about the name, description, size, type, format, and other properties of each field in a table. Data dictionaries for large corporate databases may capture additional information, such as usage, ownership (who in the organization is responsible for maintaining the data), authorization, security, and the individuals, business functions, programs, and reports that use each data element.

Data definition

Specifies the structure of the content of a database.

Data dictionary

Database tool that stores definitions of data elements and their characteristics.

Querying and Reporting

A DBMS includes tools for accessing and manipulating information in databases. For example, a specialized programming language called a **data manipulation language** is used to add, change, delete, and retrieve the data in the database. This language contains commands

Data manipulation language

A specialized programming language that is used to add, change, delete, and retrieve data in a database.

Figure 6.5 The Three Basic Operations of a Relational DBMS

The select, join, and project operations enable data from two different tables to be combined and only selected attributes to be displayed.

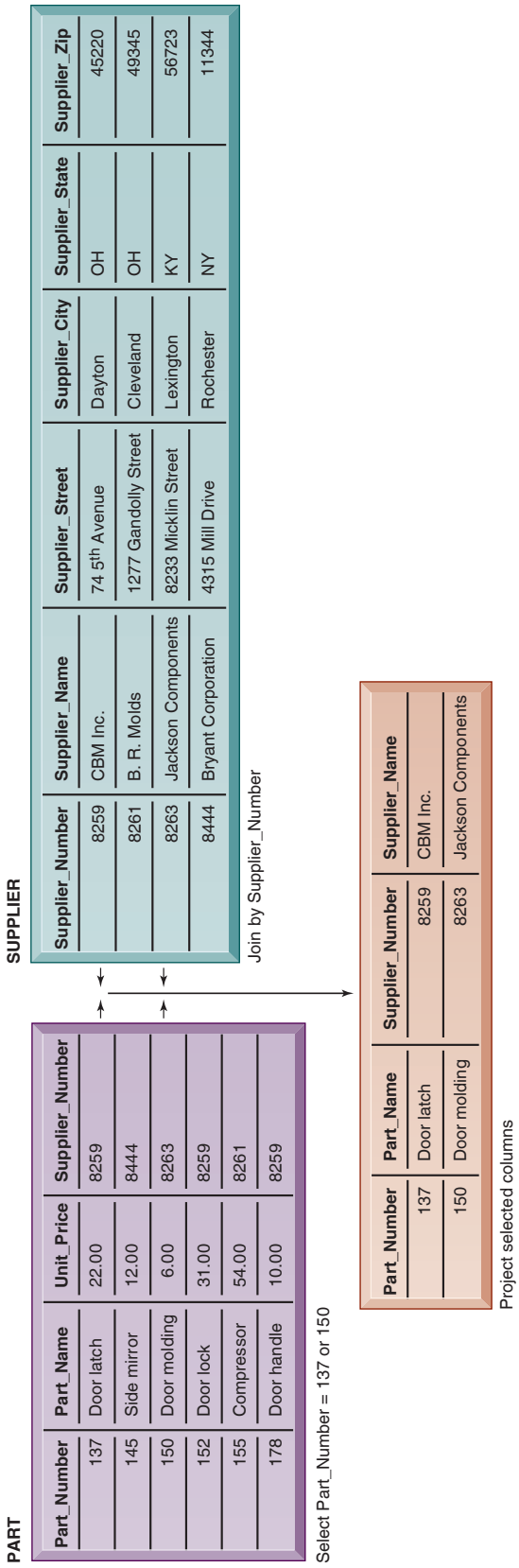


Figure 6.6 Microsoft Access Data Dictionary Features

Microsoft Access has a rudimentary data dictionary capability that displays information about the size, format, and other characteristics of each field in a database. The information displayed here is maintained in the SUPPLIER table. The small key icon to the left of Supplier_Number indicates that it is a key field.

Used with permission from Microsoft Corporation

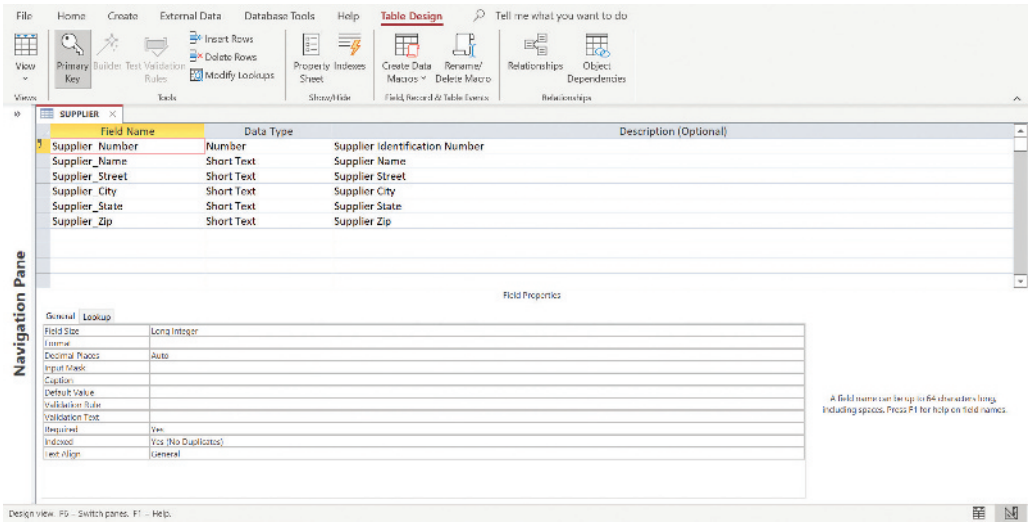


Figure 6.7 Example of an SQL Query

Illustrated here are the SQL statements for a query to select suppliers for parts 137 or 150. They produce a list with the same results as Figure 6.5.

```
SELECT PART.Part_Number, PART.Part_Name, SUPPLIER.Supplier_Number,
SUPPLIER.Supplier_Name
FROM PART, SUPPLIER
WHERE PART.Supplier_Number = SUPPLIER.Supplier_Number AND
Part_Number = 137 OR Part_Number = 150;
```

that permit end users and programming specialists to extract data from the database to satisfy information requests and develop applications. The most prominent data manipulation language today is **Structured Query Language**, or **SQL** (usually pronounced as S-Q-L but sometimes as Sequel). A **query** is a request for data from a database. Examine Figure 6.7, which illustrates the SQL query that would produce the table illustrated in Figure 6.5.

Structured Query Language (SQL)

The standard data manipulation language for relational database management systems.

Query

Request for data from a database.

Users of DBMS for large and midrange computers, such as IBM Db2, Oracle Database, or Microsoft SQL Server, can employ SQL to retrieve information they need from the database. Microsoft Access also uses SQL, but it provides its own set of user-friendly tools for querying databases and for organizing data from databases into more polished reports.

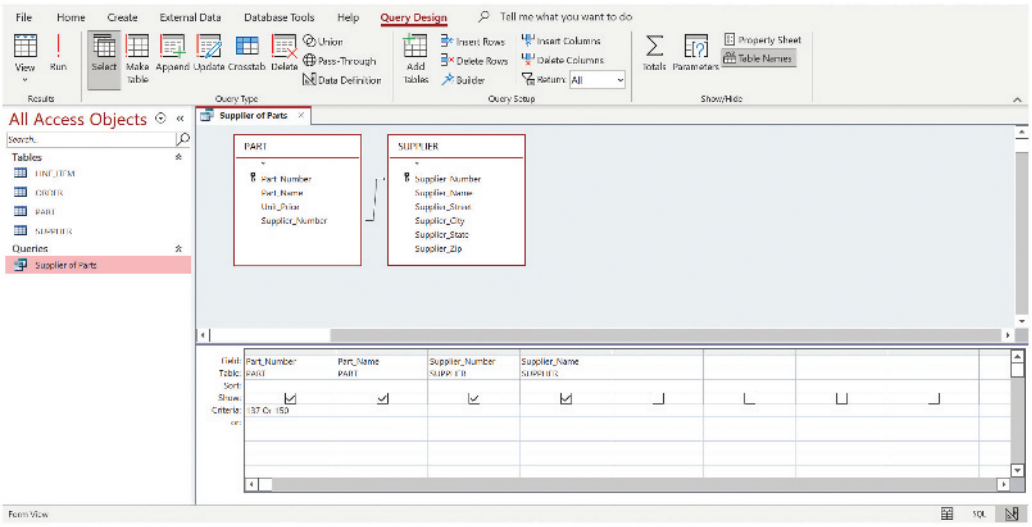
In Microsoft Access, you will find features that enable users to create queries by identifying the tables and fields they want and the results, and then selecting the rows from the database that meet specific criteria. These actions in turn are translated into SQL commands. Examine Figure 6.8, which illustrates how the same query as the SQL query to select parts and suppliers would be constructed using Microsoft Access query-building tools.

Microsoft Access and other DBMS include capabilities for report generation so that the data of interest can be displayed in a more structured and polished format than would be possible just by querying. Microsoft Access also has capabilities for developing desktop system applications. These include tools for creating data entry screens, reports, and developing the logic for processing transactions.

Figure 6.8 A Microsoft Access Query

Illustrated here is how the query in Figure 6.7 would be constructed using Microsoft Access query-building tools. It shows the tables, fields, and selection criteria used for the query.

Used with permission from Microsoft Corporation



6.3 Discuss Relational Database Design Principles

To create a database, you must understand the relationships among the data, the type of data that will be maintained in the database, how the data will be used, and how the organization will need to change to manage data from a companywide perspective. The database requires both a conceptual design and a physical design. The conceptual, or logical, design of a database is an abstract model of the database from a business perspective, whereas the physical design shows how the database is actually arranged on direct-access storage devices.

Normalization and Entity-Relationship Diagrams

Normalization

When designing a relational database, the process of streamlining complex groups of data to minimize redundant data elements and awkward many-to-many relationships.

The conceptual database design describes how the data elements in the database are to be grouped. The design process identifies relationships among data elements and the most efficient way of grouping data elements together to meet business information requirements. The process also identifies redundant data elements and the groupings of data elements required for specific application programs. Groups of data are organized, refined, and streamlined until an overall logical view of the relationships among all the data in the database emerges.

To use a relational database model effectively, complex groupings of data must be streamlined to minimize redundant data elements and awkward many-to-many relationships. This process is known as **normalization** and creates small, stable, yet flexible and adaptive data structures from complex groups of data. Examine Figures 6.9 and 6.10, which illustrate this process.

Figure 6.9 An Unnormalized Relation for ORDER

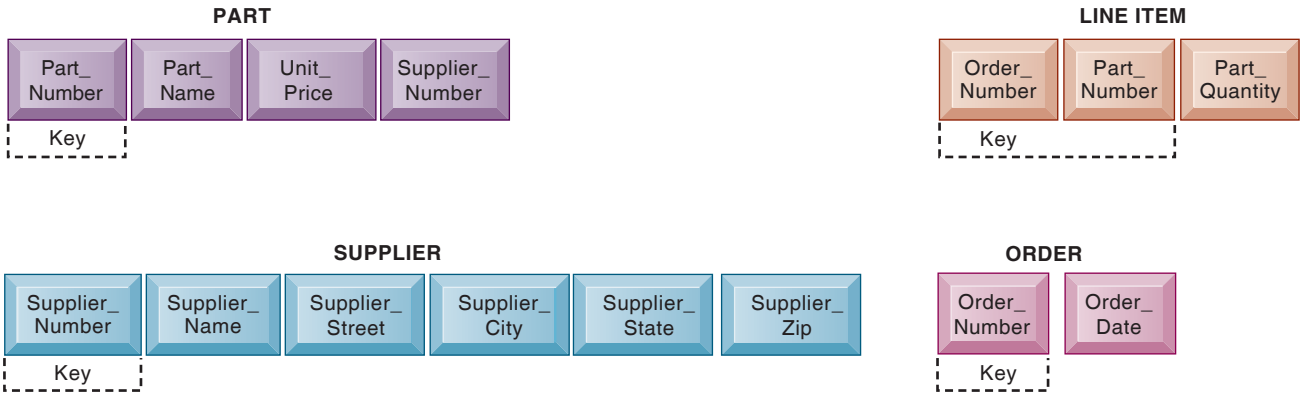
An unnormalized relation contains repeating groups. For example, there can be many parts for each order. There is only a one-to-one correspondence between Order_Number and Order_Date.

ORDER (Before Normalization)

Order_ Number	Order_ Date	Part_ Number	Part_ Name	Unit_ Price	Part_ Quantity	Supplier_ Number	Supplier_ Name	Supplier_ Street	Supplier_ City	Supplier_ State	Supplier_ Zip

Figure 6.10 Normalized Tables Created from ORDER

After normalization, the original relation ORDER has been broken down into four smaller relations. The relation ORDER is left with only two attributes, and the relation LINE_ITEM has a combined, or concatenated, key consisting of Order_Number and Part_Number.



In the particular business modeled here, an order can have more than one part, but each part is provided by only one supplier. If you build a relation called ORDER with all the fields included here, you would have to repeat the name and address of the supplier for every part on the order, even though the order is for parts from a single supplier. This relationship contains what are called repeating data groups because there can be many parts in a single order to a given supplier.

Figure 6.10 shows a more efficient way to arrange the data: by breaking down ORDER into smaller relations, each of which describes a single entity. If you go step by step and normalize the relation ORDER, you will emerge with the relations illustrated in Figure 6.10.

Relational database systems try to enforce **referential integrity** rules to ensure that relationships between coupled tables remain consistent. When one table has a foreign key that points to another table, you may not add a record to the table with the foreign key unless there is a corresponding record in the linked table. In the database examined earlier in this chapter, the foreign key Supplier_Number links the PART table to the SUPPLIER table. You may not add a new record to the PART table for a part with Supplier_Number 8266 unless there is a corresponding record in the SUPPLIER table for Supplier_Number 8266. You must also delete the corresponding record in the PART table if you delete the record in the SUPPLIER table for Supplier_Number 8266. In other words, you shouldn't have parts from nonexistent suppliers!

Referential integrity

Rules to ensure that relationships between coupled database tables remain consistent.

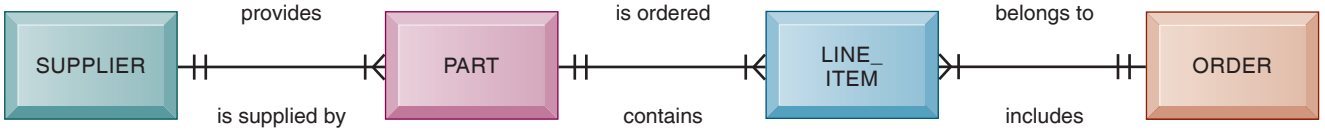
Database designers document their data model with an **entity-relationship diagram**, which illustrates the relationship among various entities in the database. Examine Figure 6.11, which illustrates the relationship between the entities SUPPLIER, PART, LINE_ITEM, and ORDER. The boxes represent entities, and the lines connecting the boxes represent relationships. A line connecting two entities that ends in two short marks designates a one-to-one relationship. A line connecting two entities that ends with a crow's foot topped by a short mark indicates a one-to-many relationship. The text above the line is read left to right (i.e., SUPPLIER provides PART and so on), while the text below the line is read right to left (i.e., ORDER includes LINE_ITEM and so on). Figure 6.11 shows that one ORDER can contain many LINE_ITEMs. (A PART can be ordered many times and appear many times as a line item in a single order.) Each PART can have only one SUPPLIER, but many PARTs can be provided by the same SUPPLIER.

Entity-relationship diagram

A methodology for documenting databases that illustrates the relationship among various entities in the database.

Figure 6.11 An Entity-Relationship Diagram

This diagram shows the relationships between the entities SUPPLIER, PART, LINE_ITEM, and ORDER that might be used to model the database in Figure 6.10.



It can't be emphasized enough: If the business doesn't get its data model right, the system won't be able to serve the business well. The company's systems will not be as effective as they could be because they'll have to work with data that may be inaccurate, incomplete, or difficult to retrieve. Understanding the organization's data and how they should be represented in a database is an important lesson you can learn from this course.

For example, *The Globe and Mail*, Canada's largest newspaper, was unable until recently to execute ambitious marketing programs because it had trouble housing and managing the data on sales prospects. Much of the required data were stored in a mainframe system where the data were not easy to access and analyze. If users needed any information, they had to extract the data from the mainframe and bring it to one of a number of local databases—including those maintained in Microsoft Access and Excel—for analysis. This practice created numerous pockets of data maintained in isolated databases for specific purposes but no central repository where the most up-to-date data could be accessed from a single place. With data scattered in so many different systems throughout the company, it was very difficult to cross-reference subscribers with prospects when developing the mailing list for a marketing campaign.

6.4 Describe Non-Relational Database, Cloud Database, and Blockchain Capabilities

Cloud computing, unprecedented data volumes, massive workloads for web services, and the need to store new types of data require database alternatives to the traditional relational model of organizing data in the form of tables, columns, and rows.

Non-Relational Databases

Non-relational database management system

Database that uses a more flexible data model and is designed for managing large data sets across many distributed computers. Also referred to as NoSQL.

Companies are turning to "NoSQL" non-relational database technologies for this purpose. A **non-relational database management system** uses a more flexible data model and is designed for managing large data sets across many distributed machines and for easily scaling up or down. They are useful for accelerating simple queries against large volumes of structured and unstructured data, including web, social media, graphics, and other forms of data that are difficult to analyze with traditional SQL-based tools. There are several different kinds of NoSQL databases, each with its own data models and technical features. Oracle NoSQL Database is one example, as is Amazon's SimpleDB, one of the Amazon Web Services that run in the cloud. SimpleDB provides a simple web services interface to create and store multiple data sets, query data easily, and return the results. There is no need to predefine a formal database structure or change that definition if new data are added later.

Leading Taiwanese mobile game developer Yile Technology Co. migrated to NoSQL MongoDB Atlas to improve customer experience and increase the speed of playing. Yile found there were performance problems when it tried to base its first prototype game Online808 on a relational database management system. The RDBMS was unable to handle the complex data for this type of application, which requires fast response times. Even when as few as 50 players connected to the game at the same time, the game response time could be as long as 3 seconds, far exceeding the 0.5-second minimum the game designers had set. Yile was able to achieve a response time of less than 0.5 seconds when it switched to MongoDB's NoSQL database for the game's customer profile data. This responsiveness helped Online808 become the second best-performing social casino game in Taiwan (Mongodb.com, 2024).

Cloud Databases and Distributed Databases

Cloud database

Database provided by a hosted cloud service provider.

A **cloud database** stores data on a hosted cloud service provider. Amazon Relational Database Service (RDS) is a managed cloud relational database service offering MySQL, Microsoft SQL Server, Oracle Database, PostgreSQL, and MariaDB as database engines. Pricing is based on usage. Oracle has its own Oracle Cloud Database, and Microsoft Azure SQL Database is a cloud-based relational database service based on the Microsoft SQLServer DBMS.

Cloud-based data management services have special appeal for online startups or small to medium-sized businesses seeking database capabilities at a lower cost than in-house database products. They are also useful for storage and transfer of the massive quantities of data used in AI applications. Public cloud vendors now offer AI outsourcing services, allowing companies to test out software and machine learning algorithms without risking their primary infrastructure. Google now offers its Spanner distributed database technology as a cloud service. A **distributed database** is one that is stored in multiple physical locations. Parts or copies of the database are physically stored in one location, and other parts or copies are maintained in other locations. Spanner makes it possible to store information across millions of machines in hundreds of data centers around the globe and to synchronize the data precisely in all of its locations and ensure the data are always consistent. Google uses Spanner to support its various cloud services and is now making the technology available to other companies that might need such capabilities to run a global business.

Distributed database
Database stored in multiple physical locations.

Blockchain

Blockchain is a distributed database system that enables firms and organizations to create and verify transactions on a network nearly instantaneously without a central authority. A blockchain stores transactions as a distributed ledger that is shared among a network of computers. The information held in the database is continually reconciled by the computers in the network.

Blockchain
Distributed database system that stores permanent and tamper-proof records of transactions as a distributed ledger that is shared among a network of computers.

The blockchain maintains a continually growing list of records called blocks. Each block contains a timestamp and a link to a previous block. Once a block of data is recorded on the blockchain ledger, it cannot be altered retroactively, which means participants can trust the data that they are presented with. When someone wants to add a transaction, participants in the network (all of whom have copies of the existing blockchain) run algorithms to evaluate and verify the proposed transaction. Legitimate changes to the ledger are recorded across the blockchain in a matter of seconds or minutes, and records are protected through cryptography. What makes a blockchain system possible and attractive to business firms is encryption and authentication of the actors and participating firms, which ensure that only legitimate actors can enter information, and only validated transactions are accepted. Once recorded, the transaction cannot be changed.

Examine Figure 6.12, which illustrates how blockchain works when fulfilling an order. The first block in the chain is the order transaction record. The block is verified by the user and then broadcast to a peer-to-peer (P2P) network of computers that run an algorithm to evaluate

Figure 6.12 How Blockchain Works

A blockchain system is a distributed database that records transactions in a peer-to-peer network of computers.

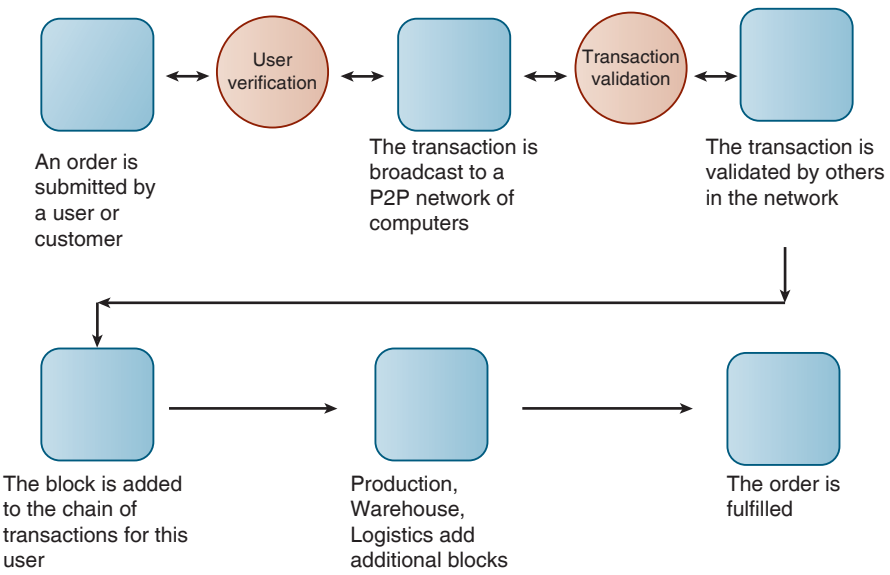


Table 6.1 Examples of Blockchain Applications

Business Specialty	Organization	Blockchain Application
Supply chain management	IBM Food Trust	IBM Food Trust, a collaborative network of growers, distributors, manufacturers, and retailers, is using blockchain to track and trace goods through the supply chain, from raw materials to finished products. The solution connects participants through a permissioned, immutable, and shared record of food provenance, transaction data, processing details and more. This can help to ensure transparency and traceability, and to reduce fraud.
Healthcare	Government of Estonia	The government of Estonia is using blockchain to create secure and tamper-proof electronic health records (EHRs). This can help to improve the quality of care and to reduce fraud.
Banking	Ripple	Blockchain can be used to make cross-border payments faster and cheaper, as it eliminates the need for intermediaries such as banks. For example, the Ripple blockchain-based payment network allows users to send and receive money in different currencies quickly and easily.
Cryptocurrency	Circle	Circle is a fintech platform that oversees the exchange of traditional and cryptocurrency payments between users. Circle's merchant payment services move money between digital currencies securely and quickly. Circle accepts crypto payments in currencies like Bitcoin, Ethereum, USD Coin and Euro Coin.
Shipping and Logistics	DHL	DHL uses blockchain to keep a digital ledger of shipments and maintain integrity of transactions.

and verify the block. Once the block is validated by the computers in the network, it is added to the chain of transactions for this user. Next, production, warehouse, and logistics add to the chain additional blocks related to the order until the order is fulfilled.

There are many benefits to firms using blockchain databases. Blockchain networks can radically reduce the cost of verifying users, validating transactions, and the risks of storing and processing transaction information across thousands of firms. Instead of thousands of firms building their own private transaction systems and then integrating them with suppliers, shippers, and financial institution systems, blockchain can provide a single, simple, low-cost transaction system for participating firms. Standardization of recording transactions is aided through the use of *smart contracts*. Smart contracts are computer programs that implement the rules governing transactions between firms, for example, what is the price of products, how they will be shipped, when the transaction will be completed, who will finance the transaction, what the financing terms are, and so on. Table 6.1 shows examples of real-world blockchain applications in supply chain management, healthcare, finance, and logistics.

The simplicity and security that blockchain offers has made it attractive for storing and securing financial transactions, supply chain transactions, medical records, and other types of data. Blockchain is a foundational technology for Bitcoin, Ethereum, and other cryptocurrencies. Chapter 8 provides more detail on securing transactions with blockchain.

6.5 Describe Tools Used to Analyze Data in Databases

Businesses use their databases to keep track of basic transactions, such as paying suppliers, processing orders, keeping track of customers, and paying employees. But they also need databases to provide information that will help the company run the business more efficiently and help managers and employees make better decisions. If a company wants to know which product is the most popular or who its most profitable customer is, the answer lies in the data.

The Challenge of Big Data

Most data collected by organizations used to be transaction data that could easily fit into rows and columns of relational database management systems. We are now witnessing an explosion of data from web traffic, email messages, and social media content (tweets, status messages), as well as machine-generated data from sensors (used in smart meters, manufacturing sensors, and electrical meters) or from electronic trading systems. These data may be unstructured or semi-structured and thus not suitable for relational database products that organize data in the form of columns and rows. We now use the term ***big data*** (see Chapter 1) to describe these data sets with volumes so huge that they are beyond the ability of typical DBMS to capture, store, and analyze.

Big data is often characterized by the "3Vs": the extreme *volume* of data, the wide *variety* of data types and sources, and the *velocity* at which data must be processed. Big data doesn't designate any specific quantity but usually refers to data in the petabyte and exabyte range—in other words, billions to trillions of records, many from different sources. Big data are produced in much larger quantities and much more rapidly than traditional data. Increasingly, two additional "V"s have been added to the characterization: value (the benefits the data provides) and veracity (the data's quality and accuracy) (Robinson and Gillis, 2023).

Businesses are interested in big data because they can reveal more patterns and interesting relationships than smaller data sets, with the potential to provide new insights into customer behavior, weather patterns, financial market activity, or other phenomena. For example, Shutterstock, the global online image marketplace, stores 300 million images, adding 200,000 more each day. To find ways to optimize the buying experience, Shutterstock analyzes its big data to find out where its website visitors place their cursors and how long they hover over an image before making a purchase. Big data is also finding many uses in the public sector. For example, city governments have been using big data to manage traffic flows and to fight crime.

Big Data and Artificial Intelligence (AI)

Big data and AI have a synergistic relationship. One of the key benefits of AI is its ability to learn and improve its performance over time. This is done through machine learning, which involves feeding large amounts of data into an algorithm and allowing it to learn and make predictions from observations and data. AI requires a massive amount of data for training, while machine learning is able to use the information provided by big data to generate more valuable business insights. AI provides a powerful tool for analyzing big data. Table 6.2 provides examples of applications of AI and big data.

Table 6.2 Big Data and AI Applications

Organization	Big Data and AI Applications
McDonalds	Uses big data and AI to analyze drive-thru traffic patterns and identify ways to improve efficiency such as determining the optimal number of order lanes and the best way to staff them. Also uses AI in voice recognition systems that can help to reduce wait times.
JP Morgan Chase	Uses big data to assess creditworthiness, detect fraud, and manage risk. The company also uses AI to develop algorithms for trading and investment.
Google Health	Uses big data and AI to accurately interpret retinal scans to detect diabetic retinopathy. A large team of ophthalmologists helped Google review more than 100,000 retinal scans to develop an AI-based application called Automated Retinal Disease Assessment.
Siemens	Advanced Artificial Intelligence Services use big data and AI to monitor factory equipment, predict downtime, and optimize production processes. Also uses AI to develop robots.
TripAdvisor	Uses AI-based sentiment analysis to process the feedback received from millions of travelers. This method enables TripAdvisor to identify essential insights within customer reviews, personalize recommendations, and enhance its services.

To derive business value from big data, organizations need new technologies and tools capable of managing and analyzing nontraditional data along with their traditional enterprise data. They also need to know what questions to ask of the data and the limitations of big data. Capturing, storing, and analyzing big data can be expensive, and information from big data may not necessarily help decision makers. It's important to have a clear understanding of the problem big data will solve for the business.

Business Intelligence Infrastructure

Suppose you wanted concise, reliable information about current operations, trends, and changes across the entire company. If you worked in a large company, the data you need might have to be pieced together from separate systems, such as sales, manufacturing, and accounting, and even from external sources, such as demographic or competitor data. Increasingly, you might need to use big data. Many companies have turned to business intelligence (BI) systems, which we previously introduced in Chapter 2, for these purposes. A contemporary business intelligence infrastructure has an array of tools for obtaining useful information from all the different types of data used by businesses today, including semi-structured and unstructured big data in vast quantities. These capabilities include data warehouses, data marts, data lakes, Hadoop, in-memory computing, and data analytics platforms. Some of these capabilities are available as cloud services.

Data Warehouses, Data Marts, and Data Lakes

Data warehouse

A database, with reporting and query tools, that stores current and historical data extracted from various operational systems and consolidated for management reporting and analysis.

The traditional tool for analyzing corporate data has been the data warehouse. A **data warehouse** is a database that stores current and historical data of potential interest to decision makers throughout the company. The data originate in core operational transaction systems, such as systems for sales, customer accounts, and manufacturing, and may include data from website transactions. The data warehouse extracts current and historical data from multiple operational systems inside the organization. These data are combined with data from external sources and transformed by correcting inaccurate and incomplete data, and then restructuring the data for management reporting and analysis before being loaded into the data warehouse.

The data warehouse makes the data available for anyone to access as needed, but the data cannot be altered. A data warehouse system also provides a range of ad hoc and standardized query tools, analytical tools, and graphical reporting facilities.

Companies often build enterprise-wide data warehouses, where a central data warehouse serves the entire organization, or they create smaller, decentralized warehouses called data marts. A **data mart** is a subset of a data warehouse in which a summarized or highly focused portion of the organization's data is placed in a separate database for a specific population of users. For example, a company might develop marketing and sales data marts to deal with customer information. Bookseller Barnes & Noble used to maintain a series of data marts—one for point-of-sale data in retail stores, another for college bookstore sales, and a third for online sales.

Data mart

A subset of a data warehouse that contains only a portion of the organization's data for a specified function or population of users.

Data lake

Repository for raw unstructured data or structured data that for the most part have not yet been analyzed.

A **data lake** is a repository for raw unstructured data or structured data that, for the most part, has not yet been analyzed, and the data can be accessed in many ways. The data lake stores these data in their native format until they are needed.

Hadoop

Relational DBMS and data warehouse products are not well-suited for organizing and analyzing big data or data that do not easily fit into columns and rows used in their data models. For handling unstructured and semi-structured data, as well as structured data, in vast quantities, organizations are using Hadoop. **Hadoop** is an open-source software framework managed by the Apache Software Foundation that enables distributed, parallel processing of very large amounts of data across many inexpensive computers. It breaks a big data problem down into subproblems, distributes them among up to thousands of inexpensive computer processing nodes, and then combines the results into a smaller data set that is easier to analyze. You've probably used Hadoop to find the best airfare on the Internet, do a search on Google, or connect with a friend on Facebook.

Hadoop

Open-source software framework that enables distributed parallel processing of very large amounts of data across many inexpensive computers.

Hadoop consists of several key services including the Hadoop Distributed File System (HDFS) for data storage and MapReduce for high-performance parallel data processing. HDFS links together the file systems on the numerous nodes in a Hadoop cluster to turn them into one big file system. HDFS is often used to store the contents across a set of clustered computer nodes in a data lake. HBase, Hadoop's non-relational database, provides rapid access to the data stored on HDFS and a platform for running high-scale, real-time applications. Hadoop can process large quantities of any kind of data, including structured transactional data, loosely structured data such as Facebook and Twitter (now known as X) feeds, complex data such as web server log files, and unstructured audio and video data.

Hadoop runs on a cluster of inexpensive servers, and processors can be added or removed as needed. Companies use Hadoop for analyzing very large volumes of data as well as for a staging area for unstructured and semi-structured data before they are loaded into a data warehouse. For instance, LinkedIn runs its big data analytics on Hadoop. It now stores more than 1 exabyte of data across all of its Hadoop clusters. Its largest cluster stores 500 petabytes of data and maintains one billion objects. Companies offering Hadoop solutions include AWS, Cloudera, Microsoft, IBM, Oracle, and HPE. Other vendors offer tools for moving data into and out of Hadoop or for analyzing data within Hadoop.

Apache Spark is a more recent open-source framework for handling big data. Spark does not have a distributed file system and uses random access memory (RAM) to store data. Therefore, Spark reduces the time required to interact with servers and can work much faster than Hadoop for smaller data sets or real-time streaming data.

In-Memory Computing

Apache Spark is an example of **in-memory computing**, which relies primarily on a computer's internal memory (RAM) for data storage. (Conventional DBMS use disk storage systems.) Users access data stored in the system's primary memory, thereby eliminating bottlenecks from retrieving and reading data in a traditional, disk-based database and dramatically shortening query response times. In-memory processing makes it possible for large sets of data, amounting to the size of a data mart or small data warehouse, to reside entirely in memory. Complex business calculations that used to take hours or days are able to be completed within seconds, and this can even be accomplished using handheld devices.

In-memory computing

Technology for very rapid analysis and processing of large quantities of data by storing the data in the computer's internal memory (RAM) rather than in secondary storage.

Chapter 5 describes some of the advances in contemporary computer hardware technology that make in-memory processing possible, such as powerful high-speed processors, multicore processing, and falling computer memory prices. These technologies help companies optimize the use of memory and accelerate processing performance while lowering costs. Leading in-memory database products include SAP HANA, Oracle Database In-Memory, Microsoft SQL Server, Amazon MemoryDB, and Teradata Intelligent Memory.

Data Analytics Platforms

Commercial database vendors have developed specialized data analytics platforms that are optimized for analyzing large data sets. A **data analytics platform** features a preconfigured hardware-software system that is specifically designed for query processing and analytics. For example, Microsoft Power BI is a cloud-based data analytics and business intelligence platform that offers a collection of software services, apps, and connectors that work together for data visualization, data modeling, and AI machine learning. Power BI is known for its ease of use and its integration with other Microsoft products, such as Excel and SharePoint. Google Looker is a cloud-based data analytics platform that is known for its ease of use and its focus on data exploration. Looker offers a variety of features to help users explore data, such as interactive dashboards, drill-down capabilities, and natural language search.

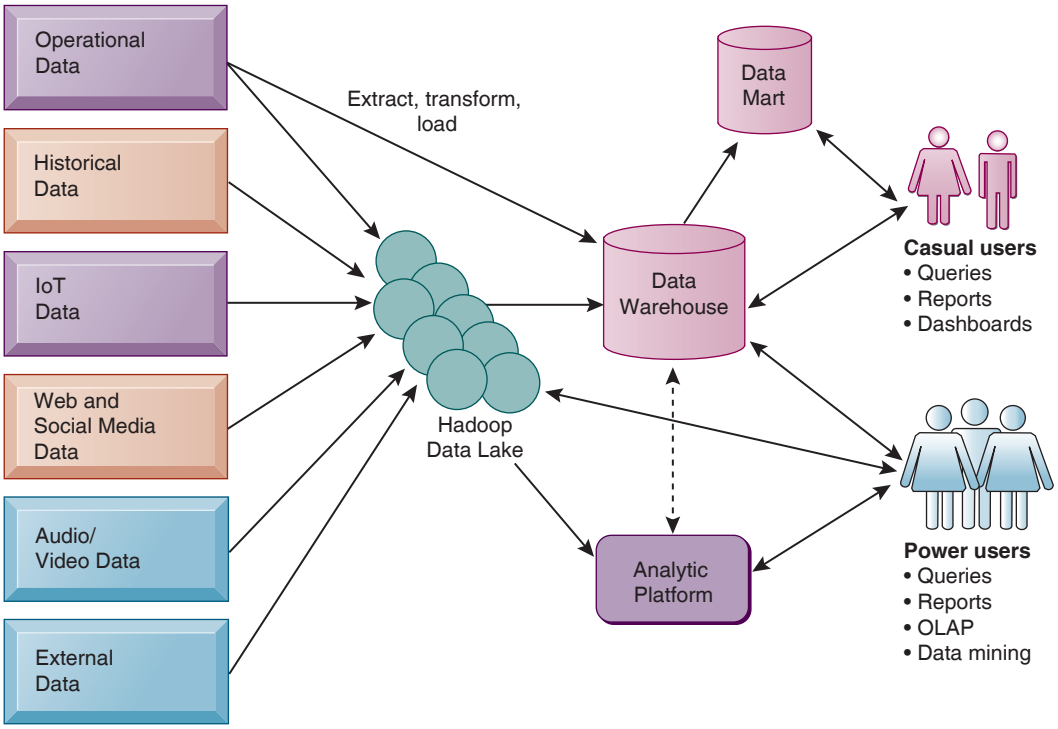
Data analytics platform

Specialized platform optimized for analyzing large data sets.

Examine Figure 6.13, which illustrates a contemporary business intelligence technology infrastructure using the technologies just described. Current operational and historical data are extracted from multiple operational systems along with Internet of Things (IoT) machine-generated data, web data and social media data, unstructured audio/visual data, and other data from external sources. All of these different types of data are poured into a Hadoop data lake, which stores them in clustered computer nodes and then processes them when they are needed for use in the data warehouse, data mart, or an analytic platform. Casual users can query the data for display on dashboards or to create reports. Power users typically

Figure 6.13 Contemporary Business Intelligence Technology Infrastructure

A contemporary business intelligence technology infrastructure features capabilities and tools to manage and analyze large quantities and different types of data from multiple sources. Easy-to-use query and reporting tools for casual business users and more sophisticated analytical toolsets for power users are included.



have additional tools or can engage in direct querying. Output includes reports and dashboards as well as query results. Chapter 12 discusses the various types of BI users and BI reporting in greater detail.

Analytical Tools: Relationships, Patterns, Trends

Once data have been captured and organized using the business intelligence technologies just described, they are available for further analysis using software for database querying and reporting, multidimensional data analysis (OLAP), data mining, and AI machine learning. This section will introduce you to these tools, with more detail about business intelligence analytics and applications in Chapter 12.

Online Analytical Processing (OLAP)

Suppose your company sells four different products—nuts, bolts, washers, and screws—in the East, West, and Central regions. If you wanted to ask a fairly straightforward question, such as how many washers were sold during the past quarter, you could easily find the answer by querying your sales database. But what if you wanted to know how many washers were sold in each of your sales regions and compare actual results with projected sales?

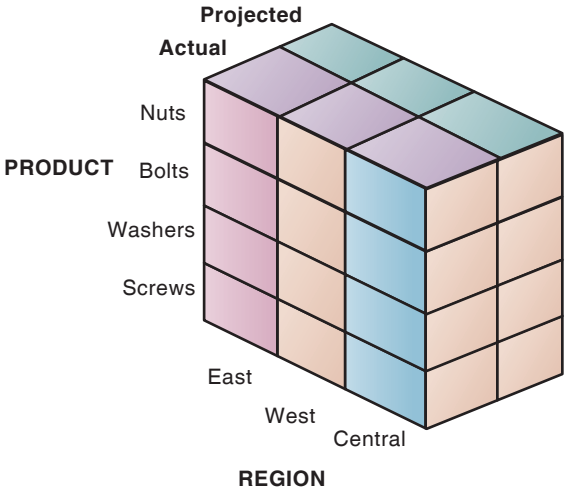
Online analytical processing (OLAP)

Supports multidimensional data analysis by providing capability for manipulating and analyzing large volumes of data from multiple perspectives.

To obtain the answer, you could use **online analytical processing (OLAP)**. OLAP supports multidimensional data analysis, enabling users to view the same data in different ways using multiple dimensions. Each aspect of information—product, pricing, cost, region, or time period—represents a different dimension. So, a product manager could use a multidimensional data analysis tool to learn how many washers were sold in the East in June, how that compares with the previous month and the previous June, and how it compares with the sales forecast. OLAP enables users to quickly obtain online answers to ad hoc questions such as these, even when the data are stored in very large databases, such as sales figures for multiple years.

Figure 6.14 Multidimensional Data Model

This view shows actual sales of each product in each region. If other views are possible.



Examine Figure 6.14, which shows a multidimensional data model that could be created to represent products, regions, actual sales, and projected sales. The model enables different views of the data. The view in the figure shows actual sales of each product (nuts, bolts, washers, and screws) in the East, West, and Central regions, while an alternative view (the "back side" of the cube that is not currently visible) would show projected sales of each product for each region. Another view would be projected versus actual sales of a particular product, such as nuts, in each region (here, represented by the top face of the cube). A fourth view would show actual and projected sales for each product in a specific region. Cubes can be nested within cubes to build even more complex views of data.

Data Mining

Traditional database queries answer such questions as, "How many units of product number 403 were shipped in November 2024?" OLAP, or multidimensional analysis, supports much more complex requests for information, such as, "Compare sales of product 403 relative to plan by quarter and sales region for the past two years." With OLAP and query-oriented data analysis, users need to have a good idea about the information they are looking for.

Data mining is more discovery-driven. Data mining provides insights into corporate data that cannot be obtained with OLAP by finding hidden patterns and relationships in large databases and inferring rules from them to predict future behavior. The patterns and rules are used to guide decision making and forecast the effect of those decisions. The types of information obtainable from data mining include associations, sequences, classifications, clusters, and forecasts.

Data mining

Analysis of large pools of data to find patterns and relationships that can be used to guide decision making and predict future behavior.

- *Associations* are occurrences linked to a single event. For instance, a study of supermarket purchasing patterns might reveal that when corn chips are purchased, a cola drink is purchased 65 percent of the time. But when there is a promotion, cola is purchased 85 percent of the time. This information helps managers make better decisions because they have learned the profitability of a promotion.
- In *sequences*, events are linked over time. For example, you might find that if a house is purchased, a new refrigerator will be purchased within two weeks 65 percent of the time, and an oven will be bought within one month of the home purchase 45 percent of the time.
- *Classification* recognizes patterns that describe the group to which an item belongs by examining existing items that have been classified and by inferring a set of rules. For

example, businesses such as credit card companies and Internet providers worry about the loss of steady customers. Classification helps discover the characteristics of customers who are likely to leave and can provide a model to help managers predict who those customers are, so that the managers can devise special campaigns to retain such customers.

- *Clustering* works in a manner similar to classification when no groups have yet been defined. A data mining tool can discover different groupings within data, such as finding affinity groups for bank cards or partitioning a database into groups of customers based on demographics and types of personal investments.
- Although these applications involve predictions, *forecasting* uses predictions in a different way. It uses a series of existing values to forecast what other values will be. For example, forecasting might find patterns in data to help managers estimate the future value of continuous variables, such as sales figures.

Data mining systems perform high-level analyses of patterns or trends, but they can also drill down to provide more detail when needed. There are data mining applications for all the functional areas of business and for government and scientific work. One popular use for data mining is to provide detailed analyses of patterns in customer data for one-to-one marketing campaigns or for identifying profitable customers. Artificial intelligence and machine learning have proved to be powerful tools for boosting data mining and helping filter out irrelevant data, making it easier to focus on the most important information.

Forbes, the global media company that publishes *Forbes* magazine, relies on revenue from advertising and subscriptions. It mines the extensive data it has collected on individual readers. Forbes claims that it can understand each person who interacts with its brand. Whether that person is a subscriber or a registered website visitor, Forbes has information on that person's demographics, values, and lifestyle as well as how that person has interacted with Forbes over the years. These details help *Forbes* advertisers target their campaigns more precisely and also help Forbes' publications increase their circulation.

Text Mining and Web Mining

Unstructured data, most in the form of text files, is believed to account for more than 80 percent of useful organizational information and is one of the major sources of big data that firms want to analyze. Email, memos, call center transcripts, survey responses, legal cases, patent descriptions, and service reports are all valuable for finding patterns and trends that will help employees make better business decisions. **Text mining** tools are now available to help businesses analyze these data. These tools can extract key elements from unstructured natural language text, discover patterns and relationships, and summarize the information.

Text mining
Discovery of patterns and relationships from large sets of unstructured data.

Businesses might turn to text mining to analyze transcripts of calls to customer service centers to identify major service and repair issues or to measure customer sentiment about their company. **Sentiment analysis** involves mining comments in an email or text message, blog, social media conversation, or survey forms to detect favorable and unfavorable opinions about specific subjects. For example, sentiment analysis helps Pizza Hut understand the attitudes conveyed about Pizza Hut, detect issues of concern, identify any key topics or particular aspects of the product users are discussing, and shed light on how users engage with Pizza Hut on Facebook or Instagram and for what purpose. This helps Pizza Hut to improve customer experience and increase sales. Sentiment analysis is a popular application of AI machine learning. Sentiment analysis uses natural language processing (NLP) and machine learning algorithms to determine the emotional tone behind conversations. NLP is a subset of machine learning that enables computers to understand, analyze, and generate human language, which we will discuss in detail in Chapter 11.

Sentiment analysis
Mining comments in an email or text message, blog, social media conversation, or survey form to detect favorable and unfavorable opinions about specific subjects.

The web is another rich source of unstructured big data for revealing patterns, trends, and insights into customer behavior. **Web mining** involves the discovery and analysis of useful patterns and information from the web. Businesses might turn to web mining to

Web mining
Discovery and analysis of useful patterns and information from the Web.

help them understand customer behavior, evaluate the effectiveness of a particular website, or quantify the success of a marketing campaign. For instance, marketers use the Google Trends service, which tracks the popularity of various words and phrases used in Google search queries, to learn what people are interested in and what they are interested in buying.

Web mining looks for patterns in data through content mining, structure mining, and usage mining. Web content mining is the process of extracting knowledge from the content of web pages, which may include text, image, audio, and video data. Web structure mining examines data related to the structure of a particular website. For example, links pointing to a document indicate the popularity of the document, while links coming out of a document indicate the richness or perhaps the variety of topics covered in the document. Web usage mining examines user interaction data recorded by a web server whenever requests for a website's resources are received. The usage data records the user's behavior when the user browses or makes transactions on the website and collects the data in a server log.

Analyzing such data can help companies determine the value of particular customers, cross-marketing strategies across products, and the effectiveness of promotional campaigns. For example, SaaS provider TechSmith used Google Analytics to identify its most important website pages. Then, it turned to Hotjar Heatmaps to identify which elements visitors clicked on the most and least and how far down the page they scrolled. This information helped TechSmith improve its product and website (HotJar, 2023).

Databases and the Web

Every time you use the web to place an order or view a product catalog you are using a website linked to an internal corporate database. Many companies now use the web to make some of the information in their internal databases available to customers and business partners.

Suppose, for example, a customer with a web browser wants to search an online retailer's database for pricing information. Examine Figure 6.15, which illustrates how that customer might access the retailer's internal database over the web. The user accesses the retailer's website over the Internet using a web browser on their client desktop computer or mobile device. The user's web browser software requests data from the organization's database, using HTML commands to communicate with the web server. Apps provide even faster access to corporate databases.

Because many back-end databases cannot interpret commands written in HTML, the web server passes these requests for data to software that translates HTML commands into SQL. This allows the commands to be processed by the DBMS working with the database. In a client/server environment, the DBMS resides on a dedicated computer called a **database server**. The DBMS receives the SQL requests and provides the required data. Middleware transfers information from the organization's internal database back to the web server for delivery in the form of a web page to the user.

Database server

A computer in a client/server environment that is responsible for running a DBMS to process SQL statements and perform database management tasks.

Figure 6.15 Linking Internal Databases to the Web

Users access an organization's internal database through the web using their desktop PC browsers or mobile apps.

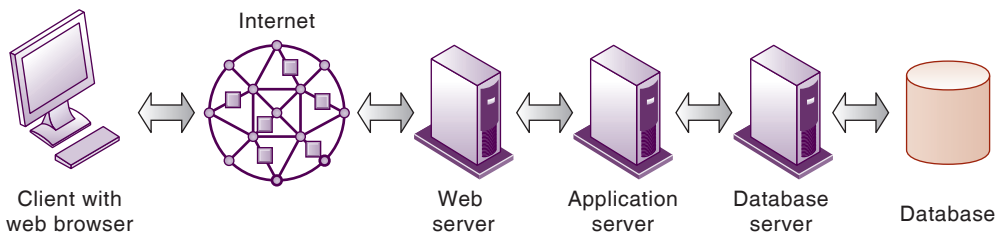


Figure 6.15 shows a dedicated computer called an application server between the web server and database server (see Chapter 5). The application server software handles all application operations, including transaction processing and data access, between browser-based computers and a company's back-end business applications or databases. The application server takes requests from the web server, runs the business logic to process transactions based on those requests, and provides connectivity to the organization's back-end systems or databases. Alternatively, the software for handling these operations could be a custom program or a script.

There are a number of advantages to using the web to access an organization's internal databases. First, web browser software is much easier to use than proprietary query tools. Second, the web interface requires few or no changes to the internal database. It costs much less to add a web interface in front of a legacy system than to redesign and rebuild the system to improve user access.

Accessing corporate databases through the web is creating new efficiencies, opportunities, and business models. For example, ThomasNet.com provides an up-to-date online directory of more than 500,000 suppliers of industrial products, such as chemicals, metals, plastics, rubber, and automotive equipment. Formerly called Thomas Register, the company used to send out huge paper catalogs with this information. Now it provides this information to users online via its website.

Other companies have created entirely new businesses based on access to large databases through the web. Facebook, which helps users stay connected with each other and meet new people, features "profiles" with information on more than 3 billion monthly active users including interests, friends, photos, and groups with which they are affiliated. Facebook maintains a very large database to house and manage all of this content. There are also many web-enabled databases in the public sector to help consumers and citizens access helpful information.

6.6 Discuss Data Governance, Data Quality Assurance, and Data Management

Data governance

Policies and procedures through which data can be managed as an organizational resource.

Setting up a database is only a start. To make sure that the data for your business remain accurate, reliable, and readily available to those who need to use the data, your business will need special policies and procedures for data governance. **Data governance** encompasses policies and procedures through which data can be managed as an organizational resource. It establishes the organization's rules for sharing, disseminating, acquiring, standardizing, classifying, and inventorying information. These include identifying which users and organizational units can share information, where information can be distributed, who is responsible for updating and maintaining the information, and how data resources should be secured (see Chapter 8). A firm's information policy might specify, for example, that only selected members of the payroll and human resources department would have the right to change or view sensitive employee data, such as an employee's salary or social security number, and that these departments are responsible for making sure that such employee data are accurate.

Assuring Data Quality

With today's organizations relying so heavily on data to drive operations and decision making, data quality assurance is especially important. What would happen if a customer's telephone number or account balance were incorrect? What would be the impact if the database had the wrong price for the product you sold? Data that are inaccurate, untimely,

or inconsistent with other sources of information create serious operational and financial problems for businesses, even with a well-designed database and information policy. When faulty data go unnoticed, they often lead to incorrect decisions, product recalls, and even financial losses. According to the Annual State of Data Quality Survey by Monte Carlo, which specializes in data quality solutions, data downtime has nearly doubled year after year, driven by a 166 percent increase in time for resolution of data quality issues. More than half of respondents reported 25 percent or more of their revenue was impacted by data quality problems (Monte Carlo, 2023). A portion of the data in large Fortune 1000 companies' databases is inaccurate or incomplete, including bad product codes and product descriptions, faulty inventory descriptions, erroneous financial data, incorrect supplier information, and incorrect employee data.

Some of these data quality problems are caused by redundant and inconsistent data produced by multiple systems. For example, the sales ordering system and the inventory management system might both maintain data on the organization's products. However, the sales ordering system might use the term *Item Number*, and the inventory system might call the same attribute *Product Number*. The sales, inventory, or manufacturing systems of a producer of bed and bath goods might use different codes to represent values for an attribute. One system might represent bath towel size as extra large, whereas the other system might use the code XL for the same purpose. During the design process for a database, data describing entities, such as a customer, product, or order, should be named and defined consistently for all business areas using the database.

Think of all the times you've received several pieces of the same direct mail advertising on the same day. This is very likely the result of having your name maintained multiple times in a database. Your name may have been misspelled, or you used your middle initial on one occasion and not on another, or the information was initially entered onto a paper form and not scanned properly into the system. Because of these inconsistencies, the database would treat you as different people! We often receive redundant mail addressed to Laudon, Lavdon, Lauden, or Landon.

If a database is properly designed and enterprise-wide data standards are established, duplicate or inconsistent data elements should be minimal. Most data quality problems, however, such as misspelled names, transposed numbers, or incorrect or missing codes, stem from errors during data input. The incidence of such errors is rising as companies move their businesses to the web and allow customers and suppliers to enter data into their websites that directly update internal systems.

Before a new database is in place, organizations need to identify and correct their faulty data and establish better routines for editing data once their database is in operation. Analysis of data quality often begins with a **data quality audit**, which is a structured survey of the accuracy and level of completeness of the data in an information system. Data quality audits can be performed by surveying entire data files, surveying samples from data files, or surveying end users for their perceptions of data quality.

Data quality audit

A structured survey and/or sample of files to determine accuracy and completeness of data in an information system.

Data cleansing, also known as *data scrubbing*, consists of processes for detecting and correcting data in a database or file that are incorrect, incomplete, improperly formatted, or redundant. Data cleansing not only corrects errors but also enforces consistency among different sets of data that originated in separate information systems. Specialized data-cleansing software is available to automatically survey data files, correct errors in the data, and integrate the data in a consistent companywide format.

Data cleansing

Processes for detecting and correcting data in a database or file that are incorrect, incomplete, improperly formatted, or redundant. Also known as data scrubbing.

Data quality problems are not just business problems; they also pose serious problems for individuals, affecting their financial condition and even their jobs. For example, inaccurate or outdated data about consumers' credit histories maintained by credit bureaus can prevent creditworthy individuals from obtaining loans or lower their chances of finding or keeping a job. Facial databases and recognition systems are not 100 percent accurate and can mistakenly identify people as criminals or terrorists. The proliferation of databases in all areas of business and everyday life threatens individual privacy. The Spotlight on Technology Case Study, *Facial Databases: Do You Know Who's Using Your Face,* explores these issues.

Spotlight on Technology Case Study

Facial Databases: Do You Know Who's Using Your Face?

A facial database is a collection of digital images of human faces. AI facial recognition systems use facial databases to train and test their algorithms for identifying individuals based on their facial features. Facial data attributes can include gender, age, facial expression, eye color, skin color, smile intensity, and head pose, along with face image quality and level of blurriness.

Facial databases are used in a variety of applications, such as security, law enforcement, and marketing. They are used in facial expression analysis, in which software can automatically recognize and analyze human emotions (useful for customer service and healthcare). Facial databases can also be used to create three-dimensional models of human faces in applications such as animations, video games, and facial reconstruction surgery.

Face recognition tools are frequently used in routine policing. Police compare mugshots of arrestees to the contents of local, state, and federal face recognition databases. Law enforcement can query these databases to identify people in photos taken from social media, traffic cameras, and closed-circuit television surveillance cameras in stores, parks, and other places. Once an arrestee's photo has been taken, the mugshot will live on in one or more databases to be scanned every time the police do another criminal search. There are systems to compare faces in real time with "hot lists" of people suspected of illegal activity. Face recognition has been used in airports, border crossings, and events such as the Olympic games.

There are a large number of facial databases in existence, with new databases created all the time. Some are public, while others are privately owned and managed. Some of these facial databases contain only a few hundred images, while others, such as DigiFace 1M or MS-Celeb 1M, contain millions. Some databases are focused on a specific demographic group or facial expression, while others are more general. Other well-known facial databases include images from more than 179 countries in the INTERPOL Face Recognition System (IFRS) and MORPH, a longitudinal facial database for age estimation with more than 200,000 facial images of 40,395 subjects.

Facial recognition systems and facial databases can make products safer and more secure. For example, face authentication can ensure that only the right person gets access to sensitive information meant just for them. They can also be used for social good: There are nonprofits using facial recognition to combat trafficking of minors. However, these systems also have limitations that can do harm as well.

Although facial recognition systems have become much more accurate than in the past, there is still a danger that a person could be misidentified. These systems have varying abilities to identify people under challenging conditions, such as poor lighting, low-quality image resolution, and suboptimal angle of view, which might occur if a photograph was taken from above looking down on an unknown person. Facial recognition software is also not as accurate when identifying racial minorities, as well as women and young people.

Hundreds of databases with people's faces are being compiled by companies and researchers, with many of the images then being shared around the world without the owners' knowledge or consent. The databases are pulled together with images from social networks, photo websites, dating services like OkCupid, and surveillance cameras placed in restaurants and on college campuses. There is no precise count of the data in these databases, and privacy activists are troubled by their proliferation and availability.

Clearview AI, which describes itself as "the world's largest facial network," has built a database that reportedly holds more than 40 billion "publicly available facial images" scraped from Facebook, YouTube, Venmo, and millions of other websites. Clearview works with customers such as law enforcement agencies to compare facial data against its database, but its facial database is also available to private businesses and individuals. U.S. federal and state law enforcement officers have used the Clearview app to help solve shoplifting, identity theft, credit card fraud, murder, and child sexual exploitation cases.

Clearview's technology has been declared illegal in Canada, Australia, and parts of Europe for violating privacy laws. Clearview has faced a provisional $22.6 million fine in the UK (which was overturned on appeal), as well as a 20 million Euro fine from Italy's data protection agency. In the United States, Clearview has recently settled a lawsuit brought by the American Civil Liberties Union and agreed to limit its facial database primarily to government agencies and not allow most American private companies or individuals to access it.

Another challenge of working with facial databases is that facial data can be difficult to protect, even though biometric authentication systems are more secure than graphical or textual passwords. If a database with your photo, name, and address is hacked, a malicious actor could use it for identity theft, robbery, or harassment. Unlike passwords and credit card information, biometric information such as facial characteristics cannot easily be changed if a data breach takes place.

SOURCES: Electronic Frontier Foundation, "Face Recognition," Eff.org, accessed August 23, 2024; Mike Swift, "Clearview AI Has Reached 'Turning Point' with US Legal Challenges, CEO Says, But Won't Return to EU, UK," Data Privacy & Security News, Mlexmarketinsight.com, January 24, 2024; Vera Lucia Raposo, "When Facial Recognition Does Not 'Recognise': Erroneous Identifications and Resulting Liabilities," *AI and Society*, January 2023; Hafiz Sheikh Adnan Ahmed, "Facial Recognition Technology and Privacy Concerns," Isaca.org, December 21, 2022.

Case Study Questions

1. What are the benefits of using facial databases? How do they help organizations improve operations and decision making? What problems can they help solve?
2. What are the disadvantages of facial databases?
3. Should businesses and individuals be allowed to use facial databases? Why or why not?

How AI Improves Data Management

Data management is crucial for creating an environment where data can be shared and utilized across the entire organization. However, managing all of the organization's data is extremely labor-intensive. As described in Table 6.3, artificial intelligence is now being used to improve data management in the following areas: classification, cataloguing, improving data quality, security, data integration, and data analysis.

Technology alone cannot replace good data management processes, such as developing data governance policies and structures, defining employee roles and responsibilities for using company data, and establishing common definitions of key terms. Human input and oversight are still required. But AI can dramatically improve both productivity and the value companies obtain from their data resources, and its ability to solve data management problems will continue to increase.

6.7 Understand How the Information in This Chapter Can Help Your Career

The information you have learned in Chapter 6 can help you advance your career by helping you to understand the importance of data that are accurate, timely, and properly organized to support organizational performance. You'll also be able to identify the typical issues and problems you are likely to encounter when using the company's data and ways of dealing with them. This chapter is a starting point. You can find additional information on these topics online.

Table 6.3 How AI Improves Data Management

Problem/Challenge	Description	AI Solution
Classification	Obtaining, extracting, and structuring data from documents, photos, handwriting, video, and other media into categories.	AI helps identify inappropriate social media postings (such as hate speech) on a large scale, including postings generated by bots. Also used for tasks such as identifying the most important data in documents such as a contract.
Cataloguing	Locating data and keeping the data current. Extremely labor-intensive.	AI can automate searches through various repositories of data and create catalogs automatically. Can also identify where data originated, how it has been modified, and where it resides.
Data Quality Assurance	Reducing errors in data	AI machine learning tools can scan data to identify values that are not allowed and can correct some errant values automatically. AI algorithms can detect missing values in the data and then can provide more accurate estimated values.
Security	Ensuring that data can't be stolen, altered, or destroyed by unauthorized parties and that data usage conforms to organizational policies.	AI machine learning solutions automate collection of security data across multiple internal and external systems and assess which threats are most credible. AI systems can predict likely attack paths based on previous attack patterns and determine whether new threats are coming from previously known actors or new ones. AI can also be used to identify internal threats of fraud.
Data Integration	Consolidating data from different sources into a unified view [single cohesive dataset] that can be analyzed for business intelligence.	Machine learning-based mastering systems use probabilistic matching techniques to decide whether records should be combined. Records that have a high probability of being the same entity are automatically merged. Records that can't be resolved by this approach can be reviewed by human subject matter experts.
Data Analysis	Ability to draw conclusions from or evaluate specific data. Data analysis is used to make purposeful discoveries, suggest conclusions, and support decision-making.	AI and machine learning can process vast amounts of data quickly-and often more accurately than humans. AI can identify complex patterns and provide predictive insights that may not be readily apparent through manual analysis. Natural language processing tools, including ChatGPT, can analyze textual data, while computer vision systems can analyze images.

Real-World Scenario

Keurig Green Mountain Embraces Data Governance

Keurig Green Mountain, best known for its single-serve K cups, had experienced meteoric growth due to surging sales and partnerships with numerous vendors such as Dunkin' Donuts and Starbucks. The company had to work with data on 80 brands and nearly 600 product varieties of hot and cold coffees, teas, cocoas, dairy-based beverages, cider, and fruit-based drinks. It could not do so effectively because it needed more appropriate tools and processes for data governance. Data quality analysts in the organization oversaw providing essential customer or vendor data and correcting inaccuracies in that data. Data corrections made downstream were not always conveyed back to the data providers. Keurig Green Mountain solved the problem by implementing a master data management program with well-defined processes and standards for the maintenance of data across the enterprise. The company chose data governance SaaS software that provided a framework of data standards, governance rules, business metrics, and business processes. Keurig also added SAP Information Steward software for discovering data characteristics and relationships, creating and running data validation rules, identifying bad data, data cleansing, cataloging data, and defining business terms for data. The two tools working in tandem made it possible for data quality analysts at Keurig to ensure more consistent companywide reporting and data management and a higher level of data quality.

Job Description and Career Path: Data Quality Analyst

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Data quality analyst	Monitors the quality of the organization's data; uses IT tools and improved standards and processes to optimize data quality and availability; works with database developers to improve systems and database design Job responsibilities include: • Identifying and assessing data quality issues • Developing and implementing data quality standards and processes • Developing and executing data cleansing measures • Monitoring and reporting on data quality metrics such as accuracy, completeness, and timeliness. • Working with stakeholders to resolve data quality issues • Collaborating with IT and other departments to improve data quality systems and processes	• Proficiency in Structured Query Language (SQL) and JavaScript • Experience with statistical software such as SAS, SPSS, and Excel • Strong analytical, interpersonal, and communication skills • Data profiling tools such as Talend and Informatica • Data quality monitoring tools such as IBM InfoSphere and SAP Information Steward • Data governance tools such as Collibra and Informatica • SQL-based data quality tools such as Oracle Data Quality and Talend Data Quality	• Entry-level Data quality analyst: $61,500 to $75,449 • Mid-level data quality analyst: $75,449 to $94,306 • Senior data quality analyst: $94,306 to $112,500 • Data quality manager: $110,000 to $170,000 • Director of data quality: $175,000 to $210,000 Salaries can vary based on factors like location, industry, and company size.

CAREER BENEFIT

Employees and managers who worked on this solution with IT specialists gained recognition for their contributions. Since data governance is a companywide issue, Keurig's data governance project provided an opportunity for data quality analysts and other participants to demonstrate their problem-solving skills to other parts of the organization and levels of management. Data governance enabled employees to perform better in their jobs because they could work with accurate, reliable data that was maintained with well-defined processes and standards.

Additional Career Resources

In MyLab MIS, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

Pursuing Sustainability with Blockchain

Unilever is one of the world's largest consumer goods companies, with more than 400 brands, such as Dove, Hellmann's Mayonnaise, and Seventh Generation, in more than 190 countries. Among large corporations, Unilever is also a leader in sustainability and has made sustainability an important objective of its business strategy. In December 2020, the Unilever Board announced a Climate Transition Action Plan (CTAP), which sets out the steps the company is taking to reduce carbon emissions to zero within its own operations by 2030 and to reduce carbon emissions to net zero across its value chain by 2039. The company is transitioning to renewable energy for its operations; finding new, low-carbon ingredients; expanding its plant-based product range; and developing fossil fuel-free cleaning and laundry products.

One of Unilever's sustainability initiatives concerns the production of palm oil, which is derived from the fruit of the oil palm tree and is used in thousands of consumer products, including detergents, cosmetics, chocolate, and biofuel. Expanding palm oil production increases deforestation as rainforests are burned to clear the way for more palm plantations. Palm oil production has more than tripled over the past two decades because palm oil is one of the most common raw materials in so many household products. Expansion of palm oil production is a leading cause of carbon dioxide emissions, rainforest destruction, and human rights problems. Moreover, even if palm oil is sustainably sourced, it can be mixed with palm oil that isn't responsibly sourced, making it difficult for companies to create sustainable supply chains.

Most of Unilever's oil palm plantations are in the Leuser rainforest in the Aceh province of Indonesia. The palm oil industry has poor production standards that have accelerated deforestation. A fall 2021 Unilever report on the environmental issues facing the palm oil industry stated that about 128,700 hectares of forests were destroyed by palm oil expansion between 2016 and 2020. (One hectare is equivalent to approximately 2.471 acres.)

Unilever made a commitment in 2020 to achieve a deforestation-free supply chain by 2023, and has largely achieved that goal, with 97.5 percent of its palm oil, paper and board, and cocoa order volumes being independently verified as deforestation-free at the end of 2023. The company has utilized technology such as satellite imaging and AI to monitor deforestation and land conversion, has also turned to blockchain to increase transparency and traceability of its palm oil supply chain.

Supply chains for palm oil are long and complex, and the material can go through dozens of transformations as it moves from company to company from the time it's harvested until it ends up as soap or shampoo. Companies want to make sure that the palm oil they're sourcing comes from responsible sources, such as plantations that don't contribute to deforestation.

Unilever started using blockchain technology from Green Token to track, verify, and report in near real time the origins and journey that palm oil takes through its long and complex supply chain. Green Token is a blockchain-based supply chain traceability and chain of custody application that allows companies to record and track facts about the raw materials they use in their products. This can include information such as raw materials' origins, specific production attributes, and certifications. Green Token is especially useful for tracking raw materials and commodities, such as palm, soy, coffee, sugar, and cocoa, which are linked to deforestation.

Other supply chain traceability solutions exist in the market, but they almost all rely on keeping tabs on batches. This approach doesn't work for tracking raw materials, which typically ship and are stored in bulk and are often commingled (mixed up) with materials from other sources. Even if batch identifiers for certain quantities are known, constant commingling generates an exponentially large number of records that need to be kept. No batch system can track this complexity. Companies that need to source raw materials and that use Green Token can accurately report on sustainable or ethical procurement even if the materials are mixed or blended. Green Token can also be used to measure the level of certain raw materials in finished goods. A consumer products producer would be able to demonstrate that it has a high volume of recycled material in the products themselves or in the packaging.

Companies can access Green Token via a SaaS dashboard. Producers can create a digital token of physical goods that can include multiple attributes, such as the original plantation and any certifications. Blockchain establishes a chain of custody for the tokens that cannot be altered as the materials move through the supply chain. Details about the commodity can also include whether it's coming from a mature plantation with no deforestation.

Responding to demands by governments and organizations such as the Rainforest Action Network (RAN) is one powerful reason for embracing sustainability. Germany's Supply Chain Due Diligence Act, which came into effect in January 2023, requires large public organizations to provide visibility of their supply chains in a digital and auditable format, and similar legislation is expected in other countries.

Another reason to embrace sustainability is to make money. A company that can demonstrate its commitment to sustainability will be rewarded by higher sales and brand loyalty. According to a report by the ocean cleanup

organization Seven Clean Seas, about 57 percent of consumers are willing to change their purchasing behavior to help the environment, and about 66 percent are willing to pay more for more sustainable brands. It is not enough for a company to manufacture green products; it must also prove to the consumer that it has done so. Using blockchain can demonstrate to environmentally conscious consumers that an item in question is what it claims to be.

Newlight Technologies, Inc. is the creator of AirCarbon, a biomaterial that is made out of greenhouse gases and that can substitute for fiber, leather, and synthetic plastic. AirCarbon is meltable into a variety of solid forms and is carbon-negative, which means its production captures or destroys more carbon dioxide than was emitted to make it. Newlight is currently using AirCarbon in fabrics, and for foodware such as compostable drinking straws and cutlery.

When it came to marketing AirCarbon, Newlight needed a way to get the sustainability message out to potential customers because there was no way for consumers to verify Newlight's environmental claims. Blockchain appeared to provide a good solution for verifying the company's environmental impact. Blockchain creates a ledger of business transactions distributed among multiple computers, making the information easily visible yet impervious to tampering, and can be used to record virtually any type of business data. Its traceability makes it especially appealing to companies concerned about promoting their environmental responsibility.

Newlight worked with IBM, which was developing blockchains for a variety of industries. The blockchain was able to assign each Newlight product a blockchain-based number that could be used to display every step in the production process, along with third-party verification of the item's carbon impact. Each product has a unique, 12-digit number printed on it during production that shows the precise time when AirCarbon was used in the production of the item. That number is then entered into Newlight's website, providing customers with details about the product's movement through the supply chain, from initial molding to the store shelf. IBM Blockchain provides visibility into not only the steps used to make each product but also the carbon impact that each specific product has on the environment. Newlight recently raised $125 million in funding. It hopes to use the investment to expand its AirCarbon manufacturing platform to manufacture decarbonized materials at a global scale.

Using blockchain to increase supply chain sustainability poses some challenges, such as how the insights will be used and how information will be shared among members of the supply chain. Companies like Unilever have had to learn how to work with suppliers to minimize deforestation while maintaining or improving the company's economic well-being. There are also challenges concerning the use, management, and security of the data collected.

A copy of a blockchain is typically held on multiple computers, creating a scalability problem as the blockchain increases in size. Each computer on the network works to confirm transactions and keep accurate records of the blockchain, storing data starting with the first block and moving to the most recent block. The redundancy creates a more secure system, but it also becomes increasingly inefficient as the network and blockchain grow.

When creating a new block on the blockchain, the computer that confirms the transactions must broadcast the new block to every other computer on the network. The other computers can then verify the transactions and add the block to the blockchain, which can consume substantial network resources as the network grows in size.

There are no universal blockchain standards to ensure interoperability, generate trust, and help ensure ease of use of the technology. Almost every implementation of blockchain technology is unique, which makes interoperability among blockchains difficult. If one company wants to share data with another company's blockchain, they'll likely need to develop additional tools to allow data to flow between the two blockchains. Nevertheless, companies that use blockchain applications correctly will be able to build stronger relationships with suppliers.

SOURCES: Greentoken.org, accessed August 23, 2024; Unilever, "Climate Transition Action Plan," Assets.unilever.com, accessed August 23, 2024; IBM.com, "Newlight Technologies, Inc: Turning Greenhouse Gas into a Resource Verified with Blockchain Technology," accessed August 23, 2024; Newlight.com, accessed August 23, 2024; Mark Segal, "Sustainable Materials Provider Newlight Raises $125 Million," Esgtoday.com, August 7, 2023; Alejandra Duenas, "How Sustainability Is Impacting Consumer Trends," Sevencleanseas.com, May 5, 2023; Joe O'Donnell, "Unilever Pursues Supply Chain Sustainability with Blockchain," Techtarget.com, March 30, 2022; "Unilever and SAP Partner for Blockchain Token Traceability Pilot," Ledgerinsights.com, March 25, 2022.

Case Study Questions

1. Why is supply chain traceability so important for sustainability?
2. Why is blockchain a useful technology for traceability?
3. What management, organizational, and technology challenges should Unilever anticipate as it deploys blockchain for supply chain management?
4. Should all businesses use blockchain? Explain your answer.

Review Summary

6.1 Describe traditional file environment data management problems.

A computer system organizes data in a hierarchy that starts with bits and bytes and progresses to fields, records, files, and databases. Traditional file management techniques make it difficult for organizations to keep track of all of the pieces of data they use in a systematic way and to organize these data so that they can be easily accessed. Different functional areas and groups were allowed to develop their own files independently. Over time, this traditional file management environment creates problems such as data redundancy and inconsistency, program-data dependence, inflexibility, poor security, and lack of data sharing and availability. A database management system (DBMS) solves these problems with software that permits centralization of data and data management so that businesses have a single consistent source for all their data needs. Using a DBMS minimizes redundant and inconsistent files.

6.2 Describe DBMS and relational DBMS capabilities.

A DBMS is software that enables an organization to centralize data, manage that data efficiently, and provide access to the stored data by application programs. The relational database has been the primary method for organizing and maintaining data in information systems because it is so flexible and accessible and relational DBMS are the most popular type of DBMS today. A relational database organizes data in two-dimensional tables called relations with rows and columns. Each table contains data about an entity and its attributes. Each row represents a record, and each column represents an attribute or field. Each table also contains a key field to uniquely identify each record for retrieval or manipulation. Relational database tables can be combined easily to deliver data required by users, provided that any two tables share a common data element.

The principal capabilities of a DBMS include a data definition capability, a data dictionary capability, and a data manipulation language. The data definition capability specifies the structure and content of the database. The data dictionary is an automated or manual file that stores information about the data in the database, including names, definitions, formats, and descriptions of data elements. The data manipulation language, such as SQL, is a specialized language for accessing and manipulating the data in the database.

6.3 Discuss relational database design principles.

Designing a database requires both a logical design and a physical design. The logical design models the database from a business perspective. The organization's data model should reflect its key business processes and decision-making requirements. The process of creating small, stable, flexible, and adaptive data structures from complex groups of data when designing a relational database is termed normalization. A well-designed relational database will not have many-to-many relationships, and all attributes for a specific entity will only apply to that entity. It will try to enforce referential integrity rules to ensure that relationships between coupled tables remain consistent. An entity-relationship diagram graphically depicts the relationship between entities (tables) in a relational database.

6.4 Describe non-relational, cloud, and blockchain database capabilities.

Non-relational databases are becoming popular for managing big data and types of data that can't be handled easily by the relational data model. They are useful for accelerating simple queries against large volumes of structured and unstructured data. Both relational and non-relational database products are available as cloud computing services. Cloud databases are useful for storing and managing large quantities of data in the cloud. A distributed database is one that is stored in multiple physical locations, including remote cloud computing centers. Parts or copies of the database are physically stored in one location, and other parts or copies are stored in other locations. Blockchain is distributed database technology that enables organizations to create and verify transactions on a network nearly instantaneously without a central authority. The information held in the blockchain database is continually reconciled by the computers in the network.

6.5 Describe tools used to analyze data in databases.

Contemporary data management technology has an array of tools for obtaining useful information from all the different types of data used by businesses today, including semi-structured and unstructured big data in vast quantities. These capabilities include data warehouses, data marts and data lakes, Hadoop, in-memory computing, and data analytics platforms. OLAP represents relationships among data as a multidimensional structure, which can be visualized as cubes of data and cubes within cubes of data, enabling more sophisticated data analysis. Data mining analyzes large pools of data, including the contents of data warehouses, to find patterns and rules that can be used to predict future behavior and guide decision making. Increasingly, this is being performed using AI machine learning. Text mining tools help businesses analyze large unstructured data sets consisting of text. Web mining tools focus on the analysis of useful patterns and information from the web, examining the structure of websites and activities of website users, as well as the contents of web pages. Conventional databases can be linked via middleware to the web or a web interface to facilitate user access to an organization's internal data.

6.6 Discuss data governance, data quality assurance, and data management.

Developing a database environment requires policies and procedures for managing organizational data, as well as a

good data model and database technology. Data governance encompasses organizational policies and procedures for the maintenance, distribution, and use of information in the organization. Data that are inaccurate, incomplete, or inconsistent creates serious operational and financial problems for businesses because they may create inaccuracies in product pricing, customer accounts, and inventory data, and lead to inaccurate decisions about the actions that should be taken by the firm. Firms must take special steps to make sure they have a high level of data quality. These include using enterprise-wide data standards, databases designed to minimize inconsistent and redundant data, data quality audits, and data cleansing software. AI is now being used to improve data management in the following areas: classification, cataloguing, improving data quality, security, data integration, and data analysis.

Key Terms

Attribute, 186
Bit, 186
Blockchain, 197
Byte, 186
Cloud database, 196
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Review Questions

6.1 Describe traditional file environment data management problems.

- List and describe each of the components in the data hierarchy.
- Define and explain the significance of entities, attributes, and key fields.
- List and describe the problems of the traditional file environment.

6.2 Describe DBMS and relational DBMS capabilities.

- Define a database and a database management system.
- Name and briefly describe the capabilities of a DBMS.
- Define a relational DBMS and explain how it organizes data.
- List and describe the three operations of a relational DBMS.

6.3 Discuss relational database design principles.

- Define and describe normalization and referential integrity, and explain how they contribute to a well-designed relational database.
- Define and describe an entity-relationship diagram and explain its role in database design.

6.4 Describe non-relational, cloud, and blockchain database capabilities.

- Define and describe the capabilities of non-relational databases and cloud databases.
- Define blockchain, explain how it works and why it is helpful for managing data.

6.5 Describe tools used to analyze data in databases.

- Define big data and describe the technologies for managing and analyzing it.
- Describe the relationship between AI and big data.
- List and describe the components of a contemporary business intelligence technology infrastructure.
- Describe the capabilities of online analytical processing (OLAP).
- Define data mining, describing how it differs from OLAP, and the types of information it provides.
- Explain how text mining and web mining differ from conventional data mining.

- Describe how users can access information from a company's internal databases through the web.

6.6 Discuss data governance, data quality assurance, and data management.

- Define data governance and explain how it helps organizations manage their data.
- List and describe the most important data quality problems.
- List and describe the most important tools and techniques for ensuring data quality.
- Describe how AI can improve data management.

Discussion Questions

6.7 It has been said, "There is no bad data, just bad management." Discuss the implications of this statement.

6.8 To what extent should end users be involved in the selection of a database management system and database design?

6.9 What are the consequences of an organization not having an information policy?

Using AI Project

6.10 AI Project: Performing Sentiment Analysis

You have been asked to analyze online reviews of a nearby Italian restaurant to see if there is anything it can do to improve the customer experience and attract more customers. Use Microsoft Copilot, Google Gemini or ChatGPT to perform a sentiment analysis of the following review for the restaurant:

"The food we ordered came out at once, with everyone passing dishes to the ends of the table to ensure each of us was able to try everything we ordered. The most memorable dishes were ravioli with spinach and cheese, linguine with seafood, and veal marsala. The restaurant gets very busy on weekends, so the service is slower at that time. When the restaurant is fully occupied, the noise level can be high. Outdoor dining is available during the warmer months but will be too cold to enjoy in winter. I personally would only come here for special occasions or a holiday because it's quite expensive. There is free parking on Sundays and holidays. I recommend making reservations or arriving at the restaurant before it opens so you don't have to wait."

Then answer the following questions:

- State how the sentiment analysis has evaluated the review. Is the overall impression positive, negative, or somewhere in between?
- Is there anything the restaurant can do to improve the customer experience?
- Is there anything the restaurant is doing that increases negative sentiment?
- What changes would you recommend for the restaurant?

Save a copy of your prompt and the output of the chat session. Submit along with the answers to the above questions to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience in establishing companywide data standards; creating a database for inventory management; using the web to search online databases; and collaboration and teamwork.

For student files please visit www.pearsonhighered.com/laudon or MyLab MIS.

6.11 Management Decision Problem

Your industrial supply company wants to create a data warehouse from which management can obtain a single, corporate-wide view of critical sales information to identify best-selling products, key customers, and current sales trends. Your sales and product information are stored in several systems: a divisional sales system running on a UNIX server and a corporate sales system running on an IBM mainframe. You would like to create a single, standard format that consolidates these data from both systems. Review the proposed format along with sample files from the two systems that would supply the data for the data warehouse, which can be found in the MIS18_Project 6.11_Student File. Then answer the following questions:

- What business problems are created by not having these data in a single, standard format?
- How easy would it be to create a database with a single, standard format that could store the data from both systems? Identify the problems that would have to be addressed.
- Should the problems be solved by database specialists or general business managers? Explain.

- Who should have the authority to finalize a single, companywide format for this information in the data warehouse?

6.12 Building a Relational Database for Inventory Management

Software skills: Database design, querying, and reporting

Business skills: Inventory management

In this exercise, you will use database software to design a database for managing inventory for a small business. Morgan's Skateboard Shop, located in Santa Cruz, California, sells skateboards for various skateboarding styles, including freestyle skateboarding, street skateboarding, park skateboarding, downhill skateboarding, and cruising. Currently, Morgan's purchases skateboards from three suppliers but plans to add new suppliers in the near future. Using the information found in the MIS18_Project 6.12 Student File, build a simple relational database to manage information about Morgan's suppliers and products. Once you have built the database, perform the following activities.

- Prepare a report that identifies the five skateboards that are the most expensive. The report should list the skateboards in descending order from most expensive to least expensive, the quantity on hand for each, and the markup percentage for each.
- Prepare a report that lists each supplier, its products, the quantities on hand, and associated reorder levels. The report should be sorted alphabetically by supplier. Within each supplier category, the products should be sorted alphabetically.
- Prepare a report listing only the skateboards that are low in stock and need to be reordered. The report should provide supplier information for the identified items.
- Write a brief description of how the database could be enhanced to improve management of the business further. What tables or fields should be added? What additional reports would be useful?

6.13 Searching Online Databases for Business Resources

Software skills: Online databases

Business skills: Researching services for overseas operations

This project develops skills in searching online, web-enabled databases for information about products and services in faraway locations.

Your company is located in Greensboro, North Carolina, and manufactures office furniture of various types. You are considering opening a facility to manufacture and sell your products in Australia. You would like to contact organizations that offer many services necessary for you to open your Australian office and manufacturing facility, including attorneys, accountants, import-export experts, and telecommunications equipment and support firms. Access the following online databases to locate companies that you would like to meet with during your upcoming trip: ShowMeLocal Australia, Austrade.gov.au, and the Nationwide Business Directory of Australia. If necessary, use search engines such as Bing or Google.

- List the companies you would contact on your trip to determine whether they can help you with these and any other functions you think are vital to establishing your office.
- Rate the databases you used for accuracy of names, completeness, ease of use, and general helpfulness.

6.14 Getting to Know Tableau

In this exercise, you will create a report or presentation about the various types of data sources (structured, semi-structured, unstructured, etc.) that can be used when working with Tableau. Begin by identifying the different types of data that fall within each of these categories of data sources (for example, data within an Excel spreadsheet is an example of structured data) and then find an example public dataset of each type of data source that is available online. Describe which data sources can be used with which versions of Tableau, and the challenges of working with each type. Use Tableau to create at least two visualizations to illustrate your findings and integrate them into your report or presentation. (If you don't have access to Tableau Desktop, ask your instructor, or alternatively, you can create a free Tableau Public account.)

6.15 Collaboration and Teamwork Project: Identifying Entities and Attributes in an Online Database

With a team of two or more other students, select an online database to explore, such as Planespotters.net, Thomasnet, or the Internet Movie Database. Explore one of these websites to see what information it provides. Then, list the entities and attributes that the company running the website must keep track of in its databases. Diagram the relationship between the entities you have identified. If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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7

Telecommunications, the Internet, and Wireless Technology

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Learning Objectives

After completing this chapter, you will be able to:

7.1 Describe key components of telecommunications networks.

7.2 Describe key digital networking technologies.

7.3 Describe the different types of networks.

7.4 Understand what the Internet is and how it works.

7.5 Explain how Internet services support business.

7.6 Discuss how the web works and how it can be used in business.

7.7 Describe the principal wireless technologies and standards.

7.8 Understand how the information in the chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** From the Football Field to the Cloud and Back Again: The NFL Partners with AWS
- **Spotlight on Technology Case Study:** Cleveland Clinic Turns to Private 5G Networks for Its Newest Hospital
- **Chapter Case Study:** AI and the Future of Search

Opening Case Study

From the Football Field to the Cloud and Back Again: The NFL Partners with AWS

The National Football League is one of the most well-known professional sports leagues in the world. The NFL has 32 teams that play an 18-week regular season, followed by playoffs that culminate in the Super Bowl.

Historically, the NFL tracked player statistics manually, but about ten years ago, it embarked on a more comprehensive approach, the Next Gen Stats program, using digital technology. It began a pilot program to track precise real-time location data generated by radio frequency identification (RFID) tags embedded into players' shoulder pads. RFID tags are sensors that use wireless technology to enable the transmission of data. The tags emit signals that are picked up by an array of receivers placed throughout the stadium. The receivers form a network that triangulates the location of each player on the field in real time. The system can capture the tiniest details of a player's movements, including the player's field position, speed, distance traveled, and acceleration.

By 2017, the NFL had expanded the pilot program to include all of its teams. The technology also advanced to include additional RFID sensors to enhance the granularity of data collected. The upgraded system enabled the measurement of player orientation, acceleration, and deceleration, enhancing the ability to analyze player biomechanics and performance.

The NFL realized that the volume and complexity of the data being generated by the program required a high-performance computing infrastructure. In 2017, the NFL partnered with Amazon Web Services (AWS) to take the Next Gen Stats program to the next level. AWS is one of the world's leading cloud computing providers and offers over 200 cloud-based services, including processing, storage, networking, database, analytics, artificial intelligence, and more. AWS provides the NFL with a robust and scalable cloud computing platform that enables the NFL to seamlessly process and store the data generated by the Next Gen Stats program in real time. As players maneuver around the field, data reflecting their every movement is transmitted to AWS servers, where the data is quickly ingested, processed, and made available for analysis. AWS provides teams with near-instantaneous access to a vast array of metrics. Generating team dashboards used to take several hours, but the process now takes only minutes.

Next Gen Stats are also being used by the NFL and Amazon for Amazon's Thursday Night Football streaming broadcast to provide users with the option to choose an alternative broadcast, Prime Vision, that features more in-depth analysis in real time.

The next frontier is using artificial intelligence (AI) and machine learning (ML). In addition to providing cloud computing services, AWS is also the NFL's official AI/ML provider. The NFL uses AI/ML to dissect the vast amounts of data generated by the Next Gen Stats system. The algorithms sift through the data to identify patterns that might escape human detection, enabling a more nuanced understanding of player performance and game dynamics.

In 2019, the NFL extended its use of AI/ML to focus on player safety. It announced that it would develop a

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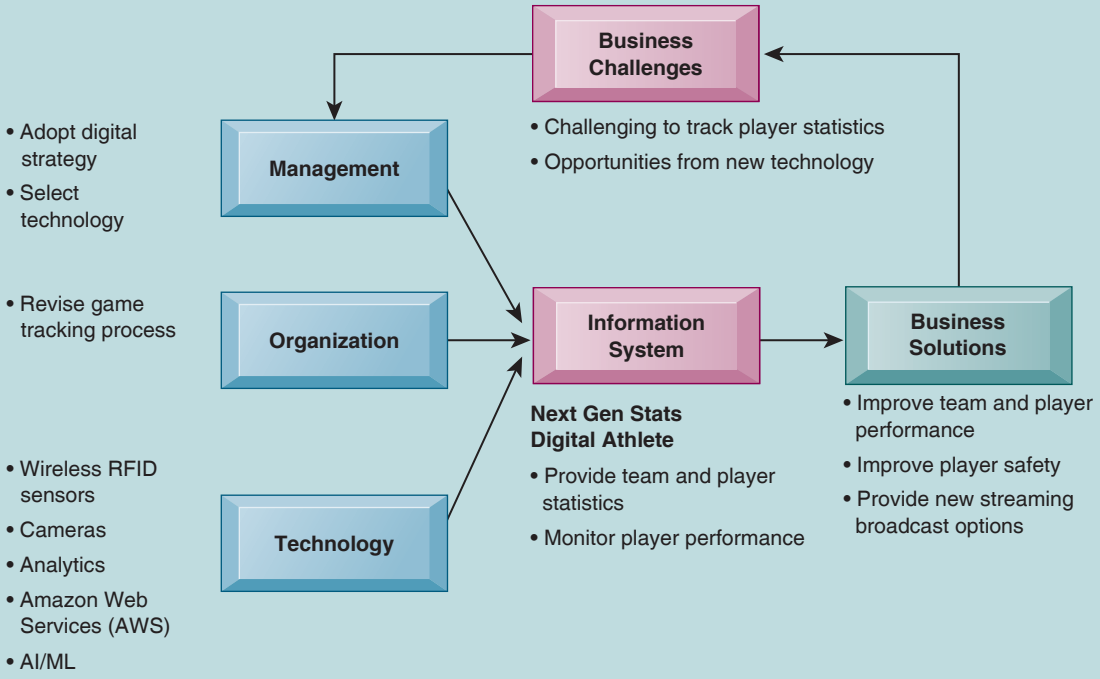


new Digital Athlete platform in conjunction with AWS that would use computer vision and machine learning to identify plays and body positions most likely to lead to player injury. The NFL piloted the new platform in 2022 and made it available to all 32 teams in 2023.

The platform captures 3D player movements with a synchronized set of 38 optical tracking cameras that can capture 5k video at 60 frames per second. The footage is then uploaded to the AWS Cloud, where it is stored and analyzed along with data from the Next Gen Stats system, as well as other data such as weather, equipment, and play type. On a weekly basis, the platform captures and processes 6.8 million video frames and documents about 100 million locations and positions of players on the field.

The Digital Athlete platform can then use the data to reconstruct the conditions of how and when an injury occurred. A Risk Mitigation Modeling tool can help determine a player's ideal training program while minimizing injury risk. Another new technology currently in development, 3D Pose Estimation, will enable teams to assess how a player's movements through space and time can lead to certain injuries, and help to develop a better understanding of player safety.

Digital Athlete data is not only helpful for creating personalized training programs for players but is also being used to drive decision making at the league level. For instance, Digital Athlete data that suggested that a reduction in kickoff returns would lead to a reduction in concussions was a key factor in the NFL's decision to implement a new fair catch rule for kickoffs in 2023.

SOURCES: Scott Thompson, "How Amazon Web Services' AI, Machine Learning Technology Is Shaping NFL's Future," Foxbusiness.com, February 17, 2024; Thor Olavsrud, "How AI Is Helping the NFL Improve Player Safety," Cio.com, February 9, 2024; Jennifer Langton, "Building a Digital Athlete: Using AI to Rewrite the Playbook on NFL Player Safety, Nfl.com, February 1, 2024; Bon Adamson, "Here's How the NFL's Next Gen Stats Technology Actually Works (And How It's Changed the Game," Slashgear.com, January 3, 2024.

The Opening Case Study illustrates some of the powerful capabilities and opportunities provided by contemporary networking technology, such as RFID tags and cloud computing services, in conjunction with new technologies such as AI/ML. The NFL now uses such technologies to track player data in real time and to use that data to enable a more nuanced understanding of player performance and game dynamics, as well as enhance player safety.

The Opening Case Study diagram above calls attention to important points raised by this case and this chapter. The NFL realized that it needed to enhance its collection and analysis of player data and did so by taking advantage of opportunities provided by RFID tags, cloud computing, and AI/ML. The NFL also realized that player safety has become an increasingly important concern and that the same technology, when coupled with new AI/ML tools, could be used to address those concerns.

Case Study Questions

1. Describe the way the technology described in the case has changed the way the NFL obtains and uses data from its games.
2. How is the NFL using AI and ML to enhance player safety?

CAREER BENEFIT

NFL's NextGen Stats and AI/ML are core IT applications that drive the business, providing metrics and visuals to improve decision making, player and team performance, and player safety. Both IT and non-IT staff who developed these systems would receive recognition throughout the entire NFL and their expertise would be very much in demand. NFL team members would be rewarded by becoming better players from using these systems, boosting their rankings and desirability.

7.1 Describe Key Components of Telecommunications Networks

If you run a business, you can't do so without networks. Networks enable your business to communicate with your customers, suppliers, and employees. Today networks are an integral part of doing business.

Firms in the past used two fundamentally different types of networks: telephone networks and computer networks. Telephone networks historically handled voice communication, and computer networks handled data traffic. During the twentieth century, telephone companies, which almost always operated as government-regulated monopolies, built telephone networks using voice transmission technologies (hardware and software). Computer companies originally built computer networks to transmit data among computers in different locations. As you learned in Chapter 1, the Internet is the world's largest and most widely used computer network.

As a result of continuing telecommunications deregulation and information technology innovation, telephone and computer networks have converged into a single digital network using shared Internet-based standards and technology. Telecommunications providers that historically provided telephone services, such as AT&T and Verizon, now offer Internet access as well as voice services over both cellular networks and traditional "land lines." Cable companies that historically offered television services, such as Comcast, Charter Spectrum, and Altice, now offer Internet access and voice services over their cable networks. Computer networks no longer transmit just data. They now also provide Internet, telephone, and video services.

Both voice and data communication networks have also become more powerful (faster), more portable (smaller and more mobile), and less expensive. For instance, in 2000, typical Internet service was only 56 Kbps (kilobits per second), but today the majority of U.S. households have broadband connections. In the United States, **broadband** is currently defined by the Federal Communications Commission as providing Internet service at a minimum of 100 megabits per second (Mbps) for downloads and 4 Mbps for uploads, although many telecommunications companies provide service with higher data transmission rates (Kan, 2024). The cost for this service has also fallen exponentially, from 50 cents per kilobit in 2000 to a tiny fraction of a cent today.

Broadband

Currently defined by Federal Communications Commission as providing Internet service at a minimum if 100 Mbps for downloads and 4 Mbps for uploads.

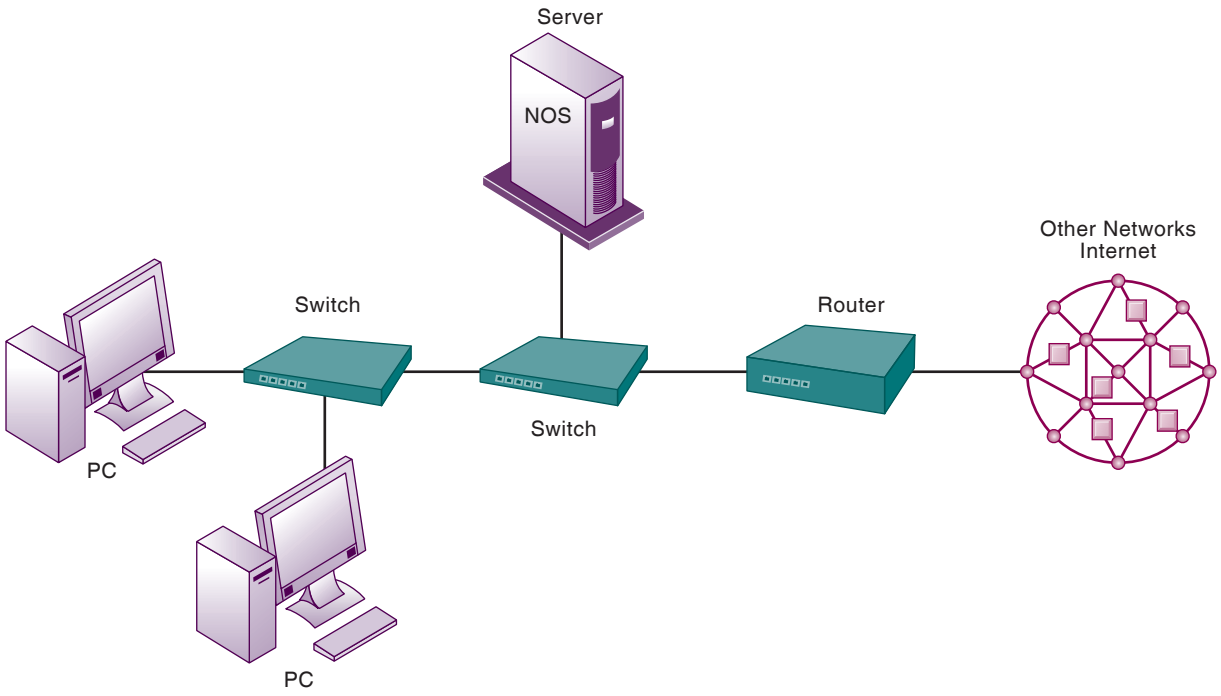
Increasingly, voice and data communication, as well as Internet access, are taking place over wireless broadband platforms using mobile devices and desktop computers in wireless networks. More than 93 percent of U.S. Internet users use smartphones and tablets to access the Internet at least some of the time (Emarketer, 2024a).

What Is a Computer Network?

If you had to connect the computers for two or more employees in the same office, you would need a computer network. In its simplest form, a computer network consists of two or more connected computers. Examine Figure 7.1, which illustrates the major hardware, software, and transmission components of a simple network connected to the Internet or another network: a client computer and a dedicated server computer, network interface controllers

Figure 7.1 Components of a Simple Computer Network

Illustrated here is a simple computer network consisting of computers, a network operating system (NOS) residing on a dedicated server computer, cable (wiring) connecting the devices, switches, and a router.



Network operating system (NOS)

Software that routes and manages communications on the network and coordinates network resources.

Hub

Simple device that connects network components, sending a packet of data to all other connected devices.

Switch

Device used to connect network components that has more intelligence than a hub and can filter and forward data to a specified destination.

Router

Communications processor that forwards packets of data from one network to another network.

Software-defined networking (SDN)

Networking approach that involves using a central control program separate from network devices to manage the flow of data on a network.

(NICs, sometimes also referred to as network interface cards), a connection medium, network operating system software, switches, and a router.

Each computer on the network contains a NIC to link the computer to the network. The connection medium for linking network components can be a telephone wire, a coaxial cable, or a radio signal (in the case of cellphone and wireless local area networks [Wi-Fi networks]).

The **network operating system (NOS)** routes and manages communications on the network and coordinates network resources. It can reside on every computer in the network or primarily on a dedicated server computer for all the applications on the network. A server is a computer on a network that performs important network functions for client computers, such as displaying web pages, storing data, and storing the network operating system (hence controlling the network). Microsoft Windows Server and Linux are the most widely used network operating systems.

Most networks also contain a switch or a hub acting as a connection point among the computers. A **hub** is a simple device that connects network components, sending a packet of data to all other connected devices. A **switch** has more intelligence than a hub and can filter and forward data to a specified destination on the network.

What if you want to communicate with another network, such as the Internet? You would need a router. A **router** is a communications processor that routes packets of data through different networks, ensuring that the packet of data sent gets to the correct address.

Network switches and routers have proprietary software built into their hardware for directing the movement of data on the network, which can create network bottlenecks and makes the process of configuring a network more complicated and time consuming. **Software-defined networking (SDN)** is a networking approach in which many of these control functions are managed by one central program, which can run on inexpensive servers that are separate from the network devices themselves. This is especially helpful in a cloud computing environment that has many pieces of hardware because SDN allows a network administrator to manage traffic loads in a flexible and more efficient manner.

Networks in Large Companies

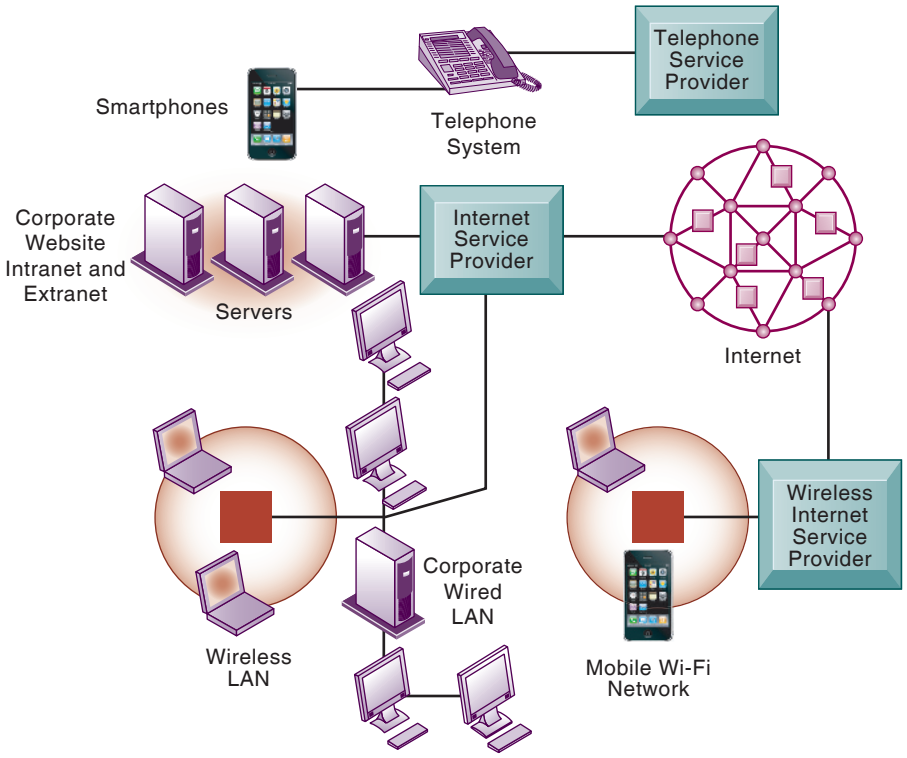
The network we've just described might be suitable for a small business, but what about large companies that have many locations and thousands of employees? As a firm grows, its small networks can be tied together into a corporate-wide networking infrastructure. The network infrastructure for a large company typically consists of a large number of these small local area networks linked to other local area networks and to firm-wide corporate networks. Powerful servers support a corporate website, a corporate intranet, and perhaps an extranet. Servers on the network may also link to other large computers supporting back-end systems.

Examine Figure 7.2, which provides an illustration of these more complex, larger-scale corporate-wide networks. Here the corporate network infrastructure supports a mobile sales force using mobile devices, mobile employees linking to the company website, and internal company Wi-Fi networks. In addition to these computer networks, the firm's infrastructure may include a separate telephone network that handles most voice data. Many firms are dispensing with their traditional telephone networks and using Internet telephones that run on their existing data networks (described later).

As you can see from this figure, a large corporate network infrastructure uses a wide variety of technologies—everything from ordinary telephone service and corporate data networks to Internet service, wireless Internet, and mobile devices. One of the major problems facing corporations today is how to integrate all these different communication networks and channels into a coherent system that enables information to flow from one part of the corporation to another and from one system to another.

Figure 7.2 Corporate Network Infrastructure

Today's corporate network infrastructure is a collection of many networks, from the public-switched telephone network to the Internet to corporate local area networks linking workgroups, departments, or office floors.



7.2 Describe Key Digital Networking Technologies

Contemporary digital networks and the Internet are based on three key technologies: client/server computing, the use of packet switching, and the development of widely used communications standards (the most important of which is Transmission Control Protocol/Internet Protocol, or TCP/IP) for linking disparate networks and computers.

Client/Server Computing

Client/server computing, introduced in Chapter 5, is a distributed computing model in which some of the processing power is located within small, inexpensive client computers, such as desktop or laptop computers or mobile devices. These clients are linked to one another through a network that is controlled by a network server computer. This server sets the rules of communication for the network and provides every client with an address so that others can find it on the network.

Client/server computing has largely replaced centralized mainframe computing, where nearly all the processing takes place on a central, large mainframe computer. Client/server computing has extended computing to departments, workgroups, factory floors, and other parts of the business that could not be served by a centralized architecture. It also makes it possible for personal computing devices such as PCs, laptops, and mobile devices to be connected to networks such as the Internet. The Internet is the largest implementation of client/server computing.

Packet Switching

Packet switching

Method of slicing digital messages into small, fixed bundles of data called packets, routing them along different communications paths as the paths become available, and then reassembling the packets once they arrive at their destination.

Packet switching is a method of slicing digital messages into small fixed bundles of data called packets, sending the packets along different communication paths as the paths become available, and then reassembling the packets once they arrive at their destinations. Prior to the development of packet switching, computer networks used leased, dedicated telephone circuits to communicate with other computers in remote locations. In circuit-switched networks, such as the telephone system, a complete point-to-point circuit is assembled, and then communication can proceed. These dedicated circuit-switching techniques were expensive and wasted available communications capacity: the circuit was maintained regardless of whether any data were being sent.

Packet switching is more efficient. Examine Figure 7.3, which illustrates how packet switching works. Messages are first broken down into small, fixed bundles of data called packets. The packets include information for directing the packet to the right address and for checking transmission errors along with the data. The packets are transmitted over various communications channels by using routers, each packet traveling independently. Packets of data originating at one source will be routed through many paths and networks before being reassembled into the original message when they reach their destinations.

TCP/IP and Connectivity

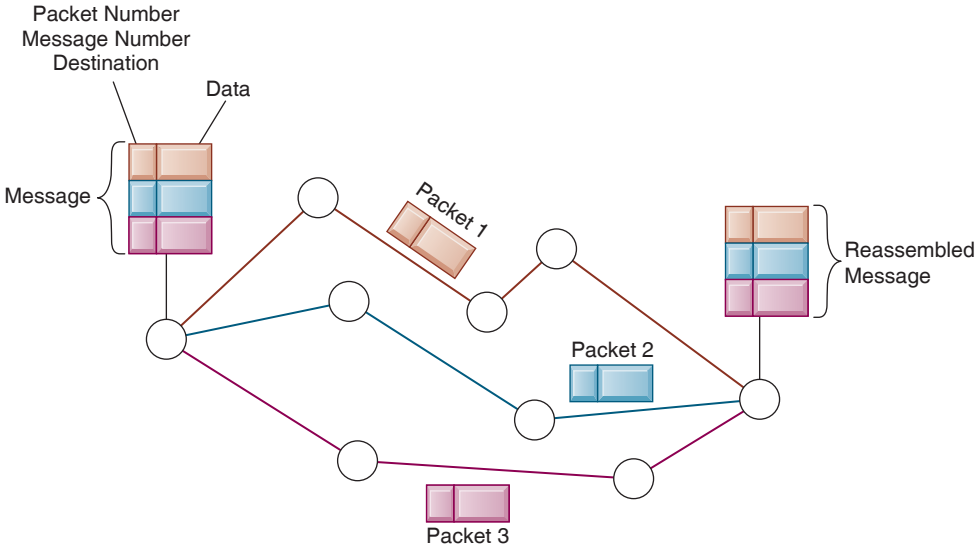
Protocol

A set of rules and procedures that govern transmission between the components in a network.

In a typical telecommunications network, diverse hardware and software components need to work together to transmit information. Different components in a network communicate with each other by adhering to a common set of rules called protocols. A **protocol** is a set of rules and procedures governing transmission of information between two points in a network.

Figure 7.3 Packet-Switched Networks and Packet Communications

Data are grouped into small packets, which are transmitted independently over various communications channels and then reassembled at their final destination.



In the past, diverse proprietary and incompatible protocols often forced business firms to purchase computing and communications equipment from a single vendor. However, today, corporate networks primarily use **Transmission Control Protocol/Internet Protocol (TCP/IP)**, a common worldwide standard. TCP/IP was developed during the early 1970s to support US Department of Defense Advanced Research Projects Agency (DARPA) efforts to help scientists transmit data among different types of computers over long distances.

Transmission Control Protocol/Internet Protocol (TCP/IP)

Dominant model for achieving connectivity among different networks. Provides a universally agreed-on method for breaking up digital messages into packets, routing them to the proper addresses, and then reassembling them into coherent messages.

TCP/IP uses a suite of protocols, the main ones being TCP (Transmission Control Protocol) and IP (Internet Protocol). TCP handles the movement of data among computers. TCP establishes a connection among the computers, sequences the transfer of packets, and acknowledges the packets sent. IP is responsible for the delivery of packets and includes the disassembling and reassembling of packets during transmission.

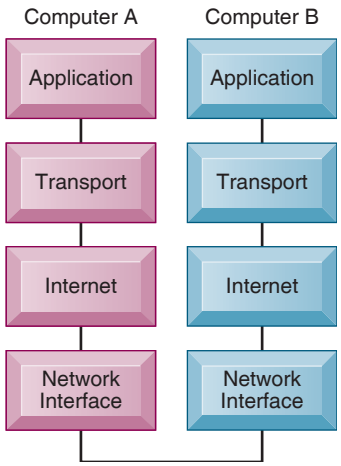
The Department of Defense's reference model conceptualizes the suite of protocols comprising TCP/IP as a series of four layers, described from top to bottom as follows:

1. **Application layer.** The Application layer enables client application programs to access the other layers and defines the protocols that the applications use to exchange data. One of these application protocols is the Hypertext Transfer Protocol (HTTP), which is used to transfer web page files.
2. **Transport layer.** The Transport layer is responsible for providing the Application layer with communication and packet services. This layer includes TCP and other protocols.
3. **Internet layer.** The Internet layer is responsible for addressing, routing, and packaging data packets called IP datagrams. The Internet Protocol is one of the protocols used in this layer.
4. **Network Interface layer.** At the bottom of the reference model, the Network Interface layer is responsible for placing packets on and receiving them from the network medium, which could be any networking technology.

Examine Figure 7.4, which illustrates how two computers using TCP/IP can communicate even if they are based on different hardware and software platforms. Data sent from one computer to the other pass downward through all four layers, starting with the sending computer's Application layer and passing through to the Network Interface layer. After the

Figure 7.4 The Transmission Control Protocol/Internet Protocol (TCP/IP) Reference Model

This figure illustrates the four layers of the TCP/IP reference model for communications.



data reach the recipient host computer, they travel up the layers and are reassembled into a format the receiving computer can use. If the receiving computer finds a damaged packet, it asks the sending computer to retransmit it. This process is reversed when the receiving computer responds.

Signals: Digital versus Analog

There are two ways to communicate a message in a network: via an analog signal or a digital signal. An *analog signal* is represented by a continuous waveform that passes through a communications medium and is used for audio communication. The most common analog devices are the traditional telephone handset, the speaker on your computer, and your iPhone earphone, all of which create analog waveforms that your ear can hear.

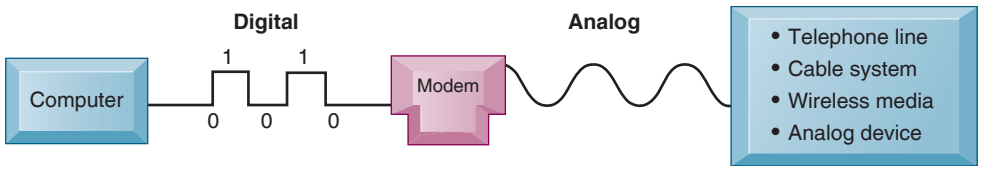
A *digital signal* is a discrete, binary waveform rather than a continuous waveform. Digital signals communicate information as strings of two discrete states: 1 bits and 0 bits, which are represented as on–off electrical pulses. Computers use digital signals and require a **modem** (which stands for "modulator-demodulator") to convert these digital signals into analog signals that can be sent over (or received from) telephone lines, cable lines, or wireless media that use analog signals. Examine Figure 7.5, which illustrates the function of a modem. Cable modems connect your computer to the Internet by using a cable network. DSL modems connect your computer to the Internet using a telephone company's landline network. Wireless modems perform the same function as traditional modems but connect your computer to a wireless network, which could be a cellular network or a Wi-Fi network.

Modem

A device for translating a computer's digital signals into analog form for transmission over analog networks or for translating analog signals back into digital form for reception by a computer.

Figure 7.5 Functions of a Modem

A modem is a device that translates digital signals into analog form (and vice versa) so that computers can transmit data over analog networks such as telephone and cable networks.



7.3 Describe the Different Types of Networks

There are many kinds of networks and ways of classifying them. One way of looking at networks is in terms of their geographic scope.

Local Area Networks

If you work in a business that uses networking, you are probably connecting to other employees and groups via a local area network. A **local area network (LAN)** is designed to connect desktop computers and other digital devices within a half-mile or 500-meter radius. LANs typically connect a few computers in a small office, all the computers in one building, or all the computers in several buildings in close proximity. LANs also are used to link to long-distance wide area networks (WANs, described later in this section) and other networks around the world, using the Internet.

Local area network (LAN)

Designed to connect computers and digital devices within a limited area (half-mile or 500-meter radius) and to link to long-distance wide area networks, such as the Internet.

Review Figure 7.1, which could serve as a model for a small LAN that might be used in an office. In this model, one computer is a dedicated network server, providing users with access to shared computing resources in the network, including software programs and data files.

The server determines who gets access to what and in which sequence. The router connects the LAN to other networks, which could be the Internet or another corporate network, so that the LAN can exchange information with networks external to it. The most common LAN operating systems are Windows and Linux.

Ethernet is the dominant LAN standard at the physical network level, specifying the physical medium to carry signals among computers, the access control rules, and a standardized set of bits that carry data over the system. Originally, Ethernet supported a data transfer rate of 10 Mbps (megabits per second). Newer versions, such as Gigabit Ethernet, support a data transfer rate of 1 Gbps (gigabits per second) and upward.

The LAN illustrated in Figure 7.1 uses a client/server architecture in which the network operating system resides primarily on a single server, and the server provides much of the control and resources for the network. Alternatively, LANs may use a **peer-to-peer** architecture. A peer-to-peer network treats all processors equally and is used primarily in small networks with 10 or fewer users. The various computers on the network can exchange data by direct access and can share peripheral devices without going through a separate server.

Peer-to-peer

Network architecture that treats all processors on the network equally; the various computers on the network exchange data by direct access and can share peripheral devices. Used primarily in small networks.

Larger LANs have many clients and multiple servers, with separate servers for specific services such as storing and managing files and databases (file servers or database servers), managing printers (print servers), storing and managing email (mail servers), or storing and managing web pages (web servers).

WANs, CANs, and MANs

A **wide area network (WAN)** spans a broad geographical distance—a region, state, continent, or even the entire globe. The most universal and powerful WAN is the Internet. Computers connect to a WAN through public networks, such as the telephone system or private cable systems, or through leased lines or satellites. A **campus area network (CAN)** is a computer network that spans a limited geographic area, up to about 1,000 meters (about 1 mile), that may be used for a corporate facility or college campus. A **metropolitan area network (MAN)** is a network that spans a metropolitan area, usually a city and its major suburbs. Its geographic scope falls between that of a WAN and that of a LAN.

Wide area network (WAN)

Telecommunications network that spans a large geographical distance–a region, state, continent, or the even the entire globe; Internet is the most universal and powerful example.

Campus area network (CAN)

Computer network that spans a limited geographic area, up to about 1,000 meters (1 mile) that may be used for a corporate facility or college campus.

Metropolitan area network (MAN)

Network that spans a metropolitan area, usually a city and its major suburbs. Its geographic scope falls between a WAN and a LAN.

Table 7.1 summarizes these different types of networks.

Transmission Media and Bandwidth

Another way to categorize networks is in terms of transmission media and bandwidth.

Table 7.1 Types of Networks

Type	Area
Local area network (LAN)	Up to 500 meters (half a mile); an office or floor of a building
Campus area network (CAN)	Up to 1,000 meters (a mile); a college campus or corporate facility
Metropolitan area network (MAN)	A city or metropolitan area
Wide area network (WAN)	A regional, transcontinental, or global area

Transmission Media

Networks use different kinds of physical transmission media, including twisted copper wire, coaxial cable, fiber-optic cable, and media for wireless transmission. Each type of transmission media has different properties, advantages, and limitations. A wide range of data transmission rates (discussed in the next section) is possible for any given medium, depending on the software and hardware configuration.

Bandwidth

Bandwidth
Measures how much data can be transferred over a communications medium within a fixed period of time and typically is expressed as bits per second (Bps), kilobits per second (Kbps), megabits per second (Mbps), gigabits per second (Gbps), and, for very advanced communications media, terabits per second (Tbps).

Bandwidth measures how much data can be transferred over a communications medium within a fixed period of time and typically is expressed as bits per second (Bps), kilobits per second (Kbps), megabits per second (Mbps), gigabits per second (Gbps), and, for very advanced communications media, terabits per second (Tbps). Bandwidth is often confused with transmission speed (particularly because Internet service providers often characterize bandwidth as speed), but the two are not the same. Bandwidth refers to the capacity of a communications link to transmit data (for example, a wide water hose allows more water to flow than a narrow one), while speed refers to the rate at which data can be transmitted (using the water hose analogy, a slow trickle of water versus a fast gush). To confuse matters further, data transmission rate (speed) is also measured in Bps/Kbps/Mbps/Gbps/Tbps.

Table 7.2 compares the different transmission media in terms of their various characteristics.

Table 7.2 Physical Transmission Media

Transmission Medium	Description	Bandwidth
Twisted copper wire	Strands of copper wire twisted in pairs for voice and data communications. Includes CAT5, CAT6, CAT7, and CAT8 cabling, with CAT6–8 capable of gigabit data transfer rates.	CAT5: 10 Mbps–100 Mbps CAT8: Up to 40 Gbps
Coaxial cable	Thickly insulated copper wire, which is capable of high-speed data transmission and is less subject to interference than twisted wire. Currently used for cable TV and for networks with longer runs (more than 100 meters).	Up to 1 Gbps
Fiber-optic cable	Strands of clear glass fiber that transmit data as pulses of light generated by lasers. Useful for high-speed transmission of large quantities of data. More expensive than other physical transmission media; used for last-mile delivery to customers and the Internet backbone.	Up to 170+ Tbps
Wireless transmission media	Based on radio signals of various frequencies. Includes terrestrial and satellite microwave systems and cellular networks. Used for long-distance communication, and Internet access.	Up to 600+ Mbps

7.4 Understand What the Internet Is and How It Works

The Internet has become an indispensable personal and business tool—but what exactly is the Internet? How does it work? Let's look at the most important Internet features.

What Is the Internet?

The Internet is the world's most extensive public communication system. It's also the world's largest implementation of client/server computing and internetworking, linking millions of individual networks all over the world. This global network of networks began in the early 1970s as a U.S. Department of Defense project to link scientists and university professors around the world.

Internet Addressing and Architecture

The Internet is based on the TCP/IP networking protocol suite described earlier in this chapter. Every device connected to the Internet (or another TCP/IP network) is assigned a unique **Internet Protocol address (IP address)** consisting of a string of numbers. There are two versions of IP addressing currently in use: IPv4 and IPv6. An IPv4 address is a 32-bit number that appears as a series of four separate numbers marked off by periods, such as 64.49.254.91. However, the world is running out of available IP addresses using IPv4 because of the explosion in the number of devices now connected to the Internet. IPv4 is now being replaced by a new version of IP addressing called **IPv6** (Internet Protocol version 6), which contains 128-bit addresses (2 to the power of 128), or more than a quadrillion possible unique addresses. IPv6 is compatible with most modems and routers sold today and will fall back to the old addressing system if IPv6 is not available on local networks. Adoption of IPv6 is still uneven and the full transition is expected to take several years.

Internet Protocol address (IP address)

Numeric address indicating a unique computer location on the Internet.

IPv6

IP addressing system using 128-bit IP addresses. Stands for Internet Protocol version 6.

The Domain Name System

Because it would be incredibly difficult for Internet users to remember long strings of numbers, an IP address can be represented by a natural language convention called a **domain name**. The **Domain Name System (DNS)** converts domain names to IP addresses. DNS servers maintain a database containing IP addresses mapped to their corresponding domain names. To access a computer on the Internet, users need only specify its domain name, such as Expedia.com.

DNS has a hierarchical structure. Examine Figure 7.6, which illustrates this hierarchical structure. At the top of the DNS hierarchy is the root domain, represented by a "dot." The child domain of the root is called a top-level domain. Top-level domains are the two- and three-character names you are familiar with from surfing the web—for example, .com, .edu, .gov, and the various country codes such as .ca for Canada, .it for Italy, and .br for Brazil. One country that has recently benefited from its country code is Anguilla, which uses the country code .ai and has seen a flurry of domain registrations in the past several years. In 2011, ICANN, the international body that governs the issuance of domains, removed nearly all restrictions on domain names, and since then has approved more than 1,200 new generic top-level domains, such as .xyz, .online, .shop, .kids, and ones that use non-English characters.

The child domain of a top-level domain is called a second-level domain. Second-level domains have two parts, which designate a top-level name and a second-level name—such as buy.com, nyu.edu, or amazon.ca. Third-level domains, which are at the bottom of the hierarchy, are used to designate the host name of a specific computer on either the Internet or a private network.

Domain name

Representation of a numeric IP address as a natural language name.

Domain Name System (DNS)

A hierarchical system of servers that maintain a database enabling the conversion of domain names to their numeric IP addresses.

Internet Architecture

Examine Figure 7.7, which illustrates some of the main physical elements of today's Internet. Internet data traffic is carried over transcontinental, high-speed backbone networks that generally operate in the gigabit range. These trunk lines are typically owned by long-distance telecommunications companies (called *Tier 1 Internet service providers [Tier 1 ISPs]*) or by

Figure 7.6 The Domain Name System

The Domain Name System is a hierarchical system with a root domain, top-level domains, second-level domains, and host computers at the third level.

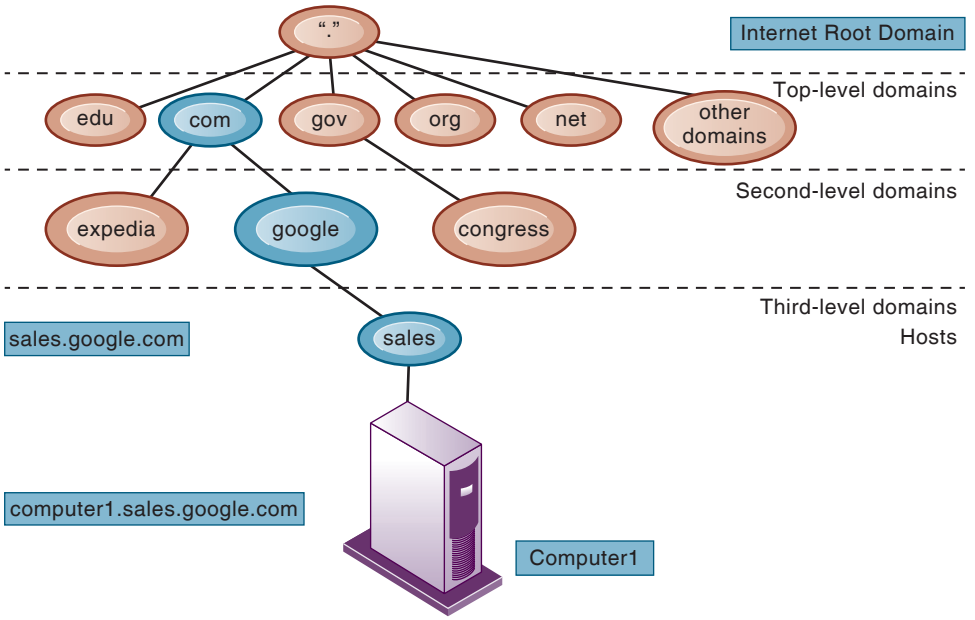
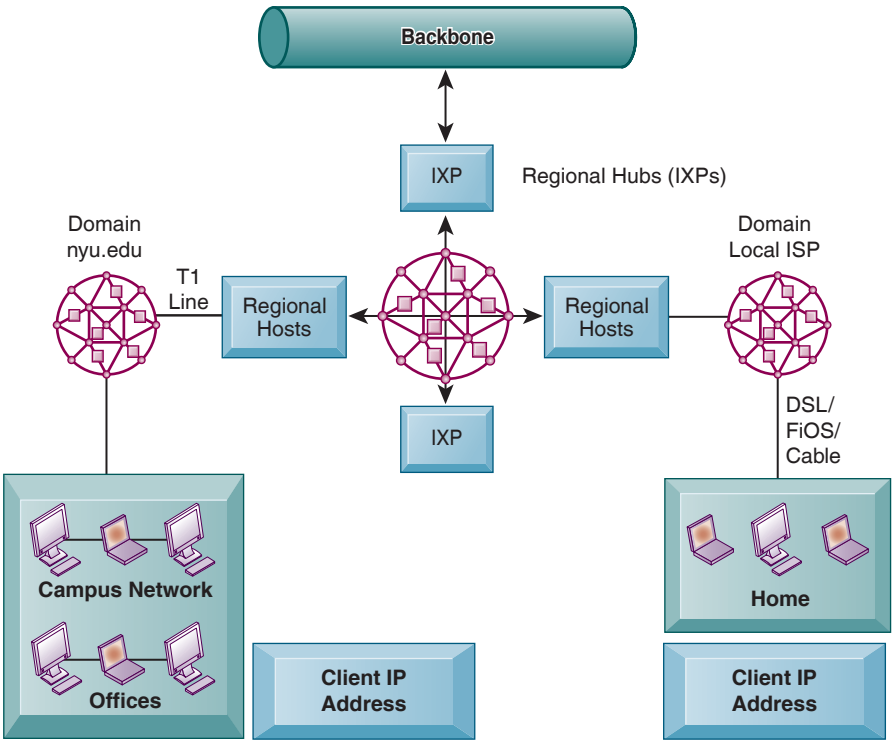


Figure 7.7 Internet Network Architecture

The Internet backbone connects to regional networks, which in turn provide access to Internet service providers, large firms, and government institutions. Internet Exchange Points (IXPs) are hubs where the backbone intersects with regional and local networks and where backbone owners connect with one another.



Internet service providers (ISPs)

Commercial organizations with a permanent connection to the Internet that sell temporary connections to subscribers.

national governments. Tier 1 ISPs have “peering” arrangements with other Tier 1 ISPs to allow Internet traffic to flow through each other’s cables and equipment without charge. Tier 1 ISPs physically connect with one another and to regional (Tier 2) ISPs at regional hubs, commonly referred to as Internet Exchange Points (IXPs). Tier 2 ISPs connect Tier 1 ISPs with Tier 3 ISPs, which deal with “the last mile of service” and provide Internet access to consumers and business. Tier 3 ISPs are popularly referred to as just **Internet service providers (ISPs)**. Major Tier 3 ISPs in the United States

include Comcast, Charter Spectrum, AT&T, Verizon, Altice (Optimum Online), CenturyLink, and Cox. (It's important to note that many Tier 3 ISPs are also Tier 1 ISPs; the two roles are not mutually exclusive.) There are also thousands of much smaller, local access ISPs.

Each organization pays for its own networks and its own local Internet connection services, a part of which is paid to the long-distance trunk line owners. Individual Internet users pay ISPs for using their service, and they generally pay a flat subscription fee no matter how much or how little they use the Internet. There are a variety of services for ISP Internet connections. Connecting via a traditional telephone line and modem, at a speed of 56.6 kilobits per second (Kbps), used to be the most common form of connection worldwide, but high-speed broadband connections have largely replaced it. Today, digital subscriber line (DSL) and fiber-optic service (FiOS), cable Internet, satellite Internet, and T lines provide these broadband services.

Digital subscriber line (DSL) technologies operate over existing telephone lines to carry voice, data, and video at transmission rates ranging from 385 Kbps to more than 100 Mbps, depending on usage patterns and distance. FiOS (Verizon's fiber-optic service) can deliver more than 900 Mbps, although most home Internet services deliver 100 Mbps. **Cable Internet connections** provided by cable providers use digital cable coaxial lines to deliver high-speed Internet access to homes and businesses. These lines can provide high-speed access to the Internet at between 20 and 600 Mbps, with high-speed cable Internet download speeds approaching 1 Gbps.

Digital subscriber line (DSL)
A group of technologies providing high-capacity voice, data, and video transmission over existing telephone lines.

Cable Internet connections
Use digital coaxial cable lines to deliver high-speed Internet access to homes and businesses.

Where DSL and cable services are unavailable, it is possible to access the Internet via satellite, although most satellite Internet connections have slower upload speeds than other broadband services have.

T1 and T3 are international telephone standards for digital communication. They are leased, dedicated lines suitable for businesses or government agencies requiring high-speed, guaranteed service levels. A **T1 line** offers guaranteed delivery at 1.54 Mbps, and a **T3 line** offers delivery at 45 Mbps. The Internet does not provide similar guaranteed service levels, but simply the best effort.

T1 line
International telephone standard for digital communication that offers guaranteed delivery at 1.54 Mbps.

T3 line
International telephone standard for digital communication that offers guaranteed delivery at 145 Mbps.

A variety of new methods are also being explored to provide Internet access to areas that are not well served by wired or cellular networks, including low earth orbit (LEO) satellites operated by firms such as Starlink.

Internet Governance

No one owns the Internet, and it has no formal management. However, worldwide Internet policies are established by a number of professional organizations and government bodies, including the Internet Architecture Board (IAB), which helps define the overall structure of the Internet; the Internet Corporation for Assigned Names and Numbers (ICANN), which manages the domain name system; and the World Wide Web Consortium (W3C), which sets Hypertext Markup Language (HTML) and other programming standards for the web.

These organizations influence government agencies, network owners, ISPs, and software developers with the goal of keeping the Internet operating as efficiently as possible. The Internet must also conform to the laws of the sovereign nation-states in which it operates, as well as to the technical infrastructures that exist within the nation-states. Although in the early years of the Internet and the web, there was very little governmental interference, this situation is changing as the Internet plays a growing role in the distribution of information and knowledge, including content that some find objectionable.

7.5 Explain How Internet Services Support Business

The Internet is based on client/server technology. Individuals using the Internet control what they do through client applications, such as web browser software, on their computers. The data, including email messages and web pages, are stored on servers. A client uses the Internet to request information from a particular web server on a distant computer, and the server sends the requested information back to the client over the Internet. Client platforms today include not only PCs and other computers but also smartphones and tablets.

Table 7.3 Major Internet Services

Capability	Functions Supported
Email	Person-to-person messaging; document sharing
Instant messaging	Interactive conversations
File transfer	The original protocol that enabled this service, File Transfer Protocol (FTP) has been mostly supplanted by more recent protocols, such as FTP over SSL (FTPS) and SSH File Transfer Protocol (SFTP).
Voice transmission	Voice over IP (VoIP) technology enables the transfer of voice over the Internet
World Wide Web (the web)	Retrieving, formatting, and displaying information (including text, audio, graphics, and video) by using hypertext links

A client computer connecting to the Internet has access to a variety of services. These services include email, instant messaging, file transfer, voice transmission, and the web. Table 7.3 provides a brief description of these services.

Each Internet service is implemented by one or more software programs. Examine Figure 7.8, which illustrates one way these services can operate within a multitiered client/server architecture. All the services may run on a single server computer (as shown in Figure 7.8) or different services may be allocated to different servers.

We look at each of these services in more detail in the following sections.

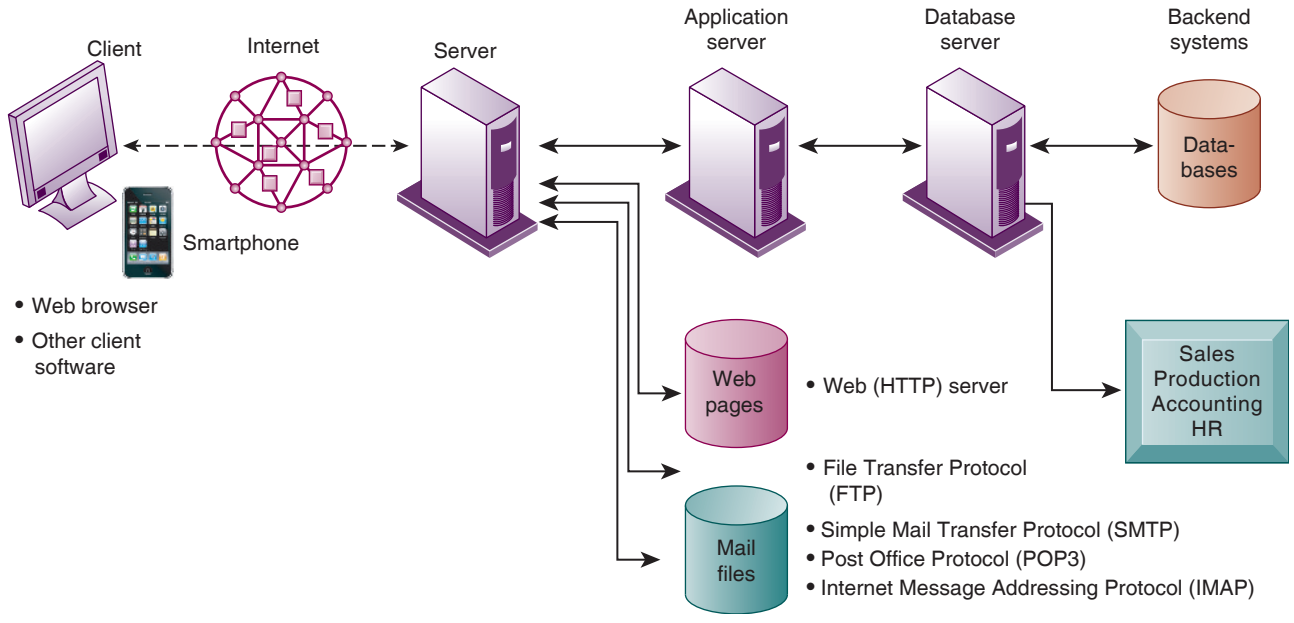
Email

Email
The computer-to-computer exchange of messages.

Email enables messages to be exchanged from computer to computer, with capabilities for routing messages to multiple recipients, forwarding messages, and attaching text documents or multimedia files to messages. Most email today is sent through the Internet. The cost of email is far lower than the equivalent voice, postal, or overnight delivery costs, and

Figure 7.8 Client/Server Computing on the Internet

Client computers running web browsers and other software can access an array of services on servers over the Internet. These services may all run on a single server or on multiple specialized servers.



email messages can arrive anywhere in the world in a matter of seconds. Internet protocols used for email include Simple Mail Transfer Protocol (SMTP), Post Office Protocol 3 (POP3), and Internet Message Access Protocol (IMAP).

Instant Messaging

Instant messaging is a type of chat service that enables participants to create their own private chat channels. The instant messaging system alerts users whenever someone on their private list is online so that the user can initiate a chat session with other individuals. Instant messaging systems for consumers include Google Chat, Facebook Messenger, and Skype. Companies concerned with security use proprietary communications and messaging systems such as HCL Sametime, Microsoft Teams, and Slack.

Instant messaging
Chat service that allows participants to create their own private chat channels so that a person can be alerted whenever someone on his or her private list is online to initiate a chat session with that particular individual.

Employee use of email, instant messaging, and the Internet is supposed to increase worker productivity, but this may not always be the case. Many companies believe they need to monitor and even regulate their employees' online activity, but is this ethical? Although there are some strong business reasons that companies may need to monitor their employees' email and web activities, what does this monitoring mean for employee privacy? The issue has become even more pressing because of the number of people now working remotely and because of businesses' desires to ensure that their employees and contractors are, in fact, working as required when they are not physically present.

Voice over IP (VoIP)

The Internet has also become a popular platform for voice transmission and corporate networking. **Voice over IP (VoIP)** technology delivers voice information in digital form using packet switching, thus avoiding charges from local and long-distance telephone networks. Examine Figure 7.9, which illustrates how VoIP works. Voice calls can be made and received with a computer equipped with a microphone and speakers, a mobile device with a VoIP app, or with a VoIP-enabled telephone. Analog voice calls that would ordinarily be transmitted over public telephone networks are instead digitized into packets and routed via a VoIP gateway over a corporate network based on the Internet protocol or over the public Internet to the IP address of the receiving device or telephone.

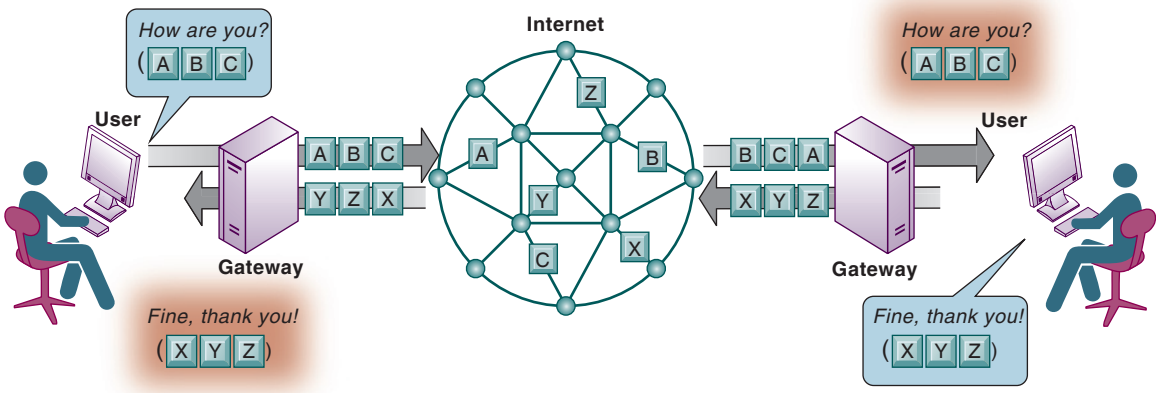
Voice over IP (VoIP)
Technology for managing the delivery of voice information using the Internet Protocol (IP).

Many telecommunications companies, such as AT&T, Verizon, Spectrum, Comcast, and Altice, provide VoIP service bundled with their high-speed Internet offerings. Skype offers free VoIP worldwide using a peer-to-peer network, and Google has its own free VoIP service.

Although up-front investments are required for an IP phone system, VoIP can reduce communication and network management costs while providing more communication

Figure 7.9 How Voice over IP Works

A VoIP phone call digitizes and breaks up a voice message into data packets that may travel along different routes before being reassembled at the final destination. A processor nearest the call's destination, called a gateway, arranges the packets in the proper order and directs them to the telephone number of the receiver or the IP address of the receiving computer.



capabilities. For example, VoIP provider Nextiva helps Titan Solar Power service its quickly growing customer base. Prior to adopting Nextiva, the Arizona-based company had no way of tracking why people were calling or the volume of calls to and from customers. Using Nextiva's tools to track these data, Titan can see the number of inbound and outbound calls and the number of calls in queue. The company can then take steps to reduce customer wait times. Additionally, Nextiva provides capabilities for recording customer calls and for directly routing customers to the right departments.

Unified Communications

Unified communications (UC)

Integrates disparate channels for voice communications, data communications, instant messaging, email, and teleconferencing into a single experience where users can seamlessly switch back and forth between different communication modes.

In the past, each of a firm's networks for wired and wireless data, voice communications, and videoconferencing operated independently of each other and had to be managed separately by the information systems department. Now, however, firms can merge disparate communications modes into a single, universally accessible service using unified communications technology. A **unified communications (UC)** system integrates disparate channels for voice communications, data communications, instant messaging, email, and teleconferencing into a single experience by which users can seamlessly switch back and forth between different communication modes. Presence technology shows whether a person is available to receive a call. Unified communications as a service (UCaaS) provides cloud-delivered unified applications.

AI functionality is now beginning to be added to UC to improve meeting workflows. The first wave of innovation includes enhanced applications for automated transcription, real-time translation, and meeting summaries. For example, a virtual assistant can manage follow-up activities, such as scheduling a meeting, providing a customized meeting summary, or generating an email, letter, or report based on the contents of a communication (Arnold, 2024).

Virtual Private Networks

Virtual private network

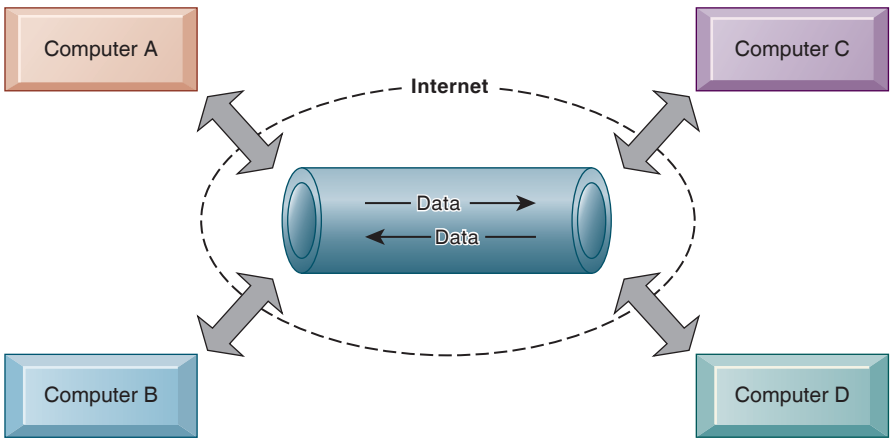
(VPN) A secure, encrypted private network that has been configured within a public network, such as the Internet.

What if you had a marketing group with members spread across the United States, charged with developing new products and services for your firm? You would want them to be able to email each other and communicate with the home office without any chance that outsiders could intercept the communications. Large private networking firms offer secure, private, dedicated networks to customers, but this type of network is expensive. A lower-cost solution is to create a virtual private network within the public Internet.

A **virtual private network (VPN)** is a secure, encrypted, private network that has been configured within a public network to take advantage of the economies of scale and the management facilities of large networks, such as the Internet. Examine Figure 7.10, which illustrates

Figure 7.10 A Virtual Private Network Using the Internet

This VPN is a private network of computers linked using a secure tunnel connection through the Internet. It protects data transmitted over the public Internet by encoding the data and wrapping them within the Internet protocol. By adding a wrapper around a network message to hide its content, organizations can create a private connection that travels through the public Internet.



how a VPN works. Computers in a private network can use VPN protocols to encrypt the data being transferred and to establish a secure "tunnel" that shields the data as they travel over the Internet. A VPN provides your firm with secure, encrypted communications at a much lower cost than the same capabilities offered by traditional non-Internet providers that use their private networks to secure communications. VPNs also provide a network infrastructure for combining voice and data networks.

A variety of protocols are used to protect data transmitted over the public Internet, including Point-to-Point Tunneling Protocol (PPTP). In a process called *tunneling*, packets of data are encrypted and wrapped inside IP packets. By adding this wrapper around a network message to hide its content, business firms create a private connection that travels through the public Internet.

7.6 Discuss How the Web Works and How It Can Be Used in Business

The web is the most popular Internet service. It's a system with universally accepted standards for storing, retrieving, formatting, and displaying information by using a client/server architecture. A **web page** is an online "page" formatted using Hypertext Markup Language (HTML) and that may contain embedded links (hyperlinks) that connect documents to one another and that also link pages to other objects, such as sound, video, or animation files (see Chapter 5). When you click a graphic and a video clip plays, you have clicked a hyperlink. A typical **website** is a collection of web pages linked to a home page.

Web page
Online "page" formatted using HTML and that may contain embedded links (hyperlinks) that connect documents to one another and also link pages to other objects, such as sound, video, or animation files.

Website
A collection of web pages linked to a home page.

Hypertext

Web pages are accessible through the Internet because the web browser software operating on your computer can request web pages stored on an Internet host server by using the **Hypertext Transfer Protocol (HTTP)**. HTTP is the communications standard that transfers pages on the web. For example, when you type a web address, such as http://www.sec.gov, in your browser, your browser sends an HTTP request to the sec.gov server requesting the home page of sec.gov.

HTTP (or HTTPS, a more secure version of HTTP that we will discuss further in Chapter 8) is the first set of letters at the start of every web address, followed by the domain name, which specifies the organization's server computer that is storing the web page. Most companies have a domain name that is the same as, or closely related to, their official corporate name. The directory path and web page name are two more pieces of information within the web address that help the browser track down the requested page. Together, the address is called a **uniform resource locator (URL)**. When typed into a browser, a URL tells the browser software exactly where to look for the information. For example, in the fictitious URL http://www.megacorp.com/content/features/082610.html, *http* names the protocol that displays web pages, www.megacorp.com is the domain name, content/features is the directory path that identifies where on the domain web server the page is stored, and 082610.html is the web page name and the name of the format it is in. (It is an HTML page.)

Hypertext Transfer Protocol (HTTP)
The communications standard used to transfer pages on the web. Defines how messages are formatted and transmitted.

Uniform resource locator (URL)
The address of a specific resource on the Internet.

Web Servers

As you learned in Chapter 5, a web server is software for locating and managing stored web pages. It locates the web page a user requests on the computer where it is stored and delivers the web page to the user's computer. Server applications usually run on dedicated computers, although these applications can all reside on a single computer in small organizations.

The leading web servers in use today are Microsoft Internet Information Services (IIS) and Apache HTTP Server. Apache is an open-source product that is free of charge and can be downloaded from the web.

Searching for Information on the Web

No one knows for sure how many web pages there really are. The surface web is the part of the web that search engines visit and about which information is recorded. According to Google, its search index contains hundreds of billions of web pages, and these web pages reflect a large portion of the publicly accessible web page population (Google, 2024). But there is a "deep web" that contains trillions of additional pages, many of them proprietary (such as the pages of *Wall Street Journal Online*, which cannot be visited without a subscription or access code), or that are stored in protected corporate databases. Facebook users can block Google and other search engines from accessing the content they post. A portion of the deep web called the **dark web** has been intentionally hidden from search engines, uses masked IP addresses, and is accessible only with a special web browser to preserve anonymity. The dark web has become a haven for criminals because it allows the buying and selling of illicit goods, including credit card and social security numbers, with complete anonymity.

Dark web

Portion of the deep web that has been intentionally hidden from search engines in order to preserve anonymity.

Search Engines

Obviously, with so many web pages, finding, nearly instantly, the specific ones that can help you or your business is an important problem. The question is, how can you find the one or two pages you really want and need out of billions of indexed web pages? A **search engine** attempts to solve the problem of finding useful information on the web nearly instantly. Searching for information is one of the web's most popular activities. Today's search engines can sift through HTML files; files of Microsoft Office applications; PDF files; and audio, video, and image files. There are hundreds of search engines in the world, but Google dominates with a more than 90 percent market share, followed by Microsoft's Bing (3 percent), Yandex (2 percent), Yahoo (1 percent), and China's Baidu (0.75 percent). Also, while we typically think of Amazon as an online store, it is also a powerful product search engine (Statcounter, 2024).

Search engine

A tool for locating specific sites or information on the Internet.

Web search engines started out in the early 1990s as relatively simple software programs that roamed the nascent web, visiting pages and gathering information about the content of each page. The first search engines were simple keyword indexes of all the pages they visited, leaving users with lists of pages that may or may not have been truly relevant to their search.

In 1994, Stanford University computer science students David Filo and Jerry Yang created a hand-selected list of their favorite web pages and called it "Yet Another Hierarchical Officious Oracle," or Yahoo. Yahoo was not initially a search engine but rather an edited selection of websites organized by categories. Yahoo relies on Microsoft's Bing for search results.

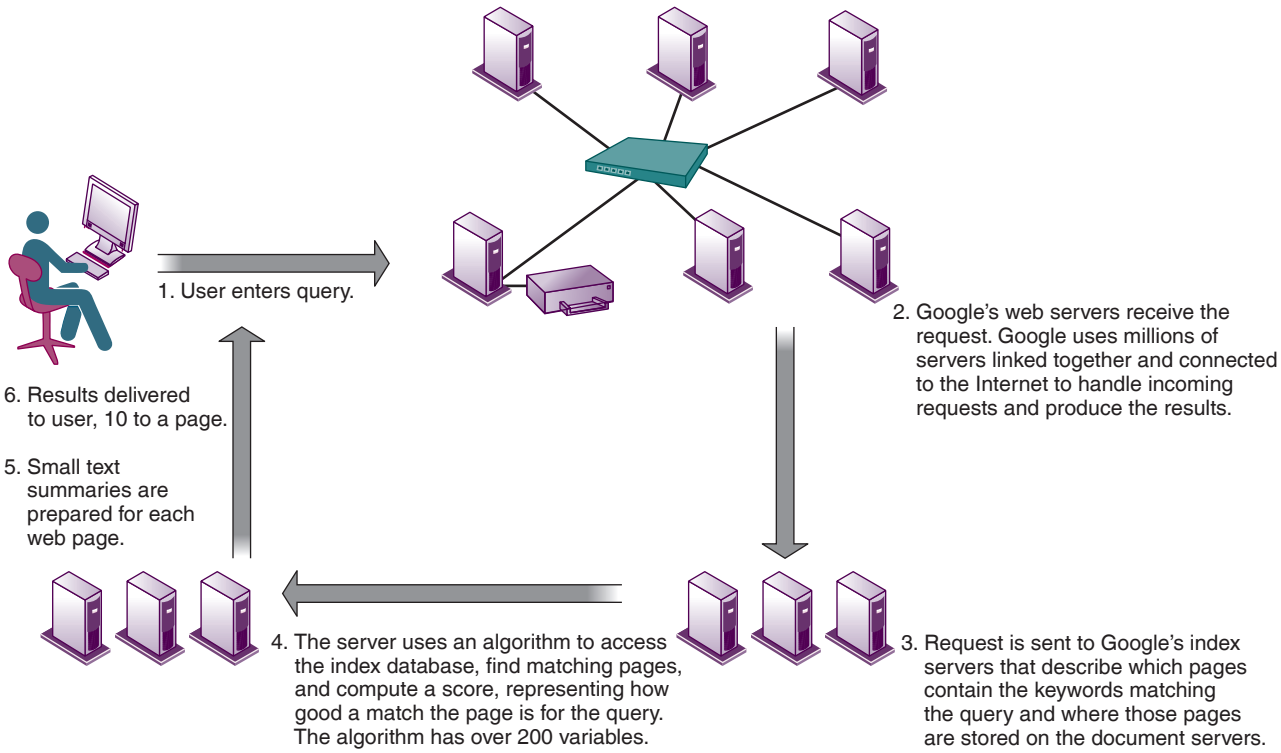
In 1998, Larry Page and Sergey Brin, two other Stanford computer science students, released their first version of Google. This search engine was different: Not only did it index each web page's words, but it also ranked search results based on the relevance of each page. Page patented the idea of a page-ranking system (called *PageRank System*), which essentially measures the popularity of a web page by calculating the number of sites that link to that page as well as the number of pages to which it links. The premise is that popular web pages are more relevant to users. Brin contributed a unique web crawler program that indexed not only keywords on a page but also combinations of words (such as the names of authors and the titles of their articles). These two ideas became the foundation of the Google search engine. Examine Figure 7.11, which illustrates what happens when you enter a search query using Google.

Mobile Search

Mobile search via smartphones and tablets makes up more than 60 percent of all searches and is expected to continue expanding rapidly in the next few years. Google, Amazon, and Yahoo have developed new search interfaces to make searching and shopping from smartphones more convenient. Google revised its search algorithm to favor sites that look good on smartphone screens. Mobile e-commerce (m-commerce) now comprises an ever-increasing percentage of e-commerce and is expected to account for almost 45 percent of retail e-commerce revenues in 2024, with almost 90 percent of those revenues derived from purchases made via smartphone (Emarketer, 2024b).

Figure 7.11 How Google's Search Engine Works

The Google search engine is continuously crawling the web, indexing the content of each page it locates, calculating each page's popularity, and storing the pages so that it can respond quickly to user requests to see a page. The entire process takes about half a second.



Search Engine Marketing

Search engines have become major advertising platforms and shopping tools by offering what is now called **search engine marketing**. Search engine marketing accounts for more than 40 percent of digital ad spending. When users enter a search term on Google, Bing, Yahoo, or any of the other sites serviced by these search engines, they receive two types of listings: sponsored links (which are usually at the top of the search results page), for which advertisers have paid to be listed, and unsponsored, organic search results. In addition, advertisers can purchase small text boxes on the side of search results pages. Paid sponsored advertisements are the fastest-growing form of Internet advertising and are powerful marketing tools that precisely match consumer interests with advertising messages at the right moment. Search engine marketing monetizes the value of the search process (Emarketer, 2023).

Search engine marketing

Use of search engines to deliver sponsored links for which advertisers have paid in search engine results.

Because search engine marketing is so effective (it has the highest click-through rate and the highest return on ad investment), companies seek to optimize their websites for search engine recognition. The better optimized a page is, the higher a ranking it will achieve in search engine result listings. **Search engine optimization (SEO)** is the process of improving the quality and volume of web traffic to a website by employing a series of techniques that help a website achieve a higher ranking with the major search engines.

Search engine optimization (SEO)

Process of improving the quality and volume of web traffic to a website by employing a series of techniques that help a website achieve a higher ranking with the major search engines.

In general, search engines have been very helpful to small businesses that cannot afford large marketing campaigns. Because shoppers frequently are looking for a specific product or service when they use search engines, they are what marketers call hot prospects—people who are looking for information and often intending to buy. Moreover, search engines charge only for click-throughs to a site. Merchants do not have to pay for ads that don't work, only for ads that receive a click-through. Consumers benefit from search engine marketing because ads for merchants appear only when consumers are looking for a specific product. Thus, search engine marketing saves consumers cognitive energy and reduces search costs (including the cost of transportation needed to search for products physically).

AI and Search

Through the years, Google and other search engines have continued to refine their search algorithms using mathematical modeling and AI/ML techniques to make responses to queries even more accurate and intelligent. For instance, rather than evaluate each word in a search separately, as it originally did, Google's current search algorithms use natural language processing to understand the context and meaning of words, phrases, and sentences used in a search query (known as **semantic search**). For instance, if your search is a phrase like *Google annual report selected financial data*, Google should be able to figure out that what you really want is Alphabet Inc.'s (Google's parent company) Form 10-K report filed with the Securities and Exchange Commission.

Semantic search

Use of natural language processing to understand the content and meanings of words, phrases, and sentences used in search queries.

Google searches also take advantage of Knowledge Graph, an effort of the search algorithm to anticipate what you might want to know more about as you search on a topic. Results of the Knowledge Graph appear on the right of the screen on many search result pages and contain more information about the topic or person you are searching for. For example, if you search "Lake Tahoe," the search engine will return basic facts about Lake Tahoe (its altitude, average temperature, and local fish, for example), a map, and hotel accommodations. Google has made **predictive search** part of most search results. This part of the search algorithm guesses what you are looking for and suggests search terms as you enter your search words.

Predictive search

Part of a search algorithm that predicts what a user query is looking for as it is entered.

Visual search involves searching for information using images instead of text-based queries. Visual search uses AI object recognition algorithms to identify elements within an image and comparison and similarity matching algorithms to identify other images that resemble the uploaded image. Examples of visual search include Google Lens, Bing Visual Search, Pinterest Lens, and Amazon StyleSnap. **Voice search** uses natural language processing and speech recognition technology to process voice queries and provide search results. The latest application of AI to search has come in the form of generative AI, which has the power to revolutionize how search is conducted. Read the Chapter Case Study, *AI and the Future of Search* to understand how generative AI may impact the future of search engines. We also cover AI/ML technologies in more depth in Chapter 11.

Visual search

Searching for information using images instead of text-based queries.

Voice search

Using natural language processing and speech recognition technology to process voice queries and provide search results.

Sharing Information on the Web

Today's websites don't just contain static content—they enable people to collaborate, share information, and create new services and content online. Today's web can support interactivity, real-time user control, social participation (sharing), and user-generated content. The technologies and services behind these features include cloud computing, mashups, apps, blogs, RSS, wikis, and social networks. We have already described cloud computing, mashups, and apps in Chapter 5 and introduced social networks in Chapter 2.

A **blog**, the popular term for a weblog, is an informal, yet structured website used by individuals and businesses to publish content online. The blog may include stories, opinions, a *blogroll* (a collection of links to other blogs) and *trackbacks* (a list of entries in other blogs that refer to a post on the first blog). Most blogs allow readers to post comments on the blog entries as well. The act of creating a blog is often referred to as blogging. Most blogs are hosted by a third-party service such as WordPress, Tumblr, Blogger, or TypePad, and blogging features have been incorporated into social networks such as LinkedIn and collaboration platforms such as Slack.

Blog

Popular term for "weblog," designating an informal yet structured website used by individuals and businesses to publish content online.

Blog pages are usually based on templates provided by the blogging service or software. Therefore, millions of people without HTML skills of any kind can post their own web pages and share content with others. Although blogs have become popular personal publishing tools, they also have business uses (see Chapters 2 and 10).

If you're an avid blog reader, you might use RSS to keep up with your favorite blogs without constantly checking them for updates. **RSS**, which stands for Really Simple Syndication or Rich Site Summary, pulls specified content from websites and feeds it automatically to users' computers. RSS reader software gathers material from the websites or blogs that you tell it to scan and brings new information from those sites to you. RSS readers have been incorporated into the major web browsers and email programs.

RSS

Technology using aggregator software to pull content from websites and feed it automatically to subscribers' computers.

Blogs allow visitors to add comments to the original content, but they do not allow visitors to change the original posted material. As you learned in Chapter 2, a wiki, in contrast, is a collaborative website that makes it easy for users to add, delete, or modify content, including the work of previous authors, without having to have any web page development or programming skills. *Wiki* comes from the Hawaiian word for "quick."

Wiki software typically provides a template that defines the layout and elements common to all pages, displays user-editable software program code, and then renders the content into an HTML-based page for display in a web browser. Some wiki software allows only basic text formatting, whereas other software allows the use of tables, images, or even interactive elements such as polls or games. Most wikis provide capabilities for monitoring the work of other users and correcting mistakes.

Because wikis make information sharing so easy, they have many business uses. Many companies use wikis for managing software development, technical documentation, knowledge bases, intranets, extranets, and developer networks. For example, Pixar Wiki is a collaborative, community wiki for publicizing the work of Pixar Animation Studios. The wiki format allows anyone to create or edit an article about a Pixar film.

Social networks enable users to build online communities of friends and professional colleagues. Members typically create a profile—a page for posting photos, videos, audio files, and text—and then share these profiles with others on the service who are identified as their friends or contacts. Social networks are highly interactive, offer real-time user control, rely on user-generated content, and are broadly based on social participation and the sharing of content and opinions. Leading social networks include Facebook, Instagram, TikTok, Pinterest, X (formerly Twitter), and LinkedIn (for professional contacts).

Social networks

Online community that enables users to connect with and expand their social or business contacts.

Social networks have radically changed how people spend their time online; how people communicate and with whom; how businesses stay in touch with their customers, suppliers, and employees; how providers of goods and services learn about their customers; and how advertisers reach potential customers. The large social networks are also application development platforms. Facebook alone has millions of apps and websites integrated into it, including applications for gaming, video sharing, and communicating with friends and family. We talk more about business applications of social networks in Chapters 2 and 10, and you can find discussions about social networks in many other chapters of this book.

The Future Internet/Web Ecosystem

The Internet and web are not static. They are constantly evolving as new technologies are developed. For instance, **Internet2** is an advanced networking consortium focused on the development of next-generation Internet technologies. More than 330 US universities, 50 government agencies, 45 regional and state education networks, 50 leading corporations, and 100 national research and education network partners from more than 100 countries participate in Internet2. To connect these communities, Internet2 developed a high-capacity, 100-Gbps network that serves as a test bed for leading-edge technologies that may eventually migrate to the public Internet, including large-scale network performance measurement and management tools, secure identity and access management tools, and capabilities such as scheduling high-bandwidth, high-performance circuits. Internet2 is currently in the process of deploying a 400-Gbps network on a coast-to-coast basis and is also testing an 800-Gbps network on a pilot basis as a part of its Next Generation Infrastructure program.

Internet2

Advanced networking consortium focused on the development of next-generation Internet technologies.

The future web will be more visible. Its key features are more tools for individuals to make sense of the billions to trillions of pages on the web and the millions of apps available for smartphones and tablets, and a visual, three-dimensional (3D) virtual reality that has been dubbed the **metaverse**. Facebook is so convinced that the metaverse will be at the center of the future web that it has rebranded itself as Meta. At the same time, ideas behind blockchain (see Chapter 6) are driving a different movement that some are calling Web3. **Web3** would be even more decentralized than the current Internet and would be controlled by its users rather than dominated by big corporations (White, 2021; Allyn, 2021).

Metaverse

Anticipated future iteration of the web featuring online 3D or virtually integrated environments that provide users with access to virtual reality and augmented reality experiences.

Web3

A more decentralized Internet based on the ideas behind blockchain that would be controlled by its users rather than dominated by big corporations.

Closer in time is a pervasive Internet/web that controls everything from a city's traffic lights and water usage to the lights in your living room and your car's rearview mirror. This is referred to as the Internet of Things (IoT), which we introduced in Chapter 1, and is based on billions of Internet-connected sensors throughout our physical world. Objects, animals, or people are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.

General Electric, IBM, HP, Oracle, and hundreds of smaller companies are building smart machines, factories, and cities through the extensive use of remote sensors and fast cloud computing. While estimates of the number of IoT devices vary widely, some experts believe that by 2030,

more than 30 billion IoT devices will be connected to the web. Over time, more and more everyday physical objects will be connected to the Internet and will be able to identify themselves to other devices, creating networks that can sense and respond as data change (Vailshery, 2023).

Apps are another element in the future Internet/web ecosystem. The growth of apps within the mobile platform is astounding. More than 90 percent of the time spent on mobile devices in the United States is spent using apps as opposed to using a mobile browser. Apps give users direct access to content and are much faster than loading a browser and searching for content (Emarketer, 2024c).

Other complementary trends leading toward the future Internet/web include more widespread use of cloud computing and software as a service (SaaS) business models, ubiquitous connectivity among mobile platforms and Internet access devices, and the transformation of the web from a network of separate, siloed applications and content into a more seamless and interoperable whole. Generative AI is also likely to play a significant role in shaping the future web by providing new tools that can be used to generate content, automate website design and development, and power chatbots and virtual assistants that enhance user interaction, among other uses. Some analysts worry that generative AI may result in the end of the web as we currently know it, filling up the web with AI-generated content of questionable quality and polluting search results (Mims, 2024).

7.7 Describe the Principal Wireless Technologies and Standards

Welcome to the wireless revolution! Smartphones, tablets, and wireless-enabled laptop computers now allow you to perform many of the computing tasks you used to do at your desk—and a whole lot more. We introduced smartphones in Chapters 1 and 5 as part of our discussion of the mobile platform. A **smartphone**, such as an iPhone or Android phone, combines the functionality of a cell phone with that of a laptop computer that has Wi-Fi capability. This makes it possible to combine music, video, Internet access, and telephone service in one device. The Internet has become a mobile, access-anywhere, broadband service for the delivery of video, audio, and web search.

Smartphone

Mobile device that combines the functionality of a cell phone with that of a Wi-Fi-enabled laptop computer, providing voice, messaging, scheduling, email, and Internet capabilities.

Cellular Systems

Today, more than 85 percent of the US population owns a mobile phone, and almost 80 percent owns a smartphone. Mobile is now the leading digital platform, with total activity on smartphones and tablets accounting for almost 50 percent of time spent per day with digital media (Emarketer, 2024d).

Earlier generations of cellular systems were designed primarily for voice and limited data transmission in the form of short text messages. A **3G network**, which has transmission speeds ranging from 144 Kbps for mobile users to more than 2 Mbps for stationary users, offers transmission speeds appropriate for email and web browsing but are too slow for videos. 3G networks have been phased out by the major telecommunications carriers in the United States. A **4G network** provides much higher speeds, up to 100 Mbps for downloading and 50 Mbps for uploading, with more than enough capacity for watching high-definition videos on your smartphone.

A newer generation of wireless communication technology, called **5G**, is designed to support transmission of much larger amounts of data in the gigabit range—with fewer transmission delays and the ability to connect many more devices (such as sensors and smart devices) at once—than existing cellular systems can support. 5G technology will aid the continued development of self-driving vehicles, smart cities, and the extensive use of the Internet of Things (IoT), as well as improve the speed and intensive data-handling ability of smartphones. T-Mobile, AT&T, and Verizon have all launched 5G networks with varying degrees of functionality. The Spotlight on Technology Case Study, *Cleveland Clinic Turns to Private 5G Network for Its Newest Hospital*, explores the Cleveland Clinic's decision to create a private 5G network for its newest hospital.

3G network

Cellular network that provides transmission speeds from 144 Kbps to 2 Mbps; have been phased out by major telecommunications carriers in the United States.

4G network

Cellular network that provides transmission speeds of up to 100 Mbps speeds for downloading and 50 Mbps for uploading.

5G

Wireless communication technology supporting transmission of very large amounts of data in the gigabit range, with fewer transmission delays and the ability to connect many more devices (such as sensors and smart devices) at once.

Spotlight on Technology Case Study

Cleveland Clinic Turns to Private 5G Network for Its Newest Hospital

Telecommunications companies first began to roll out 5G networks and devices in 2019. Although, as of 2024, 5G has perhaps not been quite as transformative from a consumer perspective as originally touted by its proponents, it is starting to have a major impact on an enterprise level. A growing list of organizations are considering how private 5G networks might be used for their wireless connectivity requirements in the coming years.

The Cleveland Clinic, the renowned healthcare organization based in Ohio, is one such organization. In July 2023, it opened the Cleveland Clinic Mentor Hospital, a community-based hospital designed to serve patients with fewer critical needs and who are expected to have shorter hospital stays and get follow-up care at home. The Mentor Hospital is a compact facility that features 34 inpatient/observation rooms, 23 outpatient rooms, 19 emergency department beds, 12 pre-/post-anesthesia care beds and four operating rooms. The hospital also has another distinguishing feature: it is the first hospital in the United States built from the ground up with a private 5G network, provided by Verizon and Ericsson. According to Matthew Kull, chief information officer for Cleveland Clinic, the new private 5G network will support its long-term vision for a fully digital hospital infrastructure. The hope is that it will enable the hospital to become more efficient, ensure better continuity of care, and enhance the overall experience for both caregivers and patients. In addition, the Cleveland Clinic intends to use the network as a test bed and platform to launch new ideas and new innovations in healthcare technology.

When most people think of 5G, they are thinking of a public 5G network. A public 5G network is usually owned by a mobile network owner who is responsible for the network's service and management. Public 5G networks are intended for use by millions of subscribers on a nationwide basis. A private 5G network is similar to a public 5G network, in that they both use the same underlying network infrastructure and the same wireless spectrums. However, there are some important differences. Unlike a public 5G network, a private 5G network is dedicated to the use of a single enterprise or organization, usually for a single location. A private 5G network allows an organization to have greater control over network configuration, performance, and security, compared to a public 5G network.

There were a number of reasons why the Cleveland Clinic chose to go with a private 5G network instead of a traditional Wi-Fi network. Private 5G networks can provide lower latency (delay in transmission time), higher data rates, and improved quality and service compared to a Wi-Fi network. Wi-Fi networks, although capable of delivering high-speed internet access, face challenges in handling large numbers of devices. Typically, a hospital's Wi-Fi network is accessible by both patients and providers, so patient care may compete with visitors' devices for network speed. Private 5G can support thousands of devices, making the network more scalable.

Private 5G networks can also provide better coverage over larger areas compared to Wi-Fi, as 5G signals can travel longer distances. Wi-Fi networks, in contrast, generally have a shorter range and may require a multitude of access points to cover large or dense areas, which can increase complexity, risk of interference, and cost. Users moving between access points may experience slow service.

Private 5G and Wi-Fi both offer robust security features, but telecommunications providers assert that 5G has an edge there as well due to its cellular architecture, SIM-based authentication, and security protocols. Modern Wi-Fi networks also employ advanced security measures but can potentially be more vulnerable to attack.

Ability to scale was an additional consideration. Rather than maintaining its own physical Wi-Fi infrastructure, a private 5G network enables the hospital to outsource connectivity to a major carrier and be able to scale up and down quickly based on demand. According to Kull, speed and scalability of a network are going to become more imperative with new technologies that are emerging.

5G-enabled features that the hospital plans to implement include 5G-augmented patient-check-in kiosks, digital monitoring, and a digital display for patients that will provide up-to-date information about care, hospital amenities, and entertainment. Although initially the network will not be used for medical devices, the goal is for the hospital to be running more advanced 5G-enabled technologies, such as augmented- and virtual-reality clinician training, within a year or two. The hospital also hopes to use the network to support the transmission of medical imaging and test results.

The Cleveland Clinic is also embracing the use of the new 5G network as a testbed. Several manufacturers have already expressed interest in partnering to explore the use of their equipment over the network. As noted by Cleveland Clinic Chief Technology Officer Shibu Thomas, providers have been waiting for device manufacturers and others to adopt 5G, and device manufacturers have been waiting for

opportunities to be supported in doing so. The Mentor Hospital is now in a position to provide that opportunity in a real-world environment.

SOURCES: Isabelle Bouquette, "Hospitals CIOs Weigh 5G as Digitized Medicine Pushes Wi-Fi to Its Limits," Wall Street Journal, February 7, 2024; Sarah Sloat, "The Famed Cleveland Clinic Just Opened a 'Smart Hospital' Where 5G May Eventually Power Everything from Check-ins to Surgery," Businessinsider.com, August 3, 2023; Giles Bruce, "Why Cleveland Clinic Went with Private 5G over Wi-Fi for Its Newest Hospital," Beckershospitalreview.com, July 31, 2023; Andrea Fox, "Cleveland Clinic's New Mentor Hospital Is a 5G mmWave Test Bed," Healthcareitnew.com, July 24, 2023; Tina Reed, "Coming to a Hospital Near You: 5G," Axios.com, May 30, 2023.

Case Study Questions

1. How does a private 5G network differ from a public 5G network?
2. Why did the Cleveland Clinic choose to create a private 5G network for Mentor Hospital rather than a traditional Wi-Fi network?
3. In what ways will the implementation of a private 5G network advance Cleveland Clinic's long-term vision for a fully digital hospital infrastructure?

The newest generation of cellular network standards is 6G. 6G networks will provide substantially higher transmission capacity. Although it is too early to estimate approximate 6G data rates, researchers have suggested that a theoretical peak data transmission rate of 1 terabyte per second may be possible. 6G will also support much lower latency (delay in transmission time), with the goal to support one microsecond latency communications. 6G research and development activities started in 2020, but industry specifications remain some years away, and the first 6G-enabled products are not likely to appear before 2030.

Wireless Computer Networks and Internet Access

An array of technologies provides high-speed wireless access to the Internet for desktop computers and mobile devices. These high-speed services have extended Internet access to numerous locations that could not be covered by traditional wired Internet services and have made anywhere, anytime computing a reality.

Bluetooth

Bluetooth

Standard for wireless personal area networks that can transmit up to 722 Kbps within a 10-meter area.

Personal area network (PAN)

Computer network used for communication among digital devices that are located within close proximity to one another.

Bluetooth is the popular name for the 802.15 wireless networking standard that is useful for creating a small **personal area network (PAN)**, a computer network used for communication among digital devices within close proximity to one another. A Bluetooth-enabled device can link up to seven other Bluetooth devices within a 10-meter area using low-power, radio-based communication and can transmit up to 722 Kbps in the 2.4-GHz band. The Bluetooth Low Energy (BLE) specification reduces the energy usage of Bluetooth peripherals. Bluetooth 5 extends the ability to use BLE to a variety of devices, such as wireless headphones, and increases the range over which these devices can communicate to up to 240 meters, as well as data transfer speeds up to 2 Mbps.

Wireless phones, pagers, computers, printers, and computing devices using Bluetooth communicate with each other and even operate each other without direct user intervention. Examine Figure 7.12, which illustrates how a person can use a smartphone or a computer's wireless keyboard and mouse to send a document file wirelessly to a printer.

Although Bluetooth lends itself to personal networking, it also has uses in large corporations. For example, FedEx drivers use Bluetooth to transmit the delivery data captured by their handheld computers to cellular transmitters, which forward the data to corporate computers. Drivers no longer need to spend time physically docking their handheld units into the transmitters, and Bluetooth has saved FedEx $20 million per year.

Wi-Fi and Wireless Internet Access

Wi-Fi

Refers to the 802.11 set of standards for wireless LANs and wireless Internet access standards.

The 802.11 set of standards for wireless LANs and wireless Internet access is also known as **Wi-Fi**. The first of these standards to be widely adopted was 802.11b, which can transmit up to 11 Mbps in the 2.4-GHz band and has an effective distance of 30 to 50 meters. The most current 802.11 standards are 802.11ac (Wi-Fi 5) and 802.11ax (Wi-Fi 6). Wi-Fi 5 provides for throughputs of 500 Mbps to 1 Gbps, while Wi-Fi 6 can operate over broader ranges of frequencies and achieve theoretical throughputs of up to 10 Gbps. Today's PCs and tablets have built-in support for Wi-Fi, as do the iPhone and other smartphones.

Figure 7.12 A Bluetooth Network (PAN)

Bluetooth enables a variety of devices, including smartphones, wireless keyboards and mice, PCs, and printers, to interact wirelessly with each other within a small, 30-foot (10-meter) area. In addition to the links shown, Bluetooth can be used to network similar devices to, for example, send data from one PC to another.

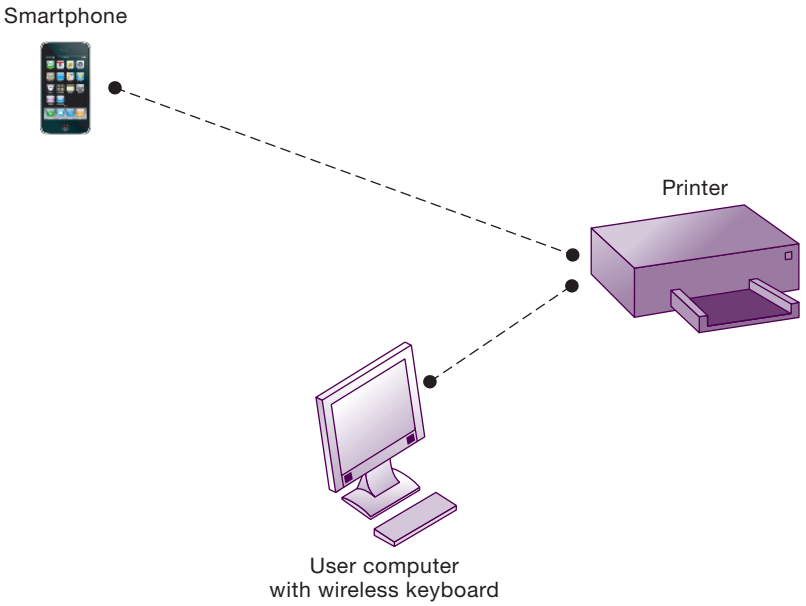
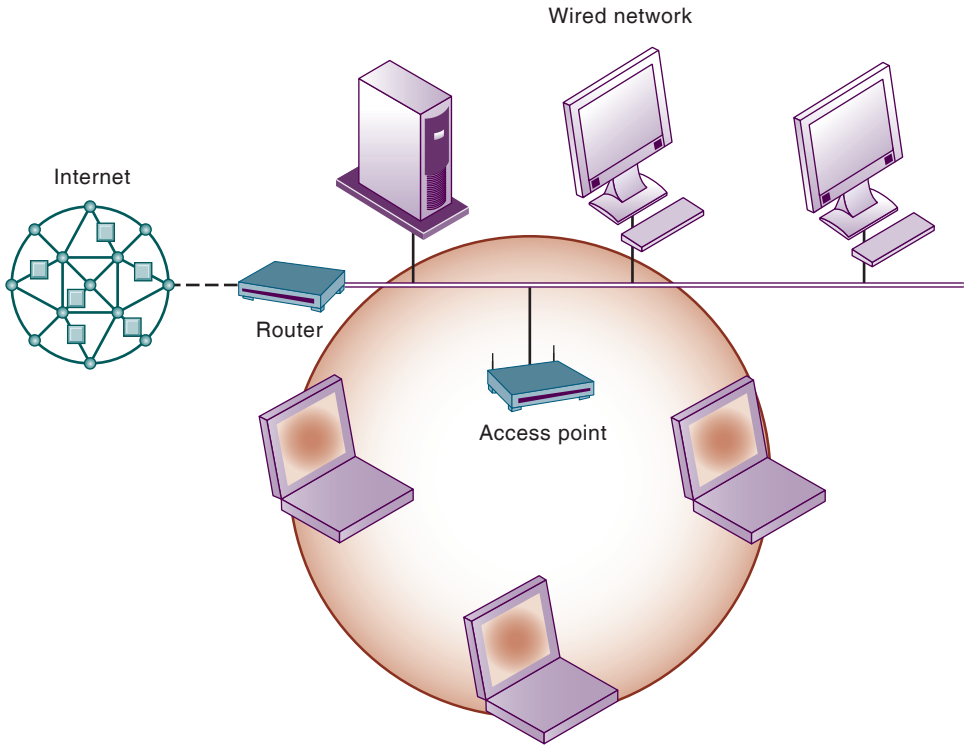


Figure 7.13 An 802.11 Wireless LAN

In a Wi-Fi network, mobile wireless devices such as laptop computers communicate with a wired LAN and the Internet via an access point. The access point uses radio waves to communicate with both the mobile devices and the wired network.



In most Wi-Fi communication, wireless devices communicate with a wired LAN using access points. An access point is a box consisting of a radio receiver/transmitter and antennas that link to a wired network, router, or hub.

Examine Figure 7.13, which illustrates an 802.11 (Wi-Fi) wireless LAN that connects laptop computers equipped with network interface adapters to a wired LAN and to the Internet. Most wireless devices in a Wi-Fi LAN are clients. The servers that the mobile client stations

need to use are on the wired LAN. The access point uses radio waves to transmit network signals from the wired network to the client adapters, which convert the signals to data that the mobile device can understand. The client adapter then transmits the data from the mobile device back to the access point, which forwards the data to the wired network.

The most popular use for Wi-Fi today is high-speed wireless Internet service. In this instance, the access point plugs into an Internet connection, which could come from a cable service, DSL or FiOS, or satellite service. Computers within range of the access point use it to link wirelessly to the Internet.

Hotspot

Location with one or more access points that provide public Wi-Fi Internet access.

A **hotspot** is a location with one or more access points providing Wi-Fi Internet access. Some hotspots are free or do not require any additional software to use; others may require activation and the establishment of a user account by providing a credit card number over the web.

Businesses of all sizes are using Wi-Fi networks to provide low-cost wireless LANs and Internet access. Wi-Fi hotspots can be found in hotels, airport lounges, libraries, cafés, and college campuses. Dartmouth College is one of many campuses where students now use Wi-Fi for research, course work, and entertainment.

Wi-Fi technology poses several challenges, however. One is Wi-Fi's weak security, which makes these wireless networks vulnerable to intruders. We provide more detail about Wi-Fi security issues in Chapter 8.

Another drawback of Wi-Fi networks is their susceptibility to interference from nearby systems operating in the same spectrum, such as microwave ovens or other wireless LANs. However, wireless networks based on the 802.11n standard solve this problem by using multiple wireless antennas in tandem to transmit and receive data and also using technology called MIMO (multiple-input multiple-output) to coordinate multiple simultaneous radio signals.

WiMax

The range of Wi-Fi systems is no more than 300 feet from the base station. To deal with this constraint, the Institute of Electrical and Electronics Engineers (IEEE) developed a family of standards known as WiMax. **WiMax**, which stands for Worldwide Interoperability for Microwave Access, is the popular term for IEEE standard 802.16. It has a wireless access range of up to 31 miles and a transmission speed of 75 Mbps (and up to 1 Gbps for fixed stations). WiMax antennas are powerful enough to beam high-speed Internet connections to rooftop antennas of homes and businesses that are miles away.

WiMax

Popular term for IEEE Standard 802.16 for wireless networking over a range of up to 31 miles with a data transfer rate of up to 75 Mbps. Stands for Worldwide Interoperability for Microwave Access.

RFID and Wireless Sensor Networks

Mobile technologies are creating new efficiencies and ways of working throughout enterprises. In addition to the wireless systems we have just described, radio frequency identification (RFID) systems and wireless sensor networks are having major impacts.

Radio Frequency Identification (RFID) and Near Field Communication (NFC)

Radio frequency identification (RFID) systems provide technology for tracking the movement of goods throughout the supply chain. To transmit radio signals over a short distance to RFID readers, RFID systems use tiny tags with embedded microchips containing data about an item and its location. The RFID readers then pass the data over a network to a computer for processing. Unlike bar codes, RFID tags do not need line-of-sight contact to be read.

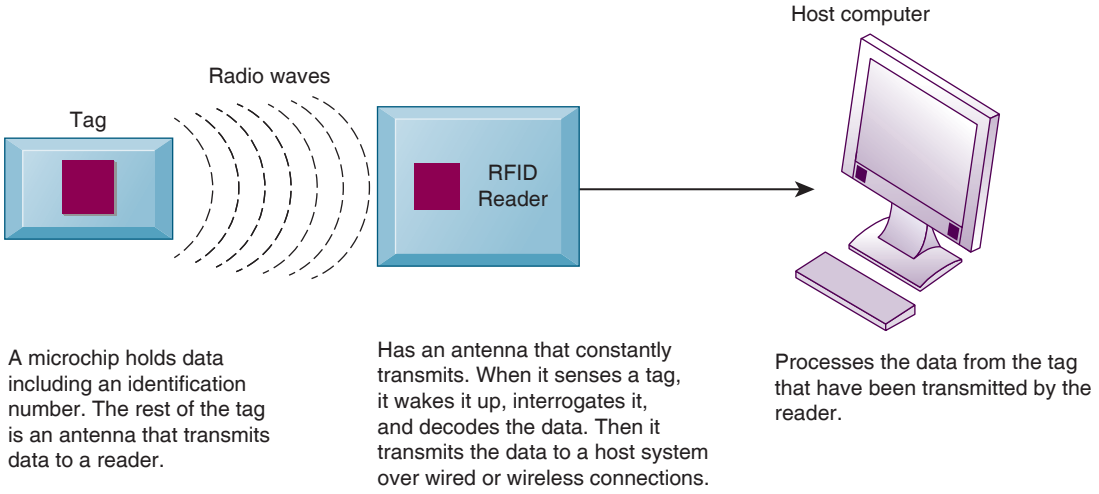
Radio frequency identification

(RFID) Technology using tiny tags with embedded microchips containing data about an item and its location; the tags transmit short-distance radio signals to special RFID readers that then pass the data on to a computer for processing.

Examine Figure 7.14, which illustrates how RFID works. The RFID tag is programmed with information that can uniquely identify an item, plus other information about the item such as its location, where and when it was made, or its status during production. An RFID reader also includes an antenna that constantly transmits radio waves in ranges anywhere from 1 inch to 100 feet. When an RFID tag comes within the range of the reader, the reader activates the tag, which starts sending data. The reader captures these data, decodes them, and sends them back over a wired or wireless network to a host computer for further processing. Both RFID tags and antennas come in a variety of shapes and sizes.

Figure 7.14 How RFID Works

RFID uses low-powered radio transmitters to read data stored in a tag at distances ranging from 1 inch to 100 feet. The reader captures the data from the tag and sends them over a network to a host computer for processing.



In inventory control and supply chain management, RFID systems capture and manage more detailed information about items in warehouses or in production than bar coding systems. If a large number of items are shipped together, RFID systems can track each pallet, lot, or even unit in the shipment. This technology helps companies improve receiving and storage operations by improving their ability to see exactly what stock is stored in warehouses and on retail store shelves.

For instance, Walmart has installed RFID readers at receiving docks at its stores to record the arrival of pallets and cases of goods shipped with RFID tags. The RFID reader reads the tags a second time just as the cases are brought onto the sales floor from backroom storage areas. Software then combines sales data from Walmart's point-of-sale systems and the RFID data regarding the number of cases brought out to the sales floor. The software program determines which items will soon be depleted and automatically generates a list of items to obtain from the warehouse in order to replenish store shelves before they are empty. This information helps Walmart reduce out-of-stock items, increase sales, and further shrink its costs.

The cost of RFID tags used to be too high for widespread use, but now the cost starts at around 7 cents per tag in the United States. As the price decreases, RFID is starting to become cost-effective for many applications.

In addition to installing RFID readers and tagging systems, companies may need to upgrade their hardware and software to process the massive amounts of data produced by RFID systems—data that could add up to tens or hundreds of terabytes.

Software is used to filter, aggregate, and prevent RFID data from overloading business networks and system applications. Applications often need to be redesigned to accept large volumes of frequently generated RFID data and to share those data with other applications. Major enterprise software vendors now offer RFID-ready versions of their supply chain management applications.

Tap-and-go services like Apple Pay and Google Wallet use an RFID-related technology called **near field communication (NFC)**. NFC is a short-range wireless connectivity standard that uses electromagnetic radio fields to enable two compatible devices to exchange data when brought within a few centimeters of each other. A smartphone or other NFC-compatible device sends out radio frequency signals that interact with an NFC tag found in compatible card readers or smart posters. The signals then create a current that flows through the NFC tag, allowing the device and the tag to communicate with one another. In most cases, the tag is passive and can only send out information, whereas the other device (such as a smartphone) is active and can both send and receive information. (There are NFC systems in which both components are active.)

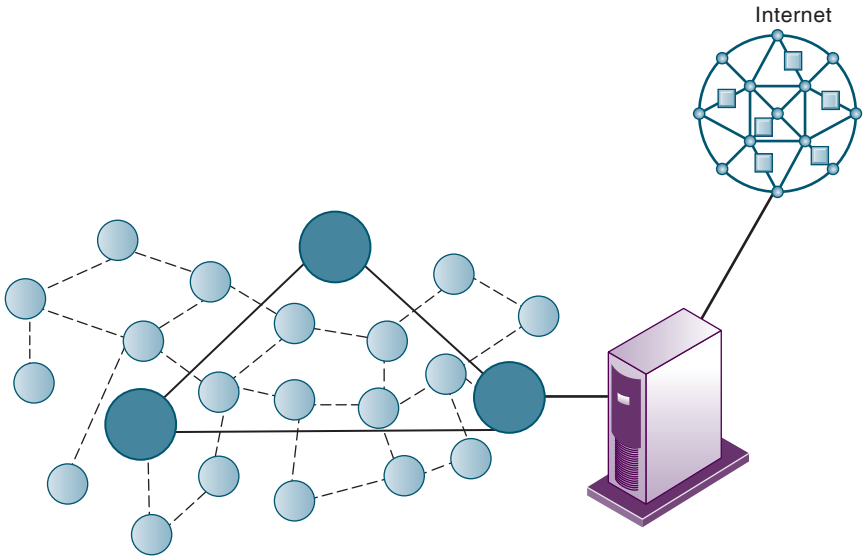
Near field communication (NFC)

Short-range wireless connectivity standard that uses electromagnetic radio fields to enable two compatible devices to exchange data when brought within a few centimeters of each other.

NFC is used for wireless payment services, for retrieving information, and even for on-the-go exchanging of videos or information with friends. You could share a website link by passing your phone over a friend's phone and then waving your phone in front of a poster or display containing an NFC tag. You could also share information about what you're viewing at a museum or exhibit.

Figure 7.15 A Wireless Sensor Network

Wireless sensor networks can contain hundreds to thousands of interconnected wireless sensors (nodes) that route the data they capture to a computer for analysis.



Wireless Sensor Networks

Wireless sensor network (WSN)

Network of interconnected wireless devices with built-in processing, storage, and radio frequency sensors and antennas that are embedded into the physical environment to provide measurements of many points over large spaces.

If your company wanted state-of-the-art technology to monitor building security or detect hazardous substances in the air, it might deploy a wireless sensor network. A **wireless sensor network (WSN)** is a network of interconnected wireless devices that are embedded in the physical environment to provide measurements of many points over large spaces. These devices have built-in processing, storage, and radio frequency sensors and antennas. They are linked into an interconnected network that routes the data that the devices capture to a computer for analysis. These networks range from hundreds to thousands of nodes. Examine Figure 7.15, which illustrates one type of wireless sensor network. The small circles in Figure 7.15 represent lower-level nodes, and the larger circles represent higher-level nodes. Lower-level nodes forward data to each other or to higher-level nodes, which transmit data to a server with greater processing power and which can also act as a gateway to a network based on Internet technology. Using only the higher-level nodes to transmit data to the server enables the data to be transmitted more rapidly and speeds up network performance.

Wireless sensor networks are valuable for uses such as monitoring environmental changes; monitoring traffic or military activity; protecting property; efficiently operating and managing machinery and vehicles; establishing security perimeters; monitoring supply chain management; and detecting chemical, biological, or radiological material.

Output from RFID systems and wireless networks is fueling the Internet of Things (IoT) (introduced in Chapter 1), in which machines such as jet engines, power plant turbines, and agricultural sensors constantly gather data and send the data over the Internet for analysis. The NFL's Next Gen Stats system, described in the chapter-opening case, is an example of an IoT application. You'll find more examples of the Internet of Things in Chapters 2 and 11.

7.8 Understand How the Information in This Chapter Can Help Your Career

The information you have learned in Chapter 7 can help you advance your career by helping you to understand the basics of today's networks, including telecommunications technology, the Internet, and wireless technology.

Real-World Scenario

Network Specialists Help Dundee Precious Metals Underground

Dundee Precious Metals is a Canadian-based, international gold mining company with operations and projects in Bulgaria, Serbia, Namibia, and Ecuador. Operations in underground mines require intensive oversight and coordination to make sure that people, materials, and equipment are available underground when and where they are needed, and that work is flowing smoothly. Tracking staff and operations manually or using older radio identification technology was extremely slow, cumbersome, and prone to error. Network specialists helped Dundee deploy the solution to this problem: underground Wi-Fi wireless networking, radio frequency identification (RFID) tagging, and AeroScout MobileView software to automate tracking of workers, equipment, and ore as they move through underground mines. The wireless network made it much easier to monitor and supervise mining activities from control rooms above ground and to work underground in deeper locations. As a result, Dundee has been able to reduce costs and increase productivity while improving the safety of its workers. Communication costs have dropped 20 percent. Dundee management believes that its ability to access real-time data will increase the company's profit per miner by 10 to 15 percent.

Job Description and Career Path: Network Specialist

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Network specialist	Job responsibilities include: • Determining company needs and coordinating the development and maintenance of network infrastructures with the IT team. • Ensuring secure and stable server connectivity and test network protocols. • Creating internet domains and optimizing intranet performance. • Testing data exchange and communication between computers, routers, modems, and servers. • Installing, updating, and configuring end-user networking accessories. • Monitoring server capacity and performance to keep up with demand in online traffic. • Running diagnostic tests and perform repairs, as well as developing backup, archiving, and data retrieval procedures. • Keeping informed of developments in network technologies and infrastructure.	• Bachelor's degree in computer science, information science, or similar • In-depth knowledge about network connectivity, technologies, and security • CompTIA, Microsoft, or Cisco network professional certifications	Entry-level: $49,560–$80,600 Mid-level: $62,970–$102,720 Senior: $80,600–$130,830 Salaries can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

AI and the Future of Search

Online search is one of the most central and important aspects of the web. Search enables people to find information they are seeking and drives the purchases of products and services, both online and offline. Search drives the digital advertising ecosystem, which is the engine that pays for so much of the content and services that currently are available online for free. Google has dominated the search landscape since the early days of the web. But that may all be starting to change, thanks to the launch of ChatGPT and other forms of generative AI. The search experience may become very different in the not-so-distant future.

As discussed in the chapter, Google has been using artificial intelligence tools and technologies for a number of years in an effort to make search more effective. Search evolved incrementally, with Google continually introducing new and improved search algorithms, often numerous times a year, to deliver relevant search results more quickly. Then in November 2022, OpenAI released its generative AI chatbot, ChatGPT, which quickly took the world by storm. Since then, the "genAI" floodgates have opened, and search (and by extension, the web itself) may never be the same.

Proponents of genAI search envision a new world where search is no longer dominated by search engines that return a listing of links that users need to wade through to find the information they are looking for. Instead, users will post a natural language question to a gen AI search engine and receive a direct answer. They can ask follow-up questions and engage in a "conversation" with the search engine. This is the model of search currently embodied by ChatGPT and Microsoft Copilot, but they are not yet full-bodied search engines.

Other companies are also jumping into the fray. OpenAI, in addition to providing AI capabilities to Microsoft Copilot, is reportedly building its own search product. Perplexity AI, backed by Jeff Bezos, is a question-based search engine that delivers direct conversational answers. Perplexity's search tool answers user queries with a detailed summary and links to the source of the information. Users are able to ask follow-up questions. The company offers both a free and subscription version that uses GPT-4, OpenAI's most advanced AI model as of March 2024. Perplexity AI has a current valuation of about $1 billion as of March 2024, a long way from Google's more than $1.7 trillion, but analysts believe it may pose a legitimate threat to Google at some point in the future.

Startup companies that have launched innovative AI-based search products include Arc Search, Consensus, and You.com. Arc Search is a free AI-powered search engine and web browser combined that identifies the best half-dozen websites with information on the topic of a query and then uses AI to "read" and summarize them. Consensus is a search startup that uses large language models to provide detailed summaries of research papers and insights about the scientific consensus on various topics. The goal of Consensus is to slice off chunks of Google's search market one piece at a time. You.com is another search engine startup that bills itself as a personal AI search assistant (YouChat).

Despite the excitement surrounding genAI search, the prospects of a search revolution based on genAI raise a number of significant issues, some of which emanate from the differences between "traditional" search and genAI search. For starters, the modus operandi of a traditional search engine is not to generate new information, but merely to report on information found. The primary goal of traditional search is to assist users in finding relevant information by guiding them to pertinent web pages. Search based on generative AI, on the other hand, is inherently very different. It is generative, i.e., "creative," rather than reportorial. It synthesizes information but does so in a way that is not necessarily apparent to the user. A traditional search engine enables the user to examine the reliability of the links it returns. In contrast, genAI's synthesis of information entails selection of certain data over others, a process that a user does not have the ability to evaluate, and which potentially could lead to wider dissemination of disinformation.

AI hallucinations are another problem. A traditional search engine typically returns a "no match found" search response to a query without relevant results. GenAI search engines, which are based on models that have been trained to predict what is the next most likely word in response to input, are more likely to guess an answer, creating what is known as an AI hallucination, a fabricated response. Currently, output from genAI search engines may on the surface appear credible to users, but cannot be fully trusted without the extra step of verification, robbing the process of any gains in efficiency and speed that it might offer. On a broader, societal level, there is the concern that users may become overly reliant on genAI search and lose the ability to critically evaluate information on their own.

There is also the issue of algorithmic bias. The data on which the models are trained, primarily from the web, are filled with biases and stereotypes, which are then learned and inadvertently incorporated into the language models.

GenAI search also has the potential to have a major impact on the overall web ecosystem. Research is already suggesting that genAI search will eliminate the need for people to click through to the websites used to produce the answer to the query. Without traffic, the business model for many publishers could collapse. For example, about 40 percent of *The Atlantic* magazine's web traffic comes from Google searches. A task force at *the Atlantic* modeled what could happen if Google integrated genAI into search and found that 75 percent of the time, AI-powered search would likely provide a full answer to a user's query and *the Atlantic*'s site would miss out on traffic it otherwise would have gotten. Publishers estimate that they may lose between 20 percent and 40 percent of their Google-generated traffic as a result. Making matters worse is the fact that many publishers claim that large language models have been trained on their content without compensation and in violation

of copyright laws, an allegation currently being tested in a number of lawsuits.

In May 2024, Google launched its new AI search tool, "AI Overviews," based on its Gemini LLM, in the United States, with plans to roll it out in more countries by the end of the year. AI Overviews provides brief AI-generated summaries highlighting key information. However, as with Google Bard, Google's initial generative AI offering, the launch of AI Overviews has been somewhat rocky. Users have noticed that AI Overviews may provide inaccurate or misleading answers to search queries. According to Google, this occurs when there is not a lot of high-quality information available on which to base a response, or because the response is drawn from posts that involve jokes or satire. Inaccurate or misleading answers can also misinterpret the sources on which it bases its answer. Although Google is continuing to tweak its algorithms to improve the quality of its responses, researchers believe that AI hallucinations are always going to be a risk.

SOURCES: Nico Grant, "Google Rolls Back A.I. Search Feature after Flubs and Flaws," *New York Times*, June 1, 2024; Rhiannon Williams, "Why Google's AI Overviews Gets Things Wrong," Technologyreview.com, May 31, 2024; Alex Heath, "Perplexity is Ready to Take on Google," Theverge.com, March 15, 2024; Berber Jin and Miles Kruppa, "AI Search Startup Perplexity Set to Double Valuation to $1 Billion," *Wall Street Journal*, March 5, 2024; Shahan Ali Memon and Jevin D. West, "Search Engines Post-ChatGPT: How Generative Artificial Intelligence Could Make Search Less Reliable," Arvix.org, February 18, 2024; Christopher Mims, "It's the End of the Web as We Know It," *Wall Street Journal*, February 16, 2024; Forrester, "Podcast: The Future of Search in a GenAI World," Forrester.com, January 11, 2024; Julia McCoy, "How AI Will Affect the Future of Search," Searchengineland.com, January 8, 2024; Danny Goodwin, "SEO Year in Review 2023: The Year of Generative AI," Searchengineland.com, December 27, 2023; Keach Hagey, Miles Kruppa, and Alexandra Bruell, "News Publishers See Google's AI Search Tool as a Traffic Destroying Nightmare," *Wall Street Journal*, December 14, 2023.

Case Study Questions

1. How does genAI search differ from traditional search?
2. What issues does genAI search raise for the future of search?
3. How might genAI search impact the web ecosystem as it currently exists?

Review Summary

7.1 Describe key components of telecommunications networks.

A simple network consists of two or more connected computers. Basic network components include computers, network interface controllers (NICs), a connection medium, network operating system software, either a hub or a switch, and a router. The networking infrastructure for a large company typically includes an array of local and wide area networks, including the Internet, mobile cellular communication, and wireless local area networks. The networking infrastructure supports applications such as a corporate website, a corporate intranet, and perhaps a corporate extranet. The firm's infrastructure may also include a separate telephone network that handles voice data.

7.2 Describe key digital networking technologies.

Contemporary digital networks and the Internet have been shaped by the rise of client/server computing, the use of packet switching, and the adoption of Transmission Control Protocol/Internet Protocol (TCP/IP) as a universal communications standard for linking disparate computers and networks, including the Internet. Protocols provide a common set of rules that enable communication among diverse components in a telecommunications network.

7.3 Describe the different types of networks.

There are many kinds of networks and ways of classifying them. One way of looking at networks is in terms of their geographic scope. Local area networks (LANs) connect desktop computers and other digital devices within a 500-meter radius and are used today for many corporate computing tasks. Wide area networks (WANs) span broad geographical distances, ranging from several miles to entire continents, and are often private networks that are independently managed. A campus area network (CAN) is a computer network that spans a limited geographic area. Metropolitan area networks (MANs) span a metropolitan urban area.

Another way to categorize networks is in terms of transmission media and bandwidth. The principal physical transmission media are twisted copper wire, coaxial cable, fiber-optic cable, and wireless transmission media. Each type of transmission media has different properties, advantages, and limitations. Bandwidth measures how much data can be transferred over a communications medium within a fixed period of time and typically is expressed as bits per second (Bps), kilobits per second (Kbps), megabits per second (Mbps), gigabits per second (Gbps), and, for very advanced communications media, terabits per second (Tbps).

7.4 Understand what the Internet is and how it works.

The Internet is a worldwide network of networks that uses the client/server model of computing and the TCP/IP network reference model. Every computer on the Internet is assigned a unique numeric IP address. The Domain Name System (DNS) converts IP addresses to more user-friendly domain names. Internet data traffic is carried over transcontinental, high-speed backbone networks. The Internet backbone connects to regional networks, which in turn provide access to Internet service providers, large firms, and government institutions. Digital subscriber line (DSL) and FiOS technologies, cable Internet connections, satellite Internet (including low earth orbit satellites), and T1 and T3 lines are often used for high-capacity Internet connections. Worldwide Internet policies are established by

organizations such as the Internet Architecture Board (IAB), the Internet Corporation for Assigned Names and Numbers (ICANN), and the World Wide Web Consortium (W3C).

7.5 Explain how Internet services support business.

Major Internet services include email, instant messaging, file transfer, voice transmission (VoIP), and the web. Firms are also using unified communications systems and virtual private networks (VPNs) as low-cost alternatives to private WANs.

7.6 Discuss how the web works and how it can be used in business.

The web is the most popular Internet service. Web pages are based on Hypertext Markup Language (HTML) and can display text, graphics, video, and audio. A website is a collection of web pages. Search engines help users locate the information they need on the web. Search engines have also become major advertising and shopping tools. Blogs, RSS, wikis, and social networks are examples of the information-sharing capabilities of the web. The future Internet/web ecosystem will feature higher-speed networks; visual, 3D virtual reality (the metaverse); the prevalence of apps; and the interconnectedness of many different devices (Internet of Things). The Web3 movement envisions a more decentralized web, controlled by its users.

7.7 Describe the principal wireless technologies and standards.

Cellular networks are evolving toward high-speed, high-bandwidth, digital packet-switched transmission. 4G networks are capable of transmission speeds of 100 Mbps, and 5G networks are capable of transmitting data in the gigabit range among many devices.

Standards for wireless computer networks include Bluetooth (802.15) for small personal area networks (PANs), Wi-Fi (802.11) for local area networks (LANs), and WiMax (802.16) for metropolitan area networks (MANs).

Radio frequency identification (RFID) systems provide technology for tracking the movements of items by using tiny tags with embedded data about an item and its location. RFID readers read the radio signals transmitted by these tags and pass the data over a network to a computer for processing. Near field communication (NFC) is a short-range wireless connectivity standard that uses electromagnetic radio fields to enable two compatible devices to exchange data when brought within a few centimeters of each other. Wireless sensor networks (WSNs) are networks of interconnected, wireless sensing and transmitting devices that are embedded in the physical environment to provide measurements of many points over large areas.

Key Terms

3G network, 240
4G network, 240
5G, 240
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Broadband, 221
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Campus area network (CAN), 227
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Wireless sensor network (WSN), 246

Review Questions

7.1 Describe key components of telecommunications networks.

- Describe the features of a simple network and the network infrastructure for a large company.

7.2 Describe key digital networking technologies.

- Define an analog signal and a digital signal.
- Name and describe the principal technologies and trends that have shaped contemporary telecommunications systems.

7.3 Describe the different types of networks.

- Distinguish among a LAN, a CAN, a MAN, and a WAN.
- Identify the principal physical transmission media and describe their basic characteristics.

7.4 Understand what the Internet is and how it works.

- Define the Internet and describe how it works.
- Explain how the Domain Name System (DNS) and the IP addressing system work.

7.5 Explain how Internet services support business.

- List and describe the principal Internet services.
- Define and describe VoIP explain how it provides value to businesses.
- Define and describe a virtual private network and explain how it provides value to business.

7.6 Discuss how the web works and how it can be used in business.

- List and describe alternative ways of locating information on the web.
- Describe how online search technologies are used for marketing.
- Identify the major ways information is shared on the web.
- Discuss the features of the future Internet/web ecosystem.

7.7 Describe the principal wireless technologies and standards.

- Define 3G, 4G, and 5G networks.
- Describe the various technologies that can be used for wireless computer networks and Internet access.
- Define RFID, explain how it works, and describe how it provides value to businesses.
- Define near field communication (NFC) and explain how it works.
- Define WSN, explain how they work, and describe the kinds of applications that use them.

Discussion Questions

7.8 It has been said that smartphones have become the single-most-important digital device we own. Discuss the implications of this statement.

7.9 Should all major retailing and manufacturing companies switch to RFID? Why or why not?

7.10 What are some of the issues to consider in determining how the Internet can provide your business with a competitive advantage?

Using AI Project

7.11 AI Project: Researching ISPs

You have been hired as a consultant by two friends who want to open a retail store selling organic pet food and treats, as well as pet toys and clothing made from organic materials. They have asked you to advise them on Internet service provider (ISP) alternatives for Internet service in the area where they wish to locate the bakery (zip code 53716). They would also like to see a PowerPoint presentation comparing two of the ISPs that were identified.

Using Microsoft's Copilot, Google's Gemini, or ChatGPT, develop a series of prompts to assist you in responding to their request. Save a copy of your prompts and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience in evaluating and selecting communications technology, using spreadsheet software to improve selection of telecommunications services, and using web search engines for business research.

7.12 Management Decision Problem

San Juan, the capital of Puerto Rico, includes the Old San Juan historic district, parts of which date back 500 years. The city was unable to provide broadband connectivity to tourists and to residents of this special district. Digging restrictions limited the amount of fiber-optic cable that could be laid down, and Old San Juan's network infrastructure was very outdated. However, the city was able to solve this problem by implementing a broadband Wi-Fi network that included technology supplied by Facebook Terragraph. Visit Terragraph.com to learn more about Facebook Terragraph and explain why it provided a good solution for

Old San Juan. Additionally, describe the management, organization, and technology issues that the city had to address when selecting a solution.

7.13 Using Spreadsheet Software to Evaluate Wireless Services

Software skills: Spreadsheet formulas, formatting

Business skills: Analyzing wireless services and costs

In this project, you'll use the web to research alternative wireless services and use spreadsheet software to calculate wireless service plan and smartphone costs.

You would like to provide your sales force of 20, based in St. Paul, Minnesota, with smartphones that have wireless service plans that include unlimited talk, text, and data. You would like to equip each sales force member with a new iPhone 15 (with 128 gigabytes of storage). Use the web to research the smartphone and service plans offered by Verizon, AT&T, and T-Mobile. Use your spreadsheet software to determine the wireless service plan and smartphone plan (purchase outright or pay monthly installments) that will offer the best pricing per user over a two-year period. For the purposes of this exercise, you do not need to consider corporate discounts.

7.14 Using Online Search Tools for Business Research

Software skills: Online search tools

Business skills: Researching new technologies

This project will help develop your online search skills for business research.

Use Google and Bing to obtain information about biofuels as alternative fuels for motor vehicles. If you wish, try some other search engines or generative AI chatbots as well. Compare the volume and quality of information you find using each search tool. Which tool is the easiest to use? Which produced the best results for your research? Why?

7.15 Collaboration and Teamwork Project: Evaluating Smartphones

Form a group with two or more of your classmates. Choose an Apple iPhone model that is currently available and compare its capabilities to the capabilities of a similar model smartphone from another vendor. Your analysis should consider the purchase cost of each device, the wireless networks where each device can operate, the plan costs, and the services available for each device. You should also consider other capabilities of each device, including available software, security features, and the ability to integrate the device with existing corporate or PC applications. Which device would you select? On what criteria would you base your selection? If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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8

Securing Information Systems

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Learning Objectives

After completing this chapter, you will be able to:

8.1 Understand why information systems are vulnerable.

8.2 Describe the most prevalent and dangerous types of malware.

8.3 Understand the dangers posed by threat actors.

8.4 Describe the security issues raised by software vulnerabilities.

8.5 Describe the security issues raised by artificial intelligence.

8.6 Describe the business value of security and control.

8.7 Describe an organizational framework for security and control.

8.8 Discuss technologies for safeguarding information resources.

8.9 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** Ransomware Attack Paralyzes U.S. Healthcare Payment Systems
- **Spotlight on Organizations Case Study:** Uptime Institute: Understanding and Preventing System Outages
- **Chapter Case Study:** Social Engineering Creates Chaos for MGM Resorts

Opening Case Study

Ransomware Attack Paralyzes U.S. Healthcare Payment Systems

In 2013, a "new" form of malware began to attract public attention. Ransomware blocks or limits access to a computer or network by encrypting files and then demands a ransom payment, typically in a cryptocurrency such as Bitcoin, in exchange for the decryption key. Ransomware has become a significant threat, and in 2024, a very high-profile ransomware attack paralyzed U.S. healthcare payment systems.

Change Healthcare is a healthcare technology company founded in 2007 that is headquartered in Tennessee. In October 2022, Change Healthcare was acquired for $8 billion by UnitedHealth Group, one of the largest health insurance companies in the United States.

Change Healthcare provides many different types of services to healthcare providers throughout the United States, as well as in Canada, Australia, the United Kingdom, the European Union, and Saudi Arabia. It helps with claims processing, payments, and revenue cycle management, and provides prescription processing services to more than 67,000 pharmacies through UnitedHealth's Optum division. It is also used by Tricare, the healthcare provider for U.S. military hospitals, clinics, and pharmacies. In addition, Change Healthcare also provides technologies to help healthcare providers make informed decisions about treatments, and patient engagement tools such as patient portals, secure messaging, and appointment scheduling. All told, Change Healthcare manages 15 billion claims a year, representing more than $1.5 trillion in payments, and is one of the largest health information exchange platforms in the United States, handling a third of all patient records in the U.S.

In February 2024, Change Healthcare publicly disclosed that it had been the target of a cyberattack. The perpetrators of the attack were reportedly an affiliate of the ransomware group known as BlackCat/ALPHV. BlackCat operates using a ransomware-as-a-service model, providing affiliates with its ransomware code, who then receive a share of any ransomware payment. Although the actual method of attack has not been publicly disclosed, sources familiar with the investigation indicated that the attackers were able to identify weakness in Change Healthcare's digital infrastructure (specifically, a remote access application that did not have multifactor authentication enabled) and used this vulnerability to gain unauthorized access to Change Healthcare's network. Once inside, they launched a ransomware attack. In the process, the group claims to have exfiltrated more than 6TB of data related to Change Healthcare's clients, including Medicare, CVS Caremark, Health Net, and TriCare, as well as the data of millions of patients, including medical records, insurance records, dental records, payment information, claims information, and patients' health data, contact information, and social security numbers.

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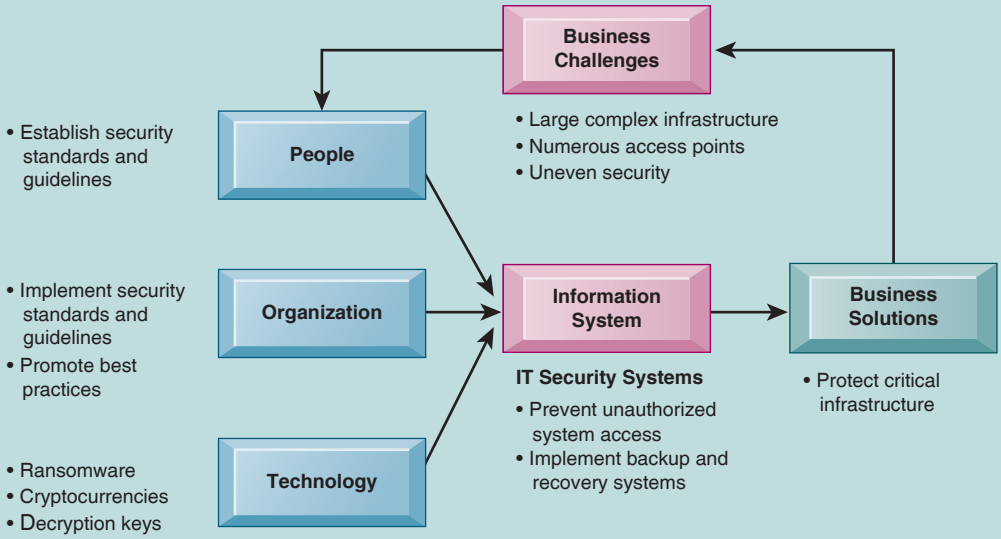
Once discovered, Change Healthcare responded to the attack by disconnecting more than 110 different services across its system to prevent further damage, disrupting key operations. The impact rippled across the United States' entire healthcare system, impacting physicians, hospitals, pharmacies, and individual patients. For example, the attack disrupted a range of services directly affecting patient care, such as eligibility verifications and medical procedure authorizations. Physicians and hospitals' ability to bill and manage claims were disrupted, with a substantial portion of claims unable to be processed. This in turn had an immediate adverse effect on hospital and physician finances, impacting their ability to offer healthcare services. Pharmacies' ability to properly fill prescriptions, and people looking to make healthcare claims, as well as fill prescriptions, were also impacted.

According to security researchers, Change Healthcare paid a $22 million ransom in Bitcoins for a decryption key and to prevent exploitation of the stolen data. However, the affiliate behind the attack claimed not to have been paid for the attack and claims to still possess a copy of the stolen data. The Blackcat/ALPHV group said the ransom was seized by law enforcement.

In response to the attack, UnitedHealth Group set up a temporary financial assistance program to help providers deal with immediate cash flow needs, advancing them more than $3 billion, although the program was criticized by the American Hospital Association and American Medical Association as falling far short of providing enough assistance. United Healthcare also moved to establish



workarounds, but the recovery process was slow, causing outages and issues that persisted for more than a month.

Cybersecurity consultants and government officials have consistently identified healthcare as one of the sectors of the U.S. economy most susceptible to attacks, and as much a part of the nation's critical infrastructure as energy and water. According to data security firm Emsisoft, ransomware attacks affected 46 hospital systems in 2023, up from 25 in 2022. Cybercriminals have also taken down companies that provide services such as medical transcription and billing in recent years. Previously, the infamous WannaCry ransomware attack locked up medical records at the National Health Service in Great Britain, leading to massive disruptions for patients.

There are numerous lessons to be learned from the attack. Cybersecurity experts suggest that Change Healthcare should have been better prepared to ensure that disruption was minimized, given the integral nature of Change Healthcare's systems within the healthcare industry. They also question whether appropriate backups and an incident response plan that had been properly tested were in place. Change Healthcare is currently facing more than two dozen proposed class-action lawsuits, as well as an investigation by the U.S. Department of Health and Human Services (HSS) focused on Change Healthcare's and UnitedHealth Group's compliance with healthcare sector privacy laws and regulations.

In the wake of the attack, HSS is also aiming to better organize its healthcare cybersecurity resources and programs. It intends to incentivize cybersecurity best practices, establish voluntary goals for the healthcare sector, and acting through CMS (Centers for Medicare and Medicaid Services), to propose new cybersecurity requirements for hospitals. Although a start, it is likely to be only a small part of the actions that need to be taken to prevent a reoccurrence of this sort of attack in the future. In fact, in April 2024, a different ransomware group, Ransom Hub, claimed that it had more than 4TB of sensitive Change Healthcare data and began leaking it in an effort to extort a second ransomware payment.

SOURCES: James Rundle, "Hackers Broke into Change Healthcare's Systems Days Before Cyberattack," *WallStreet Journal*, April 22, 2024; "UnitedHealth Says Change Healthcare Hackers Stole Health Data on'Substantial Proportion of People in America'," Techcrunch.com, April 22, 2024; Todd Doss, "ChangeHealthcare Ransomware Attack: 10 Lessons Learned," Jdsupra.com, April 2, 2024; Reed Abelson and MargotSanger-Katz, "4 Things You Need to Know about Health Care Cyberattacks," *New York Times*, March 29, 2024;Joao Correia, "A(nother) Ransomware Saga with a Twist," Securityboulevard.com, March 29, 2024; SteveAlder, "UnitedHealth Group Confirms Data Stolen in Change Healthcare Ransomware Attack,"Hipaajournal.com, March 29, 2024; Sean Michael Kerner, "The Change Healthcare Attack: How It Happened,"Techtarget.com, March 8, 2024.

The attack on Change Healthcare illustrates some of the reasons that organizations need to pay special attention to information systems security. The IT security vulnerabilities that enabled BlackCat/ALPHV to break into Change Healthcare's information systems illustrate the potential of such vulnerabilities to disrupt critical infrastructure throughout the country, paralyzing business, government, and daily life. Weak IT security has been responsible for many billions of dollars of corporate and consumer financial losses.

The Opening Case Study diagram above calls attention to important points raised by this case and this chapter. Organizations that lack or fail to effectively implement security awareness, resources, and tools create significant risk not only for their own organizations but also for their customers and for society at large.

Case Study Questions

1. In what ways did the impact of the Change Healthcare ransomware attack extend beyond Change Healthcare's immediate clients?
2. How could the ransomware attack against Change Healthcare have been prevented?
3. Is paying ransomware attackers an effective solution? Why or why not?

CAREER BENEFIT

IT specialists who were called upon to help Change Healthcare recover from the ransomware attack developed expertise in responding to a significant cybersecurity event. Given the increasingly prevalence of such events in the healthcare industry, the skills developed are likely to prove useful both in their current employment and to potential future employers.

8.1 Understand Why Information Systems Are Vulnerable

Can you imagine what would happen if you linked to the Internet without a firewall or anti-malware software? It is likely that your computer would be disabled within a few seconds, and it might take you many days to recover your computer systems and files. If you used the computer to run your business, you might not be able to sell to your customers or place orders with your suppliers while your computer was down. And you might find that your computer system had been penetrated by outsiders, who perhaps stole or destroyed valuable data, including confidential payment data from your customers. If too much data were destroyed or divulged, your business might never be able to recover!

In short, if you operate a business today, you need to make security and control top priorities. **Security** refers to the policies, procedures, and technical measures used to prevent unauthorized access, alteration, theft, or physical damage to information systems. **Controls** are the methods, policies, and organizational procedures that ensure the safety of the organization's assets, the accuracy and reliability of its records, and operational adherence to management standards.

Security

Policies, procedures, and technical measures used to prevent unauthorized access, alteration, theft, or physical damage to information systems.

Controls

All of the methods, policies, and procedures that ensure protection of the organization's assets, accuracy and reliability of its records, and operational adherence to management standards.

Why Systems Are Vulnerable

Information systems and the data stored on them are vulnerable to many kinds of threats. For instance, information systems in different locations are interconnected via communications networks. The potential for unauthorized access or damage is not limited to a single location but can occur at many access points in the network.

Examine Figure 8.1, which illustrates the most common threats against contemporary information systems. These threats can stem from technical, organizational, and environmental factors and be compounded by poor management decisions. The threats can also emanate from a **threat actor**: an individual or group of individuals seeking to exploit system vulnerabilities to intentionally cause harm. In the multitier client/server computing environment illustrated here, vulnerabilities exist at each layer and in the communications among the layers. Users at the client layer can cause harm by introducing errors or by accessing systems without authorization. It is possible for threat actors to access data flowing over networks, steal valuable data during transmission, or alter data without authorization. Electromagnetic radiation may disrupt a network at various points as well. Threat actors can introduce malware onto corporate servers, launch denial-of-service attacks aimed at servers linked to the Internet, steal data, and vandalize websites. Threat actors can also use corporate servers to access backend systems and databases to exfiltrate (steal), destroy, or alter corporate data stored in databases or files. Corporate systems are also at risk because of hardware and software failures.

Threat actor

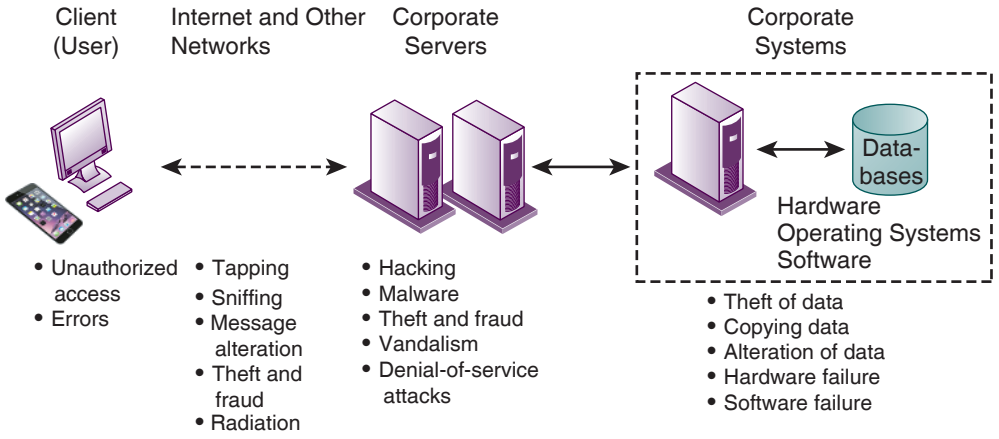
individual or group of individuals seeking to exploit system vulnerabilities to intentionally cause harm.

Systems malfunction if computer hardware breaks down, is not configured properly, or is damaged by improper use or criminal acts. Errors in programming, improper installation, or unauthorized changes cause computer software to fail. Power failures, floods, fires, or other natural disasters can also disrupt computer systems.

Domestic or offshore partnering with another company can contribute to system vulnerability if valuable information resides on networks and computers outside the organization's control. Without strong safeguards, valuable data could be lost or destroyed, or fall into the wrong hands, revealing important trade secrets or information that violates personal privacy.

Figure 8.1 Contemporary Security Challenges and Vulnerabilities

The architecture of a web-based application typically includes a web client, a server, and corporate information systems linked to databases. Each of these components presents security challenges and vulnerabilities. Floods, fires, power failures, and other electrical problems can also cause disruptions at any point in the network.



Portability makes mobile devices easy to lose or steal. Smartphones share many of the same security weaknesses as other Internet-connected devices and are vulnerable to malicious software and penetration from outsiders. Smartphones that corporate employees use often contain sensitive data such as sales figures, customer names, phone numbers, and email addresses. Intruders may also be able to access internal corporate systems via these devices.

Internet Vulnerabilities

Large public networks, such as the Internet, are more vulnerable than internal networks because they are open to virtually anyone. The Internet is so widespread that when abuses do occur, they can have a huge impact. When the Internet links to the corporate network, the organization's information systems are even more vulnerable to actions from outsiders.

Vulnerability has also increased from widespread use of email, instant messaging (IM), and peer-to-peer (P2P) file-sharing programs. Email may contain attachments or links that serve as springboards for malicious software or unauthorized access to internal corporate systems. Employees may use email messages to transmit valuable trade secrets, financial data, or confidential customer information to unauthorized recipients. Instant messaging over the Internet can in some cases be used as a back door to an otherwise-secure network. Illegally downloading copyrighted content, such as movies and television shows, from P2P networks can also transmit malicious software or expose information on either individual or corporate computers to outsiders.

A **sniffer** is a type of eavesdropping program that monitors the information traveling over a network. When used legitimately, sniffers help identify potential network trouble spots or criminal activity on networks, but when used for criminal purposes, sniffers can be damaging and very difficult to detect. Sniffers enable threat actors to steal proprietary information, including email messages, company files, and confidential reports, from anywhere on a network.

Sniffer

Type of eavesdropping program that monitors information traveling over a network.

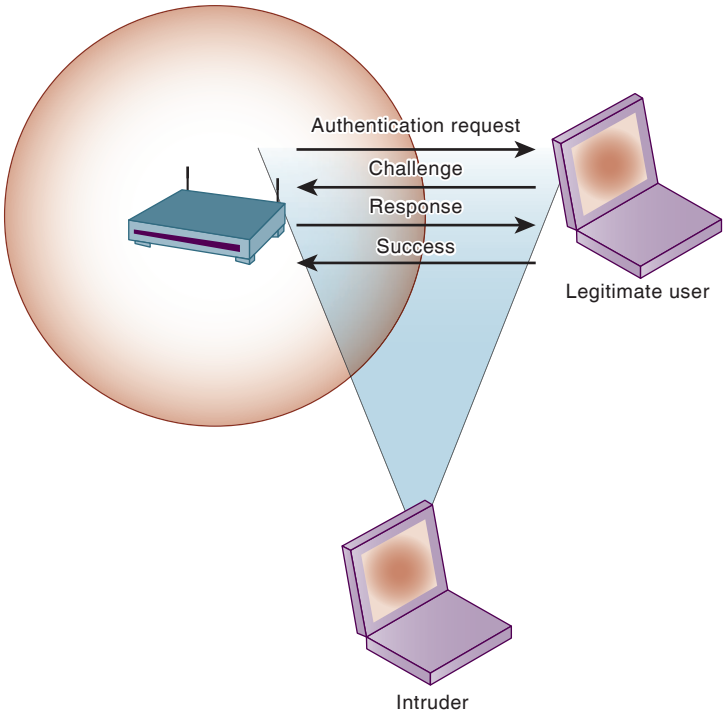
Wireless Security Challenges

Both Bluetooth and Wi-Fi networks are susceptible to hacking by eavesdroppers. Local area networks (LANs) using the 802.11 standard can be easily penetrated by outsiders armed with laptops, wireless cards, external antennas, and hacking software. Threat actors use these tools to detect unprotected networks, monitor network traffic, and, in some cases, gain access to the Internet or to corporate networks.

Wi-Fi transmission technology was designed to make it easy for devices to find and hear one another. This creates a number of security challenges. For example, examine Figure 8.2, which illustrates one such challenge. The service set identifiers (SSIDs) that identify the access points in a Wi-Fi network are broadcast multiple times and can be picked up fairly easily by

Figure 8.2 Wi-Fi Security Challenges

Many Wi-Fi networks can be penetrated easily by intruders using sniffer programs to obtain an address to access the resources of a network without authorization.



intruders' sniffer programs. Many wireless networks do not have basic protections against **war driving**, in which eavesdroppers drive by buildings or park outside them and try to intercept wireless network traffic.

War driving

Technique in which eavesdroppers drive by buildings or park outside and try to intercept wireless network traffic.

Although cellular transmission technology is considered to be more secure, 5G technology introduces fresh potential vulnerabilities. 5G increases the number and types of devices that can be connected and also involves multiple vendors and a more complex technology infrastructure, widening the potential surface for an attack (Bowman, 2024).

An intruder who has associated with an access point by using the correct SSID can access other resources on the network. For example, the intruder could use the system's network operating system to determine which other users are connected to the network, access their computers, and open or copy their files.

Intruders also use the information they have gleaned to set up rogue access points on a different radio channel in physical locations near users to force a user's radio network interface controller (NIC) to associate with the rogue access point. Once this association occurs, threat actors using the rogue access point can capture the names and passwords of unsuspecting users.

In the following sections, we will examine various types of threats to information systems. Then we'll look at the business value of security and control, discuss the components of an organizational framework for security and control, and finally, examine the most important tools and technologies for safeguarding information resources.

8.2 Describe the Most Prevalent and Dangerous Types of Malware

Malware

Malicious software programs such as viruses, worms, Trojan horses, ransomware, bots, and potentially unwanted programs.

Malicious software programs are often referred to as **malware** and include a variety of threats such as viruses, worms, Trojan horses, ransomware, bots, and potentially unwanted programs (PUPs). The incidence of malware is increasing every year. Malware is often

spread over the Internet from files of downloaded software; from files attached to email transmissions; or from compromised email messages, online ads, or instant messaging. Malware has also invaded information systems from infected external storage devices or infected devices. Especially prevalent today is the **drive-by download**, which refers to malware that a user unintentionally downloads or that is downloaded without the user's knowledge. Another method for malicious code distribution is to embed malware in the online advertising chain (known as **malvertising**), including via Google and other ad networks.

A **virus** is a rogue software program that attaches itself to other software programs or data files to be executed, usually without user knowledge or permission. Most viruses deliver a payload. The payload may be relatively benign, such as instructions to display a message or image, or it may be highly destructive—destroying programs or data, clogging computer memory, reformatting a computer's hard drive, or causing programs to run improperly. Viruses typically spread from computer to computer when humans take an action, such as sending an email attachment or copying an infected file.

A **worm** is a software program that independently copies itself from one computer to another computer over a network, and which can destroy data and programs, as well as disrupt or even halt the operation of a computer network. Unlike viruses, worms can operate on their own without attaching to other computer program files and rely less on human behavior to spread rapidly from computer to computer.

Some malware infections are designed as Trojan horses. A **Trojan horse** is a software program that appears to be legitimate but which contains hidden functionality that may cause damage. The Trojan horse is not itself a virus because it does not replicate, but it is often a way for viruses or other malicious code to be introduced into a computer system. The term is based on the Trojan Horse, a huge wooden horse that the Greeks used to trick the Trojans into opening the gates to their fortified city during the Trojan War.

An example of a modern-day Trojan horse is the ZeuS (Zbot) Trojan, which has infected millions computers worldwide over the years and still poses a threat. It has been used to steal login credentials for banking by surreptitiously capturing people's keystrokes as they use their computers. Zeus is spread mainly through drive-by downloads and phishing, and recent variants have been difficult to eradicate.

As described in the opening case, malware known as **ransomware** is proliferating on both desktop and mobile devices. Ransomware is malware that tries to extort money from users by taking control of their computers and blocking access to files. Ransomware is considered one of the most significant cyberthreats currently facing business, with the number of known attacks increasing by almost 70 percent in 2023, and the average ransom demand progressively climbing. Ransomware is increasingly being deployed in the form of "ransomware-as-a-service," a cybercrime business model in which developers of ransomware sell or rent the ransomware code to other groups, known as affiliates, in return for an upfront payment and/or share of the proceeds of an attack (Malwarebytes, 2024).

A **bot** is a type of malicious code that can be covertly installed on a computer when it is attached to the Internet, and which, once installed, responds to external commands sent by an attacker. Computers infected with bot malware may become part of a **botnet**, which is a collection of captured computers that can then be used for malicious purposes, such as distributed denial of service (DDoS) attacks, phishing campaigns, or spam, without their owners' knowledge. Although less prevalent than they once were, bots and botnets remain significant threats.

A **potentially unwanted program (PUP),** also sometimes referred to as a potentially unwanted application (PUA), can also be considered a form of malicious software. PUPs include adware, browser parasites, spyware, and other applications such as rogue security software, toolbars, and PC diagnostic tools that install themselves onto a user's computer. Some types of PUPs are especially nefarious. A **keylogger** is a form of spyware that records every keystroke made on a computer to steal serial numbers for software, to launch Internet attacks, to gain access to email accounts, to obtain passwords to protected computer systems, or to pick up personal information such as credit card or bank account numbers. The Zeus trojan described earlier uses keylogging. Other PUPs reset web browser home pages, redirect search requests, or slow performance by taking up too many computer resources.

Drive-by download

Refers to malware that a user unintentionally downloads or that is downloaded without the user's knowledge.

Malvertising

Method of malicious code distribution in which malware is embedded in the online advertising chain.

Virus

Rogue software program that attaches itself to other software programs or data files to be executed, usually without user knowledge or permission.

Worm

Software program that independently copies itself from one computer to another computer over a network, and which can destroy data and programs as well as disrupt or even halt the operation of a computer network.

Trojan horse

Software program that appears to legitimate but contains hidden functionality that may cause damage.

Ransomware

Malware that tried to extort money from users by taking control of their computers and blocking access to files.

Bot

Software application that performs repetitive pre-defined tasks over a network. Includes shopping bots, web crawlers, scrapers, monitoring bots, and malware.

Botnet

A collection of computers infected with bot malware that can then be used for malicious purposes, such as launching distributed denial-of-service (DDoS) attacks, phishing campaigns, or spam, without their owners' knowledge.

Potentially unwanted program (PUP)

Software such as adware, browser parasites, spyware, and other applications such as rogue security software, toolbars, and PC diagnostic tools that install themselves onto a user's computer; can be considered a form of malicious software.

Keylogger

A form of spyware that records every keystroke made on a computer to steal personal information or passwords or to launch Internet attacks.

Smartphones are vulnerable in most of the same ways as other Internet-connected devices: they can be infected with malicious code without user intervention, passwords and other data can be stolen, and programs can be covertly installed to monitor user actions and potentially convert the smartphone to a robot in a botnet to send email and text messages to anyone. According to IT security experts, mobile devices now pose the greatest security risks, outpacing those from computers. More than 80 percent of organizations surveyed faced some sort of attack aimed primarily at users' mobile devices in 2023, according to Verizon, while Kaspersky reported a more than 50 percent increase in attacks targeting mobile devices, with the number surging to more than 33 million. Android, which is the world's leading mobile operating system, is the mobile platform targeted by most threat actors. For instance, Malwarebytes detected Android banking trojans, disguised as regular apps like QR code scanners, fitness trackers, and even copies of popular apps like Instagram, more than 88,000 times in 2023. However, Apple devices are also at risk. For example, in February 2024, a new sophisticated mobile trojan specifically aimed at iOS users was found to be collecting facial recognition data, identity documents, and intercepting SMS. Mobile device malware poses serious threats to enterprise computing because so many wireless devices are now linked to corporate information systems (Verizon, 2023; Kaspersky, 2024; Malwarebytes, 2024).

Social networks such as Facebook, X (formerly Twitter), and LinkedIn have emerged as conduits for malware. Members are more likely to trust messages they receive from friends even if this communication is not legitimate. For example, the Facebook Messenger virus is spread by phishing messages sent from a hijacked Facebook account to people in the victim's Facebook contact list. The virus's ability to replicate contributes to its widespread and persistent nature (Danes, 2023).

Malware is also starting to be targeted at the 3D virtual reality environment known as the metaverse (see Chapter 7). The hardware needed for virtual reality and augmented reality platforms has created new endpoints to exploit. Attackers can potentially manipulate platforms to create physical dangers. Participants in the metaverse may also be subject to various forms of harassment by malicious actors. The privacy of participants and the security of their personal information are also a concern (Rosenblatt, 2023).

Artificial intelligence (AI) is creating new risks in terms of malicious code. We examine the security issues that are raised by AI in more detail in Section 8.5.

Table 8.1 provides some notable examples of malware.

Table 8.1 Notable Examples of Malware

Name	Type	Description
Emotet	Trojan/Botnet	Described by Europol as the world's most dangerous malware. Initially used to steal bank login credentials by surreptitiously capturing people's keystrokes. Later versions added malware delivery services, including other Trojans and ransomware, and were spread via botnet. In January 2021, an internationally coordinated effort took down many Emotet botnets, but by November 2021, it had once again resurfaced, indicating how difficult it is to fully eradicate. Still active as of 2024.
Cryptolocker	Ransomware/ Trojan	Hijacks users' data such as photos, videos, and text documents; encrypts them with virtually unbreakable asymmetric encryption; and demands ransom payment for them.
Conficker	Worm	First detected in November 2008 and still a problem for users of older, unpatched Windows operating systems. Uses flaws in Windows software to take over machines and link them into a virtual computer that can be commanded remotely. Had nearly 10 million computers worldwide under its control. Difficult to eradicate.
Sasser.ftp	Worm	First appeared in 2004. Spread over the Internet by attacking random IP addresses. Caused computers to continually crash and reboot and infected computers to search for more victims. Affected millions of computers worldwide and caused an estimated $14.8 billion to $18.6 billion in damages.
ILOVEYOU	Virus	First detected on May 3, 2000. Script virus written in Visual Basic script and transmitted as an email attachment with the subject line ILOVEYOU. Overwrote music, image, and other files with a copy of itself and did an estimated $10 billion to $15 billion in damage.

8.3 Understand the Dangers Posted by Threat Actors

As you learned in Section 8.1, threat actors are individuals or groups of individuals seeking to exploit system vulnerabilities to intentionally cause harm. There are numerous types of threat actors, including cybercriminals, nation-states, terrorist groups, hackers, hacktivists, and insiders.

Cybercriminals are primarily seeking financial gain. Nation-states and terrorist groups use threat actors for geopolitical purposes or ideological violence, as discussed further in the section about cyberwarfare and cyberterrorism. A **hacker** uses technical skills to identify weaknesses in the security protections that computer systems employ in order to gain unauthorized access. Organizations sometimes hire *ethical hackers* to test their own security measures. These types of hackers work under an agreement with the target firms that they will not be prosecuted for their efforts to break in. Firms such as Apple, Microsoft, Intel, HP, and many others are also often willing to pay monetary bounties ("bug bounties") to hackers who discover bugs in their software and hardware. A **hacktivist** is usually motivated to attack specific targets based on ideology. Hacktivists often engage in **cybervandalism**, the intentional disruption, defacement, or even destruction of a website or corporate information system.

Spoofing, Phishing, and Pharming

Threat actors often employ **spoofing**: the practice of tricking or deceiving others by hiding one's true identity. Spoofing may involve using a fake email address or redirecting a web link to an address different from the intended one, with the site masquerading as the intended destination. For example, customers may be redirected to a fake website that looks almost exactly like the intended site, enabling the threat actors to collect and process orders, effectively stealing business as well as sensitive customer information.

One popular form of spoofing is called **phishing**. Phishing involves setting up fake websites or sending email messages that look like those of legitimate businesses to ask users for confidential personal data. The email message instructs recipients to update or confirm records by providing social security numbers, bank and credit card information, and other confidential data by responding to the email message, entering the information at a bogus website, or calling a telephone number. Microsoft, eBay, PayPal, Amazon, Walmart, and a variety of banks have been among the top spoofed companies. In a more targeted form of phishing called *spear phishing*, messages appear to come from a trusted source, such as an individual within the recipient's own company or a friend. *Business email compromise (BEC) phishing* is another fast-growing threat. A BEC attack typically targets a high-ranking employee who is authorized to make fund transfers. The use of generative AI to create phishing attacks is also an increasing threat (Proofpoint, 2024).

Phishing attacks often involve social engineering. **Social engineering** exploits human curiosity, greed, gullibility, fear, or complacency to trick people into taking an action that enables the threat actor to gain access to a computer system or to download malware.

Phishing techniques called evil twins and pharming are especially hard to detect. An **evil twin** is a wireless network that pretends to offer a trustworthy Wi-Fi connection to the Internet. The bogus network looks identical to a legitimate public network that might be found in an airport lounge, hotel, or coffee shop. Fraudsters try to capture passwords or credit card numbers of unwitting users who log on to the network.

Pharming redirects users to a bogus web page even when the users type the correct web page address into their browser. This is possible when pharming perpetrators gain access to the IP address information that Internet service providers (ISPs) store to speed up web browsing and when flawed software on ISP servers allows the fraudsters to hack in and change those addresses.

Living off the Land and Advanced Persistent Threats

Unlike traditional malware attacks, which rely on injecting malicious code, a **Living off the Land attack (LOTL attack)** hijacks legitimate software tools, such as network administration

Hacker

Uses technical skills to identify weaknesses in security protections that computer systems employ in order to gain unauthorised access.

Hacktivist

A hacker that is usually motivated to attack specific targets based on ideology.

Cybervandalism

Intentional disruption, defacement, or destruction of a website or corporate information system.

Spoofing

The practice of tricking or deceiving others by hiding one's true identity.

Phishing

Form of spoofing that involves setting up a fake website or sending email messages that look like those of legitimate businesses to ask users for confidential personal data.

Social engineering

Exploiting human curiosity, greed, gullibility, fear, or complacency to trick people into taking an action that enables a threat actor to gain access to a computer system or download malware.

Evil twin

Wireless network that pretends to be legitimate to entice participants to log on and reveal passwords or credit card numbers.

Pharming

Phishing technique that redirects users to a bogus web page, even when an individual enters the correct web page address.

Living off the Land attack (LOTL attack)

Hijacks legitimate software tools, such as network administration tools that are already part of an organization's digital infrastructure, to carry out the attack.

tools like Powershell, Windows Management Interface, and Mimikat, that are already part of an organization's digital infrastructure, to carry out the attack. For this reason, LOTL attacks are often referred to as a "fileless" attack, since they typically do not rely on initially introducing malware into the system, although this may occur later in the attack process once the attacker has established itself. LOTL attacks are becoming more common because they are more difficult to detect with traditional security tools, which increases the likelihood of success and gives the attacker more time to escalate privileges, steal data, and set backdoors for future access.

Advanced persistent threat attack (APT attack)

A prolonged attack where an intruder infiltrates a system and remains undetected for an extended period of time, exfiltrating data, or waiting for the right time to launch a more crippling attack. Defined by its stragetic targeting and persistence, as well as the advanced tactics, techniques, and procedures it uses.

LOTL attacks are often a tactic used in conjunction with an **advanced persistent threat attack (APT attack)**. An APT attack is a prolonged attack where an intruder infiltrates the system and remains undetected for an extended period of time, exfiltrating data, or waiting for the right time to launch a more crippling attack. This type of attack is defined by its strategic targeting and persistence, as well as the advanced tactics, techniques, and procedures it uses.

Denial-of-Service Attacks

In a **denial-of-service attack (DoS attack)**, attackers flood a network server or web server with many thousands of false communications or requests for services in order to crash the network. The network receives so many queries that it cannot keep up with them and is thus unavailable to service legitimate requests. A **distributed denial-of-service attack (DDoS attack)** often employs a botnet to inundate and overwhelm a network from many launch points. One example is the Mirai botnet, which infected numerous Internet of Things (IoT) devices (such as Internet-connected surveillance cameras) and then used these devices to launch a DDoS attack against Dyn, whose servers monitor and reroute Internet traffic. The Mirai botnet overwhelmed the Dyn servers, taking down Etsy, GitHub, Netflix, Shopify, SoundCloud, Spotify, Twitter, and a number of other major websites. Numerous variants based on the source code for the original Mirai botnet remain active in 2024 (Kupchik, 2024).

Denial-of-service attack (DoS attack)

Involves flooding a network server or web server with many thousands of false communications or requests for services in order to crash the network.

Distributed denial-of-service attack (DDoS attack)

Attack that often employs a botnet to inundate and overwhelm a network from numerous launch points.

Although DoS/DDoS attacks do not destroy information or access restricted areas of a company's information systems, they often cause a website to shut down, making it impossible for legitimate users to access the site. For busy e-commerce sites, these attacks are costly. While the site is shut down, customers cannot make purchases. Especially vulnerable are small and midsize businesses whose networks tend to be less protected than those of large corporations.

Data Breaches and Identity Theft

With the growth of the Internet and e-commerce, data breaches and identity theft have become especially troubling. A **data breach** involves the loss of control over an organization's data. According to the IBM Security's 2023 Cost of a Data Breach Report, the total average cost of a data breach among the 537 companies surveyed globally was $4.45 million, a 15 percent increase over the prior three years (IBM Security, 2023). Moreover, brand damage can be significant, although hard to quantify.

Data breach

Involves the loss of control over an organization's data to outsiders.

Data breaches also enable credential stuffing attacks. A **credential stuffing attack** is a brute force attack that attackers launch via botnets and automated tools using known username and password combinations obtained from data breaches. Credential stuffing attacks are becoming increasingly common, especially in the financial services industries.

Credential stuffing attack

A brute force attack that attackers launch via botnets and automated tools using known username and password combinations obtained from data breaches.

Table 8.2 describes some major data breaches.

Data breaches are typically linked to identity theft. **Identity theft** is a crime in which an imposter obtains key pieces of personal information, such as social security numbers, driver's license numbers, or credit card numbers, to impersonate someone else. The information may be used to obtain credit, merchandise, or services in the name of the victim or to provide the thief with false credentials. Identity theft has flourished on the Internet, with credit card files a major target. More than 1 million people filed an identity theft report with the Federal Trade Commission in 2023 (Federal Trade Commission, 2024).

Identity theft

Theft of key pieces of personal information, such as social security number, driver's license numbers, or credit card numbers, in order to impersonate someone else, and obtain merchandise and services in the name of the victim or to obtain false credentials.

Table 8.2 Major Data Breaches

Data Breach	Description
AT&T	In 2024, AT&T revealed that it had discovered that the personal data of 73 million people, including 7.6 million of its current customers and 65.4 million of its former customers, had been stolen and posted them online. The personal data included social security numbers, passcodes, full names, email addresses, phone numbers, date of birth, and AT&T account numbers.
LinkedIn	In 2021, personal data, including email addresses, phone numbers, geolocation data, and social media details that were associated with more than 700 million LinkedIn users (more than 90 percent of LinkedIn's user accounts), were discovered on a Dark Web forum posted by a hacker who exploited a vulnerability in the site's application programming interface using data-scraping techniques.
Marriott	In 2018, Marriott, the world's largest hotel company, revealed that a hack in the reservation database for its Starwood properties may have exposed the personal information of up to 500 million guests. Exposed data included names, phone numbers, email addresses, passport numbers, date of birth, and credit card numbers. State-sponsored Chinese threat actors copied and encrypted the data and took steps toward removing them. In 2020, Marriott was hacked again after the login credentials of two employees were used.
Equifax	In 2017, Equifax, one of the largest major credit reporting and scoring firms, reported that threat actors had gained access to some of its systems and, potentially, the personal information of more than 140 million U.S. consumers (about 45 percent of the U.S. population), including social security numbers and driver's license numbers. Credit card numbers for more than 200,000 consumers and personal information used in disputes for more than 180,000 people were also compromised.

Global Threats: Cyberwarfare and Cyberterrorism

The cybercriminal activities we have described—launching malware, DoS/DDoS attacks, and phishing probes—are borderless. Countries that are the leading sources of malware attacks include China, Russia, Brazil, Poland, Iran, India, Nigeria, Vietnam, the United States, and Germany. The global nature of the Internet makes it possible for cybercriminals to operate—and do harm—anywhere in the world.

Internet vulnerabilities have also turned individuals and even entire nation-states into easy targets for politically motivated threat actors to conduct sabotage and espionage. **Cyberwarfare** is a state-sponsored politically motivated activity designed to harm another state or nation by damaging or disrupting its computers, information systems, or networks. **Cyberterrorism** is similar in nature, but executed by a terrorist group rather than a nation-state. For example, in February 2024, the U.S. Cybersecurity & Infrastructure Security Agency (CISA) issued an advisory warning critical infrastructure organizations in the communications, energy, transportation, water, and wastewater systems across the United States that China-backed Volt Typhon had compromised IT systems and were seeking to preposition themselves on IT networks for disruptive or destructive cyberattacks in the event of a major conflict with the United States (Cybersecurity & Infrastructure Security Agency, 2024).

Cyberwarfare

State-sponsored politically motivated activity designed to cripple and defeat another nation-state by damaging or disrupting its computers, information systems, or networks.

Cyberterrorism

Similar in nature to cyberwarfare, but executed by a terrorist group rather than a nation-state.

Cyberwarfare is more complex than conventional warfare. Although many potential targets are military, a country's power grids, dams, financial systems, communications networks, and voting systems can also be crippled. Nonstate actors such as terrorists or criminal groups can mount attacks, and it is often difficult to identify who is responsible. Therefore, nations must constantly be on the alert for new malware and other technologies that could be used against them, especially because some of these technologies are openly for sale by cybercriminals to interested governments.

Cyberwarfare attacks have become much more widespread, sophisticated, and potentially devastating. Foreign threat actors have stolen source code and blueprints to the oil and water pipelines and power grid of the United States and infiltrated the Department of Energy's networks hundreds of times. Over the years, threat actors have stolen plans for missile tracking systems, satellite navigation devices, surveillance drones, and leading-edge jet fighters.

According to U.S. intelligence, more than 30 countries, including Russia, China, Iran, and North Korea, already have or are developing offensive cyberattack capabilities. Their cyberarsenals include collections of malware for penetrating industrial, military, and critical civilian infrastructure controllers; email lists and text for phishing attacks on important targets; and algorithms for DoS/DDoS attacks. U.S. cyberwarfare efforts are concentrated in the United States Cyber Command, which coordinates and directs the operations and defense of Department of Defense information networks and prepares for military cyberspace operations. Cyberwarfare poses a serious threat to the infrastructure of modern societies because their major financial, health, government, and industrial institutions rely on the Internet for daily operations.

Internal Threats: Employees

We tend to think that the security threats to a business originate outside the organization. In fact, company insiders pose serious security problems. Studies have found that users' lack of knowledge is the single-greatest cause of network security breaches. Many employees forget the passwords they need to access computer systems or allow coworkers to use them, which compromises the system. As illustrated in the Chapter Case Study, *Social Engineering Creates Chaos for MGM Resorts,* malicious intruders often use social engineering techniques to obtain system access, for example tricking employees into revealing passwords by pretending to be legitimate members of the company in need of information. According to Ponemon Institute, non-malicious insiders accounted for 75 percent of all insider incidents in 2023.

Insiders bent on harm have also exploited their knowledge of the company to break into corporate systems, including those running in the cloud. While malicious insider incidents are far less frequent, accounting for only 25 percent of insider incidents in 2023, they cause much higher losses (Ponemon Institute/DTEX Systems, 2023). For instance, a former employee at Amazon Web Services (AWS) used her knowledge of Amazon cloud security to steal many millions of Capital One Financial customer records stored by AWS.

Computer Crime

The U.S. Congress first specifically addressed the threat of computer crime in 1986 with the Computer Fraud and Abuse Act, which makes it illegal to access a computer system without authorization. Today, most states have similar laws, and more than 156 countries around the world have enacted cybercrime legislation of some kind (UNCTAD, 2024). Computers can also be used as an instrument of crime. The U.S. Department of Justice defines computer crime as "any violations of criminal law that involve a knowledge of computer technology for their perpetration, investigation, or prosecution." Table 8.3 provides examples of the computer as both a target of and an instrument of crime.

U.S. legislation, such as the Wiretap Act, Wire Fraud Act, Economic Espionage Act, Electronic Communications Privacy Act, CAN-SPAM Act, and Protect Act, covers computer crimes involving intercepting electronic communication, using electronic communication to defraud, stealing trade secrets, illegally accessing stored electronic communications, using email for threats or harassment, and transmitting or possessing child pornography. The National Information Infrastructure Protection Act makes malware distribution and attacks to disable websites federal crimes. All 50 states, the District of Columbia, Guam, Puerto Rico, and the Virgin Islands have enacted legislation requiring private or governmental entities to notify individuals of security breaches involving personally identifiable information. In addition, in December 2023, a new U.S. Securities and Exchange Commission regulation went into effect requiring all public companies to disclose material cybersecurity incidents within four business days.

No one knows the magnitude of computer crime—how many systems are invaded, how many people engage in the practice, or the total economic damage that results from computer crime. However, according to one research firm, Cybersecurity Ventures, the estimated global cost of cybercrime in 2024 will be $9.5 trillion, and it expects the cost to grow to more than $15 trillion by 2029 (CyberSecurity Ventures, 2023).

Many companies are reluctant to report computer crimes because the crimes may involve employees or because publicizing vulnerability will hurt their reputations. The most economically damaging kinds of computer crime are DDoS attacks, activities of malicious insiders, and web-based attacks.

Table 8.3 Examples of Computer Crime

Computers As Targets of Crime
Breaching the confidentiality of protected data
Accessing a computer system without authority to do so
Knowingly accessing a protected computer to commit fraud
Intentionally accessing a protected computer and causing damage negligently or deliberately
Knowingly transmitting a program, program code, or command that intentionally causes damage to a protected computer
Threatening to cause damage to a protected computer
Computers As Instruments of Crime
Stealing trade secrets
Unauthorized copying of software or copyrighted intellectual property, such as articles, books, music, and video
Scheming to defraud
Using email or messaging for threats or harassment
Intentionally attempting to intercept electronic communication
Illegally accessing stored electronic communications, including email and voice mail
Transmitting or possessing child pornography by using a computer

8.4 Describe the Security Issues Raised by Software Vulnerabilities

Software errors pose a constant threat to information systems, causing untold losses in productivity and sometimes endangering people who use or depend on these systems. The growing complexity and size of software programs, coupled with demands for rapid delivery to markets, have contributed to an increase in software flaws or vulnerabilities.

A major problem with software is the presence of a hidden **bug**, or program code defect. Studies have shown that it is virtually impossible to eliminate all bugs from large programs. The main source of bugs is the complexity of decision-making code. Even a relatively small program of several hundred lines will contain tens of decisions leading to hundreds or even thousands of paths. Important programs within most corporations are usually much larger, containing tens of thousands or even millions of lines of code, each with many times the choices and paths of smaller programs.

Bug

Software program code defect.

Zero defects cannot be achieved in larger programs. Complete testing simply is not possible. Fully testing programs that contain thousands of choices and millions of paths would require thousands of years. Even with rigorous testing, you would not know for sure that a piece of software was dependable until the product proved itself after much operational use.

Flaws in software not only impede performance but also create security vulnerabilities that open networks to intruders. Each year security firms identify thousands of software vulnerabilities. A **zero-day vulnerability**, which is a vulnerability in software unknown to its creator, is especially troublesome. This type of vulnerability is called *zero-day* because the author of the software has zero days after learning about it to patch the code before it can be exploited in an attack. Sometimes security researchers spot the vulnerability ahead of time, but more often, it is not detected until an attack has occurred.

Zero-day vulnerability

Vulnerability in software, unknown to the creator, that hackers can exploit before the vendor becomes aware of the problem.

One recent example of a critical zero-day vulnerability was Log4Shell, in an open-source Java logging tool, Apache Log4j, that allowed attackers to take control of susceptible servers. The Log4Shell vulnerability highlighted security issues surrounding the use of open-source software, which has become a critically important part of the world's digital infrastructure (with the average application using 528 different open-source components). Open-source software is created and maintained by volunteers, who don't always have adequate resources and personnel for incident response and proactive maintenance.

SQL injection attack

Attack that exploits vulnerabilities in poorly coded SQL web applications in order to introduce malicious program code into a company's systems and networks.

An **SQL injection attack** exploits vulnerabilities in poorly coded web application software to introduce malicious program code into a company's systems and networks. These vulnerabilities occur when a web application fails to validate properly or filter data a user enters on a web page, which might occur when ordering something online. An attacker uses this input validation error to send a rogue SQL query to the underlying database to access the database, plant malicious code, or access other systems on the network. For example, the popular secure file transfer service MOVEit, used by thousands of organizations, experienced one of the largest hacks in 2023 when the Clop ransomware group was able to exploit a zero-day SQL injection vulnerability that allowed threat actors to gain unauthorized access to customer environments. The vulnerability impacted all on-premises and cloud-based versions of the service. More than 2,560 organizations and 40 million people were reportedly ultimately impacted.

Patch

Software to repair software flaws without disturbing the proper operation of the software.

To correct software flaws once they are identified, the software vendor creates software called a **patch** to repair the flaws without disturbing the proper operation of the software. It is up to users of the software to track these vulnerabilities, test the system for them, and then apply all patches. This process is called *patch management*.

Because a company's IT infrastructure is typically laden with multiple business applications, operating system installations, and other system services, maintaining patches on all devices and services that a company uses is often time-consuming and costly. Malware is being created so rapidly that companies have very little time—between the time a vulnerability and a patch are announced and the time malicious software appears to exploit the vulnerability—to respond to the vulnerability.

Software supply chain attack

Involves a threat actor infiltrating the software supply chain at the development and production stage and compromising the software before it is acquired and deployed by the customer.

Software supply chain attacks are another relatively new and menacing threat. A **software supply chain attack** involves a threat actor infiltrating the software supply chain at the development and production stage and compromising the software before it is acquired and deployed by the customer. This may occur in either newly acquired software, as part of an update, or even as part of a patch. What makes the attack so insidious is that the malicious code is inserted into a trusted piece of software via trusted mechanisms, such as the update or patch process. In addition, the hack of a single software vendor can provide threat actors with a springboard into the networks of hundreds or even thousands of the vendor's customers. In addition to compromising update servers, other techniques have been used in software supply chain attacks. Some attacks begin with the corruption of tools used to create software in the first instance. Threat actors have also targeted open-source code, hoping that unsuspecting developers will mistakenly add the blocks of malicious code into their own projects. Another technique involves undermining the code signing process, which involves applying a cryptographic signature to software components. Code signing is used to validate the identity of the software developer and the integrity of the code. Users and security tools typically trust signed code. However, threat actors have been able to undermine the process by stealing code signing keys and certificates or by breaking into signing systems, allowing them to create compromised code signing certificates impersonating those of a trusted vendor.

8.5 Describe the Security Issues Raised by Artificial Intelligence

Although much attention has been focused on the benefits of AI, it also introduces some significant challenges and risks in terms of cybersecurity. Cybercriminals are already leveraging AI technology to create sophisticated attacks, and AI algorithms can themselves be subverted for harmful purposes.

Phishing with AI

Today, a phishing email can often be identified because it contains typographical or grammatical errors or awkward phrasing. Since cybercriminals often use templates to create them, phishing emails from different sources may share common indicators, such as matching text

strings, that can be detected by security software. However, genAI tools such as WormGPT and others are increasingly being used to help cybercriminals create unique and extremely realistic phishing emails. In addition, genAI tools can be harnessed to gather information about specific targets for use in spear phishing or BEC attacks.

Deepfakes

Microsoft has recently warned that AI-generated deepfakes may be able to undermine identify solutions that currently rely on facial and voice recognition. The term **deepfake** refers to the use of AI to create highly realistic, but digitally fabricated or fake, images or audio. AI technology can be used to impersonate the voice and image of a real person, and in doing so, create a false manipulated reality. The fake content can show people doing and saying things they never actually did. Deepfakes can be used for fraud, blackmail, and the spread of misinformation and propaganda. In the past, creating deepfakes required a significant amount of data with respect to the target, as well as a significant investment of time and resources. However, today, AI enables the creation of sophisticated deepfakes with minimal technical knowledge and expense.

Deepfake

Use of AI to create highly realistic, but digitally fabricated or fake, images or audio.

Subverting LLMs

There are a number of ways cybercriminals are attempting to subvert large language models (LLM), which are machine learning models that can comprehend and generate human language text by analyzing massive data sets of language. One method is to attempt to find a way around the guardrails that the creators of LLM have installed to prevent them from being used for unethical or nefarious purposes. This approach is sometimes referred to as a prompt injection attack or jailbreaking and involves using specialized prompts designed to manipulate the LLM to produce inappropriate content, disclose sensitive information, or execute harmful code. Security researchers believe that most LLMs are susceptible to such attacks. For instance, researchers were able to trick ChatGPT into creating ransomware, as well as malware that could be used to find and exfiltrate specific documents. In another study, researchers found that they could manipulate Google's Gemini to cause it to divulge its system prompts (the instructions provided to an LLM to set up its behavior, persona, and constraints on what it can or cannot generate).

Another way cybercriminals are attempting to leverage the power of LLMs is to create their own malicious forms of genAI and make them available online to others for a price. WormGPT is an example of this approach. WormGPT is an LLM specifically designed for malicious activities. WormGPT has no pre-installed ethical boundaries or limitations and was trained with a focus on malware-related data. WormGPT can help threat actors create phishing emails and malware. FraudGPT offers similar features, but with additional functionality. DarkBART, from the creator of WormGPT and FraudGPT, is reportedly based on DarkBERT, an LLM that was originally created by a South Korean data intelligence firm with the goal of fighting cybercrime. Another tool, known as DarkGemini, is an AI-powered malicious front-end service being sold on a subscription basis. Cybercriminals can also access unrestricted AI models through open-source AI model repositories such as HuggingFace and use them to create their own custom platforms. For instance, researchers have discovered about 100 potentially malicious machine learning models available on HuggingFace.

8.6 Describe the Business Value of Security and Control

Companies have very valuable information assets to protect. Systems often house confidential information about individuals' taxes, financial assets, medical records, and job performance reviews. The systems also can contain information on corporate operations including trade secrets, new product development plans, and marketing strategies. Government systems may store information on weapons systems, intelligence operations, and military targets. These information assets have tremendous value, and the repercussions can be devastating if they are

lost, destroyed, or placed into the wrong hands. Systems that are unable to function because of security breaches, disasters, or malfunctioning technology can have permanent impacts on a company's financial health. Some experts believe that 40 percent of all businesses will not recover from application or data losses that are not repaired within three days.

Inadequate security and control may result in serious legal liability. Businesses must protect not only their own information assets but also those of customers, employees, and business partners. Failure to do so may open the firm to costly litigation for data exposure or theft. An organization can be held liable for needless risk and harm created if the organization fails to take appropriate protective action to prevent loss of confidential information, data corruption, or breach of privacy. For example, Target, a mass merchandise retailer, had to pay more than $150 million to settle various legal claims filed by banks, credit card companies, and consumers against Target related to a massive hack of Target's payment systems. Developing a sound security and control framework that protects business information assets is of critical importance to the entire enterprise, including senior management. It can no longer be limited to the IT department (Rothrock et al., 2018).

Legal and Regulatory Requirements for Electronic Records Management

Government regulations are forcing companies to take security and control more seriously by mandating the protection of data from abuse, exposure, and unauthorized access. Firms in various industries face legal obligations to retain and store electronic records, known as electronic records management (ERM).

For instance, in the healthcare industry, firms must comply with the Health Insurance Portability and Accountability Act (HIPAA). HIPAA outlines medical security and privacy rules and procedures for simplifying the administration of healthcare billing and automating the transfer of healthcare data among healthcare providers, payers, and plans. It requires members of the healthcare industry to retain patient information for six years and ensure the confidentiality of that information. It also specifies privacy, security, and electronic transaction standards for healthcare providers that handle patient information and provides penalties for breaches of medical privacy, disclosure of patient records by email, or unauthorized network access.

In the financial services industries, firms must comply with the Financial Services Modernization Act, better known as the Gramm-Leach-Bliley Act. This act requires financial institutions to ensure the security and confidentiality of customer data. Data must be stored on a secure medium, and special security measures must be enforced to protect such data on storage media and during transmittal.

Publicly traded companies must comply with the Public Company Accounting Reform and Investor Protection Act, better known as the Sarbanes-Oxley Act. This act is designed to protect investors after the financial scandals at Enron, WorldCom, and other public companies. It imposes responsibility on companies and their management to safeguard the accuracy and integrity of the financial information that is used internally and released externally. The Sarbanes-Oxley Act is fundamentally about ensuring that internal controls are in place to govern the creation and documentation of information in financial statements. Because information systems are used to generate, store, and transport such data, the legislation requires firms to consider information systems security and other controls needed to ensure the integrity, confidentiality, and accuracy of their data. Each system application that deals with critical financial reporting data requires controls to make sure the data are accurate. Controls to secure the corporate network, prevent unauthorized access to systems and data, and ensure data integrity and availability in the event of disaster or other disruption of service are essential as well.

Firms in all industries must also be aware of, and comply with, data storage and retention requirements imposed by various privacy laws and regulations, such as the European Union's General Data Protection Regulation (GDPR), which applies to all organizations that operate within the European Union, as well as various federal and state laws enacted in the United States (see Chapter 4). For instance, California's Privacy Rights Act, which went into full effect in January 2023, requires companies that operate in California to have a detailed

and transparent data retention schedule and data destruction program. Courts have ruled that simply holding onto data longer than a specified retention period violates a person's privacy, even if no other breach or use has occurred.

Electronic Evidence and Digital Forensics

Security, control, and ERM have become essential in connection with legal actions. Most of the evidence today for stock fraud, embezzlement, theft of company trade secrets, computer crime, and many civil cases is in digital form. Legal cases increasingly rely on evidence represented as digital data (including emails, text messages, instant messages, and social media) stored on mobile devices, portable storage devices, computers and/or in the cloud (referred to as electronically stored information, or ESI).

In a legal action, a firm is obligated to respond to a discovery request for ESI that may be used as evidence, and the company is required by law to produce that data. The cost of responding to a discovery request can be enormous if the company has trouble assembling the required data or the data have been corrupted or destroyed. Courts now impose severe financial and even criminal penalties for improper destruction of ESI.

An effective ESI retention policy ensures that ESI is well-organized, accessible, and neither retained too long nor discarded too soon. It also reflects an awareness of how to preserve potential evidence for digital forensics. **Digital forensics** is the scientific collection, examination, authentication, preservation, and analysis of ESI in such a way that the information can be used as evidence in a court of law. It deals with the following issues:

Digital forensics
The scientific collection, examination, authentication, preservation, and analysis of electronically stored information (ESI).

- Recovering data from digital devices while preserving evidential integrity
- Securely storing and handling recovered electronic data
- Finding significant information in a large volume of electronic data
- Presenting the information to a court of law

ESI may reside on computer storage media in the form of computer files and as *ambient data*, which are not visible to the average user. An example might be a file that has been deleted from a hard drive. Data that a computer user may have deleted on computer storage media can often be recovered through various techniques. Digital forensics experts try to recover such hidden data for presentation as evidence.

An awareness of digital forensics should be incorporated into a firm's contingency planning process. The CIO, security specialists, information systems staff, and corporate legal counsel should all work together to have a plan in place that can be executed if a legal need arises.

8.7 Describe an Organizational Framework for Security and Control

Even with the best security tools, your information systems won't be reliable and secure unless you know how and where to deploy those tools. You'll need to know where your company is at risk and what controls you must have in place to protect your information systems. You'll also need to develop a security policy and plans for keeping your business running if your information systems aren't operational.

Information Systems Controls

Information systems controls are both manual and automated and consist of general and application controls. **General controls** govern the design, security, and use of computer programs and the security of data files in general throughout the organization's information

General controls
Overall control environment governing the design, security, and use of computer programs and the security of data files in general throughout the organization's information technology infrastructure.

Table 8.4 General Controls

Type of General Control	Description
Software controls	Monitor the use of system software and prevent the unauthorized access and use of software programs, system software, and computer programs.
Hardware controls	Ensure that computer hardware is physically secure and check for equipment malfunction. Organizations that are critically dependent on their computers must also make provisions for backup or continued operation to maintain constant service.
Computer operations controls	Oversee the work of the computer department to ensure that programmed procedures are consistently and correctly applied to the storage and processing of data. They include controls over the setup of computer-processing jobs and backup and recovery procedures for processing that ends abnormally.
Data security controls	Ensure that valuable business data files maintained internally or by an external hosting service are not subject to unauthorized access, change, or destruction while they are in use or in storage.
Implementation controls	Audit the systems development process at various points to ensure that the process is properly controlled and managed.
Administrative controls	Formalize standards, rules, procedures, and control disciplines to ensure that the organization's general and application controls are properly executed and enforced.

technology infrastructure. On the whole, general controls apply to all applications and consist of a combination of hardware, software, and manual procedures that create the overall control environment.

General controls include software controls, physical hardware controls, computer operations controls, data security controls, controls over the systems development process, and administrative controls. Table 8.4 describes the functions of each of these controls.

Application controls

Specific controls unique to each application that ensure that only authorized data are completely and accurately processed by that application.

Application controls are specific controls unique to each application, such as payroll or order processing. They include both automated and manual procedures that ensure that only authorized data are completely and accurately processed by that application. Application controls can be classified as (1) input controls, (2) processing controls, and (3) output controls.

Input controls check data for accuracy and completeness when the data enter the system. There are specific input controls for input authorization, data conversion, data editing, and error handling. *Processing controls* establish that data are complete and accurate during updating. *Output controls* ensure that the results of computer processing are accurate, complete, and properly distributed.

Information systems controls should not be an afterthought. They need to be incorporated into the design of a system and should consider not only the system's performance under all possible conditions but also the behavior of the organizations and people using the system.

Risk Assessment

Before your company commits resources to security and information systems controls, it must know which assets require protection and the extent to which these assets are vulnerable. A risk assessment helps answer these questions and determine the most cost-effective set of controls for protecting assets.

Risk assessment

Determines the level of risk to a firm if a specific activity or process is not properly controlled. Assesses the probability of occurrence and the potential damage if the problem were to occur. Used to determine the cost/benefit of a control.

A **risk assessment** determines the level of risk to a firm if a specific activity or process is not properly controlled. Not all risks can be anticipated and measured, but most businesses will be able to acquire some understanding of the risks they face. Business managers working with information systems specialists should try to determine the value of information assets, points of vulnerability, the likely frequency of a problem, and the potential for damage. For example, if an event is likely to occur no more than once a year, with a maximum of a $1,000 loss to the organization, it would not be wise to spend $20,000 on the design and maintenance of a control to protect against that event. However, if that same event could occur at least once a day, with a potential loss of more than $300,000 a year, $100,000 spent on a control might be entirely appropriate.

Table 8.5 illustrates sample results of a risk assessment for an online order-processing system that processes 30,000 orders per day. The likelihood of each exposure occurring over a one-year period is expressed as a percentage. The next column shows the highest and lowest

Table 8.5 Online Order-Processing Risk Assessment

Exposure	Probability of Occurrence (%)	Loss Range/Average ($)	Expected Annual Loss ($)
Power failure	30%	$5,000–$200,000 ($102,500)	$30,750
Embezzlement	5%	$1,000–$50,000 ($25,500)	$1,275
User error	98%	$200–$40,000 ($20,100)	$19,698

possible losses that could be expected each time the exposure occurred as well as an average loss calculated by adding the highest and lowest figures and dividing by two. The expected annual loss for each exposure can be determined by multiplying the average loss by its probability of occurrence.

This risk assessment shows that the probability of a power failure occurring in a one-year period is 30 percent. Loss of order transactions while power is down could range from $5,000 to $200,000 (averaging $102,500) for each occurrence, depending on how long processing is halted. The probability of embezzlement occurring over a yearly period is about 5 percent, with potential losses ranging from $1,000 to $50,000 (and averaging $25,500) for each occurrence. User errors have a 98 percent chance of occurring over a yearly period, with losses ranging from $200 to $40,000 (and averaging $20,100) for each occurrence.

After the risks have been assessed, system builders will concentrate on the control points with the greatest vulnerability and potential for loss. In this case, controls should focus on ways to minimize the risk of power failures and user errors because anticipated annual losses are highest for these areas.

Security Policy

After you've identified the main risks to your systems, your company will need to develop a security policy for protecting the company's assets. A **security policy** consists of statements ranking information risks, identifying acceptable security goals, and identifying the mechanisms for achieving these goals. What are the firm's most important information assets? Who in the firm generates and controls this information? What existing security policies are in place to protect the information? What level of risk is management willing to accept for each of these assets? Is it willing, for instance, to lose customer credit card data once every 10 years? Or will it build a security system for credit card data that can withstand the once-in-a-hundred-years disaster? Management must estimate how much it will cost to achieve this level of acceptable risk. Companies must also review their security policies on a regular basis and revise them as necessary to take into account risks posed by new technologies and threats.

Security policy

Consists of statements ranking information risks, identifying acceptable security goals, and identifying the mechanisms for achieving these goals.

The security policy drives other policies determining the acceptable use of the firm's information resources and which members of the firm will have access to its information assets. An **acceptable use policy (AUP)** defines acceptable, as well as unacceptable, uses of the firm's information resources and computing equipment, including desktop and laptop computers, mobile devices, telephones, and the Internet, and specifies consequences for noncompliance.

Acceptable use policy (AUP)

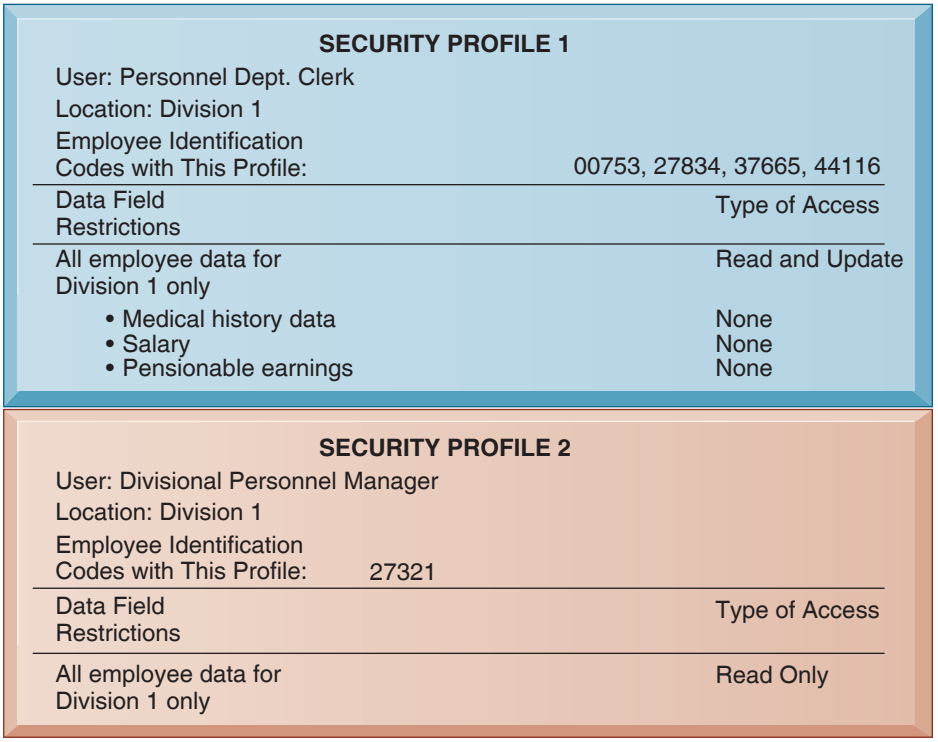
Defines acceptable, as well as unacceptable, uses of the firm's information resources and computing equipment, and specifies consequences for noncompliance.

Examine Figure 8.3, which shows one example of how an organization might specify the access rules for different levels of users in the human resources function. It specifies what portions of the human resources database each user is permitted to access, based on the information required to perform that person's job. The database contains sensitive personal information such as employees' salaries, benefits, and medical histories.

The access rules illustrated here are for two sets of users. One set of users consists of all employees who perform clerical functions, such as inputting employee data into the system. All individuals with this type of profile can update the system but can neither read nor update sensitive fields, such as salary, medical history, or earnings data. Another set of users consists of divisional managers who cannot update the system but can read all employee data fields, including medical history and salary, for their division. We provide more detail about the technologies for user authentication later in this chapter.

Figure 8.3 Access Rules for a Personnel System

These two examples represent two security profiles or data security patterns that might be found in a personnel system. Depending on the security profile, a user would have certain restrictions on access to various systems, locations, or data in an organization.



Disaster Recovery Planning and Business Continuity Planning

Disaster recovery planning
Involves creating plans for the restoration of disrupted computing and communications services.

If you run a business, you need to plan for events—such as power outages, natural disasters such as tornadoes, floods, earthquakes, and wildfires, and terrorist attacks—that may prevent your information systems and your business from operating. **Disaster recovery planning** involves creating plans for the restoration of disrupted computing and communications services. Disaster recovery plans focus primarily on the technical issues involved in keeping systems up and running, such as which files to back up and the maintenance of backup computer systems or disaster recovery services.

For example, MasterCard maintains a duplicate data center in Kansas City, Missouri, to serve as an emergency backup to its global technology and operations headquarters near St. Louis. Rather than building their own backup facilities, many firms contract with cloud-based disaster recovery services or firms that provide sites with spare computers around the country where subscribing firms can run their critical applications in an emergency.

Business continuity planning
Focuses on how the company can restore business operations after a disaster strikes.

Business continuity planning focuses on how the company can restore business operations after a disaster strikes. The business continuity plan identifies critical business processes and determines action plans for handling mission-critical functions if systems go down. For example, the Alaska USA Federal Credit Union has more than $11 billion in assets and branches in Alaska, Arizona, California, Idaho, and Washington State. Alaska USA's business continuity program consisted mainly of documents and spreadsheets that required manual updates. The process was quite time-consuming and did not provide Alaska USA with the real-time data that it required to make informed risk management decisions quickly. It decided to turn to software tools from Fusion Risk Management to remediate these issues. Doing so enabled Alaska USA to build a business continuity plan with real-time reporting capabilities, a mass notification

system to alert all of its sites to potential risks, and a methodology to break down potential silos across the organization to improve overall efficiency (Fusion Risk Management, 2023).

Business managers and information technology specialists need to work together on both types of plans to determine which systems and business processes are most critical to the company. They must conduct a business impact analysis to identify the firm's most critical systems and the impact that a systems outage would have on the business. Management must determine the maximum amount of time the business can survive with its systems down and which parts of the business must be restored first.

The Role of Auditing

How can management know that information systems security and controls are effective? To answer this question, organizations must conduct comprehensive and systematic audits. An **information systems audit** examines the firm's overall security environment as well as the controls governing individual information systems. The auditor should trace the flow of sample transactions through the system and perform tests using, if appropriate, automated audit software. The information systems audit may also examine data quality.

Information systems audit

Examines the firm's overall security environment as well as the controls that govern individual information systems.

Security audits review technologies, procedures, documentation, training, and personnel. A thorough audit will even simulate an attack or disaster to test the response of the technology, information systems staff, and business employees.

The audit lists and ranks all control weaknesses and estimates the probability of their occurrence. It then assesses the financial and organizational impact of each threat. Examine Figure 8.4, which provides a sample auditor's listing of control weaknesses for a loan system in a local commercial bank. It includes sections for listing the nature and impact of each weakness identified, the chance for error/abuse related to such weakness, and for notifying management of such weaknesses and for management's response. Management is expected to devise a plan for countering significant weaknesses in controls.

Figure 8.4 Sample Auditor's List of Control Weaknesses

This chart is a sample page from a list of control weaknesses that an auditor might find in a loan system in a local commercial bank. This form helps auditors record and evaluate control weaknesses and shows the results of discussing those weaknesses with management as well as any corrective actions management takes.

Function: Loans **Location: Peoria, IL**	**Prepared by: J. Callejas** **Date: May 15, 2024**		**Received by: T. Matoshi** **Review date: June 1, 2024**	
Nature of Weakness and Impact	Chance for Error/Abuse		Notification to Management	
	Yes/ No	Justification	Report date	Management response
User accounts with missing passwords	Yes	Leaves system open to unauthorized outsiders or attackers	5/10/24	Eliminate accounts without passwords
Network configured to allow some sharing of system files	Yes	Exposes critical system files to hostile parties connected to the network	5/10/24	Ensure only required directories are shared and that they are protected with strong passwords
Software patches can update production programs without final approval from Standards and Controls group	No	All production programs require management approval; Standards and Controls group assigns such cases to a temporary production status		

8.8 Discuss Technologies for Safeguarding Information Resources

Businesses have an array of technologies available for protecting their information resources. These technologies include tools for managing user identities, preventing unauthorized access to systems and data, ensuring system availability, addressing specific challenges raised by IoT devices, cloud computing, and mobile platform, and ensuring software quality. Increasingly, artificial intelligence is being deployed as part of many of these technologies and tools in an effort to enhance cybersecurity.

Identity and access management software (IAM software)

Software that automates the process of keeping track of users and system privileges, assigning each user a unique digital identity for accessing each system.

Zero trust

Popular cybersecurity framework based on the principle of maintaining strict access controls and not trusting anyone by default.

Authentication

The ability of know that people are who they claim to be.

Password

Secret word or string of characters that is supposed to be known only to the authorized user, and which is used to authenticate the user so the user can access a resource such as a computer system or specific files.

Security token

Physical device, similar to an identification card, that is designed to prove the identity of a single user.

Smart card

A plastic or metal about the size of a credit card that contains an embedded chip formatted with access permissions and other data. Used in conjunction with a card reader. Also used in digital payment systems.

Biometric authentication

Authentication based on the measurement of physical or behavioral characteristics, such as facial structure, fingerprints, retinal images, and voice, that makes each individual unique

Identity and Access Management and Authentication

Midsize and large companies have complex IT infrastructures and many systems, each with its own set of users. **Identity and access management software (IAM software)** software automates the process of keeping track of all these users and their system privileges, assigning each user a unique digital identity for accessing each system. It also includes tools for authenticating users, protecting user identities, and controlling access to system resources. **Zero trust** is a popular cybersecurity framework based on the principle of maintaining strict access controls and not trusting anyone or anything by default, even those behind a corporate firewall. The concept of *least privilege access* is a core part of the zero-trust framework. Least privilege access means that no user should have access to system resources beyond what is absolutely necessary to fulfill that user's specified tasks.

To gain access to a system, a user must be authorized and authenticated. **Authentication** refers to the ability to know that people are who they claim to be. Authentication is often established by using a **password**, a secret word or string of characters that is supposed to be known only to the authorized user, and which is used to authenticate the user so the user can access a resource such as a computer system or specific files. However, users often forget passwords, share them, or choose poor passwords that are easy to guess, which compromises security. Password systems that are too rigorous hinder employee productivity. When employees must change complex passwords frequently, they often take shortcuts, such as choosing passwords that are easy to guess or keeping their passwords at their workstations in plain view. Passwords can also be sniffed out if they are transmitted over a network or stolen through social engineering, as discussed in the Chapter Case Study, *Social Engineering Creates Chaos at MGM Resorts*.

Authentication technologies, such as security tokens, smart cards, and biometric authentication, overcome some of these problems. A **security token** is a physical device, similar to an identification card, that is designed to prove the identity of a single user. Security tokens are small gadgets that typically fit on key rings and display passcodes that change frequently. A **smart card** is a plastic or metal card about the size of a credit card that contains an embedded chip formatted with access permission and other data. (Smart cards are also used in digital payment systems.) A card reader interprets the data on the smart card and allows or denies access.

Biometric authentication is based on the measurement of physical or behavioral characteristics, such as facial structure, fingerprints, retinal images, and voice, that makes each individual unique. A biometric authentication system compares the person's unique characteristics against a stored profile of those characteristics to determine any differences between the characteristics and the stored profile. If the two profiles match, access is granted. Fingerprint and facial recognition technologies are now being used for many applications, with many mobile devices equipped with fingerprint identification and facial recognition software. Financial services firms such as Vanguard and Fidelity have implemented voice authentication systems for their clients. AI is increasingly being used to analyze behavioral biometrics such as typing speed and mouse movements to verify identity.

The Apple iPhone has a facial recognition system called Face ID that enables users to log into the phone as well as access various features by just looking at the screen.

© Tada Images/Shutterstock

The steady stream of incidents in which threat actors have been able to access traditional passwords highlights the need for more secure means of authentication. **Multifactor authentication (MFA)** increases security by validating users via a multistep process. Authentication credentials might include something the user knows, such as a password or personal identification number (PIN); something the user possesses, such as a smartphone or security token; and something the user "has," such as a physical characteristic embodied in biometric data like fingerprints, iris prints, or voice prints. **Two-factor authentication (2FA)** is a subset of MFA that requires two credentials. A common example of two-factor authentication is a bank card. The card itself is the physical item that the user possesses, and the PIN is the credential that goes with the physical item. AI is being used in MFA to assess risk factors and adapt MFA requirements accordingly.

Multifactor authentication (MFA)

Increases security by validating users via a multistep process.

Two-factor authentication (2FA)

Subset of MFA that validates user identity with two means of identification, one of which is typically a physical item that the user possesses, and the other of which is typically data.

Firewalls, Intrusion Detection and Prevention Systems, and Anti-Malware Software

Without protection against malware and intruders, connecting to the Internet would be very dangerous. Firewalls, intrusion detection and prevention systems, and anti-malware software have become essential business tools.

Firewalls

A **firewall** prevents unauthorized users from accessing private networks. A firewall today can either be a hardware appliance, software-based, or delivered as a service. The firewall filters communications packets and prevents some from entering or exiting the network based on security policies. Examine Figure 8.5, which illustrates how a firewall works. An outer firewall is generally placed between the organization's private internal networks and distrusted external networks such as the Internet, although inner firewalls can also be used to protect one part of a company's network from the rest of the network.

Firewall

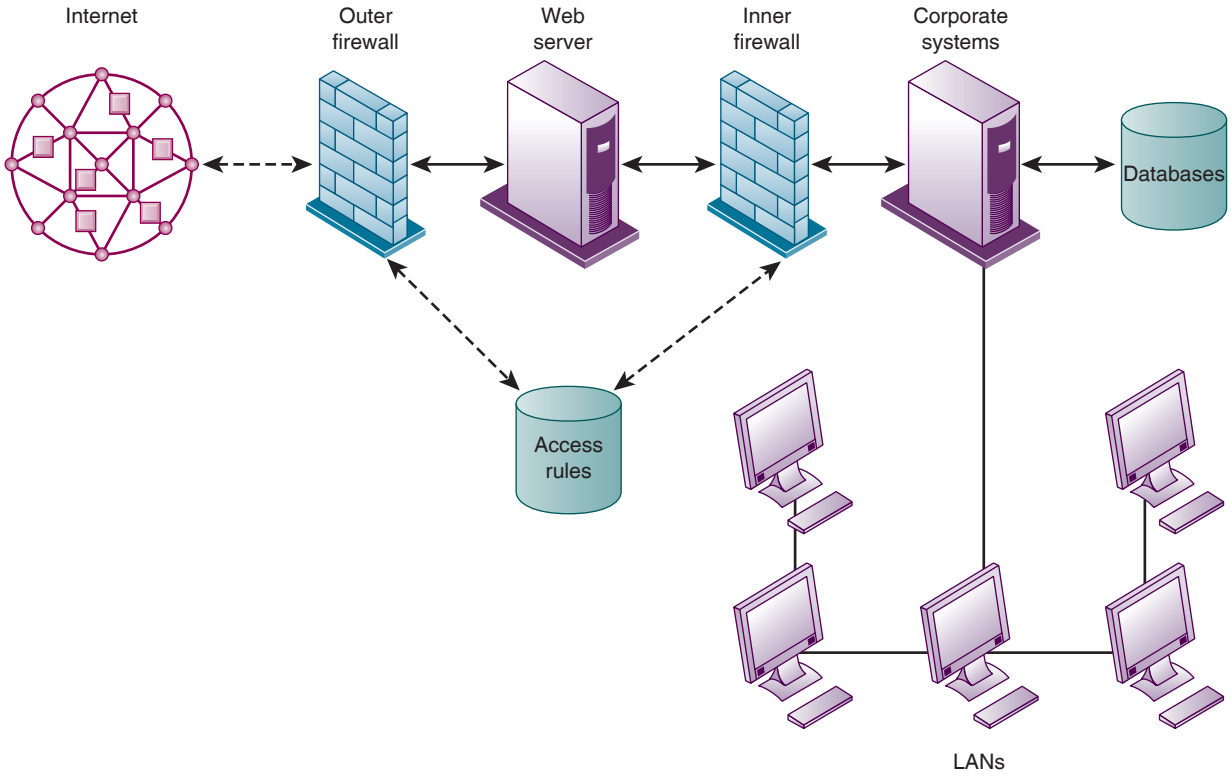
Prevents unauthorized users from accessing private networks; can be either a hardware appliance, software-based, or delivered as a service

The firewall acts like a gatekeeper that examines each user's credentials before it grants access to a network. The firewall identifies names, IP addresses, applications, and other characteristics of incoming traffic. It checks this information against the access rules that the network administrator has programmed into the system. The firewall prevents unauthorized communication going into and out of the network.

In large organizations, the firewall often resides on a specially designated computer that is separate from the rest of the network so that no incoming request directly accesses

Figure 8.5 A Corporate Firewall

The firewall is placed between the firm's private network and the public Internet or another distrusted network to protect against unauthorized traffic.



private network resources. There are a number of firewall technologies, including static packet filtering, stateful inspection, Network Address Translation, application gateways, and next-generation firewalls. They are frequently used in combination to provide enhanced protection.

Packet filtering examines selected fields in the headers of data packets flowing back and forth between the trusted network and the Internet, examining individual packets in isolation. This filtering technology can miss many types of attacks.

Stateful inspection provides additional security by determining whether packets are part of an ongoing dialogue between a sender and a receiver. It sets up state tables to track information over multiple packets. Packets are accepted or rejected based on whether they are part of an approved conversation or attempting to establish a legitimate connection.

Network Address Translation (NAT) can provide another layer of protection when static packet filtering and stateful inspection are employed. NAT conceals the IP addresses of the organization's internal host computer(s) to prevent sniffer programs outside the firewall from ascertaining the IP addresses and using that information to penetrate internal systems.

Application gateways examine the application content of packets. A proxy server stops data packets originating outside the organization, inspects them, and passes a proxy to the other side of the firewall. If a user outside the company wants to communicate with a user inside the company, the outside user first communicates with the proxy application, and the proxy application then communicates with the firm's internal computer. Likewise, a computer user inside the organization goes through the proxy to connect to computers on the outside of the organization. A *web application firewall (WAF)* is an optional component that can be added to the application gateway. A WAF inspects HTTP/S requests and helps prevent malicious attacks such as SQL injection attacks, DoS attacks, and cross-site scripting.

A *next-generation firewall (NGFW)* uses an application-centric approach to firewall control and combines the best features of a traditional firewall and an application gateway. An NGFW is able to identify applications regardless of the port, protocol, or security evasion tools used. It can also identify users regardless of device or IP address, decrypt outbound TLS traffic, and protect in real time against threats embedded in applications.

To create a good firewall, an administrator must maintain detailed internal rules identifying the people, applications, or addresses that are allowed or rejected. Firewalls can deter, but not completely prevent, network penetration by outsiders and should be viewed as only one element of an overall security plan. AI is currently being used to assist in firewall rules optimization by analyzing network traffic and suggesting rule adjustments for better security.

Intrusion Detection and Prevention Systems

In addition to firewalls, commercial security vendors now provide intrusion detection tools and services to protect against suspicious network traffic and attempts to access files and databases. An **intrusion detection system (IDS)** features full-time monitoring tools placed at the most vulnerable points or hot spots of corporate networks to detect and deter intruders continuously. The system generates an alarm if it finds a suspicious or anomalous event. Scanning software looks for patterns indicative of known methods of computer attacks such as bad passwords, checks to see whether important files have been removed or modified, and sends warnings of vandalism or system administration errors. The intrusion detection tool can also be customized to shut down a particularly sensitive part of a network if it receives unauthorized traffic. IDS are increasingly incorporating the use of AI to detect and respond to network anomalies.

Intrusion detection system (IDS)

Tools to monitor the most vulnerable points in a network to detect and deter unauthorized intruders.

An **intrusion prevention system (IPS)** has all the functionalities of an IDS, with the additional ability to take steps to prevent and block suspicious activities. For instance, an IPS can terminate a session and reset a connection, block traffic from a suspicious IP address, or reconfigure firewall or router security controls. As with IDSs, IPSs are increasingly incorporating AI that enables them to automatically respond to security incidents, block malicious IP addresses, quarantine compromised devices, and send alerts.

Intrusion prevention system (IPS)

Has all the functionalities of an IDS, with the additional ability to take steps to prevent and block suspicious activities.

Anti-Malware Software

Defensive technology plans for both individuals and businesses must include anti-malware protection for every computer. **Anti-malware software** detects and eliminates malware, including viruses, worms, Trojans, spyware, and adware. However, most anti-malware software is effective only against malware already known when the software was written. To remain effective, the software must be continually updated. Even then it is not always effective because some malware can evade detection. Organizations need to use additional malware detection tools for better protection. AI is being used by anti-malware software developers to learn from historical malware data to identify new threats. AI is also being used in anti-malware software to scan files for known malicious signatures, helping prevent infections, as well as analyze code behavior to detect zero-day exploits.

Anti-malware software

Software designed to detect, and often eliminate, malware from an information system.

Unified Threat Management Systems

To help businesses reduce costs and improve manageability, security vendors offer **unified threat management systems (UTM systems),** which combine various security tools, including firewalls, virtual private networks, intrusion detection and prevention systems, and web content filtering and anti-spam software, into a single product. UTM systems are available for all sizes of networks. Leading UTM system vendors include Fortinet, Sophos, and Check Point, and networking vendors such as Cisco Systems and Juniper Networks provide some UTM system capabilities in their products.

Unified threat management systems (UTM systems)

Comprehensive security management tool that combines multiple security tools, including firewalls, virtual private networks, intrusion detection systems, and web content filtering and anti-spam software.

Securing Wireless Networks

The initial security standard developed for Wi-Fi, called Wired Equivalent Privacy (WEP), was not very effective because its encryption keys were relatively easy to crack. WEP has been replaced by Wi-Fi Protected Access 2 (WPA2), which provides stronger security standards.

Instead of the static encryption keys used in WEP, WPA2 uses much longer keys that continually change, making them harder to crack. The most recent specification is WPA3, introduced in 2018. However, even the updated WPA3 standard has vulnerabilities that could allow attackers to recover passwords. Companies can further improve Wi-Fi security by using Wi-Fi in conjunction with virtual private network (VPN) technology when accessing internal corporate data.

Encryption and Public Key Infrastructure

Encryption
The coding and scrambling of messages to prevent their being read or accessed without authorization.

Many businesses use encryption to protect digital information that they store, physically transfer, or send over the Internet. **Encryption** is the process of transforming plain text or data into cipher text that cannot be read by anyone other than the sender and the intended receiver. Data are encrypted by using a secret numerical code, called an encryption key, that transforms plain data into cipher text. The message must be decrypted by the receiver.

Transport Layer Security (TLS)
Enables client and server computers to manage encryption and decryption activities as they communicate with each other during a secure web session; successor to Secure Sockets Layer protocol.

HTTPS
Secure version of the HTTP protocol that uses TLS for encryption and authentication.

Two methods for encrypting network traffic on the web are TLS and HTTPS. **Transport Layer Security (TLS)**, the successor to Secure Sockets Layer (SSL), enables client and server computers to manage encryption and decryption activities as they communicate with each other during a secure web session. TLS is used in conjunction with **HTTPS**, a secure version of the HTTP protocol that uses TLS for encryption and authentication. It is implemented by a server adopting the HTTP Strict Transport Security (HSTS) feature, which forces browsers to access the server using only HTTPS. Today, more than 85 percent of websites use HTTPS as their default protocol (W3techs.com, 2024).

The capability to generate secure sessions is built into Internet client browser software and servers. The client and the server negotiate what key and what level of security to use. Once a secure session is established between the client and the server, all messages in that session are encrypted.

Two methods of encryption are symmetric key encryption and public key encryption. In symmetric key encryption, the sender and the receiver establish a secure Internet session by creating a single encryption key and sending it to the receiver so that both the sender and the receiver share the encryption key. The strength of the encryption key is measured by its bit length. Today, a typical private key will be 56 to 256 bits long (a string of from 56 to 256 binary digits) depending on the level of security desired. The longer the key, the more difficult it is to break the key. The downside is that the longer the key, the more computing power it takes for legitimate users to process the information.

Public key encryption
Form of encryption that uses two keys: one shared (or public) and one private.

The problem with all symmetric encryption systems is that the key itself must be shared somehow among the senders and receivers, which exposes the key to outsiders who just might be able to intercept and decrypt the key. A more secure form of encryption called **public key encryption** uses two keys: one shared (or public) and one totally private. The keys are mathematically related so that data encrypted with one key can be decrypted using only the other key. RSA is a public key encryption algorithm that is widely used. The typical size of RSA public keys is 2,048 bits, although RSA public keys can range from 1,024 bits to 4,096 bits. Examine Figure 8.6, which illustrates how public key encryption works. To send and receive messages, communicators first create separate pairs of private and public keys. The public key is kept in a directory, and the private key must be kept secret. The sender encrypts a message with the recipient's public key. Upon receiving the message, the recipient uses their private key to decrypt it.

Digital certificate
A data file used to establish the identity of users and digital assets to protect online transactions.

A digital certificate is a data file used to establish the identity of users and digital assets to protect online transactions. A digital certificate system uses a trusted third party, known as a certificate authority (CA), to validate a user's identity. There are many CAs in the United States and around the world, including DigiCert, GoDaddy, and Sectigo.

Figure 8.6 Public Key Encryption

A public key encryption system can be viewed as a series of public and private keys that lock data when they are transmitted and unlock data when they are received. The sender locates the recipient's public key in a directory and uses the key to encrypt a message. The message is sent in encrypted form over the Internet or a private network. When the encrypted message arrives, the recipient uses their private key to decrypt the data and read the message.

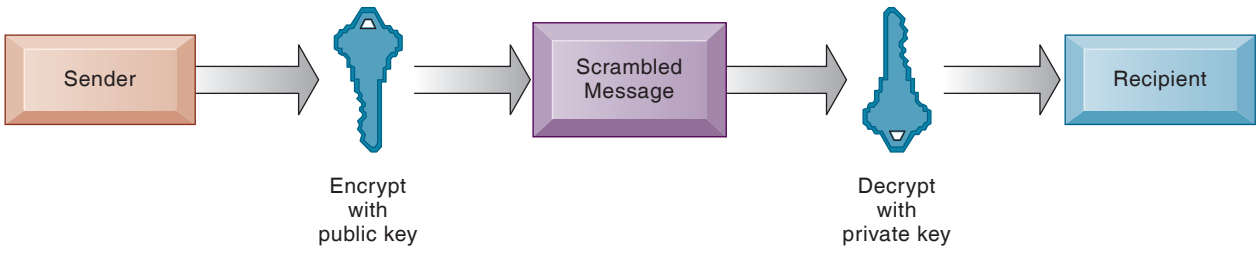
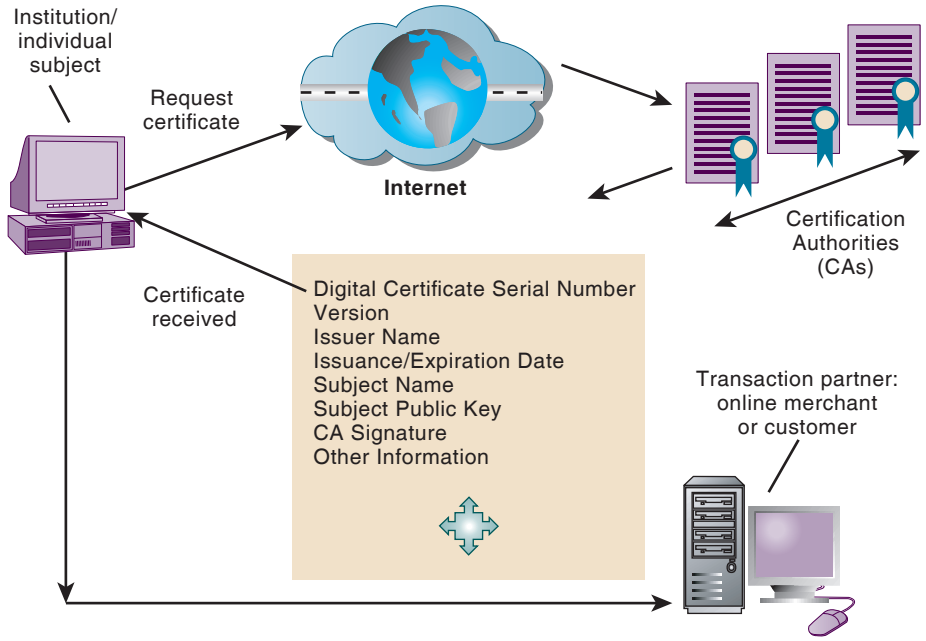


Figure 8.7 Digital Certificates

Digital certificates help establish the identity of people or digital assets. They protect online transactions by providing secure, encrypted, online communication.



Examine Figure 8.7, which illustrates how digital certificates work. An institution or an individual requests a digital certificate. The CA verifies the user's identity offline. This information is put into a CA server, which generates an encrypted digital certificate containing owner identification information and a copy of the owner's public key. The certificate authenticates that the public key belongs to the designated owner. The CA makes its own public key available either in print or perhaps on the Internet. The recipient of an encrypted message uses the CA's public key to decode the digital certificate attached to the message, verifies it was issued by the CA, and then obtains the sender's public key and identification information contained in the certificate. By using this information, the recipient can send an encrypted reply. The digital certificate system enables, for example, a credit card user and a merchant to validate that their digital certificates were issued by an authorized and trusted third party before they exchange data. **Public key infrastructure (PKI)**, the use of public key encryption working with a CA, is now widely used in e-commerce.

Public key infrastructure (PKI)

System that uses public key encryption, certificate authorities (CA,) and digital certificates for authentication.

Advances in quantum computing, introduced in Chapter 5, are raising the concern that dramatically more powerful quantum computers will have the ability to crack data encryption methods currently in use. Although this scenario is still likely a decade or more in the future, researchers are sounding the alarm now and urging the development of new cryptography methods. The rapid advance of artificial intelligence is also a complicating factor. For instance, the National Institute of Standards and Technology (NIST) released an algorithm for post-quantum cryptography in July 2022, but not long thereafter, researchers reported that they had broken the code using artificial intelligence combined with a more traditional form of attack. Analysts believe that organizations need to begin preparing now for a future in which current encryption methods no longer work (Schmieg et al, 2024; Hattar, 2023).

Securing Transactions with Blockchain

Blockchain, introduced in Chapter 6, is gaining attention as an alternative approach for securing transactions and establishing trust among multiple parties. A blockchain is a chain of digital "blocks" that contain records of transactions. Each block is connected to all the blocks before and after it, and the blockchains are continually updated and kept in sync. This makes it difficult to tamper with a single record because one would have to change the block containing that record as well as those linked to it to avoid detection.

Once recorded, a blockchain transaction cannot be changed. The records in a blockchain are secured through cryptography, and all transactions are encrypted. Blockchain network participants have their own private keys that are assigned to the transactions they create and act as a personal digital signature. If a record is altered, the signature will become invalid, and the blockchain network will know immediately that something is amiss. Because blockchains aren't contained in a central location, they don't have a single point of failure and cannot be changed from a single computer. Researchers point out, however, that blockchain is vulnerable in some of the same ways that conventional, centralized record-keeping systems are vulnerable. Blockchain systems also require the same careful attention to security and control that is given to other systems with high security requirements (Madnick, 2020).

Cryptocurrencies

Digital assets that use blockchain technology and cryptography to create a medium of exchange (currency).

Cryptocurrencies are digital assets that use blockchain technology and cryptography to create a medium of exchange (currency). Bitcoin is the most prominent example of cryptocurrency in use today, but many other alternative cryptocurrencies have emerged in recent years. The availability of cryptocurrencies is often considered to be a factor in the rise of ransomware, as it enables attackers to demand and receive payment in a form that is hard for authorities to trace.

Ensuring System Availability

As companies increasingly rely on digital networks for revenue and operations, they need to take additional steps to ensure that their systems and applications are always available. Firms such as those in the airline and financial services industries, which have critical applications requiring online transaction processing, have traditionally used fault-tolerant computer systems to ensure 100 percent availability. In online transaction processing, transactions entered online are immediately processed by the computer. Multitudinous changes to databases, reporting, and requests for information occur each instant.

Fault-tolerant computer system

System that contains redundant hardware, software, and power supply components that provide back up for a system and keep it running to prevent system failure.

A fault-tolerant computer system contains redundant hardware, software, and power supply components that create an environment that provides continuous, uninterrupted service. Fault-tolerant computers use special software routines or self-checking logic built into their circuitry to detect hardware failures and automatically switch to a backup device. In addition, parts from these computers can be removed and repaired without disruption to the computer or downtime. **Downtime** refers to periods of time in which a system is not operational.

Downtime

Periods of time when a system is not operational.

Security Operations Centers and Security Outsourcing

Security operations center (SOC)

Team of IT security professionals dedicated to monitoring an organization's entire IT infrastructure on a round-the-clock basis in order to detect, analzye and respond to security incidents in real time.

A **security operations center** (SOC, pronounced "Sock") is a team of IT security professionals dedicated to monitoring an organization's entire IT infrastructure on a round-the-clock basis, in order to detect, analyze, and respond to security incidents in real-time. SOCs are increasingly using AI tools to automate routine tasks and to analyze data from various sources to detect suspicious activities and predict emerging threats. AI tools are also being used to scan systems for vulnerabilities, prioritize them based on risk, and recommend patches or other steps to mitigate risks.

Many companies, especially small businesses, lack the resources or expertise to create a secure, high-availability computing environment or SOC on their own. However, they can outsource many, if not all, functions that would be provided by a SOC team to a **managed security service provider (MSSP)**, a company that provides security management services for subscribing clients.

Managed security service provider (MSSP)

Company that provides security management services for subscribing clients.

Achieving Digital Resiliency

Today's organizations are much more hypernetworked and interconnected than organizations were in the past, with important parts of their IT infrastructures maintained remotely in the cloud, managed by outsiders, and accessible by mobile devices. Firms are embracing the concept of digital resiliency to deal with the realities of this new digital environment.

Digital resiliency

Deals with maintaining and increasing the resilience of an organization and its business processes in an all-pervasive digital environment, not just the resiliency of the IT function.

Digital resiliency deals with maintaining and increasing the resilience of an organization and its business processes in an all-pervasive digital environment, not just the resilience of the IT function. In addition to computing, storing, and networking technologies, digital resiliency calls attention to managerial and organizational issues such as corporate policies and goals,

business processes, organizational culture, business requirements, accountability, and business risk management. These factors can affect an organization's ability to actually utilize and manage network connectivity, applications, databases, and data centers; its ability to provide 24/7 availability to its customers; and its ability to respond to changing business conditions. If resiliency has not been explicitly designed in, measured, and tested, a single weak link in this chain can cause an outage or prevent the firm from responding to new challenges and opportunities. For instance, in March 2024, McDonald's experienced a global technology outage that resulted in widespread disruptions at McDonald's locations around the world. Many restaurants could not take orders or process payments. Investigations revealed that the outage was precipitated by a third-party provider during a configuration change (a form of software system maintenance). Although the issue was resolved by the next day, McDonald's still suffered damage in terms of lost revenue and to its brand image.

The Spotlight on Organizations Case Study, *Uptime Institute: Understanding and Preventing System Outages,* illustrates how Uptime Institute, a research and advisory firm focused on data center operations, is helping companies avoid digital infrastructure failures.

Spotlight on Organizations Case Study

Uptime Institute: Understanding and Preventing System Outages

Uptime Institute (UI) is a research and advisory firm that focuses on improving the performance, efficiency, and reliability of data centers. Data centers are at the center of a company's digital infrastructure and contain the networked computers on which software applications and systems reside, storage systems, and the telecommunications that allow the systems to communicate. As you learned in Chapter 5, companies may operate their own data centers, rely on third parties, such as private or public cloud services, or use a hybrid approach.

Each year UI issues a report that examines data center digital resiliency, based on publicly reported outages, the results of UI surveys of data center operators and vendors, and a proprietary Uptime Abnormal Incident Database that provides detailed site/facility-level data shared with UI under a non-disclosure agreement. The report provides critical information to the data center industry with the goal of helping data center operators understand the causes, costs, and consequences of digital infrastructure outages. Understanding the root cause of an outage is critical for preventing a repeat occurrence and to identify areas that require greater attention and investment to mitigate risks.

UI notes that while the reliability of critical IT systems, networks, and data centers have significantly improved through the years, data center operators today are facing new challenges from increased demand, the complexity of today's digital infrastructure, and the growing number of cyberthreats. In addition, the impact, as well as the visibility, of such failures has increased due to society's increased dependence on IT.

In 2023, the number of publicly reported outages that UI categorized as either serious or severe increased by 12 percent compared to 2022. Examples included an outage at the U.S. Federal Aviation Administration due to a configuration error, when stored files in a pilot-alert system were mistakenly deleted, leading to the disruption of more than 30,000 flights and a ransomware attack on Western Digital that took its cloud services offline, locking millions of users out from their stored files.

According to UI, the most common cause of third-party IT service provider outages is software or configuration errors (63 percent), followed by networking or connectivity issues (35 percent). Malicious cyberattacks account for 20 percent of outages.

UI identifies human error, either directly or indirectly, as contributing to a significant majority of all outages. Human error may be related to a lack of training, staff failing to follow procedures, the procedures themselves being inadequate, complexity of required tasks and equipment operation, or having insufficient staff, among other factors. According to UI, nearly 80 percent of operators report that better management and procedures would have prevented their organization's most recent outages.

With human error responsible for so many data center incidents, organizations need to take a more holistic approach to staffing, organizational practices, maintenance and operations activities, management, and planning. UI's Management and Operations (M&O) assessment program is aimed at helping companies do so. The assessment measures the operational effectiveness of the teams within a data center, focusing on staffing and organization, maintenance, training, planning, coordination and management, and operating conditions. UI notes that it is important for companies to recognize the influence of management on factors such as staffing, resource allocation, and organizational culture. As it assesses a data center's staffing, organization, process and procedures, planning and management, UI also

provides recommendations with respect to industry best practices to minimize downtime. Data centers that pass the assessment receive a UI "M&O Stamp of Approval." So far, more than 350 around the world have done so.

SOURCES: Uptimeinstitute.com, accessed September 9, 2024; Denise Dubie, "Network Connectivity Are Leading Cause of IT Service Outages," Networkworld.com, April 4, 2024; Uptime Institute, "Annual Outage Analysis 2024," March, 2024; and Wade Tyler Millward, "The 15 Biggest Cloud Outages of 2023," Crn.com, December 13, 2023.

Case Study Questions

1. Why is digital resiliency so important for data centers?
2. Why are system outages likely to have a more significant impact today?
3. How does UI's Management and Operations (M&O) assessment program help make an organization's digital infrastructure more resilient?

Addressing Internet of Things, Cloud Computing, and Mobile Platform Security Challenges

Although the Internet of Things (IoT), cloud computing, and the mobile platform deliver powerful benefits, they also pose new challenges to system security and reliability. We now describe some of these challenges and how they should be addressed.

Internet of Things (IoT) Security Challenges

The Internet of Things (IoT) introduces a number of security challenges: from the Internet-linked devices themselves, their platforms and operating systems, and their communications with the systems to which they're connected. New security tools will be required to protect IoT devices and platforms from both information attacks and physical tampering, to encrypt their communications, and to address new challenges such as attacks that drain batteries. Many IoT devices, such as sensors, have simple processors and operating systems that may not support sophisticated security approaches. Table 8.6 describes some of the unique security challenges posed by IoT.

Cloud Computing Security Challenges

Using the public cloud disrupts the traditional cybersecurity models that companies may have created over the years. Although many organizations know how to manage the security for their own data center, they're often unsure of exactly what they need to do when they shift computing work to the cloud. They need new tool sets and skill sets to manage cloud security

Table 8.6 Internet of Things Security Challenges

Challenge	Implications
Many IoT devices, such as sensors, are deployed on a much wider scale than traditional Internet-connected devices are, thus creating a vast quantity of interconnected links that can be exploited.	New tools, methods, and strategies need to be developed to deal with the unprecedented scale presented by IoT devices.
Many IoT installations consist of collections of identical devices that share the same characteristics.	The potential impact of a security vulnerability is magnified.
Many IoT devices have a much longer service life than typical equipment does.	Devices may outlive their manufacturers, leaving them without long-term support, which could create persistent vulnerabilities.
Many IoT devices cannot be upgraded, or the upgrade process is difficult.	Raises the possibility that vulnerable devices cannot or will not be fixed, leaving them perpetually vulnerable.
Many IoT devices don't provide visibility into how the device works or the data being produced; nor do they alert the user when a security problem arises.	Users may believe an IoT device is functioning as intended when, in fact, it may be performing in a malicious manner.
Some IoT devices are so unobtrusive that a user may not even be aware of the device.	A security breach might persist for a long time before being noticed.

from their end to configure and launch cloud instances (a virtual server that runs workloads in the cloud), manage identity and access controls, update security controls to match configuration changes, and protect workloads and data. Companies that use a hybrid cloud model, which includes using both the company's own data centers as well as a public cloud service, face particular challenges because of the cloud's added complexity and the need to blend various components and frameworks.

In addition, there's still a misconception held by some IT departments that whatever happens in the cloud is not their responsibility. This is not the case. The category of cloud service (IaaS, PaaS, or SaaS) affects exactly how these responsibilities are shared. For IaaS, for example, the cloud service provider (CSP) typically supplies and is responsible for securing basic IT resources such as machines, storage systems, and networks. The cloud services customer is typically responsible for its operating system, applications, and corporate data placed into the cloud computing environment. This means that most of the responsibility for securing the applications and the corporate data falls on the customer. Therefore, cloud service customers should carefully review their cloud services agreement with their CSP to make sure their applications and data hosted in cloud services are secured in accordance with their own security and compliance policies.

It is essential for companies to revise their cybersecurity practices so that they can both protect critical data and fully exploit the speed and agility that cloud services provide. Basic hybrid cloud security best practices include running continuous audits to have real-time visibility into threats and any irregularities in the company's cloud environment, adopting zero-trust principles including least privilege access (which, in the context of hybrid cloud security, requires that all interactions between the company's private data center and the public cloud be as limited as possible while still being able to achieve operational goals), establishing unified security management, and embracing artificial intelligence and automation to help detect and resolve security risks.

Cloud computing also raises issues about regulatory compliance with respect to data security and protection. Public cloud applications reside in remote data centers that supply business services and data management to multiple corporate clients. To save money and keep costs low, CSPs often distribute work to data centers around the globe where the work can be accomplished most efficiently. Cloud users need to confirm that regardless of where their data are stored, they are protected at a level that meets their corporate requirements. Users should stipulate that the CSP store and process data in specific jurisdictions according to the privacy rules of those jurisdictions. Cloud clients should also find out how the CSP segregates their corporate data from those of other companies and ask for proof that encryption mechanisms are sound.

In addition, it's important to know how the CSP will respond if a disaster strikes, whether the provider will be able to restore the client's data completely, and how long this should take. Cloud users should also ask whether CSPs will submit to external audits and security certifications. These kinds of controls can be written into the service level agreement (SLA) before signing with a CSP. The Cloud Security Alliance (CSA) has created industry-wide standards for cloud security, specifying best practices to secure cloud computing.

Securing Mobile Platforms

Just like desktops and laptops, mobile devices need to be secured against malware, theft, accidental loss, unauthorized access, and hacking attempts. Mobile devices accessing corporate systems and data require special protection. Companies should make sure that their corporate security policy includes mobile devices, with additional details on how mobile devices should be supported, protected, and used. Companies will need mobile device management (MDM) tools to authorize all devices in use; to maintain accurate inventory records on all mobile devices, users, and applications; to control updates to applications; and to lock down or erase lost or stolen devices so that they can't be compromised. Data loss prevention technology can identify where critical data are saved, who is accessing the data, how data are leaving the company, and where the data are going. Firms should develop guidelines stipulating approved mobile platforms and software applications as well as the required software and procedures for remote access of corporate systems. The organization's mobile security policy should also forbid employees from using unsecured, consumer-based applications for

transferring and storing corporate documents and files or sending such documents and files to oneself by email without encryption. Companies should encrypt communication whenever possible. All mobile device users should be required to use the most advanced security features available for their particular smartphones. Mobile security has become even more important in the wake of the Covid-19 pandemic, as more of the workforce continues to work remotely, in many cases using mobile devices.

Ensuring Software Quality

In addition to implementing effective security and controls, organizations can improve system quality and reliability by employing software metrics and rigorous software testing. Software metrics are objective assessments of the system in the form of quantified measurements. Ongoing use of metrics allows the information systems department and end users to measure the performance of the system jointly and to identify problems as they occur. Examples of software metrics include the number of transactions that can be processed in a specified unit of time, the online response time, the number of payroll checks printed per hour, and the number of known bugs per hundred lines of program code. For metrics to be successful, they must be carefully designed, formal, objective, and used consistently.

Early, regular, and thorough testing will contribute significantly to system quality. Many view testing as a way to prove the correctness of the work they have done. In fact, all sizable software is riddled with errors and must be tested to uncover these errors.

Good testing begins before a software program is even written, by using a *walkthrough*—a review of a specification or design document by a small group of people carefully selected based on the skills needed for the particular objectives being tested. When developers start writing software programs, coding walkthroughs can also be used to review program code. However, code must be tested by computer runs. When errors are discovered, the source is found and eliminated through a process called *debugging*. You can find out more about the various stages of testing required to put an information system into operation in Chapter 12.

8.9 Understand How the Information in This Chapter Can Help Your Career

The information you have learned in Chapter 8 can help you advance your career by making you aware of the threats that are posed to information systems by malware, threat actors, and software vulnerabilities, as well as new security issues that are being raised by artificial intelligence. In addition, Chapter 8 also helps you understand the business value of security and controls and the components of an organizational framework for security and control, which are critical for any role in a business. Finally, you also learned about the most important tools and technologies for safeguarding information resources, something that is extremely valuable, given that cybersecurity issues are some of the most foremost challenges that all businesses face today.

Real-World Scenario

FBI Cybersecurity Specialists Help Turn the Tables on the Hive Ransomware Group

Between June 2021 and January 2023, the notorious Hive ransomware group targeted more than 1,500 victims, including hospitals, school districts, financial firms, and critical infrastructure in over 80 countries around the world. The group was able to encrypt the data in victims' systems so that it could not be accessed and extorted a number of multimillion-dollar ransom payments from victims. The U.S. Department of Justice has finally put a stop to that.

An interdisciplinary FBI team, comprised of FBI special agents and cybersecurity specialists, was able to successfully penetrate Hive's computer networks and captured its decryption keys, and offered them to victims worldwide, preventing victims from having to pay $130 million in ransom demanded. Since infiltrating Hive's network, the FBI has provided more than 300 decryption keys to Hive victims who were under attack. In addition, the FBI distributed over 1,000 additional decryption keys to previous Hive victims. In coordination with German law enforcement and the Netherlands National High Tech Crime Unit, the FBI also seized control of the servers and websites that Hive used to communicate with its members, disrupting Hive's ability to attack and extort victims. Going forward, the FBI is encouraging students to consider cyber careers with the FBI, and it is actively seeking computer scientists, cybersecurity specialists, digital forensic examiners and specialists, data analysts, and IT specialists. The FBI has also launched a Model Cyber Squad initiative that is seeking to create interdisciplinary teams in every one of its field officers similar to the one that took down the Hive ransomware network.

Job Description and Career Path: Cybersecurity/IT Security Specialist

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Cybersecurity/IT Security Specialist	• Identify and implement security measures during the development of software systems and networks. • Evaluate and identify vulnerabilities and risks in hardware and software. • Determine the most effective ways to secure IT infrastructure. • Test security strategies and defenses. • Monitor systems for intrusions and cyberattacks. • Analyze new methods that are being used by cybercriminals. • Keep up to date on new trends, strategies, and procedures to thwart attacks. • Provide training to colleagues on security protocols and best practices.	• Bachelor's degree in computer science, engineering, information science, information systems management, technology management, operations research, or a related field. • Familiarity with: intrusion detection; malware analysis and mitigation; and identity and access management tools. • Some programming knowledge (C, C++, C#, Perl, Java, etc.). • Ability to "think like a hacker". • Proficiency in using spreadsheet, database, and word processing software.	• Entry level: $60,000–$100,000 • Mid-level: $80,000–130,000 • Senior: $100,000–165,000 Salaries can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

Social Engineering Creates Chaos for MGM Resorts

On Friday September 8, 2023, MGM Resorts' IT service desk received a phone call from someone who claimed to be an employee. What service desk personnel did not realize was that the caller was a hacker.

Just a week prior to the hack, Okta, an online identity management service used by MGM Resorts and many other companies, had warned its clients of the potential for social engineering "vishing" attacks of just this type, in which the caller, impersonating an employee using information found online from sources such as company websites, social media, and the dark web, asks IT service desk personnel to reset passwords or multi-factor authentication (MFA) factors for Okta Super Administrator accounts. In this case, the warning was apparently not heeded.

According to a former MGM Resorts employee familiar with the company's cybersecurity policies, to obtain a password reset, employees only had to disclose basic information about themselves, such as their name, employee ID number, and date of birth. What was actually said in the fateful call has not been publicly disclosed, but the hacker apparently was able to convince service desk personnel that the call was a legitimate one. A few minutes later, the employee who the caller was impersonating received a notification that his password had been reset. The employee immediately notified the IT department, but by then, it was already too late. What followed was chaos that at first began slowly and then picked up speed.

On Saturday, September 9, the hackers began drilling into MGM Resorts' corporate networks, moving from one computer system to another, stealing data as they went. MGM Resorts' IT staff tried to shut down affected accounts, but the hackers, who now had super administrator access credentials and advanced privileges that gave them wide authority over internal services, always seemed to be able to stay one step ahead. On Sunday, September 10, MGM Resorts' CEO gave the order to start shutting down its systems, including its email system, to try to limit the damage. On Monday, September 11, as MGM Resorts scrambled to manage more than 37,000 hotel rooms, as well as its casino, in manual mode, the impact of the attack started to become visible to MGM Resorts visitors. For instance, about a third of MGM's guests typically check into its hotels via a mobile app and also use a mobile app as a hotel key. Those apps no longer worked. Elevators had to be operated manually. Casino gaming machines were shut down. Players couldn't cash out their winnings.

On Tuesday September 12, at 2 am, the hackers sent an email demanding a $30 million ransom payment. MGM Resorts decided to ignore the hackers' messages. According to the hackers, it was at this point that they used ransomware to encrypt more than 100 of MGM Resorts' virtualized servers. Whether or not the specific sequence of events is accurate, what's not disputed is that MGM refused to pay the ransom. Instead, they decided to rebuild the servers from backups, installing clean versions of the operating system and other software. The cost of doing so far exceeded the ransom request, but MGM Resorts decided to do it anyway. It took more than a week to restore MGM Resorts' guest-facing operations, but the behind-the-scenes effort to fully restore its systems took far longer. The hack cost the company about $100 million in lost hotel and casino revenues, as well as an unspecified one-time expense for the cybersecurity experts and lawyers it needed to hire for incident response (although the company says that this cost was less than $10 million). The effects of the attack had a substantial impact on MGM Resorts' 2023 third-quarter earnings.

However, even once the initial impact of the attack was dealt with, the ramifications are likely to continue. MGM Resorts confirmed personal data belonging to customers had been stolen during the course of the intrusion. Stolen data includes social security numbers, driver's license numbers, passport numbers, and contact details such as names, phone numbers, email addresses, postal addresses, as well as gender and dates of birth. Numerous class actions suits have been filed alleging that MGM Resorts was negligent in failing to prevent the cyberattack that exposed customers' personally identifiable information (PII), and that it failed to take adequate measures to protect its customers' PII.

Who are the hackers who perpetrated the attack? This is a surprisingly difficult question to answer, partly because there's no standardized naming system for hacking groups. According to security researchers, the hackers were a group variously identified as Scattered Spider, Oktapus, Octo Tempest, Scattered Swine, Muddled Libra, or UNC3944. The hackers are also sometimes identified as being part of a group that calls itself the Star Fraud gang, which sprang out of an online group called the Com (or Comm), short for Community. They are believed to be primarily teenagers and young adults from the United States, Canada, and the UK, who have allied themselves with the ransomware-as-a-service hacking group BlackCat/ALPHV involved in the ransomware attack on Change Healthcare, profiled in the Opening Case Study.

Some security experts have laid the blame for the MGM hack on poor identity verification methods, since it appears that the only verification required came in the form of easily phished information However, although much of the focus has been on the social engineering aspect of the attack, there are other lessons to be learned. Recommendations include using passwordless authentication systems, or at the very least, using more secure methods such as biometrics or a YubiKey (a hardware authentication device) for super administrators; limiting the number of super administrators; and requiring more stringent authentication, such as using only a trusted device, to access a company's sensitive resources.

SOURCES: Rachel Sudbeck, "What Everyone Got Wrong About the MGM Hack," Kolide.com, accessed September 9, 2024; Robert McMillan and Katherine Stone, "The Audacious MGM Hack That Brought Chaos to Las Vegas," *Wall Street Journal*, March 29, 2024; Matt Palmer, "Lessons from the MGM Cyber Attack," Securityboulevard.com, February 14, 2024; Connor Jones, "MGM Resorts Attackers Hit Personal Data Jackpot, But House Loses," Theregister.com, October 6, 2023; D. Howard Kass, "MGM, Caesars Hit with Class Action Lawsuits over Massive Breach of Personal Information," Msspalert.com, September 29, 2023; Joseph Menn, "MGM, Caesars Casino Hacks Point to an Alliance of Teens and Ransomware Gangs," Washingtonpost.com, September 22, 2023; Becky Bracken, "Okta Agent Involved in MGM Resorts Breach, Attacker Claim," Darkreading.com, September 15, 2023.

Case Study Questions

1. What steps could MGM Resorts have taken that might have prevented this attack?
2. Do you think MGM Resorts' management made the correct decision to not pay the ransom the hackers demanded? Why or why not?
3. Why are social engineering attacks so hard to prevent?

Review Summary

8.1 Understand why information systems are vulnerable.

Information systems and the data stored on them are vulnerable to destruction, misuse, error, fraud, and hardware or software failures. The Internet is designed to be an open system and makes internal corporate systems more vulnerable to actions from outsiders. Wi-Fi networks can easily be penetrated by intruders using sniffer programs to obtain an address to access the resources of the network.

8.2 Describe the most prevalent and dangerous types of malware.

Malware such as viruses, worms, Trojan horses, ransomware, bots, and potentially unwanted programs (PUPs) can disable systems and websites. Malware can be distributed and spread by a variety of different methods, including via email, drive-by-downloads, malvertising, and social networks. Mobile devices are a major target.

8.3 Understand the dangers posed by threat actors.

Threat actors are individuals or groups of individuals seeking to exploit system vulnerabilities to intentionally cause harm, and include cybercriminals, nation-states, terrorist groups, hackers, hacktivists, and insiders. Threat actors use a variety of tactics, including spoofing, various forms of phishing, and pharming. Threat actors are increasingly using Living off the Land attacks that hijack legitimate software tools that are already part of an organization's digital infrastructure. Threat actors can also unleash denial-of-service (DoS) and distributed denial-of-service (DDoS) attacks or penetrate corporate networks, causing serious system disruptions. Most hacker activities are criminal offenses, and the vulnerabilities of systems make them targets for other types of computer crime as well, such as identity theft and data breaches. Cyberterrorism and cyberwarfare have become global threats. Company employees ("insiders") can also pose serious security issues, both due to inadvertent errors and from malicious actions.

8.4 Describe the security issues raised by software vulnerabilities.

Software presents problems because software bugs may be impossible to eliminate and because software vulnerabilities can be exploited by threat actors.

8.5 Describe the security issues raised by artificial intelligence.

Cybercriminals are beginning to leverage AI technology to create sophisticated attacks, and AI algorithms can themselves be subverted for harmful purposes. For example, genAI tools are being used by cybercriminals to create unique and extremely realistic phishing emails. AI is enabling the creation of sophisticated deepfakes with minimal technical knowledge and expense. Cybercriminals are also trying to subvert LLMs by a variety of methods.

8.6 Describe the business value of security and control.

Lack of sound security and control can cause firms relying on computer systems for their core business functions to lose sales and productivity. Information assets, such as confidential employee records, trade secrets, or business plans, lose much of their value if they are revealed to outsiders or if they expose the firm to legal liability. Laws such as HIPAA, the Sarbanes-Oxley Act, and the Gramm-Leach-Bliley Act require companies to practice stringent electronic records management and adhere to strict standards for security, privacy, and control. Legal actions requiring electronic evidence and digital forensics also require firms to pay more attention to security and electronic records management.

8.7 Describe an organizational framework for security and control.

Firms need to establish a good set of both general and application controls for their information systems. A risk assessment evaluates information assets, identifies control points and control weaknesses, and determines the most cost-effective set of controls. Firms must also develop a coherent corporate security policy and plans for continuing business operations in the

event of disaster or disruption. The security policy includes policies for acceptable use and identity management. Comprehensive and systematic information systems auditing helps organizations determine the effectiveness of security and controls for their information systems.

8.8 Discuss technologies for safeguarding information resources.

Technologies for protecting businesses' information resources include tools for managing user identities and access, preventing unauthorized access to systems and data, ensuring system availability, and ensuring software quality. Many of these technologies now incorporate AI. Passwords, security tokens, smart cards, and biometric authentication are used to authenticate system users. Firewalls prevent unauthorized users from accessing a private network when it is linked to the Internet or another external network. Intrusion detection and prevention systems monitor private networks for suspicious network traffic and attempts to access corporate systems. Anti-malware software checks computer systems for infections by viruses and worms and often eliminates malicious software. Encryption, the coding and scrambling of messages, is a widely used technology for securing digital transmissions over unprotected networks. Digital certificates combined with public key encryption provide further protection of digital transactions by authenticating a user's identity. Blockchain technology enables companies to create and verify tamperproof transactions on a network without a central authority. To ensure system availability, companies can use fault-tolerant computer systems, engage managed security service providers and security operations centers, and embrace the concept of digital resiliency. Companies must also address the new challenges posed by the Internet of Things, cloud computing, and the mobile platform. Use of software metrics and rigorous software testing help improve software quality and reliability.

Key Terms

Acceptable use policy (AUP), 271
Advanced persistent threat attack (APT attack), 262
Anti-malware software, 277
Application controls, 270
Authentication, 274
Biometric authentication, 274
Bot, 259
Botnet, 259
Bug, 265
Business continuity planning, 272
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Credential stuffing attack, 262
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Cyberwarfare, 263
Data breach, 262
Deepfake, 267
Denial-of-service attack (DoS attack), 262
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Two-factor authentication (2FA), 275
Unified threat management systems (UTM systems), 277
Virus, 259
War driving, 258
Worm, 259
Zero-day vulnerability, 265
Zero trust, 274

Review Questions

8.1 Understand why information systems are vulnerable.

- Define security and controls.
- List and describe the most common threats against contemporary information systems.

8.2 Describe the most prevalent and dangerous types of malware.

- Define malware and distinguish among a virus, a worm, a Trojan horse, ransomware, a bot, and a PUP.
- Explain how malware can be spread.

8.3 Understand the dangers posed by threat actors.

- Describe the different types of threat actors and explain how they can create security problems and damage systems.
- Define spoofing and explain why phishing, evil twins, and pharming are forms of spoofing.
- Define a LOTL attack and explain how it differs from a DDoS attack.
- Explain why data breaches and identity theft are such big problems today.
- Describe the security issues that insiders can create.
- Define computer crime and provide two examples of crime in which computers are targets and two examples in which computers are used as instruments of crime.

8.4 Describe the security issues raised by software vulnerabilities.

- Explain how software defects affect system reliability and security.
- Explain how software supply chain attacks threaten security.

8.5 Describe the security issues raised by artificial intelligence.

- Explain how AI tools being used by cybercriminals for phishing attacks.
- Define a deepfake and describe the security issues it creates.
- Describe how cybercriminals are trying to subvert LLMs.

8.6 Describe the business value of security and control.

- Explain how security and control provide value for businesses.
- Describe the relationships among security and control, recent US government regulatory requirements, and digital forensics.

8.7 Describe an organizational framework for security and control.

- Define general controls and describe each type.
- Define application controls and describe each type.
- Describe the function of risk assessment and explain how it is conducted for information systems.
- Define and describe the following: security policy, acceptable use policy, and identity and access management.
- Explain how information systems auditing promotes security and control.

8.8 Discuss technologies for safeguarding information resources.

- Name and describe three authentication methods.
- Describe the roles of firewalls, intrusion detection and prevention systems, and anti-malware software in promoting security.
- Explain how encryption protects information.
- Describe the roles of encryption and digital certificates in a public key infrastructure.
- Distinguish between disaster recovery planning and business continuity planning.
- Define digital resiliency and describe its benefits for an organization.
- Identify and describe the security challenges posed by Internet of Things (IoT) devices.
- Describe measures for improving software quality and reliability.
- Describe how AI is being used to safeguard information resources.

Discussion Questions

8.9 Security isn't simply a technology issue; it's a business issue. Discuss.

8.10 If you were developing a business continuity plan for your company, where would you start? What aspects of the business would the plan address?

8.11 Your business is about to launch an e-commerce website where it will sell goods and accept credit card payments. Discuss the major security threats to this website and their potential impacts. What can be done to minimize these threats?

Using AI Project

8.12 AI Project: Devising an Information Systems Security Plan

You have been hired as a consultant by two friends who want to open a retail store selling organic pet food and treats as well as pet toys and clothing made from organic materials. They have asked you to advise them on potential information systems security problems they might face. They also want you to devise an information systems security plan that would be appropriate for a small retail business with less than 10 employees. Using Microsoft's Copilot, Google's Gemini, or ChatGPT, develop a series of prompts to assist in providing this advice and devising such a security plan. Save a copy of your prompts and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience in analyzing security vulnerabilities; using spreadsheet software for risk analysis; and using web tools to research security outsourcing services.

For student files please visit www.pearsonhighered.com/laudon or MyLab MIS.

8.13 Management Decision Problem

Scopely is a leading online gaming company offering a variety of popular games such as Scrabble Go, Star Trek Fleet Command, and Looney Tunes World of Mayhem. Although Scopely primarily focuses on the mobile platform, with Apple iOS and Android versions of its games, some games are also available for play on desktop/laptop computers. Scopely also has a corporate website and a social media presence on Facebook, Instagram, X, TikTok, and LinkedIn. Prepare a security analysis for Scopely. What kinds of threats should it anticipate? What would be their impact on the business? What steps can it take to prevent damage to its operations?

8.14 Using Spreadsheet Software to Analyze Security Events

Software skills: Spreadsheet sorting and data filtering

Business skills: Monitoring and analyzing security events

Events that occur in computer systems are commonly recorded in log files. Each operating system, including Windows, uses its own log files, and applications and hardware devices also generate logs. Security teams can use security logs to track users on the corporate network, identify suspicious activity, and detect vulnerabilities. The security log lists events related to security, including login attempts or file deletions. System administrators use their organization's audit policy to determine which events to enter into their security log.

Your company's Information Systems Department has compiled a list of security events from the company's Windows security event log that merit special attention from the organization. You will find further information in the MIS18_Project 8.14_Student File, which shows the Security Event, Event ID, and Potential Criticality of each event on that list and also additional information.

The "Event ID" is the code Windows assigns to each particular security event. The "Potential Criticality" column identifies whether the event should be considered to have Low, Medium, or High criticality in detecting attacks. A potential criticality of High means that even one occurrence of the event should be investigated. Potential criticality of Medium or Low means that these events should be investigated only if they occur unexpectedly or in numbers that significantly exceed the expected baseline in a measured period of time. All organizations should test these recommendations in their environments before creating alerts that require mandatory investigative responses. Every environment is different, and some of the events ranked with a potential criticality of High may occur because of other harmless events.

- Sort the data by Potential Criticality from High to Low.
- Filter the data to display all events of High Potential Criticality.
- Select one event of High Potential Criticality and use the web to find out more about the event's importance and significance. Write a paragraph to explain why businesses should pay close attention to this type of security event.

8.15 Evaluating Security Outsourcing Services

Software skills: Web browser and presentation software

Business skills: Evaluating business outsourcing services

This project will help develop your skills in using the web to research and evaluate security outsourcing services.

- You have been asked to help your company's management decide whether to outsource security or keep the security function within the firm. Search the web to find information to help you decide whether to outsource security and to locate security outsourcing services.
- Present a brief summary of the arguments for and against outsourcing computer security for your company.

- Select two firms that offer computer security outsourcing services and compare them and their services. Prepare a PowerPoint or other form of presentation for management, summarizing your findings. Your presentation should address whether your company should outsource computer security. If you believe your company should outsource, the presentation should identify which security outsourcing service you selected and justify your decision. If you believe the company should not outsource, you should identify the security tools it should use to ensure appropriate security.

8.16 Collaboration and Teamwork Project: Evaluating Security Software Tools

With a group of two or more students, use the web to research and evaluate two competing vendors' security products, such as anti-malware software, firewalls, or intrusion detection and prevention software. For each product, describe its capabilities, for what types of businesses it is best suited, and its cost to purchase and install. Which is the best product? Why? If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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Part Three

Key System Applications for the Digital Age

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Chapter 9: Achieving Operational Excellence and Customer Intimacy: Enterprise Applications

Chapter 10: E-commerce: Digital Markets, Digital Goods

Chapter 11: Artificial Intelligence

Chapter 12: Enhancing Decision Making with Data Analytics and Business Intelligence

PART THREE examines the core information system applications businesses are using today to improve operational excellence and decision making. These applications include enterprise resource planning systems; systems for supply chain management, and customer relationship management; artificial intelligence; e-commerce applications; and data analytics and business intelligence systems to enhance decision making. This part answers questions such as these: How can enterprise applications improve business performance? How do firms use e-commerce to extend the reach of their businesses? How can systems improve decision making and help companies benefit from artificial intelligence?

9

Achieving Operational Excellence and Customer Intimacy: Enterprise Applications

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Learning Objectives

After completing this chapter, you will be able to:

9.1 Understand ERP systems.

9.2 Understand SCM systems.

9.3 Describe global supply chain management challenges.

9.4 Understand CRM systems.

9.5 Discuss enterprise application challenges.

9.6 Describe how enterprise applications are using AI.

9.7 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** TrueBlue Speeds Ahead with Enterprise Applications
- **Spotlight on Organizations Case Study:** BYK Tames Its Supply Chains
- **Chapter Case Study:** MillerKnoll Uses Salesforce and AI to Transform Its Business Strategy

Opening Case Study

TrueBlue Speeds Ahead with Enterprise Applications

TrueBlue, Inc. based in Tacoma, Washington, is a staffing, recruiting and workforce management company that services industries such as construction, energy, manufacturing, financial services, pharmaceuticals, transportation, aviation, and energy. In 2023, TrueBlue served approximately 67,000 clients and connected approximately 464,000 people with work.

Over the years, TrueBlue grew through acquisitions as well as organic expansion. At one point, the company was relying on 20 legacy financial and human resources systems to run its business. These systems often overlapped or conflicted with one another. This meant that TrueBlue staff had to reconcile conflicting pieces of information from the various systems in order to operate. A significant number of manual workarounds had to be made just to complete payroll processing. Maintaining and upgrading all the on-premises legacy systems was becoming costly and time-consuming, reducing the company's ability to grow.

TrueBlue did not want to create complex integrations between various systems so it chose a solution based on Oracle Cloud ERP (Enterprise Resource Planning) and Oracle Cloud HCM (Human Capital Management), which enabled the company's human resource and finance systems to seamlessly "talk" to one another. Oracle Cloud HCM provides a solution for managing every stage of the employee lifecycle, from attracting talent, screening, hiring, onboarding, managing time and absence, managing payroll, compensation and benefits, managing performance, and developing talent. Oracle Cloud ERP provides data on enterprise-wide business processes to increase forecasting accuracy, shorten reporting cycles, simplify decision-making, and better manage risk and compliance. Oracle Fusion Cloud Financials is a global financial platform that connects and further automates a firm's financial management processes, including payables, receivables, fixed assets, expenses, and reporting, for a clear view of a firm's total financial health. Oracle Fusion Cloud Financials also uses artificial intelligence (AI) machine learning and analytics to reduce exception handling and quickly identify issues presenting business and audit risks.

Oracle Cloud brings together ERP and HCM on one cloud platform with a common data model. Having a single employee record across finance and HR streamlines processes, eliminates redundancy, and reduces errors. Business processes are faster and more efficient, with a week saved in back-office processes. Implementing Oracle Cloud meant that TrueBlue could also move out of its data centers, reducing its IT infrastructure costs. The cloud model improved scalability, a key requirement for future growth. Oracle Cloud applications unlocked new ways of expanding the business, such as enhancing workforce analytics that provide insights on the talent that the company has and the talent it needs.

After implementing Oracle Cloud, employees gained more time and freedom to focus on strategic initiatives, such as how to run branches better and how to deliver services faster. Today, TrueBlue is more empowered to achieve its goals.

SOURCES: Trueblue.com, accessed September 9, 2024; Oracle.com, accessed September 9, 2024; "TrueBlue Streamlines Operations with Oracle Fusion Cloud," Humanresources.report, accessed September 9, 2024.

TrueBlue's experience with incompatible legacy systems and manual business processes illustrates the critical role of enterprise resource planning and other enterprise applications in business. TrueBlue was not able to expand its business, or even run its existing business efficiently, because it lacked the requisite systems for operating and managing the enterprise. Employees and managers had to

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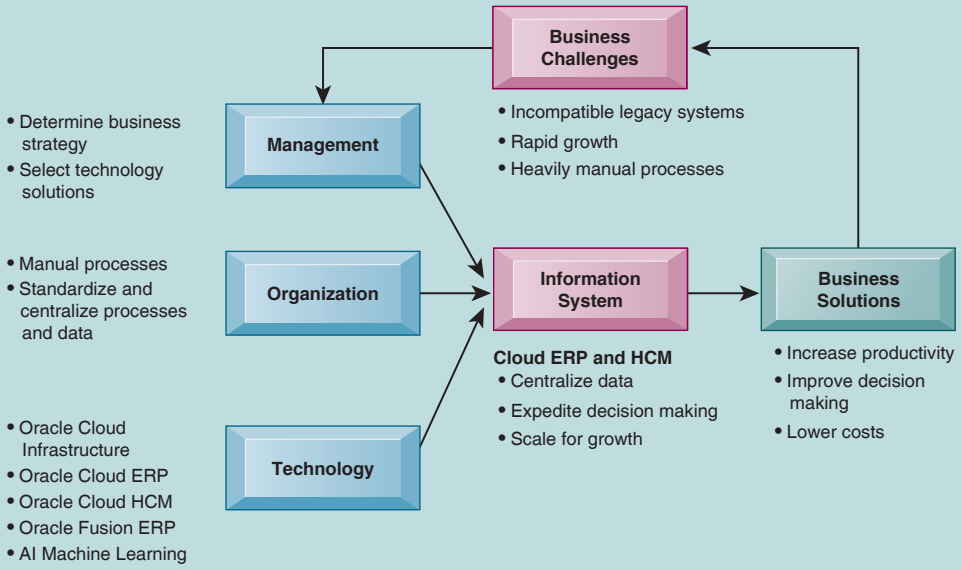


use manual workarounds to piece together the information they needed. Cloud computing helped TrueBlue lower system operating costs, and it also provided a more up-to-date platform for running enterprise applications and for analyzing big data, including AI.

The Opening Case Study diagram above calls attention to important points raised by this case and this chapter. TrueBlue had expanded rapidly and was running its business with twenty outdated and incompatible legacy systems. Too much time was being spent trying to reconcile disparate data from these systems and creating manual workarounds for them, resulting in many inefficient practices, and preventing the business from obtaining the data and capabilities needed to keep expanding. Implementing Oracle Cloud ERP, Oracle Cloud HCM and Oracle Fusion ERP Analytics enabled the company to standardize data and processes, with one single central source of data for the enterprise. It also provided TrueBlue with more sophisticated analytical tools for management decision making, including applications of AI and machine learning. The efficiencies and improved decision making, along with a new cloud computing platform, lowered operating costs.

Case Study Questions

1. How did legacy systems affect TrueBlue's business performance?
2. Why were cloud ERP, HCM, and financial applications a good solution for TrueBlue?
3. How did implementing cloud enterprise applications change operations and decision making at TrueBlue?

CAREER BENEFIT

The systems that TrueBlue wanted to upgrade were enterprise-wide and powered some of the core activities of the business. Employees, managers, and IT specialists who worked on TrueBlue's new cloud enterprise applications gained recognition for successfully implementing an important solution for the entire company and for the expertise they acquired by doing so. Employees and managers using the new systems were able to work more productively, and their careers benefited from improving the company's business performance.

9.1 Understand ERP Systems

Around the globe, companies are increasingly becoming more connected, both internally and with other companies. If you run a business, you'll want to be able to react instantaneously when a customer places a large order or when a shipment from a supplier is delayed. You may also want to know the impact of these events on the different parts of your business and how

your business is performing at any point in time, especially if you're running a large company. An enterprise resource planning (ERP) system provides the integration to make this possible. Let's look at how an ERP system works and what it can do for a firm.

What Is an ERP System?

Imagine that you had to run a business based on information from tens or even hundreds of databases and systems, none of which could communicate with one another. Imagine your company had 10 major product lines, each produced in separate factories and each with separate and incompatible sets of systems controlling production, warehousing, and distribution.

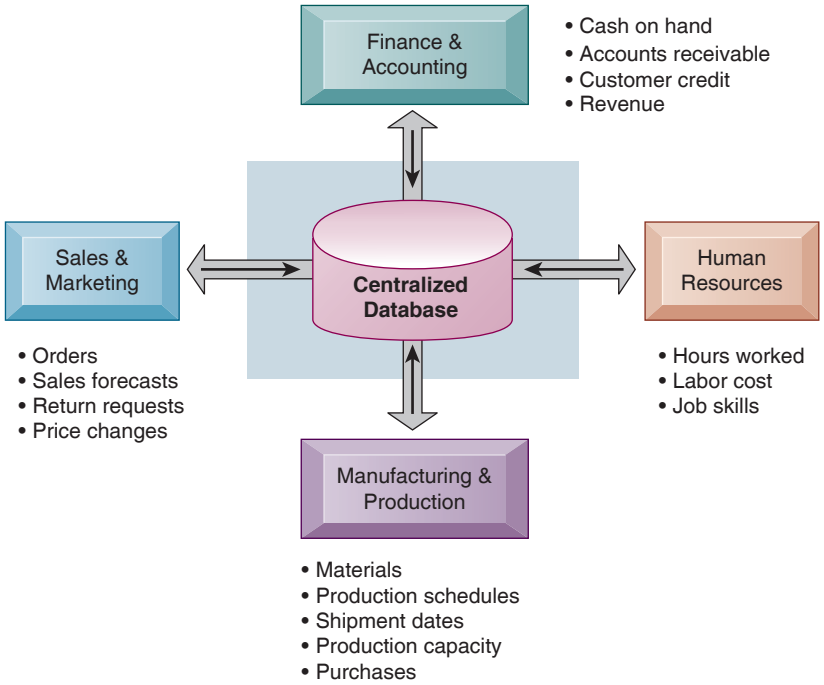
Your decision making would likely have to be based on manual, hard-copy reports that might be out of date, and it would be difficult to understand what was happening in the business as a whole. Sales personnel might not be able to tell at the time they placed an order whether the ordered items were in inventory, and manufacturing could not easily use sales data to plan for new production. You now have a good idea of why firms need a special system to integrate information.

Chapter 2 introduced ERP systems, which are based on a suite of integrated software modules and a common, central database. Examine Figure 9.1, which illustrates how ERP systems work. The database collects data from various divisions and departments in the firm and from a large number of key business processes in manufacturing and production, finance and accounting, sales and marketing, and human resources, making the data available for applications that support nearly all of the organization's internal business activities. When new information is entered by one process, the information is made immediately available to other business processes.

For instance, if a tire manufacturer is using an ERP system, when a sales representative places an order for a particular product, such as tire rims, the system verifies the customer's credit limit, schedules the shipment, identifies the best shipping route, and reserves the necessary items from inventory. If inventory stock is insufficient to fill the order, the system

Figure 9.1 How ERP Systems Work

ERP systems feature a set of integrated software modules and a central database through which business processes and functional areas throughout the enterprise can share data.



schedules the manufacture of more rims, ordering the needed materials and components from suppliers. Sales and production forecasts are immediately adjusted. General ledger and corporate cash levels are automatically updated with the revenue and cost information from the order. Users can tap into the system and find out where that particular order is at any minute, and management can obtain information at any point in time about how the business is operating. The system can also generate enterprise-wide data for management analyses of product cost and profitability.

ERP Software

ERP software is built around thousands of predefined business processes that reflect best practices. Table 9.1 describes some of the major business processes that an ERP system supports.

Companies implementing ERP software first have to select the functions of the system they wish to use and then map their business processes to the predefined business processes in the software. Configuration tables provided by the software vendor enable the firm to tailor a particular aspect of the system to the way the firm does business. For example, the firm could use these tables to select whether it wants to track revenue by product line, geographical unit, or distribution channel.

If the ERP system does not support the way the organization does business, its software can sometimes be rewritten to support the way its business processes work. However, ERP systems are very complex, and extensive customization may degrade system performance, compromising the information and process integration that are the main benefits of the system. If companies want to reap the maximum benefits from an ERP system, they must change the way they work to conform to the business processes defined by the system's software.

For example, Office Depot avoided customization when it moved from in-house systems to Oracle ERP Cloud. The retailer chose to use best practices embedded in Oracle's Supply Chain Management Cloud and in its cloud-based Human Capital Management and Enterprise Performance Management systems. By not customizing its Oracle ERP applications, Office Depot simplified its information systems and reduced the cost of maintaining and managing them.

Tasty Baking Company provides another example. To implement a new ERP system, Tasty Baking Company identified its existing business processes and then translated them into the business processes built into the SAP ERP software it had selected. To ensure that it obtained the maximum benefits from the ERP system, Tasty Baking Company deliberately planned for customizing less than 5 percent of the system and made very few changes to the SAP software itself. It used as many tools and features that were already built into the SAP software as it could.

Leading ERP software vendors include SAP, Oracle, Microsoft, Salesforce, and Blue Yonder. There are also versions of ERP software packages designed for small and medium-sized businesses and on-demand software services running in the cloud (see Section 9.4).

The Business Value of ERP Systems

ERP systems provide value by both increasing operational efficiency and providing firmwide information to help managers make better decisions. Large companies with many operating units in different locations use ERP systems to enforce standard practices and data so that everyone does business the same way worldwide.

Table 9.1 Business Processes Supported by ERP Systems

Financial and accounting processes, including general ledger, accounts payable, accounts receivable, fixed assets, cash management and forecasting, product-cost accounting, cost center accounting, asset accounting, tax accounting, credit management, and financial reporting
Human resources processes, including personnel administration, time accounting, payroll, personnel planning and development, benefits accounting, applicant tracking, time management, compensation, workforce planning, performance management, and travel expense reporting
Manufacturing and production processes, including procurement, inventory management, purchasing, shipping, production planning, production scheduling, material requirements planning, quality control, distribution, transportation execution, and plant and equipment maintenance
Sales and marketing processes, including order processing, quotes, contracts, product configuration, pricing, billing, credit checking, incentive and commission management, and sales planning

For instance, Coca-Cola implemented an SAP ERP system to standardize and coordinate important business processes in 200 countries. Lack of standard, companywide business processes had prevented the company from using its worldwide buying power to obtain lower prices for raw materials and from reacting rapidly to market changes.

ERP systems help firms respond rapidly to customer requests for information or products. Because the system integrates order, manufacturing, and delivery data, manufacturing is better able to produce only what customers have ordered, procure exactly the right number of components or raw materials to fill actual orders, stage production, and minimize the time that components or finished products are in inventory.

For example, AbbVie, a large global biopharmaceutical company with operations spanning 70 countries, had been working with more than 50 legacy systems for mission-critical processes in more than 100 worldwide locations. The company decided to standardize its business processes for all its affiliates and manufacturing facilities globally and to support those processes with a single, global SAP ERP system. The new ERP system provides a set of key metrics, such as the length of time to create new customers, vendor payments, payment terms, and order fulfillments. There are dashboards available that enable managers to look at every country, measure results, find the root cause of problems, and take corrective action more easily. Reporting from the system is also more accurate.

ERP systems provide much valuable information for improving management decision making. Corporate headquarters can access up-to-the-minute data on sales, inventory, and production and use this information to create more accurate sales and production forecasts. ERP systems include analytical tools to use the data the system captures to evaluate overall organizational performance. ERP system data have common, standardized definitions and formats that are accepted by the entire organization. Performance statistics mean the same thing across the company. ERP systems also allow senior management to find out easily and at any moment how a particular organizational unit is performing, to determine which products are most or least profitable, and to calculate costs for the company as a whole.

AI and ERP Systems

ERP system vendors are increasingly incorporating AI into their offerings. Many ERP systems now include natural language processing capabilities that enable the system to understand user queries and provide a conversational output. By integrating AI into their ERP systems, companies can improve organizational performance across various functions, including accounting, finance, human resources, manufacturing, and product development. AI-driven ERP systems can identify patterns in very large volumes of data, whether structured or unstructured. The system can then provide predictive insights that help organizations optimize workflows and find cost savings opportunities. Table 9.2 provides examples of how AI is being used in ERP applications.

9.2 Understand SCM Systems

If you manage a small firm that makes a few products or sells a few services, chances are you will have a small number of suppliers. You could coordinate your supplier orders and deliveries by using just a telephone and email. But if you manage a firm that produces more complex products and services, you will have hundreds of suppliers, and each of your suppliers will have its own set of suppliers. Suddenly, you will need to coordinate the activities of hundreds or even thousands of other firms to produce your products and services. Supply chain management (SCM) systems, which we introduced in Chapter 2, are an answer to the problems of supply chain complexity and scale.

The Supply Chain

Supply chain

Network of organizations and business processes for procuring materials, transforming raw materials into intermediate and finished products, and distributing the finished products to customers.

A firm's **supply chain** is a network of organizations and business processes that procure raw materials, transform these materials into intermediate and finished products, and distribute the finished products to customers. It links suppliers, manufacturing plants, distribution centers,

Table 9.2 Examples of AI Applications in ERP Systems

Capability	Description
Digital assistant	Use AI-powered conversational chatbots to respond to customer and employee inquiries by providing support, answering questions, capturing details, and resolving issues without human interaction.
Predictive analytics	Anticipating equipment failures, recommending preventative maintenance schedules and procedures
Anomaly detection	Analyzing ERP financial reports to identify irregularities in data entries or unusual transactions
Improve product development and innovation	Suggest new product ideas, optimize product features and assortment
Financial forecasting	Generating plausible financial scenarios based on potential economic, policy, or market changes
Optimize workflows	AI can assist with almost every function of a modern supply chain, including planning, inventory and warehouse management, transaction processing, transportation, monitoring, and inspection.

retail outlets, and customers to supply goods and services from source through consumption. Materials, information, and payments flow through the supply chain in both directions.

Goods start out as raw materials and, as they move through the supply chain, are transformed into intermediate products (also referred to as components or parts), and finally are transformed into finished products. The finished products are shipped to distribution centers and from there to retailers and customers. Returned items flow in the reverse direction, from the buyer back to the seller.

Let's look at the supply chain for Nike sneakers as an example. Nike designs, markets, and sells sneakers, socks, athletic clothing, and accessories throughout the world. Its primary suppliers are contract manufacturers with factories in China, Thailand, Indonesia, Brazil, and other countries. These companies fashion Nike's finished products.

Nike's contract suppliers do not manufacture sneakers from scratch; rather, they obtain components for the sneakers—the laces, eyelets, uppers, and soles—from other suppliers and then assemble the components into finished sneakers. These suppliers in turn have their own suppliers. For example, the suppliers of soles have suppliers of synthetic rubber, suppliers of the chemicals used to melt the rubber for molding, and suppliers of the molds into which to pour the rubber. Suppliers of laces have suppliers of their thread, dyes, and plastic lace tips.

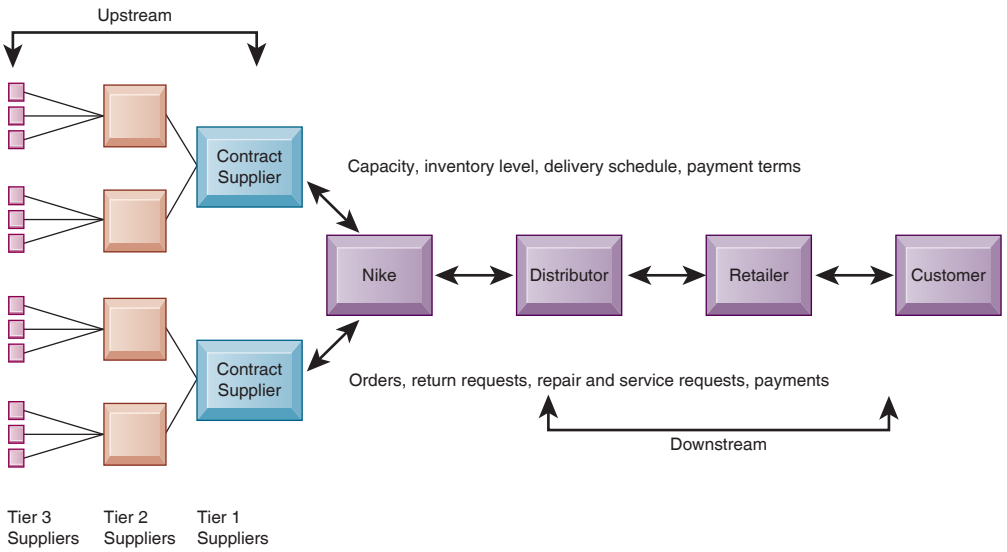
Examine Figure 9.2, which provides a simplified illustration of Nike's supply chain for sneakers. It shows the flow of information and materials among suppliers, Nike, Nike's distributors, retailers, and customers. Nike's contract manufacturers are its primary suppliers. The suppliers of soles, eyelets, uppers, and laces are the secondary (Tier 2) suppliers. Suppliers to these suppliers are the tertiary (Tier 3) suppliers.

The *upstream* portion of the supply chain includes the company's primary, Tier 2, and Tier 3 suppliers and the processes for managing relationships with them. The *downstream* portion consists of the organizations and processes for distributing and delivering the products to the final customers. Companies that manufacture, such as Nike's contract suppliers of sneakers, also manage their own *internal supply chain processes* for transforming the materials, components, and services that their suppliers furnish into finished products or intermediate products (components or parts) for their customers and for managing materials and inventory.

The supply chain illustrated in Figure 9.2 has been simplified. It shows only two contract manufacturers for sneakers and only the upstream supply chain for sneaker soles. In reality, Nike has hundreds of contract manufacturers turning out finished sneakers, socks, and athletic clothing, each with its own set of suppliers. The upstream portion of Nike's supply chain actually comprises thousands of entities. Nike also has numerous distributors and many thousands of retail stores where its products are sold, so the downstream portion of its supply chain is also large and complex.

Figure 9.2 Nike's Supply Chain

This figure illustrates the major entities in Nike's supply chain and the flow of information upstream and downstream to coordinate the activities involved in buying, making, and moving a product. Shown here is a simplified supply chain, with the upstream portion focusing only on the suppliers for sneakers and sneaker soles.



Information Systems and Supply Chain Management

Inefficiencies in the supply chain, such as parts shortages, underused plant capacity, excessive finished goods inventory, or high transportation costs, are caused by inaccurate or untimely information. For example, manufacturers may keep too many parts in inventory because they do not know exactly when they will receive their next shipments from their suppliers. Suppliers may order too few raw materials because they do not have precise information about demand. These supply chain inefficiencies waste as much as 25 percent of a company's operating costs.

If a manufacturer had perfect information about exactly how many units of product customers wanted, when they wanted the products, and when the products could be produced, it would be possible to implement a highly efficient, just-in-time strategy. A **just-in-time strategy** aims to have components arrive exactly at the moment they are needed, and finished goods shipped as soon as they are manufactured in order to minimize inventory.

Just-in-time strategy
Strategy that aims to have components arrive exactly at the moment they are needed and finished goods shipped as soon as they are manufactured in order to minimize inventory.

In a supply chain, however, uncertainties arise because many events cannot be foreseen—uncertain product demand, late shipments from suppliers, defective parts or raw materials, or production process breakdowns. To satisfy customers, manufacturers often deal with such uncertainties and unforeseen events by keeping more material or products in inventory than they think they may actually need. The *safety stock* acts as a buffer for the lack of flexibility in the supply chain. Although excess inventory is expensive, low fill rates are also costly because business may be lost from canceled orders.

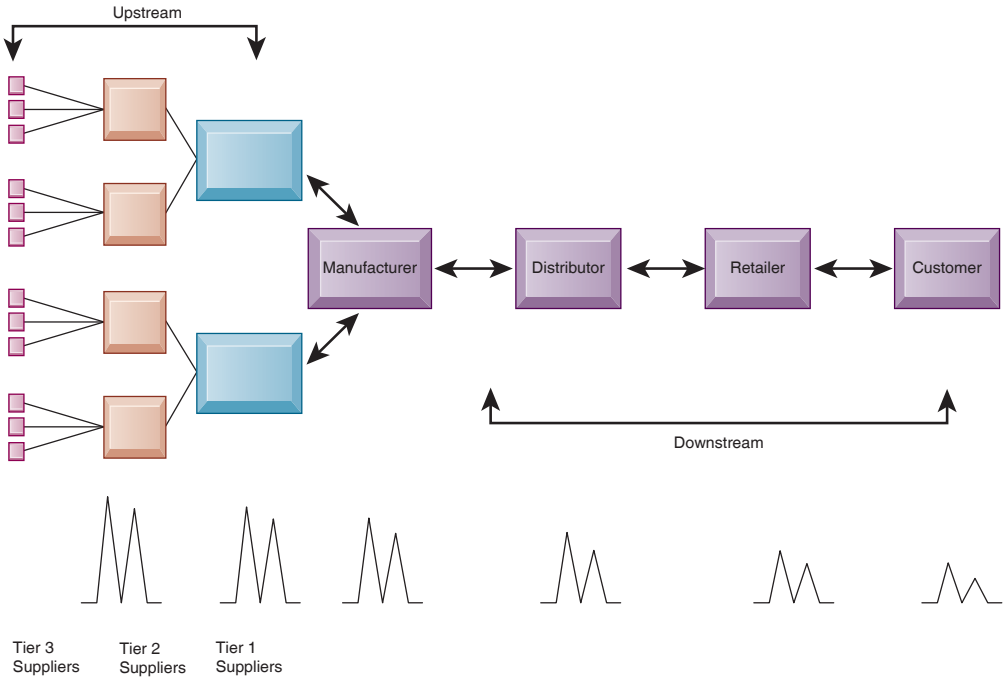
One recurring problem in supply chain management is the **bullwhip effect**, in which information about the demand for a product gets distorted as the information passes from one entity to the next across the supply chain. A slight rise in demand for an item might cause different members in the supply chain—distributors, manufacturers, suppliers, secondary suppliers (suppliers' suppliers), and tertiary suppliers (suppliers' suppliers' suppliers)—to stockpile inventory so that each has enough just in case. These changes ripple throughout the supply chain, magnifying what started out as a small change from planned orders and creating excess inventory, production, warehousing, and shipping costs. Examine Figure 9.3, which illustrates the bullwhip effect. A small change in demand from customers is amplified as it ripples backward through the supply chain.

Bullwhip effect
Distortion of information about the demand for a product as it passes from one entity to the next across the supply chain.

Some experts believe the bullwhip effect was one cause of the supply chain disruptions that emerged during and after the Covid-19 pandemic. In mid-2022, some large retailers such as Walmart, Target, and Gap overestimated the amount of inventory they would need, leading

Figure 9.3 The Bullwhip Effect

Inaccurate information can cause minor fluctuations in demand for a product to be amplified as one moves further back in the supply chain. For example, minor fluctuations in retail sales of a product can create excess inventory for distributors, manufacturers, and suppliers.



them to significantly over-order from wholesalers. The wholesalers in turn over-ordered from their own suppliers, leading to a major mismatch between consumers' actual demand and inventories on hand (McCormick, 2022).

The bullwhip effect is tamed by reducing uncertainties about demand and supply because all members of the supply chain have accurate and up-to-date information. If all supply chain members share dynamic information about inventory levels, schedules, forecasts, and shipments, they have more precise knowledge about how to adjust their sourcing, manufacturing, and distribution plans. Supply chain management systems provide the kind of information that helps members of the supply chain make better purchasing and scheduling decisions.

Supply Chain Management Software

Supply chain planning system

System that enables a firm to model its existing supply chain, generate demand forecasts for a product, and develop optimal sourcing and manufacturing plans.

Supply chain software is classified as either software to help businesses plan their supply chains (supply chain planning) or software to help businesses execute the supply chain steps (supply chain execution). A **supply chain planning system** enables the firm to model its existing supply chain, generate demand forecasts for products, and develop optimal sourcing and manufacturing plans. Such systems help companies make better decisions, such as determining how much of a specific product to manufacture in a given time period; establishing needed inventory levels of raw materials, intermediate products, and finished goods; determining where to store finished goods; and identifying the transportation mode to use for product delivery (see the Chapter Opening Case Study, *TrueBlue Speeds Ahead with Enterprise Applications*.)

Demand planning

Process by which a firm determines how much product it needs to make to satisfy all its customers' demands.

For example, if a large customer places a larger order than usual or changes that order on short notice, this change can have a widespread impact throughout the supply chain. Additional raw materials or a different mix of raw materials may need to be ordered from suppliers. Manufacturing may have to change job scheduling. A transportation carrier may have to reschedule deliveries. Supply chain planning software makes these necessary adjustments to production and distribution plans. In addition, information about changes is shared among the relevant supply chain members so that their work can be coordinated. One of the most important—and complex—supply chain planning functions is **demand planning**, a process by which a firm determines how much product it needs to make in order to satisfy all its customers' demands. Blue Yonder Software, SAP, and Oracle all offer supply chain management solutions.

A **supply chain execution system** manages the flow of products through distribution centers and warehouses to ensure that products are delivered to the right locations in the most efficient manner. These systems track the physical status of goods, the management of materials, warehouse and transportation operations, and financial information involving all parties. An example is the warehouse management system (WMS) that Haworth Incorporated uses. Haworth is a world-leading manufacturer and designer of office furniture, with distribution centers in four states. The WMS tracks and controls the flow of finished goods from Haworth's distribution centers to its customers. Acting on shipping plans for customer orders, the WMS directs the movement of goods based on immediate conditions of space, equipment, inventory, and personnel.

Supply chain execution system

System that manages the flow of products through distribution centers and warehouses to ensure that products are delivered to the right locations in the most efficient manner.

Demand-Driven Supply Chains

In addition to reducing costs, supply chain management systems facilitate efficient customer response, enabling the workings of the business to be driven more by customer demand. (We introduced efficient customer response systems in Chapter 3.)

Examine Figure 9.4, which illustrates the difference between a push-based model (also known as build-to-stock), which drove earlier supply chain management systems, and a pull-based model. In a **push-based model**, supply, production, inventory, and stock are all driven by forecasts or best guesses of demand for products, rather than what the customer orders. Instead, products are pushed to customers. With new flows of information made possible by web-based tools, supply chain management can now more easily follow a pull-based model. In a **pull-based model**, also known as a demand-driven or build-to-order model, actual customer orders or purchases trigger events in the supply chain. Transactions to produce and deliver only what customers have ordered move up the supply chain from retailers to distributors to manufacturers and eventually to suppliers. Only products that fulfill these orders move back down the supply chain to the retailer. Manufacturers use actual order demand information to drive their production schedules and the procurement of components or raw materials. Walmart's continuous replenishment system described in Chapter 3 is an example of the pull-based model.

Push-based model

Supply chain driven by forecasts or best guesses of demand for products, rather than what customers order; products are "pushed" to customers.

Pull-based model

Supply chain driven by actual customer orders or purchases. Also known as demand-driven or build-to-order model.

The Internet and Internet technology make it possible to move from sequential supply chains, where information and materials flow sequentially from company to company, to concurrent supply chains, where information flows in many directions simultaneously among members of a supply chain network. Complex supply networks of manufacturers, logistics suppliers, outsourced manufacturers, retailers, and distributors can adjust immediately to changes in schedules or orders. Examine Figure 9.5, which illustrates an Internet-driven

Figure 9.4 Push-Based Versus Pull-Based Supply Chain Models

The difference between push- and pull-based models is summarized by the slogan, "Make what we sell, not sell what we make."

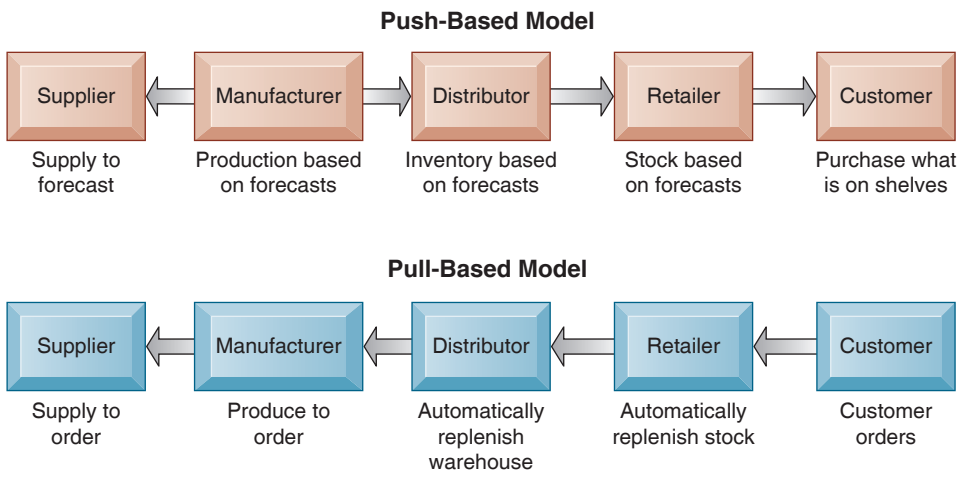
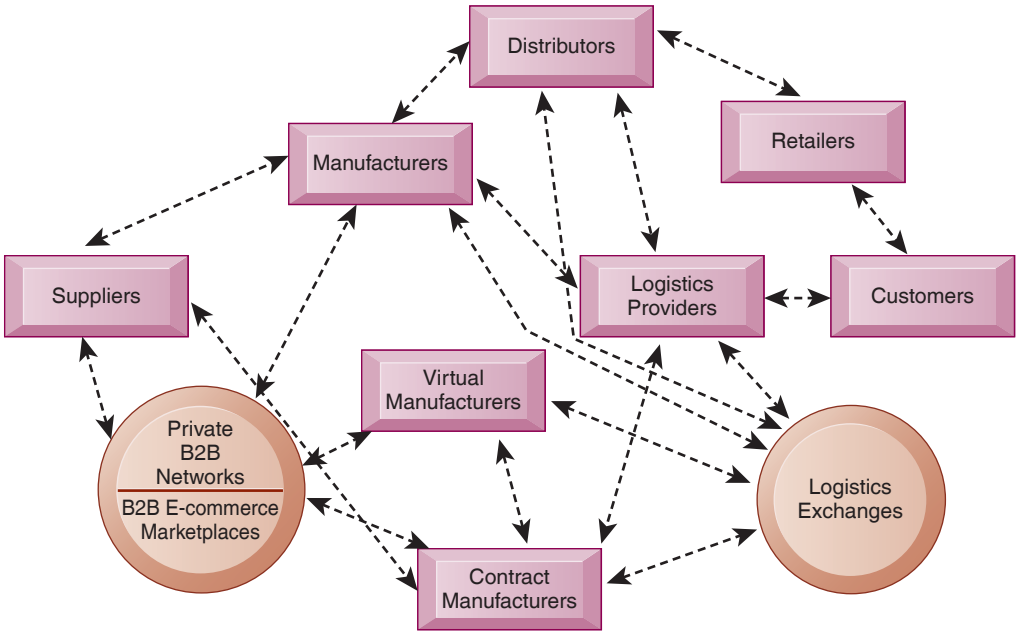


Figure 9.5 The Internet-Driven Supply Chain

The Internet-driven supply chain operates like a digital logistics nervous system. It provides multidirectional communication among firms, networks of firms, and e-marketplaces so that entire networks of supply chain partners can immediately adjust inventories, orders, and capacities.



supply chain that operates like a digital logistics nervous system. The system provides for multidimensional communication among supply chain partners and enables them to make real-time adjustments to production, inventory, and orders.

The Business Value of SCM Systems

You have just seen how supply chain management systems enable firms to streamline both their internal and external supply chain processes and provide management with more accurate information about what to produce, store, and move. By implementing a networked and integrated supply chain management system, companies match supply to demand, reduce inventory levels, improve delivery service, speed product time to market, and use assets more effectively.

Total supply chain costs represent the majority of operating expenses for many businesses and in some industries approach 75 percent of the total operating budget. Reducing supply chain costs has a major impact on firm profitability.

In addition to reducing costs, supply chain management systems help increase sales. If a product is not available when a customer wants it, the customer will often try to purchase it from someone else. More precise control of the supply chain enhances a firm's ability to have the right product available for customer purchases at the right time.

9.3 Describe Global Supply Chain Management Challenges

Before the Internet, supply chain coordination was hampered by the difficulties of making information flow smoothly among disparate internal supply chain systems for purchasing, materials management, manufacturing, and distribution. It was also difficult to share information with external supply chain partners because the systems of suppliers, distributors, or logistics providers were based on incompatible technology platforms and standards.

Today, ERP and SCM systems enhanced with Internet technology supply some of this integration. For instance, a manager can now use an online web interface to tap into suppliers' systems to determine whether inventory and production capabilities match demand for the firm's products. Business partners can use online supply chain management tools to collaborate on forecasts. Sales representatives can access suppliers' production schedules and logistics information to monitor customers' order status.

Global Supply Chain Issues

More and more companies are entering international markets, outsourcing manufacturing operations, and obtaining supplies from other countries as well as selling abroad. Their supply chains extend across multiple countries and regions, which lead to additional complexities and challenges in managing a global supply chain.

Global supply chains typically span greater geographic distances and time differences than domestic supply chains do and have participants from a greater number of countries. Performance standards may vary from region to region or from nation to nation. Supply chain management may need to reflect foreign government regulations and cultural differences.

The Internet helps companies manage many aspects of their global supply chains, including sourcing, transportation, communications, and international finance. Today's apparel industry, for example, relies heavily on outsourcing to contract manufacturers in China and other low-wage countries. Apparel companies are using the Internet to manage their global supply chain and production issues. (Review the discussion of Li & Fung in Chapter 3.)

In addition to contract manufacturing, globalization has encouraged outsourcing warehouse management, transportation management, and related operations to third-party logistics providers such as UPS Supply Chain Solutions. These logistics services offer software to give their customers a better view of their global supply chains. Customers can check a secure website or app to monitor inventory and shipments, helping them run their global supply chains more efficiently.

The Spotlight on Organizations Case Study, *BYK Tames Its Supply Chain,* describes how global specialty chemicals provider BYK uses SCM systems to deal with global supply chain challenges.

Spotlight on Organizations Case Study

BYK Tames Its Supply Chains

BYK is a leading global supplier of specialty chemicals, additives, and measuring instruments. Additives are chemicals used in small quantities to improve product properties such as scratch resistance or the gloss of surfaces, the mechanical strength or flow behavior of materials, and flame retardancy. Additives also optimize manufacturing processes. BYK's measuring and testing instruments are designed for effectively assessing appearance and physical properties. As one of the four divisions of the ALTANA Group, BYK has around 2,500 employees and is headquartered in Wesel, Germany. BYK has production sites all over the world, including in the United States (Wallingford, Chester, Columbia, Earth City, Gonzales, Louisville, Pittsford), Germany (Wesel, Kempen, Schkopau, Moosburg, Geretsried), the Netherlands (Deventer, Denekamp, Nijverdal), the UK (Widnes), and in China (Shanghai). BYK services forty different markets. The company's customers include manufacturers of coatings and printing inks, plastics, adhesives, and sealants, as well as cleaners, floor coatings, and lubricants. BYK additives are also used in the construction chemicals, oil and gas, and foundry industries.

BYK products are categorized into various groups such as adhesion promoters and coupling agents, defoamers and air release additives, processing additives, rheology additives, surface additives, viscosity reducers, wax additives, and wetting and dispersing additives. There can be hundreds of products in each group. With so many different additives and mechanical products, BYK has numerous supply chains that span the globe. Until recently, however, BYK was unable to manage these supply chains effectively. The company was saddled with disparate and isolated legacy supply chain demand planning systems that impeded forecast accuracy and made it much harder for BYK to manage growth and maintain its competitive position. BYK needed a solution that enhanced data and process transparency and established a single "source of truth" for sales, operations,

and demand-planning processes to harmonize these processes across the organization and improve forecast accuracy. The company selected SAP Integrated Business Planning for Supply Chain for this purpose.

SAP Integrated Business Planning for Supply Chain is a cloud-based solution that synchronizes an organization's supply chain planning in real time. It coordinates sales and operations planning (S&OP), forecasting and demand, response and supply, demand-driven replenishment, and inventory planning and optimization on one platform. Its capabilities include automated, tightly coordinated supply chain planning processes, "what-if" simulations, advanced AI machine learning algorithms and planning capabilities, and integration with SAP Supply Chain Control Tower and other solutions. With embedded analytics, what-if simulations, alerts, and other machine learning capabilities, SAP IBP allows businesses to quickly identify and respond to disruptions.

SAP Supply Chain Control Tower is a cloud software solution providing end-to-end visibility and control over an organization's entire supply chain network, including suppliers, manufacturers, and business partners. It collects and consolidates up-to-the-minute data in real time from various sources, such as sensors, IoT devices, and external systems, to provide a comprehensive real-time view of what's happening across an organization's entire supply chain network at any given moment. Supply chain control towers enable organizations to manage what they cannot see, plan for unknown variations, and mitigate disruptions and risk before they become major issues. In today's rapidly changing global business environment, historical data can no longer accurately predict current and future demand. With disruptions ranging from extreme weather to trade wars to global pandemics, complete visibility and instant access to up-to-date information are essential for supply chain agility and resilience. SAP Supply Chain Control Tower includes capabilities for "what-if" analysis and comparing scenarios; simulations of demand and supply changes, exception flagging and handling, and collaboration with partners across the supply chain. Organizations using SAP Supply Chain Tower are able to manage their supply chains more proactively.

For the implementation and expertise with SAP systems, BYK enlisted the services of SAP Services and Support and SAP Preferred Success, which included architecture planning, implementation guidance, and process reengineering assistance. Additionally, SAP helped BYK deploy SAP Integrated Business Planning and SAP Control Tower in a technically standardized and streamlined way.

Implementing these supply chain solutions produced major benefits for BYK. Simplified and integrated processes for operations, demand planning and sales made it easier for the company to coordinate its own supply chain activities and collaborate with its supply chain partners. BYK was able to work with a single platform and source of accurate data across the company. The company was able to increase forecast accuracy with real-time data reflecting changes to planned demand. Planning teams could make decisions based on their financial impact. BYK plans to continue work on optimizing the implemented processes.

SOURCES: Karin Fent, "How Chemicals Supplier BYK Harmonized Supply Chain Planning Processes," Sap.com, accessed September 9, 2024; Byk.com, accessed September 9, 2024.

Case Study Questions

1. Assess the importance of supply chain management for BYK.
2. Identify and describe BYK's problems in supply chain management.
3. Describe how the SAP solutions BYK implemented changed the way BYK ran its business.

AI and Supply Chain Management

Global supply chains have become increasingly complex, interconnected, and far-reaching, making them much more difficult to manage. Manufacturers of finished goods often rely on hundreds, if not thousands, of components shipped from partners around the globe to arrive in their assembly facilities on a coordinated schedule. Components and finished goods must be tracked every time they switch hands. AI is very useful for analyzing the proliferating amount of big data generated by modern global supply chains to develop more accurate forecasts, reveal operational insights, and improve the efficiency of storage and transportation processes across vast logistics networks with multiple partners. AI has proven remarkably adept at improving global supply chain planning, management, and operations. Table 9.3 provides some examples.

Table 9.3 Examples of AI Applications for SCM Systems

Capability	Description
More accurate inventory management	AI can manage inventory levels with greater precision and efficiency. AI-powered forecasting systems can use inventory information to forecast customer demand and adjust product stocking accordingly. Computer vision systems (such as cameras installed on supply chain infrastructure, vehicles, and drones) can calculate inventory levels in real time throughout the supply chain and monitor warehouse storage capacity.
More timely deliveries	Logistics companies use machine learning to train models that optimize and manage the delivery routes as components move along the supply chain. These models can prioritize shipments based on order volumes, delivery promises, contractual deadlines, customer importance, or product availability. AI can also provide all parties in distribution networks with more-accurate estimated times of arrival.
Improved supply chain sustainability	AI can drive operational efficiencies that make supply chains more sustainable, such as providing information to help organizations reduce delivery vehicle fuel consumption or wasted materials at various steps in the supply chain. More precise demand forecasts and production planning help ensure that manufacturers don't overproduce.
More precise demand planning	AI has become the gold standard for predicting demand based on both internal signals. such as sales patterns, marketing leads, economic outlook, and broader market trends. AI-infused demand planning software is capable of estimating the potential impact of external phenomena such as severe weather events, economic downturns, and political disruptions.
Simulations for supply chain optimization	AI-powered simulations provide visual insights into the operations of complex logistical networks and recognize ways to improve them. When AI selects the models and controls the workflows, these simulations become more accurate than those run with traditional computing methods. This application of AI can help engineers and production managers assess the impacts of redesigning products. Planners and operations managers can evaluate the potential impact of changing suppliers, redirecting shipping and distribution routes, or relocating storage and distribution hubs.

9.4 Understand CRM Systems

You've probably heard phrases such as "the customer is always right" and "the customer comes first." Today these words ring truer than ever. Because competitive advantage based on an innovative new product or service is often very short lived, companies are realizing that their most enduring competitive strength may be their relationships with their customers. Some say that the basis of competition has switched from who sells the most products and services to who "owns" the customer, with others stressing that customer relationships represent a firm's most valuable assets.

What Is Customer Relationship Management?

What kinds of information would you need to build and nurture strong, long-lasting relationships with customers? You'd want to know exactly who your customers are, how to contact them, whether they are costly to service and sell to, what kinds of products and services they are interested in, and how much money they have spent at your company. If you could, you'd want to make sure that you knew each of your customers well, as though you were running a small-town store. And you'd want to make your good customers feel special.

In a small business operating in a neighborhood, it is possible for business owners and managers to know their customers well on a personal, face-to-face basis, but in a large business operating on a metropolitan, regional, national, or even global basis, it is

impossible to know your customers in this intimate way. In these kinds of businesses, there are too many customers and too many ways that customers interact with the firm (over the web, the phone, email, social media, and in person). It becomes especially difficult to integrate information from all these sources and deal with the large number of customers.

A large business's processes for sales, service, and marketing tend to be highly compartmentalized, and these departments may not share much essential customer information. Some information on a specific customer might be stored and organized in terms of that person's account with the company. Other pieces of information about the same customer might be organized by products that the customer purchased. In this traditional business environment, there is no convenient way to consolidate all this information to provide a unified view of a customer across the company.

This is where customer relationship management (CRM) systems help. CRM systems, which we introduced in Chapter 2, capture and integrate customer data from all over the organization, consolidate the data, analyze the data, and then distribute the results to various systems and customer touch points across the enterprise. A **touch point** (also known as a contact point) is a method of interaction with the customer, such as through telephone, email, customer service desk, conventional mail, social network, website, mobile device, or retail store. Well-designed CRM systems provide a single enterprise view of customers that is useful for improving both sales and customer service.

Touch point

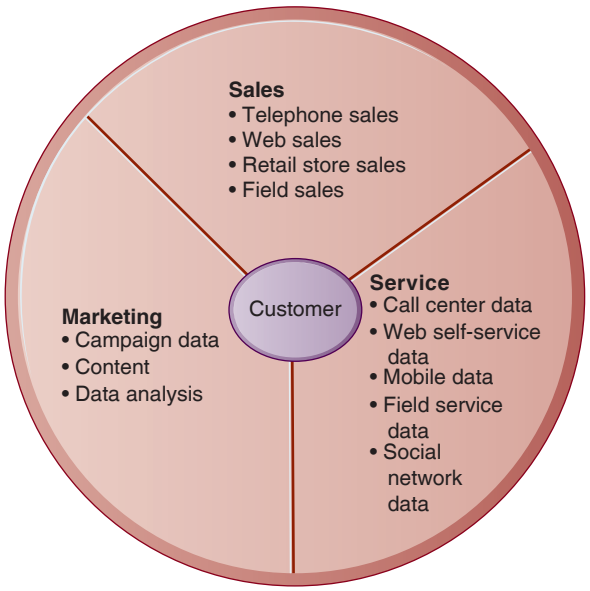
Method of interaction with a customer, such as telephone, email, customer service desk, conventional mail, social network, website, mobile device, or retail store. Also known as a contact point.

Examine Figure 9.6, which illustrates the various types of customer data that CRM systems collect: sales data such as telephone, web, retail store, and field sales; service data such as call center, web self-service, mobile, field service, and social network data; and marketing data such as data generated from campaigns, content marketing, and data analysis.

Good CRM systems provide data and analytical tools for answering questions such as these: What is the value of a particular customer to the firm over their lifetime? Who are our most loyal customers? Who are our most profitable customers? What do these profitable customers want to buy? Firms use the answers to these questions to acquire new customers, provide better service and support to existing customers, customize their offerings more precisely to customer preferences, and provide ongoing value to retain profitable customers.

Figure 9.6 CRM Systems

CRM systems examine customers from a multifaceted perspective. These systems use a set of integrated applications to address all aspects of the customer relationship, including customer service, sales, and marketing.



CRM System Software Packages

Commercial CRM system software packages range from niche tools that perform limited functions, such as personalizing websites for specific customers, to large-scale enterprise applications that capture myriad interactions with customers, analyze the interactions using sophisticated reporting tools, and link to other major enterprise applications, such as SCM and ERP systems. Major CRM system software vendors include Oracle, SAP, Salesforce, and Microsoft Dynamics CRM. CRM systems typically include tools for sales, customer service, and marketing. We briefly describe some of these capabilities.

Sales Force Automation

Sales force automation modules (SFA modules) in a CRM system help sales staff increase productivity by focusing their sales efforts on the most profitable customers, those who are good candidates for sales and services. SFA modules provide sales prospect and contact information, product information, product configuration capabilities, and sales quote generation capabilities. Such software can assemble information about a particular customer's past purchases to help the salesperson make personalized recommendations. SFA modules enable sales, marketing, and shipping departments to share customer and prospect information easily. SFA increases each salesperson's efficiency by reducing the cost per sale as well as the cost of acquiring new customers and retaining old ones. SFA modules also provide capabilities for sales forecasting, territory management, and team selling.

Sales force automation modules (SFA modules)
CRM systems module that helps staff increase productivity by focusing sales efforts on the most profitable customers.

Customer Service

CRM systems support customer service with customer service modules that provide information and tools to increase the efficiency of call centers, help desks, and customer support staff. They also have capabilities for assigning and managing customer service requests.

One such capability is an appointment or advice telephone line. When a customer calls a standard phone number, the system routes the call to the correct serviceperson, who inputs information about that customer into the system only once. When the customer's data are in the system, any service representative can handle the customer relationship. Improved access to consistent and accurate customer information helps call centers handle more calls per day and decrease the duration of each call. Thus, call centers and customer service groups achieve greater productivity, reduced transaction time, and higher quality of service at lower cost. Customers are happier because they spend less time on the phone restating their problem to customer service representatives.

CRM systems may also include online self-service capabilities. The company website can be set up to provide inquiring customers with personalized support information as well as the option to contact customer service staff by phone or online chat for additional assistance.

Marketing

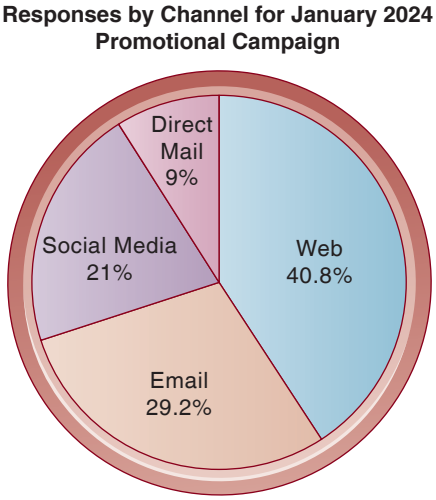
CRM systems support marketing campaigns by providing capabilities that enable companies to capture prospect and customer data, provide product and service information, qualify leads for targeted marketing, and schedule and track direct-marketing mailings or email.

Examine Figure 9.7, which illustrates a sample produced by a CRM system analyzing the results of a promotional campaign. The analysis shows that the most productive channel was the web, followed by email and social media. Direct mail was the least effective channel. Marketing modules also include tools for analyzing marketing and customer data, identifying profitable and unprofitable customers, designing products and services to satisfy specific customer needs and interests, and identifying opportunities for cross-selling. **Cross-selling** is the marketing of complementary products to customers. (For example, in financial services, a customer with a checking account might be sold a money market account or a home improvement loan.) CRM tools also help firms manage and execute marketing campaigns at all stages, from planning to determining the rate of success for each campaign.

Cross-selling
Marketing of complementary products to customers.

Figure 9.7 How CRM Systems Support Marketing

CRM systems provide a single point for users to manage and evaluate marketing campaigns across multiple channels, including the web, email, social media, and direct mail.



Examine Figure 9.8, which illustrates the most important capabilities of sales, marketing, and service processes found in major CRM systems. Like ERP systems, CRM systems area is business-process driven, incorporating hundreds of business processes thought to represent best practices in each of these areas.

To achieve maximum benefit, companies need to revise and model their business processes to conform to the best-practice business processes embedded within the CRM system. Examine Figure 9.9, which illustrates how a best practice for increasing customer loyalty through customer service might be modeled by CRM software. Directly servicing customers provides firms with opportunities to increase customer retention by singling out profitable, long-term customers for preferential treatment. CRM software can assign each customer a score based on that person's value and loyalty to the company and provide that information to help call centers route customer service requests to agents who can best handle the customers' needs. The system can automatically provide the service agent with detailed profiles of customers that include their scores for value and loyalty. The service agent can then use this information to present special offers or additional services to the customers to encourage the customers to keep transacting business with the company.

Operational and Analytical CRM

Operational CRM

Customer-facing applications, such as tools for sales force automation, call center and customer service support, and marketing automation.

Analytical CRM

Applications that analyze the customer data generated by operational CRM applications to provide information for improving business performance.

All of the applications we have just described support either the operational or the analytical aspects of customer relationship management. **Operational CRM** includes customer-facing applications, such as tools for sales force automation, call center and customer service support, and marketing automation. **Analytical CRM** includes applications that analyze the customer data generated by operational CRM applications to provide information for improving business performance.

Examine Figure 9.10, which illustrates how an analytical CRM application works. Analytical CRMs are based on customer data from operational CRM systems, customer touch points, and other sources that have been organized in data warehouses or analytic platforms for use in online analytical processing (OLAP), data mining, and other data analysis

Figure 9.8 CRM System Capabilities

The major CRM systems support business processes in sales, service, and marketing by integrating customer information from many sources.

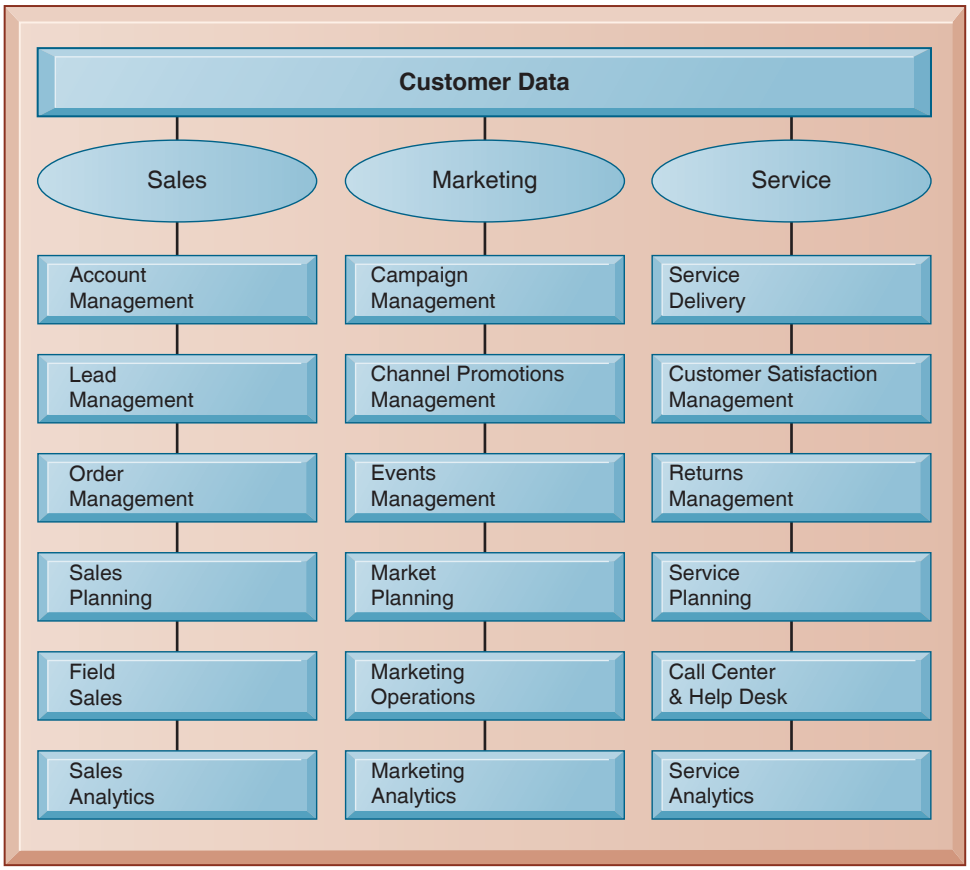
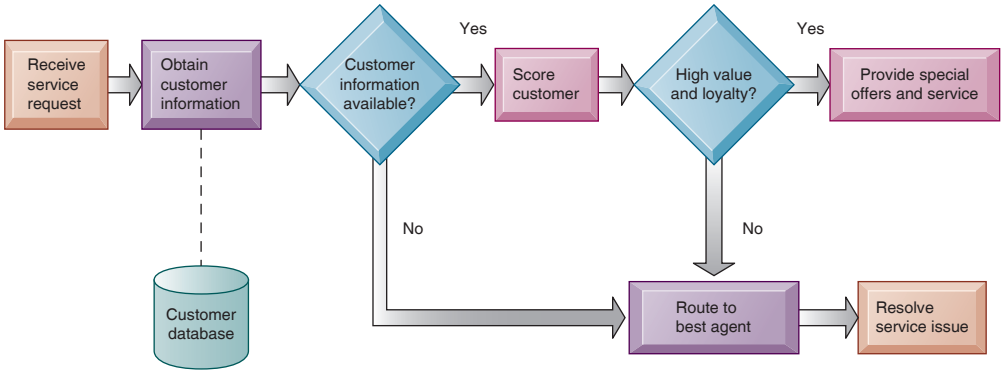


Figure 9.9 Customer Loyalty Management Process Map

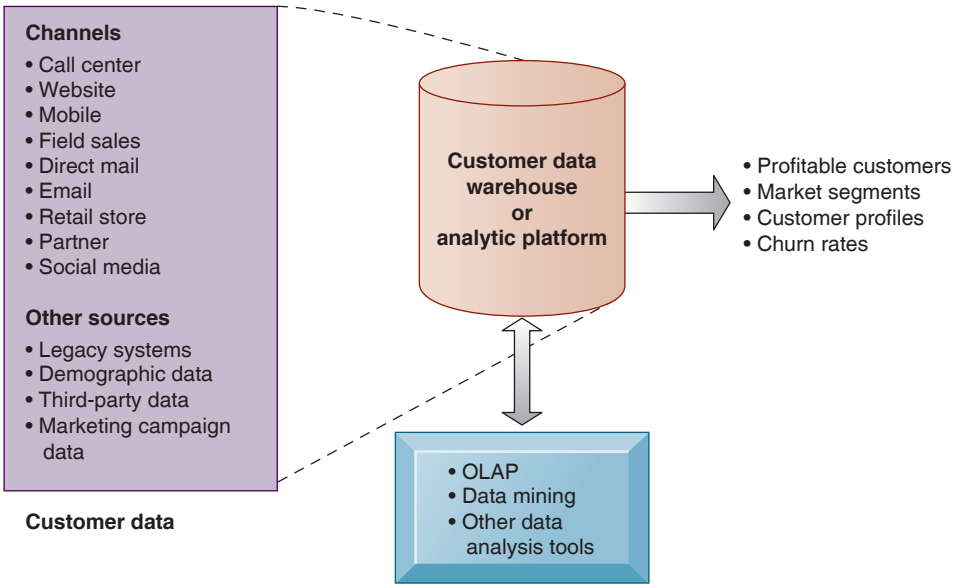
This process map shows how a best practice for promoting customer loyalty through customer service would be modeled by CRM software, helping firms identify high-value customers for preferential treatment.



techniques (see Chapter 6). Customer data collected by the organization might be combined with data from other sources, such as customer lists for direct-marketing campaigns purchased from other companies or demographic data. Such data are analyzed to pinpoint profitable and unprofitable customers, create segments for targeted marketing and customer profiles, and identify buying patterns.

Figure 9.10 Analytical CRM

Analytical CRM uses a customer data warehouse or analytic platform and tools to analyze customer data collected from the firm's customer touch points and from other sources.



Customer lifetime value (CLTV)

Based on the relationship among the revenue produced by a specific customer, the expenses incurred in acquiring and servicing that customer, and the expected life of the relationship between the customer and the company.

Another important output of analytical CRM is the customer's lifetime value to the firm. **Customer lifetime value (CLTV)** is based on the relationship among the revenue produced by a specific customer, the expenses incurred in acquiring and servicing that customer, and the expected life of the relationship between the customer and the company.

Business Value of CRM Systems

Companies with effective CRM systems realize many benefits, including increased customer satisfaction, reduced direct-marketing costs, more effective marketing, and lower costs for customer acquisition and retention. Information from CRM systems increases sales revenue by identifying the most profitable customers and segments for focused marketing and cross-selling.

Churn rate

Measurement of the number of customers who stop using or purchasing products or services from a company. Used as an indicator of the growth or decline of a firm's customer base.

Customer churn is reduced as sales, service, and marketing respond better to customer needs. The **churn rate** measures the number of customers who stop using or purchasing products or services from a company. It is an important indicator of the growth or decline of a firm's customer base.

CRM and AI

Machine learning, generative AI, and other AI technologies are being integrated into CRM systems to automate, enhance, and optimize CRM processes. AI enables businesses to analyze vast amounts of customer data in real time, offering predictive insights, automating routine tasks, and further personalizing customer interactions. AI can intelligently prioritize leads, forecast sales trends, and provide chatbot support, among other functionalities. By harnessing AI, CRM systems become more proactive, helping businesses anticipate customer needs, streamline operations, and drive more sales and customer satisfaction. Table 9.4 provides examples of these capabilities.

Table 9.4 Examples of AI in CRM Systems

Capability	Description
Digital assistant agent	Use AI-powered conversational bots to respond to customer inquiries by providing support, answering questions, capturing details, and resolving issues without human interaction.
Field service routing optimization	Use predictive AI and machine learning with individual performance pattern data to optimize field service scheduling to maximize efficient use of resources and to help service teams complete more jobs per day.
Lead conversion and sales opportunity win probability	Provide sales reps with rapid identification of their best leads and sales opportunities with predictive scoring and win probabilities. Additionally, identify sales deals that are falling behind or need more attention.
Pricing optimization	Use AI-recommended pricing and discounts to strike a balance between achieving the most revenue with the highest likelihood to close.
Predictive scoring	Use machine learning to predict which leads and acts are most likely to engage and convert to buyers.

9.5 Discuss Enterprise Application Challenges

Many firms have implemented ERP, SCM, and CRM systems because these systems are such powerful instruments for achieving operational excellence and enhancing decision making. But precisely because they are so powerful in changing the way the organization works, they are challenging to implement. Let's briefly examine some of these challenges as well as new ways of obtaining value from these systems.

Enterprise Application Challenges

Promises of dramatic reductions in inventory costs, order-to-delivery time, more efficient customer response, and higher product and customer profitability make ERP, SCM, and CRM systems very alluring. But to obtain this value, you must clearly understand how your business needs to change to use these systems effectively.

Enterprise applications involve complex pieces of software that are expensive to purchase and implement. According to a 2024 survey of 131 ERP users conducted by Panorama Consulting Group, 27.5 percent of ERP projects that completed implementation experienced cost overruns, and 5.3 percent of the projects had significant overruns. In addition, almost one-third of the projects were completed later than anticipated. Of those that were over budget, the most common reason was organizational issues (Panorama Consulting Group, 2024). Changes in project scope and additional customization work add to implementation delays and costs.

Enterprise applications require not only deep-seated technological changes but also fundamental changes in the way the business operates. Companies must make sweeping changes to their business processes to work with the software. Employees must accept new job functions and responsibilities. They must learn how to perform a new set of work activities and understand how the information they enter into the system can affect other parts of the company. This requires new organizational learning and should also be factored into enterprise application implementation costs.

SCM systems require multiple organizations to share information and business processes. Each participant in the system may have to change some of its processes and the way it uses information to create a system that best serves the supply chain as a whole.

Some firms experience enormous operating problems and losses when they first implement enterprise applications because they don't understand how much organizational change is required. Take, for example, Mission Produce, which packs, ripens, and distributes just-in-time ripe avocados all over the world. In November 2021, the company transitioned to

an ERP system to support international growth with improved operational visibility and financial reporting capabilities. Despite countless hours spent planning and preparing for conversion to the new ERP system, the company experienced significant challenges with the system's implementation. As a result, Mission no longer knew for sure how many avocados it had on hand, nor how ripe they were, leaving many unfit for sale. The company had to buy fruit from other suppliers to meet its delivery commitments, incurring large additional costs. The extent and magnitude of the disruption was much greater than Mission anticipated. Management stated that the $22.2 million drop in gross profit for the quarter following the ERP go-live was primarily due to the ERP problem.

Woolworth's Australia also encountered data-related problems when it transitioned from an antiquated home-grown ERP system to SAP. Weekly profit-and-loss reports tailored for individual stores couldn't be generated for nearly 18 months. The company then had to change its data collection procedures but failed to understand its own processes or to document these business processes properly (Fruhlinger et al., 2022).

Enterprise applications also introduce switching costs. When you adopt an enterprise application from a single vendor, such as SAP, Oracle, or others, it is very costly to switch vendors, and your firm becomes dependent on the vendor to upgrade its product and maintain your installation.

Enterprise applications are based on organization-wide definitions of data. You'll need to understand exactly how your business uses its data and how the data would be organized in a CRM, SCM, or ERP system. CRM systems typically require some data-cleansing work (see Chapter 6).

Enterprise application system vendors are addressing these problems by offering pared-down versions of their software, fast-start programs for small and medium-sized businesses, and best-practice guidelines for larger companies. Companies are also achieving more flexibility by using hybrid cloud applications for functions not addressed by basic enterprise application systems so that companies are not constrained by a single, do-it-all type of system. (See the discussion of composable ERP at the end of this section.)

Companies adopting enterprise applications can also save time and money by keeping customizations to a minimum. For example, Kennametal, a $2 billion metal-cutting tools company in Pennsylvania, had spent $10 million over 13 years maintaining an ERP system with more than 6,400 customizations. The company replaced this ERP system with a plain-vanilla, uncustomized version of SAP enterprise software and changed its business processes to conform to the software.

Enterprise Application Trends

Today, enterprise application vendors are delivering more value by becoming more flexible, user-friendly, web-enabled, mobile, and capable of integration with other systems. Stand-alone ERP, SCM, and CRM systems are becoming a thing of the past. The major enterprise application vendors are making their ERP, SCM, and CRM systems work closely with each other and link to systems of customers and suppliers.

ERP, SCM, and CRM systems harbor huge reservoirs of data, which make them strong candidates for cloud infrastructures, advanced software integrations, and embedded AI. The emphasis is on incorporating more analytics, machine learning, and generative AI into ERP, SCM, and CRM systems to improve efficiency and broaden their capabilities to handle complex data analysis, predictive modeling and real-time decision making. Additionally, cloud-based versions of these tools are becoming the platform of choice. According to Panorama Consulting's 2024 ERP Report, 21.4 percent of ERP systems it surveyed now run in the cloud. Accenture consultants found that 21 percent of cloud computing leaders it surveyed had cloud-based SCM systems. Eighty-seven percent of CRM systems are cloud-based. Cloud-based enterprise applications offer more scalability, flexibility, lower installation costs, and easier integration with other business management software.

Composable ERP

Many businesses today are looking for more flexible hybrid enterprise solutions that use best-of-breed components from different technology vendors. For enterprise resource planning, for example, a company might use SAP S/4HANA for ERP, but also use SAP SuccessFactors

for Human Resources and a combination of SAP Ariba and a third-party CRM tool such as Salesforce to manage invoices, payments, accounts, and contracts. The solutions do not rely exclusively on SAP solutions if there are better solutions available that can be tailored to firms' individual needs and easily integrated into an SAP deployment. This adaptive technology strategy is known as composable ERP. **Composable ERP** brings together disparate systems and components in a way that allows them to work together as a platform. Users can pick and choose the modules they want and install them one by one rather than having to do everything at once. "Composing" an ERP in this fashion gives an organization more flexibility to utilize different technologies and platforms to best serve their business. Composable ERP is a relatively new approach, but it is gaining prominence among businesses of all sizes. Composable tools for SCM and CRM systems are also becoming available.

Composable ERP

Brings together disparate systems and components in a way that allows them to work together as a platform.

9.6 Describe How Enterprise Applications Are Using AI

Enterprise application vendors have been adding business intelligence and AI tools to help managers obtain more meaningful information from the massive amounts of data these systems generate, including data from the Internet of Things (IoT). AI-enabled tools are capable of much more rapid and complex data analysis than traditional software, as well as capabilities for generating new content. We have already described major AI applications for ERP, SCM, and CRM earlier in this chapter. Table 9.5 provides examples of AI software tools from major enterprise application vendors.

9.7 Understand How the Information in This Chapter Can Help Your Career

The information in this chapter can help you advance your career by showing you how enterprise applications such as ERP, SCM, and CRM systems make businesses more efficient and effective. It describes the different types of systems of this sort that you are likely to encounter, and how they improve operations and decision making. Chapter 9 also describes how AI technology is making these systems more powerful. You'll also be able to identify the typical challenges and problems you are likely to encounter when using or building enterprise applications and learn ways of dealing with them.

Table 9.5 AI Functionality Provided by Major Enterprise Application Vendors

Vendor	Functions	AI
SAP	ERP, CRM, SCM, HCM	Machine learning and generative AI integrated with SAP business processes and solutions. For example, SAP Business AI is built into SAP finance applications to manage capital effectively, improve reporting accuracy, mitigate fraud risk and reduce losses with anomaly detection and prevention. Joule generative AI assistant acts as a work copilot across SAP business systems, answering questions with natural language.
Oracle	Oracle offers a range of AI solutions, including generative AI, AI services, and AI infrastructure.	AI services, including generative AI, have prebuilt machine learning models that make it easier for developers to apply AI to applications and business operations. The models can be custom trained. Oracle Cloud Infrastructure (OCI) Generative AI is a fully managed service to seamlessly integrate these versatile language models into a wide range of use cases, including writing assistance, summarization, analysis, and chat. Oracle Fusion Cloud CX, includes generative AI-assisted answer generation, assisted scheduling for field service, opportunity quality scoring, and seller engagement recommendations.
Microsoft	CRM, ERP, primarily small to medium-size companies	Microsoft Dynamics 365 AI suite includes Dynamics 365 Copilot AI assistant across all business functions. Microsoft Dynamics Customer Insights, Microsoft Dynamics Sales Insights, Microsoft Dynamics 365 Customer Service Insights are key offerings
Salesforce	CRM	Einstein suite of AI tools includes Einstein GPT Generative AI for CRM, which delivers AI-created content across every sales, service, marketing, commerce, and IT interaction.

Real-World Scenario

SAP Business ByDesign Helps Customer Relationship Managers at Skullcandy

Skullcandy is a leading maker of audio and gaming headphones, earbuds, speakers, and other accessories. The brand is a favorite of skiers, surfers, skaters, and other action sports enthusiasts and has become wildly popular worldwide. Skullcandy expanded so rapidly that it outgrew its distribution model. Skullcandy's business processes were primarily manual, with individual spreadsheets serving as the system of record for its business transactions. In order to grow into a company that could ship one million items each year, Skullcandy needed to replace its manual business processes with new set of information systems to support them. Skullcandy implemented SAP Business ByDesign, a cloud-based companywide enterprise system supporting end-to-end business processes for financial resources management, project management, supply chain management, supplier relationship management, human resources management, customer relationship management, and compliance management. The applications are integrated, and the system is the backbone for tracking all the transactions of the company. Skullcandy runs the BusinessByDesign software in SAP cloud computing centers. The new system functionality has helped the business acquire and service large customers such as Target, BestBuy, and Walmart, expand internationally, and significantly increase its customer base. Skullcandy now has 300 employees and 2023 revenue of $266.3 million. Nearly 80% of Skullcandy's transactions, including acquiring and servicing customers, are fully automated from point of sale through delivery.

Job Description and Career Path: Customer Relationship Manager

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Customer relationship manager	Job responsibilities include: • Developing and implementing strategies for building strong customer relationships. • Understanding customer needs, preferences, and feedback. • Acting as the main point of contact for customer inquiries and issue resolution. • Collaborating with sales, marketing, and support teams. • Monitoring customer account activity and providing personalized communication. • Utilizing CRM systems to manage customer interactions and track performance metrics	• Knowledge of Microsoft 365 tools: Excel, Word, PowerPoint, Teams • Experience with Salesforce CRM software	• Entry-Level: Many CRMs start in sales, customer service, or banking positions. • Early Career: Progression to roles like Customer Relationship Officer or Account Manager. • Mid- Career: Customer Relationship Manager: Increased responsibilities and strategic roles within customer relationship management. • Late Career: Further advancement, potentially to Marketing Director, Senior Project Manager, or other Executive-level position. The U.S. average annual salary for a customer relationship manager is $150,887, with a range of $129,855 to $169,052. Salary can also vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

MillerKnoll Uses Salesforce and AI to Transform Its Business Strategy

MillerKnoll, formerly known as Herman Miller, Inc., is a Michigan-headquartered company that produces innovative office furniture, equipment, and home furnishings. MillerKnoll has subsidiaries and corporate offices, independent dealers, retailers, and licensees in more than 40 countries in North America, Asia Pacific, Europe, the Middle East, and Latin America. Its products and services are sold through retail stores, e-commerce websites, catalogs, and a global dealer network of independent businesses. The company generated $3.8 billion in 2023 revenue and employed 11,600 people worldwide.

When the Covid-19 pandemic forced people all over the world to work at home, MillerKnoll's furniture was in high demand because of its ability to blend design, functionality, and comfort. E-commerce sales skyrocketed, inspiring the company to transform its business strategy and online store. The strategy called for providing a frictionless omni-channel experience that was more convenient and personalized for customers. MillerKnoll wanted to be able to engage with customers from the first moment of design inspiration; operate a user-friendly, leading-edge digital storefront; and provide exceptional customer service during and after the purchase process.

To serve customers better and manage all of its relationships with them, MillerKnoll turned to Salesforce, which features cloud-based tools for CRM and application development. Salesforce helps MillerKnoll identify key customer segments, develop closer ties to those customers, and design differentiated experiences tailored to each customer's needs. Salesforce CRM tools make it possible for MillerKnoll to have a single, consistent view of each customer across all the various channels through which that person interacts with the company.

MillerKnoll uses Salesforce Marketing Cloud to develop complex customer journeys (customer experiences with a brand across all touch points) and sophisticated email campaigns. Salesforce Marketing Cloud is a CRM platform that allows marketers to create and manage marketing relationships with customers and campaigns. The Marketing Cloud incorporates integrated solutions for customer journey management, email, mobile, social media, web personalization, advertising, content creation, content management, and data analysis. Every imaginable customer interaction and engagement is covered. The software includes predictive analytics to help make decisions such as, for example, what channel would be preferable for a given message. A component called Journey Builder helps marketers tailor campaigns to customers' behaviors and needs, demographics, and communication channel preferences.

Each of MillerKnoll's brands has a different customer base, and these bases largely do not overlap. Marketing Cloud helps the company make sure that it is treating each of these brands and customers differently, understands preferences across different channels, and uses that data to personalize customer interactions. Previously, MillerKnoll merchandising operations were manual and required too much time to put products in the right place and build interactions that had to change frequently.

The Marketing Cloud is connected to Salesforce's Sales Cloud and Service Cloud to provide a unified experience and prevent customers from being contacted separately by representatives from sales, marketing, and service groups. Service Cloud is a platform for customer service and support. Companies using Service Cloud can automate service processes, streamline workflows, and locate key articles, topics, and experts with information to help the company's 1,100 customer service agents. Service Cloud can "listen" and respond to customers across a variety of social platforms and automatically route cases to the appropriate agent. Service for Apps makes it possible to embed customer support software into mobile applications, including features for live agent video chat, screen sharing, and on-screen guided assistance. Service Cloud makes it possible to deliver service that is more personalized and convenient in whatever form each customer prefers—email, web, social media, or telephone—all from a single application. Service Cloud helped MillerKnoll handle the spike in customer service requests during the Covid-19 pandemic.

Sales Cloud is a cloud-based application designed to help salespeople sell more quickly by centralizing customer information, logging their interactions with your company, and automating many of the tasks that salespeople do every day. This means they'll spend less time on administration and more time closing deals. For sales managers, Sales Cloud gives real-time visibility into their team's activities, so forecasting sales with confidence is not as difficult. Sales Cloud is easy to use and customizable to the way salespeople work. MillerKnoll uses Sales Cloud to manage sales leads and opportunities across brands, monitor team performance, track sales insights, and automate tasks through a single dashboard.

MillerKnoll relaunched its website on Salesforce Commerce Cloud, resulting in increased site speed and a richer shopping process. Shoppers can now browse and purchase products more easily, with more tailored product recommendations, a streamlined checkout process, and tools for live video chat with sales associates.

Having unified customer data with a powerful platform to manage customer relationships helped MillerKnoll retain and

use vital customer information as the company continued to grow globally. With sales, marketing, and service teams united around the customer, MillerKnoll was able to meet customer expectations for personalized service as it expanded. According to Ben Groom, MillerKnoll's chief digital officer, Salesforce helped the company become more integrated across all of its brands, and its online shopping experience mirrors the same quality as its products.

MillerKnoll uses AI and data-driven approaches to enhance its sustainability efforts and deepen its relationship with customers. Promoting sustainability makes its products even more appealing. For example, MillerKnoll uses AI to track and manage its carbon emissions, energy use, waste, and water data. The company implemented Salesforce Net Zero Cloud, a sustainability management platform powered by Salesforce Einstein AI that is designed to help organizations manage their environmental impact and monitor their progress toward net zero carbon emissions. Companies that use Net Zero Cloud obtain a single source of data for their carbon emissions. MillerKnoll can analyze individual assets and understand impact across location and time. Net Zero Cloud also provides useful reports and executive-ready visualizations in a few clicks to facilitate tracking and management of emissions.

Net Zero Cloud is able to convert procurement spending data into a measure of total carbon dioxide output Co_2. In the past, MillerKnoll had to track carbon emissions from many different sources manually using spreadsheets. Net Zero Cloud makes it possible to track carbon emissions across the organization in real-time and eliminate unwieldy and time-consuming manual processes. The tool can even measure emissions from suppliers in its value chain and identify suppliers that have a high impact on a company's total emissions.

New Salesforce Einstein AI features for Net Zero Cloud make corporate environmental, social, and governance (ESG) reporting even easier as companies confront a rapidly evolving regulatory landscape. The EU's Corporate Sustainability Reporting Directive (CSRD) is already in effect. This directive is aimed at driving change in the business behavior of companies that operate in the EU, which includes U.S. companies with EU subsidiaries that meet certain criteria. CSRD sets out environmental, social, and governance (ESG) reporting requirements for companies and aims to hold organizations accountable for their environmental and social impacts and to accelerate the transition to a fully sustainable and inclusive economic and financial system. The directive is very complex, calling for identification of risks, opportunities, and material impacts related to social, environmental, and governance issues. Approximately 50,000 companies—including many large U.S.-based multinationals—are required to report. Net Zero Cloud's new capabilities include CSRD Report Builder, which automates report generation in alignment with CSRD requirements, helping companies identify ESG topics that are most material to their business.

By leveraging AI and data, MillerKnoll is able to make more informed decisions about their products and processes, with a strong focus on sustainability. MillerKnoll hopes that they can be good stewards of the earth's resources while making each product and experience exceptionally designed.

SOURCES: Millerknoll.com, accessed September 9, 2024; Salesforce.com, "That Chair That You Bought? See How MillerKnoll Computes Its Carbon Impact," Salesforce.com, accessed September 9, 2024; "MillerKnoll, a Company Built on Iconic Designs, Now Operates as One B2C Enterprise," Capgemeni.com, accessed September 9, 2024; "How MillerKnoll Transformed Its B2B eCommerce Capabilities with Connected Commerce," Tradecentric.com, accessed September 9, 2024.

Case Study Questions

1. What is MillerKnoll's business strategy? What is the role of customer relationship management in that strategy?
2. How did using Salesforce change the way MillerKnoll ran its business?
3. What is the role of AI in MillerKnoll's business model?
4. Give examples of two business decisions at MillerKnoll.

Review Summary

9.1 Understand ERP systems.

Enterprise resource planning (ERP) systems are based on a suite of integrated software modules and a common central database. The database collects data from and feeds the data into numerous applications that can support nearly all of an organization's internal business processes. When one process enters new information, the information is made available immediately to the other business processes. ERP systems support organizational centralization by enforcing uniform data standards and business processes throughout the company and a single, unified technology platform. ERP systems provide value by both increasing operational efficiency and providing firmwide data to help managers make better decisions. ERP systems vendors are increasingly incorporating AI into ERP software.

9.2 Understand SCM systems.

Supply chain management (SCM) systems coordinate planning, production, and logistics with suppliers. SCM systems

automate the flow of information among members of the supply chain so that they can use the information to make better decisions about when and how much to purchase, produce, or ship. More accurate information from SCM systems reduces uncertainty and the impact of the bullwhip effect. SCM software includes software for supply chain planning and for supply chain execution. In addition to reducing costs, supply chain management systems facilitate efficient customer response, enabling the workings of the business to be driven more by customer demand.

9.3 Describe global supply chain management challenges.

Global supply chains typically span greater geographic distances and time differences than domestic supply chains do and have participants from a number of countries. Performance standards may vary from region to region or from nation to nation. Supply chain management may need to reflect foreign government regulations and cultural differences. Internet technology facilitates the management of global supply chains by providing the ability of organizations in different countries to share supply chain information. Improved communication among supply chain members also facilitates efficient customer response and movement toward a demand-driven model. AI is being incorporated into many SCM systems to help manage supply chain challenges.

9.4 Understand CRM systems.

Customer relationship management (CRM) systems integrate and automate customer-facing processes in sales, customer service, and marketing, providing an enterprise-wide view of customers. Companies can use this customer knowledge when they interact with customers to provide them with better service or to sell new products and services. These systems also identify profitable or unprofitable customers or opportunities to reduce the churn rate. Operational CRM includes customer-facing applications, such as tools for sales force automation, call center and customer service support, and marketing automation. Analytical CRM includes applications that analyze the customer data generated by operational CRM applications to provide information for improving business performance. AI is being incorporated into many CRM systems to help automate, enhance, and optimize CRM processes.

9.5 Discuss enterprise application challenges.

Enterprise applications are difficult to implement. They require extensive organizational change; large, new software investments, and careful assessment of how these systems will enhance organizational performance. Enterprise applications cannot provide value if they are implemented atop flawed processes or if firms do not know how to use these systems to measure performance improvements. Employees require training to prepare for new procedures and roles. Attention to data management is essential. Enterprise applications have become more flexible, web-enabled and capable of integration with other systems. They can run in cloud infrastructures or on mobile platforms. Enterprise applications are incorporating business intelligence capabilities to analyze the large quantities of data that enterprise applications generate. Composable ERP is another major trend.

9.6 Describe how enterprise applications are using AI.

Enterprise application vendors have been adding business intelligence and AI tools to help managers obtain more meaningful information from the massive amounts of data these systems generate. The emphasis is on incorporating more analytics, machine learning, and generative AI into ERP, SCM, and CRM systems to improve efficiency and broaden their capabilities to handle complex data analysis, predictive modeling, and real-time decision making. AI-enabled tools are capable of much more rapid and complex data analysis than traditional software, as well as capabilities for generating new content.

Key Terms

Analytical CRM, 310
Bullwhip effect, 301
Churn rate, 312
Composable ERP, 315
Cross-selling, 309
Customer lifetime value (CLTV), 312
Demand planning, 302
Just-in-time strategy, 301
Operational CRM, 310
Pull-based model, 303
Push-based model, 303
Sales force automation modules (SFA modules), 309
Supply chain, 299
Supply chain execution system, 303
Supply chain planning system, 302
Touch point, 308

Review Questions

9.1 Understand ERP systems.

- Explain what an ERP system is and how it works.
- Describe how ERP systems provide value for a business.
- Discuss how AI applications are being used in ERP systems.

9.2 Understand SCM systems.

- Explain what a supply chain is and identify each of its components.
- Explain how SCM systems help reduce the bullwhip effect and how they provide value for a business.
- Define and compare supply chain planning systems and supply chain execution systems.

- Distinguish between a push-based and a pull-based model of supply chain management and explain how contemporary supply chain management systems facilitate a pull-based model.

9.3 Describe global supply chain management challenges.

- Describe the challenges of global supply chains.
- Explain how Internet technology can help companies manage global supply chains.
- Discuss how AI applications can help improve global supply chain planning, management, and operations.

9.4 Understand CRM systems.

- Define customer relationship management and explain why customer relationships are so important today.
- Describe the tools and capabilities of CRM systems for sales, marketing, and customer service.
- Distinguish between operational and analytical CRM.
- Discuss how AI applications are being used to automate, enhance, and optimize CRM processes.

9.5 Discuss enterprise application challenges.

- List and describe the challenges that enterprise applications pose.
- Explain how these challenges can be addressed.
- Describe how enterprise applications are taking advantage of cloud computing.
- Define composable ERP and explain why companies might want to use it.

9.6 Describe how enterprise applications are using AI.

- Explain the advantages of using AI for enterprise applications.
- Provide examples of AI functionality being provided by major enterprise application vendors.

Discussion Questions

9.7 Supply chain management is less about managing the physical movement of goods and more about managing information. Discuss the implications of this statement.

9.8 If a company wants to implement an enterprise application, it had better do its homework. Discuss the implications of this statement.

9.9 Which enterprise application should a business install first: ERP, SCM, or CRM? Explain your answer.

Using AI Project

9.10 AI Project: Improving Customer Experience

You have been hired as a consultant by two friends who want to open a small retail store selling organic pet food and treats, as well as pet toys and clothing. made from organic materials. They have asked you to advise them on the various customer touchpoints for such a store and how to provide the best customer experience at each of those touchpoints.

Using Microsoft Copilot, Google's Gemini, or ChatGPT, develop a series of prompts to assist you in providing them with this information. Save a copy of your prompts and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience analyzing business process problems; analyzing data to select suppliers; and using online research tools to evaluate supply chain management business services and enterprise application vendors.

For student files please visit www.pearsonhighered.com/laudon or MyLab MIS.

9.11 Management Decision Problem

Office Depot sells a wide range of office supply products and services in the United States and internationally. The company tries to offer a wider range of office supplies at a lower cost than other retailers by using just-in-time replenishment and tight inventory control systems. It uses information from a demand forecasting system and point-of-sale data to replenish its inventory in its more than 1,000 retail stores. Explain how these systems help Office Depot minimize costs and any other benefits they provide. Identify and describe other supply chain management applications that would be especially helpful to Office Depot.

9.12 Analyzing Data to Select Suppliers

Software skills: Spreadsheet date functions, data filtering, AVERAGE function

Business skills: Analyzing supplier performance and pricing

In this exercise, you'll analyze data to improve management decisions about selecting suppliers. You will use spreadsheet software to filter transactional data about suppliers based on several criteria to select the best suppliers for your company.

You work for a company that manufactures aircraft components. The company has many competitors that are trying to offer lower prices and better service to customers, and you are trying to determine whether the company can benefit from better supply chain management. The MIS18 Project 9.12_Student File for this chapter that contains a list of all the items that the firm has ordered from its suppliers during the past five months. The fields in the spreadsheet file include vendor name, vendor identification number, purchaser's order number, item identification number, and item description (for each item ordered from the vendor), cost per item, number of units of the item ordered (quantity), total cost of each order, vendor's accounts payable terms, order date, and actual arrival date for each order.

Prepare a recommendation of how you can use the data in this spreadsheet file to improve decisions about selecting suppliers. Calculate the delivery days (days required to deliver each order). Some criteria to consider for identifying preferred suppliers include the suppliers taking the least amount of time to deliver a product, the suppliers offering the best accounts payable terms, all suppliers producing a specific component such as gaskets and their pricing, and the average number of days for all suppliers to deliver. How can these reports help your company manage its supply chain?

9.13 Evaluating Supply Chain Management Services

Software skills: Online research tools and presentation software

Business skills: Evaluating supply chain management services

In addition to carrying goods from one place to another, some trucking companies provide supply chain management services and help their customers manage their information. In this project, you'll use online research tools to investigate and evaluate two of these business services, UPS and Schneider International. Examine how these companies' services can be used for supply chain management. Then prepare a presentation that responds to the following questions:

- What supply chain processes can each of these companies provide for their clients?
- How can customers use the websites of each company to help with supply chain management?
- Compare the supply chain management services that these companies provide. Which company would you select to help your firm manage its supply chain? Why?

9.14 Collaboration and Teamwork Project: Analyzing Enterprise Application Vendors

With a group of two or more other students, use online research tools to investigate and evaluate the products of two vendors of enterprise application software. You could compare, for example, the SAP and Oracle ERP systems, the SCM systems from Blue Yonder and SAP, or the CRM systems from Oracle and Salesforce. Use what you have learned to compare the software products you have selected in terms of business functions supported, technology platforms, cost, and ease of use. Which vendor would you select? Why? Would you select the same vendor for a small business (50–300 employees) as for a large one? If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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10

E-commerce: Digital Markets, Digital Goods

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Learning Objectives

After completing this chapter, you will be able to:

10.1 Discuss the evolution of e-commerce.

10.2 Describe the unique features of e-commerce.

10.3 Describe digital markets and digital goods.

10.4 Describe types of e-commerce and e-commerce business models.

10.5 Describe e-commerce revenue models.

10.6 Explain how e-commerce has transformed marketing.

10.7 Describe social networks, social marketing, and social e-commerce.

10.8 Understand how e-commerce has affected business-to-business transactions.

10.9 Describe m-commerce and m-commerce applications.

10.10 Discuss how to build an e-commerce presence.

10.11 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** Chewy Blends B2C and B2B E-commerce
- **Spotlight on Organizations Case Study:** Social Listening with Sprout Social
- **Chapter Case Study:** Lemonade Seeks to Disrupt the Insurance Industry with AI

Opening Case Study

Chewy Blends B2C and B2B E-commerce

Businesses are always on the lookout for ways to expand their revenues. Chewy is best known as an online retailer of pet food and supplies. When it began business, its primary focus was on business-to-consumer (B2C) e-commerce. But competition in the pet products business is very strong. Chewy faces competition from other online retailers, omnichannel retailers, suppliers that sell directly to consumers, traditional brick-and-mortar retailers such as local pet stores, and even veterinarians. In the face of all this competition, which Chewy expects to continue to increase, Chewy decided that it needed to diversify its business model.

When Chewy's founders started the company in 2011, they faced the specter of competing with Amazon. They planned to differentiate Chewy by providing a more personalized experience than Amazon could provide. They poured resources into call centers, live chat representatives, and prompt responses to customer email. Today, that ethos remains, with Chewy's hallmarks including 24/7 customer service provided by agents who are familiar with the company's products, handwritten notes when a customer makes a first purchase, and condolence cards and flowers when a pet passes away. Chewy grew rapidly and in 2017 was purchased by PetSmart, a brick-and-mortar retailer, for $3.35 billion. It was, at the time, the highest amount ever paid for an online retailer. Part of the deal was that Chewy would continue to be operated independently. Chewy's revenues continued to grow, and in 2019 it was spun off into its own publicly traded business. In 2020, Chewy and PetSmart officially split into two unrelated businesses.

Chewy began the process of diversifying its business model in 2018. In 2017, although its revenues had more than doubled compared to the previous year's, it had lost almost $340 million. Its first step was to establish Chewy Health, a new pet healthcare business that includes a variety of services, including an online pet pharmacy that was able to leverage Chewy's existing e-commerce platform. In 2020, Chewy launched Connect with a Vet, a proprietary telehealth platform that enables customers to connect with veterinarians. In 2021, it announced a partnership with Trupanion to offer Chewy customers an exclusive suite of pet health insurance and wellness plans.

In 2021, it took another major step, this time in the business-to-business (B2B) arena. It began the rollout of Practice Hub, a complete e-commerce solution for veterinarians that can be integrated into a vet's existing practice management solution. Chewy characterizes Practice Hub as one of the most innovative projects it has launched within Chewy Health. Using Chewy's proprietary app, veterinarians can easily create and manage preapproved prescriptions all in one place. They can earn revenue as a seller when one of their customers places an order with Chewy for those prescriptions. Chewy also benefits by being able to engage with those customers, many of whom may be new to Chewy, on its platform. Chewy handles all inventory, fulfillment, shipping, and customer service, leveraging its existing infrastructure. Chewy worked closely with a team of veterinarians in developing Practice Hub, with the vets helping to test features and provide feedback on enhancements.

Expanding further into pet healthcare is an important part of Chewy's strategy. In 2024, it announced an expansion of Chewy Health's portfolio with Chewy Vet Care, veterinary clinics that it plans to open in select

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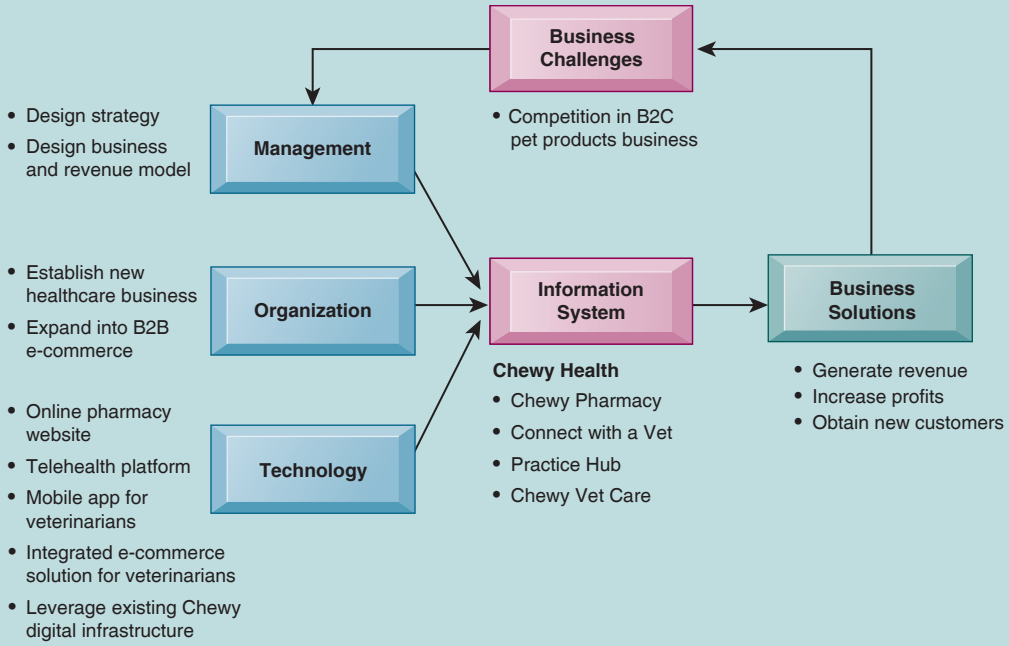


markets with high rates of pet ownership and Chewy customer density. With Chewy Vet Care, Chewy is aiming to leverage its B2C e-commerce products and services and its B2B solutions for veterinary practitioners, as well as proprietary health technology infrastructure, that it has spent years building and bundle them for use within its own vertically integrated pet healthcare ecosystem.

SOURCES: Ahmed Abdelazim, "Chewy: A Recurring Revenue Powerhouse Fueled by Pet Love," Seekingalpha.com, April 26, 2024; Fortune Editors, "Why the CEO of a $12 Billion Pet Services Company Still Responds to Customer Emails Himself," Fortune.com, April 17, 2024; Chewy, "Letter to Shareholders: Q4 and Fiscal Year 2023," March 20, 2024; Pymnts, "Chewy Health President: New Veterinary Clinics Complete Petcare Ecosystem," Pymnts.com, January 16, 2024; Brendan Meyer, "How Collaborating with Veterinarians Led to Chewy's Latest Innovative Product," Builtinboston.com, October 26, 2021; "Chewy Launches Innovative Marketplace Service for Veterinarians to Grow Clinic Revenues, Streamline Shopping Experience," Businesswire.com, September 15, 2021; Mary Hansbury, "The Cofounder of Chewy Explains How He Created a $10.2 Billion Empire Selling Pet Food to Millennials Who Treat Their Animals Like Their Firstborn Child," Businessinsider.com, November 30, 2019.

Chewy's revenues have consistently risen over the years. For the fiscal year 2023, it generated more than $11.1 billion in revenue, an increase of more than 10 percent compared to the previous year, and net income of about $40 million. It currently has a market capitalization of about $7 billion. Many analysts believe that Chewy is undervalued, pointing in part to Chewy Health and noting that this part of the company provides new opportunities for Chewy that can both accelerate its growth and increase its profit margins. Chewy Health already generates more than $3 billion in revenue annually, and analysts expect Chewy's Vet Care clinics to be a major success due to customer brand loyalty, as well as provide Chewy with major upselling and cross-selling opportunities.

Chewy is an example of a company that is using a variety of Internet-enabled e-commerce business models. When Chewy began operations, it focused on B2C e-commerce, and its primary business model was online retail. As it grew, it diversified into providing services and most recently has established a B2B offering. In addition, Chewy exemplifies the use of unique e-commerce technologies such as personalization and customization, which play a primary role in its business strategy and provide it with a competitive advantage.

The Opening Case Study diagram above calls attention to important points raised by this case and this chapter. The challenge facing Chewy is how to deal with the reality of ever-increasing competition in its primary market. One step Chewy has chosen to take is to diversify its business model, using its already-existing digital infrastructure, and expand into new areas: pet healthcare services and B2B e-commerce.

Case Study Questions

1. What management, organization, and technology issues did Chewy need to address in expanding into, first, services and then into B2B e-commerce?
2. What are the potential advantages and disadvantages of Chewy's decision to do so?
3. How does offering veterinary services fit with Chewy's strategy and business model?

CAREER BENEFIT

Diversification of Chewy's business models benefitted all employees by helping to increase Chewy revenues. Employees who participated directly in leveraging Chewy's existing B2C e-commerce platform and apps into a B2B e-commerce platform benefitted by extending their skill sets into a new e-commerce arena. Participating in expanding Chewy's B2B initiatives with Chewy Vet Care will enable employees to develop further expertise and skills in extending Chewy's pet healthcare ecosystem.

10.1 Discuss the Evolution of E-commerce

E-commerce

Use of the Internet, the web, and mobile apps and browsers running on mobile devices to transact business. More formally, can be defined as digitally enabled commercial transactions between and among organizations and individuals.

E-commerce refers to the use of the Internet, the web, and mobile apps and browsers running on mobile devices to transact business. More formally, e-commerce can be defined as digitally enabled commercial transactions between and among organizations and individuals. Commercial transactions involve the exchange of value (e.g., money) across organizational or individual boundaries in return for products and services.

The Evolution of E-commerce

Although it is difficult to pinpoint exactly when e-commerce "began," many pinpoint its real start to 1995, when one of the first Internet portals, Netscape.com, accepted the first ads from major corporations and popularized the idea that the web could be used as a new medium for advertising and sales. No one envisioned at the time what would turn out to be an exponential growth curve for e-commerce retail sales, as well as the many other types of e-commerce that would evolve.

One of the biggest changes is the extent to which e-commerce has become more social, mobile, and local. Online marketing once consisted largely of creating a corporate website, buying display ads on portal sites such as Yahoo, purchasing search-related ads on Google, and sending email messages. The workhorse of online marketing was the display ad. Display ads were based on television ads, where brand messages were flashed before millions of users who were not expected to interact with them. The primary measure of success was how many unique visitors a website produced and how many impressions a marketing campaign generated. (An impression was one ad shown to one person.) Both of these measures were carryovers from the world of television, which measures marketing in terms of audience size and ad views.

After 2007, all this changed with the rapid growth of Facebook and other social networks; the explosive growth of smartphones, beginning with the Apple iPhone; and the growing interest in social, mobile, and local marketing. What's different about the world of social-mobile-local e-commerce is the dual and related concepts of conversations and engagement. In the past, firms could tightly control their brand messaging and lead consumers down a funnel of cues that ended in a purchase. This level of control is not possible with social marketing, which instead is based on firms engaging in multiple online conversations with their customers, potential customers, and even critics. Today, brands are being talked about on the web and in social media (that's the conversation part); and marketing and building and maintaining a brand require firms to locate, identify, and participate in these conversations. Social marketing means all things social—listening, discussing, interacting, empathizing, and engaging—and is all about firms participating in and shaping this social process. Consumer purchase decisions are increasingly driven by the conversations, choices, tastes, and opinions of their social networks.

The emphasis in online marketing has shifted from a focus on unique visitors to a focus on participating in customer-oriented conversations. In this sense, social marketing is not simply an ad channel but a collection of technology-based tools for communicating with shoppers. The leading social network platforms are Facebook, Instagram, Pinterest, TikTok, X (formerly Twitter) and Threads, launched by Meta in 2023 to compete with X.

Social, mobile, and local e-commerce are connected. As mobile devices become more powerful, they are more useful for accessing social networks. As mobile devices become more widely adopted, customers can use them to find local merchants, and merchants can use them to alert customers in their neighborhood of special offers. Mobile advertising and marketing now constitute about two-thirds of all online ad spending (eMarketer, 2024e).

E-commerce Today

Today, purchasing goods and services using a desktop computer, smartphone, or tablet has become ubiquitous. In 2024, more than 240 million people in the United States (about 85 percent of the U.S. population) will shop online, and about 220 million will purchase something online, as will around 2.9 billion people worldwide. Although most purchases still take place through traditional channels, e-commerce is continuing to grow rapidly and has transformed the way many companies do business. E-commerce is composed of three major segments: retail goods, travel and other services, and online content. In 2024, U.S. e-commerce retail sales of goods and services ($1.2 trillion) and travel (about $325 billion) together will total about $1.525 trillion (eMarketer, 2024a, 2024b, 2024c, 2023d).

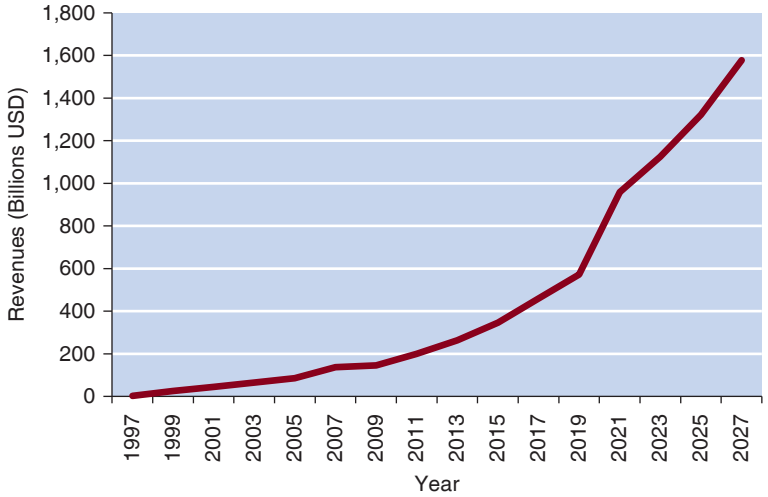
In 2024, online sales of retail goods will constitute about 16 percent of total U.S. retail sales of $7.45 trillion and are expected to grow by around 9 percent annually between 2024 and 2028 (compared to about 2 percent growth for physical retail sales). E-commerce is still a relatively small part of the much larger retail goods market that takes place in physical stores. However, e-commerce has expanded from the desktop and home computer to mobile devices, from an isolated activity into a more social form of commerce, and from commerce focused on a national audience to commerce focused on local merchants and consumers whose location is known to mobile devices. Almost 45 percent of all retail e-commerce sales involve the use of a mobile device, of which about 90 percent involve the use of a smartphone.

Examine Figure 10.1, which illustrates the growth of e-commerce from 1997 through 2027. E-commerce grew at double-digit rates from 1998 until the recession of 2008–2009, when growth slowed, but then resumed growing in the 12–15 percent range until the Covid-19 pandemic, when growth skyrocketed. Since then, e-commerce growth has dropped to a more normal rate of about 8–9 percent a year. In comparison, offline physical retail sales during the same period have only increased in the range of just 2–4 percent.

Table 10.1 highlights some of the recent trends and developments in e-commerce.

Figure 10.1 The Growth of Retail E-commerce

Retail e-commerce revenues grew rapidly until the recession of 2008–2009, when they slowed measurably. Thereafter, they resumed growing and skyrocketed in 2020 and 2021 due to the Covid-19 pandemic, before dropping back to more "normal" rates of growth rates. From 2023 through 2027, e-commerce revenues are expected to grow an estimated 8–9 percent annually.



SOURCES: Based on data from eMarketer, "Retail Ecommerce Sales, US," 2024; authors' estimates.

Table 10.1 Trends in E-commerce

Business Transformation
E-commerce remains the fastest-growing form of commerce. Mobile, social, and local e-commerce have all become major channels for e-commerce.
The breadth of e-commerce offerings continues to grow, especially in the services economy. On-demand services like Uber, Lyft, and Airbnb; meal delivery services; and pet care services have further expanded online service offerings.
E-commerce business models are constantly being refined to achieve higher levels of profitability. Traditional retail firms such as Walmart, Target, and Home Depot, have developed omnichannel business models to strengthen their dominant physical retail assets.
Small businesses and entrepreneurs are increasingly employing cloud-based computing resources and leveraging infrastructures created by industry giants such as Amazon, Apple, Google, and eBay, as well as platforms such as Shopify, BigCommerce, and Esty.
Mobile e-commerce (m-commerce) has taken off in the United States with location-based services and entertainment downloads, including movies, television, and YouTube videos, music, and e-books. Mobile retail e-commerce is expected to generate more than $540 billion in 2024.
The traditional advertising industry has been disrupted, with online advertising now comprising almost 78 percent of all ad spending. Google, Yahoo, and Facebook display trillions of ads a year.
More than two-thirds of the U.S. population has joined an online social network, created blogs, and shared photos and music. In 2024, social networks will account for an estimated 17.5 percent of time spent with digital media. Social networks have become the primary gateway to news, music, and, increasingly, products and services.
The growth of online entertainment business models has been enabled by the cooperation of the major copyright owners, as well as the movement of companies such as Netflix, Amazon, Apple, and YouTube into movie and TV production. Cable television is in decline, with many viewers cutting or reducing their cable subscriptions and relying on Internet-based streaming alternatives instead.
The media business has become more and more decentralized, with content created and distributed online by users (typically referred to as creators) becoming more and more prevalent, giving rise to what is often termed the creator economy.
Social networks such as Facebook, Instagram, TikTok, LinkedIn, X (formerly Twitter) and others have become major platforms for online marketing, and advertising, and increasingly, sales.
Technology Foundations
Wireless Internet connections (Wi-Fi, WiMax, 4G, and 5G smartphones) are continuing to expand.
A mobile platform, based on smartphones, tablet computers, wearable devices such as the Apple Watch, and mobile apps has become a reality, creating an alternative platform for online transactions, marketing, advertising, and medial viewing. In-home Internet-connected devices such as Amazon Alexa and "smart TVs" provide an additional platform.
The Internet broadband foundation becomes stronger as prices fall.
Internet-based models of computing, such as smartphone apps, cloud computing, software as a service (SaaS), and platform as a service (PaaS), reduce the costs of building and maintaining an e-commerce presence.
Artificial intelligence technologies are being increasingly employed in a variety of e-commerce related applications, such as for analyzing big data, customization and personalization, customer services, chatbots and voice assistants, as well as advertising and marketing.

AI and E-commerce: The Newest Frontier

The introduction of ChatGPT in November 2022 has thrust AI, and specifically, generative AI (GenAI), both to the forefront of public consciousness as well as high on the list of priorities for companies involved in e-commerce. Even before the release of ChatGPT, large e-commerce companies had been using AI behind the scenes for supply chain optimization, inventory management, predictive analytics, price optimization, and dynamic pricing. AI was also being employed in customer-facing applications, such as for recommendation engines used to generate personalized product recommendations, for customer service chatbots, and for enhanced search and navigation, including visual search.

The introduction of readily available GenAI tools is now taking the use of AI in e-commerce companies to the next level, making AI available to a much broader range of companies, for a wide range of purposes. GenAI can help companies generate product descriptions, product recommendations, social media posts, and other forms of content automatically. GenAI's ability to generate images can be used to create product images, as well as personalized product designs based on customer preferences. GenAI can be combined with augmented reality technology to create applications that allow customers to virtually "try on" products by overlaying images on top of individual photos or allow customers to visualize products in their real-world environments, creating a more immersive shopping experience. GenAI-powered chat bots can be used to create virtual shopping assistants to provide an even more personalized shopping experience. On the marketing front, online marketers are using AI to assist with data analysis and content creation.

Here are some real-world examples of how companies are currently using AI for e-commerce.

- Walmart has recently launched a GenAI search tool that delivers more relevant search results and product recommendations.

- eBay's Shop the Look relies on GenAI to provide users with a curated selection of outfits based on shopping histories.
- Target is using GenAI to generate product descriptions and tags to optimize search engine performance.
- Etsy is using GenAI to remove listings that violate its policies from its search results.
- BestBuy is redesigning its customer support experience with GenAI, launching a GenAI-enabled virtual assistant for self-service support that will help customers troubleshoot product issues, organize deliveries, and manage their BestBuy subscriptions.
- Klarna is using an AI assistant that reportedly does work equivalent to the work of 700 full-time customer support service agents.

10.2 Describe the Unique Features of E-commerce

Why has e-commerce grown so rapidly? The answer lies in the unique nature of the technologies underlying e-commerce that have enabled many new possibilities for marketing and selling products and services to businesses and consumers. Now, let's explore each of these unique features.

Ubiquity

In traditional commerce, a marketplace is a physical place, such as a retail store, that you visit to transact business. The Internet and web technologies underlying e-commerce enable e-commerce to be ubiquitous, meaning that it is available just about everywhere all the time. It makes it possible to shop from your desktop, at home, at work, or even in your car, using a mobile device. The result is a marketplace extended beyond traditional boundaries and removed from a temporal, geographic location, sometimes referred to as a **marketspace**.

From a consumer's point of view, ubiquity reduces **transaction costs**—the costs of participating in a market. To transact business, it is no longer necessary to spend time or money traveling to a market, and much less mental effort is required to make a purchase in a marketspace.

Marketspace

A marketplace extended beyond traditional boundaries and removed from a temporal and geographic location.

Transaction costs

The costs of participating in a market.

Global Reach

E-commerce technology permits commercial transactions to cross cultural and national boundaries far more conveniently and cost-effectively than is possible in traditional commerce. As a result, the potential market size for e-commerce merchants is roughly equal to the size of the world's online population (estimated in 2024 to be more than 4.8 billion).

In contrast, most traditional commerce is local or regional—it involves local merchants or national merchants with local outlets. Television stations, radio stations, and newspapers, for instance, are primarily local and regional institutions with limited, but powerful, national networks that can attract a national audience but not easily cross national boundaries to a global audience.

Universal Standards

One strikingly unusual feature of e-commerce technologies is that the technical standards of the Internet and, therefore, the technical standards for conducting e-commerce are universal standards. All nations around the world share them and enable any computer to link with any other computer regardless of the technology platform each computer is using. In contrast, most traditional commerce technologies differ from one nation to the next. For instance, television and radio standards differ around the world, as does cellular telephone technology.

The universal technical standards of the Internet and e-commerce greatly lower **market entry costs**—the costs merchants must pay simply to bring their goods to market. At the same time, for consumers, universal standards reduce **search costs**—the time and money spent locating a suitable product and determining the best place to purchase that product.

Market entry costs

The cost merchants must pay to bring their goods to market.

Search costs

The time and money spent locating a suitable product and determining the best place to purchase that product.

Richness

Richness
Refers to the complexity and content of a message.

Richness refers to the complexity and content of a message. Traditional markets, national sales forces, and small retail stores have great richness: They can provide personal, face-to-face service, using aural and visual cues when making a sale. The richness of traditional markets makes them powerful selling or commercial environments. Prior to the development of the web, there was a trade-off between richness and reach—the larger the audience reached, the less rich the message. But e-commerce technologies make it possible to deliver rich messages with text, audio, and video simultaneously to large numbers of people.

Interactivity

Interactivity
Allowing for two-way communication between merchant and consumer and peer-to-peer communication among individuals.

Unlike any of the commercial technologies of the twentieth century, with the possible exception of the telephone, e-commerce technologies enable **interactivity**, meaning they allow for two-way communication between merchant and consumer and peer-to-peer communication among individuals. Traditional television, for instance, cannot ask viewers any questions or enter conversations with them, and it cannot request customer information to be entered on a form. Interactivity allows an online merchant to engage a consumer in ways similar to a face-to-face experience but on a massive, global scale.

Information Density

Information density
The total amount and quality of information available to all market participants.

Price transparency
Ease with which consumers can find out the variety of prices in a market.

Cost transparency
Ability of consumers to discover the actual costs merchants pay for products.

Price discrimination
Selling the same goods, or nearly the same goods, to different targeted groups at different prices.

The Internet and the web vastly increase **information density**—the total amount and quality of information available to all market participants, consumers, and merchants alike. E-commerce technologies reduce information collection, storage, processing, and communication costs while greatly increasing the currency, accuracy, and timeliness of information.

Information density in e-commerce markets makes prices and costs more transparent. **Price transparency** refers to the ease with which consumers can find out the variety of prices in a market; **cost transparency** refers to the ability of consumers to discover the actual costs that merchants pay for products.

There are advantages for merchants as well. Online merchants can discover much more about consumers than was possible in the past. This allows merchants to segment the market into groups that are willing to pay different prices and permits the merchants to engage in **price discrimination**—selling the same goods, or nearly the same goods, to different targeted groups at different prices. For instance, an online merchant can discover a consumer's avid interest in expensive, exotic vacations and then pitch high-end vacation plans to that consumer at a premium price, knowing this person is willing to pay extra for such a vacation. At the same time, the online merchant can pitch the same vacation plan at a lower price to a more price-sensitive consumer. Information density also helps merchants differentiate their products in terms of cost, brand, and quality.

Personalization/Customization

Personalization
Ability of merchants to target marketing messages to specific individuals by adjusting the message to a person's clickstream behavior, name, interests, and past purchases.

Customization
Changing a delivered product or service based on a user's preferences or prior behavior.

E-commerce technologies permit **personalization**. Merchants can target their marketing messages to specific individuals by adjusting the message to a person's clickstream behavior, name, interests, and past purchases. The technology also permits **customization**—changing the delivered product or service based on a user's preferences or prior behavior. Given the interactive nature of e-commerce technology, much information about the consumer can be gathered in the marketplace at the moment of purchase. With the increase in information density, a great deal of information about the consumer's past purchases and behavior can be stored and used by online merchants for the purposes of personalization and customization.

The result is a level of personalization and customization unthinkable with traditional commerce technologies. For instance, you may be able to shape what you see on a traditional television by selecting a channel, but you cannot change the content of the channel you have chosen. In contrast, online news outlets such as the *Wall Street Journal Online* allow you to select the type of news stories you want to see first and also give you the opportunity to be alerted when certain events happen.

Social Technology: User Content Generation, Creators, and Social Networks

In contrast to previous technologies, the Internet and e-commerce technologies have evolved to be much more social by allowing users to create and share content with their friends (and a larger worldwide community). By using these forms of communication, users can create new social networks and strengthen existing ones.

Previous forms of mass media, such as the printing press, use a broadcast (one-to-many) model in which content is created in a central location by experts (professional writers, editors, directors, and producers), with audiences concentrated in huge numbers to consume a standardized product. E-commerce technologies empower users to create and distribute content on a large scale and permit users to program their own content consumption. E-commerce technologies provide a unique, many-to-many model of mass communication, and over the last several years, user-generated content (UGC), in the form of video, podcasts, newsletters, and more, has come to occupy an ever-increasing role in the online content landscape. People who develop and distribute such content are now typically referred to as creators, and an entire ecosystem, referred to as the creator economy, has sprung up around them.

Table 10.2 provides a summary of the unique features of e-commerce technology and each feature's business significance.

Table 10.2 Eight Unique Features of E-commerce Technology

E-commerce Technology Dimension	Business Significance
Ubiquity. The Internet/web technologies underlying e-commerce are available everywhere: at work, at home, and, elsewhere via mobile devices.	The marketplace is extended beyond traditional boundaries and is removed from a temporal and geographic location. Marketspace is created. Shopping can take place anytime, anywhere. Customer convenience is enhanced, and shopping costs are reduced.
Global Reach. E-commerce technology reaches across national boundaries and around the globe.	Commerce is enabled across cultural and national boundaries seamlessly and without modification. The marketspace includes, potentially, billions of consumers and millions of businesses worldwide.
Universal Standards. The technical standards underlying e-commerce are universal and shared around the world.	There is a common, inexpensive, global technology foundation available for all to use.
Richness. Messages containing text, audio, and video can be sent simultaneously to large numbers of people.	Video, audio, and text marketing messages are integrated into a single marketing message and consumer experience.
Interactivity. The technology enables interaction (two-way communication).	Consumers are engaged in a dialogue that dynamically adjusts the experience to the individual consumer and makes the consumer a participant in the process of delivering goods to the market.
Information Density. The technology reduces information costs and raises the total amount and quality of information available to all market participants.	Information-processing, storage, and communication costs drop dramatically, while currency, accuracy, and timeliness improve greatly. Information becomes plentiful, less expensive, and more accurate.
Personalization/Customization. The technology permits personalization and customization.	Marketing messages can be personalized and products and services customized, based on individual characteristics.
Social Technology. The technology supports user-generated content (UGC) and social networks.	Enables user content creation and distribution and supports development of social networks.

10.3 Describe Digital Markets and Digital Goods

The location, timing, and revenue models of business are based in some part on the cost and distribution of information. The Internet has created a digital marketplace where millions of people all over the world can exchange massive amounts of information directly, instantly, and free of charge. As a result, the Internet has changed the way companies conduct business and increased their global reach.

Information asymmetry

Exists when one party in a transaction has more information that is essential to the transaction than the other party; impacts the relative bargaining power of the participants.

The Internet reduces information asymmetry. An **information asymmetry** exists when one party in a transaction has more information that is important for the transaction than the other party. That information helps determine each party's relative bargaining power. In digital markets, consumers and suppliers can see the prices being charged for goods, and in that sense, digital markets are said to be more transparent than traditional markets.

For example, before companies began advertising and selling cars online, there was significant information asymmetry between auto dealers and customers. Only the auto dealers knew the manufacturers' prices, and it was difficult for consumers to shop around for the best price. Auto dealers' profit margins depended on this asymmetry of information. Today's consumers have access to a legion of websites providing competitive pricing information, and more than 90 percent of U.S. auto buyers use the Internet to research prospective car purchases. Thus, the web has reduced the information asymmetry surrounding an auto purchase. The Internet has also helped businesses seeking to purchase from other businesses reduce information asymmetries and locate better prices and terms.

Digital Markets

Menu costs

Merchants' costs of changing prices.

Dynamic pricing

Pricing of a product or service varies depending on the demand characteristics of the customer or the supply situation of the seller.

Digital markets are flexible and efficient because they operate with reduced search and transaction costs, lower **menu costs** (merchants' costs of changing prices), greater price discrimination, and the ability to change prices dynamically based on market conditions. In **dynamic pricing**, the price of a product or service varies depending on the demand characteristics of the customer or the supply situation of the seller. For instance, online retailers from Amazon to Walmart change prices on thousands of products based on time of day, demand for the product, and users' prior visits to their sites. Using big data analytics, online firms can adjust prices at the individual level based on behavioral targeting parameters such as whether the consumer is a price haggler (who will receive a lower price offer) versus a person who accepts offered prices and does not search for lower prices. Prices can also vary by zip code. Uber, along with other ride services, uses surge pricing to adjust prices of a ride based on demand (which always rises during storms and major conventions).

Digital markets such as these can either reduce or increase switching costs, depending on the nature of the product or service being sold, and they might cause some extra delay in gratification due to shipping times. Unlike a physical market, you can't immediately consume a product such as clothing that is purchased over the web (although immediate consumption is possible with digital products).

Digital markets provide many opportunities to sell directly to the consumer, bypassing intermediaries such as distributors or retail outlets. Eliminating intermediaries in the distribution channel can significantly lower purchase transaction costs. To pay for all the steps in a traditional distribution channel, a product may have to be priced as high as 135 percent of its original cost to manufacture.

Examine Figure 10.2, which illustrates how much savings result from eliminating various layers, such as distributors and retailers, in the distribution process. By selling directly to consumers or reducing the number of intermediaries, companies can raise profits while charging lower prices. The removal of organizations or business process layers responsible for intermediary steps in a value chain is called **disintermediation**. E-commerce has also given rise to a new set of new intermediaries in the retail industry such as Amazon and eBay.

Disintermediation

The removal of organizations or business process layers responsible for certain intermediary steps in a value chain.

Figure 10.2 The Benefits of Disintermediation for the Consumer

The typical distribution channel has several intermediary layers, each of which adds to the final cost of a product, such as a sweater. Removing layers lowers the final cost to the customer.

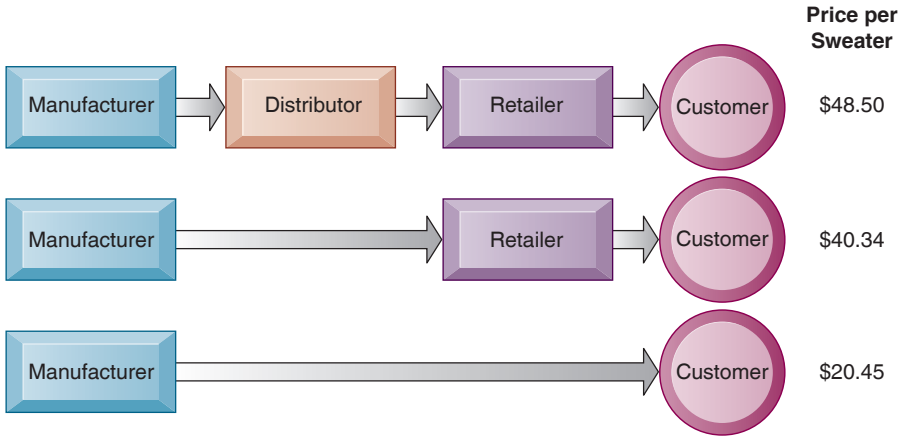


Table 10.3 Digital Markets Compared with Traditional Markets

	Digital Markets	Traditional Markets
Information asymmetry	Asymmetry reduced	Asymmetry high
Search costs	Low	High
Transaction costs	Low (sometimes virtually nothing)	High (time, travel)
Delayed gratification	High (or lower in the case of a digital good)	Lower: purchase now
Menu costs	Low	High
Dynamic pricing	Low cost, instant	High cost, delayed
Price discrimination	Low cost, instant	High cost, delayed
Market segmentation	Low cost, moderate precision	High cost, less precision
Switching costs	Higher/lower (depending on product characteristics)	High
Network effects	Strong	Weaker
Disintermediation	More possible/likely	Less possible/unlikely

Disintermediation is affecting the market for services. Airlines and hotels operating their own reservation sites online earn more per ticket because they have eliminated travel agents as intermediaries. However, as in the retail industry, e-commerce has given rise to a new set of intermediaries such as Expedia, Orbitz, and Hotels.com. Table 10.3 summarizes the differences between digital markets and traditional markets.

Digital Goods

The Internet's digital marketplace has greatly expanded sales of **digital goods**—goods that can be delivered over a digital network. Music, video, movies, software, newspapers, magazines, and books can all be expressed, stored, delivered, and sold as purely digital products. Digital goods are intellectual property, which is protected from misappropriation by copyright, patent, trademark, and trade secret laws (see Chapter 4). Today, all these products are delivered as digital streams or downloads while their physical counterparts decline in sales.

Digital goods

Goods that can be delivered over a digital network.

Table 10.4 How the Internet Changes the Market for Digital Goods

	Digital Goods	Traditional Goods
Marginal cost/unit	Zero	Greater than zero, high
Cost of production	High (most of the cost)	Variable
Copying cost	Approximately zero	Greater than zero, high
Distributed delivery cost	Low	High
Inventory cost	Low	High
Marketing cost	Variable	Variable
Pricing	More variable (bundling, random pricing)	Fixed, based on unit costs

In general, for digital goods, the marginal cost of producing another unit is about zero (it costs little to nothing to make a copy of a file). The cost of producing the original first unit, however, is relatively high—in fact, it is nearly the total cost of the product because there are few other costs of inventory and distribution. Costs of delivery over the Internet are low, marketing costs often remain the same, and pricing can be highly variable. On the Internet, the merchant can change prices as often as desired because of low menu costs.

The impact of the Internet on the market for these kinds of digital goods is nothing short of revolutionary, and we see the results around us every day. Businesses dependent on physical products for sales—such as bookstores, music stores, book publishers, music labels, and film studios—have had to deal with declining sales. Newspaper and magazine print subscriptions have declined, while online readership and digital subscriptions have expanded.

Table 10.4 describes digital goods and how they differ from traditional physical goods.

10.4 Describe Types of E-commerce and E-commerce Business Models

E-commerce is a fascinating combination of business models and information technologies. Let's start with a basic understanding of the types of e-commerce and then describe e-commerce business models.

Types of E-commerce

There are many ways to classify e-commerce transactions. One way involves looking at the nature of the participants. The three major e-commerce categories are business-to-consumer (B2C) e-commerce, business-to-business (B2B) e-commerce, and consumer-to-consumer (C2C) e-commerce. Note that many businesses, such as Chewy, profiled in the Opening Case Study, engage in both B2C and B2B e-commerce.

Business-to-consumer e-commerce (B2C e-commerce)
Type of e-commerce that involves selling products and services directly to individual consumers.

Business-to-business e-commerce (B2B e-commerce)
Type of e-commerce that involves sales of goods and services among businesses.

Consumer-to-consumer e-commerce (C2C e-commerce)
Type of e-commerce that involves consumers selling directly to other consumers.

- **Business-to-consumer e-commerce (B2C e-commerce)** involves selling products and services to individual consumers. Amazon, Walmart, and Apple Music are examples of B2C e-commerce.
- **Business-to-business e-commerce (B2B e-commerce)** involves sales of goods and services among businesses. Grainger, which focuses on the purchase and sale of maintenance, repair, and operating (MRO) products to businesses, is an example of one type of B2B e-commerce.
- **Consumer-to-consumer e-commerce (C2C e-commerce)** involves consumers selling directly to other consumers. For example, eBay enables people to sell their goods to other consumers by auctioning their merchandise off to the highest bidder or by selling it for a fixed price. eBay acts as an intermediary by creating a digital platform for peer-to-peer commerce. Craigslist is another platform that consumers use to buy from and sell directly to others.

Another way of classifying e-commerce transactions is in terms of the platforms that participants use in a transaction, such as the mobile platform or social networks. Prior to 2007 and the introduction of the smartphone, e-commerce transactions took place using a desktop computer connected to the Internet over a wired network. Since then, smartphones and tablet computers have become a popular alternative, today accounting for almost 45 percent of all retail e-commerce sales. We discuss mobile e-commerce (m-commerce) in more detail in Section 10.9. We discuss social e-commerce in more detail in Section 10.7.

E-commerce Business Models

Changes in the economics of information described earlier have created the conditions for entirely new business models, such as online retailers, to appear while disrupting, in some cases, older business models. Note that in some cases, the business models overlap. For instance, transaction brokers can also be considered service providers, as can some market creators. In addition, many businesses use more than one business model. All, in one way or another, use the Internet (including apps on mobile devices) to add extra value to existing products and services or to provide the foundation for new products and services.

Online Retailers

An **online retailer**, sometimes called an e-tailer, sells goods online. Online retailers, come in all sizes, from giant Amazon (with 2023 retail sales revenues of about $445 billion—about 40 percent of all retail e-commerce revenues) to tiny local stores that have websites. There are four main types of online retail business models: virtual merchants such as Wayfair that do not have a physical presence, omnichannel merchants such as Walmart or Target (sometimes referred to as bricks-and-clicks) that have both physical stores as well as online offerings, catalog merchants such as Lands' End or L.L. Bean, and manufacturer-direct (sometimes referred to as direct-to-consumer, or D2C/DTC), where the manufacturer of a product, such as Nike, sells directly to the consumer. Altogether, online retail (the sale of physical goods online) will generate more than $1.2 trillion in revenues in 2024. The value proposition of online retailers is to provide convenient, low-cost shopping 24/7; large selections; and consumer choice.

Online retailer

Sells goods online. Sometimes called an e-tailer.

Transaction Broker

A **transaction broker** provides online processing for transactions that previously were handled in person, by phone, or by mail. The most common industries using this model are financial services and travel services. The online transaction broker's primary value propositions are savings of money and time and providing an extraordinary inventory of financial products or travel packages in a single location. Online stockbrokers and travel booking services charge fees that are considerably less than traditional versions of these services. E*Trade and Expedia are two examples of online financial and travel service firms based on a transaction broker model.

Transaction broker

Provides online processing for transactions that previously were conducted in person, by phone, or by mail.

Market Creator

A **market creator** provides a digital environment in which buyers and sellers can meet, display products, search for products, establish prices, and engage in transactions. The value proposition of online market creators is that they provide a platform where sellers can easily display their wares and purchasers can buy directly from sellers. Online auction markets such as eBay and eBid are good examples of the market creator business model. Another example is Amazon's Merchants platform (and similar programs at eBay), where merchants set up stores on Amazon Marketplace and sell goods at fixed prices to consumers. Companies that enable on-demand services, exemplified by Uber and Airbnb, can be viewed as being based on the idea of a market creator building a digital platform where supply meets demand; for instance, spare auto or room rental capacity finds individuals who want transportation or lodging. Crowdfunding markets such as Kickstarter bring together private investors and entrepreneurs in a funding marketplace.

Market creator

Provides a digital environment in which buyers and sellers can meet, display products, search for products, establish prices, and engage in transactions.

Content Provider

Content provider

Distributes information content, such as digital video, music, audio files in the form of podcasts, photos, text, and artwork, via the Internet to computers and mobile devices.

E-commerce has increasingly become a global content channel. *Content* is defined broadly to include all forms of intellectual property. Intellectual property refers to products of the mind in which the creator claims a property right. A **content provider** distributes information content—such as digital video, music, audio files in the form of podcasts, photos, text, and artwork—via the Internet to computers and mobile devices. The value proposition of online content providers is that consumers can conveniently find and purchase a wide range of content online that can be played or viewed on multiple devices.

Providers do not have to be the creators of the content (although sometimes they are, such as Disney). Instead, many content providers act merely as a platform for the distribution of content created by others. For example, Apple Music distributes music, but it does not create or commission new music. Netflix, on the other hand, both distributes original content that it produces, as well as content produced by others.

Download

Involves transferring a file from a web server to a user's device, where it resides and can be accessed by the user at any time.

Streaming

A method of distributing audio and video files as a continuous stream of content to a user's device without it having to be stored locally on the device.

Content providers typically make content available online either via download or streaming. The **download** method involves transferring a file from a web server to the user's device, where it resides and can be accessed by the user at any time. **Streaming** is a method of distributing audio and video files as a continuous stream of content to a user's device without it having to be stored locally on the device.

Service Provider

Service provider

Offers services online.

Fintech

Start-up financial technology firms and services that use IT innovatively to compete with traditional financial services firms.

Whereas e-tailers sell products online, a **service provider** offers services online. The major service industry groups include finance, insurance, real estate, travel, professional services such as legal and accounting, business services, healthcare services, and educational services. While financial services are dominated by large firms with millions of customers, startup financial technology service firms, also known as **fintech** firms, use IT innovatively to compete with traditional financial service firms for peer-to-peer (P2P) bill payment, money transfer, lending, crowdfunding, financial advice, and account aggregation services. According to a recent survey, almost 90 percent of U.S. consumers are now using fintech solutions such as digital payment services, digital savings tools, and online investing tools (Plaid and the Harris Poll, 2023). Notable fintech companies include P2P payment services providers Venmo and Zelle (see the discussion of mobile payment app systems in Section 10.9). Fintech firms are often purchased by larger financial service firms who want to acquire the fintech's technology and customer base. Insurtech is a subset of fintech that focuses on the insurance services industry. The Chapter Case Study, *Lemonade Seeks to Disrupt the Insurance Industry with AI*, focuses on one such insurtech company, Lemonade, that is using cutting-edge AI technology to disrupt the insurance industry.

On-demand services, in addition to being viewed as an example of the market creator business model, can also be seen as a variation on the service provider business model, as are companies that offer online, cloud-based software, storage, and data backup. Google has led the way in developing online software service applications such as G Suite, Google Sites, Gmail, and online data storage services. Salesforce is a major provider of cloud-based software for customer management (see Chapters 5 and 9).

Community Providers (Social Networks)

Community provider

Creates a digital online environment where people with similar interests can transact (buy and sell goods); share interests, photos, and videos; communicate with like-minded people; and receive interest-related information.

Portal

Web interface for presenting integrated personalized business content from a variety of sources. Also refers to a website that acts as initial point of entry to the web with an integrated package of content and services.

A **community provider** creates a digital online environment where people with similar interests can transact (buy and sell goods); share interests, photos, and videos; communicate with like-minded people and receive interest-related information. Some even able members to and even play out fantasies by adopting online personalities called *avatars*. Online social networks such as Facebook, Instagram, LinkedIn, Pinterest, TikTok, X (formerly Twitter), Threads, and hundreds of other smaller, niche networks are all examples of community providers (See Section 10.7 for further information about online social networks.).

Portal

A **portal** is a website that provides an initial point of entry to the web with an integrated package of content and services. Many users set a portal site as their web browser's home page. Portals such as Google, Yahoo, MSN, and AOL offer search tools, news, email,

Table 10.5 E-commerce Business Models

Category	Description	Examples
Online retail (e-tailer)	Sells physical products directly to consumers or to individual businesses.	Amazon Wayfair
Transaction broker	Saves users money and time by processing online sales transactions and generating a fee each time a transaction occurs.	E*Trade Expedia
Market creator	Provides a digital environment where buyers and sellers can meet, search for products, display products, and establish prices for those products, generating revenue from transaction fees.	eBay eBid Etsy
Content provider	Creates revenue by providing digital content, such as news, music, photos, or video, over the web. The customer may pay to access the content, or revenue may be generated by selling advertising space.	WSJ.com Shutterfly Apple Music Scopely
Community provider/ social network	Provides an online meeting place where people with similar interests can communicate and find useful information.	Facebook Instagram TikTok
Portal	Provides initial point of entry to the web along with specialized content and other services.	Yahoo MSN AOL
Service provider	Provides online services such as financial services, career services, on-demand services, and online data storage and backup.	Uber RocketMortgage Dropbox

instant messaging, maps, calendars, shopping, music downloads, videostreaming, and more, all in one place. Today's portals hope to be a destination where users start their web searching and linger to read news, find entertainment, meet other people, and, of course, be exposed to advertising. Facebook is a different kind of portal, one that is based on social networking. Millions of users have set Facebook as their home page, and about 70 percent of U.S. adults use Facebook, including many who do so several times a day (Schaeffer, 2024). Portals generate revenue primarily by attracting very large audiences, charging advertisers for display ad placement (similar to traditional newspapers), collecting referral fees for steering customers to other sites, and charging for premium services. Although there are hundreds of portal/search engine sites, the top portals (Google, MSN, Yahoo, and AOL) gather more than 95 percent of the Internet portal traffic because of their superior brand recognition.

Table 10.5 summarizes and provides examples of these primary business models.

10.5 Describe E-commerce Revenue Models

A firm's **revenue model** describes how the firm will earn revenue, generate profits, and produce a return on investment. Although many e-commerce revenue models have been developed, most companies rely on one, or some combination, of the following major revenue models: advertising, sales, subscription, free/freemium, transaction fee, and affiliate.

Revenue model

Describes how a firm will earn revenue, generate profits, and produce a return on investment.

Advertising Revenue Model

Advertising revenue model
E-commerce revenue model that is based on generating revenue from online advertising.

The **advertising revenue model** is based on generating revenue from online advertising. An online business that offers content, services, and/or products also provides a forum for advertisements and receives fees from advertisers. The advertising model is the most widely used revenue model in e-commerce, and arguably, without advertising revenues, the web would be a vastly different experience from what it is now because people would be asked to pay for access to content. Much of the content available online today—everything from news to videos and opinions—is free because advertisers pay the production and distribution costs in return for the right to expose visitors to ads.

Companies are expected to spend more than $300 billion on online advertising in 2024, up 13 percent from 2023. About $200 billion of this will be for mobile ads, which will account for about 65 percent of all digital advertising. In 2024, online advertising is expected to constitute almost 78 percent of all advertising in the United States (eMarketer, 2024e). Websites with large numbers of unique visitors or that are able to retain user attention (stickiness) can charge higher advertising rates. Increasingly, companies that primarily use a sales revenue model, such as Amazon, Walmart, and many others, are also making their platforms available to advertisers, creating what is known as a retail media network. Retail media networks have become one of the fastest-growing channels for online advertising, and in 2024, retail media advertising is expected to constitute almost 20 percent of total digital ad spending in the United States.

Sales Revenue Model

Sales revenue model
E-commerce revenue model that is based on generating revenue from the sale of goods, content, and/or services to customers.

The **sales revenue model** is based on generating revenue from the sale of goods, content, and/or services to customers. Companies such as Target and Wayfair focus primarily on the sale of physical goods. Content providers make money by charging for downloading content such as a video (Apple TV), books (Amazon Kindle), or movies (Apple TV).

Subscription Revenue Model

Subscription revenue model
E-commerce revenue model that is based on generating revenue from subscription fees charged for access to some or all of a company's content or services.

The **subscription revenue model** is based on generating revenue from subscription fees charged for access to some or all of a company's content or services. Content providers often use this revenue model. For instance, the online version of *Consumer Reports* provides access to premium content, such as detailed ratings, reviews, and recommendations, only to subscribers for a $39 annual fee. Netflix is one of the most successful users of the subscription revenue model, with around 270 million subscribers worldwide in 2024. To be successful, the subscription model requires the content to be perceived as differentiated, having high added value, and not readily available elsewhere or easily replicated. Other companies offering content or services online on a subscription basis include Match (dating services), Ancestry (genealogy research), and Microsoft Xbox Live. A number of companies also use a subscription-based sales revenue model. For instance, Pet Plate sells premium dog food on a subscription basis. Consumers provide information about their dog, and Pet Plate creates a customized meal plan, and then ships the food to the consumer in return for a monthly subscription fee.

Free/Freemium Revenue Model

Free/freemium revenue model
E-commerce revenue model in which a business offers basic services or content for free, but generates revenue from a premium charged for advanced or special features.

In the **free/freemium revenue model**, a firm offers basic services or content for free but generates revenue from a premium charged for advanced or special features. For example, Pandora, the subscription music service, offers a free service with limited play time and advertising and a premium service with an unlimited amount of play time and without any advertising. The idea is to attract large audiences with free services and then convince some of this audience to pay a subscription for premium services. One problem with this model is convincing people to become paying customers.

Transaction Fee Revenue Model

The **transaction fee revenue model** is based on generating revenue from fees for enabling or executing transactions. For example, eBay provides an online auction marketplace and receives a small transaction fee from a seller when the seller is successful in selling an item. Online financial services, from banking to payment systems, typically rely on a transaction fee model.

Transaction fee revenue model

E-commerce revenue model based on generating revenue from fees for enabling or executing transactions.

Affiliate Revenue Model

In the **affiliate revenue model**, a business steers customers to an "affiliate," which then pays the company a referral fee or percentage of the revenue from any resulting sales. Referral fees are also sometimes referred to as lead generation fees. For example, MyPoints makes money by connecting companies to potential customers by offering special deals to MyPoints members. When MyPoints members take advantage of an offer and make a purchase, they earn points they can redeem for free products and services, and MyPoints receives a referral fee. Companies such as Yelp, which publishes crowdsourced reviews of businesses, generate much of their revenue from steering potential customers to websites where the customers make a purchase. Amazon uses affiliates that steer business to Amazon by placing links to Amazon products on their websites. Social media influencers who receive a commission from brands they represent can also be characterized as using an affiliate revenue model.

Affiliate revenue model

E-commerce revenue model in which a business steers customers to an "affiliate" which then pays the company a referral fee or percentage of revenue from any resulting sales.

10.6 Explain How E-commerce Has Transformed Marketing

Although the Internet and e-commerce have changed entire industries and enabled new business models, no industry has been more affected than marketing and marketing communications.

The Internet provides marketers with ways of identifying and communicating with millions of potential customers at costs far lower than traditional media, including search engine marketing, data mining, recommender systems, and targeted email. For instance, before the Internet, reaching a large audience was expensive, and marketers had to focus on attracting the largest number of consumers with popular, hit products, whether music, Hollywood movies, books, or cars. In contrast, the Internet allows marketers to find potential customers inexpensively for products for which demand is low (referred to as **long tail marketing**). For instance, the Internet makes it possible to sell independent music profitably to small audiences. There's always some demand for almost any product. Put a string of such long tail sales together and you have a profitable business.

Long tail marketing

Refers to the ability of firms to find potential customers inexpensively for products for which demand is low, largely because the cost of reaching very small market segments (people who fall into the long tail ends of a Bell curve) is not high.

The Internet also provides ways—often instantaneous and spontaneous—to gather information from customers, adjust product offerings, and increase customer value.

Table 10.6 describes the leading marketing and advertising formats used in e-commerce.

Behavioral Targeting

Many e-commerce marketing firms use behavioral targeting techniques to increase the effectiveness of banners, rich media, and video ads. **Behavioral targeting** (also known as interest-based advertising) refers to tracking the clickstreams (history of clicking behavior) of individuals to understand their interests and intentions and to expose them to advertisements that are uniquely suited to their online behavior. Marketers and most researchers believe this more precise understanding of the customer leads to more efficient marketing (the firm pays for ads that are directed only to those shoppers who are most interested in their products) and increased sales and revenues.

Behavioral targeting

Tracking the clickstreams (history of online clicking behavior) of individuals for the purpose of understanding their interests and intentions, and exposing them to advertisements that are uniquely suited to their interests. Also known as interest-based advertising.

Behavioral targeting takes place at two levels: at individual websites or from within apps (first-party tracking) and on various advertising networks that track users across thousands of websites (third-party tracking). All websites collect data on visitor browser activity

Table 10.6 Online Marketing and Advertising Formats

Marketing Format	Description
Search engine	Text ads targeted at precisely what the customer is looking for at the moment of shopping and purchasing. Sales oriented.
Display ads	Banner ads (pop-ups and leave-behinds) with interactive features; increasingly behaviorally targeted to individual web activity. Brand development and sales. Includes social media and blog display ads.
Video	Fastest-growing format, engaging and entertaining; behaviorally targeted, interactive. Branding and sales.
Classified	Job, real estate, and services ads; interactive, rich media, and personalized to user searches. Sales and branding.
Rich media	Animations, games, and puzzles. Interactive, targeted, and entertaining. Branding orientation.
Lead generation	Marketing firms that gather sales and marketing leads online and then sell them to online marketers for a variety of campaign types. Sales or branding orientation.
Sponsorships	Online articles, games, puzzles, and contests sponsored by firms to promote products. Sales orientation.
Email	Effective, targeted marketing tool with interactive and rich media potential. Sales oriented.

and store it in a database. They record the referring site (the site that users visited just prior to coming to the website), the specific pages visited on the site, the time spent on each page, what, if anything, the visitor purchased, the type of operating system and browser being used, and even some location data. Firms analyze this information to develop precise profiles of existing and potential customers. In addition, most popular websites have tracking programs on their home pages that are placed there by third-party data collection firms. These tracking programs are used to track your clickstream behavior across the web by following you from site to site and to retarget ads to you by showing you the same ads on different sites. The leading online advertising network is Google's Marketing Platform. Examine Figure 10.3, which illustrates the steps that occur to track website visitors, both on the site they visit and then across the web.

This information enables firms to understand how well their website is working; create unique, personalized web pages that display content or ads for products or services of special interest to each user; improve the customer's experience; and create additional value through a better understanding of the shopper.

Examine Figure 10.4, which provides several examples of personalized content that a company might display based on previous data they have collected about a user. By using personalization technology to modify the web pages presented to each customer, marketers achieve some of the benefits of using individual salespeople but at dramatically lower costs. For instance, General Motors may show a Chevrolet display ad emphasizing safety and utility to people over the age of 50, whereas young adults in their twenties may receive ads emphasizing sportiness or performance.

For large companies trying to reach millions of consumers, trying to work with hundreds, if not thousands, of websites individually would be impractical. Advertising networks solve this problem by creating a network of several thousand of the most popular websites (which millions of people visit), tracking the behavior of these users across the entire network, building profiles of each user, and then selling these profiles to advertisers

Figure 10.3 Website Visitor Tracking

Tools are available to track a shopper's every step through an online store and then across the web as shoppers move from site to site. Close examination of customer behavior at a website selling women's clothing, for example, shows what the store might learn at each step and what actions it could take to increase sales.

Click 1

The shopper clicks on the home page. The store can tell that the shopper arrived from Yahoo.com at 2:30 PM (which might help determine staffing for customer service centers) and how long the shopper lingered on the home page (which might indicate trouble navigating the site). Tracking beacons load cookies on the shopper's browser to follow her across the web.

Click 2

Click 3

Click 4

Click 5

The shopper clicks on shirts, then clicks to view a woman's blue shirt. The shopper clicks to select this item in a size 10 in blue and clicks to place it in her shopping cart. This information can help the store determine which sizes and colors are most popular. If the visitor moves to a different site, ads for blue shirts will appear from the same or a different vendor.

Click 6

From the shopping cart page, the shopper clicks to close the browser to leave the website without purchasing the shirt. This action could indicate the shopper had a problem with the website's checkout and payment process. Such behavior might signal that the website was not well designed.

Figure 10.4 Website Personalization

Firms can create unique, personalized web pages that display content or ads for products or services of special interest to individual users, improving the customer experience and creating additional value.

in a real-time bidding environment. Popular websites download dozens of web-tracking cookies, bugs, and beacons, which report users' online behavior to remote servers without the users' knowledge. Looking for young, single consumers with college degrees, living in the Northeast, in the 18–34 age range who are interested in purchasing a European car? Advertising networks can identify and deliver thousands of people who fit this profile and

expose them to ads for European cars as they move from one website to another. Estimates vary, but behaviorally targeted ads are generally 10 times more likely to produce a consumer response than a randomly chosen banner or video ad.

Examine Figure 10.5, which illustrates how an advertising network works. When a consumer requests a web page from a website that is a member of an advertising network such as Google Marketing Platform, the website's server connects to the ad network's ad server. That ad server reads the cookie on the originating website and checks its database for a profile. It then selects and serves an appropriate display ad based on the profile. All of this happens within milliseconds. The ad network follows the consumer from site to site through the use of tracking files.

It's a short step from ad networks to programmatic ad buying. Ad networks create real-time bidding (RTB) platforms, where marketers bid in an automated environment for highly targeted slots available from web publishers. Here, ad platforms can predict how many targeted individuals will view the ads, and ad buyers can estimate how much this exposure is worth to them.

In 2023, about 90 percent of spending on online display advertising and online video advertising was for targeted ads served by programmatic ad networks, while the rest depended on the context of the pages shoppers visited—the estimated demographics of visitors, or on so-called blast-and-scatter advertising—which is placed randomly on any available page with minimal targeting, such as time of day or season.

Native advertising

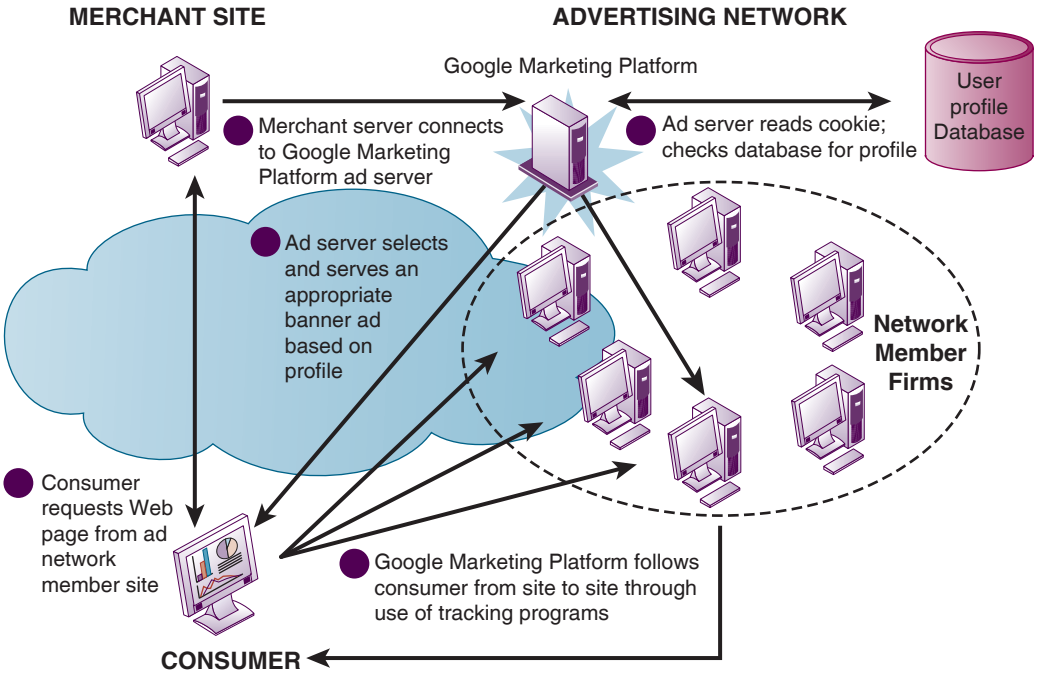
Advertising that looks similar to editorial content but typically is inconspicuously tagged as "sponsored"; often placed in social network feeds or within online magazines.

It's another short step to native advertising. **Native advertising** is advertising that looks similar to editorial content but typically is inconspicuously tagged as "sponsored". Native advertising is often placed in social network feeds or within traditional editorial content, such as an online magazine.

More than 80 percent of U.S. adults are concerned about online privacy. Backlash against the aggressive uses of personal information has grown, with increasing public and political resistance, and within the last several years, firms such as Apple and Google have begun to implement policies that prevent certain types of behavioral targeting. At the same time, more than 50 percent of consumers expect a personalized experience every time they interact with a brand. This presents significant challenges for online marketers (McClain et al., 2023; Schulz, 2022).

Figure 10.5 How an Advertising Network Works

Advertising networks and their use of tracking programs have become controversial among privacy advocates because of these programs' ability to track individual consumers across the Internet.



See Chapter 4 for a more detailed discussion of Internet challenges to privacy, as well as technological responses to issues related to behavioral tracking, such as Apple's App Tracking Transparency initiative, Google's Privacy Sandbox, and the phasing out of third-party cookies.

10.7 Describe Social Networks, Social Marketing, and Social E-commerce

One of the most significant business impacts of the Internet and e-commerce has been the development of online social networks, which are now being leveraged for both marketing and e-commerce purposes.

Social Networks

Social networks in the offline world are collections of people who voluntarily communicate with one another over an extended period of time. An **online social network**, along with other sites with social components, enables users to communicate with one another, form group and individual relationships, and share interests, values, and ideas via a website and/or app. Table 10.7 lists and describes some of the primary features of social networks.

Online social network

Enables users to commuincate with one another, form group and individual relationships, and share interests, values, and ideas, via a website and/or app.

Facebook, with more than 175 million monthly U.S. users in 2024, receives most of the public attention given to social networks. Other top social networks include Instagram, with more than 140 million U.S. monthly active users; TikTok, with around 112 million; Snapchat, with around 93 million; Pinterest, with around 85 million; LinkedIn, with around 80 million;

Table 10.7 Features of Social Networks

Social Network Feature	Description
Newsfeed	A stream of notifications from friends and advertisers that social users find on their home pages.
Timelines	A stream of photos and events in the past that create a personal history for users, one that can be shared with friends.
Social sign-on	Websites allow users to sign into their sites through their social network pages on Facebook or another social site. This allows websites to receive valuable social profile information from Facebook and use it in their own marketing efforts.
Collaborative shopping	An environment where consumers can share their shopping experiences with one another by viewing products, chatting, or texting. Friends can chat online about brands, products, and services.
Network notification	An environment where consumers can share their approval (or disapproval) of products, services, or content or share their geolocation, perhaps a restaurant or club, with friends. Facebook's ubiquitous "like" button is an example, as are tweets and followers on X (formerly Twitter).
Social search (recommendations)	An environment where consumers can ask their friends for advice on purchases of products, services, and content. Although Google can help you find things, social search can help you evaluate the quality of things by listening to the evaluations of your friends or their friends. For instance, Amazon's social recommender system can use your Facebook social profile to recommend products.

and X (formerly Twitter), with about 54 million. According to analysts, 15.5 percent of the total time spent online in the United States is spent on social networks, and using social networks is one of the most common online activities.

Social Marketing and Advertising

Social networks provide one of the fastest-growing platforms for marketing and advertising. Companies will spend an estimated $87 billion in 2024 for marketing and advertising on social networks such as Facebook, Instagram, Pinterest, TikTok, LinkedIn, and X (formerly Twitter) to reach millions of consumers who spend hours a day on social network sites and apps.

The Wisdom of Crowds

Creating sites and apps where thousands, even millions, of people can interact offers business firms new ways to market and advertise and to discover who likes (or hates) their products. Some argue that, in a theory called the **wisdom of crowds**, large numbers of people can make better decisions than a single person or even a small committee of experts.

Wisdom of crowds

The belief that large numbers of people can make better decisions under certain circumstances than a single person or even a small committee of experts.

In marketing, the wisdom of crowds concept suggests that firms should consult with thousands of their customers first as a way of establishing a relationship with them and, second, to understand better how their products and services are used and appreciated (or rejected). One of the most significant benefits of social media is the ability for marketers to understand what people are saying about a brand as well as its competitors, sometimes referred to as **social listening**. According to a recent survey, more than 60 percent of businesses have a social listening system in place. Actively soliciting the comments of your customers builds trust and sends the message to your customers that you care about what they are thinking and that you need their advice (Hutchinson, 2022). The Spotlight on Organizations Case Study, *Social Listening with Sprout Social*, examines how companies are using Sprout Social's social listening tools to build brand awareness and loyalty.

Social listening

Ability of marketers to understand what people are saying online about a brand as well as its competitors.

Spotlight on Organizations Case Study

Social Listening with Sprout Social

Social networks offer myriad opportunities for companies to engage consumers, amplify product messages, discover trends and influencers, build brand awareness, respond to customer requests and recommendations, and, increasingly, to sell their products directly to consumers. As social networks continue to become entrenched in the business and cultural landscape, an ecosystem of companies has sprung up to meet the growing demand for social marketing. Social media monitoring and social listening help companies and marketers understand more about buyers' likes, dislikes, and complaints concerning products, additional products or product modifications customers want, and what people are saying about a brand (positive or negative sentiment). Sprout Social is a market leader in this burgeoning field.

Founded in 2010, Sprout Social is based in Chicago, Illinois, and has grown from just another startup into a public company with about 1,400 employees, more than 34,000 customers in more than 100 countries, and revenue of $338 million in 2023. Sprout's rapid growth mirrors the growth of social media platforms since the company's creation. Marketers eager to tap into the emerging social media channel often lack the tools necessary to understand what techniques are working and which are not. Sprout seeks to help businesses solve this problem, and its tools offer integration with all of the major social networks, including Facebook, Instagram, TikTok, X (formerly Twitter), Threads, Pinterest, and LinkedIn.

Sprout offers a number of services to its clients via its cloud-based applications via a variety of plans. One of those services is its Social Listening tool, which enables marketers to monitor mentions of their brand and industry on social media to extract critical insights, identify trends, and stay ahead of the competition. The Social Listening service can be used for a variety of purposes. For instance, a company can track conversations about its brands to gain visibility into customer attitudes, the customer experience, and overall brand health. It provides the ability to capture and create campaign analysis reports on audience reactions to marketing campaigns. It allows companies to research customer sentiment and opinions on specific topics, track customer behavior

to stay up to date with industry trends, and engage in consumer research to identify audience demographics. It also enables companies to compare themselves to their competitors in terms of share of voice (a gauge of brand visibility) and consumer attitudes.

The Social Listening tool relies on social listening queries, which are a set of search parameters that can include keywords, hashtags, mentions, phrases, and even emojis. Sprout provides a Listening Query Builder that enables users to quickly and easily create complex, multi-faceted queries. Sprout has recently added AI functionality and its Queries by AI Assist tool automatically suggests additional related keywords based on the keywords initially provided. Sprout also offers a variety of pre-built Topic Templates that provide tips and queries to help users get started with query creation for the most common uses. The Social Listening tool can process up to 50,000 posts per second and an average volume of 600 million messages a day.

Goally, based in Denver, Colorado, is one company that has been a satisfied user of Sprout Social's Social Listening tool. Goally offers an app that helps teach neurodiverse children how to complete basic tasks in collaboration with their parents and caregivers. Goally believes that social networks offer a unique and accessible way for people to ask authentic questions and get answers. Goally began using Sprout's Social Listening tools in 2022, and the insights gained thus far are helping Goally confirm that its technology is actually helpful to parents, and also to better understand the needs of the families they aim to support. For example, social listening on TikTok helped Goally uncover a common desire of parents raising neurodiverse children: They want to see their children become independent. Sprout has also helped Goally build brand awareness by enabling it to identify relevant influencers and to connect more effectively with its target audience. Goally is also using Sprout to track follower trends on TikTok, confirming that TikTok has become a valuable channel and a place where it needs to stay active.

SOURCES: Sproutsocial.com, accessed September 30, 2024; Ronnie Gomez, "How to Create Better Social Listening Queries," Sproutsocial.com, April 16, 2024; Sprout Social, Inc., "Annual Report on Form 10-K," Sec.gov, February 23, 2024; Carly Hill, "Social Media Listening Examples that Will Help You Unearth Better Insights," Sproutsocial.com, October 18, 2023; Ronnie Gomez, "5 Stories that Showcase Social Data's Rising Influence on Business," Sproutsocial.com, August 15, 2023; "How Sprout Social Helps Goally Engage with and Learn from Parents of Neurodiverse Children," Sproutsocial.com.

Case Study Questions

1. Assess the management, organization, and technology issues for using social media technology to engage with customers.
2. What are the advantages and disadvantages of using social media for advertising, brand building, market research, and customer service?
3. What kinds of companies are best suited to using social network platforms for marketing and customer service?

Beyond merely soliciting advice, firms can be actively helped in solving some business problems by using **crowdsourcing**, which involves tapping into large online audiences for advice, market feedback, new ideas, and potential solutions to business problems. For instance, BMW's Crowd Innovation platform, launched in 2021, helps BMW tap into the wisdom of both BMW's employees as well as external innovation crowds. Kickstarter is arguably one of the most famous crowdfunding platforms where visitors can help fund creative projects such as films, music, art, books, and the creation of new, innovative products. Other crowdsourcing examples include Caterpillar working with customers to design better machinery, Lego's Lego Idea platform for developing new toys and games, and Unilever's use of a digital crowdsourcing platform to create a community of local people in Indonesia to track palm oil throughout the production process (see the Chapter 6 Chapter Case Study, *Pursuing Sustainability with Blockchain*.).

Crowdsourcing

Using large online audiences for advice, market feedback, new ideas, and solutions to business problems. Related to the "wisdom of crowds" theory.

Social E-commerce

Social e-commerce is involves the use of social networks to share knowledge about items of interest to other shoppers and enable purchases directly via the social network. Social e-commerce is based on the idea of the digital **social graph**, a mapping of all significant online social relationships. The social graph is synonymous with the idea of a social network used to describe offline relationships. You can map your own social graph (network) by drawing lines from yourself to the 10 closest people you know. If they know one another, draw lines between these people. If you are ambitious, ask these 10 friends to list and draw in the names of the 10 people closest to them. What emerges from this exercise is a preliminary

Social e-commerce

Type of e-commerce that involves the use of social networks to share knowledge about items of interest with other shoppers and enable purchases directly via the social network.

Social graph

Map of all significant online social relationships, comparable to a social network describing offline relationships.

map of your social network. Now imagine if everyone on the Internet did the same and posted the results to a large database with a website. Ultimately, you would end up with Facebook or a site like it.

According to small world theory, you are only six links away from any other person on Earth. If you entered your personal address book, which has, say, 100 names in it, into a list and sent it to your friends, and they in turn entered 50 new names of their friends, and so on, five times, the social network created would encompass 31 billion people! The social graph is therefore a collection of millions of personal social graphs (and all the people in them).

If you understand the interconnectedness of people, you will see just how important this concept is to e-commerce. The products and services you buy will influence the decisions of your friends, and their decisions will in turn influence you. If you are a marketer trying to build and strengthen a brand, you can take advantage of the fact that people are enmeshed in social networks, share interests and values, and communicate and influence one another. As a marketer, your target audience is not a million isolated people watching a TV show but the social network of people who watch the show and the viewers' personal networks. Moreover, online social networks are where the largest Internet audiences are located.

Social e-commerce has taken off within the last few years and has been growing at more than 25 percent a year. In 2024, more than 110 million consumers are expected to make a purchase via a social network, and social retail e-commerce revenues are expected to top $82 billion. All the major social networks now include social e-commerce functionality. For instance, Facebook offers both Facebook Shops and Instagram Shops, which enable retailers to post their catalogs directly on the social network. Retailers can choose to route customers to their own website or sell directly on the platform (eMarketer, 2024f, 2024g, 2023h, 2024i).

10.8 Understand How E-commerce Has Affected Business-to-Business Transactions

Before the Internet, business-to-business transactions were referred to simply as inter-firm trade or the procurement process. Today, we use the term *B2B commerce* to describe all types of inter-firm trade, involving both the purchase of inputs and the distribution of products and services. B2B commerce represents a huge marketplace. The total amount of B2B commerce in the United States in 2024 is estimated to be about $19 trillion. B2B e-commerce involves that portion of B2B commerce that is enabled by the Internet and mobile apps. B2B e-commerce is expected to generate about $8.1 trillion in 2024 and grow to about $8.9 trillion by 2028 (eMarketer, 2023j; U.S. Bureau of the Census, 2023; authors' estimates).

The process of conducting trade among business firms is complex and requires considerable human intervention; therefore, it consumes significant resources. Some firms estimate that each purchase order for support products costs, on average, at least $100 in administrative overhead, including approving purchase decisions, searching for products and arranging for purchases, shipping, and receiving the goods. Across the economy, this adds up to trillions of dollars spent annually for procurement processes that could be automated. If even just a portion of interfirm trade and the procurement process were automated, literally trillions of dollars might be released for more productive uses; consumer prices potentially would fall; productivity would increase; and the economic wealth of the nation would expand. This is the promise of B2B e-commerce. The challenge of B2B e-commerce is changing existing patterns and systems of procurement and designing and implementing new Internet and cloud-based B2B solutions.

Electronic data interchange (EDI)

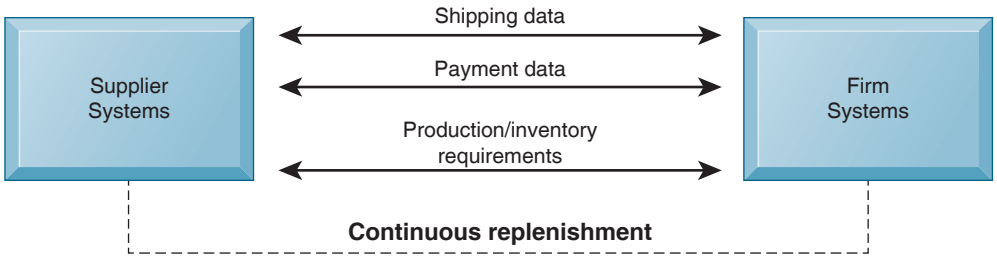
The direct computer-to-computer exchange between two organizations of standard business documents and transactions, such as orders, shipment instructions, or payments.

Electronic Data Interchange (EDI)

B2B e-commerce has evolved over a 50-year period. In the late 1970s, a new form of computer-to-computer communication called **electronic data interchange (EDI)** facilitated the sharing of standard business documents such as invoices, purchase orders, and shipping bills

Figure 10.6 Electronic Data Interchange (EDI)

Companies use EDI to automate transactions for B2B e-commerce and continuous inventory replenishment. Suppliers can automatically send data about shipments to purchasing firms. The purchasing firms can use EDI to provide production and inventory requirements and payment data to suppliers.



between two organizations. Transactions are automatically transmitted from one information system to another through a network, eliminating the printing and handling of paper at one end and the inputting of data on the other. Each major industry in the United States and much of the rest of the world has EDI standards that define the structure and information fields of digital transactions for that industry.

EDI remains an important tool for B2B e-commerce. Although many companies still use EDI for document automation, firms engaged in just-in-time inventory replenishment and continuous production use EDI as a system for continuous replenishment. Examine Figure 10.6, which illustrates the interchange of shipping data, payment data, and production/inventory requirements between supplier and purchasing firms. Suppliers have online access to selected parts of the purchasing firm's production and delivery schedules and automatically ship materials and goods to meet prespecified targets without intervention by firm purchasing agents.

Although many organizations still use private networks for EDI, these networks are increasingly web-enabled because Internet technology provides a much more flexible and low-cost platform for linking to other firms. Businesses can extend digital technology to a wider range of activities and broaden their circle of trading partners.

Procurement, for example, involves not only purchasing goods and materials but also sourcing, negotiating with suppliers, paying for goods, and making delivery arrangements. Businesses can now use the Internet to locate the lowest-cost supplier, search online catalogs of supplier products, negotiate with suppliers, place orders, make payments, and arrange transportation. They are no longer limited to partners linked by traditional EDI networks.

New Ways of B2B Buying and Selling

The Internet and web technology enable businesses to create online storefronts for selling to other businesses using the same techniques as used for B2C commerce. Alternatively, businesses can use Internet technology to create extranets or digital marketplaces for linking to other businesses for purchase and sale transactions.

A **private B2B network** is an Internet-enabled network typically employed by a large firm to link to its suppliers and other key business partners. Examine Figure 10.7, which illustrates a private B2B network. The buyer owns the network, which permits the firm and its designated suppliers, distributors, and other business partners to share product design and development, marketing, production scheduling, inventory management, and unstructured communication, including graphics and email.

Private B2B network

Internet-enabled network typically employed by a large firm to link to its suppliers and other key business partners.

An example is VW Group Supply, which links the Volkswagen Group and its suppliers. VW Group Supply handles 90 percent of all global purchasing for Volkswagen, including all automotive and parts components.

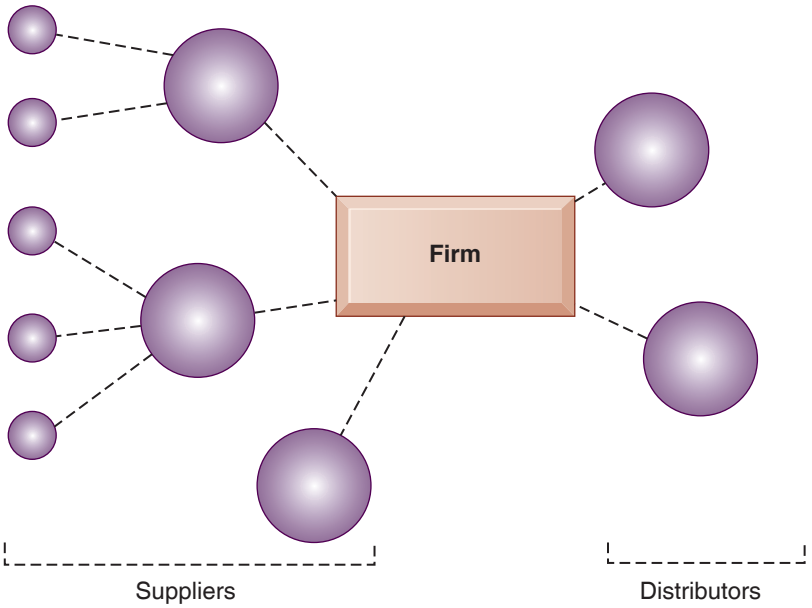
A B2B e-commerce marketplace provides a single, digital marketplace for multiple buyers and sellers to transact business. They are industry-owned or operate as independent intermediaries between buyers and sellers. B2B e-commerce marketplaces generate revenue from purchase and sale transactions and other services provided to clients.

B2B e-commerce marketplace

Provides a single, digital marketplace that allows multiple business buyers and sellers to transact business.

Figure 10.7 A Private B2B Network

A private B2B network links a firm to its suppliers, distributors, and other key business partners for efficient supply chain management and other collaborative commerce activities.



Participants in B2B e-commerce marketplaces can establish prices through online negotiations, auctions, or requests for price quotes, or they can use fixed prices. Examine Figure 10.8, which illustrates how a B2B e-commerce marketplace connects multiple suppliers and multiple buyers, providing online catalogs of goods as well as tools for sourcing, automated processing, and fulfillment.

There are many types of B2B e-commerce marketplaces and ways of classifying them. Some sell direct goods and some sell indirect goods. **Direct goods** are goods used in a production process, such as sheet steel for auto body production. **Indirect goods** are all other goods not directly involved in the production process, such as office supplies or products

Direct goods

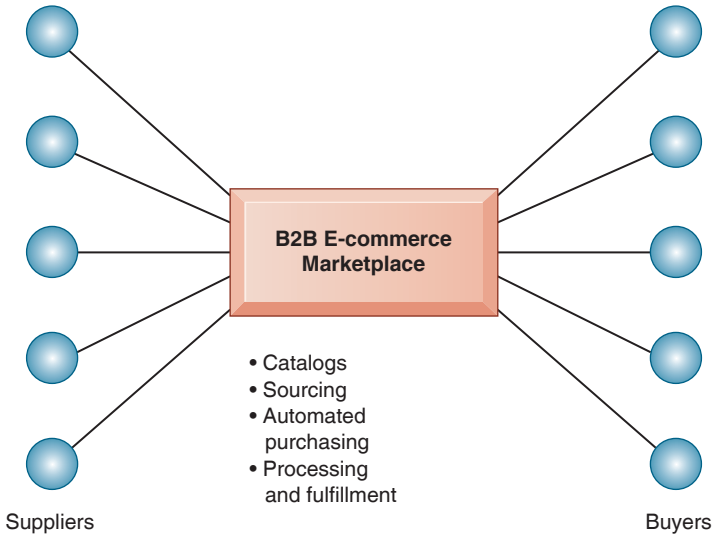
Goods used in a production process.

Indirect goods

Goods not directly used in the production process, such as office supplies.

Figure 10.8 A B2B E-commerce Marketplace

B2B e-commerce marketplaces are online marketplaces where multiple buyers can purchase from multiple sellers.



for maintenance and repair. Some B2B e-commerce marketplaces support contractual purchasing based on long-term relationships with designated suppliers, and others support short-term spot purchasing, where goods are purchased based on immediate needs, often from many suppliers.

Some B2B e-commerce marketplaces serve vertical markets for specific industries, such as automobiles, telecommunications, or machine tools, whereas others serve horizontal markets for goods and services, such as office equipment or transportation, that can be found in many industries.

SupplyOn is an example of an industry-owned B2B e-commerce marketplace, focusing on long-term contract purchasing relationships and on providing common networks and computing platforms for reducing supply chain inefficiencies. SupplyOn was founded by automotive suppliers Bosch, Continental, Schaeffler, and ZF and originally focused on the automotive industry. SupplyOn has since expanded to serve firms in the aerospace, railway, and engineering industries. More than 140,000 businesses worldwide use SupplyOn's supply chain network and solutions, and it counts among its customers such well-known companies as Airbus, BMW Group, BorgWarner, Siemens, and Thales.

An **exchange** is an independently owned third-party B2B e-commerce marketplace that is primarily transaction-oriented and that connects many suppliers and buyers for spot purchasing. Many exchanges provide vertical markets for a single industry, such as food, electronics, or industrial equipment, and they primarily deal with direct inputs. For example, Go2Paper enables a spot market for paper and packaging materials among buyers and sellers in the paper industry in more than 75 countries. Exchanges proliferated during the early years of e-commerce, but many failed. However, in recent years, exchanges have experienced a resurgence.

Exchange

Independently owned, third-party B2B e-commerce marketplace that is primarily transaction-oriented and that connects many buyers and suppliers for spot purchasing.

10.9 Describe M-commerce and M-commerce Applications

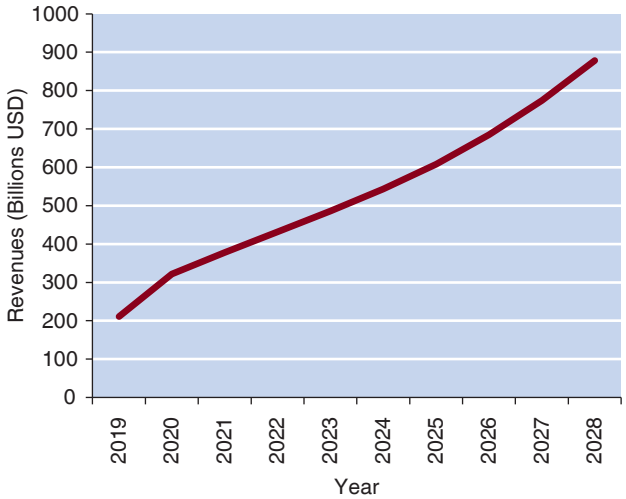
Mobile e-commerce (m-commerce) involves the sales of goods and services via mobile devices, such as smartphones, tablets, and wearables. Examine Figure 10.9, which illustrates the growth of retail m-commerce revenues from 2019 through 2028. In 2025, retail m-commerce is expected to generate more than $540 billion in revenues, which is about 45 percent of all

Mobile e-commerce (m-commerce)

Type of e-commerce that involves the sale of goods and services via mobile devices, such as a smartphone, tablet computer, or wearable.

Figure 10.9 Retail M-commerce Revenue

By 2028, retail m-commerce sales are expected to account for about $875 billion, constituting more than 50 percent of total retail e-commerce sales.



Source: Data from eMarketer chart "US Retail Mcommerce Sales," 2024.

retail e-commerce sales. Retail m-commerce is expected to continue increasing at the rate of about 12 percent per year between 2024 and 2028, when it is estimated to reach more than $875 billion (eMarketer, 2024c).

All the top retailers now have mobile websites that enable shoppers to use their mobile devices to shop and place orders. Virtually all large traditional and online retailers also have apps that enable m-commerce.

Banks and credit card companies have developed services that let customers manage their accounts from their mobile devices. JPMorgan Chase and Bank of America customers can use their smartphones to check account balances, transfer funds, and pay bills. Apple Pay for the iPhone and Apple Watch, along with Android smartphone apps, allows users to charge items to their credit card accounts with a swipe of their phone. (For more information on these types of apps, see the following section on mobile app payment systems.)

The main areas of growth in m-commerce are mass market retailing such as Amazon's; sales of digital content such as music, TV shows, movies, and e-books; and in-app sales via mobile devices. On-demand firms such as Uber and Airbnb are location-based services and examples of m-commerce as well. Larger mobile screens and more-convenient payment procedures also play roles in the expansion of m-commerce.

The mobile advertising market is the fastest-growing online ad platform, racking up an estimated $200 billion in ad revenue in 2024. Ads can be embedded in games, videos, and other mobile applications. Google is the largest mobile advertising platform, displaying ads linked to mobile searches, and generating an expected $77 billion in mobile ad revenues in 2024. Meta's Facebook/Instagram is in second place, with an expected $65 billion in mobile ad revenue (eMarketer, 2024k).

Location-based Services and Applications

Location-based services
Involves providing services to users based on the users' location; enabled by GPS, Wi-Fi, and cellular technologies.

Location-based services involve providing services to users based on the users' location. The primary technologies that enable location-based services include GPS (global positioning system) enabled within smartphones; Wi-Fi, which can estimate a user's location within a radius of known Wi-Fi access points, and cell tower location, which is used by wireless carriers to track the location of mobile devices. Some of the most well-known location-based services are on-demand firms such as Uber, Lyft, Airbnb, and hundreds more that provide services to users in local areas and are based on the user's location (or, in the case of Airbnb, the user's intended travel location).

Location-based services include geosocial, geoadvertising, and geoinformation services.

Geosocial services
Involves social networking based on users' location.

Geosocial services involve social networking based on a user's location, for instance, telling you where your friends are meeting. Foursquare is an example of a geosocial service. A geosocial service helps you find friends, or helps your friends find you, by checking in to the service and announcing your presence in a restaurant or other place. Your friends are instantly notified. Foursquare's Swarm app provides a location-based social network service that enables its users to connect with friends, update their location, and provide reviews and tips for enjoying a location. Points are awarded for checking in at designated venues. Users also earn badges by checking in at locations with certain tags, for check-in frequency, or for the time of check-in.

Geoadvertising services
Involves delivering ads to users based on their location.

Geoadvertising services involve delivering advertising to users based on their location, such as a mobile ad from a nearby Italian restaurant. Merchants buy access to these consumers when they come within range of a merchant. For instance, Best Buy uses an in-store beacon network to deliver relevant ads based on shoppers' location within a store. Users get a push notification which, if they opt to open it, turns into a full-screen ad. Shopkick is a mobile app that enables retailers to offer coupons to people when they walk into the retailers' stores. The Shopkick app automatically recognizes when the user has entered a partner retail store and offers a virtual currency called kicks, which can be redeemed for store gift cards.

Geoinformation services
Involves providing users with information on local places and things based on the users' location.

Geoinformation services involve providing users with information on local places and things based on the location of the user, such as telling you the price of a house you are looking at or about special exhibits at a museum you are passing. Waze is an example of a popular social geoinformation service. Waze is a GPS-based map and navigational app for smartphones. Waze locates the user's car on a digital map using GPS and, like other navigation programs, continuously collects information on the user's speed and

direction. What makes Waze different is that it collects traffic information from users who submit accident reports, speed traps, landmarks, street fairs, protests, and even addresses. Waze uses this information to suggest alternative routes, give travel times, and issue warnings and can even make recommendations for gas stations along the way. The Waze app is used extensively by Uber and Lyft drivers and by more than 140 million other drivers in the United States.

Mobile App Payment Systems

Mobile app payment systems use mobile apps to replace credit cards and traditional banking services. There are three main types of mobile payment apps. Near-field communication (NFC)-driven systems enable NFC-enabled smartphones and other mobile devices to make contactless payments by communicating with an NFC-enabled reader at a merchant's point-of-sale (POS) terminal in close physical proximity. (Review the discussion of radio-frequency identification [RFID] and near-field communication [NFC] in Chapter 7.) Apple Pay and Google Pay are examples. QR Code payment systems, such as Walmart Pay, use a contactless payment method. A payment is initiated by scanning a two-dimensional barcode called a QR (Quick Response) code using a mobile app on the payer's smartphone. Peer-to-peer (P2P) payment systems such as Venmo or Zelle QuickPay can be used to transfer money among individuals. Table 10.8 compares these three types of mobile app payment systems.

Table 10.8 Types of Mobile App Payment Systems

Type of Mobile App Payment System	Description	Examples
Near Field Communication (NFC)	System using technology driven by near field communication (NFC) chips in both payers' mobile devices and merchants' point-of-sale (POS) reader devices. When close together and activated, these NFC chips exchange encrypted data to complete a payment. Can be used with many different merchants if they use NFC readers and software for accepting payments.	Apple Pay, Google Pay, Samsung Pay
QR Code	Quick Response (OR) code technology uses a two-dimensional barcode in which information is encoded to perform contactless transactions using a code-scanning and generation app on a smartphone and a compatible merchant device. After a merchant enters the payment amount, the customer opens an app that displays a QR code generated for the transaction. The merchant scans the QR code, and the payment amount is deducted from the customer's mobile wallet. Alternatively, the customer opens the app and scans a QR code displayed by the merchant, enabling the app to identify the merchant. The customer then supplies the amount and completes the payment.	Starbucks, Walmart, Target, Dunkin' Donuts
Peer-to-peer (P2P) payment system	Technology allowing individuals to transfer funds from their bank accounts to other accounts on the same platform via the Internet. P2P users establish a secure account with a trusted third-party vendor, designating their bank account or credit card account to send and accept funds. Using the third-party app, users can send money to another person or to a merchant's account. Users are generally identified by their email address or mobile phone number.	Venmo, Zelle

10.10 Discuss How to Build an E-commerce Presence

Building a successful e-commerce presence requires a keen understanding of business, technology, and social issues as well as a systematic approach. Today, an e-commerce presence is not just a website but also includes a presence on social networks such as Facebook, Instagram, TikTok, Pinterest, and/or X (formerly Twitter), and mobile apps where customers can access your services. Developing and coordinating all these customer venues can be difficult. A complete treatment of the topic is beyond the scope of this text, and students should consult books devoted to just this topic (Laudon and Traver, 2024). The two most important management challenges in building a successful e-commerce presence are (1) developing a clear understanding of your business objectives and (2) knowing how to choose the right technology to achieve those objectives.

Develop an E-commerce Presence Map

E-commerce has moved from being a desktop-centric activity on the web to a mobile and tablet-based activity. Currently, more than 90 percent of Internet users in the United States use a mobile device to shop for goods and services, look up prices, enjoy entertainment, and access social networks. Your potential customers use these various devices at different times during the day and involve themselves in different conversations, depending on what they are doing—touching base with friends, tweeting, or reading a blog. Each of these is a touch point where you can meet the customer, and you have to think about how you develop a presence in these different virtual places.

Examine Figure 10.10, which provides a roadmap to the platforms and related activities you will need to think about when developing your e-commerce presence. Figure 10.10 illustrates four kinds of e-commerce presence: websites/apps, email, social media, and offline media.

Figure 10.10 E-commerce Presence Map

An e-commerce presence requires firms to consider four types of presence, with specific platforms and activities associated with each.

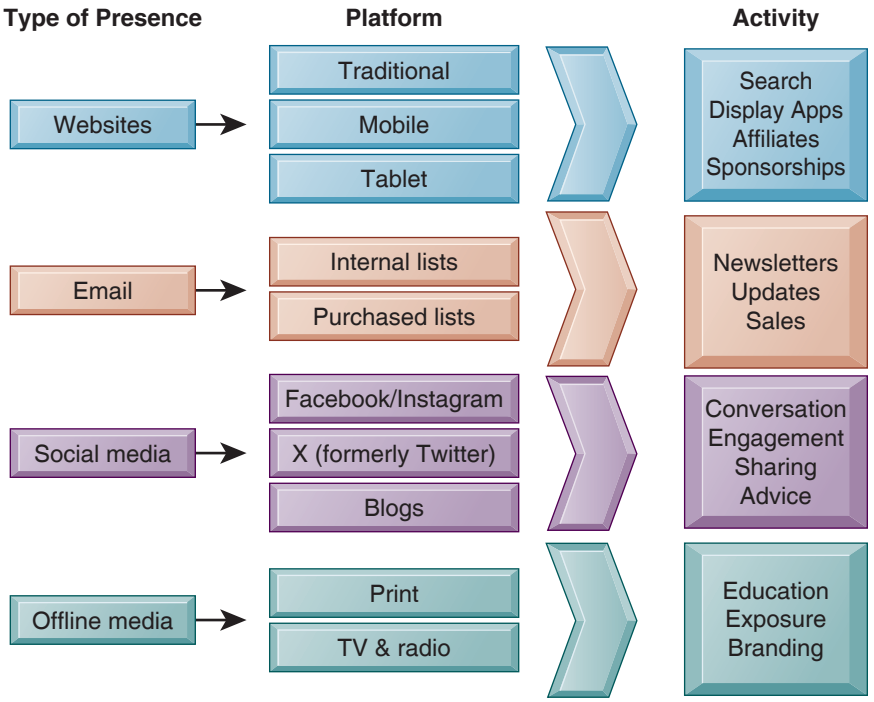


Table 10.9 E-commerce Presence Timeline

Phase	Activity	Milestone
Phase 1: Planning	Envision e-commerce presence; determine personnel.	A mission statement
Phase 2: Website development	Acquire content; develop a site design; arrange for hosting the site.	A website plan
Phase 3: Web implementation	Develop keywords and metatags; focus on search engine optimization; identify potential sponsors.	A functional website
Phase 4: Social media plan	Identify appropriate social platforms and content for your products and services.	A social media plan
Phase 5: Social media implementation	Develop Facebook/Instagram, X (formerly Twitter), TikTok, and Pinterest presence.	A Functioning social media presence
Phase 6: Mobile plan	Develop a mobile plan; consider options for porting your website to smartphones.	A mobile media plan

You must address different platforms for each of these types. For instance, in the case of website presence, there are three platforms: traditional desktop, tablets, and smartphones, each with different capabilities. Mobile apps can be in conjunction with a website or as standalone presence. Moreover, for each type of e-commerce presence, there are related activities you will need to consider. For instance, in the case of websites, you will want to engage in search engine marketing, display ads, affiliate programs, and sponsorships. Offline media, the fourth type of e-commerce presence, is included here because many firms use multiplatform or integrated marketing by which print ads refer customers to websites.

Develop a Timeline: Milestones

Where would you like to be a year from now? When you begin, it's very helpful for you to have a rough idea of the timeframe for developing your e-commerce presence. You should break your project down into a small number of phases that could be completed within a specified time. Table 10.9 illustrates a one-year timeline for the development of an e-commerce presence for a startup company devoted to fashions for teenagers.

10.11 Understand How the Information in This Chapter Can Help Your Career

The information you have learned in Chapter 10 can help you advance your career by helping you become familiar with the basic concepts underlying e-commerce, as well as the major trends that are currently impacting e-commerce today and in the future. E-commerce is the fastest-growing form of commerce and is expected to continue to be so for some time to come. Employers are looking for a wide variety of skills, and having a familiarity with the basic concepts and vocabulary underlying e-commerce can help you as you interview, and also on the job.

Real-World Scenario

eHarmony's E-commerce Analysts

Online dating website eHarmony uses big data collected from the activities of its 10 million active users on its website to "scientifically" match couples based on features of compatibility found in thousands of successful relationships. eHarmony also uses a team of e-commerce analysts to review its findings and translate them into successful predictions on how users will behave. By examining data such as how many times a user logs on to the website, how many photos they post, and the words they use to describe themselves on their eHarmony profiles, eHarmony analysts can tell who is more introverted or who is likely to be an initiator. The data also help the analysts determine whether users would be more likely to communicate with their matches at certain times of the day. E-commerce analysts then use these findings to refine the couple-matching algorithm. The matching process is highly automated, allowing eHarmony to generate matches for users in a matter of minutes most of the time. eHarmony's couple-matching algorithm varies from country to country and is regularly refined based on user behavior and new relationship research from educational institutions. E-commerce analysts also use big data analytics for marketing to determine the best times for eHarmony to send out promotions to individual users.

Job Description and Career Path: E-commerce Analyst

Job	Job Description	IT Knowledge	Career Path and Average Salaries
E-commerce analyst	An e-commerce analyst helps companies optimize online sales and customer experience, enhancing the company's online presence. Job responsibilities include: • Collecting, analyzing, and interpreting data related to website traffic, user behavior, and online sales to identify trends • Monitoring key performance indicators, such as revenue, conversion rate, and customer acquisition costs to assess the effectiveness of online strategies and marketing campaigns • Analyzing, pricing strategies, and product performance to maximize profitability • Using data to create sales forecasts and projects to guide budgeting • Analyzing customer behavior including clickthrough rates and path analysis, to optimize customer experiences • Preparing and presenting reports to communicate findings to management	• Degree in MIS, e-commerce, information science, data analytics, marketing, or business • Knowledge of Microsoft 365 tools: Excel, Word. PowerPoint, Teams • Familiarity with SQL, Tableau or other data visualization tools • Familiarity with e-commerce platforms such as Shopify, BigCommerce, Amazon, • Familiarity with web analytics tools such as Google Analytics • Familiarity with GenAI and GenAI driven tools	• Entry-level e-commerce analyst: $70,000–$90,000 • Mid-level e-commerce analyst: $90,000–125,000 • Senior e-commerce analyst/ E-commerce product manager: $125,000–150,000 • Senior product manager/ Director of e-commerce: $200,000–$300,000 Salaries can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

Lemonade Seeks to Disrupt the Insurance Industry with AI

Insurance is one of the largest industries in the world. Property, casualty, and life insurance premiums amount to approximately $5 trillion globally. In the United States, 14 of the *Fortune* 100 companies are insurance companies. Most of the leading insurance companies have been in business for more than a century.

Insurance products can be very complex, and in the past, the insurance business has been driven by person-to-person interactions between consumers and insurance agents or brokers. In addition, the leading insurance companies have years of investments in legacy IT systems as well as corporate cultures developed over many decades. The process of grafting new technologies and integrating new ways of doing business in such an environment is typically a slow one, leaving the industry ripe for a disruptive player.

Enter Lemonade. Founded in 2015, Lemonade is attempting to take the insurance industry by storm by leveraging a variety of cutting-edge technologies, such as artificial intelligence, chatbots, big data collection and analysis, and a sophisticated mobile app. Its digital platform spans marketing to customer care to claims processing, collecting and deploying data at each step along the way. Lemonade refers to itself as an AI-native and customer-centric insurance company.

Lemonade originally focused on the homeowners and renters insurance market but has since expanded into pet, term life, and car insurance on its way to becoming a one-stop shop for consumer insurance needs. In 2020, Lemonade went public, and in May 2024, it had a market valuation of more than $1.14 billion. Lemonade is also a Certified B Corp. Certified B Corps are companies certified by B Lab (an independent, nonprofit organization) as meeting rigorous standards of social and environmental performance, accountability, and transparency. Lemonade's insurance products are available throughout various states in the United States (depending on the product) and it has recently begun providing renters insurance in France, Germany, the Netherlands, and the UK.

The typical Lemonade customer is younger than 35. For more than 90 percent of these customers, Lemonade is their first experience obtaining insurance. Compared to traditional insurers, Lemonade has pioneered a fundamentally different approach to the process. For instance, the application process for a standard homeowner's policy is typically based on a form with between 20 and 50 fields (such as name, address, birthdate). In contrast, Lemonade has dispensed with forms altogether. Instead, prospective customers use their mobile app to interact with a chatbot named AI Maya. Maya uses a natural language interface and typically asks only about 13 questions before being able to give customers a personalized quote for home insurance. Lemonade is able to streamline the process by using artificial intelligence. Although Maya may only ask a limited number of questions, the interaction generates almost 1,700 data points, all of which are logged, aggregated, and mined for correlation to claims. According to Lemonade, its real advantage lies in the depth of data that it collects about each customer when the customer is onboarded. Maya provides the interface for all types of insurance that Lemonade offers.

The claims process is handled by a different chatbot, AI Jim. Jim takes the first notice of loss from a customer making a claim in almost all cases and is typically able to handle the entire claim through resolution for about one-third of the cases. In some cases, Jim can even authorize and pay a claim in as little as three seconds. Jim assigns the rest to human claims experts based on Jim's analysis of the experts' specialties, qualifications, workloads, and schedules.

Lemonade also has a chatbot platform, CX.AI, that uses natural language processing to understand and instantly resolve customer requests for assistance without human intervention. CX.AI uses natural language processing. As of the end of 2023, about one-third of all customer inquiries were being handled in this manner.

Lemonade's digital infrastructure is driven by three key proprietary applications: Forensic Graph, Blender, and Cooper. Forensic Graph uses big data, artificial intelligence, and behavioral economics to predict, detect, and block fraud throughout the customer application process. Blender is a built-from-scratch insurance management system designed to be a cohesive and streamlined management tool for Lemonade's customer experience, including underwriting, claims, growth, marketing, finance, and risk teams. Cooper is an internal AI bot that handles complex as well as repetitive tasks, from helping Lemonade's customer experience teams handle manual processes such as processing paper checks, to automatically running tens of thousands of tests on each release of its software. The latter is a particularly important task because Lemonade averages more than 40 code releases per day. The frequent refreshing of its code means that Lemonade can promptly and efficiently modify customer onboarding questions, underwriting guidelines, claims handling, and other elements of its platform, enabling it to respond to changes in customer needs and market conditions at a faster rate than its traditional insurance competitors can.

But Lemonade's approach is not without controversy. In 2021, a tweet from Lemonade's Twitter (now X) account created outrage when it appeared to boast about how Lemonade's AI system boosted its profits by automatically denying claims based on analyzing videos submitted by customers. Lemonade quickly deleted the tweet and apologized, stating that Lemonade's AI does not use physical or personal features to deny claims, and that its systems do not evaluate claims based on background, gender, appearance, skin tone, disability, or any other physical characteristic. However, in

the same statement, Lemonade confirmed that it was using facial recognition technology to flag certain claims. Not long thereafter, Lemonade was sued in class action lawsuits in both Illinois and New York, alleging that it had violated biometric privacy laws in both states by collecting such data without users' knowledge and consent. In 2022, Lemonade agreed to pay $4 million to settle some of those suits.

In addition, Lemonade's heavy reliance on data analytics and artificial intelligence creates a number of business risks. It has noted in its regulatory filings that in some states, its ability to use of external data about customer behavior may be limited. For example, in California, Lemonade's largest market for car insurance, Lemonade is currently limited in its ability to use telematics data beyond the number of miles driven to make insurance decisions. Lemonade has also noted that legislators and insurance regulators have been focusing on the use of AI, particularly in the EU, where the forthcoming EU AI Act is likely to have a material impact on the way Lemonade conducts business there.

Despite these missteps and potential issues, and even though thus far it has not shown a profit, Lemonade continues to grow. For its 2023 fiscal year, it increased annual revenues by 67 percent compared to 2022, and reduced its losses by 20 percent. It had about 2.1 million customers as of March 2024, an increase of more than 10 percent compared to the previous year, and expects to grow revenues by 20 percent in 2024 compared to 2023, while continuing to reduce losses. It has also garnered many accolades, including recognition by various publications as "best-in-class" renters, homeowners, and pet insurance.

SOURCES: Pymnts, "Lemonade Sees AI Drive Efficiency as Insurance Goes Digital-First," Pymnts.com, May 1, 2024; Lemonade, Inc., "Lemonade Shareholder Letter Q1 2024," Lemonade.com, April 30, 2024; Lemonade, Inc., "Form 10-K for the Fiscal Year Ended December 31, 2023", Sec.gov, February 28, 2024; Pymnts, "Lemonade Says AI Improved Insurance Loss Ratio," Pymnts.com, February 28, 2024; Josh Liberatore, "Insurer Lemonade Reaches $4M Deal in Ill. Biometric Data Suit," Law360.com, May 18, 2022; Carlton Fields and Michael Yaeger, "AI Insurance Company Faces Class Action for Use of Biometric Data," Jdsupra.com, February 1, 2022; Jonathan Greig, "Lemonade Insurance Faces Backlash for Claiming AI System Could Automatically Deny Claims," Zdnet.com, May 27, 2021.

Case Study Questions

1. What distinguishes the "insurtech" services described in this case from traditional insurers? Explain your answer.
2. How has Lemonade used artificial intelligence to innovate?
3. What issues are raised by Lemonade's use of artificial intelligence?
4. What factors would you consider in deciding whether to use an "insurtech" service such as Lemonade's?

Review Summary

10.1 Discuss the evolution of e-commerce.

E-commerce refers to the use of the Internet, the web, and mobile apps and browsers running on mobile devices to transact business, and more formally, involves digitally enabled commercial transactions between and among organizations and individuals. E-commerce began in 1995, and since then has grown tremendously. Today, purchasing goods and services using a desktop computer, smartphone, or tablet has become ubiquitous. One of the biggest changes is the extent to which e-commerce has become more social, mobile, and local. E-commerce has expanded from the desktop and home computer to mobile devices, from an isolated activity into a more social form of commerce, and from commerce focused on a national audience to commerce focused on local merchants and consumers whose location is known to mobile devices. AI is the newest frontier in e-commerce and is likely to have a significant impact on how e-commerce is conducted in the near future.

10.2 Describe the unique features of e-commerce.

Unique features of e-commerce technology include ubiquity, global reach, universal technology standards, richness, interactivity, information density, capabilities for personalization and customization, and social technology that enables user content generation and social networks.

10.3 Describe digital markets and digital goods.

Digital markets are said to be more transparent than traditional markets, with reduced information asymmetry, search costs, transaction costs, and menu costs along with the ability to change prices dynamically based on market conditions. Digital goods, such as music, video, software, and books, can be delivered over a digital network. Once a digital product has been produced, the cost of delivering that product digitally is extremely low.

10.4 Describe types of e-commerce and e-commerce business models.

The principal types of e-commerce are business-to-consumer (B2C) e-commerce, business-to-business (B2B) e-commerce, and consumer-to-consumer (C2C) e-commerce. Many businesses engage in more than one type. Another way of classifying e-commerce transactions is in terms of the platforms that participants use in a transaction, such as the mobile platform (mobile e-commerce) or social networks (social e-commerce). E-commerce business models include online retailers, transaction brokers, market creators, content providers, service providers, community providers (social networks), and portals. In some cases, business models overlap. In addition, many businesses use more than one business model.

10.5 Describe e-commerce revenue models.

A firm's revenue model describes how the firm will earn revenue, generate profits, and produce a return on investment. Although many e-commerce revenue models have been developed, most companies rely on one, or some combination, of the following: advertising, sales, subscription, free/freemium, transaction fee, and affiliate.

10.6 Explain how e-commerce has transformed marketing.

The Internet provides marketers with new ways of identifying and communicating with millions of potential customers at costs far lower than traditional media costs. The major online marketing and advertising formats include search engines, display ads, video, rich media, classified ads, lead generation, sponsorships, and email. Behavioral targeting techniques increase the effectiveness of display and video ads. Advertising networks and programmatic ad buying extend the reach of behavioral targeting across the web. However, concerns about online privacy have fueled a backlash against behavioral targeting and major firms such as Apple and Google have begun to implement policies that prevent certain types of targeting.

10.7 Describe social networks, social marketing, and social e-commerce.

One of the most significant business impacts of the Internet and e-commerce has been in the development of online social networks, which are now being leveraged for both marketing and e-commerce purposes. Online social networks use websites and apps that enable users to communicate with one another, form group and individual relationships, and share interests, values, and ideas. Social networks provide one of the fastest-growing platforms for marketing and advertising. One of most significant benefits of social media is the ability for marketers to understand what people are saying about a brand as well as its competitors, sometimes referred to as social listening. Social e-commerce has taken off within the last few years and has been growing at more than 25 percent a year. All the major social networks now include social e-commerce functionality.

10.8 Understand how e-commerce has affected business-to-business transactions.

B2B e-commerce generates efficiencies by enabling companies to use the Internet and other digital tools to locate suppliers, solicit bids, place orders, and track shipments in transit. Private B2B networks link a firm with its suppliers and other strategic business partners to develop highly efficient and responsive supply chains. B2B e-commerce marketplaces provide a single, digital marketplace for many buyers and sellers. Private B2B networks link a firm with its suppliers and other strategic business partners to develop highly efficient and responsive supply chains.

10.9 Describe m-commerce and m-commerce applications.

M-commerce involves the sale of goods and services via mobile devices, such as smartphones, tablets, and wearables. M-commerce is especially well-suited for location-based services and apps. Mobile devices are being used for mobile bill payment, banking, securities trading, transportation schedule updates, and downloads of digital content such as music, games, and video clips. Mobile payment systems include technologies for near-field communication, QR codes, and peer-to-peer payments. The GPS capabilities of smartphones make geoadvertising, geosocial, and geoinformation services possible.

10.10 Discuss how to build an e-commerce presence.

Building a successful e-commerce presence requires a clear understanding of the business objectives to be achieved and selecting the right platforms, activities, and timeline to achieve those objectives. An e-commerce presence includes not only a corporate website but also a presence on Facebook, Instagram, TikTok, Pinterest, X (formerly Twitter), and other social networks.

Key Terms

Advertising revenue model, 338
Affiliate revenue model, 339
B2B e-commerce marketplace, 347
Behavioral targeting, 339
Business-to-business e-commerce (B2B e-commerce), 334
Business-to-consumer e-commerce (B2C e-commerce), 334
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Wisdom of crowds, 344

Review Questions

10.1 Discuss the evolution of e-commerce.

- Define e-commerce and discuss how it has evolved.
- Describe how social, mobile, and local e-commerce are connected.
- Name and describe four business trends and three technology trends shaping e-commerce today.
- Discuss how companies are using AI in e-commerce.

10.2 Describe the unique features of e-commerce.

- List and describe the eight unique features of e-commerce.

10.3 Describe digital markets and digital goods.

- Define information asymmetry and explain how it has been reduced by the Internet.
- Define a digital market, describe its distinguishing features, and explain the benefits it provides.
- Define digital goods and explain how the Internet changes the market for digital goods.

10.4 Describe types of e-commerce and e-commerce business models.

- Describe the principal types of e-commerce.
- Describe the principal e-commerce business models.

10.5 Describe e-commerce revenue models.

- Describe the principal e-commerce revenue models and give an example of each.

10.6 Explain how e-commerce has transformed marketing.

- Describe the leading marketing and advertising formats used in e-commerce.
- Define behavioral targeting and explain how it works at individual websites and on advertising networks.

10.7 Describe social networks, social marketing, and social e-commerce.

- Describe the primary features of social networks.
- Describe the wisdom of the crowds concept and discuss how it impacts social marketing.
- Explain how the concept of the social graph impacts social e-commerce.

10.8 Understand how e-commerce has affected business-to-business transactions.

- Describe electronic data interchange (EDI) and discuss how it is used for e-commerce.
- Define and describe B2B private networks and explain how they differ from B2B e-commerce marketplaces.

10.9 Describe m-commerce and m-commerce applications.

- Describe important types of location-based services and applications.
- Describe three types of mobile app payment systems.

10.10 Discuss how to build an e-commerce presence.

- Describe four types of e-commerce presence and the platforms and activities associated with each.

Discussion Questions

10.11 How does the Internet change consumer and supplier relationships?

10.12 Marketers defend the use of behavioral targeting as a method to expose people to advertising that is uniquely suited to them. Do you agree? Why or why not?

10.13 How is AI changing e-commerce?

Using AI Project

10.14 AI Project: Using AI to Create an E-commerce Presence

You have been hired as a consultant by two friends who want to open a small retail store selling organic pet food and treats, as well as pet toys and clothing a retail made from organic materials. They have asked you to assist them in helping to establish an e-commerce presence for their store on Facebook and Instagram.

Using Microsoft Copilot, Google's Gemini, or ChatGPT, develop a series of prompts to help you create prototypes for a Facebook page, Facebook banner image, and Instagram avatar. Save a copy of your prompts and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience developing e-commerce strategies for businesses, analyzing the profitability of an e-commerce company using spreadsheet software, and using online research tools to evaluate e-commerce hosting services.

10.15 Management Decision Problem

T.J. Maxx, the giant off-price clothing retailer with nearly 1,300 stores around the world, has experienced problems implementing an e-commerce site for online sales. An off-price store like T.J. Maxx buys excess inventory and last year's fashions from a vast network of department stores and manufacturers. However, it buys these items in much smaller lots than traditional retailers such as Nordstrom or Walmart do, with much of its inventory consisting of one-time items in small quantities. T.J. Maxx has new brand-name and designer fashions arriving every week. The inventory varies a great deal from one store to the next, and you never know what you'll find when you visit a T.J. Maxx store. Shoppers are lured into the stores in the hope that they might find a really hot bargain that might be available for only a few days, and that is part of T.J. Maxx's special appeal. An online storefront can also cannibalize in-store sales. On the other hand, ignoring e-commerce can mean losing market share to competitors. Define the problem for T.J. Maxx and suggest a solution. What management, organization, and technology factors should be considered?

10.16 Using a Spreadsheet to Analyze an E-commerce Business

Software skills: Spreadsheet downloading, formatting, and formulas; online research tools; presentation tools

Business skills: Financial statement analysis

Pick an e-commerce company that is publicly traded in the United States—for example, Wayfair, Carvana, or Etsy. Visit the company's website and review the web pages that describe the company and explain its purpose and structure. Use the web to find articles that comment on the company. Then visit the Securities and Exchange Commission's website at Sec.gov to access the company's 10-K (annual report) form showing income statements and balance sheets. Select only the sections of the 10-K form containing the desired portions of financial statements that you need to examine and download them into your spreadsheet. Create simplified spreadsheets of the company's balance sheets and income statements for the past three years.

- Is the company a success, borderline business, or failure? What information provides the basis of your decision? Why? When answering these questions, pay special attention to the company's three-year trends in revenues, costs of sales, gross margins, operating expenses, and net margins.
- Prepare a presentation (with a minimum of five slides), including appropriate spreadsheets or charts, and present your work to your professor and classmates.

10.17 Evaluating E-commerce Hosting Services

Software skills: Online research tools

Business skills: Evaluating e-commerce hosting services

This project will help develop your online research skills in evaluating commercial services for hosting an e-commerce site for a small startup company.

- You would like to set up a website to sell towels, linens, pottery, and tableware from Portugal and are examining services for hosting small-business website storefronts. Your website should be able to take secure credit card payments and calculate shipping costs and taxes. Initially, you would like to display photos and descriptions of 40 products. Visit Wix, GoDaddy, and Squarespace, and compare the range of e-commerce hosting services they offer to small businesses, their capabilities, and their costs. Examine the tools they provide for creating an e-commerce site. Compare these services and decide which you would use if you were actually establishing an online store. Write a brief report indicating your choice and explaining the strengths and weaknesses of each service.

10.18 Collaboration and Teamwork Project: Performing a Competitive Analysis

Form a team with two or more of your classmates. Select two businesses that are competitors in the same industry and that use their websites for e-commerce. Visit these websites. You might compare, for example, the websites for Pandora and Spotify, PetSmart and Chewy, or E*Trade and Charles Schwab. Prepare an evaluation of each business's website in terms of its functions, user-friendliness, and ability to support the company's business strategy. Which website does a better job? Why? Can you make some recommendations to improve these websites? If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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11 Artificial Intelligence

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Learning Objectives

After completing this chapter, you will be able to:

11.1 Understand artificial intelligence.

11.2 Describe machine learning.

11.3 Describe neural networks and deep learning.

11.4 Discuss generative AI.

11.5 Discuss intelligent agents.

11.6 Understand natural language processing.

11.7 Understand computer vision systems.

11.8 Describe robotics.

11.9 Discuss genetic algorithms.

11.10 Discuss expert systems.

11.11 Describe fuzzy logic systems.

11.12 Understand the challenges of AI.

11.13 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** Computer Vision AI Helps Reddit Become More Accessible
- **Spotlight on Management Case Study:** Apple's AI Strategy: AI in Every Pocket
- **Chapter Case Study:** Is AI Coming for Your Job?

Opening Case Study

Computer Vision AI Helps Reddit Become More Accessible

Reddit is an online "community of communities" where users share news stories, photos, and other content. The content is socially curated and promoted by Reddit members through voting. Reddit is known for its vast array of communities, called "Subreddits," dedicated to virtually every topic imaginable: cars, croquet, or Beyoncé's favorite song. Reddit has hundreds of millions of visitors each month, and engagement is very high. It's an incredible source of information on pretty much every field of human knowledge and has the personal feel of a community.

There may be a Subreddit for everyone, but not everyone can use Reddit. Most of Reddit's content is in the form of text, photos, images, or videos, putting it out of reach for people who have vision issues or use low-bandwidth Internet connections. This limits accessibility, as well as the number of people who can be Reddit users. To broaden accessibility and also drive more users to its platform, Reddit decided to look for a solution that would make it easier to discover and use its content.

With millions of existing images on the Reddit platform and hundreds of thousands more uploaded each day, Reddit wanted to find an artificial intelligence (AI) solution that could quickly produce high-quality captions for use as alternative text (alt text). Alt text verbally describes the appearance or function of an image on a web page. Alt text is read aloud by programs called screen readers, which are used by people with visual impairments, different learning abilities, or difficulties viewing images online. Alt text is indexed by search engines.

Reddit investigated several AI-based caption generation products, testing caption quality on many types of images, from pictures of animals to landscapes, people, and drawings. Reddit was looking for very detailed captions, so users with screen readers can understand the images more fully. Reddit also wanted a solution that could detect both objects and actions within an image, then describe both in written form.

Reddit selected Microsoft Azure AI Vision for its solution. Reddit was impressed by the accuracy and detail of the captions generated by Azure AI Vision. Higher caption accuracy creates a better user experience. Reddit found that the captions produced by Azure AI Vision contained an impressive range of vocabulary and had a level of detail that went beyond mere object detection. Reddit also found that Azure AI Vision was easy to use, processed images quickly, and offered ample technical support, including excellent documentation. Reddit was further impressed by Microsoft's commitment to responsible AI and the careful measures taken to ensure captions are accurate and in compliance.

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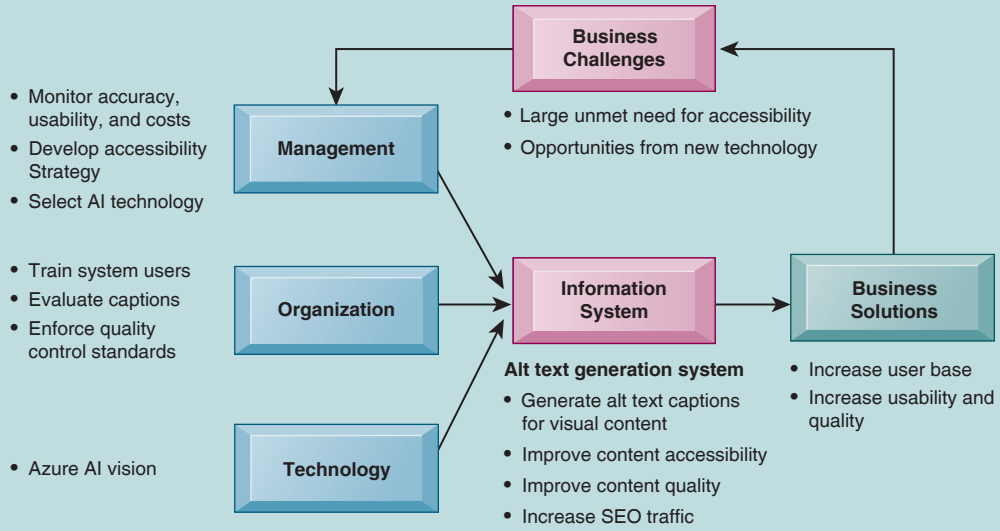
Azure AI Vision is a cloud-based unified product, and it provides developers access to advanced AI algorithms for processing images and returning information. It can analyze images, read text, and detect faces with prebuilt image tagging, text extraction with optical character recognition, and facial recognition. By uploading an image or specifying an image URL, Azure AI Vision can analyze visual content in different ways based on inputs and user choices. Software developers can incorporate vision features into their projects without needing any machine learning experience.

Since adding captions as alt text, Reddit has seen a significant upsurge in web traffic to their images. Adding alt text to images helps search engines understand them better and makes them more discoverable, driving more traffic to Reddit's platform. Reddit will continue to work with Microsoft to generate captions for new images uploaded to the platform. Accessibility has been good for business.

SOURCES: "Reddit Improves Accessibility and SEO Through Azure Cognitive Service for Vision Image and Caption Generation," Customers.microsoft.com, accessed September 17, 2024; "Azure, AI Vision," Azure.microsoft.com, accessed September 17, 2024; Reddit.com, accessed September 17, 2024; Emma Roth, "Google Cut a Deal with Reddit for AI Training Data," Theverge.com, February 22, 2024.

Reddit's experience using image processing to generate alt text captions for people with visual impairments or who lack access to high-bandwidth Internet services illustrates just some of the ways businesses and the public can benefit from AI technology. It is also a good example of how a company can create realistic, trustworthy AI applications and obtain value from them, both for the business and for its customers.

The Opening Case Study diagram calls attention to this case and this chapter. Reddit is already an extremely



popular platform. Making its content accessible to more people was both a way to grow the business and a way to open its knowledge platform to people unable to access it through conventional means. Reddit used Microsoft Azure Vision AI to generate detailed, meaningful alt text captions for images and other content. These improvements made the content on Reddit more attractive and understandable for search engines.

Case Study Questions

1. Why is Reddit concerned about accessibility?
2. What are the business and social benefits of using Microsoft Azure Vision AI to generate captions for images?

CAREER BENEFIT

Reddit's IT staff had the opportunity to develop expertise in AI image processing technology for alt text generation. This expertise helps them advance their careers at Reddit and also makes them more qualified for higher-paying jobs at other organizations. Non-IT staff benefited from contributing to the development of such an important solution for the organization, one that widened Reddit's user base while also making its knowledge, services, and sense of community accessible to an underserved population. This is a good example of how AI can benefit businesses as well as society at large.

11.1 Understand Artificial Intelligence

Artificial intelligence (AI) systems have made great strides in their ability to perform tasks normally requiring human intelligence, such as speech recognition, visual perception, recognizing patterns, making predictions, writing essays, and learning from past experience. AI systems are very much in demand right now as companies search for new solutions to help lower costs, innovate new products and services, and create value. AI is also advancing social well-being, providing some benefits for society at large. (See the Opening Case Study, *Computer Vision AI Helps Reddit Become More Accessible*.) AI is affecting so many aspects of business and everyday life that we provide AI coverage in every chapter of this book and also devote all of this chapter to this topic.

Artificial Intelligence and Human Intelligence

AI today enables machines to model, or even improve upon, some of the capabilities of the human mind, and it can often process information at a much faster rate than humans. However, AI cannot replicate everything humans do. AI systems still lack the adaptability, level of creativity, intuition, and emotional intelligence that humans possess. Human intelligence is vastly more complex than artificial intelligence.

AI and humans differ in how they learn. At present, AI systems rely on algorithms and training with large amounts of data to learn from their environment and improve their performance over time. AI is thus able to identify patterns and make predictions. In contrast, human intelligence is based on a complex interaction of genes, environment, and experience. Humans develop and refine their intelligence through exposure to different experiences, education, and social interactions.

AI and human intelligence also differ in their ability to adapt to new situations. AI systems are designed to perform specific tasks and are optimized for those tasks. Although they can learn to perform new tasks, they still require a significant amount of training to do so. Humans can quickly learn and adapt to new situations, even when they have no prior experience with them.

AI systems have made significant strides in natural language processing technology that gives computers the ability to interpret, manipulate, and comprehend human language. However, they still struggle to understand the nuances of human language and the context in which it is used. Humans can understand the meaning behind words and phrases, as well as the social and cultural context in which they are used.

AI also lags behind human intelligence in the ability to use creativity and intuition. AI systems can generate new ideas, solutions, and content based on their training data and algorithms. However, humans can come up with innovative ideas and solutions that are not based on any existing data or algorithms. They can use their intuition and experience to make connections and solve new problems beyond the reach of current AI systems.

AI also differs from humans in what is called "emotional intelligence." AI systems can recognize and respond to emotions in humans, but they do not possess emotions themselves. Humans exhibit a complex range of emotions that influence their decision-making and social interactions.

Evolution of AI

The major forces driving the rapid evolution of AI are the development of big data databases and the Internet of Things (IoT). AI technology has also benefited from the dramatic reduction in the cost of computer processing, the growing power of processors, and the affordability and availability of cloud computing services for processing AI applications. Additionally, the growth of AI has relied on the refinement of algorithms by tens of thousands of AI software engineers and university AI research centers, along with significant investment from businesses and governments.

Major Types of AI Technologies

AI systems can be classified into various categories based on their capabilities, such as machine learning, neural networks and deep learning, genetic algorithms, natural language processing, computer vision systems, robotics, intelligent agents, expert systems, and fuzzy logic. AI has numerous applications in various domains, such as healthcare, finance, education, transportation, entertainment, manufacturing, and retail. AI is rapidly evolving, with the potential to revolutionize the way we live and work.

Table 11.1 provides examples of the major branches of AI and some of their applications. You will learn more about each of these branches in the following sections.

11.2 Describe Machine Learning

Early AI systems were rule-based, using a set of predefined "if-then" rules to make decisions or find solutions to problems. Rule-based expert systems lacked the flexibility to learn, were expensive to maintain and update, relied on human input, and could only deal with a very narrow domain of human knowledge. Most AI development today involves some type of machine learning.

Table 11.1 Major Branches of AI

Branch of AI	Description	Example
Machine learning	Software that can identify patterns in very large datasets without explicit programming, although with significant human training.	Uber uses machine learning for real-time demand forecasting, route optimization, and fraud detection.
Neural networks and deep learning	Loosely based on human neurons, algorithms that can be trained to classify objects into known categories based on data inputs. Deep learning uses multiple layers of neural networks to reveal the underlying patterns in data and, in some limited cases, identify patterns without human training (unsupervised learning).	Generative AI tools such as ChatGPT, Google Gemini, and Microsoft Copilot are based on large language models that use deep learning neural networks. Finprophet provides forecasts for the most popular investment instruments, including stocks, futures, currencies, bonds, and more, for the 24 largest international financial markets.
Genetic algorithms	Algorithms based loosely on evolutionary natural selection and mutation, commonly used to generate high-quality solutions to optimization and search problems.	Boeing used genetic algorithms for wing design optimization to explore various wing shapes, sizes, and configurations. This approach helped Boeing improve aerodynamic efficiency, reduce weight, and enhance fuel efficiency in their aircraft designs.
Natural language processing	Algorithms that make it possible for a computer to understand and analyze natural human language. Often used for social media, sentiment analysis and responding to customer service requests.	Gmail uses natural language processing to identify and evaluate the content within each email, so that it can categorize them accurately.
Computer vision systems	Systems that can view and extract information from real-world images.	Automotive manufacturer Audi uses computer vision systems that can detect miniscule cracks in sheet metal parts, automatically, reliably, and in seconds, minimizing cost and operation efforts.
Robotics	Use of machines that can substitute for human movements as well as computer systems for their control and information processing.	Using computer vision technology and machine learning, Amazon's industrial robots can operate safely alongside humans. Amazon's new Containerized Storage approach to inventory management uses robots to deliver products to employees in a more ergonomically friendly manner.
Intelligent agents	Software agents that use built-in or learned knowledge to perform specific tasks or services.	Siri, Alexa and Google Assistant are examples of conversational AI agents. They use natural language processing and machine learning to understand and formulate a response to questions asked by humans.
Expert systems	Represent the knowledge of experts as a set of rules that can be programmed so that a computer can assist human decision makers. Difficult to build and update.	Purely rules-based expert systems have largely been supplanted by systems that combine some features of expert systems with more advanced technologies such as machine learning.
Fuzzy logic	Form of AI that imitates human reasoning and cognition by representing imprecision in the form of rules that use approximate or subjective values rather than just 0 or 1.	Models complex decision-making processes that involve imprecise or uncertain data. Often used in control systems that regulate the behavior of other systems or devices, such as for controlling speed in automotive systems or controlling washing machine speed and temperature.

What Is Machine Learning?

Machine learning (**ML**) uses mathematical models that help a computer learn from large quantities of data and make statistical inferences without much explicit instruction. This enables a computer system to continue learning and improving on its own, to varying degrees, based on experience. ML systems are trained on large sets of data, which enables them to identify patterns and relationships. The more data an ML algorithm is trained on, the better it will be able to perform. Machine learning is especially useful for finding patterns in very large amounts of data and predicting outcomes.

ML applications have proliferated, providing benefits to organizations of all sizes. ML can analyze very large quantities of data—often big data that would be impossible for a person to digest—much more rapidly than a human, and often more accurately than humans. Table 11.2 provides examples of ML applications.

Machine learning (ML)

Uses mathematical models that help a computer to learn from large quantities of data and make statistical inferences without much explicit instruction, enabling a computer system to continue learning and improving on its own, to varying degrees, based on experience.

Table 11.2 Examples of ML Applications

Application	Description	Example
Natural language processing	Understanding and responding to human language	People can get personalized fashion advice tailored to their individual style preferences by conversing with a Stitch Fix chatbot.
Image recognition	Software that can identify objects, places, people, writing and actions in digital images.	Pinterest is an e-commerce visual discovery platform that allows users to visually share and discover new interests. It uses image recognition software for visual search. Users can choose specific items within a photo to find similar items. For example, selecting a throw pillow on a bed will return images of more throw pillows.
Predictive analytics	Predicting future events or trends based on historical data	Alibaba, the world's largest e-commerce platform, uses AI to predict what customers might want to buy.
Recommendation system	Recommending products or services to users based on their past behavior	Used by Amazon, Netflix, Spotify, Pandora, Apple Music, and many other companies to create personalized recommendations of products and services based on the preferences and choices of their customers.

Supervised and Unsupervised Learning

Supervised learning

Machine learning algorithm trained by providing specific examples of desired inputs and outputs classified by humans in advance.

Most ML applications today are based on **supervised learning**, in which the system is "trained" by receiving specific examples of desired inputs and outputs identified by humans in advance. For example, an algorithm might be trained on images of cats, with the objective of being able to correctly identify new cat images. Over time, with tweaking by programmers and by using ever-larger computing systems and sets of data, the system will improve its performance and, in that sense, will learn. Supervised learning is one technique used to develop autonomous vehicles that need to be able to recognize objects around them, such as people, other cars, buildings, and lines on the pavement to guide them.

Unsupervised learning

Machine learning algorithm trained to use information that is neither classified nor labeled in advance and to find patterns in that information without explicit human guidance.

In **unsupervised learning**, the algorithm is trained on large data sets without known outcomes. The algorithm is left to identify patterns and relationships on its own and then report whatever patterns it finds. For example, suppose an unsupervised learning algorithm is given an input data set containing images of cats and other animals. The algorithm is never trained upon the given data set, so it has no idea about the features of the data set. The task of the unsupervised learning algorithm is to identify the cat image features on its own. The unsupervised learning algorithm will perform this task by clustering the image data set into groups according to similarities among images. Table 11.3 provides examples of unsupervised learning applications.

Table 11.3 Examples of Unsupervised Learning Applications

Application	Description	Example
Customer segmentation	AI segmentation tools group existing and potential customers based on shared characteristics and provide information for creating detailed customer profiles. This makes it possible to identify the right target audience and high-value customers.	Amazon uses unsupervised learning to segment customers based on their behavior and preferences to provide personalized product recommendations.
Medical imaging	Capabilities for medical devices, such as image detection, classification, and segmentation. Used in radiology and pathology to detect patterns quickly and accurately.	AltexSoft technology consultants created a prototype tool for chest X-ray analysis. The tool used machine learning to identify the boundaries of lungs and for lung disease classification.
Anomaly detection	Unsupervised learning can sift through large quantities of data and discover atypical data points within a dataset. These anomalies can call attention to abnormalities such as machine failures, product defects, or fraudulent credit card transactions.	JPMorgan Chase uses anomaly detection to track how customers typically use their credit cards. If a customer makes an abnormally large purchase or a purchase in a new location, the algorithm recognizes the anomaly and alerts a team member to contact the customer.
News aggregation	Gathering news information from various sources as it evolves on the web, analyzing it in real time, and organizing it into storylines.	Google News uses unsupervised learning to cluster articles on the same topic, such as World Cup soccer, from various online news outlets.

11.3 Describe Neural Networks and Deep Learning

A **neural network**, including those with deep learning, extends the powers of AI by modeling learning in a way that approximates the way the human brain works. Neural networks excel at pattern recognition and are powering some of the most advanced AI applications, such as self-driving cars, speech and image recognition, and healthcare, where they help diagnose diseases and predict patient outcomes.

Neural network
Algorithms loosely based on the processing patterns of the biological human brain that can be trained to classify objects into known categories based on data inputs; are able to find patterns and relationships in very large amounts of data that would be too complex and difficult for a human to analyze.

What Are Neural Networks

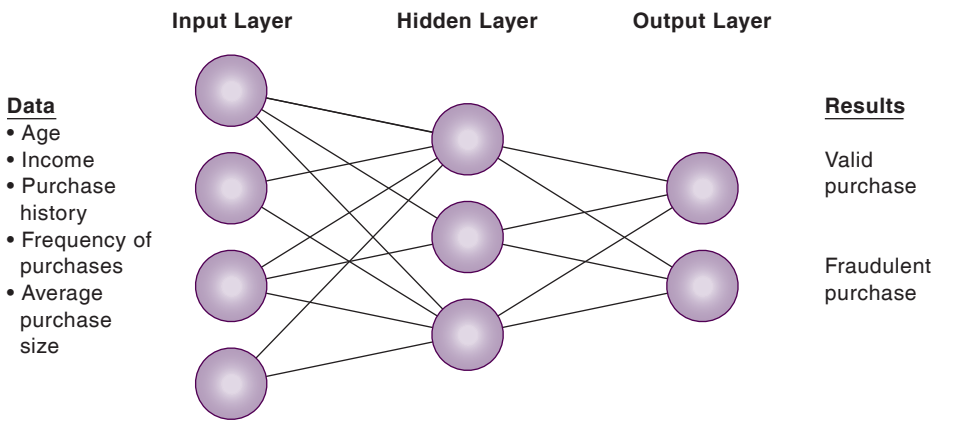
By using ML algorithms and computational models that are loosely based on how the biological human brain is thought to operate, neural networks find patterns and relationships in very large amounts of data that would be too complex and difficult for a human to analyze. A neural network is composed of interconnected units called neurons. Each neuron can take data from other neurons and transfer data to other neurons in the system. The artificial neurons are not biological, physical entities as in the human brain but instead are software programs and mathematical models that perform the input and output functions of neurons. The strength of the connections (the weight) can be controlled by researchers using a Learning Rule, an algorithm that systematically alters the strength of the connections among the neurons to produce the final, desired output, which could be identifying a picture of a cancerous tumor, fraudulent credit card transactions, or suspicious telephone calling patterns.

Neural networks are pattern detection programs that learn patterns from large quantities of data by sifting through the data and ultimately finding pathways through the network of thousands of neurons. Some pathways are more successful than others in their ability to identify objects like cars, animals, faces, and voices. There may be millions of pathways through the data. The Learning Rule algorithm identifies these successful paths and strengthens the connections among neurons in these pathways. This process is repeated thousands or millions of times until only the most successful pathways are identified. The Learning Rule identifies the best or optimal pathways through the data. At some point, after millions of pathways are analyzed, the process stops when an acceptable level of pattern recognition is reached: for instance, successfully identifying cancerous tumors about as well as humans, or even better than humans.

Examine Figure 11.1 which represents one type of neural network consisting of an input layer, a "hidden" processing layer, and an output layer. Humans train the network by inputting a set of outcomes they want the machine to learn. For instance, if the objective is to build a system that can identify patterns in fraudulent credit card purchases, the system is trained

Figure 11.1 How a Neural Network Works

A neural network "learns" from patterns in data to construct a hidden layer of logic. The hidden layer then processes inputs, classifying them based on the experience of the model. In this example, the neural network has been trained to distinguish between valid and fraudulent credit card purchases.



using actual examples of fraudulent transactions. The data set may be composed of a million examples of fraudulent transactions. The data set is then divided into two segments: a training data set and a test data set. The training data set is used to train the system. After millions of test runs, the program hopefully will identify the best path through the data. To verify the accuracy of the system, it is then used on the test data set, which the system has not analyzed before. If successful, the system will be tested on new data sets. The neural network in Figure 11.1 has learned how to identify a likely fraudulent credit card purchase.

Neural network applications in medicine, science, and business address problems in pattern classification, prediction probabilities, and control and optimization. In medicine, neural network applications are used for screening patients for coronary artery disease, for diagnosing epilepsy and Alzheimer's disease, and for performing pattern recognition of pathology images, including those of certain cancers. The financial industry uses neural networks to discern patterns in vast pools of data that might help investment firms predict the performance of equities, corporate bond ratings, or corporate bankruptcies. Visa International uses a neural network to help detect credit card fraud by monitoring all Visa transactions for sudden changes in the buying patterns of cardholders.

Deep Learning Neural Networks

Deep learning neural network

Uses multiple layers of neural networks to reveal the underlying patterns in data, and in some limited cases identify patterns without human training.

A **deep learning neural network** is more complex, with many layers of transformation of the input data to produce a target output. These extra layers allow a deep learning neural network to learn more complex patterns in data. They are often used for pattern detection on unlabeled data where the system is not told what to look for specifically but to simply discover patterns in the data. The system is expected to be self-taught.

For instance, an unsupervised ML system that could identify cats and other objects without training might use a deep learning neural network. Examine Figure 11.2, which illustrates this system. It consists of three layers of neural networks (layers 1, 2, and 3). Each of these layers has two levels of pattern detection (levels 1 and 2). Each level was developed to identify a low-level feature of photos: Layer 1 identified lines in the photos, and layer 2 identified circles. The result of the first layer may be blobs and fuzzy edges. Second and third layers refine the images emerging from the first layer, until at the end of the process the system can distinguish cats, dogs, and humans.

Figure 11.2 A Deep Learning Neural Network

A deep learning neural network consists of many layers of neural networks working in a hierarchical fashion to detect patterns. Shown here is an expanded look at layer 1. Other layers have the same structure.

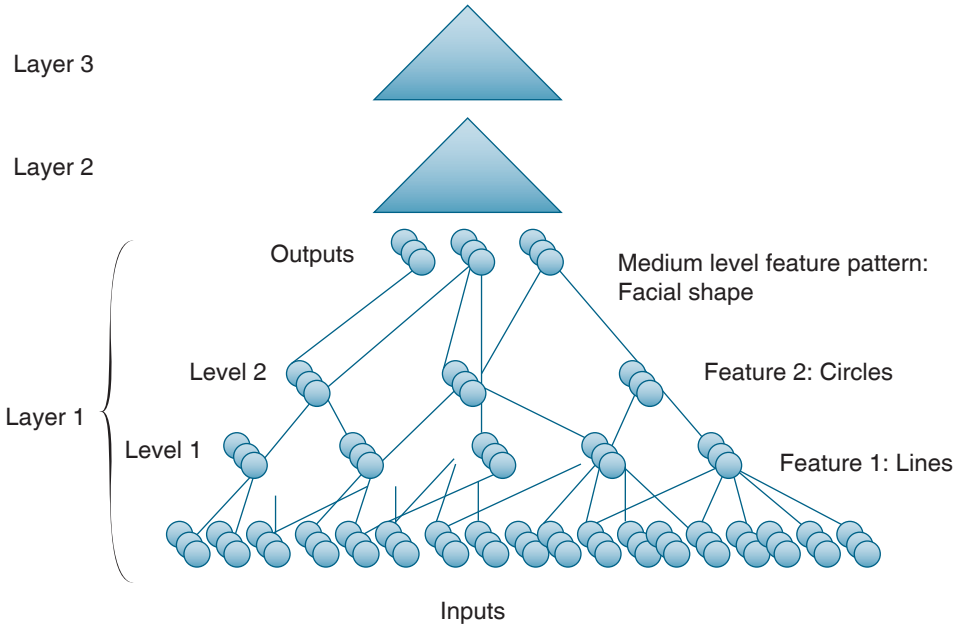


Table 11.4 Examples of Neural Networks and Deep Learning Applications

Functionality	Inputs	Process	Outputs/Application
Computer vision	Millions of digital images, videos, or sensors	Recognize patterns in images, and objects	Photo tagging; facial recognition; autonomous vehicles
Speech recognition	Digital soundtracks, voices	Recognize patterns and meaning in soundtracks and speech	Digital assistants; chatbots; help centers
Machine controls, diagnostics	Internet of Things: thousands of sensors	Identify operational status, patterns of failure	Preventive maintenance; quality control
Language translation	Millions of sentences in various languages	Identify patterns in multiple languages	Translate sentences from one language to another
Transaction analysis	Millions of loan applications, stock trades, phone calls	Identify patterns in financial and other transactions	Fraud control; theft of services; stock market predictions
Targeted online ads	Millions of browser histories	Identify clusters of consumers; preferences	Programmatic advertising

Deep learning algorithms are powering many GenAI models. The deep learning models learn the patterns, structures, and characteristics of the input data, and the generative AI model uses this information to generate new content. Deep learning is also used in applications for computer vision, natural language processing, self-driving vehicles, facial recognition, recommendation systems, and healthcare (medical imaging, drug discovery, clinical trial optimization).

Examples of Neural Networks and Deep Learning Applications

Table 11.4 provides examples of neural network and deep learning applications.

11.4 Discuss Generative AI

Generative AI (GenAI) is a machine learning model that is trained to generate content. GenAI tools can identify patterns and structures within existing data to produce new and original content, including textual responses to questions; articles, essays, fiction, and poetry; images; new software programs; musical compositions; and successful answers to test questions.

Generative AI (GenAI)
Machine learning model that is trained to generate content.

Conventional machine learning is limited to a narrow range of tasks, such as pattern recognition or making predictions. However, GenAI is actually able to perform a wide range of tasks and mimic some features of human creativity. Some believe it's a game-changer that will transform the way we create content and many other aspects of our jobs and daily lives.

GenAI starts with a prompt that could be in the form of text, an image, a video, a design, musical notes, or any input that the AI system can process. Various AI algorithms then return new content in response to the prompt.

Foundation Models, Large Language Models, and Transformers

A **foundation model** is the underlying model that enables GenAI to work. Foundation models use powerful neural networks designed to mimic how the human brain works and "learn" by finding patterns in existing data sets. Foundation models are trained on vast quantities of unstructured, unlabeled data in a variety of formats, such as text and audio, and are able to perform a wide range of tasks.

Foundation model
Underlying model that enables GenAI to work; uses a powerful neural network that is trained on vast quantities of unstructured, unlabeled data, and learns by finding patterns in data sets.

Foundation models develop these capabilities by learning patterns and relationships from the broad training data they ingest, which, for example, enables them to predict the next word in a sentence. Given the versatility of a foundation model, companies can use the same one for multiple activities and forms of content.

Large Language Models

Large language model (LLM)
Foundation model trained on immense amounts of unstructured data covering many topics; capable of understanding and generating natural language and other types of content.

A **large language model** (**LLM**) is a foundation model trained on immense amounts of unstructured data covering many topics, making them capable of understanding and generating natural language and other types of content for many language-related tasks. LLMs are able to infer from context, generate coherent and contextually relevant responses, translate languages, summarize text, answer questions, and even assist in creative writing or software coding. LLMs are revolutionizing applications in various fields, including "intelligent" chatbots and virtual assistants, content generation, research assistance, and language translation. As LLMs continue to evolve and improve, they will reshape the way we interact with technology and access information.

Transformers

Transformer
Key component of foundation models; type of neural network architecture that transforms or changes an input sequence into an output sequence by learning context and tracking relationships between sequence components.

A **transformer** is a key component of foundation models—for instance, GPT in ChatGPT stands for generative pre-trained transformer. Transformers are a type of neural network architecture that transforms or changes an input sequence into an output sequence. They do this by learning context and tracking relationships between sequence components. For example, for the input sequence: "What is the color of snow?," the transformer model uses an internal mathematical representation that identifies the relevancy and relationship between the words color, snow, and white. It uses that knowledge to generate the output: "Snow is white."

Transformers have made it possible to train increasingly larger foundation models without having to label all of the data in advance. New models can be trained on billions of pages of text, producing answers with more depth. Transformers include a capability called *attention* that enables models to track the connections between words across pages, chapters, and books rather than just in individual sentences. This capability is not limited to words. Transformers can also track connections to do things like analyze software code or write a play. With transformers, you can use AI for tasks that combine complex data sets. For instance, models like DALL-E show that transformers can generate images from textual descriptions, combining natural language processing and computer vision capabilities. With transformers, you can create AI applications that integrate different information types and mimic human understanding and creativity more closely.

Innovations in what has become known as multimodal AI make it possible to generate content across multiple types of media, including text, graphics, and video. AI tools such as DALL-E use this capability to automatically create images from a text description or generate text captions from images.

Given the breadth of knowledge and versatility exhibited by GenAI, it can appear amazingly "intelligent." However, it is not always accurate. ML systems are prone to "hallucination," meaning that the system can generate answers that are misleading, nonsensical, or factually incorrect. GenAI chatbots have no morals and can write essays and songs about things that humans might find distasteful. The technology doesn't understand anything it's saying—it's just creating new patterns based on human language. There is a difference between piecing together words and piecing together ideas.

Examples of GenAI Tools

Table 11.5 lists and describes some of the leading GenAI tools being used.

11.5 Discuss Intelligent Agents

Intelligent agent
Software program that uses a built-in or learned knowledge base to carry out specific tasks for an individual user, business process, or software application.

An **intelligent agent** is a software program that works in the background without direct human intervention to carry out specific tasks for an individual user, business process, or software application. The agent uses built-in or learned knowledge to accomplish tasks or make decisions on the user's behalf, such as deleting junk email, scheduling appointments, or finding the cheapest airfare to London. There are many intelligent agent applications today in operating systems, application software, email systems, mobile computing software, and network tools, with varying degrees of "intelligence." Intelligent agents include

Table 11.5 GenAI Tools

GenAI Tool	Description
ChatGPT	ChatGPT, first released to the general public in November 2022, is a GenAI chatbot created by OpenAI. ChatGPT can respond in multiple languages to complex questions and present comprehensive, essay-length answers on virtually any topic. ChatGPT made history by calling public attention to the potential of GenAI to perform a wide range of tasks and use natural language to communicate with users. ChatGPT showed that AI was capable of becoming more genuinely useful and more "human." ChatGPT can produce human-like textual responses to questions; summarize long texts; write articles, essays, fiction, and poetry; write and debug software; compose music; and successfully answer test questions. ChatGPT shares some of the limitations of GenAI and is not considered as up to date as Google Gemini and Microsoft Copilot. In September 2024, OpenAI released a new LLM, called OpenAI o1, which OpenAI says is able to reason through math and science problems much as a human does.
Microsoft Copilot	Microsoft Copilot, formerly Microsoft Bing Chat, is an AI chatbot that can answer questions and complete tasks using GenAI. Copilot users can engage in real-time conversations with Copilot and receive AI-generated responses that are not limited to text-based results; Copilot can also access and summarize information from the Internet using real-time data indexed by Bing, making it very up to date. Copilot can create content, including suggesting new ideas for new content, writing introductions, suggesting titles, and rephrasing paragraphs to match specific conversational styles, such as creative, balanced, or precise. Copilot is integrated with OpenAI's image generator DALL-E 3, which allows Copilot users to generate images based on their prompts and to create custom visuals for their content. Copilot includes citations, allowing users to see where its answers have been sourced from. These responses are not limited to text-based results; Copilot can also summarize information from the Internet, making it a versatile tool for answering various types of queries. In September 2024, Microsoft added Copilot Voice, enabling users to interact with Copilot using natural speech.
Google Gemini	Google Gemini is a family of large language models (LLMs) trained on massive amounts of data to understand and generate human language. They can create text, translate languages, write different kinds of creative content, and provide informed answers to questions based on current information. Gemini is multimodal, and can seamlessly understand, operate across, and combine different types of information, including text, images, video, audio, and software code. Gemini comes in three different versions, making it suitable for mobile devices as well as enterprise data centers. The Gemini chatbot replaced Google's Bard and is the primary way for users to interact with the capabilities of the Gemini large language models.
DALL-E	DALL-E is a GenAI tool developed by OpenAI that generates images from textual descriptions using natural language processing, machine learning, and computer vision technologies. Users can provide text prompts consisting of short phrases or detailed descriptions that DALL-E will understand and use to create images representing the prompt. DALL-E also can determine whether the final picture is accurate. DALL-E was first released in 2021. In 2022, OpenAI released a more powerful version, DALL-E 2. DALL-E 2 enables users to generate imagery in multiple styles driven by user prompts. The most current version of DALL-E as of September 2024 is DALL-E 3. Compared to previous versions, the images that DALL-E 3 can create feature greater detail with sharper lighting, textures, and more detailed backgrounds. DALL-E is an example of a multimodal AI application that identifies connections across multiple media, such as images, text, and audio. DALL-E has many uses, with significant potential in creative industries, such as graphic design, advertising, and gaming, where it can expedite the ideation process and reduce manual effort.

smart thermostats and other control systems, driverless cars, virtual assistants, chatbots, and spam filters. Intelligent agent bots can search for information on the Internet and make purchase recommendations.

Chatbots

A **chatbot** is a software agent designed to simulate a conversation with one or more human users via textual or auditory methods. Chatbots try to understand what you type or say and respond by answering questions or executing tasks. They provide automated conversations that allow users to do things like check the weather, manage personal finances, shop online, and receive help when they have questions for customer service. Chatbots are widely used for customer support, where they can handle many questions typically posed to customer service representatives, 24/7, thus reducing labor costs and waiting times to receive help. This facilitates better customer engagement and satisfaction. Chatbots also assist with personalization, information gathering, and user website navigation.

Chatbot

Software agent designed to simulate a conversation with one or more human users via textual or auditory methods.

For example, Pizza Hut's chatbot helps customers order food through Facebook Messenger and X (formerly Twitter). The chatbot uses natural language processing to understand the customer's order and provide real-time updates on the order status. The chatbot also allows customers to track and change their orders.

Chatbots can be either rule-based or AI-powered. Rule-based chatbots rely on predefined rules and scripted responses to manage user interactions. When a bot detects user input it analyzes it for certain keywords or phrases and then refers to its database for an appropriate response. For example, when you ask, "Check my credit card balance," a rule-based chatbot would use the words "account balance" and "credit card" to retrieve a preprogrammed response. Rule-based chatbots are quite inflexible and unable to handle complex scenarios and many different types of problems. More up-to-date chatbots are much more powerful and sophisticated, based on more leading-edge AI technology.

AI-powered chatbots use AI technologies like ML and natural language processing to deliver more conversational experiences. Such chatbots are trained on large datasets, enabling them to handle more complex user interactions and a much wider range of scenarios. AI-powered chatbots can analyze various situations. They also keep track of context, creating more natural and precise conversational experiences. An example of a prompt for an AI-powered chatbot is, "When will I receive my package if I order today?"

Many chatbots are now using GenAI technology. For example, the online one-stop travel platform Expedia integrated ChatGPT into its mobile app to make it easier for travelers to plan a trip. The app's new conversational trip planning feature lets travelers chat with a GenAI bot to get recommendations, explore activities, and find travel assistance, with fast, detailed answers and 24/7 availability. Such chatbots are capable of some degree of problem-solving and can generate original content. The generative agent understands the context, runs an analysis, and provides a detailed human language response.

Virtual Assistants

Virtual assistant

Software agent that can perform a range of tasks or services for a user based on user input such as commands or questions.

A **virtual assistant** is a software agent that can perform a range of tasks or services for a user based on user input such as commands or questions. Such technologies often incorporate online chat capabilities to facilitate interaction with users. The interaction may be via text, graphical interface, or voice. Some virtual assistants can interpret human speech and respond via synthesized voices.

In many cases virtual assistants can respond to users' questions, control home automation devices and media playback, and manage other basic tasks such as email, to-do lists, and calendars—all with verbal commands. Amazon Alexa, Apple's Siri, Google Assistant, and Samsung's Bixby are examples of widely used virtual assistants for direct consumer support.

Although both virtual assistants and chatbots use conversational AI interfaces, they have different capabilities and applications. Virtual assistants are better suited for more complex interactions than chatbots and are optimized to deal with a wider variety of scenarios. Virtual assistants often use more advanced technology and can understand and execute a wider range of commands. They are designed for completing specific tasks and engaging in relatively short conversations with users. Chatbots are designed to have extended conversations rather than focusing on a particular task.

Virtual assistants are also acquiring GenAI capabilities. Apple is updating its product line to incorporate GenAI and make Siri more conversational, versatile, and capable of understanding context (see the Spotlight on Management Case Study *Apple's AI Strategy: AI in Every Pocket*). Siri's underlying technology will include a new GenAI system that will allow it to chat rather than respond to questions one at a time. The focus will be on making Siri better at handling tasks that it already does, including setting timers, creating calendar appointments, and adding items to a grocery list. It will also be able to summarize text messages.

Bots

Bot

Software application that performs repetitive pre-defined tasks over a network. Includes shopping bots, web crawlers, scrapers, monitoring bots, and malware.

A **bot** is a software application that performs repetitive pre-defined tasks over a network. The bot can run independently without human intervention. For example, bots can interact with websites, chat with site visitors, or scan through content. Although most bots are useful, some are used for malicious purposes (see Chapter 8). Table 11.6 provides examples of the different kinds of bots.

Table 11.6 Types of Bots

Type of Bot	Description
Shopping bot	Virtual assistants that help shoppers during the buying and checkout process. Can also search the web for products and make price comparisons.
Web crawler	Search engine bot that scans and indexes web pages on the Internet; extracting data to understand the structure and relevance of web content.
Scraper	Web scraping crawlers, scan and download specific content on the internet. E-commerce businesses use scrapers to monitor live product prices on different retail platforms. Marketers use scrapers with natural language capabilities to run sentiment analysis on social media feeds
Monitoring bot	Constantly scanning systems for bugs and malicious software. They alert users to unusual web activity by collecting and analyzing user interaction data and web traffic.
Malware	Malicious bot that that can disable a system and steal data, or perform other harmful activities.

11.6 Understand Natural Language Processing

Human language is not always precise. It is often ambiguous, and the meanings of words can depend on complex variables such as slang, regional dialects, and social context. **Natural language processing** (**NLP**) makes it possible for a computer to analyze natural language—language that humans instinctively use, not language specially formatted to be understood by computers. NLP applies both to written text and speech and can be used with all human languages.

Natural language processing (NLP)

AI technique for enabling a computer to understand and analyze natural language as opposed to language formatted to be understood by computers.

NLP algorithms are typically based on machine learning, including deep learning, which can learn how to identify a speaker's intent from many examples. For example, Akershus University Hospital in Norway used NLP and IBM Watson Explorer to sift through thousands of medical records with unstructured textual data expressed in everyday language like natural speech. The algorithms could read text on a medical record and classify its meaning. Akershus staff used this information to verify whether they had made correct treatment decisions. You can also see natural language processing at work in leading search engines such as Google, voice assistants, language translation applications, spam filtering systems, and text-mining sentiment analysis discussed in Chapter 6.

Examples of NLP Applications

Earlier approaches to NLP were more rule-based. Simpler algorithms were told what words and phrases to look for in text and given prespecified responses when those phrases appeared. These simpler rule-based systems have largely been supplanted by ones based on deep learning algorithms that are more flexible, intuitive, and capable of identifying speakers' intent from many examples. Table 11.7 provides examples of some popular NLP applications.

NLP has made great strides and keeps improving every day but has not been perfected. Human speech is often ambiguous and affected by variables such as regional dialects, slang, and social context, which require a deeper understanding of the words being used and their context in a conversation. A sentence can change meaning depending on which word or syllable the speaker puts stress on. NLP algorithms can miss subtle but important tone changes in a person's voice when performing speech recognition. NLP is also challenged by the fact that language and the way people use it are continually changing, and keeping up with such changes is very difficult.

11.7 Understand Computer Vision Systems

A **computer vision system** uses machine learning and deep learning neural networks to emulate the human visual system to view and extract information from real-world images. These systems are able to understand and identify objects and people in images and videos, with input coming from sensing devices such as cameras and interpreting devices such as

Computer vision system

Uses machine learning and deep learning neural networks to emulate the human visual system to view and extract information from real-world images.

Table 11.7 Examples of NLP Applications

NLP Application	Description
Conversational agents	NLP is the core technology behind AI chatbots and virtual assistants, such as Siri or Alexa. When you pose questions to these assistants, NLP is what enables them to not only understand the user's request, but to also respond in natural language. Conversational AI tools are frequently used for customer service in companies such as H&M, Domino's Pizza, and Bank of America to help customers find answers to common questions faster, reducing the load on customer service representatives.
Sentiment analysis	Sentiment analysis enables businesses to analyze customer sentiment towards brands, products, and services using online conversations or direct feedback. It is often used for analyzing social media content and conversations.
Text classification	By using NLP, text classifiers can automatically analyze text and then assign a set of pre-defined tags or categories based on its content. Once an algorithm is sufficiently trained on email textual data, it can successfully identify, categorize and label emails as regular emails, spam emails or malicious emails.
Automatic foreign language translation of text or speech	Tools such as Google Translate, DeepL, and Translate Me can translate text, audio and documents into another language. Netflix uses Google Translate to tailor content and subtitles to local audiences.
Text summarization	NLP tools can automatically read textual content and generate a summary. Text summarization informs users without requiring them to read every single detail, thus improving user productivity. The summary can be a news headline or a paragraph or several paragraphs of text. News aggregator websites and apps such as Apple News use NLP for text summarization.
Natural language content generation	NLP enables computers to generate text or speech that is natural-sounding and realistic enough to substitute for human communication. The generated language might be used to create drafts of reports, computer code, letters, memos or tweets. This requires on advanced language models trained on vast amounts of diverse data. These models, such as OpenAI's GPT, can generate coherent and contextually relevant sentences or paragraphs.

computers to process visual data and extract meaning. Computer vision applications run on algorithms that are trained on massive amounts of visual data or images. They recognize patterns in this visual data and use those patterns to determine the content of other images.

Examples of Computer Vision Systems

Computer vision systems have a wide range of applications, including autonomous vehicles (drones and self-driving cars), industrial machine vision systems (e.g., inspecting bottles), military applications, and robotic tools. Table 11.8 provides examples.

Table 11.8 Computer Vision System Applications

Computer Vision Application	Example
Facial recognition	Facial recognition uses computer vision to analyze a face in an image or video and compare it to a database of known faces to identify a person. Verkada security cameras with facial recognition capabilities allow businesses to identify authorized personnel and potential security threats.
Self-driving cars	Helps self-driving cars navigate their surroundings: Tesla relies on computer vision for its self-driving car technology, using cameras to detect objects and navigate the road.
Manufacturing fault detection	Inspects products on assembly lines for defects: Siemens utilizes computer vision for automated quality control, with systems inspecting products for defects on assembly lines.
Medical imaging	Analyze medical images, such as X-rays and MRIs, to help diagnose diseases: Stanford ML Group's CheXNet can detect pneumonia from chest X-rays with an accuracy exceeding that of practicing radiologists
Sports performance analysis	Identify diverse poses and actions of players to produce detailed position and action tracking models. SportVU 2.0 uses cameras and computer vision technology to capture detailed movement of both the ball in soccer or basketball and the players. Besides improving performance, football teams can use computer vision technology to prevent injuries.
Precision agriculture	Drones equipped with computer vision technology can identify specific areas infested with pests and diseases, allowing for targeted pesticide application. Ensures that only the necessary amount of chemicals is used, minimizing environmental impact and reducing costs. Acme Agritech offers drone and data analysis services to farmers, including drones equipped with specialized spraying mechanisms. These drones collected data to pinpoint problem areas and deliver precise amounts of fertilizer, pesticides, or herbicides only where needed.

11.8 Describe Robotics

Robotics deals with the design, construction, operation, and use of movable machines that can substitute for humans, along with computer systems for their control, sensory feedback, and information processing. Robots cannot substitute entirely for people but are programmed to perform a specific series of actions automatically. They are often used in dangerous environments (such as bomb detection and deactivation), manufacturing processes, military operations (drones), and medical procedures (surgical robots).

Robotics
Involves the design, construction, operation, and use of movable machines that can substitute for humans, as well as computer systems for their control, sensory feedback, and information processing.

Robotics Applications

Substantial research is being devoted to using AI to expand robot functionality. Commercially available applications include the use of AI to enable robots to sense and respond to their environment. This vastly increases the range of functions robots can perform. Many employees now wonder whether robots and other forms of AI will replace people entirely and take away their jobs, as discussed in the Chapter Case Study *Is AI Coming for Your Job?*.

The most widespread use of robotic technology has been in manufacturing and logistics. For example, automobile assembly lines employ robots to do heavy lifting, welding, applying glue, and painting. People still do most of the final assembly of cars, especially when installing small parts or wiring that needs to be guided into place. Amazon, FedEx, and Walmart use robots to speed up package deliveries. Amazon deploys more than 750,000 robots to help with deliveries. The company has been setting up Amazon Robotic Fulfillment Centers strategically located across the United States and Europe. Although the Amazon Robotic Fulfillment Centers have human employees, they are in large part warehouses operated by different kinds of robots. The robots are assigned tasks such as heavy lifting, pulling carts of plastic totes from one side of a warehouse to another, and carrying inventory pods to picking stations. Human workers do the picking and perform other tasks that are too complex for the robots to handle.

Types of Robots

There are many different types of robots and ways of classifying them. We describe the most common types of robots you are likely to encounter in business settings.

Industrial Robots

An **industrial robot** is a heavy-duty machine designed to automate portions of the manufacturing process. They are typically used in factories to perform repetitive or dangerous tasks such as welding, painting, product assembly, picking, sorting, and packaging. Industrial robots are usually large, powerful, and very precise. These robots are built to either handle extremely heavy loads (like when assembling a vehicle or a part of a plane) or to be extremely precise and fast (like when building electronics or packaging small products). For example, Foxconn, a Taiwanese electronics manufacturer that operates in China, currently uses over 300,000 robots for assembling its products, which include smartphones, tablets, and gaming consoles.

Industrial robot
Heavy-duty machine designed to automate portions of the manufacturing process.

Collaborative Robots (Cobots)

A **collaborative robot** (**cobot**) is designed to function collaboratively alongside or directly interact with humans in a shared workspace. While most other types of robots perform their tasks independently, or in strictly isolated work areas, cobots can safely share spaces with workers to help them accomplish more. They're often used to eliminate manual, dangerous, or strenuous tasks from day-to-day workflows. In some cases, cobots can operate by responding to and learning from human movements, and cobots equipped with sensors can detect human presence.

Collaborative robot (cobot)
Robot that is designed to function collaboratively alongside or directly interact with humans in a shared workspace.

Service Robots

A **service robot** is designed to perform service-related tasks, assisting human employees in daily tasks so that they can focus on more valuable customer and patient-centric work. They can be found in a variety of settings, including homes, hospitals, and offices. Some common examples of service robots include robotic vacuum cleaners, lawn mowers, and delivery robots.

Service robot
Robot that is designed to perform service-related tasks.

Humanoid Robots

Humanoid robot
Robot that resembles and acts like a human.

A **humanoid robot** is a robot that resembles and acts like a human. Typically engineered to imitate authentic human expressions, interactions, and movements, these robots are often outfitted with an array of cameras, sensors, and, more recently, AI and ML technologies. You can find humanoid robots used in inspection, maintenance, and disaster response at power plants to relieve human workers of laborious and dangerous tasks. Similarly, humanoids are prepared to take over routine tasks for astronauts in space travel.

Ford is using Agility Robotics' Digit humanoid robot in a factory setting. This headless humanoid has two legs, two arms, and a torso, and it can pick up and stack boxes weighing up to 40 pounds. It has sensors that allow it to navigate various obstacles such as stairs and varying terrains. In addition to storing and moving inventory in a warehouse, Digit will be able to deliver packages to customers, automating the entire delivery process.

Medical Robots

Medical robot
Robotic machine utilized in health sciences.

A **medical robot** is a robotic machine utilized in health sciences. Medical robots include surgical robotic devices, diagnostic and drug delivery devices, assistive and rehabilitation devices, and robots mimicking parts of the human body, including prostheses. Surgical robots are now used in a variety of surgical procedures because they provide more precise control. They can be used to perform minimally invasive surgery (such as laparoscopy), which can lead to faster recovery times and less pain for patients, but some also assist surgeons with complex procedures. For example, the Da Vinci surgical system helps a surgeon perform surgery by using instruments that he or she guides via a console. The Da Vinci system translates the surgeon's hand movements at the console in real time, bending and rotating the instruments while performing the procedure. Robots are also used for more mundane medical tasks. For example, the TUG autonomous mobile robot, made specifically for hospitals, delivers linens, medications, and meals, and helps keep hospitals clean.

Drones

Drone
Type of pilotless aircraft that can be guided remotely or autonomously. Also known as an unmanned aerial vehicle (UAV).

A **drone**, also known as an unmanned aerial vehicle (UAV), is a type of pilotless aircraft that can be guided remotely or autonomously. Drones have many applications today and are widely used in the military for surveillance and as weapons platforms. Drones are also used for traffic monitoring, weather monitoring, agriculture, firefighting, aerial photography, and package delivery. For example, Amazon Prime Air drones deliver eligible merchandise purchased on Amazon, including medications ordered through Amazon Pharmacy. The National Oceanic and Atmospheric Administration (NOAA) uses drones to make data gathering more efficient and safer, enabling NOAA users to observe difficult-to-access areas.

Autonomous Cars

Autonomous car
Vehicle that is capable of operating with reduced or no human input. Also known as a self-driving car, driveless car, robotaxi, robotic car, or robo-car.

An **autonomous car**, also known as a self-driving car, driverless car, robotaxi, robotic car, or robo-car, is capable of operating with reduced or no human input. Self-driving cars are responsible for all driving activities, such as perceiving the environment, monitoring important systems, and controlling the vehicle, which includes navigating from origin to destination. These autonomous car systems have yet to achieve full automation. Waymo was the first to offer rides in self-driving taxis to the public in limited geographic areas of Phoenix, Los Angeles, San Francisco, and Austin, Texas. Its Waymo One vehicle is fully autonomous.

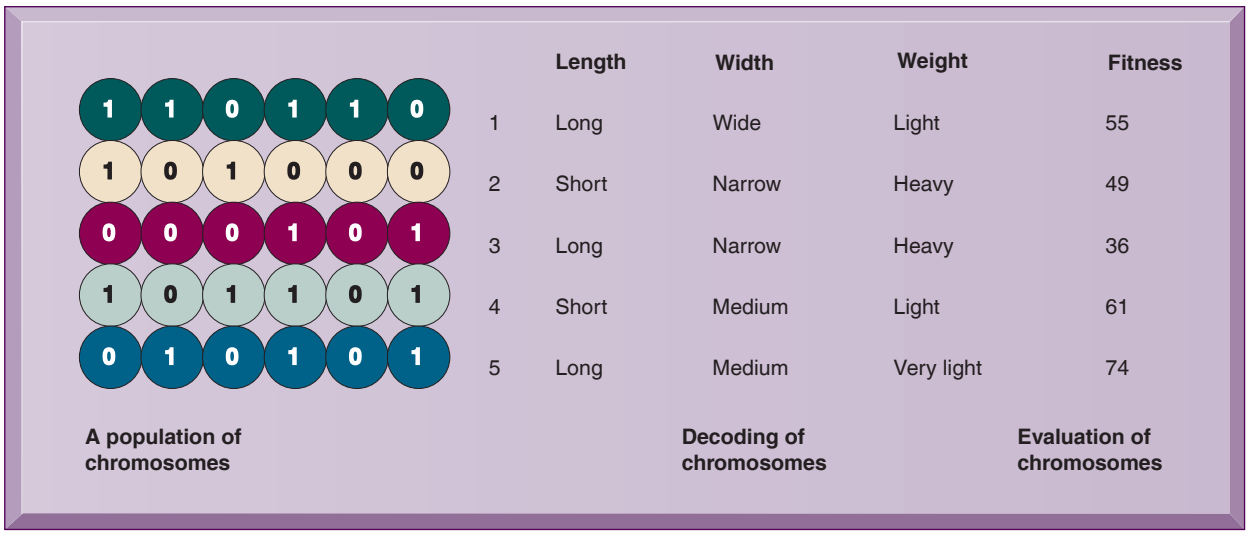
11.9 Discuss Genetic Algorithms

Genetic algorithm
A form of machine learning that is used to find the optimal solution for a specific problem based on ideas inspired by evolutionary biology.

A **genetic algorithm** is another form of machine learning. Genetic algorithms are useful for finding the optimal solution for a specific problem by examining a very large number of alternative solutions to that problem. Their method of solving problems is based on ideas inspired by evolutionary biology such as inheritance, mutation, selection, and crossover (recombination).

Figure 11.3 The Components of a Genetic Algorithm

This example illustrates an initial population of "chromosomes," each representing a different solution. The genetic algorithm uses an iterative process to refine the initial solutions so that the better ones, those with the higher fitness, are more likely to emerge as the best solution.



A genetic algorithm works by searching a population of randomly generated strings of binary digits to identify the string representing the best possible solution for the problem. As solutions alter and combine, the worst ones are discarded, and the better ones survive to go on to produce even better solutions.

Examine Figure 11.3, which illustrates the components of a genetic algorithm. Each string corresponds to one of the variables in the problem. One applies a test for fitness, ranking the strings in the population according to their level of desirability as possible solutions. After the initial population is evaluated for fitness, the algorithm then produces the next generation of strings, consisting of strings that survived the fitness test plus offspring strings produced from mating pairs of strings, and tests their fitness. The process continues until a solution is reached.

Examples of Genetic Algorithm Applications

Genetic algorithms are used to solve problems that are dynamic and complex and involve hundreds or thousands of variables or formulas. The problem must be one whose range of possible solutions can be represented genetically and for which criteria can be established for evaluating fitness. Genetic algorithms expedite the solution because they can evaluate many solution alternatives quickly to find the best one. Genetic algorithms have been widely used for problems of optimization where there are a very large number of variables to consider. Table 11.9 describes how businesses and other organizations have used genetic algorithms for solutions.

Table 11.9 Examples of Genetic Algorithm Applications

Organization	Genetic Algorithm Application
Toyota	Used genetic algorithms to optimize production schedules, logistics routes, and inventory management, improving global supply chain efficiency and reducing costs.
Ford Motor Company	Used genetic algorithms to determine optimal routes for delivery vehicles, considering factors such as traffic conditions, package sizes, and delivery deadlines. Optimization helped streamline logistics operations, reduce delivery times, and improve overall efficiency.
Autodesk	Computer-aided design (CAD) and engineering software company incorporated genetic algorithms into its products. This enabled Autodesk software users to employ genetic algorithms for optimization problems, such as finding optimal shapes for mechanical components or generating efficient 3D structures.
NVIDIA	Used genetic algorithms to explore and fine-tune the design parameters of graphics processing units, improving performance and energy efficiency in AI and gaming applications.

11.10 Discuss Expert Systems

Expert system

Captures the expertise of individual experts in an organization and represents that knowledge as sets of rules that are then converted into computer code in the form of IF-THEN rules.

Expert systems were developed in the 1970s and were the first large-scale applications of AI in businesses and other organizations. An **expert system** captures the knowledge of individual experts in an organization and represents that knowledge as sets of rules. These rules are then converted into computer code in the form of IF-THEN rules. Such programs are often used to develop apps that walk users through a process of decision-making. Expert systems rely on explicit domain knowledge and rules provided by experts, while ML models learn patterns and relationships directly from data through training. ML models are completely data-driven, whereas an expert system is based on rules, facts, and domain-specific knowledge provided by human experts. Expert systems are often used when human expertise is critical and rules are well-understood, while machine learning excels in tasks where large amounts of data can provide insights and predictions.

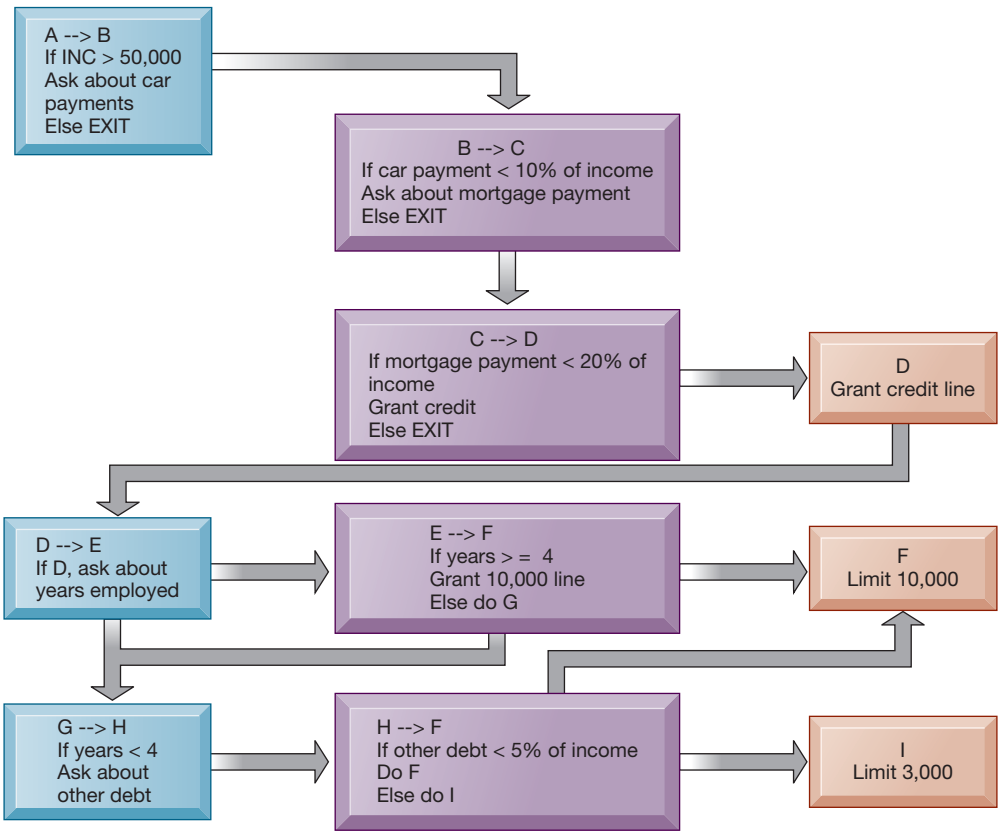
Expert systems have provided benefits such as improved decisions, reduced errors, reduced costs, reduced training time, and better quality and service. They have been used in applications for making decisions about granting credit and for diagnosing equipment problems, as well as in medical diagnostics, legal research, civil engineering, building maintenance, drawing up building plans, and educational technology (personalized learning and responsive testing).

How Expert Systems Work

Examine Figure 11.4, which illustrates an expert system used to decide whether a person should be granted a credit line and the amount that should be granted, based on the answers to a series of related questions. The rules are interconnected, the number of outcomes is known

Figure 11.4 Rules in an Expert System

An expert system contains a number of rules to be followed. The rules are interconnected, the number of outcomes is known in advance and is limited, there are multiple paths to the same outcome, and the system can consider multiple rules at a single time. The rules illustrated are for simple credit-granting expert systems.



in advance and is limited, there are multiple paths to the same outcome, and the system can consider multiple rules at a single time.

Expert systems model human knowledge as a set of rules that collectively are called the **knowledge base**. Expert systems can have from a handful to many thousands of rules, depending on the complexity of the decision-making problem. The strategy used to search through the collection of rules and formulate conclusions is called the **inference engine**. The inference engine works by searching through the rules and firing those rules that are triggered by facts the user gathers and enters.

Knowledge base

Model of human knowledge that is used by expert systems.

Inference engine

The strategy used to search through the rule base in an expert system.

Expert systems have a number of limitations. Even experts can't explain how they make decisions: They know more than they can say. People drive cars, for instance, but are challenged to say how they do it. The knowledge base can become chaotic as the number of rules can reach into the many thousands. In rapidly changing environments, say, medical diagnosis, the rules change and need to be continually updated. Expert systems are not useful for dealing with unstructured problems that managers and employees typically encounter, and these systems do not use real-time data to guide their decisions. Expert systems do not scale well to the kinds of very large data sets produced by the Internet and the Internet of Things (IoT), and they are expensive to build. For these reasons, expert system development has largely been supplanted by more advanced data-driven AI technologies such as machine learning. Today systems representing knowledge using rules are not limited to capturing just the knowledge of experts but also the broader array of rules driving business processes and are described as "rule-based" systems. For example, business application vendors such as SAP and Oracle use a rule-based approach to represent some of the business rules in their workflows.

11.11 Describe Fuzzy Logic Systems

Most people do not think in terms of traditional IF-THEN rules or precise numbers. Humans tend to categorize things imprecisely, using rules for making decisions that may have many shades of meaning. For example, a man or a woman can be *strong* or *intelligent*. A company can be *large*, *medium*, or *small* in size. Temperature can be *hot*, *cold*, *cool*, or *warm*. These categories represent a range of values.

Fuzzy logic is a form of AI that imitates human reasoning and cognition by representing imprecision in the form of rules that use approximate or subjective values rather than just 0 or 1. It can describe a particular phenomenon or process linguistically and then represent that description in a small number of flexible rules.

Fuzzy logic

Form of AI that imitates human reasoning and cognition by representing imprecision in the form of rules that use approximate or subjective values rather than just 0 or 1.

Let's look at the way fuzzy logic would represent various temperatures in a computer application to control room temperature automatically. The terms (known as *membership functions*) are imprecisely defined. Examine Figure 11.5, which shows that *cool* is between 45 degrees and 70 degrees, although the temperature is most clearly cool between about 60 degrees and 67 degrees. Note that *cool* is overlapped by *cold* or *norm*. To control the room environment using this logic, the programmer would develop similarly imprecise definitions for humidity and other factors, such as outdoor wind and temperature. The rules might include one that says: "If the temperature is *cool* or *cold* and the humidity is low while the outdoor wind is high and the outdoor temperature is low, raise the heat and humidity in the room." The computer would combine the membership function readings in a weighted manner and, using all the rules, raise and lower the temperature and humidity.

Fuzzy logic allows incremental changes in inputs to produce smooth changes in outputs instead of discontinuous ones, making it useful for consumer electronics and engineering applications. For example, fuzzy logic is widely used in washing machines and dryers. Fuzzy logic-powered washing machines feature sensors that continually monitor temperature fluctuations, and the system modifies the controls and processes. Table 11.10 describes other fuzzy logic applications.

Figure 11.5 Fuzzy Logic for Temperature Control

The membership functions for the input called temperature are in the logic of the thermostat to control the room temperature. Membership functions help translate linguistic expressions such as warm into numbers that the computer can manipulate.

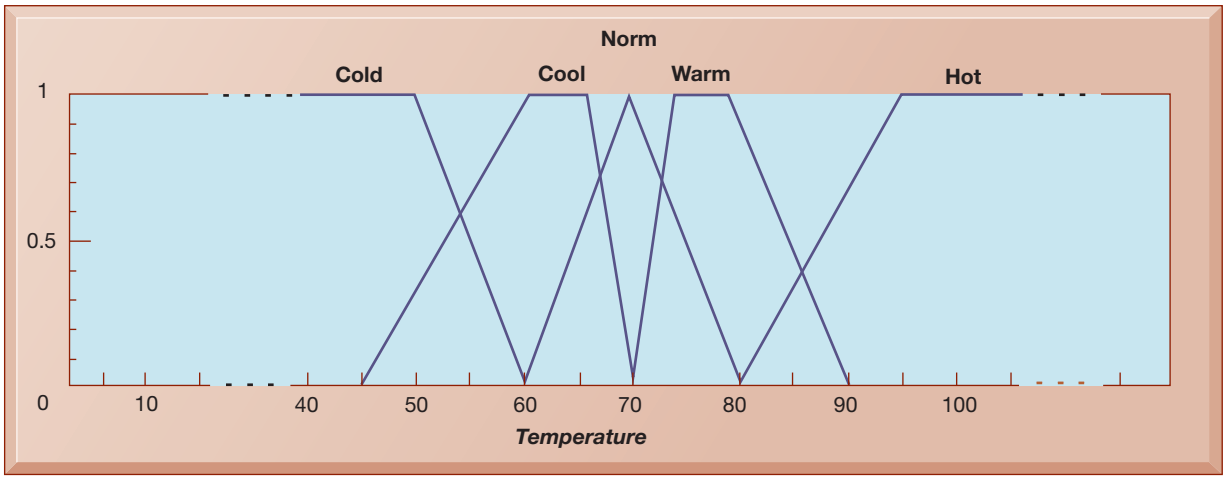


Table 11.10 Fuzzy Logic Applications

Fuzzy logic application	Description
Automobiles	Fuzzy logic is used for gear selection and is based on factors such as engine load, road conditions and style of driving
Copy machines	Fuzzy logic is used to adjust drum voltage based on factors such as humidity, picture density and temperature
Environmental control systems	In air conditioners and heaters, fuzzy logic determines output based on factors such as current temperature and target temperature
Aerospace	Fuzzy logic is used to manage altitude control for satellites and spacecrafts based on environmental factors
Sendai subway system speed controls	Fuzzy logic controls make it possible for subway cars to accelerate very smoothly

11.12 Understand the Challenges of AI

AI has many benefits, as you have learned in this chapter. However, AI is also a very challenging technology to control. In this section, we review some of these challenges: hallucination, lack of transparency, AI bias, cybersecurity problems, job loss and inequality, environmental impact, misinformation and disinformation, loss of privacy, and copyright and intellectual property challenges.

Hallucination

Hallucination

An incorrect or misleading result generated by an LLM that is presented as fact.

An AI **hallucination** is an incorrect or misleading result generated by an LLM that is presented as fact. Hallucinations can be caused by a variety of factors, including insufficient training data, incorrect assumptions made by the model, or biases in the data used to train the model. AI hallucinations are especially problematic when they occur in AI systems used for important decisions such as medical diagnoses or financial trading.

Common types of hallucinations include incorrect predictions, false positives, and false negatives. An AI model may predict that an event will occur when it is unlikely to take place. For example, an AI model used for weather forecasting may predict heavy rainstorms when there is no rain in the forecast. An AI model may identify something as a threat or problem when it is

not. For example, an AI model used to detect fraud may flag a financial transaction as fraudulent when it is not. An AI model could also fail to identify something as being a threat when it is. For example, an AI model used to detect cancer may fail to identify a cancerous tumor.

Analysts have estimated that by 2023, GenAI chatbots hallucinated as much as 27 percent of the time, with factual errors present in 46 percent of their responses. The frequency of hallucinations calls into question the reliability and trustworthiness of LLMs and is a major obstacle for organizations that want to rely on LLMs to solve real-world problems.

Lack of Transparency

Transparency in AI refers to the ability to examine the workings of an AI model and understand how it reaches its decisions. AI transparency includes the data the AI model is trained on, the set of tools and practices used to understand the model, categorizing the types and frequency of errors and biases, and the ways of communicating these issues to developers and users. The question of AI transparency has come to the forefront as AI models have grown increasingly larger and complex. More powerful models are often harder—if not impossible—to understand since the inner workings are so opaque.

AI transparency is about clearly explaining the reasoning behind the output of an AI system. Without transparency, AI systems could inadvertently perpetuate harmful biases, make inscrutable decisions, or even produce undesirable or dangerous outcomes. Enabling all stakeholders to clearly understand the workings of an AI system, including how it makes decisions and processes data, builds trust in AI. AI transparency also involves being open about a model's limitations, data handling, potential biases, and the context of its usage. Better transparency is one of the requirements of new AI regulations such as the EU AI Act, described in Chapter 4, and also one of the principles of Responsible AI, described later in this section.

AI Bias

AI bias refers to the occurrence of biased results due to human biases that skew the original training data or AI algorithm—leading to distorted outputs and potentially harmful outcomes. AI bias can impact an organization's success and hinder people's ability to participate in the economy and society. Bias reduces AI's accuracy, and therefore its potential as a solution. Biased results can also lead to unfair treatment of women, people of color, people with disabilities, and other marginalized groups. For example, computer-aided diagnosis systems have been found to return lower accuracy results for Black patients than for white patients. Predictive policing tools using AI in the criminal justice system that supposedly identify areas where crime is likely to occur often rely on historical arrest data, which can reinforce existing patterns of racial profiling and disproportionate targeting of minority communities.

AI bias

Refers to the occurrence of biased results due to human biases that skew the original training data or AI algorithm, leading to distorted outputs and potentially harmful outcomes.

There are multiple sources of AI bias. They include training data bias, algorithmic bias, and cognitive bias. AI systems learn to make decisions based on training data, and bias can be embedded in training data. For example, training data for a facial recognition algorithm that over-represents white people may create errors when attempting facial recognition for people of color. Amazon stopped using a hiring algorithm after finding it favored applicants based on words like "executed" or "captured," which were more commonly found on male resumes. Using flawed training data can cause algorithms to produce errors, unfair outcomes, or amplify the bias inherent in the flawed data. Algorithmic bias can also be caused by programming errors, such as a developer unfairly weighting factors in algorithm decision-making based on their own conscious or unconscious biases. People are deeply influenced by their own experiences and preferences and may build these biases into AI systems through the selection of data or how the data are weighted. For example, cognitive bias could lead to an AI developer favoring datasets gathered from Americans rather than sampling from a range of populations around the globe.

Cybersecurity Problems

AI has helped organizations improve cybersecurity. For example, AI systems can make automated threat detection systems more effective because they are able to quickly analyze vast amounts of data and identify patterns indicative of attacks such as phishing

attempts or malware. AI also accelerates incident response, allowing faster identification and mitigation of cyber-attacks.

At the same time, however, AI has made it easier for hackers and cybercriminals to subvert cybersecurity protection systems and cause harm. Cybercriminals are already leveraging AI technology to create more sophisticated attacks, and AI algorithms can themselves be subverted for harmful purposes. The advent of GenAI tools facilitates the creation of more personalized and authentic-looking spear phishing messages to deceive employees and circumvent traditional security measures. GenAI tools can be harnessed to gather information about specific targets for use in spear phishing or other attacks.

AI-generated deepfakes (artificial images, videos, and/or audio that have been digitally manipulated to replace one person's likeness and/or voice convincingly with that of another) may be able to undermine facial and voice recognition systems by replicating a person's face or voice so precisely that it is indistinguishable from that person's real face or voice. Most LLMs are thought to be vulnerable to prompt injection attacks, which use nefarious prompts to manipulate LLMs to produce inappropriate content, disclose sensitive information, or execute harmful code. Another way cybercriminals exploit LLMs is to create their own malicious forms of GenAI, such as WormGPT, and make them available online to others for a price. There are few built-in security features for AI models. You can find more details about AI cybersecurity issues in Chapter 8.

Job Loss and Inequality

AI-driven job loss is a major concern as the technology is adopted more widely, affecting not only low-skilled repetitive jobs in manufacturing and food service but also white-collar jobs. White-collar jobs in fields such as bookkeeping, accounting, law, and medicine have many tasks that AI can take over, and many of these tasks involve higher-level skills. The Chapter Case Study, *Is AI Coming for Your Job?,* discusses the impact of AI on jobs in more detail. Recent studies indicate that AI is likely to create disproportionate job losses among female, Black, and Hispanic workers. Although AI is projected to create 97 million new jobs by 2025, many employees won't have the skills or technical expertise needed for these new roles and could get left behind if companies don't provide their workforces with more education and training.

AI could also affect income and wealth inequality, with workers who can take advantage of AI seeing an increase in their productivity and wages—and those who cannot falling behind. AI may prove more beneficial to younger workers because it may be easier for them to adapt to technological change than older workers. The effect on income will largely depend on the extent to which AI will complement jobs rather than eliminate them and how those jobs are distributed in the labor force. If AI significantly complements higher-income jobs, it may lead to a disproportionate increase in their income. Moreover, gains in productivity from firms that adopt AI will likely boost capital returns, which may also favor high earners. Both of these phenomena could exacerbate inequality. In most scenarios, AI will likely worsen overall inequality. Governments and businesses must proactively take measures to prevent AI technology from amplifying existing inequalities. It is crucial for countries to establish comprehensive social safety nets and offer retraining programs for vulnerable workers.

Environmental Impact

AI has been both a plus and a minus in terms of environmental impact. On the plus side, AI systems are being used to help maximize the utilization of renewable energy sources like wind and solar, resulting in more environmentally friendly farming methods as well as carbon reductions for daily living. AI is also being used to optimize logistics and reduce waste in supply chains, to monitor and enforce environmental regulations, and to optimize data center operations through ML algorithms that dynamically adjust temperature settings, workload distribution, and server utilization. AI significantly boosts efficiency and productivity by optimizing processes and reducing the time and resources required to complete tasks. AI systems can

analyze data, predict outcomes, and suggest improvements, allowing businesses to streamline operations and eliminate bottlenecks. This leads to faster production cycles, reduced operational costs, and higher output quality. AI systems include tools to predict weather, monitor carbon emissions, and identify pollution.

On the minus side, AI systems consume much more energy and cooling water than traditional systems, and the amount of carbon they are throwing into the atmosphere negates many of their benefits. Google Search receives an average of 150 million search queries per hour. Each search that uses AI might require five to ten times as much power as a traditional one. Data center power consumption has already risen because of AI, and McKinsey global management consultants predict that data center electricity use will more than double by 2030.

Data centers use large amounts of water for cooling so that servers don't overheat. Data center water consumption has already impacted areas of the world where water is in short supply, and AI makes data centers even more "thirsty." Global AI demand could cause data centers by 2030 to suck up 450 million gallons of water per day.

By 2030, the world is expected to be littered with 74 million metric tons of e-waste, much of which winds up as landfill, due to the rising demand for the most up-to-date computing tools. Higher demand for AI CPUs, GPUs, and memory chips will also worsen the world's e-waste problem.

Misinformation and Disinformation

Although AI has great potential for improving people's lives worldwide, not all of its effects are benign. One of the most pernicious areas could be AI's ability to spread misinformation or disinformation at an unprecedented pace and scale. **Misinformation** is false information that is spread by people who think it's true, while **disinformation** is information that has been deliberately falsified. AI has provided a series of facilitating tools for media and journalism, such as those for voice recognition transcription and automatically generated content. However, AI technology also facilitates the creation of synthetic media, such as fake videos or deepfakes and synthetic audio content. According to researchers at Google and several fact-checking organizations, fake images generated by AI have proliferated so quickly that they're nearly as common as those manipulated by text or traditional editing tools like Photoshop. GenAI tools, which are freely available and largely unregulated, have exacerbated misinformation. It is now possible for almost anyone to generate false information and fake content in vast quantities. These include imitating the voices of real people and creating photos and videos that are indistinguishable from real ones.

Misinformation

False information that is spread by people who think the information is true.

Disinformation

Information that has been deliberately falsified.

Misinformation created by AI is also casting a shadow on political life, as evidenced by the uproar over false news during the 2016 election and thereafter. Governments and political actors worldwide are using AI to manipulate public opinion, censor critical online content, and create misinformation-ridden text, images, and videos. This false content looks or sounds plausible but often lacks accuracy. For example, Venezuelan state media used AI-generated videos of non-existent news anchors to spread pro-government propaganda. In the United States, AI-manipulated videos and images of political leaders have circulated on social media. AI systems can also recycle conspiracy theories and other misinformation found on the open web. Fake news sites now outnumber legitimate news sites, and social media is littered with false news. Some studies suggest that AI-generated misinformation can be more persuasive than false content created by humans.

Loss of Privacy

As previously noted in Chapter 4, AI systems raise a number of privacy concerns. AI models are trained using massive datasets that may inadvertently contain unauthorized personal and sensitive data collected without people's knowledge of how the data are being used. Such data may not be secure from other users when supplied to AI systems. Additionally, personal data is at risk when users of AI models input their own data in the form of queries, as this data could become part of the model's future training dataset. These data can

then show up as outputs for other users' queries. AI vendors' data storage policies are not always transparent, making it difficult to find out about where, how, and how long they store user data. The vendors often store data for lengthy periods of time or use the data in ways that do not promote privacy.

Surveillance systems using AI facial recognition technology have become increasingly accurate. Although they have beneficial uses in security, airports, border crossings, and law enforcement, many of these systems use facial data collected without the face owner's knowledge or consent, and such unauthorized use of facial data in surveillance systems poses serious privacy problems. For example, Clearview AI had compiled a facial database with more than 50 billion images collected from the Internet, including social media, for use by law enforcement and the government. The company's algorithm tries to match faces to this image database without the face owner's knowledge or permission. Clearview's practices have led to fines and bans by European Union (EU) nations for violating privacy laws and are under investigation in the United States and other countries. In 2022, Clearview reached a settlement with the American Civil Liberties Union (ACLU), in which they agreed to restrict U.S. market sales of facial recognition services to government entities. The *New York Times* had reported in 2020 that Clearview AI had gathered billions of social media photos without their owners' consent. Clearview withdrew from the United Kingdom and several other European markets. You can learn more about Clearview and facial databases in the Spotlight on Technology Case Study, *Facial Databases: Do You Know Who's Using Your Face*, in Chapter 6.

Copyright and Intellectual Property Challenges

As previously discussed in Chapter 4, the development of GenAI challenges the current copyright regimen for protecting intellectual property. One of the main challenges is defining the level of human involvement required for AI-generated works to qualify for copyright protection. It may be difficult to determine ownership rights and responsibilities when both people and machines are involved in the creation process. Traditional copyright laws typically attribute ownership to human creators. However, AI can autonomously create works without direct human involvement, raising questions about who should be considered the creator and, therefore, the copyright owner. Additionally, AI models pull training data from all over the web, and many AI vendors either don't realize or don't care when they use someone else's copyrighted artwork, text content, or other intellectual property without their consent.

The problems of using unauthorized copyrighted products and IP are magnified as AI models are trained, retrained, and fine-tuned over time. Many of today's AI models are so complex that it is impossible to identify what data are being used, where they originated, and who legally has access to them. As AI advances, it is critical that policymakers, legal experts, and content creators work together to develop specific guidelines or standards for AI-generated works. These guidelines could outline the level of human involvement required, clarify ownership rights, and propose solutions that balance innovation and copyright protection.

Web Scraping

Web scraping

Used by AI developers and vendors to scrape or extract content from publicly available sources on the Internet in order to obtain training data for LLMs.

Web scraping is often used by AI developers and vendors to "scrape," or extract, content from publicly available sources on the Internet, such as wikis, digital libraries, and third-party websites. AI tools take advantage of this technology to collect massive amounts of training data for their LLMs quickly and inexpensively. Some of the "scraped" information may be copyrighted data or personal data that has found its way into AI training sets and applications inadvertently and without authorization but continues to be used.

With time, we may find solutions to some of these problems or better ways of handling them. Alternatively, these problems could worsen. Will AI be harder to control as it becomes more ubiquitous? The Spotlight on Management Case Study, *Apple's AI Strategy; AI in Every Pocket*, describes Apple's business strategy to put an AI processor in the iPhone and other products.

Spotlight on Management Case Study

Apple's AI Strategy: AI in Every Pocket

On September 9, 2024, Apple introduced its new iPhone16, which includes Apple Intelligence, a suite of AI features it plans to start incorporating into its new operating systems and the iPhone starting in Fall 2024. The company is injecting AI tools throughout the newest versions of its operating systems: iOS 18, iPadOS 18, and macOS Sequoia. Apple Intelligence includes a new and improved context-aware Siri you can text with and new ways to generate images and emojis on the fly. Apple also announced a new partnership with OpenAI to bring ChatGPT into Apple apps. Apple's AI strategy is less concerned with having the most advanced AI tools but rather its integration of AI into the apps and Apple products that people already use every day. Apple Intelligence also puts a big emphasis on privacy—one of its longstanding strengths—and also on processing information locally on the device rather than in the cloud whenever possible.

One of the most transformative aspects of Apple's AI strategy is its emphasis on on-device processing. This approach stands in stark contrast to the traditional model of cloud-based AI, which requires extensive server resources and raises significant privacy concerns. By focusing on on-device capabilities, Apple iPhones or MacBooks become the powerhouses for running AI models, reducing the need for external server farms and massive energy consumption. While smaller models running on a phone might not match the sheer power of cloud-based systems, they offer sufficient capabilities for many everyday tasks and consume less power.

Apple started to roll out its Apple Intelligence features in the fall of 2024, with some of the more ambitious ones to debut in 2025. Not all iPhone models will have Apple Intelligence features right now, only the ones with sufficient memory and processing capacity to handle resource-hungry AI tasks, such as the iPhone 15 Pro, iPhone 15 Max, powered by Apple's powerful A17 chip, and the iPhone 16, along with iPads and Mac computers with M1 chips containing dedicated neural network hardware capable of executing 11 trillion operations per second.

The key differentiator in Apple's approach is its focus on on-device processing and privacy. Apple's emphasis on keeping data processing local on the device sets it apart from competitors that put AI processing in the cloud, and also addresses growing concerns over data privacy and security. Users will have confidence that their personal information remains under their control. This strategy aligns with Apple's long-standing commitment to user privacy and data protection.

Apple Intelligence capabilities are seamlessly woven into Apple applications. For example, Apple Intelligence can generate text for emails and text messages, creating personalized content based on users' communication styles. It can also generate images within Apple apps, making it easier to enhance creative projects with minimal effort and integrate information from a variety of different sources. Third-party developers are able to benefit from this integration to enhance their own applications for the Apple ecosystem.

Siri provides the main interface for voice and text interactions. It has context-awareness and can handle more complex queries and tasks. For example, if Siri is asked to play a podcast from a text message, Apple Intelligence can quickly locate and play the correct file without requiring additional input. Apple has also integrated OpenAI's technology into Siri. This means that while remaining the default intelligent assistant, Siri can pull in capabilities from OpenAI when needed, enhancing its ability to provide detailed and sophisticated responses.

Apple's introduction of Apple Intelligence highlights its exceptional ability to understand and cater to its audience, especially those who prioritize ease of use and privacy. By branding its AI as "for the rest of us," Apple positions itself as an inclusive and accessible technology leader, appealing to a broad range of users who seek reliable, uncomplicated, and straightforward solutions that are relevant to their everyday lives.

How much will Apple benefit from its AI strategy, and how will it affect society at large? Opinion is divided. Some experts believe that there will be a huge surge in demand for AI-enabled Apple products and that AI will reinvigorate the Apple brand. AI will be widely adopted as an indispensable personal tool. Apple products will democratize and popularize AI by bringing it further into the mainstream. Apple will seamlessly deliver a set of useful AI tools on user-friendly personal devices and provide solutions to everyday problems. Other experts are more cautious. One pundit has gone so far as to call the iPhone a "generated AI Trojan horse," normalizing a technology that, while frequently helpful, remains prone to glitches and is of uncertain value to users. When *The Washington Post* asked Apple CEO Tim

Cook about his confidence that Apple Intelligence will not "hallucinate," or assert false information as true to users, Cook admitted that "It's not 100 percent."

SOURCES: Tripp Mickle, "Apple Unveils iPhones with Built-in Artificial Intelligence," *New York Times*, September 9, 2024; Sandy Carter, "Apple's Bold Entry into the AI Arena: The Launch of Apple Intelligence," Forbes.com, June 14, 2024; Philip Michaels, "Artificial Intelligence Could Spark Interest in the iPhone But Upgrades Will Come Later," Tomsguide.com, June 16, 2024; and Matteo Wong and Charlie Warzel, "The iPhone Is Now an AI Trojan Horse," *The Atlantic*, June 10, 2024.

Case Study Questions

1. What is the role of AI in Apple's business strategy?
2. Describe the most important capabilities of Apple Intelligence.
3. Should everyone have an Apple device with Apple Intelligence? Explain your answer.

Table 11.11 Responsible AI Practices

Responsible AI Practice	Description
Fairness	AI systems should treat all individuals and groups fairly and without bias.
Accountability	Developers and organizations responsible for creating and deploying AI systems should be accountable for their decisions and actions.
Transparency	AI systems should be transparent, meaning that their inner workings are understandable and explain how AI functions operate the way they do.
Privacy and Security	Privacy and data security require special attention in AI systems because their accuracy and reliability depend on having data that have not been compromised. AI systems should comply with privacy laws that require transparency for the collection, storage, and use of data along with appropriate controls to ensure how personal data are used.
Safety	Developers and users of AI systems should do their utmost to prevent unintended consequences and potential harm. AI systems should be able to operate as originally designed, respond safely to unanticipated conditions, and resist harmful manipulation.

Responsible AI

Responsible AI

Set of practices to ensure that AI is designed, deployed, and used in an ethical and legal way.

AI systems are the product of many decisions made by those who develop and deploy them. **Responsible AI** is a set of practices to ensure that AI is designed, deployed, and used in an ethical and legal way. These practices lead toward more beneficial and equitable outcomes. Table 11.11 lists and describes five key principles associated with responsible AI.

11.13 Understand How the Information in This Chapter Can Help Your Career

The information you have learned in this chapter can help you advance your career by deepening your understanding of AI technology and how it impacts businesses and individuals. The AI concepts and vocabulary you will learn in this chapter will help prepare you for a future in which understanding AI and familiarity with AI tools will be essential requirements for doing business.

Real-World Scenario

ParentLab's Prompt Engineer

ParentLab is an AI app that guides parents to reflect on their children and parenting through an engaging conversational interface. For example, the app might check-in with a parent by saying, "I remember you mentioned Johnny's problem finishing his homework. Have you noticed any changes or improvements?" These personalized prompts encourage the parent to contemplate what they are doing as parents more closely. ParentLab's AI prompt engineer is Bene Foley, who was a teacher for 16 years and then worked in real estate for 4 years, building her expertise in understanding people. Foley tries to get the AI to talk like a person would. Foley's role is to craft the tone of the app's AI voice and teach it to respond appropriately across diverse scenarios. This requires extensive, iterative testing.

Job Description and Career Path: AI Prompt Engineer

Job	Job Description	IT Knowledge	Career Path and Average Salaries
AI prompt engineer	Specializes in developing, refining, and optimizing AI-generated prompts. • Designs and writes AI prompts. • Ensures prompts are accurate and produce the output. • Analyzes AI performance. • Works with cross-functional teams across the organization to refine prompt processes and AI system performance.	• Background in computer science with knowledge of machine learning and natural language processing or experience with SEO and digital marketing. • Knowledge of Python and programming languages for AI. • Familiarity with deep learning frameworks such as TensorFlow and PyTorch. • Familiarity with ChatGPT, Copilot, Gemini, and other AI tools. • Ability to use precise language to create very clear prompts.	Prompt engineer is a new career path leading to many promising opportunities in AI. • Entry-Level: $60,000 to $80,000 annually. • Mid-Career: $80,000 to $120,000. • Senior Positions: $120,000+ Salaries can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

Is AI Coming for Your Job?

Olivia Lipkin, a 25-year-old copywriter for a San Francisco tech startup, didn't think too much about ChatGPT when it first came out in late November 2022. Then over the next few months, Lipkin's assignments began to dwindle. In April 2023 Olivia was let go without explanation. When she found managers writing about how using ChatGPT was cheaper than paying a writer, the reason for her layoff seemed clear: Olivia had lost her job because of AI.

Olivia is not alone. Losing one's job because of AI has become a major topic of conversation and a source of worry. Whenever the topic of AI comes up, the first thing that comes to many people's minds is, "Will AI take away and make me lose my job?" How many people have lost their jobs to AI like Olivia? Even more important, how many jobs will disappear because of AI in the future?

AI has permeated many aspects of business and everyday life, and many tasks that used to be performed by humans are now partially or fully automated. For example, fast food chains such as White Castle, McDonald's, and Taco Bell are using voice-assisted AI for drive-through ordering. The technology has accelerated ordering, and it has also eliminated some of the other jobs at these restaurants. These companies and others are hoping to fully automate food ordering and preparation in the near future, potentially eliminating the need for human workers entirely. Other jobs vulnerable to elimination due to AI include customer service workers, receptionists, salespeople, accountants/bookkeepers, insurance underwriters, and warehouse workers.

The jobs threatened by AI involve not only low-level routine work but also include high-level workers such as writers, lawyers, scientists, and engineers. It used to be thought that the jobs replaced by AI were confined primarily to low-level routine tasks, such as fulfilling fast food orders or stocking shelves in a warehouse. That is no longer the case. ChatGPT and other GenAI tools have advanced algorithms that can augment or replace high-skill jobs previously thought immune to automation.

According to a report by investment bank Goldman Sachs, AI could replace the equivalent of 300 million full-time jobs, amounting to one-fourth of work tasks in the United States and Europe. The report also found that two-thirds of jobs in the United States and Europe are exposed to some degree of AI automation, and around a quarter of all jobs could be performed by AI entirely. Researchers from the University of Pennsylvania and OpenAI found that educated white-collar workers earning up to $80,000 a year are the most likely to be affected by workforce automation. The report also identified other areas with the highest exposure to automation, including administrative positions (46 percent) and tasks in the legal profession (44 percent). Not surprisingly, jobs least likely to be affected tend to be in physically intensive areas such as construction (6 percent) and maintenance.

On the other hand, Goldman Sachs further noted that AI could also be the source of new jobs and a productivity boom, eventually increasing the total annual value of goods and services produced globally by 7 percent. Some jobs will be eliminated, but many won't and will actually improve by using AI to assist them.

According to a January 2024 International Monetary Fund (IMF) report, "Gen-AI: Artificial Intelligence and the Future of Work," workers who harness the capabilities of AI are likely to see an increase in their productivity and wages, while those who don't will likely fall behind. The World Economic Forum concluded that while GenAI could eliminate 85 million jobs globally by 2025, it would also generate 97 million new jobs in fields ranging from big data and machine learning to information security and digital marketing.

How much, then, does exposure to AI actually result in job losses? According to the Pew Research Foundation, the answer is unclear. Because AI could be used either to replace or complement what workers do, it is not known exactly which or how many jobs will actually be lost. For that reason, Pew's analysis of U.S. government data sets aside the question of whether AI will lead to jobs lost or jobs gained, focusing instead on the level of exposure jobs have to AI. Pew considers jobs to be more exposed to AI if AI can either perform their most important activities entirely or help with them.

For example, in the case of customer service agents, AI chatbots could be used to enhance agent productivity or to replace agents entirely. AI may also create new types of jobs for more skilled workers, much as the expansion of the Internet age generated new classes of jobs such as web developers. Another way AI-related developments might increase employment levels is by giving a boost to the economy by elevating productivity and creating more jobs overall. Just because higher-paying, white-collar jobs will see a fair amount of exposure to the technology doesn't mean these jobs will disappear.

The Pew study found that about one-fifth of all U.S. workers have high AI-exposure jobs, especially women, along with Asian, college-educated, and higher-paid workers. However, those in the most exposed industries are also more likely to say that AI will help more than hurt them personally. For instance, 32 percent of workers in information technology said that AI will help more than hurt them personally, compared with 11 percent who say it will hurt more than it helps.

Pew's analysis did not examine the role of AI-enabled machines or robots that may perform mechanical or physical tasks. Recent evidence suggests that industrial robots may reduce both employment and wages. Moreover, jobs

held by low-wage workers, those without a high school diploma, and younger men are more exposed to the effects of industrial robots. The world has lost a total of 1.7 million manufacturing jobs to industrial robots since 2000. The number of robots taking over jobs is expected to climb to 20 million in the next 10 years.

If you're worried about AI taking your job, what can you do about it? Experts in the field offer some guidance. The World Economic Forum reports that the top skills employers will be seeking include critical thinking and problem-solving. Daniel Zhao, a senior economist at Glassdoor, recommends that workers build skills that are complementary to technology and sharpen soft skills that are hard for machines to replace. Mark Muro, senior fellow and policy director at the Brookings Institution, suggests that workers should seek work that has a creative, face-to-face, human element and notes that personalized work, teamwork, and creative work will last. Ravin Jesuthasan, a global transformation leader at Mercer, believes that there will always be demand for work requiring judgment, creativity, and empathy—in short, the things that make people human.

Christine Kiefer, a Senior Vice President at Experis, a global IT staffing and professional services firm, encourages people to individually experiment with "outsourcing" portions of their job functions to AI, then review results by asking questions like: What worked well? What still requires human intervention? What tasks does AI perform better or faster? Which tasks are better suited to humans? Your chances of staying ahead will be much greater if you are capable of constantly learning and adapting to new technologies and ways of working.

SOURCES: "Jobs Lost to Automation Statistics in 2024," Teamstage.io, accessed September 17, 2024; "Gen-AI: Artificial Intelligence and the Future of Work," International Monetary Fund (IMF), Imf.org, January 2024; Mark C. Perna, "The Steps You Can Take Today to Protect Your Job from AI," Forbes.com, September 19, 2023; Rakesh Kochhar, "Which U.S. Workers Are More Exposed to AI on Their Jobs?" Pew Research Center, Pewresearch.org, July 26, 2023; McKinsey Global Institute, "Generative AI and the Future of Work in America," Mckinsey.com, July 2023; Darina L., "Rise of Robots - Jobs Lost to Automation Statistics in 2023," Leftronic.com, March 7, 2023.

Case Study Questions

1. How does automating jobs using AI pose an ethical dilemma? Who are the stakeholders? Identify the options that can be taken and the consequences of each.
2. If you owned a factory and were considering using AI, what management, organizational, and technological factors should you consider?
3. What role has AI technology played in job losses? Explain your answer.

Review Summary

11.1 Understand artificial intelligence.

AI systems are able to perform tasks normally requiring human intelligence, such as speech recognition, visual perception, recognizing patterns, making predictions, writing essays, and learning from past experience. However, AI systems lack the breadth of knowledge, adaptability, level of creativity, intuition, and emotional intelligence that humans possess. Human intelligence is vastly more complex than artificial intelligence. The major branches of AI are machine learning, neural networks, intelligent agents, natural language processing, computer vision systems, robotics, genetic algorithms, expert systems, and fuzzy logic systems.

11.2 Describe machine learning.

Machine learning uses software that can identify patterns in very large amounts of data without explicit programming, although with significant human training. ML uses mathematical models that help a computer learn from large quantities of data and make statistical inferences without much explicit instruction. This enables a computer system to continue learning and improving on its own, to varying degrees, based on experience. ML systems are trained on large sets of data, which enables them to identify patterns and relationships. There are two kinds of machine learning: supervised and unsupervised.

11.3 Describe neural networks and deep learning.

Neural networks consist of hardware and software that attempt to mimic the thought processes of the human brain. Neural networks are notable for their ability to learn on their own with some training and to recognize patterns that cannot be easily identified by humans. Deep learning neural networks use multiple layers of neural networks to reveal the underlying patterns in data and, in some limited cases, identify patterns without human training.

11.4 Discuss generative AI.

GenAI is a machine learning model that is trained to generate content. GenAI tools can identify patterns and structures within existing data to produce new and original content, including textual responses to questions; articles, essays, fiction, and poetry; images; new software programs; musical compositions; and successful answers to test questions.

11.5 Discuss intelligent agents.

Intelligent agents are software programs with built-in or learned knowledge bases that carry out specific tasks for an individual user, business process, or software application. Intelligent agents can be programmed to navigate through large amounts of data to locate useful information and, in

some cases, to act on that information on behalf of the user. Chatbots are software agents designed to simulate a conversation with one or more human users via textual or auditory methods. More technologically advanced conversational AI agents use natural language processing and machine learning to understand and formulate a response to questions posed by humans. Virtual assistants that specialize in performing tasks for users also use NLP, machine learning, and GenAI.

11.6 Understand natural language processing.

Natural language processing technology makes it possible for a computer to understand human language and to process that information. NLP is used for sentiment analysis, text summarization, text classification, foreign language translation, and for conversational agents.

11.7 Understand computer vision systems.

Computer vision systems deal with how computers can emulate the human visual system to view and extract information from real-world images. These systems can be found in autonomous vehicles (self-driving cars and drones), medical imaging, precision agriculture, and defect-spotting systems in manufacturing.

11.8 Describe robotics.

Robotics deals with the design, construction, operation, and use of movable machines that can substitute for some human actions. There are many types of robots, including industrial robots, service robots, humanoid robots, collaborative robots, medical robots, self-driving cars, and drones.

11.9 Discuss genetic algorithms.

Genetic algorithms develop solutions to particular problems using genetically based processes such as fitness, crossover, and mutation. Genetic algorithms are useful for solving problems involving optimization, where many alternatives or variables must be evaluated to generate an optimal solution.

11.10 Discuss expert systems.

Expert systems capture the knowledge of individual experts in an organization and represent that knowledge as sets of rules. Expert systems rely on explicit domain knowledge and rules provided by experts, while machine learning models learn patterns and relationships directly from data through training.

11.11 Describe fuzzy logic systems.

Fuzzy logic is a form of AI that imitates human reasoning and cognition by representing imprecision in the form of rules that use approximate or subjective values rather than just 0 or 1. Fuzzy logic allows incremental changes in inputs to produce smooth changes in outputs instead of discontinuous ones, making it useful for consumer electronics and engineering applications with control functions.

11.12 Understand the challenges of AI.

Using AI poses many challenges, including hallucination, AI bias, lack of transparency, misinformation and disinformation, job loss, invasion of privacy, ambiguous intellectual property rights, cybersecurity problems, and environmental harm. Responsible AI practices ensure that AI is designed, deployed, and used in an ethical and legal way.

Key Terms

AI bias, 381
Autonomous car, 376
Bot, 372
Chatbot, 371
Collaborative robot (cobot), 375
Computer vision system, 373
Deep learning neural network, 368
Disinformation, 383
Drone, 376
Expert system, 378
Foundation model, 369
Fuzzy logic, 379
Genetic algorithm, 376
Generative AI (GenAI), 369
Hallucination, 380
Humanoid robot, 376
Industrial robot, 375
Inference engine, 379
Intelligent agent, 370
Knowledge base, 379
Large language model (LLM), 370
Machine learning (ML), 365
Medical robot, 376
Misinformation, 383
Natural language processing (NLP), 373
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Robotics, 375
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Supervised learning, 366
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Unsupervised learning, 366
Virtual assistant, 372
Web scraping, 384

Review Questions

11.1 Understand artificial intelligence.

- Define artificial intelligence.
- Compare artificial intelligence with human intelligence.

11.2 Describe machine learning.

- Explain what machine learning is and how it works.
- Compare supervised and unsupervised learning.
- Provide three examples of machine learning applications.

11.3 Describe neural networks and deep learning.

- Explain what a neural network is and how it works.
- Define deep learning and explain why it is so powerful.
- Provide examples of three neural network and deep learning applications.

11.4 Discuss generative AI.

- Define GenAI.
- Explain the role of transformers and large language models in GenAI.
- List and describe the capabilities of GenAI tools such as ChatGPT, Microsoft Copilot, Google Gemini, and DALL-E.

11.5 Discuss intelligent agents.

- Explain what an intelligent agent is.
- Compare the capabilities of a virtual assistant and a chatbot.
- Explain what a bot is and provide examples of three types of bots.

11.6 Understand natural language processing.

- Define natural language processing.
- Provide examples of three natural language processing applications.

11.7 Understand computer vision systems.

- Explain what a computer vision system is.
- Provide three examples of computer vision applications.

11.8 Describe robotics.

- Define robotics.
- List and describe the different kinds of robots.

11.9 Discuss genetic algorithms.

- Explain what a genetic algorithm is and how it works.
- Provide examples of three applications that use genetic algorithms.

11.10 Discuss expert systems.

- Explain what an expert system is and how it works.
- Describe the limitations of expert systems.

11.11 Describe fuzzy logic systems.

- Explain what a fuzzy logic system is and how it works.
- Provide examples of three fuzzy logic applications.

11.12 Understand the challenges of AI.

- List and describe the challenges posed by AI technology.
- Define responsible AI and explain how it can help mitigate AI's challenges.

Discussion Questions

11.13 How "intelligent" is artificial intelligence today?

11.14 What is the relationship between AI and big data?

11.15 How should AI technology and AI systems be regulated?

Using AI Project

11.16 AI Project: Writing a Press Release

You have been hired as a consultant by two friends who want to open a retail store selling 100 percent organic pet food and treats, as well as pet toys and clothing made from 100 percent organic materials. Everything the store will sell is 100 percent organic. They have asked you to prepare a press release announcing the opening of the store. For the purpose of this exercise, the store name is the 100 Percent Organic Pet Shop. Write a press release announcing that this store will open on November 1, 2024, and will be open Monday through Saturday from 9:30 AM to 5:30 PM, closed on holidays. Include contact information: The store's address is 100 Silver Maple Lane, Ithaca, New York. Telephone: 607-111-1111. The press release should be dated November 1, 2024. Using Microsoft's Copilot, Google's Gemini, or ChatGPT, develop a series of prompts to help you with this task. Save a copy of your prompts and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience analyzing an AI-related business problem, using GenAI software skills; using shopping bots to research products for sale on the web; and analyzing the role of AI and analytics in sports.

11.17 Management Decision Problem

In November 2023, the online magazine *Futurism* reported that *Sports Illustrated* was publishing articles generated by AI. According to *Futurism*, anonymous

sources were involved to create content, and Sports Illustrated published a lot of fake authors, with some articles under those fake authors' bylines generated by AI as well. Futurism also found that the author headshots in question were listed on a site that sells AI-generated portraits. The Arena Group, publisher of *Sports Illustrated*, said the articles in question were licensed content from a third party, AdVon Commerce. Arena Group said that it continually monitors its partners and that AdVon had assured the company that all of the articles in question were written and edited by humans. Arena Group further stated that AdVon writers used pen names or pseudonyms in certain articles, and that Arena Group doesn't condone those actions. Arena Group removed the articles in question from the *Sports Illustrated* website. Define the problem here and identify the management, organization, and technology factors that contributed to the problem. How could this problem be avoided?

11.18 Using GenAI Software

Software skills: GenAI and traditional search software

Business skills: Improving web searches

You want to learn how to search for topics using GenAI tools such as Google Gemini and Microsoft Copilot and how this differs from using traditional search tools such as Google or Microsoft Bing. Choose a topic to search on, such as parallelogram, Order of the Arrow, spotted lanternfly, or a topic of your choice. Search on the topic you have selected using Microsoft Bing and then Microsoft Copilot or using Google Gemini and then Google search. Compare the output from using the two types of search. How did they differ? Write several paragraphs on what you found—the differences in output, which was more comprehensive, which had more detail, which was easier to use. Print out the output to show your instructor.

11.19 Using Shopping Bots

Software skills: Web browser and shopping bot software

Business skills: Product evaluation and selection

This project will give you experience using shopping bots to search online for products, find product information, and find the best prices and vendors. Select a digital camera you might want to purchase, such as the Canon PowerShot V10 or the Olympus Tough TG-7. Visit PriceGrabber, BizRate.com, and Google Shopping to do price comparisons for you. Evaluate these shopping sites in terms of their ease of use, number of offerings, speed in obtaining information, thoroughness of information offered about the product and seller, and price selection. Which site or sites would you use and why? Which camera would you select and why? How helpful were these sites in making your decision?

11.20 Collaboration and Teamwork: Investigating the Use of AI

With two or more of your classmates, select an industry, such as the healthcare industry, financial services industry, legal industry, or one chosen by your group. Use the web to research how various companies in the industry are using AI. If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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12

Enhancing Decision Making with Data Analytics and Business Intelligence

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Learning Objectives

After completing this chapter, you will be able to:

12.1 Describe the different types of business decisions.

12.2 Describe the decision-making process.

12.3 Describe the different roles managers play.

12.4 Understand the value of improved decision making.

12.5 Understand why information systems do not always result in better decisions.

12.6 Discuss how data analytics, BI, and AI support decision making.

12.7 Describe the capabilities of BI systems.

12.8 Understand the different types of BI users.

12.9 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** Fulton County Schools Power Up Data-Driven Decision Making with Power BI
- **Spotlight on Technology Case Study:** Land O'Lakes Seeks to Reshape the Future of Farming with Predictive Analytics and AI
- **Chapter Case Study:** Big Data Analytics and AI: A New Way of Combating Natural Disasters

Opening Case Study

Fulton County Schools Power Up Data-Driven Decision Making with Power BI

Founded in 1871, Fulton County Schools (FCS) is one of the oldest school districts in Georgia. Located in the Atlanta metropolitan region, today FCS is the fourth largest school system in Georgia with around 11,000 full- and part-time employees, and around 7,000 teachers and other certified personnel. During the 2023-24 school year, almost 90,000 students attended classes in 59 elementary schools, 19 middle schools, 18 high schools, 7 start-up charter organizations, and 1 full-time virtual school.

FCS has used an annual "balanced scorecard" methodology to monitor performance and deliver information to its stakeholders (school leaders, educators, and the public) for more than 20 years. The balanced scorecard approach, which you will learn more about later in the chapter, was originally designed as a way for businesses to look at performance based on four dimensions of firm performance: financial, business process, customer, and learning and growth. FCS modified the framework to better align with educational rather than business goals.

Originally, FCS delivered the scorecard in a PDF format, with visualizations created using Microsoft Excel, and based on data gathered from more than 70 different information systems. Collecting and analyzing the data and generating the scorecard report was a labor-intensive process requiring hundreds of hours of staff time. Although FCS had been trying to make data-driven decisions for years, these methods were hindering its ability to do so. The data were not always reliable and valid, and individual schools and administrators had their own frameworks for analyzing the data. FCS wanted to modernize the approach so that it could have a holistic, near real-time view that it could use to inform resource allocation decisions, address educational challenges, and optimize operational efficiency. It also wanted to be able to showcase student performance more effectively.

FCS already had a long-standing relationship with Microsoft and turned to Microsoft's Power BI business intelligence software for a solution. FCS began the transformation by consolidating its many disparate data sets into a data warehouse. At the same time, the district implemented a set of uniform standards by which student, teacher, and operational data would be analyzed.

SOURCES: Fulton County Schools, "Balanced Scorecard Overview," Fultonschools.org, accessed September 17, 2024; "Fulton County Schools Transforms Data-driven Decision-Making by Using More than 400 Power BI Visualizations to Improve Student Outcomes," Customers.microsoft.com, May 23, 2024; Emily Bell and Heather Trotter Clark, "Review: Microsoft Power BI Makes Data Actionable in K-12," Edtechmagazine.com, March 15, 2024.

The next step was to develop visualizations of that data, culminating in an interactive balanced scorecard

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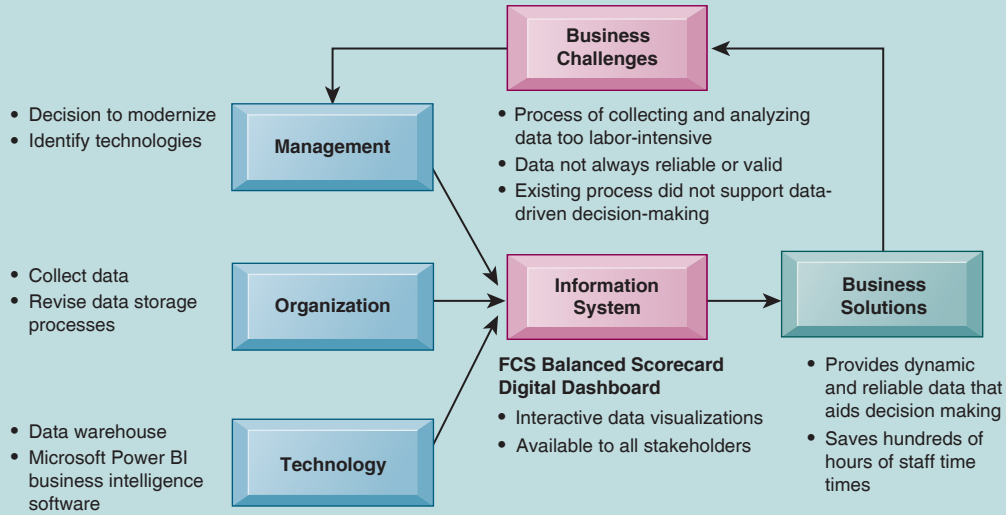
presented in the form of a digital dashboard. The scorecard is organized into four topical areas (Learning & Growth, Finance, Business Processes, and Stakeholder Input), with 18 subcategories and more than 400 different visualizations, and is available online for public view. Power BI gives the district the ability to show data from different systems in a single visualization, for example, from a school's student information system as well as an instruction-based prescriptive-diagnostic platform. Sample visualizations include real-time student enrollment and demographics, per-pupil and programmatic funding breakdowns, and the status of facilities work orders at buildings throughout the district. Despite the level of detail available, users report that they find the dashboard easily accessible and easy to navigate.

The new system provides dynamic and reliable data that helps the Fulton County Board of Education make decisions more quickly and with more confidence. The new digital scorecard has also created new opportunities for community engagement. The decision to publicly provide interactive data and visualizations has built credibility for both the Board and school district. Parents and teachers are actively using the data to ask informed questions and determine how to improve student outcomes. Principals and teachers now have access to timely data insights that enable them to identify challenges and areas for improvement at the classroom and school level and take proactive actions. The new system is also enabling FCS to save hundreds of hours of staff time by automating data collection, storage, and analysis.

Fulton County Schools' experience in adopting Microsoft Power BI is a powerful illustration of how information systems can dramatically improve decision making. In the past, school districts used manual data



collection and reporting systems to provide information to school leaders and the public using a balanced scorecard format. Fulton County Schools decided it wanted to modernize its approach so that stakeholders could get more specific, targeted, real-time information to enable better informed decisions.

The Opening Case Study diagram above calls attention to important points raised by this case and this chapter. Fulton County Schools wanted to modernize the approach it was using to generate a balanced scorecard so that it could have a holistic, more timely view of the data that it could use to inform resource allocation decisions, address educational challenges, and optimize operational efficiency.

Microsoft Power BI business intelligence software enables the Fulton County Board of Education to make decisions more quickly and with more confidence, creates new opportunities for community engagement, builds credibility for both the Board and school district, and gives principals and teachers access to timely data insights that enables them to identify challenges and areas for improvement at the classroom and school level and take proactive actions. The new system is also enabling FCS to save hundreds of hours of staff time.

Case Study Questions

1. How is business intelligence changing decision making and the way FCS provides information to its stakeholders?
2. Give an example of a type of decision that educators at FCS might make that could be improved by using business intelligence.

CAREER BENEFIT

FCS IT staff benefit by successfully identifying and implementing an appropriate solution. Staff further benefit from learning how to use Power BI business intelligence software, which saves them time and also creates skills that they can use for their current or future jobs. End users (educators and the Board of Education) benefit in having reliable and timely data that can help them make better decisions.

12.1 Describe the Different Types of Business Decisions

Decision making in businesses used to be limited to management. Today, lower-level employees are responsible for some of these decisions, as information systems make information available to lower levels of the business.

Types of Business Decisions

Chapters 1 and 2 described the different levels in an organization, such as senior management, middle management, and operational management. Each of these levels has different information requirements for decision support and responsibility for different types of decisions. Decisions can be classified as structured, semistructured, and unstructured. Examine Figure 12.1, which illustrates the types of decisions each level of management typically faces.

An **unstructured decision** is one that is novel, important, and nonroutine, and there is no well-understood or agreed-on procedure for making it. The decision maker must provide judgment, evaluation, and insight to solve the problem.

A **structured decision**, by contrast, is repetitive and routine, and it involves a definite procedure for handling the decision so that it does not have to be addressed each time as if it was new. A **semistructured decision** has elements of both types: only part of the problem has a clear-cut answer provided by an accepted procedure. In general, structured decisions are more prevalent at lower organizational levels, whereas unstructured problems are more common at higher levels of the firm.

Senior executives face many unstructured decision situations, such as establishing the firm's five-year or 10-year goals or deciding which new markets to enter. Answering the question "Should we enter a new market?" requires access to news, government reports, and industry views as well as high-level summaries of firm performance. However, the answer also requires senior managers to use their own best judgment and poll other managers for their opinions.

Middle management faces more structured decision scenarios, but their decisions may include unstructured components. A typical middle-level management decision might be, "Why is the reported order fulfillment report showing a decline over the past six months at a distribution center in Minneapolis?" This middle manager might obtain a report from the firm's enterprise system or distribution management system on order activity and operational efficiency at the Minneapolis distribution center. This is the structured part of the decision.

But before arriving at an answer, this middle manager might interview employees and gather more unstructured information from external sources about local economic conditions or sales trends.

Unstructured decision

Decision that is novel, important, and nonroutine and for which there is no agreed-upon procedure for making the decision; the decision maker must provide judgment, evaluation, and insights to solve the problem.

Structured decision

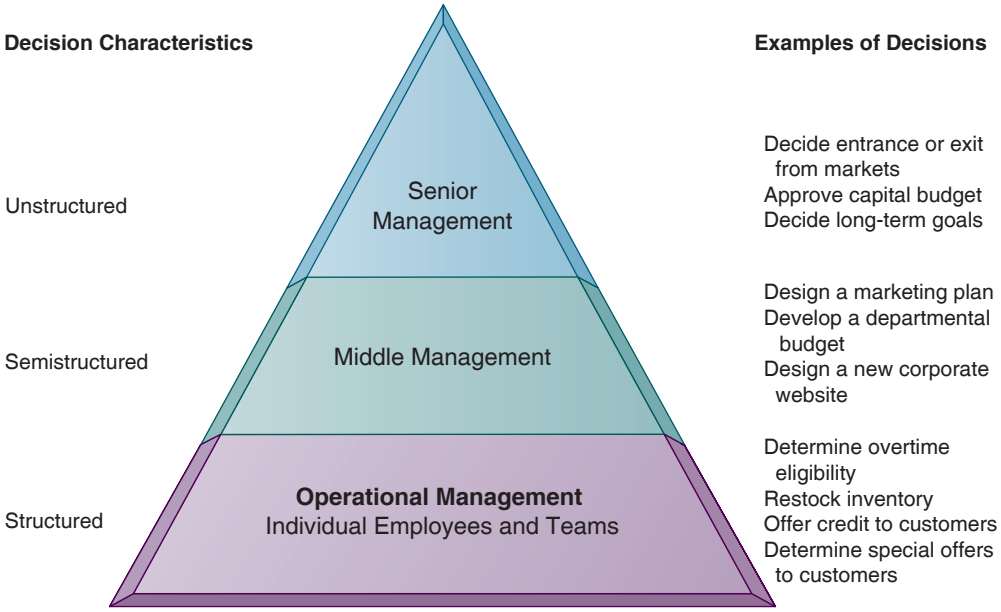
Decision that is repetitive and routine and has a definite procedure for how it should be handled.

Semistructured decision

Decision in which only part of the problem has a clear-cut answer provided by an accepted procedure.

Figure 12.1 Information Requirements of Key Decision-Making Groups in a Firm

Senior managers, middle managers, operational managers, and employees have different types of decisions and information requirements.



Operational management and rank-and-file employees tend to make more structured decisions. For example, a supervisor in a manufacturing plant might have to decide whether an hourly paid worker is entitled to overtime pay. If the employee worked more than eight hours on a particular day, the supervisor would routinely grant overtime pay for any time beyond eight hours that was clocked on that day.

A sales account representative may have to make decisions about extending credit to customers by consulting the firm's customer database, which contains credit information. If the customer meets the firm's prespecified criteria for granting credit, the account representative would grant that customer credit to make a purchase. In both instances, the decisions are highly structured and are routinely made hundreds, or even thousands, of times each day in some large firms. The answer has been preprogrammed into the firm's payroll and accounts receivable systems.

Intelligence
The first of Simon's four stages of decision making; involves discovering, identifying, and understanding problems occurring in the organization.

Design
Simon's second stage of decision making; involves identifying and exploring possible alternative solutions to a problem.

Choice
Simon's third stage of decision making; consists of choosing among various solution alternatives.

Implementation
All the organizational activities surrounding the adoption, management, and routinization of an innovation, such as a new information system.

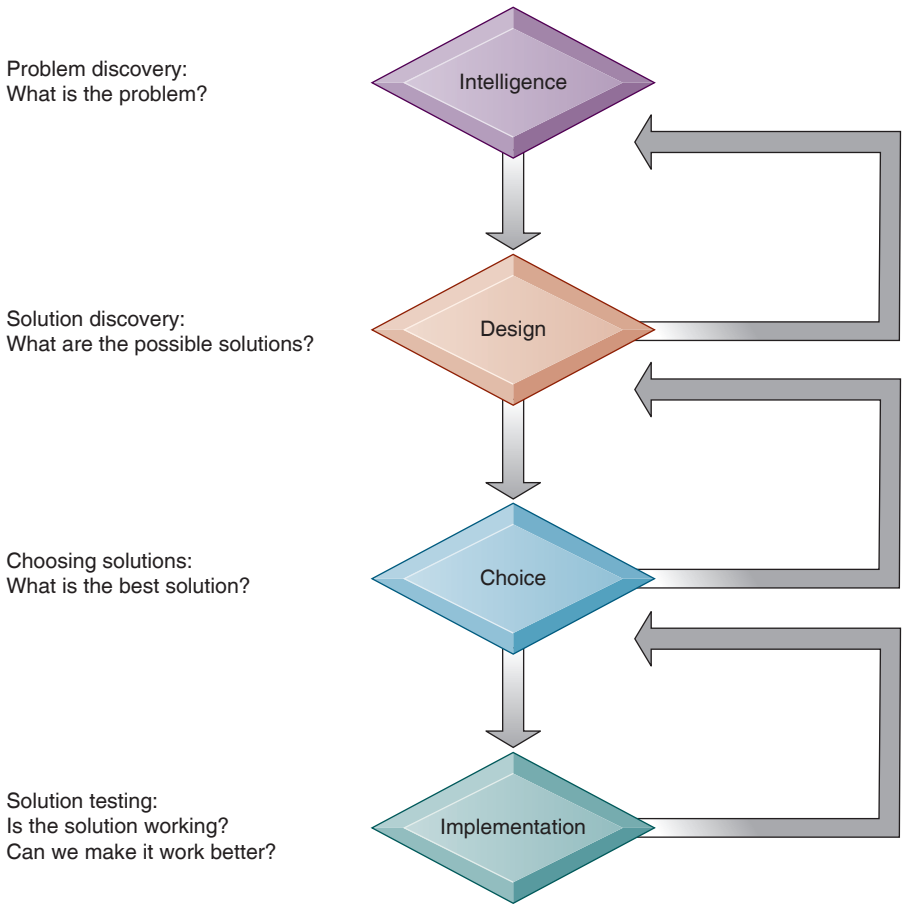
12.2 Describe the Decision-Making Process

Making a decision is a multistep process. Simon (1960) described four different stages in decision making: intelligence, design, choice, and implementation. **Intelligence** in this context consists of discovering, identifying, and understanding the problems occurring in the organization—why a problem exists, where, and what effects it is having on the firm. **Design** involves identifying and exploring various solutions to the problem. **Choice** consists of choosing among solution alternatives. **Implementation** involves making the chosen alternative work and continuing to monitor how well the solution is working.

Examine Figure 12.2, which illustrates these stages.

Figure 12.2 Stages in Decision Making

The decision-making process is broken down into four stages.



What happens if the solution you have chosen doesn't work? As shown in Figure 12.2, you can return to an earlier stage in the decision-making process and repeat it if necessary. For instance, in the face of declining sales, a sales management team may decide to pay the sales force a higher commission for making more sales to spur on the sales effort. If this does not produce sales increases, managers would need to investigate whether the problem stems from poor product design, inadequate customer support, or a host of other causes that call for a different solution.

12.3 Describe the Different Roles Managers Play

Systems to support decision making produce better decision making by managers and employees, above-average returns on investment for the firm, and ultimately higher profitability. However, information systems cannot improve every decision taking place in an organization. Let's examine the role of managers and decision making in organizations to see why this is so.

Managerial Roles

Managers play key roles in organizations. Their responsibilities range from making decisions, to writing reports, to attending meetings, to arranging birthday parties. We are able to better understand managerial functions and roles by examining classical and contemporary models of managerial behavior.

The **classical model of management**, originally developed by Henry Fayol and other early writers, identifies the five functions of managers as planning, organizing, coordinating, deciding, and controlling. This description of management activities dominated management thought for a long time, and it is still popular today.

Classical model of management

Traditional description of management that focuses on its formal functions of planning, organizing, coordinating, deciding, and controlling.

The classical model describes formal managerial functions but does not address exactly what managers do when they plan, decide things, and control the work of others. For this, we must turn to the work of contemporary behavioral scientists who have studied managers in daily action. The **behavioral model of management** argues that the actual behavior of managers appears to be less systematic, more informal, less reflective, more reactive, and less well organized than the classical model would have us believe.

Behavioral model of management

Model of management that argues that the actual behavior of managers is less systematic, more informal, less reflective, more reactive, and less well organized than posited by the classical model of management.

Observers find that managerial behavior actually has five attributes that differ greatly from the classical description. First, managers perform a great deal of work at an unrelenting pace—studies have found that managers engage in more than 600 different activities each day, with no break in their pace. There is never enough time to do everything for which a CEO is responsible (Porter and Nohria, 2018). Second, managerial activities are fragmented; most activities last for less than nine minutes, and only 10 percent of the activities exceed one hour in duration. Third, managers prefer current, specific, and ad hoc information (printed information often will be too old). Fourth, they prefer oral forms of communication to written forms because oral media provide greater flexibility, require less effort, and bring a faster response. Fifth, managers give high priority to maintaining a diverse and complex web of contacts that acts as an informal information system and helps them execute their personal agendas and short- and long-term goals.

Analyzing managers' day-to-day behavior, Henry Mintzberg found that it could be classified into 10 managerial roles (Mintzberg, 1971). **Managerial roles** are expectations of the activities that managers should perform in an organization. Mintzberg found that these managerial roles fell into three categories: interpersonal, informational, and decisional.

Managerial roles

Expectations of the activities that managers should perform in an organization.

Interpersonal Roles

In their **interpersonal role**, managers act as figureheads for the organization when they represent their companies to the outside world and perform symbolic duties, such as giving out employee awards. Managers act as leaders, attempting to motivate, counsel, and support subordinates. Managers also act as liaisons between various organizational levels; within each of these levels, they serve as liaisons among the members of the management team. Managers provide time and favors, which they expect to be returned.

Interpersonal role

Category of managerial roles identified by Mintzberg in which managers act as figureheads and leaders for the organization.

Informational Roles

Informational role
Category of managerial roles identified by Mintzberg in which managers act as the nerve centers of their organizations, receiving and disseminating critical information.

In their **informational role**, managers act as the nerve centers of their organizations, receiving the most concrete, up-to-date information and redistributing it to those who need to be aware of it. Managers are therefore information disseminators and spokespersons for their organizations.

Decisional Roles

Decisional role
Category of managerial roles identified by Mintzberg in which managers initiate activities, handle disturbances, allocate resources, and negotiate conflicts.

Managers make decisions. In their **decisional role**, they act as entrepreneurs by initiating new kinds of activities, they handle disturbances arising in the organization, they allocate resources to staff members who need them, and they negotiate conflicts and mediate between conflicting groups.

Information Systems Supporting Various Managerial Roles

Table 12.1 looks at the ways different information systems can help managers in the various roles they may play. Note that information systems are now capable of supporting most, but not all, areas of managerial life.

12.4 Understand the Value of Improved Decision Making

One of the main contributions of information systems has been to improve decision making for both individuals and groups. What does it mean to a business to make better decisions? What is the monetary value of improved decision making? Table 12.2 attempts to measure the monetary value of improved decision making for a small U.S. manufacturing firm with $280 million in annual revenue and 140 employees. The firm has identified a number of key decisions where new system investments might improve the quality of decision making. The table provides selected estimates of annual value (in the form of cost savings or increased revenue) from improved decision making in selected areas of the business.

Table 12.1 Managerial Roles and Supporting Information Systems

Authors and Mintzberg, Henry. "Managerial Work: Analysis from Observation." Management Science 18 (October 1971).

Role	Behavior	Support Systems
Interpersonal Roles	Interpersonal	
Figurehead		Telepresence systems
Leader		Telepresence, social networks
Liaison		Smartphones, social networks
Informational Roles	Information processing	
Nerve center		Management information systems, executive support systems
Disseminator		Texting, email, social networks
Spokesperson		Webinars, telepresence
Decisional Roles	Decision making	
Entrepreneur		None exist
Disturbance handler		None exist
Resource allocator		Business intelligence, decision-support system
Negotiator		None exist

Table 12.2 Business Value of Improved Decision Making

Example Decision	Decision Maker	Number of Annual Decisions	Estimated Value to Firm of a Single Improved Decision	Annual Value
Allocate support to most valuable customers	Accounts manager	12	$100,000	$1,200,000
Predict call center daily demand	Call center management	4	$150,000	$600,000
Decide parts inventory levels daily	Inventory manager	365	$5,000	$1,825,000
Identify competitive bids from major suppliers	Senior management	1	$2,000,000	$2,000,000
Schedule production to fill orders	Manufacturing manager	150	$10,000	$1,500,000
Allocate labor to complete a job	Production floor manager	100	$4,000	$400,000

You can see from Table 12.2 that decisions are made at all levels of the firm and that some of these decisions are common, routine, and numerous. Although the value of improving any single decision may be small, improving hundreds of thousands of "small" decisions can add up to a large annual value for the business.

12.5 Understand Why Information Systems Do Not Always Result in Better Decisions

As you learned in the preceding section, information systems are not helpful for all managerial roles. And in those managerial roles where information systems might improve decisions, investments in information technology do not always produce positive results. Poor decision making is estimated to cost firms on average at least 3 percent of profits each year. The costs of poor decision making may also extend beyond financial results and inflict reputational and regulatory costs (Purdy and Williams, 2023). There are three main reasons: information quality, management filters, and organizational culture (see Chapter 3).

Information Quality

High-quality decisions require high-quality information. Table 12.3 describes information quality dimensions that affect the quality of decisions.

If the output of information systems does not meet these quality criteria, decision making will suffer. Chapter 6 describes how corporate databases and files have varying levels of inaccuracy and incompleteness, which in turn will degrade the quality of decision making.

Table 12.3 Information Quality Dimensions

Quality Dimension	Description
Accuracy	Do the data represent reality?
Integrity	Are the structure of data and relationships among the entities and attributes consistent?
Consistency	Are data elements consistently defined?
Completeness	Are all the necessary data present?
Validity	Do data values fall within defined ranges?
Timeliness	Are data available when needed?
Accessibility	Are the data accessible, comprehensible, and usable?

Management Filters

Confirmation bias

Involves the tendency to give more credence to evidence that fits with one's existing beliefs, particularly in the face of conflicting information.

Even with timely, accurate information, managers often make bad decisions. Managers (like all human beings) absorb information through a series of filters to make sense of the world around them. Management decision making is often impacted by **confirmation bias**, which involves the tendency to give more credence to evidence that fits with one's existing beliefs, particularly in the face of conflicting information. Cognitive scientists, behavioral economists, and neuroeconomists have also found that managers, like other humans, are poor at assessing risk and are risk averse; perceive patterns where none exist; and make decisions based on intuition, feelings, and the framing of the problem as opposed to empirical data (Kahneman, 2011; Tversky and Kahneman, 1986).

Organizational Inertia and Politics

Organizations are bureaucracies with limited capabilities and competencies for acting decisively. When environments change and businesses need to adopt new business models to survive, strong forces within organizations resist making decisions calling for major change. Decisions taken by a firm often represent a balancing of the firm's various interest groups rather than the best solution to the problem.

Studies of business restructuring find that firms tend to ignore poor performance until threatened by falling stock prices, activist shareholders, or outside takeovers, and they systematically blame poor performance on external forces beyond their control—such as economic conditions (the economy), foreign competition, and rising prices—rather than blaming senior or middle management for poor business judgment. When the external business environment is positive and firm performance improves, managers typically credit themselves for the improved performance rather than the positive environment.

High-Velocity Automated Decision Making

Today, many decisions made by organizations are not made by managers—or any humans. For example, firms engaged in high-frequency stock trading execute trades within nanoseconds. Humans are eliminated from the decision chain because they are too slow.

In these high-speed automated decisions, the intelligence, design, choice, and implementation parts of the decision-making process have been captured by software algorithms. The people who designed the software identified the problem, identified a method for finding a solution, defined a range of acceptable solutions, and implemented a solution. In these situations, organizations are making decisions faster than managers can monitor or control, and great care needs to be taken to ensure the proper operation of these systems to prevent harm.

Risks Introduced by Using AI in Decision Making

Although AI can help managers make faster and better decisions by providing improved predictions based on real-time data and acting as a "copilot" and sounding board, using AI for management decision making also presents a number of challenges. Risks include the potential for ingrained bias, concerns about the data upon which the decisions are being based, accuracy, lack of transparency as to how decisions are arrived at, and potential ethical issues. Organizations must understand the potential weaknesses and limitations of the AI systems they are using. Chapter 11 provides more detail about these challenges.

12.6 Discuss How Data Analytics, BI, and AI Support Decision Making

Chapter 2 introduced you to the different types of systems used for supporting management decision making. The foundation for all of these systems are a data analytics and business intelligence (BI) infrastructure that supplies the data and the analytic tools for supporting decision making. AI functionality is also increasingly being integrated into BI tools.

Data Analytics, BI, and AI

Data analytics
Involves collecting, storing, inspecting, cleaning, modeling, and querying data.

The terms data analytics and BI are often used synonymously, but although they share many similarities, they are not the same thing. **Data analytics** involves collecting, storing, inspecting, cleaning, modeling, and querying data. Think databases, data warehouses, data marts, data lakes, Hadoop, and the data analytic platforms described in Chapter 6.

BI is a term used by hardware and software vendors and information technology consultants to refer to a collection of methods, systems, and tools (or infrastructure) to convert unprocessed data into actionable business insights. Data analytics provides the foundational underpinning for BI systems. The term business intelligence is also used to describe the output of a BI system: i.e., the insights produced. BI systems are aimed at helping businesses make informed decisions based on past and present data. They are primarily used by management and non-technical personnel and tend to focus on bigger picture, strategic questions. Data analytics, on the other hand, tends to be used by more technical personnel who use advanced statistical and computational methods to explore data and extract meaningful insights.

Business analytics (BA) is also a vendor-defined term and can be viewed as a combination or subset of both data analytics and BI. BA focuses on tools and techniques for analyzing and understanding business data. Think online analytical processing (OLAP), statistics, models, and data mining, also introduced in Chapter 6. Predictive analytics, discussed further in Section 12.7, can be considered a form of BA.

BI and BA are essentially about integrating all the information streams produced by a firm into a single, coherent, enterprise-wide set of data and then using modeling, statistical analysis tools, and data mining tools to make sense out of all these data so managers can make better decisions and plans.

As noted above, AI is increasingly being integrated into data analytics and BI tools. AI is being used to make data analytics and BI more efficient, effective, and accessible to a wider range of users. AI can automate many of the manual tasks involved in data analytics and BI, such as data preparation, cleansing, and analysis. Natural language processing can help users by enabling them to use natural language queries to explore data. Machine learning can be used to create more powerful predictive models that can be used to more accurately forecast future trends. AI can also be used to create BI dashboards that are more personalized and interactive. It is important to remember that BI and BA are products defined by technology vendors and consulting firms. For example, Microsoft Fabric, an end-to-end software as a service (SaaS) suite for data analytics, brings together components from Microsoft's various business intelligence tools, such as Microsoft Power BI, Azure Synapse Analytics, and Azure Data Factory, into an all-in-one integrated platform. Microsoft Fabric is built around a data lake named OneLake and provides tools for data movement, data storage, data engineering, data integration, data science, real-time analytics, and business intelligence, as well as data security, governance, and compliance. Fabric is infused with Azure OpenAI Service at every layer, enabling developers to leverage the power of generative AI and assisting business users in finding insights in their data. Other leading providers of BI and BA products include Oracle, SAP, IBM, and SAS.

The Business Intelligence and Analytic Environment

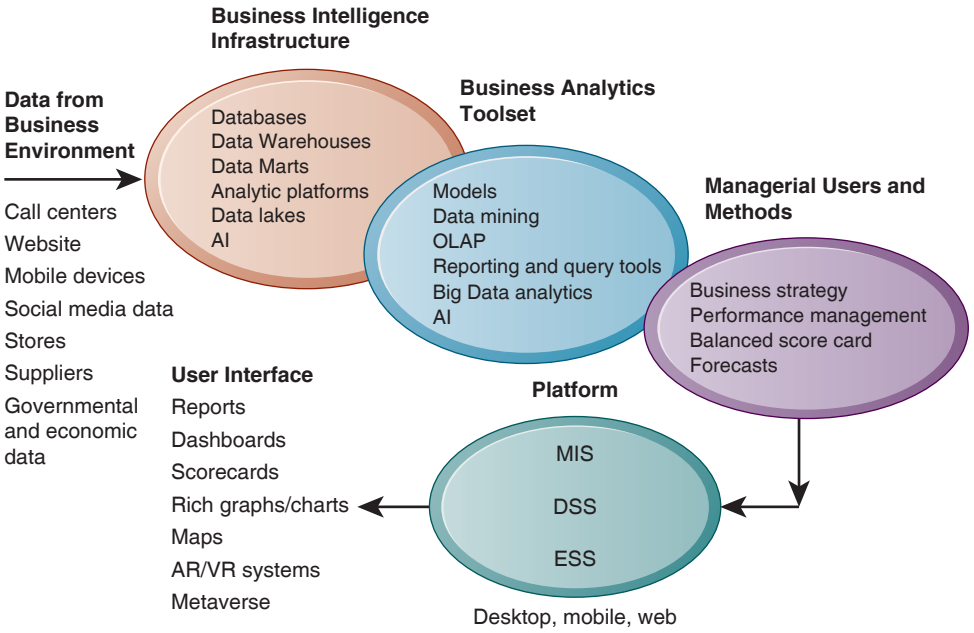
Examine Figure 12.3, which provides an overview of the business intelligence and analytics environment, highlighting the kinds of hardware, software, and management capabilities that the major vendors offer and that firms develop over time.

There are six elements in this business intelligence and analytic environment:

- **Data from the business environment:** Businesses must deal with both structured and unstructured data from many different sources, such as call centers, websites, mobile devices, social media data, stores, suppliers, and governmental and economic data, including big data. The data need to be integrated and organized so that they can be analyzed and used by human decision makers.
- **BI infrastructure:** The underlying foundation of BI is a powerful database system that captures all the relevant data to operate the business. The data may be stored in transactional databases or combined and integrated into an enterprise data warehouse, a series of interrelated data marts, or analytic platforms. Most analytic platforms also now feature AI functionality.

Figure 12.3 BI and Analytics for Decision Support

BI and analytics require a strong database foundation, a set of analytic tools, and an involved management team that can ask intelligent questions and analyze data.



- **BI and BA toolsets:** A set of software tools, such as models, data mining, OLAP, reporting and query tools, and big data analytics, is used to analyze data and produce reports, respond to questions posed by managers, and track the progress of the business by using key indicators of performance. Many BI and BA toolsets now also feature AI functionality.
- **Managerial users and methods:** BI hardware and software are only as intelligent as the humans who use them. Managers impose order on the analysis of data using a variety of managerial methods that define strategic business goals and specify how progress will be measured. These include enterprise performance management and balanced scorecard approaches that focus on key performance indicators and industry strategic analyses focusing on changes in the general business environment, with special attention to competitors. Without strong senior management oversight, BI and BA can produce a great deal of information, reports, and online screens that focus on the wrong matters and divert attention from the real issues.
- **Delivery platform—MIS, DSS, ESS:** The results from BI and BA are delivered to managers and employees in a variety of ways, depending on what they need to know to perform their jobs. MIS, DSS, and ESS, introduced in Chapter 2, deliver information and knowledge to different people and levels in the firm—operational employees, middle managers, and senior executives. In the past, these systems could not easily share data and operated as independent systems. Today, BI and BA tools can integrate all this information and bring it to managers' desktops or mobile platforms.
- **User interface:** Businesspeople often learn quicker from a visual representation of data than from a dry report with columns and rows of information. Today's BI and BA software suites feature a variety of tools, such as rich graphs, charts, dashboards, maps, and increasingly augmented reality and virtual reality functionality, that can be used for **data visualization**, which helps users see patterns and relationships in large amounts of data by presenting the data in graphical form.

Data visualization

Technology for helping users see patterns and relationships in large amounts of data by presenting the data in graphical form.

Tableau

Tableau Software is a leading BI and data visualization tool. Now owned by Salesforce, Tableau helps users create different charts, graphs, maps, dashboards, and stories for visualizing and analyzing data to help make business decisions. Nontechnical users are able to quickly and

Data visualization tools facilitate the creation of graphs, charts, dashboards, and maps to make it easier for users to obtain insights from data.

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easily create and share customized, interactive dashboards to provide business insights from a broad spectrum of data, including data from spreadsheets, corporate databases, and the web. Tableau can query relational databases, online analytical processing cubes, cloud databases, and spreadsheets to generate graph-type data visualizations, and it can join and blend different datasets.

Global restaurant and food service supplier Sysco uses Tableau to enable customers to self-serve their inventory needs instead of relying on assistance from a sales representative. For customer orders placed across multiple tracking systems for sales accounting, Tableau can visualize order activity more clearly and in greater detail, providing more insights for the sales and sales leadership teams. A sales-facing dashboard shows all orders placed by customers and internal associates. The dashboard helps the company promote its self-service model because the company can see by account or by geographic area where Sysco's e-commerce platform is being used, which customers were engaging with it, and where they need help from sales reps (Tableau, 2024).

Virtual Reality, Augmented Reality, and Metaverse Tools for BI and Data Visualization

Virtual reality (VR) technology provides powerful capabilities for three-dimensional data visualization. VR systems use interactive graphics software and hardware to create computer-generated simulations that are so close to reality that users almost believe they are participating in a real-world situation. In many VR systems, the user dons special clothing, headgear, and equipment, depending on the application. The clothing contains sensors that record the user's movements and immediately transmit that information back to the computer.

Virtual reality (VR)

Technology that provides powerful capabilities for three-dimensional data visualization using interactive graphics software and hardware to create computer-generated simulations that provide sensations that emulate real-world activities.

VR applications are currently found in entertainment, retail, and manufacturing, where an immersive experience can help customers visualize products or teach factory workers how to use complex equipment. Volkswagen Group has used VR systems to speed up vehicle design and development and to identify potentially costly design problems earlier in the development cycle. Volkswagen has been able to cut out costly physical prototypes and replace them with immersive, 360-degree views of digitally constructed interior and exterior components of a vehicle using virtual reality HTC Vive headsets. Virtual components of a car, including interior and exterior parts such as buttons, lights, or consoles, can be switched out and replaced easily with a few lines of software code during the design process.

Augmented reality (AR) is a related technology for enhancing visualization by overlaying digital data and images onto a physical real-world environment. The digital technology provides additional information to enhance the perception of reality, making the surrounding

Augmented reality (AR)

Technology for enhancing visualization by overlaying digital data and images onto a physical real-world environment.

real world of the user more interactive and meaningful. The yellow first-down markers shown on televised football games are examples of AR technology, as are medical procedures like image-guided surgery, where data acquired from computerized tomography (CT) and magnetic resonance imaging (MRI) scans or from ultrasound imaging are superimposed on the patient in the operating room.

Other industries where AR has caught on include military training, engineering design, robotics, and consumer design. Leading retail brands like IKEA, Amazon, and Sephora provide e-commerce shopping experiences using AR mobile apps or wearable glasses to help shoppers make buying decisions in real time. For example, the IKEA Place mobile app takes pictures of a living room, measures the space, and recommends furniture that fits the space. Users can thus virtually "place" furnishings into their space.

The VR and AR applications attracting much attention today are those for the metaverse, which we previously described in Chapters 2, 4, and 7. The metaverse is conceived as a graphically rich, virtual, 3D space, with some degree of verisimilitude, where people can work, shop, socialize, and play—basically, doing the things humans like to do together in real life but using digital avatars. Metaverse proponents often focus on the concept of "presence" as a defining factor: feeling like you're really there and feeling like other people are really there with you, too. The infrastructure underpinning the metaverse, including VR glasses and AR software, and increasingly, AI, relies on data showing how users interact with their surroundings in virtual worlds, digital workplaces, and elsewhere.

Most public attention has thus far been focused on the use of metaverse technologies in the consumer sector, with applications involving social networking, entertainment, and marketing. However, metaverse technologies are also being deployed for enterprise and industrial purposes. The enterprise metaverse involves virtualizing work, learning, communication, and collaboration activities. The industrial metaverse focuses on how data and AI can be used to optimize interactions among people, production assets, processes and operations, and industrial environments. The aim is to transform analyses of past activities and future activities by mirroring and simulating machines, factories, and other highly complex systems in a fully immersive, real-time, interactive representation and simulation of the real world. The automotive, energy, aerospace, and defense industries are currently leading the way in industrial metaverse investments and activities (IEEE Digital Reality, 2024; ITBriefcase, 2024; World Economic Forum, 2024).

12.7 Describe the Capabilities of BI Systems

BI and BA promise to deliver correct, nearly real-time information to decision makers, and the analytic tools help them quickly understand the information and take action.

Analytic Functionalities of BI Systems

There are six primary analytic functionalities that BI systems deliver to achieve these ends:

- **Production reports:** These are predefined reports based on industry-specific requirements (see Table 12.4).
- **Parameterized reports:** Users enter several parameters as in a pivot table to filter data and isolate the impacts of parameters. For instance, you might want to enter the region and time of day to understand how sales of a product vary by region and time. A national company such as Starbucks, for example, might discover that customers in the East buy most of their coffee in the morning, whereas in the Northwest, customers buy coffee throughout the day. This finding might lead to different marketing and ad campaigns in each region. (See the discussion of pivot tables in Section 12.8.)
- **Dashboards/scorecards:** These are visual tools for presenting performance data defined by users.
- **Ad hoc query/search/report creation:** These allow users to create their own reports based on queries and searches.

Table 12.4 Examples of BI Predefined Production Reports

Business Functional Area	Production Reports
Sales	Sales forecasts; sales team performance; cross-selling; sales cycle times
Service/call center	Customer satisfaction; service cost; resolution rates; churn rates
Marketing	Campaign effectiveness; loyalty and attrition; market basket analysis
Procurement and support	Direct and indirect spending; off-contract purchases; supplier performance
Supply chain	Backlog; fulfillment status; order cycle time; bill of materials analysis
Financial	General ledger; accounts receivable and payable; cash flow; profitability
Human resources	Employee productivity; compensation; workforce demographics; retention

- **Drill down:** This is the ability to move from a high-level summary to a more detailed view.
- **Forecasts, scenarios, models, recommendations:** These include the ability to perform linear forecasting and what-if scenario analysis and analyze data using standard statistical tools, as well as recommendations as to actions to take.

Predictive and Prescriptive Analytics

An important capability of BA is the ability to model future events and behaviors, such as the probability that a customer will respond to an offer to purchase a product. **Predictive analytics** uses statistical analysis, data mining techniques, historical data, and assumptions about future conditions to predict future trends and behavior patterns. Variables that can be measured to predict future behavior are identified. For example, an insurance company might use variables such as age, gender, and driving record as predictors of driving safety when issuing auto insurance policies. A collection of such predictors is combined into a predictive model for forecasting future probabilities with an acceptable level of reliability.

Predictive analytics
The use of statistical analysis, data mining techniques, historical data, and assumptions about future conditions to predict future trends and behavior patterns.

Netflix uses predictive analytics models in its recommendation engine to predict user preferences for TV shows and movies based on the user's past viewing habits. Royal Dutch Shell PLC is using the Microsoft Azure cloud platform and the C3 IoT platform-as-a-service (PaaS) application development platform to monitor and predict where and when maintenance is needed for compressors, valves, and other equipment.

Predictive analytics is being incorporated into numerous BI applications for sales, marketing, finance, fraud detection, and healthcare. One of the most well-known applications is credit scoring, which is used throughout the financial services industry. When you apply for a new credit card, scoring models process your credit history, loan application, and purchase data to determine your likelihood of making future credit payments on time. Healthcare insurers have been analyzing data for years to identify which patients are most likely to generate high costs.

Many companies employ predictive analytics to predict responses to marketing campaigns and other efforts to cultivate customers. By identifying customers more likely to respond, companies can lower their marketing and sales costs by focusing their resources on customers who have been identified as more promising. For instance, FedEx has been using predictive analytics to develop models that predict how customers will respond to price changes and new services, which customers are most at risk of switching to competitors, and how much revenue will be generated by new storefront or drop-box locations. The accuracy rate of FedEx's predictive analytics system ranges from 65 to 90 percent.

Prescriptive analytics takes predictive analytics one step further. **Prescriptive analytics** differs from predictive analytics in not only predicting future outcomes but also recommending actions or decisions to achieve desired outcomes or prevent undesirable ones.

Prescriptive analytics
Form of analytics that not only predicts future outcomes, but also recommends actions or decisions to achieve desired outcomes or prevent undesirable ones.

Organizations are using prescriptive analytics for customer segmentation, churn prediction, fraud detection, risk assessment, demand forecasting, prescriptive maintenance, and personalized recommendations. In addition to using many statistical techniques used for predictive modeling, today's prescriptive analytics tools are increasingly using AI machine language algorithms.

The Spotlight on Technology Case Study, *Land O'Lakes Seeks to Reshape the Future of Farming with Predictive Analytics and AI*, focuses on Land O'Lakes' use of predictive and prescriptive analytics and AI to reshape the practice of agriculture in the twenty-first century.

Spotlight on Technology Case Study

Land O'Lakes Seeks to Reshape the Future of Farming with Predictive Analytics and AI

Land O'Lakes, a renowned agricultural cooperative based in Arden Hills, Minnesota, has been in existence for more than 100 years. Companies involved in agriculture are not often at the leading edge of innovation. But Land O' Lakes is seeking to break that mold by harnessing the power of data analytics and AI to reshape how farming is conducted in the twenty-first century. Land O'Lakes believes that providing hyperlocal customized insights will help farmers and other agricultural producers be more productive and make better decisions.

Land O'Lakes has several different lines of business. The most well-known includes its dairy products business: butter, cheese, and other dairy products that are sold under the Land O'Lakes brand. Land O'Lakes also sells farm animal feeds under the Purina Animal Nutrition brand name. Its WinField United division operates a wholesale distribution business involving seeds and plant nutrition and protection. Finally, Land O'Lakes has also recently launched a new business, Truterra, focused on sustainability. These various lines of business generated nearly $17 billion in annual revenue in 2023, landing Land O'Lakes the 245th spot on the Fortune 500 list.

In 2018, Teddy Bekele was appointed Chief Technology Officer for Land O'Lakes, the first time in the organization's history that it had someone in that role. It was an acknowledgment of the importance that technology is now playing in farming. Since that time, Bekele has been leading a digital transformation effort at each of Land O'Lakes' four business units. According to Bekele, technology in the agricultural industry is transitioning from being just an enabler to being a driver and data is at the center of that transition. The success of a farming business revolves around 40 major decisions, such as what and when to plant, when and how to fertilize, feed, and harvest, and how to market and sell. According to Bekele, these are all questions that data and technology can help farmers answer.

As a case in point, Land O'Lakes WinField United division is now deploying predictive analytics and prescriptive analytics with AI functionality. The division has 115 research plots, called "Answer Plots," dispersed in all the areas where WinField does business. WinField captures all of the data about how the seed it sells performs in a given soil type, topography, and climate. At the WinField Innovation Center, wind tunnels and chambers that can mimic the conditions in certain geographic areas, such as South Dakota on a hot day, or Indiana on a cool day, enable WinField to gather data that relates to crop growth under different conditions.

WinField then uses the data to help farmers make the best decisions for their fields. It uses AI to identify seed profiles that best match where the farmer's fields are located as well as the best practices that the farmer should follow to achieve the best results. Currently, these recommendations are presented within a physical 400-page crop protection guide that is provided to local agronomists (crop scientists) who use it to develop recommendations. Land O'Lakes is now looking to digitize the guide and in partnership with Microsoft, to develop generative AI chatbot that can act as a virtual agent that can provide crop insights and answer questions. For example, a farmer could share a picture of an unidentified weed growing in a field and get an identification as well as a recommendation about how to combat it.

Land O'Lakes Truterra division is also technology-driven. Truterra has developed a data-centric program to help farmers generate and sell soil-based carbon assets. Truterra makes cash payments to farmers who sequester carbon, which involves capturing and storing greenhouse gases naturally produced by plants. The carbon assets are then purchased by corporate customers with sustainability goals. In its first three years, Truterra has paid more than $21 million to farmers for the sequestration of more than 1.1 million metric tons of carbon.

The program is delivered through Truterra's network of local agronomists, who work directly with farmers to help them make the best decisions for their fields. Truterra works with farmers to gather local data at the field level using sensors and drones, and then uses that data to power an on-farm stewardship digital platform.

The platform lets farmers and their agronomists run what-if scenarios and "plug and play" with different combinations of land stewardship practices (for example, adding cover crops and reducing the number of acres tilled) to identify the practices that maximize return on investment while at the same time protecting the land. Geospatial data analysis is critical to understanding the impact of agricultural operations on environmental sustainability with respect to water quality, soil health, greenhouse gasses, and more. Truterra is using Databricks, Apache Sedona, and R to analyze spatial data at scale and then educate and promote practices that optimize profitability, sustainability, and stewardship outcomes at the farm based on the spatial insights it develops.

Despite these advances, there are still numerous challenges. Although Land O'Lakes has been using predictive analytics for a number of years, and is increasingly deploying AI tools, it is not yet able to model exactly what will happen at the specific field-level. To do so, an enormous amount of data would be required. While the newest farm equipment is typically equipped with sensors and sophisticated software, many farmers are still using older equipment that is not. Reliable Internet service in rural areas also remains a challenge. Despite these changes, Bekele believes that agriculture is on the cusp of revolution, switching from decision making based on gut instinct and intuition to decisions enabled by data and technology.

SOURCES: John Kell, "Why Land O'Lakes Now Has a CTO after 97 Years Without One," Fortune.com, June 26, 2024; Neil C. Hughes, "2796: Cultivating Tomorrow's Farms: Land O'Lakes Tech-Driven Agriculture Revolution," The Tech Talks Daily Podcast, Youtube.com, February 7, 2024; Alicia Wallace, "As the CEO of Land O'Lakes, She's Changing the Rules of American Farming." Cnn.com, November 21, 2023; MITSloan Management Review "Big Data in Agriculture: Land O'Lakes Teddy Bekele." Me, Myself, and AI Podcast, Episode 503 Sloanreview.mit.edu, August 30, 2022; Teddy Bekele, "How Data Will Drive Profitable Sustainable Farming in 2022," Landolakes.com, April 18, 2022

Case Study Questions

1. How is Land O'Lakes helping farmers make better decisions?
2. What are some of the challenges that farmers face in using data and technology?

Big Data Analytics

Predictive and prescriptive analytics tools are now able to use big data from both private and public sectors, including data from social media, customer transactions, and output from sensors and machines. In e-commerce, many online retailers have capabilities for making personalized online product recommendations to help stimulate purchases and guide decisions about what merchandise to stock. In the past, most of these product recommendations have been based on the behaviors of similar groups of customers, such as those with incomes under $50,000 or whose ages are between 18 and 25 years. Now some retailers are starting to analyze the tremendous quantities of online and in-store customer data they collect along with social media data to make these recommendations more individualized. These efforts are translating into higher customer spending and customer retention rates. Table 12.5 provides examples of companies using big data analytics.

In the public sector, the use of big data analytics has been driving the movement toward smart cities, which make intensive use of digital technology and public record data stores to make better decisions about running cities and serving their residents. Municipalities are

Table 12.5 What Big Data Analytics Can Do

EHarmony	Online dating website analyzes personal and behavioral data on 10 million users to match couples based on features of compatibility found in thousands of successful relationships. Processes more than 15 million matches per day.
Rolls Royce	Uses big data to reduce the amount of carbon its aircraft engines produce while optimizing maintenance to prolong the life of an engine. Monitors how each engine flies, the conditions under which it is flying, and how pilots use it. Customizes maintenance regime for individual engines.
Hertz	Uses data from web surveys, email, text messages, website traffic patterns, and data gathered at all of its 8,500 rental car locations to adjust staffing levels at individual locations in order to accommodate times of the day when demand peaks or falls.
Levi Strauss & Co,	Levi's uses SAS Analytics to analyze millions of consumer demand signals to create a supply chain geared to the preferences of individual consumers. The system enables Levi's to create demand forecasts for each style, color, and size of product that it offers.

adding more data captured through sensors and location data from mobile devices. Predictive modeling programs now inform public policy decisions on utility management, transportation operation, healthcare delivery, and public safety. For example, irrigation systems built into the city of Barcelona's parks monitor soil moisture and turn on sprinklers when water is needed. The city expects to reduce its water bill by 25 percent per year after installing sensors in local parks.

Operational Intelligence and Analytics

Operational intelligence
Form of business intelligence and business analytics that delivers insights about activities inside and outside an organization from streams of big data as they are generated in real time.

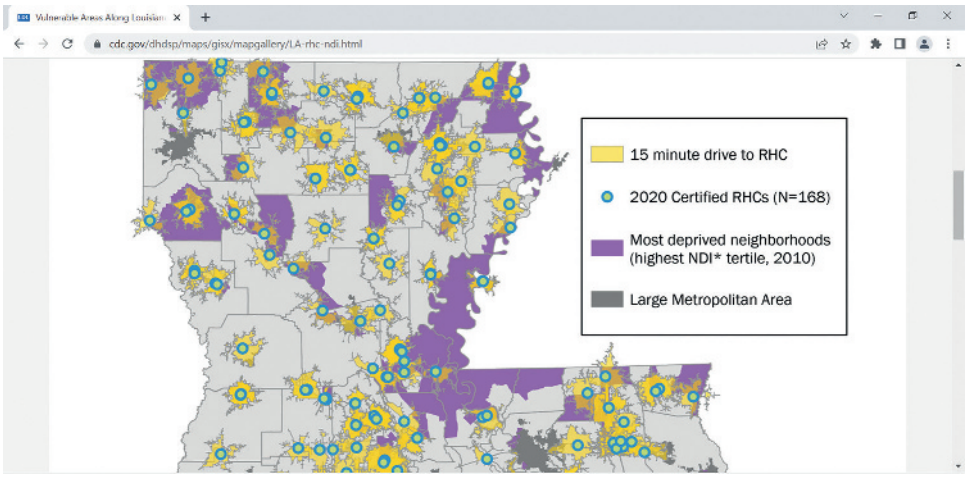
The Internet of Things is creating huge streams of data from web activities, smartphones, sensors, gauges, and monitoring devices. **Operational intelligence** is a form of BI and BA that can be used to deliver insights about activities inside and outside the organization from these streams of big data as they are generated in real time. Companies can set trigger alerts for particular events or have them fed into live dashboards to help managers with their decisions.

One example of operational intelligence is the use of data generated by sensors on trucks, trailers, and intermodal containers owned by Schneider National, one of North America's largest trucking, logistics, and intermodal services providers. The sensors monitor location, driving behaviors, fuel levels, and whether a trailer or container is loaded or empty. Data from fuel tank sensors help Schneider identify the optimal location at which a driver should stop for fuel based on how much is left in the tank, the truck's destination, and fuel prices en route.

Location Analytics and Geographic Information Systems

Location analytics
Ability to gain insights from the location (geographic) component of data, including location data from mobile devices, output from sensors or scanning devices, and data from maps.

Decisions are also based on location data. BI includes **location analytics**, the ability to gain business insight from the location (geographic) component of data, including location data from mobile devices, output from sensors or scanning devices, and data from maps. For example, location analytics might help a marketer determine which people to target with mobile ads about nearby restaurants and stores or quantify the impact of mobile ads on in-store visits. Location analytics can help a utility company view and measure outages and their associated costs as related to customer location to help prioritize marketing, system upgrades, and



This map shows the locations of vulnerable areas along Louisiana's Mississippi Delta that lack geographic access to Rural Health Clinics (RHCs). RHCs provide primary care access points for rural area residents. Although many rural areas have access to nearby RHCs, there are still rural neighborhoods along the Mississippi Delta with no nearby clinics.

Source: Anna L. Hamilton, Well-Ahead Louisiana. Accessed from the Centers for Disease Control and Prevention's Chronic Disease GIS Exchange https://www.cdc.gov/dhdsp/maps/gisx/mapgallery/LA-rhc-ndi.html

customer service efforts. UPS's package tracking and delivery-routing systems described in Chapter 1 use location analytics, as does an application used by Starbucks to determine where to open new stores. The Starbucks system analyzes very large amounts of location-based data and demographic data to determine the best places to open more stores without harming sales at other Starbucks locations. The system generates a map that shows local trade areas, retail clusters, demographics, traffic and transportation nodes, and locations with new offices that might be important sources of customers.

The Starbucks and utility company applications are examples of a geographic information system. A **geographic information system (GIS)** provides tools to help decision makers visualize problems that benefit from mapping. GIS software ties location data about the distribution of people or other resources to points, lines, and areas on a map. Some GIS have modeling capabilities for changing the data and automatically revising business scenarios. A GIS might be used to help state and local governments calculate response times to natural disasters and other emergencies, to help banks identify the best location for new branches or ATM terminals, or to help police forces pinpoint locations with the highest incidence of crime.

Geographic information system (GIS)

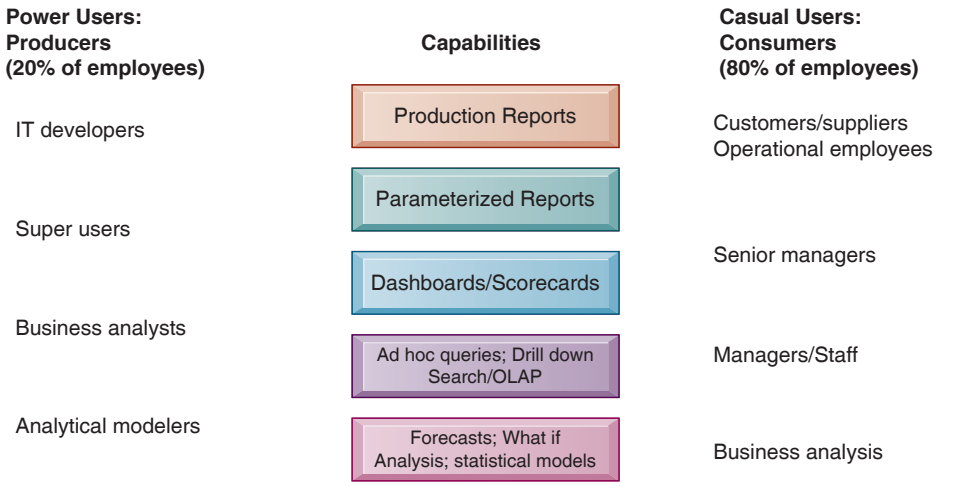
System with software that can analyze and display data using digitized maps to enhance planning and decision-making.

12.8 Understand the Different Types of BI Users

Earlier in this text and in this chapter, we described the different information constituencies in business firms—from senior managers to middle managers, analysts, and operational employees. This also holds true for BI systems. Examine Figure 12.4, which illustrates the most common users of BI and the purposes for which they use it. The vast majority (more than 80 percent) of the audience for BI consists of "casual" users. Senior executives tend to use BI to monitor firm activities using visual interfaces like dashboards and scorecards. Middle managers and analysts are much more likely to be immersed in the data and software, entering queries and slicing and dicing the data along different dimensions. Operational employees will, along with customers and suppliers, be looking mostly at prepackaged reports. The other 20 percent of BI users tend to be "power" users, who actually produce the various reports, models, and forecasts consumed by the "casual" users.

Figure 12.4 BI Users

Casual users are consumers of BI output, while intense power users are the producers of reports, new analyses, models, and forecasts.



Decision Support for Operational and Middle Management

Operational and middle management are generally charged with monitoring the performance of key aspects of the business, ranging from the downtime of machines on a factory floor to the daily or even hourly sales at franchise food stores to the daily traffic at a company's website. Most of the decisions these managers make are fairly structured. Middle managers typically use MIS, introduced in Chapter 2, to support this type of decision making. Increasingly, middle managers receive MIS reports online and are able to interactively query the data to find out why events are happening. Managers at this level often turn to exception reports, which highlight only exceptional conditions, such as when the sales quotas for a specific territory fall below an anticipated level or employees have exceeded their spending limits in a dental care plan. Table 12.6 provides some examples of MIS applications for business intelligence.

Support for Semistructured Decisions

Some managers are "super users" who want to create their own reports and use more sophisticated analytics and models to find patterns in data, to model alternative business scenarios, or to test specific hypotheses. A DSS is the BI delivery platform for this category of users, providing the ability to support semistructured decision making.

A DSS relies more heavily on modeling than an MIS does, using mathematical or analytical models to perform what-if or other kinds of analysis. "What-if" analysis, working forward from known or assumed conditions, allows the user to vary certain values in test results to predict outcomes if changes occur in those values. What happens if we raise product prices by 5 percent or increase the advertising budget by $1 million?

Sensitivity analysis

Form of modeling that asks "what-if" questions repeatedly to determine the impact of changes in one or more factors on the outcomes.

Sensitivity analysis is a form of modeling that involves asking what-if questions repeatedly to predict a range of outcomes when one or more variables are changed multiple times. Examine Figure 12.5, which illustrates a sample sensitivity analysis that examines the effect of changing the sales price and cost per unit of a specific product on the product's break-even point. Backward sensitivity analysis helps decision makers with goal seeking: the business wants to sell 1 million product units next year, how much must it reduce the price of the product?

Pivot table

Spreadsheet tool that enables reorganizing and summarizing two or more dimensions of data into a tabular format, and which can be employed to identify and understand patterns in data that may be useful for semistructured decision making.

Chapter 6 described multidimensional data analysis and OLAP as key business intelligence technologies. Spreadsheets have a similar feature for multidimensional analysis called a **pivot table**, which is a tool that enables reorganizing and summarizing two or more dimensions of data into a tabular format and which can be employed to identify and understand patterns in data that may be useful for semistructured decision making.

Examine Figure 12.6, which depicts a Microsoft Excel pivot table that examines order transactions for a company selling online management training products. It shows the relationship between two dimensions: the sales region and the source of contact (web ad or e-mail)

Table 12.6 Examples of MIS Applications

Company	MIS Application
Baylis & Harding	Wholesale distributor specializing in luxury toiletries and gift sets adopted BI tools from Phocas to enable better visibility into financial, customer and sales data. Managers and executives use the tools to create standard and ad hoc MIS reports based on data from the company's NetSuite ERP system. The reports provide instant access to the business data managers need to act proactively.
Black & Veatch	Global engineering, procurement, consulting, and consulting company uses MIS to track construction costs for various projects across the United States.
Chipotle Mexican Grill	Restaurant chain with more than 3,200 restaurants worldwide Implemented a BI system to track operational effectiveness. By standardizing reporting, the company was able to develop uniform KPIs for benchmarking.

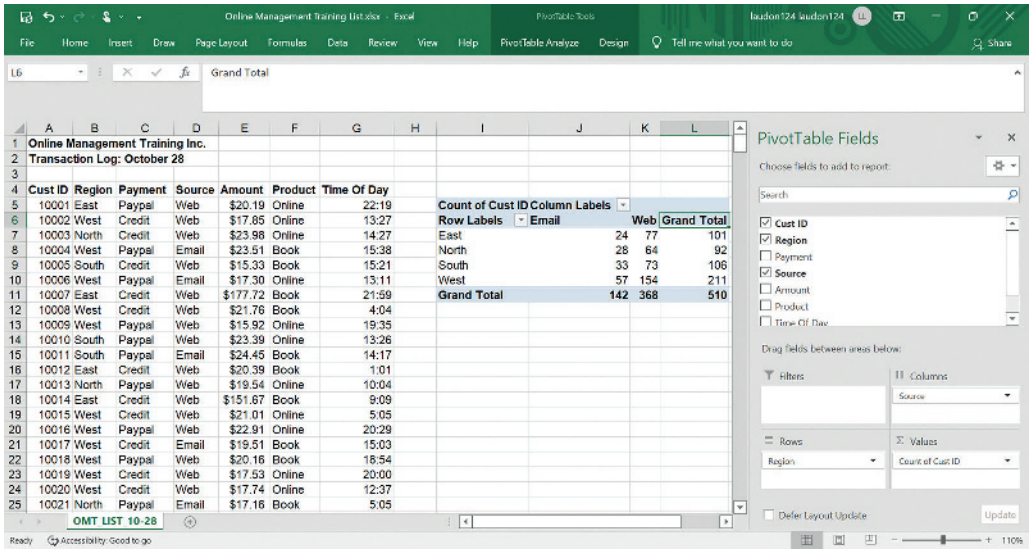
Figure 12.5 Sensitivity Analysis

This table displays the results of a sensitivity analysis of the effect of changing the sales price of a beach towel and the cost per unit on the product's break-even point. It answers the question, "What happens to the break-even point if the sales price and the cost to make each unit increase or decrease?"

Total fixed costs	**19000**					
Variable cost per unit	**3**					
Average sales price	**17**					
Contribution margin	**14**					
Break-even point	**1357**					
		Variable Cost per Unit				
Sales Price	**1357**	**2**	**3**	**4**	**5**	**6**
	14	1583	1727	1900	2111	2375
	15	1462	1583	1727	1900	2111
	16	1357	1462	1583	1727	1900
	17	1267	1357	1462	1583	1727
	18	1188	1267	1357	1462	1583

Figure 12.6 A Pivot Table That Examines Customer Regional Distribution and Advertising Source

This pivot table enables you to examine where an online management training company's customers come from in terms of region and advertising source.



Used with permission from Microsoft

for each customer order. It answers the question: Does the source of the customer make a difference in addition to the region? The pivot table in this figure shows that most customers come from the West and that advertising on the web produces most of the customers in all the regions.

In the past, much of this modeling was done with spreadsheets and small stand-alone databases. Today, these capabilities are incorporated into BI systems that are able to analyze data from large databases. BI analytics include tools for intensive modeling, some of which we described earlier. Such capabilities helped Monster.com, which provides job search, career advice, and hiring services, develop a more targeted, multichannel approach to advertising. Most of Monster's revenue comes from employers who pay to post job listings and to search its résumé database. Monster had been sending generic messages about its services to large groups of companies, but Monster now tries to find new employers and candidates using much more personalized email, direct mail, social engagement, and prioritized

telemarketing. SAS statistical modeling software identifies existing and prospective customers who are most likely to purchase job listings and other services in a specific quarter. A marketing database maintains data on when the email campaign ran, which email recipients responded to the email messages, and who clicked through. By analyzing these data, Monster is able to generate mailing lists based on criteria such as past response behavior and an "opportunity score."

Decision Support for Senior Management: Balanced Scorecard and Business Performance Management Methods

The purpose of an executive support system (ESS), introduced in Chapter 2, is to help C-level executive managers focus on the really important performance information that affects the overall profitability and success of the firm. There are two parts to developing an ESS. First, you need a methodology for understanding exactly what is "the really important performance information" for a specific firm that executives need, and second, you need to develop systems capable of delivering this information to the right people in a timely fashion.

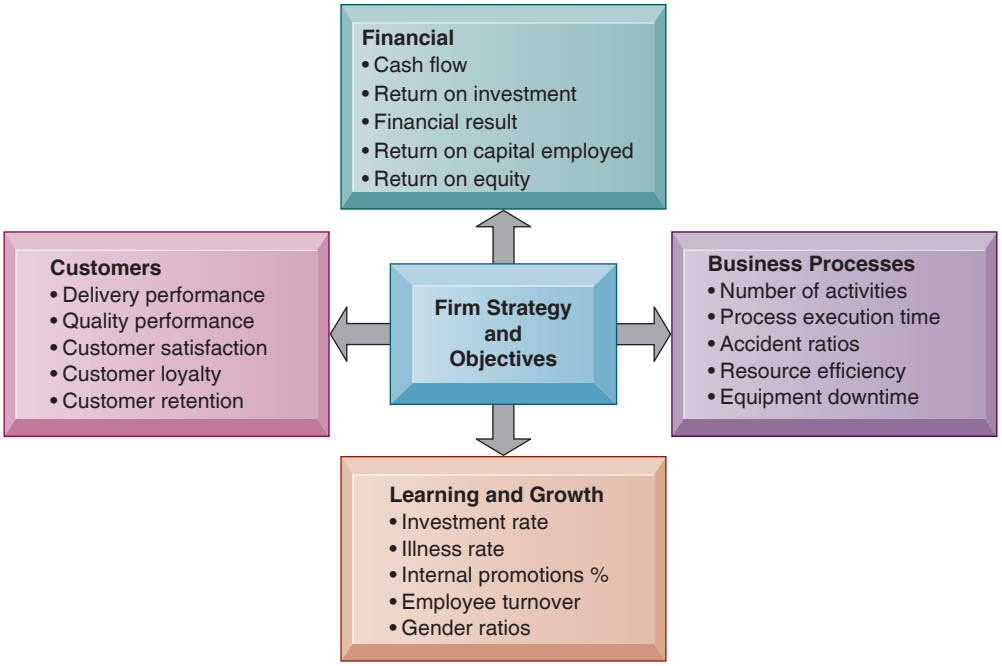
Balanced scorecard method

A leading methodology for understanding the really important information needed by an organization's executives; provides a framework for operationalizing a firm's strategic plan by focusing on measurable financial, business process, customer, and learning and growth outcomes of firm performance.

A leading methodology for understanding the really important information needed by an organization's executives is called the **balanced scorecard method** (Tawse and Tabesh, 2023; Kaplan and Norton, 1992). The balanced scorecard method is a framework for operationalizing an organization's strategic plan by focusing on measurable outcomes on four dimensions of firm performance: financial, business process, customer, and learning and growth. Examine Figure 12.7, which illustrates the balanced scorecard framework. In this figure, a firm's strategic plan is operationalized by focusing on measurable outcomes of four dimensions of firm performance: financial, business process, customer, and learning and growth.

Figure 12.7 The Balanced Scorecard Framework

In the balanced scorecard framework, an organization's strategic objectives are operationalized along four dimensions: financial, business process, customer, and learning and growth. Each dimension is measured using several KPIs.



Performance on each dimension is measured using **key performance indicators (KPIs)**, which are the measures proposed by senior management for understanding how well the firm is performing along any given dimension. Look at each of the dimensions shown in Figure 12.7 for a list of various KPIs related to that dimension. For instance, one key indicator of how well an online retail firm is meeting its customer performance objectives is the average length of time required to deliver a package to a consumer. If your firm is a bank, one KPI of business process performance is the length of time required to perform a basic function like creating a new customer account.

Key performance indicators (KPIs)

Measures proposed by senior management for understanding how well the firm is performing along specified dimensions.

The balanced scorecard framework is thought to be "balanced" because it causes managers to focus on more than just financial performance. In this view, financial performance is past history—the result of past actions—and managers should focus on the things they are able to influence today, such as business process efficiency, customer satisfaction, and employee training. Once a scorecard is developed by consultants and senior executives, the next step is automating a flow of information to executives and other managers for each of the key performance indicators.

Apple used the balanced scorecard to provide managers with a framework that they could apply to develop a holistic strategy rather than optimizing just one part of the business. Apple used five performance indicators: customer satisfaction, core competencies, employee commitment and alignment, market share, and shareholder value (Tudorache, 2021). As you learned in the Opening Case Study, the balanced scorecard method has also been adapted for use by educational organizations.

Another closely related management methodology is **business performance management (BPM)**. BPM attempts to systematically translate a firm's strategies (e.g., differentiation, low-cost producer, market share growth, and scope of operation) into operational targets. Once the strategies and targets are identified, a set of KPIs are developed that measure progress toward the targets. The firm's performance is then measured with information drawn from the firm's enterprise database systems. BPM uses the same ideas as the balanced scorecard but with a stronger strategy flavor.

Business performance management (BPM)

Management methodology that attempts to systematically translate a firm's strategies (e.g., differentiation, low-cost producer, market share growth, and scope of operation) into operational targets.

Data for a firm's ESS are supplied by the firm's existing enterprise applications (enterprise resource planning, supply chain management, and customer relationship management). ESS typically also provide access to news services, financial market databases, economic information, and other external data that senior executives require. ESS also have significant drill-down capabilities if managers need more detailed views of data.

A well-designed ESS helps senior executives monitor organizational performance, track activities of competitors, recognize changing market conditions, and identify problems and opportunities. Employees lower down in the corporate hierarchy also use these systems to monitor and measure business performance in their areas of responsibility. For these and other business intelligence systems to be truly useful, the information must be "actionable"—it must be readily available and also easy to use when making decisions. If users have difficulty identifying critical metrics within the reports they receive, employee productivity and business performance will suffer.

12.9 Understand How the Information in This Chapter Can Help Your Career

The information you have learned in Chapter 12 can help you advance your career by helping you to understand the different types of decisions that need to be made in business, how the decision-making process works, the value of improved decision making, as well as understanding why information systems do not always result in better decisions. It also will help you by introducing you to the basic technologies underlying business intelligence systems, and helping you to understand how companies are using such systems to support decision making.

Real-World Scenario

Data Analysts Help KPMG Offer New Services and Deeper Insights

KPMG is a global leader in audit, tax, and advisory services. KPMG has thousands of clients, including governments, industries, and not-for-profits, that collectively generate a huge volume of data from a myriad of sources that KPMG needs to manage and analyze. The sheer volume and complexity of KPMG's sprawling data landscape, coupled with the need for global accessibility and cost efficiency, demanded a solution that went beyond conventional data management tools. KPMG turned to Microsoft Fabric, an AI-powered analytics solution for enterprises that provides services ranging from data movement to data science, analytics, and business intelligence to meet this challenge.

Data analysts at KPMG are using Microsoft Power BI tools, which are embedded within Microsoft Fabric, to create data visualizations for analytics, dashboards, and reports. In addition, they are using Power BI's Copilot to generate narratives and create simple questions and answers for prebuilt prompts. AI is helping to expedite the development of visualizations, freeing up team members for other strategic priorities. The enhanced data analytics capability has enabled KPMG to offer new services and deliver deeper insights into operational efficiencies, market opportunities, and strategic planning. KPMG clients benefit from tailored, data-informed guidance that addresses their unique business demands.

Job Description and Career Path: Data Analyst

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Data analyst	A data analyst collects, cleans, models, and interprets data sets in order to answer a question or solve a problem. Data analysts are employed in many different industries. Job responsibilities include: • Performing data analysis based on data stored in various databases, external websites, and business intelligence platforms to produce reports for stakeholder groups and decision makers. • Eliciting data and reporting requirements using interviews, document analysis, requirements workshops, site visits, use cases, data analysis, and workflow analysis • Working with staff on the design, maintenance, and distribution of reports and incorporation of reports into balanced scorecards. • Analyzing, testing, and modifying databases and reports to meet end user specifications and quality assurance procedures • Assisting in enhancing business intelligence reporting tools, dashboards, and mobile BI to improve usability, increase user adoption, and streamline support.	• Bachelor's degree in data science, MIS, or statistics • Familiarity with Microsoft Excel and Access, SQL, R or Python, and business intelligence tools such as SAS BI, Microsoft Power BI, Tableau, etc.	• Entry-Level/Early Career: $50,00–70,000 • Mid-Career: $75,000–$100,000 • Senior: $115,000–$175,000 Salary can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

Big Data Analytics and AI: A New Way of Combating Natural Disasters

Today, there seems to be a natural disaster happening somewhere in the world almost every day. Wildfires, hurricanes, earthquakes, torrential downpours and floods, landslides, tornados and derechos, volcanoes, extreme heat and drought, extreme cold and blizzards: the list goes on and on. The frequency and severity of natural disasters appear to be increasing significantly and are costing billions of dollars a year, not to mention the cost of human lives. Examples from just the first six months of 2024 include wildfires in California and Texas; Hurricane Beryl, the earliest forming Atlantic hurricane; earthquakes in Taiwan; unprecedented flooding in the Middle East; volcanic eruptions in Iceland; and deadly tornados and derechos in the United States.

Governments are increasingly turning to technology, and particularly big data analytics, real-time data analysis, and AI, as a method of predicting, preparing for, and responding to natural disasters. Sources of big data include satellite imagery, Internet of Things (IoT) sensors, and devices that collect data on weather conditions, seismic and volcanic activity, water levels, and other environmental conditions, as well as social media and crowd-sourced data, which can provide additional insights on local conditions during a disaster. Real-time analysis of big data involves the collection, processing, and interpretation of the data as it is generated. Real-time analysis coupled with AI enables quicker identification of patterns and anomalies than humans can do and can provide enhanced accuracy in predictions about the location, magnitude, and impact of a natural disaster.

Technologies being employed include geospatial analysis, which involves the utilization of geographic information systems (GIS) to analyze and visualize spatial data, and which helps identify vulnerable areas and optimize evacuation. Predictive analytics and modeling help in estimating the impact and planning resource allocation. Social media analytics, such as sentiment analysis and keyword extraction, provide real-time information about the impact and needs of affected communities. The newest technologies include AI and machine learning, which help process large volumes of real-time data and identify complex patterns that humans might miss, allowing for faster decision making.

Using real-time big data analysis and AI enables the development of better early-warning systems and emergency response systems. By integrating real-time data into emergency response systems, first responders can gain critical insights into the situation as it unfolds. Emergency management agencies can identify trends, patterns, and hotspots for emergencies. This information allows for better allocation of resources and emergency services.

There are numerous examples of these technologies now in action. For instance, NASA's Fire Information for Resource Management System (FIRMS), launched in January 2021, was developed through an agreement between NASA and the USDA Forest Service's Geospatial Technology and Applications Center (GTAC). FIRMS distributes near real-time active data about fires and thermal anomalies within 3 hours of observation by various satellites. The data are displayed on the system's interactive global and U.S./Canada Fire Maps, a user-friendly interface that allows users to visualize and access data about the location, extent, and impact of wildfires with minimal delay.

California is a hotbed of extreme fires. The state has established a Wildfire Threat and Intelligence Integration Center, which integrates real-time National Oceanic and Atmosphere Administration (NOAA) satellite data, weather information from military flights, radar, computer-generated maps showing a 100-year history of previous burns in the area, as well as the current fuel load and its combustibility, and airplane and drone surveillance data to create a real-time picture of conditions conducive to sparking or spreading fires. Fire behavior analysts run the data through AI-enabled models that can predict what the fire is likely to do. That information is synthesized and then can be quickly relayed to fire bosses equipped with laptops and hand-held computers on the scene.

NOAA is also employing real-time big data analysis and AI to combat hurricanes. In 2023, NOAA launched a new hurricane model called the Hurricane Analysis and Forecast System (HAFS). HAFS is as good as NOAA's existing hurricane models when forecasting storm intensity but is better at predicting rapid intensification. For example, a pilot program using HAFS was the first model to accurately predict that Hurricane Ian would undergo secondary rapid intensification as the storm moved off the coast of Cuba and barreled toward southwest Florida. HAFS is also 10 to 15 percent better at predicting a hurricane's path. NOAA also uses machine learning algorithms to analyze historical data to predict the path and intensity of hurricanes.

The U.S. Geological Survey (USGS) is deploying a machine learning-based earthquake early warning system called ShakeAlert. ShakeAlert monitors for significant earthquakes and issues alerts to warn that strong shaking is expected imminently. The ShakeAlert system is being

developed to cover California, Oregon, and Washington. The U.S. Department of Defense's xView2 is an open-source project sponsored and developed by the Pentagon's Defense Innovation Unit and Carnegie Mellon's Software Engineering Institute in collaboration with research partners including Microsoft and UC Berkeley. xView2 uses ML algorithms in conjunction with satellite imagery to identify building and structural damage from earthquakes and other natural disasters and categorize severity much faster than is possible with other methods. xView2 has been used in the past by the California National Guard and in Australia in response to wildfires and most recently in Nepal after flooding and in Turkey after earthquakes. The algorithms employ a technique similar to object recognition, called "semantic segmentation," which evaluates each individual pixel of an image and its relationship to adjacent pixels to draw conclusions. xView2 can create a shared map of the affected area in minutes, which helps organizations coordinate and prioritize responses—saving time and lives.

Despite all these advances, a number of challenges still remain. One of the primary challenges is integrating disparate data sources into a single platform. Systems need to be able to collect, analyze, and visualize data from various sources, ensuring interoperability and compatibility across different technologies. While data is abundant, ensuring its quality and accuracy is crucial. Inaccurate or incomplete data can lead to poor decision making. Implementing mechanisms to validate and verify data integrity is essential. In addition, with the abundance of data being collected, privacy and security concerns are paramount. Robust protocols need to be implemented to protect sensitive data and ensure compliance with data protection regulations. Finally, getting on-the-ground organizations to use and trust AI solutions may be difficult. First responders are typically very traditional. The goal is not to replace their own time-tested intuition and experience, but to augment and enhance it.

SOURCES: Shakealert.org, accessed September 17, 2024; IEEE Public Safety Technology, "Hurricane Tracking Technology: Advancements and Opportunities," Publicsafety.ieee.org, accessed September 17, 2024; Joseph M. Smith, "Data Tool in Focus: Fire Information for Resource Management System," Earthdata.nasa.gov, June 24, 2024; "The Role of Data Science in the Management and Prediction of Natural Disasters," Moldstud.com, February 6, 2024; Julie Cart, "Drones, Satellites, and AI: How California Fights Its Unpredictable Wildfires with Analytics," Calmatters.org, July 12, 2023; Tate Ryan-Mosley, "How AI Can Actually Be Helpful in Disaster Response," MIT Technology Review, Technologyreview.com, February 20, 2023.

Case Study Questions

1. How does real-time analysis of big data help combat natural disasters?
2. How does using AI enhance real-time analysis of big data?
3. What are some of the challenges in using big data analysis and AI for these purposes?

Review Summary

12.1 Describe the different types of business decisions.

The different levels in an organization (strategic, management, operational) have different decision-making requirements. Decisions can be structured, semistructured, or unstructured, with structured decisions clustering at the operational level of the organization and unstructured decisions at the strategic level. Decision making can be performed by individuals or groups and includes employees as well as operational, middle, and senior managers.

12.2 Describe the decision-making process.

There are four stages in decision making: intelligence, design, choice, and implementation.

12.3 Describe the different roles managers play.

Early classical models of managerial activities stress the functions of planning, organizing, coordinating, deciding, and controlling. Contemporary research looking at the actual behavior of managers has found that managers' real activities are highly fragmented, varied, and brief in duration and that managers shy away from making grand, sweeping policy decisions.

Information technology provides tools for managers to carry out their various roles, enabling them to monitor, plan, and forecast with more precision and speed than ever before and to respond more rapidly to the changing business environment. Information systems have been most helpful to managers by providing support for their roles in disseminating information, providing liaisons between organizational levels, and allocating resources.

12.4 Understand the value of improved decision making.

One of the main contributions of information systems has been to improve decision making for both individuals and groups. Decisions are made at all levels of the firm. Some of these decisions are common, routine, and numerous. Although the value of improving any single decision

may be small, improving hundreds of thousands of "small" decisions can add up to a large annual value for the business.

12.5 Understand why information systems do not always result in better decisions.

Information systems are not as successful at supporting unstructured decisions. Where information systems are useful, information quality, management filters, and organizational culture can degrade decision making. High-velocity automated decision making eliminates humans from the decision chain, and great care needs to be taken to ensure the proper operation of these systems to prevent harm. Although AI can help managers make faster and better decisions by providing improved prediction based on real-time data and acting as a "copilot" and sounding board, using AI for management decision making also presents a number of risks.

12.6 Discuss how data analytics, BI, and AI support decision making.

Business intelligence (BI) is a term used by hardware and software vendors and information technology consultants to refer to a collection of methods, systems, and tools (or infrastructure) to convert unprocessed data into actionable business insights. Data analytics forms the underpinning of BI. Business analytics (BA) is also a vendor-defined term and can be viewed as a combination or subset of both data analytics and BI. BA focuses on tools and techniques for analyzing and understanding business data. A BI environment consists of data from the business environment, the BI infrastructure, a BA toolset, managerial users and methods, a BI delivery platform (MIS, DSS, or ESS), and the user interface.

12.7 Describe the capabilities of BI systems.

BI and BA promise to deliver correct, nearly real-time information to decision makers, and analytic tools help them quickly understand the information and take action. There are six primary analytic functionalities that BI systems deliver to achieve these ends: predefined production reports, parameterized reports, dashboards and scorecards, ad hoc queries and searches, the ability to drill down to detailed views of data, and the ability to model scenarios, create forecasts, and provide recommendations for action. Predictive analytics, prescriptive analytics, big data analytics, operational intelligence, and location analytics are important BI capabilities.

12.8 Understand the different types of BI users.

Operational and middle management are generally charged with monitoring the performance of their firm. Most of the decisions they make are fairly structured. An MIS producing routine production reports is typically used to support this type of decision making. For making unstructured decisions, middle managers and analysts typically use a DSS with analytics and modeling tools, including spreadsheets and pivot tables. Senior executives making unstructured decisions typically use an ESS that provides dashboards and visual interfaces that can display key performance information affecting the overall profitability, success, and strategy of the firm. The balanced scorecard and business performance management are two methodologies used in designing an ESS.

Key Terms

Augmented reality (AR), 405
Balanced scorecard method, 414
Behavioral model of management, 399
Business performance management (BPM), 415
Choice, 398
Classical model of management, 399
Confirmation bias, 402
Data analytics, 403
Data visualization, 404
Decisional role, 400
Design, 398
Geographic information system (GIS), 411
Implementation, 398
Informational role, 400
Intelligence, 398
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Predictive analytics, 407
Prescriptive analytics, 407
Semistructured decision, 397
Sensitivity analysis, 412
Structured decision, 397
Unstructured decision, 397
Virtual reality (VR), 405

Review Questions

12.1 Describe the different types of business decisions.

- List and describe the different types of business decisions.
- Describe the different levels of decision making and decision-making constituencies in an organization and how their decision-making requirements differ.

12.2 Describe the decision-making process.

- List and describe the stages in decision making.

12.3 Describe the different roles managers play.

- Compare the descriptions of managerial behavior in the classical and behavioral models.

- Identify the specific managerial roles that can be supported by information systems.

12.4 Understand the value of improved decision making.

- Explain how the value of improved decision making can be determined.

12.5 Understand why information systems do not always result in better decisions.

- Discuss the main reasons why information systems do not always result in better decisions.
- Describe some of the risks introduced by using AI for decision making.

12.6 Discuss how data analytics, BI, and AI support decision making.

- Explain the role that data analytics plays in BI systems.
- Define and describe BI and BA.
- Describe how AI is being used for data analytics and BI tools.
- List and describe the elements of a business intelligence environment.

12.7 Describe the capabilities of BI systems.

- List and describe the analytic functionalities provided by BI systems.
- Define predictive analytics and prescriptive analytics and explain how they differ.
- Discuss how companies are using big data analytics and provide two examples.
- Explain the purpose of location analytics and GIS.

12.8 Understand the different types of BI users.

- List each of the major decision-making constituencies in an organization and describe the types of decisions each makes.
- Describe how MIS, DSS, or ESS provides decision support for each of these groups.
- Define and describe the balanced scorecard method and business performance management.

Discussion Questions

12.9 As a manager or user of information systems, what would you need to know to participate in the design and use of a DSS or an ESS? Why?

12.10 If businesses used DSS and ESS more widely, would managers and employees make better decisions? Why or why not?

12.11 How much can business intelligence and business analytics help companies refine their business strategy? Explain your answer.

Using AI Project

12.12 AI Project: Identifying Business Decisions

You have been hired as a consultant by two friends who want to open a small retail store selling organic pet food and treats, as well as pet toys and clothing made from organic materials. They would like to know the kinds of decisions they will need to make in order to operate the store. They would also like to know which of these decisions are likely to be structured (routine with clear-cut procedures for making them) and which are likely to be unstructured (complex and without clear-cut procedures).

Using Microsoft's Copilot, Google's Gemini, or ChatGPT, develop a prompt to assist you in answering their questions. Save a copy of your prompt and the output of the chat session. Submit it to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience identifying opportunities for DSS, using a pivot table to analyze data, and using Tableau to create data visualizations.

For student files please visit www.pearsonhighered.com/laudon or MyLab MIS.

12.13 Management Decision Problem

Samsung sells around 80 million new mobile phones every quarter. Samsung has marketing and analytics teams with access to the massive quantities of complex customer data flowing into its systems from millions of active devices. The team was using standard business intelligence (BI) and dashboarding tools. However, these traditional business intelligence tools couldn't keep up with the volume and complexity of Samsung's data. It took too long to analyze these data with hundreds of variables in order to understand user preferences across demographics, location, mobile device

profiles, carrier loyalty, past interactions with Samsung products, and customers to be targeted by a new product launch. Which customers are likely to upgrade to a new device? What factors influence an upgrade decision? It might take weeks to answer a single question. There was no way the team could reliably check every possible factor in the data. Identify Samsung's problem and suggest a solution. Identify the management, organizational, and technology issues to be addressed by the solution.

12.14 Data Analytics: Using Pivot Tables

Software skills: Pivot tables

Business skills: Analyzing sales data

This project gives you an opportunity to learn how to use Excel's PivotTable feature to analyze a database or data list. Use the MIS18_Project 12.14 Student File, which contains a list of the sales transactions for Online Management Training Inc. (OMT) for one day. Use Excel's PivotTable feature to help you answer the following questions:

- Where are the average purchases higher? The answer might tell managers where to focus marketing and sales resources, or pitch different messages to different regions.
- What form of payment is the most common? The answer could be used to emphasize in advertising the most preferred means of payment.
- Are there any times of day when purchases are most common? Do people buy more products while at work (likely during the day) or at home (likely in the evening)?
- What's the relationship between region, type of product purchased, and average sales price?

12.15 Using Tableau for Data Visualization

Software skills: Tableau software

Business skills: Data visualization

You have been hired as a consultant to develop data-driven insights about local government employment and payroll. Access the MIS18_Project 12.15 Student File for data about local government employment and payroll by function in the United States on a state-by-state as well as regional basis. The file also contains aggregated data for the United States as a whole. Using Tableau, create a series of at least five visualizations, such as bar charts, pie charts, scatter plots, heat maps, treemaps, or others of your own choosing that you combine into a dashboard that will help you develop insights into the data. For example, what are the top 25 states in terms of total local government employment? What are the bottom 25 states? Is there any correlation to the region and regional division in which those states are located? Across the United States, which ten functions have the highest total payroll costs? How do the metrics for each region for those functions compare to the national average? Are they higher or lower, and if so, what might be some potential reasons?

12.16 Collaboration and Teamwork Project: Data-Driven Analytics in Sports

With two or more of your classmates, select a sport, such as football, baseball, basketball, soccer, or another of your own choice. Use online research tools to examine how the sport uses data analytics to improve team performance or increase ticket sales to events. If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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Part Four
Building and Managing Systems

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Chapter 13: Building Information Systems

Chapter 14: Making the Business Case for Information Systems and Managing Projects

Chapter 15: Managing Global Systems

PART FOUR shows how to use the knowledge acquired in earlier chapters to analyze and design information system solutions to business problems. This part answers questions such as these: How can I develop a solution to an information system problem that provides genuine business benefits? How can the firm adjust to the changes introduced by the new system solution? What alternative approaches are available for building system solutions? What is the impact of artificial intelligence on systems development? How can companies manage and derive the most benefit from global systems?

13

Building Information Systems

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Learning Objectives

After completing this chapter, you will be able to:

13.1 Explain how new information systems can promote organizational change.

13.2 Understand how IT can be used to manage and redesign business processes.

13.3 Describe the core activities in the systems development process.

13.4 Discuss methods for building and acquiring information systems.

13.5 Describe methods that enable rapid systems development.

13.6 Discuss the increasing democratization of software development.

13.7 Understand the impact of AI on software development.

13.8 Discuss approaches for building mobile applications.

13.9 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** Signet Jewelers Sparkles with New Virtual Sales Systems
- **Spotlight on Management Case Study:** Meet the Citizen Developers
- **Chapter Case Study:** Process Mining with Celonis

Opening Case Study

Signet Jewelers Sparkles with New Virtual Sales System

Purchasing an engagement ring or other diamond jewelry has traditionally been a very high-touch, in-person process, requiring a customer to visit a physical jewelry store. A salesperson pulls out pieces of interest from a display case for the customer to examine and perhaps try on. If the customer chooses to purchase a particular ring, the customer might have to wait one or two weeks for the ring to be properly sized, or even longer if the ring has to be custom-made. Once the ring is ready, the buyer completes payment and either picks up the ring in the store or has it shipped to the buyer's home.

In 2018, Signet Jewelers, the world's largest diamond retailer, embarked on a "Path to Brilliance" transformation plan. A primary focus of the plan was to enhance the company's e-commerce and omnichannel capabilities, which at the time only accounted for about 8 percent of its revenues. The company began by inventorying its various business processes and examining how they could be improved. Then the Covid-19 pandemic hit. Signet was forced to close the physical stores operated by its various brands, such as Kay Jewelers, Zales, and Jared, and accelerate the transformation of the jewelry the shopping process. To do so, Signet deployed information technology and created new systems so that customers could research and purchase engagement rings and other fine jewelry online from their homes.

Signet introduced virtual consultations that enable customers to communicate with a sales consultant via text, voice, or video to ask questions. Potential buyers can use computers or a mobile device to book one-hour virtual appointments with a consultant by visiting the website of a Signet brand such as Zales.

Signet deployed digital photographic tools that let buyers enlarge the photo of a diamond 40 times its real size to see the diamond's cut or check for imperfections. Signet also added visual search and collaborative browsing tools. Customers can upload a picture of a necklace or ring that they like to find that product or something similar online. Browsing technology allows sales or service agents to access a customer's browser and navigate it in real time. With joint navigation of a webpage, agents can interact and collaborate online with customers, facilitating customer support or sales assistance. After making a selection, the customer can complete the purchase online, and the product will be shipped to the customer's home. Signet also revised its appointment application to support virtual appointments.

It was a bit of a challenge for Signet to get its 2,000+ sales consultants, who were accustomed to showing off merchandise in stores, to replicate the same experience virtually. Signet had already started moving

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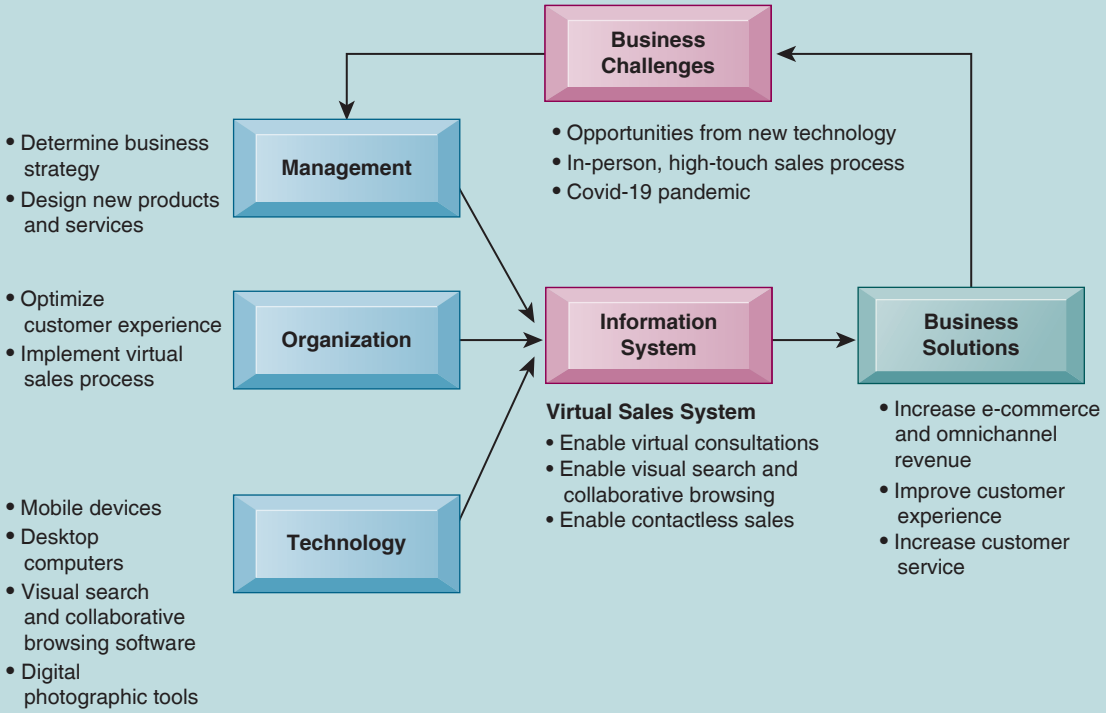
into e-commerce sales, and its virtual sales during the pandemic strengthened the company's digital sales channel. Despite the pandemic, Signet's 2020 holiday sales were the best in nine years, and since then, sales from all channels have continued to grow. The company's business strategy has shifted from TV-centric marketing and expanding the number of physical stores to an omnichannel marketing mix and more e-commerce retail sales.

Signet is also experimenting with artificial intelligence (AI). It recently launched a forecasting and replenishment project that uses AI and machine learning. Since fine jewelry is high value, Signet typically stocks only one unit of each piece per store. Advanced algorithms will enable the company to reduce lost sales resulting from out-of-stock merchandise. It also has plans to use AI and data-driven content to create more personalized shopping experiences.

As Signet's stores reopened, the virtual showcasing and contactless sales processes were so popular that Signet added more than 700 full-time digital consultants to continue the virtual shopping consultations. Jewelry consultants working in physical stores have continued to use virtual consultations with customers. More than two-thirds of Signet's total sales now include both in-store and online engagement.

SOURCES: Signetjewelers.com, accessed September 22, 2024; Accenture, "Making a Jewelry Retailer Shine," Accenture.com, accessed September 22, 2024; Signet Jewelers, "August 2023 Investor Presentation," Signetjewelers.com, August 2023; Pam Danziger, "Virginia Drosos Inspires Brilliance at Signet," Therobinreport.com, May 19, 2023; "How Tech Is Transforming the Jewelry Shopping Experience: A Q&A with Signet Jewelers' CIO, Howard Melnick," Signetjewlers.com, May 3, 2023; Kenneth Hein, "How Diamond Retailer Signet Jewellers Got Its E-commerce Channels to Sparkle," Thedrum.com, March 21, 2021.

Signet Jewelers' experience illustrates how a company can benefit from new business processes and systems. Signet used IT to create virtual sale processes that kept sales going remotely during the Covid-19 pandemic and thereafter, even when the company returned to in-person



shopping. Signet now does business as an omnichannel retailer offering a seamless mix of mobile, digital, virtual, and in-person engagement.

The Opening Case Study diagram above calls attention to important points raised by this case and this chapter. The traditional process for selling fine jewelry was highly people-intensive, requiring considerable person-to-person interaction between jewelry sales staff and the customer in physical stores. Signet realized it needed to make the sales process more digital, both to succeed in e-commerce and to keep operating during the Covid-19 pandemic. It was able to create a solution by redesigning its sales processes and implementing new systems so that sales could take place virtually yet still offer all the benefits of a person-to-person experience and more.

Selecting and implementing this solution also illustrates some of the steps required to design and build new information systems. Creating an effective virtual sales channel entailed analyzing the organization's problems with existing systems, assessing information requirements, selecting appropriate technology, and redesigning business processes and jobs. Management had to oversee the systems-building effort and evaluate benefits and costs. The information requirements were incorporated into the design of the new system, which represented a process of planned organizational change.

Case Study Questions

1. Describe Signet's traditional business strategy and business model. How did they change when the company embraced e-commerce and a virtual sales process?
2. Compare the in-store process of buying a diamond ring with Signet's virtual sales process. Diagram or list the steps in each process.

CAREER BENEFIT

Signet Jewelers' sales process redesign and virtual sales system were critical to the growth and survival of the company during a prolonged crisis that shut down most physical retail businesses. The system also provided the technical foundation for an omnichannel business model. IT specialists, managers, and sales staff who helped design and build the system were rewarded within the company. Doing so also allowed them to add these skills to their resumes, which will benefit them in the event they choose to look for a position with a different company in the future. Sales staff benefit from having an additional sales channel that provides the company with more revenue.

13.1 Explain How New Information Systems Can Promote Organizational Change

Building a new information system is one kind of planned organizational change. The introduction of a new information system involves much more than new hardware and software. It also includes changes in jobs, skills, management, and organization. When we design a new information system, we are redesigning the organization. Systems builders must understand how a system will affect specific business processes and the organization as a whole.

Systems Development and Organizational Change

Information technology can promote various degrees of organizational change, ranging from incremental to far-reaching. Examine Figure 13.1, which shows four kinds of structural organizational change that are enabled by information technology: (1) automation, (2) rationalization, (3) business process redesign, and (4) paradigm shifts. Each carries different risks and rewards.

The most common form of IT-enabled organizational change is **automation**, which involves using information technology to assist employees with performing their tasks more efficiently and effectively. Calculating paychecks and payroll registers, giving bank tellers instant access to customer deposit records, and developing a nationwide reservation network for airline ticket agents are all examples of early automation.

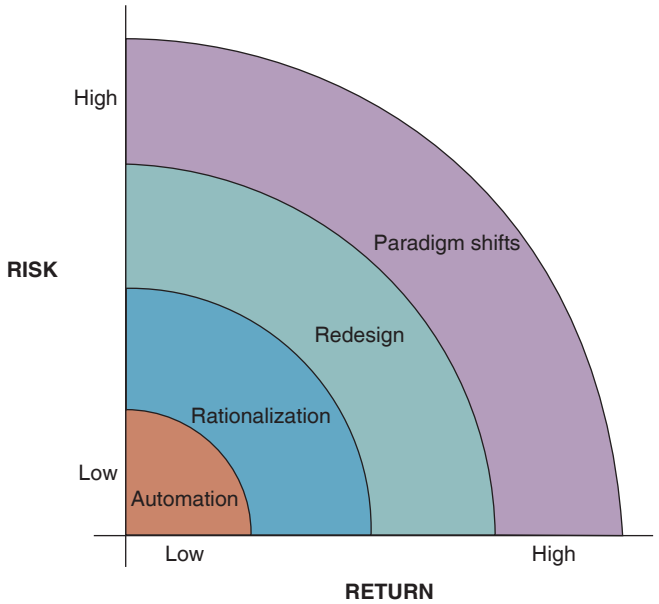
Automation
Using information technology to assist employees with performing their tasks more efficiently and effectively.

A deeper form of organizational change—one that follows quickly from early automation—is rationalization of procedures. Automation frequently reveals new bottlenecks in production and makes the existing arrangement of procedures and structures painfully cumbersome. **Rationalization of procedures** involves the streamlining of standard operating procedures. For example, Signet Jewelers' new virtual sales system is effective not only because it uses digital technology but also because the company simplified its business processes for this function. Fewer manual steps are required.

Rationalization of procedures
The streamlining of standard operating procedures.

Figure 13.1 Organizational Change Carries Risks and Rewards

The most common forms of organizational change are automation and rationalization of procedures. These relatively slow-moving and slow-changing strategies present modest returns but little risk. Faster and more comprehensive change—such as redesign and paradigm shifts—carries high rewards but offers substantial chances of failure.



Rationalization of procedures is often found in programs for making a series of continuous quality improvements in products, services, and operations, such as total quality management (TQM) and Six Sigma. **Total quality management (TQM)** makes achieving quality an end in itself and a responsibility shared by all people and functions within an organization. TQM derives from concepts developed by American quality experts such as W. Edwards Deming and Joseph Juran, but it was popularized by Japanese organizations. **Six Sigma** is a specific measure of quality, representing 3.4 defects per million opportunities. Most companies cannot achieve this level of quality but use Six Sigma as a goal for driving ongoing quality improvement programs.

Total quality management (TQM)
A concept that makes achieving quality an end in itself and a responsibility shared by all people and functions within an organization.

Six Sigma
A specific measure of quality, representing 3.4 defects per million opportunities; used to designate a set of methodologies and techniques for improving quality and reducing costs.

A more powerful type of organizational change is **business process redesign (BPR),** in which business processes are analyzed, simplified, and redesigned. Business process redesign reorganizes workflows, combining steps to cut waste and eliminate repetitive, paper-intensive tasks. (Sometimes the new design eliminates jobs as well.) It is much more ambitious than rationalization of procedures, requiring a new vision of how the process is to be organized.

Business process redesign (BPR)
Type of organizational change in which business processes are analyzed, simplified, and redesigned.

Rationalizing procedures and redesigning business processes are limited to specific parts of a business. New information systems can ultimately affect the design of the entire organization by transforming how the organization carries out its business or even the nature of the business. For instance, the long-haul trucking and transportation firm Schneider National used new information systems to change its business model. Schneider created a new business managing logistics for other companies. This more radical form of organizational change is called a **paradigm shift**. A paradigm shift involves rethinking the nature of the business and the nature of the organization.

Paradigm shift
Radical form of organizational change that involves the nature of the business and the nature of the organization.

Paradigm shifts and business process redesign often fail because extensive organizational change is so difficult to orchestrate (see Chapter 14). Why, then, do so many organizations contemplate such radical change? Because the rewards are equally high, as shown in Figure 13.1. In many instances, firms seeking paradigm shifts and pursuing reengineering strategies achieve stunning, order-of-magnitude increases in their returns on investment (or productivity). Some of these success stories, and some failure stories, are included throughout this book.

13.2 Understand How IT Can Be Used to Manage and Redesign Business Processes

Like Signet Jewelers, described in the Opening Case Study, many businesses today are trying to use information technology to improve their business processes. Some of these systems entail incremental process change, but others require more far-reaching redesign of business processes. To deal with these changes, organizations have turned to business process management. **Business process management (BPM)** provides a variety of tools and methodologies to analyze existing processes, design new processes, and optimize those processes. BPM is never concluded because process improvement requires continual change. Companies practicing business process management go through the following steps:

Business process management (BPM)
Tools and methodologies to analyze existing processes, design new processes, and optimize those processes.

1. **Identify processes for change:** One of the most important strategic decisions that a firm can make is not deciding how to use technology to improve business processes but understanding what business processes need improvement. When systems are used to strengthen the wrong business model or business processes, the business can become more efficient at doing what it should not do. As a result, the firm becomes vulnerable to competitors who may have discovered the right business model. Considerable time and cost may also be spent improving business processes that have little impact on overall firm performance and revenue. Managers need to determine what business processes are the most important and how improving these processes will help business performance.
2. **Analyze existing processes:** Existing business processes should be modeled and documented, noting inputs, outputs, resources, and the sequence of activities. The process design team identifies redundant steps, paper-intensive tasks, bottlenecks, and other inefficiencies.

3. **Design the new process:** Once the existing process is mapped and measured in terms of time and cost, the process design team will try to improve the process by designing a new one. A new streamlined "to-be" process will be documented and modeled for comparison with the old process.
4. **Implement the new process:** Once the new process has been thoroughly modeled and analyzed, it must be translated into a new set of procedures and work rules. New information systems or enhancements to existing systems may have to be implemented to support the redesigned process. The new process and supporting systems are rolled out into the business organization. As the business starts using this process, problems are uncovered and addressed. Employees working with the process may recommend improvements.
5. **Continuous measurement:** Once a process has been implemented and optimized, it needs to be continually measured. Why? Processes may deteriorate over time as employees fall back on old methods, or they may lose their effectiveness if the business experiences other changes.

Examine Figure 13.2, which illustrates the "as-is" process for purchasing a book from a physical bookstore. A customer visits a physical bookstore and searches its shelves for a book. If the customer finds the book, the customer takes it to the checkout counter and pays for it via credit card, cash, or check. If the customer is unable to locate the book, the customer must ask a bookstore clerk to search the shelves or check the bookstore's inventory records to see if it is in stock. If the clerk finds the book, the customer purchases it and leaves. If the book is not available locally, the clerk inquires about ordering it for the customer from the bookstore's warehouse or from the book's distributor or publisher. Once the ordered book arrives at the bookstore, a bookstore employee telephones the customer with this information. The customer would have to go to the bookstore again to pick up the book and pay for it. If the bookstore is unable to order the book for the customer, the customer would have to try another bookstore. You can see that this process has many steps and might require the customer to make multiple trips to the bookstore.

In contrast, examine Figure 13.3, which illustrates how the book-purchasing process can be redesigned by taking advantage of the Internet. The customer accesses an online bookstore over the Internet from the customer's computer. The customer searches the bookstore's online

Figure 13.2 As-Is Business Process for Purchasing a Book from a Physical Bookstore

Purchasing a book from a physical bookstore requires many steps to be performed by both the seller and the customer.

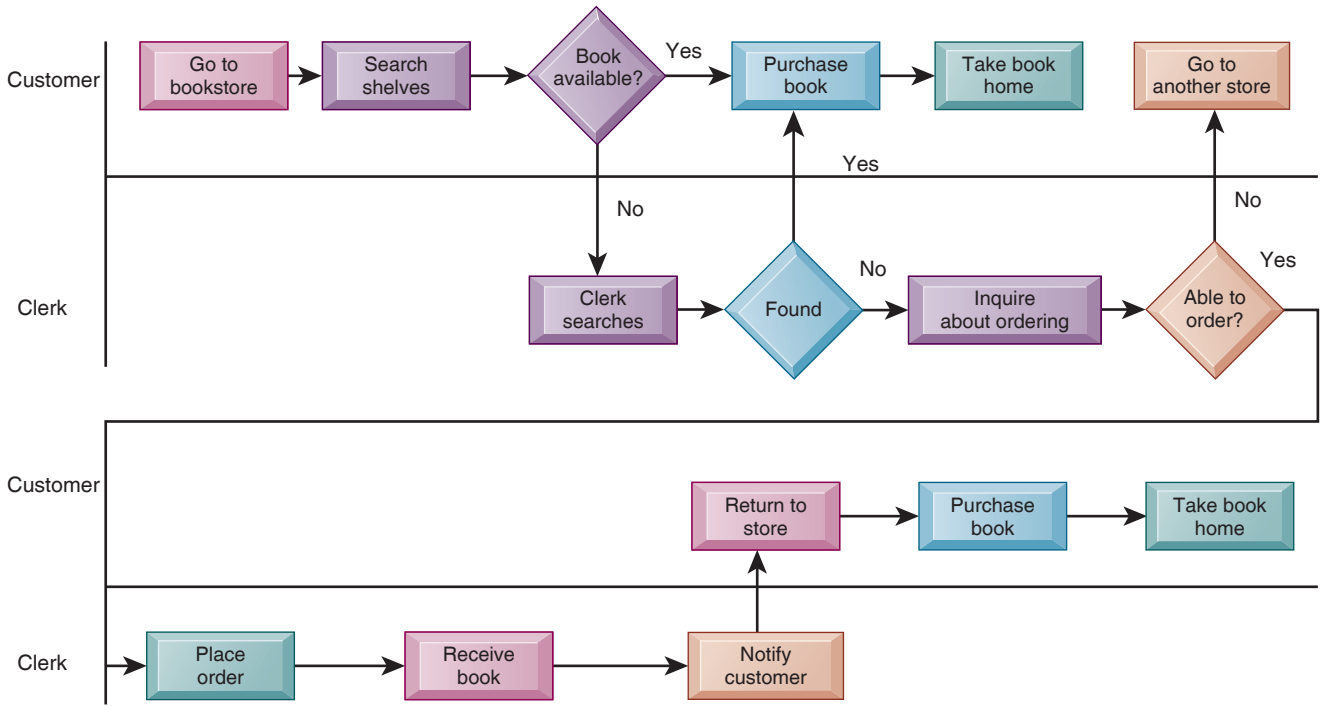
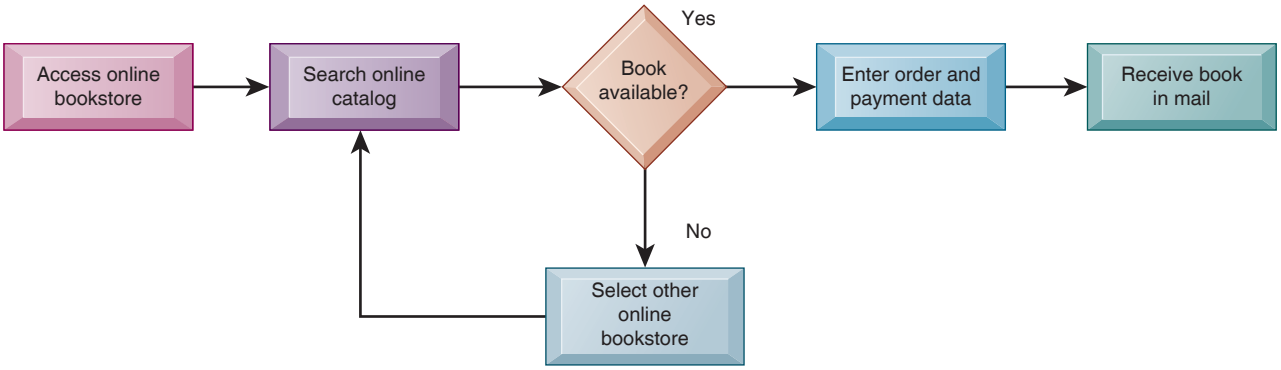


Figure 13.3 Redesigned Process for Purchasing a Book Online

Using Internet technology makes it possible to redesign the process for purchasing a book so that it requires fewer steps and consumes fewer resources.



catalog for the book the customer wants. If the book is available, the customer orders the book online, supplying credit card and shipping address information, and the book is delivered to the customer's home. If the online bookstore does not carry the book, the customer selects another online bookstore and searches for the book again. This process has far fewer steps than those for purchasing the book in a physical bookstore, requires much less effort on the part of the customer, and requires less sales staff for customer service. The new process is therefore much more efficient and saves the customer time.

The new process design needs to be justified by showing how much it reduces time and cost or enhances customer service and value. Management first measures the time and cost of the existing process as a baseline. In our example, the time required for purchasing a book from a physical bookstore might range from 15 minutes (if the customer immediately finds the desired book) to 30 minutes if the book is in stock but has to be located by sales staff. If the book has to be ordered from another source, the process might take one or two weeks and another trip to the bookstore for the customer. If the customer lives far away from the bookstore, the time to travel to the bookstore would have to be factored in. The bookstore will have to pay the costs for maintaining a physical store and keeping the book in stock, for sales staff on site, and for shipment costs if the book has to be obtained from another location.

The new process for purchasing a book online might only take several minutes, although the customer might have to wait several days or a week to have the book delivered and might have to pay a shipping charge. However, the customer saves time and money by not having to travel to the bookstore or make additional visits to pick up the book. Booksellers' costs are lower because they do not have to pay for a physical store location or for local inventory.

Although many business process improvements are incremental and ongoing, there are occasions when more radical change must take place. Our example of a physical bookstore redesigning the book-purchasing process so that it can be carried out online is an example of this type of radical, far-reaching change. When properly implemented, business process redesign produces dramatic gains in productivity and efficiency and may even change the way the business is run. In some instances, it drives a "paradigm shift" that transforms the nature of the business itself.

This actually happened in book retailing when Amazon challenged traditional physical bookstores with its online retail model and Kindle e-book reader. By radically rethinking the way a book can be published, purchased, and sold, Amazon and other online book retailers have achieved remarkable efficiencies, cost reductions, and a whole new way of doing business.

BPM poses challenges, however. Executives report that the largest single barrier to successful business process change is organizational culture. Employees do not like unfamiliar routines and often try to resist change. This is especially true of projects where organizational changes are very ambitious and far-reaching. Managing change is neither simple nor intuitive, and companies committed to extensive process improvement need a good change management strategy (see Chapter 14).

Tools for Business Process Management and Process Mining

There are many software tools for various aspects of BPM, including IBM Workflow Automation, Oracle Business Process Management, and TIBCO BPM Enterprise. These tools help businesses identify and document processes requiring improvement, create models of improved processes, capture and enforce business rules for performing processes, and integrate existing systems to support new or redesigned processes. BPM software tools also provide analytics for verifying that process performance has been improved and for measuring the impact of process changes on key business performance indicators.

Process mining can extract even more value from business processes. **Process mining** analyzes data from a company's event logs to identify problems with certain processes and figure out ways to streamline them. Enterprise information systems, such as enterprise resource planning (ERP) or customer relationship management (CRM) software, create audit trails of work processes that are recorded as log data. Process mining takes that log data and creates models that show the visualization of a process from end to end. The process map uncovers the details and all the variations that a process can take. Sometimes what process mining uncovers does not match up with the company's definition of that particular process. Process mining detects variations and prioritizes areas for improvement. The Chapter Case Study on Celonis, a leading process mining technology company, provides more detail on this topic.

Process mining

Analyzes data from a company's event logs to identify problems with processes and figure out ways to streamline them.

13.3 Describe the Core Activities in the Systems Development Process

New information systems are an outgrowth of organizational problem solving. A new information system is built as a solution to some type of problem or set of problems the organization perceives it is facing. The problem may be one in which managers and employees realize that the organization is not performing as well as expected or that the organization should take advantage of new opportunities to perform more successfully.

The activities that go into producing an information system solution to an organizational problem or opportunity are called **systems development**. Systems development is a structured kind of problem solved with distinct activities. These activities consist of systems analysis, systems design, programming, testing, conversion, and production and maintenance.

Systems development

The activities that go into producing an information systems solution to an organizational problem or opportunity.

Examine Figure 13.4, which illustrates the systems development process. The systems development activities depicted usually take place in sequential order. However, some of the activities may need to be repeated, and some may take place simultaneously, depending on the approach to systems building that is being employed (see Section 13.4).

Systems Analysis

Systems analysis is the analysis of a problem that a firm tries to solve with an information system. It consists of defining the problem, identifying its causes, specifying the solution, and identifying the information requirements that must be met by a system solution.

Systems analysis

The analysis of a problem that the organization will try to solve with an information system.

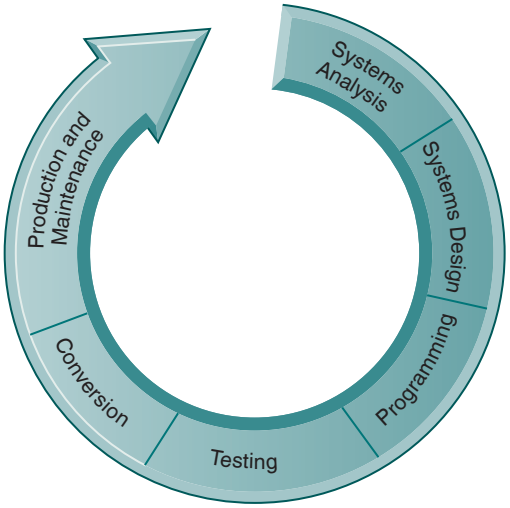
The systems analyst creates a road map of the existing organization and systems, identifying the primary owners and users of data along with existing hardware and software. The systems analyst then details the problems of existing systems. By examining documents, work papers, and procedures, observing system operations, and interviewing key users of the systems, the analyst can identify the problem areas and objectives a solution would achieve. Often, the solution requires building a new information system or improving an existing one. The systems analysis also includes a **feasibility study** to determine whether that solution is feasible, or achievable, from a financial, technical, and organizational standpoint. The feasibility study

Feasibility study

Conducted as part of the systems analysis process to determine whether the solution is achievable, given the organization's resources and constraints.

Figure 13.4 The Systems Development Process

Building a system can be broken down into six core activities.



determines whether the proposed system is expected to be a good investment, whether the technology needed for the system is available and can be handled by the firm's information systems specialists, and whether the organization can handle the changes introduced by the system.

Normally, the systems analysis process identifies several alternative solutions that the organization can pursue and assesses the feasibility of each. A written systems proposal report describes the costs and benefits, as well as the advantages and disadvantages, of each alternative. It is up to management to determine which mix of costs, benefits, technical features, and organizational impacts represents the most desirable alternative.

Establishing Information Requirements

Information requirements

A detailed statement of the information needs that a new system must satisfy; identifies who needs what information, and when, where, and how the information is needed.

Perhaps the most challenging task of the systems analyst is to define the specific information requirements that must be met by the chosen system solution. At the most basic level, the **information requirements** of a new system involve identifying who needs what information, where, when, and how. Requirements analysis carefully defines the objectives of the new or modified system and develops a detailed description of the functions that the new system must perform. Faulty requirements analysis is a leading cause of systems failure and high systems development costs (see Chapter 14). A system designed around the wrong set of requirements will either have to be discarded because of poor performance or will need to undergo major modifications. Section 13.4 describes alternative approaches to eliciting requirements that help minimize this problem.

Some problems do not require an information system solution but instead need an adjustment in management, additional training, or refinement of existing organizational procedures. If the problem is information-related, systems analysis may still be required to diagnose the problem and arrive at the proper solution.

Systems Design

Systems design

Details how a system will meet the information requirements identified by the systems analysis.

Systems analysis describes what a system should do to meet information requirements, and **systems design** details how the system will fulfill this objective. The design of an information system is the overall plan or model for that system. Like the blueprint of a building or house, it consists of all the specifications that give the system its form and structure.

The systems designer designates the system specifications that will deliver the functions identified during systems analysis. These specifications should address all of the managerial, organizational, and technological components of the system solution. Table 13.1 lists the types of specifications that would be produced during systems design.

Table 13.1 Systems Design Specifications

Output	Processing	Documentation
Medium	Computations	Operations documentation
Content	Program modules	Systems documentation
Timing	Required reports	User documentation
Input	Timing	**Conversion**
Origins	**Manual Procedures**	Data conversion rules
Flow	What activities	Testing method
Data entry	Who performs them	Conversion strategy
User Interface	When	**Training**
Simplicity	How	Training techniques
Efficiency	Where	Training modules
Logic	**Controls**	**Organizational Changes**
Feedback	Input controls (characters, limit, reasonableness)	Task redesign
Errors	Processing controls (consistency, record counts)	Job design
Database Design	Output controls (totals, samples of output)	Process design
Logical data model	Procedural controls (passwords, special forms)	Organization structure design
Volume and speed requirements	**Security**	Reporting relationships
Record specifications	Access controls	
	Catastrophe plans	
	Audit trails	

Like houses or buildings, information systems may have many possible designs. Each design represents a unique blend of technical and organizational components. What makes one design superior to others is the ease and efficiency with which it fulfills user requirements within a specific set of technical, organizational, financial, and time constraints.

The Role of End Users

User information requirements drive the entire system-building effort. Users must have sufficient control over the design process to ensure that the system reflects their business priorities and information needs, not the biases of the technical staff. Working on design increases users' understanding and acceptance of the system. As described in Chapter 14, insufficient user involvement in the design effort is a major cause of system failure. However, some systems require more user participation in design than others, and Section 13.4 shows how alternative systems development methods address the user participation issue.

Completing the Systems Development Process

The remaining steps in the systems development process translate the solution specifications established during systems analysis and design into a fully operational information system. These concluding steps consist of programming, testing, conversion, production, and maintenance.

Programming

During the **programming** stage, system specifications that were prepared during the design stage are translated into software program code. Today, many organizations no longer do their own programming for new systems. Instead, they purchase the software that meets the requirements for a new system from external sources such as software packages from a commercial software vendor, software services from a software service provider, or outsourcing firms that develop custom application software for their clients (see Section 13.4).

Programming
The process of translating the system specifications prepared during the design stage into software program code.

Testing

Testing
Process that determines whether the system produces the desired results under known conditions.

Exhaustive and thorough **testing** must be conducted to ascertain whether the system produces the right results. Testing answers the question: Will the system produce the desired results under known conditions? Some companies are starting to use cloud computing services for this work.

The amount of time needed to answer this question has been traditionally underrated in systems project planning (see Chapter 14). Testing is time-consuming: Test data must be carefully prepared, results reviewed, and corrections made in the system. In some instances, parts of the system may have to be redesigned. The risks resulting from glossing over this step are enormous. Testing an information system can be broken down into three types of activities: unit testing, system testing, and acceptance testing. **Unit testing**, or program testing, consists of testing each program separately in the system. It is widely believed that the purpose of such testing is to guarantee that programs are error-free, but this goal is realistically impossible. Testing should be viewed instead as a means of locating errors in programs, by focusing on finding all the ways to make a program fail. Once they are pinpointed, problems can be corrected.

Unit testing
The process of testing each program separately in the system. Sometimes called program testing.

System testing
Tests the functioning of the information system as a whole in order to determine if discrete modules will function together as planned.

System testing tests the functioning of the information system as a whole. It tries to determine whether discrete modules will function together as planned and whether discrepancies exist between the way the system actually works and the way it was conceived. Among the areas examined are performance time, capacity for file storage and handling peak loads, recovery and restart capabilities, and manual procedures.

Acceptance testing
Provides the final certification that the system is ready to be used in a production setting.

Acceptance testing provides the final certification that the system is ready to be used in a production setting. Systems tests are evaluated by users and reviewed by management. When all parties are satisfied that the new system meets their standards, the system is formally accepted for installation. The systems development team works with users to devise a systematic test plan. The **test plan** includes all of the preparations for the series of tests just described. Examine Figure 13.5, which shows an example of a test plan. The general condition being tested is a record change. The documentation consists of a series of test plan screens maintained on a database (perhaps a PC database) that is ideally suited to this kind of application.

Test plan
Prepared by the development team in conjunction with the users; it includes all of the preparations for the series of tests to be performed on the system.

Conversion

Conversion
The process of changing from the old system to the new system.

Conversion is the process of changing from the old system to the new system. Four main conversion strategies can be employed: the parallel strategy, the direct cutover strategy, the pilot study strategy, and the phased approach strategy.

Figure 13.5 A Sample Test Plan to Test a Record Change

When developing a test plan, it is imperative to include the various conditions to be tested, the requirements for each condition tested, and the expected results. Test plans require input from both end users and information systems specialists.

Procedure	Address and Maintenance "Record Change Series"		Test Series 2		
	Prepared by:		Date:	Version:	
Test Ref.	Condition Tested	Special Requirements	Expected Results	Output On	Next Screen
2.0	Change records				
2.1	Change existing record	Key field	Not allowed		
2.2	Change nonexistent record	Other fields	"Invalid key" message		
2.3	Change deleted record	Deleted record must be available	"Deleted" message		
2.4	Make second record	Change 2.1 above	OK if valid	Transaction file	V45
2.5	Insert record		OK if valid	Transaction file	V45
2.6	Abort during change	Abort 2.5	No change	Transaction file	V45

In a **parallel strategy**, both the old system and its potential replacement are run together for a time until everyone is assured that the new one functions correctly. This is the safest conversion approach because, in the event of errors or processing disruptions, the old system can still be used as a backup. However, this approach is very expensive, and additional staff or resources may be required to run the extra system.

The **direct cutover strategy** replaces the old system entirely with the new system on an appointed day. It is a very risky approach that can potentially be more costly than running two systems in parallel if serious problems with the new system are found. There is no other system to fall back on. Dislocations, disruptions, and the cost of corrections may be enormous.

The **pilot study strategy** introduces the new system to only a limited area of the organization, such as a single department or operating unit. When this pilot version is complete and working smoothly, it is installed throughout the rest of the organization, either simultaneously or in stages. The **phased approach strategy** introduces the new system in stages, either by functions or by organizational units. If, for example, the system is introduced by function, a new payroll system might begin with hourly workers who are paid weekly, followed six months later by adding salaried employees (who are paid monthly) to the system. If the system is introduced by organizational unit, corporate headquarters might be converted first, followed by outlying operating units four months later.

Moving from an old system to a new one requires that end users be trained to use the new system. Detailed **documentation** showing how the system works from both a technical and end-user standpoint is finalized during conversion time for use in training and everyday operations. Lack of proper training and documentation contributes to system failure, so this portion of the systems development process is very important.

Production and Maintenance

After the new system is installed and conversion is complete, the system is said to be in **production**. During this stage, the system will be reviewed by both users and technical specialists to determine how well it has met its original objectives and to decide whether any revisions or modifications are required. In some instances, this review will be conducted as a part of a formal **post-implementation audit**. After the system has been fine-tuned, it must be maintained. **Maintenance** encompasses changes in hardware, software, documentation, or procedures to a production system to correct errors, meet new requirements, or improve processing efficiency. Routine maintenance consumes a large percentage of many firms' IT budgets but can be reduced significantly through the use of up-to-date systems-building practices and technology. Table 13.2 summarizes the systems development activities.

Table 13.2 Systems Development

Core Activity	Description
Systems analysis	Identify problem(s) Specify solutions Establish information requirements
Systems design	Create design specifications
Programming	Translate design specifications into program code
Testing	Perform unit testing Perform systems testing Perform acceptance testing
Conversion	Plan conversion Prepare documentation Train users and technical staff
Production and maintenance	Operate the system Evaluate the system Modify the system

Parallel strategy
A safe and conservative conversion approach where both the old system and its potential replacement are run together for a time until everyone is assured that the new one functions correctly.

Direct cutover strategy
A risky conversion approach where the new system completely replaces the old one on an appointed day.

Pilot study strategy
A conversion strategy in which the new system is introduced to a limited area of the organization until it is proven to be fully functional; only then does the conversion to the new system take place across the entire organization.

Phased approach strategy
A conversion strategy in which the new system is introduced in stages, either by functions or by organizational units.

Documentation
Descriptions of how an information system works from both a technical and end-user standpoint.

Production
The stage after the new system is installed and the conversion is complete; during this time the system is reviewed by users and technical specialists to determine how well it has met its original goals and whether any revisions or modifications are required.

Post-implementation audit
Formal review process conducted after a system has been placed in production to determine how well the system has met its original objectives.

Maintenance
Changes in hardware, software, documentation, or procedures to a production system to correct errors, meet new requirements, or improve processing efficiency.

13.4 Discuss Methods for Building and Acquiring Information Systems

Systems differ in terms of their size and technological complexity and in terms of the organizational problems they are meant to solve. A number of systems-building approaches have been developed to deal with these differences. This section describes these alternative methods: the traditional systems development life cycle, prototyping, application software packages and cloud software services, and outsourcing. Systems builders may use multiple methodologies for different aspects of the system,

Traditional Systems Development Life Cycle

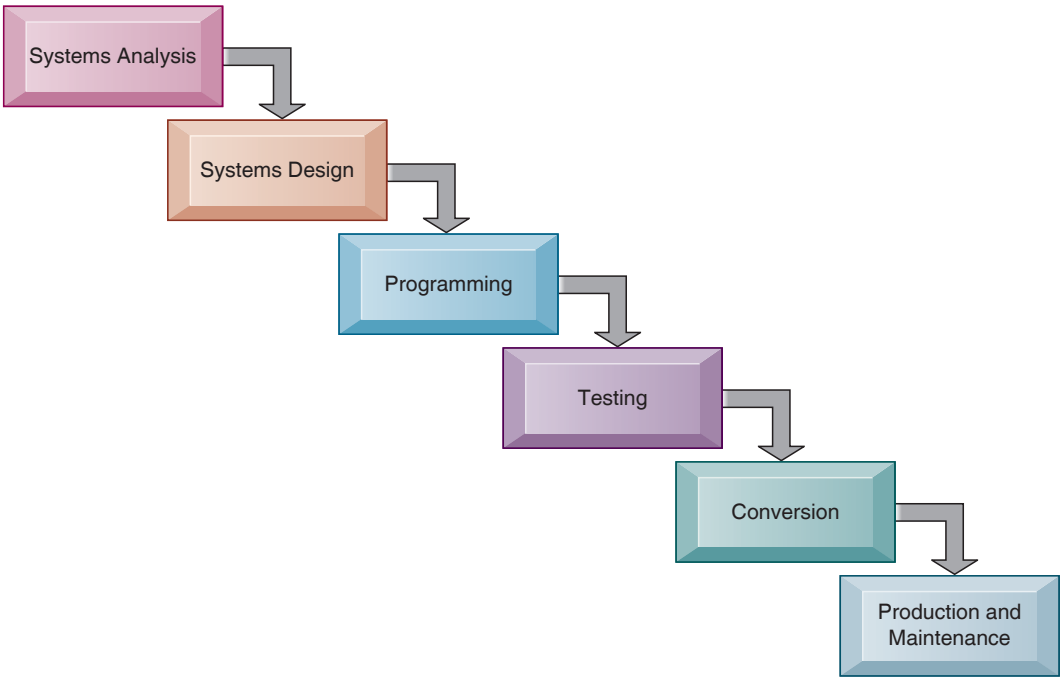
Systems development life cycle (SDLC)
A traditional method for building information systems; involves using a phased approach to building a system, dividing systems development into a series of sequential formal stages: systems analysis, systems design, programming, testing, conversion, and production and maintenance.

The **systems development life cycle (SDLC)** is a traditional method for building information systems. Examine Figure 13.6, which illustrates the SDLC. The SDLC methodology uses a phased approach to building a system, dividing systems development into a series of sequential formal stages: systems analysis, systems design, programming, testing, conversion, and production and maintenance. Although systems development specialists have different opinions on how to partition the systems-building stages, they roughly correspond to the stages of systems development just described. Note that the acronym SDLC is also used to refer to the software development lifecycle, which focuses specifically on the development and testing of software components of an information system. The systems development life cycle is a broader process that encompasses the hardware, software, people, and processes needed for a complete information system.

The SDLC approach maintains a formal division of labor between end users and information systems specialists. Technical specialists, such as systems analysts and programmers, are responsible for much of the systems analysis, design, and implementation work. End users are limited to providing information requirements and reviewing the technical staff's work. The SDLC approach also emphasizes formal specifications and paperwork, so many documents are generated during the course of a systems project.

Figure 13.6 The Traditional Systems Development Life Cycle

The systems development life cycle partitions systems development into formal stages, with each stage requiring completion before the next stage can begin.



The SDLC approach is still used for building large, complex systems that require a rigorous and formal requirements analysis, predefined specifications, and tight controls over the systems-building process. For example, civil engineering company Svevia, which maintains and manages more than 50 percent of Sweden's roads and bridges, still uses an SDLC approach for large projects (Lindstrom, 2023). However, this approach can be costly, time-consuming, and inflexible. Although systems builders can go back and forth among stages in the life cycle, the SDLC is predominantly a "waterfall" approach in which tasks in one stage are completed before work for the next stage begins. Activities can be repeated, but volumes of new documents must be generated and steps retraced if requirements and specifications need to be revised. This encourages freezing of specifications relatively early in the development process. The SDLC approach is also not suitable for many small systems, which tend to be less structured and more individualized.

Prototyping

Prototyping consists of building an experimental system rapidly and inexpensively for end users to evaluate. By interacting with the prototype, users can get a better idea of their information requirements. The prototype endorsed by the users can be used as a template to create the final system.

The **prototype** is a working version of an information system or part of the system, but it is meant to be only a preliminary model. Once operational, the prototype will be further refined until it conforms precisely to users' requirements. Once the design has been finalized, the prototype can be converted to a polished production system.

The process of building a preliminary design, trying it out, refining it, and trying again is known as an **iterative approach** to systems development because the steps required to build a system can be repeated over and over again. Prototyping is more explicitly iterative than the conventional SDLC approach, and it actively promotes system design changes. It has been said that prototyping replaces unplanned rework with planned iteration, with each version more accurately reflecting users' requirements.

Prototyping
The process of building an experimental system quickly and inexpensively for demonstration and evaluation so that users can better determine information requirements.

Prototype
The preliminary working version of an information system for demonstration and evaluation purposes.

Iterative approach
A process of repeating the steps to build a system over and over, with each version more accurately reflecting user requirements.

Steps in Prototyping

Examine Figure 13.7, which illustrates a four-step model of the prototyping process. The process consists of the following:

- ***Step 1:*** *Identify the user's basic requirements.* The systems designer (usually an information systems specialist) works with the user only long enough to capture the user's basic information needs.
- ***Step 2:*** *Develop an initial prototype.* The systems designer creates a working prototype quickly, using tools for rapidly generating software.
- ***Step 3:*** *Use the prototype.* The user is encouraged to work with the system to determine how well the prototype meets his or her needs and to make suggestions for improving the prototype.
- ***Step 4:*** *Revise and enhance the prototype.* The systems builder notes all changes the user requests and refines the prototype accordingly. After the prototype has been revised, the cycle returns to Step 3. Steps 3 and 4 are repeated until the user is satisfied.

When no more iterations are required, the approved prototype then becomes an operational prototype that furnishes the final specifications for the application. Sometimes the prototype is adopted as the production version of the system.

Advantages and Disadvantages of Prototyping

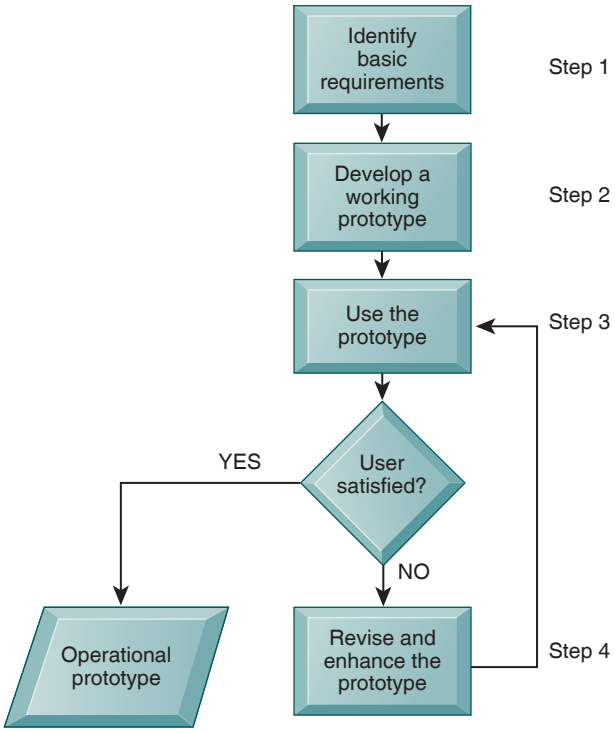
Prototyping is most useful when there is some uncertainty about requirements or design solutions and is often used for designing an information system's **end-user interface** (the part of the system with which end users interact, such as online display and data entry screens and commands.). Because prototyping encourages intense end-user involvement throughout the systems development life cycle, it is more likely to produce systems that fulfill user requirements.

However, rapid prototyping can gloss over essential steps in systems development. If the completed prototype works reasonably well, management may not see the need for

End-user interface
The part of an information system with which the end user interacts, such as online display and data entry screens and commands.

Figure 13.7 The Prototyping Process

The process of developing a prototype can be broken down into four steps. Because a prototype can be developed quickly and inexpensively, systems builders can go through several iterations, repeating steps 3 and 4, to refine and enhance the prototype before arriving at the final operational one.



reprogramming, redesign, or full documentation and testing to build a polished production system. Some of these hastily constructed systems may not easily accommodate large quantities of data or a large number of users in a production environment.

Application Software Packages, Software Services, and Outsourcing

As previously noted in Chapter 5, much of today's software is not developed in-house but is purchased from external sources. Firms can license ("rent") the software from an online software service provider. They can purchase the software from a commercial vendor as a package to run in-house, or they can have a custom application developed by an outside outsourcing firm.

Application Software Packages and Cloud Software Services

Today many systems are based on commercially available application software packages or cloud software as a service (SaaS). For example, companies can choose to implement Oracle enterprise resource planning, supply chain management, or human capital management software in-house or pay to use this software running on the Oracle Cloud platform. Microsoft desktop productivity software comes in both desktop and cloud (Microsoft 365) versions. Many applications are common to all business organizations. For example, payroll, accounts receivable, general ledger, or inventory control. For such universal functions with standard processes that do not change a great deal over time, a more generic system will fulfill the requirements of many organizations.

If a commercial software package or cloud software service can fulfill most of an organization's requirements, the company does not have to write its own software. The company can save time and money by using the prewritten, predesigned, pretested software programs from the software vendor. Package and SaaS vendors supply much of the ongoing maintenance and

support for the system, including enhancements to keep the system in line with ongoing technical and business developments. When a package or SaaS solution is pursued, end users will be responsible for supplying the business information requirements for the system, and information systems specialists will provide technical requirements. In large organizations, the number of SaaS subscriptions can quickly balloon, presenting its own challenges. For instance, according to the CIO of Carhartt, a workwear apparel brand, the company is now paying for more than 120 different vendors for SaaS products, up from around 60 five years ago, and just 20 ten years ago (Bousquette, 2024).

If an organization has unique requirements that the package does not meet, these tools include capabilities for customization. **Customization** involves modifying a commercial software package or cloud-based software to meet an organization's unique requirements without destroying the integrity of the software. If a great deal of customization is required, additional programming and customization work may become so expensive and time-consuming that they negate many of the advantages of software packages and services.

Customization

The modification of a commercial software package to meet an organization's unique requirements without destroying the package software's integrity.

When a system is developed using an application software package or a cloud software service, systems analysis will include a formal evaluation of the software package or service in which both end users and information systems specialists will participate. The most important evaluation criteria are the functions provided by the software, flexibility, user-friendliness, hardware requirements, database requirements, installation and maintenance efforts, documentation, vendor quality, and cost. The package or software service evaluation process often is based on a **request for proposal (RFP)**, which is a detailed list of questions submitted to software vendors to determine how well the vendor's product can meet the organization's specific requirements. When software from an external source is selected, the organization no longer has total control over the systems design process. Instead of tailoring the systems design specifications directly to user requirements, the design effort will consist of trying to mold user requirements to conform to the features of the package or software service. If the organization's requirements conflict with the way the package or software service works and this software cannot be customized, the organization will have to adapt to the package or software service and change its procedures.

Request for proposal (RFP)

A detailed list of questions submitted to vendors of software or other services to determine how well the vendor's product can meet the organization's specific requirements.

Outsourcing

If a firm does not want to use its internal resources to build information systems, it can outsource the work to an external organization that specializes in providing these services. Cloud computing and SaaS providers, described in Chapter 5, are one form of outsourcing. Subscribing companies use the software and hardware provided by the service as the technical platform for their systems. In another form of outsourcing, a company hires an external vendor to design and create the software for its system, but the company operates the system on its own computers. The vendor might be located domestically (in the United States) or outside of the United States (**offshore outsourcing**).

Offshore outsourcing

Outsourcing systems development work or maintenance of existing systems to external vendors outside of the United States.

Domestic outsourcing is driven primarily by the fact that outsourcing firms possess skills, resources, and assets that their clients do not have. Installing a new supply chain management system in a very large company might require hiring an additional 30 to 50 people with specific expertise in supply chain management software licensed from a vendor. Rather than hire permanent new employees, most of whom would need extensive training in the new software, and then release them after the new system is built, it makes more sense, and is often less expensive, to outsource this work for a 12-month period.

In the case of offshore outsourcing, the decision is much more cost-driven. According to Payscale, the average salary for a software engineer in India is around 664,000 rupees per year (approximately $9,000), while in the United States, the average salary is around $90,000 per year (Payscale, 2024). The Internet and low-cost communications technology have drastically reduced the expense and difficulty of coordinating the work of global teams in offshore locations. In addition to cost savings, many offshore outsourcing firms offer world-class technology assets and skills. Wage inflation outside the United States has recently eroded some of these advantages, and some jobs have moved back to the United States. Firms generally do not outsource the conception, systems analysis, and design of IT systems to offshore firms, but often do outsource programming, testing, maintenance, and daily operation of IT systems.

Figure 13.8 Total Cost of Offshore Outsourcing

If a firm spends $10 million on offshore outsourcing contracts, that company will actually spend 15.2 percent in extra costs even in the best-case scenario. In the worst-case scenario, where there is a dramatic drop in productivity along with exceptionally high transition and layoff costs, a firm can expect to pay up to 57 percent in extra costs on top of the $10 million outlay for an offshore contract.

TOTAL COST OF OFFSHORE OUTSOURCING

Cost of outsourcing contract $10,000,000

Hidden Costs	Best Case	Additional Cost ($)	Worst Case	Additional Cost ($)
1. Vendor selection	.02%	20,000	2%	200,000
2. Transition costs	2%	200,000	3%	300,000
3. Layoffs & retention	3%	300,000	5%	500,000
4. Lost productivity/cultural issues	3%	300,000	27%	2,700,000
5. Improving development processes	1%	100,000	10%	1,000,000
6. Managing the contract	6%	600,000	10%	1,000,000
Total additional costs		1,520,000		5,700,000
	Outstanding Contract ($)	Additional Cost ($)	Total Cost ($)	Additional Cost
Total cost of outsourcing (TCO) best case	10,000,000	1,520,000	11,520,000	15.2%
Total cost of outsourcing (TCO) worst case	10,000,000	5,700,000	15,700,000	57.0%

A firm is most likely to benefit from outsourcing if it takes the time to evaluate all the risks and to make sure outsourcing is appropriate for its specific needs. Any company that outsources its applications must thoroughly understand the project, including its requirements, method of implementation, anticipated benefits, cost components, and metrics for measuring performance. Many firms underestimate costs for identifying and evaluating vendors of information technology services, for transitioning to a new vendor, for improving internal software development methods to match those of outsourcing vendors, and for monitoring vendors to make sure they are fulfilling their contractual obligations. Companies will need to allocate resources for documenting requirements, sending out RFPs, handling travel expenses, negotiating contracts, and project management. Experts claim it takes from three months to a full year to fully transfer work to an offshore partner and make sure the vendor thoroughly understands your business. Offshore outsourcing incurs additional costs for coping with cultural differences that drain productivity and dealing with human resources issues, such as terminating or relocating domestic employees. All of these hidden costs undercut some of the anticipated benefits from outsourcing. Firms should be especially cautious when using an outsourcer to develop or to operate applications that give it some type of competitive advantage.

Examine Figure 13.8, which shows the best- and worst-case scenarios for the total cost of an offshore outsourcing project. It shows how much hidden costs affect the total project cost. The best case reflects the lowest estimates for additional costs, and the worst case reflects the highest estimates for these costs. As you can see, hidden costs increase the total cost of an offshore outsourcing project by an extra 15 percent to 57 percent. Even with these extra costs, many firms will benefit from offshore outsourcing if they manage the work well.

13.5 Describe Methods That Enable Rapid Systems Development

Technologies and business conditions are changing so rapidly that companies are adopting shorter, more informal systems development processes, including those for mobile applications. In addition to using software packages and online software services, businesses are relying more heavily on fast-cycle techniques such as object-oriented design (OOD), rapid application development (RAD), joint application development (JAD), agile development, and DevOps.

Object-Oriented Design (OOD)

Object-oriented design (OOD) uses an object as the basic unit of systems analysis and design. An **object** combines data and the specific processes that operate on the data. Data encapsulated in an object can be accessed and modified only by the operations, or methods, associated with that object. Instead of passing data to procedures, programs send a message for an object to perform an operation that is already embedded in it. The system is modeled as a collection of objects and the relationships among them. Because processing logic resides within objects rather than in separate software programs, objects must collaborate with each other to make the system work.

Object-oriented design (OOD)
Uses an object as the basic unit of systems analysis and design.

Object
Software building block that combines data and the procedures acting on the data.

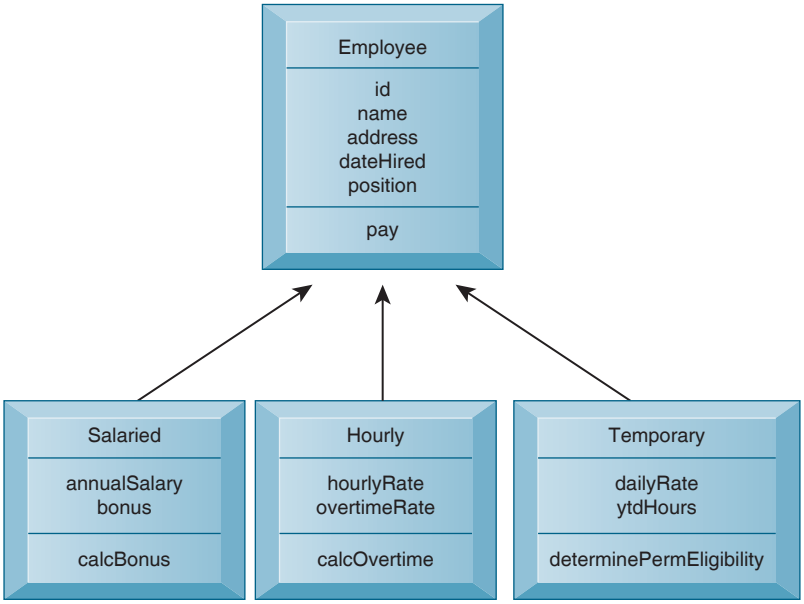
OOD is based on the concepts of class and inheritance. Objects belonging to a certain class, or general category of similar objects, have the features of that class. Classes of objects in turn can inherit all the structure and behaviors of a more general class and then add variables and behaviors unique to each object. New classes of objects are created by choosing an existing class and specifying how the new class differs from the existing class instead of starting from scratch each time.

Examine Figure 13.9, which shows how class and inheritance work. It depicts the relationships among classes concerning employees and how they are paid. Employee is the common ancestor, or superclass, for the other three classes. Salaried, Hourly, and Temporary are subclasses of Employee. The class name is in the top compartment, the attributes for each class are in the middle portion of each box, and the list of operations is in the bottom portion of each box. The features that are shared by all employees (ID, name, address, date hired, position, and pay) are stored in the Employee superclass, whereas each subclass stores features that are specific to that particular type of employee. Specific to hourly employees, for example, are their hourly rates and overtime rates. A solid line from the subclass to the superclass is a generalization path showing that the subclasses Salaried, Hourly, and Temporary have common features that can be generalized into the superclass Employee.

Object-oriented development is more iterative and incremental than traditional structured development. During analysis, systems builders document the functional requirements of the system, specifying its most important properties and what the proposed system must do. Interactions between the system and its users are analyzed to identify objects, which include both data and processes. The object-oriented design phase describes how the objects will behave and how they will interact with one another. Similar objects are grouped together to form a class, and classes are grouped into hierarchies in which a subclass inherits the attributes and methods from its superclass.

Figure 13.9 Class and Inheritance

This figure illustrates how classes inherit the common features of their superclass.



The information system is implemented by translating the design into program code, reusing classes that are already available in a library of reusable software objects, and adding new ones created during the object-oriented design phase. Implementation may also involve the creation of an object-oriented database. The resulting system must be thoroughly tested and evaluated.

Because objects are reusable, object-oriented development can potentially reduce the time and cost of creating software. Organizations can reuse software objects that have already been created as building blocks for other applications. New systems can be created by using some existing objects, changing others, and adding new objects. Object-oriented frameworks are also available that provide reusable, semi-complete applications that an organization can further customize into finished applications.

Rapid Application Development (RAD)

Rapid application development (RAD)
Process for developing workable systems in a very short time period with some flexibility to adapt as the project evolves; includes use of visual programming and other tools for building graphical user interfaces, iterative prototyping, automation of program code generation, and close teamwork among end users and information systems specialists.

The term **rapid application development (RAD)** refers to the process of creating workable systems in a very short period of time with some flexibility to adapt as a project evolves. RAD includes the use of visual programming and other tools for building graphical user interfaces, iterative prototyping of key system elements, automation of program code generation, and close teamwork among end users and information systems specialists. Simple systems often can be assembled from prebuilt components. The process does not have to be sequential, and key parts of development can occur simultaneously.

Joint Application Development (JAD)

Joint application development (JAD)
Methodology to ensure that the client, consumer, IT specialists, and stakeholders are all involved in designing and developing a technology application.

Joint application development (JAD) is a methodology to ensure that the client, consumer, IT specialists, and stakeholders are all involved in designing and developing a technology application. JAD speeds up the generation of information requirements by bringing end users and information systems specialists together to freely discuss the system's design from their various perspectives and come up with the best solution agreed upon by all members. Properly prepared and facilitated, JAD sessions can significantly speed up the design phase and increase user involvement during the system-building process. Joint application development is often used in agile development, which emphasizes quick completion of tasks and quality outcomes.

Agile Development

Agile development
Rapid delivery of working software by breaking a large project into a series of small sub-projects that are completed in short periods of time using iteration and continuous feedback.

Agile development focuses on rapid delivery of working software by breaking a large project into a series of small subprojects that are completed in short periods of time using iteration, continuous feedback, and ongoing user involvement. Iterations, called sprints, are short time frames that typically last from 1–4 weeks. Each mini-project is worked on by a cross-functional team (consisting of programmers, testers, a user representative, and other people required by the project) and regularly released to the client. Improvement or addition of new functionality takes place within the next iteration as developers clarify requirements.

Features to be developed are assigned to the sprints on a priority basis, with the "critical path" through the program developed first. This makes it possible for a barebones but functioning version of a program to be demonstrated early on. As additional features are developed, the code is added to the code already developed, a process known as "continuous integration." Testing occurs early and often throughout the entire development process. Agile methods emphasize face-to-face communication, encouraging people to collaborate and make decisions quickly and effectively, rapid and flexible response to change, and the production of working software rather than elaborate documentation. Instead of a separate testing phase, as in the traditional waterfall approach, testing occurs continuously in agile development during each sprint, and testing is encouraged. This often means that more development time will be spent on testing than when using traditional approaches.

Although essential, continuous testing can be very time-consuming, especially since developing test scripts, executing test steps, and evaluating the results used to be largely manual. Automated testing tools are now available to meet this need. Automated testing tools

perform examinations of the software, report outcomes, and compare results with earlier test runs. For example, automated testing helped security software company McAfee use an agile methodology in its projects. Manual processes could not keep up with all the continuous testing that was going on. The company implemented WorkSoft Certify, a leading automated testing solution for automated testing and business process automation, saving 2500 hours of manual work, equivalent to $200,000 (Bonneau, 2024).

DevOps

DevOps builds on agile development principles as an organizational strategy to create a culture and environment that further promotes rapid and agile development practices. *DevOps* stands for "development and operations" and emphasizes close collaboration between the software developers who create applications and the IT operational staff who run and maintain the applications. Traditionally, in a large enterprise, an application development team would be in charge of gathering business requirements for an application, designing the application, and writing and testing the software. The operations team would run and maintain the software once it was put into production. Problems arise when the development team is unaware of operational issues that prevent the software from working as expected, requiring additional time and rework to fix the software.

DevOps
Organizational strategy to create a culture and environment to promote rapid and agile development practices by emphasizing close collaboration between software developers and the IT operational staff. Stands for "development and operations".

DevOps tries to change this relationship by promoting better and more frequent communication and collaboration between systems development and operations groups, and a fast and stable workflow throughout the entire application development life cycle. With this type of organizational change along with agile techniques, standardized processes, and more powerful automated software creation and testing tools, it is possible to build, test, and release applications more rapidly and more frequently. For example, DevOps helps developers at Netflix make hundreds of software changes each day.

13.6 Discuss the Increasing Democratization of Software Development

One of the most significant trends impacting how information systems are built has been the increasing democratization of software development. The use of open-source, low-code, and no-code software as well as the rise of citizen developers have all accelerated the process by which software and the systems they power can be created, as well as broadened those who participate in the process.

What Is Open-Source Software?

Open-source software (OSS), introduced in Chapter 5, is distributed with its source code, which is the part of the software that programmers manipulate to control how a program behaves. Programmers who have access to source code can modify a program by adding to it or changing it. OSS has moved into the mainstream as a major software development approach. It's a way of using publicly available software free of charge (or at low cost) as building blocks for new applications. OSS users are able to save time and money by repurposing existing software code.

Open-source code is usually stored in a public repository and shared publicly. Anyone can access the repository to use the code independently or to improve the design and functionality of a piece of software. OSS also includes a distribution license that defines how developers can use, modify, and distribute the software. When source code is changed, the OSS user must identify what was altered as well as the methods involved. Depending on the license terms, the software resulting from these modifications may or may not be required to be made available for free.

Examples of OSS include the Linux operating system, Apache HTTP Server, Mozilla Firefox web browser, LibreOffice productivity suite, VLC Media Player, WordPress content management system, and Python programming language. Open source makes up over 80 percent of the software code in modern applications, and 78 percent of businesses use some form of open-source software (Lindner, 2023). Much of the technology we use every day runs on OSS, including mobile devices, cars, airplanes, and AI.

Examples of Open-Source Software Development

Today, OSS powers systems and applications for tech giants like Google, Facebook, and even Microsoft. Although Microsoft software products like Windows or Microsoft 365 are proprietary, Microsoft is one of the world's largest open-source contributors. The Android mobile operating system is based on a modified version of Linux and other open-source software, designed primarily for touchscreen mobile devices. Apple's iOS for the iPhone and iPad is proprietary, but some of its components are open source. Big data software tools such as Hadoop, Spark, Cassandra, Kafka, and the MongoDB NoSQL database management system are all examples of open-source software.

Amazon Web Services (AWS) is a leader in cloud computing technology, but it's also a major contributor to open-source software. AWS employees have over 1,200 projects on GitHub, an online developer platform that allows users to create, store, manage, and share their code. Many of AWS's most popular tools and services are also based on open source, including its Cloud Development Kit (CDK). AWS CDK lets users build custom cloud applications using familiar programming languages and preconfigured components.

Walmart is an example of a large company using open source software to develop applications and systems. Walmart uses open source software in its own applications, and has also released some of its own open source software to the marketplace. Walmart's research and development unit, Walmart Labs, used open-source software to develop Electrode, the application platform for Walmart.com.

TensorFlow is an example of an AI application that uses open source software. It was developed by Google Brain Team to bolster cloud-hosted AI services as an end-to-end platform with tools, libraries, and community resources for building state-of-the-art machine learning applications. TensorFlow can develop machine learning models for a wide range of tasks such as image and voice recognition, and it has been adopted by numerous businesses and research institutions.

Open-source software development can also be of benefit to smaller organizations. Take, for example, a small retail e-commerce company with limited resources that wants to improve its website performance. By implementing an open-source e-commerce platform with capabilities for customization, the company was able to modify the code to meet its unique requirements. The company was thus able to streamline its workflow, improve site navigation, and boost customer engagement.

Advantages and Disadvantages of Open-Source Software Development

Table 13.3 compares the advantages and disadvantages of open-source software development.

Open-Source Software Security

Although open-source software has many benefits, it introduces a whole new level of software risks. OSS is not completely bug-free (nor is commercial software). Security bugs and code flaws may still exist and pose significant risk. Organizations using OSS or planning to do so must be able to clearly understand what is in the OSS code so they can effectively manage and secure it. On the plus side, OSS source code is open for review by anyone and everyone, and many believe that widespread OSS use and review identifies software bugs more quickly.

Table 13.3 Advantages and Disadvantages of Open-Source Software Development

Advantages of Open-Source Software Development
• Open-source software is free or often available at a lower cost than proprietary software.
• Open source is flexible; developers can freely make changes to better fit an application's unique requirements. Customization is often easier than with proprietary software.
• Programmers can use pre-existing code to improve the software and innovate.
• Open source comes with a built-in community that continuously inspects, modifies, and improves the source code.
Disadvantages of Open-Source Software Development
• Open-source software can be difficult to set up and use due to insufficient documentation or the lack of user-friendly interfaces.
• Open source can introduce compatibility issues. Programming proprietary hardware with open-source software may require specialized software drivers to enable the operating system and devices to communicate and are typically only available from the hardware manufacturer.
• Open-source software can pose liability issues. Open source rarely contains any warranty, liability, or infringement indemnity protection. This leaves the consumer of the OSS responsible for maintaining compliance with legal obligations.
• Open-source software implementation may incur unexpected costs in training users and IT staff, importing data, and setting up required hardware.

Low-Code and No-Code Development

Low-code development is a software development approach that enables the delivery of workable applications faster and with minimal hand-coding by using visual modeling in a graphical interface to assemble and configure applications. Such tools may produce entirely operational applications or require a small amount of additional coding. Low-code development platforms reduce the amount of hand-coding for creating workable software, so that business applications can be created more rapidly by a wider range of people (including business end users), not just those with formal programming skills. Microsoft Power Platform is a leading low-code platform that features a number of tools to quickly automate processes and build apps. Its low-code toolset includes drag-and-drop functions and AI features for business users, but it also connects to professional development tools.

Low-code development
Software development approach that enables the delivery of applications faster and with minimal hand coding.

No-code development involves using tools that enable the creation of software without having to write any program code and is even easier for non-IT businesspeople to use. Everything the software vendor thinks the user needs to create an app is already built into the tool, with no coding required. However, many no-code tools were built to solve simple business problems and have limited functionality. For example, Goodgigs is a job board that allows anyone to find jobs with positive social impact. Its web app was built using Bubble, a no-code web and mobile app builder that allows users to build fully-functional applications from scratch without the need for deep coding expertise.

No-code development
Involves using tools that enable the creation of software without having to write any program code.

Citizen Development

For many years, the "application backlog" has been a serious problem for organizations that depend on information technology. The more firms depend on software to power daily operations and decision making, the more they want. Demand for new software applications has outstripped the supply of technical talent for meeting them. Some of this backlog can be reduced by having applications developed by end users themselves. The availability of low-code/no-code and generative AI tools has empowered a new type of **citizen developer:** a business user who builds new applications or modifies existing ones to create applications on their own with IT approval or with minimal help from IT staff.

Citizen developer
Business user who builds new applications or modifies existing ones to create applications of their own with IT approval or with minimal help from IT staff.

In the past, there was some end user development using earlier generations of user-friendly software (such as query, reporting, website development, graphics, and PC software tools), but much of it occurred outside the purview of the organization's IT department, and the range of applications suitable for end user development was very limited. This phenomenon is called

"shadow IT," where end users develop and run applications on their own, often without IT oversight. Examples of shadow IT include sharing work files on a personal cloud storage account, creating an unofficial group chat channel without IT approval, or maintaining an unauthorized personal database of client contacts that is not shared with the organization. There were insufficient formal mechanisms for monitoring and controlling "shadow IT," which made CIOs hesitant to move application development outside of the IT function.

Today's citizen developers are different: they work outside the IT department but collaborate more closely with IT using IT-approved software, and they have better tools available. They use no-code and low-code application development tools to build new business applications for their own use and the use of fellow employees. Employees familiar with an organization or department's business processes have the business knowledge to create solutions that support their daily work and also know the organization's overall objectives. Some IT organizations not only specify appropriate citizen tools but also provide or support guidelines, training, and code and data exchanges to build citizen capabilities. Citizen projects can make an impact on both the departmental and enterprise levels, as described in the Spotlight on Management Case Study, *Meet the Citizen Developers.*

Spotlight on Management Case Study

Meet the Citizen Developers

One of the primary barriers to end user development has been resistance from IT groups. This resistance is fading, as IT sees that it can work more closely with citizen developers to create effective and manageable solutions that benefit many parts of the organization. Citizen development expands the pool of people who can build new business processes, applications, and other IT solutions.

For example, Chevron Corporation, one of the largest U.S. oil and natural gas producers, has focused on making it easy for anyone, regardless of technical skills, to responsibly develop solutions to the problems they understand best. When flash flooding impacted some of the homes of Chevron's workforce in Africa, a citizen developer was able to quickly create an application for rapid damage reporting and resolution tracking. Chevron provides its citizen developers with foundational technologies that ensure secure, reliable, and efficient design and access to data. The company also provides the IT department with data and performance metrics to monitor what is being developed and used. Technical controls are in place to manage the development process and ensure security and privacy. There is no more "shadow IT". Chevron also believes it is important to champion citizen developers and encourage others in the organization to do so.

Chevron was an early adopter of Microsoft's Power Platform, which provides tools that can be used by citizen developers to build low-code applications. Chevron looks for application opportunities that have not yet been addressed and then supports citizen developers with four months of instruction and hands-on coaching.

Other companies have recruited citizen developers capable of working on more complex machine learning (ML) and generative AI applications. Amsterdam-based ING bank, with operations in 40 countries, wanted to build more ML models to put into production, but lacked employees with sufficient data science expertise. It turned to citizen developers as a way of alleviating this shortage. The company is providing citizen developers that have some data science background with more training and hands-on experience and also identifying appropriate ML applications, such as determining the probability that customers will click on an app message or respond to an email campaign.

AT&T is promoting employee efforts to analyze data with machine-learning models. The goal is to support all aspects of the ML pipeline, including building the model, deploying the model, monitoring and refining its performance over time, and governing it effectively. The company's citizen developers can choose from among more than 26,000 model-building features. AT&T's Chief Data Office (CDO) has mobilized 3000 "citizen data scientists" from almost every department across the company. They are deploying AI applications for a wide range of uses, including marketing campaigns, network design and management, and calculating the most efficient routes for their field service technicians serving customers.

The Shell Global group of energy and petrochemical companies, with more than 103,000 employees in over 70 countries, is at the forefront of the global citizen development trend. Its "DIY" program has empowered more than 4,000 citizen developers through a federated governance model with a centralized Center of Expertise (COE) and hundreds of DIY coaches in place throughout the organization. (Federated data governance entails distributing data governance responsibilities and decision-making across different business groups or entities while maintaining a level of centralized oversight and standards.) Citizen development goes through a four-step process which includes working with team members to analyze existing business processes in need of improvement, designing new improved processes, using digital tools to streamline workflow, and evaluating progress.

One successful Shell citizen developer project focused on manual process automation with data visualization at Shell Deer Park Refinery in Texas. The refinery relied heavily on manual paper-based processes for pump repairs, making evaluation of pump performance, maintenance records, and operating conditions difficult and time-consuming. Refinery engineers used low-code tools to create an app to automate the workflows, with a goal of reducing annual repair time by 50 percent. The improved processes provide accurate data in a centralized database, simplifying maintenance management. With a single source of data, engineers have been able to increase pump efficiency and also reduce the risk and cost associated with unplanned downtime.

Many of Shell's customers, like other businesses, have made reduction of greenhouse gas emissions a high priority. Another Shell DIY project was a Product Carbon Footprint (PCF) Calculator. The app is designed to give Shell customers a precise measurement of their carbon footprint and suggest next steps customers can take to reduce carbon emissions. All customers need to do is input information on the make-up of their fleet, average distances and capacities, and any fleet management solutions they already use. The app then produces a carbon footprint calculation tailored to their fleet, helping customers choose Shell products that reduce their carbon footprint.

Shell's DIY program has generated significant business value as well as improvements in organizational culture. Shell believes its recipe for success includes governance rules for use, cultivation of local communities, and committed management leadership. Shell is planning to double the number of citizen developers, expand what those developers can deliver, and make DIY a high priority in the firm's IT budget.

SOURCES: "Citizen Developer," Mendix.com, accessed September 22, 2024; Team Kissflow, "Citizen Development Statistics and Trends to Watch in 2024 and Beyond," Kissflow.com, March 11, 2024; Bob Violino, "Empowering Citizen Developers for Real Business Impact," Cio.com, July 19, 2023; Thomas H. Davenport, "MISQE Insight: On the Inevitability of Citizen Development," *MIS Quarterly Executive*, Volume 22, Issue 4 (December 2023); Kevin Shuler, "Unleashing Citizen Development: Eye-Opening Use Cases," Quandarycg.com, May 16, 2023.

Case Study Questions

1. What is the difference between shadow IT and citizen development?
2. Why were the organizations described in this case so successful with their citizen development projects? Explain your answer.
3. How might citizen development differ in a small and a large organization?
4. What management problems can citizen development help solve?

Table 13.4 describes some of the benefits of citizen development.

Table 13.4 Benefits of Citizen Development

Benefit	Description
Cost savings	Citizen developers that share resources, tools, and technical know-how help reduce the need for IT involvement and expenditures.
Faster development	Citizen development helps lighten the application backlog since non-IT team members can build and deploy solutions largely on their own.
Increased flexibility	Citizen developers are able to make updates and add new features to the applications they build.
Innovation	Citizen developers free up professional developers to focus on innovating new solutions and optimizing existing processes.

13.7 Understand the Impact of AI on Software Development

AI has significantly impacted software development by enabling developers to automate routine tasks, streamline workflows, and improve the accuracy and efficiency of software development processes. AI-powered tools and frameworks enable developers to write code faster and more accurately, identify and fix bugs more quickly, and optimize software performance. AI can also help developers test software more effectively, ensure better security, and make more informed decisions about software design and architecture.

Table 13.5 describes some of the ways AI helps developers create, optimize, and maintain software.

Table 13.5 How AI Facilitates Software Development

Function	AI Capabilities
Automated code generation	AI tools assist software developers in writing code more efficiently by analyzing patterns, learning from existing software code, and generating pieces of code. Especially useful for speeding up tedious and repetitive tasks.
Automated testing	AI increases the accuracy and effectiveness of the software testing process. AI can generate test cases, execute the tests, and evaluate software behavior. AI can make the software testing process faster, easier, and less time-consuming. AI-aided automated testing identifies bugs and creates reports without any human intervention. By simulating various scenarios, AI ensures better test coverage and identifies exceptional conditions that human testers might overlook.
Automated software bug detection	AI algorithms can identify bugs by analyzing code and detecting anomalies. The algorithms learn from historical bug data and can predict potential issues. When a bug is found, AI tools can suggest fixes or automatically correct the code.
Natural language processing (NLP)	AI NLP tools can understand and process user information requirements expressed in natural language and translate them into code specification. Advanced software agents using NLP can collaborate with users during software development. providing real-time progress updates, accepting feedback, and working together to make design choices.
Improving IT security	AI can analyze software code to identify potential threats and security vulnerabilities. It can also recommend secure coding practices and automate many aspects of security testing.
Optimizing software performance	AI models can analyze historical data to predict software performance, user behavior, and resource utilization, enabling developers to optimize software based on these predictions.
Automated documentation	AI tools can quickly generate software documentation by extracting relevant information from code comments, function names, and usage patterns. They can ensure that the documentation is always up-to-date, accurate, and in line with the latest code changes.

Challenges of Developing AI Software

The implementation of AI in software development poses several challenges. One of the biggest challenges is the need for more software developer expertise in AI, which is currently in short supply. Another challenge is the need for very large amounts of high-quality data to train AI models. Without the right data, AI models will not work properly. Training AI models is very resource-intensive, requiring very large amounts of computing power as well as training data that are out of reach for many organizations.

Some organizations that do not have these resources may be able to take advantage of pre-trained AI models. These models have been pre-trained on massive amounts of data and can be customized to perform specific tasks or cater to specific industries. For example, NVIDIA, a world supplier of AI semiconductors and solutions, offers over 600 pretrained AI models built by NVIDIA researchers and engineers using public and proprietary datasets for domain-specific tasks. The models enable developers to build AI applications for computer vision, speech language understanding, molecule generation, and more, that can be customized for specific tasks. For example, biotech startup Evozyne used a pretrained NVIDIA AI model to create two proteins with significant potential benefits for healthcare and clean energy. According to NVIDIA, pretrained models can cut AI application development time by up to a year and lead to cost savings of hundreds of thousands of dollars.

Another option is to use **AI as a Service (AIaaS)**, cloud-based machine learning platforms that offer scalable infrastructure and pre-built tools and frameworks for AI model development. AIaaS makes AI tools and technologies more accessible, scalable, and cost-effective for many applications. Typical AI cloud service benefits include scalability, subscription-based or pay-as-you-go pricing, AI-optimized hardware, pre-built AI models and tools, and data management and storage. Coca Cola Bottlers Japan used Google's Vertex AI platform to optimize product distribution in its 700,000 vending machines. The company developed a predictive model that determines optimal vending machine locations, the right product lineup for each machine, pricing strategy, and expected sales volume.

AI as a Service (AIaaS)

Cloud-based machine learning platform that offers scalable infrastructure and pre-built tools and frameworks for AI model development.

13.8 Discuss Approaches for Building Mobile Applications

Today, employees and customers expect, and even demand, to be able to use a mobile device of their choice to obtain information or perform a transaction anywhere and at any time. To meet these needs, companies need to develop mobile websites, mobile web apps, native apps and/or hybrid apps, as well as traditional information systems.

Once an organization decides to develop mobile apps, it has to make some important choices, including the technology it will use to implement these apps (whether to write a native app or mobile web app) and what to do about a mobile website. A **mobile website** is a version of a regular website that is scaled down in content and navigation for easy access and search on a small mobile screen. (Access Amazon's website from your computer and then from your smartphone to see the difference from a regular website.)

A **native app** is a standalone application designed to operate using a specific mobile operating system and device. Native apps can connect to the Internet to download and upload data, and they can also operate on these data even when not connected to the Internet. For example, an e-book reading app such as Kindle software can download a book from the Internet, disconnect from the Internet, and present the book for reading. Native mobile apps provide fast performance and a high degree of reliability. They are also able to take advantage of a mobile device's particular capabilities, such as its camera or touch features. However, native apps are expensive to develop because multiple versions of an app must be programmed for different mobile operating systems and hardware such as Android and Apple's iOS.

Mobile website

Version of a regular website that is scaled down in content and navigation for easy access and search on a small mobile screen.

Native app

Standalone application designed to operate using a specific mobile operating system and device.

Mobile web app
An Internet-enabled application with specific functionality for mobile devices and which is designed to be accessed via the mobile device's web browser.

Hybrid app
Mobile application that has many features of both a native app and mobile web app.

A **mobile web app** is an Internet-enabled application with specific functionality for mobile devices and which is designed to be accessed via the mobile device's web browser. The web app resides primarily on a server, is accessed via the Internet, and doesn't need to be installed on the device. The same application can be used by most devices that can surf the web, regardless of their brand.

A **hybrid app** has many features of both a native app and a mobile web app. It is built with web technologies such as HTML, CSS, and JavaScript, but like a native app, it runs within the mobile device itself and is able to access many of the device's features that are not accessible by a mobile web app.

Designing for a Multiscreen World

Developing applications for mobile platforms is quite different from development for desktop computers and their much larger screens. The reduced size of mobile devices makes using fingers and multitouch gestures much easier than typing and using keyboards. Mobile apps need to be optimized for the specific tasks they are to perform, they should not try to carry out too many tasks, and they should be designed for usability. The user experience for mobile interaction is fundamentally different from using a desktop or laptop computer. Saving resources—bandwidth, screen space, memory, processing, data entry, and user gestures—is a top priority.

When a full website created for the desktop shrinks to the size of a smartphone screen, it is difficult for the user to navigate through the site. The user must continually zoom in and out and scroll to find relevant material. Therefore, companies need to design websites specifically for mobile interfaces and create multiple mobile sites to meet the needs of smartphones, tablets, and desktop browsers. This equates to at least three sites with separate content, maintenance, and costs. Currently, websites know what device you are using because your browser will send this information to the server when you log on. Based on this information, the server will deliver the appropriate screen.

Responsive web design (RWD)
Ability of a website to automatically change screen resolution and image size as a user switches to devices of different sizes, such as a laptop, tablet computer, or smartphone. Eliminates the need for separate design and development work for each new device.

One solution to the problem of having multiple websites is to use **responsive web design (RWD)**. RWD enables websites to change layouts automatically according to the visitor's screen resolution, whether on a desktop, laptop, tablet, or smartphone. RWD tools include HTML5 and CSS3. RWD typically works well for sites with relatively simple functionality. However, RWD can be costly, and responsive websites still have the size and complexity of a traditional desktop site, sometimes making it slow to load and perform on a mobile device. Another technique, known as **adaptive web design (AWD)**, has been developed to deal with this issue. With AWD, the server hosting the website detects the attributes of the device making the request and loads a version of the site optimized for the device. AWD offers faster load times and typically a better user experience.

Adaptive web design (AWD)
Techniques that enable a server hosting a website to detect the attributes of the device making the request for service and load a version of the site optimized for the device; offers faster load times and typically a better user experience.

13.9 Understand How the Information in This Chapter Can Help Your Career

The information you have learned in Chapter 13 can help advance your career by deepening your knowledge of business processes and different tools and approaches for building and acquiring information systems solutions.

Real-World Scenario

Process Analysts Help Carter's Redesign Its Business Processes

Carter's is the largest U.S. branded marketer of apparel exclusively for babies and young children. Carter's merchandise is sold online, in 980 company stores in the United States and Canada, and in nearly 20,000 department and specialty stores. Carter's financial systems handle hundreds of thousands of transactions from all of these sources each day. The systems Carter's had used to process these transactions were heavily manual and could no longer keep pace with the company's growth. This created bottlenecks that slowed down processing and also increased the chances of human error. Carter's management wanted to transform the role of the finance function from preoccupation with transaction processing to focusing more on analyzing financial data and guiding decision making. To accomplish this goal, the company needed to streamline and simplify financial processes. Carter's replaced outdated systems with a centralized enterprise resource planning (ERP) system for financial processes. Implementing the ERP system provided the opportunity to transform older and inefficient processes into modern processes reflecting best practices for its line of business and its industry. Anywhere along the process life cycle, users can view all information related to a specific transaction on a single screen.

Job Description and Career Path: Process Analyst

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Process Analyst	• Responsible for identifying areas for improvement in current business processes • Collaborating with stakeholders to ensure successful adoption of new improved processes • Developing and implementing optimized processes and procedures to enhance efficiency, productivity and customer satisfaction. • Monitoring and evaluating the effectiveness of implemented process changes and continuous improvement. • Staying abreast of industry trends, emerging technologies and process improvement methodologies.	• Bachelor's degree in business, information technology, or related field • 1+ years of relevant experience in project management and business analysis • Knowledge of Celonis, Business Process Modeling Notation (BPMN) or other process mining tools desirable	• Entry Level: Business Process Analyst • Mid-Level: Senior Business Analyst • Senior Level: Manager • Business Process Analyst salaries currently range between $61,000 and $82,000, with top earners making $119,000 annually across the United States. Salary can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

Process Mining with Celonis

Celonis is a U.S.-German information technology company that creates "process mining" software to visualize and improve business processes. It is headquartered in Munich, Germany and New York City, with over 3,000 employees in 20 offices around the world. More than 1,400 companies use Celonis to continuously find improvement opportunities within and across business processes, and to execute targeted actions to enhance process performance. This optimization lowers costs, radically improves customer experience, and reduces carbon emissions.

Business processes are not static, nor do they always follow the path defined for them. Without attention to continuous process improvement and business process management, these deviations can become the rule, degrading organizational performance. Dynamic markets also force change. Customer expectations, new product lines, acquisitions, changing geographies, and other events can impact the ability of a process to be performed at full capacity.

In the past, companies would bring in human consultants to identify inefficiencies. This requires a significant investment in high-priced consultants and quantitative analysts and also takes a great deal of time. Using Celonis software, companies are able to easily identify inefficiencies in a matter of four to eight weeks at much lower cost. Process mining enables organizations to find and fix inefficiencies in their business processes by giving them complete real-time visibility.

Celonis offers solutions by system (Salesforce and SAP) as well as core company functions including accounts payable, accounts receivable, procurement, order management, inventory management, and opportunity management. The software is available in both on-premise and online SaaS versions. Celonis process mining software works on top of existing systems.

Celonis process mining software helps companies better understand how workflows through various company systems. Celonis software helps identify bottlenecks in production, loops in logistics, and inefficiencies in purchasing that drag down performance and generate excess operational costs. Celonis tools identify inefficiencies in real time in areas where company performance is suffering.

For example, Celonis helped Deutsche Telekom Services Europe (DTSE) save more than €66 million (about $70 million) by identifying execution gaps and inefficiencies in its Procure-to-Pay process that produced duplicate payments and cash discount losses. (Procure to Pay, or P2P, is the full cycle of actions and events that a business engages in to obtain goods or services from an outside supplier.) DTSE's P2P process handles over 2 million order items and processes almost 9 million invoices a year. Problems with this process created many millions of dollars in losses.

Process mining with Celonis can now focus on more than one process at a time. Celonis Execution Graph technology enables companies to easily evaluate trade-offs as well as optimize different processes simultaneously. Celonis Execution Graph uses graph data structures to tie together data from across business processes to see how they influence each other across systems and suppliers. Another feature, the Multi-Event Log, spots inefficiencies that cross multiple processes.

Celonis has expanded beyond process mining into execution management systems (EMS), which help businesses manage and improve their business processes. Process mining pinpoints where there are process inefficiencies and silos, as well as untapped value inside workflows. Execution management takes action on that information to fix those problems and derive value from unused sources. Execution management works from within the same platform as process mining but takes it a step further by showing what actions can fix problems. It then allows users to take those actions. Employees are able to accept recommended improvements and make changes directly through an easy-to-use question-and-answer-based interface.

Celonis tools helped Avnet, a global leader in electronic components and services, free up more than €6 million (about $6.3 million) in working capital and reduce costs by finding and fixing early supplier deliveries and making adjustments in Accounts Payable. Celonis software issues real-time alerts to buyers so they can cancel their orders if a product is withdrawn from the market. Avnet reported revenues of $25.6 billion in fiscal 2023 and has more than 15,000 employees around the globe. The company has integrated Celonis into many departments, including Procurement, Sales, and Finance, to deliver €10–15 million (about $10.5–$15.75 million) of value year-on-year in working capital impact, productivity gains, and revenue increases.

Food and beverage giant Kraft Heinz initially adopted Celonis EMS to analyze Accounts Receivable (A/R) activity and improve how cash discounts were offered and managed across the company's customer base. The company gained more visibility into customer payment behavior, which has helped it renegotiate many discounts, and optimize how new discounts are offered, recovering a great deal of lost money in the process. Overdue payments have been reduced by 30 percent.

Celonis believes that deriving business value from AI depends on effective process optimization. Celonis has been infusing AI into its tools for years. Process mining functions well suited for AI include crunching massive volumes of data, spotting patterns to determine how processes should function, and identifying discrepancies or anomalies relative

to optimized processes. Celonis makes AI work for the enterprise by feeding it the firm-specific process intelligence AI needs to understand how work flows through the organization.

Celonis gives customers access to its Machine Learning Workbench (MLWB), a fully hosted and managed machine learning tool for building custom ML models and driving more intelligent process-related decision-making. Large language models (LLMs) used by Celonis power a conversational chat tool that makes it possible for even nontechnical users to obtain business insights via intuitive natural language prompts. The LLMs are integrated with PQL generation. PQL is the Celonis Process Query Language that creates and powers everything customers see in the Celonis EMS. PQL enables users to translate process-related business questions into queries, which are then executed by a custom-built query engine. Celonis partnered with Rollio, a developer of online virtual AI assistants, to jointly develop this generative AI tool.

In November 2023, Celonis launched a generative AI innovation called Celonis Process Copilot, which acts as a trusted digital assistant to help customers find value opportunities in their business processes. For example, Celonis Process Copilot lets customers view multiple processes displayed on a subway-style map and ask questions about what they see. This helps them understand how their processes work and from there, understand various ways to improve performance and take action, such as sending follow-up emails or flagging a common process variation with another team.

Process Copilot has a natural language processing (NLP) interface that can infer what users mean in their prompts, even if their queries are not perfect. When users start working with Process Copilot, they are presented with three initial insights about their top-line metrics based on key performance monitoring and identified opportunities. Process Copilot recommendations may include keywords, follow-up questions, opportunities, and insights. BMW is among the companies working with Celonis on AI-enhanced process monitoring. Celonis Process Copilot's natural language querying capabilities will enable BMW to expand its population of end users that have access to process mining.

Celonis is also focusing on helping customers make Celonis data more accessible for the LLMs inside their companies. It is also making it easier for other third-party partners or customers to build applications on top of the process data stored in the Celonis platform, and to process the variety of data types being tracked inside Celonis. This can be challenging for a LLM, especially when each customer might have different ways to describe elements of the same type of process.

The ability of Celonis to capture all the objects and events in the business makes it possible to create a "digital twin" that replicates the business as a single source of truth with up-to-date process data across the enterprise. Celonis customers and partners can use this intelligence to enhance their applications, orchestrate automation, or as a source for large language models and other AI solutions. Process intelligence produces more "intelligent" large language models that make AI work better, faster, and more precisely.

SOURCES: Celonis.com, accessed September 22, 2024; IBM, "Process Mining and Execution Management: An X-ray Diagnosis and Treatment for Business Performance," Ibm.com, accessed September 22, 2024; Tom Smith, "BMW and Celonis Co-Create Process Mining and AI Innovations for the Auto Industry," Accelerationeconomy.com, April 11, 2024; Ron Miller, "Celonis Adds an AI Copilot to Ask Questions about a Process Map," Techcrunch.com, November 14, 2023.

Case Study Questions

1. How can using Celonis provide a competitive advantage for a business?
2. What kinds of companies would benefit from using Celonis tools? What management, organizational, and technology factors should be addressed when making this decision?
3. How does a business benefit from using Celonis process mining tools with AI?

Review Summary

13.1 Explain how new information systems can promote organizational change.

Building a new information system is a form of planned organizational change. Four kinds of technology-enabled change are (1) automation, (2) rationalization of procedures, (3) business process redesign, and (4) paradigm shift. Each carries different risks and rewards.

13.2 Understand how IT can be used to manage and redesign business processes.

Many organizations are using business process management (BPM) to redesign workflows and business processes in the hope of achieving dramatic productivity breakthroughs. Companies practicing BPM go through a five-step process: identify processes for change; analyze

existing processes; design the new process; implement the new process; and continuous measurement. The largest single barrier to successful business process change is organizational culture. There are a number of software tools for various aspects of BPM, including process mining. Process mining analyzes data from a company's event logs to identify problems with certain processes and figure out ways to improve them.

13.3 Describe the core activities in the systems development process.

The core activities in systems development are systems analysis, systems design, programming, testing, conversion, production, and maintenance. Systems analysis is the study and analysis of problems of existing systems and the identification of requirements for their solutions. Systems design provides the specifications for an information system solution, showing how its technical and organizational components fit together. The remaining steps in the process translate the solution specifications into a fully operational information system.

13.4 Discuss methods for building and acquiring information systems.

The traditional method for building systems is the systems development life cycle (SDLC), which requires that information systems be developed in formal stages. The stages proceed sequentially and have defined outputs. Each requires formal approval before the next stage can commence. The SDLC approach is useful for large projects that need formal specifications and tight management control over each stage of systems building, but it is very rigid and costly.

Prototyping consists of building an experimental system rapidly and inexpensively for end users to interact with and evaluate. Prototyping encourages end-user involvement in systems development and iteration of design until specifications are captured accurately. The rapid creation of prototypes can result in systems that have not been completely tested or documented, or that are technically inadequate for a production environment.

Using a software package or online software services (SaaS) reduces the amount of design, programming, testing, installation, and maintenance work required to build a system. Application software packages or SaaS are helpful if a firm does not have the internal information systems staff or financial resources to custom develop a system. To meet an organization's unique requirements, packages may require extensive modifications that can substantially raise development costs.

Outsourcing consists of using an external vendor to build (or operate) a firm's information systems instead of the organization's internal information systems staff. Outsourcing can save application development costs or enable firms to develop applications without an internal information systems staff. However, firms risk losing control over their information systems and becoming too dependent on external vendors. Outsourcing also entails hidden costs, especially when the work is sent offshore.

13.5 Describe methods that enable rapid systems development.

Object-oriented design uses the object as the basic unit of systems analysis and design. An object combines data and the specific processes that operate on those data. Objects belonging to a certain class, or general category of similar objects, have the features of that class. Classes of objects in turn can inherit all the structure and behaviors of a more general class and then add variables and behaviors unique to each object. Because objects are reusable, object-oriented development can potentially reduce the time and cost of creating software.

Rapid application development (RAD) uses object-oriented software, visual programming, prototyping, and tools for very rapid creation of systems. Joint application development (JAD) is a methodology to ensure that the client, consumer, IT specialists, and stakeholders are all involved in designing and developing a technology application. Agile development breaks a large project into a series of small subprojects that are completed in short periods of time using iteration and continuous feedback. DevOps emphasizes close collaboration between the software developers who create applications and the IT operational staff who run and maintain the applications.

13.6 Discuss the increasing democratization of software development.

Open-source software has become a major software development tool. It uses publicly available software free of charge (or at low cost), as building blocks for new applications. Low-code development is a software development approach that enables the delivery of applications faster and with minimal hand-coding using visual modeling in a graphical interface to assemble and configure applications. No-code development tools are even easier for non-IT businesspeople to use. Everything the software vendor thinks the user needs to create an app is already built into the tool, with no coding required. Citizen developers are business users who build new applications or modify existing ones to create applications on their own with IT approval or with minimal help from IT staff.

13.7 Understand the impact of AI on software development.

AI is enabling developers to automate routine tasks, streamline workflows, and improve the accuracy and efficiency of software development processes. AI-powered tools and frameworks enable developers to write code faster and more accurately, identify and fix bugs more quickly, and optimize software performance.

Developing AI software has a number of challenges, for instance the need for more software developer expertise in AI, which is currently in short supply, and the need for very large amounts of high-quality

data to train AI models. Organizations that do not have the necessary resources may be able to use pre-trained AI models or cloud-based machine learning platforms that offer scalable infrastructure and pre-built tools and frameworks for model development (AI as a Service, or AIaaS).

13.8 Discuss approaches for building mobile applications.

Mobile application development must pay attention to simplicity, usability, and the need to optimize tasks for tiny screens. Responsive web design enables websites to change layouts automatically for different screen sizes and mobile devices.

Key Terms

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Review Questions

13.1 Explain how new information systems can promote organizational change.

- Describe each of the four kinds of organizational change that can be promoted with information technology.

13.2 Understand how IT can be used to manage and redesign business processes.

- Define business process management and describe the steps required to carry it out.
- Define process mining and explain how businesses can use it to extract more value from business processes.

13.3 Describe the core activities in the systems development process.

- Distinguish between systems analysis and systems design and describe the activities for each.
- Define information requirements and explain why they are difficult to determine correctly.
- Explain why the testing stage of systems development is so important, and name and describe the three stages of testing for an information system.
- Describe the role of programming, conversion, production, and maintenance in systems development.

13.4 Discuss methods for building and acquiring information systems.

- Define the traditional systems development life cycle and describe its advantages and disadvantages for systems building.
- Define information system prototyping and describe its benefits and limitations. List and describe the steps in the prototyping process.
- Discuss the advantages and disadvantages of developing information systems based on software packages and cloud software services.
- Describe the advantages and disadvantages of using outsourcing for building information systems.

13.5 Describe methods that enable rapid systems development.

- Define object-oriented design and explain how it enables more rapid systems development.
- Define rapid application development (RAD) and joint application development (JAD) and explain how they enable more rapid systems development.
- Define agile development and explain how it enables more rapid systems development.
- Define DevOps and explain how it enables more rapid systems development.

13.6 Discuss the increasing democratization of software development.

- Define open-source software and discuss the advantages and disadvantages of open-source software development.
- Define low-code and no-code software development and discuss how it has accelerated the process by which software can be created.
- Define citizen developers and describe how today's citizen development is different from end-user development in the past.

13.7 Discuss the impact of AI on software development.

- Describe how the use of AI has impacted software development.
- Discuss the challenges of developing AI software.

13.8 Discuss approaches for building mobile applications.

- Explain the features of mobile application development and responsive web design.
- Define and compare a mobile website, native app, mobile web app, and hybrid app.

Discussion Questions

13.9 Why is selecting a systems development approach an important business decision? Who should participate in the selection process?

13.10 Some have said that the best way to reduce systems development costs is to use application software packages, SaaS, or user-friendly tools, such as low-code or no-code tools. Do you agree? Why or why not?

13.11 Why is it so important to understand how a business process works when trying to develop a new information system?

Using AI Project

13.12 AI Project: Identifying Business Processes and Process Efficiencies

You have been hired as a consultant by two friends who want to open a retail store selling organic pet food and treats, as well as pet toys and clothing made from organic materials. You have advised them to implement a point of sale (POS) system that has online ordering functionality. Previous retailers that they have worked at have not had such a system, so they would like to know how the system would change business processes and how these changes would create value for the business.

Using Microsoft Copilot, Google's Gemini, or ChatGPT, develop a series of prompts to assist you in providing them with this information. Save a copy of your prompts and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience in analyzing business processes; using a database to clarify business strategy; and analyzing website information requirements.

For student files please visit www.pearsonhighered.com/laudon or MyLab MIS.

13.13 Management Decision Problem

Management at your cosmetic products company has been dissatisfied with production planning. Production plans are created using best guesses of demand for each product, which are based on how much of each product has been ordered in the past. If a customer places an unexpected order or requests a change to an existing order after it has been placed, there is no way to adjust production plans. The company may have to tell customers it can't fill their orders, or it may run up extra costs maintaining additional inventory to prevent stock-outs.

At the end of each month, orders are totaled and manually keyed into the company's production planning system. Data from the past month's production

and inventory systems are manually entered into the firm's order management system. Analysts from the sales department and from the production department analyze the data from their respective systems to determine what the sales targets and production targets should be for the next month. These estimates are usually different. The analysts then get together at a high-level planning meeting to revise the production and sales targets to take into account senior management's goals for market share, revenues, and profits. The outcome of the meeting is a finalized production master schedule.

The entire production planning process takes 17 business days to complete. Nine of these days are required to enter and validate the data. The remaining days are spent developing and reconciling the production and sales targets and finalizing the production master schedule.

- Draw a diagram of the existing production planning process.
- Analyze the problems this process creates for the company.
- How could an enterprise system solve these problems? In what ways could it lower costs? Diagram what the production planning process might look like if the company implemented enterprise software.

13.14 Using a Database to Clarify Business Strategy

Software skills: Database querying and reporting; querying a multi-table database

Business skills: Reservation systems; customer analysis

In this exercise, you'll use database software to analyze the reservation transactions for a hotel and use that information to fine-tune the hotel's business strategy and marketing activities.

The MIS18_Project 13.14 Student File contains a database for hotel reservation transactions with information about Fair Winds Inn in Sanibel, Florida. At the Inn, 10 rooms overlook side streets, 10 rooms have windows with limited views of the ocean, and the remaining 10 rooms in the front of the hotel face the ocean. Room rates are based on the type of room.

The owners currently use a manual reservation and bookkeeping system, which cannot provide management with immediate data about the hotel's daily operations and revenue. They need to know which room and type of room has been assigned to each guest along with the guest's name and start and end dates for that guest's stay; the average number of visitors per room type; and the average length of visit per room type. Create queries and reports from the database to answer these questions. Then write a brief report on how the new database system solves some of the problems created by the old, manual reservation system and how the database can help Fair Winds Inn refine its business strategy.

13.15 Analyzing Website Design and Information Requirements

Software skills: Web browser software

Business skills: Information requirements analysis, website design

Visit a website of your choice and explore it thoroughly. Prepare a report analyzing the various functions provided by that website and its information requirements. Your report should answer these questions: What functions does the website perform? What data does it use? What are its inputs, outputs, and processes? What are some of its other design specifications? Does the website link to any internal systems or systems of other organizations? What value does the website provide for the firm?

13.16 Collaboration and Teamwork Project: Preparing Website Design Specifications

With two or more of your classmates, select a system described in this text that uses the web. Review the website for the system you select. Use what you have learned from the website and the description in this book to prepare a report describing some of the design specifications for the system you select. If possible, use Google Docs, Google Drive, or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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14

Making the Business Case for Information Systems and Managing Projects

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Learning Objectives

After completing this chapter, you will be able to:

14.1 Explain how to build a business case.

14.2 Describe methods for evaluating IT investments.

14.3 Describe project management methodologies and variables.

14.4 Understand the importance of effective project management.

14.5 Discuss project risk factors.

14.6 Understand how project risks can be managed.

14.7 Understand how the information in this chapter can help your career.

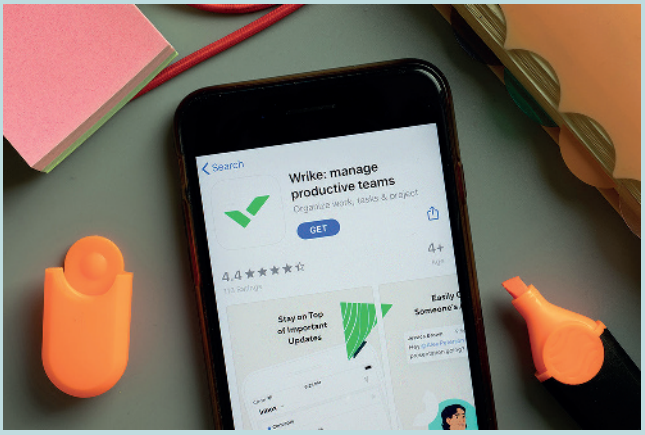
Chapter Case Studies

- **Opening Case Study:** BigCommerce Turns to Wrike for Better Project Management Tools
- **Spotlight on Management Case Study:** Happy Feet International Overcomes Change Management and Implementation Issues
- **Chapter Case Study:** Southwest Airlines Invests in Digital Transformation to Navigate Future Turbulence

Opening Case Study

BigCommerce Turns to Wrike for Better Project Management Tools

BigCommerce is a global e-commerce technology company with headquarters in Austin, Texas, and offices throughout the United States, as well as in the UK, Ukraine, and Australia. BigCommerce provides a cloud-based software-as-a-service (SaaS) e-commerce platform to companies large and small. Its platform enables retailers to build B2C, B2B, and omnichannel e-commerce solutions with the flexibility to fit their unique businesses and products. BigCommerce offers a comprehensive platform that includes online store design, catalog management, hosting, checkout, order management, reporting, and integration with third-party payment, shipping and fulfillment, point of sale, and marketing, as well as other back-office services, such as accounting. It provides access to its platform on a subscription basis, with plans tailored to customers' needs.



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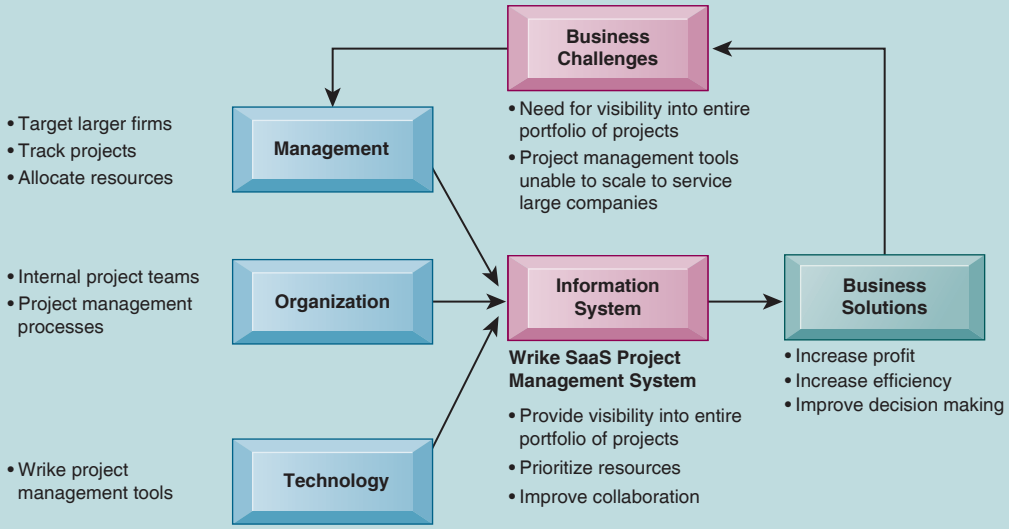
BigCommerce also offers a number of development and professional services to help its clients. For example, it provides a variety of "launch services" such as solutions architecting services, data migration services, implementation project management, and enterprise launch project management.

In the past, BigCommerce had used spreadsheets, email, and other collaboration tools such as Google Drive and Google Docs to manage the delivery of its development and professional services projects, but management believed that such tools were not going to be sufficient to help it work with the larger companies that it wished to target for growth. To scale its efforts, BigCommerce wanted a project management tool that would enable it to see its whole portfolio of projects, which often numbered more than 100 globally at any given time. It also wanted to be able to measure metrics and data across the portfolio and enable project managers to see what was on schedule and what was not. Finally, it wanted a solution that could help project managers manage interactions with customers.

To address this challenge, BigCommerce went looking for a project management tool that would help it standardize its processes, improve reporting, and give its customers more visibility into the progress of the project, all integrated into a single platform. It investigated a number of possibilities, including Smartsheet, Microsoft Project, Asana, Klient, and Jira, among others, before settling on Wrike.

Wrike is an integrated SaaS project management tool that is available both as a web-based tool as well as an iOS and Android app. It is available in ten languages (English, French, German, Italian, Spanish, Portuguese, Russian, Japanese, Chinese, and Korean). Wrike provides all of the functionalities that BigCommerce was seeking. It enables companies to coordinate projects across teams with a single tool and gives them visibility into project progress, resources, and costs. It also helps improve accountability of project team members by making it easy to assign tasks, understand who is responsible for what, and identify bottlenecks. Wrike is secure and easy to set up and integrate into existing infrastructure and applications, such as Slack and Salesforce. Wrike is also able to accommodate different project management approaches, such as agile project management, as well as a traditional waterfall approach.

According to BigCommerce, using Wrike has proven to be extremely helpful to its managers. Having all projects organized in one place has enabled project managers to become much more efficient, as well as more effective in communicating with customers. Wrike's reporting and analytics have helped BigCommerce make more informed business decisions and to be able to better predict its costs and resourcing needs. It can now better understand how much time is spent on projects versus how much is being charged. Wrike has also given BigCommerce's leadership the visibility to understand the progress of projects and to be able to act proactively as necessary to get projects back on track. For instance, BigCommerce is experimenting with Wrike's ability to analyze team efficiency by project phase, to determine which aspects of each project are the most time consuming. Using Wrike has enabled BigCommerce to improve both "time to value" (a metric that measures the amount of time it takes a BigCommerce customer to get value from its products or services) and customer satisfaction, which are two of BigCommerce's key performance indicators (KPIs).



SOURCES: Bigcommerce.com, accessed September 24, 2024; Wrike.com, accessed September 24, 2024; "BigCommerce Streamlines Systems to Scale Services," Wrike.com, accessed September 24, 2024; Richard Blatcher, "Enterprise Project Management: Comparing 5 Options," Wrike.com, February 14, 2024.

One of the principal challenges posed by information systems is ensuring they deliver genuine business benefits. Many information systems projects fail because organizations do not manage the organizational change surrounding the introduction of new technology effectively or have incorrectly assessed its business value. Projects to build or improve information systems, such as those being implemented by BigCommerce on behalf of its customers, require special managerial and organizational techniques to make them effective.

BigCommerce's management realized this when it decided to replace outdated project management tools with Wrike, an integrated cloud SaaS-based tool. Implementing Wrike involved changes to project management business processes as well as new software. BigCommerce succeeded with this project because its management clearly understood that strong project management and attention to organizational change were essential to success.

The Opening Case Study diagram above calls attention to important points raised by this case and this chapter. BigCommerce's business processes and project management tools for project management posed obstacles to accurate resource allocation and management's ability to have an enterprise-wide overview of all of the company's projects. To deal with these challenges, management decided to implement a new state-of-the-art cloud-based project management system. Implementing Wrike improved BigCommerce's resource efficiency, collaboration, and ability to manage its portfolio of projects.

Case Study Questions

1. What were the challenges that BigCommerce faced in project management?
2. Why did BigCommerce choose Wrike instead of other project management tools?
3. How did Wrike help BigCommerce solve the project management challenges it was facing?

CAREER BENEFIT

Managers who recognized that BigCommerce's existing project management system would be insufficient to achieve BigCommerce's business goals, and who then chose Wrike, benefitted by enabling the company to become more efficient in managing its professional services projects and enhancing customer value. Project managers using Wrike project management tools benefitted by becoming skilled in using a state-of-the-art project managementsystem and being able to better serve their clients, benefitting them while at BigCommerce and also making them more attractive to future potential employers.

14.1 Explain How to Build a Business Case

Project

Planned series of related activities for achieving a specific business objective.

Companies typically are presented with many alternatives for solving problems and improving their performance, including the development of new information systems or enhancement of existing ones. A **project** is a planned series of related activities for achieving a specific business objective. Information systems projects include the development of new information systems, enhancement of existing systems, or upgrade or replacement of the firm's information technology (IT) infrastructure. There are far more ideas for systems projects than there are resources. Your company will need to select the systems projects investments that promise the greatest benefit to the business. And you will need to make the business case for why the solution you select provides the greatest value to the firm compared to other solutions.

Business case

Proposal to management seeking approval for an investment by describing its costs, benefits, and business value.

A **business case** is a proposal to management seeking approval for an investment. The business case for an IT investment describes the problem facing the organization that can be solved by investing in a proposed system solution. It provides an analysis of all the costs, benefits, and risks associated with that investment and the justification for that proposed course of action. The business case describes the rationale for proceeding with an investment and shows how the investment supports the firm's strategic goals and business objectives and how it fits in with the overall information systems plan of the firm. It also provides the information necessary to make an informed decision about whether to proceed with the investment and in what form. The business case explains how this investment will provide value for the business and identifies any risks that could negatively affect outcomes. The business case identifies alternative solutions, along with the deciding factors for selecting the preferred option. A good business case will also describe how the proposed solution may require changes in organizational culture, systems, processes, and jobs (Gartner, 2024).

Examine Figure 14.1, which summarizes the seven major factors that are used in making the business case for a specific new system. (Review the discussion of business drivers of information systems in Chapter 1.) These factors are: (1) long-term strategic (lowering production costs, differentiation of products and services, increasing the scope of the firm

Figure 14.1 Factors to Consider in Making the Business Case

There are seven factors that should be addressed when making the business case for a new information system.

[e.g., global expansion,] and matching or exceeding competitor capabilities); (2) improved decision making; (3) customer and supplier relationships; (4) survival (required by the market and ESG goals); (5) new products and services; (6) financial rationale; and (7) fitting with the long-term IT plan of the firm. Smaller systems that focus on a single problem will focus on just a few of these elements, such as "improved decision making," "customer relationships," and "lowering costs," whereas larger system projects may well include all of these factors in making the business case.

The Information Systems Plan

In order to identify the information systems projects that will deliver the most business value, organizations need an enterprise-wide information systems plan that supports their overall business plan, with strategic systems incorporated into top-level planning. An **information systems plan** serves as a road map indicating the direction of systems development (the purpose of the plan), the rationale, the state of current systems, new developments to consider, the management strategy, the implementation plan, and the budget (see Table 14.1). The plan is typically developed under the supervision of the firm's Chief Information Officer (CIO) and is approved annually by the CEO and often the Board of Directors. Without a comprehensive firm-wide information systems plan, it is difficult, if not impossible to evaluate the worth of proposals for developing specific individual systems. You cannot make the case for a specific new system without understanding the larger context of all the many systems in the firm.

Information systems plan

Road map indicating the direction of systems development (the purpose of the plan), the rationale, the current situation, the management strategy, the implementation plan, and the budget.

The plan contains a statement of organizational goals and specifies how information technology will support the attainment of those goals. It explains how general goals will be achieved by specific systems projects. It identifies specific target dates and milestones that can be used later to evaluate the plan's progress in terms of how many objectives were actually attained in the time frame specified in the plan. The plan indicates the key management decisions, technology, and required organizational change.

In order to plan effectively, firms will need to inventory and document all of their information system applications, IT infrastructure components, and long- and short-term information requirements. For projects in which benefits involve improved decision making, managers should try to identify the decision improvements that would provide the greatest additional value to the firm (see Chapter 12). They should then develop a set of metrics to quantify the value of more timely and precise information on the outcome of the decision.

The plan should describe organizational changes, including management and employee training requirements; changes in business processes; and changes in authority, structure, or management practice. When you are making the business case for a new information system project, you show how the proposed system fits into that plan.

14.2 Describe Methods for Evaluating IT Investments

There are a variety of methods that can be used to evaluate a proposed IT investment, including portfolio analysis, scoring models, capital budgeting models, as well as software tools.

Portfolio Analysis

Once strategic analyses have determined the overall direction of systems development in the firm by establishing a firm-wide information systems plan, **portfolio analysis** is one tool you can use to evaluate alternative systems projects. Portfolio analysis inventories all of the organization's information systems projects and assets, including infrastructure, outsourcing contracts, and licenses. This portfolio of information systems investments can be described as having a certain profile of risk and benefit to the firm similar to a financial portfolio. Each information systems project carries its own set of risks and benefits. (Section 14.3 describes the factors that increase the risks of systems projects.) Firms try to improve the return on their portfolios of IT assets by balancing the risk and return from their systems investments (Singh, 2023).

Portfolio analysis

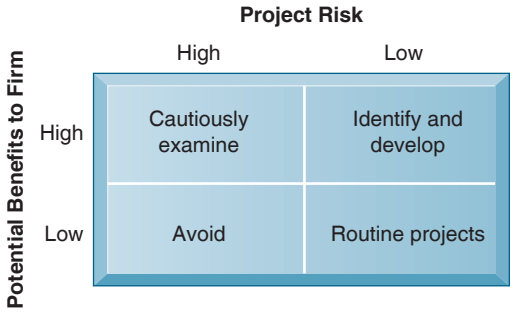
Tool that can be used to evaluate systems projects; inventories all of an organization's information systems projects and assets, and analyzes them to determine their respective risks and benefits.

Table 14.1 Information Systems Plan

1.	Purpose of the Plan
	Current business organization and future organization
	Key business processes
2.	Strategic Business Plan Rationale
	Current situation
	Current business organization
	Changing environments
	Major goals of the business plan
	Firm's strategic plan
3.	Current Systems
	Major systems supporting business functions and processes
	Current infrastructure capabilities
	Hardware
	Software
	Database
	Networking and Internet
	Cloud services
	Security
	Difficulties meeting business requirements
	Anticipated future demands
4.	New Developments
	New system projects
	Project descriptions
	Business rationale
	Applications' role in strategy
	New infrastructure capabilities required
	Hardware
	Software
	Database
	Networking and Internet
	Cloud services
	Security
	AI
5.	Management Strategy
	Acquisition plans
	Organizational realignment
	Management controls
	Major training initiatives
	Human resources strategy
6.	Implementation of the Plan
	Anticipated difficulties in implementation
	Progress reports and milestones
7.	Budget Requirements
	Resources
	Potential savings
	Financing
	Acquisition cycle

Figure 14.2 A System Portfolio

Companies should examine their portfolio of projects in terms of potential benefits and likely risks. Certain kinds of projects should be avoided altogether, and others developed rapidly. There is no ideal mix. Companies in different industries have different profiles.



Examine Figure 14.2, which illustrates how firms can use a portfolio analysis matrix to evaluate whether the firm should undertake a project. You begin first by focusing on systems of high benefit and low risk. These promise early returns and low risks. Second, high-benefit, high-risk systems should be examined; low-benefit, high-risk systems should be totally avoided; and low-benefit, low-risk systems should be re-examined for the possibility of rebuilding and replacing them with more desirable systems having higher benefits. By using portfolio analysis, management can determine the optimal mix of investment risk and reward for their firms, balancing riskier, high-reward projects with safer, lower-reward ones.

Scoring Models

Scoring model

A method for deciding among alternative systems solutions when many criteria must be considered; assigns weights to various features of a system and then calculates weighted totals.

Another method for evaluating alternative systems solutions is a scoring model. A **scoring model** is useful for selecting projects where many criteria must be considered. It assigns weights to various features of a system and then calculates the weighted totals. Table 14.2 shows an example of a scoring model that can help a firm decide among two alternative

Table 14.2 Example of a Scoring Model for an ERP System

Criteria	Weight	ERP System A %	ERP System A Score	ERP System B %	ERP System B Score
1.0 Order Processing					
1.1 Online order entry	4	67	268	73	292
1.2 Online pricing	4	81	324	87	348
1.3 Inventory check	4	72	288	81	324
1.4 Customer credit check	3	66	198	59	177
1.5 Invoicing	4	73	292	82	328
Total Order Processing			1,370		1,469
2.0 Inventory Management					
2.1 Production forecasting	3	72	216	76	228
2.2 Production planning	4	79	316	81	324
2.3 Inventory control	4	68	272	80	320
2.4 Reports	3	71	213	69	207
Total Inventory Management			1,017		1,079
3.0 Warehousing					
3.1 Receiving	2	71	142	75	150
3.2 Picking/packing	3	77	231	82	246
3.3 Shipping	4	92	368	89	356
Total Warehousing			741		752
Grand Total			3,128		3,300

enterprise resource planning (ERP) systems. The first column lists the criteria that decision makers will use to evaluate the systems. These criteria are usually the result of lengthy discussions among the decision-making group. Often the most important outcome of a scoring model is not the score but agreement on the criteria used to judge a system (Betadam, 2020).

Table 14.2 shows that this particular company attaches the most importance to capabilities for sales order processing, inventory management, and warehousing. The second column in Table 14.2 lists the weights that decision makers attached to the decision criteria. Columns 3 and 5 show the percentage of requirements for each function that each alternative ERP system can provide. Each alternative's score can be calculated by multiplying the percentage of requirements met for each function by the weight attached to that function. ERP System B has the highest total score.

Determining Solution Costs and Benefits

As we pointed out earlier, the business case for a system solution includes an assessment of whether each solution represents a good investment for the company.

Even if a systems project supports a firm's strategic goals and meets user information requirements, it needs to be a good investment for the firm. The value of systems from a financial perspective essentially revolves around the issue of return on invested capital. Does a particular information system investment produce sufficient returns to justify its costs?

Information System Costs and Benefits

Tangible benefits

Benefits that can be quantified and assigned a monetary value.

Intangible benefits

Benefits that cannot be immediately quantified, such as more efficient customer service or enhanced decision making, but that can lead to quantifiable gains in the long run.

Table 14.3 lists some of the more common costs and benefits of systems. **Tangible benefits** can be quantified and assigned a monetary value. **Intangible benefits**, such as more efficient customer service or enhanced decision making, cannot be immediately quantified but may lead to quantifiable gains in the long run. Transaction and clerical systems that displace labor and save space always produce more measurable, tangible benefits than management information systems, decision-support systems, and systems for collaborative work (see Chapter 2).

Chapter 5 introduced the concept of total cost of ownership (TCO), which is designed to identify and measure the components of information technology expenditures beyond the initial cost of purchasing and installing hardware and software. TCO analysis, however, provides only part of the information needed to evaluate an information technology investment because it typically does not deal with benefits, cost categories such as complexity costs, and "soft" and strategic factors discussed later in this section.

Using a Capital Budgeting Model to Evaluate Information Systems Projects

Capital budgeting models

Method of measuring the value of investing in long-term information systems projects.

To determine the benefits of a particular project, you'll need to calculate all of its costs and all of its benefits. Obviously, a project where costs exceed benefits should be rejected. But even if the benefits outweigh the costs, additional financial analysis is required to determine whether the project represents a good return on the firm's invested capital. **Capital budgeting models** are one way to measure the value of investing in long-term information systems projects.

Capital budgeting models rely on measures of cash flows into and out of the firm; capital projects generate those cash flows. The investment cost for information systems projects is an immediate cash outflow caused by expenditures for hardware, software, and labor. In subsequent years, the investment may cause additional cash outflows that will be balanced by cash inflows resulting from the investment. Cash inflows take the form of increased sales of more products (for reasons such as new products, higher quality, or increasing market share) or reduced costs in production and operations. The difference between cash outflows and cash inflows is used for calculating the financial worth of an investment. Once the cash flows have been established, several alternative methods are available for comparing different projects and deciding about the investment.

The principal capital budgeting models for evaluating IT projects are the payback method, the accounting rate of return on investment (ROI), net present value (NPV), and the internal rate of return (IRR). Examine Figure 14.3, which illustrates part of the capital budgeting analysis for a mobile sales system.

Table 14.3 Costs and Benefits of Information Systems

Costs
Hardware
Networking
Software
Services
Personnel
Tangible Benefits (Cost Savings)
Increased productivity
Lower operational costs
Reduced workforce
Lower computer expenses
Lower outside vendor costs
Lower clerical and professional costs
Reduced rate of growth in expenses
Reduced facility costs
Intangible Benefits
Improved asset utilization
Improved resource control
Improved organizational planning
Increased organizational flexibility
More timely information
Improved customer experience
Increased organizational learning
Legal requirements attained
Enhanced employee goodwill
Increased job satisfaction
Improved decision making
Improved operations
Higher client satisfaction
Better corporate image

Limitations of Financial Models

The traditional focus on the financial and technical aspects of an information system tends to overlook the social and organizational dimensions of information systems that may affect the true costs and benefits of the investment. Many companies' information systems investment decisions do not adequately consider costs from organizational disruptions created by a new system, such as the cost to train end users, the impact that users' learning curves for a new system have on productivity, or the time managers need to spend overseeing new system-related changes. Intangible benefits such as more timely decisions from a new system or enhanced employee learning and expertise may also be overlooked in a traditional financial analysis.

Software to Help Make Technology Investment Decisions

Today, information technology executives are increasingly embracing **technology business management (TBM)**, a framework aimed at improving business outcomes by giving organizations a consistent method of translating technology investments, such as the development of a new information system, into business value. Spending on **IT financial management tools (ITFM tools)** that automate TBM and help organizations analyze, plan, and control costs

Technology business management (TBM)

Framework aimed at improving business outcomes by giving organizations a consistent method of translating technology investments into business value.

IT financial management tools (ITFM tools)

Tools that automate TBM and help organizations analyze, plan, and control costs associated with IT projects.

Figure 14.3 Capital Budgeting for an Information System Investment

This spreadsheet illustrates a simplified capital budgeting analysis for a mobile sales system.

Used with permission from Microsoft

	A	B	C	D	E	F	G	H
1	Estimated Costs & Benefits - Mobile Online Ordering System							
2	Year		0	1	2	3	4	5
3			2024	2025	2026	2027	2028	2029
4	Costs							
5	Hardware							
6	50 iPads @$500		$25,000					
7	Cloud IaaS		$4,000	$2,000	$2,000	$2,000	$2,000	$2,000
8	Networking		$1,500	$1,500	$1,500	$1,500	$1,500	$1,500
9	Software							
10	Mobile ordering app		$35,000					
11	Integration with ERP		$25,000					
12	Human Resources							
13	Business Staff		$10,000					
14	IT Staff + Consultants		$45,000	$2,000	$2,000	$2,000	$2,000	$2,000
15	Training		$7,000	$1,000	$1,000	$1,000	$1,000	$1,000
16	Maintenance and Support			$5,000	$5,000	$5,000	$5,000	$5,000
17		Annual Costs	$1,52,500	$11,500	$11,500	$11,500	$11,500	$11,500
18		Total Costs	$2,10,000					
19	Benefits							
20	Reduced labor costs			$52,000	$52,000	$52,000	$52,000	$52,000
21	Reduced errors and returns			$70,000	$70,000	$70,000	$70,000	$70,000
22		Annual Net Cash Flow	-$1,52,500	$1,10,500	$1,10,500	$1,10,500	$1,10,500	$1,10,500
23		Total Benefits	$4,00,000					
24								
25		Net Present Value	$2,68,407					
26		ROI	4.1%					
27		Internal Rate of Return	17.0%					

associated with IT projects, has been growing rapidly. ITFM tools enable organizations to create and maintain a comprehensive IT budget, track expenses, and perform cost-benefit analyses of IT investments. IBM Apptio is a leading ITFM tool that provides a "single source of truth" for making technology spending decisions, with functionality for managing technology costing, planning, billing, and benchmarking.

14.3 Describe Project Management Methodologies and Variables

Project management

Application of knowledge, skills, tools, and techniques to achieve specific targets within specified budget and time constraints.

Project management refers to the application of knowledge, skills, tools, and techniques to achieve specific targets within specified budget and time constraints. Project management activities include planning the work, assessing risk, estimating resources required to accomplish the work, organizing the work, acquiring human and material resources, assigning tasks, directing activities, controlling project execution, reporting progress, and analyzing the results.

Project Management Methodologies

Waterfall project management

Traditional format for project management; uses a somewhat rigid and linear approach that typically advances in direct phases.

Agile project management

Project management approach that allows for more adaptability and flexibility; uses "sprints" where the project team works to complete a set amount of work within a short period.

As with systems development, there are various project management methodologies. The two most common are the waterfall methodology and the agile approach. The **waterfall project management** approach is the traditional format for project management. The waterfall method, like the SDLC approach to systems development, uses a somewhat rigid and linear approach that typically advances in direct phases, from determining requirements, to implementation, to deployment. More recently, agile project management has become increasingly popular. **Agile project management** allows for more adaptability and flexibility, and like agile software development, uses "sprints", where the project team works to complete a set amount of work within a short, specified time period. Although the agile method typically includes documentation, it does not emphasize it as much as the waterfall method, where documentation is often very detailed (Wells, 2023).

Project Management Variables

As in other areas of business, project management for information systems must deal with five major variables: scope, time, cost, quality, and risk, as well as the necessity for trade-offs between these variables.

Scope defines what work is or is not included in a project. For example, the scope of a project for a new order processing system might be to include new modules for inputting orders and transmitting them to production and accounting but not any changes to related accounts receivable, manufacturing, distribution, or inventory control systems. Project management defines all the work required to complete a project successfully and should ensure that the scope of a project does not expand beyond what was originally intended.

Scope
Defines what work is and is not included in a project.

Time is the amount of time required to complete the project. Project management typically establishes the amount of time required to complete major components of a project. Each of these components is further broken down into activities and tasks. Project management tries to determine the time required to complete each task and establish a schedule for completing the work.

Cost is based on the time to complete a project multiplied by the cost of human resources required to complete the project. Information systems project costs also include the cost of hardware, software, and workspace. Project management develops a budget for the project and monitors ongoing project expenses.

Quality is an indicator of how well the end result of a project satisfies the objectives specified by management. The quality of information systems projects usually boils down to improved organizational performance and decision making. Quality also considers the accuracy and timeliness of information produced by the new system and ease of use.

Risk refers to potential problems that would threaten the success of a project. These potential problems might prevent a project from achieving its objectives by increasing time and cost, lowering the quality of project outputs, or preventing the project from being completed altogether. Section 14.4 describes the most important risk factors for information systems.

Technical debt (also sometimes called tech debt or code debt) results from the trade-offs between time, cost, and quality that often occur in the development of an information system. Project teams may trade-off quality by taking short cuts or creating workarounds to meet delivery deadlines. Sometimes, these are prudent choices, based on a reasoned decision that time is the highest priority. In other instances, these short-term actions may be either reckless or inadvertent and have a long-term impact in terms of the performance, scalability, resiliency, or other functionality of the information system. As time passes, the organization may be required to spend an increasing amount of time and money dealing with accumulated technical debt issues and be less capable of focusing on growth. According to management consulting firm Protiviti, technical debt is the leading obstacle to innovation for nearly 70 percent of the organizations it surveyed. According to McKinsey, which has developed a metric to quantify technical debt, firms that are in the top 80th percentile in terms of working to manage and reduce their technical debt have revenue growth that is 20 percent higher than firms in the bottom 20th percentile (Pratt, 2023; McKinsey Digital, 2022).

Technical debt
Results from the trade-offs between time, cost, and quality that often occur in the development of an information system. Sometimes also called tech debt or code debt.

14.4 Understand the Importance of Effective Project Management

There is a very high failure rate among information systems projects that have not been properly managed. In nearly every organization, information systems projects take much more time and money to implement than originally anticipated, or the completed system does not work properly. Firms may have incorrectly assessed the business value of the new system or be unable to manage the organizational change the new technology requires. When an information system does not meet expectations or costs too much to develop, companies may not realize any benefit from their information system investment, and the system may not be able

to solve the problems for which it was intended. The development of a new system must be carefully managed and orchestrated, and the way a project is executed is likely to be the most important factor influencing its outcome. That's why it's essential to have some knowledge about managing information systems projects and the reasons why they succeed or fail.

Runaway Projects and System Failure

How badly are projects managed? On average, private sector projects are underestimated by half in terms of budget and time required to deliver the complete system promised in the system plan. Many projects are delivered with missing functionality (promised for delivery in later versions). A joint study by McKinsey and Oxford University found that large software projects on average run 66 percent over budget and 33 percent over schedule. Another McKinsey study found that the average digital transformation project stands a 45 percent chance of delivering less profit than expected, with the chance of surpassing project expectations only one in 19. Similarly, a Forrester Consulting study of firms that had invested at least $500 million in digital transformation initiatives found that more than 50 percent of those firms felt that less than half of the amount invested in information systems projects was translated into tangible business value. Overall, only 20 percent of the firms believed that their organization's project had been successful in achieving the desired business outcomes (Bughin, Deakin, and O/Beirne, 2019; Forrester Consulting, 2024).

Examine Figure 14.4, which illustrates the consequences that a systems development project without proper management will most likely suffer, such as:

- Costs that vastly exceed budgets
- Unexpected time slippage
- Technical performance that is less than expected
- Failure to obtain anticipated benefits

The systems produced by failed information projects are often not used in the way they were intended or are not used at all. Users often have to develop parallel manual systems to make these systems work.

The actual design of the system may fail to capture essential business requirements or improve organizational performance. Information may not be provided quickly enough to be helpful, it may be in a format that is impossible to digest and use, or it may represent the wrong pieces of data.

The way in which nontechnical business users must interact with the system may be excessively complicated and discouraging. A system may be designed with a poor user interface. The **user interface** is the part of the system with which end users interact. For example, an online input form or data entry screen may be so poorly arranged that no one wants to submit data or request information. System outputs may be displayed in a format that is too difficult to comprehend.

User interface

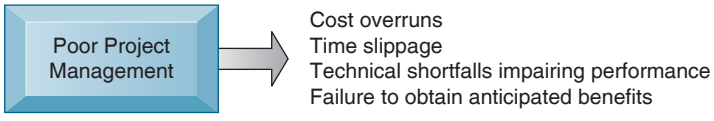
The part of the information system with which the end user interacts; in the context of IT infrastructure encompasses the type of hardware and the series of on-screen commands and responses required for a user to work with the system.

Websites may discourage visitors from exploring further if the web pages are cluttered and poorly arranged, if users cannot easily find the information they are seeking, or if it takes too long to access and display the web page on the user's computer.

Additionally, the data in the system may have a high level of inaccuracy or inconsistency. The data in certain fields may be erroneous or ambiguous, or it may not be organized properly for business purposes. Information required for a specific business function may be inaccessible because the data are incomplete.

Figure 14.4 Consequences of Poor Project Management

Without proper management, a systems development project takes longer to complete and most often exceeds the allocated budget. The resulting information system may not be able to demonstrate any benefits to the organization.



14.5 Discuss Project Risk Factors

We have already introduced the topic of information systems risks and risk assessment in Chapter 8. In this section, we describe the specific risks involved in developing information systems projects and show what can be done to manage them effectively.

Dimensions of Project Risk

Systems differ dramatically in their size, scope, level of complexity, and organizational and technical components. Some systems development projects are more likely to create the problems we have described earlier or to suffer delays because they carry a much higher level of risk than others. The level of project risk is influenced by project size, project structure, and the level of technical expertise of the information systems staff and project team.

- *Project size.* The larger the project—as indicated by the dollars spent, the size of the implementation staff, the time allocated for implementation, and the number of organizational units affected—the greater the risk. Very large-scale systems projects have a much higher failure rate than that for other projects because such projects are complex and difficult to control. For instance, according to Gartner, 55 percent to 75 percent of all ERP projects fail to meet their objectives. The organizational complexity of the system—how many units and groups use it and how much it influences business processes—contributes to the complexity of large-scale systems projects just as much as technical characteristics, such as the number of lines of program code, length of project, and budget. In addition, there are few reliable techniques for estimating the time and cost to develop large-scale information systems (Wu and Misra, 2023).
- *Project structure.* Some projects are more highly structured than others. Their requirements are clear and straightforward, so outputs and processes can be easily defined. Users know exactly what they want and what the system should do; there is almost no possibility of the users changing their minds. Such projects run a much lower risk than those with relatively undefined, fluid, and constantly changing requirements; with outputs that cannot be fixed easily because they are subject to users' changing ideas; or with users who cannot agree on what they want.
- *Experience with technology.* The project risk rises if the project team and the information system staff lack the required technical expertise. If the team is unfamiliar with the hardware, system software, application software, or database management system proposed for the project, it is highly likely that the project will experience technical problems or take more time to complete because of the need to master new skills.

Although the difficulty of the technology is one risk factor in information systems projects, the other factors are primarily organizational, dealing with the complexity of information requirements, the scope of the project, and how many parts of the organization will be affected by a new information system.

Change Management and the Concept of Implementation

The introduction or alteration of an information system has a powerful behavioral and organizational impact. Changes in the way that information is defined, accessed, and used to manage the organization's resources often lead to new distributions of authority and power. This internal organizational change breeds resistance and opposition and can lead to the demise of an otherwise good system.

Change management

Managing the impact of organizational change associated with an innovation, such as a new information system.

A very large percentage of information systems projects stumble because the process of organizational change surrounding system building was not properly addressed. Successful system building requires careful **change management**: management of the organizational change associated with an innovation, such as the introduction of a new system.

The Concept of Implementation

Implementation
All the organizational activities surrounding the adoption, management, and routinization of an innovation, such as a new information system.

Change agent
The individual acting as the catalyst during the change process to ensure successful organizational adaptation to a new system or innovation; in the context of the implementation process, this role is often played by the systems analyst.

Effective change management begins with the process of implementation. **Implementation** refers to all organizational activities working toward the adoption, management, and routinization of an innovation, such as a new information system. In the implementation process, the systems analyst is a **change agent**, the catalyst for the entire change process and responsible for ensuring that all parties involved accept the changes created by a new system. The analyst not only develops technical solutions but also redefines the configurations, interactions, job activities, and power relationships of various organizational groups. The change agent communicates with users, mediates between competing interest groups, and ensures that the organizational adjustment to such changes is complete.

The Role of End Users

System implementation generally benefits from high levels of user involvement and management support. User participation in the design and operation of information systems has several positive results. First, if users are heavily involved in systems design, they have more opportunities to mold the system according to their priorities and business requirements and more opportunities to control the outcome. Second, they are more likely to react positively to the completed system because they have been active participants in the change process. Incorporating user knowledge and expertise leads to better solutions.

The relationship between users and information systems specialists has traditionally been a problem area for information systems implementation efforts. Users and information systems specialists tend to have different backgrounds, interests, and priorities. This is referred to as the **user-designer communications gap**. These differences lead to divergent organizational loyalties, approaches to problem solving, and vocabularies.

User-designer communications gap
The difference in backgrounds, interests, and priorities that impede communications and problem solving between end-users and information systems specialists.

Information systems specialists, for example, often have a highly technical, or machine, orientation to problem solving. They look for elegant and sophisticated technical solutions in which hardware and software efficiency is optimized at the expense of ease of use or organizational effectiveness. Users prefer systems that are oriented toward solving business problems or facilitating organizational tasks. Often the orientations of both groups are so at odds that they appear to speak in different tongues.

These differences are illustrated in Table 14.4, which depicts the typical concerns of end users and technical specialists (information systems designers) regarding the development of a new information system. Communication problems between end users and designers are a major reason why user requirements are not properly incorporated into information systems and why users are driven out of the implementation process.

Systems development projects run a very high risk of failure when there is a pronounced gap between users and technical specialists and when these groups continue to pursue different goals. Under such conditions, users are often driven away from the project. Because they cannot comprehend what the technicians are saying, users conclude that the entire project is best left in the hands of the information specialists alone.

Management Support and Commitment

If an information systems project has the backing and commitment of management at various levels, it is more likely to be perceived positively by both users and the technical information

Table 14.4 The User-Designer Communications Gap

User Concerns	Designer Concerns
Will the system deliver the information we need for our work?	What demands will this system put on our servers?
Can we access the data on our mobile devices and desktop computers?	What kind of programming demands will this place on our group?
What new procedures do we need to enter data into the system?	Where will the data be stored? What's the most efficient way to store the data?
How will the operation of the system change our daily routines?	What technologies should we use to secure the data?

services staff. Both groups will believe that their participation in the development process will receive higher-level attention and priority. They will be recognized and rewarded for the time and effort they devote to implementation. Management backing also ensures that a systems project receives sufficient funding and resources to be successful. Furthermore, to be enforced effectively, all the changes in work habits and procedures and any organizational realignments associated with a new system depend on management backing. If a manager considers a new system a priority, the system will more likely be treated that way by employees. Having executive sponsors who are actively engaged is a leading factor in project success (Anderson, 2022).

Change Management Challenges for Business Process Management, Enterprise Applications, and Mergers and Acquisitions

Given the challenges of innovation and implementation, it is not surprising to find a very high failure rate among enterprise application and business process management projects which require extensive organizational change, and which may require replacing old technologies and legacy systems that are deeply rooted in many interrelated business processes. As noted previously, a number of studies have indicated that a significant percentage of large-scale projects fail to deliver promised benefits. Likewise, a high percentage of enterprise applications fail to be fully implemented or to meet the goals of their users even after several years of work.

Many enterprise application and ambitious business process management projects have been undermined by poor implementation and change management practices that failed to address employees' concerns about change. Dealing with fear and anxiety throughout the organization, overcoming resistance by key managers, and changing job functions, career paths, and recruitment practices have posed greater threats to business process management than the difficulties companies faced visualizing and designing breakthrough changes to business processes. All of the enterprise applications require tighter coordination among different functional groups as well as extensive business process change (see Chapter 9). The Spotlight on Management Case Study, *Happy Feet International Overcomes Change Management and Implementation Issues,* focuses on this issue.

Spotlight on Management Case Study

Happy Feet International Overcomes Change Management and Implementation Issues

Happy Feet International is a distributor of luxury vinyl plank and tile flooring products. Headquartered in Ringgold, Georgia, Happy Feet began in 2012 as a small, family-owned business with limited quantities of a single product and one warehouse. Fast forward to 2024, and it now has more than 300 employees, carries 23 unique product lines, and holds 29 million square feet of flooring in inventory. It has 13 different warehouses located in Tennessee, Georgia, Arizona, California, Colorado, Texas, and Washington, which service more than 1,000 dealers and distributors across the lower 48 states.

When current CIO Nick Schwartz first joined Happy Feet in 2019, the flooring industry, in general, was not very advanced from a technology standpoint. For instance, many of Happy Feet's salespeople were still more comfortable conducting business over the phone, rather than email. Time-consuming, manual processes were also hampering Happy Feet's ability to make on-time freight deliveries to its customers and the company was spending more on shipping than it should have. Happy Feet was also attempting to manage its business with an old in-house ERP system. Glitches in the system were making it almost impossible for Happy Feet to accurately track the inventory it had on hand at any given time. The inventory issues arose during a period of global supply chain disruption, adding more chaos to the mix. To deal with these issues, Happy Feet needed to introduce some new information systems.

The first step was adopting the Kuebix TMS platform, a cloud-based transportation management system. The Kuebix system enables Happy Feet to easily compare rates for different carriers and shipping points to find the best pricing, resulting in more than $1 million in savings with improved day-to-day shipping decisions. The second step was a new ERP system from Oracle NetSuite. The implementation of this system proved a bit more challenging.

According to Schwartz, change management was one of the biggest struggles that Happy Feet faced in implementing the new ERP system. One of the primary challenges Happy Feet faced was deciding what parts of the old system and processes, if any, needed to be carried over to the new system. Making the decision even more complicated was the fact that the appropriate processes had not always been documented. In addition, it took time and working with employees, especially those who had been with the company for a long time, to get them to accept that they needed to adapt to the new system. Employees were used to doing things a certain way, had been doing them that way for many years, and resisted doing them a different way. According to Schwartz, Happy Feet worked through these struggles by engaging in company-wide conversations with employees about the concepts of change and change management. It also provided numerous training sessions to help employees learn to use the new system and tried to simplify the experience as much as possible for them.

The Kuebix TMS platform is now integrated with the NetSuite ERP system, which has enabled Happy Feet to streamline operations and eliminate manual order data entry. What used to take five steps and up to an hour to accomplish now takes just a few short minutes. The simpler processes also reduce training time and costs. In addition, integration of Kuebix into the NetSuite ERP system allows for better visibility into shipments, which also improves customer service. Prior to the implementation of Kuebix, employees needed to field numerous customer calls each day on shipment status and then obtain that information from the carriers manually. Instead, today, the Kuebix system uses proactive notifications to identify issues like missed pickups and late arrivals and also enables Happy Feet to see when delivery guarantees are not met, enabling it to hold carriers accountable for service failures.

The NetSuite ERP system also solved many of Happy Feet's inventory issues. It can now do inventory just twice a year, instead of monthly, resulting in significant savings. Overall, Happy Feet is faster, leaner, and more scalable.

SOURCES: Happyfeetinternational.com, accessed September 24, 2024; Jen Miller, "How CIOs can Prepare for the Stumbling Blocks of ERP Migration," Ciodive.com, February 28, 2024; Linda Rosencrance. "6 BI Challenges IT Teams Must Address," Cio.com, December 21, 2022; Netsuite, "Happy Feet CIO Nick Schwartz on NetSuite Advanced Customer Support (ACS)," Youtube.com, October 11, 2022; Kuebix, "Happy Feet Realizes Savings & Productivity Improvements with Kuebix TMS and Netsuite ERP," Kuebix.com, 2022.

Case Study Questions

1. What management, organization, and technology challenges did Happy Feet face?
2. How did Happy Feet address those challenges?
3. Why was change management one of the biggest issues that Happy Feet faced in implementing its new systems?

Projects related to mergers and acquisitions have a similar failure rate. Mergers and acquisitions are deeply affected by the organizational characteristics of the merging companies as well as by their IT infrastructures. Combining the information systems of two different companies usually requires considerable organizational change and complex systems projects to manage. If the integration is not properly managed, firms can emerge with a tangled hodgepodge of inherited legacy systems built by aggregating the systems of one firm after another. Without a successful systems integration, the benefits anticipated from the merger cannot be realized, or, worse, the merged entity cannot execute its business processes effectively.

14.6 Understand How Project Risks Can Be Managed

Various project management, requirements gathering, and planning methodologies have been developed for specific categories of implementation problems. Strategies have also been devised for ensuring that users play appropriate roles throughout the implementation period and for managing the organizational change process. Not all aspects of the

implementation process can be easily controlled or planned. However, anticipating potential implementation problems and applying appropriate corrective strategies can increase the chances for system success.

The first step in managing project risk involves identifying the nature and level of risk confronting the project. Implementers can then handle each project with the tools and risk management approaches geared to its level of risk. Not all risks are identifiable in advance, but with skillful project management, most are. Frequent communication and a culture of collaboration will help project teams adapt to unforeseen problems that arise.

Managing Technical Complexity

The success of projects involving challenging and complex technology depends on how well their technical complexity is managed. Project leaders need both technical and administrative experience. They must be able to anticipate problems and develop smooth working relationships among a predominantly technical team. The team should be under the leadership of a manager with a strong technical and project management background, and team members should be highly experienced. Team meetings should take place frequently. Essential technical skills or expertise not available internally should be secured from outside the organization.

Formal Planning and Control Tools

Large projects benefit from appropriate use of formal planning and control tools for documenting and monitoring project plans. Two commonly used tools of this type are Gantt charts and PERT charts. Examine Figure 14.5, which illustrates a Gantt chart. A **Gantt chart** lists project activities and their corresponding start and completion dates. The Gantt chart visually represents the timing and duration of different tasks in a development project as well as their human resource requirements. It shows each task as a horizontal bar whose length is proportional to the time required to complete it.

Gantt chart

Chart that visually represents the timing, duration, and resource requirements of project tasks, with each task represented as a horizontal bar whose length is proportional to the time required to complete it.

Although Gantt charts show when project activities begin and end, they don't depict task dependencies, how one task is affected if another is behind schedule, or how tasks should be ordered. That is where a PERT chart is useful. *PERT* stands for "Program Evaluation and Review Technique." A **PERT chart** graphically depicts project tasks and their interrelationships. The PERT chart lists the specific activities that make up a project and the activities that must be completed before a specific activity can start. Examine Figure 14.6, which shows a simplified PERT chart used for the creation of a simple website. The PERT chart portrays a project as a network diagram consisting of numbered nodes (either circles or rectangles, as shown in Figure 14.6) representing project tasks. Each node is numbered and shows the task, its duration, the starting date, and the completion date. The direction of the arrows on the lines indicates the sequence of tasks and shows which activities must be completed before the commencement of another activity. In Figure 14.6, the tasks in nodes 2, 3, and 4 are not dependent on each other and can be undertaken simultaneously, but each is dependent on completion of the first task.

PERT chart

Chart that graphically depicts project tasks and their interrelationships, showing the specific activities that must be completed before others can start.

A **Kanban board** is a visual tool (which can be either digital or physical) that uses "cards" (such as sticky notes or their digital equivalent) to represent specific tasks, and columns to represent each stage (such as "to do/backlog," "in progress," "review," and "completed") of the project process. Kanban boards are often used as a part of agile project management. Figure 14.7 illustrates what a digital Kanban board might look like.

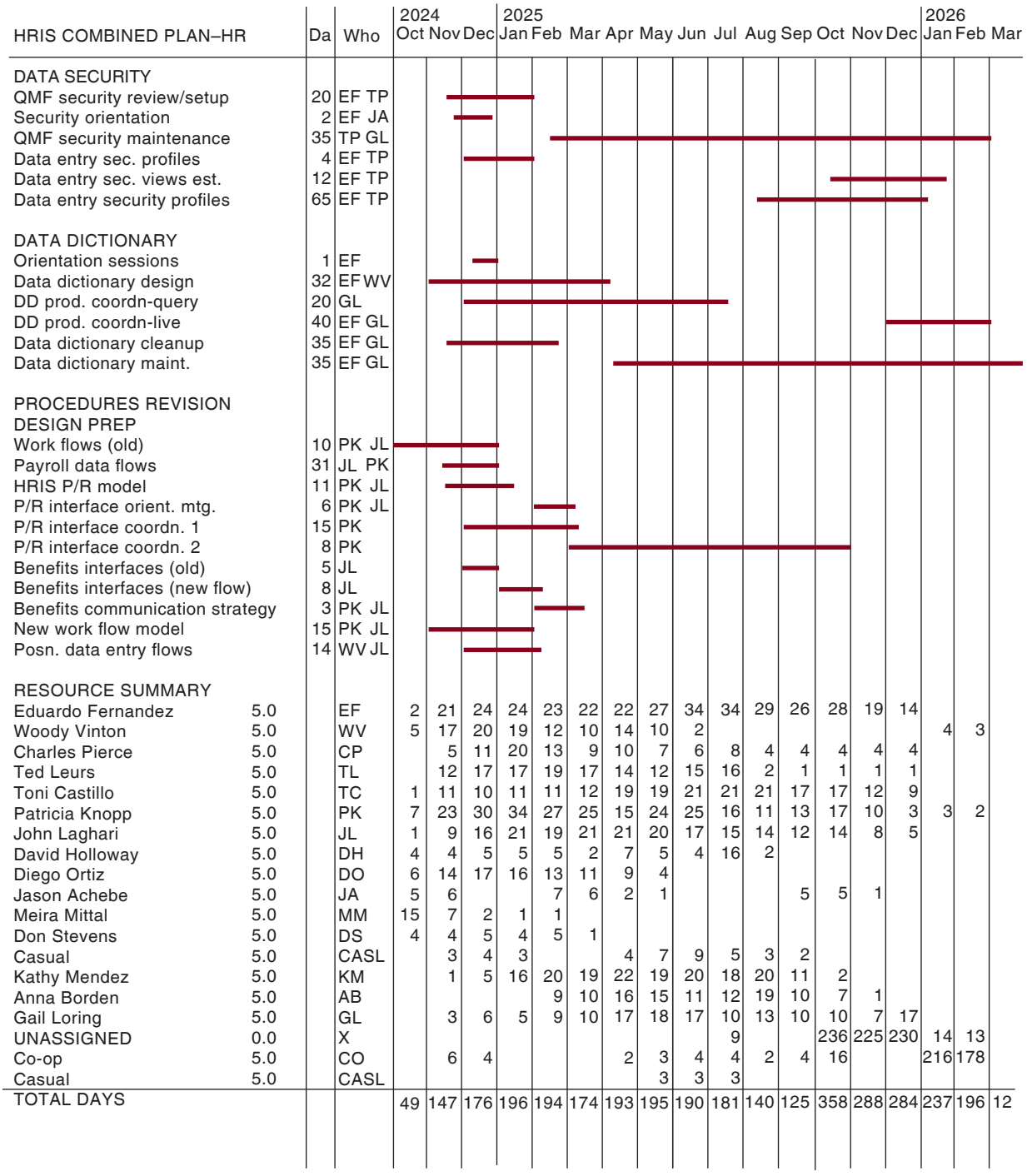
Kanban board

Visual tool (either digital or physical) that uses "cards," such as sticky notes or their digital equivalent, to represent specific tasks, and columns to represent each stage of the project process.

These project management techniques can help managers identify bottlenecks and determine the impact that problems will have on project completion times. They can also help systems developers partition projects into smaller, more manageable segments with defined, measurable business results. Standard control techniques can successfully chart the progress of the project against budgets and target dates, so deviations from the plan can be spotted.

Figure 14.5 A Gantt Chart

The Gantt chart in this figure shows the task, person-days, and initials of each responsible person as well as the start and finish dates for each task. The resource summary provides a good manager with the total person-days for each month and for each person working on the project to manage the project successfully. The project described here is a data management project for Human Resources.



HRIS COMBINED PLAN–HR	Da	Who
DATA SECURITY		
QMF security review/setup	20	EF TP
Security orientation	2	EF JA
QMF security maintenance	35	TP GL
Data entry sec. profiles	4	EF TP
Data entry sec. views est.	12	EF TP
Data entry security profiles	65	EF TP
DATA DICTIONARY		
Orientation sessions	1	EF
Data dictionary design	32	EF WV
DD prod. coordn-query	20	GL
DD prod. coordn-live	40	EF GL
Data dictionary cleanup	35	EF GL
Data dictionary maint.	35	EF GL
PROCEDURES REVISION DESIGN PREP		
Work flows (old)	10	PK JL
Payroll data flows	31	JL PK
HRIS P/R model	11	PK JL
P/R interface orient. mtg.	6	PK JL
P/R interface coordn. 1	15	PK
P/R interface coordn. 2	8	PK
Benefits interfaces (old)	5	JL
Benefits interfaces (new flow)	8	JL
Benefits communication strategy	3	PK JL
New work flow model	15	PK JL
Posn. data entry flows	14	WV JL

RESOURCE SUMMARY			Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Eduardo Fernandez	5.0	EF	2	21	24	24	23	22	22	27	34	34	29	26	28	19	14			
Woody Vinton	5.0	WV	5	17	20	19	12	10	14	10	2							4	3	
Charles Pierce	5.0	CP		5	11	20	13	9	10	7	6	8	4	4	4	4	4			
Ted Leurs	5.0	TL		12	17	17	19	17	14	12	15	16	2	1	1	1	1			
Toni Castillo	5.0	TC	1	11	10	11	11	12	19	19	21	21	21	17	17	12	9			
Patricia Knopp	5.0	PK	7	23	30	34	27	25	15	24	25	16	11	13	17	10	3	3	2	
John Laghari	5.0	JL	1	9	16	21	19	21	21	20	17	15	14	12	14	8	5			
David Holloway	5.0	DH	4	4	5	5	5	2	7	5	4	16	2							
Diego Ortiz	5.0	DO	6	14	17	16	13	11	9	4										
Jason Achebe	5.0	JA	5	6			7	6	2	1				5	5	1				
Meira Mittal	5.0	MM	15	7	2	1	1													
Don Stevens	5.0	DS	4	4	5	4	5	1												
Casual	5.0	CASL		3	4	3			4	7	9	5	3	2						
Kathy Mendez	5.0	KM		1	5	16	20	19	22	19	20	18	20	11	2					
Anna Borden	5.0	AB					9	10	16	15	11	12	19	10	7	1				
Gail Loring	5.0	GL		3	6	5	9	10	17	18	17	10	13	10	10	7	17			
UNASSIGNED	0.0	X										9			236	225	230	14	13	
Co-op	5.0	CO		6	4				2	3	4	4	2	4	16			216	178	
Casual	5.0	CASL								3	3	3								
TOTAL DAYS			49	147	176	196	194	174	193	195	190	181	140	125	358	288	284	237	196	12

Increasing User Involvement and Overcoming User Resistance

External integration tools

Tools that links the work of the implementation team to that of users at all organizational levels.

Projects with relatively little structure and many undefined requirements must involve users fully at all stages. Users must be mobilized to support one of many possible design options and to remain committed to a single design. **External integration tools** consist of ways to link the work of the implementation team to users at all organizational levels. For instance,

Figure 14.6 A Pert Chart

This is a simplified PERT chart for creating a simple website. It shows the ordering of project tasks and the relationship of a task with preceding and succeeding tasks.

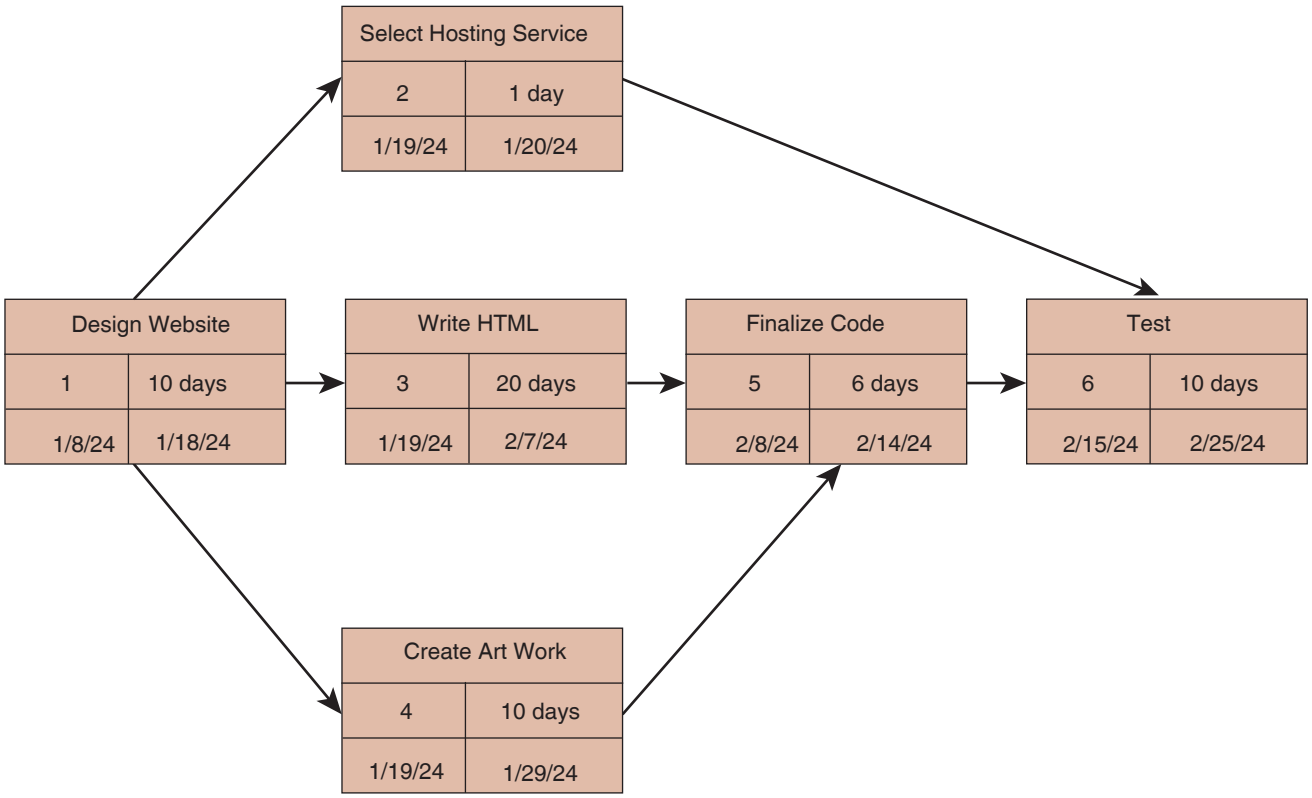
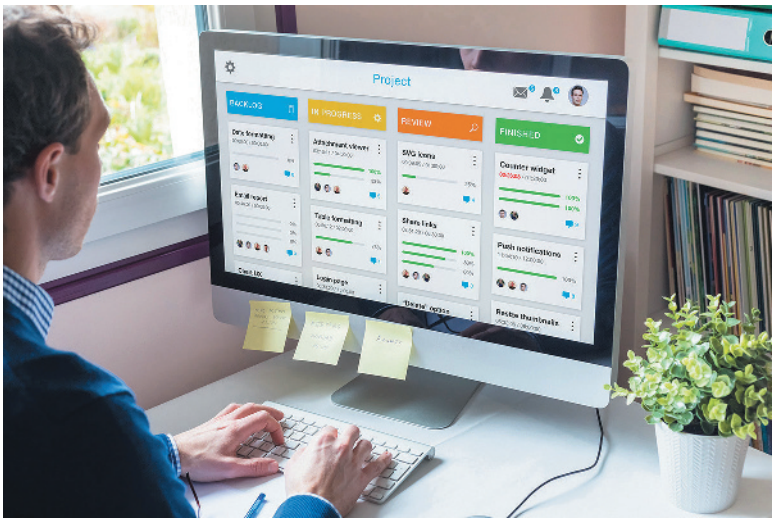


Figure 14.7 A Kanban Board

An example of a digital Kanban board used in agile project management. Tasks are represented as cards in columns that represent various project phases, in this case: Backlog, In Progress, Review, and Completed.



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users can become active members of the project team, take on leadership roles, and take charge of installation and training. The implementation team can demonstrate its responsiveness to users, promptly answering questions, incorporating user feedback, and showing their willingness to help.

Participation in implementation activities may not be enough to overcome the problem of user resistance to organizational change. Different users may be affected by the system in different ways. Whereas some users may welcome a new system because it brings changes they perceive as beneficial to them, others may resist these changes because they believe the shifts are detrimental to their interests.

If the use of a system is voluntary, users may choose to avoid it; if use is mandatory, resistance will take the form of increased error rates, disruptions, turnover, and even sabotage. Therefore, the implementation strategy must not only encourage user participation and involvement, but it must also address the issue of counterimplementation. **Counterimplementation** is a deliberate strategy to thwart the implementation of an information system or an innovation in an organization.

Counterimplementation

A deliberate strategy to thwart the implementation of an information system or an innovation in an organization.

Strategies to overcome user resistance include user participation (to elicit commitment as well as to improve design), user education and training, management edicts and policies, and better incentives for users who cooperate. The new system can be made more user-friendly by improving the end-user interface. Users will be more cooperative if organizational problems are solved prior to introducing the new system (Jalo and Pirkkalainen, 2024)

Designing for the Organization

Because the purpose of a new system is to improve the organization's performance, information systems projects must explicitly address the ways in which the organization will change when the new system is installed, including installation of mobile and web applications. In addition to procedural changes, transformations in job functions, organizational structure, power relationships, and the work environment should be carefully planned.

Areas where users interface with the system require special attention, with sensitivity to ergonomics issues. **Ergonomics** refers to the interaction of people and machines in the work environment. It considers the design of jobs, health issues, and the end-user interface of information systems. Table 14.5 lists the organizational dimensions that must be addressed when planning and implementing information systems.

Ergonomics

The interaction of people and machines in the work environment, including with respect to the design of jobs, health issues, and the end-user interface of information systems.

Although systems analysis and design activities are supposed to include an organizational impact analysis, this area has traditionally been neglected. An **organizational impact analysis** explains how a proposed system will affect organizational structure, attitudes, decision making, and operations. To integrate information systems successfully with the organization, thorough and fully documented organizational impact assessments must be given more attention in the development effort.

Organizational impact analysis

Study of the way a proposed system will affect organizational structure, attitudes, decision making, and operations.

Sociotechnical Design

One way of addressing human and organizational issues in information systems projects is to incorporate **sociotechnical design**, which aims to produce information systems that blend technical efficiency with sensitivity to organizational and human needs, resulting in higher job satisfaction. Designers set forth separate sets of technical and social design solutions.

Sociotechnical design

Design which aims to produce information systems that blend technical efficiency with sensitivity to organizational and human needs.

Table 14.5 Organizational Factors in Systems Planning and Implementation

Employee participation and involvement
Job design
Standards and performance monitoring
Ergonomics (including equipment, user interfaces, and the work environment)
Employee grievance resolution procedures
Health and safety
Government regulatory compliance

The social design plans explore different workgroup structures, allocation of tasks, and the design of individual jobs. The proposed technical solutions are compared with the proposed social solutions. The solution that best meets both social and technical objectives is selected for the final design.

Project Management Software Tools

Commercial software tools that automate many aspects of project management facilitate the project management process. Project management software typically features capabilities for defining and ordering tasks, assigning resources to tasks, establishing starting and ending dates to tasks, tracking progress at both individual and team levels, and facilitating modifications to tasks and resources. Many automate the creation of Gantt and PERT charts and provide communication, collaboration, and social tools.

Some of these tools are complex sophisticated programs for managing very large projects, dispersed work groups, and enterprise functions. These high-end tools can manage very large numbers of tasks and activities and complex relationships. The most widely used project management tool today is Microsoft Project, but there are also lower-cost tools for smaller projects and small businesses. Many project management applications are now cloud-based to enable project team members to access project management tools and their data wherever they are working. The Opening Case Study describes some capabilities of Wrike, a cloud-based project management tool.

While project management software helps organizations track individual projects, the resources allocated to them, and their costs, **project portfolio management software (PPM software)** helps organizations manage portfolios of projects and dependencies among them. Project portfolio management software helps managers compare proposals and projects against budgets and resource capacity levels to determine the optimal mix and sequencing of projects that best achieves the organization's strategic goals.

Project portfolio management software (PPM software)
Helps organizations manage portfolios of projects and the dependencies among them.

AI and Project Management Software Tools

Project management software tools are increasingly incorporating AI functionality to enhance efficiency and project workflow. AI tools can help automate routine project management tasks, such as data entry, scheduling, creating meeting summaries, sending notifications, and updating project status. AI tools can also help analyze project data to help identify potential project risks, as well as suggest ways to mitigate those risks or avoid them entirely. AI functionality is also being used to enhance project team collaboration by providing tools that facilitate better communication, task delegation, and document sharing. For example, ClickUp includes an AI virtual assistant called ClickUp Brain that can be used to create content, manage and automate work tasks as project updates, and get answers to work questions from tasks, documents, and fellow team members.

14.7 Understand How the Information in This Chapter Can Help Your Career

The information in this chapter can help your career by familiarizing you with various methods that are used to build a business case to support investments in information systems and by helping you to understand the importance of effective project management in developing information systems. This chapter also introduced you to various project management methodologies, the principal risk factors in developing information system projects, and methods and tools that can be used to manage those risks.

Real-World Scenario

Project Managers Help Intuit Launch New Cloud Platform

Intuit is a leading provider of financial management software and tools for consumers and business professionals such as TurboTax and QuickBooks. Selling these tools and providing customer service generates a great deal of valuable data. Until recently, the data were maintained in isolated silos that could not be utilized across the entire organization. Intuit launched the Intuit Analytics Cloud (IAC) project to use data more effectively to improve operations and also the experiences of Intuit customers. Intuit's Analytics Cloud was an enterprise-wide platform serving the entire company for business units to use in any way they chose. The Intuit IAC project required cooperation across the entire company because it incorporated all of the company's enterprise data, its product data, and third-party data, into a single platform. IAC project leaders moved a number of functional teams from Intuit's data engineering group to production work for the project.

The project team was very sensitive to deadlines and the project budget. One key step was breaking down the project into easy-to-handle pieces. The IAC project was a multi-year endeavor. A "big bang" approach to implementation, in which an entire system is implemented all at once in a single step, was not operationally or politically feasible. Creating a series of smaller project deliverables made the project more manageable for the project teams. The success of the smaller project "bites" meant that project leaders didn't have to work as hard to convince business unit leaders to participate in the project. Organizing a large project into "bite-sized" pieces also made it possible to deliver demonstrable results as the project progressed, enabling executives to tie the authorization of project expenditures more easily to those results. Intuit's IAC project was successful and Intuit's business units have been eager to use the new enterprise analytics cloud.

Job Description and Career Path: IT Project Manager

Job	Job Description	IT Knowledge	Career Path and Average Salaries
IT project manager	Job Responsibilities include: • Developing project plans • Coordinating project phases and schedules • Analyzing each project phase to identify potential roadblocks and areas for improvement • Communicating with and acting as a liaison between all stakeholders involved with the project • Using project management tools to document and manage budget, time and projected outcomes of project • Track progress of project and adjust plans as needed	• Knowledge of Microsoft 365 tools: Excel, Word. PowerPoint, Teams • Familiarity with project management tools such as Microsoft Project, Wrike, Asana, Klient, Jira • Certifications available from various organizations including Project Management Institute (PMI), CompTIA, IBM, Microsoft, others	• Entry-Level [Project coordinator/ assistant project manager/junior project manager]: $60,000–$75.000 • Early Career [IT project manager]: $125,000 • Mid-Career [Senior IT project manager]: $150,000 • Late Career [Senior project manager; further advancement, potentially to director or other executive-level position: $200,000–$225,000 Salary can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, pre-pare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

Southwest Airlines Invests in Digital Transformation to Navigate Future Turbulence

Southwest Airlines is one of the United States' major passenger airlines. Southwest began operations in Dallas, Texas, in 1971 and grew rapidly from a small carrier into what is now the third largest in the United States based on revenue. It currently serves more than 120 destinations in the United States, as well as additional destinations in Mexico and various countries in the Caribbean. Historically, Southwest has also been one of the most beloved airline brands, known for its high regard for customer service. Through 2022, Southwest had ranked either first or second in J.D. Power's airline customer satisfaction ratings for over a decade.

However, during the period from December 21 through December 30, 2022, Southwest experienced one of the largest operational disruptions in aviation history, leading to the cancellation of nearly 17,000 flights during its busiest time of year. At the center of the meltdown: outdated software systems and poor managerial decisions with respect to investments in IT systems. Most of Southwest's technology initiatives and spending in the preceding years had been directed to an upgrade of its revenue management system as well as outward-facing customer experience improvements, with Southwest making those experiences the heart of a $2 billion five-year technology investment plan announced in 2022. Modernizing internal operational systems and processes were not a priority for Southwest's leadership, with the company reportedly putting familiarity with existing solutions ahead of the benefits that upgrades and new technology might bring.

Although bad weather was the immediate catalyst for the meltdown, it was not the underlying reason. Other airlines experienced the same bad weather and recovered much more rapidly, quickly reassigning crew, rerouting passengers, and rebooking flights. But unlike those airlines, Southwest was relying on a twenty-year-old crew scheduling software system called SkySolver. SkySolver's inadequacies were not a secret within Southwest. Its shortcomings had contributed to previous, smaller-scale meltdowns and Southwest pilots had reportedly been asking Southwest since at least 2015 to update the antiquated system. But despite the clear warning signs, Southwest had done little to address the issues with SkySolver other than to make piecemeal patches and updates. Then, on December 20, 2022, winter storm Elliott arrived, with high winds, snowfall, and record cold temperatures across much of the United States. Flight delays and cancellations due to the weather snowballed even further as SkySolver became overwhelmed and unable to handle the task of matching crew members with flights. Flight crews were forced to contact Southwest over the telephone, and Southwest crew schedulers were left with trying to manually match flight crews with available planes, all while navigating federal regulations about crew flight time. Flight crew were often on hold with schedulers for hours. Thousands of holiday travelers were left stranded at airports around the country.

This problem—relying on older or deficient software that needs updating—is known as technical debt. Southwest's CEO, Bob Jordan, admitted that Southwest had dropped the ball in failing to pay attention to this issue. Analysts were even harsher. While legacy applications, proprietary solutions, and amassed technical debt are rife in the aviation industry, according to analyst firm Gartner, Southwest was a standout compared to the other major U.S. airlines in terms of its delinquency in "paying down" (i.e., taking steps to fix) that technical debt. Doug Laney, a data expert and innovation fellow at consulting firm West Monroe, noted that there had been little mention by Southwest of data and analytics as a key competency in the previous ten years of annual reports, and that companies that have failed to adopt advanced analytics are poorly positioned to compete. Southwest, like many other airlines, had implemented dashboard capabilities in recent years, but some of the operational dysfunction related to missing data, not a lack of tools. CEO Jordan admitted that as well, noting that while the airline had good systems in some areas, those systems still needed better integration to enable them to share data. For instance, an optimization system developed by Southwest to automate disruption recovery and select flights to cancel needed "better visibility" into its crew-scheduling systems. In addition to unleashing a huge public relations problem, the meltdown also cost Southwest an estimated $825 million and triggered a $140 million fine from the U.S. Department of Transportation.

Southwest, to its credit, immediately began to take steps to rectify the problem. In January 2023, the company announced that it was accelerating existing plans to enhance processes and IT capabilities and allocated $1.3 billion of its operating budget to IT upgrades and improving operational resiliency. The company pledged to enhance crew engagement technology, update crew recovery systems, and establish supplemental staffing as stopgap measures to reduce the risk of future operational disruptions. As part of the modernization push, Southwest chose Amazon Web Services as its preferred cloud provider in March 2023. Cloud migration is central to Southwest's modernization effort and part of a long-term resiliency plan, according to a company spokesperson.

To upgrade the crew reassignment software, Southwest CIO Lauren Woods assembled a project team to build a tool called Crew and Aircraft Integrated Recovery and Optimizer

(CAIRO). CAIRO enables Southwest to automate, prioritize, and reset the crew and aircraft simultaneously and identify flights that might need to be canceled, while minimizing the potential impact on crew members. Woods gave credit to Southwest management for helping the team successfully accelerate the work, noting that she had the full support of the executive team, who were open to hearing what needed to be modernized, what needed to be improved, and where the biggest investments needed to be made.

In January 2024, Southwest announced that it was spending an additional $1.7 billion on technology and cloud migration. It had already shifted about 50 percent of its "tech estate" to the cloud and said that its goal was to shift a lot more. According to Jordan, the shift was as much a resiliency effort and a modernization of code base as it was a cost-saving effort. Southwest is hoping that these efforts will go a long way to paying down its technical debt and enable it to deal with extreme weather events much more successfully in the future.

SOURCES: Sarah White, "How Southwest's CIO Modernized the Airline Through Turbulence," Cio.com, April 19, 2024; Matt Ashare, "Southwest Airlines to Spend $1.7B on Tech, Cloud Migration This Year," Ciodive.com, January 29, 2024; Matt Ashare, "Southwest Airlines Targets More than IT in December Meltdown Analysis," Ciodive.com, April 4, 2023; Matt Ashare, "3 Technology Lessons from Southwest's IT Failure," Ciodive.com, February 2, 2023; Andrew Paul, "You Can Blame SouthWest Airlines' Holiday Catastrophe on Outdated Software," Popsci.com, January 3, 2023; Belle Lin, "Southwest Meltdown Shows Airlines Need Tighter Software Integration," *Wall Street Journal*, January 3, 2023; Zeynep Tufekci, "The Shameful Open Secret Behind Southwest's Failure," *New York Times*, December 31, 2022; Southwest Airlines Co., "Southwest Airlines Outlines Two Billion Dollar Plan to Transform the Customer Experience, Bringing More Simplicity and Speed from the Ground Up," Prnewswire.com, May 11, 2022.

Case Study Questions

1. Why did Southwest Airlines have such a hard time recovering from the disruptions caused by winter storm Elliot?
2. What actions did Southwest Airlines take to deal with its technical debt issues in 2023 and 2024?

Review Summary

14.1 Explain how to build a business case.

A project is a planned series of related activities for achieving a specific business objective. Information systems projects include the development of new information systems, enhancement of existing systems, or upgrade or replacement of the firm's information technology (IT) infrastructure.

The business case for an IT investment describes the problem facing the organization that can be solved by investing in a proposed system solution. It provides an analysis of whether an information system project is a good investment by calculating its costs and benefits. Tangible benefits are quantifiable, and intangible benefits cannot be immediately quantified but may provide quantifiable benefits in the future. Benefits that exceed costs should be analyzed using capital budgeting methods to make sure they represent a good return on the firm's invested capital. Organizations should develop an information systems plan that describes how information technology supports the company's overall business plan and strategy.

14.2 Describe methods for evaluating IT investments.

There are a variety of methods that can be used to evaluate a proposed IT investment, including portfolio analysis, scoring models, capital budgeting models, as well as software tools. In order to effectively evaluate a potential IT investment, it is also important to be able to have an understanding of the costs as well as both the tangible and intangible benefits of the proposed system.

14.3 Describe project management methodologies and variables.

Project management refers to the application of knowledge, skills, tools, and techniques to achieve specific targets within specified budget and time constraints. Project management activities include planning the work, assessing the risk, estimating and acquiring resources required to accomplish the work, organizing the work, directing execution, and analyzing the results. The two most common project management methodologies are the waterfall methodology and the agile approach. Project management must deal with five major variables: scope, time, cost, quality, and risk. Technical debt results from the trade-offs between time, cost, and quality that often occur in the development of an information system.

14.4 Understand the importance of effective project management.

There is a very high failure rate among information systems projects that have not been properly managed. When an information system does not meet expectations or costs too much to develop, companies may not realize any benefit from their information system investment, and the system may not be able to solve the problems for which it was intended. The development of a new system must be carefully managed and orchestrated, and the

way a project is executed is likely to be the most important factor influencing its outcome. That's why it's essential to have some knowledge about managing information systems projects and the reasons why they succeed or fail.

14.5 Discuss project risk factors.

The level of risk in an information systems development project is determined by (1) project size, (2) project structure, and (3) experience with technology. IS projects are more likely to fail when there is insufficient or improper user participation in the systems development process, lack of management support, and poor management of the implementation process. There is a very high failure rate among projects involving business process management, enterprise applications, and mergers and acquisitions because they require extensive organizational change.

14.6 Understand how project risks can be managed.

Implementation refers to the entire process of organizational change surrounding the introduction of a new information system. User support and involvement and management support and control of the implementation process are essential, as are mechanisms for dealing with the level of risk in each new systems project. Project risk factors can be brought under some control by a contingency approach to project management. The risk level of each project determines the appropriate mix of external integration tools, internal integration tools, formal planning tools, and formal control tools to be applied. Project management software helps organizations track individual projects and project portfolio management software helps them manage portfolios of projects and dependencies among them.

Key Terms

Agile project management, 468
Business case, 462
Capital budgeting models, 466
Change agent, 472
Change management, 471
Counterimplementation, 478
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External integration tools, 476
Gantt chart, 475
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Technical debt, 469
Technology business management (TBM), 467
User-designer communications gap, 472
User interface, 470
Waterfall project management, 468

Review Questions

14.1 Explain how to build a business case.

- Describe types of information systems projects.
- Define and describe the components of a business case for a proposed investment.
- List and describe the major components of an information systems plan.

14.2 Describe methods for evaluating IT investments

- Explain the difference between tangible and intangible benefits.
- List six tangible benefits and six intangible benefits of an IT investment.
- Describe how portfolio analysis, scoring models, and capital budgeting models can be used to establish the worth of systems.
- Discuss the limitations of using financial models to evaluate IT investments.
- Describe software that is available to help make technology investment decisions.

14.3 Describe project management methodologies and variables.

- Define project management.
- List and describe the project management activities and variables addressed by project management.
- Define technical debt and explain how it arises.

14.4 Understand the importance of effective project management.

- Describe some of the problems that can arise as a result of poor project management when developing an information system.

14.5 Discuss project risk factors.

- Identify and describe each of the principal risk factors in information systems projects..
- Explain why builders of new information systems need to address implementation and change management.

14.6 Understand how project risks can be managed.

- Explain why eliciting support of management and end users is so essential for successful implementation of information systems projects.
- Identify and describe the strategies for controlling project risk.
- Identify the organizational considerations that should be addressed by project planning and implementation.
- Explain how project management software tools contribute to successful project management.

Discussion Questions

14.7 How much does project management impact the success of a new information system?

14.8 It has been said that most systems fail because systems builders ignore organizational behavior problems. Why might this be so?

14.9 What is the role of end users in information systems project management?

Using AI Project

14.10 AI Project: Identifying Change Management Problems

You have been hired as a consultant by two friends who want to open a small retail store selling organic pet food and treats, as well as pet toys and clothing made from organic materials. They have asked you to help prepare them for challenges associated with implementing a Point of Sale (POS) system. They are particularly concerned about employee resistance to using the system.

Using Microsoft Copilot, Google's Gemini, or ChatGPT, develop a series of prompts to identify and describe these change management problems. Save a copy of your prompts and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience evaluating information systems projects, using spreadsheet software to perform capital budgeting analyses for new information systems investments, and using web tools to analyze the financing for a new home. Visit MyLab MIS to access the files needed for certain of this chapter's Hands-On MIS Projects.

For student files please visit www.pearsonhighered.com/laudon or MyLab MIS.

14.11 Management Decision Problems

Caterpillar is the world's leading maker of earth-moving machinery and supplier of agricultural equipment. Caterpillar wants to end its support for its Dealer Business System (DBS), which it licenses to its dealers to help them run their businesses. The software in this system is becoming outdated, and senior management wants to transfer support for the hosted version of the software to Accenture Consultants so it can concentrate on its core business. Caterpillar never required its dealers to use DBS, but the system had become a de facto standard for doing business with the company. The majority of the 50 Cat dealers in North America use some version of DBS, as do about half of the 200 or so Cat dealers in the rest of the world. Before Caterpillar turns the product over to Accenture, what factors and issues should it consider? What questions should it ask? What questions should its dealers ask?

14.12 Using Spreadsheets for Capital Budgeting

Software skills: Spreadsheet formulas and functions

Business skills: Capital budgeting

This project provides you with an opportunity to use spreadsheet software to use the Return on Investment (ROI) profitability model to analyze the return on an investment for a new computer-aided design (CAD) system.

Your company would like to invest in a CAD system that requires purchasing hardware, software, and networking technology as well as expenditures for installation, training, and support. The MIS18_Project 14.12 Student File provides you with tables showing each cost component for the new system as well as annual maintenance costs over a five-year period. You believe the new system will reduce the amount of labor required to generate designs and design specifications, thereby increasing your firm's annual cash flow.

- Using the data provided in these tables, create a worksheet that calculates the costs and benefits of

the investment over a five-year period and analyzes the Return on Investment (ROI). The ROI ratio helps measure the amount of return from an investment in relation to its overall cost. The higher the ROI, the more profitable the project. To calculate ROI, divide the investment's net benefit (Total Benefits minus Total Costs) by the total cost of the investment. You should consider one-time and recurring benefits and costs from Year 0 and Years 2021 to 2025. For purposes of this exercise, you do not have to consider depreciation. Is this investment worthwhile? Why or why not?

14.13 Using Online Research Tools for Buying and Financing a Home

Software skills: Online research tools

Business skills: Financial planning

This project will develop your skills using online research tools for searching for a home and calculating mortgage financing for that home.

You would like to purchase a home in Fort Collins, Colorado. Ideally, it should be a single-family house with at least three bedrooms and one bathroom that costs between $350,000 and $450,000 and can be financed with a 30-year fixed-rate mortgage. You can afford a down payment that is 20 percent of the value of the house. Before you purchase a house, you would like to find out what homes are available in your price range, find a mortgage, and determine the amount of your monthly payment. Use the Realtor.com site or app to help you with the following tasks:

- Locate homes in Fort Collins, Colorado that meet your specifications.
- Find a mortgage for 80 percent of the list price of the home. Compare rates from at least three potential lenders.
- After selecting a mortgage, calculate your closing costs and the monthly payment.

When you are finished, evaluate the whole process. For example, assess the ease of use of the site or app and your ability to find information about houses and mortgages, the accuracy of the information you found, and the breadth of choice of homes and mortgages.

14.14 Collaboration and Teamwork Project: Identifying Implementation Problems

With two or mor of your classmates, select a system described in one of the case studies in the text. Write a description of the problems you might expect to encounter in implementing the system and an analysis of the steps you would take to solve or prevent these problems. If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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15

Managing Global Information Systems

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Learning Objectives

After completing this chapter, you will be able to:

15.1 Discuss factors impacting development of global information systems.

15.2 Discuss how to develop effective global information systems architecture.

15.3 Discuss strategies and organizational structures for global businesses and information systems.

15.4 Discuss management challenges in developing global information systems.

15.5 Discuss management solutions for global information systems development challenges.

15.6 Discuss technology issues and alternatives in developing global information systems.

15.7 Understand how the information in this chapter can help your career.

Chapter Case Studies

- **Opening Case Study:** Hormel Foods Implements Oracle Cloud Applications to Unify Its Systems Worldwide
- **Spotlight on Organizations Case Study:** The Rise of Digital Nationalism
- **Chapter Case Study:** Wyndham Centralizes Its Information Systems to Better Serve Its Worldwide Hotels

Opening Case Study

Hormel Foods Implements Oracle Cloud Applications to Unify Its Systems Worldwide

Hormel Foods Corporation, founded by George A. Hormel in 1891, has grown over its more than 130 years in business to become a global provider of food products, with iconic brands such as SPAM, Skippy Peanut Butter, Dinty Moore, Planters Peanuts, and Applegate generating more than $12 billion in annual revenue.

Initially, Hormel Foods conducted most of its international business on an export basis. But over the past few decades, the company has transitioned to a multinational strategy focused on building its presence through subsidiaries, joint ventures, and licensing agreements around the world. Currently, Hormel Foods operates or partners in numerous facilities around the globe, including Europe, Asia, South America, and Australia, with more than 40 major brands sold in over 80 countries. Asia has become a vital growth area with operations in South Korea, Japan, China, and the Philippines.

One way that Hormel Foods has fueled its global growth has been through acquisitions of other companies. Although beneficial in many ways, this strategy also resulted in a patchwork of different information systems that did not work well together, leaving Hormel Foods without a clear view of the company's performance and ultimately impacting its ability to compete on a global scale. In addition, the systems were difficult to maintain, in some instances depending on a single person who understood how the system worked. The company realized that it needed to make a major upgrade to its IT systems. Its goal was not just incremental improvement. According to Jim Sheehan, executive vice president and chief financial officer, Hormel was looking to make a generational shift to allow it to leapfrog its competitors.

To do so, in 2019, Hormel Foods embarked on a multi-year strategic effort to modernize its core business systems and improve data quality. It turned to Oracle, a major provider of cloud-based enterprise systems, to implement a suite of Oracle Cloud applications to support Hormel's human resources, enterprise performance management, and supply chain management. Hormel Foods went live with Oracle Fusion Cloud HCM (human capital management) in January 2020, Fusion Cloud EPM (enterprise performance management) and Fusion Cloud ERP (enterprise resource planning) in June 2020, and later began phasing in Oracle Supply Chain Management Cloud. Oracle's suite of applications also included tools that allowed Hormel to quickly connect applications and data. Hormel also turned to Oracle Analytics to obtain better insights into its data.

Hormel Foods' efforts have begun to pay off. Today, Hormel Foods has a single platform that provides visibility into all of its brands from the same point of view. As Sheehan notes, when its various brands report their financial results, such as cash flow, income statement, or balance sheet items, they are all presented the same way. Using Oracle Analytics has enabled Hormel Foods to better pinpoint growth opportunities across its many brands. According to Jana Haynes, vice president and controller, having a centralized platform enables Hormel Foods to notice things that previously were not apparent.

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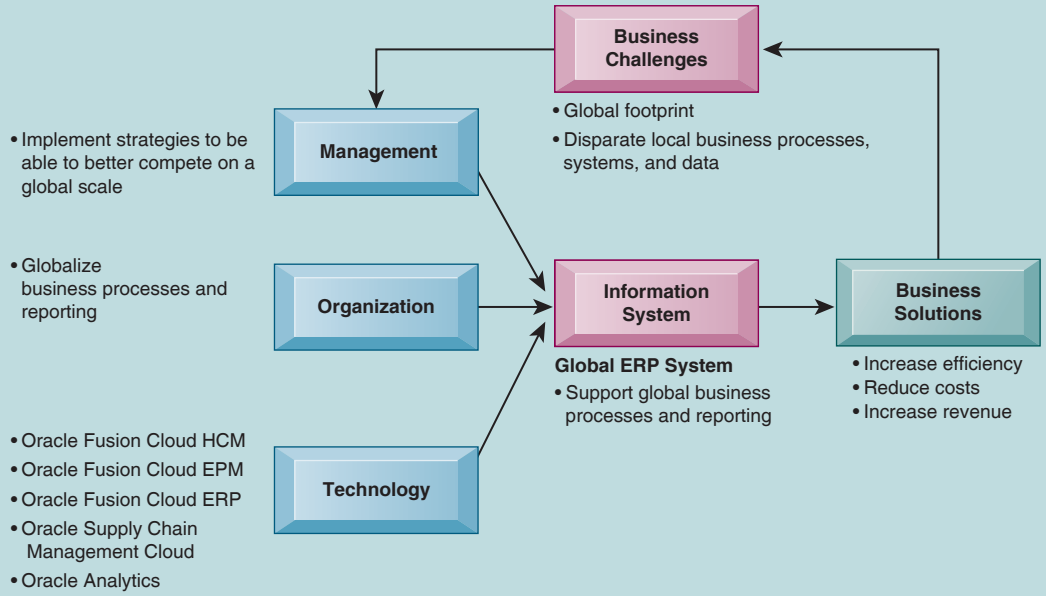
Oracle has also enabled Hormel Foods to achieve a similar overarching view of global procurement data. With all procurement except for China and Brazil now on Oracle Cloud, it has become easier for Hormel Foods to analyze vendors. For instance, it discovered that some of its brands paid more than others for the same item. It has also become easier for Hormel Foods to identify vendors that are the sole source of items its brands rely on. Knowing this allows the company to explore alternative sources and be ready should a vendor prove unable to meet demand. As noted by Sheehan, the ability to do this type of analysis provides Hormel Foods with a competitive advantage and allows it to make smarter decisions.

SOURCES: Hormelfoods.com, accessed September 29, 2024; Hormel Foods Corporation Annual Report on Form 10-K for the fiscal year ended October 29, 2023, filed with the Securities and Exchange Commission, December 6, 2023; Alba Munoz Saiz and Ethan Watters, "Unleashing Innovation in Global Markets," Hormelfoods.com, September 6, 2023; Mark Jackly, "3 ERP Implementation Case Studies," Oracle.com, January 2023; Oracle, "Hormel Unifies Finance and HR with Oracle Cloud Applications," Oracle.com, September 3, 2021.

Hormel Foods' efforts to create integrated global information systems identify some of the issues that truly global organizations need to consider if they want to operate worldwide. Like many large, multinational firms, Hormel Foods has numerous operating units in many different countries. These units had their own systems, business processes, and reporting standards. As a result, Hormel Foods was unable to effectively coordinate global



operations or manage its financial reporting controls across multiple countries and regions. Management was unable to obtain a clear view into Hormel Foods' performance.

The Opening Case Study diagram above calls attention to important points raised by this case and this chapter. To solve its global management and business challenges, Hormel Foods turned to Oracle Cloud to streamline its business processes on a global level. The company implemented an integrated suite of Oracle Cloud applications based on one platform worldwide. Hormel Foods' new global systems provide the firm with enterprise-wide information on firm operations and financial performance so that the company can be more easily managed and coordinated from a global perspective.

Case Study Questions

1. What issues did Hormel Foods have in dealing with its legacy systems?
2. How did Hormel Foods use information technology to improve operations and management decision making?
3. How did Hormel Foods' new Oracle Cloud applications help it become a more global organization?

CAREER BENEFIT

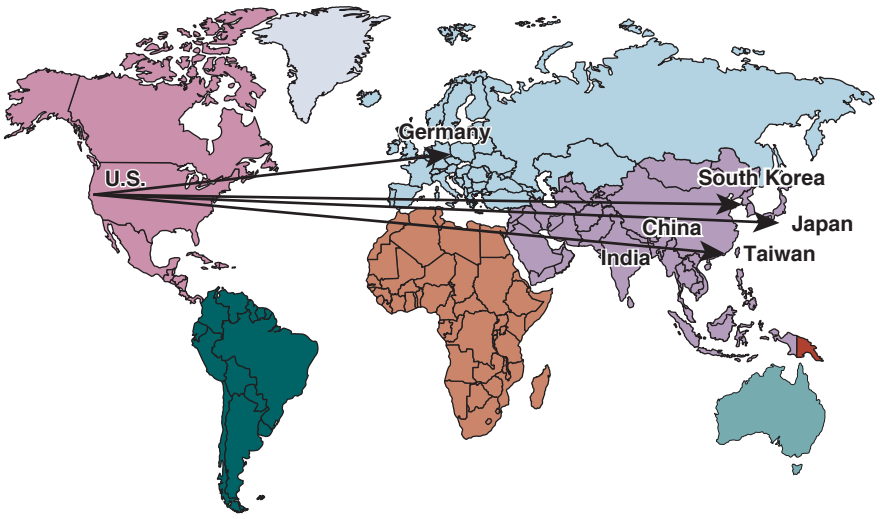
Working on this project provided Hormel Food's IT staff with opportunities to develop expertise in cloud computing tools and applications for enterprise-wide information integration, enabling them to advance their careers and making them more qualified for higher paying jobs at other global companies. Non-IT staff also benefited from contributing to the development of such an important solution for the entire organization and from being able to dramatically improve Hormel Food's global business performance.

15.1 Discuss Factors Impacting Development of Global Information Systems

In previous chapters, we described the emergence of a global economic ecosystem driven by the development of advanced telecommunications networks and information systems. These advances, along with changes in trade policy of various nations, helped drive cross-border trade as well as the development of global value chains, in which some stages of production

Figure 15.1 Apple iPhone's Global Supply Chain

Apple designs the iPhone in the United States and relies on suppliers in the United States, Germany, Japan, South Korea and various other countries throughout the world for its parts. In years past, almost all iPhones were assembled in mainland China. However, Apple has now also begun to assemble iPhones in India as well.



are outsourced and take place in other countries. The growth of international trade has radically altered domestic economies around the globe. Many localized firms have been replaced by networked corporations that transcend national boundaries. Today, world trade constitutes almost 60 percent of overall global economic activity (World Trade Organization, 2023a).

As a case in point, examine Figure 15.1, which illustrates the path to market for an Apple iPhone. The iPhone is designed by Apple engineers in the United States, but its more than 200 high-tech components are sourced from suppliers around the world. For instance, various companies in Taiwan, South Korea, Japan, Germany, and the United States all provide components such as the case, camera, processors, accelerator, gyroscope, electronic compass, power management chip, touch screen controller, and high-definition display screen. Foxconn, a Chinese division of Taiwan's Hon Hai Group, is in charge of the iPhone's manufacturing and assembly, which still occurs primarily in mainland China. Apple then sells the finished iPhones around the world. The path to market for an iPhone is just one example of the globalization of business.

Global Business Drivers

Table 15.1 lists business drivers (forces in the environment to which business must respond and that influence the direction of the business) leading to the development of global information systems.

Global business drivers can be divided into two groups: general cultural factors and specific business factors. General cultural factors include information, communication, and transportation technologies that have worked together to create a *global village* in which

Table 15.1 Global Business Drivers

General Cultural Factors	Specific Business Factors
Global information, communication, and transportation technologies	Global markets
Development of global culture	Global production and operations
Global knowledge base	Global coordination
	Global workforce
	Global economies of scale

communication (by telephone, television, radio, or computer network) around the globe is no more difficult than communication down the block. The cost of moving goods and services to and from geographically dispersed locations has fallen dramatically.

The development of global communications has created a global village in a second sense. A **global culture** created by the Internet and other globally shared media, such as television and movies, now permits different cultures and peoples to develop common expectations, shared artifacts, and social norms.

Global culture

The development of common expectations, shared artifacts, and social norms among different cultures and peoples.

Another factor to consider is the growth of a global knowledge base. At the end of World War II, knowledge, education, science, and industrial skills were highly concentrated in North America, Western Europe, and Japan. This is no longer true. Many other regions in the world, such as China, Southern Asia, Eastern Europe, and Latin America, have developed powerful educational, industrial, and scientific centers, resulting in a much more democratically and widely dispersed knowledge base.

In addition to these general cultural factors, there are also specific business factors that affect most industries. The growth of powerful communications technologies and the emergence of global cultures provided the groundwork for *global markets*—consumers in different parts of the world interested in consuming similar products. Coca-Cola, American sneakers (made in Korea but designed in the United States), and Starbucks coffee are now sold worldwide.

Responding to this demand, global production and operations have emerged with precise online coordination between far-flung production facilities and central headquarters thousands of miles away. For instance, Maersk, a major global shipping company based in Copenhagen, Denmark, offers its customers a variety of tools to track the shipment of their products, such as Logistics Hubs, a centralized dashboard that includes an AI-powered estimated time of arrival (ETA) system, and an interactive shipping route map that tracks ship locations with real-time information. The Maersk App puts similar information into the palm of a customer's hand.

Global markets and pressure toward global production and operation have driven new capabilities for global coordination. As discussed in the Opening Case Study, production, accounting, marketing and sales, human resources, and systems development (all the major business functions) can be coordinated on a global scale. These new levels of global coordination permit the location of business activity according to comparative advantage. Design should be located where it is best accomplished, as should marketing, production, and finance.

Finally, global markets, production, and administration create conditions for powerful, sustained global economies of scale. Production driven by worldwide global demand can be concentrated where it can best be accomplished. Fixed resources can be allocated over larger production runs, and production runs in larger plants can be scheduled more efficiently and precisely estimated. Production can be shifted to regions that offer lower costs. The result is a powerful strategic advantage to firms that can organize globally. These general and specific business drivers have greatly enlarged world trade and commerce. However, the Covid-19 pandemic has also illustrated the dangers of dependence on global supply chains where sourcing is concentrated in a single country or supplier.

Not all industries are similarly affected by these trends. Initially, manufacturing was much more affected than services, but that has changed. The localism of services is breaking down in general business, finance, entertainment, content, and many other service industries, as services are increasingly able to be delivered digitally. Global exports of digitally delivered services have risen by more than 8 percent per year on average over the last decade, far outpacing the increase in global trade of physical goods (World Trade Organization, 2023b). Clearly, firms that can understand the globalization of their industry and respond appropriately are positioned to reap substantial gains.

Global Business Challenges

Although the possibilities of globalization for business success are significant, and the development of a global economy and ecosystem since the end of the Second World War has been remarkably successful, there are still a number of fundamental forces operating to inhibit a global economy and to disrupt international business.

Particularism

Making judgments and taking action on the basis of narrow or personal characteristics, in all its forms (religious, nationalistic, ethnic, regionalism, geopolitical position).

At a cultural level, **particularism**, making judgments and taking action on the basis of narrow or personal characteristics, in all its forms (religious, nationalistic, ethnic, regionalism, geopolitical position), rejects the very concept of a shared global culture and rejects the penetration of domestic markets by foreign goods and services. Language remains a significant barrier. Although English often acts as the standard business language at higher levels of a company, this is typically not the case throughout the middle and lower ranks.

Differences among cultures produce differences in social expectations. For instance, in some countries, consumers expect domestic name-brand products to be built domestically and are disappointed to learn that much of what they thought of as domestically produced is in fact foreign-made.

Different cultures produce different political and legal systems. For instance, different countries have different laws governing the movement of information, information privacy of their citizens, origins of software and hardware in systems, and radio and satellite telecommunications. Even the hours of business and the terms of business trade can vary greatly. These differences complicate global business and must be considered when building global information systems.

Transborder data flow

The movement of information across international boundaries in any form.

For example, European countries have different laws concerning data flow and privacy from those in the United States. **Transborder data flow** is defined as the movement of information across international boundaries in any form. The General Data Protection Regulation (GDPR), which went into effect in 2018, applies to all data produced by European Union (EU) citizens, whether or not the company collecting the data in question is located within the EU, as well as all people whose data are stored within the EU, whether or not they are actually EU citizens. (Review the discussion of GDPR in Chapter 4.)

Cultural, social, and political/legal differences profoundly affect organizations' business processes and applications of information technology. A host of specific barriers arise from such differences, including technological standards, varying reliability of phone networks, and the shortage of skilled consultants. Software may have to be built with local language interfaces before a new information system can be successfully implemented.

There have also been a number of "shocks" to the forward march of globalism and global business over the last decade and a half, including the 2008–2009 global financial crisis, various environmental disasters, geopolitical crises and conflicts, and the Covid-19 pandemic. These shocks have raised concerns about how dependent companies have become on other countries for critical supplies and technologies; how disruptions can reverberate across complex and integrated supply chains; and how interconnectivity and interdependence can seem to place companies at risk. In the face of these shocks, there has been a trend toward what is alternatively termed "deglobalization" or "regionalization." Some companies have reacted to these trends by "nearshoring" (shifting manufacturing and production operations closer to their main markets) or "friend-shoring/ally-shoring" (focusing operations and supply chain networks on countries regarded as political and economic allies).

These inhibiting factors must be taken into account when you are designing and building international systems for your business. Companies trying to implement systems spanning national boundaries typically underestimate the time, expense, and logistical difficulties of making goods and information flow freely across different countries.

Table 15.2 summarizes the challenges to the development of global information systems.

Table 15.2 Global Business Challenges

Global	Specific
Cultural particularism: Nationalistic, religious, language differences	Standards: Different Electronic Data Interchange (EDI), email, telecommunications standards
Social expectations: Brand-name expectations, work hours	Reliability: Telecommunications networks not uniformly reliable
Political and legal: Transborder data and privacy laws, commercial regulations	Speed: Different data transfer speeds, many slower than United States
Deglobalization and regionalization trends driven by "shocks" such as the Covid-19 pandemic	Personnel: Shortages of skilled consultants

15.2 Discuss How to Develop Global Information Systems Architecture

In this section, we'll examine the process that should be followed to develop an information systems architecture suitable for supporting a company's global strategy. An **international information systems architecture** consists of the basic information systems required by organizations to coordinate worldwide trade and other activities. One might think, given the opportunities for achieving competitive advantages as outlined previously and the interest in future applications, that most international companies have rationally developed international systems architectures. Nothing could be further from the truth. Many companies have inherited patchwork international systems, often based on outdated concepts of information processing, with reporting from independent foreign divisions to corporate headquarters, manual entry of data from one legacy system to another, and little online control and communication. Businesses in this situation increasingly face powerful competitive challenges in the marketplace from firms that have rationally designed truly international systems. Still other companies have recently built technology platforms for international systems but lack global strategy. As it turns out, there are significant difficulties in building appropriate international information systems architectures.

International information systems architecture

The basic information systems required by organizations to coordinate worldwide trade and other activities.

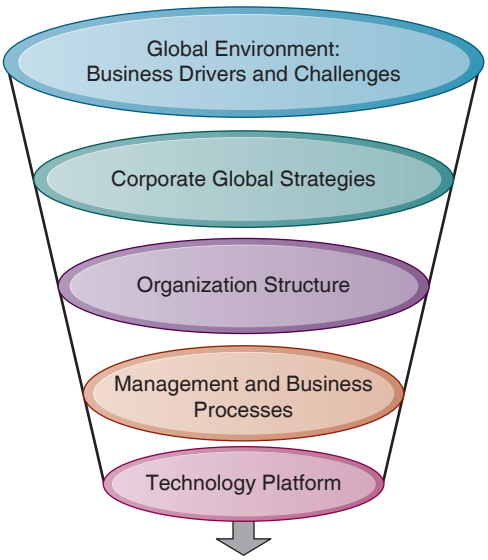
Examine Figure 15.2, which illustrates the major dimensions that need to be considered when developing an effective international information systems architecture.

The first step in the process is to understand the global environment in which the firm is operating (refer to Section 15.1). This means understanding the overall market forces, or business drivers, that are pushing the firm's industry toward global competition. Likewise, you need to carefully examine the factors that may create *management challenges*—factors that could scuttle the development of a global business.

Once you have examined the global environment, you will need to consider the best strategy for competing in that environment. How will your firm respond? You could ignore the global market and focus on domestic competition only, sell internationally from a domestic base, or organize production and distribution around the globe. There are also many in-between choices. We further discuss various strategies that the organization might choose to adopt in Section 15.3.

Figure 15.2 International Information Systems Architecture

The major dimensions for developing an international information systems architecture involve considering the global environment, corporate global strategies, the structure of the organization, management issues and business processes, and the technology platform.



After you have developed a strategy, you need to consider how to structure the organization so it can pursue the strategy. How will you accomplish a division of labor across a global environment? Where will production, administration, accounting, marketing, and human resource functions be located? Who will handle the systems function? Various organizational structures that could be adopted are discussed further in Section 15.3.

Next, you must consider the management issues in implementing the strategy. Key here will be the design of business processes. How can you discover and manage user requirements? How can you induce change in local units to conform to international requirements? How can you reengineer on a global scale, and how can you coordinate systems development? Various management issues are discussed further in Section 15.4.

The last issue to consider is the technology platform (see Section 15.5). Although changing technology is a key driving factor leading toward global markets, you need to have a corporate strategy and structure before you can choose the right technology.

15.3 Discuss Strategies and Organizational Structures for Global Businesses and Information Systems

Three organizational issues face businesses seeking a global presence: choosing a strategy, organizing the business, and organizing the systems management area. These three issues are closely connected.

Global Strategies and Business Organization

Firms seeking to operate on a global scale typically choose one of the following strategies: domestic exporter, multinational, franchiser, and transnational. Each of these strategies is pursued with a specific business organizational structure (see Table 15.3). For simplicity's sake, we describe three kinds of organizational structure or governance: centralized (in the home country), decentralized (to local foreign units), and coordinated (all units participate as equals). Other types of governance patterns can be observed in specific companies (e.g., dominance by one unit, a confederacy of equals, a federal structure balancing power among strategic units, and so forth).

Domestic exporter strategy
Global firm business strategy characterized by heavy centralization of business activities in the home country of origin.

The **domestic exporter strategy** is characterized by heavy centralization of business activities in the home country of origin. Production, finance/accounting, sales/marketing, human resources, and strategic management are set up to optimize resources in the home country, although some operations may also take place in other countries. International sales are typically accomplished using distribution agreements or subsidiaries, but even here, foreign marketing normally relies on the domestic home base for marketing themes and strategies. Nearly all companies that sell products or services internationally begin this way, but often move on to other strategies as they grow. Leatherman Tool Group, a U.S. manufacturer of

Table 15.3 Global Business Strategy and Structure

Business Function	Domestic Exporter	Multinational	Franchiser	Transnational
Production	Centralized	Dispersed	Coordinated	Coordinated
Finance/accounting	Centralized	Centralized	Centralized	Coordinated
Sales/marketing	Mixed	Dispersed	Coordinated	Coordinated
Human resources	Centralized	Centralized	Coordinated	Coordinated
Strategic management	Centralized	Centralized	Centralized	Coordinated

multitools and knives, with its headquarters and manufacturing facility located in Portland, Oregon, is an example of a company using the domestic exporter strategy. All of Leatherman Tool Group's products are designed and made in the United States and are sold in approximately 90 countries around the world via authorized distributors.

The **multinational strategy** concentrates on financial management and control out of a central home base while decentralizing production, sales, and marketing operations to units in other countries. The products and services available in different countries are sometimes adapted to suit local market conditions. The organization becomes a far-flung confederation of production and marketing facilities in different countries. Many firms, such as Hormel Foods, profiled in the Opening Case Study, *Hormel Foods Implements Oracle Cloud Applications to Unify Its Systems Worldwide*, use this strategy. Hormel's corporate headquarters and 90 percent of its employees are located in the United States. However, it has regional teams around the world that have the autonomy to adapt and iterate based on local needs and preferences, while also benefiting from the broader knowledge and resources of the global company.

Multinational strategy

Global firm business strategy that concentrates financial management and control out a central home base while decentralizing production, sales, and marketing operations to units in other countries.

A **franchiser strategy** uses an interesting mix of strategies. On the one hand, the product is created, designed, financed, and initially produced in the home country, but for product-specific reasons, typically must rely heavily on foreign personnel for further production, marketing, and human resources. Food franchisers such as McDonald's and Starbucks fit this pattern. McDonald's continues to rely largely on the United States for development of new products, strategic management, and financing. Nevertheless, because the product must be produced locally—it is perishable—extensive coordination and dispersal of production, local marketing, and local recruitment of personnel are required.

Franchiser strategy

Global firm business strategy characterized by use of a mix of strategies: Product iscreated, designed, financed, and initially produced in the home country, but for product-specific reasons, typically must rely heavily on foreign personnel for further production, marketing, and human resources.

Transnational firms are globally managed firms. Transnational firms typically do not have a single national headquarters, but instead have many regional headquarters and perhaps a world headquarters. In a **transnational strategy**, nearly all the value-adding activities are managed from a global perspective without reference to national borders, optimizing sources of supply and demand wherever they appear, and taking advantage of any local competitive advantages. Transnational firms view the globe, not the home country, as their management frame of reference. The governance of these firms has been likened to a federal structure in which there is a strong central management core of decision making but considerable dispersal of power and financial muscle throughout the global divisions. Few companies have fully attained transnational status. Unilever is an example of a company that employs a transnational strategy. Unilever manufactures and sells a wide range of consumer goods products under more than 400 different brands. It has a world headquarters in the United Kingdom and operates in more than 190 countries around the world. Nestle S.A. and Eli Lilly are examples of other companies pursuing a transnational strategy.

Transnational strategy

Global firm business strategy in which nearly all value-adding activities are managed from a global perspective without reference to national borders; the globe, not the home country, is viewed as the management frame of reference.

Information technology and improvements in global telecommunications are giving firms more flexibility to shape their global strategies. Government trade policies that create barriers to the import of certain products, thereby incentivizing their manufacture in the country in which they are to be sold, coupled with a need to serve local markets and local regulations, encourage companies to localize operations. At the same time, the drive to achieve economies of scale, the desire for simplification and integration of processes and operations across regions, and the ability to deliver a consistent customer experience move companies toward global standardization. Hence, there are forces of decentralization and dispersal as well as forces of centralization and global coordination. Increasingly, the most successful strategy is one that balances global efficiency with local needs, with a central core of systems and local layers that allow for selective deployment of capabilities in local markets and units (Bossert et al., 2023).

Global Systems Configurations to Fit the Strategy

Information technology and improvements in global telecommunications are giving international firms more flexibility to shape their global strategies. The configuration, management, and development of systems tend to follow the global strategy chosen. Examine Figure 15.3, which depicts typical systems configurations. By *systems configurations*, we mean the full range of activities involved in building and operating information systems: conception and

Figure 15.3 Global Strategy and Systems Configurations

The large X's show the dominant patterns, and the small x's show the emerging patterns. For instance, domestic exporters rely predominantly on centralized systems, but there is some development of decentralized systems in local marketing regions.

SYSTEMS CONFIGURATION	Strategy			
	Domestic Exporter	Multinational	Franchiser	Transnational
Centralized	X			
Duplicated			X	
Decentralized	x	X	x	
Networked		x		X

alignment with the strategic business plan, systems development, and ongoing operation and maintenance. For the sake of simplicity, we consider four types of systems configurations. *Centralized systems* are those in which systems development and operation occur totally at the domestic home base. *Duplicated systems* are those in which development occurs at the home base, but operations are handed over to autonomous units in foreign locations. *Decentralized systems* are those in which each foreign unit designs its own unique solutions and systems. *Networked systems* are those in which systems development and operations occur in an integrated and coordinated fashion across all units.

As can be seen in Figure 15.3, domestic exporters tend to have highly centralized systems in which a single domestic systems development staff develops worldwide applications. Multinationals offer a direct and striking contrast: Here, foreign units devise their own systems solutions based on local needs with few, if any, applications in common with headquarters (the exceptions being financial reporting and some telecommunications applications). Franchisers have the simplest systems structure: Like the products they sell, franchisers develop a single system usually at the home base and then replicate it around the world. Each unit, no matter where it is located, has identical applications. Lastly, the most ambitious form of systems development is found in transnational firms: Networked systems are those in which there is a solid, singular global environment for developing and operating systems. This usually presupposes a powerful telecommunications backbone, a culture of shared applications development, and a shared management culture that crosses cultural barriers. The networked systems structure is the most visible in financial services where the homogeneity of the product—money and money instruments—seems to overcome cultural barriers.

Reorganizing the Business

How should a firm organize itself for doing business on a global scale? To develop a global company and information systems support structure, a firm needs to follow these principles:

1. Organize value-adding activities along lines of comparative advantage. For instance, marketing/sales functions should be located where they can best be performed for the least cost, maximum impact, and reliability. Likewise with production (with safeguards for supply chain resiliency), finance, human resources, and information systems.
2. Develop and operate systems units at each level of corporate activity—regional, national, and international. To serve local needs, there should be *host country systems units* of some magnitude. *Regional systems units* should handle telecommunications and systems development across national boundaries that take place within major geographic regions (European, Asian, North American, South American, Global South, and so forth). *Transnational systems units* should be established to create linkages across major regional areas and coordinate the development and operation of international telecommunications and systems development.
3. Establish a single office responsible for the development of global information systems—a global chief information officer (CIO) position.

Many successful companies have devised organizational systems and structures based on these principles. The success of these companies relies not only on the proper organization of activities, but also on a key ingredient: a management team that can understand the risks and benefits of international systems, and that can devise strategies for overcoming the risks. We will turn to these management topics next.

15.4 Discuss Management Challenges in Developing Global Information Systems

Table 15.4 lists the principal management challenges involved in developing global information systems. It is interesting to note that many of these problems are the same difficulties managers experience in developing domestic information systems as well, but these are enormously complicated in the global environment.

A Typical Scenario: Disorganization on a Global Scale

Let's look at a common scenario. A traditional multinational consumer-goods company based in the United States and operating in Europe would like to expand into Asian markets and wants to develop a transnational strategy and a supportive information systems structure. Like most multinationals, it has dispersed production and marketing to regional and national centers while maintaining a headquarters and strategic management in the United States. Historically, it has allowed each of the subsidiary foreign divisions to develop its own systems. The only centrally coordinated system is financial controls and reporting. The central systems group in the United States focuses only on domestic functions and production.

The result is a hodgepodge of hardware, software, and telecommunications. Each production facility uses a different MRP (manufacturing resources planning) system (or a different version of the same ERP system) and different marketing, sales, and human resource systems. Hardware and database platforms are wildly different. Communications between different sites may be difficult at times due to local conditions.

What do you recommend to the senior management leaders of this company who now want to pursue a transnational strategy and develop an information systems architecture to support a highly coordinated global information systems environment? Consider the problems you face by examining Table 15.4. The foreign divisions may resist efforts to agree on common user requirements; they may have never thought about much other than their own units' needs. The U.S. systems groups, which have been enlarged recently and told to focus on local needs, may not easily accept guidance from anyone recommending a transnational strategy. It may be difficult to convince local managers throughout the world that they should change their business procedures to align with other units in the world, especially if this might interfere with their local performance. Typically, local managers are rewarded in this company for meeting local objectives of their division or plant. Finally, it may be difficult to coordinate development of projects around the world and therefore difficult to encourage local users to take on ownership in the systems developed.

Table 15.4 Management Challenges in Developing Global Information Systems

Agreeing on common user requirements
Introducing changes in business processes
Coordinating applications development
Coordinating software releases
Encouraging local users to support global information systems

15.5 Discuss Management Solutions for Global Information Systems Development Challenges

In this section, we will first consider how to devise solutions for the challenges posed in developing global information systems, and then consider the issues involved in actually implementing those solutions.

Devising Management Solutions for Global Information Systems Development Challenges

Core systems
Systems that support functions that are absolutely critical to the organization.

When devising a strategy for global information systems, first consider that not all systems should be coordinated on a transnational basis. Only some core systems are truly worth sharing from a cost and feasibility point of view. **Core systems** support functions that are absolutely critical to the organization. Other systems should be partially coordinated because they share key elements, but they do not have to be totally common across national boundaries. For such systems, a good deal of local variation is possible and desirable. A final group of systems is peripheral and needed to suit local requirements only.

Define the Core Business Processes

How do you identify core systems? The first step is to define a short list of critical core business processes. Business processes are defined and described in Chapters 2, 3, and 13, which you should review. Briefly, business processes are sets of logically related tasks to produce specific business results, such as shipping out correct orders to customers or delivering innovative products to the market. Each business process typically involves many functional areas, communicating and coordinating work, information, and knowledge. Core business processes are candidates for transnational systems.

The way to identify these core business processes is to conduct a business process analysis. How are customer orders taken? What happens to them once they are taken? Who fills the orders, and how are they shipped to the customers? What about suppliers? Do they have access to enterprise resource planning systems so that supply is automatic? You should be able to identify and set priorities in a short list of 10 business processes that are critical for the firm. You will have a clearer understanding of core business processes and how to proceed if you can also specify your organization's regional and local business processes. Table 15.5 provides some examples.

Table 15.5 Examples of Core, Regional, and Local Business Processes

Type of Business Process	Examples
Core business processes	Managing global supply chains
	Maintaining a global brand image
	Establishing global quality standards
Regional business processes	Operating regional distribution centers
	Running a regional marketing campaign
	Evaluating regional sales district performance
	Providing region-specific customer service
Local business processes	Taking customer orders
	Customizing food product labels for different languages and local tastes
	Training local employees
	Hiring local office staff

Next, can you identify centers of excellence for these processes? Is the customer order fulfillment superior in the United States, manufacturing process control superior in Germany, and human resources superior in Asia? You should be able to identify some areas of the company for some lines of business where a division or unit stands out in the performance of one or several business functions.

When you understand the business processes of a firm, you can rank them in order. You can then decide which processes should be core applications, centrally coordinated, designed, and implemented around the globe, and which should be regional and local. At the same time, by identifying the critical business processes, you have gone a long way toward defining a vision of the future that you should be working toward.

Identify the Core Systems to Coordinate Centrally

By identifying the critical core business processes, you can begin to see opportunities for transnational systems. The second strategic step is to define which systems should be transnational. The financial and political costs of defining and implementing transnational systems are extremely high. Therefore, keep the list to an absolute minimum, letting experience be the guide and erring on the side of minimalism. By dividing off a small group of systems as absolutely critical, you divide opposition to a transnational strategy. At the same time, you can appease those who oppose the central worldwide coordination implied by transnational systems by permitting peripheral systems development to progress unabated with the exception of some technical platform requirements.

Choose an Approach: Incremental, Grand Design, Evolutionary

A third step is to choose an approach. Avoid piecemeal approaches. These surely will fail for lack of visibility, opposition from all who stand to lose from transnational development, and lack of power to convince senior management that the transnational systems are worth it. Likewise, avoid grand design approaches that try to do everything at once. These also tend to fail because of an inability to focus resources. Nothing gets done properly, and opposition to organizational change is needlessly strengthened because the effort requires extraordinary resources. An alternative approach is to evolve transnational applications incrementally from existing applications with a precise and clear vision of the transnational capabilities the organization should have in five years. This is sometimes referred to as the "salami strategy" or one slice at a time.

Make the Benefits Clear

What is in it for the company? One of the most important situations to avoid is building global information systems for the sake of building global systems. From the beginning, it is crucial that senior management at headquarters and foreign division managers clearly understand the benefits that will come to the company as well as to individual units. Although each system offers unique benefits to a particular budget, the overall contribution of global information systems lies in four areas.

First, global information systems—truly integrated, distributed, and transnational systems—contribute to superior management and coordination. They include the ability to switch suppliers on a moment's notice from one region to another in a crisis, the ability to move production in response to natural and man-made disasters, and the ability to use excess capacity in one region to meet demand in another.

A second major contribution of global information systems is the vast improvement in production, operation, and supply and distribution. A global value chain with global suppliers and a global distribution network will enable senior managers to locate value-adding activities in regions where they are most economically and reliably performed (with more emphasis now on supply chain resiliency).

Third, global information systems mean global customers and global marketing. Fixed costs around the world can be amortized over a much larger customer base. This will unleash new economies of scale at production facilities.

Last, global information systems enable companies to optimize the use of corporate funds over a much larger capital base. This means, for instance, capital in a surplus region can be moved efficiently to expand production of capital-starved regions. Cash can be managed more effectively within the company and put to use more effectively.

Implementing Management Solutions

We can now consider how to implement solutions to the most vexing problems described in Table 15.4 that face managers in developing global information systems architectures.

Agreeing on Common User Requirements

Establishing a short list of the core business processes and core support systems will begin a process of rational comparison across the many divisions of the company. It will also develop a common language for discussing the business and naturally lead to an understanding of common elements as well as the unique qualities that must remain local.

Introducing Changes in Business Processes

Legitimacy
The extent to which one's authority is accepted on grounds of competence, vision, or other qualities.

Your success as a change agent will depend on your legitimacy, your authority, and your ability to involve users in the change design process. **Legitimacy** is defined as the extent to which your authority is accepted on grounds of competence, vision, or other qualities. The selection of a viable change strategy, which we have defined as evolutionary but with a vision, should assist you in convincing others that change is feasible and desirable. Involving people in change, assuring them that change is in the best interests of the company and their local units, is a key tactic.

Coordinating Applications Development

Choice of change strategy is critical for this problem. At the global level, there is far too much complexity to attempt a grand design strategy of change. It is far easier to coordinate change by making small incremental steps toward a larger vision. Imagine a five-year plan of action rather than a two-year plan of action and reduce the set of transnational systems to a bare minimum to reduce coordination costs.

Coordinating Software Releases

Firms can institute procedures to ensure that all operating units convert to new software updates at the same time, so that everyone's software is compatible.

Encouraging Local Users to Support Global Information Systems

Co-optation
Bringing the opposition into the process of designing and implementing a solution without giving up control of the direction and nature of the change.

The key to this problem is to involve users in the creation of the design without giving up control over the development of the project to parochial interests. The overall tactic for dealing with resistant local units in a transnational company is co-optation. **Co-optation** is defined as bringing the opposition into the process of designing and implementing the solution without giving up control over the direction and nature of the change. As much as possible, raw power should be avoided. Minimally, however, local units must agree on a short list of transnational systems, and raw power may be required to solidify the idea that transnational systems of some sort are truly required.

How should co-optation proceed? Several alternatives are possible. One alternative is to permit each country unit the opportunity to develop one transnational application first in its home territory and then throughout the world. In this manner, each major country systems group is given a piece of the action in developing a transnational system, and local units feel a sense of ownership in the transnational effort. On the downside, this assumes the ability to develop high-quality systems is widely distributed and that a German team, for example, can successfully implement systems in France and Italy. This will not always be the case.

A second tactic is to develop new transnational centers of excellence, or a single center of excellence. There may be several centers around the globe that focus on specific business processes. These centers draw heavily from local national units, are based on multinational teams, and must report to worldwide management. Centers of excellence perform the initial identification and specification of business processes, define the information requirements, perform the business and systems analysis, and accomplish all design and testing. Implementation, however, and pilot testing are rolled out to other parts of the globe. Recruiting a wide range of local groups to transnational centers of excellence helps send the message that all significant groups are involved in the design and will have an influence.

Even with the proper organizational structure and appropriate management choices, it is still possible to stumble over technology issues. Choices of technology platforms, networks, hardware, and software are the final elements in building transnational information systems architectures.

15.6 Discuss Technology Issues and Alternatives in Developing Global Information Systems

Once firms have defined a global business model and systems strategy, they must select hardware, software, and networking standards along with key system applications to support global business processes. Hardware, software, and networking pose special technical challenges in an international setting.

One major challenge is finding some way to standardize a global computing platform when there is so much variation from operating unit to operating unit and from country to country. Another major challenge is finding specific software applications that are user-friendly and that truly enhance the productivity of international work teams. The universal acceptance of the Internet around the globe has greatly reduced networking problems. But the mere presence of the Internet does not guarantee that information will flow seamlessly throughout the global organization because not all business units use the same applications, and the quality of Internet service can be highly variable. For instance, German business units may use an open-source collaboration tool to share documents and communicate, which is incompatible with U.S. headquarters teams, which use Microsoft solutions. Overcoming these challenges requires systems integration and connectivity on a global basis.

Computing Platforms and Systems Integration

The development of a transnational information systems architecture based on the concept of core systems raises questions about how the new core systems will fit in with the existing suite of applications developed around the globe by different divisions and different people, and for different kinds of computing hardware. The goal is to develop global, distributed, and integrated systems to support digital business processes spanning national boundaries. Briefly, these are the same problems faced by any large domestic systems development effort. However, the problems are magnified in an international environment. Just imagine the challenge of integrating systems based on Windows, Linux, Unix, or proprietary operating systems running on various forms of hardware in many different operating units in many different countries!

Moreover, having all sites use the same hardware and operating system does not guarantee integration. Some central authority in the firm must establish data standards as well as other technical standards with which sites are to comply. For instance, technical accounting terms such as the beginning and end of the fiscal year must be standardized (review the earlier discussion of the cultural challenges to building global businesses) as well as the acceptable interfaces between systems, communication speeds and architectures, and network software.

What are the most important software applications? Many international systems focus on basic transaction and management reporting systems. Increasingly, firms are turning to supply chain management and enterprise resource planning systems to standardize their business processes on a global basis and to create coordinated supply chains and workforces. However, these cross-functional systems are not always compatible with differences in languages, cultural heritages, and business processes in other countries. Company units in countries that are not technically sophisticated may also encounter problems trying to manage the technical complexities of enterprise applications.

Electronic Data Interchange (EDI) systems and supply chain management systems are widely used by manufacturing and distribution firms to connect to suppliers on a global basis. Collaboration and enterprise social networking systems, email, and video conferencing are especially important worldwide tools for knowledge- and data-based firms, such as advertising firms, research-based firms in medicine and engineering, and graphics and publishing firms.

Table 15.6 Challenges of International Networks

Quality of service
Security
Costs and tariffs
Network management
Installation delays
Poor quality of international service
Regulatory constraints
Network capacity

Connectivity

Truly integrated global information systems must have connectivity—the ability to link together the systems and people of a global firm into a single integrated network just like the phone system but capable of voice, data, and image transmissions. The Internet has provided an enormously powerful foundation for providing connectivity among the dispersed units of global firms. However, many issues remain. The public Internet does not guarantee any level of service (even in the United States). Few global corporations trust the security of the public Internet and generally use private networks to communicate sensitive data and Internet-based virtual private networks (VPNs) for communications that require less security. Not all countries support even basic Internet service that requires obtaining reliable circuits, coordinating among different carriers and the regional telecommunications authority, and obtaining standard agreements for the level of telecommunications service provided. Table 15.6 lists the major challenges posed by international networks.

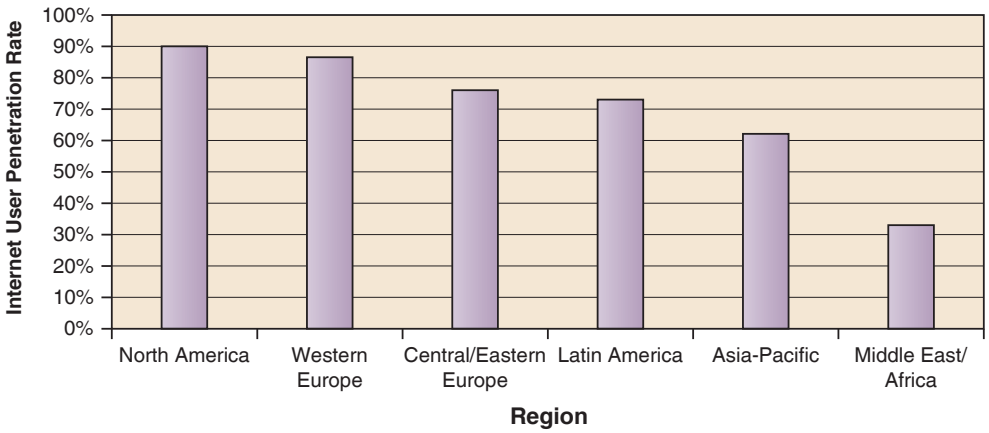
While private networks have guaranteed service levels and better security than the Internet, the Internet is the primary foundation for global corporate networks when lower security and service levels are acceptable. Companies can create global intranets for internal communication or extranets to exchange information more rapidly with business partners in their supply chains. They can use the public Internet to create global networks using VPNs from Internet service providers, which provide many features of a private network using the public Internet (see Chapter 7). However, VPNs may not provide the same level of quick and predictable response as private networks, especially during times of the day when Internet traffic is very congested, and they may not be able to support large numbers of remote users.

Examine Figure 15.4, which illustrates the percentage of Internet users in various regions of the world. As the graph shows, the percentage of the population that is accessing the Internet is limited in many developing countries, particularly in the Middle East, Africa, and certain parts

Figure 15.4 Internet User Penetration Rate in Regions Around the World

The percentage of the total population using the Internet in developing regions of the world, such as the Middle East and Africa, is much lower than in North America and Europe.

Source: Based on data from eMarketer, August 2023.



of Asia-Pacific, where Internet infrastructure often lacks bandwidth capacity and is unreliable in part due to power grid issues. The lack of purchasing power of many people in developing countries also makes access to Internet services very expensive in local currencies, although inexpensive mobile devices and low-cost data plans are becoming more widely available.

On the other hand, the rate of growth in the Internet population has been faster in Asia, Africa, and the Middle East than in North America and Europe. In the future, Internet connectivity will be much more widely available and reliable in developing regions of the world, and it will play a significant role in integrating these economies with the world economy. Much of the world's population accesses the Internet using a mobile device, so apps must be built for mobile platforms, tiny screens, and low bandwidth.

There are also other issues. Governments in many countries, such as China, Iran, Turkey, and Saudi Arabia monitor Internet traffic and block access to websites they consider offensive. Moreover, nation-states all over the world have been asserting their authority over Internet use and claiming various forms of digital sovereignty (see the Spotlight on Organizations Case Study, *The Rise of Digital Nationalism*).

Spotlight on Organizations Case Study

The Rise of Digital Nationalism

The Internet began as a network of networks designed to foster the free and open exchange of information anywhere in the world. It has been a major driver of globalization, embodying a public sphere beyond the control of sovereign governments. As discussed in the chapter, many companies have built a competitive advantage by standardizing their information systems across geographic boundaries. But today, this is changing. All over the world, nation-states have been asserting their authority over Internet use and claiming various forms of digital sovereignty. The Internet now has become subject to firewalls, shutdowns, and data localization laws that have fundamentally changed its character as a unified global infrastructure. These changes are forcing global companies to change their approach, and in many instances, think locally rather than globally.

Authoritarian governments have long sought to rein in the Internet, with China being one of the first. Its Great Firewall (known in China as the "Golden Shield") restricts what people can read and do online, and has served as a model for promoting Chinese "digital sovereignty." China's efforts have demonstrated to other authoritarian regimes that the Internet can be effectively controlled.

A number of countries have followed the Chinese example. To control contact with the outside world and suppress dissident content, Iran has set up an isolated, domestic version of the Internet that it calls the "National Information Network". In 2022, Iran intermittently restricted Internet service and blocked WhatsApp and Instagram, the only international social media platforms that had been accessible in the country. In Russia, Vladimir Putin signed a "sovereign Internet" law in 2019 to set up a self-sufficient Runet. The law also enables the Russian government to shut off the global network to Russian users. Russia's invasion of Ukraine in 2022 was accompanied by a major clampdown on access to the global Internet within Russia. According to the *New York Times*, at least a quarter of the world's governments have temporarily shut down the Internet within their country at some point during the recent past.

In recent years, digital nationalism has also taken the form of data localization (or data protectionism) laws. Some countries require that data on their citizens (or certain types of data, such as medical or financial data) be physically stored and processed on servers within their borders. Others allow the data to leave their borders but insist on a copy remaining domestically. Complying with these sorts of regulations requires companies to build a separate infrastructure and support team for each of the countries in which they operate.

These laws have legitimate uses, including privacy protection, national security, and promoting local economies. At least 75 percent of all countries now have some version of data localization requirements in place, and they are not limited to authoritarian states such as China and Russia. Australia, Canada, New Zealand, South Korea, and Switzerland are among the countries that now restrict cross-border flows of data.

The European Union's General Data Protection Regulation (GDPR—see Chapter 4) is not specifically about localization, but it imposes stringent restrictions that make it difficult for companies to move data across borders. For instance, in 2023, Meta, the owner of Facebook and Instagram, was fined $1.3 billion for transferring EU users' personal data to U.S. servers, violating the GDPR. Cloud providers, such as Microsoft, Google, and Amazon's AWS, have been forced to take steps to exclusively store and process all personal data from their European users within the EU.

Many of these laws are ostensibly targeted at Western multinationals, but larger companies are generally better able to meet these requirements than smaller ones, which lack the necessary resources. Developing countries, often

at the forefront of digital nationalism, might also suffer. For example, India and the Philippines have large numbers of outsourcing firms that rely on a unified global information network. These countries' efforts to set up roadblocks on that network could come back to haunt them.

The great risk is that digital nationalism will splinter the Internet, breaking it up into a patchwork of incompatible and irreconcilable fiefdoms. This scenario, sometimes referred to as the Splinternet, is already affecting Internet-based content and services. China's population does not have access to Wikipedia, Facebook, and most of Google. When the EU's GDPR first came into effect, many American media companies were ready to stop offering their content to European consumers, at least temporarily.

Digital nationalism could also reshape the Internet's underlying technical infrastructure. Over the last decade, several countries, citing cultural sensitivities, have considered banning or otherwise restricting the .xxx top-level domain name (generally used for pornography), raising the prospect that the Internet's naming system could eventually fracture. After Edward Snowden's revelations about U.S. spying, Brazil moved to build a separate undersea cable link to the EU to bypass existing Internet infrastructure. At the Internet Engineering Task Force (IETF), a key Internet standards committee, representatives who wanted to maintain a backdoor for government agencies clashed with those pushing for more robust encryption.

What can be done to curb digital nationalism? Experts recommend finding ways to restore a sense of inclusiveness and fair play among Internet users and to remind them of the original principles of universality and inclusiveness that made the network so effective in the first place. Another recommendation is to develop "zone"-based approaches for running the Internet, where interconnecting blocs of member-nations would commit to uphold principles such as free trade, privacy, and freedom of expression. This approach would not produce a single global network, but a "coalition of the willing" might be the best way to counter growing Internet fragmentation and keep the network relatively open and free.

SOURCES: Dev Kundaliya, "Microsoft, Google and UK Make Moves on Sovereign Data Storage," Computing.co.uk, January 15, 2024, and "AWS Announces 'European Sovereign Cloud' for Government and Regulated Industries," Computing.co.uk, October 26, 2023; Andrew Sullivan, "Misguided Policies the World Over Are Slowly Killing the Open Internet," Internetsociety.org, March 1, 2023; Adam Satariano and Valerie Hopkins, Satyajit Parekh, et al., "Localization of Data Privacy Regulations are Creating Competitive Opportunities," Mckinsey.com, June 30, 2022; "Russia, Blocked from the Global Internet, Plunges into Digital Isolation," *New York Times*, March 7, 2022; Lily Hay Newman, "How the Iranian Government Shut Off the Internet," Wired.com, November 17, 2019; Akash Kapur, "The Rising Threat of Digital Nationalism," *Wall Street Journal*, November 1, 2019.

Case Study Questions

1. What is digital nationalism? Give two examples.
2. What problems does digital nationalism pose for conducting business globally?

Software Localization

The development of core systems poses unique challenges for application software. How will the old systems interface with the new? Entirely new interfaces must be built and tested if old systems are kept in local areas, which is common. These interfaces can be costly and messy to build. If new software must be created, another challenge is to build software that can be realistically used by multiple business units from different countries, given that business units are accustomed to their unique business processes and definitions of data.

Aside from integrating the new with the old systems, there are problems of human interface design and functionality of systems. For instance, to be truly useful for enhancing productivity of a global workforce, software interfaces must be easily understood and mastered quickly. When international systems involve knowledge workers only, English may be the assumed international standard. But as international systems penetrate deeper into management and clerical groups, a common language cannot be assumed, and human interfaces must be built to accommodate different languages and even conventions. The entire process of converting software to operate in a second language is called **software localization**. Artificial intelligence tools are increasingly being used for this purpose, enabling companies to save time and money (Sharma, 2023).

Software localization
Process of converting software to operate in a second language.

15.7 Understand How the Information in This Chapter Can Help Your Career

The information you have learned in Chapter 15 can help you advance your career by helping you to understand the opportunities and challenges involved with developing global information systems.

Real-World Scenario

Fiat Learns How to Coordinate Global Operations

Fiat Group Automobiles S.p.A., which merged with Chrysler Group and became a subsidiary of the Stellantis multinational automotive conglomerate, has been one of the world's leading automakers. Its brands (such as Alfa Romeo, Fiat, Chrysler, Jeep, and Maserati) and operating units had their own legacy systems, business processes, and reporting standards. As a result, Fiat Group was unable to effectively coordinate global operations or manage its brands across multiple countries and regions. Management was unable to see how Fiat was performing on an enterprise-wide basis. The solution was to develop new information systems that could provide enterprise-wide data for management reporting and analysis. Fiat implemented a new reporting platform based on Oracle Hyperion for enterprise performance management. The new system provides users with pre-aggregated data across countries, legal entities, and business functions. For example, regional reports break down the data by brand and geographical location to provide management with more precise information about pricing, profit and loss, and dealer incentives. Controllers use the reports to analyze sales activity among dealers as well as fleet, rental car, and government sales channels. Fiat's global supply chain analysts use the system to ensure that Fiat's brands have the right products in inventory in the right place and at the right time. Fiat's finance managers use the system to track profit margins by brand, generate balance sheets, and monitor cash flow across the entire Fiat organization.

Job Description and Career Path: Global Supply Chain Analyst

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Global Supply Chain Analyst	A global supply chain analyst plays a vital role in ensuring a company has the right products, at the right places, at the right times. Job Responsibilities include: • Managing inventory ordering plans and execution for all product lines to ensure proper inventory levels are maintained at all times • Executing all tactical responsibilities associated with purchase order management and buying materials and finished good product from global suppliers • Building strong relationships with internal teams and both internal and external suppliers to further understand supply chain processes and business needs • Analyzing data and recommending enhancements to various process management needs • Negotiating with core supply partners on current and future production plans to optimize inventory levels and fulfill customer demands • Identifying supply chain disruptions, including inventory shortages or capacity conditions that adversely affect customer satisfaction	• Microsoft 365 • Experience with SQL databases • Experience with business intelligence products such Tableau • Experience with SAP ERP systems a plus	• Entry level global supply chain analyst: $75,000. • Mid-level global supply chain analyst: $70,000–$87,000 • Senior supply chain analyst: $105,000. Salaries can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

Wyndham Centralizes Its Information Systems to Better Serve Its Worldwide Hotels

Many people don't realize that the brand-name hotel that they are staying in is actually owned by a franchisee, an independent owner who is operating the hotel under a franchise agreement. Wyndham Hotel & Resorts is one of the most prominent global hotel companies that uses a franchise business strategy. Through the years, Wyndham has acquired the rights to some of the most iconic hotel brands around the world, including Howard Johnson, Ramada, Days Inn, Super 8, Travelodge, Microtel, La Quinta, and its namesake brand, Wyndham, among others. Today it has become the world's largest hotel franchising company, with 6,000 franchisees that operate 9,100 hotels in more than 95 countries under one of Wyndham's 24 global brands.

As part of their franchise agreements, Wyndham provides operational and technical support for the hotels that are part of their network. Like many other global companies that have grown via acquisitions, Wyndham, for many years, had what the company's chief information officer, Scott Strickland, characterized as a "dog's breakfast" of systems, including five different reservation systems, multiple web platforms, and 12 different property management systems. Wyndham realized that this was not an optimal way to do business on a global scale. So, around five years ago, the company began to invest in standardizing the company's platforms, an investment that has now topped $275 million.

Wyndham began with an investment in Salesforce technology that has now evolved into what Wyndham calls the Wyndham Community, a franchise owner engagement platform. According to Strickland, the Wyndham Community gives Wyndham a 360-degree view of its franchisees. Lisa Checchio, Wyndham's chief marketing officer, notes that the system was built so that all its users, including its field operations teams, its revenue managers, and its IT support teams, are using the same interface and able to see what their franchisees are seeing. The interface was built over an 18-month period with feedback from Wyndham's field operation staff and franchisees. The platform rolled out in September 2023 and had 12,000 logins on the first day. Daily logins have since grown to an average of 18,000, along with 100,000 dashboard views. Franchise owners can access real-time insights into key business metrics across their portfolio, from anywhere and any device. The platform also provides a direct connection to operational support while serving as franchise owners' primary destination for news and updates. Strickland expects that the adoption rate of the technology will ultimately be in the 90 percent range. Franchisees can use the system to pay Wyndham, resolve issues, and work with the company's field operations.

Wyndham is also in the process of winnowing down its 12 different property management systems to two cloud-based systems: Sabre's SynXis Property Hub (running on Google Cloud) is typically being used for smaller hotels in North America with less complicated needs, and Oracle Hospitality's OPERA Cloud Property Management System (running on Amazon Web Services) for all its franchisees outside North America and also for larger hotels and those with amenities like restaurants and meeting facilities within North America. The fact that the systems are cloud-based means that hotels do not need to manage and maintain on-site hardware.

Wyndham was one of the first hotel chains to adopt the OPERA Cloud Property Management System. The system allows hotels to manage reservations, check-ins, billing, and reporting. Wyndham began piloting the system internationally in 2020 and started rolling it out to about 800 of its full-service hotels in mid-2021. In September 2023, it announced plans to expand the system by the end of 2024 to more than 2,000 additional hotels, mostly those near the top of its mid-tier segment, such as Ramada, LaQuinta, and Wingate. Strickland notes that it can migrate hotels onto OPERA Cloud in a matter of days, and they are seeing immediate benefits, including lower costs, higher efficiencies, and the ability to deliver better guest experiences. According to Strickland, OPERA Cloud enables Wyndham to deliver new innovations such as room upselling, integrated revenue management, and mobile housekeeping management, among others, on a global scale. OPERA Cloud is designed to meet financial reporting requirements across more than 200 countries, and the system can be customized to operate in 20 languages.

OPERA Cloud also includes a number of additional tools, such as OPERA Reporting and Analytics, which enable hotels to unlock insights needed to boost performance at the department, hotel, and chain level. OPERA Sales and Event Management helps hotels maximize revenue by providing detailed, real-time information on room and event offerings. Wyndham franchisees can also use the Oracle Hospitality Integration Platform within OPERA Cloud to integrate with RevIQ, Wyndham's advanced revenue management system. Developed in collaboration with IDeaS, a provider of hotel revenue management software, RevIQ is tailored specifically for the needs of Wyndham franchisees.

Wyndham has also recently debuted a new guest engagement platform powered by Canary Technologies. The platform offers a number of mobile-centric tools, including AI-driven property messaging, which enables guests to text hotels directly for any needs during their stay; smart mobile check-in; dynamic upselling that enables hotels to offer enhancements like early check-in, late checkout, and room upgrades; and smart mobile checkout.

Though centralizing its platforms clearly has provided Wyndham and its franchisees with a number of benefits, having all the company's data in cloud-based systems also raises cybersecurity and privacy concerns. To alleviate these concerns, Wyndham has been working with San Francisco-based Okta to layer a security system on top of its platforms. Wyndham first implemented Okta Workforce Identity Cloud for its internal employees and then expanded it to the more than 100,000 employees of its franchisees. Building on that success, Wyndham is now rolling out Customer Identity Cloud to its more than 100 million loyalty members. Okta is also helping Wyndham meet data privacy requirements because, often, property management systems need to be housed in the localities they are serving.

AI is the next frontier for Wyndham. In addition to AI-driven property messaging, Wyndham is currently using AI for real-time coaching for its call center agents, for example, letting agents know that they are speaking too quickly. Wyndham is also using AI to mine the data of its more than 100 million loyalty members. For instance, Wyndham learned that its loyalty members' number one concern was Wi-Fi. This has enabled Wyndham to prove with data that Wi-Fi is important to guests and that it's worth a significant investment by its franchisees. Investors believe that Wyndham is on the right track. Although its stock price plummeted in the wake of the Covid-19 pandemic, it has since rebounded and its recent financial results show increasing revenue and profits, driven in large part by international growth.

SOURCES: Okta, "Wyndham Hotels & Resorts Gives 100M Users an Unforgettable Digital Experience with Okta," Okta.com, September 8, 2023; Rob Schneider: "How Wyndham's Technology Improvements Can Upgrade Its Franchisees' ROI," Hotelinvestmenttoday.com, September 15, 2023; Adam Perkowsky, "HB on the Scene: Wyndham Launches New Owner and Guest Engagement Platforms," Hotelbusiness.com, September 14, 2023; "Ep 24: The Power of AI at Wyndham Hotels & Resorts with EVP & CIO Scott Strickland," Enterprisesoftware.blog, June 14, 2023; Greg Travarez, "Next-Level Hospitality: Oracle Cloud Revolutionizing Wyndham Portfolio, Cloud-computing.tmcnet.com, May 16, 2023; Oracle, "Oracle Teams with Wyndham to Bring OPERA Cloud to 2,000 Additional Hotels," Oracle.com, May 11, 2023; Sean O'Neill, "Wyndham Signs on for Next-Gen Technology from Oracle Hospitality," Skift.com, May 18, 2021.

Case Study Questions

1. Identify the business challenges that Wyndham Hotels & Resorts faced. How did it use technology to address those challenges?
2. What benefits does the OPERA Cloud platform provide for Wyndham and its franchisees?
3. How is Wyndham using AI?

Review Summary

15.1 Discuss factors impacting development of global information systems.

The development of advanced telecommunications networks and information systems, along with changes in trade policy of various nations, has helped drive cross-border trade as well as the development of global value chains. There are both general cultural factors, such as the development of global culture and a global knowledge base, and specific business factors, such as the development of global markets, global production and operations, global coordination, and global economies of scale, acting as global business drivers. There are also continuing challenges to globalization, including cultural, social, and political/legal differences; trends toward deglobalization and regionalization driven by shocks such as the Covid-19 pandemic; and a host of specific technical challenges, such as different technical standards throughout the world.

15.2 Discuss how to develop effective global information systems architecture.

The first step in the process of developing an effective global information system is to understand the global environment in which the firm is operating. The next step is to consider the best strategy for competing in that environment. The third step is to consider how to structure the organization so it can pursue its selected strategy. Next, management issues in implementing the strategy need to be considered. The last issue to consider is the technology platform.

15.3 Discuss strategies and organizational structures for global businesses and information systems.

There are four basic strategies for developing a global business: domestic exporter, multinational, franchiser, and transnational. The choice of strategy is a function of the type of business and product.

There is a connection between firm strategy and information systems design. Transnational firms must develop networked systems configurations and permit considerable decentralization of development and operations. Franchisers almost always duplicate systems across many countries and use centralized financial controls. Multinationals typically rely on decentralized independence among foreign units with some movement toward development of networks. Domestic exporters typically are centralized in domestic headquarters with some decentralized operations permitted.

15.4 Discuss management challenges in developing global information systems.

Global information systems pose challenges because cultural, political, and language diversity magnifies differences in organizational culture and business processes and encourages proliferation of disparate local information systems that are difficult to integrate. Typically, international systems have evolved without a conscious plan.

15.5 Discuss management solutions for global information systems development challenges.

Not all systems should be coordinated on a transnational basis. Managers should define a small subset of core business processes and focus on building systems to support these processes. Tactically, managers will have to co-opt widely dispersed foreign units to participate in the development and operation of these systems, being careful to maintain overall control.

15.6 Discuss technology issues and alternatives in developing global information systems.

Implementing a global system requires an implementation strategy that considers both business design and technology platforms. The main hardware and telecommunications issues are systems integration and connectivity. The choices for integration are to go with a proprietary architecture or with open systems technology. Global networks are extremely difficult to build and operate. Firms can build their own global networks or they can create global networks based on the Internet (intranets, extranets, or virtual private networks). The main software issues concern building interfaces to existing systems and selecting applications that can work with multiple cultural, language, and organizational frameworks.

Key Terms

Co-optation, 500
Core systems, 498
Domestic exporter strategy, 494
Franchiser strategy, 495
Global culture, 491
International information systems architecture, 493
Legitimacy, 500
Multinational strategy, 495
Particularism, 492
Software localization, 504
Transborder data flow, 492
Transnational strategy, 495

Review Questions

15.1 Discuss factors impacting development of global information systems.

- Describe the general cultural factors leading to growth in global business and the four specific business factors. Also, describe the interconnection among these factors.
- List and describe the major challenges to the development of global information systems.

15.2 Discuss how to develop effective global information systems architecture.

- Explain why some firms have not planned for the development of international systems.
- List and describe the five major steps for developing an international information systems architecture.

15.3 Discuss strategies and organizational structures for global businesses and information systems.

- Describe the four main strategies for global business and organizational structure.
- Describe the four different systems configurations that can be used to support different global strategies.

15.4 Discuss management challenges in developing global information systems.

- List and describe the major management issues in developing international systems.

15.5 Discuss management solutions for global information systems development challenges.

- Identify and describe three principles to follow when organizing the firm for global business.
- Identify and describe three steps of a management strategy for developing and implementing global information systems.
- Define co-optation and explain how it can be used in building global information systems.

15.6 Discuss technology issues and alternatives in developing global information systems.

- Describe the main technical issues involved in the development of global information systems.
- Identify some technologies that can help firms develop global information systems.

Discussion Questions

15.7 If you were a manager in a company that operates in many countries, what criteria would you use to determine whether an application should be developed as a global application or as a local application?

15.8 Describe ways the Internet can be used in global information systems.

Using AI Project

15.9 AI Project: Understanding the Impact of Global Supply Chain Issues

You have been hired as a consultant by two friends who want to open a retail store selling organic pet food and treats as well as pet toys and clothing made from organic materials. They have asked you to advise them on whether they need to be concerned about the possibility of global supply chain issues, given that they primarily sell products made in the United States and if so, what steps they can take to mitigate those issues.

Using Microsoft's Copilot, Google's Gemini, or ChatGPT, develop a prompt to help you provide advice on this topic. Save a copy of your prompt and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience in conducting international market research; analyzing international systems issues for an expanding business; and building a job posting database and web page for an international company.

15.10 Management Decision Problem

United Parcel Service (UPS) has been expanding its package delivery and logistics services in mainland China, serving both multinational companies and local businesses. UPS drivers in China need to use UPS systems and tools, such as its handheld DIAD (Delivery Information Acquisition Device), for capturing package delivery data. UPS wants to make its WorldShip and other shipping-management services accessible to Chinese and multinational customers via the web. What are some of the global information systems issues UPS must consider in order to operate successfully in China?

15.11 Building a Job Database and Web Page for an International Consulting Firm

Software skills: Database and web page design

Business skills: Human resources internal job postings

Companies with many overseas locations need a way to inform employees about available job openings in these locations. In this project, you'll use database software to design a database for posting internal job openings and a web page for displaying this information.

KTP Consulting operates in various locations around the world. KTP specializes in designing, developing, and implementing enterprise systems for medium- to large-sized companies. KTP offers its employees opportunities to travel, live, and work in various locations throughout the United States, Europe, and Asia. The firm's human resources department has a simple database that enables its staff to track job vacancies. When employees are interested in relocating, they contact the human resources department for a list of KTP job vacancies. KTP also posts its employment opportunities on the company website.

What type of data should be included in the KTP job vacancies database? What information should not be included in this database? Based on your answers to these questions, build a job vacancies database for KTP. Populate the database with at least 20 records. You should also build a simple web page that incorporates job vacancy data from your newly created database. Submit a copy of the KTP database and web page to your professor.

15.12 Conducting International Marketing and Pricing Research

Software skills: Online research tools

Business skills: International pricing and marketing

In this project, you'll use online research tools to learn about overseas distributors and customs regulations and to calculate prices in foreign currencies.

You are in charge of marketing for a U.S. manufacturer of furniture that has decided to enter the international market. You want to test the market by contacting office furniture retailers located in the UK to offer a specific desk that you wish to sell at about $195. Using the web, identify an office furniture retailer in the UK and then find out how much you would get for the chair in the current market. Consider using a universal currency converter website, which determines the

value of one currency expressed in other currencies. Obtain both the information needed to contact the firm you selected and the price of your chair in its local currency. Then locate and obtain customs and legal restrictions on the products you will export from the United States and import into the UK. Finally, locate a company that can represent you as a customs agent and gather information on shipping costs.

15.13 Collaboration and Teamwork Project: Identifying Technologies for Global Business Strategies

With two or more of your classmates, identify an area of information technology and explore how this technology might be useful for supporting global business strategies. For instance, you might choose email, smartphones, virtual private networks, enterprise systems, collaboration software, or the web. It will be necessary to identify a business scenario to discuss the technology. You might choose an automobile parts franchise or a clothing franchise, such as Express, as example businesses. Which applications would you make global, which core business processes would you choose, and how would the technology be helpful? If possible, use Google Docs, Google Drive, or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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Glossary

3G network Cellular network that provides transmission speeds from 144 Kbps to 2 Mbps; have been phased out by major telecommunications carriers in the United States.

4G network Cellular network that provides transmission speeds of up to 100 Mbps speeds for downloading and 50 Mbps for uploading.

5G Wireless communication technology supporting transmission of very large amounts of data in the gigabit range, with fewer transmission delays and the ability to connect many more devices (such as sensors and smart devices) at once.

Acceptable use policy (AUP) Defines acceptable, as well as unacceptable, uses of the firm's information resources and computing equipment, and specifies consequences for noncompliance.

Acceptance testing Provides the final certification that the system is ready to be used in a production setting.

Accountability Feature of systems and social institutions; mechanisms for assessing responsibility for decisions made and actions taken.

Adaptive web design (AWD) Techniques that enable a server hosting a website to detect the attributes of the device making the request for service and load a version of the site optimized for the device; offers faster load times and typically a better user experience.

Advanced persistent threat attack (APT attack) A prolonged attack where an intruder infiltrates a system and remains undetected for an extended period of time, exfiltrating data, or waiting for the right time to launch a more crippling attack. Defined by its stragetic targeting and persistence, as well as the advanced tactics, techniques, and procedures it uses.

Advertising revenue model E-commerce revenue model that is based on generating revenue from online advertising.

Affiliate revenue model E-commerce revenue model in which a business steers customers to an "affiliate" which then pays the company a referral fee or percentage of revenue from any resulting sales.

Agency theory Economic theory that views the firm as a nexus of contracts among self-interested individuals who must be supervised and managed.

Agile development Rapid delivery of working software by breaking a large project into a series of small sub-projects that are completed in short periods of time using iteration and continuous feedback.

Agile project management Project management approach that allows for more adaptability and flexibility; uses "sprints" where the project team works to complete a set amount of work within a short period.

AI as a Service (AIaaS) Cloud-based machine learning platform that offers scalable infrastructure and pre-built tools and frameworks for AI model development.

AI bias Refers to the occurrence of biased results due to human biases that skew the original training data or AI algorithm, leading to distorted outputs and potentially harmful outcomes.

Analytical CRM Applications that analyze the customer data generated by operational CRM applications to provide information for improving business performance.

Android Open-source operating system for mobile devices such as smartphones and tablet computers, developed by the Open Handset Alliance led by Google; currently the most popular smartphone operating system worldwide.

Anti-malware software Software designed to detect, and often eliminate, malware from an information system.

App Software program that can be accessed via the Internet.

Application controls Specific controls unique to each application that ensure that only authorized data are completely and accurately processed by that application.

Application server Software that handles all application operations between a browser-based client and a company's back-end business applications or databases. Note that the term application server is also sometimes used to refer to the computer on which the application server software resides.

Artificial intelligence (AI) Effort to develop computer systems that are able to perform tasks normally requiring human intelligence.

Attribute A piece of information describing a particular entity.

Augmented reality (AR) Technology for enhancing visualization by overlaying digital data and images onto a physical real-world environment.

Authentication The ability of know that people are who they claim to be.

Automation Using information technology to assist employees with performing their tasks more efficiently and effectively.

Autonomous car Vehicle that is capable of operating with reduced or no human input. Also known as a self-driving car, driveless car, robotaxi, robotic car, or robo-car.

B2B e-commerce marketplace Provides a single, digital marketplace that allows multiple business buyers and sellers to transact business.

Balanced scorecard method A leading methodology for understanding the really important information needed by an organization's executives; provides a framework for operationalizing a firm's strategic plan by focusing on measurable financial, business process, customer, and learning and growth outcomes of firm performance.

Bandwidth Measures how much data can be transferred over a communications medium within a fixed period of time and typically is expressed as bits per second (Bps), kilobits per second (Kbps), megabits per second (Mbps), gigabits per second (Gbps), and, for very advanced communications media, terabits per second (Tbps).

Behavioral model of management Model of management that argues that the actual behavior of managers is less systematic, more informal, less reflective, more reactive, and less well organized than posited by the classical model of management.

Behavioral targeting Tracking the clickstreams (history of online clicking behavior) of individuals for the purpose of understanding their interests and intentions, and exposing them to advertisements that are uniquely suited to their interests. Also known as interest-based advertising.

Benchmarking Involves comparing efficiency and effectiveness of business processes to strict standards and then measuring organizational performance against those standards.

Best practices The most successful solutions or problem-solving methods for consistently and effectively achieving a business objective that have been developed by a specific organization or industry.

BI dashboard Displays on a single screen graphs and charts of key performance indicators for managing a company.

Big data Data sets typically characterized by the 3Vs (the extreme volume of data, the wide variety of data, and the velocity at which data must be processed) and which are beyond the ability of a typical DBMS to capture, store, and analyze.

Biometric authentication Authentication based on the measurement of physical or behavioral characteristics, such as facial structure, fingerprints, retinal images, and voice, that makes each individual unique.

Bit A binary digit representing the smallest unit of data a computer can handle. It can only have one of two states, representing a value of 0 or 1.

Blockchain Distributed database system that stores permanent and tamper-proof records of transactions as a distributed ledger that is shared among a network of computers.

Blog Popular term for "weblog," designating an informal yet structured website used by individuals and businesses to publish content online.

Bluetooth Standard for wireless personal area networks that can transmit up to 722 Kbps within a 10-meter area.

Bot Software application that performs repetitive pre-defined tasks over a network. Includes shopping bots, web crawlers, scrapers, monitoring bots, and malware.

Botnet A collection of computers infected with bot malware that can then be used for malicious purposes, such as launching distributed denial-of-service (DDoS) attacks, phishing campaigns, or spam, without their owners' knowledge.

Broadband Currently defined by Federal Communications Commission as providing Internet service at a minimum if 100 Mbps for downloads and 4 Mbps for uploads.

Bug Software program code defect.

Bullwhip effect Distortion of information about the demand for a product as it passes from one entity to the next across the supply chain.

Business case Proposal to management seeking approval for an investment by describing its costs, benefits, and business value.

Business continuity planning Focuses on how the company can restore business operations after a disaster strikes.

Business ecosystem Loosely coupled but interdependent networks of suppliers, distributors, outsourcing firms, transportation service firms, and technology manufacturers.

Business functions Specialized tasks performed in a business organization, such as manufacturing and production, sales and marketing, finance and accounting, and human resources.

Business intelligence (BI) Applications and technologies to help users make more informed and thus better business decisions.

Business model Describes how a company produces, delivers, and sells a product or service in order to create value.

Business performance management (BPM) Management methodology that attempts to systematically translate a firm's strategies (e.g., differentiation, low-cost producer, market share growth, and scope of operation) into operational targets.

Business process management (BPM) Tools and methodologies to analyze existing processes, design new processes, and optimize those processes.

Business process redesign (BPR) Type of organizational change in which business processes are analyzed, simplified, and redesigned.

Business processes The unique ways in which organizations coordinate and organize work activities, information, and knowledge to produce a product or service.

Business-to-business e-commerce (B2B e-commerce) Type of e-commerce that involves sales of goods and services among businesses.

Business-to-consumer e-commerce (B2C e-commerce) Type of e-commerce that involves selling products and services directly to individual consumers.

BYOD Stands for "bring your own device" and refers to employees using their own personal computing devices in the workplace.

Byte A group of bits used to represent a single character, which can be a letter, number, or other symbol.

Cable Internet connections Use digital coaxial cable lines to deliver high-speed Internet access to homes and businesses.

Campus area network (CAN) Computer network that spans a limited geographic area, up to about 1,000 meters (1 mile) that may be used for a corporate facility or college campus.

Capital budgeting models Method of measuring the value of investing in long-term information systems projects.

Carpal tunnel syndrome (CTS) Type of RSI in which pressure on the median nerve through the wrist's bony carpal tunnel structure produces pain.

Change agent The individual acting as the catalyst during the change process to ensure successful organizational adaptation to a new system or innovation; in the context of the implementation process, this role is often played by the systems analyst.

Change management Managing the impact of organizational change associated with an innovation, such as a new information system.

Chatbot Software agent designed to simulate a conversation with one or more human users via textual or auditory methods.

Chief AI officer (CAIO) Responsible for overseeing a company's overall strategy, acquisition, implementation, and monitoring of AI technology.

Chief data officer (CDO) Responsible for enterprise-wide governance and utilization of information to maximize the value the organization can realize from its data.

Chief information officer (CIO) Senior manager in charge of the information technology and information systems in the firm.

Chief knowledge officer (CKO) Senior executive in charge of the organization's knowledge management program.

Chief privacy officer (CPO) Responsible for ensuring the company complies with existing data privacy laws and regulations.

Chief security officer (CSO) In charge of information systems security and responsible for enforcing the firm's information security policy.

Choice Simon's third stage of decision making; consists of choosing among various solution alternatives.

Chrome OS Google's lightweight operating system for cloud computing using a connected computer such as Google's Chromebook laptop and tablet computers.

Churn rate Measurement of the number of customers who stop using or purchasing products or services from a company. Used as an indicator of the growth or decline of a firm's customer base.

Citizen developer Business user who builds new applications or modifies existing ones to create applications of their own with IT approval or with minimal help from IT staff.

Classical model of management Traditional description of management that focuses on its formal functions of planning, organizing, coordinating, deciding, and controlling.

Client Various types of devices that can request access to services or resources in a client/server network.

Client/server computing A model of computing in which a client is connected in a network with one or more servers that perform common functions that the clients on the network need.

Cloud computing Model of computing in which computer processing, storage, software, and other services are provided as a shared pool of virtualized resources over a network, primarily the Internet.

Cloud database Database provided by a hosted cloud service provider.

Co-optation Bringing the opposition into the process of designing and implementing a solution without giving up control of the direction and nature of the change.

Collaboration Working with others to achieve shared and explicit goals.

Collaborative robot (cobot) Robot that is designed to function collaboratively alongside or directly interact with humans in a shared workspace.

Community provider Creates a digital online environment where people with similar interests can transact (buy and sell goods); share interests, photos, and videos; communicate with like-minded people; and receive interest-related information.

Competitive forces model Most widely used model for understanding competitive advantage. Views the strategic position of the firm as being determined not only by competition with its traditional direct competitors, but also four other forces in the firm's environment: new market entrants, substitute products and services, customers, and suppliers.

Complementary assets Additional assets required to derive value from a primary investment.

Composable ERP Brings together disparate systems and components in a way that allows them to work together as a platform.

Computer abuse The commission of acts involving a computer that may not be illegal but are considered unethical.

Computer crime The commission of illegal acts through the use of a computer or against a computer system.

Computer hardware Physical equipment used for input, processing, and output activities in an information system.

Computer literacy Knowledge about information technology and understanding how computer-based technologies work.

Computer software Detailed, preprogrammed instructions that control and coordinate the work of computer hardware components in an information system.

Computer vision syndrome (CVS) Eyestrain condition related to display screen use; symptoms include headaches, blurred vision, and dry and irritated eyes.

Computer vision system Uses machine learning and deep learning neural networks to emulate the human visual system to view and extract information from real-world images.

Confirmation bias Involves the tendency to give more credence to evidence that fits with one's existing beliefs, particularly in the face of conflicting information.

Consumer-to-consumer e-commerce (C2C e-commerce) Type of e-commerce that involves consumers selling directly to other consumers.

Consumerization of IT New information technology originating in the consumer market that spreads to business organizations.

Content provider Distributes information content, such as digital video, music, audio files in the form of podcasts, photos, text, and artwork, via the Internet to computers and mobile devices.

Controls All of the methods, policies, and procedures that ensure protection of the organization's assets, accuracy and reliability of its records, and operational adherence to management standards.

Conversion The process of changing from the old system to the new system.

Cookies Small text file placed on a user's computer or mobile device when the user visits a website; used to identify the user and track visits to the website.

Copyright Protects creators of certain types of intellectual property against copying by others for a certain period of time, depending on several factors.

Core competency Activity at which a firm excels as an industry leader.

Core systems Systems that support functions that are absolutely critical to the organization.

Cost transparency Ability of consumers to discover the actual costs merchants pay for products.

Counterimplementation A deliberate strategy to thwart the implementation of an information system or an innovation in an organization.

Credential stuffing attack A brute force attack that attackers launch via botnets and automated tools using known username and password combinations obtained from data breaches.

Cross-selling Marketing of complementary products to customers.

Crowdsourcing Using large online audiences for advice, market feedback, new ideas, and solutions to business problems. Related to the "wisdom of crowds" theory.

Cryptocurrencies Digital assets that use blockchain technology and cryptography to create a medium of exchange (currency).

Culture Fundamental set of assumptions, values, and ways of doing things that has been accepted by most members of an organization.

Customer experience management Management of all the interactions between a customer and a company throughout their entire business relationship.

Customer lifetime value (CLTV) Based on the relationship among the revenue produced by a specific customer, the expenses incurred in acquiring and servicing that customer, and the expected life of the relationship between the customer and the company.

Customer relationship management system (CRM system) Information system that enables a firm to coordinate all the business processes that deal with customers in sales, marketing, and service to optimize revenue, customer satisfaction, and customer retention.

Customization The modification of a commercial software package to meet an organization's unique requirements without destroying the package software's integrity. In e-commerce, changing a delivered product or service based on a user's preferences or prior behavior.

Cyberterrorism Similar in nature to cyberwarfare, but executed by a terrorist group rather than a nation-state.

Cybervandalism Intentional disruption, defacement, or destruction of a website or corporate information system.

Cyberwarfare State-sponsored politically motivated activity designed to cripple and defeat another nation-state by damaging or disrupting its computers, information systems, or networks.

Dark web Portion of the deep web that has been intentionally hidden from search engines in order to preserve anonymity.

Data Streams of raw facts representing events occurring in organizations or the physical environment before they have been organized and arranged into a form that people can understand and use.

Data analytics Involves collecting, storing, inspecting, cleaning, modeling, and querying data.

Data analytics platform Specialized platform optimized for analyzing large data sets.

Data breach Involves the loss of control over an organization's data to outsiders.

Data center Facility that houses computer systems and associated components such as telecommunications, storage, security systems, and backup power supplies.

Data cleansing Processes for detecting and correcting data in a database or file that are incorrect, incomplete, improperly formatted, or redundant. Also known as data scrubbing.

Data definition Specifies the structure of the content of a database.

Data dictionary Database tool that stores definitions of data elements and their characteristics.

Data governance Policies and procedures through which data can be managed as an organizational resource.

Data inconsistency The presence of different values for the same attribute, when the attribute is stored in multiple systems.

Data lake Repository for raw unstructured data or structured data that for the most part have not yet been analyzed.

Data management technology Software governing the organization of data on physical storage media.

Data manipulation language A specialized programming language that is used to add, change, delete, and retrieve data in a database.

Data mart A subset of a data warehouse that contains only a portion of the organization's data for a specified function or population of users.

Data mining Analysis of large pools of data to find patterns and relationships that can be used to guide decision making and predict future behavior.

Data quality audit A structured survey and/or sample of files to determine accuracy and completeness of data in an information system.

Data redundancy The presence of duplicate data in multiple data files.

Data visualization Technology for helping users see patterns and relationships in large amounts of data by presenting the data in graphical form.

Data warehouse A database, with reporting and query tools, that stores current and historical data extracted from various operational systems and consolidated for management reporting and analysis.

Data workers People such as clerks or bookkeepers who assist with administrative work at all levels of a firm.

Database A group of related files.

Database management system (DBMS) Software used to create and maintain a database and enable individual business applications to extract the data they need without having to create separate files or data definitions in the applications.

Database server A computer in a client/server environment that is responsible for running a DBMS to process SQL statements and perform database management tasks.

Decision-support system (DSS) Information system that combines data and sophisticated analytical models or data analysis tools to support decision making.

Decisional role Category of managerial roles identified by Mintzberg in which managers initiate activities, handle disturbances, allocate resources, and negotiate conflicts.

Deep learning neural network Uses multiple layers of neural networks to reveal the underlying patterns in data, and in some limited cases identify patterns without human training.

Deepfake Use of AI to create highly realistic, but digitally fabricated or fake, images or audio.

Demand planning Process by which a firm determines how much product it needs to make to satisfy all its customers' demands.

Denial-of-service attack (DoS attack) Involves flooding a network server or web server with many thousands of false communications or requests for services in order to crash the network.

Design Simon's second stage of decision making; involves identifying and exploring possible alternative solutions to a problem.

DevOps Organizational strategy to create a culture and environment to promote rapid and agile development practices by emphasizing close collaboration between software developers and the IT operational staff. Stands for "development and operations".

Digital certificate A data file used to establish the identity of users and digital assets to protect online transactions.

Digital divide Significant disparity in access to computers and the Internet, as well as many other information resources, among different racial, ethnic, and income-level groups and in rural locations.

Digital firm Organization where nearly all significant business processes and relationships with customers, suppliers, and employees are digitally enabled, and key corporate assets are managed through digital means.

Digital forensics The scientific collection, examination, authentication, preservation, and analysis of electronically stored information (ESI).

Digital goods Goods that can be delivered over a digital network.

Digital Millennium Copyright Act (DMCA) Makes it illegal to make, distribute, or use devices that circumvent technology-based protections of copyrighted materials.

Digital resiliency Deals with maintaining and increasing the resilience of an organization and its business processes in an all-pervasive digital environment, not just the resiliency of the IT function.

Digital subscriber line (DSL) A group of technologies providing high-capacity voice, data, and video transmission over existing telephone lines.

Digital transformation Large-scale change efforts to capture the benefits of digital technology, driving fundamental changes in how businesses operate and how they deliver value to customers.

Direct cutover strategy A risky conversion approach where the new system completely replaces the old one on an appointed day.

Direct goods Goods used in a production process.

Disaster recovery planning Involves creating plans for the restoration of disrupted computing and communications services.

Disinformation Information that has been deliberately falsified.

Disintermediation The removal of organizations or business process layers responsible for certain intermediary steps in a value chain.

Disruptive technologies Technologies that substitute for and significantly reduce the market share of existing products, services, and business models.

Distributed database Database stored in multiple physical locations.

Distributed denial-of-service attack (DDoS attack) Attack that often employs a botnet to inundate and overwhelm a network from numerous launch points.

Documentation Descriptions of how an information system works from both a technical and end-user standpoint.

Domain name Representation of a numeric IP address as a natural language name.

Domain Name System (DNS) A hierarchical system of servers that maintain a database enabling the conversion of domain names to their numeric IP addresses.

Domestic exporter strategy Global firm business strategy characterized by heavy centralization of business activities in the home country of origin.

Download Involves transferring a file from a web server to a user's device, where it resides and can be accessed by the user at any time.

Downtime Periods of time when a system is not operational.

Drive-by download Refers to malware that a user unintentionally downloads or that is downloaded without the user's knowledge.

Drone Type of pilotless aircraft that can be guided remotely or autonomously. Also known as an unmanned aerial vehicle (UAV).

Due process A process in which laws are known and understood and there is an ability to appeal to higher authorities to ensure that laws are applied correctly.

Dynamic pricing Pricing of a product or service varies depending on the demand characteristics of the customer or the supply situation of the seller.

E-commerce Use of the Internet, the web, and mobile apps and browsers running on mobile devices to transact business. More formally, can be defined as digitally enabled commercial transactions between and among organizations and individuals.

E-waste (electronic waste) All types of end-discarded electronic equipment.

Edge artificial intelligence (Edge AI) Refers to the deployment of AI algorithms and AI models directly on local edge devices such as smartphones, sensors or IoT devices, allowing computations to be performed close to where data is actually collected without constant reliance on central cloud infrastructure.

Edge computing Method of optimizing cloud computing systems by performing some data processing on a set of linked servers at the edge of the network, near the source of the data.

Efficient customer response system Information system that directly links consumer behavior back to distribution and production, and supply chains.

Electronic data interchange (EDI) The direct computer-to-computer exchange between two organizations of standard business documents and transactions, such as orders, shipment instructions, or payments.

Email The computer-to-computer exchange of messages.

Encryption The coding and scrambling of messages to prevent their being read or accessed without authorization.

End users Representatives of departments outside the information systems group for whom applications are developed.

End-user interface The part of an information system with which the end user interacts, such as online display and data entry screens and commands.

Enterprise applications Systems that can coordinate activities, decisions, and knowledge across many different functions, levels, and business units in a firm. Include enterprise systems, supply chain management systems, customer relationship management systems, and knowledge management systems.

Enterprise content management system (ECM system) Information system that helps organizations collect and manage structured, semi-structured and unstructured types of information, providing a unified repository of documents, reports, presentations, and best practices.

Enterprise system Integrated enterprise-wide information systems that enable firms to coordinate key business processes in manufacturing and production, finance and accounting, sales and marketing and human resources. Also known as an enterprise resource planning (ERP) system.

Enterprise-wide knowledge management systems General-purpose, firmwide systems that collect, store, distribute, and apply digital content and knowledge.

Entity A person, place, thing, or event about which information is kept.

Entity-relationship diagram A methodology for documenting databases that illustrates the relationship among various entities in the database.

Ergonomics The interaction of people and machines in the work environment, including with respect to the design of jobs, health issues, and the end-user interface of information systems.

Ethical no-free-lunch rule Assumption that virtually all tangible and intangible objects are owned by someone else, unless there is a specific declaration otherwise, and that the creator wants compensation for this work.

Ethics Principles of right and wrong that can be used by individuals acting as free moral agents to make choices to guide their behavior.

EU AI Act World's first broad-based law governing the use of AI.

Evil twin Wireless network that pretends to be legitimate to entice participants to log on and reveal passwords or credit card numbers.

Exchange Independently owned, third-party B2B e-commerce marketplace that is primarily transaction-oriented and that connects many buyers and suppliers for spot purchasing.

Executive support system (ESS) Information system designed to help senior managers make nonroutine decisions that require judgment, evaluation and insight.

Expert system Captures the expertise of individual experts in an organization and represents that knowledge as sets of rules that are then converted into computer code in the form of IF-THEN rules.

Explicit knowledge Knowledge that has been documented.

Extensible Markup Language (XML) A more powerful and flexible markup language than HTML; provides the foundation technology for web services.

External integration tools Tools that links the work of the implementation team to that of users at all organizational levels.

Extranet Private intranet that is accessible to authorized outsiders.

Fair Information Practices (FIP) A set of principles originally set forth in 1973 that governs the collection and use of information about individuals and forms the basis of most U.S. and European privacy laws.

Fault-tolerant computer system System that contains redundant hardware, software, and power supply components that provide back up for a system and keep it running to prevent system failure.

Feasibility study Conducted as part of the systems analysis process to determine whether the solution is achievable, given the organization's resources and constraints.

Feedback Output that is returned to the appropriate members of the organization to help them evaluate or correct input.

Field A grouping of characters into a word, a group of words, or a complete number, such as a person's name or age.

File A group of records of the same type.

Fintech Start-up financial technology firms and services that use IT innovatively to compete with traditional financial services firms.

Firewall Prevents unauthorized users from accessing private networks; can be either a hardware appliance, software-based, or delivered as a service.

Foreign key Field in a database table that enables users to find related information in another database table.

Foundation model Underlying model that enables GenAI to work; uses a powerful neural network that is trained on vast quantities of unstructured, unlabeled data, and learns by finding patterns in data sets.

Franchiser strategy Global firm business strategy characterized by use of a mix of strategies: Product iscreated, designed, financed, and initially produced in the home country, but for product-specific reasons, typically must rely heavily on foreign personnel for further production, marketing, and human resources.

Free/freemium revenue model E-commerce revenue model in which a business offers basic services or content for free, but generates revenue from a premium charged for advanced or special features.

Fuzzy logic Form of AI that imitates human reasoning and cognition by representing imprecision in the form of rules that use approximate or subjective values rather than just 0 or 1.

Gantt chart Chart that visually represents the timing, duration, and resource requirements of project tasks, with each task represented as a horizontal bar whose length is proportional to the time required to complete it.

General controls Overall control environment governing the design, security, and use of computer programs and the security of data files in general throughout the organization's information technology infrastructure.

General Data Protection Regulation (GDPR) Updated framework for protecting personally identifiable information; unifies data privacy laws across the European Union, focusing on making businesses more transparent and expanding the privacy rights of data subjects.

Generative AI (GenAI) Machine learning model that is trained to generate content.

Genetic algorithm A form of machine learning that is used to find the optimal solution for a specific problem based on ideas inspired by evolutionary biology.

Geoadvertising services Involves delivering ads to users based on their location.

Geographic information system (GIS) System with software that can analyze and display data using digitized maps to enhance planning and decision-making.

Geoinformation services Involves providing users with information on local places and things based on the users' location.

Geosocial services Involves social networking based on users' location.

Global culture The development of common expectations, shared artifacts, and social norms among different cultures and peoples.

Globalization The flow of goods, services, capital, people, and ideas across international boundaries.

Golden Rule Ethical principle that involves putting oneself in the place of others and thinking of yourself as the object of a decision.

Green computing Refers to practices and technologies for designing, manufacturing, using, and disposing of computers and other IT equipment in order to minimize impact on the environment. Also referred to as green IT.

Hacker Uses technical skills to identify weaknesses in security protections that computer systems employ in order to gain unauthorised access.

Hacktivist A hacker that is usually motivated to attack specific targets based on ideology.

Hadoop Open-source software framework that enables distributed parallel processing of very large amounts of data across many inexpensive computers.

Hallucination An incorrect or misleading result generated by an LLM that is presented as fact.

Hotspot Location with one or more access points that provide public Wi-Fi Internet access.

HTML5 Newest evolution of HTML, which makes it possible to embed images, audio, video, and other elements directly into a document without add-on software.

HTTPS Secure version of the HTTP protocol that uses TLS for encryption and authentication.

Hub Simple device that connects network components, sending a packet of data to all other connected devices.

Human-AI collaboration Refers to interactions between humans and artificial intelligence systems that combine their respective complementary strengths to achieve shared goals.

Humanoid robot Robot that resembles and acts like a human.

Hybrid app Mobile application that has many features of both a native app and mobile web app.

Hybrid cloud Computing model where firms use both their own IT infrastructure for their most essential core activities and public cloud computing for less critical systems or for additional processing capacity.

Hypertext Markup Language (HTML) Page description language for specifying how text, graphics, video and audio are placed on a web page and for creating dynamic links to other web pages and objects.

Hypertext Transfer Protocol (HTTP) The communications standard used to transfer pages on the web. Defines how messages are formatted and transmitted.

Identity and access management software (IAM software) Software that automates the process of keeping track of users and system privileges, assigning each user a unique digital identity for accessing each system.

Identity theft Theft of key pieces of personal information, such as social security number, driver's license numbers, or credit card numbers, in order to impersonate someone else, and obtain merchandise and services in the name of the victim or to obtain false credentials.

Immanuel Kant's categorical imperative Ethical principle that states that if an action is not right for everyone to take it is not right for anyone.

Implementation All the organizational activities surrounding the adoption, management, and routinization of an innovation, such as a new information system.

In-memory computing Technology for very rapid analysis and processing of large quantities of data by storing the data in the computer's internal memory (RAM) rather than in secondary storage.

Indirect goods Goods not directly used in the production process, such as office supplies.

Industrial robot Heavy-duty machine designed to automate portions of the manufacturing process.

Industry cloud platform Cloud computing model in which specialized cloud services are provided to meet the specific requirements of a particular industry such as healthcare, finance, or manufacturing. Also known as a vertical cloud platform.

Inference engine The strategy used to search through the rule base in an expert system.

Information Data that have been shaped into a form that is meaningful and useful to humans.

Information asymmetry Exists when one party in a transaction has more information that is essential to the transaction than the other party; impacts the relative bargaining power of the participants.

Information density The total amount and quality of information available to all market participants.

Information requirements A detailed statement of the information needs that a new system must satisfy; identifies who needs what information, and when, where, and how the information is needed.

Information rights The rights that individuals and organizations have with respect to information that pertains to them.

Information system (IS) Interrelated components working together to collect, process, store, and distribute information to support decision making, coordination, control, analysis, and visualization in an organization.

Information systems audit Examines the firm's overall security environment as well as the controls that govern individual information systems.

Information systems department The formal organizational unit that is responsible for information technology services within the organization.

Information systems literacy Broad-based understanding of information systems that includes behavioral knowledge about organizations and individuals using information systems as well as technical knowledge about computers.

Information systems managers Leaders of the teams of various specialists in the information systems department.

Information systems plan Road map indicating the direction of systems development (the purpose of the plan), the rationale, the current situation, the management strategy, the implementation plan, and the budget.

Information technology (IT) All the hardware and software technologies a firm needs to achieve its business objectives.

Information technology infrastructure (IT infrastructure) Computer hardware, software, data, storage technology, and networks providing a portfolio of shared IT resources for the organization.

Informational role Category of managerial roles identified by Mintzberg in which managers act as the nerve centers of their organizations, receiving and disseminating critical information.

Informed consent Consent given with knowledge of all the facts needed to make a rational decision.

Infrastructure as a service (IaaS) Computing infrastructure provisioned from a cloud service provider, which maintains and operates it; users pay only for the amount of computing and storage resources they actually use.

Input The capture or collection of raw data from within the organization or from its external environment for processing in an information system.

Instant messaging Chat service that allows participants to create their own private chat channels so that a person can be alerted whenever someone on his or her private list is online to initiate a chat session with that particular individual.

Intangible benefits Benefits that cannot be immediately quantified, such as more efficient customer service or enhanced decision making, but that can lead to quantifiable gains in the long run.

Intellectual property Products of the mind created by individuals or corporations and which may be protected by copyright, patent, trademark and trade secret law.

Intelligence The first of Simon's four stages of decision making; involves discovering, identifying, and understanding problems occurring in the organization.

Intelligent agent Software program that uses a built-in or learned knowledge base to carry out specific tasks for an individual user, business process, or software application.

Intelligent techniques Technologies such as data mining and AI tools that aid human decision makers by capturing individual and collective knowledge, discovering patterns and behaviors in large quantities of data, and generating solutions to problems that are too large and complex for humans to solve on their own.

Interactivity Allowing for two-way communication between merchant and consumer and peer-to-peer communication among individuals.

International information systems architecture The basic information systems required by organizations to coordinate worldwide trade and other activities.

Internet Global network of networks using universal standards to connect millions of different networks.

Internet of Things (IoT) Network of physical objects (things) embedded with sensors, software, and other technologies enabling them to connect and exchange data with other devices and systems via the Internet.

Internet Protocol address (IP address) Numeric address indicating a unique computer location on the Internet.

Internet service providers (ISP) Commercial organizations with a permanent connection to the Internet that sell temporary connections to subscribers.

Internet2 Advanced networking consortium focused on the development of next-generation Internet technologies.

Interorganizational system Information system that automates the flow of information across organizational boundaries and links a company to its customers, distributors, or suppliers.

Interpersonal role Category of managerial roles identified by Mintzberg in which managers act as figureheads and leaders for the organization.

Intranet An internal network based on Internet technology and standards.

Intrusion detection system (IDS) Tools to monitor the most vulnerable points in a network to detect and deter unauthorized intruders.

Intrusion prevention system (IPS) Has all the functionalities of an IDS, with the additional ability to take steps to prevent and block suspicious activities.

iOS Apple's operating system for the iPhone, iPad, and Apple Watch.

IPv6 IP addressing system using 128-bit IP addresses. Stands for Internet Protocol version 6.

IT financial management tools (ITFM tools) Tools that automate TBM and help organizations analyze, plan, and control costs associated with IT projects.

IT governance Strategy and policies for using information technology within an organization, specifying the decision rights and framework for accountability to ensure that the use of information technology supports the organization's strategies and objectives.

Iterative approach A process of repeating the steps to build a system over and over, with each version more accurately reflecting user requirements.

Java Operating system-independent, processor-independent, object-oriented programming language that has become a leading interactive programming environment for the web.

JavaScript Core technology for making web pages more dynamic and interactive.

Joint application development (JAD) Methodology to ensure that the client, consumer, IT specialists, and stakeholders are all involved in designing and developing a technology application.

Just-in-time strategy Strategy that aims to have components arrive exactly at the moment they are needed and finished goods shipped as soon as they are manufactured in order to minimize inventory.

Kanban board Visual tool (either digital or physical) that uses "cards," such as sticky notes or their digital equivalent, to represent specific tasks, and columns to represent each stage of the project process.

Key field A field in a record that uniquely identifies instances of that record so that it can be retrieved, updated, or sorted.

Key performance indicators (KPIs) Measures proposed by senior management for understanding how well the firm is performing along specified dimensions.

Keylogger A form of spyware that records every keystroke made on a computer to steal personal information or passwords or to launch Internet attacks.

Knowledge base Model of human knowledge that is used by expert systems.

Knowledge management The set of processes developed in an organization to create, gather, store, maintain, and disseminate the firm's knowledge.

Knowledge management system (KMS) Information system that supports the creation, capture, storage, and dissemination of firm expertise and knowledge.

Knowledge work system (KWS) Information system that aids knowledge workers in the discovery, creation and integration of new knowledge into the organization.

Knowledge workers People such as engineers, scientists, or architects who design products or services and create knowledge for the organization.

Large language model (LLM) Foundation model trained on immense amounts of unstructured data covering many topics; capable of understanding and generating natural language and other types of content.

Learning management system (LMS) Tools for the management, delivery, tracking, and assessment of various types of employee learning.

Legacy system Generally, an older system that continues to be used to avoid the high cost of replacing or redesigning it.

Legitimacy The extent to which one's authority is accepted on grounds of competence, vision, or other qualities.

Liability Body of laws that permit individuals to recover the damages done to them by other actors, systems, or organizations.

Linux Open-source operating system that is an offshoot of UNIX; is scalable, reliable, can run on many different hardware platforms and is available free or at very low cost.

Living off the Land attack (LOTL attack) Hijacks legitimate software tools, such as network administration tools that are already part of an organization's digital infrastructure, to carry out the attack.

Local area network (LAN) Designed to connect computers and digital devices within a limited area (half-mile or 500-meter radius) and to link to long-distance wide area networks, such as the Internet.

Location analytics Ability to gain insights from the location (geographic) component of data, including location data from mobile devices, output from sensors or scanning devices, and data from maps.

Location-based services Involves providing services to users based on the users' location; enabled by GPS, Wi-Fi, and cellular technologies.

Long tail marketing Refers to the ability of firms to find potential customers inexpensively for products for which demand is low, largely because the cost of reaching very small market segments (people who fall into the long tail ends of a Bell curve) is not high.

Low-code development Software development approach that enables the delivery of applications faster and with minimal hand coding.

Machine learning (ML) Uses mathematical models that help a computer to learn from large quantities of data and make statistical inferences without much explicit instruction, enabling a computer system to continue learning and improving on its own, to varying degrees, based on experience.

Mainframe High-performance, large-capacity computer that can process high volumes of data very quickly.

Maintenance Changes in hardware, software, documentation, or procedures to a production system to correct errors, meet new requirements, or improve processing efficiency.

Malvertising Method of malicious code distribution in which malware is embedded in the online advertising chain.

Malware Malicious software programs such as viruses, worms, Trojan horses, ransomware, bots, and potentially unwanted programs.

Managed security service provider (MSSP) Company that provides security management services for subscribing clients.

Management information systems (MIS) The study of information systems, focusing on their use in business and management. Also refers to specific category of information system serving middle management.

Managerial roles Expectations of the activities that managers should perform in an organization.

Market creator Provides a digital environment in which buyers and sellers can meet, display products, search for products, establish prices, and engage in transactions.

Market entry costs The cost merchants must pay to bring their goods to market.

Marketspace A marketplace extended beyond traditional boundaries and removed from a temporal and geographic location.

Mass customization The capacity to offer individually tailored products or services on a large scale.

Medical robot Robotic machine utilized in health sciences.

Menu costs Merchants' costs of changing prices.

Metaverse Anticipated future iteration of the web featuring online 3D or virtually integrated environments that provide users with access to virtual reality and augmented reality experiences.

Metropolitan area network (MAN) Network that spans a metropolitan area, usually a city and its major suburbs. Its geographic scope falls between a WAN and a LAN.

Microsoft Windows operating system Microsoft's operating systems for Windows desktop computers and tablets.

Microsoft Windows Server Leading operating system for corporate servers in a Windows-based IT infrastructure.

Middle management People in the middle of an organization's hierarchy and who are responsible for carrying out the plans and goals of senior management.

Misinformation False information that is spread by people who think the information is true.

Mobile device management (MDM) Software that monitors, manages, and secures mobile devices that are deployed across multiple mobile service providers and multiple mobile operating systems used in the organization.

Mobile e-commerce (m-commerce) Type of e-commerce that involves the sale of goods and services via mobile devices, such as a smartphone, tablet computer, or wearable.

Mobile web app An Internet-enabled application with specific functionality for mobile devices and which is designed to be accessed via the mobile device's web browser.

Mobile website Version of a regular website that is scaled down in content and navigation for easy access and search on a small mobile screen.

Modem A device for translating a computer's digital signals into analog form for transmission over analog networks or for translating analog signals back into digital form for reception by a computer.

Multifactor authentication (MFA) Increases security by validating users via a multistep process.

Multinational strategy Global firm business strategy that concentrates financial management and control out a central home base while decentralizing production, sales, and marketing operations to units in other countries.

Multitiered client/server architecture Client/server network in which the work of the entire network is balanced over multiple levels of servers, depending on the kind of service being requested. Often referred to as N-tier client/server architecture.

Multitouch Interface that enables the use of one or more finger gestures to manipulate lists or objects on a screen without using a mouse or keyboard.

Native advertising Advertising that looks similar to editorial content but typically is inconspicuously tagged as "sponsored"; often placed in social network feeds or within online magazines.

Native app Standalone application designed to operate using a specific mobile operating system and device.

Natural language processing (NLP) AI technique for enabling a computer to understand and analyze natural language as opposed to language formatted to be understood by computers.

Near field communication (NFC) Short-range wireless connectivity standard that uses electromagnetic radio fields to enable two compatible devices to exchange data when brought within a few centimeters of each other.

Network The linking of two or more computers to share data or resources, such as a printer.

Network economics Refers to market situations where the economic value being produced depends in large part on the number of people using a product, and where the marginal costs of adding another participant are about zero but can create much larger marginal gains.

Network operating system (NOS) Software that routes and manages communications on the network and coordinates network resources.

Network operating system software (NOS software) Software that connects multiple devices and computers on a network and allows them to share resources on the network.

Networking and telecommunications technology Physical devices and software that link various computer hardware components and transfer data from one physical location to another.

Neural network Algorithms loosely based on the processing patterns of the biological human brain that can be trained to classify objects into known categories based on data inputs; are able to find patterns and relationships in very large amounts of data that would be too complex and difficult for a human to analyze.

No-code development Involves using tools that enable the creation of software without having to write any program code.

Non-relational database management system Database that uses a more flexible data model and is designed for managing large data sets across many distributed computers. Also referred to as NoSQL.

Nonobvious relationship awareness (NORA) Data analysis technology that can find obscure hidden connections among people and entities by analyzing information from many different sources to correlate relationships.

Normalization When designing a relational database, the process of streamlining complex groups of data to minimize redundant data elements and awkward many-to-many relationships.

Object Software building block that combines data and the procedures acting on the data.

Object-oriented design (OOD) Uses an object as the basic unit of systems analysis and design.

Offshore outsourcing Outsourcing systems development work or maintenance of existing systems to external vendors outside of the United States.

On-demand computing The purchase of computing services from remote providers and paying only for computing resources actually used.

Online analytical processing (OLAP) Supports multidimensional data analysis by providing capability for manipulating and analyzing large volumes of data from multiple perspectives.

Online retailer Sells goods online. Sometimes called an e-tailer.

Online social network Enables users to commuincate with one another, form group and individual relationships, and share interests, values, and ideas, via a website and/or app.

Open-source software Software that provides free access to its program code, allowing users to modify the program code to make improvements or fix errors.

Operating system System software that manages and controls the activities of the computer.

Operational CRM Customer-facing applications, such as tools for sales force automation, call center and customer service support, and marketing automation.

Operational intelligence Form of business intelligence and business analytics that delivers insights about activities inside and outside an organization from streams of big data as they are generated in real time.

Operational management People who are responsible for monitoring the day-to-day activities of an organization.

Opt-in Model of informed consent permitting prohibiting an organization from collecting any personal information unless the individual specifically takes action to approve information collection and use.

Opt-out Model of informed consent permitting the collection of personal information until the consumer specifically requests that the data not be collected.

Organization A stable, formal social structure that takes resources from the environment and processes them to produce outputs.

Organizational impact analysis Study of the way a proposed system will affect organizational structure, attitudes, decision making, and operations.

Output The distribution of processed information to the people who will use it or to the activities for which it will be used.

Packet switching Method of slicing digital messages into small, fixed bundles of data called packets, routing them along different communications paths as the paths become available, and then reassembling the packets once they arrive at their destination.

Paradigm shift Radical form of organizational change that involves the nature of the business and the nature of the organization.

Parallel strategy A safe and conservative conversion approach where both the old system and its potential replacement are run together for a time until everyone is assured that the new one functions correctly.

Particularism Making judgments and taking action on the basis of narrow or personal characteristics, in all its forms (religious, nationalistic, ethnic, regionalism, geopolitical position).

Password Secret word or string of characters that is supposed to be known only to the authorized user, and which is used to authenticate the user so the user can access a resource such as a computer system or specific files.

Patch Software to repair software flaws without disturbing the proper operation of the software.

Patent Grants the owner an exclusive monopoly on the ideas behind an invention for a certain period of time, typically 20 years; designed to ensure that inventors of new machines or methods are rewarded for their labor while enabling the widespread use of their inventions.

Peer-to-peer Network architecture that treats all processors on the network equally; the various computers on the network exchange data by direct access and can share peripheral devices. Used primarily in small networks.

Personal area network (PAN) Computer network used for communication among digital devices that are located within close proximity to one another.

Personalization Ability of merchants to target marketing messages to specific individuals by adjusting the message to a person's clickstream behavior, name, interests, and past purchases.

PERT chart Chart that graphically depicts project tasks and their interrelationships, showing the specific activities that must be completed before others can start.

Pharming Phishing technique that redirects users to a bogus web page, even when an individual enters the correct web page address.

Phased approach strategy A conversion strategy in which the new system is introduced in stages, either by functions or by organizational units.

Phishing Form of spoofing that involves setting up a fake website or sending email messages that look like those of legitimate businesses to ask users for confidential personal data.

Pilot study strategy A conversion strategy in which the new system is introduced to a limited area of the organization until it is proven to be fully functional; only then does the conversion to the new system take place across the entire organization.

Pivot table Spreadsheet tool that enables reorganizing and summarizing two or more dimensions of data into a tabular format, and which can be employed to identify and understand patterns in data that may be useful for semistructured decision making.

Platform Information systems, technologies, and services provided by a keystone firm in a business ecosystem that thousands of other firms in different industries use to enhance their own capabilities.

Platform as a service (PaaS) Platform that provides infrastructure and programming tools that enable customers to develop, run, and manage applications without having to build and maintain the infrastructure typically associated with developing an application.

Portal Web interface for presenting integrated personalized business content from a variety of sources. Also refers to a website that acts as initial point of entry to the web with an integrated package of content and services

Portfolio analysis Tool that can be used to evaluate systems projects; inventories all of an organization's information systems projects and assets, and analyzes them to determine their respective risks and benefits.

Post-implementation audit Formal review process conducted after a system has been placed in production to determine how well the system has met its original objectives.

Potentially unwanted program (PUP) Software such as adware, browser parasites, spyware, and other applications such as rogue security software, toolbars, and PC diagnostic tools that install themselves onto a user's computer; can be considered a form of malicious software.

Predictive analytics The use of statistical analysis, data mining techniques, historical data, and assumptions about future conditions to predict future trends and behavior patterns.

Predictive search Part of a search algorithm that predicts what a user query is looking for as it is entered.

Prescriptive analytics Form of analytics that not only predicts future outcomes, but also recommends actions or decisions to achieve desired outcomes or prevent undesirable ones.

Price discrimination Selling the same goods, or nearly the same goods, to different targeted groups at different prices.

Price transparency Ease with which consumers can find out the variety of prices in a market.

Primary activities Activities most directly related to the production and distribution of a firm's products or services.

Primary key Unique identifier for all the information in any row of a database table.

Privacy Claim of individuals to be left alone, free from surveillance or interference from other individuals or organizations, including the state; component of information rights.

Private B2B network Internet-enabled network typically employed by a large firm to link to its suppliers and other key business partners.

Private cloud Cloud computing model in which cloud infrastructure is operated solely for the benefit of a single organization.

Process mining Analyzes data from a company's event logs to identify problems with processes and figure out ways to streamline them.

Processing The conversion, manipulation, and analysis of raw input into a form that is more meaningful to humans.

Product differentiation Competitive strategy for creating brand loyalty by developing new and unique products and services that are not easily duplicated by competitors.

Production The stage after the new system is installed and the conversion is complete; during this time the system is reviewed by users and technical specialists to determine how well it has met its original goals and whether any revisions or modifications are required.

Production and service workers People who actually produce the products or services of the organization.

Profiling The use of computers to combine data from multiple sources and create electronic dossiers of detailed information on individuals.

Program-data dependence Refers to the coupling of data stored in files and the specific programs required to update and maintain those files such that changes in the program require changes to the data.

Programmers Highly trained technical specialists who write computer software instructions.

Programming The process of translating the system specifications prepared during the design stage into software program code.

Project Planned series of related activities for achieving a specific business objective.

Project management Application of knowledge, skills, tools, and techniques to achieve specific targets within specified budget and time constraints.

Project portfolio management software (PPM software) Helps organizations manage portfolios of projects and the dependencies among them.

Protocol A set of rules and procedures that govern transmission between the components in a network.

Prototype The preliminary working version of an information system for demonstration and evaluation purposes.

Prototyping The process of building an experimental system quickly and inexpensively for demonstration and evaluation so that users can better determine information requirements.

Public cloud Cloud computing model in which the cloud infrastructure is owned and maintained by a cloud service provider and made available to organizations and the general public.

Public key encryption Form of encryption that uses two keys: one shared (or public) and one private.

Public key infrastructure (PKI) System that uses public key encryption, certificate authorities (CA,) and digital certificates for authentication.

Pull-based model Supply chain driven by actual customer orders or purchases. Also known as demand-driven or build-to-order model.

Push-based model Supply chain driven by forecasts or best guesses of demand for products, rather than what customers order; products are "pushed" to customers.

Quantum computing Use of principles of quantum mechanics to represent data and perform operations on the data.

Query Request for data from a database.

Radio frequency identification (RFID) Technology using tiny tags with embedded microchips containing data about an item and its location; the tags transmit short-distance radio signals to special RFID readers that then pass the data on to a computer for processing.

Ransomware Malware that tried to extort money from users by taking control of their computers and blocking access to files.

Rapid application development (RAD) Process for developing workable systems in a very short time period with some flexibility to adapt as the project evolves; includes use of visual programming and other tools for building graphical user interfaces, iterative prototyping, automation of program code generation, and close teamwork among end users and information systems specialists.

Rationalization of procedures The streamlining of standard operating procedures.

Record A group of related fields.

Referential integrity Rules to ensure that relationships between coupled database tables remain consistent.

Relational database A type of logical database model that that organizes data into two-dimensional tables with columns and rows. It can relate data stored in one table to data in another as long as the two tables share a common data element.

Relational DBMS (RDBMS) DBMS system designed to work with relational databases.

Repetitive stress injury (RSI) Injury that occurs when muscle groups are forced through repetitive actions with high-impact loads or thousands of repetitions with low-impact loads.

Request for proposal (RFP) A detailed list of questions submitted to vendors of software or other services to determine how well the vendor's product can meet the organization's specific requirements.

Responsibility Accepting the potential costs, duties, and obligations for the decisions one makes.

Responsible AI Set of practices to ensure that AI is designed, deployed, and used in an ethical and legal way.

Responsive web design (RWD) Ability of a website to automatically change screen resolution and image size as a user switches to devices of different sizes, such as a laptop, tablet computer, or smartphone. Eliminates the need for separate design and development work for each new device.

Revenue model Describes how a firm will earn revenue, generate profits, and produce a return on investment.

Richness Refers to the complexity and content of a message.

Risk assessment Determines the level of risk to a firm if a specific activity or process is not properly controlled. Assesses the probability of occurrence and the potential damage if the problem were to occur. Used to determine the cost/benefit of a control.

Risk aversion principle Ethical principle that one should take the action that produces the least harm or the least potential cost.

Robotics Involves the design, construction, operation, and use of movable machines that can substitute for humans, as well as computer systems for their control, sensory feedback, and information processing.

Router Communications processor that forwards packets of data from one network to another network.

Routines Precise rules, procedures, and practices that have been developed to cope with expected situations. Sometimes called standard operating procedures.

RSS Technology using aggregator software to pull content from websites and feed it automatically to subscribers' computers.

Safe harbor Private self-regulating policy and enforcement mechanism that meets the objectives of government regulations but does not involve government regulation or enforcement.

Sales force automation modules (SFA module) CRM systems module that helps staff increase productivity by focusing sales efforts on the most profitable customers.

Sales revenue model E-commerce revenue model that is based on generating revenue from the sale of goods, content, and/or services to customers.

Scalability The ability of a computer, product, or system to expand to serve a larger number of users without breaking down.

Scope Defines what work is and is not included in a project.

Scoring model A method for deciding among alternative systems solutions when many criteria must be considered; assigns weights to various features of a system and then calculates weighted totals.

Search costs The time and money spent locating a suitable product and determining the best place to purchase that product.

Search engine A tool for locating specific sites or information on the Internet.

Search engine marketing Use of search engines to deliver sponsored links for which advertisers have paid in search engine results.

Search engine optimization (SEO) Process of improving the quality and volume of web traffic to a website by employing a series of techniques that help a website achieve a higher ranking with the major search engines.

Security Policies, procedures, and technical measures used to prevent unauthorized access, alteration, theft, or physical damage to information systems.

Security operations center (SOC) Team of IT security professionals dedicated to monitoring an organization's entire IT infrastructure on a round-the-clock basis in order to detect, analzye and respond to security incidents in real time.

Security policy Consists of statements ranking information risks, identifying acceptable security goals, and identifying the mechanisms for achieving these goals.

Security token Physical device, similar to an identification card, that is designed to prove the identity of a single user.

Semantic search Use of natural language processing to understand the content and meanings of words, phrases, and sentences used in search queries.

Semistructured decision Decision in which only part of the problem has a clear-cut answer provided by an accepted procedure.

Senior management People occupying the topmost position in an organization's hierarchy and who are responsible for making long-range decisions.

Sensitivity analysis Form of modeling that asks "what-if" questions repeatedly to determine the impact of changes in one or more factors on the outcomes.

Sentiment analysis Mining comments in an email or text message, blog, social media conversation, or survey form to detect favorable and unfavorable opinions about specific subjects.

Server A computer that is dedicated to performing common functions that clients in a client/server network need, such as file storage, software applications, printing and Internet access.

Service level agreement (SLA) Formal contract between customers and their service providers that defines the specific responsibilities of the service provider and the level of service expected by the customer.

Service provider Offers services online.

Service robot Robot that is designed to perform service-related tasks.

Six Sigma A specific measure of quality, representing 3.4 defects per million opportunities; used to designate a set of methodologies and techniques for improving quality and reducing costs.

Slippery slope rule Ethical principle that if an action cannot be taken repeatedly, it is not right to take at all.

Smart card A plastic or metal about the size of a credit card that contains an embedded chip formatted with access permissions and other data. Used in conjunction with a card reader. Also used in digital payment systems.

Smartphone Mobile device that combines the functionality of a cell phone with that of a Wi-Fi-enabled laptop computer, providing voice, messaging, scheduling, email, and Internet capabilities.

Sniffer Type of eavesdropping program that monitors information traveling over a network.

Social business Use of social network platforms, such as Facebook, X (formerly Twitter), as well as internal corporate social tools, to engage employees, customers, and suppliers.

Social e-commerce Type of e-commerce that involves the use of social networks to share knowledge about items of interest with other shoppers and enable purchases directly via the social network.

Social engineering Exploiting human curiosity, greed, gullibility, fear, or complacency to trick people into taking an action that enables a threat actor to gain access to a computer system or download malware.

Social graph Map of all significant online social relationships, comparable to a social network describing offline relationships.

Social listening Ability of marketers to understand what people are saying online about a brand as well as its competitors.

Social network Online community that enables users to connect with and expand their social or business contacts.

Sociotechnical design Design which aims to produce information systems that blend technical efficiency with sensitivity to organizational and human needs.

Sociotechnical view Viewing systems as being composed of both technical and social elements.

Software as a service (SaaS) Software hosted by a vendor on the vendor's cloud infrastructure and delivered as a service on a subscription fee basis to users who access the software via a web browser; data and software are maintained on the vendors' remote servers.

Software localization Process of converting software to operate in a second language.

Software outsourcing Contracting custom software development or maintenance to an outside firm.

Software package A prewritten, precoded, commercially available set of programs that eliminates the need to write software programs for certain functions.

Software supply chain attack Involves a threat actor infiltrating the software supply chain at the development and production stage and compromising the software before it is acquired and deployed by the customer.

Software-defined networking (SDN) Networking approach that involves using a central control program separate from network devices to manage the flow of data on a network.

Software-defined storage (SDS) Storage architecture that separates the software for managing data storage from the physical storage hardware.

Spam Unsolicited commercial email.

Spoofing The practice of tricking or deceiving others by hiding one's true identity.

SQL injection attack Attack that exploits vulnerabilities in poorly coded SQL web applications in order to introduce malicious program code into a company's systems and networks.

Streaming A method of distributing audio and video files as a continuous stream of content to a user's device without it having to be stored locally on the device.

Structured decision Decision that is repetitive and routine and has a definite procedure for how it should be handled.

Structured knowledge Knowledge in the form of structured documents and reports.

Structured Query Language (SQL) The standard data manipulation language for relational database management systems.

Subscription revenue model E-commerce revenue model that is based on generating revenue from subscription fees charged for access to some or all of a company's content or services.

Supervised learning Machine learning algorithm trained by providing specific examples of desired inputs and outputs classified by humans in advance.

Supply chain Network of organizations and business processes for procuring materials, transforming raw materials into intermediate and finished products, and distributing the finished products to customers.

Supply chain execution system System that manages the flow of products through distribution centers and warehouses to ensure that products are delivered to the right locations in the most efficient manner.

Supply chain management system (SCM system) Information system that automates the flow of information between a firm and its suppliers in order to optimize the planning, sourcing, manufacturing, and delivery of products and services.

Supply chain planning system System that enables a firm to model its existing supply chain, generate demand forecasts for a product, and develop optimal sourcing and manufacturing plans.

Support activities Activities that make the delivery of a firm's primary activities possible. Consist of actitivies related to the firm's organizational infrastructure, human resources, technology, and procurement.

Switch Device used to connect network components that has more intelligence than a hub and can filter and forward data to a specified destination.

Switching costs The expense incurred in lost time and expenditure of resources when changing from one supplier or system to a competing supplier or system.

System testing Tests the functioning of the information system as a whole in order to determine if discrete modules will function together as planned.

Systems analysis The analysis of a problem that the organization will try to solve with an information system.

Systems analysts Specialists to translate business problems and requirements into information requirements and systems, acting as liaison between the information systems department and the rest of the organization.

Systems design Details how a system will meet the information requirements identified by the systems analysis.

Systems development The activities that go into producing an information systems solution to an organizational problem or opportunity.

Systems development life cycle (SDLC) A traditional method for building information systems; involves using a phased approach to building a system, dividing systems development into a series of sequential formal stages: systems analysis, systems design, programming, testing, conversion, and production and maintenance.

T1 line International telephone standard for digital communication that offers guaranteed delivery at 1.54 Mbps.

T3 line International telephone standard for digital communication that offers guaranteed delivery at 145 Mbps.

Tacit knowledge Expertise and experience of organizational members that has not been formally documented.

Tangible benefits Benefits that can be quantified and assigned a monetary value.

Taxonomy Method of classifying things according to a predetermined system.

Team Formal group whose members collaborate to achieve specific goals.

Technical debt Results from the trade-offs between time, cost, and quality that often occur in the development of an information system. Sometimes also called tech debt or code debt.

Technology business management (TBM) Framework aimed at improving business outcomes by giving organizations a consistent method of translating technology investments into business value.

Telepresence Technology that allows people to give the appearance of being present at a location other than their actual physical location.

Test plan Prepared by the development team in conjunction with the users; it includes all of the preparations for the series of tests to be performed on the system.

Testing Process that determines whether the system produces the desired results under known conditions.

Text mining Discovery of patterns and relationships from large sets of unstructured data.

Threat actor individual or group of individuals seeking to exploit system vulnerabilities to intentionally cause harm.

Total cost of ownership (TCO) Designates the actual total cost of owning technology resources, including the initial cost of acquiring and installing hardware and software, as well as ongoing adminisration costs for hardware and software upgrades, maintenance, technical support, training and even utility and real estate costs for running and housing the technology.

Total quality management (TQM) A concept that makes achieving quality an end in itself and a responsibility shared by all people and functions within an organization.

Touch point Method of interaction with a customer, such as telephone, email, customer service desk, conventional mail, social network, website, mobile device, or retail store. Also known as a contact point.

Trade secret Any intellectual work product used for a business purpose, provided it is not based on information in the public domain.

Trademarks Marks, symbols, or images used to distinguish products in the marketplace.

Transaction broker Provides online processing for transactions that previously were conducted in person, by phone, or by mail.

Transaction cost theory Economic theory that firms and individuals seek to economize on transaction costs. Traditionally this was done vertical integration, with firms becoming larger, but today information technology enables firms to do so while decreasing the number of employees.

Transaction costs The costs of participating in a market.

Transaction fee revenue model E-commerce revenue model based on generating revenue from fees for enabling or executing transactions.

Transaction processing system (TPS) Information system that performs and records the daily routine transactions necessary to conduct business.

Transborder data flow The movement of information across international boundaries in any form.

Transformer Key component of foundation models; type of neural network architecture that transforms or changes an input sequence into an output sequence by learning context and tracking relationships between sequence components.

Transmission Control Protocol/Internet Protocol (TCP/IP) Dominant model for achieving connectivity among different networks. Provides a universally agreed-on method for breaking up digital messages into packets, routing them to the proper addresses, and then reassembling them into coherent messages.

Transnational strategy Global firm business strategy in which nearly all value-adding activities are managed from a global perspective without reference to national borders; the globe, not the home country, is viewed as the management frame of reference.

Transport Layer Security (TLS) Enables client and server computers to manage encryption and decryption activities as they communicate with each other during a secure web session; successor to Secure Sockets Layer protocol.

Trojan horse Software program that appears to legitimate but contains hidden functionality that may cause damage.

Two-factor authentication (2FA) Subset of MFA that validates user identity with two means of identification, one of which is typically a physical item that the user possesses, and the other of which is typically data.

Unified communications (UC) Integrates disparate channels for voice communications, data communications, instant messaging, email, and teleconferencing into a single experience where users can seamlessly switch back and forth between different communication modes.

Unified threat management systems (UTM system) Comprehensive security management tool that combines multiple security tools, including firewalls, virtual private networks, intrusion detection systems, and web content filtering and anti-spam software.

Uniform resource locator (URL) The address of a specific resource on the Internet.

Unit testing The process of testing each program separately in the system. Sometimes called program testing.

UNIX Older operating system for servers, mobile devices, and embedded systems that has been largely supplanted by Linux.

Unstructured decision Decision that is novel, important, and nonroutine and for which there is no agreed-upon procedure for making the decision; the decision maker must provide judgment, evaluation, and insights to solve the problem.

Unsupervised learning Machine learning algorithm trained to use information that is neither classified nor labeled in advance and to find patterns in that information without explicit human guidance.

User interface The part of the information system with which the end user interacts; in the context of IT infrastructure encompasses the type of hardware and the series of on-screen commands and responses required for a user to work with the system.

User-designer communications gap The difference in backgrounds, interests, and priorities that impede communications and problem solving between end-users and information systems specialists.

Utilitarian principle Ethical principle that assumes one can put values in rank order of utility and understand the consequences of various courses of action.

Value chain model Model that views the firm as a series or chain of basic activities that add a margin of value to a firm's products or services and identifies where information systems can best be applied to achieve a competitive advantage.

Value web Collection of independent firms that use information technology to coordinate their value chains to collectively produce a product or service for a market.

Virtual assistant Software agent that can perform a range of tasks or services for a user based on user input such as commands or questions.

Virtual company Organization using networks to link people, assets, and ideas, enabling it to ally with other companies to create and distribute products and services without being limited to traditional organizational boundaries or physical location.

Virtual private network (VPN) A secure, encrypted private network that has been configured within a public network, such as the Internet.

Virtual reality (VR) Technology that provides powerful capabilities for three-dimensional data visualization using interactive graphics software and hardware to create computer-generated simulations that provide sensations that emulate real-world activities.

Virtual worlds Online 3D environments populated by residents who have built graphical representations of themselves known as avatars.

Virtualization A method of presenting a set of computing resources so that they can be accessed in ways that are not restricted by physical configuration or geographic location.

Virus Rogue software program that attaches itself to other software programs or data files to be executed, usually without user knowledge or permission.

Visual search Searching for information using images instead of text-based queries.

Voice over IP (VoIP) Technology for managing the delivery of voice information using the Internet Protocol (IP).

Voice search Using natural language processing and speech recognition technology to process voice queries and provide search results.

War driving Technique in which eavesdroppers drive by buildings or park outside and try to intercept wireless network traffic.

Waterfall project management Traditional format for project management; uses a somewhat rigid and linear approach that typically advances in direct phases.

Web beacon Tiny image invisibly embedded in email messages and web pages that is designed to monitor the behavior of the user visiting a website or sending email; also called a web bug or tracking file.

Web browser An easy-to-use software tool with a graphical user interface for displaying web pages and accessing the web and other Internet resources.

Web hosting service Service provider that maintains websites for fee-paying clients.

Web mining Discovery and analysis of useful patterns and information from the web.

Web page Online "page" formatted using HTML and that may contain embedded links (hyperlinks) that connect documents to one another and also link pages to other objects, such as sound, video, or animation files.

Web scraping Used by AI developers and vendors to scrape or extract content from publicly available sources on the Internet in order to obtain training data for LLMs.

Web server Software that is responsible for locating and managing stored web pages and delivering them to a client in response to a request for service. Note that the term web server is also sometimes used to refer to the computer on which the web server software resides.

Web services Set of loosely coupled software components that exchange information with each other using universal web communication standards and languages. Used for linking systems of different organizations or for linking disparate systems within the same organization.

Web3 A more decentralized Internet based on the ideas behind blockchain that would be controlled by its users rather than dominated by big corporations.

Website A collection of web pages linked to a home page.

Wi-Fi Refers to the 802.11 set of standards for wireless LANs and wireless Internet access standards.

Wide area network (WAN) Telecommunications network that spans a large geographical distance–a region, state, continent, or the even the entire globe; Internet is the most universal and powerful example.

Wiki Collaborative website that makes it easy for users to add, delete, or modify content, without any knowledge of web page development or programming techniques.

WiMax Popular term for IEEE Standard 802.16 for wireless networking over a range of up to 31 miles with a data transfer rate of up to 75 Mbps. Stands for Worldwide Interoperability for Microwave Access.

Wireless sensor network (WSN) Network of interconnected wireless devices with built-in processing, storage, and radio frequency sensors and antennas that are embedded into the physical environment to provide measurements of many points over large spaces.

Wisdom of crowds The belief that large numbers of people can make better decisions under certain circumstances than a single person or even a small committee of experts.

World Wide Web A Internet service that uses universally accepted standards for storing, retrieving, formatting, and displaying information in a page format on the Internet; usually referred to as just the "web".

Worm Software program that independently copies itself from one computer to another computer over a network, and which can destroy data and programs as well as disrupt or even halt the operation of a computer network.

Zero trust Popular cybersecurity framework based on the principle of maintaining strict access controls and not trusting anyone by default.

Zero-day vulnerability Vulnerability in software, unknown to the creator, that hackers can exploit before the vendor becomes aware of the problem.

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